
1st International Conference on Business Intelligence, Computational Mathematics, and Data Analytics, IBCD 2024

Item Type Journal Article

Editor Gupta V.B.

Editor Shandilya S.K.

Editor Ortiz-Rodríguez F.

Editor Martinez-Rodriguez J.L.

Abstract The proceedings contain 16 papers. The special focus in this conference is on Business Intelligence, Computational Mathematics, and Data Analytics. The topics include: Facial Expression Tracking and Analysis in Video Through Motion with SqueezeNet and SGD; AI and Machine Learning: Revolutionizing the Fight Against Online Child Sexual Abuse Material (CSAM); modelling and Evaluation of Land Use Land Cover Changes Through Satellite Imagery Data for Ujjain District of Malwa Plateau Region Using Maximum Likelihood Classification and Support Vector Machine; computational Approaches to Calculate Stresses on Polycarbonate Specimen of Rectangular Shape and Compare with Trapezoidal Specimen with Experimental Validation; balancing Privacy and Utility in E-Healthcare Data Analysis: A Comprehensive Review; deep Learning Methods for Network Intrusion Detection Systems; investigating the Dissemination of Deepfake Conversations on Social Media via Tweeting; data Analytics for Fake Content Detection on Web; a Study of The Effect of Data-Driven, Personalized Digital Marketing Campaigns on the Online Sales of Mid-Size Food-Tech Aggregator; sentiment Analysis on Election Data Using Machine Learning and Ensemble Algorithms: An Ample Study; the Impact of Migration on Brand Consumption in India: A Qualitative Study; a Comparative Analysis of the Robustness to Noise of Machine Learning Classifiers; optimization Accuracy of Intrusion Detection System Based on Multilayered Neural Network; leveraging Long-Context Large Language Models for Multi-document Understanding and Summarization in Enterprise Applications.

Date 2025

URL <https://www.scopus.com/pages/publications/105005252869?origin=resultslist>

Archive Scopus

Language English

Volume 2413 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Communications in Computer and Information Science

ISSN 18650929 (ISSN); 978-303187510-6 (ISBN)

Journal Abbr Commun. Comput. Info. Sci.

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EID: 2-s2.0-105005252869

Export Date: 20 September 2026; Cited By: 0; Conference name: 1st International Conference on Business Intelligence, Computational Mathematics, and Data Analytics, IBCD 2024; Conference date: 18 August 2024 through 19 August 2024; Conference code: 331429

1st International Conference on Intelligent Multilingual Information Processing, IMLIP 2024

Item Type Journal Article

Editor Zhang H.

Editor Shang J.

Editor Su J.

Abstract The proceedings contain 30 papers. The special focus in this conference is on Intelligent Multilingual Information Processing. The topics include: Leveraging Pre-trained Language Models for Mongolian Intent Classification and Slot Filling; MIPS: Multilingual Inference Paradigm in Zero-Shot Cross-Lingual Stance Detection with Contrastive Learning; tibInkTrans: A Top-k Integrated Transformer for Enhancing Tibetan Document Images Affected by Ink Bleed; a Multi-dialect Tibetan-Chinese Speech Translation Dataset Based on Speech Synthesis; A Study of Central Nodes in the Interlanguage Syntactic Dependency Networks of Japanese CFL Learners; addressing Syntactic Divergence in Low-Resource Neural Machine Translation via Language Independent Word Reordering; character-Level Generative Network for Vietnamese Spelling Error Correction; cognitive Mechanism of Focus and Pauses in Mandarin Declaratives; dataset Construction for Learner-Based Chinese Answering Questions Scoring; domain-Oriented Chinese Named Entity Recognition Based on Enhanced Word-Word Relation Classification; ERT:Emotion in Real-Scene Transformer for Speech Emotion Recognition; EmpLLM: Enhancing Empathy in LLMs Through Psychologist Simulation; enhancing Chinese-Turkish Neural Machine Translation Through Bilingual Dictionary and Contrastive Learning; evaluating Large Language Models on Twitter Based on Hashtag Dynamics and Scaling Properties; face-Threatening Contexts on the Understanding of Sometimes; From Bytes to Words: Enhancing Multilingual NMT Through Explicit Byte Integration; Instruction Fine-Tuning Guidance: How PEFT Methods Impact the Generation of Language Model via Different Attributions; Japanese Author Attribution Using BERT Finetuning with Stylometric Features; leveraging Domain Expertise: The Impact of Media Experts in Developing Large Language Models; mining Positive Examples from Negatives for Enhanced Knowledge Graph Completion; research on Optimizing Speaker Diarization Tasks Based on the Semantic Capabilities of Large Language Models; MM-Eval: A Hierarchical Benchmark for Modern Mongolian Evaluation in LLMs; research on Small Object Segmentation of Tibetan Newspapers Using DeepLabV3+ at High Resolution; the Formation and Application of the Contradictory Meanings of Several Words; Ti-MSUM: Tibetan Image and Text Summarization Dataset.

Date 2025

URL <https://www.scopus.com/pages/publications/105008377497?origin=resultslist>

Archive Scopus

Language English

Volume 2395 CCIS

Publisher Springer Science and Business Media Deutschland GmbH
Publication Communications in Computer and Information Science
ISSN 18650929 (ISSN); 978-981965122-1 (ISBN)
Journal Abbr Commun. Comput. Info. Sci.
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Notes:

EID: 2-s2.0-105008377497

Export Date: 20 September 2026; Cited By: 0; Conference name: 1st International Conference on Intelligent Multilingual Information Processing, IMLIP 2024; Conference date: 16 November 2024 through 17 November 2024; Conference code: 332419

1st International Conference on Robust Argumentation Machines, RATIO 2024

Item Type Journal Article

Editor Cimiano P.

Editor Frank A.

Editor Kohlhasse M.

Editor Stein B.

Abstract The proceedings contain 21 papers. The special focus in this conference is on Robust Argumentation Machines. The topics include: Argument Mining of Attack and Support Patterns in Dialogical Conversations with Sequential Pattern Mining; cluster-Specific Rule Mining for Argumentation-Based Classification; automatic Analysis of Political Debates and Manifestos: Successes and Challenges; PAKT: Perspectivized Argumentation Knowledge Graph and Tool for Deliberation Analysis ; polArg: Unsupervised Polarity Prediction of Arguments in Real-Time Online Conversations; are Large Language Models Reliable Argument Quality Annotators?; the Impact of Argument Arrangement on Essay Scoring; Finding Argument Fragments on Social Media with Corpus Queries and LLMs; enhancing Abstract Argumentation Solvers with Machine Learning-Guided Heuristics: A Feasibility Study; ranking Transition-Based Medical Recommendations Using Assumption-Based Argumentation; argumentation-Based Probabilistic Causal Reasoning; from Networks to Narratives: Bayes Nets and the Problems of Argumentation; enhancing Argument Generation Using Bayesian Networks; “Do Not Disturb My Circles!” Identifying the Type of Counterfactual at Hand (Short Paper); BEA: Building Engaging Argumentation; deciphering Personal Argument Styles – A Comprehensive Approach to Analyzing Linguistic Properties of Argument Preferences; extending the Comparative Argumentative Machine: Multilingualism and Stance Detection; objective Argument Summarization in Search; argServices: A Microservice-Based Architecture for Argumentation Machines.

Date 2024

URL <https://www.scopus.com/pages/publications/85200679567?origin=resultslist>

Archive Scopus

Language English

Volume 14638 LNAI

Publisher Springer Science and Business Media Deutschland GmbH
Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)
ISSN 03029743 (ISSN); 978-303163535-9 (ISBN)
Journal Abbr Lect. Notes Comput. Sci.
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EID: 2-s2.0-85200679567

Export Date: 20 September 2026; Cited By: 0; Conference name: 1st International Conference on Robust Argumentation Machines, RATIO 2024; Conference date: 5 June 2024 through 7 June 2024; Conference code: 315709

2nd International Conference on Artificial Intelligence and its Practical Applications in the Age of Digital Transformation, I2COMSAPP 2025

Item Type Journal Article

Editor Elhadj Y.M.

Editor Koubaa A.

Editor Meziane F.

Editor Deriche M.

Editor Ould-Elhassen Aoueileyine M.

Abstract The proceedings contain 47 papers. The special focus in this conference is on rtificial Intelligence and its Practical Applications in the Age of Digital Transformation. The topics include: A Systematic Review of Enhanced Deep Convolutional Neural Network Techniques for Skin Cancer Classification; evolving Temporal Logic Formulas with Dynamic Grammar-Based Learning; A Meta-learning Approach for Algorithm Selection in the Multi-Skill Resource-Constrained Project Scheduling Problem (MSRCPSP); enhanced Ant Colony Optimization Variants for Efficient Task Scheduling in Cloud Environments; a Hybrid Search Strategy Algorithm for Mining Frequent Itemsets; machine Learning Models for Predicting Bus Arrival Times at Stops; An Optimized LSTM for Moroccan Darija Sentiment Analysis on Social Media; To Publish or Not to Publish AI-Generated Research Articles in Scholarly Journals: A Perspective from Editors and Publishers; image-to-Sequence Models for the Recognition of Text Lines of Maghrebin Handwritten Scripts: New Dataset and New Results; a Low-Resource English–Hassaniya Parallel Corpus with Neural Machine Translation Benchmarks; Trustworthy Arabic LLMs: A Multidimensional Survey of Risks, Realities, and Resilience; Transformer-Based Quran Insight Representation Framework for ICBIR; Enhancing Legal Text Processing with AI: Inconsistency Detection, Translation, and Access; Authentiscore: A Comparative Study of Fine-Tuned Transformers for AI Text Detection; Explainable Text-to-SQL Generation: An Overview of Recent Trends and Developments; a Rule-Based Approach for Arabic Poem Meter Classification Using Pairwise Alignment Matching; mujeeb: A Hybrid Arabic Chatbot to Enhance Customer Support in Financial Administration Services; A Review of Textual Data Augmentation

Approaches for Arabic NLP; topic Modeling of User-Generated Arabic on Social Media: A Systematic Literature Review; fine-Tuning Transformer Model with a High-Quality French-Arabic Parallel Corpus of Medical Prescriptions for Domain-Specific Machine Translation.

Date 2026

URL <https://www.scopus.com/pages/publications/105044033898?origin=resultslist>

Archive Scopus

Language English

Volume 1916 LNNS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Networks and Systems

ISSN 23673370 (ISSN); 978-303222031-8 (ISBN)

Journal Abbr Lect. Notes Networks Syst.

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EID: 2-s2.0-105044033898

Export Date: 20 September 2026; Cited By: 0; Conference name: 2nd International Conference on Artificial Intelligence and its Practical Applications in the Age of Digital Transformation, I2COMSAPP 2025; Conference date: 20 October 2025 through 22 October 2025; Conference code: 355369

2nd International Conference on Computing and Machine Learning, CML 2025

Item Type Journal Article

Editor Bansal J.C.

Editor Borah S.

Editor Hussain S.

Editor Salhi S.

Abstract The proceedings contain 46 papers. The special focus in this conference is on Computing and Machine Learning. The topics include: AI-Supported Predictive System for Global Healthcare Services; advancements in Aspect-Based Sentiment Analysis: A Review of Models, Techniques, and Challenges; a Systematic Development of Stopwords for Manipuri Language Processing; Comparative Analysis of Deep Learning Techniques for Isolating Interictal Activity from EEG Artifacts; exploring Research Trends and Publications in Gift-Giving Behavior: A Bibliometric Analysis Using Data Mining Techniques; enhancing Security and Privacy in 6G V2X Communication: Leveraging Homomorphic-Encryption for Confidentiality-Preserving Computation; detecting Cyberbullying in the Sentiments Over Social Media; A Parallel Approach to Partial Periodic Pattern Mining on Temporal Datasets Using MPI; improving Untreated Pneumothorax Classification in Chest X-Ray Analysis Using Auxiliary Chest Tube Detection; enhancing Digital Integrity: Clickbait Detection Using Machine Learning; Evaluating CNN-Based Approaches for Diabetic Retinopathy Detection and Classification; Hybrid Integration of Machine Learning and Deep Learning Approaches for Accurate

Anomaly Detection with the BETH Data; application of Deep Learning in Healthcare: A Review; video-Based System for Detecting Falls in Elderly Individuals; enhanced Melanoma Recognition: Leveraging Deep Learning for Accurate Classification and Early Diagnosis; attention-Based Deep Learning Approach for Sentiment Analysis in the Low-Resource Assamese Language; Investigating the Prevalence of Depression in Academia Amid Covid-19 Using Multimodal Signals and Bi-LSTM Architecture; Deep Learning-Based Tumor Detection in Head and Neck Cancer Using PET Scans; Classification of Lung Cancer on CT Scans Using MobileNetV3, MobileNetV4, and EfficientNetV2; Enhancing Dialogue Systems with Attention Mechanisms: Conversational AI Using Seq2Seq; bladder Cancer Prediction on Urinary Biomarkers Using Ensemble Learning.

Date 2026

URL <https://www.scopus.com/pages/publications/105029702069?origin=resultslist>

Archive Scopus

Language English

Volume 1613 LNNS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Networks and Systems

ISSN 23673370 (ISSN); 978-981952874-5 (ISBN)

Journal Abbr Lect. Notes Networks Syst.

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EID: 2-s2.0-105029702069

Export Date: 20 September 2026; Cited By: 0; Conference name: 2nd International Conference on Computing and Machine Learning, CML 2025; Conference date: 22 March 2025 through 23 March 2025; Conference code: 346179

2nd International Conference on Smart Medical, IoT and Artificial Intelligence, ICSMAI 2025

Item Type Journal Article

Editor Ghoumid K.

Editor Elmiad A.K.

Editor Lazaar M.

Editor Haddouch K.

Editor Elmazria O.

Abstract The proceedings contain 24 papers. The special focus in this conference is on Smart Medical, IoT and Artificial Intelligence. The topics include: AI-Powered Energy Management Diagnostics: Expert System Integration with LLM for Predictive Maintenance; SVHRSP From Scorpion Venom for Skin Ageing: Bioinformatics Rationale and a Microneedle-Delivered Cosmetic Strategy; AI in Aesthetic Medicine: From Subjective Judgment to Algorithmic Precision; NLP Based Exploratory Study of Interviews on Blood Donation Awareness Campaigns: Evidence From the

University of Oujda (2023–2024); modeling the Impact of Patient Experience Attributes on Satisfaction: Contributions of Generalized Additive Models; few-Shot Learning Hybrid Approach for Visual Quality Inspection with Limited Data; sentiment Analysis in Social Media with Artificial Intelligence; A Dual-Band Reconfigurable Antenna Using a PIN Diode for 2.4 GHz and 5 GHz IoT Applications; Unified AI Energy Management Diagnostics: Expert System Integration with LLM for Predictive Maintenance; smartRoute+: Toward Intelligent Road Maintenance Ecosystems Through Deep Learning and Mobile Crowdsensing in Morocco; BAI-PLF: A Blockchain-AI-IoT Framework for Energy-Efficient and Predictive Logistics and Healthcare in Africa; Automated Role Classification in Security Policies: A Comparative Study of NLP Approaches; Adaptive AI Pipeline for Real-Time Cybersecurity Alert Classification: A Multi-dataset Approach; Towards a Paperless Environment: AI for Smart Invoice Processing; AI and Big Data in Marketing Decision-Making: A Systematic Review; similarity Redefined: A Robust Metric for Predicting Links in Complex Networks; travel Time Optimization in the Old Medina of Fez; An Empirical Comparison of Topic Models: LDA, NMF, and BERTopic Applied to Hotel Customer Reviews; an Extensive Categorization of Triplestores; artificial Intelligence in Hospital Management: Balancing Strategic Imitation and Contextual Innovation in Moroccan Private Hospitals.

Date 2026

URL <https://www.scopus.com/pages/publications/105046060096?origin=resultslist>

Archive Scopus

Language English

Volume 1845 LNNS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Networks and Systems

ISSN 23673370 (ISSN); 978-303218896-0 (ISBN)

Journal Abbr Lect. Notes Networks Syst.

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EID: 2-s2.0-105046060096

Export Date: 20 September 2026; Cited By: 0; Conference name: 2nd International Conference on Smart Medical, IoT and Artificial Intelligence, ICSMAI 2025; Conference date: 23 October 2025 through 25 October 2025; Conference code: 353619

2nd International Conference on Speech and Language Technologies for Low-Resource Languages, SPELLL 2023

Item Type Journal Article

Editor Chakravarthi B.R.

Editor B B.

Editor García Cumbreras M.A.

Editor Jiménez Zafra S.M.

Editor Subramanian M.

Editor Shanmugavadeivel K.

Editor Nakov P.

Abstract The proceedings contain 33 papers. The special focus in this conference is on Speech and Language Technologies for Low-resource Languages. The topics include: SamPar: A Marathi Hate Speech Dataset for Homophobia, Transphobia; l3Cube-MahaNews: News-Based Short Text and Long Document Classification Datasets in Marathi; creation and Classification of Kannada Meme Dataset: Exploring Domain and Troll Categories; the Impact of Tamil Python Programming in Wikisource; natural Language Processing for Tulu: Challenges, Review and Future Scope; depBoost-TransNet: Boosted Transformer Network for Depression Classification; Optimized BERT Model for Question Answering System on Mobile Platform; A Comparative Analysis of Pretrained Models for Sentiment Analysis on Restaurant Customer Reviews (CAPM-SARCR); lightweight Language Agnostic Data Sanitization Pipeline for Dealing with Homoglyphs in Code-Mixed Languages; textGram: Towards a Better Domain-Adaptive Pretraining; abusive Social Media Comments Detection for Tamil and Telugu; sales Forecasting from Group Conversation Using Natural Language Processing; hands in Harmony: Empowering Communication Through Translation; offensive Text Detection for Tamil Language; Telugu-English Abusive Comment Detection Using XLMRoBERTa and mBERT; a Knowledge Engineering Framework Addressing High Incidence of Farmer Suicides; event Categorization from News Articles Using Machine Learning Techniques; from Words to Emotions: Identifying Depression Through Social Media Insights; text Summarisation for Low-Resourced Languages, A Review; multi Speaker Activity Detection Using Spectral Centroids; spoken Language Identification System Using Convolutional Neural Networks; exploring the Role of Entropy in Music Classification; spectral Features Based Spoken Dialect Identification for Punjabi Language; hate Speech Detection Using Audio in Portuguese Language; an Empirical Analysis of the Consonantal Phonemic Patterns and Characteristics of English Spoken in India; Multilingual Fake News Detection in Low-Resource Languages: A Comparative Study Using BERT and GPT-3.5; optimized Latent-Dirichlet-Allocation Based Topic Modeling–An Empirical Study.

Date 2024

URL <https://www.scopus.com/pages/publications/85192390623?origin=resultslist>

Archive Scopus

Language English

Volume 2046 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Communications in Computer and Information Science

ISSN 18650929 (ISSN); 978-303158494-7 (ISBN)

Journal Abbr Commun. Comput. Info. Sci.

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Notes:

EID: 2-s2.0-85192390623

Export Date: 20 September 2026; Cited By: 0; Conference name: 2nd International Conference on Speech and Language Technologies for Low-Resource Languages, SPELLL 2023; Conference date: 6 December 2023 through 8 December 2023; Conference code: 311549

3rd Annual Meeting of the ELRA-ISCA Special Interest Group on Under-Resourced Languages, SIGUL 2024 at LREC-COLING 2024 - Workshop Proceedings

Item Type Journal Article

Editor Melero M.

Editor Sakti S.

Editor Soria C.

Abstract The proceedings contain 50 papers. The topics discussed include: a bit of a problem: measurement disparities in dataset sizes across languages; a novel corpus for automated sexism identification on social media; assessing pre-built speaker recognition models for endangered language data; beyond error categories: a contextual approach of evaluating emerging spell and grammar checkers; bidirectional English-Nepali machine translation(MT) system for legal domain; BK3AT: Bangsamoro K-3 children's speech corpus for developing assessment tools in the Bangsamoro languages; developing infrastructure for low-resource language corpus building; evaluating Icelandic sentiment analysis models trained on translated data; exploring text classification for enhancing digital game-based language learning for Irish; and forget NLI, use a dictionary: zero-shot topic classification for low-resource languages with application to Luxembourgish.

Date 2024

URL <https://www.scopus.com/pages/publications/85195294635?origin=resultslist>

Archive Scopus

Language English

Publisher European Language Resources Association (ELRA)

Publication 3rd Annual Meeting of the ELRA-ISCA Special Interest Group on Under-Resourced Languages, SIGUL 2024 at LREC-COLING 2024 - Workshop Proceedings

ISSN 978-249381429-6 (ISBN)

Journal Abbr Annu. Meet. ELRA-ISCA Spec. Interest Group Under-Resour. Lang., SIGUL LREC-COLING - Workshop Proc.

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EID: 2-s2.0-85195294635

Export Date: 20 September 2026; Cited By: 0; Conference name: 3rd Annual Meeting of the ELRA-ISCA Special Interest Group on Under-Resourced Languages, SIGUL 2024; Conference date: 21 May 2024 through 22 May 2024; Conference code: 199845

3rd International Conference on Language Processing and Knowledge Management, LPKM 2024

Item Type Journal Article

Editor Belguith L.H.

Editor Shaalan K.

Abstract The proceedings contain 29 papers. The special focus in this conference is on Language Processing and Knowledge Management. The topics include: Machine Learning Serving the School Orientation Process; improving Arabic Fake News Detection Across Context-Aware Attention Deep Model Based on Natural Language Processing; towards a Maude-Based Approach for Formal Modeling Deep Neural Networks; Anti-pattern Based IoRT-Aware Business Process Structure Verification Approach; opinion Analysis Based on a Sentiment Lexical Ontology and Deep Learning Models: Tunisian Dialect Case; disfluent-to-Fluent Tunisian Dialect Speech Translation with Fine-Tuning Pre-trained Language Models; BERT-Based Model for Sarcasm Detection in Arabic Texts; typology of Event Data Imperfections; towards Sentiment Analysis for Libyan Dialect; text Categorization Can Enhance Domain-Agnostic Stopword Extraction; normalized Orthography for Tunisian Arabic; deep Learning Approach for Early Prediction of Depression on Social Network; adapting Large Language Models to Biomedical Domain: A Survey of Techniques and Approaches; tunisian Arabic Understanding: Resources Analysis and Evaluation; from Data to Decisions: An Ontology-Driven Method for Opinion Mining; LLMs for Cyberbullying Detection in Political Social Media; Assessing BERT Models for Arabic Named Entity Recognition in a Multi-dialectal Context; tunisian Normalized Pronunciation; traffScOnto: Ontology for Traffic Management in the Context of Smart City Domain; SERTUS Dataset Collection from Spontaneous Environments; an Agricultural Sentiment Dataset for Pest Control and Crop Diseases; deep Learning Approach to Identify and Classify Arabic Verbal Multi-word Expressions; a Design Pattern-Based Approach for Analyzing MapReduce Applications; analyzing the Impact of Big Data in Mental Health; The Impact of AI on Knowledge Management; a Rule-Based System for Translating Libyan Dialect Dual Forms to Modern Standard Arabic.

Date 2025

URL <https://www.scopus.com/pages/publications/105002011373?origin=resultslist>

Archive Scopus

Language English

Volume 1303 LNNS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Networks and Systems

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Journal Abbr Lect. Notes Networks Syst.

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EID: 2-s2.0-105002011373

Export Date: 20 September 2026; Cited By: 0; Conference name: 3rd International Conference on Language Processing and Knowledge Management, LPKM 2024; Conference date: 3 June 2024 through 5 June 2024; Conference code: 329189

3rd International Conference on Speech and Language Technologies for Low-Resource Languages, SPELLL 2024

Item Type Journal Article

Editor Chakravarthi B.R.

Editor B B.

Editor Rajiakodi S.

Editor García Cumbreras M.A.

Editor Jiménez Zafra S.M.

Editor Kovács G.

Editor Eger S.

Editor Wahyu Pamungkas E.

Editor Dobrovoljc K.

Abstract The proceedings contain 44 papers. The special focus in this conference is on Speech and Language Technologies for Low-Resource Languages. The topics include: Towards End-to-End Speech Synthesis for Tigrinya Language; An Effective LSTM – Autoencoder Approach with Acoustic Features in Indian Classical Music Raga Recognition; automatic Assessment of Pathological Voice Quality Using Machine Learning Approaches; audio to Text Conversion Using Deep Learning; comparative Analysis of Speaker Diarization Results on Multi-lingual Multi Speaker Data and Single Speaker Data; enhancing End-to-End Malayalam Automatic Speech Recognition with Language Model Augmentation; developing Bodo Speech Recognition with Kaldi; a Deep Learning Based Approach to Detect Paraphrase in Healthcare; Enhancing Machine Translation Performance: A Comparative Study of Finetuning the MBart-mmt Model; translation of Sanskrit (Shlokas)-Malayalam Using Deep Learning Techniques; A Novel Multilingual Human-AI Collaborative Writing Design for Indian Languages; tamilFacts: A Comprehensive Multimodal Dataset of Fact-Checked Social Media Content in Tamil Language; a Comparative Study of Sentiment Analysis Using Transformer Models and Text Augmentation Techniques for Bangla Text; towards Automated Sanskrit Writing Correction: Evaluation on Large Language Models; Optimizing Tamil News Headline Generation with LoRA Techniques: Insights and Challenges; corpus Creation for Racial Hoax in Code-Mixed Hindi-English Low Resource Text; Machine Learning Approaches for Tamil POS Tagging and Dependency Parsing; Empowering Consumers Through Advanced Technology: A Streamlit Web App Enabled Solution Leveraging OCR and BERT for Understanding Packaged Food Ingredients; captioning-Based Zero-Shot Visual Question Answering System; Enhancing Facial Emotion Recognition Ecosystem Through Information System (IS) Guided Fine-Tuning Vision Transformer Models; Optimizing OCR Model Performance : A Comparative Study of Backbone Architectures and Hyperparameter Tuning; dravLangGuard: A Multimodal Approach for Hate Speech Detection in Dravidian Social Media.

Date 2026

URL <https://www.scopus.com/pages/publications/105021804402?origin=resultslist>
Archive Scopus
Language English
Volume 2656 CCIS
Publisher Springer Science and Business Media Deutschland GmbH
Publication Communications in Computer and Information Science
ISSN 18650929 (ISSN); 978-303205854-6 (ISBN)
Journal Abbr Commun. Comput. Info. Sci.
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Notes:

EID: 2-s2.0-105021804402

Export Date: 20 September 2026; Cited By: 0; Conference name: 3rd International Conference on Speech and Language Technologies for Low-Resource Languages, SPELLL 2024; Conference date: 4 December 2024 through 6 December 2024; Conference code: 342699

5th International Conference on Intelligent Systems and Pattern Recognition, ISPR 2025

Item Type Journal Article

Editor Ensari T.

Editor Bennour A.

Editor Bouaziz B.

Editor Mahdi W.

Editor Rida I.

Abstract The proceedings contain 59 papers. The special focus in this conference is on Intelligent Systems and Pattern Recognition. The topics include: A U-Net GNN Hybrid Approach for MRI-Based Brain Tumor Segmentation; HeteroGAT: A Heterogeneous Graph Attention Network for Multimodal Fake News Detection in Chinese Social Media; an Event-Based Data Generation Framework for Autonomous Driving Scenarios; improving Low-Resource Dialect Identification through Audio Data Augmentation: A Case Study on Algerian Language; From Farm to Warehouse: Toward a Semantic and AI-Based Architecture for Wheat Lifecycle; spatio-temporal Transformers for High-Accuracy Detection of Embryo Developmental Transitions; prompting Recovery: Filling Missing Diabetes Data with Large Language Models; regularized Continual Learning for Generative Crowd Counting; real-Time Defect Detection: A Lightweight Deep Learning Framework for Industrial Applications; Robust 1D CNN-Based Approach with Data Augmentation for Incipient and Severe Stator Fault Classification in Induction Motors; A Comparative Study of Arabic Embedding Models in RAG-Based Fatwa Retrieval; Sentiment Classification of COVID-19 Tweets: From Machine Learning and Deep Learning to BERT; maintenance-Oriented Test Generators and Test Teams Simulation for Demand-Controlled Ventilation and Heating Systems; CASCADE: Engineering a Secure, Real-Time, and Collaborative Code Execution for Multi-tenant Learning Platforms; a Review of E-Document Analysis with Multi-biometric

Technique for Secure Verification; NeuroYOLO: Data-Efficient Fine-Tuning of YOLOv11 for Vision-Based Rehabilitation Monitoring; machine Learning-Based Autism Detection Using Oriented Basic Image Features; preface; spine 3-D Representation from 2-D X-Ray Images and Measurement for Scoliosis Patient Follow-up; stacked Deep Learning Models Leveraging Denoising Autoencoder-Based Features for Student Performance Prediction; A Deep Attention-BLSTM Pipeline for Human Action Recognition; balancing Accuracy and Efficiency in Deep Violence Detection Models for Surveillance Applications; comparative Analysis of Deep Learning and Classical Machine Learning for Deepfake Image Detection.

Date 2026

URL <https://www.scopus.com/pages/publications/105039946308?origin=resultslist>

Archive Scopus

Language English

Volume 2859 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Communications in Computer and Information Science

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Journal Abbr Commun. Comput. Info. Sci.

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Notes:

EID: 2-s2.0-105039946308

Export Date: 20 September 2026; Cited By: 0; Conference name: 5th International Conference on Intelligent Systems and Pattern Recognition, ISPR 2025; Conference date: 25 September 2025 through 27 September 2025; Conference code: 352269

8th Chinese Conference on Pattern Recognition and Computer Vision, PRCV 2025

Item Type Journal Article

Editor Kittler J.

Editor Xiong H.

Editor Yang J.

Editor Chen X.

Editor Lu J.

Editor Lin W.

Editor Yu J.

Editor Zheng W.

Abstract The proceedings contain 580 papers. The special focus in this conference is on Pattern Recognition and Computer Vision,. The topics include: TMFit: Enhancing Fashion Image Editing Precision via Text-Driven Mask Generation; dreamDancer: Music-Driven Dance Video Intelligent Generation; VSE-MQA: A Semantic Grouping and Multi-modal Approach for Accurate Video Quality Assessment;

enhancing Compositional Reasoning in Multimodal Large Language Models; latentEdit: Adaptive Latent Control for Consistent Semantic Editing; clinCoCoOp: An Interpretable Prompt Learning Framework with Clinical Concept Guidance for Context Optimization; all-in-Focus Seeing Through Occlusions with Event and Frame; capturing Cross-Modal Semantics by Generating Comments for Image-Text Contents; false Negatives Do Matter: A Novel Soft Label and Reranking Based Plug-in Method for Image-Text Retrieval; uncertainty-Aware Deep Anchor Graph Learning for Multimodal Remote Sensing Image Clustering; CFANet: Cross-Frequency Adaptive Fusion with Frequency-Modulated Attention for Multi-focus Image Fusion; adaptive Multimodal Fusion for Graph Learning in Brain Disease Prediction; SentiMM: A Multimodal Multi-agent Framework for Sentiment Analysis in Social Media; Bias-Unlearning in MABSA: A Causal Framework with Cross-Modal Counterfactual Inference; mapping Semantic, Unmasking Falsehoods: Topic-Driven Hierarchical Graph Network for Short Video Fake News Detection; revisiting Symmetric Teacher-Student Network Distillation for Anomaly Detection; user-Adjustable Image Cropping Based on Visual Semantic Awareness; multi-source Domain Adaptation Image Steganalysis for Cover Source Mismatch; PhoneGuide-SLAM: Low-Cost Smartphone Navigation for the Visually Impaired Using Visual Semantic SLAM; fairMoE: Decoupled Expert Learning for Unbiased Customized Face Generation; strategyAdapter: One-Shot Learning for Unseen-Domain Procedural Sequence Generation; divide, Weight, and Route: Difficulty-Aware Optimization with Dynamic Expert Fusion for Long-Tailed Recognition; c3F: A Coarse-to-Fine Feature Fusion Approach for Scene Script Identification; secure and Scalable Face Retrieval via Cancelable Product Quantization; few-Shot Font Generation via Attribute-Guided Diffusion with Style Contrastive Learning; rethinking the Evaluation of Scene Graph Generation; probabilistic Visual Prompt Tuning.

Date 2026

URL <https://www.scopus.com/pages/publications/105028409403?origin=resultslist>

Archive Scopus

Language English

Volume 16277 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-981955678-6 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

Date Added 20/09/2026, 8:35:15 pm

Modified 20/09/2026, 8:35:15 pm

Notes:

EID: 2-s2.0-105028409403

Export Date: 20 September 2026; Cited By: 0; Conference name: 8th Chinese Conference on Pattern Recognition and Computer Vision, PRCV 2025; Conference date: 15 October 2025 through 18 October 2025; Conference code: 345989

9th International Visual Informatics Conference, IVIC 2025

Item Type Journal Article

Editor Badioze Zaman H.

Editor Robinson P.

Editor Smeaton A.F.

Editor Shih T.K.

Editor Jørgensen B.N.

Editor Huajing Z.

Editor Xiaoping L.

Editor Mohamad Ali N.

Editor Mat Surin E.S.

Abstract The proceedings contain 36 papers. The special focus in this conference is on Visual Informatics. The topics include: Design and Development of Multi-party Quantum Key Distribution Simulation Framework Based on Optical Frequency Division Multiplexing in the Fiber Optics Using OptiSystem™; Data Quality Assessment Framework for Predictive Maintenance (PdM) in Industrial AI: Review; exploring the Impact of Surface Temperature on Soil Moisture and Thermal Dynamics; a Review of Phatic Communication in Enhancing Social Engagement Among Older Adults; design Elements in Brain-Computer Interface for Working Memory: A Decade Review; effects of Alarm Sound Type on Sleep Inertia: Exploring Links with Individual Differences; visualization of Physics Concepts and Investigation of Educational Effects Using Virtual Reality Teaching Materials with Object Tracking; Assessing the Effects of Virtual Teaching Tools on ESL Primary Students' Vocabulary Learning; social Media Analytics for Understanding Public Perception of Green Energy in Malaysia: Trends, Challenges, and Insights; RetinaNet vs. SSDMobileNetV2: A Comparative Study on Smartphone-Based Classification of Indonesian Rice Varieties; Scam Surveillance Model Integrating ELA-Based ResNeXt and BERT-Linguistic Fusion for Misinformation Detection; LoRA-Based Efficient Fine-Tuning of Transformer Models for Short Text Classification; Enhancing Legal Text Classification Performance by Using DistilBERT; a Review and Multi-Task Learning Approach for Emotion Identification in Cyber Sexual Harassment Using Large Language Models; Comparison of Clustering Method for LiDAR Loop-Closure Detection; EDAIN-Enhanced Machine Learning for Robust Carbon Price Forecasting in Volatile Market; visual Analysis and Scientometric Mapping of Deep Learning Techniques for Grapevine Disease Classification; deep Learning-Based Emotion Recognition in Children with Real-Time Web Application Deployment.

Date 2026

URL <https://www.scopus.com/pages/publications/105021220645?origin=resultslist>

Archive Scopus

Language English

Volume 16215 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-981953807-2 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

Date Added 20/09/2026, 8:35:05 pm

Modified 20/09/2026, 8:35:05 pm

Notes:

EID: 2-s2.0-105021220645

Export Date: 20 September 2026; Cited By: 0; Conference name: 9th International Visual Informatics Conference, IVIC 2025; Conference date: 12 November 2025 through 13 November 2025; Conference code: 342189

10th International Conference on ICT for Sustainable Development, ICT4SD 2025

Item Type Journal Article

Editor Fong S.

Editor Dey N.

Editor Joshi A.

Abstract The proceedings contain 51 papers. The special focus in this conference is on ICT for Sustainable Development. The topics include: Tracking of Material Management on Site by Using Analogy Based System; Lung Cancer Detection Using CNN; employee Promotion Prediction Model Using Machine Learning; dynamic Stock Price Forecasting Using Machine Learning and Real-Time Data Integration; AI-Enhanced Fashion Assistant Featuring Image Recommendation - Glambot; process of Imprecise Data Using New Methods of Neutrosophic Set; integrating Social Media's Role in Elections with the Theory of Planned Behavior: A Comprehensive Examination of Voter Intention; understanding Unsupervised Learning Using Hierarchical Clustering; smart Crop Health Analytics System; customized Convolutional Neural Network for Accurate Human Motion Forecasting; CFD-Based Blood Flow Simulation in Microfluidic Device; secure Authentication Using Biometric and Behavioral Analysis; solar Powered Multipurpose Agricultural Robot; SKYASSIST: Conversational AI-Driven Revenue Management System for Airlines; emerging Trends and Innovations in Sentiment Analysis: A Comprehensive Review; Zero-Shot Classification with NLI DeBERTa V3: Evaluating MultiNLI for NLP Tasks; AI and Technology Adoption as Drivers of Firm Performance: BERTopic Modeling and Text Mining Approach; joint Feature Learning and Hashing for Multi-modal Data Via Cosine Normalization; web Application Firewall Using Machine Learning and Features Engineering; Fraudlens: Deepfake Intelligence with ML; An Artificially Intelligent System to Strategize and Predict Employee Attrition and Retention in HR Management; advancing Crop Cultivation Estimation with Aerial Imaging and Artificial Intelligence: A Comprehensive Review; UrbanFix AI: Smart Reporting System for Road Safety and Management; Emoji Prediction for Sentiment Analysis: A Comparative Study of LSTM and BERT Models; AI Driven-Virtual Mouse Using Hand Gestures.

Date 2026

URL <https://www.scopus.com/pages/publications/105023200228?origin=resultslist>

Archive Scopus

Language English

Volume 1654 LNNS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Networks and Systems

ISSN 23673370 (ISSN); 978-303206696-1 (ISBN)

Journal Abbr Lect. Notes Networks Syst.

Date Added 20/09/2026, 8:35:14 pm**Modified** 20/09/2026, 8:35:14 pm**Notes:**

EID: 2-s2.0-105023200228

Export Date: 20 September 2026; Cited By: 0; Conference name: 10th International Conference on ICT for Sustainable Development, ICT4SD 2025; Conference date: 17 July 2025 through 19 July 2025; Conference code: 341139

14th International Conference on Pattern Recognition Applications and Methods, ICPRAM 2025

Item Type Journal Article**Editor** Castrillon-Santana M.**Editor** De Marsico M.**Editor** Fred A.

Abstract The proceedings contain 89 papers. The special focus in this conference is on Pattern Recognition Applications and Methods. The topics include: Assessment of Training Progression on a Surgical Simulator Using Machine Learning and Explainable Artificial Intelligence Techniques; an Easy-to-Use System for Tracking Robotic Platforms Using Time-of-Flight Sensors in Lab Environments; rethinking Model Selection Beyond ImageNet Accuracy for Waste Classification; computation of 2D Discrete Geometric Moments on Quadrees; hyperspectral Image Compression Using Implicit Neural Representation and Meta-Learned Based Network; multi-View Skeleton Analysis for Human Action Segmentation Tasks; stance Detection in Twitter Conversations Using Reply Support Classification; LAST: Utilizing Synthetic Image Style Transfer to Tackle Domain Shift in Aerial Image Segmentation; A Comparative Analysis of Hyperparameter Effects on CNN Architectures for Facial Emotion Recognition; non Contact Stress Assessment Based on Deep Tabular Method; action Recognition in Law Enforcement: A Novel Dataset from Body Worn Cameras; anomalous Event Detection in Traffic Audio; silhouette Segmentation for Near-Fall Detection Through Analysis of Human Movements in Surveillance Videos; single Hyperspectral Image Super-Resolution Utilizing Implicit Neural Representations; game State and Spatio-Temporal Action Detection in Soccer Using Graph Neural Networks and 3D Convolutional Networks; distribution Controlled Clustering of Time Series Segments by Reduced Embeddings; partition Tree Ensembles for Improving Multi-Class Classification; FFAD: Fixed-Position Few-Shot Anomaly Detection for Wire Harness Utilizing Vision-Language Models; DNN Layers Features Reduction for Out-of-Distribution Detection; occlusion Detection for Face Image Quality Assessment; lifelong Learning Needs Sleep: Few-Shot Incremental Learning Enhanced by Sleep; Improving Floating Wind Turbine Stability with Evolutionary Computation for TMD Optimization.

Date 2025**URL** <https://www.scopus.com/pages/publications/105002379932?origin=resultslist>**Archive** Scopus**Language** English**Volume** 1

Publisher Science and Technology Publications, Lda
Publication International Conference on Pattern Recognition Applications and Methods
ISSN 21844313 (ISSN); 978-989758730-6 (ISBN)
Journal Abbr Int. Conf. Pattern. Recognit. Appl. Method.
Date Added 20/09/2026, 8:35:38 pm
Modified 20/09/2026, 8:35:38 pm

Notes:

EID: 2-s2.0-105002379932

Export Date: 20 September 2026; Cited By: 0; Conference name: 14th International Conference on Pattern Recognition Applications and Methods, ICPRAM 2025; Conference date: 23 February 2025 through 25 February 2025; Conference code: 328979

15th International Conference of the CLEF Association, CLEF 2024

Item Type Journal Article

Editor Goeuriot L.

Editor Mulhem P.

Editor Quénot G.

Editor Schwab D.

Editor Di Nunzio G.M.

Editor Faggioli G.

Editor Ferro N.

Editor Soulier L.

Editor Galuščáková P.

Editor García Seco de Herrera A.

Abstract The proceedings contain 32 papers. The special focus in this conference is on CLEF Association. The topics include: Assessing Document Sanitization for Controlled Information Release and Retrieval in Data Marketplaces; the Impact of Web Search Result Quality on Decision-Making; improving Laypeople Familiarity with Medical Terms by Informal Medical Entity Linking; mapping the Media Landscape: Predicting Factual Reporting and Political Bias Through Web Interactions; under-Sampling Strategies for Better Transformer-Based Classifications Models; classification of Social Media Hateful Screenshots Inciting Violence and Discrimination; SessionPrint: Accelerating kNN via Locality-Sensitive Hashing for Session-Based News Recommendation; Who Will Evaluate the Evaluators? Exploring the Gen-IR User Simulation Space; de-noising Document Classification Benchmarks via Prompt-Based Rank Pruning: A Case Study; best of Touché 2023 Task 4: Testing Data Augmentation and Label Propagation for Multilingual Multi-target Stance Detection; Leveraging LLM-Generated Data for Detecting Depression Symptoms on Social Media; from Sentence Embeddings to Large Language Models to Detect and Understand Wordplay; replicability Measures for Longitudinal Information Retrieval Evaluation; SimpleText Best of Labs in CLEF-2023: Scientific Text Simplification Using Multi-prompt Minimum Bayes Risk Decoding; large Language Model Cascades and Persona-Based In-Context Learning for Multilingual Sexism Detection.

Date 2024
URL <https://www.scopus.com/pages/publications/85205392488?origin=resultslist>
Archive Scopus
Language English
Volume 14959 LNCS
Publisher Springer Science and Business Media Deutschland GmbH
Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)
ISSN 03029743 (ISSN); 978-303171907-3 (ISBN)
Journal Abbr Lect. Notes Comput. Sci.
Date Added 20/09/2026, 8:35:27 pm
Modified 20/09/2026, 8:35:27 pm

Notes:

EID: 2-s2.0-85205392488

Export Date: 20 September 2026; Cited By: 0; Conference name: 15th International Conference of the CLEF Association, CLEF 2024; Conference date: 9 September 2024 through 12 September 2024; Conference code: 319229

15th International Conference of the CLEF Association, CLEF 2024

Item Type Journal Article

Editor Goeuriot L.

Editor Mulhem P.

Editor Quénot G.

Editor Schwab D.

Editor Di Nunzio G.M.

Editor Faggioli G.

Editor Ferro N.

Editor Soulier L.

Editor Galuščáková P.

Editor García Seco de Herrera A.

Abstract The proceedings contain 32 papers. The special focus in this conference is on CLEF Association. The topics include: Assessing Document Sanitization for Controlled Information Release and Retrieval in Data Marketplaces; the Impact of Web Search Result Quality on Decision-Making; improving Laypeople Familiarity with Medical Terms by Informal Medical Entity Linking; mapping the Media Landscape: Predicting Factual Reporting and Political Bias Through Web Interactions; under-Sampling Strategies for Better Transformer-Based Classifications Models; classification of Social Media Hateful Screenshots Inciting Violence and Discrimination; SessionPrint: Accelerating kNN via Locality-Sensitive Hashing for Session-Based News Recommendation; Who Will Evaluate the Evaluators? Exploring the Gen-IR User Simulation Space; de-noising Document Classification Benchmarks via Prompt-Based Rank Pruning: A Case Study; best of Touché 2023 Task 4: Testing Data Augmentation and Label Propagation for Multilingual Multi-

target Stance Detection; Leveraging LLM-Generated Data for Detecting Depression Symptoms on Social Media; from Sentence Embeddings to Large Language Models to Detect and Understand Wordplay; replicability Measures for Longitudinal Information Retrieval Evaluation; SimpleText Best of Labs in CLEF-2023: Scientific Text Simplification Using Multi-prompt Minimum Bayes Risk Decoding; large Language Model Cascades and Persona-Based In-Context Learning for Multilingual Sexism Detection.

Date 2024

URL <https://www.scopus.com/pages/publications/85205093494?origin=resultslist>

Archive Scopus

Language English

Volume 14958 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

ISSN 03029743 (ISSN); 978-303171735-2 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

Date Added 20/09/2026, 8:35:37 pm

Modified 20/09/2026, 8:35:37 pm

Notes:

EID: 2-s2.0-85205093494

Export Date: 20 September 2026; Cited By: 0; Conference name: 15th International Conference of the CLEF Association, CLEF 2024; Conference date: 9 September 2024 through 12 September 2024; Conference code: 319229

16th International Conference of the Pacific Association for Computational Linguistics, PACLING 2019

Item Type Journal Article

Editor Nguyen L.-M.

Editor Tojo S.

Editor Phan X.-H.

Editor Hasida K.

Abstract The proceedings contain 42 papers. The special focus in this conference is on Pacific Association for Computational Linguistics. The topics include: Learning to Evaluate Neural Language Models; recommending the Workflow of Vietnamese Sign Language Translation via a Comparison of Several Classification Algorithms; Document Classification by Word Embeddings of BERT; deep Domain Adaptation for Low-Resource Cross-Lingual Text Classification Tasks; multi-task Learning for Aspect and Polarity Recognition on Vietnamese Datasets; evaluating Classification Algorithms for Recognizing Figurative Expressions in Japanese Literary Texts; model-Driven Web Page Segmentation for Non Visual Access; Update Frequency and Background Corpus Selection in Dynamic TF-IDF Models for First Story Detection; a Pilot Study on Argument Simplification in Stance-Based Opinions; split First and

Then Rephrase: Hierarchical Generation for Sentence Simplification; automatic Approval of Online Comments with Multiple-Encoder Networks; is the Simplest Chatbot Effective in English Writing Learning Assistance?; towards Task-Oriented Dialogue in Mixed Domains; timing Prediction of Facilitating Utterance in Multi-party Conversation; evaluating Co-reference Chains Based Conversation History in Conversational Question Answering; multiple Linear Regression of Combined Pronunciation Ease and Accuracy Index; Rap Lyrics Generation Using Vowel GAN; emotion Recognition for Vietnamese Social Media Text; effects of Soft-Masking Function on Spectrogram-Based Instrument - Vocal Separation; japanese Predicate Argument Structure Analysis with Pointer Networks; Abstractive Text Summarization Using LSTMs with Rich Features; an Experimental Study on Constituency Parsing for Vietnamese; antonyms-Synonyms Discrimination Based on Exploiting Rich Vietnamese Features; Towards a UMLS-Integratable Vietnamese Medical Terminology; Vietnamese Word Segmentation with SVM: Ambiguity Reduction and Suffix Capture; towards Computing Inferences from English News Headlines.

Date 2020

URL <https://www.scopus.com/pages/publications/85088530661?origin=resultslist>

Archive Scopus

Language English

Volume 1215 CCIS

Publisher Springer

Publication Communications in Computer and Information Science

ISSN 18650929 (ISSN); 978-981156167-2 (ISBN)

Journal Abbr Commun. Comput. Info. Sci.

Date Added 20/09/2026, 8:35:41 pm

Modified 20/09/2026, 8:35:41 pm

Notes:

EID: 2-s2.0-85088530661

Export Date: 20 September 2026; Cited By: 0; Conference name: 16th International Conference of the Pacific Association for Computational Linguistics, PACLING 2019; Conference date: 11 October 2019 through 13 October 2019; Conference code: 241929

17th International Conference on Social Networks Analysis and Mining, ASONAM 2025

Item Type Journal Article

Editor An A.

Editor Cuzzocrea A.

Editor Hu H.

Abstract The proceedings contain 33 papers. The special focus in this conference is on Social Networks Analysis and Mining. The topics include: Enhancing Large Language Models for Arabic Dialects Using Knowledge-Based Rethinking and Contrastive Learning; analysis of User Temperament and Personality Traits in Social Media Through Complex Networks; a Scalable Approach to Marketing Funnel Modeling;

Cross-Industry Insights from LinkedIn; trendScope: A Temporal Hypergraph Framework for Food Trend Discovery; Discovering Root Causes of Risks Using Counterfactual Knowledge Graphs (CKG); enumeration of Subgraph of Interest Based on Pruning; TrackGAE: Tracking Dynamic Community Evolution with Graph Autoencoders; social Engineering and Information System Security-A Survey on the Necessity of Prevention; data-Driven Social Signal Mining for Stock Return Modeling via LinkedIn Networks; LLM-MAD: Multi-agent LLM Reasoning for Multi-modal Shilling Attack Detection in Online Platforms; Prompt-Augmented LLMs with RAG for Addressing Cold-Start and Sparsity in Online Recommender Systems; TempHypE: Time-Aware Hyperbolic Neural Ordinary Differential Equation (ODEs) Knowledge Graph Embeddings For Dynamic Link Prediction; Empowering Recommender Systems with Agentic AI: Towards Adaptive Online Personalization; Benchmarking GNN and Graph Transformer Models for Dynamic Link Prediction; session-Based Recommender Systems Enhanced with Anomaly Detection: A Comparative Study; Position Bias in LLMs for Critical Decision Support: A Case Study on Multiple Casualty Triage; automated Definition Generation for Online Jargon Analysis; outsmarting Willful-Thinking Opponents: Bayesian Belief Revision for Adversarial Reasoning in Large Language Models; preface; GraphRAG-V: Fast Multi-hop Retrieval via Text-Chunk Communities; PATHS: Agent-Based Modeling of Homelessness Pathways; public Sentiment Analysis Toward the Department of Education: A Social Media Study Using Topic Modeling and Sentiment Analysis; detection of Suicidal Risk on Social Media: A Hybrid Model; fediverse Sharing: Cross-Platform Interaction Dynamics Between Threads and Mastodon Users; fact-Checking with Large Language Models via Cost-Effective First-Order Logic Reformulation.

Date 2026

URL <https://www.scopus.com/pages/publications/105029924249?origin=resultslist>

Archive Scopus

Language English

Volume 16324 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-303214106-4 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

Date Added 20/09/2026, 8:35:01 pm

Modified 20/09/2026, 8:35:01 pm

Notes:

EID: 2-s2.0-105029924249

Export Date: 20 September 2026; Cited By: 0; Conference name: 17th International Conference on Social Networks Analysis and Mining, ASONAM 2025; Conference date: 25 August 2025 through 28 August 2025; Conference code: 346829

17th International Conference on Social Networks Analysis and Mining,
ASONAM 2025

Item Type Journal Article**Editor** An A.**Editor** Cuzzocrea A.**Editor** Hu H.

Abstract The proceedings contain 33 papers. The special focus in this conference is on Social Networks Analysis and Mining. The topics include: Dynamic and Overlapping Community Detection in Link Streams Through Formal Concept Analysis; boosting Attributed Network Embeddings with Clustering; developing a Commenter Behavior-Based Framework for Characterizing YouTube Channels; evaluating Structural Attractors and Retainers in YouTube Recommendation Networks; adaptiSent: Context-Aware Adaptive Attention for Multimodal Aspect-Based Sentiment Analysis; efficient Influence Maximization in Signed Networks with Positive Influence Increasing and Negative Influence Decreasing Simultaneously via Edge-View Influence Estimation; simulating Hashtag Dynamics with Networked Groups of Generative Agents; modeling Cross-Platform Narrative Diffusion: A Multiplex Approach to Information Spread in Social Media Ecosystems; a Network-Based Covariate Augmented Factorization Approach for Modeling Facebook Common Knowledge Experiments; contagious Rhythms: A Wave-Based Epidemic Approach for Music Virality on Social Platforms; Graph-Based Approaches to Utilizing LLMs for Generation of Individualized Student Learning Segments; Temporal Motifs for Financial Networks: A Study on Mercari, JPMC, and Venmo Platforms; say the Task, Build the Team: Prompt-Based Team Formation; identification of Authoritative Nodes and Dismantling of Illicit Networks Using a Novel Metric for Measuring Strength of a Graph; GraphRAG-Based NLP at Risk: Graphemic Dot-Level Adversarial Attack on Arabic Sentiment and LLM Retrieval-Augmented Models; LLM-Based Community Surveys for Operational Decision Making in Interconnected Utility Infrastructures; Can LLMs Reliably Label YouTube Videos? A Committee-Based Evaluation; Multiview Commonsense Reasoning Using LLMs for Understanding Crime Drama Series; Beliefs in Motion: Simulating Opinion Dynamics via LLM-Powered Community Reactions; from Non-overlapping to Overlapping Communities; temporal Motif Participation Profiles for Analyzing Node Similarity in Temporal Networks; decentralized and Self-adaptive Core Maintenance on Temporal Graphs; empathy Between Neighboring Nations: Distance Matters; Enhancing Regional Airbnb Trend Forecasting Using LLM-Based Embeddings of Accessibility and Human Mobility; discovering Linkages Among Multiple Disease Networks by Joint Clustering; preface.

Date 2026**URL** <https://www.scopus.com/pages/publications/105028924993?origin=resultslist>**Archive** Scopus**Language** English**Volume** 16322 LNCS**Publisher** Springer Science and Business Media Deutschland GmbH**Publication** Lecture Notes in Computer Science**ISSN** 03029743 (ISSN); 978-303213512-4 (ISBN)**Journal Abbr** Lect. Notes Comput. Sci.**Date Added** 20/09/2026, 8:35:04 pm**Modified** 20/09/2026, 8:35:04 pm**Notes:**

EID: 2-s2.0-105028924993

Export Date: 20 September 2026; Cited By: 0; Conference name: 17th International Conference on Social Networks Analysis and Mining, ASONAM 2025; Conference date: 25 August 2025 through 28 August 2025; Conference code: 346829

17th International Symposium on Distributed Computing and Artificial Intelligence, DCAI 2020

Item Type Journal Article

Editor Dong Y.

Editor Herrera-Viedma E.

Editor Matsui K.

Editor Omatsu S.

Editor González Briones A.

Editor Rodríguez González S.

Abstract The proceedings contain 30 papers. The special focus in this conference is on Distributed Computing and Artificial Intelligence. The topics include: Collaborative recommendations in online judges using autoencoder neural networks; natural language inference in ordinary and support verb constructions; comparative analysis between different automatic learning environments for sentiment analysis; context-aware music recommender system based on automatic detection of the user's physical activity; classification of chest diseases using deep learning; mobile device-based speech enhancement system using lip-reading; mobile networks and internet of things: Contributions to smart human mobility; design and implementation of a system to determine property tax through the processing and analysis of satellite images; smartsec4cop: Smart cyber-grooming detection using natural language processing and convolutional neural networks; multi-step ultraviolet index forecasting using long short-term memory networks; multispectral image analysis for the detection of diseases in coffee production; photograph classification based on main theme and multiple values by deep neural networks; in-vehicle violence detection in carpooling: A brief survey towards a general surveillance system; multiagent systems and role-playing games applied to natural resources management; a systematic review to multiagent systems and regulatory networks; the reversibility of cellular automata on trees with loops; virtual reality tool for learning sign language in spanish; data augmentation using gaussian mixture model on csv files; sentiment analysis in twitter: Impact of morphological characteristics; improving bert with focal loss for paragraph segmentation of novels; the use of artificial intelligence for clinical coding automation: A bibliometric analysis; a machine learning platform for stock investment recommendation systems; preface.

Date 2021

URL <https://www.scopus.com/pages/publications/85089751153?origin=resultslist>

Archive Scopus

Language English

Volume 1237 AISC

Publisher Springer

Publication Advances in Intelligent Systems and Computing

ISSN 21945357 (ISSN); 978-303053035-8 (ISBN)

Journal Abbr Adv. Intell. Sys. Comput.

Date Added 20/09/2026, 8:35:45 pm

Modified 20/09/2026, 8:35:45 pm

Notes:

EID: 2-s2.0-85089751153

Export Date: 20 September 2026; Cited By: 0; Conference name: 17th International Symposium on Distributed Computing and Artificial Intelligence, DCAI 2020; Conference date: 17 June 2020 through 19 June 2020; Conference code: 243089

18th International Conference on Knowledge Science, Engineering and Management, KSEM 2025

Item Type Journal Article

Editor Zhu T.

Editor Zhou W.

Editor Zhu C.

Abstract The proceedings contain 14 papers. The special focus in this conference is on Knowledge Science, Engineering and Management. The topics include: M3Net: Multimodal-Feature-Masked Networks for Fake News Detection; TB-DML4HS: A Task-Based Modeling and Causal Effect Estimation Method Using DML for Heterogeneous UAV Swarm; WebGCN: Web Information Extraction Algorithm Based on Graph Neural Networks; federated Rank Learning with Dimensionality Reduction and Clustering for Electricity Load Forecasting; Context-Aware Vectors: A New Method Integrating Personality Into LLMs for Enhanced Sentiment Analysis; research on Detection and Reconstruction of Multiple Types of Anomalies in Wind Speed-Power Data of Wind Farms; An LLM-Enabled Data Augmentation Framework for Low-Resource Scenarios; Lightweight Remote Sensing Tiny Object Detection Model Based on YOLOv8n Architecture; Causal Encoding Generative Model Based on Attention and KAN; TRVP: Transformer-VAE Framework for 3D Point Cloud Instance Segmentation; a Metapath-Based Neighborhood Reconstruction Network for Graph Anomaly Detection; darkFusionNet: A Fusion Neural Network Based Architecture for Darknet Text Classification; key Nodes Evaluation for Human Proximity Networks Based on Gravity Model; SRViT-MCNet: An IoT Malware Classification Model; MultiTEmb: Multi-scale Embeddings for Temporal KG Completion; education Distillation: Let the Model Learn in the School; A Multi-source Temporal Graph Approach for Reliable Market Forecasting with LLM Synergy; memory-Enhanced Transformer Adaptive Graph Convolutional Recurrent Network for Traffic Flow Forecasting; investigation into Auto-scaling Mechanisms in Cloud Computing; multi-receptive-Field Feature Fusion Knowledge Graph Embedding for Link Prediction; RAG with Visual Alert: Boosting Multimodal Language Models for Enhanced Visual Question Answering; dynamic Heterogeneous Graph Neural Network for Personality Detection in Chinese Social Media Texts; DualCBR: Cross-Modal Collaborative Filtering with Bidirectional Alignment for Long-Tail Recommendation; semantic Information Extraction with Language Models for Zero-Day Attack Detection; GeoER: A Challenging Benchmark

for Geometric Element Recognition.

Date 2026

URL <https://www.scopus.com/pages/publications/105023198655?origin=resultslist>

Archive Scopus

Language English

Volume 15923 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-981953060-1 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

Date Added 20/09/2026, 8:35:01 pm

Modified 20/09/2026, 8:35:01 pm

Notes:

EID: 2-s2.0-105023198655

Export Date: 20 September 2026; Cited By: 0; Conference name: 18th International Conference on Knowledge Science, Engineering and Management, KSEM 2025; Conference date: 4 August 2025 through 7 August 2025; Conference code: 340899

19th International Conference on Computational Linguistics and Intelligent Text Processing, CICLing 2018

Item Type Journal Article

Editor Gelbukh A.

Abstract The proceedings contain 68 papers. The special focus in this conference is on Computational Linguistics and Intelligent Text Processing. The topics include: Towards Automated Fiqh School Authorship Attribution; stance and Gender Detection in Spanish Tweets; natural-Annotation-Based Malay Multiword Expressions Extraction and Clustering; self-adaptive Privacy Concern Detection for User-Generated Content; complexity of Russian Academic Texts as the Function of Syntactic Parameters; an Experimental Approach for Information Extraction in Multi-party Dialogue Discourse; automatic Matching and Expansion of Abbreviated Phrases Without Context; relation Extraction Datasets in the Digital Humanities Domain and Their Evaluation with Word Embeddings; one World - Seven Thousand Languages (Best Paper Award, Third Place); prediction of Cryptocurrency Market; analysis of Speeches in Indian Parliamentary Debates; enrichment of OntoSenseNet: Adding a Sense-annotated Telugu Lexicon; a Neural Network Classifier Based on Dependency Tree for English-Vietnamese Statistical Machine Translation; the Utility of Hierarchical Phrase-Based Model Machine Translation for Low Resource Languages; automatic Method to Build a Dictionary for Class-Based Translation Systems; Addressing the Issue of Unavailability of Parallel Corpus Incorporating Monolingual Corpus on PBSMT System for English-Manipuri Translation; on the Influence of Machine Translation on Language Origin Obfuscation; error Classification and Evaluation of Machine Translation Evaluation Metrics for Hindi as a Target Language; unsupervised Learning of Word Segmentation: Does Tone Matter?; POS, ANA and LEM: Word

Embeddings Built from Annotated Corpora Perform Better (Best Paper Award, Second Place); i-vectors and Deep Convolutional Neural Networks for Language Identification in Clean and Reverberant Environments; automatic Normalization of Word Variations in Code-Mixed Social Media Text; experiments on Deep Morphological Inflection; identification of Lemmatization Errors Using Neural Models; full Inflection Learning Using Deep Neural Networks; semi-supervised Textual Entailment on Indonesian Wikipedia Data; arabic Named Entity Recognition Using Clustered Word Embedding; argumentation in Text: Discourse Structure Matters.

Date 2023

URL <https://www.scopus.com/pages/publications/85149684850?origin=resultslist>

Archive Scopus

Language English

Volume 13396 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-303123792-8 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

Date Added 20/09/2026, 8:35:40 pm

Modified 20/09/2026, 8:35:40 pm

Notes:

EID: 2-s2.0-85149684850

Export Date: 20 September 2026; Cited By: 0; Conference name: 19th International Conference on Computational Linguistics and Intelligent Text Processing, CICLing 2018; Conference date: 18 March 2018 through 24 March 2018; Conference code: 291039

19th International Conference on Computational Linguistics and Intelligent Text Processing, CICLing 2018

Item Type Journal Article

Editor Gelbukh A.

Abstract The proceedings contain 68 papers. The special focus in this conference is on Computational Linguistics and Intelligent Text Processing. The topics include: Towards Automated Fiqh School Authorship Attribution; stance and Gender Detection in Spanish Tweets; natural-Annotation-Based Malay Multiword Expressions Extraction and Clustering; self-adaptive Privacy Concern Detection for User-Generated Content; complexity of Russian Academic Texts as the Function of Syntactic Parameters; an Experimental Approach for Information Extraction in Multi-party Dialogue Discourse; automatic Matching and Expansion of Abbreviated Phrases Without Context; relation Extraction Datasets in the Digital Humanities Domain and Their Evaluation with Word Embeddings; one World - Seven Thousand Languages (Best Paper Award, Third Place); prediction of Cryptocurrency Market; analysis of Speeches in Indian Parliamentary Debates; enrichment of OntoSenseNet: Adding a Sense-annotated Telugu Lexicon; a Neural Network Classifier Based on Dependency Tree for English-Vietnamese Statistical

Machine Translation; the Utility of Hierarchical Phrase-Based Model Machine Translation for Low Resource Languages; automatic Method to Build a Dictionary for Class-Based Translation Systems; Addressing the Issue of Unavailability of Parallel Corpus Incorporating Monolingual Corpus on PBSMT System for English-Manipuri Translation; on the Influence of Machine Translation on Language Origin Obfuscation; error Classification and Evaluation of Machine Translation Evaluation Metrics for Hindi as a Target Language; unsupervised Learning of Word Segmentation: Does Tone Matter?; POS, ANA and LEM: Word Embeddings Built from Annotated Corpora Perform Better (Best Paper Award, Second Place); i-vectors and Deep Convolutional Neural Networks for Language Identification in Clean and Reverberant Environments; automatic Normalization of Word Variations in Code-Mixed Social Media Text; experiments on Deep Morphological Inflection; identification of Lemmatization Errors Using Neural Models; full Inflection Learning Using Deep Neural Networks; semi-supervised Textual Entailment on Indonesian Wikipedia Data; arabic Named Entity Recognition Using Clustered Word Embedding; argumentation in Text: Discourse Structure Matters.

Date 2023

URL <https://www.scopus.com/pages/publications/85149952632?origin=resultslist>

Archive Scopus

Language English

Volume 13397 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-303123803-1 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

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Notes:

EID: 2-s2.0-85149952632

Export Date: 20 September 2026; Cited By: 0; Conference name: 19th International Conference on Computational Linguistics and Intelligent Text Processing, CICLing 2018; Conference date: 18 March 2018 through 24 March 2018; Conference code: 291039

21st Annual Workshop of the Australasian Language Technology Association, ALTA 2023

Item Type Journal Article

Abstract The proceedings contain 28 papers. The special focus in this conference is on Australasian Language Technology Association. The topics include: Chat Disentanglement: Data for New Domains and Methods for More Accurate Annotation; enhancing Bacterial Infection Prediction in Critically Ill Patients by Integrating Clinical Text; predicting Empathic Accuracy from User-Designer Interviews; CRF-based recognition of invasive fungal infection concepts in CHIFIR clinical reports; BanglaClickBERT: Bangla Clickbait Detection from News Headlines

using Domain Adaptive BanglaBERT and MLP Techniques; the uncivil empathy: Investigating the relation between empathy and toxicity in online mental health support forums; Overview of the 2023 ALTA Shared Task: Discriminate between Human-Written and Machine-Generated Text; a Prompt in the Right Direction: Prompt Based Classification of Machine-Generated Text Detection; automatic Detection of Machine-Generated Text Using Pre-Trained Language Models; An Ensemble Based Approach To Detecting LLM-Generated Texts; Feature-Level Ensemble Learning for Robust Synthetic Text Detection with DeBERTaV3 and XLM-RoBERTa; Stacking the Odds: Transformer-Based Ensemble for AI-Generated Text Detection; natural Language Processing for Clinical Text; story Co-telling Dialogue Generation based on Multi-Agent Reinforcement Learning and Story Highlights; Using C-LARA to evaluate GPT-4's multilingual processing*; introduction; exploring Causal Directions through Word Occurrences:Semi-supervised Bayesian Classification Framework; the sub-band cepstrum as a tool for local spectral analysis in forensic voice comparison; right the docs: Characterising voice dataset documentation practices used in machine learning; MCASP: Multi-Modal Cross Attention Network for Stock Market Prediction; catching Misdiagnosed Limb Fractures in the Emergency Department Using Cross-institution Transfer Learning.

Date 2023

URL <https://www.scopus.com/pages/publications/85201263186?origin=resultslist>

Archive Scopus

Language English

Volume 2

Publisher Australasian Language Technology Association

Publication Proceedings of the Australasian Language Technology Workshop

ISSN 18347037 (ISSN)

Journal Abbr Proc. Australas. Lang. Technol. Workshop.

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EID: 2-s2.0-85201263186

Export Date: 20 September 2026; Cited By: 0; Conference name: 21st Annual Workshop of the Australasian Language Technology Association, ALTA 2023; Conference date: 29 November 2023 through 1 December 2023; Conference code: 316129

22nd Annual Workshop of the Australasian Language Technology Association, ALTA 2024

Item Type Journal Article

Editor Baldwin T.

Editor Rodríguez Méndez S.J.

Editor Kuo N.I-H.

Abstract The proceedings contain 21 papers. The special focus in this conference is on Australasian Language Technology Association. The topics include: GenABSA-Vec: Generative Aspect-Based Sentiment Feature Vectorization for Document-Level Sentiment Classification; A Closer Look at Tool-based Logical Reasoning with LLMs: The Choice of Tool Matters; generating bilingual example sentences with large language models as lexicography assistants; MoDEM: Mixture of Domain Expert Models; simultaneous Machine Translation with Large Language Models; which Side Are You On? Investigating Politico-Economic Bias in Nepali Language Models; Advancing Community Directories: Leveraging LLMs for Automated Extraction in MARC Standard Venue Availability Notes; lesser the Shots, Higher the Hallucinations: Exploration of Genetic Information Extraction using Generative Large Language Models; “Is Hate Lost in Translation?”: Evaluation of Multilingual LGBTQIA+ Hate Speech Detection; personality Profiling: How informative are social media profiles in predicting personal information?; rephrasing Electronic Health Records for Pretraining Clinical Language Models; comparison of Multilingual and Bilingual Models for Satirical News Detection of Arabic and English; breaking the Silence: How Online Forums Address Lung Cancer Stigma and Offer Support; Truth in the Noise: Unveiling Authentic Dementia Self-Disclosure Statements in Social Media with LLMs; Overview of the 2024 ALTA Shared Task: Detect Automatic AI-Generated Sentences for Human-AI Hybrid Articles; Advancing LLM detection in the ALTA 2024 Shared Task: Techniques and Analysis; Simple models are all you need: Ensembling stylometric, part-of-speech, and information-theoretic models for the ALTA 2024 Shared Task.

Date 2024

URL <https://www.scopus.com/pages/publications/105003197782?origin=resultslist>

Archive Scopus

Language English

Volume 22

Publisher Australasian Language Technology Association

Publication Proceedings of the Australasian Language Technology Workshop

ISSN 18347037 (ISSN)

Journal Abbr Proc. Australas. Lang. Technol. Workshop.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 22nd Annual Workshop of the Australasian Language Technology Association, ALTA 2024; Conference date: 2 December 2024 through 4 December 2024; Conference code: 329959

22nd International Conference on Intelligent Computing, ICIC 2026

Item Type Journal Article

Editor Huang D.-S.

Editor Zhang C.

Editor Chen W.

Editor Pan Y.

Editor Premaratne P.

Abstract The proceedings contain 287 papers. The special focus in this conference is on Intelligent Computing. The topics include: HSE-GTA: Generative Augmentation Based Multimodal Entity Alignment; hybrid Expert Architecture with Character-Level Feature Enhancement and Graph Reasoning for Personality Detection; HUAL-PAF: Boosting Multimodal Sentiment Analysis with Harmless Unimodal Auxiliary Learning and Progressive Attention Fusion; CalibLoop: Self-Calibrating Pseudo-Label Learning for Noise-Robust Source-Free QA Domain Adaptation; backtrack-on-Graph: Efficient Path Navigation via Candidate Stack and Role-Based Adaptive Update; a Low-Resource Neural Machine Translation Data Augmentation Method Based on Keyword Exchange; genetic Evolution of Domain Expertise with Multi-agent Collaboration; A Transformer-Based Model for Detection of AI-Generated Fake News: A Multi-perspective Exploration; PRON: Proposition-Oriented Repair and Operator Normalization for UMR Parsing; steering into Weights: Persistent Behavioral Editing for Targeted Harm Categories; Hybrid KBQA with Tree-Structured Sub-Question Decomposition; ALP: Adaptive Layerwise Pruning in Large Language Models; Multi-Stage Think-on-Graph: A Structured Reasoning Framework for LLM over Knowledge Graphs; iterative Mutual Refinement Network for Implicit Emotion Cause Extraction; RAVE-MNER: Relevance-Aware Visual Evidence for Multimodal Named Entity Recognition; MLiteBart: Lightweight Text-to-Gloss Generation via Adapter Tuning and Multitask Learning; BRACE: A Benign-History Robust Anchored Calibration and Evaluation Framework for Large Language Model Refusal; BATL: A Boundary-Aware Toxic Content Localization Method for Chinese Social Media; task-Dependent Algorithmic Equivalence of In-Context Learning; E-GCG: An Efficient Jailbreaking Method of Aligned Language Models via Weighted Dual-Objective Optimization; OASIS: Optimizing Attacker Sequences for Hard-Label Black-Box Text Attacks; A Dual-Agent Mental Health Counseling System with RAG and PEFT; RetailSQL: A Chinese Text-to-SQL Dataset for Retail Industry.

Date 2027

URL <https://www.scopus.com/pages/publications/105046889180?origin=resultslist>

Archive Scopus

Language English

Volume 16665 LNAI

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-981923434-9 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

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EID: 2-s2.0-105046889180

Export Date: 20 September 2026; Cited By: 0; Conference name: 22nd International Conference on Intelligent Computing, ICIC 2026; Conference date: 22 July 2026 through 26 July 2026; Conference code: 356579

23rd International Conference on Text, Speech, and Dialogue, TSD 2020

Item Type Journal Article

Editor Sojka P.

Editor Kopecek I.

Editor Pala K.

Editor Horák A.

Abstract The proceedings contain 57 papers. The special focus in this conference is on Text, Speech, and Dialogue. The topics include: Finest bert and crosloengual bert: Less is more in multilingual models; employing sentence context in czech answer selection; grammatical parallelism of russian prepositional localization and temporal constructions; costra 1.1: An inquiry into geometric properties of sentence spaces; next step in online querying and visualization of word-formation networks; evaluating a multi-sense definition generation model for multiple languages; combining cross-lingual and cross-task supervision for zero-shot learning; reading comprehension in czech via machine translation and cross-lingual transfer; measuring memorization effect in word-level neural networks probing; multilingual dependency parsing from universal dependencies to sesame street; semi-supervised induction of morpheme boundaries in czech using a word-formation network; interpreting word embeddings using a distribution agnostic approach employing hellinger distance; verb focused answering from cord-19; adjusting bert's pooling layer for large-scale multi-label text classification; recognizing preferred grammatical gender in russian anonymous online confessions; attention to emotions: Detecting mental disorders in social media; cross-lingual transfer for hindi discourse relation identification; authorship verification with personalized language models; a semantic grammar for augmentative and alternative communication systems; assessing unintended memorization in neural discriminative sequence models; multimodal fake news detection with textual, visual and semantic information; investigating the impact of pre-trained word embeddings on memorization in neural networks; investigating the corpus independence of the bag-of-audio-words approach; developing resources for te reo Māori text to speech synthesis system; acoustic characteristics of vot in plosive consonants produced by parkinson's patients; speaker-dependent bilstm-based phrasing.

Date 2020

URL <https://www.scopus.com/pages/publications/85091159155?origin=resultlist>

Archive Scopus

Language English

Volume 12284 LNAI

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-303058322-4 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

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EID: 2-s2.0-85091159155

Export Date: 20 September 2026; Cited By: 0; Conference name: 23rd International Conference on Text, Speech, and Dialogue, TSD 2020; Conference date: 8 September 2020 through 11 September 2020; Conference code: 244789

23rd International Symposium on Knowledge and Systems Sciences, KSS 2024

Item Type Journal Article

Editor Tang X.

Editor Huynh V.N.

Editor Xia H.

Editor Bai Q.

Abstract The proceedings contain 23 papers. The special focus in this conference is on Knowledge and Systems Sciences. The topics include: Construction and Evaluation of a Subjects Synergistic Network for Cross-Regional Major Infectious Disease Emergency Response; evolution of Cumulative Reciprocity in Structured Populations; Simulating Social Network with LLM Agents: An Analysis of Information Propagation and Echo Chambers; How Public Opinion Risks in Social Hot Events Are Generated: A fsQCA Perspective; vulnerability Measurement of Social Media Users to Online Public Opinion in Emergency Context; analyzing Replies and Interactions Among Users with Different Stances: A Case Study of the Russia-Ukraine Conflict; explainable Machine Learning-Based Research on Key Factors in the Formation of Public Opinion on Similar Events; UMCap: User Memory Augmented Method for Personalized Image Descriptions; research on the Construction Method of Island Chain Knowledge Graph; a Knowledge and Data Driven Method for Air Combat Intention Recognition; Model-Based Systems Engineering Supporting Architecture Modeling of Air Traffic Management System and Model Verifying Based on SMT; Mission Modeling for the Perseverance Rover Based on KARMA Language; social Media Oriented Fake News Detection Based on Social Context and Cascade Graph; data Augmentation Using Large Language Model for Fake Review Identification; digital Transformation Mechanisms for Emergency Management in Chemical Enterprises: An Industrial Agglomeration Perspective; research on the Credibility Evaluation Method of Online Medical Community Answer Content Based on Domain Knowledge Graph; a Comprehensive Framework for Sentiment Analysis and Cold-Start Recommendations in Vietnam Hospitality Sector; mining Complementary Relationships of Items for Diversified Recommendation; a Novel Two-Stage Approach for Customer Satisfaction Analysis; predicting the Decision-Making Performance Based on Self-attention and Long-Short Term Memory Network.

Date 2025

URL <https://www.scopus.com/pages/publications/85210246299?origin=resultslist>

Archive Scopus

Language English

Volume 2269 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Communications in Computer and Information Science

ISSN 18650929 (ISSN); 978-981960177-6 (ISBN)

Journal Abbr Commun. Comput. Info. Sci.

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EID: 2-s2.0-85210246299

Export Date: 20 September 2026; Cited By: 0; Conference name: 23rd International Symposium on Knowledge and Systems Sciences, KSS 2024; Conference date: 16 November 2024 through 17 November 2024; Conference code: 322919

24th International Conference on Informatics in Economy, IE 2025

Item Type Journal Article

Editor Ciurea C.

Editor Pocatilu P.

Editor Filip F.G.

Abstract The proceedings contain 39 papers. The special focus in this conference is on Informatics in Economy. The topics include: Blockchain Integration in Trust Service Providers: Trends, Challenges, and Future Directions; ZKFlag: A Privacy-First, Open-Source Platform for Feature Flags Using Zero-Knowledge Proofs; ARDL Insights into Financial Contagion: Assessing the Impact of Crude Oil and Bitcoin on European Banks; mapping the Evolution of Fuzzy Real Options and Decision-Making: A Bibliometric Analysis; exploring the Innovation Claudia Curve and Load Capacity Utilization in Finland: An Autoregressive Distributed Lag Approach; a Bibliometric Analysis of Evacuation Simulation Research: Trends, Collaboration, and Thematic Evolution (2002–2024); Ranking EU Happiness: TOPSIS and VIKOR Classification; circular Economy and the Implementation of the Green Supply Chain. An Analysis of the Implications on the Actual Economy; LSTM Deep Learning Network Applied to Stock Market; romanian News Article Classification: A Multi-Model Comparison with Class Balancing; market Volatility Prediction Under Powerful Geopolitical Events. A Case Study on the Latest European Conflict; a Bibliometric Analysis of the Bibliometric Papers in Transportation Sector; a Solution to University Course Timetabling Problem Using Genetic Algorithms; Unveiling IT Job Market Sentiment and Trends: A Reddit-Based Study Using Sentiment Analysis and Named Entity Recognition; Dealing with Imbalanced Data for Loan Default Prediction: A Comparative Study Across Balancing Techniques for CNN, LSTM, RF, and SVM Models; assessing the Static Malware Analysis Capabilities of Large Language Models; assessing the Trustworthiness of Large Language Models: A Two-Stage Framework Using Ragas and LlamaIndex; ethics of Artificial Intelligence in Business: Challenges, Applications and Pathways for Responsible Innovation; preface.

Date 2026

URL <https://www.scopus.com/pages/publications/105039581271?origin=resultslist>

Archive Scopus

Language English

Volume 471 SIST

Publisher Springer Science and Business Media Deutschland GmbH

Publication Smart Innovation, Systems and Technologies
ISSN 21903018 (ISSN); 978-981956135-3 (ISBN)
Journal Abbr Smart Innov. Syst. Technol.
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EID: 2-s2.0-105039581271

Export Date: 20 September 2026; Cited By: 0; Conference name: 24th International Conference on Informatics in Economy, IE 2025; Conference date: 15 May 2025 through 16 May 2025; Conference code: 351829

24th International Conference on Information Integration and Web Intelligence, iiWAS 2022, held in conjunction with the 20th International Conference on Advances in Mobile Computing and Multimedia Intelligence, MoMM 2022

Item Type Journal Article

Editor Pardede E.

Editor Delir Haghighi P.

Editor Khalil I.

Editor Kotsis G.

Abstract The proceedings contain 53 papers. The special focus in this conference is on Information Integration and Web Intelligence. The topics include: GPU Accelerated Parallel Implementation of Linear Programming Algorithms; Towards Explaining DL Non-entailments by Utilizing Subtree Isomorphisms; Query Optimization in NoSQL Databases Using an Enhanced Localized R-tree Index; handling Exit Node Vulnerability in Onion Routing with a Zero-Knowledge Proof; an Accurate, Flexible and Private Trajectory-Based Contact Tracing System on Untrusted Servers; bilateral Bargaining for Healthcare Data Sharing; modifying Neo4j's Object Graph Mapper Queries for Access Control; a Recommendation Method for Recipes Containing Unskillful Elements Using Naïve Bayes Classifier to Improve Cooking Skills; extraction of Complementary Topics Based on Phrase Importance and Co-occurrence in Technical Blogs; sharing Semantic Knowledge for Autonomous Robots: Cooperation for Social Robotic Systems; feature Relevance Analysis of Product Reviews to Support Online Shopping; annotation System for Dialogue Datasets of Older Adult's with Photos and Storytelling; evaluation Axes for Automatically Generated Product Descriptions; CISQA: Corporate Smart Insights Question Answering System; symmetry Metrics for Pairwise Entity Similarities; a New Method to Measure Similarity of Words in Japanese Twitter Based on Related Images; Hybrid Phishing URL Detection Using Segmented Word Embedding; a Scheme for News Article Classification in a Low-Resource Language; expertise Computation for Automatic Reviewer Assignment; detection of Hot Topics Using Multi-view Text Clustering; exploiting Named Entity Recognition for Information Extraction from Italian Procurement Documents: A Case Study; interactive Visualization of Comic Character Correlation Diagrams for Understanding Character Relationships and Personalities; two Case Studies on Guiding Human Behavior Using

Virtualization Technologies in Distributed Virtual Shopping Malls; central Figures in the Climate Change Discussion on Twitter; e-Tracer: A Smart, Personalized and Immersive Digital Tourist Software System; bNaming: An Intelligent Application to Assist Brand Names Definition; preface.

Date 2022

URL <https://www.scopus.com/pages/publications/85145105280?origin=resultslist>

Archive Scopus

Language English

Volume 13635 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Journal Abbr Lect. Notes Comput. Sci.

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EID: 2-s2.0-85145105280

Export Date: 20 September 2026; Cited By: 0; Conference name: 24th International Conference on Information Integration and Web Intelligence, iiWAS 2022, held in conjunction with the 20th International Conference on Advances in Mobile Computing and Multimedia Intelligence, MoMM 2022; Conference date: 28 November 2022 through 30 November 2022; Conference code: 287349

25th International Conference on Business Information Systems, BIS 2025

Item Type Journal Article

Editor Wecel K.

Abstract The proceedings contain 23 papers. The special focus in this conference is on Business Information Systems. The topics include: Identification of Challenges in the Economic Evaluation of AI Projects; LumiViz: Automating Business Data Visualization with Generative AI; AI-Driven Video Avatar for Academic Support; think Outside the Box – and Between Them: A Visual Guide to Smarter Technological Development; Interview Study to Determine the Potential of Large Language Model Usage in IT-Consultancy; sentiment Analysis of Wikipedia Articles About Companies: A Comparison of Different Models; AI-Driven Patent Claim Construction: Opportunity or the Dawn of Inventive Cannibalism?; Political Narratives and Misinformation During the COVID-19 Pandemic: A Comparative Analysis of Polish Political Parties on Twitter; can Users Trust Process Mining?; do not Stay Behind: Embracing Process Mining in the Italian Public Sector; exploring the Role of Visualization in Process Mining: A Systematic Literature Review; a Data-Driven Framework for Retail Inventory Optimization: Integrating Object-Centric Process Mining and Mathematical Models; cryptocurrency Wallet Security in the Era of Big Data and Quantum Computing: Brute Force Attacks and Blockchain Analysis with Google BigQuery; Compliance On-Chain Framework for Decentralized Finance to Meet AML/CTF Requirements While Preserving Privacy, Security and

Decentralization; integrating Genetic Algorithms and Ensemble Learning for Improved and Transparent Credit Scoring; architectures and Implementations of Data Lakehouses: Case Studies from Industrial Practice; reducing Risks of Continuous Deployments for SaaS Excellence: A Hybrid Risk-Model Matrix Approach Implementation of Continuous Deployments; The Influence of Agile IT Departments: Appreciation as a Catalyst for Organizational Transformation; Activity-Based Cost & GHG Accounting: Conceptualization, Design and Implementation; call Detail Record Generator for Modeling Real-World Telecommunication Scenarios; automating Water Management Using an Online Planning Approach.

Date 2025

URL <https://www.scopus.com/pages/publications/105009410600?origin=resultslist>

Archive Scopus

Language English

Volume 554 LNBIP

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Business Information Processing

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Journal Abbr Lect. Notes Bus. Inf. Process.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 25th International Conference on Business Information Systems, BIS 2025; Conference date: 25 June 2025 through 27 June 2025; Conference code: 333899

26th International Conference on Applications of Natural Language to Information Systems, NLDB 2021

Item Type Journal Article

Editor Métais E.

Editor Meziane F.

Editor Horacek H.

Editor Kapetanios E.

Abstract The proceedings contain 33 papers. The special focus in this conference is on Applications of Natural Language to Information Systems. The topics include: The Importance of Character-Level Information in an Event Detection Model; sequence-Based Word Embeddings for Effective Text Classification; BERT-Capsule Model for Cyberbullying Detection in Code-Mixed Indian Languages; multiword Expression Features for Automatic Hate Speech Detection; semantic Text Segment Classification of Structured Technical Content; on the Generalization of Figurative Language Detection: The Case of Irony and Sarcasm; extracting Facts from Case Rulings Through Paragraph Segmentation of Judicial Decisions; Detection of Misinformation About COVID-19 in Brazilian Portuguese WhatsApp Messages; multi-Step Transfer

Learning for Sentiment Analysis; scaling Federated Learning for Fine-Tuning of Large Language Models; improving Sentiment Classification in Low-Resource Bengali Language Utilizing Cross-Lingual Self-supervised Learning; human Language Comprehension in Aspect Phrase Extraction with Importance Weighting; exploring Summarization to Enhance Headline Stance Detection; predicting Vaccine Hesitancy and Vaccine Sentiment Using Topic Modeling and Evolutionary Optimization; sentiment Progression Based Searching and Indexing of Literary Textual Artefacts; argument Mining in Tweets: Comparing Crowd and Expert Annotations for Automated Claim and Evidence Detection; authorship Attribution Using Capsule-Based Fusion Approach; on the Explainability of Automatic Predictions of Mental Disorders from Social Media Data; using Document Embeddings for Background Linking of News Articles; let's Summarize Scientific Documents! A Clustering-Based Approach via Citation Context; overcoming the Knowledge Bottleneck Using Lifelong Learning by Social Agents; cross-Active Connection for Image-Text Multimodal Feature Fusion; profiling Fake News Spreaders: Personality and Visual Information Matter; comparing MultiLingual and Multiple MonoLingual Models for Intent Classification and Slot Filling.

Date 2021

URL <https://www.scopus.com/pages/publications/85111418311?origin=resultslist>

Archive Scopus

Language English

Volume 12801 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-303080598-2 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 26th International Conference on Applications of Natural Language to Information Systems, NLDB 2021; Conference date: 23 June 2021 through 25 June 2021; Conference code: 261479

28th International Conference on Applications of Natural Language to Information Systems, NLDB 2023

Item Type Journal Article

Editor Métais E.

Editor Meziane F.

Editor Manning W.

Editor Reiff-Marganiec S.

Editor Sugumaran V.

Abstract The proceedings contain 45 papers. The special focus in this conference is on Applications of Natural Language to Information Systems. The topics include: Prompt and Instruction-Based Tuning for Response Generation in Conversational Question Answering; IndQNER: Named Entity Recognition Benchmark Dataset from the Indonesian Translation of the Quran; comparing Object Recognition Models and Studying Hyperparameter Selection for the Detection of Bolts; morphosyntactic Evaluation for Text Summarization in Morphologically Rich Languages: A Case Study for Turkish; building Knowledge Graphs in Heliophysics and Astrophysics; Text to Image Synthesis Using Bridge Generative Adversarial Network and Char CNN Model; evaluation of Transformer-Based Models for Punctuation and Capitalization Restoration in Spanish and Portuguese; sentence-to-Label Generation Framework for Multi-task Learning of Japanese Sentence Classification and Named Entity Recognition; could KeyWord Masking Strategy Improve Language Model?; how Challenging is Multimodal Irony Detection?; regularization, Semi-supervision, and Supervision for a Plausible Attention-Based Explanation; node-Weighted Centrality Ranking for Unsupervised Long Document Summarization; characterization of the City of the Future from a Science Fiction Corpus; on the Rule-Based Extraction of Statistics Reported in Scientific Papers; GRAM: Grammar-Based Refined-Label Representing Mechanism in the Hierarchical Semantic Parsing Task; expanding Domain-Specific Knowledge Graphs with Unknown Facts; knowledge Graph Representation Learning via Generated Descriptions; lonXplain: Lonesomeness as a Consequence of Mental Disturbance in Reddit Posts; a Comparative Study of Evaluation Metrics for Long-Document Financial Narrative Summarization with Transformers; effective Information Retrieval, Question Answering and Abstractive Summarization on Large-Scale Biomedical Document Corpora; less is More: A Prototypical Framework for Efficient Few-Shot Named Entity Recognition; abstractive Summarization Based Question-Answer System For Structural Information; adversarial Capsule Networks for Romanian Satire Detection and Sentiment Analysis; a Few-Shot Approach to Resume Information Extraction via Prompts; decoding Strategies for Code Conciseness and Efficiency in Transformer-Generated Programs; SP-BERT: A Language Model for Political Text in Scandinavian Languages; improving Context-Awareness on Multi-Turn Dialogue Modeling with Extractive Summarization Techniques; cross-Domain Toxic Spans Detection.

Date 2023

URL <https://www.scopus.com/pages/publications/85164697954?origin=resultslist>

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Language English

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Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Export Date: 20 September 2026; Cited By: 0; Conference name: 28th International Conference on Applications of Natural Language to Information Systems, NLDB 2023; Conference date: 21 June 2023 through 23 June 2023; Conference code: 296599

29th International Conference on Natural Language and Information Systems, NLDB 2024

Item Type Journal Article

Editor Rapp A.

Editor Di Caro L.

Editor Meziane F.

Editor Sugumaran V.

Abstract The proceedings contain 72 papers. The special focus in this conference is on Natural Language and Information Systems. The topics include: Efficient Knowledge Distillation: Empowering Small Language Models with Teacher Model Insights; text Role Classification in Scientific Charts Using Multimodal Transformers; LaFiCMIL: Rethinking Large File Classification from the Perspective of Correlated Multiple Instance Learning; unveiling Depression on Social Media: Active Learning with Human-in-the-Loop Labeling for Mental Health Data Annotation and Analysis; all-Words Pronunciation Estimation of Japanese Homographs Using Automatically Tagged Data; I've Got the "answer"! : Interpretation of LLMs Hidden States in Question Answering; Improving DRS-to-Text Generation Through Delexicalization and Data Augmentation; large Language Models for Few-Shot Automatic Term Extraction; data Augmentation Method Utilizing Template Sentences for Variable Definition Extraction; Semi-supervised Named Entity Recognition for Low-Resource Languages Using Dual PLMs; improved Models for Media Bias Detection and Subcategorization; grounding Toxicity in Real-World Events Across Languages; unveiling the Hate: Generating Faithful and Plausible Explanations for Implicit and Subtle Hate Speech Detection; the Influence of Iconicity in Transfer Learning for Sign Language Recognition; s3: A Simple Strong Sample-Effective Multimodal Dialog System; body-Shaming Detection and Classification in Italian Social Media; adaptive Greedy Layer Pruning: Iterative Layer Pruning with Subsequent Model Repurposing; toward Automatic Group Membership Annotation for Group Fairness Evaluation; REA: Refine-Estimate-Answer Prompting for Zero-Shot Relation Extraction; enhancing Romanian Offensive Language Detection Through Knowledge Distillation, Multi-task Learning, and Data Augmentation; automating Gender-Inclusive Language Modification in Italian University Administrative Documents; relation Extraction Techniques in Cyber Threat Intelligence; shact: Disentangling and Clustering Latent Syntactic Structures from Transformer Encoders; maskPure: Improving Defense Against Text Adversaries with Stochastic Purification; SERIEMA: A Framework to Enhance Clustering Stability, Compactness, and Separation by Fusing Multimodal Data; exploiting Graph Embeddings from Knowledge Bases for Neural Biomedical Relation Extraction.

Date 2024

URL <https://www.scopus.com/pages/publications/85205453261?origin=resultslist>

Archive Scopus

Language English

Volume 14763 LNCS
Publisher Springer Science and Business Media Deutschland GmbH
Publication Lecture Notes in Computer Science
ISSN 03029743 (ISSN); 978-303170241-9 (ISBN)
Journal Abbr Lect. Notes Comput. Sci.
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Export Date: 20 September 2026; Cited By: 0; Conference name: 29th International Conference on Applications of Natural Language to Information Systems, NLDB 2024; Conference date: 25 June 2024 through 27 June 2024; Conference code: 319819

29th International Conference on Natural Language and Information Systems, NLDB 2024

Item Type Journal Article

Editor Rapp A.

Editor Di Caro L.

Editor Meziane F.

Editor Sugumaran V.

Abstract The proceedings contain 72 papers. The special focus in this conference is on Natural Language and Information Systems. The topics include: Efficient Knowledge Distillation: Empowering Small Language Models with Teacher Model Insights; text Role Classification in Scientific Charts Using Multimodal Transformers; LaFiCMIL: Rethinking Large File Classification from the Perspective of Correlated Multiple Instance Learning; unveiling Depression on Social Media: Active Learning with Human-in-the-Loop Labeling for Mental Health Data Annotation and Analysis; all-Words Pronunciation Estimation of Japanese Homographs Using Automatically Tagged Data; I've Got the "answer"! : Interpretation of LLMs Hidden States in Question Answering; Improving DRS-to-Text Generation Through Delexicalization and Data Augmentation; large Language Models for Few-Shot Automatic Term Extraction; data Augmentation Method Utilizing Template Sentences for Variable Definition Extraction; Semi-supervised Named Entity Recognition for Low-Resource Languages Using Dual PLMs; improved Models for Media Bias Detection and Subcategorization; grounding Toxicity in Real-World Events Across Languages; unveiling the Hate: Generating Faithful and Plausible Explanations for Implicit and Subtle Hate Speech Detection; the Influence of Iconicity in Transfer Learning for Sign Language Recognition; s3: A Simple Strong Sample-Effective Multimodal Dialog System; body-Shaming Detection and Classification in Italian Social Media; adaptive Greedy Layer Pruning: Iterative Layer Pruning with Subsequent Model Repurposing; toward Automatic Group Membership Annotation for Group Fairness Evaluation; REA: Refine-Estimate-Answer Prompting for Zero-Shot Relation Extraction; enhancing Romanian Offensive Language Detection Through Knowledge Distillation, Multi-task Learning, and Data Augmentation; automating Gender-

Inclusive Language Modification in Italian University Administrative Documents; relation Extraction Techniques in Cyber Threat Intelligence; shact: Disentangling and Clustering Latent Syntactic Structures from Transformer Encoders; maskPure: Improving Defense Against Text Adversaries with Stochastic Purification; SERIEMA: A Framework to Enhance Clustering Stability, Compactness, and Separation by Fusing Multimodal Data; exploiting Graph Embeddings from Knowledge Bases for Neural Biomedical Relation Extraction.

Date 2024

URL <https://www.scopus.com/pages/publications/85205353768?origin=resultslist>

Archive Scopus

Language English

Volume 14762 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

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29th International Conference on Technologies and Applications of Artificial Intelligence, TAAI 2024

Item Type Journal Article

Editor Chu W.-T.

Editor Shen C.-Y.

Editor Shuai H.-H.

Abstract The proceedings contain 24 papers. The special focus in this conference is on Technologies and Applications of Artificial Intelligence. The topics include: Spectator Support Systems for Mahjong Novices by Displaying Candidate Tiles for Winning; effects of Online News Comments on Attitude Formation of Readers; QCI Robot with Generative AI Knowledge Graph for Taiwanese/Japanese Co-Learning Model Application; optimal Assignment of Immediate Tasks in Multi-agent Pickup and Delivery; enhancement of the Speech Ability of an Intelligent Dialogue Agent by Distributed Text Representation and Sentiment Analysis; comicFaves: Organizing Shelves for Favorite Comics Based on User's Affection; procedurally Generating Natural-Looking Villages in Minecraft with Ant Colony Optimization Algorithms; proposal of Modeling Personal Values Using Large Language Models for Extracting Mentions of Item Attributes and Evaluation Polarity from Review Texts; an Eye Pupil Detection-Based Review Support System for Online Video-Based Learning; A Study of LLM Generated Pseudo-Data for Improving Small-Scale

Models in Human Values Estimation; Exploring Optimal Color Filter Array Patterns for Demosaicing with SwinIR; proposal for a New Evaluation Index for Human Fallibility in Shogi; systematic Selection of N-Tuples for Game 2048 Using Neural-Network Function Approximator; a User Interactive Shift Scheduling and Timetabling System Based on Evolutionary Computation for Private Tutoring Schools; fish Robot Locomotion Control Using Central Pattern Generator Based on Kuramoto Oscillator Model; a Crow Search Algorithm for Solving Virus Disease Transmission; physics-Informed Neural Network for Shock Absorber Design; exploring Compromised Accounts in Social Media by Change Point Detection; a Comparison of Deep Learning Methods for Multi-Class Classification of Bloodstain Patterns; genetic Algorithms for Variational Quantum Eigensolvers with Non-orthogonal Quantum State Encoding; AI-Based Microsection Measurement Framework Using ComfyUI Workflow for Flexible Printed Circuit Board; detecting Unknown Attacks: Transformer-Based Image Classifiers with an Out-of-Distribution Detector.

Date 2025

URL <https://www.scopus.com/pages/publications/105003640531?origin=resultslist>

Archive Scopus

Language English

Volume 2415 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Communications in Computer and Information Science

ISSN 18650929 (ISSN); 978-981964595-4 (ISBN)

Journal Abbr Commun. Comput. Info. Sci.

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EID: 2-s2.0-105003640531

Export Date: 20 September 2026; Cited By: 0; Conference name: 29th International Conference on Technologies and Applications of Artificial Intelligence, TAAI 2024; Conference date: 6 December 2024 through 7 December 2024; Conference code: 330309

29th International Conference on Technologies and Applications of Artificial Intelligence, TAAI 2024

Item Type Journal Article

Editor Chu W.-T.

Editor Shen C.-Y.

Editor Shuai H.-H.

Abstract The proceedings contain 24 papers. The special focus in this conference is on Technologies and Applications of Artificial Intelligence. The topics include: Spectator Support Systems for Mahjong Novices by Displaying Candidate Tiles for Winning; effects of Online News Comments on Attitude Formation of Readers; QCI Robot with Generative AI Knowledge Graph for Taiwanese/Japanese Co-Learning Model Application; optimal Assignment of Immediate Tasks in Multi-agent

Pickup and Delivery; enhancement of the Speech Ability of an Intelligent Dialogue Agent by Distributed Text Representation and Sentiment Analysis; comicFaves: Organizing Shelves for Favorite Comics Based on User's Affection; procedurally Generating Natural-Looking Villages in Minecraft with Ant Colony Optimization Algorithms; proposal of Modeling Personal Values Using Large Language Models for Extracting Mentions of Item Attributes and Evaluation Polarity from Review Texts; an Eye Pupil Detection-Based Review Support System for Online Video-Based Learning; A Study of LLM Generated Pseudo-Data for Improving Small-Scale Models in Human Values Estimation; Exploring Optimal Color Filter Array Patterns for Demosaicing with SwinIR; proposal for a New Evaluation Index for Human Fallibility in Shogi; systematic Selection of N-Tuples for Game 2048 Using Neural-Network Function Approximator; a User Interactive Shift Scheduling and Timetabling System Based on Evolutionary Computation for Private Tutoring Schools; fish Robot Locomotion Control Using Central Pattern Generator Based on Kuramoto Oscillator Model; a Crow Search Algorithm for Solving Virus Disease Transmission; physics-Informed Neural Network for Shock Absorber Design; exploring Compromised Accounts in Social Media by Change Point Detection; a Comparison of Deep Learning Methods for Multi-Class Classification of Bloodstain Patterns; genetic Algorithms for Variational Quantum Eigensolvers with Non-orthogonal Quantum State Encoding; AI-Based Microsection Measurement Framework Using ComfyUI Workflow for Flexible Printed Circuit Board; detecting Unknown Attacks: Transformer-Based Image Classifiers with an Out-of-Distribution Detector.

Date 2025

URL <https://www.scopus.com/pages/publications/105003882174?origin=resultslist>

Archive Scopus

Language English

Volume 2414 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

Publication Communications in Computer and Information Science

ISSN 18650929 (ISSN); 978-981964588-6 (ISBN)

Journal Abbr Commun. Comput. Info. Sci.

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EID: 2-s2.0-105003882174

Export Date: 20 September 2026; Cited By: 0; Conference name: 29th International Conference on Technologies and Applications of Artificial Intelligence, TAAI 2024; Conference date: 6 December 2024 through 7 December 2024; Conference code: 330309

30th Pacific-Asia Conference on Knowledge Discovery and Data Mining, PAKDD 2026

Item Type Journal Article

Editor Wong R.C.-W.

Editor Kwok J.
Editor Tong H.
Editor Lu H.
Editor Salim F.
Editor Song Y.
Editor Yiu M.

Abstract The proceedings contain 184 papers. The special focus in this conference is on Knowledge Discovery and Data Mining. The topics include: From Moderation to Mediation: Can LLMs Serve as Mediators in Online Flame Wars?; a Regime-Aware Fusion Framework for Time Series Classification; on Imbalanced Regression with Hoeffding Trees; Multi-RAG: A Multimodal Retrieval-Augmented Generation System for Adaptive Video Understanding; hybrid Time Series Diffusion Model and Conformal Prediction for Incomplete Time Series; nCLEAR: Nonlinear Coupled Latent Embedding for Cross-Sectional Asset Returns; Facet-Level Persona Control by Trait-Activated Routing with Contrastive SAE for Role-Playing LLMs; AI-Generated Image Detection Based on the Most Suspicious Patch; GROFF: A Generative Restoration Model for Occluded Face Filling; q-Matrix Completeness and Annotation Strategies: A Comparative Study of Knowledge Tracing Architectures; HSGTCN: A Hybrid Semantic-Guided Transformer-CNN Network for Learned Image Compressed Sensing; STEPS: Scene Text Editing with Preserved Style Using Diffusion and Contrastive Style Encoding; Semantic-Enhanced Representation Architecture for Automatic ICD Coding via Hyperbolic Graph Neural Networks; approximately Bisubmodular Regret Minimization in Billboard and Social Media Advertising; what Helps—and What Hurts: Bidirectional Explanations for Vision Transformers; Adaptive PII Detection via Evolutionary Orchestration and Multi-agent Debate; cloud-Edge Collaborative Anomaly Detection for Water Treatment Systems; CONAN: CONcept-Oriented ANomaly Detector for Interpretable Zero-Shot Detection; Automated Multiple-Choice Question Generation: A Survey from a KDDM Perspective; Graph Neural Networks in EEG-Based Emotion Recognition: A Survey; generative Artificial Intelligence for Project Management: A Review of Architectures, Applications, and Governance; A Review of Human-Centric Evaluation of Cultural Biases in Indic Languages in LLMs: Rethinking Research Directions; foreword; PathoLLM: Accurate Endometrial Cancer Subtyping Using Clinically-Aligned Language Models; Efficient Choice-Based Facility Location: A Hybrid AHP/DEA-Greedy Approach; Fourier Neural Operator on Implicit Points for PDE Surrogate Modeling; bridging Statistical Seasonality and Diffusion-Based Deep Learning for Multivariate Time Series Forecasting.

Date 2027

URL <https://www.scopus.com/pages/publications/105043681924?origin=resultslist>

Archive Scopus

Language English

Volume 16618 LNAI

Publisher Springer Science and Business Media Deutschland GmbH

Publication Lecture Notes in Computer Science

ISSN 03029743 (ISSN); 978-981921946-9 (ISBN)

Journal Abbr Lect. Notes Comput. Sci.

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EID: 2-s2.0-105043681924

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31st Computational Linguistics in the Netherlands Conference , CLIN 2021

Item Type Journal Article

Abstract The proceedings contain 16 papers. The topics discussed include: keeping up with the neighbors - an agent-based simulation of the divergence of the standard Dutch pronunciations in the Netherlands and Belgium; a hybrid ASR system for southern Dutch; a logic-based framework for natural language inference in Dutch; probing for Dutch relative pronoun choice; evaluating the impact of word classes on cross-domain age detection models' performance; aspect-based sentiment analysis for German: analyzing 'talk of literature' surrounding literary prizes on social media; vaccinpraat: monitoring vaccine skepticism in Dutch twitter and Facebook comments; and can pictograph translation technologies facilitate communication and integration in migration settings?.

Date 2021

URL <https://www.scopus.com/pages/publications/85130350382?origin=resultslist>

Archive Scopus

Language English

Volume 11

Publisher Computational Linguistics in the Netherlands

Publication Computational Linguistics in the Netherlands Journal

ISSN 22114009 (ISSN)

Journal Abbr Comput. Linguist. Netherlands J.

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EID: 2-s2.0-85130350382

Export Date: 20 September 2026; Cited By: 0; Conference name: 31st Computational Linguistics in the Netherlands Journal, CLIN 2021; Conference code: 179127

31st International Conference on Natural Language and Information Systems, NLDB 2026

Item Type Journal Article

Editor Cabrio E.

Editor Monteiro E.

Abstract The proceedings contain 22 papers. The special focus in this conference is on Natural Language and Information Systems. The topics include: What Do Claim Verification Datasets Actually Test? A Reasoning Trace Analysis; using Text Simplification in Norwegian News Summarization; if I Could Turn Back Time: Temporal Reframing as a Historical Reasoning Task for Large Language Models; ontology-Augmented Prompt Engineering for Aspect-Based Sentiment Classification; consp2VecD: A Dataset Based on Emotional Dynamics Expressed by Reddit Conspiracy Groups for Information Disorder Analysis; adaptive Filtering for Large Language Models; BEADS: Bias Evaluation Across Domains; mitigating Gender Bias in English to Romanian Machine Translation; Zoom In Disparities in Healthcare LLM Q&A; tokenizations for Austronesian Language Models: Study on Languages in Indonesia Archipelago; efficient Error-Type Transfer for Grammatical Error Detection via Embedding Alignment; towards Robust Uzbek Neural Dependency Parsing: Cross-Treebank Training; Adapting GPT for Egyptian Arabic-English Code-Switched Sentiment Analysis Through Prompting, Retrieval, and Sentiment-Guided Fine-Tuning; Evaluating Noisy Optimization in Finetuning LMs for Neural Ranking; Evaluating LLM-Generated Wikipedia Content: Political Topics in a French Setting; When Words Move Markets: Interpretable Behavioural and Robustness Analysis of LLMs for Financial Sentiment Reasoning via Local Perturbation Explanations; Attention-Pruned SHAP: Accelerating SHAP-Based Explainability with Attention-Guided Feature Pruning; Temporal Structure in LLM Reasoning: Analyzing Hidden States and Semantic Embeddings; Expansion Is Not Enough: Revisiting Dynamically Expandable Networks for NLP Tasks; Automated ICD-10 Coding Approaches with UMLS Integration and Model Interpretability Support.

Date 2027

URL <https://www.scopus.com/pages/publications/105045130948?origin=resultslist>

Archive Scopus

Language English

Volume 16696 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

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Journal Abbr Lect. Notes Comput. Sci.

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39th IFIP International Conference on ICT Systems Security and Privacy Protection, SEC 2024

Item Type Journal Article

Editor Pitropakis N.

Editor Katsikas S.

Editor Furnell S.

Editor Markantonakis K.

Abstract The proceedings contain 34 papers. The special focus in this conference is on ICT Systems Security and Privacy Protection. The topics include: Reduce to the MACs - Privacy Friendly Generic Probe Requests; PRIDA: PRIVacy-Preserving Data Aggregation with Multiple Data Customers; keep Your Memory Dump Shut: Unveiling Data Leaks in Password Managers; DryJIN: Detecting Information Leaks in Android Applications; bruteware: A Novel Family of Cryptoviral Attacks; towards Practical Hardware Fingerprinting for Remote Attestation; examining the Strength of Three Word Passwords; can Synthetic Data Preserve Manifold Properties?; "Alexa, How Do You Protect My Privacy?" A Quantitative Study of User Preferences and Requirements About Smart Speaker Privacy Settings; putting Authorization Servers on User-Owned Devices in User-Managed Access; identification of Cyber Threats and Vulnerabilities in Norwegian Distribution Networks; chain of Trust: Unraveling References Among Common Criteria Certified Products; lightArmor: A Lightweight Trusted Operating System Isolation Approach for Mobile Systems; satellite: Effective and Efficient Stack Memory Protection Scheme for Unsafe Programming Languages; Neurosymbolic Learning in the XAI Framework for Enhanced Cyberattack Detection with Expert Knowledge Integration; Transforming EU Governance: The Digital Integration Through EBSI and GLASS; malicious Insider Threat Detection Using Sentiment Analysis of Social Media Topics; a Structural-Semantic Approach Integrating Graph-Based and Large Language Models Representation to Detect Android Malware; Session Replication Attack Through QR Code Sniffing in Passkey CTAP Registration; GAD: A Real-Time Gait Anomaly Detection System with Online Adaptive Learning; queuing Theoretic Analysis of Dynamic Attribute-Based Access Control Systems; IPEQ: Querying Multi-attribute Records with Inner Product Encryption; hiding Your Awful Online Choices Made More Efficient and Secure: A New Privacy-Aware Recommender System; cognition Behind Access Control: A Usability Comparison of Rule- and Category-Based Mechanisms; obfuscating Code Vulnerabilities Against Static Analysis in Android Apps.

Date 2024

URL <https://www.scopus.com/pages/publications/85200775445?origin=resultslist>

Archive Scopus

Language English

Volume 710

Publisher Springer Science and Business Media Deutschland GmbH

Publication IFIP Advances in Information and Communication Technology

ISSN 18684238 (ISSN); 978-303165174-8 (ISBN)

Journal Abbr IFIP Advances in Information and Communication Technology

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Export Date: 20 September 2026; Cited By: 0; Conference name: 39th IFIP International Conference on ICT Systems Security and Privacy Protection, SEC 2024; Conference date: 12 June 2024 through 14 June 2024; Conference code: 316289

45th European Conference on Information Retrieval, ECIR 2023

Item Type Journal Article

Editor Kamps J.

Editor Goeuriot L.

Editor Crestani F.

Editor Maistro M.

Editor Joho H.

Editor Davis B.

Editor Gurrin C.

Editor Caputo A.

Editor Kruschwitz U.

Abstract The proceedings contain 45 papers. The special focus in this conference is on Information Retrieval. The topics include: Towards Detecting Interesting Ideas Expressed in Text; towards Linguistically Informed Multi-objective Transformer Pre-training for Natural Language Inference; dirichlet-Survival Process: Scalable Inference of Topic-Dependent Diffusion Networks; consumer Health Question Answering Using Off-the-Shelf Components; MOO-CMDS+NER: Named Entity Recognition-Based Extractive Comment-Oriented Multi-document Summarization; fragmented Visual Attention in Web Browsing; Weibull Analysis of Item Visit Times; don't Raise Your Voice, Improve Your Argument: Learning to Retrieve Convincing Arguments; learning Query-Space Document Representations for High-Recall Retrieval; investigating Conversational Search Behavior for Domain Exploration; evaluating Humorous Response Generation to Playful Shopping Requests; joint Span Segmentation and Rhetorical Role Labeling with Data Augmentation for Legal Documents; trigger or not Trigger: Dynamic Thresholding for Few Shot Event Detection; The Impact of a Popularity Punishing Hyperparameter on ItemKNN Recommendation Performance; neural Ad-Hoc Retrieval Meets Open Information Extraction; augmenting Graph Convolutional Networks with Textual Data for Recommendations; utilising Twitter Metadata for Hate Classification; topic-Enhanced Personalized Retrieval-Based Chatbot; evolution of Filter Bubbles and Polarization in News Recommendation; capturing Cross-Platform Interaction for Identifying Coordinated Accounts of Misinformation Campaigns; improving the Generalizability of the Dense Passage Retriever Using Generated Datasets; segmentCodeList: Unsupervised Representation Learning for Human Skeleton Data Retrieval; knowing What and How: A Multi-modal Aspect-Based Framework for Complaint Detection; preface; Extractive Summarization of Financial Earnings Call Transcripts: Or: When GREP Beat BERT; what Is Your Cause for Concern? Towards Interpretable Complaint Cause Analysis; DeCoDE: Detection of Cognitive Distortion and Emotion Cause Extraction in Clinical Conversations; domain-Aligned Data Augmentation for Low-Resource and Imbalanced Text Classification; privacy-Preserving Fair Item Ranking; multimodal Geolocation Estimation of News Photos; topics in Contextualised Attention Embeddings; Probing BERT for Ranking Abilities.

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Language English
Volume 13982 LNCS
Publisher Springer Science and Business Media Deutschland GmbH
Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)
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45th European Conference on Information Retrieval, ECIR 2023

Item Type Journal Article

Editor Kamps J.

Editor Goeuriot L.

Editor Crestani F.

Editor Maistro M.

Editor Joho H.

Editor Davis B.

Editor Gurrin C.

Editor Caputo A.

Editor Kruschwitz U.

Abstract The proceedings contain 45 papers. The special focus in this conference is on Information Retrieval. The topics include: Towards Detecting Interesting Ideas Expressed in Text; towards Linguistically Informed Multi-objective Transformer Pre-training for Natural Language Inference; dirichlet-Survival Process: Scalable Inference of Topic-Dependent Diffusion Networks; consumer Health Question Answering Using Off-the-Shelf Components; MOO-CMDS+NER: Named Entity Recognition-Based Extractive Comment-Oriented Multi-document Summarization; fragmented Visual Attention in Web Browsing; Weibull Analysis of Item Visit Times; don't Raise Your Voice, Improve Your Argument: Learning to Retrieve Convincing Arguments; learning Query-Space Document Representations for High-Recall Retrieval; investigating Conversational Search Behavior for Domain Exploration; evaluating Humorous Response Generation to Playful Shopping Requests; joint Span Segmentation and Rhetorical Role Labeling with Data Augmentation for Legal Documents; trigger or not Trigger: Dynamic Thresholding for Few Shot Event

Detection; The Impact of a Popularity Punishing Hyperparameter on ItemKNN Recommendation Performance; neural Ad-Hoc Retrieval Meets Open Information Extraction; augmenting Graph Convolutional Networks with Textual Data for Recommendations; utilising Twitter Metadata for Hate Classification; topic-Enhanced Personalized Retrieval-Based Chatbot; evolution of Filter Bubbles and Polarization in News Recommendation; capturing Cross-Platform Interaction for Identifying Coordinated Accounts of Misinformation Campaigns; improving the Generalizability of the Dense Passage Retriever Using Generated Datasets; segmentCodeList: Unsupervised Representation Learning for Human Skeleton Data Retrieval; knowing What and How: A Multi-modal Aspect-Based Framework for Complaint Detection; preface; Extractive Summarization of Financial Earnings Call Transcripts: Or: When GREP Beat BERT; what Is Your Cause for Concern? Towards Interpretable Complaint Cause Analysis; DeCoDE: Detection of Cognitive Distortion and Emotion Cause Extraction in Clinical Conversations; domain-Aligned Data Augmentation for Low-Resource and Imbalanced Text Classification; privacy-Preserving Fair Item Ranking; multimodal Geolocation Estimation of News Photos; topics in Contextualised Attention Embeddings; Probing BERT for Ranking Abilities.

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Volume 13981 LNCS

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45th European Conference on Information Retrieval, ECIR 2023

Item Type Journal Article

Editor Kamps J.

Editor Goeuriot L.

Editor Crestani F.

Editor Maistro M.

Editor Joho H.

Editor Davis B.

Editor Gurrin C.
Editor Caputo A.
Editor Kruschwitz U.

Abstract The proceedings contain 45 papers. The special focus in this conference is on Information Retrieval. The topics include: Towards Detecting Interesting Ideas Expressed in Text; towards Linguistically Informed Multi-objective Transformer Pre-training for Natural Language Inference; dirichlet-Survival Process: Scalable Inference of Topic-Dependent Diffusion Networks; consumer Health Question Answering Using Off-the-Shelf Components; MOO-CMDS+NER: Named Entity Recognition-Based Extractive Comment-Oriented Multi-document Summarization; fragmented Visual Attention in Web Browsing; Weibull Analysis of Item Visit Times; don't Raise Your Voice, Improve Your Argument: Learning to Retrieve Convincing Arguments; learning Query-Space Document Representations for High-Recall Retrieval; investigating Conversational Search Behavior for Domain Exploration; evaluating Humorous Response Generation to Playful Shopping Requests; joint Span Segmentation and Rhetorical Role Labeling with Data Augmentation for Legal Documents; trigger or not Trigger: Dynamic Thresholding for Few Shot Event Detection; The Impact of a Popularity Punishing Hyperparameter on ItemKNN Recommendation Performance; neural Ad-Hoc Retrieval Meets Open Information Extraction; augmenting Graph Convolutional Networks with Textual Data for Recommendations; utilising Twitter Metadata for Hate Classification; topic-Enhanced Personalized Retrieval-Based Chatbot; evolution of Filter Bubbles and Polarization in News Recommendation; capturing Cross-Platform Interaction for Identifying Coordinated Accounts of Misinformation Campaigns; improving the Generalizability of the Dense Passage Retriever Using Generated Datasets; segmentCodeList: Unsupervised Representation Learning for Human Skeleton Data Retrieval; knowing What and How: A Multi-modal Aspect-Based Framework for Complaint Detection; preface; Extractive Summarization of Financial Earnings Call Transcripts: Or: When GREP Beat BERT; what Is Your Cause for Concern? Towards Interpretable Complaint Cause Analysis; DeCoDE: Detection of Cognitive Distortion and Emotion Cause Extraction in Clinical Conversations; domain-Aligned Data Augmentation for Low-Resource and Imbalanced Text Classification; privacy-Preserving Fair Item Ranking; multimodal Geolocation Estimation of News Photos; topics in Contextualised Attention Embeddings; Probing BERT for Ranking Abilities.

Date 2023

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Language English

Volume 13980 LNCS

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Publication Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Export Date: 20 September 2026; Cited By: 0; Conference name: 45th European Conference on Information Retrieval, ECIR 2023; Conference date: 2 April 2023 through 6 April 2023; Conference code: 292029

2021 5th International Conference on Natural Language Processing and Information Retrieval, NLPPIR 2021

Item Type Journal Article

Abstract The proceedings contain 26 papers. The topics discussed include: aspect-based sentiment analysis of social media data with pre-trained language models; semantic preserving siamese autoencoder for binary quantization of word embeddings; annotation and evaluation of utterance intention tag for interview dialogue corpus; topic segmentation for interview dialogue system; feature extraction technique based on Conv1D and Conv2D network for Thai speech emotion recognition; architecture-based semantic description framework for model transformation; examination of the quality of Conceptnet relations for PubMed abstracts; prediction of number of likes and retweets based on the features of tweet text and images; scored and error-annotated essay dataset of Chinese EFL/ESL learners; pandemic rumor identification on social networking sites: a case study of COVID-19; research on judgment reasoning using natural language inference in Chinese medical texts; and a study of predicting the sincerity of a question asked using machine learning.

Date 2021

URL <https://www.scopus.com/pages/publications/85127249799?origin=resultslist>

Archive Scopus

Language English

Publisher Association for Computing Machinery

Publication ACM International Conference Proceeding Series

ISSN 978-145038735-4 (ISBN)

Journal Abbr ACM Int. Conf. Proc. Ser.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 5th International Conference on Natural Language Processing and Information Retrieval, NLPPIR 2021; Conference date: 17 December 2021 through 20 December 2021; Conference code: 177546

2023 5th International Conference on Computer Communication and the Internet, ICCCI 2023

Item Type Journal Article

Abstract The proceedings contain 45 papers. The topics discussed include: FSD: fully-specialized detector via neural architecture search; in-the-wild deepfake detection using adaptable CNN models with visual class activation mapping for improved accuracy; deep learning based efficient pedestrian age group estimation and tracking techniques for autonomous mover; a heterogeneous ensemble learning approach for leather species prediction; a comparative sentiment analysis about HIV and AIDS on twitter tweets using supervised machine learning approach; combining data resampling and DRL algorithm for intrusion detection; strengthening spatial relations to multi-scale features for few-shot learning; a deep learning-based white shrimp body length and bait detection methodology for turbid underwater aquaculture environment; cattle behavior classification based on integration of acceleration sensor and position information; and an efficient sharing of docker environments with updating and expanding over a hybrid peer-to-peer network.

Date 2023

URL <https://www.scopus.com/pages/publications/85170102215?origin=resultslist>

Archive Scopus

Language English

Publisher Institute of Electrical and Electronics Engineers Inc.

Publication 2023 5th International Conference on Computer Communication and the Internet, ICCCI 2023

ISSN 979-835032695-6 (ISBN)

Journal Abbr Int. Conf. Comput. Commun. Internet, ICCCI

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Export Date: 20 September 2026; Cited By: 0; Conference name: 5th International Conference on Computer Communication and the Internet, ICCCI 2023; Conference date: 23 June 2023 through 25 June 2023; Conference code: 191737

2025 11th International Conference on Web Research, ICWR 2025

Item Type Journal Article

Abstract The proceedings contain 93 papers. The topics discussed include: chatbot for student discipline handbook-related queries: a RAG-Based LLM using Llama-3 approach; gender bias mitigation of Persian stereotypical words using FastText; a comprehensive review on frequent pattern mining algorithms in data mining; a hybrid architecture for out of domain intent detection and intent discovery; employing attention-based BERT-BiRNN-CNN for Twitter sentiment analysis - a sentiment analysis and deep neural networks approach; retraction citation network analysis of Iranian computer scientists; assessing large language models as agile scrum masters: a comparative study of project planning efficiency; and user intent detection in Persian text-based chatbots: a comprehensive review of methods and challenges.

Date 2025

URL <https://www.scopus.com/pages/publications/105011583044?origin=resultslist>
Archive Scopus
Language English
Publisher Institute of Electrical and Electronics Engineers Inc.
Publication International Conference on Web Research, ICWR
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Issue 2025
Journal Abbr Int. Conf. Web Res., ICWR
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EID: 2-s2.0-105011583044

Export Date: 20 September 2026; Cited By: 0; Conference name: 11th International Conference on Web Research, ICWR 2025; Conference date: 16 April 2025 through 17 April 2025; Conference code: 209076

2025 Global Conference in Emerging Technology, GINOTECH 2025

Item Type Journal Article
Abstract The proceedings contain 311 papers. The topics discussed include: maximum degree diameter bounded subgraph problem for square-octagonal lattice; a packaging design approach based on the visual characteristics of the human eye; analysis and design of human resource management system based on data mining technology; enhanced medical image super-resolution using GANs and multi-modal fusion; deep learning-driven smart pedagogy and assessment: a quality framework for outcome based education in higher education; Internet of Things (IoT) devices encryption approaches: enabling secure communication possible in low-resource settings; and comparing random forest and XGBoost for sentiment classification of student social media posts: a case study on pre-board exam stress.
Date 2025
URL <https://www.scopus.com/pages/publications/105013638434?origin=resultslist>
Archive Scopus
Language English
Publisher Institute of Electrical and Electronics Engineers Inc.
Publication 2025 Global Conference in Emerging Technology, GINOTECH 2025
ISSN 979-833150775-6 (ISBN)
Journal Abbr Glob. Conf. Emerg. Technol., GINOTECH
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EID: 2-s2.0-105013638434

Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 IEEE International Global Conference in Emerging Technology, GINOTECH 2025; Conference date: 9 May 2025 through 11 May 2025; Conference code: 210563

2026 2nd International Conference on Artificial Intelligence Systems, AIS 2026

Item Type Journal Article

Editor O'Brien M.J.

Editor Alsmadi I.

Editor Jararweh Y.

Abstract The proceedings contain 76 papers. The topics discussed include: FedLoRA-C: communication-efficient federated fine-tuning of large language models via selective layer compression; layer adaptive error-bounded compression for federated learning on edge devices; intelligent privileged access management using predictive AI models for enterprise anomaly detection; enabling AI-driven personalization through unified engagement data at streaming scale; comparative analysis of action recognition techniques: exploring LSTM, zero-shot and one-shot learning models; sentiment classification of social media hashtags related to global conflicts using machine learning; sealed tokens, open throughput: LLM serving performance under hopper-class GPU enclaves; a data-driven machine learning framework for performance optimization in digital advertising analysis; and employing artificial intelligence technologies to enhance national identity among first cycle students.

Date 2026

URL <https://www.scopus.com/pages/publications/105046961825?origin=resultslist>

Archive Scopus

Language English

Publisher Institute of Electrical and Electronics Engineers Inc.

Publication 2026 2nd International Conference on Artificial Intelligence Systems, AIS 2026

ISSN 979-831950746-4 (ISBN)

Journal Abbr Int. Conf. Artif.l Intell. Syst., AIS

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EID: 2-s2.0-105046961825

Export Date: 20 September 2026; Cited By: 0; Conference name: 2nd International Conference on Artificial Intelligence Systems, AIS 2026; Conference date: 12 May 2026 through 15 May 2026; Conference code: 226532

A Benchmark Dataset for Cricket Sentiment Analysis in Bangla Social Media Text

Item Type Conference Paper

Author T. Mahmud

Author R. Karim

Author R. Chakma

Author T. Chowdhury

Author M.S. Hossain

Author K. Andersson

Editor Shakshuki E.

Abstract This study introduces a novel benchmark dataset designed for Cricket Sentiment Analysis on Bangla social media posts, emphasizing a low-resource setting. The dataset was meticulously curated through manual collection across diverse social media platforms, ensuring comprehensive representation of user sentiments. Annotations validated dataset quality, achieving a remarkable Cohen Kappa score of 0.97. Experimentation with machine learning (ML) models revealed challenges, with traditional approaches yielding modest RNN accuracy of 0.5239. However, deep learning (DL) models showcased significant performance enhancements. The LSTM model achieved 0.897 accuracy, while the BiLSTM model surpassed expectations at 0.952. These findings highlight DL's efficacy in capturing nuanced sentiments in Bangla cricket-related social media posts, contributing a high-quality benchmark dataset and insights into DL's suitability for sentiment analysis in low-resource linguistic contexts. © 2024 Elsevier B.V.. All rights reserved.

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Language English

Extra Journal Abbreviation: Procedia Comput. Sci.

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Conference Name Procedia Computer Science

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Learning systems, Machine-learning, Sentiment analysis, Machine learning, Language processing, Natural language processing, Natural languages, Social networking (online), Benchmarking, Social media, Sentiment Analysis, Long short-term memory, Machine Learning (ML), Benchmark datasets, Quality control, Low-resource settings, Polarity, Natural Language Processing(NLP)

Notes:

EID: 2-s2.0-85199563174

Export Date: 20 September 2026; Cited By: 42; Correspondence Address: T. Mahmud; Department of Computer Science and Engineering, Rangamati Science and Technology University, Bangladesh; email: tanjim_cse@yahoo.com; Conference name: 15th International Conference on Ambient Systems, Networks and Technologies Networks, ANT 2024 / The 7th International Conference on Emerging Data and Industry 4.0, EDI40 2024; Conference date: 23 April 2024 through 25 April 2024; Conference code: 200877

A Bert-based Model with Tsk Fuzzy Neural Network for Ai Comments Detection

Item Type Conference Paper**Author** T. Liu**Author** Q. Zhang

Abstract With the advancement of large language models (LLMs) and chatbot technologies, the application of AI-generated text on social media has become increasingly popular. Therefore, the detection of AI comments has become important. Traditional AI text detection methods are mainly applied to long texts like academic writings. To make up for this deficiency, our study focuses on comments, which are relatively short texts. We propose two BERT-based models with Takagi-Sugeno-Kang fuzzy neural network (TSK FNN) to detect AI-generated comments on social media platforms. The first model is just a TSK FNN (BERT-TSK for short), which uses BERT features extracted from the hidden layers and the sentiment value of each comment (obtained by the sentiment analysis model from Hugging Face) as inputs, and the second is a BERT-TSK hybrid model (BERT-TSK-H for short), which embeds TSK FNN into BERT hidden layers for jointly training. The study compares the performance of our proposed two models with the fine-tuned BERT model on Chinese and English comment datasets. The experimental results show that the proposed hybrid model BERT-TSK-H outperforms BERT-TSK and the fine-tuned BERT model in terms of most of the metrics, Accuracy, Precision, Recall and F1-score on the two datasets. © 2025 IEEE.

Date 2025**DOI** 10.1109/AANN66429.2025.11257727**URL** <https://www.scopus.com/pages/publications/105031094391?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Int. Conf. Adv. Algorithms Neural Networks, AANN**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833159736-8 (ISBN)**Pages** 298-303**Proceedings Title** Int. Conf. Adv. Algorithms Neural Networks, AANN**Conference Name** 2025 5th International Conference on Advanced Algorithms and Neural Networks, AANN 2025**Date Added** 20/09/2026, 8:35:24 pm

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Tags:

Language model, Sentiment analysis, Social networking (online), Chatbots, BERT, Fuzzy systems, Hybrid model, Error detection, AI discourse detection, Fuzzy inference, Fuzzy inference systems, fuzzy neural network, Fuzzy neural networks, Fuzzy-neural-networks, Hidden layers, Takagi-sugeno-kang fuzzy neural networks, TSK fuzzy inference system

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A bilingual Malay-English social media dataset for binary hate speech detection

Item Type Journal Article

Author J.-C. Tan

Author L.-Y. Ong

Author M.-C. Leow

Abstract In recent years, online hate speech has posed a growing threat to user safety, social harmony, and community cohesion especially on social media platforms. However, most existing hate speech datasets are monolingual and resource-rich, which leave Southeast Asia languages such as Malay underrepresented in natural language processing research. The aim of this dataset is to handle this gap by providing a balanced and quality-controlled resource that supports machine learning applications in multilingual settings. Binary classification is selected as a foundation task because it simplifies practical deployment in real-world and early-stage detection systems. It is beneficial in low-resource languages where detailed or multi-label annotations are always unavailable or inconsistent. This dataset presents 26,985 bilingual Malay-English social media texts curated from five public sources for binary hate speech detection. It combines human-annotated and filtered through controlled pseudo-labelling to retain only high-confidence, quality-controlled texts. The dataset is provided in UTF-8 encoded CSV format with 13,609 English and 13,376 Malay-language texts. Each entry includes the social media post, binary label (0 = non-hate, 1 = hate), language identifier (en or ms), and data source information. The dataset meets clear practical demands, including training multilingual transformer-based classifiers, benchmarking cross-lingual NLP models, and developing effective hate speech detection systems and educational NLP resources for English and Malay-speaking communities. © 2025 The Author(s).

Date 2025

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Language English
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Speech communication, Speech processing, Artificial intelligence, Classification (of information), Learning systems, Learning algorithms, Natural language processing systems, Social networking (online), Social media, Labelings, Linguistics, Speech analysis, Speech detection, Low resource languages, Bilinguals, Hate speech detection, Low-resource languages, Quality control, Pseudo-labeling, Community cohesions, Detection system, Malay-english bilingual data, Malay-English bilingual data, Pseudo-labelling

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Export Date: 20 September 2026; Cited By: 3; Funding details: Telekom Malaysia Research and Development, (RDTC/241125, MMUE/240066); Correspondence Address: L.-Y. Ong; Faculty of Information Science and Technology, Multimedia University, Melaka, 75450, Malaysia; email: lyong@mmu.edu.my

A Comparative Analysis of BiLSTM, CNN-BiLSTM, and Transformer Models for Bangla Sentiment Analysis

Item Type Conference Paper

Author M.M.I. Hasan

Author A. Das

Author S. Tasnim

Author T. Tarik

Author S.B. Sikto

Editor Shah M.S.

Abstract The morphology of Bangla, lack of labeled data and use of informal and code-mixed text in social media platforms make the task of sentiment analysis especially difficult. Although traditional machine learning methods and isolated deep learning architectures have both been tried on this task, they are usually afflicted by the problems of overfitting, poor generalization on unbalanced data, and a lack of the ability to simultaneously fit both local feature patterns and

sequential contextual influences. To overcome these shortcomings, this paper is a comparative analysis of three state-of-the-art deep learning architectures, namely a BiLSTM network with attention mechanism, a hybrid CNN-BiLSTM model, and a Transformer-CNN architecture. All models are applied to a processing pipeline that consists of a complete processing of the SentNoB dataset of noisy Bangali texts. The suggested systems were strictly tested by means of real experimental protocols. Transformer-CNN model had the highest results in terms of test accuracy (92.46%) and macro F1-score (92.47%), and performed better than the BiLSTM with Attention (91.94%) and CNN-BiLSTM Hybrid (90.90%) models. Moreover, the Transformer-CNN model was found to have the best discriminative ability with maximum ROC AUC scores (0.9880 micro, 0.9873 macro). The present work is informative about the issue of architectural choice and maximization to sentiment analysis in low-resource language environments. © 2026 IEEE.

Date 2026

DOI 10.1109/QPAIN69676.2026.11545819

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Language English

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Conference Name 2026 IEEE 2nd International Conference on Quantum Photonics, Artificial Intelligence and Networking, QPAIN 2026

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Tags:

Deep learning, Architecture, Learning systems, Sentiment analysis, Network architecture, Data mining, Deep Learning, Sentiment Analysis, Transformer, Comparative analyzes, Class imbalance, Labeled data, Pipeline processing systems, BiLSTM, Learning architectures, Bangla NLP, Class Imbalance, CNN-BiLSTM, Electric transformer testing, Sentnob dataset, SentNoB Dataset

Notes:

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A Comparative Analysis of Transformer and Traditional ML Models for Cyberbullying Detection on Twitter (now X)

Item Type Conference Paper

Author M. Abusaqer

Author J. Saquer

Editor Shahriar H.

Editor Alam K.S.

Editor Ohsaki H.

Editor Cimato S.

Editor Capretz M.

Editor Ahmed S.

Editor Ahamed S.I.

Editor Majumder A.J.A.

Editor Haque M.

Editor Yoshihisa T.

Editor Cuzzocrea A.

Editor Takemoto M.

Editor Sakib N.

Editor Elsayed M.

Abstract Cyberbullying on social media poses critical risks to mental health and public safety. This paper investigates advanced computational approaches for detecting and classifying cyberbullying in a large Twitter dataset of 47,692 tweets, labeled into six categories. We compare three transformer-based models (GPT-3.5, BERT, and RoBERTa) against three traditional machine learning algorithms (Naïve Bayes, SVM, and Random Forest), evaluating accuracy, precision, recall, F1-score, and computational efficiency. Our results indicate that RoBERTa achieves the highest overall performance (87-88% accuracy) but at a higher computational cost (13 hours on CPU), while Random Forest offers a strong balance between speed and performance (85.36% accuracy in 83 seconds). In contrast, the experiment using GPT-3.5 in a batched, zero-shot configuration achieved accuracy of 25.41%, an F1-score of 23.61%, and elapsed time of 5.14 hours, highlighting the challenges of applying generative models to cyberbullying detection without fine-tuning. These findings inform model selection for real-world deployment of cyberbullying detection systems, illuminating the trade-offs between transformer-based and traditional methods for automated social media monitoring. © 2025 IEEE.

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DOI 10.1109/COMPSAC65507.2025.00216

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Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Abusaqer; Minot State University, Department of Mathematics & Computer Science, Minot, United States; email: muhammad.abusaqer@minotstateu.edu; Conference name: 49th Annual IEEE Computers, Software, and Applications Conference, COMPSAC 2025; Conference date: 8 July 2025 through 11 July 2025; Conference code: 211715

A Comparative Sentiment Analysis of Airline Customer Reviews Using Bidirectional Encoder Representations from Transformers (BERT) and Its Variants

Item Type Journal Article

Author Z. Li

Author C. Yang

Author C. Huang

Abstract The applications of artificial intelligence (AI) and natural language processing (NLP) have significantly empowered the safety and operational efficiency within the aviation sector for safer and more efficient operations. Airlines derive informed decisions to enhance operational efficiency and strategic planning through extensive contextual analysis of customer reviews and feedback from social media, such as Twitter and Facebook. However, this form of analytical endeavor is labor-intensive and time-consuming. Extensive studies have investigated NLP algorithms for sentiment analysis based on textual customer feedback, thereby underscoring the necessity for an in-depth investigation of transformer architecture-based NLP models. In this study, we conducted an exploration of the large language model BERT and three of its derivatives using an airline sentiment tweet dataset for downstream tasks. We further honed this fine-tuning by adjusting the hyperparameters, thus improving the model's consistency and precision of outcomes. With RoBERTa distinctly emerging as the most precise and overall effective model in both the binary (96.97%) and tri-class (86.89%) sentiment classification tasks and persisting in outperforming others in the balanced dataset for tri-class sentiment classification, our results validate

the BERT models' application in analyzing airline industry customer sentiment. In addition, this study identifies the scope for improvement in future studies, such as investigating more systematic and balanced datasets, applying other large language models, and using novel fine-tuning approaches. Our study serves as a pivotal benchmark for future exploration in customer sentiment analysis, with implications that extend from the airline industry to broader transportation sectors, where customer feedback plays a crucial role. © 2023 by the authors.

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machine learning, natural language processing, sentiment analysis, airline customer service

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A Comparative Social Media Sentiment Analysis of Indonesia's Free Nutritious Meal Program (MBG)

Item Type Conference Paper

Author D. Irfansyah

Author A.A. Gozali

Author A.D. Ciftady

Editor Wibowo F.W.

Abstract The Free Nutritious Meal Program (MBG) program is a strategic national initiative introduced by the Indonesian government to improve student nutrition. Its expansion in late 2025 triggered extensive public discourse across social media. However, previous sentiment analysis studies on MBG have largely relied on small datasets and single platforms, utilizing traditional methods that struggle with informal Indonesian text. This study addresses these limitations by

performing a comparative analysis of public sentiment on X (Twitter), YouTube, and TikTok using a massive dataset of 360,809 comments. The methodology utilizes the Indonesian RoBERTa Base Sentiment Classifier for sentiment inference and BERTopic for thematic extraction. This study is among the first to systematically compare MBG sentiment across text-based and video-centric platforms simultaneously. Results show distinct sentiment profiles: X (Twitter) exhibits the highest positive sentiment \$(38.74 \%)\$ driven by organized support, while YouTube displays the highest negative sentiment (30.13%) due to critical reactions. BERTopic reveals that negative sentiment stems from corruption concerns, while positive sentiment focuses on presidential support and inclusivity. These findings demonstrate the value of multi-platform monitoring using language-specific LLMs, enabling policymakers to formulate targeted responses to societal concerns. © 2026 IEEE.

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Language English

Extra Journal Abbreviation: Proceeding - ISIBER : Int. Semin. Intell. Bus. Edge-Comput. Res.

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Public policy, Information systems, Sentiment analysis, Social networking (online), Social media, Information use, sentiment analysis, YouTube, Bertopic, BERTopic, Indonesia, Negative sentiments, indonesian roberta, Indonesian robertum, Large language modeld, large language modelds, MBG program, National initiatives, public policy

Notes:

EID: 2-s2.0-105037592851

Export Date: 20 September 2026; Cited By: 0; Conference name: 2026 International Seminar on Intelligent Business and Edge-Computing Research, ISIBER 2026; Conference code: 222384

A Comparative Study of Lexicon-Based, Machine Learning, Deep Learning, and LLM Methods for Sentiment Analysis on Standard and Dialectal Arabic Texts

Item Type Conference Paper

Author C. Ferroud

Author M. Maghfour

Author A. Elouardighi

Editor Motahhir S.

Editor Bossoufi B.

Editor Guerrero J.M.

Abstract Social media has gained substantial interest and emerged as the main platforms for sharing information and expressing opinions. Getting insight from such opinions became one major concern. In this context, sentiment analysis is one application that studies opinions and feelings, which are mainly expressed in natural language. This paper proposes a comparative study of sentiment analysis of Arabic texts that are expressed in both modern standard Arabic and Moroccan dialect. Our methodology encapsulates four different approaches, namely, lexicon-based, Machine learning using two ensemble learning methods: Random Forest and XGBoost, a deep learning model based on the Bi-LSTM network and a large language model using BERT. This comprehensive methodology has several objectives. First, to illustrate the role of the preprocessing tasks adopted in each approach to address linguistic complexities. Second, to highlight the steps of feature representation, which is a crucial transitional step from unstructured text to structured data. Finally, to evaluate the performance of each model in sentiment classification based on the same datasets. The experimental results demonstrate the superiority of the advanced models, BERT and BI-LSTM compared to the classic Machine learning and lexicon methods. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Notes:

EID: 2-s2.0-105028353030

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: C. Ferroud; LM2CE Laboratory, FEG, Hassan 1st University, Settati, Morocco; email: c.ferroud@uhp.ac.ma; Conference name: 5th International Conference on Digital Technologies and Applications, ICDTA 2025; Conference date: 17 April 2025 through 18 April 2025; Conference code: 345159

A Comparative Study of Sentiment Classification Models for Greek Reviews

Item Type Journal Article

Author P.D. Michailidis

Abstract In recent years, people have expressed their opinions and sentiments about products, services, and other issues on social media platforms and review websites. These sentiments are typically classified as either positive or negative based on their text content. Research interest in sentiment analysis for text reviews written in Greek is limited compared to that in English. Existing studies conducted for the Greek language have focused more on posts collected from social media platforms rather than on consumer reviews from e-commerce websites and have primarily used traditional machine learning (ML) methods, with little to no work utilizing advanced methods like neural networks, transfer learning, and large language models. This study addresses this gap by testing the hypothesis that modern methods for sentiment classification, including artificial neural networks (ANNs), transfer learning (TL), and large language models (LLMs), perform better than traditional ML models in analyzing a Greek consumer review dataset. Several classification methods, namely, ML, ANNs, TL, and LLMs, were evaluated and compared using performance metrics on a large collection of Greek product reviews. The empirical findings showed that the GreekBERT and GPT-4 models perform significantly better than traditional ML classifiers, with BERT achieving an accuracy of 96% and GPT-4 reaching 95%, while ANNs showed similar performance to ML models. This study confirms the hypothesis, with the BERT model achieving the highest classification accuracy. © 2024 by the author.

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Export Date: 20 September 2026; Cited By: 16; Correspondence Address: P.D. Michailidis; Department of Balkan, Slavic and Oriental Studies, University of Macedonia, Thessaloniki, 54636, Greece; email: pmichailidis@uom.edu.gr

A Comparative Study of Three Sentiment Classification Models for Romanized Kannada Using Stacking Ensembles

Item Type	Conference Paper
Author	T.S. Shobana
Author	K.B. Rashmi
Author	A.N. Rajatha
Author	A.N. Rohit
Author	S. Namitha
Abstract	The sentiment analysis of Indian regional languages puts forward significant challenges, because of romanization, code-mixing, and limited availability of datasets which are already labelled. In this study, we are going to investigate the sentiment classification of Romanized Kannada text. To do this, we will use ensemble machine learning approaches. A multi-strategy framework which incorporates three-class, binary and hierarchical classification models. These are supported by advanced stacking ensemble techniques along with statistical feature engineering and TF-IDF vectorization. Our study involves a corpus of 5,353 Romanized Kannada social media texts samples, categorized into Positive, Negative, and Mixed Feelings (Neutral) categories. Highest accuracy of 88.96 % is achieved by the hierarchical two-stage model. It outperforms all the conventional approaches. The binary classification model has achieved an accuracy of 82.62 %, whereas the direct three-class model reached 72.16 %. The proposed framework leads to low-resource language processing by removing the need for native script datasets. This is particularly suitable for social media analytics where romanization is prevalent. © 2026 IEEE.
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Extra	Journal Abbreviation: Proc. Int. Conf. Sentim. Anal. Deep Learn., ICSADL
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A Comparative Study of Transformer-Based Models and Machine Learning Techniques for Enhancing Bangla Sentiment Analysis

Item Type Conference Paper
Author M.S.I.K. Hamim
Author I. Ahmed
Author T.I. Sazid
Author F. Haque
Author M.N. Hossain
Author S.R. Pal
Author M.S. Tahosin
Author M.A. Sheakh
Abstract Social media monitoring alongside business intelligence needs accurate sentiment analysis because the digital activity of Bangla-speaking communities keeps expanding. The current classification models do not deliver effective Bangla context analysis mainly because of insufficient annotated datasets when processing rare languages. The research examines sentiment classification through the combination of deep learning transformers, Flair, TARS (TaskAware Representation of Sentences), and GPT-2 (Generative Pre-training Transformer) with traditional machine learning algorithms such as LR (Logistic Regression), SVC (Support Vector Classifier), RF (Random Forest), XGB (XGBoost), DT (Decision Tree), KNN (K-Nearest Neighbors), and NB (Naive Bayes). During tests on our Bangla dataset, the Flair model implementing Bangla BERT Base

embeddings produced the best performance of 94.21% accuracy, which surpassed both TARS at 88.5% and traditional classifiers. Embedded into contexts selects complex sentiment patterns efficiently, which proves integral to boosting low-resource NLP (Natural Language Processing) task performance. For better sentiment classification in practical settings, researchers will concentrate on enhancing the efficiency and diversity of datasets and implementing crosslingual capabilities. © 2025 IEEE.

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A Comparative Study of X Data About the NHS Using Sentiment Analysis

Item Type Journal Article

Author S.U. Rehman

Author O.O. Blessing

Author A. Ali

Abstract This study investigates sentiment analysis of X data about the National Health Service (NHS) during a politically charged period, using lexicon-based, machine learning, and deep learning approaches, as well as topic modelling and aspect-based

sentiment analysis (ABSA). This study is distinct in its comparative evaluation of sentiment analysis techniques on NHS-related tweets during a politically sensitive period, offering insights into public opinion shaped by political discourse. A dataset of 35,000 tweets collected and analysed using various techniques, including VADER, TextBlob, Naive Bayes, Support Vector Machines, Logistic Regression, Ensemble Learning, and BERT. Unlike previous studies that focus on structured feedback or general sentiment, this research uniquely explores unstructured public discourse during an election period, capturing real-time political sentiment towards NHS policies. The sentiment distribution from lexicon-based methods depicted that the presence of stop words could affect model performance. While all models achieved high accuracy on the validation dataset, challenges such as class imbalance and limited labelled data impacted performance, with signs of overfitting observed. Topic modelling identified nine topic clusters, with “waiting list,” “service,” and “immigration” carrying negative sentiments. At the same time, words like “thank,” “support,” “care,” and “team” had the most positive sentiments, reflecting public delight in these areas. ABSA identified positive sentiments towards aspects like “useful service”. This study contributes a comparative framework for evaluating sentiment analysis techniques in politically contextualised healthcare discourse, offering insights for policymakers and researchers. The study underscores the importance of data quality in sentiment analysis. Future research should consider incorporating multilingual datasets, extending data collection periods, optimising deep learning models, and employing hybrid approaches to enhance performance. © 2025 by the authors.

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EID: 2-s2.0-105020194302

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: A. Ali; Faculty of Science and Engineering, Swansea University, Swansea, Bay Campus, SA1 8EN, United Kingdom; email: anwar.ali@swansea.ac.uk

A Comparison of ChatGPT and Fine-Tuned Open Pre-Trained Transformers (OPT) Against Widely Used Sentiment Analysis Tools: Sentiment Analysis of COVID-19 Survey Data

Item Type Journal Article

Author J.A. Lossio-Ventura

Author R. Weger

Author A.Y. Lee

Author E.P. Guinee

Author J. Chung

Author L. Atlas

Author E. Linos

Author F. Pereira

Abstract Background: Health care providers and health-related researchers face significant challenges when applying sentiment analysis tools to health-related free-text survey data. Most state-of-the-art applications were developed in domains such as social media, and their performance in the health care context remains relatively unknown. Moreover, existing studies indicate that these tools often lack accuracy and produce inconsistent results. Objective: This study aims to address the lack of comparative analysis on sentiment analysis tools applied to health-related free-text survey data in the context of COVID-19. The objective was to automatically predict sentence sentiment for 2 independent COVID-19 survey data sets from the National Institutes of Health and Stanford University. Methods: Gold standard labels were created for a subset of each data set using a panel of human raters. We compared 8 state-of-the-art sentiment analysis tools on both data sets to evaluate variability and disagreement across tools. In addition, few-shot learning was explored by fine-tuning Open Pre-Trained Transformers (OPT; a large language model [LLM] with publicly available weights) using a small annotated subset and zero-shot learning using ChatGPT (an LLM without available weights). Results: The comparison of sentiment analysis tools revealed high variability and disagreement across the evaluated tools when applied to health-related survey data. OPT and ChatGPT demonstrated superior performance, outperforming all other sentiment analysis tools. Moreover, ChatGPT outperformed OPT, exhibited higher accuracy by 6% and higher F-measure by 4% to 7%. Conclusions: This study demonstrates the effectiveness of LLMs, particularly the few-shot learning and zero-shot learning approaches, in the sentiment analysis of health-related survey data. These results have implications for saving human labor and improving efficiency in sentiment analysis tasks, contributing to advancements in the field of automated sentiment analysis. © 2024 JMIR Publications Inc.. All rights reserved.

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A Comparison of Lexicon-based and Transformer-based Sentiment Analysis on Code-mixed of Low-Resource Languages

Item Type Conference Paper
Author C. Tho
Author Y. Heryadi
Author I.H. Kartowisastro
Author W. Budiharto
Abstract Sentiment analysis from code-mixed texts has been gaining wide attention in the past decade from researchers and practitioners from various communities motivated, among others, by the increasing popularity of social media resulted in a huge volume of code-mixed texts. Sentiment analysis is an interesting problem in Natural Language Processing with wide potential applications, among others, to understand public concerns or aspirations toward some issues. This paper presents experimentation results aim to compare performance of lexicon-based

and Sentence-BERT as sentiment analysis models from code-mixed of low-resources texts as input. In this study, some code-mixed texts of Bahasa Indonesia and Javanese language are used as sample of low-resource code-mixed languages. The input dataset are first translated to English using Google Machine Translation. The Sentiwordnet and VADER are two English lexicon label datasets used in this study as basis for predicting sentiment category using lexicon-based sentiment analysis method. In addition, a pretrained Sentence-BERT model is used as classification model from the translated input text to English. In this study, the dataset is categorized into positives and negative categories. The model performance was measured using accuracy, precision, recall, and F1 score. The experimentation found that the combined Google machine translator and Sentence-BERT model achieved 83 % average accuracy, 90 % average precision, 76 % average recall, and 83 % average F1 Score. © 2021 IEEE.

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A Comprehensive Review on Sentiment Analysis in Low-Resource and Code-Mixed Languages

Item Type Conference Paper

Author G. Suresh

Abstract As the world is getting digital day by day, sentiment analysis in data scarce and code-mixed languages has become crucial since more people are expressing their opinions on various digital platforms, along with mixing English with their local languages. Since there are very limited dataset, repeated transliteration on these regional languages, producing a sentiment classification would be more difficult compared to resource-rich languages. Because of these reasons, currently existing models face problems to maintain robust performance, especially in social media, feedback of customers where code-mixing is usual. This paper therefore reviews 20 papers on sentiment analysis for low-resource and code-mixed languages, focusing on different datasets used, various model building approaches and their evaluation strategies. A growing trend towards neural and transformer-based architectures, as well as problems occurred due to low training data availability in some languages, insufficient labelled corpora, differences of sentiment across different contexts were discovered after study of these papers. The primary objective of this review is to consolidate the current findings to address methodological gaps while providing future directions to advance sentiment analysis across neglected linguistic contexts. © 2026 IEEE.

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A Context-Aware Text Sentiment Classification Framework Leveraging Pretrained Large Language Models and Transfer Learning for Large-Scale Opinion Mining

Item Type Journal Article

Author R.Y. Totare

Author V.N. Patil

Author R. Golhar

Author C. Patil

Abstract The proliferation of user-generated digital text across social media platforms, e-commerce ecosystems, and online discourse forums has established large-scale opinion mining as a critical capability for organisations seeking actionable intelligence from unstructured data. Classical lexicon-driven and shallow machine learning approaches to sentiment classification are demonstrably insufficient for capturing the compositional, context-dependent, and often irony-laden polarity signals embedded in contemporary natural language. The advent of self-supervised pretrained transformer architectures—most notably the BERT family and its successors—has redefined the performance ceiling for sentiment analysis; however, significant gaps persist in exploiting contextual disambiguation at the document level, adapting general-purpose language representations to domain-specific opinion vocabularies, and preserving classification performance under low-resource conditions. Despite impressive benchmark scores, existing transfer-learning frameworks for sentiment classification encounter four persistent limitations: (i) shallow context integration that fails to model long-range sentiment dependencies spanning multiple clauses; (ii) domain mismatch between pretraining corpora and target-domain lexicons resulting in systematic misclassification of domain-specific sentiment expressions; (iii) poor calibration under distributional shift, manifesting as overconfident incorrect predictions on adversarial and out-of-domain inputs; and (iv) prohibitive computational cost of full fine-tuning for large-scale transformer models in resource-constrained deployment environments. This dissertation introduces the Context-Aware Transfer Language Model (CAT-LLM) framework, integrating DeBERTa-v3-Large as the contextual encoder with three complementary components: a Context-Aware Position Encoding (CAPE) module that augments disentangled attention with document-level positional signals; a parameter-efficient Low-Rank Adaptation (LoRA) fine-tuning strategy with layer-wise learning-rate decay; and a Monte Carlo Dropout uncertainty estimation head providing calibrated confidence scores alongside sentiment predictions. Evaluated across six benchmark datasets—SST-2, IMDb, Yelp-5, Amazon Reviews, Twitter SemEval, and SemEval 2017—the CAT-LLM framework achieved macro F1-scores of 0.962, 0.958, 0.924, 0.941, 0.887, and 0.903 respectively, surpassing all baselines including RoBERTa-Large and XLNet-Large by margins of 1.5–7.5%. AUC-ROC reached 0.984. With only 10 training examples, CAT-LLM achieved 71.2% accuracy—12.7 percentage points above BiLSTM—demonstrating exceptional few-shot transferability. The CAT-LLM framework establishes a new state of the art for large-scale opinion mining, reconciling contextual expressiveness, parameter efficiency, and uncertainty quantification within a unified, production-deployable sentiment analysis pipeline. © 2026, Nexora Academic Press. All rights reserved.

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A Dataset for Addressing Patient's Information Needs related to Clinical Course of Hospitalization

Item Type Journal Article

Author S. Soni

Author D. Demner-Fushman

Abstract Patient's unique information needs about their hospitalization can be addressed using clinical evidence from electronic health records (EHRs) and artificial intelligence (AI). However, robust datasets to assess the factuality and relevance of AI-generated responses are lacking and, to our knowledge, none capture patient information needs in the context of their EHRs. To address this gap, we introduce ArchEHR-QA, an expert-annotated dataset of 134 cases from intensive care unit and emergency department settings. Cases comprise patient questions from public health forums, clinician-interpreted versions, relevant clinical note excerpts with sentence-level relevance annotations, and clinician-authored answers. To establish benchmarks for grounded EHR question answering (QA), we evaluated three open-weight large language models (Llama 4, Llama 3, and Mixtral) across three prompting strategies. We assessed performance on two dimensions: Factuality (overlap between cited and ground truth evidence) and Relevance (similarity to reference answers). © This is a U.S. Government work and not under copyright protection in the US; foreign copyright protection may apply 2026.

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A Deep Learning–Based Framework for Dataset Creation and Sentiment Classification of English–Bengali Code-Mixed Texts

Item Type Journal Article

Author D. Barua

Author T.S. Walia

Abstract Datasets form the foundation for most Natural Language Processing (NLP) operations, such as sentiment analysis, summarization, and translation. Lack of big and diverse datasets poses a critical problem for lowresource languages like Bengali. The problem is aggravated in code-mixed environments where users tend to intermix Bengali and English on web platforms like social media, video comments, and product reviews on e-commerce websites. It is very difficult to conduct sentiment analysis of such Bengali–English codemixed reviews, as there are no high-quality, large-scale datasets available. To address this challenge, the present research introduces a novel synthetic English–Bengali code-mixed product review dataset, specifically curated for sentiment analysis in the Bangladeshi e-commerce domain. The dataset was generated through a new deep learning–driven pipeline integrating part-of-speech tagging, translation, transliteration, and a two-tier ensemble-based sentiment annotation framework to ensure linguistic diversity and contextual realism in code-mixed expressions. The dataset, En–Bn–Code–Mixed–Two–Class–

Sentiment–Dataset–, is available on the Hugging Face repository, with a total of 100,000 reviews, balanced for both positive and negative sentiments. The quality and reliability of the generated dataset were validated through quantitative, linguistic, and statistical analyses, including translation, transliteration, and sentiment analysis evaluation metrics. The proposed two-tier ensemble sentiment annotation approach, implemented through deep learning models, meta-learning models, and majority voting, achieved 93.00% accuracy, 92.16% precision, 94.00% recall, and 93.07% F1-score. This work not only provides a useful resource for code-mixed Bengali–English NLP but also establishes a scalable methodology for low-resource and multilingual text processing, opening up further possibilities of more comprehensive and inclusive studies on sentiment analysis. © The authors

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A deep Natural Language Inference predictor in Italian without Italian training data

Item Type Conference Paper

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Author A. Manenti

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Editor Via G. Moruzzi 1, Pisa Institute of Information Science and Technologies "A. Faedo" of the National Research Council of Italy

Abstract We present an application in the sphere of Natural Language Processing to tackle the problem of inference relation (NLI) between pairs of sentences in Italian without an Italian NLI training dataset. The model has been developed making use of the Knowledge Distillation technique. The model may be exploited in industrial scenarios to analyse any kind of user-generated content. Specifically, applications may include product or service review analysis, brand reputation analysis, or social media sentiment analysis. To support this claim, we evaluate the proposed architecture over the native Italian ABSITA dataset on different tasks, such as Aspect-Based Sentiment Analysis, Sentiment Analysis, and Topic Recognition. We balanced the dataset used for each task to obtain a 50/50 division, and we respectively achieve the performances in terms of accuracy of 94.03%, 88.12%, and 71.19%. By applying the model over different scenarios, we empirically demonstrate the generality and exploitability of the NLI technique. While the developed model has still room for improvement, it is suitable to operate unsupervised free-text information extraction - mainly over reviews - in a production environment. © 2022 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

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A deep neural network based multi-task learning approach to hate speech detection

Item Type Journal Article

Author P. Kapil

Author A. Ekbal

Abstract With the advent of the internet and numerous social media platforms, citizens now have enormous opportunities to express and share their opinions on various societal and political issues. This phenomenal growth of the internet, social media networks, and messaging platforms provide plenty of opportunities for building intelligent systems, but these are also being heavily misused by certain groups who often disseminate offensive, racial, and hate speeches. Hence, detecting hate speech at the right time plays a crucial role as its spread might affect social fabrics. In recent times, although a few benchmark datasets have emerged for hate speech detection, these are limited in volume and also do not follow any uniform annotation schema. In this paper, a deep multi-task learning (MTL) framework is proposed to leverage useful information from multiple related classification tasks in order to improve the performance of the individual task. The proposed multi-task model is based on the shared-private scheme that assigns shared and private layers to capture the shared-features and task-specific features from five classification tasks. Experiments¹ on the 5 datasets show that the proposed framework attains encouraging performance in terms of macro-F1 and weighted-F1. © 2020 Elsevier B.V.

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A Fine-Grained Emotion Prediction Method Based on Large Language Models

Item Type Conference Paper**Author** F. Li**Author** Y. Qu**Author** C. Feng**Author** X. Dou**Author** L. Li

Abstract Every day, much multimodal data emerges on social media. Previous studies have focused on the problem of fine-grained sentiment analysis in multimodal data. However, many multimodal data must be annotated for specific emotions to train fine-grained emotion prediction models. To efficiently train fine-grained emotion risk prediction models for multimodal data, we propose a few-shot learning-based method for multimodal fine-grained sentiment analysis. This model captures sentiment polarity in text using a pre-trained large language model, annotates a small amount of data, and generates a training dataset. Then, utilizing an intermediate layer fusion strategy, it combines representation vectors of text, images, videos, and audio to ultimately create a fine-grained emotion prediction model based on multimodal data. Experimental work on a real multimodal dataset, compared with random fine-grained sentiment analysis methods, shows that our proposed method has significant advantages in handling multimodal fine-grained sentiment analysis tasks and practical value. © 2024 IEEE.

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A Framework for Constructing Thai Sentiment Corpus using the Cosine Similarity Technique

Item Type Conference Paper
Author N. Khamphakdee
Author P. Seresangtakul
Abstract Unstructured data is growing rapidly due to the increase in various social media that allow individuals to express and write reviews in either a formal or informal style. Thus, in sentiment analysis, it becomes difficult to identify and analyze both positive and negative reviews. Thai is a low-resource language with few resources with which to conduct NLP research and the lack of a sentiment corpus. The main objective of this paper is to present a framework in which to construct a Thai sentiment corpus and sentiment polarity classification utilizing the cosine similarity technique. The proposed framework consists of three main steps: data

collection; data preprocessing; and sentiment similarity measurement. Initially, data collection generates a sub-step that is a manual sentiment that classifies polarity into positive and negative. Data preprocessing is then applied. Moreover, we also created a special database in which to store text tokenization, convert abbreviations, check spelling errors, and conduct stop-word removal. Lastly, sentiment similarity measurement was conducted between two reviews with a combination of TF-IDF and the cosine similarity technique. We evaluated our framework by employing training data incorporating 3, 129 reviews and testing data comprised of 1, 000 reviews. The experiment results demonstrated that the proposed framework achieved an accuracy of 81.2%. We further observed that the data preprocessing step herein significantly affected the accuracy of the sentence similarity measurements of the two reviews. © 2021 IEEE.

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A Framework for Identifying Opinion Holders and Sentiment Triggers in Code-Mixed Multilingual Text

Item Type Conference Paper

Author E. Francis

Author P. Praveen

Abstract The growing phenomena of code-mixed communication in multiple languages online and via social media have created great challenges for sentiment analysis systems, especially when the sentiment is implicit and context dependent. Conventional approaches for sentiment analysis are predominantly focused on the polarity classification and do not tend to consider who is expressing opinion and what are the linguistic/cultural factors that prompt sentiment. Such limitations make the interpretability lower and limit the usefulness of the sentiment analysis in a real-world multilingual scenario. Addressing the above challenges involves a well-defined approach that is able to model the cause-and-effect of opinion and sentiment from noise in informal text. A well-defined structure is presented for facilitating fine-grained sentiment understanding in code-mixed multilingual text through identifying the opinion holders and sentiment trigger words. The framework combines multilingual contextual representations with structured prediction modules in order to jointly extract sources of opinion together with sentiment-inducing elements such as implicit cues for which there does not exist an explicit sentiment-bearing word. Task-specific annotations to support structured task evaluation are created from publicly available code-mixed sentiment datasets. Experimental evaluation shows that structured modeling helps to improve performance when compared with the baseline multilingual and sequence labeling systems. The framework scores higher in precision, recall and F1-scores for identifying opinion holders and extracting sentiment triggers with significant improvements in implicit expressions of sentiment. Comparative analysis goes further by making a more explicit link between expressions of sentiment and sources and causes. The outcomes show the benefits of using structured opinion modelling as a more transparent and effective alternative to flat sentiment classification, which is amorous for applications anticipating explainable sentiment reasoning in multilingual and code-mixed settings. © 2026 IEEE.

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A GAN-Driven Transformer-based Approach for Sentiment Analysis of Social Media Posts

Item Type Conference Paper

Author J. Hossain

Author M.M. Hoque

Abstract Recently, there has been an influx of interest in sentiment analysis because of its many uses in various sectors and research domains. Understanding sentiment is vital for market research, public opinion analysis, and social impact assessment. However, existing sentiment analysis tools are primarily designed for high-resource languages (HRLs), limiting their efficacy in low-resource languages (LRLs), including Bengali. This work proposes a comprehensive framework that leverages the Generative Adversarial Network (GAN) to handle dataset imbalance, along with transformer-based models, to classify sentiment into positive, negative, and neutral categories. Machine learning (ML), deep learning (DL), and transformer-based baselines with GAN models are exploited to classify sentiment in Bengali social media posts. The evaluation results reveal that the GAN+Bangla-BERT-1 model achieved the highest micro F1-score of 70.52. © 2024 IEEE.

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A Generative AI Framework for Privacy-Aware Audience Perception Analysis: Leveraging Large Language Models for Comment Understanding and Content Profiling

Item Type Conference Paper

Author Q. Liu

Author J. Wu

Author R. Deng

Abstract Generative artificial intelligence has shown remarkable potential in natural language understanding, yet its application to user-generated content analysis raises important considerations regarding data privacy and content security. This paper proposes an end-to-end generative AI framework for privacy-aware audience perception analysis that processes user comments without requiring personally identifiable information. The framework leverages large language models (LLMs) across all analytical stages: temporal-context-enriched data collection, privacy-aware paraphrase augmentation, sentiment classification through prompt engineering, structured topic extraction, and audience profiling. A three-tier privacy design ensures data locality through anonymization, paraphrase-based text replacement, and support for on-premise open-weight model deployment. We collect anonymized comments and danmaku (with playback timestamps) from six Chinese documentaries on Bilibili spanning five categories. Three LLM variants (GPT-4o, Qwen-Max, DeepSeek-V3) are compared against strong supervised baselines (RoBERTa-wwm-ext, TextCNN, LDA) across four prompt strategies. GPT-4o with chain-of-thought prompting achieves 86.2% accuracy and 0.848 macro-F1, outperforming RoBERTa-wwm-ext by 4.9 percentage points. LLM-based topic extraction produces 61% higher coherence scores than LDA. Cost-benefit analysis reveals that open-weight

DeepSeek-V3 achieves comparable accuracy at 18× lower cost than GPT-4o, offering a practical deployment path for production scenarios. © 2026 IEEE.

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A Glimpse to Scalable LLM Architectures: Building Real-Time Sentiment Analyser with Kafka and FastAPI

Item Type Conference Paper

Author U. Çetin

Author Y.E. Gündoğmuş

Editor Adali E.

Abstract In today's digital world, the real-time generation and analysis of data, particularly on social media platforms, are becoming increasingly important. This study aims to design and evaluate a scalable architecture for a scenario where multiple users simultaneously perform sentiment analysis. Two different architectures were examined in the study: the Basic Architecture and the Proposed Architecture. The Basic Architecture processes and responds to user requests directly using FastAPI and Gunicorn. In contrast, the Proposed Architecture employs FastAPI, Kafka, and Redis to create a more complex and distributed structure. Test results demonstrate that the Proposed Architecture offers higher throughput and lower response times under high demand and user loads. This study presents significant findings on how to establish a scalable and high-performance infrastructure for large language models. © 2024 IEEE.

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A global–local relational graph attention networks for aspect-level sentiment analysis

Item Type Journal Article

Author M. Zhang

Author X. Li

Author X. Chen

Author Y. Du

Author D. Huang

Author S. Wang

Author Y. Fan

Abstract Aspect-level sentiment analysis pertains to a meticulous sentiment classification that aims to identify aspect terms within online reviews and forecast their associated sentiment polarities. Nevertheless, this undertaking presents difficulties primarily attributable to numerous aspect words within a sentence or intricate dependency relationships. In order to tackle these obstacles, we put forth a global–local relational graph attention network (GLR-GAT) for aspect-level sentiment analysis. This model achieves a comprehensive syntactic analysis of sentences by syntactic dependency trees. In order to improve the accuracy of sentiment analysis by capturing the interactive extraction and dependencies of aspects in a sentence, we propose a framework that considers the syntactic dependencies of a sentence, the interrelationships among aspect words, and the relationships between aspect words and the sentence, respectively. Moreover, we apply a multi-head attention mechanism to correlate these three types of relationships and highlight the important ones. Comparison experiments show that the proposed GLR-GAT model performs more remarkably than existing baseline models. The ablation study confirms that each part of the GLR-GAT model plays an important role. Compared with large language models, the GLR-GAT model also shows significant advantages on a Twitter dataset. Copyright © 2025. Published by Elsevier Ltd.

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A guard against ambiguous sentiment for multimodal aspect-level sentiment classification

Item Type Journal Article

Author Y. Wang

Author K. Sun

Author B. Shi

Author H. Wu

Author K. Zhang

Author B. Dong

Abstract Recent advances in multimodal learning have achieved state-of-the-art results in aspect-level sentiment classification by leveraging both text and image data. However, images can sometimes contain contradictory sentiment cues or convey complex messages, making it difficult to accurately determine the sentiment expressed in the text. Intuitively, we should only use image data to complement the text if the latter contains ambiguous sentiment or leans toward the neutral polarity. Therefore, instead of trying to forcefully use images as done in prior work, we develop a Guard against Ambiguous Sentiment (GAS) for multimodal aspect-level sentiment classification (MALSC). Built on a pretrained language model, GAS is equipped with a novel “ambiguity learning” strategy that focuses on learning the degree of sentiment ambiguity within the input text. The sentiment ambiguity then serves to determine the extent to which image information should be utilized for accurate sentiment classification. In our experiments with two benchmark twitter datasets, we found that GAS achieves a performance gain of up to 0.98% in macro-F1 score compared to recent methods in the task. Furthermore, we explore the efficacy of large language models (LLMs) in the MALSC task by employing the core ideas behind GAS to design tailored prompts. We show that multimodal LLMs such as LLaVA, when provided with GAS-principled prompts, yields a 2.4% improvement in macro-F1 score for few-shot learning on the MALSC task. © 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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A Human–AI Collaborative Pipeline for Decision Support in Urban Development Projects Based on Large-Scale Social Media Text Analysis

Item Type Journal Article

Author A.A. Kharlamov

Author M. Pilgun

Abstract The rapid growth of digital communication platforms has generated vast volumes of user-generated textual data and digital footprints, creating growing demand for scalable artificial intelligence systems capable of supporting evidence-based decision-making. This study proposes and evaluates a human–AI collaborative analytical pipeline for multi-class sentiment and aggression analysis of large-scale social media data ($N = 15,064$ messages) related to an urban infrastructure project. The proposed framework integrates standard NLP preprocessing, machine learning-based classifiers, temporal aggregation, and controlled large language model (LLM)-assisted classification within a structured analytical workflow that incorporates expert validation and oversight. A stratified manual validation procedure ($n = 301$) demonstrated substantial inter-annotator agreement ($\kappa = 0.70$) and stable multi-class classification accuracy (80%). The results indicate that combining sentiment polarity and aggression detection as complementary linguistic indicators improves sensitivity to shifts in discourse dynamics and enables early identification of emerging social tension. The study demonstrates the potential of human–AI collaborative analytical

frameworks for transparent, interpretable, and predictive large-scale social media analysis in decision-support contexts. © 2026 by the authors.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Pilgun; Department of General and Comparative-Historical Linguistics, Lomonosov Moscow State University, Moscow, 119991, Russian Federation; email: pilgunm@yandex.ru

A Hybrid Deep Learning-Machine Learning Stacking Model for Yemeni Arabic Dialect Sentiment Analysis

Item Type Journal Article

Author A.A.H. Brihi

Author M. Ghurab

Abstract With the rise of online communities, Yemeni Arabic has gained increasing exposure to written social media content. Nevertheless, sentiment analysis studies have largely centered on Modern Standard Arabic (MSA) and other regional varieties (e.g., Egyptian, Levantine, Gulf), leaving the Yemeni dialect understudied, with insufficient specialized resources. This study bridges this gap by introducing the largest sentiment-annotated corpus for the Yemeni Arabic dialect, consisting of 45,862 manually labeled Facebook comments collected from the Facebook pages of the main telecommunications companies in Yemen (Yemen Telecom, Yemen Mobile, YOU, and Sabafon). These comments comprise user feedback on public service matters related to these companies. Multiple reviewers annotated the data to ensure reliability, with disagreements resolved through discussion. Moreover, we constructed a novel Yemeni dialect sentiment lexicon, classifying words/phrases according to polarity. We systematically evaluate sentiment analysis approaches, including lexicon-based, machine learning, deep learning, and hybrid, in terms of user sentiment regarding Yemeni telecom services. This study implemented different advanced deep-learning models in our dataset to demonstrate superior performance compared to other methods. The experimental results indicate a hybrid approach that combines advanced deep learning models with a stacked approach, leveraging deep networks for hierarchical feature extraction and RF/LinearSVC as meta-learners, achieving the highest accuracies of 94.71% and 94.28%, respectively. This study provides a preliminary reference for sentiment analysis in low-resource Arabic dialects such as Yemeni. It highlights the effectiveness of stacked feature engineering and meta-learning in dialectal NLP tasks. © IEEE. 2013 IEEE.

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A Hybrid Feature-Enhanced IndoBERT Framework with Controlled Semi-Supervised Learning for Low-Resource Indonesian Hate Speech Detection

Item Type Journal Article

Author S. Saifullah

Author R. Dreżewski

Abstract Low-resource hate speech detection remains a challenging task for Indonesian social media due to limited labeled annotations, highly informal linguistic expressions, and substantial lexical variability. Under such conditions, purely supervised transformer models often suffer from unstable semantic generalization, while conventional pseudo-labeling methods are vulnerable to noisy unlabeled sample propagation. To address these limitations, this study proposes a hybrid feature-enhanced IndoBERT framework integrated with a controlled semi-supervised learning strategy. The proposed model combines contextual IndoBERT embeddings with abusive lexicon cues, handcrafted linguistic indicators, and TF-IDF–SVD statistical representations through a lightweight concatenation–projection feature fusion mechanism, while unlabeled data are incorporated via adaptive confidence thresholding and class-balanced pseudo-label selection to improve pseudo-label reliability. Extensive experiments were conducted under realistic low-resource supervision settings using only 5%, 10%, and 20% labeled data, and the proposed framework was systematically compared against representative baselines, including sparse lexical machine learning models, shallow neural architectures, multilingual transformers, IndoBERTweet, naive pseudo-labeling, and LLM-based prompting. The results show that model effectiveness is strongly supervision-dependent. Under the most extreme low-resource setting, compact statistical augmentation provides the most stable complementary signal, whereas under moderate low-resource supervision, the full hybrid representation combined with controlled semi-supervised learning yields the strongest and most consistent gains. The proposed Hybrid IndoBERT + controlled SSL framework outperforms all baselines at the 20% labeled setting, reaching an accuracy of 0.8654, Macro-F1 of 0.8633, and ROC-AUC of 0.9334. Additional analyses of pseudo-label reliability, calibration behavior, computational efficiency, and qualitative error patterns further show that the proposed framework improves low-resource robustness while maintaining comparable inference-time efficiency. These findings demonstrate that low-resource hate speech detection benefits most from the staged integration of contextual semantic modeling, interpretable linguistic cues, global lexical–statistical structure, and carefully regulated unlabeled data exploitation. Additional experiments using GPT-4o-mini and Llama-3.1-8B further demonstrate that the proposed framework remains competitive against general-purpose large language model prompting approaches under low-resource Indonesian hate speech detection scenarios. The proposed framework provides a practical and reproducible direction for hate speech detection in annotation-constrained social media environments. © 2026 by the authors.

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A Hybrid Lexicon–Transformer Framework for Sentiment, Emotion, and Context Classification in Moroccan Darija (TriLex-Darija)

Item Type Journal Article

Author S. El Ouahabi

Author S. El Ouahabi

Author E.W. Dadi

Abstract This article introduces TriLex-Darija, a large-scale affective lexicon suite and a hybrid lexicon–transformer framework for analyzing Moroccan Arabic (Darija) social media text across three complementary dimensions: sentiment, emotion, and pragmatic context. The resource is constructed from a corpus of 288,709 manually annotated comments and consists of three unigram lexicons, each mapping 147,565 words to normalized probability distributions over task-specific labels. We first evaluate a symbolic lexicon-based classifier (without machine learning) based on word-level score aggregation to assess the intrinsic quality of the proposed TriLex-Darija resource. Despite the absence of contextual modeling, this approach achieves competitive performance, demonstrating that corpus-derived lexical knowledge captures substantial affective information in Moroccan Darija. To further improve performance, we propose a unified hybrid framework that combines TriLex-Darija features with contextual embeddings extracted from MARBERT. All models are trained using a consistent LinearSVC classifier to ensure fair comparison and reproducibility. In addition to the symbolic model, we evaluate a lexicon-feature-based LinearSVC model, allowing a clear distinction between symbolic, feature-based, and hybrid approaches. Experimental results show that the hybrid model consistently outperforms both BERT-only and lexicon-feature-based baselines across all tasks. For sentiment classification, the hybrid model achieves a macro F1-score of 72.96%, compared to 59.15% for BERT-only and 67.94% for the lexicon-feature-based model. For emotion classification, it reaches 92.92%, outperforming BERT-only (79.17%) and lexicon-feature-based (89.55%) models. For pragmatic context classification, the hybrid model achieves 91.35%, compared to 75.93% for BERT-only and 87.72% for the lexicon-feature-based model. Bootstrap confidence intervals (95%) and McNemar’s tests confirm that all improvements are statistically significant ($p < 0.001$). Overall, TriLex-Darija demonstrates that combining lexical knowledge with contextual embeddings leads to robust, interpretable, and statistically validated affective models for Moroccan Darija in low-resource settings. © 2026 Copyright held by the owner/author(s).

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A Hybrid ML Algorithm Trading Model combining Sentiment, Technical Indicator and Risk Management

Item Type Conference Paper
Author S. Shrestha
Author C. Sanin
Author M. Hasan
Author V. Sebastian Ramirez
Author E. Szczerbicki
Editor Chen Y.-W.
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Abstract Large language models (LLMs) and sentiment-aware machine learning are examples of technological developments that have influenced algorithmic trading using non-traditional data sources such as social media. The unpredictable, non-

stationary nature of financial data makes stock market prediction difficult. Machine learning has emerged as a powerful paradigm for dynamic market adaptation. The machine learning model's ability to reduce trading risk and maximise profits has been limited by the current methodologies' frequent lack of regard for structured risk management. In this paper, we propose a hybrid model combining sentiment analysis and technical indicators along with market data to predict the trend in the market. We also explore the returns obtained from the model and compare them with our risk strategy and traditional buy-and-hold (B&H) strategy. The results show that machine learning can decisively outperform traditional methods if paired with efficient risk management, which contributes to our understanding of machine learning-driven trading. © 2025 Elsevier B.V.. All rights reserved.

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A Hybrid NER–Sentiment Model for Uzbek Texts: Integrating Lexical, Deep Learning, and Entity-Based Approaches

Item Type Journal Article

Author B. Saidov

Author V. Barakhnin

Author R. Saparbaev

Author Z. Narmuratov

Author R. Manzura

Author R. Zilolakhon

Author A. Atajanova

Abstract This work proposes a hybrid Uzbek sentiment analysis model (sometimes referred to as tonality analysis in the local literature) that integrates contextual text representations with named-entity information from an NER module and emoji-based emotional cues that are common in short online messages. To provide a comprehensive baseline comparison, we evaluate seven approaches—SVM, LSTM, mBERT, XLM-RoBERTa-base, mDeBERTa-v3, LaBSE, and the proposed hybrid model—covering both classical machine learning and modern multilingual transformer architectures for low-resource sentiment tasks. The overall pipeline begins with Uzbek-specific text normalization to reduce noise from informal spellings, transliteration variants, and inconsistent apostrophe usage. In parallel, the system performs explicit emoji extraction to capture affective signals that are often expressed non-verbally in social media texts. Next, we construct three complementary feature streams: a context encoder for sentence-level semantics, NER-driven entity features that encode entity mentions and types, and an emotion module that models emoji priors and their interaction with contextual meaning. These streams are fused into a unified representation and fed to a final classifier to predict sentiment polarity. Experiments on an Uzbek test set demonstrate that the hybrid model reaches an F1-score of 0.92, consistently outperforming text-only baselines. The results indicate that entity-aware and emoji-informed features improve robustness under sarcasm/irony, mixed sentiment with multiple targets, and orthographic noise, making the approach suitable for social media analytics, public opinion monitoring, customer feedback triage, and recommendation-oriented text mining. © 2026 by the authors.

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A hybrid Persian sentiment analysis framework: Integrating dependency grammar based rules and deep neural networks

Item Type Journal Article

Author K. Dashtipour

Author M. Gogate

Author J. Li

Author F. Jiang

Author B. Kong

Author A. Hussain

Abstract Social media hold valuable, vast and unstructured information on public opinion that can be utilized to improve products and services. The automatic analysis of such data, however, requires a deep understanding of natural language. Current sentiment analysis approaches are mainly based on word co-occurrence frequencies, which are inadequate in most practical cases. In this work, we propose a novel hybrid framework for concept-level sentiment analysis in Persian language, that integrates linguistic rules and deep learning to optimize polarity detection. When a pattern is triggered, the framework allows sentiments to flow from words to concepts based on symbolic dependency relations. When no pattern is triggered, the framework switches to its subsymbolic counterpart and leverages deep neural networks (DNN) to perform the classification. The proposed framework outperforms state-of-the-art approaches (including support vector machine, and logistic regression) and DNN classifiers (long short-term memory, and Convolutional Neural Networks) with a margin of 10–15% and 3–4% respectively, using benchmark Persian product and hotel reviews corpora. © 2019

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A Hybrid, Transparent Trust and Risk Assessment Framework for Cryptocurrency Exchanges

Item Type Conference Paper

Author B. Mawhayi

Author J. Botha

Author L. Leenen

Editor Furnell S.

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Abstract Cryptocurrency exchanges act as critical intermediaries within the digital asset ecosystem, yet users currently rely on largely opaque, platform-defined trust scores to assess their reliability and risk. Existing industry frameworks, notably those produced by CoinGecko and CoinMarketCap, provide useful signals related to liquidity and volume integrity but suffer from limited transparency, fixed weighting schemes, and the absence of sentiment-based assessment. This paper presents HTREx (Hybrid Trust and Risk Evaluation framework for exchanges), a semi-automated, modular trust and risk assessment framework for cryptocurrency exchanges that addresses these limitations. The framework integrates five

dimensions of exchange integrity: user sentiment, regulatory compliance, technical security, transparency, and incident history. Sentiment is quantified using transformer-based natural language processing applied to user-generated content from mobile application reviews and online forums. Compliance is assessed through structured extraction of regulatory and operational disclosures from Terms of Service documents using large language models. Security, transparency, and incident history are evaluated through a combination of publicly verifiable indicators, third-party assessments, and a recency-weighted incident scoring model. All components are normalised and aggregated into a composite score using user-adjustable weights, enabling personalised risk prioritisation while retaining a defensible default configuration for comparative analysis. The framework is demonstrated using four prominent exchanges-Kraken, Coinbase, Binance, and Uniswap-highlighting clear differences between centralised and decentralised platforms and illustrating how sentiment, compliance, and historical incidents materially influence overall trust assessments. The results suggest that transparent, extensible, and user-configurable scoring models can provide a more interpretable and context-sensitive evaluation of exchange risk than existing monolithic trust scores, with direct relevance for both retail and institutional participants. © 2026 Curran Associates Inc.. All rights reserved.

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A Large Language Model-Based Approach for Multilingual Hate Speech Detection on Social Media

Item Type Journal Article

Author M. Usman

Author M. Ahmad

Author G. Sidorov

Author I. Gelbukh

Author R.Q. Tellez

Abstract The proliferation of hate speech on social media platforms poses significant threats to digital safety, social cohesion, and freedom of expression. Detecting such content—especially across diverse languages—remains a challenging task due to linguistic complexity, cultural context, and resource limitations. To address these challenges, this study introduces a comprehensive approach for multilingual hate speech detection. To facilitate robust hate speech detection across diverse languages, this study makes several key contributions. First, we created a novel trilingual hate speech dataset consisting of 10,193 manually annotated tweets in English, Spanish, and Urdu. Second, we applied two innovative techniques—joint multilingual and translation-based approaches—for cross-lingual hate speech detection that have not been previously explored for these languages. Third, we developed detailed hate speech annotation guidelines tailored specifically to all three languages to ensure consistent and high-quality labeling. Finally, we conducted 41 experiments employing machine learning models with TF-IDF features, deep learning models utilizing FastText and GloVe embeddings, and transformer-based models leveraging advanced contextual embeddings to comprehensively evaluate our approach. Additionally, we employed a large language model with advanced contextual embeddings to identify the best solution for the hate speech detection task. The experimental results showed that our GPT-3.5-turbo model significantly outperforms strong baselines, achieving up to an 8% improvement over XLM-R in Urdu hate speech detection and an average gain of 4% across all three languages. This research not only contributes a high-quality multilingual dataset but also offers a scalable and inclusive framework for hate speech detection in underrepresented languages. © 2025 by the authors.

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A large-scale annotated Urdu corpus and deep learning benchmark for mental health classification

Item Type Journal Article

Author M. Fatima

Author M.S. Khan

Author M.S. Faisal

Author T. Shahzad

Author M.A. Iqbal

Author S.K. Kim

Abstract Millions of people worldwide are affected by mental health disorders, particularly anxiety and depression, which pose a growing health concern globally. Early detection and intervention are critical for improving treatment outcomes. Conventional clinical evaluation methods face significant barriers including limited accessibility, social stigma, and diagnostic delays. Recent advances in artificial intelligence and natural language processing have demonstrated promising capabilities for automated mental health detection through text analysis of social media. However, most existing research focuses exclusively on high-resource languages such as English, leaving a critical gap for low-resource languages. Our study addresses this limitation by developing a comprehensive deep learning framework for mental health classification in Urdu. More than 230 million people speak Urdu, but it is severely underrepresented in computational mental health research. This research provides the first large-scale, publicly released Urdu dataset specifically designed for multi-class anxiety, depression, and neutral classification, and establishes performance benchmarks for future work. We created and validated a novel annotated dataset of 36,000 Urdu tweets classified into anxiety, depression, and neutral categories through systematic translation, manual annotation, and automated labeling with rigorous quality validation. Three established deep learning architectures were adapted and systematically evaluated in this study. The CNN+BiLSTM model achieved an accuracy of 79.08%, while the CNN+BiGRU

model attained 78.25%. Among all evaluated approaches, UrduBERT delivered the best performance with an accuracy of 81.71%, along with superior precision, recall, and F1-score values across all categories. These findings highlight the advantage of transformer-based contextual representations in modeling complex linguistic structures in morphologically rich languages. Overall, the proposed framework provides a strong basis for scalable and culturally sensitive mental health monitoring systems, with the potential to support early detection of at-risk individuals in linguistically underrepresented communities. © The Author(s) 2026.

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A longitudinal consumer feedback analytics approach for theory building and testing: Examining consumer satisfaction and loyalty in buy now, pay later (BNPL) platforms

Item Type Journal Article

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Author S.A. Alzaeemi

Author M. Albashrawi

Author Y.K. Dwivedi

Author R. Alahmad

Author I. Chae

Abstract This study investigates the post-adoption factors influencing consumer satisfaction and loyalty in Buy Now, Pay Later (BNPL) platforms, with a focus on the culturally specific and religiously regulated context of Saudi Arabia. A longitudinal dataset of 61,631 cleaned and validated bilingual user-generated evaluations from prominent BNPL applications, collected between 2018 and 2025, was analyzed to capture temporal shifts in consumer perceptions. A mixed-methods approach was employed, combining topic modelling, sentiment analysis, and large language models to identify underlying experiential themes and map them to theoretical constructs. Confirmatory analysis using Elastic Net regression and Random Forest, with 80/20 train–test splits, 10-fold cross-validation, and bootstrap resampling, demonstrates that Information Quality and Perceived Security are the strongest predictors of Consumer Satisfaction, while Sharia Compliance functions as a cultural boundary condition and significant driver of Consumer Loyalty. Perceived Financial Risk consistently exerts a detrimental effect on both outcomes. The findings extend retailing and consumer behaviour as well as IS/FinTech literature by incorporating culturally specific constructs and by demonstrating how user-generated content (UGC) analyzed with machine learning, providing unsolicited, ecologically valid, and large-scale insights into post-adoption behaviour. Practical implications highlight the importance of transparency, ethical adherence, and financial clarity in enhancing the BNPL user experience, while also emphasising cost–benefit considerations, regulatory compliance, and consumer segment differences in implementation strategies. © 2025 Elsevier Ltd

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A Machine Learning based Approach to Identify User Interests from Social Data

Item Type Conference Paper

Author R. Tahir

Author M. Asif Naeem

Abstract Social media platforms like Twitter, Facebook, Instagram, etc., are considered a common source of extracting information about individuals, such as their needs, interests, and opinions. Our major contribution in this paper is to identify user interests and desires related to the fashion industry in Pakistan. Since people in Pakistan mostly write tweets and reviews in Roman Urdu, the dataset we focused on in this research was comprised of Roman Urdu Tweets and Google Map reviews. From the literature, we observed that not much effort has been done on Roman Urdu tweets and reviews because of its being a low resource language. In terms of methodology, we applied LDA, LSA, and BERT for topic modeling; Vader combined with TextBlob and DistilBert for sentiment analysis; and K-Means for identifying user clusters with similar interests. In our experiments, we used 15000 tweets and 6000 Google reviews. We were able to create five distinct clusters for each brand. These clusters were further used to track the users based on their interests. We evaluated the performance of our approach and validated it empirically based on Cohen's Kappa score, and achieved a score of 0.45 that shows moderate agreement between human and machine. © 2022 IEEE.

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A multi-agent large language model framework for intelligent vendor evaluation and risk-aware procurement decisions

Item Type Journal Article

Author A. Joshi

Author M. Singh

Author D. Parashar

Author M. Singh

Abstract The process of vendor selection is a multi-dimensional and intricate task that requires analyzing the financial health, amount of risk, industry standards, and the mood in the marketplace. The use of traditional methods that consider individual data points or fixed metrics usually leads to less than optimal procurement decision-making. The current paper introduces a new Multi-Agent Large Language Model (LLM) framework that will improve the process of vendor evaluation based on the analysis of both structured and unstructured data. Its framework uses specialized agents that target different aspects of vendor evaluation: financial analysis, risk profiling, sentiment monitoring, and industry benchmarking. Structured indicators that financial agents consider include liquidity, profitability and solvency. Exposure to geopolitical, operational, and compliance risks is assessed by risk agents. Sentiment agents identify sentiment based on the news articles, reviews, and social media to determine the perception of the people and stakeholders. Agents in benchmarking compare vendors to industry standards to determine outliers and best-in-class vendors. The combination of these insights provides a data-driven yet context-aware procurement recommendation generated by the framework. The qualitative assessments are used to complement quantitative metrics, allowing procurement teams to make informed and holistic decisions. The system does not only help in identifying the most appropriate vendors but also assists in managing relationships with suppliers in the long term whereby the system alerts of any risk and performance deviations in real time. The method enables organizations to concentrate vendor selection in tandem with strategic objectives, reduce risk, and enhance the long-term value. The suggested multi-Agent LLM scheme is a major development in smart procurement systems, which enhance more resilient and value-oriented supply chains. © The Author(s) 2026.

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A Multi-Layer Hybrid Deep Learning Model for Toxic Comment Detection in Assamese

Item Type Journal Article

Author M. Neog

Author N. Baruah

Abstract The present research proposes a highly precise technique for detecting toxic comments in Assamese, a low-resource language that is a member of the Eastern Indo-Aryan family. As a low-resource language, Assamese encounters specific challenges such as dialectal differences, incorrect spellings, frequent code-mixing with English, and a lack of resources with annotations. A custom-built dataset of 100,000 Assamese social media comments was gathered and split equally between toxic and non-toxic categories in order to solve the lack of labelled data. The article presents a multi-layer hybrid deep learning model that integrates multi-scale Convolutional Neural Networks (CNN), bidirectional LSTM (BiLSTM), and bidirectional GRU (BiGRU). A thorough grid search utilising several optimisers and activation functions was used to create and optimise two hybrid architectures: CNN-BiLSTM-BiGRU and BiGRU-BiLSTM-CNN. Among them, the CNN-BiLSTM-BiGRU model outperformed present approaches for Assamese and other structurally related languages, reaching 94.25% accuracy and an F1- score of 0.93%. This work not only

produces strong analytical findings, but it also makes a novel contribution in developing and evaluating models for Assamese toxic comment detection under real-world linguistic and resource constraints, providing a solid foundation and valuable insights for furthering NLP research in other under resourced languages. © 2013 IEEE.

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A Multi-layered Large Language Model Framework for Disease Prediction

Item Type Conference Paper

Author M. Mohamed

Author R. Emad

Author A. Hamdi

Editor Yang X.-S.

Editor Sherratt S.

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Editor Joshi A.

Abstract Social telehealth has made a breakthrough in healthcare by allowing patients to share their symptoms and have medical consultations remotely. Users frequently post symptoms on social media and online health platforms, creating a huge repository of medical data that can be leveraged for disease classification and symptom severity assessment. Large language models (LLMs) like LLAMA3, GPT-3.5 Turbo, and BERT process complex medical data, enhancing disease classification. This study explores three Arabic medical text preprocessing techniques: text summarization, text refinement, and Named Entity Recognition (NER). Evaluating CAMEL-BERT, AraBERT, and Asafaya-BERT with LoRA, the best performance was achieved using CAMEL-BERT with NER-augmented text (83% Type classification, 69% Severity assessment). Non-fine-tuned models performed poorly (13–20% Type classification, 40–49% Severity assessment). Embedding LLMs in social telehealth enhances diagnostic accuracy and treatment outcomes. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Mohamed; Faculty of Computer Science, MSA University, 6th of October, Egypt; email: malak.mohamed17@msa.edu.eg; Conference name: 10th International Congress on Information and Communication Technology, ICICT 2025; Conference date: 18 February 2025 through 21 February 2025; Conference code: 339419

A Multi-Model Deep Learning Framework for Hate Speech Detection in Code-Mixed Low-Resource Languages

Item Type Conference Paper

Author M. Mohanasundari

Author S.S. Boopathy

Author C. Kotteswari

Author M. Manikandan

Author K.S. Aswin

Author S. Kishore

Abstract Rapid expansion of social media platforms has magnified ethical, social, and cultural issues, allowing for a concurrent rise in inappropriate and harmful material, specifically hate speech, in regional languages. Detection of hate speech in Dravidian languages - in particular, Tamil (TA-AT-v2), Telugu, and Malayalam (ML-AT-v2) - is extremely difficult given the high level of linguistic variability, code-mixing of English with regional languages, and critical lack of annotated data. This paper addresses this gap by undertaking a comparison of deep learning techniques by applying three state-of-the-art multilingual transformer models: mDeBERTa-v3-base, MPNet (paraphrase-multilingual-mpnet-base-v2), and the lightweight MiniLM-L6v2 (Minimal Language Model). To address the severe class imbalance and scarcity (for example, the Tamil dataset has 514 samples with only 33 samples of a Political hate speech category), an efficient data augmentation strategy to increase the usable dataset Tamil to 740 samples. Investigation made on these different architectures' abilities to create an efficient, languagesensitive systems and show that combining their complementary abilities within a final Optimized Multi-Model Classification / Prediction / Analysis (MCPA) Framework, creates better and more reliable performance within this low-resource environment. Final result contributes robust, scalable system architecture to build a safer online experience for users of Dravidian languages. © 2026 IEEE.

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A Multi-Output BERT Framework for Abusive Comment Detection and Sentiment Analysis on Low-Resource Language

Item Type Journal Article

Author M. Yagnik

Author M. Hashmi

Author D. Kumar

Author K. Jain

Author E. Grover

Author J.D. Hemanth

Abstract In the modern digital world, social media has become essential for interpersonal interaction by promoting the interchange of ideas and points of view. But there are difficulties in this digital environment, especially concerning rude behavior and offensive remarks. To address both problems at once, the research focuses on sentiment analysis and abusive comment detection in social media interactions. The dataset contains Hate Speech and Offensive Content Identification (HASOC) data from 2019 to 2021 to identify hate speech in Hindi on various social media platforms. To categorize comments into abusive and non-abusive groups, several BERT models, including mBERT, DistilBERT, RoBERTa, HateBERT, and IndicBERT, have been utilized. Additionally, a comprehensive sentiment analysis of the derogatory comments has been performed. The research presents a stacked ensemble framework for binary (abusive and non-abusive) and multiclass (hate, offensive, and profane) classification that integrates predictions from mBERT, HateBERT, and IndicBERT models. Further, the study provides an integrated approach for providing abusive comment detection and sentiment analysis using a multi-output model. The proposed ensemble model achieves 94% accuracy in binary classification, with precision, recall, and F1 scores all approaching 94%. Multiclass ensemble models yield an accuracy of 93% and associated precision, recall, and F1 scores of 91%, 92%, and 92%, respectively. A comparative analysis using several state-of-the-art techniques has been generated to verify the efficacy of the suggested methodology. © 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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A Multi-Stage BERT-Based Cognitive Framework for Conversational Business Intelligence in SMEs

Item Type Conference Paper

Author D. Baviskar

Author H. Halwai

Author A. Patekar

Author A. Kachewar

Abstract Social media is becoming a more important tool for small and medium-sized businesses (SMBs) to watch competitors, evaluate customer sentiment, and identify new market opportunities. However, it is challenging to convert raw signals into understandable, actionable insights due to the unstructured, fast-moving nature of cross-platform data. A conversational business intelligence layer and a multi-platform trend analytics pipeline were established in earlier work on this project. The multi-stage family of BERT-based models that serve as that system's cognitive foundation is the particular subject of this research. The

approach uses Sentiment-BERT for polarity-sensitive refinement, NER-BERT for business-entity extraction, SBERT for semantic retrieval over trend-aware embeddings, and Intent-BERT for query categorization. Within a single pipeline, each model maintains compatibility while addressing a different reasoning layer. We explain how these models are modified for SMB use scenarios, how they work together to facilitate end-to-end conversational intelligence, and how they open up new avenues for real-time decision support, low-resource fine-tuning, and domain adaptability. This paper focuses specifically on the cognitive foundation of that system, presenting a unified multistage architecture built on BERT-based models. © 2026 IEEE.

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A Multi-Task Benchmark for Abusive Language Detection in Low-Resource Settings

Item Type Conference Paper

Author F. Gaim

Author H. Song

Author H. Lee

Author C. Ko

Author E.J. Hwang

Author J.C. Park

Abstract Content moderation research has recently made significant advances, but remains limited in serving the majority of the world's languages due to the lack of resources, leaving millions of vulnerable users to online hostility. This work presents a large-scale human-annotated multi-task benchmark dataset for abusive language detection in Tigrinya social media with joint annotations for three tasks: abusiveness, sentiment, and topic classification. The dataset comprises 13,717 YouTube comments annotated by nine native speakers, collected from 7,373 videos with a total of over 1.2 billion views across 51 channels. We developed an iterative term clustering approach for effective data selection. Recognizing that around 64% of Tigrinya social media content uses Romanized transliterations rather than native Ge'ez script, our dataset accommodates both writing systems to reflect actual language use. We establish strong baselines across the tasks in the benchmark, while leaving significant challenges for future contributions. Our experiments demonstrate that small fine-tuned models outperform prompted frontier large language models (LLMs) in the low-resource setting, achieving 86.67% F1 in abusiveness detection (7+ points over best LLM), and maintain stronger performance in all other tasks. The benchmark is made public to promote research on online safety. © 2025 Neural information processing systems foundation. All rights reserved.

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A Multi-Task LSTM Approach for Comprehensive Detection of Emotion, Hate Speech, and Violence in Text

Item Type Conference Paper

Author S. Satapathy

Author K.M. Gopal

Author B. Kar

Abstract Increase in the frequency of availability of user-generated content on digital platforms is hampering the detection of harmful and sensitive language. This paper it presents an approach by which we can address this challenge by employing a multi-task model based on LSTM network, a type of RNN particularly designed to process and retain the information over multiple time steps. The multi-task NLP model is designed to classify three categories: emotions, hate speech and violence. Utilizing the LTSM networks we train the model on distinct text-based dataset for each task. This approach will process the different dataset for each task separately through a shared architecture. This model will employ shared layers, including LSTM, Pooling, Embedding and Dropout. Multi-task framework is designed to capture the common linguistic features between task by sharing the core layers while maintaining the task specified output layer for the independent predictions. This will enable the model to generalize well across the task, by learning from the common patterns in the text while maintaining the ability to produce accurate task specific outputs. The performance of the model is evaluated by using the different datasets for each task, that achieve high accuracy in classifying emotions, identifying violent language, and detecting hate speech, which demonstrate the effectiveness of LSTM architecture in managing diverse labeling schemes while maintaining task specific precision. This research highlights the potential of multi-task learning by improving the efficiency and accuracy of text classification model like content moderation and sentimental analysis. © 2025 IEEE.

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A multidimensional cybercrime annotation dataset for a closed-access Russian–English underground hacking forum

Item Type Journal Article**Author** M. Mischinger**Author** S. Pastrana**Author** G. Suarez-Tangil

Abstract This dataset provides a multidimensional cybercrime annotation layer for XIN, a closed-access Russian–English underground hacking forum comprising approximately 1.3 million post annotations across roughly 200,000 threads and 52 sub-forums, posted between February 2005 and August 2025. Every post is annotated along four dimensions drawn from established underground-forum coding schemes: post type (the communicative function of the post), author intent (the stance or social purpose), crime type (the category of cybercrime discussed), and supply-chain category (the role of the traded product or service). Labels were assigned through a codebook-based prompting procedure that applied a predefined scheme of category definitions, decision rules, and anonymized examples. We release the codebooks and the per-dimension label distributions over the full corpus. The annotations enable supervised and zero- or few-shot classification, benchmarking of automated labeling against a documented scheme, cyber threat intelligence extraction, and cross-lingual analysis of a community that is underrepresented in public corpora. © 2026 The Authors.

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A Multilingual Transformer-Based Approach for Sarcasm Detection and Sentiment Analysis in Hindi-Marathi Social Media

Item Type Conference Paper

Author S. Suryawanshi

Author C. Patil

Author V.K. Pachgare

Abstract Understanding sarcasm and sentiment in Indian social media text is challenging due to multilingual usage, code-mixing, informal expressions, and contextual ambiguity. This paper presents a multilingual transformer-based framework for sarcasm detection and sentiment analysis in Hindi-Marathi social media content. The proposed approach fine-tunes a pre-trained Indic language transformer model on real-world social media data to effectively capture linguistic and contextual cues across languages. To address the problem of class imbalance commonly observed in sarcasm-related data, a class-weighted training strategy is employed. Experimental evaluation indicates that the proposed framework performs consistently across sarcasm detection and sentiment classification tasks, with particularly strong effectiveness in identifying sarcastic expressions in Marathi text. The results demonstrate that Indic language transformer models provide a robust and reliable foundation for sarcasm-aware sentiment analysis in multilingual and low-resource social media settings. © 2026 IEEE.

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A New Direction in Stance Detection: Target-Stance Extraction in the Wild

Item Type Conference Paper

Author Y. Li

Author K. Garg

Author C. Caragea

Abstract Stance detection aims to detect the stance toward a corresponding target. Existing works have achieved promising progress on stance detection tasks in which the goal is to predict the stance given both a target and a text. However, they all work under the assumption that the target is known in advance, which is often not the case in the wild. Given a text from social media platforms, the target information is often unknown due to implicit mentions in the source text and it is infeasible to have manual target annotations at a large scale. Therefore, in this paper, we propose a new task Target-Stance Extraction (TSE) that aims to extract the (target, stance) pair from the text. We benchmark the task by proposing a two-

stage framework that first identifies the relevant target in the text and then detects the stance given the predicted target and text. Specifically, we first propose two different settings: Target Classification and Target Generation, to identify the potential target from a given text. Then we propose a multitask approach that takes target prediction as the auxiliary task to detect the stance toward the predicted target. We evaluate the proposed framework on both in-target stance detection in which the test target is always seen in the training stage and zero-shot stance detection that needs to detect the stance for the unseen target during the inference stage. The new TSE task can facilitate future research in the field of stance detection. We publicly release our code. © 2023 Association for Computational Linguistics.

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A Novel Approach for Emotion Detection and Sentiment Analysis for Low Resource Urdu Language Based on CNN-LSTM

Item Type Journal Article

Author F. Ullah

Author X. Chen
Author S.B.H. Shah
Author S. Mahfoudh
Author M.A. Hassan
Author N. Saeed

Abstract Emotion detection (ED) and sentiment analysis (SA) play a vital role in identifying an individual's level of interest in any given field. Humans use facial expressions, voice pitch, gestures, and words to convey their emotions. Emotion detection and sentiment analysis in English and Chinese have received much attention in the last decade. Still, poor-resource languages such as Urdu have been mostly disregarded, which is the primary focus of this research. Roman Urdu should also be investigated like other languages because social media platforms are frequently used for communication. Roman Urdu faces a significant challenge in the absence of corpus for emotion detection and sentiment analysis because linguistic resources are vital for natural language processing. In this study, we create a corpus of 1021 sentences for emotion detection and 20,251 sentences for sentiment analysis, both obtained from various areas, and annotate it with the aid of human annotators from six and three classes, respectively. In order to train large-scale unlabeled data, the bag-of-words, term frequency-inverse document frequency, and Skip-gram models are employed, and the learned word vector is then fed into the CNN-LSTM model. In addition to our proposed approach, we also use other fundamental algorithms, including a convolutional neural network, long short-term memory, artificial neural networks, and recurrent neural networks for comparison. The result indicates that the CNN-LSTM proposed method paired with Word2Vec is more effective than other approaches regarding emotion detection and evaluating sentiment analysis in Roman Urdu. Furthermore, we compare our based model with some previous work. Both emotion detection and sentiment analysis have seen significant improvements, jumping from an accuracy of 85% to 95% and from 89% to 93.3%, respectively. © 2022 by the authors.

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Correspondence Address: S.B.H. Shah; School of Computing and Informatics, Dar Al-Hekma
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A novel approach to cross-linguistic transfer learning for hope speech detection in Tamil and Malayalam

Item Type Journal Article

Author J.P. V.

Author A.A.V. S.

Abstract In the field of Natural Language Processing (NLP), accurately identifying hope speech in low-resource languages such as Tamil and Malayalam poses significant challenges. This research introduces the Sentimix Transformer (SentT), a novel transformer-based model designed for detecting hope speech in YouTube comments composed in Tamil and Malayalam, two linguistically rich but computationally low-resource languages. The SentT model innovatively combines multilingual BERT (mBERT) embeddings with specialized cultural and code-mixing adaptations to effectively process the linguistic diversity and complexities inherent in code-mixed data. This approach allows SentT to capture nuanced expressions of hope by integrating domain-specific knowledge into the transformer framework. Our methodology extends traditional transformer architectures by incorporating a unique ensemble of embeddings that encapsulate linguistic, cultural, and code-mixing attributes, significantly enhancing the model's sensitivity to context and cultural idioms. We validate our approach using the Hope Speech dataset for Equality, Diversity, and Inclusion (HopeEDI), which includes diverse comments from social media. The SentT model achieves an impressive accuracy of 93.4%, a precision of 92.7%, and a recall of 94.1% outperforming existing models and demonstrating its efficacy in handling the subtleties of hope speech in multilingual contexts. The model's architecture and the results of extensive evaluations not only underscore its effectiveness but also its potential as a scalable solution for similar tasks in other low-resource languages. Through this research, we contribute to the broader field of sentiment analysis by demonstrating the potential of tailored, context-aware models in enhancing digital communication's positivity and inclusiveness. © 2025 Elsevier Ltd

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A novel large language model enhanced joint learning framework for fine-grained sentiment analysis on drug reviews

Item Type Journal Article

Author H. Zou

Author Y. Wang

Abstract Patient feedback on drug reviews extracted from social media platforms and online forums provides genuine sentiment information regarding post-medication usage. The insights are invaluable for prospective patients seeking appropriate medical guidance and stakeholders within the biomedical industry aiming to improve products. This paper presents a novel joint learning framework for fine-grained aspect-based sentiment analysis on drug reviews. By leveraging prior biomedical knowledge from the domain-specific pre-trained large language model, we address the challenge of fine-grained aspect-based sentiment analysis by collaboratively integrating both coarse and fine-grained contextual features within the text content, capturing precise biomedical aspect terms and the corresponding sentiment polarities. To the best of our knowledge, this work pioneers the initial introduction of a joint learning framework based on the fine-tuned biomedical large language model for fine-grained aspect-based sentiment analysis within drug reviews. By conducting extensive experiments on publicly available drug review datasets and comparing the constructed architecture with state-of-the-art techniques, the joint learning framework outperforms baseline competitors across evaluation metrics. © 2025 Elsevier B.V.

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A Parallel Corpus-Based Approach to the Crime Event Extraction for Low-Resource Languages

Item Type Journal Article

Author N. Khairova

Author O. Mamyrbayev

Author N. Rizun

Author M. Razno

Author Y. Galiya

Abstract These days, a lot of crime-related events take place all over the world. Most of them are reported in news portals and social media. Crime-related event extraction from the published texts can allow monitoring, analysis, and comparison of police or criminal activities in different countries or regions. Existing approaches to event extraction mainly suggest processing texts in English, French, Chinese, and some other resource-rich and well-annotated languages. This paper presents a parallel corpus-based approach that follows a closed-domain event extraction methodology to event extraction from web news articles in low-resource languages. To identify the

event, its arguments, and the arguments' roles in the source-language part of the corpus we utilize an enhanced pattern-based method that involves the multilingual synonyms dictionary with knowledge about crime-related concepts and logic-linguistic equations. The event extraction from the target-language part of the corpus uses a cross-lingual crime-related event extraction transfer technique that is based on supplementary knowledge about the semantic similarity patterns of the considered pair of languages. The presented approach does not require a preliminarily annotated corpus for training making it more attractive to low-resource languages and allows extracting TRANSFER, CRIME, and POLICE types of events and their seven subtypes from various topics of news articles simultaneously. Implementation of our approach for the Russian-Kazakh parallel corpus of news portals articles allowed obtaining the F1-measure of crime-related event extraction of over 82% for the source language and 63% for the target language. © 2013 IEEE.

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A quantum-like zero-shot approach for sentiment analysis in finance

Item Type Journal Article

Author X. Yang

Author J. Zhu

Author P. De Meo

Abstract Sentiment analysis has become an indispensable tool across various domains, including political communication, marketing analytics, and finance. In the financial sector, sentiments such as confidence or fear play a pivotal role in shaping market dynamics, influencing supply and demand, and precipitating significant price fluctuations. The timely extraction of sentiment from sources like financial news articles and social media posts is crucial for devising informed investment strategies. Despite its importance, current approaches to sentiment analysis in finance have not fully harnessed the potential of emerging technologies, such as large language models (LLMs) and quantum computing models. These cutting-edge technologies have demonstrated remarkable success in other areas of natural language processing (NLP), but their application in financial sentiment analysis remains limited. This paper aims to bridge this gap by introducing a novel quantum model for text representation, specifically designed for the financial domain. Our approach integrates this quantum representation with a vector-based representation generated by a pre-trained LLM specialized in finance. The resulting fusion of these two representations yields a more comprehensive input for an LLM, leading to a significant enhancement in the accuracy of the sentiment analysis task. We evaluate our approach through extensive experimental tests on a publicly available dataset and demonstrate that our quantum-based method outperforms state-of-the-art models on a widely recognized financial sentiment analysis benchmark. Our results highlight the potential of integrating quantum computing principles with traditional NLP methods for more accurate and effective sentiment analysis in finance. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024.

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A Review of Large Language Models for Stock Price Forecasting from a Hedge-Fund Perspective

Item Type	Conference Paper
Author	O. Zhang
Author	Z. Zhang
Abstract	Large language models (LLMs) are increasingly deployed in quantitative finance for stock price forecasting. This review synthesizes recent applications of LLMs in this domain, including extracting sentiment from financial news and social media, analyzing financial reports and earnings-call transcripts, tokenizing or symbolizing stock price series, and constructing multi-agent trading systems. Particular attention is paid to practical pitfalls that are often understated in the literature, such as fragility in sentiment analysis, dataset and horizon design, performance evaluation metrics, data leakage, illiquidity premia, and limits of stock price predictability. Organized from a hedge-fund perspective, the review is intended to guide both academic researchers and hedge fund managers in integrating LLMs into real-world trading pipelines and in stress-testing their robustness under realistic market frictions. © 2026 IEEE.
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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: O. Zhang; Hockaday School, Dallas, 11600 Welch Rd, 75229, United States; email: ozhang30@hockaday.org; Conference name: 4th IEEE Conference on Artificial Intelligence, CAI 2026; Conference date: 8 May 2026 through 10 May 2026; Conference code: 224402

A Review on Sentiment Analysis Using Transformers and Ensemble methods

Item Type Conference Paper**Author** Y.R. Devi**Author** A. Bharthepudi**Author** A. Govindarajula

Abstract Sentiment Analysis is the field of study that examines people's opinions, sentiments, valuations, appraisals, and attitudes and emotions from written language. It is one of the most active research areas in natural language processing and is also widely studied in data mining, web mining, and text mining. Sentiment analysis has become almost an essential task for companies to keep track of their brand or product success on the social media market. Sentiment analysis can help one gain insights about how people feel about things, i.e., their opinions. This paper presents a comprehensive review of sentiment analysis using transformer architectures and Large Language Models (LLMs), and their ensembles with traditional machine learning models for various sentiment analysis tasks. The performances of sentiment analysis tasks can be significantly improved by transformers and LLM-based approaches. The models that are covered in paper, including SAE, LSTM, BERT, RoBERTa, MPNet, GRU, XLNet and others are trained on different publicly available benchmark datasets used are IMDB Reviews and Twitter Airline. Reviews and so on to measure the performance of surveyed models, the metrics primarily considered are accuracy and F1-score. The main objective of this survey is to provide an overall understanding on transformer-based approaches for sentiment analysis, including advantages and Problems. Future scopes are depicted at the end. Index Terms - Sentiment Analysis, Transformers, SAE, LSTM, BERT, RoBERTa, MPNet, GRU, XLNet. © 2025 IEEE.

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A scalable hybrid framework for sentiment analysis of COVID-19 tweets using transformer embeddings and lightweight classifiers

Item Type Journal Article**Author** Z. Rezaei**Author** S.S. Samghabadi**Author** Y.M. Banad

Abstract During the COVID-19 pandemic, Twitter became a key platform for expressing public sentiment. This study explores a simple and scalable way to analyze the sentiment of COVID-19 tweets without the heavy computational cost of full transformer fine-tuning. Pretrained transformer models—Bidirectional Encoder Representations from Transformers (BERT), Distilled Bidirectional Encoder Representations from Transformers (DistilBERT), and Robustly Optimized BERT Pretraining Approach (RoBERTa)—were used in frozen mode to generate sentence embeddings, which were then classified using lightweight models: Support Vector Machine (SVM) and Logistic Regression (LR). On a balanced set of 7,500 labeled tweets from April–June 2020, the best hybrid model DistilBERT embeddings with

Logistic Regression reached 0.64 accuracy and 0.62 F1-score (harmonic mean of precision and recall). For comparison, fully fine-tuned transformers on the larger 143,902-tweet dataset achieved up to 0.973 accuracy, zero-shot classifiers scored only 0.21–0.33, VADER reached 0.73, and TextBlob 0.45. The results show that combining frozen transformer embeddings with classical classifiers offers a practical middle ground: it delivers reasonable performance using only a central processing unit (CPU) and minimal training time, making it suitable for research groups or public-health teams with limited computational resources. Copyright 2026 Rezaei et al. Distributed under Creative Commons CC-BY 4.0

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Y.M. Banad; Gallegly College of Engineering, University of Oklahoma, Norman, United States; email: bana@ou.edu

A Social Cognitive-Inspired Social Media Sentiment Classification Model

Item Type Journal Article

Author N. Natesh

Author M.V. Vijayakumar

Abstract Social cognition, encompassing the perception, interpretation, and response to social stimuli, plays a vital role in understanding human interactions. Nowadays, with the surge of social media platforms, analyzing social cognition through digital text has become an imperative task. Traditional cognitive architectures like ACT-R and SOAR

face challenges in processing large-scale, unstructured data, such as those produced from social media platforms. This study addresses this problem by proposing a fine-tuned BERT-based model for understanding social cognition and emotions. The objective was to enhance social cognitive emotion classification accuracy and contextual understanding of social interactions. The methodology involved fine-tuning BERT with adaptations inspired by cognitive models, enabling the classification of emotions (neutral, positive, and negative) and the analysis of social interaction patterns. The model was evaluated on a Twitter dataset, achieving an accuracy of 83.1%, outperforming existing models such as RoBERTa (80.54%) and standard BERT (79.25%). The results underscore the effectiveness of task-specific fine-tuning in capturing social and emotional cognition. This work's novelty lies in integrating transformer-based models with cognitive principles, providing a robust, scalable framework for analyzing social cognition. (c) by the authors

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A Study of Large Model Sentiment Classification Based on LoRA Fine-Tuning

Item Type Conference Paper

Author C. Wu

Author F. Wan

Abstract Sentiment classification is an important task in natural language processing, which is widely used in many fields such as public opinion monitoring, commodity evaluation, and social media analysis. In recent years, pre-trained large language models have performed prominently in the task of sentiment categorization, but large language models still have certain deficiencies in dealing with the complexity and polysemy of textual sentiments as well as long textual reasoning. With the emergence of deepseek model, this paper, based on the Chinese microblog sentiment classification dataset, adopts LoRA (Low-Rank Adaptation) fine-tuning technique with the help of deepseek model to improve its performance of sentiment classification, and compares it with the effect of the model before fine-tuning. The experimental results show that the LoRA fine-tuning method exhibits excellent results in improving the model in sentiment classification tasks, especially in processing complex text and binary sentiment labeling tasks with stronger robustness. The fine-tuned model improves 14.10 %, 12.82 % and 12.85 % in Accuracy, F1-Score and ROC-AUC metrics, respectively, compared with the baseline model, which effectively improves the accuracy and diversity of complex text sentiment classification. ©2025 IEEE.

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A survey and study impact of tweet sentiment analysis via transfer learning in low resource scenarios

Item Type Journal Article

Author M.V. dos Santos Neto

Author N.F.F. da Silva

Author A. da Silva Soares

Abstract Sentiment analysis (SA) is a study area focused on obtaining contextual polarity from the text. Currently, deep learning has obtained outstanding results in this task. However, much annotated data are necessary to train these algorithms, and obtaining this data is expensive and difficult. In the context of low-resource scenarios, this problem is even more significant because there are little available data. Transfer learning (TL) can be used to minimize this problem because it is possible to develop some architectures using fewer data. Language models are a way of applying TL in natural language processing (NLP), and they have achieved competitive results. Nevertheless, some models need many hours of training using many computational resources, and in some contexts, people and organizations do not have the resources to do this. In this paper, we explore the models BERT (Pretraining of Deep Bidirectional Transformers for Language Understanding), MultiFiT (Efficient Multilingual Language Model Fine-tuning), ALBERT (A Lite BERT for Self-supervised Learning of Language Representations), and RoBERTa (A Robustly Optimized BERT Pretraining Approach). In all of our experiments, these models obtain better results than CNN (convolutional neural network) and LSTM (Long Short Term Memory) models. To MultiFiT and RoBERTa models, we propose a pretrained language model (PTLM) using Twitter data. Using this approach, we obtained competitive results compared with the models trained in formal language datasets. The main goal is to show the impacts of TL and language models comparing results with other techniques and showing the computational costs of using these approaches. © The Author(s), under exclusive licence to Springer Nature B.V. 2023.

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A Survey on Sentiment Analysis in Persian: a Comprehensive System Perspective Covering Challenges and Advances in Resources and Methods

Item Type Journal Article**Author** Z. Rajabi**Author** M.R. Valavi

Abstract Social media has been remarkably grown during the past few years. Nowadays, posting messages on social media websites has become one of the most popular Internet activities. The vast amount of user-generated content has made social media the most extensive data source of public opinion. Sentiment analysis is one of the techniques used to analyze user-generated data. The Persian language has specific features and thereby requires unique methods and models to be adopted for sentiment analysis, which are different from those in English language. Sentiment analysis in each language has specified prerequisites; hence, the direct use of methods, tools, and resources developed for English language in Persian has its limitations. The main target of this paper is to provide a comprehensive literature survey for state-of-the-art advances in Persian sentiment analysis. In this regard, the present study aims to investigate and compare the previous sentiment analysis studies on Persian texts and describe contributions presented in articles published in the last decade. First, the levels, approaches, and tasks for sentiment analysis are described. Then, a detailed survey of the sentiment analysis methods used for Persian texts is presented, and previous relevant works on Persian Language are discussed. Moreover, we present in this survey the authentic and published standard sentiment analysis resources and advances that have been done for Persian sentiment analysis. Finally, according to the state-of-the-art development of English sentiment analysis, some issues and challenges not being addressed in Persian texts are listed, and some guidelines and trends are provided for future research on Persian texts. The paper provides information to help new or established researchers in the field as well as industry developers who aim to deploy an operational complete sentiment analysis system. © 2021, The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature.

Date 2021**DOI** 10.1007/s12559-021-09886-x**URL** <https://www.scopus.com/pages/publications/85111472402?origin=resultlist>**Archive** Scopus**Language** English**Volume** 13**Publisher** Springer**Pages** 882-902**Publication** Cognitive Computation

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A Survey on the Impact of Pre-Trained Language Models in Sentiment Classification Task

Item Type Journal Article**Author** H. Gautam**Author** A. Gaur**Author** D.K. Yadav

Abstract The evolution of pre-trained language models (PLMs) has significantly transformed the landscape of sentiment analysis, particularly in handling complex, noisy, informal, and short-text commonly found on social media. While numerous surveys have explored PLMs and sentiment analysis separately, few provide a focused evaluation of large language models (LLMs) in the context of sentiment classification across diverse, real-world datasets. This survey addresses that gap by systematically comparing traditional sentiment analysis techniques—such as lexicon-based, machine learning, and deep learning approaches—with modern large pre-trained language models, using text-based unimodal and multimodal benchmark dataset. Our survey offers a detailed knowledge of how transformer-based architectures, including models like BERT, RoBERTa, GPT-3.5, GPT-4, and LLaMA-2, are leveraged for sentiment classification tasks. Beyond analyzing performance improvements, this paper highlights practical aspects such as fine-tuning techniques, and domain adaptability. Furthermore, it sheds light on ongoing limitations, such as computational cost, interpretability issues, domain sensitivity, bias, fairness, and ethical concerns. Through a comprehensive and comparative approach, this study offers a valuable resource to identify the most appropriate tools and methods for sentiment classification in the dynamic and data-rich environment. © The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025.

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A Survey on the Use of Large Language Models (LLMs) in Fake News

Item Type Journal Article

Author E. Papageorgiou

Author C. Chronis

Author I. Varlamis

Author Y. Himeur

Abstract The proliferation of fake news and fake profiles on social media platforms poses significant threats to information integrity and societal trust. Traditional detection methods, including rule-based approaches, metadata analysis, and human fact-checking, have been employed to combat disinformation, but these methods often fall short in the face of increasingly sophisticated fake content. This review article

explores the emerging role of Large Language Models (LLMs) in enhancing the detection of fake news and fake profiles. We provide a comprehensive overview of the nature and spread of disinformation, followed by an examination of existing detection methodologies. The article delves into the capabilities of LLMs in generating both fake news and fake profiles, highlighting their dual role as both a tool for disinformation and a powerful means of detection. We discuss the various applications of LLMs in text classification, fact-checking, verification, and contextual analysis, demonstrating how these models surpass traditional methods in accuracy and efficiency. Additionally, the article covers LLM-based detection of fake profiles through profile attribute analysis, network analysis, and behavior pattern recognition. Through comparative analysis, we showcase the advantages of LLMs over conventional techniques and present case studies that illustrate practical applications. Despite their potential, LLMs face challenges such as computational demands and ethical concerns, which we discuss in more detail. The review concludes with future directions for research and development in LLM-based fake news and fake profile detection, underscoring the importance of continued innovation to safeguard the authenticity of online information. © 2024 by the authors.

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“A Tale of Two Movements”: Identifying and Comparing Perspectives in #BlackLivesMatter and #BlueLivesMatter Movements-related Tweets using Weakly Supervised Graph-based Structured Prediction

Item Type Conference Paper

Author S. Roy

Author D. Goldwasser

Abstract Social media has become a major driver of social change, by facilitating the formation of online social movements. Automatically understanding the perspectives driving the movement and the voices opposing it, is a challenging task as annotated data is difficult to obtain. We propose a weakly supervised graph-based approach that explicitly models perspectives in #BlackLivesMatter-related tweets. Our proposed approach utilizes a social-linguistic representation of the data. We convert the text to a graph by breaking it into structured elements and connect it with the social network of authors, then structured prediction is done over the elements for identifying perspectives. Our approach uses a small seed set of labeled examples. We experiment with large language models for generating artificial training examples, compare them to manual annotation, and find that it achieves comparable performance. We perform quantitative and qualitative analyses using a human-annotated test set. Our model outperforms multitask baselines by a large margin, successfully characterizing the perspectives supporting and opposing #BLM. © 2023 Association for Computational Linguistics.

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A Transformer-Based Approach for Sentiment Analysis of Marathi News Headlines

Item Type Conference Paper

Author M. Shaikh

Author M. Ghule

Author K. Pal

Author A. Save

Abstract Marathi news headlines remain largely underexplored in sentiment analysis research, even though Marathi is widely used in Indian digital news platforms. Existing research focuses largely on social media text and rely on generic multilingual transformer models, which are not optimized for the complex linguistic structure of news headlines in low resource languages. This paper presents a sentiment classification system specifically for Marathi news headlines, which employs three state of art transformer based models -XLM-R, Baseline MahaBERT, and a fine-tuned MahaBERT model which is obtained by optimizing task-specific hyperparameters to improve contextual representation and sentiment classification performance. Experiments conducted on the news headlines dataset demonstrates that the SA-MahaBERT(Sentiment-aware MahaBERT) achieves best accuracy of 83.18%, outperforming both the XLM-R and baseline MahaBERT. The results highlight the effectiveness of language specific optimization and provide a benchmark for future research in Marathi NLP. © 2026 IEEE.

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A transformer-based deep learning model for Persian moral sentiment analysis

Item Type Journal Article

Author B. Karami

Author F. Bakouie

Author S. Gharibzadeh

Abstract Moral expressions in online communications can have a serious impact on framing discussions and subsequent online behaviours. Despite research on extracting moral sentiment from English text, other low-resource languages, such as Persian, lack enough resources and research about this important topic. We address this issue using the Moral Foundation theory (MFT) as the theoretical moral psychology paradigm. We developed a Twitter data set of 8000 tweets that are manually annotated for moral foundations and also we established a baseline for computing moral sentiment from Persian text. We evaluate a plethora of state-of-the-art machine learning models, both rule-based and neural, including distributed dictionary representation (DDR), long short-term memory (LSTM) and bidirectional encoder representations from transformer (BERT). Our findings show that among different models, fine-tuning a pre-trained Persian BERT language model with a linear network as the classifier yields the best results. Furthermore, we analysed this model to find out which layer of the model contributes most to this superior accuracy. We also proposed an alternative transformer-based model that yields competitive results to the BERT model despite its lower size and faster inference time. The proposed model can be used as a tool for analysing moral sentiment and framing in Persian texts for downstream social and psychological studies. We also hope our work provides some resources for further enhancing the methods for computing moral sentiment in Persian text. © The Author(s) 2023

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A Transformer-Based Perspective to Bangla-English Code-Mixed Sentiment Analysis

Item Type Conference Paper

Author S.D. Talukder

Author D. Das

Author M.H. Badhan

Author R. Ahmod

Author M.S. Islam

Author I.M. Saiful

Editor Shah M.S.

Abstract Despite Bangla being the seventh most spoken language worldwide, it remains a low-resource language in Natural Language Processing, particularly concerning Bangla-English (BE) code-mixed text. While Banglish (transliterated Bangla) has been studied, sentiment analysis of text containing native Bangla script alongside English words remains underexplored. This paper addresses this gap by introducing BE-CodeMix-3, a manually annotated sentiment dataset of 8,179 social media comments categorized into three sentiment classes. We provide a comprehensive evaluation across various architectures, including traditional Recurrent Neural Networks (LSTM, BiLSTM, GRU) and transformer-based models (mBERT, XLMRoBERTa, and BanglaBERT). Experimental results demonstrate that language-specific pretraining is critical for capturing the nuances of code-mixed text. The BanglaBERT model significantly outperforms

all baselines, achieving a peak accuracy and F1-score of 0.90. Our findings highlight that while multilingual models provide a strong foundation, transformers pretrained specifically on large-scale Bangla corpora offer superior semantic and syntactic encoding for real-world, codemixed applications. © 2026 IEEE.

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A Unified Framework for Analyzing Textual Context and Intent in Social Media

Item Type Journal Article

Author V.J. Prakash

Author S.A.A. Vijay

Abstract In the realm of natural language processing, tasks like emotion recognition, irony detection, hate speech detection, offensive language identification, and stance detection are pivotal for understanding user-generated content. While several task-

specific and multitask learning models have been proposed, there remains a need for a unified framework that can effectively address these tasks simultaneously. This research introduces a novel unified framework designed to tackle multiple NLP tasks concurrently, aiming to outperform existing task-specific and multitask models in terms of accuracy, F1-score, and AUC-ROC. We compared our proposed framework against several baseline models, including task-specific models like SVM, RF, LSTM, CNN, and BERT, as well as multitask learning frameworks such as Hard Parameter Sharing, Soft Parameter Sharing, Cross-stitch Networks, MMoE, and T5. The performance was evaluated across various tasks, and statistical significance was assessed using the Wilcoxon signed-rank test. Additionally, an ablation study was conducted to determine the contribution of individual components within our proposed method. The proposed framework consistently outperformed other models across all tasks. For instance, in emotion recognition, our model achieved an accuracy of 0.899, F1-score of 0.883, and AUC-ROC of 0.971, surpassing all baseline models. The Wilcoxon signed-rank test further confirmed the statistical superiority of our model over the baselines across all datasets. © 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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AbCoRD: Exploiting multimodal generative approach for Aspect-based Complaint and Rationale Detection

Item Type Conference Paper

Author R. Jain

Author A. Singh

Author V. Gangwar

Author S. Saha

Abstract Valuable feedback can be found in customer reviews about a product or service, but it can be difficult to identify specific complaints and their underlying causes from the text. While various methods have been used to detect and analyze complaints, there has been little research on examining complaints at the aspect level and the reasons behind these complaints. To address this, we added rationale annotation for aspect-based complaint classes to a publicly available benchmark multimodal complaint dataset (CESAMARD) covering five domains (books, electronics, edibles, fashion, and miscellaneous). Current methods treat these tasks as classification problems and do not use external knowledge. Our study proposes a knowledge-infused, hierarchical, multimodal generative approach for aspect-based complaint and rationale detection that reframes the multitasking problem as a multimodal text-to-text generation task. Our approach achieved benchmark performance in the aspect-based complaint and rationale detection task through an extensive evaluation. We demonstrated that our model consistently outperformed all other baselines and state-of-the-art models in both full and few-shot settings. Our study contributes to the development of more accurate and efficient methods for extracting valuable insights from customer reviews to improve products and service 1 © 2023 ACM.

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Abstractive Text Summarization for the Urdu Language: Data and Methods

Item Type Journal Article**Author** M. Awais**Author** R. Muhammad Adeel Nawab

Abstract The task of abstractive text summarization aims to automatically generate a short and concise summary of a given source article. In recent years, automatic abstractive text summarization has attracted the attention of researchers because large volumes of digital text are readily available in multiple languages on a wide range of topics. Automatically generating precise summaries from large text has potential application in the generation of news headlines, a summary of research articles, the moral of the stories, media marketing, search engine optimization, financial research, social media marketing, question-answering systems, and chatbots. In literature, the problem of abstractive text summarization has been mainly investigated for English and some other languages. However, it has not been thoroughly explored for the Urdu language despite having a huge amount of data available in digital format. To fulfill this gap, this paper presents a large benchmark corpus of 2,067,784 Urdu news articles for the Urdu abstractive text summarization task. As a secondary contribution, we applied a range of deep learning (LSTM, Bi-LSTM, LSTM with attention, GRU, Bi-GRU, and GRU with attention), and large language models (BART and GPT-3.5) on our proposed corpus. Our extensive evaluation on 20,000 test instances showed that GRU with attention model outperforms the other models with ROUGE-1=46.7 , ROUGE-2=24.1 , and ROUGE-L =48.7. To foster research in Urdu, our proposed corpus is publically and freely available for research purposes under the Creative Common Licence. © 2013 IEEE.

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Abusive and Hate speech Classification in Arabic Text Using Pre-trained Language Models and Data Augmentation

Item Type Journal Article

Author N. Badri

Author F. Koubi

Author A. Habacha Chaibi

Abstract Hateful content on social media is a worldwide problem that adversely affects not just the targeted individuals but also anyone whose content is accessible. The majority of studies that looked at the automatic identification of inappropriate content addressed the English language, given the availability of resources. Therefore, there are still a number of low-resource languages that need more attention from the community. This article focuses on the Arabic dialect, which has several specificities that make the use of non-Arabic models inappropriate. Our hypothesis is that leveraging pre-trained language models (PLMs) specifically designed for Arabic, along with data augmentation techniques, can significantly enhance the detection of hate speech in Arabic mono- and multi-dialect texts. To test this hypothesis, we conducted a series of experiments addressing three key research questions: (RQ1) Does text augmentation enhance the final results compared to using an unaugmented dataset? (RQ2) Do Arabic PLMs outperform other models utilizing techniques such as fastText and AraVec word embeddings? (RQ3) Does training and fine-tuning models on a multilingual dataset yield better results than training them on a monolingual dataset? Our methodology involved the comparison of PLMs based on transfer learning, specifically examining the performance of DziriBERT, AraBERT v2, and BERT-base-arabic models. We implemented text augmentation techniques and evaluated their impact on model performance. The tools used included fastText and AraVec for word embeddings, as well as various PLMs for transfer learning. The results demonstrate a notable improvement in classification accuracy, with augmented datasets showing an increase in performance metrics (accuracy, precision, recall, and F1-score) by up to 15-21% compared to non-augmented datasets. This underscores the potential of data augmentation in enhancing the models' ability to generalize across the nuanced spectrum of Arabic dialects. © 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Abusive content detection in transliterated Bengali-English social media corpus

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Author S. Sazzed
Editor Solorio T.
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Abstract Abusive text detection in low-resource languages such as Bengali is a challenging task due to the inadequacy of resources and tools. The ubiquity of transliterated Bengali comments in social media makes the task even more involved as monolingual approaches cannot capture them. Unfortunately, no transliterated Bengali corpus is publicly available yet for abusive content analysis. Therefore, in this paper, we introduce an annotated corpus of 3000 transliterated Bengali comments categorized into two classes, abusive and non-abusive, 1500 comments for each. For baseline evaluations, we employ several supervised machine learning (ML) and deep learning-based classifiers. We find support vector machine (SVM) classifier shows the highest efficacy for identifying abusive content. We make the annotated corpus publicly available for the researchers to aid abusive content detection in Bengali social media data. © 2021 Association for Computational Linguistics.

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Accuracy of a large language model in distinguishing anti- and pro-vaccination messages on social media: The case of human papillomavirus vaccination

Item Type Journal Article

Author S. Kim

Author K. Kim

Author C. Wonjeong Jo

Abstract Objective: Vaccination has engendered a spectrum of public opinions, with social media acting as a crucial platform for health-related discussions. The emergence of artificial intelligence technologies, such as large language models (LLMs), offers a novel opportunity to efficiently investigate public discourses. This research assesses the accuracy of ChatGPT, a widely used and freely available service built upon an LLM, for sentiment analysis to discern different stances toward Human Papillomavirus (HPV) vaccination. Methods: Messages related to HPV vaccination were collected from social media supporting different message formats: Facebook (long format) and Twitter (short format). A selection of 1,000 human-evaluated messages was input into the LLM, which generated multiple response instances containing its classification results. Accuracy was measured for each message as the level of concurrence between human and machine decisions, ranging between 0 and 1. Results: Average accuracy was notably high when 20 response instances were used to determine the machine decision of each message: .882 (SE = .021) and .750 (SE = .029) for anti- and pro-vaccination long-form; .773 (SE = .027) and .723 (SE = .029) for anti- and pro-vaccination short-form, respectively. Using only three or even one instance did not lead to a severe decrease in accuracy. However, for long-form messages, the language model exhibited significantly lower accuracy in categorizing pro-vaccination messages than anti-vaccination ones. Conclusions: ChatGPT shows potential in analyzing public opinions on HPV vaccination using social media content. However, understanding the characteristics and limitations of a language model within specific public health contexts remains imperative. © 2024

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Artificial intelligence, Article, human, Social media, ChatGPT, social media, public health, sentiment analysis, public opinion, health behavior, large language model, Large Language Models, vaccination, anti-vaccination movement, Vaccination, clinical assessment, clinical research, clinical evaluation, public health message, clinical classification, health service, Human papilloma virus vaccine, Human Papillomavirus, measurement accuracy, Public discourse, Wart virus

Notes:

EID: 2-s2.0-85190816975

Export Date: 20 September 2026; Cited By: 16; Correspondence Address: S. Kim; Davis, 1 Shields Ave, Kerr Hall 361, 95616, United States; email: sjokim@ucdavis.edu

ACOPI-QGen: Implicitness-aware joint quintuple extraction for aspect-based sentiment analysis via contrastive and graph-based modeling

Item Type Journal Article

Author L. Dewangan

Author C.K. Maurya

Abstract In social media and online reviews, sentiments are often expressed implicitly through contextual, figurative, or indirect language, making them difficult to capture using traditional Aspect-Based Sentiment Analysis (ABSA) techniques. Many prior studies on ABSA treat implicit aspects and opinions as auxiliary cases or evaluation subsets, limiting their role during learning. To address this limitation, we propose ACOPI-QGen, a joint structured framework that extracts a five-element sentiment quintuple (Aspect Term, Aspect Category, Opinion Term, Polarity, Implicitness). Rather than treating implicitness as a post-hoc attribute, we model it as a supervised four-class variable: EAEO, EAIO, IAEO, and IAIO, where A/O denote aspect/opinion and E/I denote explicit/implicit realization. The proposed framework combines BERT-based contextual encoding, Supervised Contrastive Implicit Learning, and a Relational Graph Attention Network guided by a Surrogate Aspect-Opinion Dependency Tree to support implicit sentiment modeling when aspect or opinion spans are missing. A non-autoregressive set decoder uses parallel tuple queries to predict multiple candidate sentiment quintuples while preserving aspect-opinion alignment. We curate three datasets for quintuple extraction (Restaurant-ACOPI: 2584; Laptop-ACOPI: 4376; and Shoes-ACOPI: 3896 instances). ACOPI-QGen achieves Exact Match F1 scores of 55.68, 48.43, and 60.49 on Restaurant-ACOPI, Laptop-ACOPI, and Shoes-ACOPI, respectively, with corresponding improvements of 2.92, 2.53, and 5.45 absolute F1 points over the best-performing state-of-the-art models across the three domains. These results show that explicit modeling of implicitness improves joint quintuple extraction in implicit sentiment settings. © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Journal Abbr Inf. Process. Manage.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: L. Dewangan; Department of Computer Science and Engineering, Indian Institute of Technology Indore, Madhya Pradesh, India; email: phd2101201004@iiti.ac.in; CODEN: IPMAD

Adapting GPT for Egyptian Arabic-English Code-Switched Sentiment Analysis Through Prompting, Retrieval, and Sentiment-Guided Fine-Tuning

Item Type Conference Paper

Author A. Sherif

Author R. Hussein

Author C. Sabty

Editor Cabrio E.

Editor Monteiro E.

Abstract Arabic–English code-switching is common in Egyptian social media, where users naturally mix Arabic and English in short, informal comments that often include emojis and non-standard spelling. This setting poses challenges for sentiment analysis, while publicly available resources for Egyptian code-switched text remain limited. In this paper, we investigate GPT-based sentiment classification for Egyptian Arabic–English code-switched text using the EESA corpus, a labelled dataset of YouTube comments annotated for three-way sentiment. We evaluate three adaptation strategies: prompt-based inference, retrieval-augmented generation (RAG), and fine-tuning. For prompt-only inference, the best configuration combines GPT-4o with annotation guidelines and sentiment hints, achieving an F1-score of 90.8%. RAG further improves performance by conditioning predictions on dynamically retrieved in-domain labelled examples. The strongest RAG configuration, based on the text-embedding-3-large encoder and a modified retrieval setup with explanations, reaches an F1-score of 93.6%. We additionally propose enriching fine-tuning data with sentiment hints, lightweight textual cues that guide the model toward the correct polarity during training. Fine-tuning GPT-4o-mini with sentiment hints achieves an F1-score of 95.5%, establishing a new state of the art on EESA and surpassing the previous best result by 2.8 points. Overall, the findings show that structured prompting provides a strong baseline, retrieval augmentation offers substantial gains without

parameter updates, and lightweight fine-tuning with sentiment hints yields the highest performance. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2027.

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Adaptive Multimodal Deepfake Detection for Social Media: Robust Fusion of Textual and Visual Artifact

Item Type Conference Paper

Author G. Deshmukh

Author A. Momin

Author T. Abhang

Author A. Dharmadhikari

Author A. Pathak

Abstract Deepfake news is rapidly proliferating on social media, endangering digital identity, misinformation management, and public trust. Despite the high accuracy of current deepfake detection techniques in offline settings, their practical application in real world browsing situations is still restricted by computational limitations and a lack of contextual awareness. A context gated, near real time deepfake detection framework for social media analysis is presented in this study. The proposed solution combines a browse- based monitoring pipeline, a FastAPI backend for efficient inference, and a fine tuned EfficientNet B0 image classifier. A zero shot BART MNLI text classifier is used to analyze webpage context to selectively activate detection. The RVF10k dataset, which includes both real images from FFHQ and fake images generated using StyleGAN, is used to train and evaluate the image classifier. The model achieves a validation accuracy of 98 percent with balanced precision and recall across classes. The possibility of implementing deepfake detection models directly within user browsing environments is shown by experimental results, which demonstrate that contextual gating dramatically increases inference efficiency while maintaining high detection performance. © 2026 IEEE.

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Adaptive multitask emotion recognition and sentiment analysis using resource-constrained MobileBERT and DistilBERT: an efficient approach for edge devices

Item Type Journal Article

Author M. Hussain

Author C. Chen

Author M. Anwar

Author S.A. Ghorashi

Author A. Ahmed

Author M.S.A. Malik

Author I. Yamin

Abstract Emotion recognition and sentiment analysis are crucial tasks in natural language processing, enabling machines to understand human emotions and opinions. However, the complex, nuanced relationship between emotions and sentiment in conversation poses significant challenges to accurate emotion recognition, as sentiment cues can be easily misinterpreted. Deploying emotion recognition and sentiment analysis tasks on edge devices poses substantial challenges due to computational resource constraints. We present an adaptive multitask learning approach that jointly leverages resource-constrained Mobile Bidirectional Encoder Representations from Transformers (MobileBERT) and Distilled BERT (DistilBERT) models to optimise emotion recognition and sentiment analysis. Our proposed approach utilises prototypical networks to learn effective representations of emotions and sentiment, while a focal weighted loss function effectively mitigates the class imbalance. We adaptively fine-tune the learning process to balance task importance and resource utilisation, resulting in better performance and efficiency. Our experimental results demonstrate the efficacy of our method, achieving the best results on MELD and IEMOCAP benchmark datasets while keeping a compact model size. Despite limited computational demands, our solution demonstrates that emotion and sentiment analysis can deliver performance comparable to resource-intensive large language models (LLMs). Facilitating various applications in human-computer interaction, affective computing, social media, dialogue conversion, and healthcare. Copyright © 2025 Hussain et al. Distributed under Creative Commons CC-BY 4.0

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Adaptive Real-Time Context-Aware Cyberbullying Detection and Intervention Framework

Item Type Conference Paper

Author K. Bhole

Author T. Ugale

Abstract The growing prevalence of cyberbullying on digital platforms demands intelligent, real-time systems capable of detecting harmful interactions with contextual sensitivity. Traditional approaches often rely on surface-level sentiment analysis or keyword filtering, resulting in high false positives and missing abusive language veiled through sarcasm or code-switching. This study presents a novel framework that leverages large language models to understand both the emotional tone and intent of messages. The architecture employs DeepSeek R1 Distill Qwen 7B to extract emotional cues and generate conversational summaries, emulating human-like interpretation. These insights are passed to a finetuned Mistral 7B classifier, which evaluates the message and issues a confidence-based verdict. Low-confidence outputs are escalated via a human-in-the-loop layer to maintain moderation accuracy and ethical oversight.

Additionally, a SEAL-based self-adaptive loop enables the system to autonomously expand its training dataset using edge-case interactions, reducing reliance on manual updates. Real-time intervention mechanisms issue escalating warnings and alerts to mitigate ongoing abuse. The system is trained on a multilingual, emotionally diverse dataset combining Reddit, Meta, and Kaggle data, achieving 93.38% accuracy and strong F1-scores. It also reports a BLEU score of 0.72, reflecting high lexical consistency in summarization. The framework is scalable, language-aware, and deployable via APIs for broad platform integration. © 2025 IEEE.

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Adopting Pre-trained Large Language Models for Regional Language Tasks: A Case Study

Item Type Conference Paper

Author H. Gaikwad

Author A. Kiwelekar

Author M. Laddha
Author S. Shahare
Editor Choi B.J.
Editor Singh D.
Editor Tiwary U.S.
Editor Chung W.-Y.

Abstract Large language models have revolutionized the field of Natural Language Processing. While researchers have assessed their effectiveness for various English language applications, a research gap exists for their application in low-resource regional languages like Marathi. The research presented in this paper intends to fill that void by investigating the feasibility and usefulness of employing large language models for sentiment analysis in Marathi as a case study. The study gathers a diversified and labeled dataset from Twitter that includes Marathi text with opinions classified as positive, negative, or neutral. We test the appropriateness of pre-existing language models such as Multilingual BERT (M-BERT), IndicBERT, and GPT-3 ADA on the obtained dataset and evaluate how they performed on the sentiment analysis task. Typical assessment metrics such as accuracy, F1 score, and loss are used to assess the effectiveness of sentiment analysis models. This research paper presents additions to the growing area of sentiment analysis in languages that have not received attention. They open up possibilities for creating sentiment analysis tools and applications specifically tailored for Marathi-speaking communities. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

Date 2024

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Language English

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Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: H. Gaikwad; Department of Computer Engineering, Dr. Babasaheb Ambedkar Technological University, Lonere, Maharashtra, 402 103, India; email: harsha.gaikwad@dbatu.ac.in; Conference name: 15th International Conference on Intelligent Human Computer Interaction, IHCI 2023; Conference date: 8 November 2023 through 10 November 2023; Conference code: 308899

Advanced deep learning and large language models for suicide ideation detection on social media

Item Type Journal Article

Author M. Qorich

Author R. El Ouazzani

Abstract Recently, suicide ideations represent a worldwide health concern and pose many anticipation challenges. Actually, the prevalence of expressing self-destructive thoughts especially on forums and social media requires effective monitoring for suicide prevention, and early intervention. Meanwhile, deep learning techniques and Large Language Models (LLMs) have emerged as promising tools in diverse Natural Language Processing (NLP) tasks, including sentiment analysis and text classification. In this paper, we propose a deep learning model incorporating triple models of word embeddings, as well as various fine-tuned LLMs, to identify suicidal thoughts in Reddit posts. In effect, we implemented a Bidirectional Long Short-Term Memory (BiLSTM), and a Convolutional Neural Network (CNN) model to categorize posts associated with non-suicidal and suicidal thoughts. Besides, through the combination of Word2Vec, FastText and GloVe embeddings, our models learn intricate patterns and prevalent nuances in suicide-related language. Furthermore, we employed a merged version of CNN and BiLSTM models, entitled C-BiLSTM, and several LLMs, including pre-trained Bidirectional Encoder Representations from Transformers (BERT) models and a Generative Pre-training Transformer (GPT) model. The analysis of all our proposed models shows that our C-BiLSTM model with triple word embedding and our GPT model got the best performance compared to deep learning and LLMs baseline models, reaching accuracies of 94.5% and 97.69%, respectively. In fact, our best model's capacity to extract meaningful interdependencies among words significantly promotes its classification performance. This analysis contributes to a deeper understanding of the psychological factors and linguistic markers indicative of suicidal thoughts, thereby informing future research and intervention strategies. © Springer-Verlag GmbH Germany, part of Springer Nature 2024.

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ISSN 21926352 (ISSN)**Issue** 2**Journal Abbr** Prog. Artif. Intell.**Date Added** 20/09/2026, 8:35:35 pm**Modified** 20/09/2026, 8:35:35 pm**Tags:**

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Notes:

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Export Date: 20 September 2026; Cited By: 21; Correspondence Address: M. Qorich; IMAGE Laboratory, ISNET Team, School of Technology, Moulay Ismail University of Meknes, Meknes, Morocco; email: mohamedqorich@gmail.com

Advancements in Aspect-Based Sentiment Analysis: A Review of Models, Techniques, and Challenges

Item Type Conference Paper**Author** M. Ashokkumar**Author** R.M. Koushiik**Author** T. Mummudi Cholan**Author** I.S. Theveesh**Author** P. Jegathesh**Editor** Bansal J.C.**Editor** Borah S.**Editor** Hussain S.**Editor** Salhi S.

Abstract Aspect-Based Sentiment Analysis (ABSA) has emerged as a vital domain within Natural Language Processing (NLP) that extracts and classifies sentiment for specific aspects within text. With the surge in user-generated content, ABSA is widely applied in customer feedback analysis, social media monitoring, and market research. Traditional sentiment analysis often fails to capture aspect-specific nuances, necessitating advanced machine learning and deep learning techniques. This paper reviews the evolution of ABSA from early machine learning models (e.g., SVM, Naive Bayes) to deep learning approaches (e.g., RNNs, LSTMs, CNNs) and state-of-the-art transformers (e.g., BERT, ALBERT,

RoBERTa). Transformer models have shown a 12–18% accuracy improvement over traditional methods in benchmark datasets. Key challenges in multilingual adaptability, sarcasm detection, and domain-specific ABSA are discussed. Emerging trends, including hybrid models, zero-shot learning, and explainability techniques, demonstrate performance improvements. Recent studies (2023–2024) indicate that hybrid approaches enhance F1 scores by 5–10%, while zero-shot learning enables sentiment analysis in unseen domains with minimal labeled data. This review consolidates recent advancements, offering insights into ABSA methodologies, applications, and future directions, with a focus on model interpretability, efficiency, and multilingual scalability. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Advancements in Text Subjectivity Analysis: From Simple Approaches to BERT-Based Models and Generalization Assessments

Item Type Conference Paper

Author M. Antal

Author K. Buza

Author S. Nemes

Editor Nguyen N.-T.

Editor Kozierekiewicz A.

Editor Franczyk B.

Editor Ludwig A.

Editor Nunez M.

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Editor Vossen G.

Abstract Text subjectivity is an important research topic due to its applications in various domains such as sentiment analysis, opinion mining, social media monitoring, clinical research and patient feedback analysis. While rule-based approaches dominated this field at the beginning of the 21st century, contemporary works rely on transformers, a specific neural network architecture designed for language modeling. This paper explores the performance of various BERT-based models, including our fine-tuned BERT (Bidirectional Encoder Representations from Transformer) model, and compares them with pre-built models. To assess the generalization abilities of the models, we evaluated the models on benchmark datasets. Additionally, the models underwent evaluation on two synthetic datasets created using large language models. To ensure reproducibility, we have made our implementation publicly available at <https://github.com/margital68/TextSubjectivity>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Advances in Deep Learning for Affective Intelligence: Language Models, Multimodal Trends, and Research Frontiers

Item Type Journal Article

Author D.A. Andrade-Segarra

Author J.C. Santillán-Lima

Author M. Duque-Vaca

Author F.T. Molina-Granja

Abstract The accelerated growth of digital content and the increasing presence of emotional expressions, polarized opinions, and toxic behaviors in social media have driven the development of advanced Affective Analysis techniques. This study presents a broad and up-to-date review of recent studies covering Sentiment Analysis, Emotion Recognition, Hate Speech Detection, cyberbullying, and multimodal approaches grounded in deep learning. The review provides a comparative analysis of the architectures employed—including Transformer-based Models, multimodal frameworks, and variants designed for low-resource languages—along with their metrics, performance outcomes, and emerging patterns. The findings reveal a clear consolidation of Transformer-based Models as the dominant standard, significant progress in multimodality for affective interpretation, and growing attention to multilingual models adapted to diverse cultural contexts. Furthermore, persistent challenges are identified, including limitations related to data availability and quality, Explainable AI (XAI), computational efficiency, and robustness in cross-domain generalization. This review synthesizes current trends, limitations, and opportunities in the field, offering a structured perspective that can serve as a reference for researchers and practitioners involved in the development of more accurate, efficient, and culturally responsible affective systems. © (2026), (Science and Information Organization). All Rights Reserved.

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Advancing performance measurement in public healthcare organizations through artificial intelligence: evidence from action research

Item Type Journal Article

Author G. Noto

Author M. Barraco

Author F. De Domenico

Author G. Barresi

Abstract Digital transformation is reshaping healthcare management by introducing advanced technologies that improve processes, services, and decision-making. Among these, Artificial Intelligence (AI) represents a promising tool for enhancing performance measurement (PM). This study investigates the potential of AI to support PM by focusing on outcomes perceived at the community level, a dimension rarely measured yet important to assess the public value recognized by community. Adopting an action-research approach, the study applies Sentiment Analysis (SA) to nine public hospitals in the Sicilian regional health system to develop a PM dashboard. Online mentions collected from social media, news sites, blogs, and forums were analysed through AI-based tool to generate structured indicators of sentiment and thematic insights. Results were synthesized in interactive dashboards, enabling benchmarking across hospitals and offering managers actionable information to improve

communication, accountability, and service strategies. The findings highlight how AI-driven sentiment and topic analysis can support traditional PM by capturing non-investigated dimensions of perceived value at the societal level and enhancing dialogic accountability. The study advances theoretical knowledge on digital transformation in healthcare PM and offers practical implications for managers seeking to integrate AI into value recognition and strategic decision-making. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: G. Noto; Department of Economics, University of Messina, Messina, Italy; email: gnoto@unime.it

Advancing sentiment analysis for low-resourced african languages using pre-trained language models

Item Type Journal Article

Author K.R. Mabokela

Author M. Primus

Author T. Celik

Abstract While sentiment analysis systems excel in high-resource languages, most African languages facing limited resources, remain under-represented. This gap leaves a significant portion of the world's population without access to technologies in their native languages. However, multilingual pre-trained language models (PLM) offer a promising approach for sentiment analysis in low-resource languages. Although the absence of large data in African languages poses a challenge for developing PLMs, fine-tuning and task adaptation of existing multilingual PLMs is an alternative solution. This paper explores the use of multilingual PLMs for sentiment analysis in

five Southern African languages: Sepedi, Sesotho, Setswana, isiXhosa, and isiZulu. We leverage existing PLMs and fine-tune them for this specific task, avoiding training the models from scratch. Our work expands on the SAfriSenti corpus, a Twitter sentiment dataset for these languages. We employ various annotation techniques to create a labelled dataset and perform benchmark experiments utilising various multilingual PLMs. Our findings demonstrate the effectiveness of multilingual PLM, particularly for closely-related languages (Sotho-Tswana), where the ensemble PLMs method achieved an average weighted F1 score above 63%. In particular, Nguni closely-related languages achieved an even higher average weighted F1 score, exceeding 77%, highlighting the potential of PLMs for sentiment analysis in South African languages. © 2025 Mabokela et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Advancing Sentiment Analysis with LLM: Comparative Study with Traditional ML on Social Media Reviews

Item Type Conference Paper

Author J. Kamala Kanth

Author P. Vidhya Saraswathi

Author T. Radha

Abstract This research develops a sentiment classification model using machine learning to analyze textual data. It follows three stages: data collection, preprocessing, and model development. Using the NLTK dataset, text mining techniques extract and refine features. Supervised machine learning classifies tweets as positive or negative, with testing across two frameworks yielding higher accuracy than prior studies. Given Twitter's prominence, this study examines whether sentiment analysis models can match human evaluators in educational research. Sentiment ratings from 333 students on their learning experiences were analyzed. Nine machine learning models were tested under five experimental setups, alongside two non-ML methods. Comparisons with human raters' sentiment ratings revealed that the Naïve Bayes model achieved a 98% accuracy rate after excluding neutral sentiments. Challenges in identifying neutral sentiments were noted, with a word-sentiment association method attaining 87% accuracy without model training, enhancing generalization and adaptability. Expanding educational datasets is crucial for improving sentiment analysis models, as AI and machine learning facilitate automated classification. Businesses leverage this technology to analyze consumer emotions, monitor social media discussions, evaluate reviews, assess employee sentiment, and detect harmful content. Traditional models like SVM excel in processing short text, while GPT-4 outperforms them in analyzing context-rich texts, achieving superior precision, recall, and F1 scores. Additionally, GPT-4 is more effective in detecting mixed sentiments. Findings suggest that large language models enhance sentiment classification accuracy, providing valuable insights for businesses extracting meaningful information from textual data. © 2025 IEEE.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: J. Kamala Kanth; Kalasalingam Academy of Research and Education, Department of Computer Applications, India; email: kamalkanth7781@gmail.com; Conference name: 11th International Conference on Communication and Signal Processing, ICCSP 2025; Conference date: 5 June 2025 through 7 June 2025; Conference code: 210861

Adversarial contrastive representation training with external knowledge injection for zero-shot stance detection

Item Type Journal Article**Author** Y. Ding**Author** Y. Lei**Author** A. Wang**Author** X. Liu**Author** T. Zhu**Author** Y. Li

Abstract Zero-shot stance detection (ZSSD) is a task that involves identifying the author's perspective on specific issues in text, particularly when the target topic has not been encountered during the model training process, to address rapidly evolving topics on social media. This paper introduces a ZSSD framework named KEL-CA. To enable the model to more effectively utilize transferable stance features for representing unseen targets, the framework incorporates a multi-layer contrastive learning and adversarial domain transfer module. Unlike traditional contrastive or adversarial learning, our framework captures both correlations and distinctions between invariant and specific features, as well as between different stance labels, and enhances the generalization ability and robustness of the features. Subsequently, to address the problem of insufficient information about the target context, we designed a dual external knowledge injection module that uses a large language model (LLM) to extract external knowledge from a Wikipedia-based local knowledge base and a Chain-of-Thought (COT) process to ensure the timeliness and relevance of the knowledge to infer the stances of unseen targets. Experimental results demonstrate that our approach outperforms existing models on two benchmark datasets, thereby validating its efficacy in ZSSD tasks. © 2024 Elsevier B.V.

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Adversarial Learning-Based Stance Classifier for COVID-19-Related Health Policies

Item Type Conference Paper

- Author** F. Xie
- Author** Z. Zhang
- Author** X. Zhao
- Author** H. Wang
- Author** J. Zou
- Author** L. Tian
- Author** B. Zhou
- Author** Y. Tan
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Abstract The ongoing COVID-19 pandemic has caused immeasurable losses for people worldwide. To contain the spread of the virus and further alleviate the crisis, various health policies (e.g., stay-at-home orders) have been issued which spark heated discussions as users turn to share their attitudes on social media. In this paper, we consider a more realistic scenario on stance detection (i.e., cross-target

and zero-shot settings) for the pandemic and propose an adversarial learning-based stance classifier to automatically identify the public's attitudes toward COVID-19-related health policies. Specifically, we adopt adversarial learning that allows the model to train on a large amount of labeled data and capture transferable knowledge from source topics, so as to enable generalize to the emerging health policies with sparse labeled data. To further enhance the model's deeper understanding, we incorporate policy descriptions as external knowledge into the model. Meanwhile, a GeoEncoder is designed which encourages the model to capture unobserved background factors specified by each region and then represent them as non-text information. We evaluate the performance of a broad range of baselines on the stance detection task for COVID-19-related health policies, and experimental results show that our proposed method achieves state-of-the-art performance in both cross-target and zero-shot settings. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Adversarial neural network with sentiment-aware attention for detecting adverse drug reactions

Item Type Journal Article

Author T. Zhang

Author H. Lin

Author B. Xu

Author L. Yang

Author J. Wang

Author X. Duan

Abstract Adverse drug reaction (ADR) detection is an important issue in drug safety. ADRs are health threats caused by medication. Identifying ADRs in a timely manner can reduce harm to patients and can also assist doctors in the rational use of drugs. Many studies have investigated potential ADRs based on social media due to the openness and timeliness of this resource; however, they have ignored the fine-grained emotional expression in social media text. In addition, the benchmark datasets from social media are usually small, which can result in the problem of over-fitting. In this paper, we propose the Adversarial Neural Network with Sentiment-aware Attention (ANNSA) model, which enhances the sentimental element in social media and improves the performance of neural networks via data augmentation. Specifically, a sentiment-aware attention mechanism is proposed to extract the word-level sentiment features associated with sentiment words and learn task-related information by optimizing a task-specific loss. For low-resource datasets, we use an adversarial training approach to generate perturbations of the word embeddings via an implicit regularization technique. ANNSA was tested on three social media ADR detection datasets, namely, Twitter, TwiMed (Twitter) and CADEC. The experimental results indicated the ability to achieve F1 values of 48.84%, 64.18% and 83.06%, respectively, comparable to the best results reported for state-of-the-art methods. Our study demonstrates that sentiment words are highly correlated with ADRs and that word-level sentiment features can assist in detecting ADRs from social media datasets. © 2021

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AfroXLMR-Social: Adapting Pre-trained Language Models for African Languages Social Media Text

Item Type Conference Paper

Author T.D. Belay

Author I.A. Azime

Author I.S. Ahmad

Author D.I. Adelani

Author I. Abdulmumin

Author A.A. Ayele

Author S.H. Muhammad

Author S.M. Yimam

Editor Christodoulopoulos C.

Editor Chakraborty T.

Editor Rose C.

Editor Peng V.

Abstract Language models built from various sources are the foundation of today's NLP progress. However, for many low-resource languages, the diversity of domains is often limited, more biased to a religious domain, which impacts their performance when evaluated on distant and rapidly evolving domains such as social media. Domain adaptive pre-training (DAPT) and task-adaptive pre-training (TAPT) are popular techniques to reduce this bias through continual pre-training for BERT-based models, but they have not been explored for African multilingual encoders. In this paper, we explore DAPT and TAPT continual pre-training approaches for African languages social media domain. We introduce AfriSocial, a large-scale social media and news domain corpus for continual pretraining on several African languages. Leveraging AfriSocial, we show that

DAPT consistently improves performance (from 1% to 30% F1 score) on three subjective tasks: sentiment analysis, multi-label emotion, and hate speech classification, covering 19 languages. Similarly, leveraging TAPT on the data from one task enhances performance on other related tasks. For example, training with unlabeled sentiment data (source) for a fine-grained emotion classification task (target) improves the baseline results by an F1 score ranging from 0.55% to 15.11%. Combining these two methods (i.e. DAPT + TAPT) further improves the overall performance. The data and model resources are available at HuggingFace1 ©2025 Association for Computational Linguistics.

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AFuNet: an attention-based fusion network to classify texts in a resource-constrained language

Item Type Journal Article**Author** M.R. Hossain**Author** M.M. Hoque**Author** M.A.A. Dewan**Author** E. Hoque**Author** N. Siddique

Abstract In the era of widespread Internet use and extensive social media interaction, the digital realm is accumulating vast amounts of unstructured text data. This unstructured data often contain undesirable information, necessitating time-consuming manual classification efforts. An intelligent text classification system capable of automatically categorizing digitized texts based on semantic meaning is crucial. However, this task is particularly challenging for low-resource languages like Bengali due to a shortage of annotated corpora, issues with out-of-vocabulary words, lack of domain-specific hyperparameter tuning, limited ability to extract generalized text features, and class imbalances within the corpus. AFuNet: an attention-based fusion network to classify texts in a resource-constrained language. AFuNet undergoes a comprehensive four-phase experimental process, including baseline model evaluation and hyperparameter tuning, late fusion and model selection, attention-based early fusion and model identification, and an ablation study with impact analysis. Fine-tuned based on five Bengali text classification corpora, AFuNet achieves impressive accuracies: 96.60 ± 0.2 (BTCC11), 85.37 ± 0.2 (OSBC), 97.35 ± 0.2 (BARD), 93.74 ± 0.2 (IndicNLP), and 96.51 ± 0.2 (ProthomAlo). In comparison with previous state-of-the-art models on these corpora, AFuNet demonstrates significant accuracy improvements ranging from 0.54% to 4.49%, showcasing its effectiveness in advancing text classification capabilities for the Bengali language. © The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2025.

Date 2025**DOI** 10.1007/s00521-024-10953-1**URL** <https://www.scopus.com/pages/publications/85217165684?origin=resultslist>**Archive** Scopus**Language** English**Volume** 37**Publisher** Springer Science and Business Media Deutschland GmbH**Pages** 6725-6748**Publication** Neural Computing and Applications**ISSN** 09410643 (ISSN)**Issue** 9**Journal Abbr** Neural Comput. Appl.**Date Added** 20/09/2026, 8:35:09 pm**Modified** 20/09/2026, 8:35:09 pm**Tags:**

Classification (of information), Language processing, Natural language processing, Natural language processing systems, Natural languages, Semantics, Metadata, Hyper-parameter, Text classification, Text mining, Low resource languages, Low-resource languages, Multi-head attention, Bengalis, Early fusion, Transformer-based learning

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AI and social science: Automatic classification tools for big data analysis in sociological research

Item Type Journal Article**Author** A. Nucita**Author** A. Penna**Author** A. Cava**Author** G. Iannizzotto**Author** M. Mucciardi

Abstract This study examines the use of Social Network Sites for public institutional communication through a sociological, data-driven lens, focusing on the challenges and potential of automated classification tools for data analysis. Although Large Language Models are increasingly used to process social media data, a key research gap remains: few studies systematically assess whether AI-based categorizations are as reliable as human coding, especially when categories are semantically ambiguous. The research addresses the following questions: How reliable are AI-generated classifications compared to those made by human experts? Is human-machine agreement comparable to the level of agreement observed among human coders? To experimentally test this approach, we conducted a case study on Facebook posts published by two Italian universities (March 2020–March 2023), classified into eight categories of public institutional communication. Three researchers independently annotated the dataset. Human annotations are used as a benchmark to assess agreement patterns and to compare them with classifications produced by AI-based systems. Results show substantial interpretive ambiguity across several categories, mirrored by variability among human coders. Nonetheless, automated models achieve agreement with human classifications that is broadly comparable to inter-coder agreement. Overall, the findings support integrating AI as an additional coder within hybrid workflows to enable scalable and transparent sociological analysis of complex social media data. © 2026 Nucita et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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AI in Cybersecurity: How Large Language Models Classify Risk and Opportunity

Item Type Conference Paper**Author** A.R. Caciulescu**Author** A. Baraitaru**Author** A. Radovici**Editor** Dumitrache I.**Editor** Moisescu M.**Editor** Pop F.**Editor** Copot D.**Editor** Popescu D.

Abstract This study examines the convergence and divergence of large language models (LLMs) in detecting risk and opportunity framings of artificial intelligence (AI) within cybersecurity discourse. Using a structured measurement framework of 18 binary-coded indicators, we prompted seven LLMs to analyze three high-engagement Reddit threads discussing AI in cybersecurity. All models received identical input and were instructed to apply the framework without inference or contextual extrapolation. The analysis shows that while models tend to agree on broadly discussed operational claims, such as AI's capacity to automate tasks or enable attacks, they diverge on more context-dependent indicators, including

institutional reliability, auditability, and human oversight. These findings show that LLMs reflect stabilized discursive patterns but vary in interpreting embedded or contested evaluations. The study contributes to methodological research on LLM-based content analysis and highlights the need for caution when using automated annotation tools in domains involving evaluative and normative judgments. © 2025 IEEE.

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AI perception in digital consumption: A multi-layered analysis of evaluation, discourse, and engagement

Item Type Journal Article

Author I. Bayram

Author N.G. Ugur

Abstract Artificial intelligence (AI) is increasingly visible in digital consumption environments, yet how consumers respond to AI-mediated interactions remains insufficiently understood. Accordingly, this study aims to examine how AI-related cues inferred from naturally occurring digital discourse shape multiple layers of consumer response in platform-based consumption settings. Prior research largely assumes a direct link between consumer evaluation and downstream behavioral outcomes, often treating AI perception as a uniform construct. This study challenges these assumptions by conceptualizing consumer response to AI as a multi-layered system comprising evaluative, discursive, and engagement-based components. Drawing on the stimulus–organism–response (S-O-R) framework, we model AI perception as a multidimensional stimulus that shapes consumer sentiment (evaluation layer), discursive response indicators (trust-related and purchase-related expressions), and observable engagement outcomes. Empirically, the study analyzes a large-scale dataset of over 1.6 million YouTube comments using a multi-stage computational pipeline combining weak supervision, large language model-assisted pre-labeling, and transformer-based classification. The findings reveal a multi-layered and partially fragmented response structure. AI perception is strongly associated with negative evaluative expression, with substantial variation across perception types. However, these negative evaluations do not translate into comparable increases in trust-related or purchase-related discourse, pointing to a divergence between evaluative expression and discursive consumer response in AI-mediated digital environments. At the same time, engagement outcomes exhibit a distinct platform-mediated dynamic, with AI-perceived content associated with lower interaction intensity at the content level. These results suggest that consumer responses to AI do not follow a uniform evaluation–response pathway, but instead unfold across partially independent evaluative, discursive, and engagement layers shaped by digital platform affordances. The study contributes to the retailing and consumer services literature by introducing a multi-layered perspective on consumer response to AI, refining the conceptualization of AI perception as a multidimensional construct, and providing ecologically valid insights based on large-scale naturalistic data. Managerially, the findings highlight the importance of AI communication design, authenticity signaling, and disclosure strategies in AI-mediated consumer interactions. Importantly, the study does not measure transactional behavior directly, but instead examines discursive and engagement-based indicators of consumer response within ecologically valid digital environments. © 2026 Elsevier Ltd

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Ai Powered Sentiment Insights from Tweets Using Mahasbert Embedding with the Bilstm Architecture

Item Type Conference Paper

Author S. Dhumal

Author K. Attarde

Author S. Mehta

Author G. Shinde

Author G. Bobhate

Author V. Patil

Abstract As social media exploded with growth, people made more and more comments in regional languages such as Marathi, but low-resource environments, complicated morphology, and code mixing present major challenges for the task of automated sentiment analysis. We build a balanced dataset of 16,365 Marathi tweets from different domains (movies, politics, general commentary) and use preprocessing, tokenization, and downsampling to achieve class balance. We compared seven embedding approaches: static (Word2Vec, Indic FastText) and contextual (MuRIL, XLM-RoBERTa, MahaAlBERTa, DistilBERT, and the domain-adapted MahaSBERT) with three deep learning classifiers: Convolutional Neural Networks (CNN), unidirectional LSTMs, and Bidirectional LSTMs (BiLSTM). Our experiments show how contextual embeddings significantly outperform static embeddings, with the best model MahaSBERT + BiLSTM yielding a top accuracy of 85 % for three-class sentiment classification. Fine-grained error analysis showcases the advantages of language-specific transformer embeddings and bidirectional sequence modeling in understanding fine-grained sentiment in Marathi tweets. We end by suggesting fine-tuning transformer models specifically on classification problems, incorporating attention mechanisms, and applying the approach to code-mixed settings and other Indic languages. © 2025 IEEE.

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AI-Based Sentiment Analysis of E-Commerce Customer Feedback: A Bilingual Parallel Study on the Fast Food Industry in Turkish and English

Item Type Journal Article

Author E. Kahya Özyirmidokuz

Author B. Molu Elmas

Author E.A. Stoica

Abstract Across digital platforms, large-scale assessment of customer sentiment has become integral to brand management, service recovery, and data-driven marketing in e-commerce. Still, most studies center on single-language settings, with bilingual and culturally diverse environments receiving comparatively limited attention. In this study, a bilingual sentiment analysis of consumer feedback on X (formerly Twitter) was conducted for three global quick-service restaurant (QSR) brands—McDonald's, Burger King, and KFC—using 145,550 English tweets and 15,537 Turkish tweets. After pre-processing and leakage-safe augmentation for low-resource Turkish data, both traditional machine learning models (Naïve Bayes, Support Vector Machines, Logistic Regression, Random Forest) and a transformer-based deep learning model, BERT (Bidirectional Encoder Representations from Transformers), were evaluated. BERT achieved the highest performance (macro-F1 ≈ 0.88 in Turkish; ≈ 0.39 in temporally split English), while Random Forest emerged as the strongest ML baseline. An apparent discrepancy was observed between pseudo-label agreement (Accuracy > 0.95) and human-label accuracy (EN: 0.75; TR: 0.49), indicating the limitations of lexicon-derived labels and the necessity of human validation. Beyond

methodological benchmarking, linguistic contrasts were identified: English tweets were more polarized (positive/negative), whereas Turkish tweets were overwhelmingly neutral. These differences reflect cultural patterns of online expression and suggest direct managerial implications. The findings indicate that bilingual sentiment analysis yields brand-level insights that can inform strategic and operational decisions. © 2025 by the authors.

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AI-Driven Analysis of Users' reactions to deep-fake imagery in Archaeological Misinformation

Item Type Conference Paper

Author G.J. Pezzullo

Author S. Venticinque

Author B. Di Martino

Author A. Amato

Abstract Fake news and deep-fakes are introducing information disorder even in the digital humanities, where misleading narratives and manipulated images can distort the collective perception in the context of history and art. This study proposes a depth analysis of the phenomenon analyzing the users' reactions to elements of misinformation related to cultural field. Using an approach based on Deep

Learning and Large Language Models (LLMs), we classified and validated comments collected from social media to estimate the level of acceptance or skepticism concerning different fake news and deep-fakes. In addition classification techniques were tested, comparing methodologies based on deterministic assignment and strategies with uncertainty thresholds to improve the accuracy of the analysis. The results, presented and discussed in the paper, confirm the widespread misinformation in digital platforms. The performance of the developed model shows good classification ability, however, open challenges remain, including the difficulty in recognizing sarcasm and linguistic ambiguities, aspects that may still affect the accuracy of classification. © 2025 IEEE.

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AI-Driven Topic Modeling and Sentiment Analysis of Systemic Lupus Erythematosus Discussions on Social Media: Cross-Platform Study

Item Type Journal Article
Author J. Tang
Author H. Jiang
Author J. Peng

Author H. Zhang

Author P. Qin

Author Z. Huang

Abstract Background: Systemic lupus erythematosus (SLE) is a multifactorial autoimmune disease influenced by genetic, epigenetic, ecological, and environmental factors, with a global prevalence of 7.7 to 13 per 100,000, and standardized mortality rates of 2.4% to 5.9%. Between 14% and 75% of patients experience psychiatric comorbidities such as anxiety and depression, which impair treatment adherence and health-related quality of life. Social media has become an important channel for patients to express health concerns and seek support. Reddit and Weibo, as mainstream platforms globally and in China, respectively, host large volumes of user-generated content; however, no prior study has examined SLE-related discourse across both cultural and platform contexts. Objective: This study aimed to characterize SLE-related discussions on Reddit and Weibo. Specifically, we sought to identify and hierarchically categorize discussion topics, compare platform-specific differences in patient concerns, assess sentiment polarity across topics and thematic groups, and characterize public misconceptions related to SLE. Methods: This observational, computational content analysis examined all SLE-related discussions on Reddit (r/lupus) and Weibo up to November 30, 2024. An AI-driven pipeline first embedded the discussions using the Multilingual-E5-base BERT (bidirectional encoder representations from transformers) model, then applied uniform manifold approximation and projection (UMAP) for dimensionality reduction and hierarchical density-based spatial clustering of applications with noise (HDBSCAN) for density-based clustering to identify fine-grained topics, with keywords extracted via class-based term frequency-inverse document frequency (c-TF-IDF). Topics were further grouped into higher-level thematic domains through spectral clustering, with labels and definitions generated using GPT-5 Thinking via prompt engineering. Sentiment polarity (positive, neutral, and negative) was classified using a fine-tuned multilingual BERT model. Two clinical pharmacists independently validated the topic modeling and sentiment results. Results: We analyzed 11,318 discussions from 3649 unique authors on Reddit and 33,628 discussions from 21,509 authors on Weibo. We identified 99 fine-grained topics on Reddit and 189 on Weibo, grouped into 6 thematic domains per platform. Reddit discussions centered on diagnostic journeys, treatment experiences, and symptom management, whereas Weibo discussions emphasized social news, charitable activities, and traditional Chinese medicine (eg, artemisinin). The overall sentiment on Reddit was positive (mean 0.42, SD 0.86; 95% CI 0.40-0.43), whereas that on Weibo was neutral (mean 0.01, SD 0.91; 95% CI 0.00-0.02). Sentiment was positive in 7493, neutral in 1032, and negative in 2793 Reddit discussions, whereas on Weibo, it was positive, neutral, and negative in 14,026, 5768, and 13,834 discussions, respectively. Treatment-related misconceptions and unverified remedies recurred on both platforms. Conclusions: This cross-platform analysis reveals both shared and platform-specific concerns among people with SLE, highlighting distinct informational and emotional needs across cultural and platform contexts. Recurring treatment-related misconceptions and the substantial volume of negative-sentiment discussions warrant targeted public health communication and psychological support. The findings may help clinicians, public health authorities, and patient support organizations identify unmet needs, while the AI-driven approach offers a scalable, real-time tool for monitoring patient perspectives. © Jian Tang, Huifang Jiang, Jie Peng, Hao Zhang, Ping Qin, Zikun Huang.

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Language English**Volume** 28**Publisher** JMIR Publications Inc.**Publication** Journal of Medical Internet Research**ISSN** 14394456 (ISSN)**Journal Abbr** J. Med. Internet Res.**Date Added** 20/09/2026, 8:35:13 pm**Modified** 20/09/2026, 8:35:13 pm**Tags:**

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Institute of Hospital Administration, NHC, China, (NIHAYSZX2543)

AI-Powered Fake News Detection Using LLMs and Knowledge Graphs

Item Type Conference Paper**Author** M. Shah**Author** A. Sharma**Author** S. Shawl**Author** A. Singhal

Abstract The social media platforms experience an urgent requirement for automatic fact-checking systems that operate with high accuracy because they face problems from unverified misleading information that spreads rapidly throughout their networks. The paper presents Epistemic Multimodal Veracity Evaluation (EMVE), a multimodal fact verification engine that evaluates the truthfulness of statements by processing textual, visual, and contextual information. The system uses large language models to perform semantic reasoning while Neo4j and Wikidata knowledge graphs enable relationship validation and computer vision methods to handle image and video content analysis for the verification process. The system uses real-time data from trustworthy online platforms which include news websites and authenticated social media profiles to maintain current information during the verification process. The experimental results show that EMVE achieves an accuracy of 95.2%, outperforming established fact-checking tools while it effectively detects false and manipulated information. The research

shows that using language reasoning together with structured knowledge systems and multimodal information leads to better verification results. © 2026 IEEE.

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An Active Learning Framework for Continuous Rapid Rumor Detection in Evolving Social Media

Item Type	Conference Paper
Author	Z. Shao
Author	G. Cai
Author	Q. Liu
Author	Y. Shang
Abstract	The widespread of social media has led to the propagation of rumors, resulting in detrimental effects on society. Existing rumor detection methods rely on labeled data and face challenges in obtaining sufficient data, particularly in the early stages of rumor propagation. Active learning has been proposed as a solution to

reduce labeling costs by selecting significant samples for training. However, applying active learning to rumor detection presents challenges, such as sensitivity to initial sets and performance loss due to sample reduction. Inspired by Active learning, Contrastive learning, and Prompt tuning, we propose a novel Rumor Detection framework named ACP-RD in this paper. ACP-RD addresses the limitations of active learning by leveraging the benefits of contrastive learning and prompt tuning. Our framework reduces labeling costs, improves rumor detection performance in zero-shot scenarios, aligns active query criteria with downstream tasks through unsupervised contrastive learning, and optimizes word embeddings using supervised contrastive learning. Experimental results demonstrate that ACP-RD achieves superior performance compared to fine-tuning with the whole labeled dataset, using only 10% of the labeled data. Overall, our proposed framework presents a novel and efficient approach to rumor detection, with potential for broader applications in natural language classification tasks. © 2024 IEEE.

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An adaptive multimodal semantic knowledge enhanced framework for sarcasm detection

Item Type Journal Article

Author J. Dong

Author Y. Sui

Author Q. Zhang

Author H. Fang

Author G. Schaefer

Author R. Liu

Author P. Yi

Author X. Fang

Abstract Multimodal sarcasm detection (MSD) has become an important research topic for understanding sentiments on social media, while various recent MSD approaches extract high-level semantic knowledge from images to improve performance. However, some key semantic information, such as emotions expressed in images, is still neglected, limiting reliable sentiment understanding. To address this issue, we propose an adaptive multimodal semantic knowledge enhanced framework for sarcasm detection. We first design an adaptive processing pipeline to extract emotion-aware visual semantics as an auxiliary modality to enhance multimodal feature representations. Enabled by two attention mechanisms, bidirectional cross-modal attention and graph attention, interactions between modalities are analysed to improve MSD performance. Extensive experiments are conducted on two public multimodal sarcasm detection datasets, MMSD and MMSD 2.0, comprising approximately 19,000 tweet samples. Our proposed approach achieves consistent improvements in both sarcasm detection accuracy and F1-score compared to strong baseline models such as DIP and KnowleNet. Built upon a ViT-based architecture, the fine-tuned model offers competitive performance with lower computational overhead, highlighting its potential for practical deployment. © 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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An aspect-based sentiment analysis model for Arabic game reviews based on hybrid transformers models

Item Type Journal Article

Author M. Hammad

Author N. AbuEnnab

Author M. Al-Refai

Abstract Aspect-based sentiment analysis (ABSA) is a natural language processing (NLP) technique to determine the various sentiments of a customer in a single comment regarding different aspects. The increasing online data content generated by interested customers and reviewers motivated researchers and data scientists to conduct ABSA. ABSA has become increasingly popular in recent years due to its versatility in e-commerce, social media, and customer feedback analysis. However, ABSA faces several significant challenges, including determining the aspects and their sentiment polarities (positive, negative, or neutral) in a given text. Moreover, ABSA faces particular challenges in non-English languages such as Arabic due to the lack of resources and mature models. Typically, ABSA tackles one or more of the ABSA research tasks: (T1) aspect term extraction, (T2) aspect term polarity, (T3) aspect category identification, and (T4) aspect category polarity. To identify the aspects and their corresponding sentiment polarities in a given text, accurate and efficient NLP techniques are required. Despite growing interest in Arabic ABSA, the lack of annotated datasets and pre-trained models has hindered its development. In this research, we have collected a dataset of Arabic game reviews and annotated them using three annotators, and then we trained an ABSA deep learning model based on the BERT pre-trained model combined with zero-shot learning (ZSL) to tackle all the four aforementioned tasks. Our best performing model achieved a high accuracy on all four tasks with an accuracy of 91.61% on T1, 90.99% on T2, 79.08% on T3, and 88.17% on T4. Finally, we compared our model's accuracy with the state-of-the-art Arabic-based ABSA models on different datasets. © The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2025.

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: M. Hammad; Artificial Intelligence Research Center (AIRC), IT Department, Ajman University, Ajman, United Arab Emirates; email: m.hammad@ajman.ac.ae

An Automatic Sentiment Analysis Method for Short Texts Based on Transformer-BERT Hybrid Model

Item Type Journal Article

Author H. Xiao

Author L. Luo

Abstract Sentiment analysis towards short texts is always facing challenges, because short texts only contain limited semantic characteristics. As a result, this paper constructs a specific large language structure to deal with this issue. In all, a novel automatic sentiment analysis method for short texts based on Transformer-BERT hybrid model is proposed by this paper. Firstly, BERT structure is utilized to extract word vectors, and is integrated with topic vectors to improve textual feature expression ability. Then, the fused word vectors are input into a Bidirectional Gated Recurrent Unit (Bi-GRU) structure to learn contextual features. In this part, a Transformer structure is applied behind the Bi-GRU and combined with the previous module to output sentiment analysis results. In addition, Accuracy, Precision, Recall and F1 indexes were collected from real-world Twitter datasets and shopping data to evaluate the

performance of the proposed method. The experimental results show that the method performs well in many indexes. Compared with traditional method, it has achieved remarkable performance improvement, and this method achieves higher accuracy and efficiency in sentiment analysis of short texts, and has good generalization ability. © 2013 IEEE.

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An effective convolutional cross-lingual language model for Vietnamese online social media mining

Item Type Journal Article

Author T.M. Nguyen

Author T.M. Nguyen

Author T.V. Huynh

Author K.V. Nguyen

Abstract With the fast expansion rate of online social media platforms, tons and tons of user-generated content, such as comments, blogs, or messages, are uploaded or sent every hour. This makes monitoring the virtual space impossible to do manually. Machine learning models can be used to integrate sentiment analysis, emotion recognition, and

spam detection into business applications. While recent studies are mostly centered on how pre-trained models or particular methods are used to improve the accuracy and the classification of comments on Vietnamese social media, there are not many studies directed explicitly at how their effectiveness can be improved. Additionally, there is a limited availability of models for the Vietnamese language to solve the problem of categorizing social media comments. Moreover, the series of studies focused on low-resource languages has yet to address many challenges in such languages as Vietnamese. In addition to this, we face additional challenges as code-mixed commentaries begin to appear on Vietnamese social media platforms. Extracting relevant features from short social media comments may give the models a better understanding and classification capabilities over time. To address these problems, we present a strategy that leverages the cross-lingual language model with 1D-CNN layers for social media text processing. In addition, we propose a sentiment classifier to effectively capture and represent relevant features for classification and a custom loss function to deal with data imbalance. In addition, we also created the first dataset (ViCM) related to code-mixed Vietnamese, which contains almost 5415 code-mixed hate speech comments. Our method has demonstrated significant improvements over previous studies in these NLP tasks, achieving state-of-the-art performance on benchmark social media datasets such as VSMEC, VSFC, and ViSpam. Additionally, we have made our ViCM dataset(ViCM dataset repository: <https://github.com/tiennho2608/ViCM>) publicly available to support further research and advancements in this field. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026.

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An Empirical Study on Sentiment Intensity Analysis via Reading Comprehension Models

Item Type Conference Paper

Author L. Zhong

Author L. Li

Author X. Zhong

Abstract Sentiment intensity analysis, a significant branch of natural language processing (NLP), has garnered widespread attention due to its potential to extract valuable insights from user-generated content on various internet platforms. This information can aid individuals and consumers in making informed decisions. Applications of this technology span multiple domains, including public opinion analysis, where it monitors and controls sentiment trends, and stock market analysis, which often reflects economic conditions and capital flows. Current methods for sentiment intensity analysis encompass lexicon-based methods, traditional machine learning techniques, deep learning models, and large language models. With the rapid advancement of deep learning technologies and the availability of extensive digital datasets, sentiment intensity analysis has achieved a relatively mature state. However, there has been little work comparing how reading comprehension models perform on sentiment intensity analysis tasks. In this paper, we explore the implementation of sentiment intensity analysis through reading comprehension modeling. Specifically, we experiment with five models, including pretrained, extractive, and generative models. We conduct extensive experiments on two distinct sentiment intensity analysis datasets: movie scripts and restaurant reviews. Our findings shed light on the feasibility and adaptability of reading comprehension models for sentiment analysis tasks, offering valuable insights for the continued development and application of sentiment intensity analysis. © 2024 Copyright held by the owner/author(s).

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An Ensemble Classification Approach in A Multi-Layered Large Language Model Framework for Disease Prediction

Item Type Conference Paper

Author A. Hamdi

Author M. Mohamed

Author R. Emad

Author K. Shaban

Abstract Social telehealth has made remarkable progress in healthcare by allowing patients to post symptoms and participate in medical consultations remotely. Users frequently post symptoms on social media and online health platforms, creating a huge repository of medical data that can be leveraged for disease classification. Large language models (LLMs) such as LLAMA3 and GPT-3.5, along with transformer-based models like BERT, have demonstrated strong capabilities in processing complex medical text. In this study, we evaluate three Arabic medical text preprocessing methods such as summarization, refinement, and Named Entity Recognition (NER) before applying finetuned Arabic transformer models (CAMELBERT, AraBERT, and AsafayaBERT). To enhance robustness, we adopt a majority voting ensemble that combines predictions from original and preprocessed text representations. This approach achieved the best classification accuracy of 80.56%, thus showing its effectiveness in leveraging various text representations and model predictions to improve the understanding of medical

texts. To the best of our knowledge, this is the first work that integrates LLM-based preprocessing with fine-tuned Arabic transformer models and ensemble learning for disease classification in Arabic social telehealth data. © 2025 IEEE.

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An Ensemble Scoring System for User Comment Sentiment Analysis

Item Type Conference Paper

Author F.-S. Sun

Editor Arabnia H.R.

Editor Deligiannidis L.

Editor Shenavarmasouleh F.

Editor Amirian S.

Editor Ghareh Mohammadi F.

Abstract Analyzing user comments on social media platforms is essential for understanding user responses and preferences, as well as for enhancing future media production. In this preliminary research, we proposed an ensemble sentiment scoring system that integrates three subsystems: a dictionary-based system, a large language model (LLM)-based system, and a local classification system. By aggregating votes from each subsystem and applying dynamic weighting criteria, a final sentiment score is reached. We conducted a small experiment to evaluate the effectiveness and accuracy of the system. Based on our current design and findings, we discussed several potential directions for future research. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: F.-S. Sun; Department of Computer Science, Ball State University, Muncie, 47304, United States; email: fsun@bsu.edu; Conference name: 11th International Conference on Computational Science and Computational Intelligence, CSCI 2024; Conference date: 11 December 2024 through 13 December 2024; Conference code: 332229

An Experimental Comparison of Large Language Models for Emotion Recognition in Italian Tweets

Item Type Conference Paper

Author C. Diamantini

Author A. Mircoli

Author D. Potena

Author S. Vagnoni

Editor Bena N.

Editor Di Martino B.

Editor Maratea A.

Editor Sperduti A.

Editor Di Nardo E.

Editor Ciaramella A.

Editor Montella R.

Editor Ardagna C.A.

Abstract In recent years, the advent of Large Language Models (LLMs), which are task-agnostic models trained on huge amounts of textual data, has given momentum to a wide variety of NLP applications, ranging from chatbots to sentiment classifiers. Currently, many LLMs are publicly available, each with different features and performance, and the selection of the best LLM for a specific task may be challenging. In this work, we focus on the task of emotion recognition in Italian social media content and we present an experimental comparison among three of the most popular LLMs: Google Bidirectional Encoder Representations from Transformers (BERT), OpenAI Generative Pre-trained Transformer 3 (GPT-3) and GPT-3.5. Model specialization in emotion recognition has been achieved by using two different approaches, namely fine-tuning and prompt engineering with few-shot task transfer. The experimentation has been performed on TwIT, a corpus of about 3100 Italian tweets annotated with respect to six emotions. The results show that fine-tuning GPT-3 leads to the best performance on the considered dataset, achieving a remarkable $F1=0.90$. © 2023 Copyright for this paper by its authors.

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An Explainable Artificial Intelligence Text Classifier for Suicidality Prediction in Youth Crisis Text Line Users: Development and Validation Study

Item Type Journal Article

Author J. Thomas

Author A. Lucht

Author J. Segler

Author R. Wundrack

Author M. Miché

Author R. Lieb

Author L. Kuchinke

Author G. Meinlschmidt

Abstract Background: Suicide represents a critical public health concern, and machine learning (ML) models offer the potential for identifying at-risk individuals. Recent studies using benchmark datasets and real-world social media data have demonstrated the capability of pretrained large language models in predicting suicidal ideation and behaviors (SIB) in speech and text. Objective: This study aimed to (1) develop and implement ML methods for predicting SIBs in a real-world crisis helpline dataset, using transformer-based pretrained models as a foundation; (2) evaluate, cross-validate, and benchmark the model against traditional text classification approaches; and (3) train an explainable model to highlight relevant risk-associated features. Methods: We analyzed chat protocols from adolescents and young adults (aged 14-25 years) seeking assistance from a German crisis helpline. An ML model was developed using a transformer-based language model architecture with pretrained weights and long short-term memory layers. The model predicted suicidal ideation (SI) and advanced suicidal engagement (ASE), as indicated by composite Columbia-Suicide Severity Rating Scale scores. We compared model performance against a classical word-vector-based ML model. We subsequently computed discrimination, calibration, clinical utility, and explainability information using a Shapley Additive

Explanations value-based post hoc estimation model. Results: The dataset comprised 1348 help-seeking encounters (1011 for training and 337 for testing). The transformer-based classifier achieved a macroaveraged area under the curve (AUC) receiver operating characteristic (ROC) of 0.89 (95% CI 0.81-0.91) and an overall accuracy of 0.79 (95% CI 0.73-0.99). This performance surpassed the word-vector-based baseline model (AUC-ROC=0.77, 95% CI 0.64-0.90; accuracy=0.61, 95% CI 0.61-0.80). The transformer model demonstrated excellent prediction for nonsuicidal sessions (AUC-ROC=0.96, 95% CI 0.96-0.99) and good prediction for SI and ASE, with AUC-ROCs of 0.85 (95% CI 0.97-0.86) and 0.87 (95% CI 0.81-0.88), respectively. The Brier Skill Score indicated a 44% improvement in classification performance over the baseline model. The Shapley Additive Explanations model identified language features predictive of SIBs, including self-reference, negation, expressions of low self-esteem, and absolutist language. Conclusions: Neural networks using large language model-based transfer learning can accurately identify SI and ASE. The post hoc explainer model revealed language features associated with SI and ASE. Such models may potentially support clinical decision-making in suicide prevention services. Future research should explore multimodal input features and temporal aspects of suicide risk. ©Julia Thomas, Antonia Lucht, Jacob Segler, Richard Wundrack, Marcel Miché, Roselind Lieb, Lars Kuchinke, Gunther Meinlschmidt.

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An explainable ensemble machine learning approach for multi-domain, multiclass sentiment analysis in Amazon product reviews

Item Type Journal Article

Author K. Mokgwatjane

Author T. Paepae

Abstract Sentiment analysis (SA) of online reviews is pivotal for e-commerce platforms yet challenges such as massive user-generated content volumes and class imbalances hinder accurate multiclass predictions and model interpretability. This study introduces a novel explainable ensemble learning framework for multiclass SA (positive, neutral, negative) across three Amazon product domains: appliances, groceries, and clothing. The framework integrates diverse supervised classifiers in a stacking ensemble, with SHapley Additive exPlanations (SHAP) innovatively employed not only to elucidate feature contributions but also to rank and interpret the individual impacts of base classifiers on ensemble predictions, a pioneering application in domain-specific SA, as it enables global insights into model dynamics and base model selection, addressing gaps in prior studies that relied on local explanations like LIME (Local Interpretable Model-agnostic Explanations). Evaluated using imbalance-sensitive metrics (weighted/macro F1-score, Matthews Correlation Coefficient, Cohen's Kappa, Geometric Mean), the ensemble surpasses individual classifiers and demonstrates higher macro F1 and G-Mean than the transformer-based ALBERT model, while ALBERT excels in weighted F1, MCC, and Cohen's Kappa. Extra Trees notably excelled in the G-Mean for minority classes. SHAP analysis uncovers domain-specific drivers and base model roles, enhancing transparency. The results underscore the framework's efficacy in delivering robust performance and actionable insights for trust modelling, automated analytics, and personalized recommendations. This work lays the groundwork for extensions to low-resource domains, multimodal data, and finer rating scales, advancing interpretable SA in e-commerce. © 2025 The Author(s).

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An Exploration and Analysis of Generative AI in Enhancing Personalized Treatment Plans for Chronic Disease Management

Item Type Conference Paper

Author Q.U.A. Naheed

Author K. Sanyour

Author O. Hammad

Editor Duffy V.G.

Abstract Generative AI (GenAI) shows a growing potential for personalized treatment planning in chronic disease management. Although Large Language Models (LLMs) are capable of processing clinical text and generating treatment-related recommendations, their adoption in real clinical practice remains limited due to concerns around trust, safety, bias, privacy, and accountability. From a Human–Computer Interaction (HCI) perspective, these concerns directly shape whether GenAI systems can be trusted, and used responsibly in high-stakes medical environments. In this study we verify the specific trust-related concerns that shape clinicians’ willingness to engage with GenAI. Using a mixed-methods approach, we first analyzed public discourse on GenAI and healthcare taken from social media platforms, professional forums, news outlets, and academic databases. Topic modeling and sentiment analysis were applied to identify recurring patterns, resulting in six core categories of concern. These concerns were subsequently validated through a survey with practicing clinicians to assess their relevance in real clinical contexts. The results subsequently indicated low clinician comfort with GenAI systems, reinforcing the need for cautious, transparent, and human-in-the-loop system behavior. This study contributes an empirically validated framework that establishes a foundation for future design

and evaluation of safer, more inclusive GenAI systems for chronic disease management. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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An Exploratory Study of Conventional Machine Learning and Large Language Models for Sentiment Analysis

Item Type Conference Paper

Author C. Zou

Author J. Cai

Author L. Chen

Author F.F.-H. Nah

Editor Degen H.

Editor Ntoa S.

Abstract Sentiment analysis is the use of natural language processing to identify affective states and determine people's opinions in various analytical applications such as customer reviews and social media analyses. Large language models (LLMs) such as GPT-4o demonstrate impressive performance in text generation tasks. Despite numerous studies in the extant literature, few have compared the performance of conventional machine learning models with LLMs for sentiment analysis. This study aims to fill this gap by conducting an evaluation of these models using a balanced dataset of 2,000 IMDb movie reviews. Our study shows that GPT-4o achieves the highest performance, while GPT-3.5 and FLAN-T5 models also show strong performance, being slightly below that of GPT-4o. Advanced LLMs outperform conventional machine learning models. Our findings highlight the advanced capabilities and user-friendliness of LLMs compared to conventional machine learning models. This research underscores the rapid evolution of LLMs for sentiment analysis. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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An Integration Framework of Remote Sensing and Social Media for Dynamic Post-Earthquake Impact Assessment

Item Type Journal Article

Author Z. Ren

Author T. Yang

Author G. Li

Author S. Hu

Author N. Mou

Author Z. Chen

Abstract Effective post-disaster management requires continuous and reliable monitoring of the evolving disaster situation. While remote sensing provides objective measurements of ground deformation, social media data offer dynamic insights into public perception and disaster progression. However, integrating these complementary data sources to achieve sustained monitoring of disaster remains a challenge. To address this, we propose a novel framework that combines Sentinel-1 SAR data with Sina Weibo posts to improve dynamic earthquake impact assessment. Physical damage was quantified using D-InSAR-derived deformation. Disaster-related locations were identified using a fine-tuned pre-trained language model, and public sentiment was inferred through prompt-based few-shot learning with a large language model. Spatiotemporal analysis was performed to examine the relationship between sentiment dynamics and varying levels of physical damage, followed by an analysis of topic transitions within regional semantic networks to compare discussion patterns across areas. A case study of the 2023 Jishishan earthquake demonstrates the framework's capability to continuously track disaster evolution: regions experiencing severe physical damage exhibit clear concentrations of negative sentiment, whereas increases in positive sentiment coincide with areas where rescue operations are effectively underway. These findings indicate that integrating the two data sources improves continuous disaster monitoring and situational awareness, thereby supporting emergency response. © 2025 by the authors.

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An Intelligent Sentiment Analysis Model Based on Fine-Tuned BERT Architecture for Context-Aware Opinion Mining and Emotional Polarity Classification

Item Type Journal Article

Author P.S. Aher

Author B.L. Gunjal

Abstract The rapid expansion of user generated text through social media, online reviews, and feedback mechanisms has made consequent and rapid development of sentiment analysis as a key challenge in the field of Natural Language Processing (NLP) the social media text, online reviews and feedback. In the automated decision support systems, sentiment analysis, as a part of the system, brand/ product/ service monitoring systems, and public sentiment systems/analyst applications, picking positive/ negative emotion from the text is a must. Although a lot of research has been done, the earlier Machine Learning (ML) and Deep Learning (DL) models are still facing issues of contextual dependence, polysemy, and lack of resilience in various linguistic phenomena. To overcome the above research gaps, this paper presents an adaptive intelligent-aware framework for sentiment analysis using fine-tuned Transformers models/architectures whereby RoBERTa and DistilBERT are suggested as main/primary models. The proposed methodology will include data preprocessing, subword embedding tokenization, and sentiment polarity classification through supervised fine-tuning of the pre-trained language models. The use of RoBERTa will be since RoBERTa is known for its pre-training and contextual representation. DistilBERT will be used to provide the balance of other models, and to allow the framework to be designed for use in time-constrained and low-resource computing environments. The proposed system will be tested using the available datasets for sentiment analysis using the standard metrics, and to evaluate its performance, the accuracy, precision, recall, and the F1 score will be tested. The results show that the fine-tuned RoBERTa model can outperform the baseline BERT-based and traditional deep learning models, in terms of classification accuracy and managing context-dependent sentiment expressions. Even though DistilBERT is less

accurate, it has a lower inference time and memory usage, creating a balance between performance and efficiency. The comparative analysis exhibits the proposed models' scalability and robustness, despite differing lengths of text and sentiment distributions. This study advocates the proven utility of the transformer-based language models for sentiment classification and recognizes RoBERTa and DistilBERT as intelligent, efficient, and accurate sentiment analysis models. The proposed framework can be adapted for future work, such as sentiment analysis for multiple languages or opinion mining in specific domains. © 2026, Nexora Academic Press. All rights reserved.

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An LLM-Enabled Knowledge Elicitation and Retrieval Framework for Zero-Shot Cross-Lingual Stance Identification

Item Type Conference Paper

Author R. Zhang

Author Y. Tian

Author P. Wei

Author D. Zeng

Author W. Mao

Editor Al-Onaizan Y.

Editor Bansal M.

Editor Chen Y.-N.

Abstract Stance detection aims to identify the attitudes toward specific targets from text, which is an important research area in text mining and social media analytics. Existing research is mainly conducted in monolingual setting on English datasets. To tackle the data scarcity problem in low-resource languages, cross-lingual stance detection (CLSD) transfers the knowledge from high-resource (source) language to low-resource (target) language. The CLSD task is the most challenging in zero-shot setting when no training data is available in target language, and transferring stance-relevant knowledge learned from high-resource language to bridge the language gap is the key for improving the performance of zero-shot CLSD. In this paper, we leverage the capability of large language model (LLM) for stance knowledge acquisition, and propose KEAR, a knowledge elicitation and retrieval framework. The knowledge elicitation module in KEAR first derives different types of stance knowledge from LLM's reasoning process. Then, the knowledge retrieval module in KEAR matches the target language input to the most relevant stance knowledge for enhancing text representations. Experiments on multilingual datasets show the effectiveness of KEAR compared with competitive baselines as well as the CLSD approaches trained with labeled data in target language. © 2024 Association for Computational Linguistics.

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AN NLP-BASED APPROACH FOR SARCASM DETECTION IN MARATHI LANGUAGE WITHIN MULTILINGUAL ENVIRONMENTS

Item Type Journal Article

Author P. Patil

Author S. Kolhe

Abstract The expansion of social media platforms has required the development of complex natural language processing (NLP) methods for sentiment analysis and sarcasm detection, especially for low-resource languages. This research presents a novel, ensemble-based NLP framework for sarcasm detection in Marathi-English code-mixed text, addressing a significant gap in multilingual sentiment analysis. The study confronts the challenges inherent to the Marathi language, including its rich morphology and a paucity of annotated corpora, by constructing a dedicated dataset of 2,400 tweets. Through a hybrid annotation strategy, this corpus was refined to contain 647 sarcastic, 1,477 non-sarcastic, and 276 unsure instances. The primary aim of this study is to develop and evaluate an ensemble-based model that enhances sarcasm detection accuracy in Marathi-English code-mixed text through the integration of traditional machine learning and neural network approaches. The proposed methodology employs comprehensive feature engineering—incorporating TF-IDF, word embedding, sentiment lexicons, and code-switching indicators—and an ensemble model architected to synergize the strengths of a Multinomial Naïve Bayes classifier and a Spiking Neural Network (SNN). Empirical evaluation demonstrates that the ensemble model achieves superior performance, attaining an accuracy of 94.2%, a precision of 90.5%, a recall of 92.1%, and an F1-score of 91.3%. These results signify a substantial improvement over established baseline models, including XGBoost (87% accuracy), SVM (85% accuracy), BERT-base Multilingual (92% accuracy), and XLNet (90% accuracy). This work validates the efficacy of ensemble learning and context-sensitive feature extraction for sarcasm detection, providing a scalable and robust paradigm for analogous low-resource linguistic environments. © Little Lion Scientific

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Analysis of Public Sentiment in Azerbaijani News and Social Media: Exploring Public Opinion Trends in Social Media and News Articles for Azerbaijani Language

Item Type Conference Paper

Author N. Guliyev

Author Z. Rustamov

Author S. Rustamov

Abstract This study delves into the analysis of public sentiment within Azerbaijani social media comments and news articles employing Natural Language Processing methodologies and sentiment analysis techniques. The research explores distinct sentiment narratives in social media and news contexts, drawing from a dataset comprising 38,000 comments and 12,000 news articles. In our investigation, we evaluated a spectrum of models, including BERT, GPT-3.5-Turbo, LSTM, SVM, and LR which are complemented with Bag of Words, TF-IDF, and FastText vectorizers, alongside word embedding models. Additionally, we employ various preprocessing techniques, such as eliminating links and mentions (@ or 'at'), transforming emojis into a readable format, and performing stemming and stop word removal on news articles. Our trained models demonstrate superior performance compared to the Large Language Models which suggests the efficacy of our tailored approach in capturing polarity within Azerbaijani social media and news data. The findings underscore the superiority of SVM over other classification models across both social media and news articles. Moreover, it contributes valuable insights into sentiment analysis in diverse linguistic contexts, with implications for understanding public sentiment dynamics in Azerbaijani

media. © 2024 ACM.

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Archive Scopus

Language English

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Deep learning, Support vector machines, Public opinions, Social aspects, Classification (of information), Learning systems, machine learning, Machine-learning, Sentiment analysis, Social networking (online), Social media, deep learning, Long short-term memory, Transformer, sentiment analysis, News media, News articles, Public sentiments, transformers, Social news

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Analysis of Sentiment and Topic Evolution in Multi-Platform User Comments based on LoRA-Fine-Tuned Large Language Models

Item Type Journal Article

Author M. Chen

Author X. Lu

Author Y. Zhou

Author W. Li

Abstract With the rapid development of domestic large language models (LLMs) in China, understanding public sentiment toward these technologies is essential for technical optimization and policy-making. This study collected user comments about Chinese LLMs from three major social media platforms—Douyin, Rednote, and Tieba—through web scraping and performed data cleaning and preprocessing. We conducted a comparative performance evaluation of several representative LLMs and selected Qwen3-4b as the core model. The model was further fine-tuned using the LoRA approach, and its performance in sentiment classification tasks was compared with

various baseline methods. Experimental results show that the fine-tuned Qwen3-4b model achieved the best performance in multi-class sentiment analysis. The overall public sentiment was found to be predominantly negative or neutral, with a relatively low proportion of positive attitudes. Moreover, sentiment distribution varied across platforms: Douyin and Rednote users focused more on model stability and user experience, showing a higher level of negative sentiment, while Tieba users engaged in more rational discussions centered on technological innovation and application prospects. Limitations of this study include the relatively small sample size, the subjective nature of user comments, ambiguity in sentiment expression, and limited model adaptability in detecting sarcastic emotions. Copyright ©World Academic Press, World Academic Union

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BERTopic model, LoRA fine-tuning sentiment analysis, LSTM-Transformer, Sentiment-topic evolution

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Analyzing Public Perceptions of Mobility Electrification in Germany and China Through Social Media with Large Language Models

Item Type Journal Article

Author K.U. Bulut

Author H. Mostofi

Abstract This study investigates cross-cultural differences in public perception of mobility electrification by applying natural language processing (NLP) techniques to social media discourse in Germany and China. Using a large language model (LLM), this study conducted sentiment analysis and zero-shot text classification on over 10,000

posts to explore how citizens in each country engage with the topic of electric mobility. Results reveal that while infrastructure readiness is a dominant concern in both contexts, German discourse places greater emphasis on environmental impact, often reflecting skepticism toward sustainability claims. On the other hand, Chinese discussions highlight technological advancement and infrastructure expansion, with comparatively limited focus on environmental concerns. These findings show the importance of culturally tailored policy and communication strategies in supporting the public acceptance of electric mobility. By demonstrating how artificial intelligence-driven large-scale social media data analysis can be used to analyze public sentiment across linguistic and cultural contexts, this study contributes methodologically to the emerging field of computational social science and offers practical insights for mobility policy in diverse national settings. © 2026 by the authors.

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Publication Vehicles

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Mostofi; Department of Work, Technology and Participation, Technische Universität Berlin, Berlin, Marchstraße 23, 10587, Germany; email: mostofidarbani@tu-berlin.de

Analyzing public sentiment in Iranian presidential elections on Twitter using large language models

Item Type Journal Article

Author M. Askarizade

Author E. Davoodijam

Abstract This study investigates the dynamics of public sentiment surrounding the 2024 Iranian presidential election by analyzing Persian-language tweets. We introduce the IranElectionTweet dataset, a comprehensive collection of 111,386 election-related tweets enriched with textual content, user metadata, and engagement indicators. Due to the sensitive political context and privacy considerations, the full dataset is not publicly released; instead, we provide a manually annotated subset of 500 tweets (Tweet IDs and dates) for benchmarking, along with reconstruction instructions and analysis code. To conduct sentiment analysis, we fine-tuned GPT-4 on a publicly available Persian sentiment dataset, adapting it to the linguistic and cultural nuances of Persian political discourse. In parallel, we evaluated three cutting-edge large language models, Claude Sonnet 3.7, DeepSeek-V3, and Grok-4, using a few-shot learning framework due to the unavailability of fine-tuning access at the time of experimentation. All models were benchmarked on a manually annotated subset of 500 tweets. DeepSeek-V3 attained the highest weighted F1-score and overall accuracy, indicating stronger performance on the majority classes and was selected as the primary model for sentiment classification. The final sentiment analysis was applied to the full dataset, capturing hourly and daily variations in sentiment and candidate mentions throughout the election period. The results reveal distinct patterns in public opinion corresponding to key political events, offering valuable insights into the real-time evolution of electoral sentiment on social media. This research highlights the effectiveness of advanced multilingual language models in low-resource settings and contributes to the broader understanding of political behavior in digital environments. © The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd. 2025.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: E. Davoodijam; Department of Computer Engineering, Faculty of Engineering, Arak University, Arak, Iran; email: e-davoudijam@araku.ac.ir

Analyzing the Performance of Sentiment Analysis Using BERT, DistilBERT, RoBERTa and GPT-2

Item Type Conference Paper

Author S. Imtiaz

Author E.K. Hashi

Abstract Sentiment analysis, also known as opinion mining, employs natural language processing (NLP) techniques to determine, extract, and measure the emotional undertones within a text. This approach is valuable for measuring public sentiment and interests across various domains, including social media discussions, product and movie reviews, among others. Techniques like Linear Regression, Support Vector Machines, Convolutional Neural Networks (CNN), Recurrent Neural Networks (RNN), and Long Short-Term Memory (LSTM) are utilized in analyzing sentiments. However, these are no comparison to the capabilities of the pre-trained transformer models such as BERT, DistilBERT, RoBERTa, and the recent large language model (LLM) GPT-2, which offer significantly enhanced performance and flexibility. This study employs the Sentiment140 dataset to evaluate sentiments expressed in tweets, utilizing four advanced transformer-based models: BERT, DistilBERT, RoBERTa, and GPT-2. The sentiment analysis conducted within this research framework focuses on accuracy as the primary performance metric. The findings of the analysis indicate that the GPT-2 model outperforms its counterparts by achieving an impressive accuracy of 94.13%, thereby demonstrating its superior capability in handling sentiment analysis tasks. This study not only reinforces the efficacy of transformer-based models in processing complex language data but also highlights the cutting-edge potential of GPT-2 to revolutionize sentiment analysis in social media contexts. Furthermore, the integration of these technologies aligns with the principles of Industry 5.0 by enhancing human-machine collaboration, thus driving more personalized, efficient, and sustainable technological advancements. By leveraging such sophisticated models, businesses can harness real-time sentiment analysis to foster a more responsive and sustainable operational framework, ultimately contributing to smarter and more sustainable technology deployments. © 2024 IEEE.

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Annotated Lexicon for Sentiment Analysis in the Bosnian Language

Item Type Journal Article

Author S. Jahić

Author J. Vičić

Abstract The paper presents the first sentiment-Annotated lexicon of the Bosnian language. The annotation process and methodology are presented along with a usability study, which concentrates on language coverage. The composition of the starting base was done by translating the Slovenian annotated lexicon and later manually checking the translations and annotations. The language coverage was observed using two reference corpora. The Bosnian language is still considered a low-resource language. A reference corpus comprised of automatically crawled web pages is available for the Bosnian language, but the authors had a hard time sourcing any corpora with a clear time frame for the text contained therein. A corpus of contemporary texts was constructed by collecting news articles from several Bosnian web portals. Two language coverage methods were used in this experiment. The first used a frequency list of all words extracted from two reference Bosnian language corpora, and the second ignored the frequencies as the main factor in counting. The computed coverage using the first presented method for the first corpus was 19.24%, while the second corpus yielded 28.05%. The second method yielded 2.34% coverage for the first corpus and 6.98% for the second corpus. The results of the study present a language coverage that is comparable to the state of the art in the field. The usability

of the lexicon was already proven in a Twitter-based comparison. © 2023 Ljubljana University Press, Faculty of Arts. All rights reserved.

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Anthropology-informed AI: modeling the spiritual-first hierarchy of illness explanation in Arabic-speaking patient communities

Item Type Journal Article

Author M. Mehdoui

Author A. Fraisse

Author W.M. El Hadi

Author M. Zrigui

Abstract Introduction – AI-mediated health communication lacks scalable, evidence-based models of cultural tailoring derived from authentic patient narratives. Methods – We developed and validated a hybrid human-AI methodology. A large-scale analysis of 41, 393 posts from Arabic-language online mental health forums was conducted. Manual annotation using Kleinman's Explanatory Model (EM) was followed by fine-tuning a large language model (LLM) to automate this interpretive framework. Results – The analysis revealed a distinct, culturally grounded hierarchy: Arabic-speaking patients consistently prioritized divine/spiritual causes first (e.g., God's will, divine testing, evil eye, jinn), followed by moral/ethical considerations (sin, guilt, repentance), social/familial impact (honor, stigma, reputation), psychological factors,

and finally biomedical symptoms. This “spiritual-first” pattern was experimentally tested by comparing spiritually-attuned AI messages versus standard secular biomedical messages. Discussion – Spiritually-aligned messages significantly outperformed secular counterparts across all engagement metrics and qualitative indicators of trust, relevance, and intent to seek care, validating that communication respecting patients’ spiritual explanatory models fosters greater resonance and therapeutic alliance. A key limitation is the exclusive focus on mental health narratives; future work should investigate extension to physical illness contexts and clinical outcomes. © 2026 Mehdoui, Fraisse, El Hadi and Zrigui.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Mehdoui; GERiiCo Laboratory, University of Lille, Lille, France; email: mouheb.mehdoui.etu@univ-lille.fr

Application of Fuzzy c-Means Clustering Algorithm and NLP for Social Media Post Distribution Based on Expert-Defined Topics

Item Type Conference Paper

Author G. Guskov

Author N. Yarushkina

Author M. Novichkova

Author P. Dudarin

Editor Kovalev S.

Editor Kotenko I.

Editor Sukhanov A.

Abstract This article explores the application of Natural Language Processing (NLP) techniques and the Fuzzy c-means clustering algorithm for the automated categorization of social media posts based on expert-defined topics. The study analyzes data from Avito, VKontakte communities, including Volga-IT, UISTU Students Community, Ulyanovsk State Technical University, and RAIL Summer School. The initial stage performs text preprocessing to extract keywords, terms, and named entities, then employs a large language model to identify thematic categories. The Fuzzy c-means algorithm is then applied to cluster posts, accounting for their partial membership in multiple topics, which reflects the inherent ambiguity of social media content. The results of automated clustering are compared with expert-labeled data to evaluate accuracy and consistency. The proposed approach can benefit SMM specialists by providing a tool for automated content analysis, reducing manual categorization efforts, and improving community management efficiency. Key advantages of this method include its flexibility in handling vaguely defined topics and compatibility with modern NLP technologies. The study demonstrates the potential of combining fuzzy logic and machine learning for enhancing social media analytics, offering a practical solution for digital content classification. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Application of Large Language Models for Aspect-Based Sentiment Analysis on Social Media Data: A Case Study of the Thai Telecommunications Industry

Item Type Conference Paper

Author K. Limseesawan

Author P. Netisopakul

Author S. Chotipant

Author W. Voravuthikunchai

Author T. Sirivorachodphokin

Abstract Social media is a valuable source of consumer opinion data, particularly in Thailand's highly competitive telecommunications market among AIS, TRUE, and DTAC. This study compares three LLM-based approaches for Aspect-Based Sentiment Analysis (ABSA) on 3,508 Thai-language messages from Pantip.com and YouTube.com: (1) Prompt Engineering, (2) Domain Adaptation & Fine-Tuning, and (3) Multi-Agent Debate Framework. Results show that Domain Adaptation & Fine-Tuning achieves the best accuracy-latency trade-off, with Qwen2.5-1.5B-Instruct exceeding 87% average accuracy in under 4 seconds per message. Multi-Agent Debate achieves the highest Category accuracy (82.82%) but at a latency cost of 10-16 seconds per message. Gemini-2.5-Flash-Lite provides the best speed-accuracy balance for Prompt Engineering without additional training. Critically, small open-source models (1B-1.5B parameters) subjected to domain-specific fine-tuning can approach or surpass proprietary large models on this task, suggesting domain alignment may outweigh raw parameter scale for narrow, well-defined ABSA tasks. © 2026 IEEE.

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Language model, Large language model, Economic and social effects, Sentiment analysis, Social networking (online), Social media, Fine tuning, Fine-Tuning, Multi agent, Prompt engineering, Large Language Models, Multi agent systems, Aspect-based sentiment analyze, Intelligent agents, Aspect-Based Sentiment Analysis, Domain adaptation, Domain Adaptation, Multi-agent debate, Multi-Agent Debate, Prompt Engineering, Telecommunication industry

Notes:

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Application of Nepali Large Language Models to Improve Sentiment Analysis

Item Type Conference Paper

Author S. Pudasaini

Author S. Ghimire

Author P. Ale

Author A. Shakya

Author P. Paudel

Author B. Joshi

Abstract With the rise in internet usage, Nepali individuals have left a flood of opinionated comments in their language on YouTube and other social media sites. Such remarks can be subjected to sentiment analysis, which can be useful for both research and business purposes. Such sentiment analysis models can be extremely useful in understanding the user's expectations towards the product which can uplift the business of any organization. Similarly, with the rise of Large Language models in the NLP space, there are several large language models pre-trained on the BERT architecture upon the Nepali text corpus. This research focuses on developing a benchmarking dataset for sentiment analysis in the Nepali language and demonstrating how large Nepali language models can be used to improve the results on downstream NLP tasks like sentiment analysis on such benchmark datasets. This paper describes an approach to how proper embeddings for a Nepali sentence can be extracted from the pre-trained Nepali language models. The comparison of transfer learning applied to the dataset on different machine learning and deep learning algorithms has been done in this study. From this experimentation, a state-of-the-art sentiment analysis model in the Nepali language with an F-score of 0.88 has been developed. © 2024 ACM.

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DOI 10.1145/3647782.3647804**URL** <https://www.scopus.com/pages/publications/85198143680?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: ACM Int. Conf. Proc. Ser.**Publisher** Association for Computing Machinery**ISBN** 979-840071665-2 (ISBN)**Pages** 144-150**Proceedings Title** ACM Int. Conf. Proc. Ser.**Conference Name** ACM International Conference Proceeding Series**Date Added** 20/09/2026, 8:35:30 pm**Modified** 20/09/2026, 8:35:30 pm**Tags:**

Language model, Large language model, Deep learning, Sentiment analysis, Natural Language Processing, Language processing, Natural language processing, Natural language processing systems, Natural languages, Machine learning algorithms, Benchmarking, Large datasets, Transfer learning, Transfer Learning, Adversarial machine learning, Contrastive Learning, Sentiment Analysis, Machine Learning Algorithms, Large Language Models, Distribution transformers, Finetuning, Bidirectional encoder for representational transformer, Bidirectional Encoders for Representational Transformers

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AquaticCLIP: A Vision-Language Foundation Model and Dataset for Underwater Scene Analysis

Item Type Journal Article**Author** B. Alawode**Author** I. Iyappan Ganapathi**Author** S. Javed**Author** M. Bennamoun**Author** A. Mahmood

Abstract The preservation of aquatic biodiversity is critical in mitigating the effects of climate change. Aquatic scene understanding plays a pivotal role in aiding marine scientists in their decision-making processes. In this article, we introduce AquaticCLIP, a novel contrastive language-image pretraining (CLIP) model tailored for aquatic scene understanding. AquaticCLIP presents an underwater domain-specific learning

framework that aligns images and texts in aquatic environments, enabling tasks such as segmentation, classification, detection, and object counting. By leveraging our large-scale underwater image-text paired dataset without the need for ground-truth (GT) annotations, our model enriches existing vision-language models (VLMs) in the aquatic domain. For this purpose, we construct a 2-million underwater image-text paired dataset using heterogeneous resources, including YouTube, Netflix, National Geographic (NatGeo), etc. To fine-tune AquaticCLIP, we propose a prompt-guided vision encoder (PGVE) that progressively aggregates patch features via learnable prompts, while a vision-guided mechanism enhances the language encoder by incorporating visual context. The model is optimized through a contrastive pretraining loss to align visual and textual modalities. AquaticCLIP achieves notable performance improvements in zero-shot settings across multiple underwater computer vision tasks, outperforming existing methods in both accuracy and robustness. Our model sets a new benchmark for vision-language applications in underwater environments. The code and dataset for AquaticCLIP are publicly available on GitHub at: <https://github.com/BasitAlawode/AquaticCLIP> © 2012 IEEE.

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Arabic Dialect Identification on Social Media: Mini Review

Item Type Conference Paper

Author E.Y. Alqulaity

Author W.M.S. Yafooz

Author A.A. Alsaeedi

Editor Jusman Y.

Abstract Arabic dialect identification plays a crucial role in natural language processing. It serves as a foundational step in various language processing applications like machine translation, Sentiment Analysis, and cross-language text generation. Therefore, in the last decade, interest has increased in addressing the problem of Arabic dialect identification. In the context of social media Arabic dialect identification poses unique challenges due to the close relationship and shared vocabulary among the dialects. This paper provides a comprehensive overview of challenges in this domain, exploring common benchmarks in existing literature and offering insights into ongoing studies on identifying Arabic dialects in written texts. The literature review highlights the evolving methodologies for Arabic dialect identification, including traditional machine learning, advanced deep learning models like BERT and LSTM, and hybrid approaches. The focus on finer dialect distinctions and collaborative tasks like NADI has driven innovation. Challenges include handling low-resource dialects, adapting to evolving language on social media, and addressing subtleties within closely related dialects. Techniques like character N-grams and pre-trained models, particularly BERT, enhance identification accuracy. The outcomes of this systematic review offer valuable insights into the current trends in Arabic dialect identification research and serve as an essential reference for scholars in the field of Arabic Natural Language Processing. © 2023 IEEE.

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Proceedings Title Proceeding - Int. Conf. Artif. Intell. Robot., Signal Image Process., AIRoSIP

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Deep learning, machine learning, Machine-learning, Sentiment analysis, Language processing, Natural languages, Deep reinforcement learning, Social media, Federated learning, Adversarial machine learning, Contrastive Learning, deep learning, Processing applications, Translation (languages), Arabic natural language processing, Machine translations, Arabic dialects, Arabic dialect, Arabic Natural Language Processing, Dialect identification

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Arabic machine-generated text detection: Stylometric analysis and cross-model evaluation

Item Type Journal Article

Author M.S. Al-Shaibani

Author M. Ahmed

Abstract Large Language Models (LLMs) have achieved unprecedented capabilities in generating human-like text, posing subtle yet significant challenges for information integrity across critical domains. This challenge becomes severe, particularly in under-explored and low-resource languages like Arabic. This paper presents a comprehensive investigation of Arabic machine-generated text detection, examining outputs from diverse LLMs across academic and social media domains using multiple generation methods. Through systematic stylometric analysis, we identify distinctive linguistic patterns that differentiate human-written from machine-generated Arabic text, revealing domain-specific characteristics that vary notably across contexts. These insights inform our development of BERT-based detection models, which achieve exceptional performance in formal contexts (up to 99 % F1-score). However, cross-domain analysis reveals generalization challenges, consistent with prior literature. Robustness evaluation demonstrates that detectors maintain notable performance even with more than 90 % random text dropping. This work provides a comprehensive investigation of Arabic machine-generated text to date, uniquely integrating multiple generation methods, diverse LLMs, and in-depth stylometric analysis across varied domains to establish a foundation for robust, linguistically-informed detection systems essential for preserving information integrity in Arabic-language contexts. © 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Computational linguistics, Language model, Large language model, Artificial intelligence, Classification (of information), Natural language processing systems, Generalisation, Large language models, Cross-domain, Text classification, Text mining, Cross-domain generalization, Arabic natural language processing, Machine-generated text detection, Machine-generated texts, Stylometric analysis, Stylometric analyze, Stylometrics, Text detection

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Arabic Short-Text Dataset for Sentiment Analysis of Tourism and Leisure Events

Item Type Journal Article

Author S. Basabain

Author A. Al-Dubai

Author E. Cambria

Author K. Alomar

Author A. Hussain

Abstract The focus of this study is to present the detailed process of collecting a dataset of Arabic short-text in the tourism context and annotating this dataset for the task of sentiment analysis using an automatic zero-shot labelling technique utilising transformer-based models. This is benchmarked against a baseline manual annotation approach utilising native Arab human annotators. This study also introduces an approach exploiting both manual/handcrafted and automatically generated

annotations of the dataset tweets for the task of sentiment analysis as part of a cross-domain approach using a model trained on sarcasm labels and vice versa. The total collected corpus size is 2293 tweets; after annotation, these tweets were labelled in a three-way classification approach as either positive, negative or neutral. We run different experiments to provide benchmark results of Arabic sentiment classification. Comparative results on our dataset show that the highest performing baseline model when utilising manual labels was MARBERT, with an accuracy of up to 87%, which was pre-trained for Arabic on a massive amount of data. It should be noted that this model enhanced its performance additionally after pre-training on a dialectical Arabic and modern standard Arabic corpus. On the other hand, zero-shot automatically generated labels achieved an 84% accuracy rate in predicting sarcasm classes from sentiment labels. © 2025 The Author(s). Expert Systems published by John Wiley & Sons Ltd.

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ARC-NLP at ClimateActivism 2024: Stance and Hate Speech Detection by Generative and Encoder Models Optimized with Tweet-Specific Elements

Item Type Conference Paper

Author A.K. Kaya

Author O. Ozcelik

Author C. Toraman

Editor Hurriyetoglu A.

Editor Tanev H.

Editor Thapa S.

Editor Uludogan G.

Abstract Social media users often express hate speech towards specific targets and may either support or refuse activist movements. The automated detection of hate speech, which involves identifying both targets and stances, plays a critical role in event identification to mitigate its negative effects. In this paper, we present our methods for three subtasks of the Climate Activism Stance and Hate Event Detection Shared Task at CASE 2024. For each subtask (i) hate speech identification (ii) targets of hate speech identification (iii) stance detection, we experiment with optimized Transformer-based architectures that focus on tweet-specific features such as hashtags, URLs, and emojis. Furthermore, we investigate generative large language models, such as Llama2, using specific prompts for the first two subtasks. Our experiments demonstrate better performance of our models compared to baseline models in each subtask. Our solutions also achieve third, fourth, and first places respectively in the subtasks. © 2024 Association for Computational Linguistics.

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Language English

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Are LLMs Good for Synthetic Data? A Systematic Synthetic Dataset Validation

Item Type Conference Paper

Author I.A. Akinifev

Author S.S. Bodrunova

Editor Coman A.

Editor Vasilache S.

Abstract Effectively tracking and understanding changes in public opinion poses a complex challenge for social science, market analysis, and political forecasting. Traditional research relies on large-scale, longitudinal datasets that are labor-intensive and expensive to annotate. Large Language Models (LLMs) are seen today as new paradigm-shifting data generators that produce synthetic text to fill critical data gaps: however, validity of simulations based on synthetic data remains dubious and under-investigated. This study evaluates the robustness of synthetic datasets for modelling complex social phenomena such as public opinion drift, consensus, etc. We investigate whether synthetic data is able to accurately replicate real-world dynamics or if it remains constrained by hidden artefacts and insufficient variability inherited from both LLM architecture and training corpora. Our experimental design compares three generation strategies: (1) a Zero-Shot approach based only on prompt instructions, (2) a Retrieval-Augmented Generation (RAG) framework that utilizes external context, and (3) a Parameter-Efficient Fine-Tuning (PEFT) method using LoRA adapters. These synthetic cohorts are validated against a “Gold Standard” dataset of authentic Reddit discourse using a multi-dimensional framework of Rationality, Emotionality, and Consensus metrics. Statistical similarity is assessed via the two-sample Kolmogorov-Smirnov test and delta-mean analysis. Our results reveal that targeted fine-tuning of smaller models (1.5B) significantly outperforms larger models (3B) using prompt-based methods. We conclude that deep parameter adaptation is the most reliable method for capturing the nuanced markers of discussions such as opinion drift, shifts in sentiment polarity, etc. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Pages 237-247**Proceedings Title** Lect. Notes Comput. Sci.**Conference Name** Lecture Notes in Computer Science**Date Added** 20/09/2026, 8:35:02 pm**Modified** 20/09/2026, 8:35:02 pm**Tags:**

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Are LLMs Good Zero-Shot Classifiers? Re-Evaluating the Performance of LLMs on Arabic Sentiment Analysis

Item Type Conference Paper**Author** M. Alkaoud

Abstract Recent evaluations of Large Language Models (LLMs) in Arabic Natural Language Processing have suggested their inferior performance compared to traditional state-of-the-art (SOTA) models in zero-shot learning scenarios [1]. This paper presents a critical re-examination of this assertion, focusing specifically on Arabic sentiment analysis. We found that inconsistent class definitions in prior evaluations skewed performance comparisons between LLMs and SOTA models. While SOTA models were evaluated on a simplified three-class sentiment classification task, LLMs were tested on a more complex four-class problem. By standardizing the evaluation framework and aligning class definitions, we demonstrate that LLMs, particularly Claude Sonnet 3.5 and GPT-4o, achieve competitive performance in zero-shot settings, with F1-scores within 2.3-3.7% of the fine-tuned SOTA model. Our study also provides the first evaluation of Grok's capabilities in Arabic sentiment analysis, revealing its relatively lower performance compared to other LLMs. Additionally, we investigate the impact of prompt language (Arabic vs. English) on model performance, finding varying effects across different LLMs. These findings challenge previous conclusions about LLMs' limitations in Arabic sentiment

analysis and emphasize the critical importance of standardized evaluation methodologies in comparative model assessments. © 2025 IEEE.

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Language model, Large language model, Economic and social effects, Classification (of information), Sentiment analysis, Natural Language Processing, Language processing, Natural language processing, Natural language processing systems, Natural languages, Social media, Contrastive Learning, GPT, Sentiment Analysis, Large Language Models, Zero-shot learning, Arabic NLP, Model evaluation, Zero-Shot Learning, Dataless classification, Dataless Classification, Large language model evaluation, LLM Evaluation, Social Media Sentiment Analysis, Social medium sentiment analyze

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Are LLMs Really Underperforming in Stance Detection? Identifying Patterns of Challenging Instances in Zero-Shot Stance Detection

Item Type Conference Paper

Author A. Gheorghe

Author S. Ruseti

Author M. Dascalu

Author C. Caragea

Abstract Stance Detection (SD) is gaining broad adoption in social media analytics, as it is a refined form of sentiment analysis that focuses on determining whether a text expresses a position in favor of, against, or neutral with a given target. Despite significant progress, current SD approaches still suffer from instructions and labeling inconsistencies, especially for the neutral class, and the influence of annotator bias. In this paper, we conduct a comprehensive analysis of these challenges by combining training dynamics with instruction-based reflective reasoning using Large Language Models. This hybrid approach allows us to characterize a set of difficulties that pose challenges during the annotation process, and also to identify a subset of highly probable mislabeled instances. Our results highlight frequent disagreements between model predictions and crowdsourced labels, particularly on examples involving semantic ambiguity, target misinterpretation and mixed sentiment, along with flawed annotation instructions. To address these issues, we propose strategies that include confidence-guided data filtering and reasoning-based clustering, and generate natural language explanations that clarify the sources of confusion related to either the text or the target. This study presents a novel perspective on diagnosing annotation errors and semantic difficulty in zero-shot stance detection and opens promising directions for integrating LLM reasoning into dataset curation and evaluation workflows. We release the results and the corresponding code as open-source: <https://anonymous.4open.science/r/stance-under-the-scope-2643>. © 2025 IEEE.

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Artificial intelligence in anesthesia: comparison of the utility of ChatGPT v/s google gemini large language models in pre-anesthetic education: content, readability and sentiment analysis

Item Type Journal Article

Author P. Sharma

Author B. Sidhu

Author A. Reddy

Author P. Saxena

Author R. Gupta

Author R.K. Gill

Author A. Ray

Author R. Chauhan

Abstract Background: Large Language Models (LLMs) such as ChatGPT and Google Gemini are increasingly explored for their potential in patient education, particularly in the perioperative setting. As text-based tools trained on extensive datasets, they can generate detailed responses to common medical queries. However, their utility in delivering reliable, understandable, and emotionally appropriate preanesthetic information remains unclear. Methods: We conducted a prospective observational study comparing ChatGPT and Google Gemini in generating educational content for patients undergoing laparoscopic cholecystectomy. From 68 patient questions submitted by anesthesiologists, 13 high-relevance items were selected. Responses from both models were independently rated by 20 anesthesiologists using a 5-point Likert scale across four domains: accuracy, comprehensiveness, clarity, and safety. Mixed-effects ordinal regression with random intercepts for rater and question estimated odds ratios (OR) and 95% confidence intervals (CI) for ChatGPT versus Gemini. Readability was assessed using standard linguistic indices, and sentiment analysis was performed. Inter-rater reliability was evaluated using Krippendorff's α . Results: ChatGPT had significantly higher odds of receiving better scores for accuracy (OR 2.32, 95% CI 1.62–3.32, $p < 0.001$) and comprehensiveness (OR 2.38, 95% CI 1.67–3.37, $p < 0.001$), with no differences for clarity (OR 1.05, 95% CI 0.75–1.47) or safety (OR 1.01, 95% CI 0.72–1.43). Gemini generated text with greater readability, demonstrated by a lower Flesch-Kincaid Grade level ($p = 0.04$) and higher Flesch Reading Ease score ($p = 0.04$). Sentiment analysis showed Gemini responses contained a wider emotional range, while ChatGPT responses were more neutral overall. Inter-rater reliability was low across domains (Krippendorff's α 0.23–0.46). Conclusion: ChatGPT produced more accurate and comprehensive perioperative anesthesia information, whereas Gemini offered greater readability and emotional expressiveness. Both models may serve as adjuncts in preanesthetic patient education but are not substitutes for clinician counselling. Larger, multi-center studies incorporating direct patient testing are warranted to validate these findings. © The Author(s) 2025.

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Aspect Based Abusive Sentiment Detection in Nepali Social Media Texts

Item Type Conference Paper
Author O.M. Singh
Author S. Timilsina
Author B.K. Bal
Author A. Joshi
Editor Atzmuller M.
Editor Coscia M.

Editor Missaoui R.

Abstract With the increase in internet access and the ease of writing comments in the Nepali language, fine-grained sentiment analysis of social media comments is becoming more and more pertinent. There are a number of benchmarked datasets for high-resource languages (English, French, and German) in specific domains like restaurants, hotels or electronic goods but not in low-resource languages like Nepali. In this paper, we present our work to create a dataset for the targeted aspect-based sentiment analysis in the social media domain, set up a dataset benchmark and evaluate using various machine learning models. The dataset comprises of code-mixed and code-switched comments extracted from Nepali YouTube videos. We present convincing baselines using a multilingual BERT model for the Aspect Term Extraction task and BiLSTM model for the Sentiment Classification Task achieving 57.978% and 81.60% F1 score respectively. © 2020 IEEE.

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Aspect sentiment learning for Aspect-Level Sentiment Classification

Item Type Journal Article**Author** Z. Jian**Author** J. Li**Author** M. Wang**Author** J. Yao**Author** Q. Wu

Abstract Aspect-Level Sentiment Classification (ALSC) is a fine-grained Sentiment Analysis (SA) task that aims to determine the sentiments of a sentence toward different aspects. Despite their significant success, most existing methods derive aspect sentiment semantics from individual sentences, overlooking the interrelationships among relevant sentences that could provide a more comprehensive understanding of aspect sentiment semantics. To this end, we propose AspLearn, an aspect-learning method to optimize aspect sentiment semantics and generate more robust aspect-specific sentence features for the ALSC task. In a nutshell, AspLearn employs the Aspect-aware Contrastive Learning (AspCL) to mine valuable aspect-related knowledge from aspect-relevant samples, thereby optimizing aspect sentiment semantics and enhancing the model's performance. AspLearn is a simple yet effective method, with its superior aspect learning capabilities confirmed through extensive experiments on three benchmarks. Notably, AspLearn, using DeBERTa as the backbone, achieves Macro F1 score improvements of 3.13%, 0.76%, and 1.07% over the second-best results on the Laptops, Restaurants, and Twitter datasets, respectively. Furthermore, AspLearn's mechanism can retrieve the most relevant demonstrations for Large Language Models (LLMs), enhancing their sentiment recognition capabilities. © 2025

Date 2025**DOI** 10.1016/j.neunet.2025.107758**URL** <https://www.scopus.com/pages/publications/105010209676?origin=resultslist>**Archive** Scopus**Language** English**Volume** 191**Publisher** Elsevier Ltd**Publication** Neural Networks**ISSN** 08936080 (ISSN)**Journal Abbr** Neural Netw.**Date Added** 20/09/2026, 8:35:01 pm**Modified** 20/09/2026, 8:35:01 pm**Tags:**

Computational linguistics, Language model, Large language model, Article, human, Humans, Classification (of information), Learning systems, machine learning, Sentiment analysis, Large Language Model, Language processing, Natural language processing, Natural languages, Large datasets, Data mining, Semantics, Contrastive Learning, Machine Learning, social media, controlled study, Social Media, artificial neural network, sentiment analysis, Learning methods, benchmarking, Neural Networks, Computer, Fine grained, aspect level sentiment, aspect sentiment learning, Aspect-level sentiment classification, Contrastive learning, learning, semantics, Sentiment classification

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: J. Yao; Institute of Artificial Intelligence, Xiamen University, Fujian, 422, Siming South Road, Xiamen, 361005, China; email: yao0010@xmu.edu.cn; CODEN: NNETE

Aspect-Based Multimodal Mining: Unveiling Sentiments, Complaints, and Beyond in User-Generated Content

Item Type Conference Paper**Author** G.V. Singh**Author** D.R. Kori**Author** A. Ekbal

Abstract Sentiment analysis and complaint identification are key tools in mining user preferences by measuring the polarity and breach of expectations. Recent works on complaint identification identify aspect categories and classify them into complaint or non-complaint classes. However, aspect category-based complaint identification provides high-level information about the features of products. In addition, it is also observed that the user sometimes does not complain about a specific aspect but expresses concern about specific aspects in a respectful way. Currently, unimodal and multimodal studies do not differentiate this thin line between complaint and concern. In this work, we propose the task of multimodal aspect term-based analysis beyond sentiments and complaints. It comprises of two sub-tasks, viz (i) classification of the given aspect term into one of the four classes, viz. praise, concern, complaint, and others, (ii) identification of the cause of praise, concern, and complaint classes. We propose a first benchmark explainable multimodal corpus annotated for aspect term-based complaints, praises, concerns, their corresponding causes, and sentiments. Further, we propose an effective technique for the joint learning of aspect term-based complaint/concern/praise identification and cause extraction tasks (primary tasks) where sentiment analysis is used as a secondary task to assist primary tasks and establish them as baselines for further research in this direction. The dataset has been made available on <https://www.iitp.ac.in/~ai-nlp-ml/resources.html> and at Github repository: <https://github.com/20118/MASpectX>. © 2024 ACM.

Date 2024**DOI** 10.1145/3664647.3681703**URL** <https://www.scopus.com/pages/publications/85209785258?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: MM - Proc. ACM Int. Conf. Multimed.**Publisher** Association for Computing Machinery, Inc**ISBN** 979-840070686-8 (ISBN)**Pages** 6433-6442**Proceedings Title** MM - Proc. ACM Int. Conf. Multimed.**Conference Name** MM 2024 - Proceedings of the 32nd ACM International Conference on Multimedia

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Language model, Classification (of information), Sentiment analysis, Data mining, Multi-modal, User's preferences, User-generated, language models, High level languages, Sentiment, sentiment, Aspect term, aspect terms, Complaint, complaints, few shot prompting, Few shot prompting, Primary task

Notes:

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Aspect-Based Sentiment Analysis (ABSA): A Review of Techniques and Applications

Item Type Conference Paper
Author A. Hussain
Author S. Samant
Author A. Singh
Author R. Chhabra
Author V. Arora
Author A. Kapoor
Abstract Aspect-Based Sentiment Analysis (ABSA) is a specialized subfield of sentiment analysis that identifies opinions toward specific aspects or attributes within text. Unlike traditional sentiment analysis, which provides only overall polarity, ABSA offers fine-grained insights by linking sentiments to distinct elements such as 'battery life' or 'service quality.' This review synthesizes advances in ABSA from rule-based and machine learning models to deep learning and transformer-based architectures. Benchmark datasets such as SemEval, Amazon reviews, and Twitter corpora are analyzed alongside applications in e-commerce, healthcare, education, and social media monitoring. Results from recent studies highlight the superiority of transformer-based models, which achieve state-of-the-art performance across multiple benchmarks. Key challenges include implicit aspect recognition, sarcasm detection, domain adaptation, and the scarcity of multilingual datasets. The paper concludes with recommendations for future research on explain ability, cross-domain adaptability, and few-shot or zero-shot learning, providing both academics and practitioners with a comprehensive foundation for advancing ABSA. © 2025 IEEE.
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Aspect-Based Sentiment Analysis for Arabic Business Insights with Generative Language Models

Item Type Conference Paper

Author A. Shayea

Author M.A. Mangoud

Abstract With the growth of online reviews across social media and e-commerce platforms, businesses face increasing difficulty in monitoring and leveraging customer feedback. Traditional sentiment analysis lacks the depth required for actionable insights. Research on Aspect-Based Sentiment Analysis (ABSA), with generative large language models (LLMs), and efficient fine-tuning for the Arabic language remains limited. This paper presents a scalable Arabic ABSA framework and introduces a practical approach to fine-tune LLMs such as T5, GPT, DeepSeek, and Allam using Low-Rank Adaptation (LoRA) and prompt-based optimization to achieve high accuracy on limited Arabic datasets with reduced computational cost. Among the evaluated models, GPT-4o-mini achieved the best performance in aspect extraction (F1-score 0.80), while Allam excelled

in sentiment classification (F1-score 0.83). Finally, the paper demonstrates how these results enable businesses to extract actionable insights through a real-world analytics platform built upon the proposed framework. © 2025 IEEE.

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Language English

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Computational linguistics, Language model, Information systems, Sentiment analysis, Language processing, Natural language processing, Natural Language Processing (NLP), Natural languages, Social networking (online), Fine tuning, Low-rank adaptation, Low-Rank Adaptation (LoRA), Electronic commerce, GPT, Extraction, Parameter-efficient fine-tuning, Tuning, Aspect-based sentiment analyze, DeepSeek, Deepseek, Allam, Aspect polarity classification, Aspect Polarity Classification (APC), Aspect term extraction, Aspect Term Extraction (ATE), Aspect-Based Sentiment Analysis (ABSA), Generative large language model, Generative Large Language Models, mBART, MBART, mT5, Mt5, Parameter-Efficient Fine-Tuning, Polarity classification, Term extraction

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Aspect-Based Sentiment Analysis of Multilingual Hotel Reviews in Jakarta Using Multilingual Transformer Models: A Comparative Study of Multilingual BERT and CrossLingual RoBERTa

Item Type Journal Article

Author A.W. Suryabrata

Author H.L.H. Spits Warnars

Author M.K. Muyeba

Abstract Online reviews have become a critical source of information for evaluating service quality in the hospitality industry. However, extracting fine-grained insights from multilingual user-generated content remains challenging due to linguistic variability and the presence of code-mixed expressions. Aspect-based sentiment analysis (ABSA) provides an effective framework for identifying customer opinions toward specific service attributes. This study investigates the effectiveness of multilingual transformer models for ABSA on online reviews of five-star hotels in Jakarta. A large-scale dataset comprising more than 96,000 reviews collected from TripAdvisor and Google Reviews was analyzed. The proposed framework adopts a sentence-pair classification strategy that reformulates ABSA as a natural language inference task, enabling transformer models to capture aspect–sentiment relationships more effectively. Two multilingual pretrained language models—multilingual Bidirectional Encoder Representations from Transformers (mBERT) and Cross-lingual Language Model-Robustly Optimized BERT Pretraining Approach (XLM-RoBERTa)—were fine-tuned and systematically compared. Experimental results show that XLM-RoBERTa achieved the best performance with an accuracy of 97.20% and an F1-score of 0.9729, slightly outperforming mBERT while requiring higher computational resources. In contrast, mBERT demonstrated greater stability across validation folds. Aspect-level sentiment analysis further revealed that cleanliness, facilities, and service are the most positively perceived aspects of five-star hotels in Jakarta, while pricing remains the primary source of negative sentiment. These findings demonstrate the effectiveness of transformerbased ABSA for multilingual hospitality reviews and provide actionable insights for data-driven decision-making in the luxury hotel sector. © 2026 The authors. This article is published by IIETA and is licensed under the CC BY 4.0 license (<http://creativecommons.org/licenses/by/4.0/>).

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A.W. Suryabrata; Computer Science Department, Bina Nusantara University, Jakarta, Jakarta, 11480, Indonesia; email: arghanta.suryabrata@binus.ac.id

Assessing accuracy and completeness of drug information in mobile apps compared with Micromedex®: An AI-based study in Thai online communities

Item Type Journal Article

Author Y. Meepradist

Author S. Chokchaitam

Author N. Teerakulkittipong

Author P. Sukpornsawan

Abstract Background Ensuring access to accurate and complete drug information is fundamental to rational medication use. Mobile medical applications (MMAs) are increasingly used by healthcare providers; however, their quality compared with institutional databases remains underexplored, especially in non-English and resource-limited settings. Natural Language Processing (NLP), particularly using Thai-language transformer models such as WangchanBERTa, enables automated screening and classification of real-world drug-related queries derived from public online communities. Objectives This study aimed to compare the accuracy and completeness of drug information from three MMAs—Lexicomp®, Medscape®, and Epocrates®—against the institutional gold standard, Micromedex®, using AI-classified Thai-language clinical questions. Methods A total of 1500 Thai-language questions about drug therapy were collected from online health forums (Pharmacafe, Pantip, Reddit). Using WangchanBERTa for text classification and stratified sampling, 194 representative questions were mapped to 13 pharmacoinformatic domains. Each question was answered using the three MMAs and Micromedex®. Three licensed pharmacists independently scored each response for accuracy and completeness using a validated binary checklist (1 = correct/complete; 0 = incorrect/incomplete). Inter-rater consensus was achieved through group discussion. Accuracy and completeness were expressed as percentages and analyzed via one-way ANOVA with Tukey HSD post-hoc testing. Results Micromedex® demonstrated the highest accuracy (55.7 %) and completeness (53.2 %), significantly outperforming Epocrates® ($p < 0.05$). Among MMAs, Lexicomp® showed superior performance (accuracy = 32.3 %; completeness = 29.4 %), whereas Medscape® (accuracy = 31.6 %; completeness = 28.8 %) and Epocrates® (accuracy = 20.7 %; completeness = 18.0 %) ranked lower. The weighted composite score used previously (60 % accuracy + 40 % completeness) was removed for simplicity and clarity. Conclusions While Lexicomp® demonstrates potential as a practical alternative to Micromedex® in ambulatory and community-based pharmacy environments, none of the MMAs achieved equivalent reliability. Real-world Thai-language data analyzed through NLP pipelines provide a reproducible framework for pharmacoinformatic benchmarking. This approach supports rational medication use and guides digital-tool selection in low-resource healthcare systems. © 2025 Elsevier Inc.

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Assessing AI-Based Large Language Models (ChatGPT, Google Gemini, and DeepSeek) for Common Parent Questions about Autism: Acceptability, Readability, and Accuracy

Item Type Journal Article

Author A.A. Almulla

Author M.A.S. Khasawneh

Abstract With growing reliance on AI chatbots for parenting support, this study presents the first evaluation of large language models (LLMs) in addressing common autism-related questions. It compared ChatGPT, Google Gemini, and DeepSeek based on the accuracy, clarity, and usefulness of their responses. The findings aim to inform parents and clinicians about the strengths and limitations of using AI tools in early ASD care. Twenty common questions about Autism Spectrum Disorder (ASD) were identified through content analysis of social media, Google Trends, and ASD forums.

These questions were refined by two educational psychologists, and standardized benchmark answers were created by a panel of pediatric neurodevelopment specialists. Two blinded pediatric autism experts then evaluated the AI-generated responses based on quality, as well as usefulness and reliability. GPT-4 achieved the highest mean quality score ($M = 4.85$, $SD = 0.36$), followed by Gemini and DeepSeek (both $M = 4.55$, $SD = 0.51$; $p > 0.05$). For usefulness, GPT-4 scored $M = 6.40$ ($SD = 0.75$), Gemini $M = 6.10$ ($SD = 0.85$), and DeepSeek $M = 6.05$ ($SD = 0.82$; $p > 0.05$). In reliability ratings, Gemini led with $M = 6.40$ ($SD = 0.82$), GPT-4 $M = 6.25$ ($SD = 0.71$), and DeepSeek $M = 5.95$ ($SD = 0.94$; $p > 0.05$). Findings indicated that AI-based chatbots, by providing rapid, comprehensible, and evidence-based guidance on early signs, interventions, and family support, demonstrate significant potential in bridging the information gap for parents—especially when access to specialists is limited. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025.

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Assessing large language model responses to pediatric depression FAQs: a cross-sectional study on readability, accuracy, and sentiment

Item Type Journal Article

Author R. Jiao

Author M. Chen

Author J. Zhang

Abstract Background – Pediatric depression shows age-specific symptoms that hinder recognition and delay care, while parents and adolescents increasingly turn to online sources, including large language models, for mental health information and guidance. The quality of such information depends on readability, factual accuracy, completeness, and emotional tone. This study compared responses from 3 contemporary large language models (LLMs) to frequently asked questions about pediatric depression to assess their suitability as informational tools. Methods – A cross-sectional analytical study design was used. 15 standardized frequently asked questions covering definition, causes, clinical features, diagnosis, prevention, treatment, and prognosis of pediatric depression were submitted to ChatGPT-5, Microsoft Copilot GPT-5 in Smart Research mode, and DeepSeek 3.1V. Responses were collected verbatim. Readability was assessed using seven established indices. Accuracy and completeness were independently scored on a 0 to 6 scale using a predefined rubric. Sentiment was measured with sentiment scores. One-way analysis of variance (ANOVA) with Tukey post hoc statistical analysis was performed. Results – Readability was different among the various models. DeepSeek 3.1V achieved the highest Flesch Reading Ease Score of 54 to 55 and the lowest Flesch–Kincaid Grade Level of about 9.5 thus indicating easier comprehension. ChatGPT-5 showed intermediate readability with scores of 49 to 50 and grade level about 10.5. Copilot-5 had the lowest Reading Ease score of 43 to 44 and the highest grade level near 10.8. Accuracy on a 0 to 6 scale was highest for Copilot-5. ChatGPT-5 showed the greatest completeness, whereas other models had variable coverage in detailed clinical items. Conclusion – Large language models (LLMs) provide information on pediatric depression but show varying levels of readability, accuracy, and completeness. DeepSeek 3.1V provides greater linguistic accessibility, Microsoft Copilot GPT-5 shows stronger factual consistency, and ChatGPT-5 provides more comprehensive coverage. These artificial intelligence (AI) chatbot systems require human understanding before use in pediatric mental health education or guidance. Copyright © 2026 Jiao, Chen and Zhang.

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Assessing Large Language Models for annotating data in Dementia-Related texts: A Comparative Study with Human Annotators

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Author T. Stoev

Author S. Konow

Author K. Yordanova

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Editor Postfach 3251, Wiesbaden Hochschule RheinMain

Abstract As the aging population increases, there is a sharp hike in the increasing number of people who have dementia. Domain-based Natural Language Processing (NLP) models are necessary to extract domain-specific information that holds valuable insights into dementia. Training this NLP model demands a large amount of annotated data. Human-annotated data, while accurate, is also time-consuming and expensive. A large language model (LLM) offers a potential solution that could be a faster and more cost-effective option for annotating data in various NLP tasks. However, LLMs may struggle to understand complex domain-specific contexts, which could lead to incorrect annotations. This paper compares LLM-based annotated data with human-annotated data to investigate whether LLMs can annotate words/phrases in domain-specific ambiguous texts. Following the annotation scheme, we ask the LLM and the raters to annotate a textual corpus of informal texts collected from forums of family carers of people

with dementia (PwD). The inter-rater agreement score highlights a moderate overlapping between the LLM and the expert rater and the LLM found almost twice as many instances as the expert rater. Data annotation is time-consuming, and relying on LLM is only partially feasible for the complex domain. Therefore, by refining LLM-generated data through experience rater, the workload on human raters can be reduced, enabling the rapid creation of labeled datasets. © 2024 Gesellschaft für Informatik (GI). All rights reserved.

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Assessing Large Language Models in Building a Structured Dataset From AskDocs Subreddit Data: Methodological Study

Item Type Journal Article

Author Q. Snell

Author C. Westhoff

Author J. Westhoff

Author E. Low

Author C.L. Hanson

Author E.S.N. Tass

Abstract Background: In an era marked by a growing reliance on digital platforms for health care consultation, the subreddit r/AskDocs has emerged as a pivotal forum. However, the vast, unstructured nature of forum data presents a formidable challenge; the extraction and meaningful analysis of such data require advanced tools that can navigate the complexities of language and context inherent in user-generated content. The emergence of large language models (LLMs) offers new tools for the extraction of health-related content from unstructured text found in social media platforms such as Reddit. Objective: This methodological study aimed to evaluate the use of LLMs to systematically transform the rich, unstructured textual data from the AskDocs subreddit into a structured dataset, an approach that aligns more closely with human cognitive processes than traditional data extraction methods. Methods: Human annotators and LLMs were used to extract data from 2800 randomly sampled r/AskDocs subreddit posts. For human annotation, at least 2 medical students extracted demographic information, type of inquiry (diagnosis, symptom, or treatment), proxy relationship, chronic condition, health care consultation status, and primary focus topic. For LLM data extraction, specially engineered prompts were created using JavaScript Object Notation and few-shot prompting. Prompts were used to query several state-of-the-art LLMs (eg, Llama 3, Genna, and GPT). Cohen κ was calculated across all human annotators, with this dataset serving as the gold standard for comparison with LLM data extraction. A high degree of human annotator reliability was observed for the coding of demographic information. Lower reliability was seen in coding the health-related content of the posts. Results: The highest performance scores compared with the gold standard were achieved by Llama 3 70B with 7 few-shot prompt examples (average accuracy=87.4) and GPT-4 with 2 few-shot prompt examples (average accuracy=87.4). Llama 3 70B excelled in coding health-related content while GPT-4 performed better coding demographic content from unstructured posts. Conclusions: LLMs performed comparably with human annotators in extracting demographic and health-related information from the AskDocs subreddit unstructured posts. This study validates the use of LLMs for analyzing digital health care communications and highlights their potential as reliable tools for understanding online behaviors and interactions, shifting toward more sophisticated methodologies in digital research and practice. © Quinn Snell, Chase Westhoff, John Westhoff, Ethan Low, Carl L Hanson, E Shannon Neeley Tass.

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Assessing Social Media's Impact on Stock Market Predictions: Financial Sentiment Embedding Approaches

Item Type Conference Paper**Author** V. Ivanenko

Abstract In the digital age, social media platforms like Twitter have become significant channels for disseminating information that can influence financial markets. However, the challenge lies in sifting through the vast volume of tweets to extract valuable insights for stock market prediction. This paper presents a robust methodology to address this challenge, focusing on four key objectives: Tweet Validation, Stock Social Embedding, Stock Market Social Embedding, and Social Data Effectiveness. Utilizing a mathematical model and objective functions, the paper employs Zero-Shot Learning for tweet validation and introduces nuanced embeddings to capture market sentiments. The research employs an experimental setup comparing a baseline ARIMAX model with an enhanced model incorporating social media data. The enhanced model demonstrated a significant improvement in predictive accuracy, achieving a 65.7% accuracy rate compared to the baseline model's 60.1%. The findings have profound implications for investors, financial institutions, and regulators, offering a more reliable and accurate tool for stock market prediction. Future work could extend the dataset and explore other social media platforms to further validate the effectiveness of the proposed methodology. © 2023 IEEE.

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Assessing Text Classification Methods for Cyberbullying Detection on Social Media Platforms

Item Type Journal Article

Author A.G. Philipo

Author D. Sebastian Sarwatt

Author J. Ding

Author M. Daneshmand

Author H. Ning

Abstract Cyberbullying significantly impacts mental health by adversely affecting victims' psychological well-being. It is a prevalent issue on social media platforms, necessitating effective real-time detection systems to identify harmful content. However, current detection systems face challenges related to performance, dataset quality, time efficiency, and computational costs. This study compares existing text classification techniques for cyberbullying detection, evaluating their effectiveness on social media platforms. Large language models such as BERT, RoBERTa, XLNet, DistilBERT, and GPT-2.0 are assessed for their suitability. Results show that BERT achieves optimal performance, with 95% accuracy, precision, recall, and F1 score; a 5% error rate; 0.053 seconds inference time; 35.28 MB RAM usage; 0.4% CPU/GPU utilization; and 0.000263 kWh energy consumption. These findings highlight that while generative AI models are powerful, fine-tuned models often outperform them

when adapted to specific datasets and tasks. © IEEE. 2005-2012 IEEE.

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Assessing the Emotional Impact of Fake News on Social Media: An Ensemble Approach to Sentiment Analysis

Item Type Conference Paper

Author D. Kumar

Author R. Panigrahi

Editor Chhikara R.

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Editor Singal P.

Abstract The progress of Web 2.0 has radically transformed how we interact in the present day. Although it has enhanced connectedness and interpersonal communication, it has its disadvantages including sharing of fake information and the effects of the same on mental health. With the current influence of social media in our lives, it is important to thoughtfully weigh the benefits that it has brought to us against the

potential harm that it can cause. This research undertaking attempts to examine the sentiment aroused by fake news shared in leading global social media sites like Twitter (X). A hard voting ensemble classifier was used, and it was able to classify tweets into phony and authentic with a remarkable 94.21% accuracy rate. Moreover, sentiment analysis with the same classifier resulted in an accuracy rate of 84.11%. The psychological implications of sharing misinformation with the general population were then examined using these models. The research also considers the significant correlation between fake news and emotions and depicts that misleading information is directly associated with negative mental health conditions. This establishes a strong relationship between the false news and attitudes. The results of the proposed approach were compared to the traditional machine learning algorithms. A larger multilingual dataset could justifiably be considered an improvement of the future study to test false tweets and real-time human emotions. As generative AI and large language models (LLMs) that can create highly realistic and misleading content, such as deepfakes; grow more popular, the issue of emotional reactions to false information has gained even greater significance. © 2025 IEEE.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: D. Kumar; Bharati Vidyapeeth's College of Engineering, Department of Computer Science and Engineering, New Delhi, India; email: deepika.kumar@bharatividyaapeeth.edu; Conference name: 2nd International Conference on Intelligent Systems for Cybersecurity, ISCS 2025; Conference date: 14 November 2025 through 15 November 2025; Conference code: 220454

Assessing the Level of Toxicity Against Distinct Groups in Bangla Social Media Comments: A Comprehensive Investigation

Item Type Conference Paper

Author M.B. Moin

Author P. Debnath

Author U.A. Rifa

Author R.B. Anis

Editor Ullah A.

Editor Anwar S.

Abstract Social media platforms have a vital role in the modern world, serving as conduits for communication, the exchange of ideas, and the establishment of networks. However, the misuse of these platforms through toxic comments, which can range from offensive remarks to hate speech, is a concerning issue. This study focuses on identifying toxic comments in the Bengali language targeting three specific groups: transgender people, indigenous people, and migrant people, from multiple social media sources. The study delves into the intricate process of identifying and categorizing toxic language while considering the varying degrees of toxicity—high, medium, and low. The methodology involves creating a dataset, manual annotation, and employing pre-trained transformer models like BanglaBERT, BanglaBERT-base, distil-BERT, and Bert-base-multilingual-case for classification. Diverse assessment metrics such as accuracy, recall, precision, and F1 score are employed to evaluate the model's effectiveness. The experimental findings reveal that BanglaBERT surpasses alternative models, achieving an F1 score of 0.8903. This research exposes the complexity of toxicity in Bangla social media dialogues, revealing its differing impacts on diverse demographic groups. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Assessing the Performance of ChatGPT-4, Fine-tuned BERT and Traditional ML Models on Moroccan Arabic Sentiment Analysis

Item Type Conference Paper

Author M. Hannani

Author A. Soudi

Author K. Van Laerhoven

Editor Hamalainen M.

Editor Ohman E.

Editor Miyagawa S.

Editor Alnajjar K.

Editor Bizzoni Y.

Abstract Large Language Models (LLMs) have demonstrated impressive capabilities in various natural language processing tasks across different languages. However, their performance in low-resource languages and dialects, such as Moroccan Arabic (MA), requires further investigation. This study evaluates the performance of ChatGPT-4, different fine-tuned BERT models, FastText as text representation, and traditional machine learning models on MA sentiment analysis. Experiments were done on two open source MA datasets: an X(Twitter) Moroccan Arabic corpus (MAC) and a Moroccan Arabic YouTube corpus (MYC) datasets to assess their capabilities on sentiment text classification. We compare the performance of fully fine-tuned and pre-trained Arabic BERT-based models with ChatGPT-4 in zero-shot settings. © 2024 Association for Computational Linguistics.

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Assessing the Representation of Suicidal Ideation in Social Media Datasets Relative to Suicide Notes

Item Type Journal Article

Author M. Kavut

Author A. Dzafic

Author U. Bayram

Abstract Clinical records are inaccessible due to legal restrictions, and genuine suicide notes are scarce, limiting the availability of verified suicidal expression data. Consequently, most Natural Language Processing (NLP) studies rely on widely accessible social media datasets containing expert annotations or heuristic labels. However, it remains unclear how expressions of suicidal ideation in these datasets relate to the ideation in genuine suicide notes. In this study, we address this gap by comparing manually annotated social media datasets in English and Turkish, as well as automatically labeled English datasets, directly against genuine suicide notes using a combination of linguistic and statistical analyses, classification models, and zero-shot embedding-based representations. Results show that expert-labeled social media data exhibit limited overlap with suicide notes across both languages while remaining distinguishable, whereas automatically labeled data diverge substantially from the notes in all analyses. These findings indicate that the annotation method critically shapes the signals learned by the computational models and that social media data capture different stages or forms of ideation than those in suicide notes. Overall, our findings emphasize the importance of careful dataset evaluation and caution against misinterpreting model performance in suicide risk detection research, given the shared objective: supporting global suicide prevention efforts. © 2010-2012 IEEE.

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Attacking Misinformation Detection Using Adversarial Examples Generated by Language Models

Item Type Conference Paper
Author P. Przybyla
Author E. McGill
Author H. Saggion
Editor Christodouloupoulos C.
Editor Chakraborty T.
Editor Rose C.
Editor Peng V.

Abstract Large language models have many beneficial applications, but can they also be used to attack content-filtering algorithms in social media platforms? We investigate the challenge of generating adversarial examples to test the robustness of text classification algorithms detecting low-credibility content, including propaganda, false claims, rumours and hyperpartisan news. We focus on simulation of content moderation by setting realistic limits on the number of queries an attacker is allowed to attempt. Within our solution (TREPAT), initial rephrasings are generated by large language models with prompts inspired by meaning-preserving NLP tasks, such as text simplification and style transfer. Subsequently, these modifications are decomposed into small changes, applied through beam search procedure, until the victim classifier changes its decision. We perform (1) quantitative evaluation using various prompts, models and query limits, (2) targeted manual assessment of the generated text and (3) qualitative linguistic analysis. The results confirm the superiority of our approach in the constrained scenario, especially in case of long input text (news articles), where exhaustive search is not feasible. © 2025 Association for Computational Linguistics.

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Attention LSTM Framework for Aspect-Enabled Sentiment Analysis: Integrating Multi-Component Attention Graph Convolutional Neural Network Optimized with Gradient Jellyfish Search Algorithm for Enhanced Tamil Text Understanding

Item Type Conference Paper

Author R. Pushpakumar

Author K. Seethalakshmi

Abstract The proliferation of user-generated content in low-resource languages such as Tamil has intensified the need for sophisticated sentiment analysis frameworks capable of extracting nuanced opinions across diverse domains. In this research, we propose a novel Attention-Long Short-Term Memory (LSTM) framework for aspect-enabled sentiment analysis, meticulously designed for Tamil text. The core of our methodology is the integration of a Multi-component Attention Graph Convolutional Neural Network (MA-GCN) that captures intricate semantic relationships and contextual dependencies among words and aspect terms. In order to further boost the performance and having optimum model convergence, we propose the Gradient Jellyfish Search Algorithm (GJSA) which is a dynamic parameter tuning and architecture optimization algorithm. The combination of the aspect-focused attention processes with the use of graph-based relations modeling enables our model to effectively identify the sentiment polarities, even in the presence of the rich morphological structures and the code-mixed situations, so common in Tamil language data. Considerable experiments done on several benchmark Tamil sentiment datasets show the much-needed boost in classification accuracy, precision, recall, and F1-score compared to the traditional LSTM, CNN and the generalized GCN methods. The outcomes emphasize the effectiveness of our Attention LSTM-MA-GCN-GJSA framework, which can be described as a robust, scalable solution to aspect-based sentiment analysis of underrepresented languages and is the first step towards its implementation in multilingual social media surveillance and analysis of customer feedback and more. © 2025 IEEE.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: R. Pushpakumar; Vel Tech Rangarajan Dr.Sagunthala R&d Institute of Science and Technology, Department of Information Technology, Chennai, Tamilnadu, 600062, India; email: pushpakumar.btech@gmail.com; Conference name: 2025 3rd International Conference on Sustainable Computing and Smart Systems, ICSCSS 2025; Conference date: 20 August 2025 through 22 August 2025; Conference code: 223914

Attention-augmented hybrid CNN-LSTM model for social media sentiment analysis in cryptocurrency investment decision-making

Item Type Journal Article

Author D. Tiwari

Author B.S. Bhati

Author B. Nagpal

Author N. Alturki

Author L. Bayisenge

Abstract Cryptocurrencies have emerged miraculously all over the globe due to their legitimacy, transparency, immutability, and the traceability that blockchain technology provides. However, the benefits it provides are dwarfed by how unpredictable and extremely price-volatile the cryptocurrencies are. That makes it really tough for investors to find their profitable opportunities in such volatile markets. Social media sources, like Twitter and Reddit, have evolved as crucial tools of sentiment estimation above the explosively volatile price movements of decentralized currencies. Here we introduce an attention-based hybrid CNN-LSTM model optimized for social media sentiment analysis to use them towards investment decisions in a broad portfolio of cryptocurrencies. The existing Convolutional Neural Network (CNN) effectively extracts the essential features, and Long Short-Term Memory (LSTM) has the potential to capture the long dependencies between phrases. Although these models can process massive textual data, they limit treating all the features equally important. Therefore, the proposed model induces the attention mechanism into hybrid CNN-LSTM for emphasizing more or fewer weights on different words according to their contributions and optimizes the parameters of employed neural networks using grid search. In our pipeline, the attention-augmented CNN-LSTM first transforms each tweet/review into a 512-dimensional task-specific embedding; a calibrated radial-basis SVM then serves as the final decision layer, refining the margin for classes that the neural network alone tends to blur. This sequential ('deep-features-plus-SVM') architecture boosts F1 by 3.2 pp over a pure

Softmax head while adding only 0.4 ms of inference time. Extensive experiments conducted on cryptocurrency-related tweets and Reddit reviews reveal the outperformance of the proposed model over existing Deep Neural Networks (DNNs) and state-of-the-art models. Trained on 9.9 k crypto-tweets and 33 k Reddit comments, AEH attains 98.7% accuracy, 0.987 F1, and $\kappa = 0.94$, outperforming strong baselines (pure LSTM + 8.3 pp; pure CNN + 19.3 pp) and the widely-used VADER toolkit (+ 11.8 pp). On the forecasting side, a complementary GRU regressor trained on eight-year price series yielded MAE = 0.0315, MAPE = 5.95%, and MSE = 0.0022 for Bitcoin, beating an ARIMA benchmark at $p < 0.001$. The primary objective of the proposed hybrid model attributed to processing huge social sentiments with an attention mechanism to break the dilemma of cryptocurrency investors. © The Author(s) 2025.

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Audience perception and decoding of Chinese cultural content in global cross-media context based on LLM-enhanced temporal topic modeling

Item Type Journal Article**Author** R. Xu**Author** H. Guo**Author** K. Wu**Author** Y. Zhuang**Author** M. Zhou

Abstract Methodological limitations in cross-cultural social media research—temporal blindness, semantic misalignment, and data sparsity—constrain understanding of how global audiences decode Chinese cultural content. This study integrates Large Language Models with temporally-weighted topic modeling to examine non-Chinese audience reception of Chinese cultural videos on YouTube within an algorithmically-mediated environment. Using an LLM-enhanced Seeded-LDA model ($k = 11$), we analyzed 24, 847 non-Chinese comments from high-engagement videos (2014–2025). DeepSeek-v4-flash facilitated multilingual preprocessing, comment filtering, and sentiment analysis. Temporal evolution was tracked via Dynamic Topic Modeling, and reply-thread networks were analyzed to detect politicization spillover. Eleven topics emerged, organized into four decoding positions: algorithmic empathy (travel and food content, >60% positive), value friction, civilizational comparison, and ideological contestation (narrative framing, >50% negative). Network analysis revealed platform-mediated politicization spillover in one-third of high-engagement threads. Three temporal regimes were identified: aesthetic curiosity (2014–2018), polarization (2018–2022), and complex bifurcation (2023–2025). Platform affordances and source attribution cues appear systematically linked to divergent decoding patterns, with self-media triggering empathetic responses and official media activating ideological contestation. The proposed computational framework advances large-scale analysis of platform-mediated cultural communication, though observational constraints limit causal claims. © 2026 Xu, Guo, Wu, Zhuang and Zhou.

Date 2026**DOI** 10.3389/fcomm.2026.1850055**URL** <https://www.scopus.com/pages/publications/105043683245?origin=resultslist>**Archive** Scopus**Language** English**Volume** 11**Publisher** Frontiers Media SA**Pages** 1-13**Publication** Frontiers in Communication**ISSN** 2297900X (ISSN)**Journal Abbr** Front. Commun.**Date Added** 20/09/2026, 8:35:03 pm**Modified** 20/09/2026, 8:35:03 pm**Tags:**

large language models, topic modeling, platform studies, YouTube, algorithmic culture, computational communication, cross-cultural communication, ultramediated society

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Auditing the impact of social media's policy shift on anti-vaccine discourse: A large language model-driven empirical study

Item Type Journal Article**Author** Y. Li**Author** T. Chen**Author** Y. Zhao**Author** W. Ke**Author** P. Pang**Author** D. McKay**Author** S. Chang**Author** N. Baxter

Abstract The sudden termination of X's (formerly Twitter) misinformation policy on November 23, 2022, provides an opportunity to assess the effects of lifting content moderation restrictions on vaccine-related discourse. This study examines changes in the prevalence, thematic composition, and engagement of anti-vaccine discourse following X's policy shift, analyzing tweets from a seven-day period before and after the policy termination (November 16–30, 2022), excluding the announcement date itself from regression analyses. Using GPT-4o for stance classification, thematic categorization, and stance consistency assessment, with validation through external benchmarks and cross-annotator agreement, we find that anti-vaccine tweets increased significantly post-policy (OR = 1.60, 95% CI: 1.50–1.72), particularly via retweets, suggesting content amplification. Sensitivity analyses excluding highly retweeted content revealed that the policy change was also associated with increased creation of new anti-vaccine content. Thematically, health concerns over vaccination became more prominent, while conspiracy-related and anti-mandate narratives declined in relative prevalence. Stance consistency in quote tweets increased, indicating reinforced ideological alignment in anti-vaccine discourse. These results suggest that content moderation policies may constrain both the volume and amplification of anti-vaccine content, with policy removal associated with rapid shifts in discourse patterns. © 2026 Li et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Augmentation and Classification of Requests in Moroccan Dialect to Improve Quality of Public Service: A Comparative Study of Algorithms

Item Type Journal Article**Author** H. Zaidani**Author** R. Koulali**Author** A. Maizate**Author** M. Ouzzif

Abstract Moroccan Law 55.19 aims to streamline administrative procedures, fostering trust between citizens and public administrations. To implement this law effectively and enhance public service quality, it is essential to use the Moroccan dialect to involve a wide range of people by leveraging Natural Language Processing (NLP) techniques customized to its specific linguistic characteristics. It is worth noting that the Moroccan dialect presents a unique linguistic landscape, marked by the coexistence of multiple texts. Though it has emerged as the preferred medium of communication on social media, reaching wide audiences, its perceived difficulty of comprehension remains unaddressed. This article introduces a new approach to addressing these challenges. First, we compiled and processed a dataset of Moroccan dialect requests for public administration documents, employing a new augmentation technique to enhance its size and diversity. Second, we conducted text classification experiments using various machine learning algorithms, ranging from traditional methods to advanced large language models (LLMs), to categorize the requests into three classes. The results indicate promising outcomes, with an accuracy of more than 80% for LLMs. Finally, we propose a chatbot system architecture for deploying the most

efficient classification algorithm. This solution also contains a voice assistant system that can contribute to the social inclusion of illiterate people. The article concludes by outlining potential avenues for future research. © 2025 by the authors.

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Export Date: 20 September 2026; Cited By: 3; Correspondence Address: A. Maizate; RITM Laboratory, Higher School of Technology Casablanca, CED ENSEM, Hassan II University Casablanca, Casablanca, 20430, Morocco; email: abderrahim.maizate@univh2c.ma

Augmenting Sentiment Analysis with a Hierarchical Pre-Attention Framework and Strategic LLM Fine-Tuning

Item Type Journal Article

Author T. Xia

Author X. Liu

Author Y. Deng

Author F. Qiu

Abstract Sentiment analysis is significant for exploiting countless opinion-rich data from social media. However, it faces known challenges, such as informal expression and data sparsity. While large language models (LLMs) excel at dynamic contextual disambiguation, such deep learning models neglect to leverage prior knowledge calculated statistically and globally based on dataset-wide category tendencies to guide the encoding process. To overcome these challenges, this study proposes a pre-attention framework that includes a modified gated recurrent unit hierarchically integrated with a dual-level pre-attention mechanism (PA-GRU) and an LLM-PA-GRU model equipped with a strategic LLM fine-tuning method. The pre-attention mechanism extracts inter-category and intra-text statistical priors from the training set. Trainable coefficient matrices are proposed to adaptively fuse these priors within a unified formulation, enabling flexible allocation of global and local statistical signals. The proposed PA-GRU cell features a modified gating structure and a computation flow that integrates these extracted priors as explicit statistical anchors during sequence encoding. Moreover, we implement an LLM-PA-GRU model that connects the LLM's deep feature representations to the PA-GRU. This strategic LLM fine-tuning approach is intended to mitigate gradient instability frequently encountered in models with heterogeneous architectures. Finally, a prototype-based alignment loss is employed to enforce feature consistency across modules. Extensive experiments on benchmark datasets demonstrate that the proposed approach achieves competitive performance compared with recent literature-reported models. © 2026 by the authors.

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AustroTox: A Dataset for Target-Based Austrian German Offensive Language Detection

Item Type Conference Paper

Author P. Pachinger

Author J. Goldzycher

Author A.M. Planitzer

Author W. Kusa

Author A. Hanbury

Author J. Neidhardt

Editor Ku L.-W.

Editor Martins A.

Editor Srikumar V.

Abstract Model interpretability in toxicity detection greatly profits from token-level annotations. However, currently such annotations are only available in English. We introduce a dataset annotated for offensive language detection sourced from a news forum, notable for its incorporation of the Austrian German dialect, comprising 4,562 user comments. In addition to binary offensiveness classification, we identify spans within each comment constituting vulgar language or representing targets of offensive statements. We evaluate fine-tuned language models as well as large language models in a zero- and few-shot fashion. The results indicate that while fine-tuned models excel in detecting linguistic peculiarities such as vulgar dialect, large language models demonstrate superior performance in detecting offensiveness in AustroTox. We publish the data and code. © 2024 Association for Computational Linguistics.

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Automated Claim Matching with Large Language Models: Empowering Fact-Checkers in the Fight Against Misinformation

Item Type Conference Paper

Author E.C. Choi

Author E. Ferrara

Abstract In today's digital era, the rapid spread of misinformation poses threats to public well-being and societal trust. As online misinformation proliferates, manual verification by fact checkers becomes increasingly challenging. We introduce FACT-GPT (Fact-checking Augmentation with Claim matching Task-oriented Generative Pre-trained Transformer), a framework designed to automate the claim matching phase of fact-checking using Large Language Models (LLMs). This framework identifies new social media content that either supports or contradicts claims previously debunked by fact-checkers. Our approach employs LLMs to generate a labeled dataset consisting of simulated social media posts. This data set serves as a training ground for fine-tuning more specialized LLMs. We evaluated FACT-GPT on an extensive dataset of social media content related to public health. The results indicate that our fine-tuned LLMs rival the performance of larger pre-trained LLMs in claim matching tasks, aligning closely with human annotations. This study achieves three key milestones: it provides an automated framework for enhanced fact-checking; demonstrates the potential of LLMs to complement human expertise; offers public resources, including datasets and codes, to further research and applications in the fact-checking domain. © 2024 Copyright held by the owner/author(s).

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Automated Detection of Human Values in Texts: ML Challenges and Performance Benchmarks

Item Type Conference Paper

- Author** O. Rink
- Author** A. Maysuradze
- Author** A. Fedorov
- Author** R. Ischenko
- Author** A. Korchagina
- Author** A. Tabachenkov
- Author** I. Tsybanov
- Author** K. Vorontsov
- Editor** Coman A.
- Editor** Vasilache S.

Abstract Automated content analysis involves applying AI to insufficiently formalized areas of social studies and humanities. In 2023–2024, we researched the landscape of human values, represented in Russian social media writings, compiled a classifier, and labeled the collected data by 105 human values and their tonality. This paper discusses our development pipeline for generalized text markup structures, the challenges of machine learning to automatically identify human values in texts, own benchmark performance, the Human Values dataset in Russian, and evaluation procedures. The data set is used to create a RoBERTa-based model and a prompt-based model with Qwen 2.5 14B Instruct. These new models automatically categorize texts and focus on multi-annotator assessments. Benchmarks are created to compare the performance of these ML models with human annotators and a Mixture of Experts strategy. Both models demonstrate

relatively high performance in detecting groups of human values, and sentiment classification does not significantly degrade it. The RoBERTa-based model exhibits a higher average performance (F1 0.753) in detecting parent classes in our hierarchical classifier compared to the Qwen-based model (F1 0.739); both indicate a good balance between precision and recall. In recognizing human values at child levels in the classifier RoBERTa (F1 0.274) performs significantly less effectively than the prompt-based strategy (F1 0.537). The dependence of the F1 score, obtained in the validation stage, on the number of annotated instances per class, used in training the RoBERTa-based model, demonstrates an almost linear relationship between the performance achieved and the number of annotated instances. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: O. Rink; MSU Institute for Artificial Intelligence, Moscow, Lomonosovsky Avenue 27s1, 119192, Russian Federation; email: olga.rink@phystech.edu; Conference name: 17th International Conference on Social Computing and Social Media, SCSM 2025, held as part of the 27th HCI International Conference, HCII 2025; Conference date: 22 June 2025 through 27 June 2025; Conference code: 332669

Automated Reddit Data Annotation with Large Language Models

Item Type Conference Paper

Author S. Vuruma

Author D. Wu

Author S.S. Gupta

Author L. Aust

Author V. Lookingbill

Author W. Bellamy

Author Y. Ren

Author E. Kasson

Author L.-S. Chen

Author P. Cavazos-Rehg

Author D. Hu

Author H. Liu

Author M. Huang

Abstract In recent years, the United States has witnessed a significant surge in the popularity of vaping or e-cigarette use, leading to a notable rise in cases of e-cigarette and vaping use associated lung injury (EVALI) that caused hospitalizations and fatalities during the EVALI outbreak in 2019, highlighting the urgency to comprehend vaping behaviors and develop effective strategies for cessation. Due to the ubiquity of social media platforms, over 4.7 billion users worldwide use them for connectivity, communications, news, and entertainment with a significant portion of the discourse related to health, thereby establishing social media data as an invaluable organic data resource for public health research. In this study, we extracted a sample dataset from one vaping sub-community on Reddit to analyze users' quit-vaping intentions. We leveraged latest Large Language Models (LLMs) such as OpenAI's GPT-4, Meta's LLAMA 3.1, Microsoft's Phi 3.5 and Google's Gemini 1.5 & Gemma 2, in addition to traditional machine learning and deep learning models such as BERT, CNN, SVM and XGBoost for sentence-level quit vaping intention detection. This study compares the outcomes of these models against layman and clinical expert annotations. Using different prompting strategies such as zero-shot, one-shot, few-shot and chain-of-thought prompting, and a custom variable called 'detail,' we developed 8 prompts to explain the task to the LLMs and evaluated the performance of these strategies against each other. Our preliminary findings emphasize the potential of using Generative AI tools in social media data analysis, especially in identifying users' subtle intentions that may elude human detection. © 2025 IEEE.

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AUTOMATED SARCASM RECOGNITION USING APPLIED LINGUISTICS DRIVEN DEEP LEARNING with LARGE LANGUAGE MODEL

Item Type Journal Article

Author A.Q.A. Hassan

Author S.D. Alotaibi

Author W.B. Subait

Author A.S. Al-Dobaian

Author H.A. Sultan

Author M. Almanea

Author R. Allafi

Author M. Alshammeri

Abstract Posting sarcastic comments on social media has become popular in the modern era. Sarcasm is a linguistic expression that typically conveys the contrary meaning of what has already been said, making it challenging for machines to find the literal meaning. It depends mainly on context, making it a tedious process for computational analysis. It is well known for its modulation with spoken words and an irony undertone. In addition, sarcasm conveys negative sentiment using positive words, which easily confuses sentiment analysis (SA) models. Sarcasm detection is a natural language processing (NLP) process and is prevalent in SA, human-machine dialogue, and other NLP applications due to sarcasm's ambiguities and complex nature. Concurrently, the advancement of machine learning (ML) techniques makes it easier to develop robust sarcasm detection methods. This paper presents an automated

sarcasm recognition using applied linguistics-driven deep learning with a large language model (ASR-ALDL3M) technique. The purpose of the ASR-ALDL3M technique is to focus on recognizing the sarcastic data using the DL model. In the ASR-ALDL3M technique, the initial data preprocessing phase is utilized, and glove word embedding is applied. Next, the sarcasm recognition procedure is applied using the long short-term memory (LSTM) model. Moreover, the hyperparameter selection of the LSTM model is performed using the fractals monarch butterfly optimization (MBO) technique. At last, a large language model (LLM) is utilized to enhance the sarcastic recognition process. A comprehensive result analysis is made to validate the outcomes of the ASR-ALDL3M technique. The performance evaluation outcomes stated that the ASR-ALDL3M method gains better performance over other models. © 2024 The Author(s).

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Automated stance detection in complex topics and small languages: The challenging case of immigration in polarizing news media

Item Type Journal Article

Author M. Mets

Author A. Karjus

Author I. Ibrus

Author M. Schich

Abstract Automated stance detection and related machine learning methods can provide useful insights for media monitoring and academic research. Many of these approaches require annotated training datasets, which limits their applicability for languages where these may not be readily available. This paper explores the applicability of large language models for automated stance detection in a challenging scenario, involving a morphologically complex, lower-resource language, and a socio-culturally complex topic, immigration. If the approach works in this case, it can be expected to perform as well or better in less demanding scenarios. We annotate a large set of pro- and anti-immigration examples to train and compare the performance of multiple language models. We also probe the usability of GPT-3.5 (that powers ChatGPT) as an instructable zero-shot classifier for the same task. The supervised models achieve acceptable performance, but GPT-3.5 yields similar accuracy. As the latter does not require tuning with annotated data, it constitutes a potentially simpler and cheaper alternative for text classification tasks, including in lower-resource languages. We further use the best-performing supervised model to investigate diachronic trends over seven years in two corpora of Estonian mainstream and right-wing populist news sources, demonstrating the applicability of automated stance detection for news analytics and media monitoring settings even in lower-resource scenarios, and discuss correspondences between stance changes and real-world events. © 2024 Mets et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Automatic Analysis of Political Discourse: A Comparative Study of Multilingual and Large Language Models

Item Type Journal Article

Author A. Sairanbekova

Author A. Nazyrova

Author G. Bekmanova

Author L. Zhetkenbay

Author B. Yergesh

Author Z. Lamasheva

Abstract This paper proposes the growing importance of automated analysis of political discourse in low-resource languages, using the Kazakh language as a case study. As political communication in Kazakhstan has increasingly moved online between 2019 and 2023, the need for accurate tools to evaluate political sentiment has grown. However, limited linguistic resources in Kazakh have hindered tool development. This paper introduces the first annotated corpus of political discourse in Kazakh, comprising 3,022 sentences selected from official statements, televised debates, policy documents, and social media publications. Each text was manually annotated for political sentiment by expert linguists and political scientists, with inter-annotator agreement measured to confirm reliability. Two main methodological approaches were employed for automatic sentiment classification: adapting multilingual neural network models to the Kazakh corpus and testing advanced generative language models in scenarios with minimal training examples. Performance was evaluated using standard classification procedures. The inclusion of pragmatic features such as code-switching, rhetorical emphasis, and discursive context led to notable improvements in classification accuracy. Experimental results demonstrate that models adapted to multilingual input achieved high classification quality, with fine-tuned multilingual transformer models reaching F₁-scores of up to 0.90, while large language models reached an F₁-score of 0.94 in few-shot settings. Explicit modeling of code-switching and pragmatic features yielded an improvement of approximately 4

percentage points in F1. This research contributes a practical resource and a methodological framework for analyzing political sentiment in underrepresented languages, highlighting the feasibility of developing high-quality automated tools for political text analysis without extensive training data. © Authors retain all copyrights.

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Export Date: 20 September 2026; Cited By: 0; Funding details: Ministry of Education and Science of the Republic of Kazakhstan, (AP19679847)

AUTOMATIC DETECTION OF MORAL VALUES IN MUSIC LYRICS

Item Type Journal Article

Author V. Preniqi

Author I. Ghinassi

Author J. Ive

Author K. Kalimeri

Author C. Saitis

Abstract Moral values play a fundamental role in how we evaluate information, make decisions, and form judgements around important social issues. The possibility to extract morality rapidly from lyrics enables a deeper understanding of our music-listening behaviours. Building on the Moral Foundations Theory (MFT), we tasked a set of transformer-based language models (BERT) fine-tuned on 2,721 synthetic lyrics generated by a large language model (GPT-4) to detect moral values in 200 real

music lyrics annotated by two experts. We evaluate their predictive capabilities against a series of baselines including out-of-domain (BERT fine-tuned on MFT-annotated social media texts) and zero-shot (GPT-4) classification. The proposed models yielded the best accuracy across experiments, with an average F1 weighted score of 0.8. This performance is, on average, 5% higher than out-of-domain and zero-shot mod-els. When examining precision in binary classification, the proposed models perform on average 12% higher than the baselines. Our approach contributes to annotation-free and effective lyrics morality learning, and provides useful insights into the knowledge distillation of LLMs regard-ing moral expression in music, and the potential impact of these technologies on the creative industries and musical culture. © V. Preniqi, I. Ghinassi, J. Ive, K. Kalimeri, and C. Saitis.

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Archive Scopus

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Automatic Identification of COVID-19-related Narratives in German Telegram Channels and Chats

Item Type Conference Paper

Author P. Heinrich

Author A. Blombach

Author B.M.D. Dang

Author L. Zilio

Author L. Havenstein

Author N. Dykes

Author S. Evert

Author F. Schäfer

Editor Calzolari N.

Editor Kan M.-Y.

Editor Hoste V.

Editor Lenci A.

Editor Sakti S.

Editor Xue N.

Abstract We are concerned with mapping the discursive landscape of conspiracy narratives surrounding the COVID-19 pandemic. In the present study, we analyse a corpus of more than 1,000 German Telegram posts manually tagged with 14 conspiracy and conspiracy-related narrative labels by three independent annotators. Since emerging narratives on social media are short-lived and notoriously hard to track, we experiment with different state-of-the-art approaches to few-shot and zero-shot text classification. We report performance in terms of ROC-AUC and in terms of optimal F1, and compare fine-tuned methods with off-the-shelf approaches and human performance. © 2024 ELRA Language Resource Association: CC BY-NC 4.0.

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Automating Alternative Generation in Decision-Making

Item Type Conference Paper

Author Y. Kostiuk

Author C. Seyfried

Author C. Reed

Editor Christodoulopoulos C.

Editor Chakraborty T.

Editor Rose C.

Editor Peng V.

Abstract In decision making, generating alternative solutions is crucial for solving a problem. However, cognitive biases can impede this process by constraining individual decision makers' creativity. To address this issue, we introduce a new task for automatically generating alternatives, inspired by the process of human "brainstorming". We define alternative options based on atomic action components and present a dataset of 106 annotated Reddit r/Advice posts containing unique alternative options extracted from users' replies. We also introduce new metrics to assess the quality of generated components, including distinctiveness, creativity, upvote-weighted, crowd intersection, and final commit intersection scores. As a baseline, we evaluated the large language models (LLMs) LLaMa3:8b, LLaMa3.1:8b, and Gemma 2:9b on the alternative component generation task. On the one hand, models demonstrated high creativity (ability to generate options beyond what Reddit users suggested) and performed well at proposing distinct alternatives. A subset of generated components was manually evaluated and found overall useful. This indicates that LLMs might be used to extend lists of alternative options, helping decision makers consider a problem from different perspectives. On the other hand, LLMs' outputs often failed to align with human suggestions, implying that they still tend to miss important components. The code and annotation guidelines can be found in the project's GitHub repository¹. The dataset is available via the link² © 2025 Association for Computational Linguistics.

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Notes:

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Autonomous Aspect-Image Instruction A2II: Q-Former Guided Multimodal Sentiment Classification

Item Type Conference Paper

Author J. Feng

Author M. Lin

Author L. Shang

Author X. Gao

Editor Calzolari N.

Editor Kan M.-Y.

Editor Hoste V.

Editor Lenci A.

Editor Sakti S.

Editor Xue N.

Abstract Multimodal aspect-oriented sentiment classification (MABSC) task has garnered significant attention, which aims to identify the sentiment polarities of aspects by combining both language and vision information. However, the limited multimodal data in this task has become a big gap for the vision-language multimodal fusion. While large-scale vision-language pretrained models have been adapted to multiple tasks, their use for MABSC task is still in a nascent stage. In this work, we present an attempt to use the instruction tuning paradigm to MABSC task and leverage the ability of large vision-language models to alleviate the limitation in the fusion of textual and image modalities. To tackle the problem of potential irrelevance between aspects and images, we propose a plug-and-play selector to autonomously choose the most appropriate instruction from the instruction pool, thereby reducing the impact of irrelevant image noise on the final sentiment classification results. We conduct extensive experiments in various scenarios and our model achieves state-of-the-art performance on benchmark datasets, as well as in few-shot settings. © 2024 ELRA Language Resource Association: CC BY-NC 4.0.

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Awakened at EXIST2025: Adaptive Mixture of Transformers

Item Type	Conference Paper
Author	A. Petrescu
Author	E.-S. Apostol
Author	C.-O. Truică
Editor	Faggioli G.
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Editor	Rosso P.
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Abstract	This paper presents an adaptive Mixture of Transformers architecture developed for the EXIST 2025 Lab, targeting the detection of sexist content in social media text. The proposed system combines nine transformer-based models—spanning both English-specific and multilingual variants—each specialized by language, platform, or task. A dynamic weighting mechanism automatically adjusts the contribution of each model in the ensemble based on the detected language and performance metrics, enabling robust and context-aware classification across

diverse linguistic settings. Experimental results demonstrate that the 2025 architecture achieves competitive performance compared to previous years, surpassing the 2023 and 2024 iterations in several Subtasks and delivering notable improvements in cross-lingual and task-specific detection. However, results for certain Subtasks indicate areas for further optimization. The findings highlight the effectiveness of adaptive model selection and weighting in ensemble architectures for harmful content detection and suggest promising directions for future research in multilingual and context-sensitive text classification. © 2025 Copyright for this paper by its authors.

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Balancing Sentiment Analysis Datasets Through Representative-Word-Guided Synthetic Review Generation: A Case Study on Mexican Spanish Tourism Reviews

Item Type Journal Article

Author A. Díaz-Pacheco

Author A.B. García-Gutiérrez
Author A.Y. Rodríguez-González
Author R. Aranda
Author M.Á. Álvarez-Carmona

Abstract Class imbalance remains one of the most challenging problems in sentiment analysis, particularly in tourism review datasets where positive opinions substantially outnumber neutral and negative comments. This issue is especially critical because minority classes often contain the most valuable information regarding customer dissatisfaction, service failures, and opportunities for improvement. In this work, we propose a hybrid balancing methodology for sentiment analysis in Mexican Spanish tourism reviews that combines undersampling and Large Language Model (LLM)-based oversampling. The proposed framework first extracts representative words from each sentiment class using Mutual Information, then enriches them through dictionary-based or embedding-based lexical substitutions, and finally generates synthetic reviews using GPT-4o-mini guided by these representative terms. Experiments were conducted on a corpus that contains approximately 300,000 tourism reviews collected from TripAdvisor, exhibiting severe sentiment imbalance. Three undersampling strategies and multiple oversampling configurations were evaluated across six traditional machine learning classifiers and one Transformer-based model (BETO). Results show that random undersampling consistently outperformed centroid-based and K-means-based alternatives while also requiring the lowest computational cost. The best overall performance was obtained by BETO, achieving a Macro-F1 score of 0.57 compared to 0.51 on the original imbalanced dataset, representing an improvement of 11.8%. Significant gains were also observed for minority classes, with improvements exceeding 16% for the most underrepresented category. Furthermore, the proposed methodology consistently outperformed direct prompt-based generation using GPT-4o-mini, Gemini 2.5 Flash, and Llama 3.3 70B. These findings suggest that guiding synthetic review generation through representative words effectively preserves domain-specific lexical and semantic patterns of Mexican Spanish tourism reviews, resulting in more balanced datasets and improved sentiment classification performance. © 2026 by the authors.

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BAN-SentiFuse: A Feature Fusion Technique for Bangla Sentiment Analysis

Item Type Conference Paper

Author A. Debnath

Editor Shah M.S.

Abstract Sentiment analysis of Bangla text has drawn more interest recently because of its numerous applications in assessing popular opinion on digital platforms, social media, and entertainment. In contrast to high-resource languages, Bangla's linguistic complexity and the absence of reliable techniques that can capture its semantic richness have hindered development in this area. This study proposed BAN-SentiFuse, a feature fusion framework for Bangla sentiment classification. The technique constructs 3072-dimensional semantic features by concatenating the [CLS] token outputs from BanglaBERT's final four transformer layers to extract contextual embeddings. These are utilized with TF-IDF vectorized 2000-dimensional statistical representations to create a complete 5072-dimensional feature space. A multilayer perceptron network (MLP) and popular machine learning classifiers were employed to utilize the fused features. Experimental results on a benchmark dataset of Bangla drama and TV series reviews demonstrate that BAN-SentiFuse frequently surpasses conventional ML approaches and even a deep learning-based stacking ensemble when evaluated on the same dataset. With an F1-score of 0.83 and an accuracy of 83.52%, the MLP classifier demonstrated outstanding performance. These results exhibit how well semantic and statistical characteristics can be combined, providing a strong foundation for sentiment analysis in low-resource languages like Bangla. © 2026 IEEE.

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Bangla Advice Text Classification: A Novel Dataset and Comparative Study of Machine Learning and Deep Learning Models

Item Type Conference Paper
Author P. Sinha
Author A.K. Shil
Author A. Debnath
Abstract Advice is a significant type of interaction driver for the formation of decisions in areas such as education, health, finance, technology and daily life. Vast amount of such advice is exchanged in Bangladesh as well as worldwide, and increased volume is making it difficult for users to find a way towards the answer effectively. This paper attempts to explain decision making by structuring advice in terms of suggestive, directive and opinion-based so that the users can search for useful information quickly. For this purpose, we created a corpus of 7,010 Bengali advice texts from various sources such as Quora, Reddit, social media, books and online forums. A number of models were investigated, from traditional ML to deep learning and transformers. The findings demonstrate the potential of transformer-based architectures such as BanglaBERT in learning fine-grained

patterns of advisory language, and sheds light on the role played by machine learning mechanisms ranging from Logistic Regression to Support Vector Machines. This work has the potential to improve NLP for Bengali by filling a low-resource gap, providing a systematically annotated dataset, and demonstrating strategies for the practical application of advice classification, such as in recommendation systems, decision-support tools, and intelligent assistants.
© 2025 IEEE.

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Bangla Feminism Sentiment Detection on Social Media Using Transformer-Based Deep Learning Models

Item Type Conference Paper

Author S. Rahman

Author M. Tanzid

Author S.M.S. Farsi

Author M.K. Tonmoy

Author M.S. Rana

Author M. Rahman

Editor Shah M.S.

Abstract The social awareness of feminism on social media and the analysis of the gender discourse is important to understand the social sentiment regarding the feminism issue that serves as the basis of making policy decisions in low-resource languages such as Bangla. Therefore, This work includes an indepth discussion of Bangla feminist sentiment through Transformer-based models. Training and evaluation were done using a manually assembled data of 4,640 comments of Bangla social media posts that had been annotated with positive and negative sentiments. BanglaBERT-BUET and BanglaBERT, Multilingual BERT (mBERT), XLM-RoBERTa (XLM-R), DistilBERT, RemBERT, and MiniLM-L12 were fine-tuned and compared. The experimental performance indicates that BanglaBERT-BUET performed best in the overall performance (Accuracy =94.94%, F1-score = 94.93%, AUC =0.98%). RemBERT demonstrated the best recall (93.88 %), whereas MiniLM-L12 had a stable performance on a relatively low cost of computation, which is appropriate to be used in real time. © 2026 IEEE.

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Bangla-ToCo: A context-aware dataset for Bengali toxic comment detection

Item Type Journal Article

Author S.A. Rupa

Author M.M. Anwar

Author N.A. Ritu

Author K.N. Arifin

Abstract In today's online world, abusive and toxic comments on social media have become a growing concern, particularly for languages like Bengali, where there are currently few reliable automatic moderation systems. In this study, we introduce a context-aware dataset for identifying toxic comments in Bengali by considering how a comment's meaning changes when we read it alongside the relevant information. We present a dataset of 1004 Bengali news comments obtained from the official Facebook pages of two widely read Bangladeshi news portals such as ProthomAlo and News24. Each data record includes six attributes: the news title, metadata of the article, the target comment, its predecessor comment, its successor comment, and the commenter's username. Unlike existing Bengali corpora that present isolated text, this dataset preserves conversational context, enabling more accurate interpretation of ambiguous or short comments. The dataset has been annotated into two categories: Toxic and Non-Toxic. Annotation was performed independently by three human annotators, and the final label for each comment was assigned through majority voting to ensure consistency and reduce individual bias. This balanced class distribution makes the dataset particularly suitable for supervised learning and benchmarking. To the best of our knowledge, this is the first context-aware Bengali toxic comment dataset. Although modest in size, it provides one of the first systematically annotated resources for Bengali, a low-resource language in NLP. It has potential applications in abusive content detection, sentiment and stance classification, conversational modelling, and social discourse analysis, making it a valuable contribution for both computational linguistics and social sciences. © 2025 The Author(s).

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BanglaASTE: A Novel Framework for Aspect-Sentiment-Opinion Extraction in Bangla E-commerce Reviews Using Ensemble Deep Learning

Item Type Conference Paper

Author A. Islam

Author M.R. Hossen

Author A. Ahmed

Author B.M. Taslimul Haque

Abstract Aspect-Based Sentiment Analysis (ABSA) has emerged as a critical tool for extracting fine-grained sentiment insights from user-generated content, particularly in e-commerce and social media domains. However, research on Bangla ABSA remains significantly underexplored due to the absence of comprehensive datasets and specialized frameworks for triplet extraction in this language. This paper introduces BanglaASTE, a novel framework for Aspect Sentiment Triplet Extraction (ASTE) that simultaneously identifies aspect terms, opinion expressions, and sentiment polarities from Bangla product reviews. Our contributions include: (1) creation of the first annotated Bangla ASTE dataset containing 3,345 product reviews collected from major e-commerce platforms including Daraz, Facebook, and Rokomari; (2) development of a hybrid classification framework that employs graph-based aspect-opinion matching with semantic similarity techniques; and (3) implementation of an ensemble model combining BanglaBERT contextual embeddings with XGBoost boosting algorithms for enhanced triplet extraction performance. Experimental results demonstrate that our ensemble approach achieves superior performance with 89.9% accuracy and 89.1% F1-score, significantly outperforming baseline models across all evaluation metrics. The framework effectively addresses key challenges in Bangla text processing including informal expressions, spelling variations, and data sparsity. This research advances the state-of-the-art in low-resource language sentiment analysis and provides a scalable solution for Bangla e-commerce

analytics applications. © 2025 IEEE.

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BanglaNLP at BLP-2023 Task 2: Benchmarking different Transformer Models for Sentiment Analysis of Bangla Social Media Posts

Item Type Conference Paper

Author S. Saha

Author A. Nanda

Editor Sadeque F.

Editor Amin R.

Editor Kar S.

Editor Chowdhury S.A.

Editor Alam F.

Abstract Bangla is the 7th most widely spoken language globally, with a staggering 234 million native speakers primarily hailing from India and Bangladesh. This morphologically rich language boasts a rich literary tradition, encompassing diverse dialects and language-specific challenges. Despite its linguistic richness and history, Bangla remains categorized as a low-resource language within the natural language processing (NLP) and speech community. This paper presents our submission to Task 2 (Sentiment Analysis of Bangla Social Media Posts) of the BLP Workshop. We experimented with various Transformer-based architectures to solve this task. Our quantitative results show that transfer learning helps in better learning of the models in this low-resource language scenario. This becomes evident when we further finetuned a model that had already been finetuned on Twitter data for sentiment analysis task and that finetuned model performed the best among all other models. We also performed a detailed error analysis, finding some instances where ground truth labels need to be looked at. We obtained a micro-F1 of 67.02% on the test set and our performance in this shared task is ranked at 21 in the leaderboard. © 2023 Association for Computational Linguistics.

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BanglaSAPM: A Deep Learning Model for Suicidal Attempt Prediction Using Social Media Content in Bangla

Item Type Conference Paper

Author S. Islam

Author M.S. Alam Forhad

Author H. Murad

Abstract Nowadays, people are constantly expressing their emotions, thoughts, opinions, and daily activities on social media. Therefore, social media posts have become a powerful tool among psychiatrists for the early detection of suicidal tendencies. However, the automatic detection of suicidal posts has become a challenging problem among researchers in the field of Natural Language Processing (NLP). A significant number of previous works have been found in the literature for the automatic detection of suicidal posts in different languages such as English. However, little effort has been devoted to automatically detecting suicidal posts in low-resource languages like Bangla due to the lack of available datasets. In this study, we have created a noble Bangla suicidal posts dataset named BanglaSPD and compared the performance of various machine learning and deep learning models for suicide attempt prediction by training and evaluating with the dataset. Finally, we have found that a deep learning-based model with CNN+BiLSTM has outperformed with 0.61 F1-Score in Fasttext word embedding methods. © 2022 IEEE.

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Based on RuNewsCorp: Improving Accuracy in Long Text Classification

Item Type Conference Paper

Author S. Lina

Author K.A. Aksyonov

Author W. Shiyong

Abstract Text classification is an extremely important task in the field of Natural Language Processing (NLP), and sentiment analysis is a subfield of text classification that identifies and categorizes the emotional tendencies within text. This is crucial for market research, brand monitoring, and customer service. However, current sentiment analysis is predominantly based on short English reviews posted on Twitter. There is a scarcity of sentiment analysis tasks for long Russian texts. This paper introduces the construction of a sentiment classifier for Russian sports news, based on a large language model-BERT, using fine-tuning methods on the self-built dataset Ru-News-Corp-Sen. Furthermore, the paper explores and compares various methods for improving the accuracy of sentiment classification in long texts. © 2024 IEEE.

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BCLSA: Advancing Bangla sentiment analysis with concept-level reasoning and efficiency

Item Type Journal Article

Author M.A. Ullah

Abstract Accurate sentiment analysis in Bangla remains a significant research challenge due to limited annotated corpora, complex morphology, insufficient linguistic resources, and the absence of interpretable concept-level knowledge bases. Existing approaches often struggle to capture context-dependent sentiment, idiomatic expressions, and domain adaptability, further constrained by the low-resource nature of the language. To address these limitations, this study proposes the Bangla Concept-Level Sentiment Analysis (BCLSA) framework, introducing two dedicated algorithms: a Bangla-specific concept extraction method and the Concept-Level Sentiment Analysis for Bangla (CLSA-Bn) weighted scoring algorithm. The first extracts sentiment-bearing concepts through syntactic pattern recognition, multiword expression detection, and affective lexicon mapping, while the second refines polarity estimation via negation handling, modifier scaling, and weighted aggregation for interpretable classification. To mitigate data scarcity and morphological variation, BCLSA applies language-specific preprocessing, including Unicode normalization, phonetic correction, and lemmatization. Evaluations on 10,243 formal news articles and 12,084 informal social media comments show that CLSA-Bn outperforms the Bi-LSTM and SVM baselines, achieving 90.2 % Accuracy, 90 % Macro-F1, 85 % Matthews Correlation Coefficient (MCC), and 94 % Area Under the Curve (AUC) for formal text, and 86.8 % Accuracy, 86 % Macro-F1, and 91 % AUC for informal text. The proposed Concept-Level Polarity Accuracy (CLPA) metric confirmed semantic fidelity above 88 %. Efficiency analysis revealed that CLSA-Bn requires only 30 s initialization, 5 ms inference, and a 50 MB model. Error rate analysis further confirmed robustness with the lowest misclassification ratios (9.8 % formal, 13.2 % informal), demonstrating balanced improvement in performance and error minimization. Copyright © 2025. Published by Elsevier Ltd.

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Benchmarking Arabic Emotion Detection: Transformers Evaluation using a Unified Dataset

Item Type Journal Article

Author N. Vanetik

Author M. Litvak

Author C. Liebeskind

Abstract Emotion detection in Arabic text is inadequately supported and inconsistent, with current datasets exhibiting significant variability in labeling schemes, dialect representation, annotation uniformity, and data availability. These differences, along with the predominance of datasets accessible solely as Twitter IDs and the misalignment among principal emotion theories, obstruct systematic benchmarking and the advancement of resilient Arabic emotion detection models. In this work, we provide a comprehensive, data-driven survey and analysis of Arabic emotion datasets with publicly accessible text. We examine their label distributions, linguistic characteristics, and cross-dataset similarities, revealing substantial annotation inconsistencies, class imbalance, and variability in lexical and stylistic features. Building on these findings, we propose a principled unification pipeline that maps heterogeneous labeling schemes to Ekman's six basic emotions with an additional neutral category. The resulting unified dataset contains over 31K single-labeled texts spanning diverse domains and dialects. Using this resource, we conduct an extensive

evaluation of state-of-the-art Arabic and multilingual transformer models in both zero-shot and fine-tuned settings. Results show broad performance disparities across emotions, notable instability in zero-shot predictions, and a clear advantage for Arabic-language models over multilingual ones. Our study demonstrates that the unification of Arabic emotion datasets is both achievable and advantageous, establishing a replicable basis for future research and presenting an extensive benchmark for emotion detection in Arabic. © 2010 IEEE.

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Benchmarking Large Language Models for Sarcasm Detection in Hindi Romanized Texts

Item Type Conference Paper

Author A.M. Rahi

Author N. Ismam

Author M.A. Rusho

Author S. Hossain

Author J.F. Maliha

Author M.A. Hossain

Author M. Arman

Author M.K. Jahan

Abstract Sarcasm detection is a critical challenge in natural language processing (NLP) due to its intricate reliance on context, tone, and implicit meaning. Accurately identifying sarcasm is essential for sentiment analysis, opinion mining, and various downstream NLP applications, including social media monitoring, customer feedback analysis, and chat-bot development. With the rise of large language models (LLMs), significant advancements have been made in many NLP tasks, ranging from machine translation to text generation. However, the capability of these models in detecting sarcasm, particularly in Hindi Romanized text a widely used informal writing style among Hindi speakers remains underexplored. This study presents a comprehensive comparative analysis of twelve pretrained LLMs, assessing their effectiveness in sarcasm detection using a Hindi Romanized dataset. The evaluation highlights notable performance variations across models, emphasizing the complexities associated with processing Romanized Hindi text. Factors such as transliteration inconsistencies, code mixing, and the lack of explicit contextual markers contribute to the unique challenges faced by these models. Additionally, the findings offer insights into the strengths and limitations of different LLM architectures in handling nuanced language constructs like sarcasm. By identifying key challenges and model specific performance trends, this research aims to bridge the gap in sarcasm detection for Hindi Romanized text, providing valuable implications for future model development and dataset curation in multilingual NLP. © 2025 IEEE.

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Benchmarking Lightweight Transformers and Small Language Models for Climate Change Sentiment Analysis

Item Type Conference Paper

Author B. Rachman

Abstract Understanding public sentiment toward climate change is essential for guiding communication, policy, and public awareness. This study benchmarks Transformer-based and Small Language Model (SLM) paradigms for climate change sentiment classification using the publicly available Twitter Climate Change Sentiment Dataset. Evaluated models include fine-tuned Transformers (BERT-tiny, DistilBERT, RoBERTa-base, RoBERTa-large, and a custom Transformer), a drift-aware streaming variant, and zero-shot and few-shot SLMs (Phi-3, Mistral 7B, and Qwen2 7.6B). Experiments measured macro-F1, accuracy, and inference efficiency across uniform tweet samples. Results show that RoBERTa-large fine-tuned for 20 epochs on GPU achieved the highest performance (macro-F1 =0.8217), followed by RoBERTabase (0.7874) and DistilBERT (0.7444), outperforming BERT-tiny (0.6767). SLMs performed poorly (macro-F1 =0.1667-0.5070) and exhibited high latency (>2.5 s / sample), while the driftaware Transformer achieved moderate performance (macro F1=0.5526, accuracy =0.5527) with resilience to concept drift in streaming conditions. These findings demonstrate that lightweight fine-tuned Transformers remain the most effective and efficient models for domain-specific sentiment analysis, whereas current SLMs are not yet practical for real-time or fine-grained classification without adaptation. Future work will explore prompt optimization, instruction tuning, and adaptive streaming integration to improve the robustness of environmental Natural Language Processing (NLP) systems. © 2025 IEEE.

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Benchmarking Machine Learning Techniques in Under-Resourced Contexts: Analysis of Public Perceptions of Government Policies from South Karnataka Reddit Discourse

Item Type Conference Paper

Author V. Sharma

Author A. Tripathi

Author K.M. Kavitha

Editor Fantidis J.

Abstract Understanding public perception is crucial for the successful implementation of government policies. This study employs established Natural Language Processing (NLP) techniques to gauge public sentiment of government policies in Indian context and identify key themes related to policies in the South Karnataka region of India. Traditional machine learning techniques such as Support Vector Machine (SVM), Decision Tree (DT), Naïve Bayes (NB), K-Nearest Neighbors (KNN) and Random Forest (RF) along with deep learning techniques such as DistilBERT were explored in uncovering public opinion on government policies under very low training data conditions. Latent themes in the public discourse were identified using K-Means Clustering and BERTopic. Our findings reveal a clear performance trade-off for the highly nuanced task of perception mining, as all the chosen models under-performed under lowdata settings with SVM emerging as the best performing model with a 41.5% accuracy slightly higher than the deep learning model DistilBERT with an accuracy of 40.4%. For thematic clustering, K-Means produced highly coherent and separable topics with 94.6% accuracy, demonstrating its effectiveness in identifying keyword-driven themes, whereas BERTopic identified more semantically nuanced topics that were less distinct. © 2025 IEEE.

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Benchmarking sentiment analysis of algerian arabic dialect on X (Twitter)

Item Type Journal Article

Author L. Ouchene

Author S. Bessou

Abstract The rise of Arabic dialects on social media has made sentiment analysis essential for understanding regional opinions. However, most existing research focuses on high-resource languages, leaving low-resource dialects like Algerian Arabic underexplored. To address this, we introduced a publicly available dataset of 18,589 Algerian dialect tweets and presented a customized preprocessing pipeline tailored to the dialect's unique linguistic features. We further conducted a comprehensive

evaluation of sentiment analysis models, where our best-performing MARBERT-LSTM model achieved 91.23% accuracy, setting a new benchmark. This work provides both a valuable resource and a strong baseline for future research in dialectal Arabic NLP. © The Author(s) 2025.

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Benchmarking Zero-Shot Text Classification for Dutch

Item Type Conference Paper

Author L. De Langhe

Author A. Maladry

Author B. Vanroy

Author L. De Bruyne

Author P. Singh

Author E. Lefever

Author O. De Clercq

Abstract The advent and popularisation of Large Language Models (LLMs) have given rise to prompt-based Natural Language Processing (NLP) techniques which eliminate the need for large manually annotated corpora and computationally expensive supervised training or fine-tuning processes. Zero-shot learning in particular presents itself as an attractive alternative to the classical train-

development-test paradigm for many downstream tasks as it provides a quick and inexpensive way of directly leveraging the implicitly encoded knowledge in LLMs. Despite the large interest in zero-shot applications within the domain of NLP as a whole, there is often no consensus on the methodology, analysis and evaluation of zero-shot pipelines. As a tentative step towards finding such a consensus, this work provides a detailed overview of available methods, resources, and caveats for zero-shot prompting within the Dutch language domain. At the same time, we present centralised zero-shot benchmark results on a large variety of Dutch NLP tasks using a series of standardised datasets. These tasks vary in subjectivity and domain, ranging from more social information extraction tasks (sentiment, emotion and irony detection for social media) to factual tasks (news topic classification and event coreference resolution). To ensure that the benchmark results are representative, we investigated a selection of zero-shot methodologies for a variety of state-of-the-art Dutch Natural Language Inference models (NLI), Masked Language models (MLM), and autoregressive language models. The output on each test set was compared to the best performance achieved using supervised methods. Our findings indicate that task-specific fine-tuning delivers superior performance in all but one (emotion detection) task. In the zero-shot settings it could be observed that large generative models through prompting seem to outperform NLI models, which in turn perform better than the MLM approach. Finally, we note several caveats and challenges tied to using zero-shot learning in application settings. These include, but are not limited to, properly streamlining evaluation of zero-shot output, parameter efficiency compared to standard finetuned models and prompt optimization. © 2024 Loic De Langhe, Aaron Maladry, Bram Vanroy, Luna De Bruyne, Pranaydeep Singh, Els Lefever & Orphée De Clercq.

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BengSentiLex and BengSwearLex: creating lexicons for sentiment analysis and profanity detection in low-resource Bengali language

Item Type Journal Article

Author S. Sazzed

Abstract Bengali is a low-resource language that lacks tools and resources for various natural language processing (NLP) tasks, such as sentiment analysis or profanity identification. In Bengali, only the translated versions of English sentiment lexicons are available. Moreover, no dictionary exists for detecting profanity in Bengali social media text. This study introduces a Bengali sentiment lexicon, BengSentiLex, and a Bengali swear lexicon, BengSwearLex. For creating BengSentiLex, a cross-lingual methodology is proposed that utilizes a machine translation system, a review corpus, two English sentiment lexicons, pointwise mutual information (PMI), and supervised machine learning (ML) classifiers in various stages. A semi-automatic methodology is presented to develop BengSwearLex that leverages an obscene corpus, word embedding, and part-of-speech (POS) taggers. The performance of BengSentiLex compared with the translated English lexicons in three evaluation datasets. BengSentiLex achieves 5%-50% improvement over the translated lexicons. For identifying profanity, BengSwearLex achieves documentlevel coverage of around 85% in an document-level in the evaluation dataset. The experimental results imply that BengSentiLex and BengSwearLex are effective resources for classifying sentiment and identifying profanity in Bengali social media content, respectively. Subjects Computational Linguistics, Data Mining and Machine Learning, Data Science, Natural Language and Speech, Network Science and Online Social Networks © 2021 Sazzed. All Rights Reserved.

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Export Date: 20 September 2026; Cited By: 13; Correspondence Address: S. Sazzed; Department of Computer Science, Old Dominion University, Norfolk, United States; email: ssazz001@odu.edu

BERT-Based Knowledge Graph Construction from Social Media

Item Type Conference Paper**Author** A. Gueddes**Author** W. Fathallah**Author** M.A. Mahjoub

Abstract Social media platforms serve as massive repositories of textual data, reflecting diverse human interactions and preferences. However, the unstructured nature of this content poses significant challenges for extracting semantically rich insights. This paper introduces a novel methodology for the automated construction of knowledge graphs (KGs) from social media discourse, specifically focusing on Twitter tweets. Our approach synergistically integrates large language models (LLMs), specifically a fine-tuned BERT model, with an ontology-driven framework. First, we define a detailed ontology of online communication concepts. The pre-trained BERT model is then fine-tuned using a multi-task learning approach on a curated dataset of anonymized and segmented Twitter discussions, thereby aligning its semantic representations with the predefined ontology. The fine-tuned LLM is leveraged for several critical tasks including entity and relation extraction, sentiment analysis, intention classification and the inference of contextual information and discussion styles. Furthermore, a mechanism is introduced to infer inter-user relationships and shared interests using graph neural networks (GNNs), analyzing patterns in interaction and language use. This multi-faceted extracted and inferred data is subsequently employed to build a knowledge graph, stored and queried via the Neo4j graph database management system. This study presents several contributions such as the integration of a ontology with an LLM method and the innovative user relationship and shared interest extraction using graph neural networks. The proposed methodology was rigorously evaluated using a real-world dataset of Twitter discussions, showcasing its ability to capture semantic content, and

elucidate inter-user relationships effectively and revealing shared interests of the users involved. Furthermore, an ablation study is included which further demonstrates each of the method component contribution and demonstrates the importance of such integrations. Our findings highlight the potential for various downstream applications such as in community structure analysis and sentiment analysis to improve information management within online social networks. © 2025 IEEE.

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BERT-Enhanced Bi-LSTM with weighted cross-entropy for multilingual sentiment classification

Item Type Journal Article

Author M.M. Siddique

Author S. Kumar

Abstract With the increasing volume of multilingual user-generated content across social media platforms, effective sentiment analysis (SA) becomes crucial, especially for low-resource languages. However, traditional models relying on context-independent embeddings, such as Word2Vec, GloVe, and fastText, struggle to handle the complexity of multilingual sentiment classification. To address this, we propose an Automatic Multilingual Sentiment Detection (AMSD) framework that leverages the contextual capabilities of BERT for feature extraction and a Bidirectional Long Short-Term Memory (Bi-LSTM) network for classification. Our method, termed Elite Opposition Cross-Entropy Weighted Bi-LSTM (EOCEWBi-LSTM), integrates elite opposition-based learning to optimize hyperparameters and enhance classification accuracy. A weighted cross-entropy loss function further refines the model's sensitivity to class imbalance, thereby improving its performance. The model is trained and evaluated on the NEP_EDUSET corpus, comprising 45,434 tweets in English, Hindi, and Tamil. Experimental results demonstrate notable improvements in precision, recall, F1-score, and accuracy, highlighting the effectiveness of EOCEWBi-LSTM in multilingual sentiment analysis, especially across both high-resource and low-resource languages. The experimental results show that the proposed EOCEWBi-LSTM achieves a high F1-score ratio of 93.83% and an accuracy ratio of 93.83% compared to other existing methods. EOCEWBi-LSTM provides an effective solution for multilingual sentiment analysis, especially for languages with limited resources. © 2025 The Author(s).

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BESSTIE: A Benchmark for Sentiment and Sarcasm Classification for Varieties of English

Item Type Conference Paper

Author D. Srirag

Author A. Joshi

Author J. Painter

Author D. Kanojia

Editor Che W.

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Abstract Despite large language models (LLMs) being known to exhibit bias against non-standard language varieties, there are no known labelled datasets for sentiment analysis of English. To address this gap, we introduce BESSTIE, a benchmark for sentiment and sarcasm classification for three varieties of English: Australian (en-AU), Indian (en-IN), and British (en-UK). We collect datasets for these language varieties using two methods: location-based for Google Places reviews, and topic-based filtering for Reddit comments. To assess whether the dataset accurately represents these varieties, we conduct two validation steps: (a) manual annotation of language varieties and (b) automatic language variety prediction. Native speakers of the language varieties manually annotate the datasets with sentiment and sarcasm labels. We perform an additional annotation exercise to validate the reliance of the annotated labels. Subsequently, we fine-tune nine large language models (LLMs) (representing a range of encoder/decoder and mono/multilingual models) on these datasets, and evaluate their performance on the two tasks. Our results show that the models consistently perform better on inner-circle varieties (i.e., en-AU and en-UK), in comparison with en-IN, particularly for sarcasm classification. We also report challenges in cross-variety generalisation, highlighting the need for language variety-specific datasets such as ours. BESSTIE promises to be a useful evaluative benchmark for future research in equitable LLMs, specifically in terms of language varieties. The BESSTIE dataset is publicly available at: <https://huggingface.co/datasets/unswnlporg/BESSTIE>. © 2025 Association for Computational Linguistics.

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Beyond Accuracy: A Mixed-Methods Audit of Chain-of-Thought Failures in LLM-Based COVID-19 Vaccine Stance Detection

Item Type Journal Article

Author A. Praschk

Author V. Fischill-Neudeck

Author T. Caspari

Author H.-P. Wiesinger

Abstract This mixed-methods study assessed whether reasoning-enabled large language models (LLMs) can classify stances towards COVID-19 vaccination on X (formerly Twitter) and whether model-generated chain-of-thought (CoT) summaries contain reasoning failures relevant to transparent and auditable public health applications. Zero-shot stance classification by o4-mini and Gemini 2.5 Flash (Gemini) was evaluated on 3,060 rehydrated COVID-19 vaccination tweets against human-annotated labels (positive, negative, neutral). We reported accuracy and macro-F1, measured CoT availability, and qualitatively analysed dual-error cases (tweets misclassified by both models) using Mayring's content analysis guided by the FUTURE-AI framework. At each model's best-performing setting, both models reached macro-F1 around 0.8, with o4-mini outperforming Gemini (accuracy 0.819 vs. 0.799, McNemar $p = 0.0015$; Δ macro-F1 = 0.020, 95% CI 0.008–0.032). Under the reasoning-intensive settings, CoT availability differed: Gemini returned a reasoning summary for all tweets, whereas o4-mini did so for 64.7%. Among 1,981 tweets with CoTs from both models, 295 (14.9%) were dual-errors; in 88.8%, both models produced the same wrong label, suggesting shared failure modes. Qualitatively, both models showed the same errors: target confusion (policy vs. vaccine), literal readings of sarcasm, and label–rationale mismatches, recurring across models despite their markedly different CoT lengths. Reasoning LLMs can therefore classify stance accurately, but their readiness for transparent public health applications depends on whether a CoT is available at all and whether it is coherent with the label it accompanies (label–rationale coherence). CoT availability, label–rationale coherence, and safeguards against systematic reasoning failures offer candidate explainability-readiness metrics, alongside accuracy, for trustworthy digital epidemiology. © The Author(s) 2026.

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Article, human, Humans, prevention and control, social media, Large language models, qualitative analysis, Social Media, Explainability, COVID-19, coronavirus disease 2019, large language model, Large Language Models, vaccination, Chain-of-thought, Stance detection, automated reasoning, chain of thought prompting, clinical audit, COVID-19 Vaccines, Mixed-methods study, reading, SARS-CoV-2 vaccine, Vaccine hesitancy, zero shot prompting

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Beyond accuracy: assessing the information quality of large language models for systematic literature reviews

Item Type Journal Article

Author H. Lai

Author Y. Li

Abstract Systematic literature reviews (SLRs) are increasingly supported by large language models (LLMs), but existing evaluations rarely go beyond accuracy and overlook broader information quality dimensions. This study provides an Information Quality (IQ)-driven evaluation of LLM support for SLR tasks in communication research, focused on diagnosing failure modes and their implications for responsible use. Using 1,325 articles on cross-platform social media research, we design a seven-step evaluation pipeline that mirrors core SLR decisions and maps them onto contextual, intrinsic, and representational IQ dimensions. Two leading LLMs, GPT-4o and Claude 3.7 Sonnet, classify or extract information about article type and empirical status (empirical vs. non-empirical), data sources (e.g., media types and platform

names), data collection methods, and the number and names of social media platforms. Validated human-coded annotations serve as ground truth, and mixed-method insights are reported from both quantitative performance analysis and qualitative error analysis. Both models achieve near-perfect accuracy for document type identification, yet their performance on tasks requiring boundary-setting judgments, methodological interpretation, or exact structured extraction is lower and more variable. Qualitative error analysis reveals that both models rely on surface-level cues rather than holistic reasoning; they apply narrow rules mechanically and fail to reconcile conflicting signals across different parts of a paper. The two models also exhibit task-specific error profiles rather than stable opposite decision biases: for example, GPT-4o is more conservative in digital trace detection but more expansive in platform-count extraction, while Claude 3.7 Sonnet shows stronger performance on grounded platform extraction. We conclude that current LLMs can reliably assist, but not replace, human judgment in SLRs. Effective deployment requires making model reasoning visible, designing prompts that account for the full range of methods in a field, and using models with complementary task-specific error profiles to flag ambiguous cases for human review. © 2026 The Author(s)

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Lai; School of Information Studies, School of Information Studies, Syracuse University, Syracuse, 343 Hinds Hall, 13244, United States; email: hlai12@syr.edu; CODEN: TEINE

Beyond fear go viral: A machine learning study on infodemic detection during covid-19 pandemic

Item Type Conference Paper

Author T. Thaipisutikul

Author T.K. Shih

Author A. Enkhbat

Author W. Aditya

Author H.-C. Shih

Author P. Mongkolwat

Abstract With the restrictions in our daily life activities under the current situation of the covid-19 pandemic worldwide, billions of people rely on social media platforms to share and obtaining covid-19 related news information. This made social media platforms easily be used as a source of myths and disinformation, which can cause severe public risks. It is thus of vital importance to constraint the spread of misinformation to the public. Although many works have shown promising results on the misinformation detection problem, only a few studies focus on the infodemic detection during the covid-19 pandemic, especially in the low resource language like Thai. Therefore, in this paper, we conduct extensive experiments on the real-world social network datasets to detect misinformation about covid-19 targeting both English and Thai languages. In particular, we perform an exploratory data analysis to get the statistic and characteristics of real and fake content. Also, we evaluate a series of three feature extraction, seven traditional machine learning, and eleven deep learning methods in detecting the fabricated content on social media platforms. The experimental results demonstrate that the transformer-based model significantly outperforms other deep learning and traditional machine learning methods in all metrics, including accuracy and F-measure. © 2022 IEEE.

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Extra Journal Abbreviation: KST - Int. Conf. Knowl. Smart Technol.

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Proceedings Title KST - Int. Conf. Knowl. Smart Technol.

Conference Name KST 2022 - 2022 14th International Conference on Knowledge and Smart Technology

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Deep learning, Feature extraction, Classification (of information), machine learning, Social networking (online), Social media, social media, deep learning, Current situation, Social media platforms, Text processing, Low resource languages, text classification, Fake detection, covid-19, Covid-19, Detection problems, Daily life activities, infodemic detection, Infodemic detection, News information

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Export Date: 20 September 2026; Cited By: 5; Conference name: 14th International Conference on Knowledge and Smart Technology, KST 2022; Conference date: 26 January 2022 through 29 January 2022; Conference code: 177713

Beyond Polarity: Unsupervised Sentiment Analysis in Turkish Social Media Texts

Item Type	Conference Paper
Author	Z. Bozkurt
Author	A. Karatas
Editor	Adali E.
Abstract	The ability to extract sentiment from social media data has become increasingly critical for understanding public opinion, tracking emotional dynamics, and informing timely decision-making across various domains. This study proposes a novel unsupervised and integrated methodology for extracting nuanced emotional and thematic insights from unstructured Turkish social media texts. Addressing the limitations of polarity-based sentiment classification, our proposed method enables multiclass sentiment detection - like joy, sadness, anger, and excitement - without relying on manually labeled data. It combines three key components: BERTurk-based contextual text embeddings, noise-resistant clustering using the HDBSCAN algorithm, and zero-shot sentiment analysis grounded in Natural Language Inference. This combination allows for the automatic identification of dominant discussion topics and their associated emotional tones across large-scale, unlabeled datasets. To ensure interpretive validity, sentiment classifications are independently reviewed by human evaluators. The findings show that the proposed method effectively captures both semantic structures and emotional nuances within the dynamic and morphologically rich landscape of Turkish social media content. This research offers a practical contribution to the fields of Turkish NLP but also offers a practical solution for monitoring public sentiment and deriving actionable insights for policy-makers, businesses, and researchers alike. © 2025 IEEE.
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DOI	10.1109/UBMK67458.2025.11207009
URL	https://www.scopus.com/pages/publications/105030844698?origin=resultslist
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Language	English
Extra	Journal Abbreviation: Int. Conf. Comput. Sci. Eng., UBMK
Publisher	Institute of Electrical and Electronics Engineers Inc.

ISBN 25211641 (ISSN)**Pages** 551-556**Proceedings Title** Int. Conf. Comput. Sci. Eng., UBMK**Conference Name** International Conference on Computer Science and Engineering, UBMK**Issue** 2025**Date Added** 20/09/2026, 8:35:19 pm**Modified** 20/09/2026, 8:35:19 pm**Tags:**

Behavioral research, Public opinions, Social aspects, Decision making, Sentiment analysis, Social networking (online), Social media, Large datasets, Semantics, BERT, Clustering algorithms, Automatic identification, Inference engines, Labeled data, Emotion classification, Sentiment classification, Topic clustering, Turkish Social Media, Turkish social medium, Turkishs, Zero-shot Sentiment Analysis, Zero-shot sentiment analyze

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Export Date: 20 September 2026; Cited By: 0; Conference name: 10th International Conference on Computer Science and Engineering, UBMK 2025; Conference date: 17 September 2025 through 21 September 2025; Conference code: 214243

Beyond text: Multimodal stance detection in Arabic tweets

Item Type Journal Article**Author** N. AlShenaifi**Author** N. Alangari

Abstract Despite the growing importance of multimodal signals on social media, Arabic stance detection has remained largely text-only, overlooking the visual context that often accompanies user posts. To bridge this gap, we present MAWQIF-MM, the first publicly available Arabic multimodal stance detection corpus of tweet-image pairs annotated with three stance labels: Favor, Against, and Neutral. Building on this resource, we propose a novel attention-based cross-modal fusion model that jointly encodes text and images. Textual content is processed using AraBERT v2, a transformer-based language model optimized for Arabic, while visual features are extracted using BLIP with a ViT-B backbone, a state-of-the-art vision-language model. These two modalities are integrated via multi-head cross-attention to capture cross-modal interactions. Experimental results demonstrate the effectiveness of our approach: on a held-out test set, the model achieves 88% accuracy, outperforming a text-only AraBERT baseline by 12 percentage points and an image-only BLIP baseline by 4 points. To further probe large vision-language models (VLMs) in low-resource settings, we benchmark Gemini 2.5 Flash and GPT-4o under zero-shot and few-shot prompting. While these models show promising generalization, they struggle with nuanced stances without fine-tuning, underscoring the value of domain-specific supervised training. Copyright © 2025. Published by Elsevier Ltd.

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Computational linguistics, Language model, Visual languages, Learning systems, Machine learning, Computer vision, Learning algorithms, Natural language processing systems, Social media, Data mining, Multi-modal, Interactive computer systems, Multi-modal learning, Multimodal learning, Stance detection, Tweets, Arabic natural language processing, Attention-based fusion, Image fusion, Signal-on, Vision-language model, Vision-language models, Visual context

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: N. AlShenaifi; King Saud University, College of Computer and Information Sciences, Department of Computer Science, Riyadh, 11543, Saudi Arabia; email: noalshenaifi@ksu.edu.sa

Bias and Cyberbullying Detection and Data Generation Using Transformer Artificial Intelligence Models and Top Large Language Models

Item Type Journal Article

Author Y. Kumar

Author K. Huang

Author A. Perez

Author G. Yang

Author J.J. Li

Author P. Morreale

Author D. Kruger

Author R. Jiang

Abstract Despite significant advancements in Artificial Intelligence (AI) and Large Language Models (LLMs), detecting and mitigating bias remains a critical challenge, particularly on social media platforms like X (formerly Twitter), to address the

prevalent cyberbullying on these platforms. This research investigates the effectiveness of leading LLMs in generating synthetic biased and cyberbullying data and evaluates the proficiency of transformer AI models in detecting bias and cyberbullying within both authentic and synthetic contexts. The study involves semantic analysis and feature engineering on a dataset of over 48,000 sentences related to cyberbullying collected from Twitter (before it became X). Utilizing state-of-the-art LLMs and AI tools such as ChatGPT-4, Pi AI, Claude 3 Opus, and Gemini-1.5, synthetic biased, cyberbullying, and neutral data were generated to deepen the understanding of bias in human-generated data. AI models including DeBERTa, Longformer, BigBird, HateBERT, MobileBERT, DistilBERT, BERT, RoBERTa, ELECTRA, and XLNet were initially trained to classify Twitter cyberbullying data and subsequently fine-tuned, optimized, and experimentally quantized. This study focuses on intersectional cyberbullying and multilabel classification to detect both bias and cyberbullying. Additionally, it proposes two prototype applications: one that detects cyberbullying using an intersectional approach and the innovative CyberBulliedBiasedBot that combines the generation and detection of biased and cyberbullying content. © 2024 by the authors.

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transformer models, cyberbullying detection, bias data generator, bias detection tokens, CyberBulliedBiasedBot, intersectional cyberbullying, multilabel text classification, synthetic cyberbullying data

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Bias in the Machine: Cross-Model Evaluation of ChatGPT-4 and Qwen in Hotel Booking Review Annotation

Item Type Conference Paper

Author M.C. Voutsas

Author N. Tsapatsoulis

Author C. Djouvas

Author C. Andreou

Abstract Large language models (LLMs) are increasingly adopted by everyday users, entrepreneurs, practitioners, and researchers to implement complex methodologies that enable knowledge acquisition and decision making. In consumer analytics, LLMs are also used as annotators of reviews, yet GPT-family models have been shown to exhibit systematic biases, particularly neutrality bias. This study benchmarks ChatGPT-4 and Qwen, an open bilingual LLM, against a human expert in hotel review data. Using Cohen's κ , we find that annotation biases are model-contingent: ChatGPT-4 defaults to neutrality, while Qwen polarizes in closer alignment with humans. Both models converge on concrete topics, but diverge on abstract ones. We introduce model-contingent polarity bias and argue that cross-model triangulation improves annotation reliability for semantic content modeling and personalization. © 2025 IEEE.

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Language English

Extra Journal Abbreviation: Proc. - Int. Workshop Semantic Soc. Media Adapt. Pers., SMAP

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Proceedings Title Proc. - Int. Workshop Semantic Soc. Media Adapt. Pers., SMAP

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M.C. Voutsas; Cyprus University of Technology, Faculty of Communication and Media Studies, Cyprus; email: maria.voutsas@cut.ac.cy; Conference name: 20th International Workshop on Semantic and Social Media Adaptation and Personalization, SMAP 2025; Conference date: 27 November 2025 through 28 November 2025; Conference code: 218734

Bias mitigation in text classification through cGAN and LLMs

Item Type Journal Article

Author G. Kumar

Author J.P. Singh

Abstract Investigating bias, fairness, and ethics in AI/ML models is essential in the digital era, as automated systems are increasingly used to enhance people's daily lives. Evaluating these systems in terms of fairness has become crucial. Our proposed work focuses on identifying and mitigating gender bias in text classification models. Specifically, we are evaluating a hate speech classification (HSC) model using fairness metrics and mitigating bias through pre- and in-processing debiasing techniques. In the pre-debiasing stage, we address bias by applying gender swap techniques and removing gender-related terms from the dataset. For in-processing debiasing, we use conditional generative adversarial networks (cGAN) and large language models (LLMs) to generate text similar to that of minority classes, helping balance the dataset and reduce bias during model training. We assess the impact of both methods on bias reduction in the HSC model, with each targeting bias at different stages of model development. While the classification performance of the HSC model showed a slight decrease, fairness evaluation metrics improved by 3.03% and 3.42%, demonstrating the methods' effectiveness. Addressing gender bias in HSC models applied to social media posts is vital for fostering a healthier digital society. © Indian National Science Academy 2024.

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BLP-2023 Task 2: Sentiment Analysis

Item Type Conference Paper

Author Md.A. Hasan

Author F. Alam

Author A. Anjum

Author S. Das

Author A. Anjum

Editor Sadeque F.

Editor Amin R.

Editor Kar S.

Editor Chowdhury S.A.

Editor Alam F.

Abstract We present an overview of the BLP Sentiment Shared Task, organized as part of the inaugural BLP 2023 workshop, co-located with EMNLP 2023. The task is defined as the detection of sentiment in a given piece of social media text. This task attracted interest from 71 participants, among whom 29 and 30 teams submitted systems during the development and evaluation phases, respectively. In total, participants submitted 597 runs. However, a total of 15 teams submitted system description papers. The range of approaches in the submitted systems spans from classical machine learning models, fine-tuning pre-trained models, to leveraging Large Language Model (LLMs) in zero- and few-shot settings. In this paper, we provide a detailed account of the task setup, including dataset development and evaluation setup. Additionally, we provide a brief overview of the systems submitted by the participants. All datasets and evaluation scripts from the shared task have been made publicly available for the research community, to foster further research in this domain. © 2023 Association for Computational Linguistics.

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Language English

Extra Journal Abbreviation: BLP - Workshop Bangla Lang. Process., Proc. Workshop

Publisher Association for Computational Linguistics (ACL)

ISBN 979-889176058-5 (ISBN)**Pages** 72-84**Proceedings Title** BLP - Workshop Bangla Lang. Process., Proc. Workshop**Conference Name** BLP 2023 - 1st Workshop on Bangla Language Processing, Proceedings of the Workshop**Date Added** 20/09/2026, 8:35:31 pm**Modified** 20/09/2026, 8:35:31 pm**Tags:**

Computational linguistics, Language model, Sentiment analysis, Social media, Fine tuning, Machine learning models, Research communities, Co-located, Development phasis, Evaluation phase, System description

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Blunt Giants, Sharp Specialists: Decoding Stress and Triggers Beyond General-Purpose LLMs

Item Type Journal Article**Author** S. Ghosh**Author** G.V. Singh**Author** K. Pathak**Author** A. Ekbal

Abstract The United Nations sustainable development goals emphasize mental health as essential for inclusive and sustainable development, making stress, an important factor affecting well-being and societal participation, a critical area of study. Leveraging social media data, we propose the emotion-aware stress cause analysis framework (ESCAF), which jointly models stress detection (SD) and stress cause identification (SCI). ESCAF employs a contextual attention block (CAB) to prioritize stress-relevant context and incorporates sentence-level emotion information for enriched contextual understanding. A key contribution is the stress and cause recognition dataset, derived from the Dreaddit corpus, providing utterance-level annotations for emotion and stress-triggering sentences within each post. ESCAF surpasses in-domain state-of-the-art methods with average F1 gains of +5.23% for SD and +6.57% for SCI. For comprehensive evaluation, we also assess fine-tuned and zero-shot performances of leading large language models (LLMs); ESCAF outperforms larger models including Llama, Mistral, Qwen, and Phi, while maintaining a smaller parameter footprint. These findings highlight the need for task-specific architectures requiring nuanced emotional understanding and caution against over-reliance on generic LLMs in sensitive applications. © 2014 IEEE.

Date 2026

DOI 10.1109/TCSS.2026.3686713**URL** <https://www.scopus.com/pages/publications/105040922685?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Institute of Electrical and Electronics Engineers Inc.**Publication** IEEE Transactions on Computational Social Systems**ISSN** 2329924X (ISSN)**Journal Abbr** IEEE Trans. Computat. Soc. Syst.**Date Added** 20/09/2026, 8:35:01 pm**Modified** 20/09/2026, 8:35:01 pm**Tags:**

Computational linguistics, Language model, Large language model, Behavioral research, Mental health, Learning systems, Sustainable development, Affective Computing, Emotion Recognition, Modeling languages, Affective computing, computational mental health, Computational mental health, emotion-aware modeling, Emotion-aware modeling, large language models (LLMs), multitask learning, Multitask learning, Psychology computing, stress cause, Stress cause, Stress detection, stress detection (SD), Stress trigger, stress triggers, Stresses, Sustainable development goal, Sustainable development goals, sustainable development goals (SDGs)

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Ghosh; Fondazione Bruno Kessler (FBK), Trento, 38123, Italy; email: ghosh.soumitra2@gmail.com

BMT-Net: Broad Multitask Transformer Network for Sentiment Analysis

Item Type Journal Article**Author** T. Zhang**Author** X. Gong**Author** C.L.P. Chen

Abstract Sentiment analysis uses a series of automated cognitive methods to determine the author's or speaker's attitudes toward an expressed object or text's overall emotional tendencies. In recent years, the growing scale of opinionated text from social networks has brought significant challenges to humans' sentimental tendency mining. The pretrained language model designed to learn contextual representation achieves better performance than traditional learning word vectors. However, the existing two basic approaches for applying pretrained language models to downstream tasks, feature-based and fine-tuning methods, are usually considered separately. What is more, different sentiment analysis tasks cannot be handled by the single task-specific contextual representation. In light of these pros and cons, we strive to propose a broad multitask transformer network (BMT-Net) to address these problems. BMT-Net takes

advantage of both feature-based and fine-tuning methods. It was designed to explore the high-level information of robust and contextual representation. Primarily, our proposed structure can make the learned representations universal across tasks via multitask transformers. In addition, BMT-Net can roundly learn the robust contextual representation utilized by the broad learning system due to its powerful capacity to search for suitable features in deep and broad ways. The experiments were conducted on two popular datasets of binary Stanford Sentiment Treebank (SST-2) and SemEval Sentiment Analysis in Twitter (Twitter). Compared with other state-of-the-art methods, the improved representation with both deep and broad ways is shown to achieve a better $F1$ -score of 0.778 in Twitter and accuracy of 94.0% in the SST-2 dataset, respectively. These experimental results demonstrate the abilities of recognition in sentiment analysis and highlight the significance of previously overlooked design decisions about searching contextual features in deep and broad spaces. © 2013 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 127; Correspondence Address: C.L.P. Chen; South China University of Technology, School of Computer Science and Engineering, Guangzhou, 510006, China; email: philip.chen@ieee.org

Borrowing Human Senses: Comment-Aware Self-Training for Social Media Multimodal Classification

Item Type Conference Paper

Author C. Xu

Author J. Li

Editor Goldberg Y.

Editor Kozareva Z.

Editor Zhang Y.

Abstract Social media is daily creating massive multimedia content with paired image and text, presenting the pressing need to automate the vision and language understanding for various multimodal classification tasks. Compared to the commonly researched visual-lingual data, social media posts tend to exhibit more implicit image-text relations. To better glue the cross-modal semantics therein, we capture hinting features from user comments, which are retrieved via jointly leveraging visual and lingual similarity. Afterwards, the classification tasks are explored via self-training in a teacher-student framework, motivated by the usually limited labeled data scales in existing benchmarks. Substantial experiments are conducted on four multimodal social media benchmarks for image-text relation classification, sarcasm detection, sentiment classification, and hate speech detection. The results show that our method further advances the performance of previous state-of-the-art models, which do not employ comment modeling or self-training. © 2022 Association for Computational Linguistics.

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Conference Name Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing, EMNLP 2022

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Computational linguistics, Classification (of information), Social networking (online), Social media, Semantics, Multi-modal, Text processing, Personnel training, Classification tasks, Self-training, Language understanding, Image texts, Human sense, Multimedia contents, Pressung, Vision understanding

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Brand-Level Sentiment Analysis in the Telecom Domain Using Large Language Models

Item Type Conference Paper

Author B. Kaya

Author A.B. Kanburoglu

Abstract In recent years, sentiment analysis has become a critical tool for understanding public perception of brands, particularly in dynamic industries such as telecommunications. This study explores brand-level sentiment analysis in the telecom domain using both open-source and closed-source large language models (LLMs). We constructed a domain-specific dataset by collecting user-generated content from social media platforms, focusing on telecom-related discussions. To ensure real-time applicability, we prioritized small-scale models in the closedsource category due to their lower inference latency. We finetuned the open-source models on the telecom dataset, resulting in significantly improved performance compared to their closedsource counterparts. Among the fine-tuned models, Gemma3-1B achieved the highest F1-score, reaching 87.67 %, demonstrating its effectiveness in capturing sentiment in telecom-specific contexts. Our findings highlight the potential of domain-adapted open-source LLMs for efficient and accurate sentiment analysis in industry-specific applications. © 2025 IEEE.

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Bridging Lexical and Contextual Semantics for Adolescents' Health Sentiment Analysis: A Cross-Platform Study Using Large Language Models

Item Type Conference Paper

Author A. Shifa

Author G.G. Md. Nawaz Ali

Author D. Henricks

Abstract To understand, how adolescents emotionally engage with online health-related content, we require both contextual reasoning and semantic interpretation. This study presents a hybrid semantic framework that bridges classic and modern AI techniques for sentiment analysis of obesity-related discourse on social media platforms like TikTok and Instagram. The approach integrates lexicon-based polarity estimation with large language model's (e.g., GPT-4) contextual refinement to enhance the derivation and integration of affective semantics from user-generated text. The dataset comprises of approximately 21,000 public comments related to adolescence obesity, healthy diet etc. categorized across five different content source types, including, healthcare professionals, fitness coaches, influencers, food and beverage creators, and others. Comparative results show that Instagram content exhibits cohesive, positive sentiment, while TikTok demonstrates greater emotional polarization and variability. By combining symbolic sentiment modeling with generative AI reasoning, this work contributes to semantic computing for health communication and highlights pathways for bridging lexical semantics and contextual intelligence in real-world social media analysis. The results also underscore the value of integrating large language models in health sentiment research and highlight the need for platform-sensitive, credibility-aware communication strategies when addressing adolescent audiences online. © 2026 IEEE.

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Language English

Extra Journal Abbreviation: Proc. - IEEE Int. Conf. Semant. Comput., ICSC

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Pages 360-365**Proceedings Title** Proc. - IEEE Int. Conf. Semant. Comput., ICSC**Conference Name** Proceedings - IEEE International Conference on Semantic Computing, ICSC**Date Added** 20/09/2026, 8:35:07 pm**Modified** 20/09/2026, 8:35:07 pm**Tags:**

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Notes:

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Bridging the Maghrebi Dialect Gap: Multi-Dialect Named Entity Recognition for North African Arabic Using Dialect-Aware Active Learning and Cross-Dialectal Transfer

Item Type Conference Paper**Author** H. Mahdhaoui**Author** A. Mars**Author** M.J. Hazar**Author** A.A.M. Abid Ali**Author** S. Zrigui**Author** M. Zrigui**Editor** Rocha A.P.**Editor** Wahde M.**Editor** van den Herik H.J.

Abstract Named Entity Recognition (NER) for Maghrebi Arabic dialects presents formidable challenges due to substantial linguistic variations across North African varieties (Tunisian, Moroccan, Algerian, Libyan), pervasive French-Arabic code-switching phenomena, Berber substrate influences, absence of orthographic standardization, and severe scarcity of annotated dialectal corpora.

While Modern Standard Arabic (MSA) NER has achieved performance exceeding 90% F1-score, these systems demonstrate catastrophic degradation when applied to dialectal text, with F1-scores dropping to 55-65%. In this work, we present a comprehensive multi-dialect NER framework specifically architected for Maghrebi Arabic varieties. Our contributions comprise: (1) a Maghrebi dialect detection module achieving 89.7% classification accuracy, (2) a novel neural architecture with Common Maghrebi Features Layer handling French/Berber code-switching, (3) dialect-aware active learning reducing annotation costs by 43%, (4) effective cross-dialectal transfer learning (13.4% improvement for low-resource Libyan), and (5) integration of large language models for auto-correction. We construct and release the first large-scale multi-dialect Maghrebi NER benchmark comprising 330,000 annotated sentences. Our model achieves macro-averaged F1-score of 87.34% across all Maghrebi dialects, substantially outperforming MSA-only baselines (61.2% F1), standard multi-dialect approaches (80.9% F1), and existing dialectal Arabic NER systems. This work establishes the first comprehensive foundation for Maghrebi Arabic NER with direct applications to social media analytics, government digital services, and business intelligence. © 2026, Science and Technology Publications, Lda. All rights reserved.

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Language English

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Volume 2

Publisher Science and Technology Publications, Lda

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Bridging the Socioeconomic Gap in Education: A Hybrid AI and Human Annotation Approach

Item Type Conference Paper

Author N. Abdelgaber

Author L. Jahan

Author A.V. Doshi

Author R. Suri

Author H.R. Pavel

Author J. Zhang

Editor Boleda G.

Editor Roth M.

Abstract Students' academic performance is influenced by various demographic factors, with socioeconomic class being a prominently researched and debated factor. Computer Science research traditionally prioritizes computationally definable problems, yet challenges such as the scarcity of high-quality labeled data and ethical concerns surrounding the mining of personal information can pose barriers to exploring topics like the impact of socioeconomic status (SES) on students' education. Overcoming these barriers may involve automating the collection and annotation of high-quality language data from diverse social groups through human collaboration. Therefore, our focus is on gathering unstructured narratives from Internet forums written by students with low-SES using machine learning models and human insights. We developed a hybrid data collection model that semi-Automatically retrieved narratives from the Reddit website and created a dataset five times larger than the seed dataset. Additionally, we compared the performance of traditional ML models with recent large language models (LLMs) in classifying narratives written by low-SES students, and analyzed the collected data to extract valuable insights into the socioeconomic challenges these students encounter and the solutions they pursue. © 2025 Association for Computational Linguistics.

Date 2025

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Language English

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Tags:

Computational linguistics, Learning systems, Demographic factors, Data collection, Ethical concerns, Economics, Data concerns, Students, High quality, Labeled data, Human annotations, Education computing, Academic performance, Computer science research, Socio-economic status, Socio-economics

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: N. Abdelgaber; Southern Methodist University, Dallas, 75205, United States; email: nabdelgaber@smu.edu; Conference name: 29th Conference on Computational Natural Language Learning, CoNLL 2025; Conference date: 31 July 2025 through 1 August 2025; Conference code: 223474

BRoverbs-Measuring how much LLMs understand Portuguese proverbs

Item Type Journal Article

Author T.S. Almeida

Author G.K. Bonás

Author J.G.A. Santos

Abstract Large Language Models (LLMs) exhibit significant performance variations depending on the linguistic and cultural context in which they are applied. This disparity signals the necessity of mature evaluation frameworks that can assess their capabilities in specific regional settings. In the case of Portuguese, existing evaluations remain limited, often relying on translated datasets that may not fully capture linguistic nuances or cultural references. Meanwhile, native Portuguese-language datasets predominantly focus on structured national exams or sentiment analysis of social media interactions, leaving gaps in evaluating broader linguistic understanding. To address this limitation, we introduce BRoverbs, a dataset specifically designed to assess LLM performance through Brazilian proverbs. Proverbs serve as a rich linguistic resource, encapsulating cultural wisdom, figurative expressions, and complex syntactic structures that challenge the model comprehension of regional expressions. BRoverbs aims to provide a new evaluation tool for Portuguese-language LLMs, contributing to advancing regionally informed benchmarking. The benchmark is available at <https://huggingface.co/datasets/Tropic-AI/BRoverbs>. © 2025, Brazilian Computing Society. All rights reserved.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T.S. Almeida; Institute of Computing, University of Campinas, Campinas, Av. Albert Einstein, 1251, SP, 13083-852, Brazil; email: t224732@dac.unicamp.br

Building lexicon-based sentiment analysis model for low-resource languages

Item Type Journal Article**Author** I. Mohammed**Author** R. Prasad

Abstract Natural Language Processing (NLP) has transformed machine translation, sentiment analysis, information retrieval, and conversation systems. NLP applications rely on complete linguistic resources, which might be difficult for low-resource languages. NLP solutions for every language require a language-specific dataset. Dataset in a language is essential for NLP solution creation. Over 7000 languages are spoken worldwide. Only around 20 languages have text corpora for NLP applications. English has the most datasets, then Chinese and Spanish. Japanese has several Western European language datasets. For an accurate NLP system, most Asian and African languages lack training datasets. To address this challenge, we propose a methodology for building a lexicon-based sentiment analysis model for languages with limited resources. The Hausa language was used as training and evaluation language. The methodology combines lexicon creation; augmentation, annotation, and fine-tuning model, and has been tested on a corpus of Hausa tweets achieving an accuracy of 98 %. The results suggest that our proposed model is a promising tool for sentiment analysis in a variety of applications, such as social media monitoring, customer service, and market research. Our methodology can be used for any low-resource language. The outline of the work done in this paper can be shown as follows: • We propose a methodology for building a lexicon-based sentiment analysis model for languages with limited resources, using the Hausa language as a case study. • The methodology combines lexicon creation, augmentation, annotation, and fine-tuning model, and achieves an accuracy of 98 % on a corpus of Hausa tweets. • The results suggest that the proposed model is a promising tool for sentiment analysis in a variety of applications for low-resource languages. © 2023 The Author(s)

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human, human experiment, Sentiment analysis, marketing, article, natural language processing, sentiment analysis, Fine-tuning, Low-resource languages, social media monitoring, Hausa language, information retrieval, conversation, Lexicon dictionary, Western European

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Export Date: 20 September 2026; Cited By: 15; Correspondence Address: I. Mohammed; Computer Science Department, African University of Science and Technology, Abuja, Nigeria; email: mohammed@aust.edu.ng

C-MTCSD: A Chinese Multi-Turn Conversational Stance Detection Dataset

Item Type Conference Paper

Author F. Niu

Author Y. Yang

Author X. Fu

Author G. Dai

Author B. Zhang

Abstract Stance detection has become an essential tool for analyzing public discussions on social media. Current methods face significant challenges, particularly in Chinese language processing and multi-turn conversational analysis. To address these limitations, we introduce C-MTCSD, the largest Chinese multi-turn conversational stance detection dataset, comprising 24, 264 carefully annotated instances from Sina Weibo, which is 4.2 times larger than the only prior Chinese conversational stance detection dataset. Our comprehensive evaluation using both traditional approaches and large language models reveals the complexity of C-MTCSD: even state-of-the-art models achieve only 64.07% F1 score in the challenging zero-shot setting, while performance consistently degrades with increasing conversation depth. Traditional models particularly struggle with

implicit stance detection, achieving below 50% F1 score. This work establishes a challenging new benchmark for Chinese stance detection research, highlighting significant opportunities for future improvements. © 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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C&D SENTINEL: AN ENTERPRISE-GRADE MULTIMODAL CONTENT MODERATION SYSTEM USING HYBRID TWO-STAGE NLP AND COMPUTER VISION

Item Type Journal Article

Author K. Shailaja

Author T. Jalaja

Author M. Chaitanya

Author V. Darpad Sai

Abstract The rapid increase of user-generated content through digital means has resulted in an urgent requirement for scalable, accurate and relevant content moderation systems. The C&D Sentinel content moderation pipeline solves this issue with an enterprise-class hybrid architecture consisting of locally deployed machine learning models and cloud-based large language model (LLM) reasoning. The system identifies toxic, hateful and unsafe content across a range of formats, including text, still images, and video, while providing low latency. Traditional moderation systems rely primarily on manual keyword filtering or opaque "black box" classifiers that cannot interpret contextual differences between words; these methods often mislabel harmless items – such as satire, slang, and artistic forms – as dangerous, resulting in numerous false positive instances. This paper demonstrates that using a Hybrid Two-Stage Architecture combining light weight local models and advanced reasoning capabilities provided by cutting-edge LLMs will resolve these limitations. DistilBERT, an example of such an efficient model, can perform quick text classification, while BLIP (Bootstrapping Language-Image Pretraining) will create captions to image content effectively converting visual data into an appropriate textual form for semantic analysis. The Llama 3 70B model is utilized within the system to provide fast inference with low latency with integration onto the Groq LPU (host machine) allowing for improved detection of malicious behaviour and amending false positives compared to previous methodologies for detection of malicious behaviour in context. One of the implementations of this system is its video moderation, which is processed in a dual-stream framework where images are captured using OpenCV for visual features and where audio is converted to text using Whisper. These two streams are analysed together to allow a greater understanding of what the video is and therefore whether or not the video contains any HA. A second component of the system is an RLHF (reinforcement learning from human feedback) feedback loop, which allows for continual improvement of the content detection capabilities through a human-in-the-loop process. RLHF will allow for continual enhancement of detecting new types of language and harmful content as they emerge. The system was evaluated against the Facebook Hateful Memes Challenge data set, where it achieved 90% accuracy, far better than a zero-shot baseline using CLIP that achieved only 14% accuracy. The system is also deployed to Groq LPU hardware which ensures that the system can perform real-time analyse with less than 200 milliseconds of inference latency, making the system well-suited for enterprise content moderation. Overall, the C&D Sentinel pipeline has demonstrated a significant improvement in terms of accuracy, lower false positive rates and real-time performance capabilities to develop a robust and scalable platform for next-generation content moderation. © Little Lion Scientific

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C3NLP 2024 - 2nd Workshop on Cross-Cultural Considerations in NLP, Proceedings of the Workshop

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Editor	Prabhakaran V.
Editor	Dev S.
Editor	Benotti L.
Editor	Hershcovich D.
Editor	Cabello L.
Editor	Cao Y.
Editor	Adebara I.
Editor	Zho L.
Abstract	The proceedings contain 9 papers. The topics discussed include: CDEval: a benchmark for measuring the cultural dimensions of large language models; conformity, confabulation, and impersonation: persona inconstancy in multi-agent LLM collaboration; synchronizing approach in designing annotation guidelines for multilingual datasets: a COVID-19 case study using English and Japanese tweets; CRAFT: extracting and tuning cultural instructions from the wild; does cross-cultural alignment change the commonsense morality of language models?; do multilingual large language models mitigate stereotype bias?; sociocultural considerations in monitoring anti-LGBTQ+ content on social media; are generative language models multicultural? a study on Hausa culture and emotions using ChatGPT; and computational language documentation: designing a modular annotation and data management tool for cross-cultural applicability.
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URL	https://www.scopus.com/pages/publications/85204359482?origin=resultslist
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Language	English
Publisher	Association for Computational Linguistics (ACL)
Publication	C3NLP 2024 - 2nd Workshop on Cross-Cultural Considerations in NLP, Proceedings of the Workshop
ISSN	979-889176146-9 (ISBN)

Journal Abbr C3NLP - Workshop Cross-Cult.Considerations NLP, Proc. Workshop

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Cache-Augmented Multimodal Sentiment Detection Using Embedding Systems

Item Type Conference Paper

Author V. Srilakshmi

Author G. Uday Kiran

Author S. Sai Sreeja

Author V. Varun Reddy

Author V. Karthik

Author P. Sridar Chowdary

Abstract The exponential growth of multimodal content across digital platforms has exposed the limitations of traditional sentiment analysis systems that rely solely on text-based methods. This work presents a cache-augmented multimodal sentiment detection framework that integrates DistilBERT text embeddings (768D) and CLIP image embeddings (512D) through FAISS similarity search and LRU (Least Recently Used) caching mechanisms. The system addresses computational redundancy in real-time applications by implementing a retrieval-first pipeline that reuses previous computations before fallback MLP (Multi-Layer Perceptron) classification. Evaluated on MVSA-Single dataset (374 image-text pairs) and Twitter/Reddit corpus (600 text samples), the approach achieves 81.3% accuracy and 81.1% macro-F1 score. Key contributions include: (1) a FAISS-based retrieval pipeline reducing inference latency by 60% for repeated inputs, (2) adaptive LRU cache mechanism with 1000-entry capacity for computational efficiency, and (3) late fusion architecture combining normalized text and image embeddings for robust multimodal sentiment classification. © 2025 IEEE.

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Language English

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Conference Name 8th World Engineering Conference on Contemporary Technologies, WECON 2025: NextGen Sustainable Technologies

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Language model, Large language model, Classification (of information), Sentiment analysis, Computational efficiency, Data mining, Multi-modal, Information retrieval, Embeddings, Interactive computer systems, Search engines, Large Language Models, Multimodal sentiment analyze, Multimodal Sentiment Analysis, Cache mechanism, Cache Mechanism, CLIP embedding, CLIP embeddings, Digital image storage, FAISS indexing, FAISS Indexing, Real-time inference, Real-time Inference

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: V. Srilakshmi; B. V. Raju Institute of Technology, Department of CSE (Artificial Intelligence and Machine Learning), Narsapur, Telangana, India; email: srilakshmi.v@bvrit.ac.in; Conference name: 8th World Engineering Conference on Contemporary Technologies, WECON 2025; Conference date: 10 October 2025 through 11 October 2025; Conference code: 221142

CALD-pseudo 2024 - Workshop on Computational Approaches to Language Data Pseudonymization, Proceedings of the Workshop

Item Type Journal Article

Editor Volodina E.

Editor Alfter D.

Editor Dobnik S.

Editor Tiedemann T.L.

Editor Sanchez R.M.

Editor Szawerna M.I.

Editor Vu X.-S.

Abstract The proceedings contain 10 papers. The topics discussed include: handling name errors of a BERT-based de-identification system: insights from stratified sampling and Markov-based pseudonymization; assessing authenticity and anonymity of synthetic user-generated content in the medical domain; automatic detection and labelling of personal data in case reports from the ECHR in Spanish: evaluation of two different annotation approaches; PSILENCE: a pseudonymization tool for international law; deidentifying a Norwegian clinical corpus - an effort to create a privacy-preserving Norwegian large clinical language model; extending off-the-shelf NER systems to personal information detection in dialogues with a virtual agent: findings from a real-life use case; detecting personal identifiable information in Swedish learner essays; data anonymization for privacy-preserving large language

model fine-tuning on call transcripts; when is a name sensitive? eponyms in clinical text and implications for de-identification; when is a name sensitive? eponyms in clinical text and implications for de-identification; and did the names I used within my essay affect my score? diagnosing name biases in automated essay scoring.

Date 2024

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Language English

Publisher Association for Computational Linguistics (ACL)

Publication CALD-pseudo 2024 - Workshop on Computational Approaches to Language Data Pseudonymization, Proceedings of the Workshop

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Can AI writing be salvaged? Mitigating Idiosyncrasies and Improving Human-AI Alignment in the Writing Process through Edits

Item Type Conference Paper

Author T. Chakrabarty

Author P. Laban

Author C.-S. Wu

Abstract LLM-based applications are helping people write, and LLM-generated text is making its way into social media, journalism, and our classrooms. However, the differences between LLM-generated and human-written text remain unclear. To explore this, we hired professional writers to edit paragraphs in several creative domains. We first found these writers agree on undesirable idiosyncrasies in LLM-generated text, formalizing it into a seven-category taxonomy (e.g. clichés, unnecessary exposition). Second, we curated the LAMP corpus: 1,057 LLM-generated paragraphs edited by professional writers according to our taxonomy. Analysis of LAMP reveals that none of the LLMs used in our study (GPT4o, Claude-3.5-Sonnet, Llama-3.1-70b) outperform each other in terms of writing quality, revealing common limitations across model families. Third, building on existing work in automatic editing we evaluated methods to improve LLM-generated text. A large-scale preference annotation confirms that although experts largely prefer text edited by other experts, automatic editing methods show promise in improving alignment between LLM-generated and human-written text. © 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Can GPT-4 Identify Propaganda? Annotation and Detection of Propaganda Spans in News Articles

Item Type Conference Paper
Author M. Hasanain
Author F. Ahmed
Author F. Alam
Editor Calzolari N.
Editor Kan M.-Y.
Editor Hoste V.
Editor Lenci A.
Editor Sakti S.
Editor Xue N.
Abstract The use of propaganda has spiked on mainstream and social media, aiming to manipulate or mislead users. While efforts to automatically detect propaganda techniques in textual, visual, or multimodal content have increased, most of them

primarily focus on English content. The majority of the recent initiatives targeting medium to low-resource languages produced relatively small annotated datasets, with a skewed distribution, posing challenges for the development of sophisticated propaganda detection models. To address this challenge, we carefully develop the largest propaganda dataset to date, ArPro, comprised of 8K paragraphs from newspaper articles, labeled at the text span level following a taxonomy of 23 propagandistic techniques. Furthermore, our work offers the first attempt to understand the performance of large language models (LLMs), using GPT-4, for fine-grained propaganda detection from text. Results showed that GPT-4's performance degrades as the task moves from simply classifying a paragraph as propagandistic or not, to the fine-grained task of detecting propaganda techniques and their manifestation in text. Compared to models fine-tuned on the dataset for propaganda detection at different classification granularities, GPT-4 is still far behind. Finally, we evaluate GPT-4 on a dataset consisting of six other languages for span detection, and results suggest that the model struggles with the task across languages. We made the dataset publicly available for the community. © 2024 ELRA Language Resource Association: CC BY-NC 4.0.

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Can Large Language Models Grasp Abstract Visual Concepts in Videos? A Case Study on YouTube Shorts about Depression

Item Type Conference Paper

Author J. Liu

Author Y. Su

Author P. Seth

Abstract Large language models (LLMs) are increasingly used to assist computational social science research. While prior efforts have focused on text, the potential of leveraging multimodal LLMs (MLLMs) for online video studies remains underexplored. We conduct one of the first case studies on MLLM-assisted video content analysis, comparing AI's interpretations to human understanding of abstract concepts. We leverage LLaVA-1.6 Mistral 7B to interpret four abstract concepts regarding video-mediated self-disclosure, analyzing 725 keyframes from 142 depression-related YouTube short videos. We perform a qualitative analysis of MLLM's self-generated explanations and found that the degree of operationalization can influence MLLM's interpretations. Interestingly, greater detail does not necessarily increase human-AI alignment. We also identify other factors affecting AI alignment with human understanding, such as concept complexity and versatility of video genres. Our exploratory study highlights the need to customize prompts for specific concepts and calls for researchers to incorporate more human-centered evaluations when working with AI systems in a multimodal context. © 2025 Copyright held by the owner/author(s).

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Proceedings Title Conf Hum Fact Comput Syst Proc

Conference Name Conference on Human Factors in Computing Systems - Proceedings

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Export Date: 20 September 2026; Cited By: 3; Funding details: University of Texas at Austin; National Science Foundation, NSF, (IIS2107524); National Science Foundation, NSF; Correspondence Address: J. Liu; School of Information, University of Texas at Austin, Austin, United States; email: jiaingliu@utexas.edu; Conference name: 2025 CHI Conference on Human Factors in Computing Systems, CHI EA 2025; Conference date: 26 April 2025 through 1 May 2025; Conference code: 208611

Can LLMs assist humans in assessing online misogyny? Experiments with GPT-3.5

Item Type	Conference Paper
Author	C. Morbidoni
Author	A. Sarra
Editor	Torrielli F.
Editor	Department of Computer Science, Corso Svizzera 185, Torino University of Torino
Editor	Di Caro L.
Editor	Department of Computer Science, Corso Svizzera 185, Torino University of Torino
Editor	Rapp A.
Editor	Department of Computer Science, Corso Svizzera 185, Torino University of Torino
Abstract	Today's social media landscape is flooded with unfiltered content, which can range from hate speech to cyberbullying and cyberstalking. As a result, locating and eliminating such toxic language presents a significant challenge and is an active current research area. In this paper we focus on detecting hate speech against women, i.e. misogyny, exploiting a “prompt-based learning” paradigm with the aim of providing a first assessment of recent developed LLM (OpenAI's GPT-3.5-turbo). We experiment with a benchmark dataset of Reddit posts and evaluate different prompts types w.r.t. response stability, classification accuracy and inter-annotator agreement. Our experiments show that zero-shot detection GPT capabilities - against human annotations - outperform supervised baselines on our evaluation dataset and that ensembling different prompts possibly further improve the accuracy up to 91%. We also found that responses to specific prompts is quite stable, while slightly more variation and less agreement is observed when asking the questions in different ways. © 2022 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).
Date	2023
URL	https://www.scopus.com/pages/publications/85180126587?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: CEUR Workshop Proc.
Volume	3571
Publisher	CEUR-WS

ISBN 16130073 (ISSN)**Pages** 31-43**Proceedings Title** CEUR Workshop Proc.**Conference Name** CEUR Workshop Proceedings**Date Added** 20/09/2026, 8:35:20 pm**Modified** 20/09/2026, 8:35:20 pm**Tags:**

Language model, Classification (of information), Social media, GPT, E-learning, Text classification, Text processing, prompt-based learning, Prompt-based learning, text classification, Zero-shot learning, pre-trained language model, Pre-trained language model, Cyber bullying, Active current, Cyberstalking, online misogyny detection, Online misogyny detection

Notes:

EID: 2-s2.0-85180126587

Export Date: 20 September 2026; Cited By: 6; Correspondence Address: C. Morbidoni; Università degli Studi G. d'Annunzio, Pescara, Italy; email: christian.morbidoni@unich.it; Conference name: 2023 Workshop on GENerative, Explainable and Reasonable Artificial Learning, GENERAL 2023; Conference date: 20 September 2023 through 22 September 2023; Conference code: 194815

Can social media provide early warning of retraction? Evidence from critical tweets identified by human annotation and large language models

Item Type Journal Article**Author** E.-T. Zheng**Author** H.-Z. Fu**Author** M. Thelwall**Author** Z. Fang

Abstract Timely detection of problematic research is essential for safeguarding scientific integrity. To explore whether social media commentary can serve as an early indicator of potentially problematic articles, this study analyzed 3815 tweets referencing 604 retracted articles and 3373 tweets referencing 668 comparable non-retracted articles. Tweets critical of the articles were identified through both human annotation and large language models (LLMs). Human annotation revealed that 8.3% of retracted articles were associated with at least one critical tweet prior to retraction, compared to only 1.5% of non-retracted articles, highlighting the potential of tweets as early warning signals of retraction. However, critical tweets identified by LLMs (GPT-4o mini, Gemini 2.0 Flash-Lite, and Claude 3.5 Haiku) only partially aligned with human annotation, suggesting that fully automated monitoring of post-publication discourse should be applied with caution. A human–AI collaborative approach may offer a more reliable and scalable alternative, with human expertise helping to filter out tweets critical of issues unrelated to the research integrity of the articles. Overall, this study provides insights into how social media signals, combined with generative AI technologies, may support efforts to strengthen research integrity. © 2025 The

Author(s). Journal of the Association for Information Science and Technology published by Wiley Periodicals LLC on behalf of Association for Information Science and Technology.

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Can You Label Less by Using Out-of-Domain Data? Active & Transfer Learning with Few-shot Instructions

Item Type Conference Paper

Author R. Kocielnik

Author S. Kangaslahti

Author S. Prabhumoye

Author M. Hari

Author R.M. Alvarez

Author A. Anandkumar

Editor Albalak A.

Editor Zhou C.

Editor Raffel C.

Editor Ramachandran D.

Editor Ruder S.

Editor Ma X.

Abstract Labeling social-media data for custom dimensions of toxicity and social bias is challenging and labor-intensive. Existing transfer and active learning approaches meant to reduce annotation effort require fine-tuning, which suffers from overfitting to noise and can cause domain shift with small sample sizes. In this work, we propose a novel Active Transfer Few-shot Instructions (ATF) approach which requires no fine-tuning. ATF leverages the internal linguistic knowledge of pretrained language models (PLMs) to facilitate the transfer of information from existing pre-labeled datasets (source-domain task) with minimum labeling effort on unlabeled target data (target-domain task). Our strategy can yield positive transfer achieving a mean AUC gain of 10.5% compared to no transfer with a large 22b parameter PLM. We further show that annotation of just a few target-domain samples via active learning can be beneficial for transfer, but the impact diminishes with more annotation effort (26% drop in gain between 100 and 2000 annotated examples). Finally, we find that not all transfer scenarios yield a positive gain, which seems related to the PLMs initial performance on the target-domain task. © 2023 Proceedings of Machine Learning Research. All rights reserved.

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Extra Journal Abbreviation: Proc. Mach. Learn. Res.

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Proceedings Title Proc. Mach. Learn. Res.

Conference Name Proceedings of Machine Learning Research

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Caste-Based Hate Speech Detection in Low-resource Hindi Language

Item Type Conference Paper

Author S. Gupta

Author S.P. Muthusamy Chinnan

Author S. Rajiakodi

Author R. Rajalakshmi

Author R. Ponnusamy

Author B.R. Chakravarthi

Abstract In the digital era, social media discourse significantly shapes societal attitudes, frequently perpetuating entrenched hierarchies such as caste in India. Despite legal measures against caste-based discrimination, caste divisions continue to thrive online, fostering exclusion and hostility. To address this critical issue, we introduce the first curated Hindi-language dataset focused on caste related hate speech. Our annotation strategy employs a hybrid human-in-the-loop approach: expert annotators first labeled a subset of the data, which then served as a seed for large language model (LLM)-driven annotation of the remaining corpus. This methodology ensures both accuracy and scalability while capturing essential contextual and cultural nuances. We further benchmark a range of state-of-the-art hate speech detection models on our dataset, providing rigorous evaluation and insights into their performance on this challenging domain. Notably, we find that MuRIL, a multilingual language model tailored for Indian languages, achieves the best results on our test set, which consists of 761 non-hate speech and 712 hate speech comments. MuRIL attains an F1-score of 0.82 for hate speech, 0.84 for non-hate speech, and an overall accuracy of 0.83. These findings underscore the necessity of culturally and linguistically informed approaches for effective hate speech detection in low-resource settings. Our dataset, methodology and model insights offer a valuable foundation for further research in this domain, paving the way for the development of more nuanced and context-aware moderation tools. ContentWarning: This article contains illustrative examples of hateful content. © 2025 Owner/Author.

Date 2025

DOI 10.1145/3746275.3762203

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Archive Scopus

Language English

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Proceedings Title DHOW - Proc. Int. Workshop Diffus. Harmful Content Online Web, Co-Located with MM

Conference Name DHOW 2025 - Proceedings of the 2nd International Workshop on Diffusion of Harmful Content on Online Web, Co-Located with MM 2025

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Tags:

Computational linguistics, Language model, Speech communication, Learning systems, Natural language processing systems, Social media, Statistical tests, Human-in-the-loop, few-shot learning, Few-shot learning, Speech analysis, Speech detection, hate speech detection, Hate speech detection, In contexts, Context learning, In-context learning, Digital era, human-in-the-loop, in-context learning

Notes:

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Centering Patient Voices in Lupus Pain: A Biopsychosocial Analysis of Reddit Narratives Using Large Language Models

Item Type Journal Article

Author A. Walker

Author J. Leung

Author A. Alagappan

Author S. Rajwal

Author S. Lakamana

Author T. Park

Author N. Le

Author A. Irani

Author A. Sarker

Author T. Falasinnu

Author S. Bozkurt

Abstract Objective: Patients with chronic illness share their experiences in online communities and generate rich data on pain management. This study applied natural language processing methods, including large language models (LLMs), to Reddit discussions from lupus communities to characterize multidimensional pain experiences framed in the biopsychosocial model. Methods: We extracted Reddit posts from the r/Lupus and r/LupusSupport subreddits posted from June 9, 2010, through December 31, 2023. Pain-related posts were identified using a clinically informed pain lexicon. Topic modeling was used to identify thematic patterns, which were then compared with

structured summaries generated by LLM instructions that were fine-tuned using the biopsychosocial model of pain. Two reviewers conducted content analysis of the LLM-generated summaries, evaluating thematic accuracy and coverage. Results: Data from Reddit included 31,785 posts from 10,857 authors. We identified common pain complaints, management strategies, and sociocultural, affective, and nociplastic dimensions of pain. Instruction fine-tuned LLMs produced structured summaries with an average thematic accuracy score of 3.1 of 4 (kappa =.09) and content coverage score of 2.9 of 4 (kappa =.38). Sociocultural features presented in 123 posts (33.8%), including peer support and validation (n = 106) and provider interactions or access issues (n = 35). Nociplastic pain presented in 205 posts (56.3%). Conclusion: Natural language processing methods can be used to extract rich, multidimensional insights into pain experiences from online communities focused on lupus. These approaches highlight the psychological, social, and cultural facets of pain that may be underrepresented in clinical settings, supporting more patient-centered approaches to care in rheumatology. © 2025 American College of Rheumatology.

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adult, Article, female, human, Humans, male, psychology, Natural Language Processing, reproducibility, social media, procedures, anxiety, diagnosis, Social Media, cognition, natural language processing, thematic analysis, content analysis, large language model, Large Language Models, accuracy, depression, abdominal cramp, analgesia, arthralgia, biopsychosocial model, complication, emotional stress, etiology, hallucination, headache, Likert scale, Lupus Erythematosus, Systemic, Lupus Pain, myalgia, Narration, nausea, neuralgia, nociplastic pain, nonsteroid antiinflammatory agent, pain, Pain, Pain Management, peer support, rash, systemic lupus erythematosus, tetrahydrocannabinol, thorax pain, unsupervised machine learning, verbal communication, voice

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Chain of Thought Guided Few-Shot Fine-Tuning of LLMs for Multimodal Aspect-Based Sentiment Classification

Item Type Conference Paper

Author H. Wu

Author D. Yang

Author P. Liu

Author X. Li

Editor Ide I.

Editor Kompatsiaris I.

Editor Xu C.

Editor Yanai K.

Editor Chu W.-T.

Editor Nitta N.

Editor Riegler M.

Editor Yamasaki T.

Abstract With the proliferation of multimodal information on social media, the task of multimodal aspect-based sentiment classification (MA-BSC) gained significant attention. However, existing methods for this task heavily relies on the large-scale supervised data, which is costly and time-consuming to obtain. Recent efforts have shifted towards few-shot MABSC, where small-scale pretrained models (e.g. BERT, BART) are fine-tuned using a limited samples (1% of the training set). While large language models (LLMs) have shown strong zero-shot and few-shot capabilities across various NLP tasks, their exploration in few-shot MABSC remains insufficient. To address this gap, we propose a novel fine-tuning framework for LLMs in few-shot MABSC, incorporating three key modules: few-shot sampling, label-introduced Chain-of-Thought (CoT) Generation and CoT enhanced fine-tuning. Specifically, we first construct a few-shot training set via selecting informative samples based on the LLM's zero-shot capabilities. Moreover, we generate CoT reasoning through the LLM's zero-shot reasoning to guide fine-tuning, enhancing both performance and interpretability. Extensive experiments on two benchmark datasets (Twitter-2015 and Twitter-2017) demonstrate the superiority of our approach. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

Date 2025

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Language model, Large language model, Large Language Model, Social networking (online), Social media, Fine tuning, Multi-modal, Sentiment classification, Tweets, Zero-shot learning, Multi-modal information, Few-shot MABSC, Chain of thought, Chain of Thought, Training sets

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Challenges of Code-Switching and Code-Mixing for Sentiment Analysis in Low-Resource Languages

Item Type Conference Paper
Author B. Prasad
Author P. Manjunath
Author C. Nukala
Author M. Shalini
Abstract The digital landscape is shaped by user-generated content, with online reviews serving as a critical pulse for businesses, policymakers, and public discourse. Accurately interpreting sentiments in these data streams is vital for understanding consumer experiences, improving services, and shaping socio-political narratives. In linguistically diverse regions like the Philippines, the prevalence of code-switching and code-mixing complicates sentiment analysis. These phenomena, where speakers alternate or blend languages within sentences or words, pose barriers to conventional natural language processing (NLP) models trained predominantly on monolingual datasets. Misinterpretations can lead to skewed

insights, misinformed decisions, and misunderstanding of public opinion. This study investigates how code-switching and code-mixing affect the performance of six sentiment analysis tools: VADER, Filipino Sentiment Analyzer (FSA), and four large language models (LLMs)-ChatGPT-4o, DeepSeek-R1, Gemini 2.0 Flash and Perplexity Auto. We also compare review and sentence level performances to examine how granularity comes to play in multilingual contexts. Using 1,000 Shopee Philippines and Google Maps reviews, 2,589 sentences were segmented and annotated for sentiment (positive, negative, neutral, or mixed) and linguistic features (English/Filipino; monolingual/multilingual; and code-switching/code-mixing). Results reveal code-switching and code-mixing degrading performance, with F1-scores dropping by around 35% for VADER, 8% for FSA, and 9% for LLMs across both review and sentence levels. © 2025 IEEE.

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Language English

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Changes in perceptions of cultural ecosystem services in urban coastal park during different phases of COVID-19: Integrating the prompt engineering and aspect-based sentiment analysis with a large language model

Item Type Journal Article

Author Z. Zhao

Author P. Li

Author B. Chen

Author W. Yu

Author W. Hu

Author Z. Ma

Author J. Liao

Abstract Understanding changes in cultural ecosystem services (CES) perceptions across different COVID-19 pandemic phases is vital for human well-being. While social media big data are increasingly used for CES evaluation, few studies harness AI's intelligent advantages to process massive data and reveal changes across pre-, during, and post-COVID-19 phases. This study integrates prompt engineering and aspect-based sentiment analysis (ABSA) into large language models (LLMs). Four LLMs are compared with three baseline models trained on different datasets using Chinese RoBERTa; the best-performing LLM (DeepSeek-V3.2-Reasoner) is selected, and a topic model is applied to analyze changes in public concern, based on social media comments from Wuyuan Bay, China. Results show that in three ABSA tasks—aspect identification, sentiment score, and sentiment polarity—LLMs outperform baselines with limited training data. When data increase to 30000, three LLMs perform slightly lower than the baseline models on the aspect identification task, but still outperform them on remaining tasks. Nine CES categories were identified, with recreation, aesthetic, social relation, and health/wellness showing higher frequencies and positive sentiment (>3.0). Across phases, perception frequencies and sentiment scores of recreation and social relation continuously increased, while aesthetic and health/wellness first increased then declined. Public concern content included four topics; parent-child leisure proportion increased continuously. Overall, compared to baselines, all prompt engineering-based LLMs demonstrated data efficiency advantages. Meanwhile, they exhibited cross-model generalizability in certain tasks. This study demonstrates the relative advantages of the method in CES evaluation and can offer recommendations for further optimizing urban coastal parks. © 2026 Elsevier Ltd.

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Characterizing and comparing sensory and hedonic characteristics of products through free-comment responses to open-ended questions

Item Type Journal Article

Author M. Visalli

Author B. Mahieu

Abstract The Free-Comment (FC) is a rapid, easy-to-implement and flexible method that allows respondents—typically consumers—to express their perceptions or experiences in their own words, offering more authentic and personalized insights than predefined list-based methods. This chapter explores how the method can be applied to characterize and compare products based on their sensory and hedonic attributes. While the focus here is on these data types for illustrative purposes, the approach could also be extended to other data relevant in sensory and consumer science (e.g., emotional, situational, perceptual, attitudinal, behavioral). It covers different approaches for collecting FC data, key techniques for preprocessing text data, and statistical tools for comparing products and identify drivers of liking. A case study based on in-lab controlled experiments illustrates its practical application, while noting that similar insights can also be derived from sources such as online reviews or social media. The chapter concludes by exploring future directions and opportunities to further develop and refine the FC method. © 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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EID: 2-s2.0-105042839258

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Characterizing nature-based recreation preferences in a Mediterranean small island environment using crowdsourced data

Item Type Journal Article

Author L. Costadone

Author M.V. Balzan

Abstract Nature-based recreation is a key ecosystem service that contributes to positive physical and mental welfare but, at the same time, nature-based recreational activities can increase human pressure and impacts on natural areas and biodiversity. Understanding people's preference for visiting natural settings is challenging due to data and methodological limitations. Social media data can be used to map nature-based recreation. However, variation in popularity of platforms and limitations to data accessibility are highlighting the importance of exploring and using different data sources. We analyzed complementary crowdsourced data using an automated content analysis refined by manual identification to assess nature-based recreation ecosystem services across the Maltese archipelago. A content analysis of images uploaded to Flickr between 2015 and 2021 was performed using the Google Vision machine learning algorithm to identify nature-based interactions and nature visitation patterns were modeled based on landscape characteristics, environmental variables and socio-economic parameters. Flickr data were compared and complemented with publicly available geolocated data from the iNaturalist platform. Significant difference was found between the spatial distribution of Flickr and iNaturalist data. Generalized linear models identified coastal areas, protected areas, natural habitats and accessibility via the road network as significant predictors of nature-based recreational visits. Localities with a higher percentage of people receiving old age and unemployment benefits were also positively correlated with users' preference for nature-based recreation. Finally, we discussed how the low resource methodology

developed here to identify people's nature-based recreational preferences can be used to assess which natural areas should be prioritized for ecological restoration efforts. © 2023 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

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machine learning, accessibility, ecosystem service, cultural ecosystem services, Flickr, Google Vision, iNaturalist, island, landscape change, landscape ecology, Nature-based recreation, spatial distribution

Notes:

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Export Date: 20 September 2026; Cited By: 5; Funding details: National Science Foundation, NSF, (1927167, 1927468); Correspondence Address: L. Costadone; Finnish Environment Institute, Helsinki, Finland; email: laura.costadone2@gmail.com

CheckGuard: Advancing Stolen Check Detection with a Cross-Modal Image-Text Benchmark Dataset

Item Type Conference Paper

Author F. Zhao

Author J. Chen

Author B. Huang

Author C. Zhang

Author G. Warner

Abstract The prevalence of check fraud, particularly with stolen checks sold on platforms such as Telegram, creates significant challenges for both individuals and financial institutions. This underscores the urgent need for innovative solutions to detecting and preventing such fraud on social media platforms. While deep learning techniques show great promise in detecting objects and extracting information from images, their effectiveness in addressing check fraud is hindered by the lack

of comprehensive, open-source, large training datasets specifically for check information extraction. To bridge this gap, this paper introduces "CheckGuard," a large labeled image-to-text cross-modal dataset designed for check information extraction. CheckGuard comprises over 7,000 real-world stolen check image segments from more than 15 financial institutions, featuring a variety of check styles and layouts. These segments have been manually labeled, resulting in over 50,000 samples across seven key elements: Drawer, Payee, Amount, Date, Drawee, Routing Number, and Check Number. This dataset supports various tasks such as visual question answering (VQA) on checks and check image captioning. Our paper details the rigorous data collecting, cleaning, and annotation processes that make CheckGuard a valuable resource for researchers in check fraud detection, machine learning, and multimodal large language models (MLLMs). We not only benchmark state-of-the-art (SOTA) methods on this dataset to assess their performance but also explore potential enhancements. Our application of parameter-efficient fine-tuning (PEFT) techniques on the SOTA MLLMs demonstrates significant performance improvements, providing valuable insights and practical approaches for enhancing model efficacy on this task. As an evolving project, CheckGuard will continue to be updated with new data, enhancing its utility and driving further advancements in the field. Our PEFT-based MLLM code is available at: <https://github.com/feizhao19/CheckGuard>. For data access, researchers are required to contact the authors directly. © 2024 Owner/Author.

Date 2024

DOI 10.1145/3627673.3679155

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Language English

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Conference Name International Conference on Information and Knowledge Management, Proceedings

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Tags:

Language model, Deep learning, Crime, machine learning, Machine-learning, Benchmarking, Multi-modal, Adversarial machine learning, Cross-modal, Labeled data, Fraud detection, check fraud detection, Check fraud detection, cross-modal generation, Cross-modal generation, Financial institution, multi-modal large language model, Multi-modal large language model, stolen check, Stolen check

Notes:

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ClaimPT: A Portuguese Dataset of Annotated Claims in News Articles

Item Type Conference Paper

Author R. Campos

Author R. Sequeira

Author S. Nerea

Author I. Cantante

Author D. Folques

Author L.F. Cunha

Author J. Canavilhas

Author A. Branco

Author A. Jorge

Author S. Nunes

Author N. Guimarães

Author P. Silvano

Editor Campos R.

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Editor Bauer C.

Editor MacAvaney S.

Editor Anand A.

Editor Bai N.

Editor Mansoury M.

Editor Ren Z.

Editor Verberne S.

Abstract Fact-checking remains a demanding and time-consuming task, still largely dependent on manual verification and unable to match the rapid spread of misinformation online. This is particularly important because debunking false information typically takes longer to reach consumers than the misinformation itself; accelerating corrections through automation can therefore help counter it more effectively. Although many organizations perform manual fact-checking, this approach is difficult to scale given the growing volume of digital content. These limitations have motivated interest in automating fact-checking, where identifying claims is a crucial first step. However, progress has been uneven across languages, with English dominating due to abundant annotated data. Portuguese, like other languages, still lacks accessible, licensed datasets, limiting research, Natural Language Processing (NLP) developments, and applications. In this paper, we introduce ClaimPT, a dataset of European Portuguese news articles

annotated for factual claims, comprising 1,308 articles and 6,875 individual annotations. Unlike most existing resources based on social media or parliamentary transcripts, ClaimPT focuses on journalistic content, collected through a partnership with LUSA, the Portuguese News Agency. To ensure annotation quality, two trained annotators labeled each article, with a curator validating all annotations according to a newly proposed scheme. We also provide baseline models for claim detection, establishing initial benchmarks and enabling future NLP and Information Retrieval (IR) applications. By releasing ClaimPT, we aim to advance research on low-resource fact-checking and enhance understanding of misinformation in news media. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Conference Name Lecture Notes in Computer Science

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Language processing, Natural languages, Benchmarking, Misinformation, Low resource languages, News articles, Annotated datasets, Fact-checking, Annotated Dataset, Claim detection, Claim Detection, Digital contents, Low-resource Languages, News Articles, Time-consuming tasks

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: R. Campos; University of Beira Interior, Covilhã, Portugal; email: ricardo.campos@ubi.pt; Conference name: 48th European Conference on Information Retrieval, ECIR 2026; Conference date: 29 March 2026 through 2 April 2026; Conference code: 349479

Classification of Hacker's Posts Based on Zero-Shot, Few-Shot, and Fine-Tuned LLMs in Environments with Constrained Resources

Item Type Journal Article

Author T. Giannilias

Author A. Papadakis

Author N. Nikolaou

Author T. Zahariadis

Abstract This paper investigates, applies, and evaluates state-of-the-art Large Language Models (LLMs) for the classification of posts from a dark web hackers' forum into four cyber-security categories. The LLMs applied included Mistral-7B-Instruct-v0.2, Gemma-1.1-7B, Llama-3-8B-Instruct, and Llama-2-7B, with zero-shot learning, few-shot learning, and fine-tuning. The four cyber-security categories consisted of "Access Control and Management", "Availability Protection and Security by Design Mechanisms", "Software and Firmware Flaws", and "not relevant". The hackers' posts were also classified and labelled by a human cyber-security expert, allowing a detailed evaluation of the classification accuracy per each LLM and customization/learning method. We verified LLM fine-tuning as the most effective mechanism to enhance the accuracy and reliability of the classifications. The results include the methodology applied and the labelled hackers' posts dataset. © 2025 by the authors.

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Language English

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Publisher Multidisciplinary Digital Publishing Institute (MDPI)

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Notes:

EID: 2-s2.0-105006437304

Export Date: 20 September 2026; Cited By: 14; Correspondence Address: A. Papadakis; Synelixis Solutions S.A., Chalkida, Farmakidou 10, 34100, Greece; email: papadakis@synelixis.com

Classifying and Understanding Mental Health Discourse on Social Media: A Hybrid Topic Modelling Approach

Item Type Conference Paper

Author B. Kaur

Author D. Chen

Abstract The growing prevalence of mental health discourse on social media presents both an opportunity and a challenge for timely detection and nuanced understanding of conditions such as anxiety, depression, bipolar disorder, suicidal ideation, stress, and personality disorders. Identifying critical signals remains taxing due to noisy data and context-dependent language. This study presents an integrated approach combining transformer-based classification with topic modelling to analyse mental health discourse on social media. DistilBERT, a transformer-based Large Language Model, has been fine-tuned for multi-class mental health classification, which has demonstrated a robust contextual understanding as compared to traditional deep learning models, such as Long-Short Term Memory, Recurrent Neural Network, Gated Recurrent Unit, etc. To enhance classification with interpretable insights, BERTopic, a transformer-embedding-based topic modelling technique, is utilised to reveal predominant themes within each mental health category. The resultant topics have revealed coherent clusters that reflect personal struggles, medication experiences, social stigma, and coping strategies. Coherence scores range from 0.36 to 0.47, indicating a meaningful separation of topics in short, informal tweets. The proposed approach highlights the potential of combining powerful language models with unsupervised topic modelling for scalable mental health monitoring. This dual integrated method facilitates early risk identification and provides profound insights into the online discourse surrounding mental health disorders, hence guiding therapies, digital mental health technologies, and policy formulation. © 2025 IEEE.

Date 2025

DOI 10.1109/CPSI66656.2025.11343977

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Archive Scopus

Language English

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Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-833159961-4 (ISBN)

Proceedings Title Int. Conf. Cyber-Phys. Soc. Intell., CPSI

Conference Name 2025 International Conference on Cyber-Physical Social Intelligence, CPSI 2025

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Tags:

Computational linguistics, Language model, Mental health, Learning systems, Social networking (online), Health risks, Social media, BERT, Topic Modeling, topic modelling, Modeling languages, Long short-term memory, Transformer, Modeling approach, Text classification, Text processing, mental health, Bertopic, BERTopic, text classification, Diseases, Electric transformers, transformers, Distilbert, distilBERT

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Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 International Conference on Cyber-Physical Social Intelligence, CPSI 2025; Conference date: 7 November 2025 through 10 November 2025; Conference code: 219473

Classifying Sentiments on Social Media Texts: A GPT-4 Preliminary Study

Item Type Conference Paper

Author L.L. Maceda

Author J.L. Llovido

Author M.B. Artiaga

Author M.B. Abisado

Abstract In today's digital age, social media has become a hub for people to express their thoughts and feelings. Sentiment classification discerns public opinions and trends to understand their sentiments towards a certain topic. Often, achieving accurate sentiment classifications in large datasets necessitate the use of human-Annotated training data which can be costly and time-consuming. Large Language Models (LLMs) like the Generative Pre-Trained models by OpenAI have surged in popularity due to its capabilities in understanding the given tasks. In this preliminary study, we report the performance of the latest OpenAI GPT-4 using zero-and one-shot learning approaches on classifying sentiments when fed with social media dataset. Notably, the latter approach written in English which mimics the instructions designed for human annotators, achieved a substantial agreement ($k = 0.77$) with human annotations, displaying high accuracy, precision, and recall accordingly even without explicit training data. Meanwhile, the fine-Tuned mBERT resulted to lower evaluation scores than the GPT-4. Our findings provide foundational insights into the strengths and limitations of GPT-4 for sentiment classification in a social media dataset, setting the groundwork for broad future research in this field. © 2023 ACM.

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Language model, Public opinions, Classification (of information), Social networking (online), Social media, Large datasets, Digital age, Sentiment classification, Zero-shot learning, GPT-4, Social Media Data, Social media datum, Large language model prompting, LLM Prompting, Sentiment annotation, Sentiment Annotation

Notes:

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Export Date: 20 September 2026; Cited By: 12; Correspondence Address: L.L. Maceda; Bicol University, Legazpi City, Philippines; email: llmaceda@bicol-u.edu.ph; Conference name: 7th International Conference on Natural Language Processing and Information Retrieval, NLPPIR 2023; Conference date: 15 December 2023 through 17 December 2023; Conference code: 197810

Classifying the Information Needs of Survivors of Domestic Violence in Online Health Communities Using Large Language Models: Prediction Model Development and Evaluation Study

Item Type Journal Article

Author S. Guan

Author V. Hui

Author G. Stiglic

Author R.E. Constantino

Author Y.J. Lee

Author A.K.C. Wong

Abstract Background: Domestic violence (DV) is a significant public health concern affecting the physical and mental well-being of numerous women, imposing a substantial health care burden. However, women facing DV often encounter barriers to seeking in-person help due to stigma, shame, and embarrassment. As a result, many survivors of DV turn to online health communities as a safe and anonymous space to share their experiences and seek support. Understanding the information needs of survivors of DV in online health communities through multiclass classification is crucial for providing timely and appropriate support. Objective: The objective was to develop a fine-tuned large language model (LLM) that can provide fast and accurate predictions of the information needs of survivors of DV from their online posts, enabling health care professionals to offer timely and personalized assistance. Methods: We collected 294 posts from Reddit subcommunities focused on DV shared by women aged ≥ 18 years who self-identified as experiencing intimate partner violence. We identified 8 types of information needs: shelters/DV centers/agencies; legal; childbearing; police; DV report procedure/documentation; safety planning; DV knowledge; and communication. Data augmentation was applied using GPT-3.5 to expand our dataset to 2216 samples by generating 1922 additional posts that imitated the existing data. We adopted a progressive training strategy to fine-tune GPT-3.5 for multiclass text classification using 2032 posts. We trained the model on 1 class at a time, monitoring performance closely. When suboptimal results were observed, we generated additional samples of the misclassified ones to give them more attention. We reserved 184 posts for internal testing and 74 for external validation. Model performance was

evaluated using accuracy, recall, precision, and F1-score, along with CIs for each metric. Results: Using 40 real posts and 144 artificial intelligence-generated posts as the test dataset, our model achieved an F1-score of 70.49% (95% CI 60.63%-80.35%) for real posts, outperforming the original GPT-3.5 and GPT-4, fine-tuned Llama 2-7B and Llama 3-8B, and long short-term memory. On artificial intelligence-generated posts, our model attained an F1-score of 84.58% (95% CI 80.38%-88.78%), surpassing all baselines. When tested on an external validation dataset (n=74), the model achieved an F1-score of 59.67% (95% CI 51.86%-67.49%), outperforming other models. Statistical analysis revealed that our model significantly outperformed the others in F1-score ($P=.047$ for real posts; $P<.001$ for external validation posts). Furthermore, our model was faster, taking 19.108 seconds for predictions versus 1150 seconds for manual assessment. Conclusions: Our fine-tuned LLM can accurately and efficiently extract and identify DV-related information needs through multiclass classification from online posts. In addition, we used LLM-based data augmentation techniques to overcome the limitations of a relatively small and imbalanced dataset. By generating timely and accurate predictions, we can empower health care professionals to provide rapid and suitable assistance to survivors of DV. ©Shaowei Guan, Vivian Hui, Gregor Stiglic, Rose Eva Constantino, Young Ji Lee, Arkers Kwan Ching Wong.

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Notes:

EID: 2-s2.0-105004823759

Export Date: 20 September 2026; Cited By: 9; Correspondence Address: V. Hui; Centre for Smart Health, School of Nursing, The Hong Kong Polytechnic University, The Hong Kong Polytechnic University, Hung Hom, HJ 544, 11 Yuk Choi Rd, 000000, Hong Kong; email: vivianc.hui@polyu.edu.hk

CLEF 2024 JOKER Lab: Automatic Humour Analysis

Item Type Conference Paper

Author L. Ermakova

Author A.-G. Bosser

Author T. Miller

Author T. Thomas

Author V.M.P. Preciado

Author G. Sidorov

Author A. Jatowt

Editor Goharian N.

Editor Tonellotto N.

Editor He Y.

Editor Lipani A.

Editor McDonald G.

Editor Macdonald C.

Editor Ounis I.

Abstract The JOKER Lab at the Conference and Labs of the Evaluation Forum (CLEF) aims to foster research on automated processing of verbal humour, including tasks such as retrieval, classification, interpretation, generation, and translation. Despite the heady success of large language models, humour and wordplay automatic processing are far from being a solved problem. JOKER brings together experts from the social and computational sciences and encourages them to collaborate on shared tasks with quality-controlled annotated datasets. In 2024, we will offer entirely new shared tasks on humour-aware information retrieval, as well as fine-grained sentiment analysis and classification of humour for conversational agents. As in the past JOKER Labs, we will also make our data available for an unshared task that solicits novel use cases. In this paper, we provide a brief retrospective on the JOKER Labs, with a focus on the results and lessons learnt from last year's iteration, and we preview the tasks to be held at JOKER 2024. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Language English

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Language model, Classification (of information), Sentiment analysis, Information retrieval, Quality control, Automated processing, Humor, Humor classification, Humor generation, Humor-aware information retrieval, Humour, Humour classification, Humour generation, Humour-aware information retrieval, Laboratories, Pun, Puns, Retrieval classifications, Wordplay

Notes:

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Export Date: 20 September 2026; Cited By: 12; Correspondence Address: T. Miller; Department of Computer Science, University of Manitoba, Winnipeg, Canada; email: tristan.miller@umanitoba.ca; Conference name: 46th European Conference on Information Retrieval, ECIR 2024; Conference date: 24 March 2024 through 28 March 2024; Conference code: 309699

CLPsych 2024 - 9th Workshop on Computational Linguistics and Clinical Psychology, Proceedings of the Workshop

Item Type Journal Article

Editor Yates A.

Editor Desmet B.

Editor Prud'hommeaux E.

Editor Zirikly A.

Editor Bedrick S.

Editor MacAvaney S.

Editor Bar K.

Editor Ireland M.

Editor Ophir Y.

Editor Ophir Y.

Abstract The proceedings contain 28 papers. The topics discussed include: assessing motivational interviewing sessions with AI-generated patient simulations; delving into the depths: evaluating depression severity through BDI-biased summaries; how can client motivational language inform psychotherapy agents?; linguistic markers of schizophrenia: a case study of Robert Walser; therapist self-disclosure as a natural language processing task; ethical thematic and topic modelling analysis of sleep concerns in a social media derived suicidality dataset; automatic annotation of dream report's emotional content with large language models; explainable depression

detection using large language models on social media data; analyzing relevance of discourse structure for improved mental health estimation; and using daily language to understand drinking: multi-level longitudinal differential language analysis.

Date 2024

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Language English

Publisher Association for Computational Linguistics (ACL)

Publication CLPsych 2024 - 9th Workshop on Computational Linguistics and Clinical Psychology, Proceedings of the Workshop

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CMANet:Sarcasm Detection with LLMs and Cross-Modal Adaptive Attention Networks

Item Type Conference Paper

Author N. Zhao

Author D. Sun

Author C. Meng

Author N. Li

Author Y. Wang

Author H. Yang

Editor Awan I.

Editor Younas M.

Editor Zhang Y.

Editor Barhamgi M.

Editor Wang H.

Abstract With the widespread popularity of social media, sarcastic expressions have become increasingly common in online communication. Accurately detecting such expressions is crucial for understanding user intentions, conducting sentiment analysis, and maintaining a healthy online environment. However, sarcasm detection faces numerous challenges, such as the subtlety of text semantics and the complexity of multi-modal information fusion. To effectively address these issues, this paper proposes a sarcasm detection method that integrates Large Language Models (LLMs) with a cross-modal adaptive attention network. In terms of text processing, the powerful language capabilities of large

language models are utilized to perform in-depth semantic representation of the input text, capturing rich language patterns, contextual relationships, and implied semantic information, thereby providing high-quality textual features. Meanwhile, considering the close association between multi-modal information and text, a cross-modal adaptive attention network is constructed to achieve effective fusion of text and multi-modal data. This network dynamically focuses on key information through an adaptive attention mechanism, automatically adjusting the weights of each modality to fully leverage the complementarity between modalities. In experiments, multiple representative social media datasets are selected for validation, and the results demonstrate that, compared with traditional methods, the proposed method significantly improves evaluation metrics such as accuracy and recall. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

Date 2026

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Notes:

EID: 2-s2.0-105040530103

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: D. Sun; Computer Network Information Center, Chinese Academy of Sciences, Beijing, China; email: sundegang@iie.ac.cn; Conference name: 26th International Conference on Web Information Systems Engineering, WISE 2025; Conference date: 15 December 2025 through 17 December 2025; Conference code: 352629

Co-Designer: A Human-AI Collaborative Framework for Deriving Physical Product Concept Design Opportunities from User-Generated Content(UGC) in Manufacturing

Item Type Conference Paper

Author R. Cai

Author M. Guo

Author S. Yang

Author R. Li

Author X. Wu

Author J. Li

Editor Chua T.-S.

Editor Wong L.-K.

Editor Chan C.S.

Editor Tang J.

Editor Ngo C.-W.

Editor Schoeffmann K.

Editor Liu J.

Editor Ho Y.-S.

Abstract For designers in the manufacturing industry, discovering design opportunities from large-scale User-Generated Content (UGC) is crucial in physical product conceptual design, yet manual analysis is inefficient. This problem limits product designers' ability and hinders the manufacturing industry's ability to produce products that do not easily meet users' needs. This study proposes Co-Designer, a human-AI collaboration framework that helps designers efficiently discover opportunities from UGC. To ensure the development of software that designers need, this study proposed a formative study with eight professional designers, who identified workflow pain points and requirements for AI support. Based on these insights, Co-Designer applies Aspect-Based Sentiment Analysis (ABSA) to transform UGC into a structured "aspect-sentiment" knowledge base, enabling an AI design partner that adapts to designer profiles and dynamically acts as an "analyst" or "creative teammate." Following the Double Diamond design model, it can invoke workflows to translate insights into design strategies. To evaluate the software's system availability, this study used a mixed-methods approach with 10 designers, complemented by expert evaluation, which showed that Co-Designer significantly improves efficiency, insight, creativity, and planning. This work introduces a new paradigm positioning AI as a strategic collaborator and creativity catalyst in physical product concept design. © 2025 Copyright held by the owner/author(s).

Date 2025

DOI 10.1145/3743093.3771654

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Language English

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Publisher Association for Computing Machinery, Inc

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Proceedings Title Proc. ACM Int. Conf. Multimed. Asia, MMAAsia

Conference Name Proceedings of the 7th ACM International Conference on Multimedia in Asia, MMAsia 2025

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Tags:

Language model, Large language model, Product design, Sentiment analysis, Knowledge based systems, Conceptual design, Large Language Models (LLMs), User-generated, User-generated content, Design ideation, Design Ideation, Human-AI collaboration, Human-AI Collaboration, Manufacturing industries, Physical product concept design, Physical Product Concept Design, Physical products, Product concept design, User-Generated Content(UGC)

Notes:

EID: 2-s2.0-105025133289

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Guo; Harbin Institute of Technology, Shenzhen, Shenzhen, China; email: guomengyao@hit.edu.cn; Conference name: 7th ACM International Conference on Multimedia in Asia, MMAsia 2025; Conference date: 9 December 2025 through 12 December 2025; Conference code: 215361

Co-training semi-supervised deep learning for sentiment classification of MOOC forum posts

Item Type Journal Article

Author J. Chen

Author J. Feng

Author X. Sun

Author Y. Liu

Abstract Sentiment classification of forum posts of massive open online courses is essential for educators to make interventions and for instructors to improve learning performance. Lacking monitoring on learners' sentiments may lead to high dropout rates of courses. Recently, deep learning has emerged as an outstanding machine learning technique for sentiment classification, which extracts complex features automatically with rich representation capabilities. However, deep neural networks always rely on a large amount of labeled data for supervised training. Constructing large-scale labeled training datasets for sentiment classification is very laborious and time consuming. To address this problem, this paper proposes a co-training, semi-supervised deep learning model for sentiment classification, leveraging limited labeled data and massive unlabeled data simultaneously to achieve performance comparable to those methods trained on massive labeled data. To satisfy the condition of two views of co-training, we encoded texts into vectors from views of word embedding and character-based embedding independently, considering words' external and internal information. To promote the classification performance with limited data, we propose a double-check strategy sample selection method to select samples with high confidence to augment the training set iteratively. In addition, we propose a mixed

loss function both considering the labeled data with asymmetric and unlabeled data. Our proposed method achieved a 89.73% average accuracy and an 93.55% average F1-score, about 2.77% and 3.2% higher than baseline methods. Experimental results demonstrate the effectiveness of the proposed model trained on limited labeled data, which performs much better than those trained on massive labeled data. © 2019 by the authors.

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CODAI 2024 - Proceedings of the 1st Workshop on Countering Disinformation with Artificial Intelligence, co-located with European Conference on Artificial Intelligence 2024, ECAI 2024

Item Type Journal Article

Editor Sharma R.

Editor Penas A.

Editor Sabir A.

Abstract The proceedings contain 7 papers. The topics discussed include: analysis of climate change misleading information in TikTok; diachronic political content analysis: a comparative study of topics and sentiments in echo chambers and beyond; factoring in context for the automatic detection of misrepresentation; are misinformation

propagation models holistic enough? identifying gaps and needs; detecting fake news using twitter social information; on the categorization of corporate multimodal disinformation with large language models; and automated fact-checking based on large language models: an application for the press.

Date 2024

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Archive Scopus

Language English

Volume 3782

Publisher CEUR-WS

Publication CEUR Workshop Proceedings

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Journal Abbr CEUR Workshop Proc.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 1st Workshop on Countering Disinformation with Artificial Intelligence, CODAI 2024; Conference code: 203184

Code-Mixed Sentiment Analysis in Social Media: A Critical Review of Recent Methods, Trends, and Open Challenges

Item Type Conference Paper

Author V. Lad

Author R.J. Deshmukh

Author C.J. Awati

Author R. Dhabarde

Abstract The analysis of sentiment in code-mixed social media text has gained increasing attention in recent years, driven by the growing prevalence of informal multilingual communication in online platforms. This review examines thirty peer-reviewed journal articles published recently, with the aim of providing a structured and critical synthesis of recent progress in the field. Rather than presenting a purely descriptive overview, the surveyed literature is systematically organized according to key methodological dimensions, including representation learning strategies, model architectures, learning paradigms, and considerations related to explainability and deployment. The review highlights a clear transition toward multilingual transformer-based models, alongside emerging interest in prompt-based learning and parameter-efficient adaptation techniques such as adapters. At the same time, persistent challenges related to data scarcity, benchmark inconsistency, generalization across domains, and limited interpretability remain evident. By explicitly situating recent developments with respect to earlier surveys and identifying underexplored directions, this work aims to provide a coherent foundation for future research on robust and

practically deployable sentiment analysis models for code-mixed social media content. © 2026 IEEE.

Date 2026

DOI 10.1109/ICICDS70526.2026.11604839

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Int. Conf. Innov. Cybersecur. Data Science, ICICDS

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-831953130-8 (ISBN)

Pages 1237-1242

Proceedings Title Proc. Int. Conf. Innov. Cybersecur. Data Science, ICICDS

Conference Name Proceedings of the 3rd International Conference on Innovations in Cybersecurity and Data Science, ICICDS 2026

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Tags:

Codes (symbols), Artificial intelligence, Learning systems, Sentiment analysis, Explainable artificial intelligence, Language processing, Natural languages, Social networking (online), Social media, Data mining, Transformer, Electric transformers, (NLP), Code-mixed sentiment analysis, Code-mixed sentiment analyze, explainable Artificial Intelligence (AI), Low-resource natural language processing., low-resource Natural Language Processing. (NLP), multilingual social media, Multilingual social medium, transformers

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EID: 2-s2.0-105046128658

Export Date: 20 September 2026; Cited By: 0; Conference name: 3rd International Conference on Innovations in Cybersecurity and Data Science, ICICDS 2026; Conference date: 25 June 2026 through 27 June 2026; Conference code: 226105

CoFFEE-Qwen: A Large Language Model for Chinese Financial Sentiment Analysis Using the Contrastive Learning and Fine-Tuning Paradigm

Item Type Journal Article

Author W. Feng

Author C. Yang

Author M. Zhao

Author Z. Xia

Author F. Wang

Abstract Financial sentiment analysis is increasingly recognized as a pivotal component of stock market research, providing a more accurate and efficient means of quantifying and interpreting market sentiment. However, stock commentaries—one of the most prevalent forms of financial text on social media—are often unstructured and rife with typographical errors and colloquial expressions. To mitigate the lack of publicly available Chinese financial sentiment analysis datasets, we constructed a dedicated expert-annotated corpus and further validated the model's robustness on the large-scale Eastmoney Guba benchmark. To address challenges such as misclassification arising from the non-standard nature of stock commentary, we introduce CoFFEE-Qwen (Contrastive Fine-tuned Financial Embedding for Qwen), a novel framework that integrates supervised contrastive learning with parameter-efficient fine-tuning of large language models (LLMs). This approach fully exploits the transfer learning capabilities of LLMs. Furthermore, the CoFFEE mechanism enables the incorporation of heterogeneous domain data, thereby improving the model's representational capacity for financial texts with limited samples, alleviating the scarcity of social media financial data, and enhancing the overall quality of learned feature representations. In addition, to handle typographical errors and colloquial expressions commonly found in stock commentaries, we design a homophonic character perturbation mechanism that improves the model's robustness and encoding effectiveness when processing noisy input text. Experimental results show that CoFFEE-Qwen consistently outperforms baseline models across multiple metrics—including accuracy, F1-score, precision, and recall—demonstrating its clear superiority in sentiment analysis tasks involving stock commentaries with typographical noise and informal language. © 2026, International Association of Engineers. All rights reserved.

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Pages 2526-2539

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ISSN 1819656X (ISSN)

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Language model, Large language model, Commerce, Sentiment analysis, Social networking (online), Social media, Large datasets, Fine tuning, Transfer learning, Electronic trading, Contrastive Learning, LLM, Embeddings, Financial markets, Market Research, Financial sentiment analysis, Financial sentiment analyze, Coffee, CoFFEE-Qwen, Contrastive, Contrastive fine-tuned financial embedding for qwen, Homophonic perturbation mechanism, Homophonic Perturbation mechanism

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Collaborative Stance Detection via Small-Large Language Model Consistency Verification

Item Type Conference Paper

Author Y. Yan

Author S. Sun

Author Z. Tang

Author T. Liu

Author M. Liu

Editor Zhu F.

Editor Lim E.-P.

Editor Yu P.S.

Editor Nadamoto A.

Editor Shim K.

Editor Ding W.

Editor Zhang B.

Abstract Stance detection on social media aims to identify attitudes expressed in tweets towards specific targets. Current studies prioritize Large Language Models (LLMs) over Small Language Models (SLMs) due to the overwhelming performance improving provided by LLMs. However, heavily relying on LLMs for stance detection, regardless of the cost, is impractical for real-world social media monitoring systems that require vast data analysis. To this end, we propose Collaborative Stance Detection via Small-Large Language Model Consistency Verification (CoVer) framework, which enhances LLM utilization via context-shared batch reasoning and logical verification between LLM and SLM. Specifically, instead of processing each text individually, CoVer processes texts batch-by-batch, obtaining stance predictions and corresponding explanations via LLM reasoning in a shared context. Then, to exclude the bias caused by context noises, CoVer introduces the SLM for logical consistency verification. Finally, texts that repeatedly exhibit low logical consistency are classified using consistency-weighted aggregation of prior LLM stance predictions. Our experiments show that CoVer outperforms state-of-the-art methods across multiple benchmarks in the zero-shot setting, achieving 0.54 LLM queries per tweet while significantly enhancing performance. Our CoVer offers a more practical solution for LLM deploying for social media stance detection. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

Date 2026

DOI 10.1007/978-981-95-4158-4_1

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Proceedings Title Lect. Notes Comput. Sci.
Conference Name Lecture Notes in Computer Science
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Tags:

Language model, 'current, Distributed computer systems, Performance, Social networking (online), Benchmarking, Social media, Stance detection, Tweets, Stance Detection, Collaborative interference, Collaborative Interference, Consistency verifications, Logical consistency, Model consistency, Small and large language model, Small and Large Language Model

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Liu; Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China; email: liumin@ict.ac.cn; Conference name: 30th International Conference on Database Systems for Advanced Applications, DASFAA 2025; Conference date: 26 May 2025 through 29 May 2025; Conference code: 344929

Combining LLM-Generated Knowledge Graphs with RAG for Financial Sentiment Extraction

Item Type Conference Paper
Author Z. Huang
Author K. Du
Author X. Zhang
Author R. Mao
Author E. Cambria
Abstract This research proposes a novel approach to financial sentiment analysis by integrating knowledge graphs with Retrieval-Augmented Generation (RAG) techniques. The core innovation lies in transforming traditional financial texts into structured knowledge graph representations, capturing entities, relationships, and attributes inherent in financial data. These knowledge graphs are leveraged within a RAG framework powered by large language models, enhancing the system's ability to retrieve relevant information and generate accurate sentiment assessments. Utilizing datasets such as PhraseBank and Twitter Financial News, the proposed method effectively captures nuanced and hidden sentiments often overlooked by traditional techniques. Experimental observations indicate that this

approach significantly enhances both reliability and interpretability of sentiment analysis compared to using models such as GPT-4omini. The system relies on the structured knowledge within the knowledge graphs, enabling more context-aware and evidence-based sentiment evaluations rather than depending solely on the model's internal representations. Moreover, this method improves retrieval efficiency and accuracy without requiring additional training processes, saving computational resources, and facilitating easier deployment in real-world applications. By avoiding the need for extensive fine-tuning or large-scale parameter updates, this approach offers a sustainable and scalable solution for financial sentiment analysis. The expected outcome is a robust, contextaware sentiment analysis system that provides comprehensive and reliable sentiment assessments, enhancing transparency and explainability in the financial decision-making process. This can ultimately aid investors, analysts, and policymakers by providing deeper insight into market sentiments and trends. © 2025 IEEE.

Date 2025

DOI 10.1109/ICDMW69685.2025.00250

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Archive Scopus

Language English

Extra Journal Abbreviation: IEEE Int. Conf. Data Mining Workshops, ICDMW

Publisher IEEE Computer Society

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Proceedings Title IEEE Int. Conf. Data Mining Workshops, ICDMW

Conference Name IEEE International Conference on Data Mining Workshops, ICDMW

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Tags:

Context-Aware, Decision making, Sentiment analysis, Investments, LLM, Information retrieval, Reliability analysis, NLP, Knowledge management, Knowledge graph, Knowledge graphs, Retrieval-augmented generation, Graphic methods, RAG, FinTech, Financial sentiment analyze, financial sentiment analysis, Entity-relationship, Generation techniques, Graph representation, Structured knowledge

Notes:

EID: 2-s2.0-105035380539

Export Date: 20 September 2026; Cited By: 2; Conference name: 25th IEEE International Conference on Data Mining Workshops, ICDMW 2025; Conference date: 12 November 2025 through 15 November 2025; Conference code: 221413

CommentAgent: a LLM-powered agent framework for automated comment generation and opinion understanding

Item Type Journal Article**Author** X. Han**Author** J. Sun**Author** S. Yu**Author** L. Qin**Author** H. Luo**Author** Y. Li

Abstract High-quality Chinese datasets for public opinion analysis are scarce, with most resources based on outdated English-language platforms like Twitter and Reddit, which are not relevant to Chinese social media. To address this, we introduce CommentAgent, a novel simulation framework that generates high-quality Chinese datasets for sentiment analysis, sarcasm detection, and stance detection. CommentAgent simulates the Weibo environment using a multi-agent architecture that models user behavior, information flow, and personality evolution, allowing for realistic and diverse comment generation. Unlike traditional data augmentation or basic generation methods, CommentAgent generates topic-driven data from scratch. Three benchmark datasets are released for the above tasks, with extensive experiments demonstrating that CommentAgent outperforms existing methods in data quality and generalizability. Moreover, lightweight models trained on these datasets achieve performance comparable to large language models (LLMs). CommentAgent offers a scalable solution for creating up-to-date Chinese datasets for robust public opinion analysis. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025.

Date 2025**DOI** 10.1007/s11227-025-07957-6**URL** <https://www.scopus.com/pages/publications/105018668785?origin=resultslist>**Archive** Scopus**Language** English**Volume** 81**Publisher** Springer**Publication** Journal of Supercomputing**ISSN** 09208542 (ISSN)**Issue** 15**Journal Abbr** J Supercomput**Date Added** 20/09/2026, 8:35:05 pm**Modified** 20/09/2026, 8:35:05 pm**Tags:**

Language model, Large language model, Behavioral research, Multiagent systems (MASs), Public opinions, Social aspects, Sentiment analysis, Social networking (online), Benchmarking, Large datasets, Data mining, High quality, English languages, Multi agent systems, Intelligent agents, Agent Framework, Data generation, Multi-agent system, Opinion analysis, Resources based

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Community Perspectives on ChatGPT: Sentiment Analysis in Educational Forum

Item Type Journal Article

Author Y. Lee

Abstract This study investigates community sentiments and opinions regarding the use of ChatGPT in education within online user forums. Large Language Model (LLM)-based sentiment analysis was conducted on the posts and comments submitted to Teachers Subreddit, an online community for educators. The analysis revealed that the members of Teachers Subreddit expressed more negative sentiments towards the use of ChatGPT in education compared to those who shared their opinions on X (formerly known as Twitter). Additionally, LLM-based topic modeling was performed on the positive and negative opinions identified during the sentiment analysis phase. Seven positive topics emerged: (1) ChatGPT as a content generation tool, (2) enhanced student engagement, (3) embracing changes, (4) need for effective professional development, (5) educational reform, (6) plagiarism-detection software, and (7) utilizing ChatGPT in writing classes. Seven negative topics were also identified: (1) challenges in education, (2) increase in plagiarism, (3) ineffective professional development, (4) academic dishonesty, (5) negative effects on the workforce, (6) curriculum relevance and meaning of education, and (7) lack of emphasis on critical/higher-order thinking skills. © Association for Educational Communications & Technology 2024.

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DOI 10.1007/s11528-024-01012-6

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Social media, Data science applications in education, Improving classroom teaching

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: Y. Lee; Department of Learning Technologies, University of North Texas, Denton, 3940 North Elm Street (G157), 76207, United States; email: Youngjin.Lee@unt.edu

Comparative Analysis of LLM-based Market Prediction and Human Expertise with Sentiment Analysis and Machine Learning Integration

Item Type	Conference Paper
Author	M. Abdelsamie
Author	H. Wang
Abstract	This study conducts a comparative analysis of market prediction accuracy between Large Language Model (LLM)-based systems and human expertise within the financial analysis domain. Leveraging Quantum, an advanced LLM specialized for financial forecasting, we evaluate its predictive performance against human analysts and general-purpose LLMs, including GPT-3, GPT-4, FinGPT, and FinBERT. Employing a dataset of historical financial data, news headlines, and social media sentiment, we systematically assess predictive accuracy, response efficiency, and interpretability across models. The integration of sentiment analysis and machine learning further strengthens prediction reliability. Results reveal that Quantum's specialized model demonstrates superior accuracy and speed in financial forecasting compared to human predictions and generalized LLMs, particularly in fast-moving, data-rich contexts. Nevertheless, limitations in nuanced contextual understanding and adaptability persist, highlighting the enduring value of human expertise. This research reinforces the potential of LLMs as robust tools for financial decision-making while identifying key areas for refinement to enhance synergy with human analytical insights. https://chatgpt.com/g/g-bS4Q76v0I-quantum © 2024 IEEE.
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Archive	Scopus
Language	English
Extra	Journal Abbreviation: Proc. - Int. Conf. Data Sci. Inf. Technol., DSIT
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Language model, Large language model, Machine-learning, Sentiment analysis, Machine learning, Prediction models, Predictive analytics, Adversarial machine learning, Contrastive Learning, Machine Learning, Sentiment Analysis, Comparative analyzes, Large Language Models, Financial prediction, Financial Prediction, Human expertise, Market forecasting, Market Forecasting, Market prediction, Quantum AI

Notes:

EID: 2-s2.0-86000192406

Export Date: 20 September 2026; Cited By: 5; Correspondence Address: M. Abdelsamie; Zhejiang University, International Business School, China; email: Abdelsamie.23@intl.zju.edu.cn; Conference name: 7th International Conference on Data Science and Information Technology, DSIT 2024; Conference date: 20 December 2024 through 22 December 2024; Conference code: 207065

Comparative Analysis of Machine Learning and Deep Learning Models for Sentiment Analysis in Somali Language

Item Type Journal Article

Author A.A. Abdirahman

Author A.O. Hashi

Author U.M. Dahir

Author M.A. Elmi

Author O.E.R. Rodriguez

Abstract Understanding and analysing sentiment in user-generated content has become crucial with the increasing use of social media and online platforms. However, sentiment analysis in less-resourced languages like Somali poses unique challenges. This paper presents the performance of three ML algorithms (DTC, RFC, XGB) and two DL models (CNN, LSTM) in accurately classifying sentiment in Somali text. The CC100-Somali dataset, comprising 78M monolingual Somali texts from the Common crawl snapshots, is utilized for training and evaluation. The study employed rigorous evaluation techniques, including train-test splits and cross-validation, to assess classification accuracy and performance metrics. The results demonstrated that DTC achieved the highest accuracy among ML algorithms, 87.94%, while LSTM achieved the highest accuracy among DL models, 88.58%. This study's findings contribute to sentiment analysis in less-resourced languages, specifically Somali, and provide valuable insights into the performance of ML and DL techniques. Moreover, the study highlights the potential of leveraging both ML and DL approaches to analyze sentiment in Somali text effectively. The results and evaluation metrics benchmark future research in sentiment analysis for Somali and other low-resource languages. © 2023 Seventh Sense Research Group.

Date 2023

DOI 10.14445/23488379/IJEEE-V10I7P104

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Deep learning, Sentiment analysis, Machine learning, Somali dataset, Somali language

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: A.A. Abdirahman; Faculty of Computing, SIMAD University, Mogadishu, Somalia; email: aaayare@simad.edu.so

Comparative Analysis of Pretrained Models for Text Classification, Generation and Summarization: A Detailed Analysis

Item Type Conference Paper
Author P. Pathak
Author P.S. Rana
Editor Antonacopoulos A.
Editor Chaudhuri S.
Editor Chellappa R.
Editor Liu C.-L.
Editor Bhattacharya S.
Editor Pal U.

Abstract The exponential growth in natural language processing (NLP) technologies has been propelled by the emergence of pretrained models, which have demonstrated remarkable efficacy across a spectrum of tasks including text classification, generation, and summarization. Drawing upon the WikiText dataset as a standard benchmark, we meticulously assess the performance of a diverse array of pre-trained models, focusing on critical metrics such as classification accuracy, text generation quality, and summarization effectiveness. Our study extends beyond mere performance measurement by leveraging a suite of sophisticated evaluation metrics including BERTScore, ROGUE Score, Jaccard Similarity, among others, to provide a nuanced understanding of the models' capabilities across different tasks. Additionally, we employ the Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS) method to aggregate the disparate performance metrics into a unified ranking framework, facilitating a comprehensive compar-

ison of the pretrained models. The findings of this study offer valuable insights into the nuanced strengths and limitations of pretrained models in addressing the multifaceted challenges of text processing tasks. Moreover, by elucidating the comparative performance of various models, our analysis contributes to advancing the scholarly discourse surrounding NLP technologies. For our Wikitext Dataset, GPT-3.5 trumps all the other models for all the 3 tasks, with Facebook's Llama-65B and Twitter's Roberta Base Sentiment coming close in some of the tasks. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

Date 2025

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Notes:

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Comparative Analysis of Transformer Models for Sentiment Analysis in Low-Resource Languages

Item Type Journal Article**Author** Y. Aliyu**Author** A. Sarlan**Author** K.U. Danyaro**Author** A.S.B.A. Rahman

Abstract The analysis of sentiments expressed on social media platforms is a crucial tool for understanding user opinions and preferences. The large amount of the texts found on social media are mostly in different languages. However, the accuracy of sentiment analysis in these systems faces different challenges in multilingual low-resource settings. Recent advancements in deep learning transformer models have demonstrated superior performance compared to traditional machine learning techniques. The majority of preceding works are predominantly constructed on the foundation of monolingual languages. This study presents a comparative analysis that assesses the effectiveness of transformer models, for multilingual low-resource languages sentiment analysis. The study aims to improve the accuracy of the existing baseline performance in analyzing tweets written in 12 low-resource African languages. Four widely used start-of-the-art transformer models were employed. The experiment was carried out using standard datasets of tweets. The study showcases AfriBERTa as a robust performer, exhibiting superior sentiment analysis capabilities across diverse linguistic contexts. It outperformed the established benchmarks in both SemEval-2023 Task 12 and AfriSenti baseline. Our framework achieves remarkable results with an F1-score of 81% and an accuracy rate of 80.9%. This study provides validation of the framework's robustness in the domain of sentiment analysis across a low-resource linguistics context. our research not only contributes a comprehensive sentiment analysis framework for low-resource African languages but also charts a roadmap for future enhancements. Emphasize the ongoing pursuit of adaptability and robustness in sentiment analysis models for diverse linguistic landscapes. © (2024), (Science and Information Organization). All Rights Reserved.

Date 2024**DOI** 10.14569/IJACSA.2024.0150437**URL** <https://www.scopus.com/pages/publications/85192071707?origin=resultslist>**Archive** Scopus**Language** English**Volume** 15**Publisher** Science and Information Organization**Pages** 353-364**Publication** International Journal of Advanced Computer Science and Applications**ISSN** 2158107X (ISSN)**Issue** 4**Journal Abbr** Intl. J. Adv. Comput. Sci. Appl.**Date Added** 20/09/2026, 8:35:35 pm**Modified** 20/09/2026, 8:35:35 pm**Tags:**

Deep learning, Learning systems, Sentiment analysis, Transformer modeling, Social networking (online), Embeddings, Transformer, Comparative analyzes, Social media platforms, Linguistics, Low resource languages, multilingual, Multilingual, low-resource languages, African languages, Word-embedding, transformer, word-embedding

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Comparative Evaluation of Large Language Models for Sentiment Analysis

Item Type	Conference Paper
Author	K.K. Agarwal
Author	S. Tiwari
Author	V. Jain
Abstract	Sentiment analysis is an important task in the processing of natural language which is necessary for understanding public opinion and emotions in the fields such as social media, feedback from customers and market analysis. Despite notable progress, a comprehensive evaluation of different models (LLMs) is needed to understand the strengths and weaknesses of these models in the tasks of sentiment analysis. This research performs a comparative analysis of two prominent LLMs: BERT, a well-established model best known for text classification, and LLaMA 3.2, a newly released model by Meta Technologies. While previous research has thoroughly used models like BERT, there is still a gap in the literature regarding the performance and efficiency of LLaMA 3.2, especially in the context of sentiment analysis. By fine-tuning these models on a dataset consisting of 12 distinct sentiment classes, this research seeks to fill this gap by providing new findings into the comparative evaluation of these LLMs. The outcomes of this study are significant as they offer a comprehensive analysis of LLaMA 3.2's capabilities relative to established models, helping researchers and practitioners select the most suitable model for sentiment analysis tasks. It is essential to solve this problem to strengthen the state-of-the-art in sentiment analysis, resulting in more accurate and better applications across various industries. © 2025 IEEE.
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DOI	10.1109/ICCSAI64074.2025.11064177
URL	https://www.scopus.com/pages/publications/105012117086?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Int. Conf. Commun., Secur., Artif. Intell. ICCSAI
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	979-833153607-7 (ISBN)
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Tags:

Computational linguistics, Language model, Large language model, Social aspects, Classification (of information), Sentiment analysis, Language processing, Natural language processing, Natural Language Processing (NLP), Natural languages, Benchmarking, BERT, Sentiment Analysis, Large Language Models (LLMs), Text classification, Text Classification, LLaMA, Comparative evaluations, Comparative Evaluation

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Export Date: 20 September 2026; Cited By: 0; Conference name: 3rd International Conference on Communication, Security, and Artificial Intelligence, ICCSAI 2025; Conference date: 4 April 2025 through 6 April 2025; Conference code: 210436

Comparative Study of BERT and ChatGPT for Multi-Label Cyberbullying Detection with Intensity Scoring

Item Type Conference Paper
Author N.X.R. Tay
Author H.N. Chua
Author M.B. Jasser
Author B. Issa
Author G.S. Aldharhani
Abstract Cyberbullying on social media often involves subtle, context-dependent language, posing challenges for automated detection. Traditional machine learning approaches, reliant on handcrafted features and shallow context, struggle to capture such nuances. Transformer-based Large Language Models (LLMs) like BERT and ChatGPT offer deeper semantic understanding and zero-shot learning capabilities, yet their comparative effectiveness in multi-label cyberbullying classification and severity estimation remains underexplored. This study evaluates BERT (zero-shot classification) and ChatGPT (prompt-based classification) against a reference-standard dataset of human-annotated online comments, covering overlapping categories of abuse. Performance was measured using BLEU, ROUGE, macro-F1, and weighted-F1 metrics, with an intensity scoring framework to quantify severity. Results show BERT aligns closely with human annotations across all metrics, detecting both common and rare categories, while ChatGPT performs well for high-frequency categories but struggles with nuanced cases. This study's contributions include: (i) a reproducible evaluation framework for LLM-based multi-label abuse detection, (ii) operationalization of

intensity scoring for severity-aware moderation, and (iii) comparative evidence on strengths and limitations of discriminative versus generative LLMs in ethically sensitive Natural Language Processing (NLP) tasks. © 2025 IEEE.

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Tags:

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Notes:

EID: 2-s2.0-105032055584

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Comparative Study of Machine Learning, Transformer, and Generative Models for Sentiment Analysis

Item Type Conference Paper
Author A.W. Hasnan
Author B. Chaya
Author A.A.S. Gunawan
Author J.J. Tedjasulaksana
Abstract The increasing adoption of artificial intelligence (AI) in surveillance systems has intensified public debate regarding privacy, ethics, and civil liberties. Social media platforms such as Twitter (X) provide large-scale textual data that can be utilized to analyze public sentiment toward AI-based surveillance technologies.

This study presents a comparative evaluation of three sentiment analysis approaches: a traditional machine learning model, a transformer-based discriminative model, and a generative artificial intelligence (GenAI) model. A secondary dataset consisting of 6,860 labeled tweets with positive, neutral, and negative sentiments is used. Linear Support Vector Machine (SVM) with TF-IDF, Bidirectional Encoder Representations from Transformers (BERT), and a prompt-based Large Language Model (LLaMA 3.0) are evaluated using accuracy, precision, recall, and F1-score. Experimental results show that BERT achieves the highest overall performance with an accuracy of 0.83, outperforming SVM which reaches 0.74. Meanwhile, the GenAI model demonstrates limited performance under a few-shot prompting setting without fine-tuning, highlighting its constraints for multi-class sentiment classification. These findings indicate that transformer-based models demonstrate superior performance, while generative models show potential for rapid sentiment analysis, their effectiveness remains highly dependent on prompting strategy and task-specific adaptation. © 2026 IEEE.

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Publisher Institute of Electrical and Electronics Engineers Inc.

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Tags:

Security systems, Network security, Performance, Support vector machines, Support vectors machine, Learning systems, Sentiment analysis, Social networking (online), BERT, Social sciences computing, Generative AI, Intelligence models, Machine learning models, Generative artificial intelligence, Sentiment Analysis, Bidirectional encoder representation from transformer, Artificial intelligence surveillance, Artificial Intelligence Surveillance, Generative model, LLaMA, Support Vector Machine

Notes:

EID: 2-s2.0-105047902401

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A.W. Hasnan; Bina Nusantara University, School of Computer Science, Computer Science Department, Jakarta, Indonesia; email: adiartha.hasnan@binus.ac.id; Conference name: 2026 International Conference on Advancement in Data Science, E-Learning and Information System, ICADEIS 2026; Conference date: 23 June 2026 through 24 June 2026; Conference code: 226864

Comparative Study of Pre-Trained BERT Models for Code-Mixed Hindi-English Data

Item Type Conference Paper

Author A. Patil

Author V. Patwardhan

Author A. Phaltankar

Author G. Takawane

Author R. Joshi

Abstract The term "Code Mixed" refers to the use of more than one language in the same text. This phenomenon is predominantly observed on social media platforms, with an increasing amount of adaptation as time goes on. It is critical to detect foreign elements in a language and process them correctly, as a considerable number of individuals are using code-mixed languages that could not be comprehended by understanding one of those languages. In this work, we focus on low-resource Hindi-English code-mixed language and enhancing the performance of different code-mixed natural language processing tasks such as sentiment analysis, emotion recognition, and hate speech identification. We perform a comparative analysis of different Transformer-based language Models pre-trained using unsupervised approaches. We have included the code-mixed models like HingBERT, HingRoBERTa, HingRoBERTa-Mixed, mBERT, and non-code-mixed models like ALBERT, BERT, and RoBERTa for comparative analysis of code-mixed Hindi-English downstream tasks. We report state-of-the-art results on respective datasets using HingBERT-based models which are specifically pre-trained on real code-mixed text. Our HingBERT-based models provide significant improvements thus highlighting the poor performance of vanilla BERT models on code-mixed text. © 2023 IEEE.

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Notes:

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Export Date: 20 September 2026; Cited By: 25; Correspondence Address: A. Patil; Pune Institute of Computer Technology, L3Cube Pune, Thane, India; email: aryanpatil1503@gmail.com; Conference name: 8th IEEE International Conference for Convergence in Technology, I2CT 2023; Conference date: 7 April 2023 through 9 April 2023; Conference code: 188943

Comparative Study of Zero-Shot and Supervised Learning for Tamil Tweet Sentiment Analysis

Item Type Conference Paper

Author R. Jain

Author M. Kesavan

Author V.K. Jayaraman

Editor Yadav R.

Editor Nanda S.

Editor Prasad M.

Editor Saraswat M.

Abstract Sentiment analysis in low resource, morphologically rich languages like Tamil remains a significant challenge due to limited annotated data, code-mixed usage, and orthographic variability in social media content. Addressing this gap is crucial for inclusive NLP research and regional language applications such as opinion mining and public sentiment tracking. The study evaluates the effectiveness of pretrained multilingual models in both zero-shot and supervised learning settings for sentiment analysis of Tamil tweets, focusing on the difficulty of dealing with an agglutinative, low-resource language for Tamil tweet sentiment classification. We assess encoder-only transformers (e.g., XLM-RoBERTa, DeBERTa-v3, MuRIL, IndicBERT, mBERT) and decoder-only LLMs (e.g., LLaMA 3.1, Gemma 3, Mixtral, Mistral) on two datasets: Tamil Tweet Binary Classification Dataset and a filtered binary subset of the ACTSEA dataset. In the zero-shot setting, XLM-RoBERTa-XNLI achieves an F1 score of 0.9697 on ACTSEA using natural language inference (NLI) prompting. Supervised fine-tuning of MuRIL yields an F1 score of 0.9101, while LaBSE embeddings combined with SVM reach 0.8914. Our findings highlight that NLI-finetuned encoder models and cross-lingual embeddings offer strong generalization in low-resource scenarios. This work emphasizes the importance of task framing, multilingual representation learning, and hybrid modeling strategies for effective sentiment analysis in underrepresented languages like Tamil. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: V.K. Jayaraman; FLAME University, Pune, Maharashtra, India; email: jayaraman.vk@flame.edu.in; Conference name: 6th International Conference on Data Science and Applications, ICDSA 2025; Conference date: 16 July 2025 through 18 July 2025; Conference code: 345349

Comparative Study on the Efficiency of Using PaLM and CAMELBERT for Arabic Entity Sentiment Classification

Item Type Conference Paper
Author S. Almutrash
Author S. Abudalfa
Abstract Sentiment classification, which is usually employed to categorize emotions in social media text, has been extensively studied for public opinion monitoring. For example, it can be used to examine feedback from customers on online shopping platforms to assess their satisfaction levels. Sentiment classification achieves high accuracy with the English language. In contrast, sentiment classification in Arabic falls behind, mainly due to the intricate nature of Arabic text, which poses difficulties for contemporary machine learning methods. This study evaluates the effectiveness of the Pathways Language Model (PaLM) and CAMELBERT to show efficiency of large language models in detecting sentiment in Arabic social

media. The results indicate that a pre-trained model using the BERT (CAMELBERT) architecture outperformed PaLM, achieving an accuracy of 63.87% compared to 61.01% with PaLM. These results underscore the ongoing difficulties that modern language models encounter in accurately interpreting the subtleties of Arabic sentiment classification. © 2024 1st Saudi Conference on Information Systems, SaudiCIS 2024. All rights reserved.

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Conference Name 1st Saudi Conference on Information Systems, SaudiCIS 2024

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Export Date: 20 September 2026; Cited By: 3; Conference name: 1st Saudi Conference on Information Systems, SaudiCIS 2024; Conference date: 19 November 2024 through 21 November 2024; Conference code: 210241

Comparing Fine-Tuning and Prompt Engineering for Multi-Class Classification in Hospitality Review Analysis

Item Type Journal Article

Author I. Botunac

Author M. Brkić Bakarić

Author M. Matetić

Abstract This study compares the effectiveness of fine-tuning Transformer models, specifically BERT, RoBERTa, DeBERTa, and GPT-2, against using prompt engineering in LLMs like ChatGPT and GPT-4 for multi-class classification of hotel reviews. As the hospitality industry increasingly relies on online customer feedback to improve

services and strategize marketing, accurately analyzing this feedback is crucial. Our research employs a multi-task learning framework to simultaneously conduct sentiment analysis and categorize reviews into aspects such as service quality, ambiance, and food. We assess the capabilities of fine-tuned Transformer models and LLMs with prompt engineering in processing and understanding the complex user-generated content prevalent in the hospitality industry. The results show that fine-tuned models, particularly RoBERTa, are more adept at classification tasks due to their deep contextual processing abilities and faster execution times. In contrast, while ChatGPT and GPT-4 excel in sentiment analysis by better capturing the nuances of human emotions, they require more computational power and longer processing times. Our findings support the hypothesis that fine-tuning models can achieve better results and faster execution than using prompt engineering in LLMs for multi-class classification in hospitality reviews. This study suggests that selecting the appropriate NLP model depends on the task's specific needs, balancing computational efficiency and the depth of sentiment analysis required for actionable insights in hospitality management. © 2024 by the authors.

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Export Date: 20 September 2026; Cited By: 15; Correspondence Address: I. Botunac; Faculty of Informatics and Digital Technologies, University of Rijeka, Rijeka, Radmile Matejčić 2, 51000, Croatia; email: ive.botunac@student.uniri.hr

Comparing traditional natural language processing and large language models for mental health status classification: a multi-model evaluation

Item Type Journal Article

Author T. Kallstenius

Author A.J. Capusan

Author G. Andersson

Author A. Williamson

Abstract The substantial increase in mental health disorders globally necessitates scalable, accurate tools for detecting and classifying these conditions in digital environments. This study addresses the critical challenge of automated mental health classification by comparing three distinct computational approaches: (1) Traditional Natural Language Processing (NLP) with advanced feature engineering, (2) Prompt-engineered large language models (LLMs), and (3) Fine-tuned LLMs. The dataset consisted of over 51,000 publicly available text statements from social media platforms, tagged with seven mental health conditions: Normal, Depression, Suicidal, Anxiety, Stress, Bipolar Disorder, and Personality Disorder. The dataset was stratified into training, validation, and test sets for model evaluation. The primary outcome was classification accuracy across these seven mental health conditions. Additional metrics like precision, recall, and F1-score were analyzed. We compared the results of the three computational approaches and overfitting was monitored through validation loss across epochs for the fine-tuned LLM. The NLP model with advanced feature engineering achieved an overall accuracy of 95%, surpassing both the prompt-engineered LLM (65%) and the fine-tuned LLM (91%). This model performed exceptionally well in terms of accuracy and precision. While fine-tuning for three epochs yielded optimal results, further training led to overfitting and decreased performance. This study demonstrates the significant benefits of applying advanced text preprocessing and feature engineering techniques to traditional NLP models, alongside fine-tuning LLMs, such as GPT-4o-mini, for mental health classification tasks. The results clearly indicate that off-the-shelf LLM chatbots using prompt engineering are inadequate for mental health classification, performing 30% points below specialized NLP approaches. Despite the popularity of general-purpose LLMs, specialized approaches remain superior for critical healthcare applications like mental health classification. © The Author(s) 2025.

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Comparing Zero-Shot Text Classification and Rule-Based Matching in Identifying Cyberbullying Behaviors on Social Media

Item Type Conference Paper

Author W.J. Chong

Author H.N. Chua

Author M.F. Gan

Abstract The incidences of cyberbullying have skyrocketed due to the continuous expansion of social media users. Therefore, proactive efforts are necessary to address cyberbullying, including countermeasures for handling the various cyberbullying behaviors. Nevertheless, due to the large volume of social media texts being generated persistently, it is challenging to identify cyberbullying behaviors in a social text and not scalable in using the manual approach of human annotation. Most previous studies adopt the human annotation approach to determine whether a text is a cyberbully or non-cyberbully. Therefore, this paper aims to experiment with approaches that can improve the efficiency of recognizing the different cyberbullying behaviors through textual data using zero-shot classification and rule-based matching, and compare how they perform in classifying cyberbullying behaviors. This study uses techniques such as topic modelling, zero-shot text classification, and information extraction with rule-based matching to identify and classify the cyberbully behaviors underlying a cyberbully comment. The human annotation approach serves as the benchmark to compare the performance of both models in identifying cyberbullying behaviors. Our results show that zero-shot classification performed better accuracy in categorizing cyberbullying behaviors. Among the behaviors, the zero-shot model we generated presents a better accuracy rate in recognizing the flaming behavior, but it achieves a lower accuracy rate in identifying the other cyberbully behaviors. © 2022 IEEE.

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Comparison of Artificial Intelligence Tools With Human Coding for Sentiment, Topic, and Thematic Analysis Tasks of Public Health Datasets During the COVID-19 Pandemic in Australia: Case Study

Item Type Journal Article

Author D. Hutchinson

Author L. Lee

Author H. Stone

Author A. Moa

Author H. Seale

Author C.R. Macintyre

Abstract Background: Public opinion, which may be influenced by personal experiences, news, and social media, can impact compliance with public health measures (PHMs) during health emergencies. Artificial intelligence (AI) tools offer opportunities to analyze public opinion in real time during health emergencies. However, their

performance in accurately identifying sentiment and themes in health-related online content remains unclear. Objective: This study aimed to evaluate the performance of natural language processing-based and large language model (LLM)-based AI tools when compared to human coding for sentiment analysis, topic modeling, and thematic analysis of public health datasets. Tools were selected to reflect those available to public health analysts and decision-makers. Methods: Data were collected via Google Alerts (GA) and social media posts from X (formerly known as Twitter) relevant to COVID-19 mitigation PHMs from December 2022 to February 2023. Following relevance screening, the sentiment of the complete datasets was analyzed by a human rater, with descriptive statistics used to summarize the overall sentiment profile. Subsets of 400 GA articles and 400 tweets were manually coded for sentiment by 2 human raters. Results were compared with outputs from 5 AI tools, including VADER (Valence Aware Dictionary and Sentiment Reasoner), SentimentGI, SentimentQDAP, Microsoft Azure, and OpenAI's ChatGPT-4. Topic modeling of the GA and X datasets was conducted using latent Dirichlet allocation in R and zero-shot prompting in ChatGPT-4 and compared with manual topic summaries. Thematic analysis of positive and negative sentiment datasets was conducted by a human rater and ChatGPT-4, with outputs evaluated for proficiency and reasonableness. The sentiment of the entire datasets was analyzed by a human rater, and descriptive statistics were calculated. Results: Of 2227 GA results and 3484 tweets, 58% (n=1238) and 71% (n=2473), respectively, were relevant to PHMs. Human-coded sentiment analysis showed mostly neutral reporting in the news media, while social media expressed more polarized views. Across both datasets, AI tools demonstrated poor concordance with human-coded sentiment (Cohen $\kappa < 0.5$ for all tools and sentiment categories). Topic modeling with ChatGPT-4 aligned more closely with human-rated topics than latent Dirichlet allocation, and of the 20 LLM-generated thematic outputs, 13 were rated proficient, and 7 were rated partially proficient. LLM outputs provided coherent, high-level summaries but lacked contextual insight. Human and LLM thematic analyses both identified themes of vaccine effectiveness, debate regarding PHMs, and public trust. Conclusions: Accessible AI tools demonstrate limited reliability for sentiment classification of health-related online text but show promise for rapid thematic exploration when combined with human oversight. These tools could complement traditional qualitative research in the context of health emergencies; however, they require human review to enhance the accuracy of interpretation. Further research is needed for non-English datasets. © 2026, JMIR Publications Inc.. All rights reserved.

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Comparison of Traditional Machine Learning Models and Large Language Models for Multi-Label Text Classification on X (Twitter)

Item Type Conference Paper

Author V. Yesmaya

Author W. Budiharto

Author M.S. Anggreainy

Author S.M. Isa

Author I.H. Kartowisastro

Editor Jain L.C.

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Editor Balas V.E.

Editor Wu Q.

Editor Gaol F.L.

Editor Shi F.

Editor Dey N.

Abstract Social media platform X (Twitter) is a major source of public information across domains such as technology, education, and health. However, its short and informal text poses challenges for natural language processing (NLP). This study applies a Large Language Model (LLM)-based approach for multi-label text classification on X (Twitter) data covering six domains: Artificial Intelligence and Technology, Education, Music and Film, Sports, Health, and Finance. Data were collected via the Twitter API using keyword filters, followed by preprocessing steps including text normalization, URL removal, and language detection. The baseline models-KNN, Naïve Bayes, and Random Forest-used TF-IDF features and were compared with LLMs through zero-shot and fine-tuning approaches. Experimental results show that fine-tuned LLMs significantly improved accuracy and generalization compared to traditional models. However, among all tested models, Random Forest achieved the best performance with an Micro-F1: 0.612, Macro-F1: 0.568, Hamming Loss: 0.096, making it the most effective model in this study. These findings highlight the potential of LLMs and traditional machine learning integration for real-time public conversation analysis on social media. © 2026 IEEE.

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Compiling Linguistic Resources for Specific Purposes: Using LLMs to Collect Data From Social Media

Item Type	Conference Paper
Author	M. Panagiotou
Author	K. Gkatzionis
Author	E. Kaloudis
Editor	Vezzani F.
Editor	Di Nunzio G.M.
Editor	Loupaki E.
Editor	Meditskos G.
Editor	Papoutsoglou M.
Abstract	Social media generate vast amounts of authentic linguistic data daily, offering insights into users' conceptualizations, language use, and cultural phenomena. However, managing this data manually is challenging due to its volume and evolving algorithms. This study identifies key concepts and specific words that shape Greek consumer identity by analyzing Instagram posts on traditional and

local food consumption. Sentiment analysis was conducted using generative AI, and new computational tools were developed for automated data collection and analysis. AI-generated responses were evaluated against human assessments, and they proved to be successful in identifying and categorizing posts and hashtags according to criteria set for the task. Case studies included Mediterranean snails as a sustainable meat alternative and local cheeses from the North-Aegean islands. The methodology can be applied to specialized lexicography, linguistic and cultural studies, and terminology research, such as identifying neologisms and tracking term usage over time. © 2025 Copyright for this paper by its author.

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Computational measurement of political positions: a review of text-based ideal point estimation algorithms

Item Type Journal Article

Author P. Parschan

Author C. Jakob

Abstract This article presents the first systematic review of unsupervised and semi-supervised computational text-based ideal point estimation (CT-IPE) algorithms, methods designed to infer latent political positions from textual data. These algorithms are widely used in political science, communication, computational social science, and computer science to estimate ideological preferences from parliamentary speeches, party manifestos, and social media. Over the past two decades, their development has closely followed broader NLP trends—beginning with word-frequency models and most recently turning to large language models (LLMs). While this trajectory has greatly expanded the methodological toolkit, it has also produced a fragmented field that lacks systematic comparison and clear guidance for applied use. To address this gap, we identified 25 CT-IPE algorithms through a systematic literature review and conducted a manual content analysis of their modeling assumptions and development contexts. To compare them meaningfully, we introduce a conceptual framework that distinguishes how algorithms generate, capture, and aggregate textual variance. On this basis, we identify four methodological families—word-frequency, topic modeling, word embedding, and LLM-based approaches—and critically assess their assumptions, interpretability, scalability, and limitations. Our review offers three contributions. First, it provides a structured synthesis of two decades of algorithm development, clarifying how diverse methods relate to one another. Second, it translates these insights into practical guidance for applied researchers, highlighting trade-offs in transparency, technical requirements, and validation strategies that shape algorithm choice. Third, it emphasizes that differences in estimation outcomes across algorithms are themselves informative, underscoring the need for systematic benchmarking. © The Author(s) 2025.

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Computing recommendations from free-form text

Item Type Journal Article

Author L. Eberhard

Author K. Popova

Author S. Walk

Author D. Helic

Abstract While searching for consumer goods, users frequently ask for suggestions from their peers by writing short free-form textual requests. For example, when searching for movies users may ask for “Drama movies with a mind-bending story and a surprise ending, such as Fight Club” in one of the many online discussion boards. Despite the recent developments in large language models (LLMs) and natural language processing (NLP), modern recommender systems still struggle to process such requests. Therefore, in this paper we evaluate several approaches for annotating structured information from such short, free-form natural language user texts to calculate recommendations. We set up this evaluation as a two phase processes including (a) identification of the best NLP approach to identify key elements of users’ requests, and (b) assessment of the quality of recommendations computed with such elements. For our evaluation, we use a gold-standard reddit movie recommendation dataset consisting of annotations, manually created by crowdworkers who extracted keywords, actor names and movie titles. Using this dataset we evaluate a collection of more than 30 NLP and five recommender approaches. In addition, we perform an ablation study to assess relative annotation importance for movie recommendations. We find that domain-specific deep learning models, trained on a subset of data as well as embedding-based recommendation approaches are able to match the recommendation performance of recommendations computed from manual annotations. These promising results warrant further investigation in automatic processing of short free-form texts for computation of recommendations. Specifically, we provide insights into which NLP models and configurations work best for automatically annotating free text to compute (movie) recommendations, hence substantially reducing the search space for combinations of NLP and recommendation algorithms in the movie and potentially other domains. © 2023 The Authors

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Constructing and evaluating ArabicStanceX: a social media dataset for Arabic stance detection

Item Type Journal Article

Author A. Alkhathlan

Author F. Alahmadi

Author F. Kateb

Author H. Al-Khalifa

Abstract Arabic stance detection has attracted significant interest due to the growing importance of social media in shaping public opinion. However, the lack of comprehensive datasets has limited research progress in Arabic Natural Language Processing (NLP). To address this, we introduce ArabicStanceX, a novel and extensive Arabic stance detection dataset sourced from social media, comprising 14,477 tweets across 17 diverse topics. Utilizing the transformer-based MARBERTv2 model, we explore stance detection through Multi-Topic Single Model (MTSM) strategies, achieving a promising F1 score of 0.74 for detecting 'favor' and 'against' stances, and 0.67 overall. Our experiments highlight the model's capabilities and challenges, particularly in accurately classifying neutral stances and generalizing to unseen topics. Further investigations using zero-shot and few-shot learning demonstrate the model's adaptability to new contexts. This study significantly advances Arabic NLP, providing crucial resources and insights into stance detection methodologies and future research directions. The dataset is publicly available¹. Copyright © 2025 Alkhathlan, Alahmadi, Kateb and Al-Khalifa.

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Constructing colloquial dataset for Persian sentiment analysis of social microblogs

Item Type Journal Article

Author M. Mazoochi

Author L. Rabiei

Author F. Rahmani

Author Z. Rajabi

Abstract Microblogging websites have massed rich data sources for sentiment analysis and opinion mining. In this regard, sentiment classification has frequently proven inefficient because microblog posts typically lack syntactically consistent terms and representatives since users on these social networks do not like to write lengthy statements. Also, there are some limitations to low-resource languages. The Persian language has exceptional characteristics and demands unique annotated data and models for the sentiment analysis task, which are distinctive from text features within the English dialect. This paper first constructs a user opinion dataset called ITRC-Opinion in a collaborative environment and insource way. Our dataset contains 60,000 informal and colloquial Persian texts from social microblogs such as Twitter and Instagram. Second, this study proposes a new architecture based on the convolutional neural network (CNN) model for more effective sentiment analysis of colloquial text in social microblog posts. The constructed datasets are used to evaluate the presented architecture. Furthermore, some models, such as LSTM, CNN-RNN, BiLSTM, and BiGRU with different word embeddings, including Fasttext, Glove, and Word2vec, investigated our dataset and evaluated the results. The results demonstrate the benefit of our dataset and the proposed model (72% accuracy), displaying meaningful improvement in sentiment classification performance. © The

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Construction of a comprehensive dataset for named entity recognition and entity linking in Algerian Dialectal Arabic

Item Type Journal Article

Author W. Bouarroudj

Author A. Belbekri

Author I. Djouablia

Author H. Hanine Bouguettoucha

Abstract Algerian Arabic (Darija) dominates digital communication in North Africa yet remains severely under-resourced in Natural Language Processing (NLP), hindering the development of robust applications for social media analysis and e-commerce. This paper addresses this scarcity by presenting a systematic framework for constructing and benchmarking Named Entity Recognition (NER) and Entity Linking

(EL) resources tailored to the dialect's linguistic complexity. We introduce a large-scale, multi-script dataset constructed through a novel hybrid methodology that integrates manual annotation of authentic texts, automated knowledge graph extraction from Wikidata, and rule-based synthetic generation. This approach ensures diverse coverage across ten semantic categories while explicitly addressing the challenges of code-switching and orthographic variation (Arabizi and Arabic script). A transformer-based model (XLM-RoBERTa) fine-tuned on this resource achieves state-of-the-art performance, demonstrating significant robustness compared to existing baselines. Beyond the dataset, we provide a practical deployment interface and comprehensive evaluation metrics, establishing a crucial foundation for advancing NLP capabilities in North African dialects and facilitating downstream tasks such as content moderation and cultural heritage preservation. © The Author(s) 2026

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Context-Aware Hate and Offensive Speech Detection Using Hybrid BERT-LSTM Architecture

Item Type Conference Paper

Author S. Kalra

Author V. Gupta

Author S. Jha

Author A. Shukla

Editor Kaiser M.S.

Editor Xie J.

Editor Joshi A.

Abstract The rapid proliferation of social media platforms has accelerated the dissemination of hate and offensive discourse, creating formidable obstacles for automated content oversight. Conventional algorithms frequently fail to accommodate the interplay between contextual dependency and temporal sequence inherent in such lexicons. This article proposes a hybrid BERT-LSTM framework for the classification of hate and offensive speech, combining BERT's contextual embeddings with a bidirectional LSTM to leverage both fine-grained contextual representation and effective sequential modeling. Within a comparative evaluation including TF-IDF + XGBoost, convolutional networks, and a standalone BERT baseline, the proposed architecture secured the highest performance metric, attaining a classification accuracy of 92.1% and surpassing all comparison models in precision, recall, and F1-score. Empirical results demonstrate the design of context-driven deep learning paradigms as a preferred method for the identification of latent and subjective speech phenomena in social media. Future research will focus on multilingual cross-lingual corpus construction for hate speech, zero-shot transfer between very different corpora culturally and incorporation of explainable artificial intelligence components to address the transparency needs of sensitive larger deployments. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Context-Aware Pseudo-Label Scoring for Zero-Shot Video Summarization

Item Type Conference Paper

Author Y. Wu

Author L. Zhang

Author Y. Du

Author B. Li

Abstract With the rapid proliferation of video content across social media, surveillance, and education platforms, efficiently summarizing long videos into concise yet semantically faithful surrogates has become increasingly vital. Existing supervised methods achieve strong in-domain accuracy by learning from dense annotations but incur high labeling costs and limited cross-dataset generalization, while unsupervised approaches, though label-free, often miss high-level human semantics and fine-grained narrative cues. Recent zero-shot pipelines leverage large language models (LLMs) for training-free video summarization, yet remain highly sensitive to handcrafted prompts and dataset-specific normalization. To address these limitations, we propose a rubric-guided, pseudo-labeled prompting framework that converts a small subset of human annotations into high-confidence pseudo labels and aggregates them into structured, dataset-adaptive scoring rubrics for interpretable scene evaluation. During inference, first and last segments are scored based solely on their descriptions, whereas intermediate segments incorporate brief contextual summaries of adjacent scenes to assess narrative progression and redundancy. This contextual prompting enables the LLM to balance local salience and global coherence without parameter tuning. Our framework achieves consistent improvements across three benchmarks. On SumMe and TVSum, it reaches F1 scores of 57.58 and 63.05, surpassing the zero-shot baseline PROMPTS TO SUMMARIES (56.73 and 62.21) by +0.85 and +0.84, respectively, while approaching the performance of supervised models. On the query-focused QFVS benchmark, our method attains an average F1 of 53.79, exceeding the baseline (53.42) by +0.37 and

demonstrating stable performance across all validation videos. These results confirm that rubric-guided pseudo labeling, combined with contextual prompting, effectively stabilizes LLM-based scoring and establishes a general, interpretable zero-shot paradigm applicable to both generic and query-focused video summarization. Code is available at: <https://github.com/wuyuanli60-svg/Context-Aware-Pseudo-Label-Scoring-for-Zero-Shot-Video-Summarization>. © 2025 Copyright held by the owner/author(s).

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Context-Enriched NLP Pipelines using Large Language Models for High Accuracy Text Mining and Sentiment Analysis

Item Type Conference Paper

Author N. Mishra

Author A. Soy

Abstract The current state of Natural Language Processing (NLP) technology has accelerated the development of large language models (LLMs) such as GPT, BERT, and T5, which can now be used to substantially improve text mining and sentiment analysis. Nevertheless, despite these developments, the issue remains to enhance the accuracy of models by introducing contextual knowledge into NLP pipelines. The paper suggests the application of the context-enriched NLP pipeline, which is driven by LLCMs, to obtain high-accuracy text mining and sentiment analysis. External knowledge sources can also be added to the input data in the pipeline to allow varying amounts of contextual information, including sentence and document context, to be incorporated therein. This richness enhances the performance of this model as it is more sensitive to subtle sentiment variations and is better able to extract meaningful data out of the text data. Experiment evaluation is a set of text mining activities, sentiment classification, and entity extraction, based on publicly available benchmark datasets. The context-enriched pipeline achieves 92.3% accuracy on sentiment classification tasks (SST-2), exceeding the baseline LLM's performance (88.5%). In entity recognition (CoNLL-03), the model achieves an F1-score of 87.2, compared to the baselines' 83.4. It also demonstrates better precision (91.5% vs. 86.0%) and recall (93.0% vs. 88.0%), indicating that the context-enriched model is much more efficient than traditional LLMs and baseline models. These results suggest a significant influence of context on the performance of returning LLMs in practice, especially in social media analysis and customer feedback. The discussion of possible challenges and future directions, such as incorporating domain-specific knowledge and handling noisy or ambiguous data, concludes the paper. © 2026 IEEE.

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Contextual Emotion Retrieval in Low-Resource Languages: A Deep Neural Paradigm for Bengali Nostalgia Detection

Item Type Conference Paper**Author** B. Sing**Author** M.R. Rahman**Editor** Shah M.S.

Abstract Nostalgia is a complex emotion that connects individuals to their cultural and personal pasts, often expressed in digital platforms through comments, posts, and narratives. This study focuses on the automatic detection of nostalgic sentiment in Bengali text, a low-resource language with rich cultural depth but limited computational tools. We introduce a manually annotated dataset of 20,561 Bengali comments from different social media categorized as nostalgic or non-nostalgic. The dataset, initially imbalanced, was augmented to improve class distribution and provide a robust benchmark for machine learning and deep learning models. Our methodology includes a comparative evaluation of classical models such as Logistic Regression, SVM, and Random Forest, as well as deep learning models including CNN, LSTM, BiLSTM, and a fine-tuned BERT-based classifier. Experimental results show that the BERT Nostalgia Classifier achieves the highest accuracy of 87.63 % and an F1-score of 0.81, significantly outperforming other models. The findings confirm that deep contextual embeddings are effective for detecting nuanced emotions like nostalgia in resource-scarce languages. This work contributes a novel dataset and a scalable framework for affective computing in Bengali, with potential applications in sentiment analysis, cultural informatics, and emotionally aware recommender systems. © 2025 IEEE.

Date 2025**DOI** 10.1109/QPAIN66474.2025.11171754**URL** <https://www.scopus.com/pages/publications/105019057144?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Int. Conf. Quantum Photonics, Artif. Intell., Netw., QPAIN**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833159693-4 (ISBN)**Proceedings Title** IEEE Int. Conf. Quantum Photonics, Artif. Intell., Netw., QPAIN**Conference Name** 2025 IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking, QPAIN 2025**Date Added** 20/09/2026, 8:35:35 pm

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Contrastive Learning Based Unsupervised Sentence Embeddings for Hinglish

Item Type Conference Paper

Author A. Chaudhari

Author P. Kushare

Abstract Hinglish language is a hybrid mixture of Hindi and English text in Latin script. According to a survey by a company named milestone localization, 58 % of the population of India who are using technology prefer reading and sending social media messages in Hinglish. Sentence embeddings are basically the vector representation of that sentence in numerical form, which computers can understand. There are a lot of applications where embeddings are mandatory. The current state of the art sentence embedding model, SBERT (sentence Bert) is trained on specific English-supervised labeled datasets, STS (semantic textual similarity), and NLI (natural language inference). These datasets are in paired format and require time-consuming and expensive human annotations. So building these models for low-resourced languages like Hindi becomes hard because there aren't enough specialized datasets. The proposed work is a slight modification of unsupervised model SimCSE (Simple Contrastive Learning of Sentence Embeddings), which was introduced in 2021 by authors Tianyu Gao et al. The work revolves around finetuning an unsupervised embedding model on approximately 10,000 hinglish sentences. We evaluated a couple of transformers based on SimCSE with a Hinglish transliterated version of standard STS data. Out of all the transformer DistilBERT model performed best with a Spearman's similarity score of 0.60. A comparative analysis of the embedding quality of different transformer models is also discussed. © 2023 IEEE.

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CorEDs: a Corpus on Eating Disorders

Item Type Conference Paper
Author M. Donati
Author C. Strapparava
Editor Kokkinakis D.
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Abstract Eating disorders (EDs) constitute a widespread group of mental illnesses affecting the everyday life of many individuals in all age groups. One of the main difficulties in the diagnosis and treatment of these disorders is the interpersonal variability of symptoms and the variety of underlying psychological states that are not considered in traditional approaches. In order to gain a better understanding of these disorders, many studies have collected data from social media and analysed them from a computational perspective, but the resulting dataset were very limited and task-specific. Aiming to address this shortage by providing a dataset that could be easily adapted to different tasks, we built a corpus collecting ED-related and ED-unrelated comments from Reddit focusing on a limited number of topics (fitness, nutrition, etc.). To validate the effectiveness of the dataset, we evaluated the performance of two classifiers in

distinguishing between ED-related and unrelated comments. The high-level accuracy of both classifiers indicates that ED-related texts are separable from texts on similar topics that do not address EDs. For explorative purposes, we also carried out a linguistic analysis of word class dominance in ED-related texts, whose results are consistent with the findings of psychological research on EDs.
© European Language Resources Association (ELRA)

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Conference Name Proceedings - 4th RaPID Workshop: Resources and Processing of Linguistic, Para-Linguistic and Extra-Linguistic Data from People with Various Forms of Cognitive/Psychiatric/Developmental Impairments, as part of the 13th Edition of the Language Resources and Evaluation Conference, LREC 2022

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Corpus Creation for Racial Hoax in Code-Mixed Hindi-English Low Resource Text

Item Type Conference Paper

Author S.S. Dhawale

Author R. Ponnusamy
Author Y.R. Kale
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Editor Chakravarthi B.R.
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Editor Eger S.
Editor Wahyu Pamungkas E.
Editor Dobrovoljc K.

Abstract Warning: This paper contains derogatory language that may be offensive to some readers. As a type of misinformation, hoaxes seek to propagate incorrect information in order to gain popularity on social media. Racial hoaxes are a particular kind of hoax that is particularly harmful since they falsely link individuals or groups to crimes or incidents because of their race. This involves identifying fabricated or misleading statements that falsely implicate racial groups in negative actions. One of these applications is the study of social media video thoughts based on comments from viewers. On the other hand, social media comments frequently incorporate many languages and are written in scripts that are not native to the user. They also rarely adhere to inflexible grammar norms. Lack of code-mixed annotated data for a Low-resource languages like Code-Mixed Hindi and English make this issue more challenging. In order to address this, we generated a racial hoax-annotated, code-mixed corpus of 2,611 YouTube comment postings in Hindi-English. We outline the method of building the corpus by annotation guidelines and assigning the binary values indicating the presence of racial hoax. The study also evaluates the performance of various deep learning models, in detecting these hoaxes. This work contributes a novel dataset for low-resource, code-mixed language contexts and establishes baseline results for racial hoax classification, providing a valuable resource for future research in multilingual misinformation detection. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Language English

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Conference Name Communications in Computer and Information Science

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Codes (symbols), Deep learning, Performance, Classification (of information), Learning systems, Natural language processing systems, Social networking (online), Social media, Social Media, Linguistics, YouTube, Text processing, Multimedia systems, Low resource languages, Corpus, Binary values, Code-mixed, Code-Mixed, Fake detection, Hindi-english, Hindi-English, Racial hoax, Racial Hoax, Youtube comment, YouTube Comments

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S.S. Dhawale; School of Computer Science, University of Galway, Galway, Ireland; email: shanu.dhawale11@gmail.com; Conference name: 3rd International Conference on Speech and Language Technologies for Low-Resource Languages, SPELLL 2024; Conference date: 4 December 2024 through 6 December 2024; Conference code: 342699

Corpus Creation for Sentiment Analysis in Code-Mixed Tulu Text

Item Type Conference Paper

Author A. Hegde

Author M.D. Anusha

Author S. Coelho

Author H.L. Shashirekha

Author B.R. Chakravarthi

Editor Melero M.

Editor Sakti S.

Editor Soria C.

Abstract Sentiment Analysis (SA) employing code-mixed data from social media helps in getting insights to the data and decision making for various applications. One such application is to analyze users' emotions from comments of videos on YouTube. Social media comments do not adhere to the grammatical norms of any language and they often comprise a mix of languages and scripts. The lack of annotated code-mixed data for SA in a low-resource language like Tulu makes the SA a challenging task. To address the lack of annotated code-mixed Tulu data for SA, a gold standard trilingual code-mixed Tulu annotated corpus of 7,171 YouTube comments is created. Further, Machine Learning (ML) algorithms are employed as baseline models to evaluate the developed dataset and the performance of the ML algorithms are found to be encouraging. © European Language Resources Association (ELRA), licensed under CC-BY-NC-4.0.

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Language English

Extra Journal Abbreviation: Annu. Meet. ELRA/ISCA Spec. Interest Group Under-Resour. Lang., SIGUL - held conjunction Int. Conf. Lang. Resour. Eval., LREC - Proc.

Publisher European Language Resources Association (ELRA)
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Pages 33-40
Proceedings Title Annu. Meet. ELRA/ISCA Spec. Interest Group Under-Resour. Lang., SIGUL - held conjunction Int. Conf. Lang. Resour. Eval., LREC - Proc.
Conference Name 1st Annual Meeting of the ELRA/ISCA Special Interest Group on Under-Resourced Languages, SIGUL 2022 - held in conjunction with the International Conference on Language Resources and Evaluation, LREC 2022 - Proceedings
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Codes (symbols), Decisions makings, Decision making, Sentiment analysis, Machine learning, Learning algorithms, Social networking (online), Machine learning algorithms, Social media, Sentiment Analysis, Linguistics, YouTube, Code-mixed, Mixed data, Corpus creation, Trilingual, Tulu

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Council of LLMs: Evaluating Capability of Large Language Models to Annotate Propaganda

Item Type Conference Paper

Author V. Sharma

Author S. Jain

Author M.M. Shokri

Author S.I. Levitan

Author E. Filatova

Abstract Data annotation is essential for supervised natural language processing tasks but remains labor-intensive and expensive. Large language models (LLMs) have emerged as promising alternatives, capable of generating high-quality annotations either autonomously or in collaboration with human annotators. However their use in autonomous annotations is often questioned for their ethical take on subjective matters. This study investigates the effectiveness of LLMs in an autonomous, and hybrid annotation setups in propaganda detection. We evaluate GPT and open-source models on two datasets from different domains, namely, Propaganda Techniques Corpus (PTC) for news articles and the Journalist Media Bias on X (JMBX) for social media. Our results show that LLMs, in general, exhibit high recall but lower precision in detecting propaganda, often over-predicting persuasive content. Multi-annotator setups did not outperform the best models in single-annotator setting although it helped reasoning models boost their

performance. Hybrid annotation, combining LLMs and human input, achieved the highest overall accuracy than LLM-only settings. We further analyze misclassifications and found that LLM have higher sensitivity towards certain propaganda techniques like loaded language, name calling, and doubt. Finally, using error typology analysis, we explore the reasoning provided on misclassifications by the LLM. Our result shows that although some studies report LLM outperforming manual annotations and it could prove useful in hybrid annotation, its incorporation in the human annotation pipeline must be implemented with caution. © 2026 Association for Computational Linguistics.

Date 2026

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Extra Journal Abbreviation: WASSA - Workshop Comput. Approach. Subj., Sentim. Soc. Media Anal., Proc. Workshop

Publisher Association for Computational Linguistics (ACL)

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Pages 1-12

Proceedings Title WASSA - Workshop Comput. Approach. Subj., Sentim. Soc. Media Anal., Proc. Workshop

Conference Name WASSA 2026 - 15th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis, Proceedings of the Workshop

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CRAB: Assessing the Strength of Causal Relationships Between Real-World Events

Item Type Conference Paper

Author A. Romanou

Author S. Montariol

Author D. Paul

Author L. Laugier

Author K. Aberer

Author A. Bosselut

Editor Bouamor H.

Editor Pino J.

Editor Bali K.

Abstract Understanding narratives requires reasoning about the cause-and-effect relationships between events mentioned in the text. While existing foundation models yield impressive results in many NLP tasks requiring reasoning, it is unclear whether they understand the complexity of the underlying network of causal relationships of events in narratives. In this work, we present CRAB, a new Causal Reasoning Assessment Benchmark designed to evaluate causal understanding of events in real-world narratives. CRAB contains fine-grained, contextual causality annotations for ~ 2.7K pairs of real-world events that describe various newsworthy event timelines (e.g., the acquisition of Twitter by Elon Musk). Using CRAB, we measure the performance of several large language models, demonstrating that most systems achieve poor performance on the task. Motivated by classical causal principles, we also analyze the causal structures of groups of events in CRAB, and find that models perform worse on causal reasoning when events are derived from complex causal structures compared to simple linear causal chains. We make our dataset and code available to the research community. ©2023 Association for Computational Linguistics.

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Language English

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Conference Name EMNLP 2023 - 2023 Conference on Empirical Methods in Natural Language Processing, Proceedings

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Notes:

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Crisis communication and media influence during Nokia water contamination event

Item Type Journal Article

Author V. Kumar

Author F. Hopfgartner

Author P.M. Rossi

Author M. Oussalah

Abstract This study analyzes public narratives, stakeholder communication, and media framing during the Nokia water contamination crisis (2007–2008), Finland’s largest waterborne epidemic. Especially, using data from online forum (Suomi24), news reports, and official audits, we apply a new aspect-based sentiment analysis combined with thematic clustering over a 3-month crisis timeline period to trace the evolution of public perception in response to crisis events and stakeholder actions. The findings reveal that positive sentiment clusters were driven by visible mitigation efforts and hygiene-related practices, while negative sentiments were centered around contaminated water, institutional mistrust, and health fears. Media analysis showed minimal bias, with most coverage remaining factual, though occasional sensationalism could have amplified public anxiety. Compared to previous institutional communication research, our findings provide a data-driven perspective on citizen discourse-based analysis, offering new insights into how digital platforms can serve as real-time indicators of public trust and emotional response. The study underscores the importance of timely, transparent, and coordinated communication and highlights the potential of aspect-based sentiment analysis as a tool for adaptive crisis management and comprehending stakeholder actions. © 2025 The Authors.

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CRISIS: Classification of Response Intent via Social Media for Improved Support

Item Type Conference Paper

Author J. Schmidt

Author A. Romano

Author A. Adesokan

Author S. Madria

Abstract Social media delivers vital real-time signals during disasters; however, many crisis response models overlook communicative intent, a crucial factor in prioritizing actionable information. We present CRISIS, a transformer-based framework designed to classify disaster-related tweets into three operationally significant intent categories: Report (factual updates), Awareness (advisory or emotional messages), and Other (irrelevant or unclear content). By fine-tuning BERT with an innovative Disentangled Attention module, CRISIS effectively distinguishes between tweet semantics, sentiment, and noise without relying on cross-attention mechanisms. We enhance the FReCS dataset with intent annotations, facilitating robust evaluation across various disaster benchmarks. Our model consistently surpasses strong baselines (e.g., RoBERTa, BERTweet) across four datasets, demonstrating effective generalization even in low-resource environments. Consequently, CRISIS provides an interpretable, scalable, and domain-adaptable solution for integrating intent-aware classification into real-time disaster informatics workflows. © 2025 IEEE.

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Language English

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Pages 2880-2889

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Conference Name IEEE International Conference on Data Mining Workshops, ICDMW
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Information systems, Classification (of information), Learning systems, Transformer modeling, Transformer Models, Social networking (online), Social media, Semantics, Information management, Social media analysis, Informatics, Disasters, Social Media Analysis, Tweets, Crisis informatics, Communicative intent, Crisis Informatics, Crisis response, Disentangled attention, Disentangled Attention, Intent classification, Intent Classification, Real-time signals, Response model

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Export Date: 20 September 2026; Cited By: 0; Conference name: 25th IEEE International Conference on Data Mining Workshops, ICDMW 2025; Conference date: 12 November 2025 through 15 November 2025; Conference code: 221413

Crop the Way You Like: Personalized Image Cropping by Integrating Subjective and Objective Features

Item Type Journal Article

Author S. Zhang

Author X. Yang

Author X. Bai

Abstract Image cropping aims to find more attractive crops for users by recomposing images. With the development of mobile social media, the significance of this technology is increasingly evident. Most existing methods focus on learning from average cropping results and rely on objective image features for image cropping. However, the subjectivity of image cropping and the preferences of people are often overlooked in this paradigm. In this work, we analyze the difference between individual users and average users, proposing a new task called personalized image cropping (PIC). Compared to existing methods, the PIC model not only requires generic cropping features but also captures personalized preferences from a small set of user data. Nevertheless, as a typical few-shot learning problem, direct training is inherently challenging. To address this issue, we first construct a prior model that includes both user subjective features and image objective features. Considering the difficulty in explicitly modeling subjective features, we propose an individual preference module to embed user attributes into a continuous latent space and explore their cropping preferences. Then, the model is jointly trained on PIC tasks to integrate two features, facilitating the transfer of prior knowledge to evaluate user-specific cropping. Finally, by fine-tuning with a small set of annotations, we can obtain the individualized PIC model for each user. Extensive quantitative and qualitative experiments demonstrate the necessity of the PIC task and the effectiveness of our proposed method. ©

1999-2012 IEEE.

Date 2026**DOI** 10.1109/TMM.2026.3651046**URL** <https://www.scopus.com/pages/publications/105027322349?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Institute of Electrical and Electronics Engineers Inc.**Publication** IEEE Transactions on Multimedia**ISSN** 15209210 (ISSN)**Journal Abbr** IEEE Trans Multimedia**Date Added** 20/09/2026, 8:35:12 pm**Modified** 20/09/2026, 8:35:12 pm**Tags:**

Learning systems, User's preferences, Crops, Image cropping, Image features, Learning problem, Mobile social medias, Personalized image cropping, personalized prediction, Personalized prediction, photo composition, Photo compositions, prior model, Prior modeling, User data, user preference

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: X. Yang; Xi'an Jiaotong University, School of Computer Science and Technology, Xi'an, 710049, China; email: yxyphd@mail.xjtu.edu.cn; CODEN: ITMUF

Cross Domain Sentiment Analysis in Indonesian Using a Transformer-Based Language Model

Item Type Conference Paper**Author** M.N. Akbar**Author** A.F. Wicaksono**Author** I. Budi

Abstract Cross-domain sentiment analysis (CDSA) remains a challenging task, particularly for low-resource languages like Indonesian, where domain shifts can significantly hinder model performance. While pre-trained language models offer promising capabilities, there is a need to understand the most effective adaptation strategies for the Indonesian context. This study compares a traditional machine learning baseline (ExtraTree) against language-specific (IndoBERT) and multilingual (XLM-R) Transformer models for Indonesian CDSA. Using single and multi-source adaptation strategies across four distinct domains of web reviews and social media text, our experiments demonstrate that Transformer-based models consistently outperform the baseline. However, this higher accuracy comes with a significant computational tradeoff, as the baseline is substantially more efficient in both training and inference time. The results show that for single-source

adaptation, IndoBERT achieves the best performance with an average Macro F1-score of 0.6851. In contrast, for multi-source adaptation, XLM-R yields the top result with a macro F1-score of 0.7351. © 2025 IEEE.

Date 2025

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Cross Platform MultiModal Retrieval Augmented Distillation for Code-Switched Content Understanding

Item Type Conference Paper

Author S. Thapa

Author H. Veeramani

Author I. Razzak

Author R.K.-W. Lee

Author U. Naseem

Abstract In the era of digital communication and social media, the proliferation of multimodal content, such as code-switched memes, has become a ubiquitous form of expression. This phenomenon is especially significant for low-resource languages like Nepali, where the need for sentiment analysis and hate speech

detection remains unmet due to the unavailability of publicly available datasets. To address this gap, we provide ENeMeme, an annotated dataset of 4, 211 code-switched memes in the Nepali-English language for sentiment and hate speech. While the previous state-of-the-art methods of meme analysis particularly focus on high-resource language, they fail to perform well in low-resource language. To bridge this gap, our paper also builds on existing literature to adapt a novel multimodal model, MM-RAD, designed to understand code-switched Nepali-English memes, leveraging both textual and visual content. The model's effectiveness is analyzed across various retrieval platforms. Our proposed MM-RAD demonstrates superior performance in sentiment analysis and hate speech detection compared to individual baseline models. The dataset can be availed through <https://github.com/therealthapa/crossplatform>. © 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Speech communication, Codes (symbols), Visual languages, Digital communication systems, Sentiment analysis, Speech recognition, Natural Language Processing, Language processing, Natural language processing, Natural languages, Data mining, Multi-modal, Speech detection, Low resource languages, Code-switched language, Code-Switched Languages, Meme Analysis, Meme analyze, Multimodal processing, Multimodal Processing

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EID: 2-s2.0-105009231798

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Cross-Lingual and Multilingual Approaches to Fake News Detection in the Kazakh Language

Item Type Conference Paper

Author D. Telman

Author A. Yerimbetova

Author M. Sambetbayeva

Author E. Daiyrbayeva

Author A. Bazarkulova

Author A. Turganbayev

Author B. Bolatov

Editor Shaalan K.

Editor El-Beltagy S.R.

Abstract This paper explores the problem of fake news detection in the Kazakh language under a limited resource environment. A manually annotated corpus was developed to support analysis. The corpus contains 1106 news stories and social media posts, of which 591 are true and 515 are false. 1081 texts are written in Kazakh. 3 experimental configurations were examined. In Experiment 1, when an English fact-checking model was applied to an automatically translated Kazakh article, the F1 score decreased by 49.7%, revealing the seriousness of the semantic distortion caused by the translation. In Experiment 2, we constructed a Kazakh baseline using only TF-IDF features using logistic regression, and achieved an accuracy of 0.65 and an F1 score of 0.63. This suggests that lexical cues provide some signal for distinguishing between real and fake news, but the overall discriminatory power remains limited. The final configuration employed a multilingual contextual transformer (XLM-RoBERTa) trained directly with Kazakh data. It reproduced the reference performance (F1 = 0.97) and, at the same time, produced a modest improvement in recovery. These results demonstrate that multilingual models can effectively process Kazakh without translation, while translation processes introduce significant semantic biases. The dataset and assessment framework serve as one of the early quantitative benchmarks for fake news detection in Kazakh, offering a basis for further work on cross-language disinformation. © 2025 The Authors. Published by Elsevier B.V.

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Cross-Lingual Sentiment Analysis Using Pre-Trained Transformer Models for Multilingual Social Media Data

Item Type	Conference Paper
Author	P. Srilakshmi
Author	D.R. Babu
Author	K.R. Chaganti
Author	P. Padma
Author	S.R. Reddy
Author	S.M.P. Qubeb
Abstract	Social media content growth at a rapid pace produces an escalating need for efficient sentiment analysis systems which can transfer between languages. The standard sentiment analysis systems encounter difficulties with multilingual adoption because they need different types of linguistic formats and vocabulary which require significant labeled dataset creation for each language. This research investigates the performance of three pretrained transformer models namely mBERT and XLM-R and mT5 in order to classify sentiment across languages. Sentiment classification tasks can be performed using multilingual embeddings on these models which require limited languagespecific training. The models undergo performance evaluation across different multilingual sentiment datasets in zero-shot, fewshot and fully fine-tuned circumstances. Experimental findings show that mT5 achieved higher performance compared to mBERT and XLM-R in most cases with particular advantages for operating on low-resource languages while XLM-R balances accuracy with operational efficiency. The performance boost from fine-tuning sentiment classification reaches 10 - 15% but the process demonstrates higher computational requirements. The research shows that transformer-enabled solutions represent an efficient method for multilingual sentiment analysis which has applications in social media oversight and opinion

extraction and worldwide opinion tracking systems. Research needs to concentrate on minimizing cross-lingual transfer learning hassles while making the system quicker for real-time operations and handling sentiment categorization across specific domains. © 2025 IEEE.

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Cross-lingual sentiment analysis via multimodal transformer fusion and lightweight deep ensemble learning framework

Item Type Journal Article

Author R. Yadav

Author D.K. Yadav

Author A. Jain

Author L. Prasad

Abstract The increasing availability of multilingual and multimodal data on online platforms has created a need for the development of sophisticated models for cross-lingual sentiment analysis. Although existing sentiment analysis models are largely dependent on monolingual text data, they tend to fail in generalizing across languages and modalities, especially in low-resource and code-switching settings. This work aims to handle the most important task of combining heterogeneous modalities of text

and audio data in multiple languages for accurate and interpretable sentiment classification. X-Sent-Fuse, a cross-lingual sentiment fusion model using Transformers, combines text and audio modalities to make accurate, robust, and culturally informed sentiment predictions for a wide range of languages. The architecture incorporates a Multimodal Transformer Cross-Attentive Gated Fusion (MT-CAG2F) module to effectively align semantic and prosodic features, followed by a BiSRO-Net classifier, to ensure stable and high-performing predictions. A lightweight ensemble of multilingual large language models further improves linguistic adaptability. Extensive evaluations on four benchmark datasets, Sentiment analysis in Twitter, Amazon Reviews, mTEDx, and CMU-MOSEAS, demonstrate state-of-the-art performance, achieving up to 98.67% overall accuracy and 0.996 AUC, with consistently high precision, recall, and F1-scores. These results validate the effectiveness of X-Sent-Fuse in handling diverse and noisy cross-lingual inputs, offering a scalable solution for real-world applications such as multilingual customer feedback analysis, cross-cultural media monitoring, and affective computing in global communication platforms. © The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2026.

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Cross-Lingual Sentiment Classification in Sustainable Mobility: A Zero-Shot Domain Transfer Evaluation Framework

Item Type Journal Article

Author A. Serna

Author J.K. Gerrikagoitia

Author J. de Oña

Abstract This study evaluates zero-shot domain transfer for multilingual sentiment analysis in sustainable urban mobility using XLM-RoBERTa, a transformer pre-trained on social media data and applied to transport reviews without task- or domain-specific fine-tuning. Starting from a manually annotated English corpus of 375 transport-related user reviews, we created sentence-aligned translations in Spanish, French, German, and Italian, yielding a multilingual evaluation dataset of 1875 instances. Results show that the model assigns consistently high confidence to polarized content (mean: 0.76–0.85) and lower confidence to neutral or ambiguous expressions (0.58–0.65), with visible but preliminary cross-lingual variations that require further linguistic validation. Confidence scores are treated as diagnostic indicators of model certainty, not as evidence of correctness or calibration. A qualitative analysis of 113 categorized low-confidence predictions identifies six recurring linguistic patterns associated with model uncertainty (led by translation drift, mixed sentiment, and idiomatic expressions) with substantial inter-annotator agreement ($\kappa = 0.664$). By releasing the annotated multilingual dataset and code publicly, this work provides a reproducible exploratory evaluation framework for annotation-scarce, domain-specific multilingual NLP. © 2026 by the authors.

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Cross-Modal Rationale Transfer for Explainable Humanitarian Classification on Social Media

Item Type Conference Paper

Author T.H. Nguyen

Author K. Rudra

Author W. Nejdl

Abstract Advances in social media data dissemination enable the provision of real-time information during a crisis. The information comes from different classes, such as infrastructure damages, persons missing or stranded in the affected zone, etc. Existing methods attempted to classify text and images into various humanitarian categories, but their decision-making process remains largely opaque, which affects their deployment in real-life applications. Recent work has sought to improve transparency by extracting textual rationales from tweets to explain predicted classes. However, such explainable classification methods have mostly focused on text, rather than crisis-related images. In this paper, we propose an interpretable-by-design multimodal classification framework. Our method first learns the joint representation of text and image using a visual language transformer model and extracts text rationales. Next, it extracts the image rationales via the mapping with text rationales. Our approach demonstrates how to learn rationales in one modality from another through cross-modal rationale transfer, which saves annotation effort. Finally, tweets are classified based on extracted rationales. Experiments are conducted over CrisisMMD benchmark dataset, and results show that our proposed method boosts the classification Macro-F1 by 2-35% while extracting accurate text tokens and image patches as rationales. Human evaluation also supports the claim that our proposed method is able to retrieve better image rationale patches (12%) that help to identify humanitarian classes. Our method adapts well to new, unseen datasets in zero-shot mode, achieving an accuracy of 80%. © 2026 Owner/Author.

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Cross-Platform Sentiment Intelligence for Adolescent Health: A Modular Pipeline Approach Inspired by System-of-Systems Principles Using Large Language Models

Item Type Journal Article**Author** A. Shifa**Author** G.G.M. Nawaz Ali**Author** D.E. Henricks

Abstract Social media platforms shape social media sentiment related to adolescent health topics, including obesity. The proposed framework demonstrates how large language model (LLM)-based refinement enhances contextual interpretability in sentiment analysis. The pipeline performs collection, preprocessing, account classification into five source categories, lexicon scoring, LLM refinement applied to all cleaned comments, and visualization. We have evaluated the system on a curated corpus of approximately 21 000 public comments collected from TikTok ($n \approx 20,900$) and Instagram ($n \approx 238$). Statistical analyses including chi-square, Mann-Whitney U, and bootstrapped confidence intervals reveal platform-specific differences in sentiment polarity distributions: TikTok exhibits greater affective variability, while Instagram comments are concentrated toward moderately positive polarity. Finally, we conclude with design recommendations for platform-sensitive adolescent health communication and discuss ethical and operational considerations for deploying LLMs in public health analytics. © 2007-2012 IEEE.

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Cross-target stance detection: A survey of techniques, datasets, and challenges

Item Type Journal Article

Author P. Jamadi Khiabani

Author A. Zubiaga

Abstract Stance detection is the task of determining the viewpoint expressed in a text towards a given target. A specific direction within the task focuses on cross-target stance detection, where a model trained on samples pertaining to certain targets is then applied to a new, unseen target. With the increasing need to analyze and mine viewpoints and opinions online, the task has recently seen a significant surge in interest. This review paper examines research in cross-target stance detection over the last decade, highlighting the evolution from basic statistical methods to contemporary neural and LLM-based models. All this research has led to notable improvements in accuracy and adaptability. Innovative approaches include the use of topic-grouped attention and adversarial learning for zero-shot detection, as well as fine-tuning techniques that enhance model robustness. Additionally, prompt-tuning methods and

the integration of external knowledge have further refined model performance. A comprehensive overview of the datasets used for evaluating these models is also provided, offering valuable insights into the progress and challenges in the field. We conclude by highlighting emerging research directions and suggesting avenues for future work. © 2025 The Authors

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CueSupport-MH: An Explainable Dataset and Interpretable Framework for Mental Health Risk Detection and Response Safety Evaluation

Item Type Journal Article

Author F. Alotaibi

Author M. Sathvik

Author M. Younus

Author M. Lee

Abstract Online platforms have become an important medium for individuals to express emotional distress and seek support. Existing mental health NLP resources usually focus on risk classification alone and provide limited support for explaining risk cues or evaluating the safety of responses in conversational settings. This paper introduces CueSupport-MH, a 6,400-instance benchmark constructed from public Reddit conversations and annotated for four complementary dimensions: risk label, risk

evidence span, protective cue, and response-safety label. The revised dataset protocol specifies source selection, filtering, deduplication, anonymization, controlled augmentation, split construction, and redistribution constraints. We also propose Trace-MH, a lightweight interpretable framework for joint risk detection and response-safety classification. To avoid privileged-feature comparisons, we report both a text-only operational setting and a gold-span upper-bound setting that uses human evidence annotations only for controlled interpretability analysis. Across five random seeds, Trace-MH obtains a risk macro-F1 of 0.724 in the gold-span setting and 0.676 in the text-only setting, compared with 0.646 for MentalBERT. For response safety, Trace-MH achieves a macro-F1 of 0.752, with harmful-response detection remaining the most difficult class. Ablation studies show that evidence spans, protective cues, contextual features, and joint training each contribute measurable gains, while token-level F1 and intersection-over-union complement cosine similarity for rationale evaluation. The results support CueSupport-MH as a reproducible benchmark for explainable and safety-aware mental health NLP, while also emphasizing that the dataset and models are research tools and not clinical decision systems. © 2026 The Authors.

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CulturePark: Boosting Cross-cultural Understanding in Large Language Models

Item Type Conference Paper

Author C. Li

Author D. Teney

Author L. Yang

Author Q. Wen

Author X. Xie

Author J. Wang

Editor Globerson A.

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Editor Belgrave D.

Editor Fan A.

Editor Paquet U.

Editor Tomczak J.

Editor Zhang C.

Abstract Cultural bias is pervasive in many large language models (LLMs), largely due to the deficiency of data representative of different cultures. Typically, cultural datasets and benchmarks are constructed either by extracting subsets of existing datasets or by aggregating from platforms such as Wikipedia and social media. However, these approaches are highly dependent on real-world data and human annotations, making them costly and difficult to scale. Inspired by cognitive theories on social communication, this paper introduces CulturePark, an LLM-powered multi-agent communication framework for cultural data collection. CulturePark simulates cross-cultural human communication with LLM-based agents playing roles in different cultures. It generates high-quality cross-cultural dialogues encapsulating human beliefs, norms, and customs. Using CulturePark, we generated 41,000 cultural samples to fine-tune eight culture-specific LLMs. We evaluated these models across three downstream tasks: content moderation, cultural alignment, and cultural education. Results show that for content moderation, our GPT-3.5-based models either match or outperform GPT-4 on 41 datasets. Regarding cultural alignment, our models surpass GPT-4 on Hofstede's VSM 13 framework [Hofstede, 2013]. Furthermore, for cultural education of human participants, our models demonstrate superior outcomes in both learning efficacy and user experience compared to GPT-4. CulturePark proves an important step in addressing cultural bias and advancing the democratization of AI, highlighting the critical role of culturally inclusive data in model training. Code is released at <https://github.com/Scarelette/CulturePark>. © 2024 Neural information processing systems foundation. All rights reserved.

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Curation and Extraction of Drug-Related Entities from Reddit Platform

Item Type Conference Paper
Author Z. Wang
Author Z. Xu
Author Y. Wei
Author M. Chary
Author Y. Peng
Abstract Physicians learn primarily about illicit drugs from clinical overdose cases, limiting their understanding of real-world usage. Meanwhile, drug users share first-hand experiences online, offering insights into dosage and effects of drugs. To bridge this gap, we introduce ReDose (REddit Drug DOSe and Effect), a dataset of 6,435 Reddit posts on substance use. A board-certified toxicologist primarily annotated both the training and test sets, while two medical science students contributed to the test set, labeling DRUG, DOSE, and EFFECT entities. We benchmarked 6,267 annotations using BERT-based, large language model (LLM)-based, and Retrieval-Augmented Generation (RAG) models. BiomedBERT achieved an F1-score of 0.843 for DRUG, while Llama-3 70B outperformed GPT-4 (F1 = 0.79 vs. 0.72). EFFECT extraction remains challenging, with GPT-4 achieving a recall of 0.41. ReDose captures patient-curated narratives to advance medical data extraction from social media. © 2026 IEEE.
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CVT-FoMO: A Framework for the Association between Creator Induction Strategies and Collective FoMO in Cryptocurrency Video Subtitles

Item Type	Conference Paper
Author	K. Chen
Author	K. Jia
Abstract	To address the scarcity of fine-grained, dual-source data in cryptocurrency FoMO research, we introduce CVT-FoMO, a bilingual dataset of 16,187 transcript-comment pairs. We further propose an optimized prompting strategy integrating Role-Playing (RP), Few-Shot (FS), and Chain-of-Thought (CoT). Experiments demonstrate that this strategy achieves an 85% human win rate, significantly outperforming baselines. Further analysis reveals distinct asset-specific FoMO drivers (e.g., value anxiety for Bitcoin vs. meme-narratives for Dogecoin) and validates an "Induction-Resonance" mechanism, where creator agendas are statistically amplified by audience interactions ($k = 0.28$). This study provides a robust benchmark for sentiment analysis and platform risk governance. © 2026 Copyright held by the owner/author(s).

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Cyberbullying Detection on Social Media using Transformer-Based Language Models (BERT and RoBERTa)

Item Type Conference Paper

Author A. Wagji

Author A. Pattnaik

Author M.K. Rauniyar

Author S. Mohanty

Author S. Patra

Author D. Das

Abstract This paper proposes a BERT-based model for binary classification and a RoBERTa-based model for multiclass classification of tweet texts from a cyberbullying dataset. These approaches significantly enhance classification accuracy. Experimental results show that the BERT model achieves the highest

accuracy of 86% for binary classification compared to other models. For multiclass classification, RoBERTa outperforms LLaMA with an accuracy of 92%, primarily due to its optimized pretraining strategy and superior fine-tuning capabilities for text classification tasks. Challenges encountered during the study include managing communication overhead and ensuring consistent model updates. Overall, the study highlights the effectiveness of transformer-based models for accurate tweet text classification in cyberbullying detection. © 2026 IEEE.

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Cyberbullying victimization identification and large language model-assisted assessment: a study of cyberbullying victimization lexicon construction and validation

Item Type Journal Article

Author X. Liu

Author Y. Liao

Author F. Feng

Author Y. Tu

Author X. Kang

Author M. Liu

Author N. Han

Abstract Introduction – Cyberbullying poses a global mental health threat, yet its accurate identification remains challenging due to biases in self-reporting and help-seeking barriers. Methods – Based on large-scale social media data, the present study constructed a Chinese cyberbullying victimization lexicon from three dimensions—cyberbullying methods (the types of cyberbullying experienced by the individual), perceived harm (the harm perceived by the victim), and coping strategies (the behavioral responses adopted by the victim)—using Weibo texts, psychological lexicons, and cyberbullying questionnaires. This approach aims to improve the precision of victim identification and facilitate timely intervention. Lexicon validity was evaluated by examining correlations between word-frequency statistics derived from 500 Weibo posts and expert ratings ($n = 3$). In addition, based on 3, 442 RedNote posts, we preliminarily explored the lexicon's cross-platform applicability. To assess whether DeepSeek-R1 and GPT-4o could function as research assistants in dictionary construction, the present study replicated the manual lexicon development process—including text classification, vocabulary screening, weight assignment, and victimization severity assessment—and compared model outputs with human evaluations under simple and complex prompts using Cohen's Kappa, intraclass correlation coefficients (ICC), recall, and precision. Results – (1) The lexicon comprised 442 words across three dimensions: cyberbullying methods, perceived harm, and coping strategies. The lexicon demonstrated strong validity in identifying cyberbullying victimization expressions in social media text across each of the three sub-dimensions and the overall dimension (cyberbullying methods: $r=0.500$, $p < 0.001$; perceived harm: $r=0.408$, $p < 0.001$; coping strategies: $r=0.509$, $p < 0.001$; overall cyberbullying victimization expression: $r = 0.870$, $p < 0.001$), and effectively assessed the degree of cyberbullying victimization ($r = 0.533$, $p < 0.001$); (2) While DeepSeek-R1 performed well on small-scale text classification (Kappa = 0.775–0.781), both models showed significant limitations in large-scale processing (12, 600 entries). Vocabulary selection and weight assignment tasks revealed substantial discrepancies with human evaluation (Kappa as low as -0.874), though DeepSeek-R1 achieved moderate consistency in dimension partitioning (Kappa = 0.780). Discussion – As the first Chinese cyberbullying victimization dictionary, this research demonstrates that large language models offer preliminary utility in structured tasks but require human oversight for complex, large-scale research phases, supporting a human-machine collaborative approach for optimal outcomes. Copyright © 2026 Liu, Liao, Feng, Tu, Kang, Liu and Han.

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D-LexCan: A Dynamic Lexicon-Based Framework for Sentiment Analysis in Tarifit, a Low-Resource Multiscript Language

Item Type Journal Article

Author A. Amakssoum

Author F. Bouhafer

Author A. El Haddadi

Author A. Bahri

Abstract —Sentiment analysis for low-resource languages remains challenging due to limited annotated data, orthographic instability, informal writing practices, and the lack of dedicated linguistic resources, challenges that are particularly acute for Tarifit (Tamazight of the Rif), an under-resourced Amazigh language characterized by strong dialectal variation, pervasive multi-script usage, and highly noisy user-generated content on social media. This study introduces D-LexCan, a dynamic lexicon-based sentiment analysis framework that infers polarity directly from annotated corpus evidence without relying on predefined sentiment dictionaries or computationally intensive pretrained deep learning and transformer-based models. The framework combines deterministic multi-script normalization, unifying Arabic script, Tifinagh, and Arabizi into a single Tarifit Latin representation with automatic induction of sentiment-bearing unigrams and bigrams, while explicitly modeling

negation and amplification phenomena through linguistically motivated operators and preserving emojis as meaningful discourse-level sentiment cues. The approach is evaluated on a manually annotated social media corpus collected from multiple online platforms, where it achieves an accuracy of 0.8800 and a Macro-F1 score of 0.8798. The results outperform a static lexicon baseline with an accuracy of 0.5275, a classical machine-learning model based on TF-IDF and SVM with an accuracy of 0.8525, and neural architectures including BiLSTM with an accuracy of 0.7950. Experiments with frozen multilingual transformer encoders show accuracy ranging from 0.6725 to 0.7650. Fine-tuned multilingual transformers such as mBERT achieve competitive performance, reaching an accuracy of 0.8175. Overall, the results demonstrate that adaptive and linguistically grounded dynamic lexicon induction constitutes an effective, interpretable, and computationally efficient alternative for sentiment analysis in low-resource, noisy, and multi-script African language contexts. © (2026), (Science and Information Organization). All rights reserved.

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Export Date: 20 September 2026; Cited By: 0

DAO-LGBM: dual annealing optimization with light gradient boosting machine for advocates prediction in online customer engagement

Item Type Journal Article**Author** B. Abu-Salih**Author** S. Alotaibi**Author** R. Abukhurma**Author** M. Almiani**Author** M. Aljaafari

Abstract Social networks have modernized the way people communicate, share information, and consume content. The widespread use of social media platforms has resulted in the creation of vast amounts of user-generated content, which can be analyzed to gain valuable insights into customer behaviour, emotions, preferences, and trends. Previous studies on online customer engagement have mainly focused on brand perspective and its socially significant elements, such as brand personality, image, reputation, and loyalty. These studies have explored how these elements influence the behavioural engagement of customers, such as their purchase intentions, word-of-mouth recommendations, and repeat purchases. However, more recent research has started to shift towards a more customer-centric perspective, which acknowledges that customer engagement is a two-way process, involving both the brand and the customer. This approach considers the role of customer experiences, emotions, topics of interest, and motivations in shaping their social engagement with the brand. This paper contributes to these endeavours by developing a consolidated framework that incorporates various facets of the customer's emotional and behavioural social content. In particular, features of online customers have been extracted using various sophisticated modules that incorporate natural language inference, topic modelling, sentiment analysis, emotion detection, and the Big-Five Personality Traits. Further, a heuristic-based feature selection (FS) strategy, Dual Annealing Optimisation (DAO), is integrated with Light Gradient Boosting Machine (LGBM) to furnish a consolidated machine learning module (DAO-LGBM) that is implemented and examined to detect advocates in online customer engagement. A thorough examination of a proposed model and its utility for detecting advocates using rigorous evaluation metrics is undertaken, reported, and discussed. These findings have substantial implications for both academic research and practical applications in social media analytics. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024.

Date 2024**DOI** 10.1007/s10586-023-04220-6**URL** <https://www.scopus.com/pages/publications/85182157920?origin=resultslist>**Archive** Scopus**Language** English**Volume** 27**Publisher** Springer**Pages** 5047-5073**Publication** Cluster Computing**ISSN** 13867857 (ISSN)**Issue** 4**Journal Abbr** Cluster Comput.**Date Added** 20/09/2026, 8:35:30 pm**Modified** 20/09/2026, 8:35:30 pm**Tags:**

Optimisations, Sales, Feature extraction, Sentiment analysis, Social networking (online), BERT, Adaptive boosting, Optimization, NLP, Annealing, Customer advocacy, Dual annealing optimisation, Dual annealing optimization, Dual-annealing, Feature selection, Features selection, Gradient boosting, Light gradient boosting machine, Light gradients, Online customer engagement, Online customers

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Darijamachaaair: a large-scale multi-dimensional emotion and context dataset for Moroccan dialect social media texts

Item Type Journal Article

Author S. El Ouahabi

Author S. El Ouahabi

Author E.W. Dadi

Abstract This paper introduces DarijaMachaaair, a large-scale, multi-dimensional dataset for sentiment, emotion, and context analysis in Moroccan Darija, a widely spoken yet underexplored Arabic dialect. The dataset comprises over 117,870 social media comments collected from Facebook, YouTube, Instagram, and TikTok, covering multiple domains. Each instance was manually annotated by trained native speakers across three complementary dimensions: sentiment polarity (positive, negative, neutral), fine-grained emotions (anger, love, joy, neutral, sadness, confusion, optimism, fear, surprise, and ambiguity), and pragmatic contexts (disappointment, appeasing, neutral, sarcastic, and defensive). DarijaMachaaair stands out for supporting both Arabic and Arabizi scripts and for capturing authentic emotional expressions grounded in real-world Moroccan cultural discourse. With its combination of large scale and rich multi-dimensional annotations, the dataset provides a valuable benchmark for dialect-specific and emotion-aware natural language processing. This resource fills an important gap in Arabic NLP, particularly for North African dialects, enabling future research on emotionally intelligent applications such as chatbots, moderation tools, and computational sociolinguistic studies. © The Author(s), under exclusive licence to Springer Nature B.V. 2026.

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Data Sorting Influence on Short Text Manual Labeling Quality for Hierarchical Classification

Item Type Journal Article
Author O. Narushynska
Author V. Teslyuk
Author A. Doroshenko
Author M. Arzubov

Abstract The precise categorization of brief texts holds significant importance in various applications within the ever-changing realm of artificial intelligence (AI) and natural language processing (NLP). Short texts are everywhere in the digital world, from social media updates to customer reviews and feedback. Nevertheless, short texts' limited length and context pose unique challenges for accurate classification. This research article delves into the influence of data sorting methods on the quality of manual labeling in hierarchical classification, with a particular focus on short texts. The study is set against the backdrop of the increasing reliance on manual labeling in AI and NLP, highlighting its significance in the accuracy of hierarchical text classification. Methodologically, the study integrates AI, notably zero-shot learning, with human annotation processes to examine the efficacy of various data-sorting strategies. The results demonstrate how different sorting approaches impact the accuracy and consistency of manual labeling, a critical aspect of creating high-quality datasets for NLP applications. The study's findings reveal a significant time efficiency improvement in terms of labeling, where ordered manual labeling required 760 min per 1000 samples, compared to 800 min for traditional manual labeling, illustrating the practical benefits of optimized data sorting strategies. Comparatively,

ordered manual labeling achieved the highest mean accuracy rates across all hierarchical levels, with figures reaching up to 99% for segments, 95% for families, 92% for classes, and 90% for bricks, underscoring the efficiency of structured data sorting. It offers valuable insights and practical guidelines for improving labeling quality in hierarchical classification tasks, thereby advancing the precision of text analysis in AI-driven research. This abstract encapsulates the article's background, methods, results, and conclusions, providing a comprehensive yet succinct study overview. © 2024 by the authors.

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Data-Driven Democracy? Using Social Media and AI for Knowledge Co-production in Energy Transition Research

Item Type Conference Paper

Author I.M. Kousa

Abstract As global challenges grow increasingly difficult to address, the need for relevant knowledge in policymaking has become more critical than ever. In fields such as sustainability research, discussions about the co-production of knowledge emphasise the involvement of diverse stakeholders in generating relevant information for democratic decision-making processes. Particularly in technology-intensive areas like energy, the voices of experts have traditionally dominated, often marginalising the perspectives of citizens affected by energy policies and leading to epistemic injustices. Social media serves as a central forum for political activity and civic engagement, making it an important research environment for understanding the complexities of knowledge production and sharing. In my dissertation, I explore the opportunities and challenges of integrating citizen voices into the co-production of knowledge through social media using artificial intelligence (AI) based methods. My approach includes analysing stakeholder perspectives across social media, policymakers' speeches, and traditional media, using the discourse on energy justice in the context of sustainable energy transition as empirical material. Additionally, I conduct a comparative analysis of a rule-based ontological classification tool and a large language model (LLM) chatbot as qualitative content analysis tools. I demonstrate that social media data can reveal critical perspectives and marginalised discourses that might otherwise be overlooked. However, social media platforms are also used to spread disinformation and incite polarisation, which undermines their reliability as accurate reflections of prevailing attitudes and opinions in society. In addition, although AI methods can make information processing more effective and support the use of large datasets, it is important to consider their inherent limitations. I address the biases and ethical concerns associated with social media data and AI-based analysis tools and discuss potential solutions to mitigate these challenges. © 2025 Copyright is held by the owner/author(s). Publication rights licensed to ACM.

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Dataset creation and evaluation of aspect based sentiment analysis in telugu, a low resource language

Item Type Conference Paper

Author R.Y. Reddy

Author G.R.R. Reddy

Author R. Mamidi

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Abstract In recent years, sentiment analysis has gained popularity as it is essential to moderate and analyse the information across the internet. It has various applications like opinion mining, social media monitoring, and market research. Aspect Based Sentiment Analysis (ABSA) is an area of sentiment analysis which deals with sentiment at a finer level. ABSA classifies sentiment with respect to each aspect to gain greater insights into the sentiment expressed. Significant

contributions have been made in ABSA, but this progress is limited only to a few languages with adequate resources. Telugu lags behind in this area of research despite being one of the most spoken languages in India and an enormous amount of data being created each day. In this paper, we create a reliable resource for aspect based sentiment analysis in Telugu. The data is annotated for three tasks namely Aspect Term Extraction, Aspect Polarity Classification and Aspect Categorisation. Further, we develop baselines for the tasks using deep learning methods demonstrating the reliability and usefulness of the resource. © European Language Resources Association (ELRA), licensed under CC-BY-NC

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DCCL: Distance-coefficient guided Clustering with Contrastive Learning for Few-shot Text Classification

Item Type Conference Paper

Author H. Wang

Author C. Gu

Editor Shen W.

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Editor Luo J.

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Abstract Social media collaboration can be seen as an innovative collaborative approach that contains rich textual information, and categorizing them can make downstream collaboration tasks more efficient. Due to the limited number of samples, many previous methods fail to capture the rich semantic information. It is difficult to generate specific class vectors and predict the semantically similar samples. To address this issue, we propose a few-shot classification model based on contrastive learning. Firstly, we introduce a distance coefficient to dynamically guide the weight assignment and better divide the feature space for generating class-specific representations. Secondly, our method learns discriminative text representations via contrastive learning, optimizes the position of the samples in the metric space and alleviates the contradiction in the prediction process. We evaluate the performance of our model on three benchmark datasets and achieve an average improvement of 1.41%, 0.94%, and 0.73% compared with the baseline model. © 2024 IEEE.

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DeBERTa-GRU: Sentiment Analysis for Large Language Model

Item Type Journal Article

Author A. Assiri

Author A. Gumaedi

Author F. Mehmood

Author T. Abbas

Author S. Ullah

Abstract Modern technological advancements have made social media an essential component of daily life. Social media allow individuals to share thoughts, emotions, and ideas. Sentiment analysis plays the function of evaluating whether the sentiment of the text is positive, negative, neutral, or any other personal emotion to understand the sentiment context of the text. Sentiment analysis is essential in business and society because it impacts strategic decision-making. Sentiment analysis involves challenges due to lexical variation, an unlabeled dataset, and text distance correlations. The execution time increases due to the sequential processing of the sequence models. However, the calculation times for the Transformer models are reduced because of the parallel processing. This study uses a hybrid deep learning strategy to combine the strengths of the Transformer and Sequence models while ignoring their limitations. In particular, the proposed model integrates the Decoding-enhanced with Bidirectional Encoder Representations from Transformers (BERT) attention (DeBERTa) and the Gated Recurrent Unit (GRU) for sentiment analysis. Using the Decoding-enhanced BERT technique, the words are mapped into a compact, semantic word embedding space, and the Gated Recurrent Unit model can capture the distance contextual semantics correctly. The proposed hybrid model achieves F1-scores of 97% on the Twitter Large Language Model (LLM) dataset, which is much higher than the performance of new techniques. © 2024 Tech Science Press. All rights reserved.

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Decoding antisemitism online: linguistic and multimodal challenges in the age of AI

Item Type Journal Article

Author M.J. Becker

Author J. Blatter

Author O. Stanevich

Abstract Introduction: This article investigates the linguistic and computational challenges of detecting antisemitism in digital communication, integrating discourse-analytical and artificial intelligence (AI) perspectives. It conceptualizes antisemitic discourse as a continuum ranging from explicit incitement to implicit, coded expressions whose interpretation depends on contextual, cultural, and pragmatic knowledge. Methods: The study draws on empirical case studies from the Decoding Antisemitism project, analyzing YouTube reactions to two events: the Hamas terror attack of 7 October 2023 and the antisemitic double murder in Washington, D.C., in May 2025. Qualitative discourse analysis is combined with computational considerations related to annotation practices and model design for automated detection. Results: The analysis shows that antisemitic discourse has become normalized in mainstream digital spaces. Reactions to 7 October were characterized by open glorification of violence, whereas responses to the Washington case centered on denial, irony, and the inversion of victimhood. Together, these cases illustrate both the normalization and diversification of antisemitic communication online. Discussion: Building on these findings, the article discusses methodological and computational implications for antisemitism detection. It highlights challenges such as semantic ambiguity, pragmatic drift, multimodal signaling, and data scarcity, and evaluates emerging computational approaches, including transformer-based fine-tuning, retrieval-augmented systems, and context-engineered large language models (LLMs). The study concludes that effectively confronting digital antisemitism requires sustained

collaboration between linguists, data scientists, and policymakers to develop context-sensitive, transparent, and ethically grounded AI systems capable of reliable interpretive reasoning. Copyright © 2026 Becker, Blatter and Stanevich.

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Decoding Emotions in Text: A Comprehensive Review and a Novel Framework for Context-Aware Analysis

Item Type Conference Paper

Author J. Rawal

Author P. Nagale

Author P. Edke

Abstract The ability to computationally interpret human emotion from text is a critical frontier in artificial intelligence, underpinning the development of empathetic and socially aware systems. This paper presents a comprehensive review of the evolution of emotion detection using Natural Language Processing (NLP) and introduces a novel conceptual framework to address key limitations in current models. We systematically survey the landscape of methodologies, beginning with foundational lexiconbased and classical machine learning techniques, progressing through the deep learning era of CNNs and RNNs, and culminating in the state-of-the-art with transformer architectures and Large Language Models

(LLMs). The review further provides an essential overview of benchmark datasets and evaluation metrics that drive the field. To push beyond existing capabilities, particularly in capturing nuanced context, we propose a novel conceptual framework integrating a RoBERTa encoder with a specialized emotion-focused attention mechanism. We analyze key applications in mental health, customer experience, and social media, alongside a deep dive into persistent challenges such as sarcasm, multilinguality, and ethical considerations, including the need for Explainable AI (XAI). This work serves as both a detailed survey for researchers and a forward-looking proposal for building the next generation of emotionally intelligent systems. © 2025 IEEE.

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Decoding Hate: A Review of Identifying Hateful Content in Gujarati and Other Under-Resourced Indic Languages on Social Media

Item Type Conference Paper

Author P. Chauhan

Author P. Virparia

Editor Jain A.

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Abstract The rapid spread of hateful content on online platforms poses significant challenges to maintaining civility and safety. While extensive research has focused on high-resource languages like English, there is a lack of studies on low-resource languages. This review evaluates hate speech detection techniques specific to under-resourced languages in India, summarizing recent advancements and exploring text preprocessing, feature selection, and three categories of machine learning models: conventional machine learning, deep learning, and transformer learning. The study also addresses model generalization, available datasets, and limitations of previous research, with a particular focus on challenges faced by the Gujarati language. Additionally, it proposes solutions to enhance efforts against online hate speech highlight opportunities for future research in sentiment analysis and hate speech detection in less-resourced Indian languages specifically Gujarati language. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Deep learning and NLP in cryptocurrency forecasting: Integrating financial, blockchain, and social media data

Item Type Journal Article

Author V. Gurgul

Author S. Lessmann

Author W.K. Härdle

Abstract We introduce novel approaches to cryptocurrency price forecasting, leveraging Machine Learning (ML) and Natural Language Processing (NLP) techniques, with a focus on Bitcoin and Ethereum. By analysing news and social media content, primarily from Twitter and Reddit, we assess the impact of public sentiment on cryptocurrency markets. A distinctive feature of our methodology is the application of the BART MNLI zero-shot classification model to detect bullish and bearish trends, significantly advancing beyond traditional sentiment analysis. Additionally, we systematically compare a range of pre-trained and fine-tuned deep learning NLP models against conventional dictionary-based sentiment analysis methods. Another key contribution of our work is the adoption of local extrema alongside daily price movements as predictive targets, reducing trading frequency and portfolio volatility. Our findings demonstrate that integrating textual data into cryptocurrency price forecasting not only improves forecasting accuracy but also consistently enhances the profitability and Sharpe ratio across various validation scenarios, particularly when applying deep learning NLP techniques. The entire codebase of our experiments is available via an online repository: <https://anonymous.4open.science/r/crypto-forecasting-public>. © 2025 The Author(s)

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Deep learning and sentence embeddings for detection of clickbait news from online content

Item Type Journal Article

Author A. Muqadas

Author H.U. Khan

Author M. Ramzan

Author A. Naz

Author T. Alsahfi

Author A. Daud

Abstract With the rise of user-generated content, ensuring the authenticity and originality of online information has become increasingly challenging. Artificial intelligence (AI) and Natural Language Processing (NLP) play a crucial role in large-scale content analysis and moderation. However, the widespread use of clickbait—sensational or misleading headlines designed to maximize engagement—undermines the reliability of shared information. The existing studies focus on news clickbait detection from English content using NLP techniques. To the best of our knowledge, this study is novel to focus on news clickbait detection from Urdu language content. We propose to use state of the art deep features including sentence embeddings to be applied as input to deep learning models. The dataset is prepared from authentic online source, labelled by domain experts, and pre-processed using standard steps. In contrast, traditional models, including machine learning and ensemble learning, utilize textual features and word embedding features are used as baseline models for comparing the performance of the proposed deep learning approaches. All models are evaluated using standard performance measures, including accuracy, precision, recall, F1-score, and ROC curve analysis, to determine their effectiveness in identifying clickbait in Urdu news headlines. The results show that the Bi-LSTM model with sentence embeddings achieved the highest accuracy of 88% for clickbait identification in low resource language. © The Author(s) 2025.

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Deep learning techniques for sentiment analysis in code-switched Hausa-English tweets

Item Type Journal Article

Author Y. Aliyu

Author A. Sarlan

Author K.U. Danyaro

Author A. Sani abd Rahman

Author A.A. Muazu

Author M.Y. Abubakar

Abstract Social media serve as a crucial platform for expressing opinions and perspectives. Its texts often characterised by code-switching or mixed languages in multilingual setting. This results in a diverse and complex linguistic context, which can negatively affect the accuracy of sentiment analysis for low-resource languages such as Hausa. Prior research has predominantly concentrated on sentiment analysis within single-language data rather than code-switched data. This paper proposes an efficient hyperparameter tuning framework and a novel stemming algorithm for the Hausa language. The framework leverages word embeddings to determine the polarity scores of code-mixed tweets and enhances the accuracy of sentiment analysis models in low-resource language. The extensive experiments demonstrate the framework's efficiency and reveal a superior performance of transformer models over

conventional deep learning models. The framework achieves a balance between accuracy and computational efficiency, making it suitable for deployment in practical applications. Compared to state-of-the-art transformer models, our framework significantly reduces computational costs while maintaining competitive performance. Notably, the AfriBERTa model achieves outstanding results, with an F1-score of 0.92 and an accuracy of 0.919, surpassing current baseline standards. These findings have broad implications for social media monitoring, customer feedback analysis, and public sentiment tracking, enabling more inclusive and accessible NLP tools for underrepresented linguistic communities. © 2025 The Author(s)

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Deep Learning-Based Depression Detection from Social Media: Comparative Evaluation of ML and Transformer Techniques

Item Type Journal Article

Author B.G. Bokolo

Author Q. Liu

Abstract Detecting depression from user-generated content on social media platforms has garnered significant attention due to its potential for the early identification and monitoring of mental health issues. This paper presents a comprehensive approach for

depression detection from user tweets using machine learning techniques. The study utilizes a dataset of 632,000 tweets and employs data preprocessing, feature selection, and model training with logistic regression, Bernoulli Naïve Bayes, random forests, DistilBERT, SqueezeBERT, DeBERTa, and RoBERTa models. Evaluation metrics such as accuracy, precision, recall, and F1 score are employed to assess the models' performance. The results indicate that the RoBERTa model achieves the highest accuracy ratio of 0.981 and the highest mean accuracy of 0.97 (across 10 cross-validation folds) in detecting depression from tweets. This research demonstrates the effectiveness of machine learning and advanced transformer-based models in leveraging social media data for mental health analysis. The findings offer valuable insights into the potential for early detection and monitoring of depression using online platforms, contributing to the growing field of mental health analysis based on user-generated content. © 2023 by the authors.

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Deep-AP: An efficient multi-task architecture for Author Profiling and Depression Detection

Item Type Journal Article

Author I. Oulahbib

Author M. Benhaddi

Author S. El Hadaj

Abstract Deep Author Profiling (Deep-AP) is a unified multi-task deep learning architecture that jointly infers age and gender at the author level and depression at the publication level from social media text. Existing approaches commonly treat Author Profiling (AP) and Depression Detection (DP) as independent problems, which increases model redundancy and computational cost and limits scalable deployment. To address this gap, Deep-AP adopts a shared-encoder, multi-head formulation that enables parameter sharing across heterogeneous tasks while preserving task-specific outputs. The study follows a systematic methodology that combines large-scale Reddit data collection, automatic silver-standard annotation via regular expressions and subreddit supervision, and user-level aggregation of multiple posts to support author-centric inference. Deep-AP processes up to twelve publications per user through a single backbone encoder (GRU, BiGRU, BERT variants, or LLaMA-3.2), producing post-level representations for DP and aggregated user-level representations for demographic prediction. We also introduce Deep Author Profiling Username Enhanced (Deep-AP-uname), which incorporates username embeddings as auxiliary input to enrich demographic inference. Experiments across backbone models show that BERT-large-based Deep-AP achieves F1-scores of 0.914, 0.717, 0.881 for gender, age, and depression respectively. These findings demonstrate that joint learning can deliver strong predictive performance while improving practicality, reducing inference time by 66% and memory usage by 40% compared to parallel single-task BERT-large models. © 2026 The Authors. Published by Elsevier B.V. This is an open access article under the CC BY license. <http://creativecommons.org/licenses/by/4.0/>

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DeepSarc: A Transformer-Based Deep Learning Approach for Sarcasm Detection in Social Media Text

Item Type Journal Article

Author R. Haq

Author X. Zhang

Author S. Yitagesu

Author W. Khan

Author Z. Feng

Abstract Sarcasm detection is a critical and challenging task in sentiment analysis, particularly for low-resource languages like Urdu, where limited annotated data, linguistic complexity, and subtle contextual cues hinder accurate classification. Traditional machine learning methods often fail to capture the nuanced and often contradictory nature of sarcastic expression. To address these challenges, this paper presents a comprehensive and computationally efficient framework for Urdu sarcasm detection. We first mitigate severe class imbalance through strategic down-sampling and back-translation-based data augmentation. We then conduct extensive benchmarking of traditional deep learning architectures against fine-tuned pre-trained language models, including multilingual, monolingual, and Twitter-specific variants. Building on these insights, we propose DeepSarc, a novel hybrid model that integrates the contextual embeddings from XLM-T, the multi-scale feature extraction capabilities of dilated convolutional neural networks (DCNNs), and the sequential dependency modeling of bidirectional long short-term memory (BiLSTM) networks. While slightly more computationally intensive than simpler alternatives, DeepSarc achieves a state-of-the-art F1-score, significantly outperforming existing approaches. Our results establish a new benchmark for sarcasm detection in low-resource languages and provide a scalable, high-performance framework adaptable to diverse linguistic contexts. © 2025 John Wiley & Sons Ltd.

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DepGLM: Depression degree recognition on social media based on large language models

Item Type Journal Article

Author J. Liu

Author P. Ding

Author J. Chen

Abstract Background: Depression is recognized as one of the most prevalent mental health disorders globally. Early detection and intervention can significantly reduce its negative impact on patients. Data from social media platforms provide new opportunities for early warning, intervention, and support for individuals with depression. However, traditional discriminative methods are limited in both the accuracy and interpretability of depression identification. Most studies focus on binary classification tasks, lacking more fine-grained identification. Methods: We model depression degree recognition as a multi-level text classification task and a multi-level text generation task, exploring more fine-grained identification in the Chinese context. We construct the depression degree ((Depression Weibo) dataset and the first multi-level interpretable depression degree instruction dataset. Based on these datasets and ChatGLM3, a Chinese open-source large language model, we train DepGLM. We also compare how fine-tuned large language models perform against multiple baseline models across different tasks. Results: The results show that the text classification model, DepGLM-tc, demonstrates superior accuracy compared to current state-of-the-art (SOTA) discriminative approaches, achieving a weighted F1 score of 84%. Meanwhile, the text generation model, DepGLM-tg, approaches the accuracy of SOTA discriminative methods and is capable of generating high-quality explanations comparable to ChatGPT. Conclusion: This study demonstrates that large language models fine-tuned for specific domains more accurately identify depression degree. Moreover, our model demonstrates enhanced performance in classification tasks. © The Author(s) 2025. This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access page (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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social media, Large language models, text classification, depression degree recognition, text generation

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Depression detection with machine learning in a low-resource language: A case study in Hebrew digital health communities

Item Type Journal Article

Author R. Keinan

Author E. Margalit

Author D. Bouhnik

Abstract Depression is a debilitating global health crisis; while digital social platforms offer an avenue for early detection through user-generated text, leveraging these digital traces is essential for timely clinical intervention. However, current computational linguistics for mental health are overwhelmingly English-centric. Standard natural language processing (NLP) methodologies may fail when applied to morphologically-rich low-resource languages like Hebrew, where non-concatenative structures and dense root-patterns obscure semantic markers of psychological distress. To bridge this gap, we propose a tailored machine learning framework explicitly designed for morphologically rich languages. We rigorously evaluated 16 classical and ensemble models across more than 130,000 configurations, isolating the optimal interactions between character-level n-grams and specialized Hebrew emotion lexicons to maximize feature extraction. This methodology is empirically tested on a curated dataset from “Camoni,” Israel's largest dedicated digital health community, which provides a high-value, clinically adjacent environment free from the noise of generalist social media platforms. Our optimized Stacking ensemble framework achieved an 84.1% accuracy rate, significantly outperforming standard

baselines as well as fine-tuned, state-of-the-art Transformer models like AlephBERT. The integration of 3-to-5 character n-grams proved vital in navigating Hebrew's morphological sparsity, proving that computationally efficient, expert-engineered features can surpass deep learning in low-resource clinical domains. This study provides a generalizable validated blueprint for deploying mental health screening tools. We urge the clinical and computational integration of these NLP frameworks into digital health ecosystems to facilitate early, life-saving interventions across diverse and underrepresented linguistic landscapes. © 2026 The Authors

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DepressionEmo: A novel dataset for multilabel classification of depression emotions

Item Type Journal Article
Author A.B.S. Rahman
Author H.-T. Ta
Author L. Najjar
Author A. Azadmanesh

Author A.S. Gönül

Abstract Emotions are integral to human social interactions, with diverse responses elicited by various situational contexts. Particularly, the prevalence of negative emotional states has been correlated with negative outcomes for mental health, necessitating a comprehensive analysis of their occurrence and impact on individuals. In this paper, we introduce a novel dataset named DepressionEmo designed to detect 8 emotions associated with depression by 6037 examples of long Reddit user posts. This dataset was created through a majority vote over inputs by zero-shot classifications from pre-trained models and validating the quality by annotators and ChatGPT, exhibiting an acceptable level of inter-rater reliability between annotators. The correlation between emotions, and linguistic analysis are conducted on DepressionEmo. Besides, we provide several text classification methods classified into two groups: machine learning methods such as SVM, XGBoost, and LightGBM; and deep learning methods such as BERT, BART, GAN-BERT, and T5. Despite achieving the same F1 Macro score of 0.76 as BART, the pretrained BERT model, bert-base-uncased, stands out as the most efficient model in our experiments due to its lower number of parameters. Across all emotions, the highest F1 Macro value is achieved by suicide intent, indicating a certain value of our dataset in identifying emotions in individuals with depression symptoms through text analysis. The curated dataset is publicly available at: <https://github.com/abuBakarSiddiqurRahman/DepressionEmo>. © 2024 Elsevier B.V.

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DepreSym: A Depression Symptom Annotated Corpus and the Role of Large Language Models as Assessors of Psychological Markers

Item Type Journal Article

Author A. Pérez

Author M. Fernández-Pichel

Author J. Parapar

Author D.E. Losada

Abstract Computational methods for depression detection aim to mine traces of depression from online publications posted by Internet users. However, solutions trained on existing collections exhibit limited generalisation and interpretability. To tackle these issues, recent studies have shown that identifying specific depressive symptoms can lead to more robust and effective models. The eRisk initiative fosters research on this area and has recently proposed a new ranking task focused on developing search methods to find sentences related to depressive symptoms. This search challenge relies on the symptoms specified by the Beck Depression Inventory-II (BDI-II), a questionnaire widely used in clinical practice. It includes symptoms such as sadness, irritability or lack of sleep. Given the input submitted by systems participating in eRisk, we first apply top-k pooling over the systems' relevance rankings, obtaining a diverse set of sentences. These sentences are judged for relevance, leading to DepreSym, a dataset consisting of 21,580 sentences annotated according to their relevance to the 21 BDI-II symptoms. This dataset serves as a valuable resource for advancing the development of models that monitor depression markers. Due to the complex nature of this relevance annotation, we designed a robust assessment methodology carried out by three expert assessors, including a trained psychologist. As part of this study, we explore the potential of recent Large Language Models (ChatGPT, GPT4 and Vicuna) as assessors in this complex task. We undertake a comprehensive examination of the LLMs' performance, studying their main limitations and analysing their role as a complement or replacement for human annotators. Finally, we incorporate our dataset into the Benchmarking Information Retrieval (BEIR) framework for a thorough search evaluation. We use state-of-the-art retrieval systems, including lexical, sparse, dense and re-ranking architectures, to gain insights about the dataset's complexity and identify potential avenues for improvement. © The Author(s) 2025.

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Design of a Distributed Reactive Framework for Localized LLM-Driven Financial Sentiment Analysis

Item Type Conference Paper

Author A. Rana

Author T.A. Rashid

Author R. Kumar

Abstract The digital financial platforms have been expanding at a very fast rate leading to an unprecedented quantity of heterogeneous textual information such as news feeds, social media updates, regulatory updates, and conversations between investors. Such data must be analyzed in realtime using the best sentiment analysis because this is important in making critical decisions in volatile financial markets. The problem with the existing centralized sentiment analysis systems is however, that they are not as scalable, latent, or adaptable to regional linguistic differences. The paper outlines the design and implementation of a distributed reactive architecture of localized Large Language Model (LLM)-driven financial sentiment analysis. The solution includes proposed architecture which combines event-driven microservice, stream-processing pipelines and localized LLM inference modules to process financial text streams in a fault tolerant, scalable and low latency manner. The model has region-sensitive sentiment adaptation processes that take into consideration linguistic heterogeneity and cultural heritage in financial interpretation. Through experimental analysis, it is shown to have better throughput, lower latency, and better sentiment classification compared to conventional centralized methods. The suggested system provides multilingual and dynamic markets with real-time financial intelligence which is scalable and adaptive. © 2026 IEEE.

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Designing a novel technique for evaluation of tourism informatization in scenic spots from a big data perspective

Item Type Journal Article
Author L. Fu
Author Y. Yi
Author L. Liu
Author R. Chen

Abstract In recent years, tourism has become a significant driver of many countries' economies. To maximize revenue from tourism, it is crucial to prioritize the effective management of scenic spots and tourist attractions, and also raise awareness about these places. Social media platforms have played a pivotal role in promoting tourism, as users frequently share videos and reviews related to tourism. Analyzing and managing these reviews is essential for understanding tourists' opinions about specific destinations. In this study, we evaluated a scenic spot by analyzing tourists' sentiments. Data was collected from popular social media sites such as TripAdvisor and Twitter using web scraping and the Twitter API. The raw data was preprocessed to remove irrelevant information and redundancies and was properly annotated for further processing. We applied two approaches to analyze the sentiments of tourists. First, we vectorized the text representing the sentiment using the term frequency-inverse document frequency (TF-IDF) and utilized big data analytics to extract meaningful insights. Secondly, we employed a pre-trained large language model, bidirectional encoder representations from transformers (BERT), with a linear classifier to classify tourists' sentiments. The results of the big data analytics approaches were compared with those of BERT and previously proposed methods. BERT outperformed other machine learning models, achieving an average accuracy of 83.5% on the test set. These insights are valuable for evaluating the informatization of tourist spots, destination management, hospitality, and overall tourist attractions. © 2025 Fu et al.

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Detecting Adverse Drug Events in Social Media: A Brief Literature Review

Item Type Journal Article

Author I. Guellil

Author Y. Berrachedi

Author N.E. Chenni

Author M.-N. Abboud

Author J. Wu

Author H. Wu

Author B. Alex

Abstract Adverse drug events (ADEs) remain a significant burden to public health and a persistent challenge for pharmacovigilance. The proliferation of patient-generated discourse on social media offers a complementary, real-time signal for ADE surveillance. This article provides a concise yet comprehensive review of recent natural language processing (NLP) research on identifying ADEs in social media text. We systematically reviewed 100 peer-reviewed studies (2017–2025) on NLP/AI for detecting or analysing ADEs in social media. Searches in Google Scholar targeted English-language journal and conference papers; patents and protocols were excluded. Of 130 records screened, 6 were protocols and 24 were excluded because the full text could not be located or the item was a conference abstract lacking methodological detail (i.e., no description of approaches or experiments), yielding a final sample of 100 studies. One reviewer performed screening, with full-text eligibility verified by a second. We extracted objectives, data sources/languages, preprocessing and annotation practices, datasets, model families, evaluation metrics, and stated limitations. Studies were grouped into five task categories—classification, extraction, normalization, corpus creation, and broader analytical work—with evidence tables summarizing contributions, toolchains, datasets, and performance. Recurrent challenges include noisy/imbalanced data, multilingual and code-mixed content, and variability in annotation standards. Twitter remains the primary data source: 60% of studies analyse Twitter alone and a further 18% combine Twitter with other platforms (78% in total). English overwhelmingly dominates; only about 5% of studies draw on non-English sources (e.g., French, Chinese, Arabic). Standard pre-processing—URL removal, tokenisation, and lowercasing—is near-universal. Transformer-based models predominate, with BERT and its biomedical or “tweet” variants (e.g., RoBERTa, BioBERT, BERTweet) used in more than 60% of approaches. Persistent obstacles include severe class imbalance and ambiguous or implicit drug-event expressions. Although shared tasks such as SMM4H provide widely used benchmarks, comprehensive annotation guidelines remain uncommon (12% of papers). Recent work increasingly incorporates multimodal inputs and integrates structured biomedical knowledge, yet gaps persist in multilingual coverage, temporal/longitudinal modelling, and real-world deployment. To our knowledge, this is the first review to synthesise findings from a corpus of 100 peer-reviewed studies on ADE detection in social media using NLP. By organising the literature by task type and tracing methodological trends and limitations, it provides practical guidance for researchers and practitioners. The review also outlines actionable directions for future work, including model explainability, support for low-resource languages, and closer collaboration with regulatory authorities to enable real-world deployment. © The Author(s) 2026.

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Detecting Effects of AI-Mediated Communication on Language Complexity and Sentiment

Item Type Conference Paper
Author K. Sussman
Author D. Carter
Abstract Given the subtle human-like effects of large language models on linguistic patterns, this study examines shifts in language over time to detect the impact of AI-mediated communication (AI-MC) on social media. We compare a replicated dataset of 970, 919 tweets from 2020 (pre-ChatGPT) with 20, 000 tweets from the same period in 2024, all of which mention Donald Trump during election periods. Using a combination of Flesch-Kincaid readability and polarity scores, we analyze changes in text complexity and sentiment. Our findings reveal a significant increase in mean sentiment polarity (0.12 vs. 0.04) and a shift from predominantly neutral content (54.8% in 2020 to 39.8% in 2024) to more positive expressions (28.6% to 45.9%). These findings suggest not only an increasing presence of AI in social media communication but also its impact on language and emotional expression patterns. © 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.
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Computational linguistics, Language model, Information systems, Human engineering, Sentiment analysis, Social networking (online), Pattern recognition, Social media, ChatGPT, social media, Information use, Interactive computer systems, Tweets, AI detection, AI-detection, AI-MC, AI-mediated communication, Human like, Language complexity, Linguistic patterns, Media communications, Mediated Communication

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Detecting Emotions from Hindi Text Using Natural Language Processing

Item Type Conference Paper

Author S. Shashank

Author N. Tyagi

Author R.K. Behera

Editor Ray N.K.

Editor Udgata S.K.

Editor Puthal D.

Abstract With the ever-growing world of user-generated content in local languages, particularly on communication and social media platforms, the need for emotion analysis and interpretation in such content is on the rise. Hindi being one of the biggest global languages provides a complex and rich language structure and uniqueness that calls for emotion classification in a unique way. This paper outlines a method to categorize Hindi text into pre-established emotional classes through Natural Language Processing and Machine Learning methods. The preprocessing process includes language-dependent operations like handling Devanagari script, removing stopwords, normalization, and TF-IDF vectorization. Various machine learning models like Naïve Bayes, Logistic Regression, Extra Trees Classifier, Ridge Classifier, Random Forest Classifier, and Support Vector

Classifier was trained and their results were compared using different performance metrics. Performance metrics measured in terms of the following parameters: F1-score, recall, accuracy, precision, false positives, false negatives, true positives, and true negatives. Among the classifiers, Extra Trees Classifier had the maximum accuracy (0.8497). The research proves the efficacy and applicability of machine learning algorithms in analyzing emotional expressions in Hindi language and lays down a basis for sentiment-aware applications in low-resource languages. © 2025 IEEE.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Shashank; Amity University, Department of Computer Science and Engineering, Noida, Uttar Pradesh, 201301, India; email: salabhshashank@gmail.com; Conference name: 23rd OITS International Conference on Information Technology, OCIT 2025; Conference date: 18 December 2025 through 20 December 2025; Conference code: 221660

Detecting Homophobic Speech in Soccer Tweets Using Large Language Models and Explainable AI

Item Type Conference Paper

Author G.L. Santos

Author V.G. dos Santos

Author C. Kearns

Author G. Sinclair

Author J. Black

Author M. Doidge

Author T. Fletcher

Author D. Kilvington

Author K. Liston

Author P.T. Endo

Author T. Lynn

Editor Aiello L.M.

Editor Chakraborty T.

Editor Gaito S.

Abstract Homophobic speech is a form of hate speech. Social media enables hate speech to spread rapidly and widely through the internet, and unlike offline hate speech, can persist indefinitely, thereby prolonging its impact. Due to the adverse impact of hate speech, policymakers have called for greater action from online platforms to moderate and remove hate speech, including homophobic content. While homophobic hate speech is prevalent in online soccer discourses, there are few studies on this empirical context in general and specifically on the use of Large Language Models (LLMs) for detecting such speech. This study addresses this gap by proposing a homophobic speech text classification pipeline. We introduce H-DICT, a new general dictionary for identifying potential homophobic content in documents, and leverage this dictionary to curate and manually label an annotated dataset of homophobic and non-homophobic samples from the UEFA European Football Championships (the Euros) discourse on Twitter. We fine-tune and evaluate five large language models (LLMs) based on the BERT architecture - BERT, DistilBERT, RoBERTa, BERT Hate, and RoBERTa Offensive - and use Integrated Gradients, an explainable AI technique to explain each model's predictions. RoBERTa Offensive, an LLM fine-tuned specifically for detecting offensive language, presented the best performance when compared to the other LLMs. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Detecting Obstetric Violence Tweets from Mexico: Annotation Guidelines and Classification with LLMs

Item Type Journal Article

Author M. Vázquez-Hernández

Author H.M. Gómez-Adorno

Author N. Lerín-Hernández

Author I. Islas Barajas

Author O. Ramos-Flores

Abstract This paper presents the construction and analysis of a manually annotated corpus of tweets related to Obstetric Violence (OV) shared on Twitter (now X). The study aims to identify different types of violence experienced by women during the perinatal period, as well as activism efforts that seek to raise awareness about OV. The methodology includes data collection through keyword filtering, manual annotation guided by typologies of OV, and a descriptive analysis using BERTopic to identify themes in the data. The tweets were classified into categories such as OV, Non-OV, and Activism, and further annotated based on narrator type and type of OV violence. The study also evaluates the performance of large language models (LLMs) including ChatGPT, Copilot, and Metas LLaMA for zero-shot classification of tweets, highlighting their limitations in accurately identifying nuanced cases of OV. The research contributes a labeled dataset, a detailed annotation guide, and insights into the challenges of detecting OV in social media texts. It underscores the importance of addressing the invisibility and normalization of OV in both healthcare and NLP research. © 2003-2012 IEEE.

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Detecting pro-kremlin disinformation using large language models

Item Type Journal Article

Author M. Kramer

Author Y. Golovchenko

Author F. Hjorth

Abstract A growing body of literature examines manipulative information by detecting political mis-/disinformation in text data. This line of research typically involves highly costly manual annotation of text for manual content analysis, and/or training and validating automated downstream approaches. We examine whether Large Language Models (LLMs) can detect pro-Kremlin disinformation about the war in Ukraine, focusing on the case of the downing of the civilian flight MH17. We benchmark methods using a large set of tweets labeled by expert annotators. We show that both open and closed LLMs can accurately detect pro-Kremlin disinformation tweets, outperforming both a research assistant and supervised models used in earlier research and at drastically lower cost compared to either research assistants or crowd workers. Our findings contribute to the literature on mis-/disinformation by showcasing how LLMs can substantially lower the costs of detection even when the labeling requires complex, context-specific knowledge about a given case. © The Author(s) 2025. This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by->

nc/4.0/) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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Detecting Semantic Cues from Short-Form Content: A Data-Driven Approach for Neutral Trigger Identification

Item Type Conference Paper

Author J.-Y. Chen

Author X.-W. Yao

Author T.-Y. Xu

Abstract In the era of big data, sentiment analysis has become a critical tool for detecting public reactions and managing risk across large-scale information streams. However, most existing models rely heavily on lexicon-based features and often fail to capture the semantic complexity of emotionally neutral words that influence tone. This paper presents a data-driven sentiment modeling approach that identifies latent emotional triggers-terms not included in standard sentiment dictionaries but strongly correlated with tone fluctuations. Leveraging a large-scale English news dataset in the business domain, the proposed method applies VADER-based tone classification in conjunction with frequency and co-

occurrence analysis to extract candidate trigger words. These latent triggers are further categorized by tone direction and evaluated for their semantic role in emotional amplification. The findings indicate that triggerbased modeling enhances interpretability and introduces a lightweight yet effective feature set for tone prediction, particularly in short-text and low-resource environments such as social media headlines or early-warning alert systems. The approach contributes to the advancement of big datadriven emotion modeling by expanding feature engineering beyond lexical boundaries and offers a scalable solution for real-time sentiment detection. This aligns with ongoing research efforts in the fields of big data analytics, machine learning applications, and semantic information extraction, offering practical value for automated public opinion tracking, crisis forecasting, and intelligent communication systems. © 2025 IEEE.

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Detecting Toxic Comments in Bengali Language

Item Type Conference Paper

Author A. Mondal

Author K. Roy

Author S. Das

Author A. Dutta

Editor Das A.K.

Editor Nayak J.

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Editor Vimal S.

Editor Pelusi D.

Abstract An increasing number of users utilize public platforms to communicate. The languages used by general public are diverse and varied. Detection of the offensive words utilized by people on these online platforms is challenging. Unstructured texts, inappropriate models combined with low-resource datasets, cause the research on less prevalent languages quite arduous. In this chapter, we have attempted to address the problem of offensive comments detection in Bengali, which is a low-resource language with limited approaches. Our approach is a binary classification model that successfully classifies toxic comments from nontoxic comments. Natural Language Processing techniques have been implemented for preprocessing of the unstructured data. Deep learning model with a modified version of BERT architecture has been applied. This model has been pretrained on a large corpus of Bengali text and efficiently detects the offensive words in the social media Bengali comments. The proposed approach is performing effectively when compared with different baselines. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Development of Sentiment Analysis for Wolaita Language using Machine Learning Approaches

Item Type Conference Paper

Author A.T. Bedecho

Author R.K. Bokka

Editor Alemneh E.

Editor Mekuria F.

Editor Nigussie E.

Abstract The exponential augmentation of digitally-produced data engendered daily manifests substantial obstacles and fa-vorable prospects. With the proliferation of social media, e-commerce, and online reviews, comprehending the sentiments and viewpoints articulated by individuals has become pivotal for enterprises, policymakers, and scholars. Therefore, this study explores the development of a sentiment analysis system for the Wolaita language, which can enable social media monitoring, customer feedback analysis, and decision support systems that cater to the needs and concerns of Wolaita-speaking communities. For this reason, a Wolaita language sentiment dataset was constructed by collecting and annotating texts from divers data sources, including religious materials, textbooks, news articles and social media posts. The proposed system was evaluated using a range of machine learning models, encompassing Naive Bayes, Support Vector Machines (SVMs), Long Short-Term Memory (LSTMs), and transformer-based architectures, as well as feature engineering techniques like n-grams and sentiment lexicons to determine the most efficacious approach for Wolaita language sentiment analysis. The transformer-based models, specifically a Wolaita-language BERT model, achieved the best performance with an F1-score of 0.82 for classifying positive, negative, and neutral sentiment. The study ascertained that transfer learning from higher-resource languages assisted in surmounting the data sparsity impediments for Wolaita. This work demonstrates the feasibility of developing accurate sentiment analysis capabilities for the Wolaita language leveraging state-of-the-art machine learning techniques, despite the language's low-resource status. Prospective work will entail augmenting the dataset and exploring semi-supervised learning to enhance the performance of the system. © 2024 IEEE.

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Digital Pedagogy in the Comment Section: A Study of Teacher-Learner Interaction in Bangla Admission Coaching

Item Type Conference Paper
Author J.T. Jinia
Author M. Sakibur Rahman
Author S. Biswas
Author M. Zarafat Islam
Author M.S. Islam
Author M. Hassan Imam Bijoy
Editor Arefin M.S.
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Abstract In Bangladesh's rapidly evolving educational landscape, social media platforms like Facebook have become pivotal for informal teacher–student engagement, especially in university admission coaching. This study analyzes teacher feedback from Facebook comment sections under Bangla-language educational videos, offering insights into how digital feedback shapes learning dynamics. A new dataset, BnAdEduFeed3, was developed by scraping and manually annotating 6,010 comments across three sentiment classes: Positive, Neutral, and Negative. After preprocessing, Word2Vec-based synonym replacement was applied for augmentation to improve lexical diversity and class balance. Three transformer models: sagorsarker/bangla-bert-base, csebuetnlp/banglabert, and xlm-roberta-base, were fine-tuned, with bangla-bert-base achieving the best results (accuracy: 95.97%, precision: 93.97%, recall: 96.97%). The findings establish a benchmark for sentiment analysis in Bangla educational discourse and highlight the pedagogical potential of real-time, student-generated feedback. Ethical safeguards, including anonymization and adherence to platform guidelines, were maintained throughout. While the dataset's specificity limits generalizability, the study advocates extending research to multi-platform and multi-modal feedback in low-resource language contexts, reinforcing Facebook's role as a feedback ecosystem in modern education. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Disaster Tweet Classification Using Fine-Tuned Deep Learning Models Versus Zero and Few-Shot Large Language Models

Item Type Conference Paper

Author S. Taghian Dinani

Author D. Caragea

Author N. Gyawali

Editor Gusikhin O.

Editor Hammoudi S.

Editor Cuzzocrea A.

Abstract The task of effectively filtering and categorizing the vast volume of user-generated content on social media during disaster events is crucial for emergency management and disaster response organizations to allocate their resources efficiently. Deep learning approaches, including recurrent neural networks, Capsule Neural Networks (CapsNets) and transformer-based models, have been successfully used for text classification. However, they have not been extensively compared in the context of crisis tweet classification. Furthermore, such models have not been compared with foundational large language models (LLMs), such as the open-source Llama 2, which are known to produce good results in zero and few-shot settings. In this study, we conducted a comprehensive comparison between CapsNet models, BERT models, and Bi-LSTM, one of the widely used recurrent neural network models, in the task of classifying crisis tweets. We also compared these models with Llama 2 in 0-shot, 1-shot, and 5-shot learning settings. We considered two classification tasks: binary classification for informativeness and multi-class classification for humanitarian content. For this purpose, we used several benchmark datasets for crisis tweet classification, namely CrisisBench, CrisisNLP, and CrisisLex. Our experiments revealed that the results of the CapsNet models are comparable to those of the Bi-LSTM models across various metrics. However, BERT models consistently outperformed both Bi-LSTM and CapsNet models across different datasets for both classification tasks. Notably, Llama 2 exhibited promising performance, considering its exposure to no data instances in the 0-shot and only few instances in 1-shot and 5-shot learning settings, showcasing its potential for handling crisis tweet classification. However, it is important to highlight that the fine-tuned models, including CapsNet, BERT, and Bi-LSTM, achieved superior performance overall, emphasizing the influence of fine-tuning on model effectiveness. These findings suggest that BERT is still one of the most effective models for classifying crisis tweets. Additionally, Llama 2, with its strong performance in limited-shot scenarios that match the scenarios in the crisis domain, presents an intriguing path for further exploration in handling crisis-related content. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Distantly Supervised Explainable Stance Detection via Chain-of-Thought Supervision

Item Type Journal Article

Author D. Ding

Author G. Dai

Author C. Peng

Author X. Peng

Author B. Zhang

Author H. Huang

Abstract Investigating public attitudes on social media is crucial for opinion mining systems. Stance detection aims to predict the attitude towards a specific target expressed in a text. However, effective neural stance detectors require substantial training data, which are challenging to curate due to the dynamic nature of social media. Moreover, deep neural networks (DNNs) lack explainability, rendering them unsuitable for scenarios requiring explanations. We propose a distantly supervised explainable stance detection framework (DS-ESD), comprising an instruction-based chain-of-thought (CoT) method, a generative network, and a transformer-based stance predictor. The CoT method employs prompt templates to extract stance detection explanations from a very large language model (VLLM). The generative network learns the input-explanation mapping, and a transformer-based stance classifier is trained with VLLM-annotated stance labels, implementing distant supervision. We propose a label rectification strategy to mitigate the impact of erroneous labels. Experiments on three benchmark datasets showed that our model outperformed the compared methods, validating its efficacy in stance detection tasks. This research contributes to the advancement of explainable stance detection frameworks, leveraging distant supervision and label rectification strategies to enhance performance and interpretability. © 2024 by the authors.

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Do language families matter? Evaluating LLMs for sentiment analysis through a hierarchical cross-lingual lens

Item Type Journal Article

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Author A. Manaf

Author A.S. Imran

Author Z. Kastrati

Author S.M. Daudpota

Author M. Biba

Abstract Social media sentiment analysis has become one of the most significant instruments for understanding the opinion of the population in the spheres of healthcare, politics, and education. Yet, large language models (LLMs) remain unevenly distributed in their linguistic coverage, failing to adequately serve a large portion of the world's languages. This study evaluates five state-of-the-art LLMs: GPT-4o, Gemini 2.0 Flash, DeepSeek-V3, Mistral Large, and Claude 3.7 Sonnet on three-class sentiment classification across 36 datasets spanning 36 languages, with emphasis on low- and medium-resource settings, using zero-shot and few-shot prompting without task-specific fine-tuning. In addition to the traditional measures of performance per language, the study presents a hierarchical analysis of languages based on a genealogical tree of Indo-European, Afro-Asiatic, Niger-Congo, Turkic, Austronesian, and English Creole language families, so that it is possible to identify the systematic patterns of performance superiority and inferiority among the language families. The findings show that few-shot prompting improves the results of a vast majority of languages, with several models approaching or surpassing the performance of the state-of-the-art benchmark of task-specific models. The GPT-4o and Claude achieved the highest performance in the high-resource and medium-resource settings, and Gemini is a competent trade-off that allows balancing the performance and the computational cost. Although it has lower zero-shot performance, Mistral benefits the most from few-shot prompting and becomes highly competitive in the few-shot setting. Despite these developments, the level of performance on low-resource languages, such as Oromo, Xitsonga, Azerbaijani, and Twi, remains significantly lower, underscoring that progress in multilingual LLMs requires moving beyond English-centric evaluation toward genuinely representative and globally inclusive benchmarks. Copyright © 2026 Kastrati, Manaf, Imran, Kastrati, Daudpota and Biba.

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Do Memes Speak Hate? A Residual-Adapter Approach for Bengali Memes

Item Type Conference Paper

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Author V. Gupta

Author M. Gupta

Editor Shastri A.

Editor Singh R.

Editor Tiwary U.

Editor Singh D.

Abstract The widespread presence of social media in everyday life has led to a surge in online interactions. While it offers numerous benefits, social media is frequently misused as a platform to propagate hate and negativity, which can have detrimental effects on users' mental well-being. Although several tools exist to detect harmful content, low-resource languages like Bengali remain significantly under-resourced in such efforts. Given that contemporary social media content often spans multiple modalities like text and images, a multimodal approach offers a more robust solution for automatic content moderation. This study presents a multimodal hateful meme detection model tailored for Bengali, which simultaneously processes textual and visual information. The Res-AFFNet (Residual-Adapter Feature-level Fusion Network) architecture employs MuRIL (Multilingual representations for Indian languages) and ViT (Vision Transformer) to process text and image, respectively. Each modality's embeddings are passed through lightweight adapter units, and a residual fusion (80% original +20% adapted) is applied. The adapter outputs are fused and passed through a linear projection and classifier to predict whether the input is 'hate' or 'not-hate'. The multimodal approach outperforms unimodal approaches in terms of accuracy. Trained and evaluated on the MUTE dataset, the model achieves an improvement of 1.27% over the existing baseline. Additionally, ablation studies highlight the effectiveness of the multimodal framework compared to unimodal and reduced-component variants. Multimodal Bengali hate speech detection can be used for automatic content moderation on social networking websites and help support low-resource communities. One future scope of this study is that this framework could be extended to other low-resource languages. Fine-grained classification of hate speech can also be incorporated in this framework by expanding the dataset annotations. © The Author(s), under exclusive license to Springer Nature

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Does Your Definition Matter? LLMs Comparison Between Prompt Sensitivity and Internal Behavior for Social Media Analysis

Item Type Conference Paper

Author M.-A. Azaïs

Author M. Coustaty

Author J.-L. Guillaume

Editor De Marsico M.

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Editor Jurie F.

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Editor Ross A.

Editor Wang L.

Abstract Although large language models show promise for automating data annotation in social media, social scientists lack confidence in their outputs due to perceived limited control. In practice, social scientists classify text according to their own topic definitions, which do not necessarily align with the common evaluation approach of benchmarking models against a unique ground truth. To better simulate real-world conditions, we compared the outputs of seven open-source models from three families (Mistral, Gemini, and LLaMA) of varying sizes (8B–70B) on 2,000 realistic French hotel reviews. We asked 13 participants to provide definitions for five dominant topics in the hospitality domain and to add any missing topics with their definition. Our results show that model variability is strongly topic-dependent. Among 210 comparisons, only one showed that shared topic definitions yielded significantly higher agreement than prompt sensitivity, indicating that model choice plays a more decisive role than a detailed topic definitions. Furthermore, for all models, adding topics to the prompt influences the classification of previously present topics, providing evidence of prompt sensitivity. While models can serve as reliable assistants for data annotation, our findings underscore the importance of validating their outputs with social scientists before applying them for social media analysis. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2027.

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Dual Causes Generation Assisted Model for Multimodal Aspect-Based Sentiment Classification

Item Type Journal Article

Author R. Fan

Author T. He

Author M. Chen

Author M. Zhang

Author X. Tu

Author M. Dong

Abstract Multimodal aspect-based sentiment classification (MABSC) aims to identify the sentiment polarity toward specific aspects in multimodal data. It has gained significant attention with the increasing use of social media platforms. Existing approaches primarily focus on analyzing the content of posts to predict sentiment. However, they often struggle with limited contextual information inherent in social media posts, hindering accurate sentiment detection. To overcome this issue, we propose a novel multimodal dual cause analysis (MDCA) method to track the underlying causes behind expressed sentiments. MDCA can provide additional reasoning cause (RC) and direct cause (DC) to explain why users express certain emotions, thus helping improve the accuracy of sentiment prediction. To develop a model with MDCA, we construct MABSC datasets with RC and DC by utilizing large language models (LLMs) and visual-language models. Subsequently, we devise a multitask learning framework that leverages the datasets with cause data to train a small generative model, which can generate RC and DC, and predict the sentiment assisted by these causes. Experimental results on MABSC benchmark datasets demonstrate that our MDCA model achieves the state-of-the-art performance, and the small fine-tuned model exhibits superior adaptability to MABSC compared to large models like ChatGPT and BLIP-2. © 2024 IEEE.

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Dual perspective hierarchical alignment network for joint multimodal aspect-based sentiment analysis

Item Type Journal Article

Author J. Yang

Author W. Zou

Author X. Du

Author J. Shen

Abstract With the accelerating growth of multimodal user-generated content, effectively aligning image and text representations to establish deep interactions has become a key challenge in Multimodal Aspect-Based Sentiment Analysis (MABSA). To address the limitations in existing methods, such as the lack of cross-modal structural alignment, the introduction of potential noise by simple semantic alignment, and shallow fusion strategies, we propose a novel Dual Perspective Hierarchical Alignment (DPHA) framework for joint MABSA. Specifically, DPHA leverages the denoised image descriptions and aligns them with the textual content at the structural level. To further enhance structural alignment, an unsupervised contrastive learning is proposed to minimize the distance between the dependency structures of image descriptions and the corresponding text. Meanwhile, DPHA performs a cascaded fusion of object regions and salient patches to make a balance between sufficient visual information and potential noise. The fused visual representations are then

semantically aligned with textual information. Next, the alignment matrices computed from the two previous modules are used to guide attention flow across multiple Transformer layers, thereby achieving deep multimodal fusion. During training, a task-specific curriculum learning strategy is incorporated to enable dynamic sample scheduling to more effectively facilitate the acquisition of the model's capability. Experiments on benchmark datasets have verified that DPHA significantly outperforms existing baselines. Ablation studies have also demonstrated the effectiveness of each component in the proposed DPHA. © 2025 Elsevier B.V.

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Dual-Source Retrieval-Augmented Generation Chatbot for Women's Health (HerCare): Design and Multimethod Evaluation Study

Item Type Journal Article

Author K.T. Zaman

Author W. Ul Hasan

Author N. Ahmed

Author J. Li

Abstract Background: Conversational agents for women's health often fail to meet user needs, offering either clinically sterile advice or unreliable peer anecdotes. This limitation creates a tension between the need for factual safety and emotional resonance in sensitive health contexts. Objective: We aimed to address this gap by developing and conducting a formative evaluation of HerCare, a conversational agent built on a novel dual-source retrieval-augmented generation architecture. The system integrates expert medical knowledge with peer narratives and makes the provenance of each response visible to users, enabling trust calibration through transparent source attribution. Methods: We conducted a remote, web-based single-session field study (December 2024 to January 2025; North Dakota State University Institutional Review Board Protocol #IRB0005368) with 243 completers (from 335 eligible, consenting visitors) recruited via social media (Facebook [Meta], Reddit, and Instagram [Meta]) and university mailing lists. Eligible participants self-identified as women aged 18-45 years with English proficiency and internet access. We used a quantitative multimethod evaluation, combining standardized self-report metrics—the Chatbot Usability Questionnaire and net promoter score (NPS)—with computational linguistic analyses (VADER [Valence Aware Dictionary and Sentiment Reasoner] sentiment analysis and NRC [National Research Council] Emotion Lexicon) of 1191 conversational turns. Results: Among the 243 participants who completed the protocol, reported usability was high (Chatbot Usability Questionnaire median 78.1, IQR 65.2-87.5; mean 75.67, SD 15.50) and advocacy was strong (NPS 60.0; 171/243, 70.4% promoters, 25/243, 10.3% detractors), though this NPS reflects completers only. Postinteraction ratings were high (all facets median 4-5 on a 5-point scale; helpfulness, ease of use, and clarity median 5, IQR 4-5). Computational analysis revealed a consistent polarity shift from neutral to negative user queries (compound -0.18 to +0.15) to strongly positive agent responses (compound +0.55 to +0.83), with a recurring validate-then-redirect empathy pattern in which the agent acknowledges user distress before pivoting to constructive guidance. Conclusions: Among completers, the dual-source architecture was associated with high perceived empathy and trust, suggesting it can combine clinical accuracy with emotional support. These formative findings indicate the feasibility of weaving clinical sources with lived experiences toward safer, more resonant health AI and surface a candidate design pattern for future empathy-attuned systems that warrants controlled evaluation. © Kimia Tuz Zaman, Wordh Ul Hasan, Nova Ahmed, Juan Li.

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DuluthNLP at SemEval-2023 Task 12: AfriSenti-SemEval: Sentiment Analysis for Low-resource African Languages using Twitter Dataset

Item Type Conference Paper

Author S. Akrah

Author T. Pedersen

Editor Ojha A.K.

Editor Dogruoz A.S.

Editor Da San Martino G.

Editor Madabushi H.T.

Editor Kumar R.

Editor Sartori E.

Abstract This paper describes the DuluthNLP system that participated in Task 12 of SemEval-2023 on AfriSenti-SemEval: Sentiment Analysis for Low-resource African Languages using Twitter Dataset. Given a set of tweets, the task requires participating systems to classify each tweet as negative, positive or neutral. We evaluate a range of monolingual and multilingual pre-trained models on the Twi language dataset, one among the 14 African languages included in the SemEval task. We introduce TwiBERT, a new pretrained model trained from scratch. We show that TwiBERT, along with mBERT, generally perform best when trained on the Twi dataset, achieving an F1 score of 64.29% on the official evaluation test data, which ranks 14 out of 30 of the total submissions for Track 10. The TwiBERT model is released at <https://huggingface.co/sakrah/TwiBERT>. © 2023 Association for Computational Linguistics.

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Dynamic Emoji Perception and Bidirectional Cross-Modal Attention for Large Language Model Sentiment Analysis

Item Type Journal Article

Author B. LIU

Author F. LI

Author S. LI

Author S. LIANG

Author D. WANG

Abstract Social media platforms (such as Twitter) serve as vital conduits for user sentiment and public discourse in modern society. Their vast volumes of user-generated content carry rich personal emotions and collective opinions. The real-time nature, short- text characteristics, diversity, and unstructured nature of this data provide a rich foundation for sentiment analysis research while also presenting significant technical challenges. Traditional sentiment analysis methods based on a single text modality struggle to fully capture the complex emotional information embedded in social media. They particularly overlook the significant emotional characteristics conveyed by visual modalities (e.g., images) and symbolic modalities (e.g., Emojis), leading to significant bottlenecks in model performance. Addressing the issues of semantic dynamism and text-image heterogeneity in social media, this paper focuses on the Twitter platform and proposes an emotion analysis framework integrating large language models with dynamic Emoji perception and bidirectional cross-modal attention. This paper designs a bidirectional cross-modal attention mechanism incorporating dynamic modality quality assessment. This mechanism adaptively adjusts fusion weights based on the informational quality and reliability of text and images, outperforming simple feature concatenation or fixed-weight attention fusion. This paper also proposes a context-aware dynamic Emoji semantic parsing and text augmentation method driven by large language models. This approach leverages the context-aware capabilities of large language models to overcome the static semantic

limitations of Emoji symbols, enhance textual sentiment features, and effectively handle ironic contexts. Utilizing Llama3.1-8B as the base model, combined with advanced low-rank adaptive fine-tuning techniques and hyperparameter optimization strategies, the method achieves high accuracy in sentiment recognition while significantly reducing computational resource consumption. Experimental design involves ablation studies validating the effectiveness of multimodal fusion and dynamic Emoji processing. Comprehensive evaluations are conducted on the SemEval-2017 and Twitter-2015/2017 datasets. Experimental results demonstrate improvements across all evaluation metrics for the fine-tuned model. Particularly for samples containing sarcastic content, the model achieves notably higher recognition accuracy due to the complementary nature of multimodal information. © 2026 Science Press. All right reserved.

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Dynamic Hedging Strategies in Derivatives Markets with LLM-Driven Sentiment and News Analytics

Item Type Conference Paper

Author J. Yang

Author Y. Tang

Author Y. Li

Author L. Zhang**Author** H. Zhang

Abstract Dynamic hedging strategies are essential for effective risk management in derivatives markets, where volatility and market sentiment can greatly impact performance. This paper introduces a novel framework that leverages large language models (LLMs) for sentiment analysis and news analytics to inform hedging decisions. By analyzing textual data from diverse sources like news articles, social media, and financial reports, our approach captures critical sentiment indicators that reflect current market conditions. The framework allows for real-time adjustments to hedging strategies, adapting positions based on continuous sentiment signals. Backtesting results on historical derivatives data reveal that our dynamic hedging strategies achieve superior risk-adjusted returns compared to conventional static approaches. The incorporation of LLM-driven sentiment analysis into hedging practices presents a significant advancement in decision-making processes within derivatives trading. This research showcases how sentiment-informed dynamic hedging can enhance portfolio management and effectively mitigate associated risks. © 2025 IEEE.

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Dynamic loss function in graph attention networks for sentiment analysis on imbalanced datasets

Item Type Journal Article

Author A.F. Dehkordi

Author H. Rastegari

Author F. Safi-Esfahani

Abstract Sentiment Analysis (SA) is a technique in Natural Language Processing (NLP) that uses computational methods to analyze the emotions expressed in user reviews. Analyzing opinions' polarity can provide valuable insights into products based on user feedback, which can benefit producers. While many machine learning and deep learning models have demonstrated strong performance, they often face challenges in understanding complex relationships between entities, leading to missed nuances in real-world interactions. This limitation can impede a thorough understanding of user sentiment. In this article, we propose DLF-GAT, a novel Dynamic Loss Function Graph Attention Network architecture, which leverages XLM-RoBERTa embeddings for textual representation and is specifically designed to address the challenge of class imbalance in low-resource and multilingual datasets while effectively capturing complex semantic and relational dependencies in e-commerce platforms. Our model incorporates an innovative dynamic weighting mechanism that adaptively prioritizes underrepresented classes, and the attention mechanism within the graph network dynamically highlights the most influential nodes and relationships, enabling the model to capture intricate contextual and relational information with unprecedented precision. We evaluate our approach using two SA datasets: Twitter, which contains multilingual data, and DigiKala in the Persian language, which is classified as a low-resource language. Given the focus on Persian, the results demonstrate the effectiveness of DLF-GAT, achieving 70% accuracy on DigiKala and 83% on Twitter, thereby establishing it as a robust and novel approach for deepening sentiment understanding in complex, imbalanced, and multilingual contexts. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2026.

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Publisher Springer

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ISSN 13807501 (ISSN)

Issue 5

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Rastegari; Department of Computer Engineering, Na.C., Islamic Azad University, Najafabad, Iran; email: rastegari@iau.ac.ir; CODEN: MTAPF

Dynamic Reliable-Aware Router for Few-Shot Multimodal Sentiment Analysis**Item Type** Conference Paper**Author** J. Lin**Author** T. Cai**Author** W. Chen**Author** Z. Yang**Author** X. Liao

Abstract Few-shot multimodal sentiment analysis (MSA) aims to predict sentiment polarity in low-data regimes by integrating complementary information from different modalities. However, existing prompt-based methods often assume all visual information is inherently beneficial and indiscriminately inject all visual tokens into the language model. Consequently, they struggle to handle complex cross-modal semantic relationships, resulting in severe interference from globally contradictory images and locally aspect-irrelevant visual noise. To address these issues, we propose Reliable-Aware Router (RAR), an instance-level dynamic evidence routing framework. RAR constructs two complementary enhancement paths: Soft Token Weighting (STW) filters visual patch tokens, while Multimodal Demo Residual Enhancement (MDR) derives residual textual enhancement from retrieved demonstrations. An Adaptive Router then computes two independent reliability gates that control the STW visual enhancement and the MDR demo residual before final prediction. Experiments on Twitter-15, Twitter-17, and MVSA-S show that RAR achieves the best results among the compared methods. © 2026 IEEE.

Date 2026**DOI** 10.1109/DLCV69906.2026.11635125**URL** <https://www.scopus.com/pages/publications/105047648451?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: DLCV - IEEE Int. Conf. Deep Learn. Comput. Vis.**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833156268-7 (ISBN)**Proceedings Title** DLCV - IEEE Int. Conf. Deep Learn. Comput. Vis.**Conference Name** DLCV 2026 - 2026 IEEE 3rd International Conference on Deep Learning and Computer Vision**Date Added** 20/09/2026, 8:35:13 pm**Modified** 20/09/2026, 8:35:13 pm**Tags:**

Language model, Artificial intelligence, Visual languages, Sentiment analysis, Data mining, Semantics, Dynamics, Multi-modal, Cross-modal, few-shot learning, Few-shot learning, Multimodal sentiment analyze, prompt learning, Prompt learning, Visual information, multimodal sentiment analysis, demonstration retrieval, Demonstration retrieval, Routers, Soft tokens

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Natural Science Foundation of China, NNSF, (62476060); Correspondence Address: Z. Yang; Fuzhou University, College of Computer and Data Science, Fuzhou, China; email: yangzexian@fzu.edu.cn; Conference name: 3rd IEEE International Conference on Deep Learning and Computer Vision, DLCV 2026; Conference date: 6 June 2026 through 8 June 2026; Conference code: 226781

DziriFake: A Dataset and Comparative Benchmark of Large Language Models for Fake News Detection In Algerian Dialect

Item Type	Conference Paper
Author	K. Hemina
Author	F. Boumahdi
Author	A. Madani
Author	A. Guendouz
Author	K. Harma
Author	M. Cherif Hamadi
Author	M. Ibrahim Shapiai
Editor	Poonia R.C.
Editor	Upreti K.
Editor	Sharma S.
Editor	Hameed I.A.
Abstract	To address the growing challenge of misinformation in low-resource language contexts, we present DziriFake, a novel dataset for fake news detection in the Algerian dialect (Darija). DziriFake contains 1,110 news articles, 605 labeled as fake and 505 as real, manually verified by native speakers and collected from social media, crowd-sourced contributions, and translated content from existing English and Arabic datasets. Beyond dataset construction, we perform a comprehensive comparative evaluation of both transformer-based models and large language models (LLMs) fine-tuned for binary fake news classification. The combination of a Random Forest classifier and DziriBERT embeddings produced the highest accuracy of 90.98% among all tested methods. With 90% accuracy, Qwen3 1.7B was the best-performing model in the general-purpose LLM category, demonstrating the value of smaller LLMs when properly adjusted. These results show that in dialectal environments with limited resources, model architecture and fine-tuning approach are both critical to performance. DziriFake is made publicly available to support further research on misinformation detection and in Algerian dialect. © The Author(s), under exclusive license to Springer

Nature Switzerland AG 2027.

Date 2027

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Language English

Extra Journal Abbreviation: Smart Innov. Syst. Technol.

Volume 505 SIST

Publisher Springer Science and Business Media Deutschland GmbH

ISBN 21903018 (ISSN); 978-303229236-0 (ISBN)

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Proceedings Title Smart Innov. Syst. Technol.

Conference Name Smart Innovation, Systems and Technologies

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Computational linguistics, Language model, Large language model, Classification (of information), Social media, Large datasets, Large Language Models, Fake news detection, Fake News Detection, Text classification, Text Classification, Text processing, Low resource languages, News articles, Low-Resource Languages, Dialectal Arabic, Dialectal arabics, Algerian dialect, Algerian Dialect, Comparative evaluations

Notes:

EID: 2-s2.0-105043963189

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K. Hemina; LRDSI Laboratory, Computer Science Department, Sciences Faculty, University Saad Dahlab Blida, Soumaa, Blida, Algeria; email: hemina_karim@univ-blida.dz; Conference name: 2nd International Conference on Smart Cyber Physical Systems, ICSCPS 2026; Conference date: 23 January 2026 through 24 January 2026; Conference code: 355199

Early Prediction of Depression and Suicidal Risk in Bangla Social Media Using Deep Learning

Item Type Conference Paper

Author M.A. Rahman

Author H. Ali

Author M.F. Hossen

Author M.R. Palas

Author M.T. Hossen

Abstract Mental disorders such as depression, anxiety, and suicidal ideation are rising rapidly among young people in Bangladesh, with social media providing a prominent platform for expressing oneself and emotions. Early identification of these through automatic systems can provide timely intervention and even save lives. To address this problem, we propose a hybrid CNN-BiGRU model that combines linguistic, psychological, and sentiment attributes to model semantic meaning along with longterm contextual relations in Bangla text with a 94% accuracy. Our method, trained on a well-curated dataset, is more precise and has superior generalization compared to standard machine learning methods like Logistic Regression, Support Vector Machines, and Multinomial Naive Bayes. These results indicate the potential of deep learning towards effective mental health detection in low-resource languages. © 2025 IEEE.

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DOI 10.1109/STI69347.2025.11367594

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Language English

Extra Journal Abbreviation: IEEE Int. Conf. Sustain. Technol. Ind. 5.0, STI

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-833158310-1 (ISBN)

Proceedings Title IEEE Int. Conf. Sustain. Technol. Ind. 5.0, STI

Conference Name 2025 IEEE 7th International Conference on Sustainable Technologies For Industry 5.0, STI 2025

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Deep learning, Learning systems, Sentiment analysis, Social networking (online), Social media, Data mining, Semantics, Neural networks, CNN, Automatic identification, Logistic regression, NLP, Support vector regression, Suicidal ideation, Mental disorders, Automatic systems, Bangladesh, BiGRU, Early prediction, Sentiment-Analysis, Young peoples

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Green University, GUB; Correspondence Address: M.A. Rahman; Green University of Bangladesh, Department of Computer Science and Engineering, Dhaka, Bangladesh; email: rahmanf708@gmail.com; Conference name: 7th International Conference on Sustainable Technologies For Industry 5.0, STI 2025; Conference date: 11 December 2025 through 12 December 2025; Conference code: 219995

Editorial: Special Section Forum for Information Retrieval Evaluation (FIRE)
2024

Item Type Journal Article**Author** T. Mandl**Author** P. Majumder

Abstract The Forum for Information Retrieval Evaluation (FIRE) is an evaluation initiative focused on resources for languages of India. FIRE 2024 was the 16th edition and comprised 10 evaluation tracks which ran as shared tasks. Three contributions showcase the research contributed to these evaluation tracks of the FIRE conference. The first contribution is related spoken language retrieval for six languages of India. The second paper deals with source code retrieval for the language C and employs LLMs for the task of generating good comments. The third contribution discusses performance patterns of classifiers for hate speech collections for languages of India. Furthermore, connections to the Pattern Recognition community are discussed. © 2025 Elsevier B.V.

Date 2026**DOI** 10.1016/j.patrec.2025.10.012**URL** <https://www.scopus.com/pages/publications/105024139306?origin=resultslist>**Archive** Scopus**Language** English**Volume** 199**Publisher** Elsevier B.V.**Pages** 285-287**Publication** Pattern Recognition Letters**ISSN** 01678655 (ISSN)**Journal Abbr** Pattern Recogn. Lett.**Date Added** 20/09/2026, 8:35:09 pm**Modified** 20/09/2026, 8:35:09 pm**Tags:**

Natural language processing, Text classification, Low resource languages, Information Retrieval, Language models

Notes:

EID: 2-s2.0-105024139306

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T. Mandl; University of Hildesheim, Hildesheim, Germany; email: mandl@uni-hildesheim.de

Efficiency vs. understanding: a critical examination of ChatGPT's performance in context-sensitive annotation tasks

Item Type Journal Article**Author** C. Buder**Author** N.-S. Fritsch**Author** C. Osorio-Krauter

Author A. Philipp

Author R. Verwiebe

Author S. Weißmann

Abstract This paper examines context-sensitive annotation tasks performed by OpenAI's o3 reasoning model using content creator profiles on YouTube. It explores the inherently ambiguous and socially and culturally embedded task of annotating race. Analysing 500 annotations generated by ChatGPT, we first examine performance metrics and benchmark results against a human-annotated dataset. We further conduct a thematic analysis of the justifications provided by the model, as users increasingly rely on them making them important for understanding how the model frames social practices. Analyzing justifications also exposes inconsistencies, biases, and classification errors that remain invisible in aggregate performance metrics alone. Our findings show that despite efficiency and scalability, ChatGPT's limited cultural understanding and lack of critical reflexivity constrain its performance in complex annotation tasks. We argue that the annotation of sensitive social characteristics requires reflexive scientific practices and potentially hybrid annotation strategies to mitigate bias and preserve contextual integrity in academic research. © 2026 The Author(s). Published with license by Taylor & Francis Group, LLC.

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Publication International Journal of Sociology

ISSN 15579336 (ISSN)

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ChatGPT, large language model, YouTube, Annotating race, content creators, social media data

Notes:

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Efficient Sentiment, Emotion, and Toxic Speech Analysis on Social Media
Using Lightweight LLMs: A Practical Approach for Digital Transformation

Item Type Journal Article

Author I. Kapantaidakis

Author I. Kopanakis

Author E. Perakakis

Author K. Vassakis

Abstract Social media generates massive amounts of unstructured, multilingual textual data that contain many sentiments, emotions, and opinions from users that provide insights for enterprises undergoing digital transformation. In this paper, we evaluate a variety of commercially available, lightweight large language models (LLMs) for sentiment, emotion, aspect-based sentiment, and toxic speech detection on noisy, real-world social media datasets. These lightweight models are compared (against traditional RoBERTa-based classifiers and heavyweight “pro” LLMs) in terms of performance, latency, and cost trade-offs that are key for scalable deployment across the enterprise. Our results show that lightweight LLMs achieve good accuracy with considerably lower response times and costs, which makes them suitable as the next step in digital transformation for real-time social media analytics. Additionally, we investigate the importance of prompt engineering and show that preprocessing can be very limited for LLMs. This study provides operational guidelines regarding model selection based on latency, accuracy, and cost trade-offs as well as optimal prompt engineering techniques. Finally, it evaluates the intersection of operational performance and resource efficiency to help researchers and developers integrate LLM-based social media analytics into digitized business practices. © 2026 by the authors.

Date 2026

DOI 10.3390/asi9070145

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Notes:

EID: 2-s2.0-105045704772

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: I. Kapantaidakis; Department of Management Science and Technology, Hellenic Mediterranean University, Agios Nikolaos, 72100, Greece; email: jkapad@hmu.gr

EGov-Lens: A Multi-Dimensional Machine Learning Approach to Aspect-Based Public Feedback Analysis on Bengali e-Government Platforms

Item Type Conference Paper

Author M.M. Fuad

Author A.A. Mazid

Author M.I. Adnan

Author T. Das

Author S.S. Prapti

Author N. Mahbub

Abstract With the rapid expansion of e-services under Digital Bangladesh initiatives, citizens increasingly rely on online platforms to access government services and most often share their opinion in social media and digital forums. Existing sentiment analysis lacks of aspect-level insights in terms of egovernment related services. This paper presents eGov-Lens, a practical pipeline for aspect-based analysis of Bengali public feedback on e-government channels. We curate and annotate 34,001 comments (2020-2024) spanning twelve services and four policy-relevant aspects- sentiment, ease of find, responsiveness, and accuracy. The approach benchmarks three text representations (bag-of-words/ CountVectorizer, BanglaBERT, mBERT) with logistic regression, support vector classifier, and naïve bayes. BanglaBERT with a support-vector head attains the strongest single-aspect quality on the policy-critical Accuracy task (micro-avg accuracy ≈ 0.93), while BanglaBERT with logistic regression offers the best accuracy-latency trade-off for deployment and a classical CountVectorizer+LR baseline trains and infers fastest. An entropy-weighted TOPSIS combines quality and cost to rank alternatives, favoring CountVectorizer+LR overall with BanglaBERT+LR a close second, and providing actionable guidance for NLP deployment on resource-constrained government infrastructure. © 2025 IEEE.

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Public policy, Economic and social effects, Social aspects, Learning systems, Sentiment analysis, Social networking (online), Data mining, TOPSIS, Information services, Logistic regression, Text classification, text classification, Low-resource NLP, Aspect-based sentiment analysis (ABSA), Aspect-based sentiment analyze, Support vector regression, Bengalis, Banglabert, BanglaBERT, low-resource NLP, Bengali (bangla), Bengali (Bangla), e-government, e-Government, public feedback, Public feedback

Notes:

EID: 2-s2.0-105041621053

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M.M. Fuad; Bangladesh University of Engineering and Technology, Department of Industrial and Production Engineering, Dhaka, Bangladesh; email: 2008079@ipe.buet.ac.bd; Conference name: 28th International Conference on Computer and Information Technology, ICCIT 2025; Conference date: 19 December 2025 through 21 December 2025; Conference code: 223902

EilMoB: Emotion-aware Incongruity Learning and Modality Bridging Network for Multi-modal Sarcasm Detection

Item Type Conference Paper**Author** H. Zhao**Author** Y. Xu**Author** X. Lin**Author** J. Lu**Author** H. Xu**Author** Y. Wang

Abstract Multi-modal sarcasm detection aims to identify sarcasm by fusing textual and visual information, with incongruity serving as a central cue. Although existing incongruity-based approaches have achieved notable success, they often fail to fully leverage the rich information embedded in text and images, leading to potential misinterpretations. Specifically, text-image pairs frequently contain Emotion-aware Incongruity (EAI), which can significantly amplify sarcastic expressions, particularly when the emotional tones of the text and image are misaligned. However, the modality gap between text and images hinders effective cross-modal incongruity capture, limiting the generalization of current models. To address these challenges, we propose an Emotion-aware Incongruity Mining and Modality Bridging Network. Our approach starts with a carefully designed prompt template that utilizes Multi-modal Large Language Models (MLLMs) to extract the EAI embedded within text-image pairs. Since EAI is inherently textual, its integration with the original text does not suffer from modality gaps.

Moreover, we design a Modality Bridging Module to further mitigate the impact of the modality gap. We extract features from the original text, image, and the EAI, and introduce an Incongruity Mining Module to fully exploit the incongruity among these features. Finally, we propose a Constraint Fusion Module to fuse multi-modal features better. Extensive experiments on two benchmark datasets demonstrate that our approach outperforms state-of-the-art methods. © 2025 ACM.

Date 2025

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Publisher Association for Computing Machinery, Inc

ISBN 979-840071877-9 (ISBN)

Pages 1868-1876

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Tags:

Modal analysis, Human engineering, Sentiment analysis, Social media, Multi-modal, social media, Generalisation, Cross-modal, sentiment analysis, Text mining, Textual information, Visual information, Image pairs, multi-modal sarcasm detection, Multi-modal sarcasm detection, Text images

Notes:

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Export Date: 20 September 2026; Cited By: 2; Funding details: National Key Laboratory on Blind Signal Processing, (23007522); Correspondence Address: Y. Xu; Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China; email: xuyongxiu@iie.ac.cn; Conference name: 2025 International Conference on Multimedia Retrieval, ICMR 2025; Conference date: 30 June 2025 through 3 July 2025; Conference code: 210340

EMNLP 2023 - 2023 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Industry Track

Item Type Journal Article

Editor Wang M.

Editor Zitouni I.

Abstract The proceedings contain 77 papers. The topics discussed include: BeautifulPrompt: towards automatic prompt engineering for text-to-image synthesis; enhancing language model with unit test techniques for efficient regular expression generation; a comparative analysis of task-agnostic distillation methods for compressing transformer language models; towards effective automatic debt collection with persona awareness; efficient transformer knowledge distillation: a performance review; CDD: a large scale dataset for legal intelligence research; personalized dense retrieval on global index for voice-enabled conversational systems; and Text2Topic: multi-label text classification system for efficient topic detection in user generated content with zero-shot capabilities.

Date 2023

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Publication EMNLP 2023 - 2023 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Industry Track

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EmoBlend Fusion: Leveraging Handcrafted and Deep Features for Emotion Detection

Item Type Conference Paper

Author F. Anzum

Author M.L. Gavrilova

Editor Hou M.

Editor Falk T.H.

Editor Mohammadi A.

Editor Guerrieri A.

Editor Kaber D.

Abstract Correct detection of human emotions plays a vital role in human-machine communication and decision-making. Recent advancements in large language models (LLMs) have sparked research in opinion mining, emotion detection, and sentiment analysis. While features extracted from texts using pre-trained LLMs can capture patterns effectively, integrating domain-specific handcrafted features remains crucial for constructing predictive models using user-generated

microblogs. This paper proposes EmoBlend Fusion, a hybrid emotion detection model incorporating handcrafted features and latent deep features using pre-trained embeddings extracted from tweets. While the handcrafted features are trained using an ensemble of random forest, XGBoost, and linear SVM classifiers, a new bidirectional convolutional neural network-based model is trained using the deep features. Model predictions are then fused through a weighted ensemble mechanism for classifying emotions into one of the six distinct classes: sadness, joy, love, anger, fear, and surprise. The experimental results demonstrate that the proposed model outperforms the state-of-the-art techniques, achieving the highest recall (0.94), precision (0.95), accuracy (0.94), and F1-score (0.94) on a benchmark Twitter emotion detection dataset. © 2024 IEEE.

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Language English

Extra Journal Abbreviation: IEEE Int. Conf. Hum.-Mach. Syst., ICHMS

Publisher Institute of Electrical and Electronics Engineers Inc.

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Notes:

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Export Date: 20 September 2026; Cited By: 1; Funding details: Natural Sciences and Engineering Research Council of Canada, NSERC; Innovation for Defense Excellence and Security Network; Correspondence Address: F. Anzum; University Of Calgary, Dept. Of Computer Science, Calgary, Canada; email: fahim.anzum@ucalgary.ca; Conference name: 4th IEEE International Conference on Human-Machine Systems, ICHMS 2024; Conference date: 15 May 2024 through 17 May 2024; Conference code: 200331

Emoji Prediction of Japanese X Posts by LLMs

Item Type Conference Paper

Author Y. Hua

Author T. Utsuro

Editor Oco N.

Editor Dita S.N.

Editor Borlongan A.M.

Editor Kim J.-B.

Abstract Social media users often enhance and complement the emotions underlying in their posts by using emojis. This leads to an increase in research on sentiment analysis using text accompanied by emojis. While there are plenty of previous works for predicting emojis in English posts based on understanding the meaning of posts and classifying them into appropriate emoji categories, research on emoji prediction in Japanese is scarce. Additionally, all of those previous works utilize classification models like BERT instead of large language models. Therefore, in this paper, we utilize three large language models, ChatGPT, Claude and Gemini to predict emojis for Japanese X posts, and compare the results to pre-trained models such as XLM (Conneau and Lample, 2019), Japanese BERT and Japanese RoBERTa. The results show that Claude with 8 shots provided performs the best. © PACLIC 2024. All Rights Reserved.

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Language English

Extra Journal Abbreviation: Proc. Pac. Asia Conf. Lang., Inf. Comput. PACLIC.

Publisher Association for Computational Linguistics

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Proceedings Title Proc. Pac. Asia Conf. Lang., Inf. Comput. PACLIC.

Conference Name PACLIC - Pacific Asia Conference on Language Information and Computation

Issue 2024

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Emoji-aware multimodal representations for multi-label emotion analysis in Indonesian social media

Item Type Journal Article

Author A. Amalia

Author M.S. Lydia

Author R.P. Rangkuti

Author F.P. Marulitua

Author F. Hanif

Author M. Fitra

Author D. Gunawan

Abstract Text Emotion Analysis (TEA) has progressed rapidly, but many studies still approach emotion as a single-label problem and focus solely on text. Research on TEA in low-resource languages such as Indonesian remains relatively scarce, and most available datasets do not explicitly capture paralinguistic cues, such as emojis, even though emojis often convey strong emotional signals. To bridge this gap, we propose an emoji-aware multimodal framework for multi-label emotion classification in Indonesian social media, using the eight Plutchik emotions as labels. We constructed the dataset using a semi-automated annotation pipeline: distant supervision was used to bootstrap labels, and experts then refined them by considering both text and emoji signals. The release contains 139,414 cleaned Indonesian social media posts from YouTube, X (formerly Twitter), Reddit, and Steam. We provide (i) a mixed corpus with posts both with and without emojis and (ii) an emoji-rich subset for emoji-aware experiments, along with the annotation guidelines and fixed train/ validation/test splits. For all model variants, Indonesian Bidirectional Encoder Representations from Transformers (IndoBERT) serves as the text backbone; the only difference lies in how emoji information is incorporated. We tested three integration options for emojis: adding them early (by concatenation or tokenizer expansion), fusing them mid-stream with cross-attention, or combining predictions late (average/weighted). Compared to a text-only baseline, emoji-aware models perform better overall, reflecting the fact that emojis often convey emotional cues that text alone misses. The best scores come from tokenizer expansion (micro-averaged F1 = 0.8554; macro-averaged F1 = 0.7769), likely because emojis are processed alongside words through IndoBERT's contextual layers. Beyond model comparisons, we also introduce an Indonesian emoji-aware multi-label dataset and release fixed splits and guidelines to support reproducible research. © 2026 Y. Sarhan et al. Distributed under Creative Commons CC-BY 4.0

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Emotion Analysis for Chinese Text Using an OpenAI Embedding Model

Item Type Conference Paper

Author M. Zhang

Author Y. Adachi

Author J. Zhu

Author M. Uehara

Abstract Emotion analysis techniques, which automatically analyze the emotions expressed in text, are used to analyze online reviews, survey responses, and social media content. In emotional analysis, the detailed information of a sentence must be analyzed, not just the nouns but also the verbs, adjectives, adverbs, and negation words, and it is desirable to use a large language model based on higher dimensional embeddings. In this study, we report a fine-grained emotion analysis method that extracts 11 emotion categories from Chinese sentences using an OpenAI embedding model that generates 3072dimensional embeddings as a feature extractor. We built an emotion analyzer that uses the embeddings generated by OpenAI as input and a support vector machine or deep neural network as a feature classifier, and evaluated the performance (analysis accuracy and speed). These emotion analyzers had good analytical accuracy for emotion categories with a large number of training data, but they had poor analytical accuracy for emotion categories with few training data. When we evaluated the coarse-grained sentiment analyzer by labeling each specific group of 11 emotion categories as either positive, neutral, or negative, it showed very high accuracy. © 2025 IEEE.

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Extra	Journal Abbreviation: Int. Symp. Comput. Networking Workshops, CANDARW
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	28321324 (ISSN); 979-833155533-7 (ISBN)
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Proceedings Title	Int. Symp. Comput. Networking Workshops, CANDARW
Conference Name	International Symposium on Computing and Networking Workshops, CANDARW
Issue	2025
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Tags:

Behavioral research, Support vector machines, Support vectors machine, Classification (of information), Sentiment analysis, Data mining, Neural-networks, Embeddings, sentiment analysis, Training data, deep neural network, Psychology computing, Emotion analysis, support vector machine, Analysis techniques, Online reviews, Chinese text, emotion analysis, Network embeddings, Openai embedding model, OpenAI embedding model

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Zhang; Toyo University, Grad. Sch. of Inf. Sci. and Arts, Kawagoe, Japan; email: s3B102500166@toyo.jp; Conference name: 13th International Symposium on Computing and Networking Workshops, CANDARW 2025; Conference date: 25 November 2025 through 28 November 2025; Conference code: 218914

Emotion Analysis in Kerala's Political Discourse: Curating and Evaluating a Multilabel Code-Mixed Malayalam-English Social Media Dataset Using LLMs

Item Type	Conference Paper
Author	C.V. Anushka
Author	P.C. Nair
Editor	Vijayan K V.K.
Editor	Singh V.
Abstract	In Kerala, social media has become the primary arena for political debate, with much of this discourse expressed in Malayalam-English code-mixing, commonly known as Manglish. While this hybrid language reflects the region's cultural and linguistic realities, it creates serious hurdles for Natural Language Processing (NLP) tasks like emotion detection, where most existing tools are built for monolingual or well-resourced languages. To overcome this limitation, we present a new code-mixed Malayalam-English dataset specifically curated from YouTube comments centred on Kerala's political discussions. The dataset is manually annotated for five key political emotions-Anger, Joy/Support, Sadness/

Opposition, Fear, and Sarcasm-by native bilingual speakers, and annotation consistency is verified through Krippendorff's Alpha. We evaluate this resource through multiple baselines, including conventional machine learning(ML) algorithms, deep learning(DL) models, and transformer architectures, and extend the analysis to advanced language models like LLMs using zero-shot as well as few-shot approaches. The results reveal both the difficulties and the potential of existing models when applied to code-mixed political text. Our work represents one of the earliest systematic efforts to build and benchmark an emotion-labelled dataset for Manglish in a political context. Beyond serving as a valuable testbed for multilingual and low-resource NLP research, this dataset has practical relevance in analysing public sentiment, shaping data-driven political communication strategies, and informing policy decisions. © 2026 The Authors.

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Language English

Extra Journal Abbreviation: Procedia Comput. Sci.

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Conference Name Procedia Computer Science

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Notes:

EID: 2-s2.0-105046091030

Export Date: 20 September 2026; Cited By: 0; Conference name: 4th international conference on Machine Learning and Data Engineering, ICMLDE 2025; Conference date: 6 November 2025 through 8 November 2025; Conference code: 226014

Emotion Analytics: Unveiling the Sentiment Aspect in E-Commerce Review Platforms

Item Type Conference Paper

Author N.B.T. Nghia

Author H.K. Tu

Author N.M. Nhut

Author N.D. Thuan

Editor Quan T.T.

Editor Pham H.-A.

Editor Tran N.T.

Editor Sombattheera C.

Abstract Aspect-Based Sentiment Analysis (ABSA) has gained increasing attention due to its wide applicability in domains such as marketing, e-commerce, and social media analytics. Nevertheless, research on ABSA in Vietnamese remains limited, especially in the context of e-commerce product reviews, which restricts the development of practical sentiment analysis applications for this language. To address this gap, this study aims to develop and evaluate hybrid models that integrate PhoBERT, a pre-trained language model optimized for Vietnamese, with classical machine learning techniques to enhance aspect-level sentiment classification. Two approaches are proposed: (i) combining PhoBERT embeddings with a Naive Bayes classifier and (ii) integrating PhoBERT with a Hidden Markov Model (HMM) to capture sequential sentiment dynamics. Experiments conducted on a Vietnamese product review dataset collected from Tiki, a leading e-commerce platform, show that the PhoBERT + Naive Bayes model achieves strong performance with an accuracy of 94.54% and an F1-score of 0.95, making it well suited for real-time and interpretable sentiment analysis. In contrast, the PhoBERT + HMM model underperforms with only 32.96% accuracy, exhibiting high inference time and prediction instability. These findings demonstrate the superiority of the PhoBERT + Naive Bayes model while highlighting the limitations of probabilistic sequence models like HMM when combined with transformer-based embeddings. The study contributes to advancing ABSA in low-resource languages and underscores the potential of hybrid modeling approaches to balance accuracy, scalability, and interpretability. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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DOI 10.1007/978-981-95-4963-4_6

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Emotion detection for misinformation: A review

Item Type Journal Article

Author Z. Liu

Author T. Zhang

Author K. Yang

Author P. Thompson

Author Z. Yu

Author S. Ananiadou

Abstract With the advent of social media, an increasing number of netizens are sharing and reading posts and news online. However, the huge volumes of misinformation (e.g., fake news and rumors) that flood the internet can adversely affect people's lives, and have resulted in the emergence of rumor and fake news detection as a hot research topic. The emotions and sentiments of netizens, as expressed in social media posts and news, constitute important factors that can help to distinguish fake news from genuine news and to understand the spread of rumors. This article comprehensively reviews emotion-based methods for misinformation detection, with a particular focus on advanced fusion methods. We begin by explaining the strong links between emotions and misinformation. We subsequently provide a detailed analysis of a range of misinformation detection methods that employ a variety of emotion, sentiment and stance-based features, and describe their strengths and weaknesses. Finally, we discuss a number of ongoing challenges in emotion-based misinformation detection based on large language models, and suggest future research directions, including data collection (multi-platform, multilingual), annotation, benchmark, multimodality, and interpretability. © 2024 The Author(s)

Date 2024

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Emotion Detection in Gujarati Text: Evaluation of Machine Learning Classifiers with Bag-of-Words and TF-IDF Feature Extraction Techniques

Item Type Conference Paper

Author K. Patil

Author J. Nasriwala

Author R. Savant

Author P. Bhatt

Editor Tiwari S.

Editor Ortiz-Rodríguez F.

Editor Shivakumar A.

Editor Shimizu C.

Abstract In the model digital era, social media platforms have become central hub for an individual to express their thoughts, opinion and emotions in various form of textual data. Text classification tasks such as sentiment analysis, and emotion detection from the text is increasing day-by-day for the applications such as social media analysis, customer feedback and interpretation and mental health monitoring. This study presents an emotion detection from Gujarati text which is a low resource and morphologically rich Indian Language. It also employs four machine learning classifiers namely Naïve Bayes, Logistic Regression, Support Vector Machine (SVM), and Random Forest along with two feature extraction technique BoW and TF_IDF. The performance of the model is evaluated using confusion matrix which provides class wise as well as overall accuracy. The

outcome reveals that Random Forest achieves highest accuracy followed by SVM classifier. Overall, using this approach the accuracy of the emotion detection from Gujarati text is increased. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Notes:

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Emotion Detection in Hindi Language Using GPT and BERT

Item Type Conference Paper

Author R. Agarwal

Author N. Abbas

Editor Bramer M.

Editor Stahl F.

Abstract Emotion detection in textual data plays an important role in various NLP applications and is an important branch of sentiment analysis. There is a dearth of applications of emotion detection in low resource languages such as Hindi. There are a number of challenges in the available open-source resources in Hindi language such as imbalanced datasets and lack of understanding of textual nuances specific to Hindi. This paper presents a comprehensive study on emotion detection in Hindi text using Large Language Models (LLMs). We will be leveraging GPT-3.5-turbo API for dataset augmentation using few-shot learning and zero-shot learning techniques and further fine-tuning pretrained BERT and mBERT models. This study's key findings include generating two balanced datasets to establish baselines, fine-tuning models for emotion detection and evaluating their performance. There is a significant improvement in Accuracy and F1-score, demonstrating the effectiveness of the approach employed. We further discuss the future directions for research including multimodal emotion detection and cross lingual applications along with ethical considerations in deploying emotion detection systems. This research contributes to advancing the understanding and applications of emotion detection in Hindi text such as customer feedback assessment, social media sentiment analysis, mental health evaluation and human-computer interaction (HCI) via virtual assistants like Alexa and Siri. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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DOI 10.1007/978-3-031-77918-3_8

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Sentiment analysis, Fine tuning, Emotion Recognition, Emotion detection, Low resource languages, Synthetic data generations, Data augmentation, Zero-shot learning, Textual data, Open-source, Fine-tuning mBERT, Hindi emotion detection, Hindi Emotion detection, Synthetic data generation

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: R. Agarwal; University of Leeds, Leeds, United Kingdom; email: ritika.agarwal24@gmail.com; Conference name: 44th SGAI International Conference on Innovative Techniques and Applications of Artificial Intelligence, SGAI 2024; Conference date: 17 December 2024 through 19 December 2024; Conference code: 323609

Emotion Recognition and Analysis System (ERAS): Leveraging Advanced AI for real-time Emotion Detection in Social Media Content

Item Type Conference Paper

Author A. Siddika

Author T. Rayhan

Author M.S. Sarkar

Author M.M. Hassan

Abstract This study outlines the development of an advanced system for detecting and analyzing emotions expressed in social media content using cutting-edge artificial intelligence techniques in a web application. The application system utilizes advanced natural-language processing (NLP) and machine-learning techniques to distinguish emotions precisely in text-based content, including posts, comments, and tweets. By employing sentiment analysis, machine learning models, and emotion recognition algorithms within the developed web application Emotion Recognition and Analysis System (ERAS), the application utilizes 28 distinct emotional categories and employs 9 machine learning algorithms, such as Logistic-Regression, Linear-SVC, BERT-Classifer, Decision-Tree Classifier, AdaBoost-Classifer, Zero-shot Classifier, RNN-Classifer, Random-Forest, and Naive-Bayes Classifier including Logistic-Regression, which achieved the highest accuracy of 89%. The application itself delivers an accuracy of 90% above through a custom model, which dominates the performance of the Logistic-Regression model in some emotions like anger, fear, joy, sadness and so on. To develop this system, datasets were sourced from Kaggle and Hugging Face, ensuring comprehensive emotional coverage and alignment with research objectives. The GoEmotions dataset and selected Hugging Face datasets were primarily used for final model training due to their broad range of emotional categories. The evaluation of performance was carried out using metrics like precision, re-call, and F1-score. This application facilitates real-time thought analysis, presenting valuable insights for marketing, mental health, and social research. © 2025 IEEE.

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Artificial intelligence, Emotional intelligence, Sentiment analysis, Emotion recognition, Emotion Recognition, Sentiment Analysis, Artificial Intelligence (AI), Logistic regression, Labeled data, Emotion analysis, Tweets, Zero-shot learning, Social media analytics, Social Media Analytics, Analysis system, Societal impacts, Linear regression, Introduction, Recognition systems, Societal impact., Societal Impact. Introduction

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EID: 2-s2.0-105007682718

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: A. Siddika; World University of Bangladesh, Dept. of Cse, Dhaka, Bangladesh; email: ayesha@cse.wub.edu.bd; Conference name: 2025 International Conference on Electrical, Computer and Communication Engineering, ECCE 2025; Conference date: 13 February 2025 through 15 February 2025; Conference code: 209222

Emotional Comfort or Technological Delusion? Unveiling Public Perceptions and Attitudes Toward AI Companions Based on Video Social Medias

Item Type Journal Article

Author H. Wei

Author B. Xiang

Author X. Zheng

Author J. Yang

Abstract AI companions, driven by artificial intelligence, are increasingly embedded in users' daily lives via anthropomorphic and interactive conversational experiences. Although these systems can offer emotional support, they also raise concerns regarding algorithmic bias or data security. A comprehensive understanding of public perceptions is therefore essential for promoting healthy human-AI relationships. This paper investigates public perceptions of AI companions using 613 videos and 33,133 comments from two video platforms. Combining text mining and large language model-based sentiment analysis, it examines public topics and sentiment patterns regarding AI companions from multiple dimensions. Public attention surged after generative AI's rise in 2023, with a pronounced "East-West divide" in user engagement. Discussions cover society (role substitution, family) and technology (companionship, anthropomorphism), with notable differences in the topics between men and women, and overall sentiment is positive. Moreover, personality traits also influence attitudes. By comparing these results with existing studies, it highlights how Chinese cultural contexts shape public perceptions. Drawing on diverse user-generated data from video platforms, the study offers valuable insights for the better design, marketing, and policymaking of AI companions. © 2026 Taylor & Francis

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EmotionPrompt: Prompt-Tuned Large Language Models for Multilingual Sentiment Analysis on MELD and SemEval

Item Type Conference Paper

Author R. Utthirakumari

Author R. Harini

Author S. Deepika

Abstract Proper and efficient emotion and sentiment detection in a multilingual conversation is critical to facilitate intelligent human computer interaction in real-time and scalable content understanding. This paper presents a prompt-tuned large language model (LLM) method of multilingual sentiment and emotion recognition, particularly on dialogue data of the MELD and SemEval benchmarks. The mechanism employs the use of prompts by which light, expressive prompts can be used to control the LLM such that the underlying emotion (e.g., happiness, anger, sadness) and the sentiment (positive, negative, neutral) can be mutually determined using short and varied prompts in numerous languages. The system can also be trained through prompt-tuning instead of conventional full-model fine-tuning, which means it can train on zero-shot and

few-shot learning, which reduces the need to use large quantities of labelled data in each target language by a substantial amount. The proposed pipeline preprocesses and encodes conversation data, multi-label, context-aware prediction which is done with the prompt-tuned LLM, and has been widely tested on different MELD and SemEval datasets. The results of the experiments demonstrate that the method achieves high accuracy and saves much on adaptation costs, being more flexible than the standard fine-tuned baselines in both flexibility and multilingual support. The new paradigm of light, generalizable emotion and sentiment detection is an efficient and scalable solution to low-resource and real-life conditions such as international customer service, mental health monitoring, chatbot conversation systems, and cross-lingual social media monitoring. Finally, wider situational context, less development, and more representative AI in multilingual emotionally dense environments are offered by the application of prompt-tuned LLMs. © 2025 IEEE.

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Language model, Large language model, Prompt tuning, Speech processing, Artificial intelligence, Learning systems, Sentiment analysis, Speech recognition, Human computer interaction, Real-world, Chatbots, Emotion recognition, Emotion Recognition, Psychology computing, Low resource languages, Tuning, Multilingual Sentiment Analysis, Multilingual sentiment analyze, Prompt Tuning, Low-resource Languages, Zero-shot Learning, SemEval, Large Language Model (LLM), MELD, Real-world dialog, Real-world Dialogue, Semeval

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EmoTweetID: A dataset of Indonesian tweets for emotion classification and word embedding construction

Item Type Journal Article

Author K.S. Nugroho

Author F.A. Bachtiar

Author W.F. Mahmudy

Author M.M. Henry

Author M. Isnani

Author G. Pangestu

Author B. Pardamean

Abstract The EmoTweetID dataset is a large-scale open resource of Indonesian tweets curated for emotion classification and word embedding, addressing the scarcity of publicly available datasets for this low-resource language. Tweets were collected from the X platform (formerly Twitter) using the snsrape library, yielding >4.5 million raw tweets based on basic and derived emotion keywords from Ekman's six basic emotions. After cleaning to remove duplicates, irrelevant tweets, and sensitive elements, a corpus of 3,126,987 clean unlabeled tweets was compiled from the first scraping stage. The second stage gathered 2,243 clean tweets annotated through lexicon-based and manual methods, into six emotion classes based on Ekman's basic emotions: anger, disgust, fear, joy, sadness, and surprise. The manual annotation by three psychology students used majority voting and achieved substantial inter-annotator agreement, with a Fleiss' Kappa score of 0.7323. Two pretrained embeddings, Word2Vec and fastText, were trained on the corpus using a 300-dimensional skip-gram architecture to enrich semantic representation. Baseline evaluations using BiLSTM with fastText achieved a weighted F1-score of 0.8285 on the human-annotated set, demonstrating the practical utility of the dataset and embeddings for the downstream emotion classification task. Publicly available on Mendeley Data, the EmoTweetID dataset provides a valuable foundation for advancing Indonesian natural language processing by enabling unsupervised pre-training and supervised multi-class emotion classification. © 2026 The Author(s).

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K.S. Nugroho; Bioinformatics and Data Science Research Center, Bina Nusantara University, Jakarta, 11530, Indonesia; email: kuncahyo.nugroho@binus.edu

EmoVibe: AI-driven multimodal emotion analysis for mental health via social media dashboards

Item Type Journal Article**Author** D. Vora**Author** A. Sharma**Author** M. Garg**Author** S. Francis

Abstract Monitoring mental health via social media often utilizes unimodal approaches, such as sentiment analysis on text or single-staged image categorization, or executes early feature fusion. However, in real-world contexts where emotions are conveyed via text, emojis, and images, unimodal approach leads to obscured decision-making pathways and overall diminished performance. To overcome these limitations, we propose EmoVibe, a hybrid multimodal AI framework for emotive analysis. EmoVibe uses attention-based late fusion strategy, where text embeddings are generated from bidirectional encoder representations from transformers (BERT) and visual features are extracted by vision transformer. Subsequently, emoticon vectors linked to avatars are processed independently. Later, these independent data features are integrated at higher levels, enhancing interpretability and performance. In contrast to early fusion methods and integrated multimodal large language models (LLMs) like CLIP, Flamingo, GPT-4V, MentaLLaMA, and domain-adapted models like EmoBERTa, EmoVibe preserves modality-specific contexts without premature fusion. This architecture saves processing cost, allowing for clearer, unambiguous rationalization and explanations. EmoVibe outperforms unimodal baselines and early fusion models, obtaining 89.7% accuracy on GoEmotions, FER, and AffectNet, compared to BERT's 87.4% and ResNet-50's 84.2%, respectively. Furthermore, a customizable, real-time,

privacy-aware dashboard is created, supporting physicians and end users. This technology enables scalable and proactive intervention options and fosters user self-awareness of mental health. © 2025, Institute of Advanced Engineering and Science. All rights reserved.

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Empirical Study of BERT-Based Models for Sentiment Analysis

Item Type Conference Paper

Author B. Huang

Author F. Song

Editor Yoe H.

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Abstract The rapid expansion of online applications like social media apps and e-commerce websites has led to a large volume of reviews about different subjects, products, and services. Sentiment analysis, which is a crucial area of study in natural language processing, aims to classify the sentiments of these reviews so that the feedback can be valuable to companies, governments, and individuals in making informed decisions based on the information gathered about the public opinions. The emerging deep learning methods, such as large language models have shown great promises in text-based analysis, but their performance for sentiment analysis has not been fully understood yet. This paper presents a comparative analysis of BERT-based models for sentiment analysis, especially those using prompt-based learning techniques across five diverse datasets. We explore how dataset characteristics influence the effectiveness of hand-crafted, adaptive, and hybrid prompting strategies. Our results demonstrate that input length, class distribution, and dataset structure significantly impact model performance. © 2025 IEEE.

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Empowering Airline Route Decisions with LLM-Generated Pseudo-labels and Zero-Shot Review Prediction

Item Type Conference Paper

Author A. Alhamadani

Author K. Althubiti

Author J. He

Author S. Sarkar

Author L. Alkulaib

Author A.R. Shaik

Author S. Lee

Author M. Khan

Author C.-T. Lu

Editor Aiello L.M.

Editor Chakraborty T.

Editor Gaito S.

Abstract The airline industry has suffered a severe impact due to the COVID-19 pandemic. It resulted in significant financial losses. Strategic route planning is now an urgent need to mitigate the ongoing crisis. Motivated by the importance of customer sentiment in informing airline route decisions, this paper presents EAGLE (Enhancing Airline Groundtruth Labels and rEview rating prediction), a novel two-stage framework that leverages the power of Large Language Models (LLMs) to address the limitations of current works, which often rely on manual labeling and traditional machine learning models. In the first phase, EAGLE introduces a pseudo-labeling approach using LLMs to automatically label customer reviews to reduce the need for manual annotation and mitigate potential biases that exist in human labeling. The second phase employs a zero-shot LLM-based text classification method to predict customer sentiment and preferences from online reviews to provide a more accurate and context-aware analysis of customer feedback. Through extensive experiments, we demonstrate the effectiveness and robustness of EAGLE to demonstrate its superior performance compared to existing techniques. The proposed framework empowers airline companies to make data-driven decisions about route expansions, considering

customer preferences and sentiments. Our contribution fibs in enhancing the objectivity of sentiment analysis and providing a comprehensive and scalable solution for airline route planning in the post-pandemic era, eventually leading to improved customer satisfaction and optimized operations. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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EmptyMind at BLP-2023 Task 2: Sentiment Analysis of Bangla Social Media Posts using Transformer-Based Models

Item Type Conference Paper

Author K. Fatema

Author U. Das

Author Md.A. Mia
Author Md.S. Mowla
Author M. Yahan
Author M.D.F. Ullah
Author A. Sarker
Author H. Murad

Editor Sadeque F.
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Editor Chowdhury S.A.
Editor Alam F.

Abstract With the popularity of social media platforms, people are sharing their individual thoughts by posting, commenting, and messaging with their friends, which generates a significant amount of digital text data every day. Conducting sentiment analysis of social media content is a vibrant research domain within the realm of Natural Language Processing (NLP), and it has practical, real-world uses. Numerous prior studies have focused on sentiment analysis for languages that have abundant linguistic resources, such as English. However, limited prior research works have been done for automatic sentiment analysis in low-resource languages like Bangla. In this research work, we are going to finetune different transformer-based models for Bangla sentiment analysis. To train and evaluate the model, we have utilized a dataset provided in a shared task organized by the BLP Workshop co-located with EMNLP-2023. Moreover, we have conducted a comparative study among different machine learning models, deep learning models, and transformer-based models for Bangla sentiment analysis. Our findings show that the BanglaBERT (Large) model has achieved the best result with a micro F1-Score of 0.7109 and secured 7th position in the shared task 2 leaderboard of the BLP Workshop in EMNLP 2023. © 2023 Association for Computational Linguistics.

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Enhanced Multilingual Text Classification with Pre-Trained Language Models

Item Type Journal Article**Author** K. Narasimharao**Author** A.S. V.Jayasri

Abstract Introduction: Multilingual text classification is an important task in natural language processing (NLP), especially in the context of sentiment analysis in languages with different grammatical, semantic, and lexical properties. Labelled data sets are not available for many languages, and hence it is difficult to predict the sentiment. The use of pre-defined language models and translation-based methods is a promising solution to cross-lingual generalization. Objectives: This research has the following research objectives: (1) multilingual sentiment classification, (2) neural machine translation to translate foreign language data, (3) neural machine translation using transformer models and large language models (LLMs) combined with ensemble learning, and (4) multilingual model performance evaluation. Methods: Four languages-Arabic, Chinese, French, and Italian are translated into English using neural machine translation tools: DeepL Translator and Google Translate. The translated text was then fed into an ensemble of 3 pre-trained models: Twitter-RoBERTa-Base-Sentiment-Latest, BERT-Base-Multilingual-Uncased-Sentiment and GPT-4 from OpenAI. The sentiment predictions were aggregated output in a weighted voting mechanism, which was based on performance metrics, such as accuracy, precision, recall, and F1-score. Results: The ensemble model suggested here obtained an overall accuracy of greater than 96%, which was improved when compared to individual models. The ensemble learning with translation enhanced the understanding of context, robust across languages and showed a consistent level of performance in multilingual sentiment classification. Conclusion: The obtained results support the fact that translation-based ensemble learning using pretrained transformers and LLMs is a scalable and accurate multilingual sentiment analysis approach. This paper shows how neural translation and the best pre-trained models can overcome language barriers in NLP. © 2025 Little Lion Scientific.

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Enhanced Sentiment Analysis Using Extended Corpus and Co-Occurrence Graph

Item Type Journal Article

Author N. Romyen

Author H. Unger

Author M. Maliyaem

Abstract Lexicon-based sentiment analysis often faces challenges due to the limited coverage and adaptability of existing sentiment lexicons, particularly in domain-specific or resource-scarce contexts. Constructing new lexicons from scratch is labor-intensive and rarely feasible for large-scale applications. To address this gap, we introduce the Enhanced Sentiment Analysis using Extended Corpus and Co-Occurrence Graph (ECG) framework, which leverages co-occurrence graph construction, triadic closure theory, and PageRank-inspired weighting to propagate sentiment values from a small seed set to a broader vocabulary. Our study focuses on nouns extracted from the Harry Potter corpus, which are typically underrepresented in standard sentiment resources. Experimental results demonstrate that ECG achieves effective sentiment propagation within 1500 iterations, reaching stable convergence. Evaluation using the Coefficient of Variation (COV) indicates improved sentiment differentiation, with broader coverage compared to benchmark lexicons such as SentiWordNet, Valence Aware Dictionary and Sentiment Reasoner (VADER), and National Research Council (NRC). For instance, ECG consistently assigned interpretable sentiment values to narrative-specific entities (e.g., “Slytherin”, “Voldemort”), which were often missed by conventional lexicons. These findings highlight ECG’s scalability, interpretability, and suitability for low-resource scenarios. Beyond literary analysis, the framework can be extended to other domains such as social media, customer feedback, and specialized corpora. Overall, ECG represents a resource-efficient alternative that bridges the limitations of static lexicons and enhances sentiment inference in diverse contexts. © 2026 by the authors.

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Enhanced Sentiment Classification in Code-Mixed Texts Using Hybrid Embeddings and Synthetic Data Generation

Item Type Conference Paper

Author L.L. MacEda

Abstract Social media platforms provide a space for users to share thoughts, connect, and discuss various topics. While some communicate in a single language, others use multiple languages, reflecting the linguistic diversity of their nations. Existing research has largely focused on monolingual data, with less attention on code-mixed data. Studies have explored hybrid models, which have been shown to improve sentiment analysis accuracy compared to single models across various datasets. To enhance sentiment classification in this type of data, we employed a hybrid model combining the distilled multilingual embeddings of BERT with BiLSTM to capture sequential dependencies. Further, we address class imbalance through generating synthetic data with ChatGPT, a large language model with human language comprehension. Our approach shows significant performance improvements across various hyperparameter settings over its vanilla counterparts. Best result showed a 4.82% increase in both accuracy and macro F1 score against DistilmBERT alone. We observed that when prompting ChatGPT for generating synthetic data, every prompt should be distinct to avoid data duplication during data generation. With LIME (Local Interpretable Model-agnostic Explanations) technique, we provide interpretations for correctly

classified and misclassified instances weighing words according to their contribution to our model's prediction. A potential area of further study adapting the model to a more large, extensive, and diverse datasets, including various social media platforms and languages, could improve its generalizability and robustness. © 2024 IEEE.

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Enhancing Aspect-Based Sentiment Analysis in Tourism Reviews Through Hybrid Data Augmentation

Item Type Journal Article
Author N.M.S. Iswari
Author N. Afriliana
Abstract The increasing reliance on online reviews in tourism has made User-Generated Content (UGC) an invaluable resource for understanding visitor perceptions. However, extracting meaningful insights from these reviews remains challenging due to their unstructured nature, aspect imbalance, and the prevalence of code-mixing between languages such as Indonesian and English—particularly in multicultural destinations like Bali. Aspect-Based Sentiment Analysis (ABSA) offers a promising

solution by associating sentiment polarity with specific aspects of tourist experiences. Yet, its performance is often constrained by limited and imbalanced datasets, especially for underrepresented aspects such as sanitation and amenities. This study proposes a hybrid data augmentation framework that integrates three complementary strategies: generative augmentation using ChatGPT, semantic filtering via Sentence-BERT (SBERT), and domain refinement through Masked Language Modeling (MLM). The framework is designed to improve ABSA performance on multilingual tourism reviews by generating synthetic aspect-relevant data while preserving semantic integrity and contextual nuance. Using 398 reviews of Kuta Beach in Bali, we evaluate the effectiveness of the proposed approach across five tourism aspects: scenery, dusk, surf, amenities, and sanitation. Results show that the hybrid strategy reduces hallucination rates from 12% (using ChatGPT alone) to 3.8%, increases F1-scores for underrepresented aspects by up to 5.1%, and improves cross-lingual alignment (Cohen's $\kappa = 0.78$). These improvements demonstrate the synergy between generative and semantic augmentation in addressing real-world ABSA challenges. The proposed method not only advances the state of multilingual ABSA but also offers practical implications for tourism analytics, allowing destination managers to better understand and respond to aspect-specific visitor feedback. The framework is extensible to other low-resource domains, where linguistic diversity and data scarcity present similar limitations. © Authors retain all.

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Enhancing Bangla Sentiment Analysis Through Data Balancing and Preprocessing: A Comparative Insight from Classical and Deep Learning Approaches

Item Type Conference Paper

Author M.A.H. Labu

Author M.T. Sultana

Author M.M. Hassan

Author M.A. Chowdhury

Author A.M. Nitu

Author M.A. Ali

Abstract Sentiment analysis in Bangla remains a challenging task due to the scarcity of labeled data, class imbalance, and the noisy nature of user-generated social media text. Unlike the languages with extensive resources such as English, Bangla lacks large-scale, high-quality datasets that can support robust natural language processing models. In this study, we focus on the SentNoB dataset, which contains 12,575 comments across 13 domains, annotated into positive, negative, and neutral classes. A key limitation of this dataset is the scarcity of neutral samples, which significantly affects model performance and generalization. Initial experiments confirmed these challenges, with Support Vector Machine (SVM) achieving 66% accuracy and Bidirectional Long Short-Term Memory (BiLSTM) reaching only 57.86% of accuracy. These results highlight the adverse effects of both class imbalance and noisy textual features. To address these issues, the authors enriched the dataset with 2,000 additional neutral and 200 negative samples collected from social media, followed by extensive preprocessing that included text cleaning, normalization, stopword removal, and noise reduction. Retraining with this improved dataset yielded substantial gains, with SVM achieving 76.94% and BiLSTM reaching 78.37% accuracy. The authors findings demonstrate that dataset balancing and robust preprocessing are critical for Bangla sentiment analysis and emphasize the complementary strengths of classical machine learning and deep learning in low-resource contexts. © 2025 IEEE.

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Enhancing Content Moderation with LLMs: A Reddit Case Study on Evaluating and Refining Human Decisions

Item Type Conference Paper

Author J. He

Author Y. Wei

Author G. Tyson

Abstract Large Language Models (LLMs) offer significant potential for assisting with the design and implementation of social platform moderation. This study evaluates their efficacy as both a replacement to and an augmentation for human moderators. Using Reddit as a case study, we first demonstrate that LLMs can effectively replicate human moderation decisions, achieving 83.9% agreement. Through a mix of LLMs and human annotations, we then evaluate real moderator decisions, uncovering substantial error rates: 15.2% of removals and 13% of approvals are estimated as incorrect, primarily stemming from moderators citing the incorrect rules (84.3% of errors). This motivates us to propose RuleSharpener, a tool that uses LLMs to diagnose the root causes of moderation errors (e.g. ambiguous rules) and generate clearer, more actionable guidelines. Our evaluation shows that RuleSharpener increases the accuracy of identifying the specific rules violated by violation posts by 38.0%. Our work demonstrates how LLMs can augment human moderation, refine community policies, and reduce operational burdens, offering a better solution for platform governance on the web. © 2026 Owner/Author.

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Enhancing Cryptocurrency Sentiment Analysis with GPT-4: A Comparative Study

Item Type Conference Paper

Author J.K. Ray

Author S.S. Mim

Author D. Logofatu

Editor Maglogiannis I.

Editor Iliadis L.

Editor Papaleonidas A.

Editor Andreou A.

Abstract Attracting investors seeking distributed investing possibilities, cryptocurrencies are gradually taking front stage on financial markets. But sentiment analysis is essential for understanding market dynamics with considerable volatility molded by news, social media trends, and investor mood. This paper investigates the relevance of Large Language Models (LLMs), particularly fine-tuned GPT-4, in cryptocurrency sentiment analysis. By fine-tuning GPT-4 using a cryptocurrency news dataset, this paper compares its sentiment classification performance against other models, including FinBERT, BERT, Flan-T5, and Gemma-7B. The results

indicate higher accuracy since finely tuned LLMs show better in classifying sentiments as positive, neutral, or negative. These results highlight the need of optimizing to raise sentiment analysis capability. This paper contributes to both academic research and financial applications, offering insights into how LLMs can be leveraged for market trend predictions and risk management strategies. © IFIP International Federation for Information Processing 2025.

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Enhancing Fake News Detection through Cross-Lingual Models: A COVID-19 Case Study

Item Type Conference Paper

Author E. de Oliveira Rocha

Author C. de Souza Baptista

Author A.L.F. Alves
Author A.C. de Paiva
Editor Filipe J.
Editor Smialek M.
Editor Brodsky A.
Editor Hammoudi S.

Abstract The proliferation of health-related misinformation on social media poses a serious threat to public well-being and the integrity of information. This paper addresses how to transfer labels effectively to a target language with limited annotations, investigating how performance degrades with lexical distance in the context of COVID-19 misinformation. Using the MM-COVID dataset comprising texts in English, Spanish, and Portuguese, we systematically evaluate five cross-lingual learning strategies: Zero-Shot Transfer, Joint Learning, Cascade Learning, a JL/CL hybrid approach, and an extended JL/CL+. The experiments employ both monolingual BERT encoders (BERT-base, BETO, and BERTimbau) and multilingual models such as XLM-R and mDeBERTa. Performance is assessed using F1 and ROC-AUC metrics, complemented by statistical significance testing. The results show that progressive adaptation strategies, including Cascade Learning and JL/CL variants, consistently outperform monolingual and zero-shot baselines, achieving F1-scores as high as 98.2 percent for Portuguese. These findings suggest that lexical distance plays a role in simple transfer setups but becomes less influential under progressive adaptation schemes. Copyright © 2026 by SCITEPRESS - Science and Technology Publications, Lda.

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Enhancing food safety review classification with large language models and label embedding

Item Type Journal Article

Author S. Xiong

Author W. Tian

Author B. Wang

Author Y. Zhang

Author X. Zhang

Author Y. Che

Author L. Shi

Author H. Si

Abstract In the digital age, internet users actively contribute to food safety supervision by providing feedback on platforms like social media. This user-generated content has become a critical resource for food regulation. However, traditional text classification methods often fail to capture the complex semantic structures of such fragmented and large-scale comments. To address this challenge, we propose a novel LLM-enhanced classification framework featuring three key innovations: (1) keyword-enhanced label representation using domain-specific terminology extraction, (2) LLM-powered semantic summarization for generating category descriptions, and (3) a hybrid deep learning architecture combining BiLSTM(Bidirectional Long Short-Term Memory), BERT(Bidirectional Encoder Representations for Transformers), and self-attention mechanisms. The framework first reduces information noise through keyword extraction, then leverages large language models to generate enriched semantic summaries from keyword-related statements. Finally, it integrates label embeddings by fusing BiLSTM-derived vectors with self-attention mechanisms and BERT-based text representations. Experimental results demonstrate that our model significantly outperforms existing methods in food safety comment classification, achieving a 2% improvement in F1 score compared to baseline models. The proposed framework offers a new paradigm for domain-specific text classification by effectively combining LLM capabilities with deep learning architectures. It excels in handling specialized vocabulary and enhancing label semantic understanding, providing a robust solution for real-world applications in food safety regulation. © 2025

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Enhancing Green Practice Detection in Social Media with Paraphrasing-Based Data Augmentation

Item Type Journal Article

Author A. Glazkova

Author O. Zakharova

Abstract Detecting mentions of green waste practices on social networks is a crucial tool for environmental monitoring and sustainability analytics. Social media serve as a valuable source of ecological information, enabling researchers to track trends, assess public engagement, and predict the spread of sustainable behaviors. Automatic extraction of mentions of green waste practices facilitates large-scale analysis, but the uneven distribution of such mentions presents a challenge for effective detection. To address this, data augmentation plays a key role in balancing class distribution in green practice detection tasks. In this study, we compared existing data augmentation techniques based on the paraphrasing of original texts. We evaluated the effectiveness of additional explanations in prompts, the Chain-of-Thought prompting, synonym substitution, and text expansion. Experiments were conducted on the GreenRu dataset, which focuses on detecting mentions of green waste practices in Russian social media. Our results, obtained using two instruction-based large language models, demonstrated the effectiveness of the Chain-of-Thought prompting for text augmentation. These findings contribute to advancing sustainability analytics by improving automated detection and analysis of environmental discussions.

Furthermore, the results of this study can be applied to other tasks that require augmentation of text data in the context of ecological research and beyond. © 2025 by the authors.

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ENHANCING LLM INTERPRETATION OF APPRAISALS IN SPANISH DIGITAL DISCOURSE

Item Type Journal Article

Author S. Sikorskii

Author M.L. Carrio-Pastor

Abstract This article examines the challenges Large Language Models (LLMs) face in interpreting evaluative language in digital discourse. These models often distort the semantics of evaluative expressions, hindering their accurate linguistic interpretation. The aim of the study is to determine whether integrating specialised knowledge improves a model's ability to correctly identify and classify evaluative meanings. Methodologically, the work draws on Martin and White's Appraisal Theory and

includes an experimental evaluation of GPT-4 on a stratified corpus of posts from a social network. The analysis is conducted in two conditions – with and without external contextual knowledge – and the results are compared to expert annotations in terms of precision and recall. The findings demonstrate a substantial improvement in automatic classification: the accuracy of identifying evaluative categories increases, the range of detectable appraisal elements expands, and new patterns of meaning variation emerge. The conclusions emphasise that enriching LLMs with structured knowledge enhances the reliability of evaluative language analysis and provides deeper insight into how such meanings function in digital discourse. The proposed approach opens new avenues for improving automated methods for analysing evaluative meanings in linguistic research. © Sikorskii S., Carrio-Pastor M.L., 2025.

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Enhancing Machine-Generated Text Detection: Adversarial Fine-Tuning of Pre-Trained Language Models

Item Type Journal Article

Author D.H. Lee

Author B. Jang

Abstract Advances in large language models (LLMs) have revolutionized the natural language processing field. However, the text generated by LLMs can result in various issues, such as fake news, misinformation, and social media spam. In addition, detecting machine-generated text is becoming increasingly difficult because it produces text that resembles human writing. We propose a new method for effectively detecting machine-generated text by applying adversarial training (AT) to pre-trained language models (PLMs), such as Bidirectional Encoder Representations from Transformers (BERT). We generated adversarial examples that appeared to have been modified by humans and applied them to the PLMs to improve the model's detection capabilities. The proposed method was validated on various datasets and experiments. It showed improved performance compared to traditional fine-tuning methods, with an average reduction in the probability of misclassification of machine-generated text by about 10%. We demonstrated the robustness of the model when generated with input tokens of different lengths and under different training data ratios. We suggested future research directions for applying AT to different languages and language model types. This study opens new possibilities for applying AT to the problem of machine-generated text detection and classification and contributes to building more effective detection models. © 2013 IEEE.

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Enhancing Online Safety: Natural Language Processing Based Multi-Label Cyberbullying Classification in Bangla

Item Type Conference Paper

Author M. Saifuddin

Author M. Ahmed

Author S. Basu

Author P. Acharjee

Abstract Cyberbullying can have devastating consequences on its victims, leading to emotional distress, psychological harm, and even suicidal thoughts. Numerous research studies have been conducted on the detection and classification of cyberbullying in high-resource languages like English, but there is a significant scarcity of research in low-resource languages such as Bangla. The lack of research may leave vulnerable communities without effective tools to combat this growing online threat, potentially increasing the harm caused by cyberbullying in these regions. Therefore, it is imperative to conduct additional research in low-resource languages to establish a secure online environment for these regions. This paper provides a comprehensive study on Bangla text classification, focusing primarily on cyberbullying detection, where we explored various classifiers, with fastText Pretrained Models emerging as top performers, achieving an accuracy of 84% and 0.84 as weighted F1-score. Our approach excels in advancing cyberbullying detection in the Bangla language and surpasses previous accuracy benchmarks, offering valuable insights and tools for online safety in this context. © 2023 IEEE.

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Enhancing Sentiment Analysis Accuracy and Evaluating Task Affinity Using Large Language Models

Item Type Conference Paper

Author K. Inoshita

Abstract Sentiment analysis is a crucial natural language processing task that involves extracting emotions and opinions from text data, with applications spanning marketing, social media analysis, and enhancing user experience. Traditional approaches to sentiment analysis, such as dictionary-based and machine learning-based methods, have been predominant; however, these methods have limitations in handling complex emotional expressions and contextual dependencies. Recently, the advent of Large Language Models (LLMs) has raised expectations for improving sentiment analysis accuracy, and research exploring this potential is ongoing. In this study, we employed the Llama 2-7B model, an LLM, to thoroughly evaluate the effects of Supervised Fine-Tuning (SFT) and Few-shot Learning on various sentiment analysis subtasks. Specifically, we focused on tasks such as Sentiment Classification (SC), Aspect-Based Sentiment Analysis (ABSA), and Multifaceted Analysis of Subjective Text (MAST) to assess the affinity of LLMs for these tasks. The experimental results demonstrated that Few-shot Learning significantly improved accuracy across many tasks, with the 2-shot setting proving particularly effective. On the other hand, SFT led to a decrease in accuracy for certain tasks, suggesting the need for careful design in LLM training methods. Additionally, it was revealed that prompt design considering contextual information contributes to performance improvements in tasks such as irony and offensive language detection. These findings provide valuable insights into the design and training methods for specializing LLMs in sentiment analysis, offering important guidance for future research and practical applications. © 2025 IEEE.

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Enhancing Sentiment Analysis in Regional Languages Using Transfer Learning Models Trained on Tamil and English Tweets

Item Type Conference Paper**Author** K. Kanmani**Author** S.E. Uthirapathy**Author** P.N. Periyasamy**Author** D. Sathyanarayanan**Author** C. Sivasankar**Author** M. Agoramoorthy

Abstract This research addresses the challenges of sentiment analysis in local languages, which are often hindered by limited linguistic resources, complex morphological structures, and common code-switching patterns. A transfer learning framework was developed using pretrained transformer models to perform sentiment classification for the Tamil language. The approach involved fine-tuning both mBERT and XLM-RoBERTa on a bilingual dataset of over 25,000 annotated tweets, which included Tamil-only, English-only, and code-mixed Tamil-English texts. This hybrid training design successfully transferred semantic knowledge from high-resource English data to enhance low-resource Tamil classification tasks. Experimental results demonstrate a marked improvement in mean accuracy and F1-scores, with heightened robustness compared to single-language models, particularly in negative sentiment classification. The methodology exhibits excellent compatibility with the natural, informal text characteristic of online communication. The findings confirm that transfer learning is a highly effective technique for bridging resource and linguistic gaps in multilingual sentiment analysis. © 2025 IEEE.

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Enhancing Sentiment Analysis with GPT—A Comparison of Large Language Models and Traditional Machine Learning Techniques

Item Type Conference Paper
Author T. Obinwanne
Author P. Brandtner
Editor Nagar A.K.
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Editor Joshi A.

Abstract Sentiment analysis is the process of extracting and analyzing opinions, attitudes, and emotions expressed in text data. Due to the increased amount of user-generated content on the Internet, sentiment analysis has become an important research area, which has successfully been tackled with machine learning models such as SVMs and Random Forests. Against the backdrop of rapidly growing popularity of Generative Pre-trained Transformers (GPT), the question arises, as to how good such models perform in sentiment analysis. This research paper analyzes and compares the performance of GPT-based models with traditional machine learning models for sentiment analysis. The results paint a clear picture: GPTs are a powerful concept that are applicable in sentiment analysis. In our study, they outperform traditional models such as SVM, Random Forests, or Naïve Bayes based on F1-score, precision, accuracy, and AUC-ROC score. As more advanced versions of GPT continue to be developed, it is likely that these models will become even more effective and popular sentiment analysis. Hence, the application and evaluation of GPTs represents a promising avenue for future research. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2024.

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Enhancing sentiment and intent analysis in public health via fine-tuned Large Language Models on tobacco and e-cigarette-related tweets

Item Type Journal Article

Author S. Elmitwalli

Author J. Mehegan

Author A. Gallagher

Author R. Alebshehy

Abstract Background: Accurate sentiment analysis and intent categorization of tobacco and e-cigarette-related social media content are critical for public health research, yet they necessitate specialized natural language processing approaches. Objective: To compare pre-trained and fine-tuned Flan-T5 models for intent classification and sentiment analysis of tobacco and e-cigarette tweets, demonstrating the effectiveness of pre-training a lightweight large language model for domain specific tasks. Methods: Three Flan-T5 classification models were developed: (1) tobacco intent, (2) e-cigarette intent, and (3) sentiment analysis. Domain-specific datasets with tobacco and e-cigarette tweets were created using GPT-4 and validated by tobacco control specialists using a rigorous evaluation process. A standardized rubric and consensus mechanism involving domain specialists ensured high-quality datasets. The Flan-T5 Large Language Models were fine-tuned using Low-Rank Adaptation and evaluated against pre-trained baselines on the datasets using accuracy performance metrics. To further assess model generalizability and robustness, the fine-tuned models were evaluated on real-world tweets collected around the COP9 event. Results: In every task, fine-tuned models performed much better than pre-trained models. Compared to the pre-trained model's accuracy of 0.33, the fine-tuned model achieved an overall accuracy of 0.91 for tobacco intent classification. The fine-tuned model achieved an accuracy of 0.93 for e-cigarette intent, which is higher than the accuracy of 0.36 for the pre-trained model. The fine-tuned model significantly outperformed the pre-trained model's accuracy of 0.65 in sentiment analysis, achieving an accuracy of 0.94 for sentiments. Conclusion: The effectiveness of lightweight Flan-T5 models in analyzing tweets associated with tobacco and e-cigarette is significantly improved by domain-specific fine-tuning, providing highly accurate instruments for tracking public conversation on tobacco and e-cigarette. The involvement of domain specialists in dataset validation ensured that the generated content accurately represented real-world discussions, thereby enhancing the quality and reliability of the results. Research on tobacco control and the formulation of public policy could be informed by these findings. Copyright © 2024 Elmitwalli, Mehegan, Gallagher and Alebshehy.

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Enhancing Sentiment Classification Accuracy in Myanmar YouTube Comments Using a Hybrid CNN-BiLSTM Attention Model

Item Type Conference Paper

Author Y.T. Reddy

Author K. Suvarchala

Author N.S.C. Goud

Author V. Sarda

Abstract Comments posted on YouTube have become a valuable source of opinion mining with the swift growth of user-generated content on social media. Even though SA has already made impressive feats with the English language, computational implementation of sentiment analysis tasks with low-resource languages, including Myanmar, is still difficult, as it has a complicated agglutinative morphology and no whitespace-delimiting characters. This study suggests a strong model of classifying sentiment polarities of Myanmar posts on YouTube in terms of positive and negative polarities. Our hypothesis is that a new architecture, the Hybrid CNN-BiLSTM with Attention Mechanism (HCB-AM), will be able to surmount the distinctive peculiarities of the Myanmar script. It employs CNNs to obtain local features on n-grams and BiLSTM networks to obtain the context of long texts. It has an attention mechanism, which its aim is to concentrate on important words and not noisy words. The paper performs research on a manually annotated dataset of 9,902 comments on viral Myanmar music videos. We have done extensive preprocessing, such as syllable-based segmentation and removal of stop-words to clean up the noisy text. We made comparisons on the proposed architecture and the standard deep learning models, which are RNN, GRU, and simple LSTM. It has been demonstrated through experiment that deep learning can be highly effective in the field, and the LSTM baseline reporting an accuracy of 97%. The suggested hybrid model increases the performance even further and provides a scaling method of sentiment analysis in under-resourced languages. © 2026 IEEE.

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Enhancing Sentiments Analysis via Large Language Model Meet for Text-Centric Twitter Classification

Item Type Conference Paper
Author D. Patel
Abstract The quick and widespread creation of user views on social media platforms, especially Twitter, has made them essential tools for gauging public opinion. Sentiment analysis has become essential for understanding user sentiments because social networking sites communicate views exclusively through text. Language representation models have multiplied in natural language processing (NLP) because pretrained language models were introduced. The research establishes a goal to enhance text-centric sentiment analysis through big language model processing of Twitter sentiment data. The research improves sentiment classification through the utilization of XLNet as a transformer-based model that analyzes Twitter-focused emotional content. The suggested approach begins with cleaning up the data and then uses XLNet for classification after extracting features using the Bag-of-Words (BoW) method. The results show that XLNet outperforms BERT-base-cased and RoBERTa when evaluating models using

F1score, recall, accuracy, and precision. The results demonstrate that XLNet achieves superior performance in contextual sentiment recognition because it delivers 99.62% classification accuracy. The comparative analysis and performance metrics were provided to show that XLNet outperforms the others at refining sentiment analysis as a product of contextual learning, which establishes it as a viable solution for text-emotion detection on social media. © 2025 IEEE.

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Enhancing social media hate speech detection in low-resource languages using transformers and explainable AI

Item Type Journal Article

Author E. Fetahi

Author A. Susuri

Author M. Hamiti

Author Z. Kastrati

Author E. Canhasi

Author A. Misini

Abstract Hate speech (HS) on social media exposes public discourse and community well-being. Despite its prevalence, majority of the research and technological advances have focused on rich-resourced languages. In contrast, Albanian, a language marked by dialects, limited NLP tools, and scarcity of annotated data, remains underexplored. To address this gap, this paper explores the enhancement of automatic hate speech detection in Albanian social media using advanced deep neural techniques. We collected a real-life dataset comprising 20,860 Facebook comments manually annotated using a rigorous multi-annotator process. Various techniques, including machine learning, deep neural networks, and transformers, are tested on the collected dataset. Our findings show that character-level 4-gram TF-IDF consistently reinforces ML classifiers effective with informal spelling and slang. However, the fine-tuned transformer XLM-RoBERTa achieves the highest performance with an F1-score up to 86%. These results mark a new benchmark for Albanian HS detection and highlight the impact of thorough feature engineering and robust representation techniques in a low-resource context. We also carried out an error analysis using both manual inspection and explainable AI approaches such as SHAP and LIME. Key language challenges include dialectal writing, short sentences, and implicit insults. These insights highlight the importance of domain-aligned embedding resources, more extensive annotation, and fine-grained context handling for improved detection. This study establishes a new state-of-the-art result for Albanian hate speech detection, provides a valuable comparison of models and feature engineering, and underscores the potential of multilingual transformers in low-resource scenarios. © The Author(s) 2025.

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Enhancing stock market price prediction through FinGPT-driven sentiment analysis, technical indicators, historical price data, and attention-based hybrid deep learning models

Item Type Journal Article

Author M. Golabzaei

Author E. Taheripour

Author S.J. Sadjadi

Abstract Accurate stock price forecasting is increasingly dependent on models that can integrate heterogeneous market signals beyond traditional quantitative indicators. Classical sentiment analysis tools and lexicon-based methods often fail to capture the contextual and domain-specific nuances of financial language, limiting their predictive contribution. To address these limitations, this study introduces a multimodal forecasting framework that leverages FinGPT, an advanced financial large language model specifically trained for market-related text understanding. FinGPT is employed to extract high-resolution sentiment polarity scores from Twitter data, replacing conventional sentiment models and providing a more robust representation of investor psychology. These sentiment features are fused with historical stock prices and technical indicators to form a comprehensive input space. The integrated dataset is used to evaluate four deep learning architectures: CNN, LSTM, Attention-based LSTM (ALSTM), and a newly proposed Hybrid ALSTM-CNN model with an attention mechanism. Experiments performed on eight major NASDAQ and NYSE stocks demonstrate that incorporating FinGPT-derived sentiment significantly enhances forecasting accuracy across all models. The Hybrid model consistently achieves the best performance, effectively capturing both long-term dependencies and short-term market dynamics. Results confirm that FinGPT-driven sentiment modeling provides substantial predictive value compared with traditional sentiment approaches, enabling more realistic, behavior-aware stock price forecasting. For instance, in the case of AMD stock, it recorded the lowest MSE (46.06) and the highest R^2 (0.624), highlighting its superior ability to capture nonlinear dependencies and market dynamics. Overall, the proposed sentiment-enhanced hybrid framework provides a more robust and realistic procedure for stock price prediction by integrating quantitative and qualitative market signals. © The Author(s) 2026.

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ISSN 20452322 (ISSN)**Issue** 1**Journal Abbr** Sci. Rep.**Date Added** 20/09/2026, 8:35:15 pm**Modified** 20/09/2026, 8:35:15 pm**Tags:**

human, Deep learning, Sentiment analysis, article, social media, prediction, deep learning, sentiment analysis, forecasting, large language model, hybrid, Stock price prediction, FinGPT, Multimodal data fusion, predictive value, price, stock market, Technical analysis

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Enhancing Text-Based Deepfake Detection: A Comprehensive Evaluation of LoRA Fine-Tuned Large Language Models

Item Type Journal Article**Author** A. Mohammed**Author** R. Kora

Abstract The rapid progress of large language models (LLMs) has enabled highly convincing text-based deepfakes on social media, threatening information integrity. Existing zero-shot and few-shot detection methods suffer from unstable performance, while full-model retraining is computationally prohibitive. This paper evaluates LoRA-based parameter-efficient fine-tuning as a scalable solution. We assess thirteen state-of-the-art LLMs across four multilingual benchmarks, comparing zero-shot inference with LoRA, QLoRA, and PLoRA. Results show that LoRA fine-tuning significantly boosts detection accuracy - enabling smaller models to rival or surpass larger zero-shot counterparts while reducing computational cost. QLoRA achieves comparable performance with lower memory use; PLoRA trades slight accuracy for faster adaptation. Our findings establish LoRA-based fine-tuning as a robust, efficient, and scalable approach for multilingual and low-resource deepfake detection. © 2026 The Authors.

Date 2026**DOI** 10.1109/ACCESS.2026.3675941**URL** <https://www.scopus.com/pages/publications/105033380654?origin=resultslist>**Archive** Scopus**Language** English**Volume** 14**Publisher** Institute of Electrical and Electronics Engineers Inc.

Pages 44709-44727**Publication** IEEE Access**ISSN** 21693536 (ISSN)**Journal Abbr** IEEE Access**Date Added** 20/09/2026, 8:35:01 pm**Modified** 20/09/2026, 8:35:01 pm**Tags:**

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Enhancing the Accuracy of Nikkei 225 Forecasting Via LLM-Based Filtering of Noisy Social Media Data

Item Type Conference Paper**Author** S. Kojima**Author** Y. Sakurai

Abstract In today's information-oriented society, social networking services (SNSs) have emerged as significant sources of information that influence economic activities and investment decisions. The results of sentiment analysis of SNS posts can be used as positive or negative indicators of market sentiments. The utilization of these indicators has led to ongoing research endeavors aimed at enhancing the accuracy of stock market trend forecasts, including the Nikkei Stock Average. However, SNS posts often contain irrelevant content, such as advertisements, which degrade prediction quality. This study demonstrates that filtering the irrelevant content using a large language model (LLM) is a highly effective method for improving stock price prediction models using indicators based on SNS posts. Using Long Short-Term Memory (LSTM) and Transformer-based models, our experiments show that the proposed LLM-based filtering achieves a statistically significant improvement in prediction accuracy compared to both (1) an unfiltered baseline and (2) a conventional keyword-based filter. These results highlight that the advanced contextual understanding provided by LLMs is crucial for constructing high-quality financial datasets from noisy social media sources.

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Enhancing Toxic Comment Classification using an Active Learning-based DistilBERT-SVM Framework

Item Type Conference Paper

Author C. Zephaniah

Author R. Karunambikai

Abstract The proliferation of online media sites has caused user-generated content to grow exponentially, with much of it including toxic content in the form of insults, threats, harassment, and hate speech against gender, race, religion, or sexual identity. Even though machine learning methods have proven effective in identifying such content, ensuring consistent performance is challenging in low-resource settings where the presence of labeled data is scarce. This paper explores

the effectiveness of active learning in the context of toxic comment classification in limited conditions of annotation. In particular, random sampling, uncertainty-based sampling, and incremental human annotation are compared using the calibrated Linear Support Vector Machine that was trained on the large-scale toxic comment dataset. This is aimed at assessing the efficiency of model performance improvement while minimizing annotation effort. Based on experimental findings, uncertainty sampling performs marginally better than random sampling with a final F1-score of 0.7100 compared to 0.7079. Moreover, adding human-labeled samples selectively in small batches will raise the performance, reaching an F1-score of 0.6782 in limited-label conditions. These results indicate that active learning methods have stable and low-cost performance gains when used in low-resource settings and thus are practically applicable to scalable toxic content detectors. © 2026 IEEE.

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Enhancing trial recruitment through a social media influencer campaign in the MeMaF study: a mixed methods approach

Item Type Journal Article

Author S. Lysk

Author C.M. Witt

Author C. Canella

Author S. Wiegmann

Author R. Paprott

Author H. Neumayer

Author T. Gärtner

Author S. Konigorski

Author D. Pach

Abstract Objectives Recruitment remains a major challenge in clinical research, particularly among young populations. This study evaluated the effectiveness of a paid social media influencer campaign as a recruitment strategy for the MeMaF study (Menstrual Health and Hybrid Care Model for All Young Females), which targets young women with severe menstrual pain in Germany. Study Design and Setting This mixed methods study was nested within the MeMaF single-arm trial. Recruitment data were collected via the study app, and three recruitment periods were defined retrospectively: (1) basic recruitment period (March 8, 2024–January 8, 2025), (2) social media influencer campaign (January 9–31, 2025), and (3) postcampaign period (February 1–28, 2025). Quantitative analyses included descriptive statistics of daily enrollments, app downloads, and download-to-enrollment ratios across the three recruitment periods. Qualitative data consisted of anonymized social media comments from Instagram and TikTok related to the influencer postings. A thematic content analysis was conducted using MAXQDA, supported by large language models integrated into the software. Results Over the entire recruitment period, 3496 participants were enrolled, thereby meeting the recruitment goal. The mean daily enrollment rate during the social media influencer campaign (45.5 participants/day) was six times higher than during the basic recruitment period (6.8 participants/day) and remained elevated in the postcampaign period (13.4 participants/day). Posts from six female influencers accumulated 2.8 million views and 114,600 likes on Instagram, and 865,000 views and 100,000 likes on TikTok. Qualitative analysis of 501 comments revealed strong audience engagement, including discussion of the study's inclusion criteria, often expressing criticism of eligibility restrictions alongside appreciation for the study initiative. Comment sections also evolved into a space for sharing personal experiences with menstrual pain and healthcare access. Conclusion A paid influencer campaign can substantially enhance study recruitment among young populations. Collaboration with influencers represents a feasible and effective strategy to broaden reach, strengthen engagement, and complement traditional recruitment methods. Future studies should evaluate cost-effectiveness and transferability to other populations and platforms. Plain Language Summary Recruiting participants for clinical studies can be difficult, especially among young people. This study examined whether social media influencers, popular online personalities on platforms such as Instagram and TikTok, can help improve recruitment. We used a paid influencer campaign to promote a study on menstrual health among young women in Germany. Six influencers shared information about the study with their followers. We compared how many people joined the study before, during, and after the campaign. During the influencer campaign, the number of people enrolling in the study increased six-fold compared to the period before the

campaign. The posts also received high engagement, including many views, likes, and comments. In the comments, users discussed the study, shared personal experiences, and sometimes expressed frustration about who was eligible to participate. These findings suggest that working with social media influencers can be an effective way to reach young people and increase participation in clinical studies.
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Enhancing Vietnamese Sentiment Analysis with Extended SentiWordNet

Item Type Conference Paper

Author H.-V. Tran

Author V.-T. Bui

Author L.-Q. Tran

Editor Tu T.X.

Editor Thuy N.T.

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Editor Nghia P.T.

Editor Thai V.D.

Editor Van Nui N.

Abstract Sentiment analysis is a fundamental task in Natural Language Processing that focuses on identifying the emotional polarity of textual data. It plays an important role in extracting insights from domains such as e-commerce, customer feedback, and social media. Recent advances in pre-trained language models have significantly improved sentiment classification by enabling effective knowledge transfer through fine-tuning. In this study, we propose a novel approach to Vietnamese sentiment analysis that combines PhoBERT-V2, a state-of-the-art RoBERTa-based model for Vietnamese, with an extended version of Vietnamese SentiWordNet, a sentiment-specific lexical resource. Our proposed model, named CombViSA, integrates contextual embeddings from PhoBERT-V2 with sentiment-aware lexical features to overcome the limitations of existing sentiment lexicons. Experimental results on two benchmark datasets, VLSP 2016 and AIVIVN 2019, demonstrate that CombViSA achieves state-of-the-art performance, obtaining an F1-score of 0.95 on VLSP 2016 and significantly outperforming strong baselines. This work contributes to advancing sentiment analysis for low-resource languages and supports real-world applications such as opinion mining, brand monitoring, and customer experience analysis. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Enhancing VLM-based multimodal sentiment analysis via prompt tuning and contrastive decoding

Item Type Journal Article

Author X. Wang

Author Y. Liu

Author Y. Chen

Author J. Zhang

Author X. Lu

Abstract Multimodal Sentiment Analysis (MSA) leveraging Vision-Language Models (VLMs) has shown promise, yet accurately capturing affective cues in low-resource scenarios remains challenging. Existing methods relying on full fine-tuning are computationally prohibitive and prone to overfitting, while standard retrieval-augmented approaches often suffer from noise interference due to the loose semantic alignment in multimodal data. To bridge these gaps, we propose PTCd, a novel MSA framework that synergizes parameter-efficient prompt tuning with retrieval-augmented contrastive decoding. Specifically, we first freeze the VLM backbone and optimize lightweight soft prompts to steer the representation space toward sentiment-aware alignment. Furthermore, to address the instability of few-shot demonstrations, we introduce an adaptive Contrastive Decoding mechanism during inference. This strategy dynamically gates the influence of retrieved examples by contrasting the model's intrinsic confidence against retrieval-augmented predictions, effectively filtering out misleading noise while leveraging high-quality demonstrations. Extensive experiments on MVSA and Twitter benchmarks demonstrate that PTCd significantly outperforms state-of-the-art methods, validating that combining prompt tuning with contrastive decoding is more effective for sentiment analysis in multimodal data. © 2026 Elsevier Ltd

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Entity Sentiment Classification using Arabic Lightweight LLMs**Item Type** Conference Paper**Author** M.M. Raza**Author** S.I. Abudalfa

Abstract Arabic sentiment analysis on entity level, especially on social media, remains a challenging task due to linguistic complexity and limited resources. This study evaluates the performance of some selected lightweight LLMs using different learning settings: zero-shot, few-shot, bootstrap finetuning based on Chain-of-Thought for sentiment classification on the AT-ODTSA dataset. We investigate target-independent and target-dependent sentiment classification across three language models: Microsoft/Phi-4-mini-instruct, Qwen/Qwen3-4B, and Meta Llama 3.21B Instruct. Zero-shot results show Microsoft/Phi-4-mini-instruct achieving 56.8% accuracy in target-independent analysis, while Qwen reaches 66.2% in targetdependent scenarios. Few-shot learning demonstrates Microsoft/Phi-4-mini-instruct achieving 56.1% accuracy in target-dependent classification while Qwen achieving 69.1% accuracy in target dependent. Fine-tuning results reveal significant improvements, with Qwen achieving state-of-the-art performance of 83.8% accuracy with target independent. The study establishes new benchmarks for Arabic sentiment analysis and demonstrates the effectiveness of the method we used which is unique from previous works. © 2025 IEEE.

Date 2025**DOI** 10.1109/EngiTek68245.2025.11567535**URL** <https://www.scopus.com/pages/publications/105044422376?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Int. Conf. Eng. Comput. Technol., EngiTek**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833156047-8 (ISBN)**Proceedings Title** Int. Conf. Eng. Comput. Technol., EngiTek**Conference Name** 2025 International Conference on Engineering and Computing Technologies, EngiTek 2025**Date Added** 20/09/2026, 8:35:23 pm**Modified** 20/09/2026, 8:35:23 pm**Tags:**

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Equity and reliability of public electric vehicle charging stations in the United States

Item Type Journal Article

Author Q. Yu

Author T. Que

Author L.J. Cushing

Author G. Pierce

Author K. Shen

Author M. Kejriwal

Author Y. Yao

Author Y. Zhu

Abstract Equitable coverage and reliable operation of electric vehicle charging stations (EVCSs) are crucial for a just transition to a carbon-free future. Yet, a comprehensive national analysis of public EVCSs across different communities is lacking in the United States. Here, we utilize real-world reviews ($n = 470,142$) from a user-generated content platform to analyze public EVCSs at the census tract level. We find that disadvantaged communities (DACs) have 64% fewer public EVCSs per capita than non-DACs. This disparity rises to 73% when considering renters in multi-dwelling units. Additionally, EVCS users in DACs and urban areas experience significantly more reliability issues compared to those in non-DACs and rural areas, primarily related to hardware and technical failures. Given the limited access to home charging in DACs and their underserved public infrastructure, these findings highlight critical equity concerns and call for targeted investment in EVCS infrastructure and reliability improvements, particularly in DACs. © The Author(s) 2025.

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ESG-GPT:GPT4-based Few-Shot Prompt Learning for Multi-lingual ESG News Text Classification

Item Type Conference Paper

Author K. Tian

Author H. Chen

Editor Chen C.-C.

Editor Ma Z.

Editor Hahn U.

Abstract Environmental, Social, and Governance (ESG) factors for company assessment have gained great attention from finance investors to identify companies' risks and growth opportunities. ESG Text data regarding the company like sustainable reports, media news text, and social media text are important data sources for ESG analysis like ESG factors classification. Recently, FinNLP has proposed several ESG-related tasks. One of the tasks is Multi-Lingual ESG Issue Identification 3(ML-ESG-3) which is to determine the duration or impact level of the impact of an event in the news article regarding the company. In this paper, we mainly discussed our team: KaKa's solution to this ML-ESG-3 task. We proposed the GPT4 model based on few-shot prompt learning to predict the impact level or duration of the impact of multi-lingual ESG news for the company. The experiment result demonstrates that GPT4-based few-shot prompt

learning achieved good performance in leaderboard quantitative evaluations of ML-ESG-3 tasks across different languages. © 2024 ELRA Language Resource Association.

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Extra Journal Abbreviation: Jt. Workshop Financ. Technol. Nat. Lang. Process., Knowl. Discov. from Unstructured Data Financ. Serv. Econ. Nat. Lang. Process., FinNLP-KDF-ECONLP LREC-COLING - Workshop Proc.

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Evaluating Arabic Sentiment Analysis With GPT-4o: A Comparative Study of Raw and Preprocessed Text

Item Type Journal Article

Author A. Wali

Author A. Alahmadi

Author A. Aljohani

Author A. Alharbi

Abstract Generative Pretrained Transformer 4 Omni (GPT-4o) is a large language model that has contributed to natural language processing by modeling complex linguistic patterns. However, given the scarcity of labeled Arabic data and the growing interest in deploying large language models without task-specific fine-tuning, there is a lack of studies that evaluate the effectiveness and reliability of GPT-4o in processing Arabic texts in the domain of sentiment analysis under zero-shot settings. To address this gap, experiments on original and preprocessed texts using two Arabic datasets—restaurant reviews and multi-topic tweets—were conducted. The zero-shot performance of GPT-4o was compared with Arabic Bidirectional Encoder Representations from Transformers, Deep Bidirectional Transformers for Arabic, and Multilingual Bidirectional Encoder Representations from Transformers. Overall, GPT-4o outperformed other models, with accuracies of 85% for preprocessed restaurant reviews and 77% for original tweets. The tweet dataset was more challenging, indicating GPT-4o's performance is affected by informal and noisy social media content. These findings highlight important considerations for adoption of GPT-4o in Arabic sentiment analysis applications. © 2026 IGI Global. All rights reserved.

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DOI 10.4018/JCIT.402749

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Evaluating ChatGPT and Bard AI on Arabic Sentiment Analysis

Item Type Conference Paper

Author A. Al-Thubaity

Author S. Alkhereyf

Author H. Murayshid

Author N. Alshalawi

Author M.B. Omirah

Author R. Alateeq

Author R. Almutairi

Author R. Alsuwailem

Author M. Alhassoun

Author I. Alkhanen

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Editor Abdelali A.

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Editor Almatham R.

Abstract Large Language Models (LLMs) such as ChatGPT and Bard AI have gained much attention due to their outstanding performance on a range of NLP tasks. These models have demonstrated remarkable proficiency across various languages without the necessity for full supervision. Nevertheless, their performance in low-resource languages and dialects, like Arabic dialects in comparison to English, remains to be investigated. In this paper, we conduct a comprehensive evaluation of three LLMs for Dialectal Arabic Sentiment Analysis: namely, ChatGPT based on GPT-3.5 and GPT-4, and Bard AI. We use a Saudi dialect Twitter dataset to assess their capability in sentiment text classification and generation. For classification, we compare the performance of fully fine-tuned Arabic BERT-based models with the LLMs in few-shot settings. For data generation, we evaluate the quality of the generated new sentiment

samples using human and automatic evaluation methods. The experiments reveal that GPT-4 outperforms GPT-3.5 and Bard AI in sentiment analysis classification, rivaling the top-performing fully supervised BERT-based language model. However, in terms of data generation, compared to manually annotated authentic data, these generative models often fall short in producing high-quality Dialectal Arabic text suitable for sentiment analysis. © 2023 Association for Computational Linguistics.

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Evaluating Computational Approaches for Harmful Content Analysis: Promise, Pitfalls and Tools for Responsible Research

Item Type Journal Article

Author I. Himelboim

Author M. Baid

Abstract This manuscript develops and demonstrates a practical framework for evaluating automated classifiers used in communication research, using harmful language detection as an illustrative case. We combine (a) a structured review of documentation practices for 27 publicly available classifiers and their associated annotation processes with (b) a cross-dataset evaluation that re-tests each model beyond its original training context. Across 27 datasets, we extract and compare

reporting on construct definitions, annotator instructions, and inter-annotator agreement, and we quantify generalization by applying each model to multiple out-of-domain test sets. We also benchmark a contemporary large language model (GPT-5) under a consistent prompting protocol to illustrate how LLM-based classification compares to fine-tuned classifiers. Results show that documentation is uneven and often insufficient for theory-driven measurement, inter-annotator agreement varies widely across datasets, and cross-dataset performance frequently drops substantially relative to within-dataset evaluations. Building on these findings and existing validation guidance, we provide a reusable checklist and decision flow to help researchers select, justify, and report classifier-based measures in ways that support transparency and cumulative science. Recommendations for researchers, reviewers, and journal editors stress aligning model selection with standards of validity, reliability, and transparency. © 2026 by the authors.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: I. Himelboim; Department of Advertising and Public Relations, University of Georgia, Athens, 30602, United States; email: itai@uga.edu

Evaluating Graph Attention Networks as an Alternative to Transformers for ABSA Task in Low-Resource Languages

Item Type Conference Paper

Author G.A. Gomes

Author A.T. Bender

Author A. Cerveira

Author L.A. Freitas

Author U.B. Corrêa

Abstract Opinions toward subjects and products hold immense relevance in business to guide decision-making processes. However, due to the increase in user-generated content, manual analysis is unrealistic. Techniques such as Sentiment Analysis are paramount to understanding and quantifying human emotion expressed in text data. Aspect-Based Sentiment Analysis aims to extract aspects from an opinionated text while identifying their underlying sentiment. Graph-based text representations have been shown to bring benefits to this task, as they explicitly represent structural relationships in text. While studies have demonstrated the effectiveness of this representation for Aspect-based Sentiment Analysis using Graph Neural Networks in English, there is only sparse evidence of improvement using these techniques for low-resource languages such as Portuguese. We develop a straightforward Graph Attention Network model for the Aspect-Based Sentiment Analysis task in Brazilian Portuguese. The proposed approach achieves a Balanced Accuracy score of 0.74, yielding competitive results and ranking third place in the ABSAPT competition. Furthermore, by leveraging sparse graph connections our model is less computationally demanding than a Transformer architecture in terms of training and inference. Copyright © 2024 by the authors.

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Evaluating human-based and large language model-based content analysis: A case study on negative word-of-mouth classification in social media

Item Type Journal Article

Author Y.Y. Yang

Author C.-C. Wang

Author P.-L. Chou

Author C.-Y. Tien

Author J.-C. Wen

Author N.-H. Chen

Abstract This main purpose of this study is to examine the effectiveness of human annotation versus large language model (LLM)-based automated content analysis in classifying negative word-of-mouth (NWOM) on social media. Using ChatGPT as the representative LLM, we evaluated its performance in categorizing comments into five sentiment types: positive, neutral, negative, irony, and boycott. Through three experiments, we compared ChatGPT's classification accuracy, precision, recall, and F1 score with human annotators under varying conditions, including the provision of classification definitions and keywords, as well as parameter adjustments such as temperature and Top_p. The findings reveal that while ChatGPT demonstrated strong recall in detecting negative comments, it struggles with nuanced classifications such as irony and boycott, where human annotators exhibit higher precision. Incorporating explicit definitions and keywords significantly enhanced ChatGPT's classification accuracy and consistency, particularly in distinguishing boycott-related comments. However, adjustments to temperature and Top_p yield only marginal improvements. These results highlight the potential of integrating LLMs with human annotation to optimize large-scale content analysis while addressing the limitations of automated sentiment classification. The study provides theoretical and practical insights into AI-assisted content analysis, with implications for social media monitoring, brand crisis management, and influencer engagement strategies. © 2026

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Evaluating Instruction-Tuned Large Language Models for Automatic Annotation of Multilingual Climate Disaster-Related Social Media Messages

Item Type Conference Paper**Author** J.A. Orama**Author** M. Juvillà Garcia**Author** M. Melero**Editor** Trejo K.**Editor** Aguilo I.**Editor** Riera J.V.**Editor** Pascual J.

Abstract Analysing social media messages shared during a climate disaster to extract relevant information is a difficult task due to the high volume of messages and the subjective nature of relevance. Nevertheless, this is a necessary step when annotating texts to train models for crisis management. Traditionally, samples are annotated manually, which is time consuming and challenging to maintain consistency, especially when dealing with multilingual corpora. Instruction-Tuned generative language models are a potential solution to this problem due to their ability to follow instructions and their state of the art performance in several natural language processing tasks including text classification. This paper evaluates several open-source multilingual instructed models with the goal of seeing how well they perform as annotators or text filtering tools in a crisis. We conduct experiments using both zero-shot and few-shot prompting, exploring different ways of incorporating examples in few-shot prompting. Finally, we examine the impact of post-Annotation label cleaning and further refine results through score thresholding. © 2025 The Authors.

Date 2025**DOI** 10.3233/FAIA250578**URL** <https://www.scopus.com/pages/publications/105021083832?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Front. Artif. Intell. Appl.**Volume** 410**Publisher** IOS Press BV**ISBN** 09226389 (ISSN); 978-164368618-9 (ISBN)**Pages** 79-88**Proceedings Title** Front. Artif. Intell. Appl.**Conference Name** Frontiers in Artificial Intelligence and Applications

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Evaluating Large Language Models for Depression Detection in Text: A Comparative Analysis

Item Type Conference Paper**Author** B. Ihnaini**Author** J. Li**Author** Z. Yu**Editor** Mahmud M.**Editor** Doborjeh M.**Editor** Wong K.**Editor** Leung A.C.S.**Editor** Doborjeh Z.**Editor** Tanveer M.

Abstract Depression is a prevalent mental health problem that affects millions of people worldwide, and early detection is crucial for effective intervention and treatment. The rise of digital communication and social media provides new opportunities for monitoring mental health through text analysis. In this paper, we evaluate the performance of popular open-source Large Language Models for text classification of depression through supervised fine-tuning with LoRA and QLoRA techniques. In addition, the study provides a comprehensive comparison of traditional Machine Learning algorithms, Deep Learning models, and Large Language Models. The study shows that Mistral-7B with full 16-bit precision (0.976), 8-bit (0.977), and 4-bit (0.975) weight quantization all achieve the highest F1 score, indicating that Mistral-7B is greatly compatible with depression diagnosis tasks. The findings will contribute to further research on Large

Language Models for depression detection by providing insights into model selection, fine-tuning strategies, and the impact of quantization on performance. This will aid researchers in optimizing model architectures and adaptation techniques for improved accuracy and efficiency in mental health monitoring applications. Additionally, the findings will provide insights into the performance differences between traditional Machine Learning, Deep Learning, and LLMs for depression monitoring. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Evaluating large language models for health-related text classification tasks with public social media data

Item Type Journal Article

Author Y. Guo
Author A. Ovadjé
Author M.A. Al-Garadi
Author A. Sarker

Abstract Objectives: Large language models (LLMs) have demonstrated remarkable success in natural language processing (NLP) tasks. This study aimed to evaluate their performances on social media-based health-related text classification tasks. Materials and Methods: We benchmarked 1 Support Vector Machine (SVM), 3 supervised pretrained language models (PLMs), and 2 LLMs-based classifiers across 6 text classification tasks. We developed 3 approaches for leveraging LLMs: employing LLMs as zero-shot classifiers, using LLMs as data annotators, and utilizing LLMs with few-shot examples for data augmentation. Results: Across all tasks, the mean (SD) F1 score differences for RoBERTa, BERTweet, and SocBERT trained on human-annotated data were 0.24 (± 0.10), 0.25 (± 0.11), and 0.23 (± 0.11), respectively, compared to those trained on the data annotated using GPT3.5, and were 0.16 (± 0.07), 0.16 (± 0.08), and 0.14 (± 0.08) using GPT4, respectively. The GPT3.5 and GPT4 zero-shot classifiers outperformed SVMs in a single task and in 5 out of 6 tasks, respectively. When leveraging LLMs for data augmentation, the RoBERTa models trained on GPT4-augmented data demonstrated superior or comparable performance compared to those trained on human-annotated data alone. Discussion: The results revealed that using LLM-annotated data only for training supervised classification models was ineffective. However, employing the LLM as a zero-shot classifier exhibited the potential to outperform traditional SVM models and achieved a higher recall than the advanced transformer-based model RoBERTa. Additionally, our results indicated that utilizing GPT3.5 for data augmentation could potentially harm model performance. In contrast, data augmentation with GPT4 demonstrated improved model performances, showcasing the potential of LLMs in reducing the need for extensive training data. Conclusions: By leveraging the data augmentation strategy, we can harness the power of LLMs to develop smaller, more effective domain-specific NLP models. Using LLM-annotated data without human guidance for training lightweight supervised classification models is an ineffective strategy. However, LLM, as a zero-shot classifier, shows promise in excluding false negatives and potentially reducing the human effort required for data annotation. © The Author(s) 2024.

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Evaluating Large Language Models for Sentiment Analysis and Hesitancy Analysis on Vaccine Posts From Social Media: Qualitative Study

Item Type Journal Article

Author A. Annan

Author A.L. Eiden

Author D. Wang

Author J. Du

Author M. Rastegar-Mojarad

Author V.K. Nomula

Author X. Wang

Abstract Background: In the digital age, social media has become a crucial platform for public discourse on diverse health-related topics, including vaccines. Efficient sentiment analysis and hesitancy detection are essential for understanding public opinions and concerns. Large language models (LLMs) offer advanced capabilities for processing complex linguistic patterns, potentially providing valuable insights into vaccine-related discourse. Objective: This study aims to evaluate the performance of various LLMs in sentiment analysis and hesitancy detection related to vaccine discussions on social media and identify the most efficient, accurate, and cost-effective model for detecting vaccine-related public sentiment and hesitancy trends. Methods: We used several LLMs—generative pretrained transformer (GPT-3.5), GPT-4, Claude-3 Sonnet, and Llama 2—to process and classify complex linguistic data related to human papillomavirus; measles, mumps, and rubella; and vaccines overall from X (formerly known as Twitter), Reddit, and YouTube. The models were tested across different learning paradigms: zero-shot, 1-shot, and few-shot to determine their adaptability and learning efficiency with varying amounts of training data. We evaluated the models’ performance using accuracy, F1-score, precision, and recall. In addition, we conducted a cost analysis focused on token usage to assess the computational efficiency of each approach. Results: GPT-4 (F1-score=0.85 and accuracy=0.83) outperformed GPT-3.5, Llama 2, and Claude-3 Sonnet across various metrics, regardless of the sentiment type or learning paradigm. Few-shot learning did not significantly enhance performance compared with the zero-shot paradigm. Moreover, the increased computational costs and token usage associated with few-

shot learning did not justify its application, given the marginal improvement in model performance. The analysis highlighted challenges in classifying neutral sentiments and convenience, correctly interpreting sarcasm, and accurately identifying indirect expressions of vaccine hesitancy, emphasizing the need for model refinement. Conclusions: GPT-4 emerged as the most accurate model, excelling in sentiment and hesitancy analysis. Performance differences between learning paradigms were minimal, making zero-shot learning preferable for its balance of accuracy and computational efficiency. However, the zero-shot GPT-4 model is not the most cost-effective compared with traditional machine learning. A hybrid approach, using LLMs for initial annotation and traditional models for training, could optimize cost and performance. Despite reliance on specific LLM versions and a limited focus on certain vaccine types and platforms, our findings underscore the capabilities and limitations of LLMs in vaccine sentiment and hesitancy analysis, highlighting the need for ongoing evaluation and adaptation in public health communication strategies. © Augustine Annan, Amanda L Eiden, Dong Wang, Jingcheng Du, Majid Rastegar-Mojarad, Varun Kumar Nomula, Xiaoyan Wang.

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Evaluating large language models for user stance detection on X (Twitter)

Item Type Journal Article

Author M. Gambini

Author C. Senette

Author T. Fagni

Author M. Tesconi

Abstract Current stance detection methods employ topic-aligned data, resulting in many unexplored topics due to insufficient training samples. Large Language Models (LLMs) pre-trained on a vast amount of web data offer a viable solution when training data is unavailable. This work introduces Tweets2Stance - T2S, an unsupervised stance detection framework based on zero-shot classification, i.e. leveraging an LLM pre-trained on Natural Language Inference tasks. T2S detects a five-valued user's stance on social-political statements by analyzing their X (Twitter) timeline. The Ground Truth of a user's stance is obtained from Voting Advice Applications (VAAs). Through comprehensive experiments, a T2S's optimal setting was identified for each election. Linguistic limitations related to the language model are further addressed by integrating state-of-the-art LLMs like GPT-4 and Mixtral into the T2S framework. The T2S framework's generalization potential is demonstrated by measuring its performance (F1 and MAE scores) across nine datasets. These datasets were built by collecting tweets from competing parties' Twitter accounts in nine political elections held in different countries from 2019 to 2021. The results, in terms of F1 and MAE scores, outperformed all baselines and approached the best scores for each election. This showcases the ability of T2S, particularly when combined with state-of-the-art LLMs, to generalize across different cultural-political contexts. © The Author(s) 2024.

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Evaluating large language models on a hermeneutically complex text annotation task

Item Type Journal Article

Author A. Fodor

Author E. Katona

Author R. Németh

Abstract The study examines whether large language models (LLMs) are capable of interpreting individual framings of depression in online forums - a challenging annotation task involving categories that are neither mutually exclusive nor trivial from a hermeneutic perspective. Using a corpus annotated by two annotators, the texts and categories that posed difficulties for human interpretation were identified, and the LLMs were prompted to report their confidence in their classifications. The study addresses the following research questions: whether LLMs can perform complex, meaning-making tasks comparable to human interpretation; how model and prompt choices affect accuracy and reliability; the extent to which model confidence in its own classifications correlates with classification accuracy; whether LLMs and humans struggle with the same texts; and how these findings can inform human–LLM collaboration in annotation process. Two models, GPT-4o mini (closed-source) and Llama 3.3 70B (open-source) were evaluated using zero-shot and few-shot prompting strategies. While both models showed strong reliability in terms of run-to-run, model-to-model and prompt-to-prompt comparisons, their overall accuracy was moderate. Performance was highest on texts that were also easier for human annotators. Notably, higher model confidence was often associated with greater accuracy, suggesting that confidence scores may be useful in supporting collaborative annotation systems. The findings highlight both the potential and the interpretive limits of LLMs in social science applications, underscoring the importance of careful annotation design and the thoughtful integration of LLMs into human-centered workflows. © 2026 The Authors

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: E. Katona; Eötvös Loránd University, Faculty of Social Sciences, 1 - Research Center for Computational Social Science, 2 - Department of Social Research Methodology, Budapest, Pázmány Péter sétány. 1/A, 1117, Hungary; email: katona.eszter@tatk.elte.hu

Evaluating Large-Language Models for Dimensional Music Emotion Prediction from Social Media Discourse

Item Type Conference Paper

Author P.J. Donnelly

Author A. Beery

Editor Abbas M.

Editor Freihat A.A.

Abstract The automatic prediction of emotional responses to music is a task of inherent interest to the field of music information retrieval. These efforts are often hindered by the absence of large datasets available for this task. In this work, we investigate the use of sentiment analysis on online social media conversations as an alternate data source to train computational models to predict the emotive responses to a piece of music. Using two datasets annotated with valence and arousal values, we create a corpus of social media commentary for these songs extracted from YouTube, Twitter, and Reddit. We evaluate our approach with transformer models to predict the affective values of the 2402 songs in our dataset. We achieve a moderate Pearson's correlation of 0.62 and 0.72 for valence and arousal, respectively, for discourse from YouTube. These promising results demonstrate that discourse about music may carry semantic information useful to making determinations about the music itself. Such an approach could potentially supplement music information retrieval systems to estimate emotion for pieces of music for which the audio is restricted by copyright or otherwise unavailable. © ICNLSP 2022. All rights reserved

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Evaluating LLM Prompts for Data Augmentation in Multi-Label Classification of Ecological Texts

Item Type Conference Paper

Author A. Glazkova

Author O. Zakharova

Abstract Large language models (LLMs) play a crucial role in natural language processing (NLP) tasks, improving the understanding, generation, and manipulation of human language across domains such as translating, summarizing, and classifying text. Previous studies have demonstrated that instruction-based LLMs can be effectively utilized for data augmentation to generate diverse and realistic text samples. This study applied prompt-based data augmentation to detect mentions of green practices in Russian social media. Detecting green practices in social media aids in understanding their prevalence and helps formulate recommendations for scaling eco-friendly actions to mitigate environmental issues. We evaluated several prompts for augmenting texts in a multi-label classification task, either by rewriting existing datasets using LLMs, generating new data, or combining both approaches. Our results revealed that all strategies improved classification performance compared to the models fine-tuned only on the original dataset, outperforming baselines in most cases. The best results were obtained with the prompt that paraphrased the original text while clearly indicating the relevant categories. © 2024 IEEE.

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Language model, Large language model, Social media, Federated learning, Adversarial machine learning, Contrastive Learning, large language model, Text classification, prompt-based learning, Prompt-based learning, text classification, data augmentation, Data augmentation, Multi-label classifications, Green development, Abiotic, Ecological text, ecological texts, Instruction-based large language model, instruction-based LLM

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Evaluating Reasoning LLMs for Suicide Screening with the Columbia-Suicide Severity Rating Scale

Item Type Conference Paper
Author A. Gedhu
Author S. Tao
Author A. Patil
Abstract Suicide prevention remains a critical public health challenge. While online platforms such as Reddit's r/SuicideWatch have historically provided spaces for individuals to express suicidal thoughts and seek community support, the advent of large language models (LLMs) introduces a new paradigm - where individuals may begin disclosing ideation to AI systems instead of humans. This study evaluates the capability of LLMs to perform automated suicide risk assessment using the Columbia-Suicide Severity Rating Scale (C-SSRS). We assess the zero-shot performance of six models - including Claude, GPT, Mistral, and LLaMA - in classifying posts across a 7-point severity scale (Levels 0-6). Results indicate that Claude and GPT closely align with human annotations, while Mistral achieves the lowest ordinal prediction error. Most models exhibit ordinal sensitivity, with misclassifications typically occurring between adjacent severity levels. We further analyze confusion patterns, misclassification sources, and

ethical considerations, underscoring the importance of human oversight, transparency, and cautious deployment. Full code and supplementary materials are available at https://github.com/av9ash/llm_cssrs_code. © 2025 IEEE.

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Language model, Large language model, Artificial intelligence, Risk assessment, Information systems, Risks assessments, Mental health, Language processing, Natural language processing, Natural language processing systems, Natural languages, Health risks, Ethical aspects, Reddit, Public health, natural language processing, Online systems, Large Language Models, mental health, zero-shot learning, ethical AI, Ethical AI, C-SSRS, Columbia, Columbia-suicide severity rating scale, digital triage, Digital triage, ordinal classification, Ordinal classification, Rating scale, Suicide risk assessment

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Evaluating TF-IDF and Word2Vec in Multilingual Sentiment Analysis: A Controlled Empirical Approach

Item Type Conference Paper

Author Y. Jin

Abstract This study compares two popular text feature extraction methods - TF-IDF and Word2Vec - for multilingual sentiment classification. As social media content becomes more multilingual and informal, sentiment analysis gets trickier. We tested both methods on a multilingual Twitter dataset using the same preprocessing and a linear SVM for classification. The results show that TF-IDF outperformed Word2Vec in both multi-class and binary sentiment tasks. TF-IDF showed strong accuracy and could distinguish between different sentiments, while Word2Vec often predicted a single class. This suggests that TF-IDF is more effective and reliable in noisy, resource-limited, and diverse linguistic settings, offering valuable insights for feature selection in multilingual NLP applications. © 2025 IEEE.

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Evaluating the effectiveness of fine-tuning in financial NLP: The case of Social Trading Action Detection

Item Type Journal Article**Author** S. D'Amico**Author** A. Maurino**Author** F. Osborne**Author** G. Sperli

Abstract Financial Natural Language Processing crucially leverages social media for market insights. However, most existing methods for this purpose rely on simple sentiment analysis models, which fail to capture the concrete trading intentions expressed in these discussions. While Large Language Models (LLMs) offer a promising alternative to simplistic sentiment analysis, the actual benefits of fine-tuning across different model families remain unclear in noisy, domain-specific contexts like online forums. To address this gap, we present a comprehensive assessment of the advantages and limitations of fine-tuning for Social Trading Action Detection (STAD), a novel task that aims to classify online posts into actionable categories, namely buy, sell, or other. In addition, we introduce FinReddit-2K, a manually annotated dataset consisting of 2123 Reddit posts, designed to serve as a benchmark for this task. Our experimental analysis goes beyond standard performance metrics and identifies both the types of errors that fine-tuning can successfully mitigate and those that it may inadvertently introduce. Through a systematic evaluation of 57 models, comparing 14 traditional models with 23 zero-shot LLMs and 20 fine-tuned variants, our results show that fine-tuning yields an average F1-score improvement of +15.1%. The best-performing model, a fine-tuned Mistral-7B, achieves an F1-score of 86.0%, although our analysis reveals that fine-tuning fails to produce meaningful performance gains in several scenarios. © 2026 Published by Elsevier Ltd.

Date 2026**DOI** 10.1016/j.ipm.2026.104910**URL** <https://www.scopus.com/pages/publications/105039924575?origin=resultslist>**Archive** Scopus**Language** English**Volume** 63**Publisher** Elsevier Ltd**Publication** Information Processing and Management**ISSN** 03064573 (ISSN)**Issue** 8**Journal Abbr** Inf. Process. Manage.**Date Added** 20/09/2026, 8:35:04 pm**Modified** 20/09/2026, 8:35:04 pm**Tags:**

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Evaluating the effectiveness of large language models in detecting mental health disorders from social media

Item Type Journal Article

Author W. Zhao

Author Y. Xu

Abstract Mental health disorders affect over 25% of the global population, making scalable detection essential for early intervention. Current social media datasets often use community-assigned or platform-inferred labels, which may lack semantic clarity and category consistency. This study explores whether large language models (LLMs), such as GPT-4, can generate more reliable annotations by leveraging prompt-based reasoning aligned with standardised symptom criteria. Using a Reddit dataset of 17,159 posts, we re-annotate entries using a chain-of-thought (CoT) framework guided by symptom profiles from screening instruments. We then evaluate these LLM-generated annotations against subreddit-derived labels via two downstream tasks: 1) classification performance under supervised learning; 2) clustering under unsupervised methods. Results show that LLM-generated annotations yield higher consistency and improve downstream performance, particularly for depression and anxiety, demonstrating their potential to enhance mental health detection from online text. Copyright © 2026 Inderscience Enterprises Ltd.

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Evaluating the Reliability and Accuracy of an AI-Powered Search Engine in Providing Responses on Dietary Supplements: Quantitative and Qualitative Evaluation

Item Type Journal Article

Author M. Liu

Author T. Okuhara

Author R. Shirabe

Author Y. Nishiie

Author Y. Xu

Author H. Okada

Author T. Kiuchi

Abstract Background: The widespread adoption of artificial intelligence (AI)-powered search engines has transformed how people access health information. Microsoft Copilot, formerly Bing Chat, offers real-time web-sourced responses to user queries, raising concerns about the reliability of its health content. This is particularly critical in the domain of dietary supplements, where scientific consensus is limited and online misinformation is prevalent. Despite the popularity of supplements in Japan, little is known about the accuracy of AI-generated advice on their effectiveness for common diseases. Objective: We aimed to evaluate the reliability and accuracy of Microsoft Copilot, an AI search engine, in responding to health-related queries about dietary supplements. Our findings can help consumers use large language models more safely and effectively when seeking information on dietary supplements and support developers in improving large language models' performance in this field. Methods: We simulated typical consumer behavior by posing 180 questions (6 per supplement × 30 supplements) to Copilot's 3 response modes (creative, balanced, and precise) in Japanese. These questions addressed the effectiveness of supplements in treating 6 common conditions (cancer, diabetes, obesity, constipation, joint pain, and hypertension). We classified the AI search engine's answers as "effective," "uncertain," or "ineffective" and evaluated for accuracy against evidence-based assessments conducted by licensed physicians. We conducted a qualitative content analysis of the response texts and systematically examined the types of sources cited in all responses. Results: The proportion of Copilot responses claiming supplement effectiveness was 29.4% (53/180), 47.8% (86/180), and 45% (81/180) for the creative, balanced, and precise modes, respectively, whereas overall accuracy of the

responses was low across all modes: 36.1% (65/180), 31.7% (57/180), and 31.7% (57/180) for creative, balanced, and precise, respectively. No significant difference was observed among the 3 modes ($P=.59$). Notably, 72.7% (2240/3081) of the citations came from unverified sources such as blogs, sales websites, and social media. Of the 540 responses analyzed, 54 (10%) contained at least 1 citation in which the cited source did not include or support the claim made by Copilot, indicating hallucinated content. Only 48.5% (262/540) of the responses included a recommendation to consult health care professionals. Among disease categories, the highest accuracy was found for cancer-related questions, likely due to lower misinformation prevalence. Conclusions: This is the first study to assess Copilot's performance on dietary supplement information. Despite its authoritative appearance, Copilot frequently cited noncredible sources and provided ambiguous or inaccurate information. Its tendency to avoid definitive stances and align with perceived user expectations poses potential risks for health misinformation. These findings highlight the need for integrating health communication principles—such as transparency, audience empowerment, and informed choice—into the development and regulation of AI search engines to ensure safe public use. © Mingxin Liu, Tsuyoshi Okuhara, Ritsuko Shirabe, Yuriko Nishiie, Yinghan Xu, Hiroko Okada, Takahiro Kiuchi.

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Evaluation of ChatGPT-4 as an Online Outpatient Assistant in Puerperal Mastitis Management: Content Analysis of an Observational Study

Item Type Journal Article

Author F. Dolu

Author O.F. Ay

Author A.H. Kupeli

Author E. Karademir

Author M.H. Büyükavcı

Abstract Background: The integration of artificial intelligence (AI) into clinical workflows holds promise for enhancing outpatient decision-making and patient education. ChatGPT, a large language model developed by OpenAI, has gained attention for its potential to support both clinicians and patients. However, its performance in the outpatient setting of general surgery remains underexplored. Objective: This study aimed to evaluate whether ChatGPT-4 can function as a virtual outpatient assistant in the management of puerperal mastitis by assessing the accuracy, clarity, and clinical safety of its responses to frequently asked patient questions in Turkish. Methods: Fifteen questions about puerperal mastitis were sourced from public health care websites and online forums. These questions were categorized into general information (n=2), symptoms and diagnosis (n=6), treatment (n=2), and prognosis (n=5). Each question was entered into ChatGPT-4 (September 3, 2024), and a single Turkish-language response was obtained. The responses were evaluated by a panel consisting of 3 board-certified general surgeons and 2 general surgery residents, using five criteria: sufficient length, patient-understandable language, accuracy, adherence to current guidelines, and patient safety. Quantitative metrics included the DISCERN score, Flesch-Kincaid readability score, and inter-rater reliability assessed using the intraclass correlation coefficient (ICC). Results: A total of 15 questions were evaluated. ChatGPT's responses were rated as "excellent" overall by the evaluators, with higher scores observed for treatment- and prognosis-related questions. A statistically significant difference was found in DISCERN scores across question types ($P=.01$), with treatment and prognosis questions receiving higher ratings. In contrast, no significant differences were detected in evaluator-based ratings (sufficient length, understandability, accuracy, guideline compliance, and patient safety), JAMA benchmark scores, or Flesch-Kincaid readability levels ($P>.05$ for all). Interrater agreement was good across all evaluation parameters ($ICC=0.772$); however, agreement varied when assessed by individual criteria. Correlation analyses revealed no significant overall associations between subjective ratings and objective quality measures, although a strong positive correlation between literature compliance and patient safety was identified for one question ($r=0.968$, $P<.001$). Conclusions: ChatGPT demonstrated adequate capability in providing information on puerperal mastitis, particularly for treatment and prognosis. However, evaluator variability and the subjective nature of assessments highlight the need for further optimization of AI tools. Future research should emphasize iterative questioning and dynamic updates to AI knowledge bases to enhance reliability and accessibility. © Fatih Dolu, Oğuzhan Fatih AY, Aydın Hakan Kupeli, Enes Karademir, Muhammed Huseyin Büyükavcı.

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: O.F. Ay; Department of General Surgery, Necip Fazıl City Hospital Merkez, Kahramanmaras, Erkenez Mh., Recep Tayyip Erdoğan Bulvarı 12., Km Kahramanmaras, 46050, Turkey; email: droguzhanf.ay@gmail.com

Evaluation of LLMs, BERT, and Ensemble Techniques for Analyzing Online Vaccine Sentiment

Item Type Conference Paper
Author A. Parsa
Author A. Dubey
Abstract The widespread use of social media platforms during the COVID-19 pandemic has highlighted the importance of understanding public sentiment towards vaccines. This paper presents an evaluation of large language models (LLMs), a fine-tuned BERT model, and ensemble methods for analyzing social media posts for vaccine sentiment. We introduce a novel dataset comprising annotated social media posts and evaluate the performance of various models in sentiment classification. Our results demonstrate that LLMs can accurately classify sentiments, open-source LLMs have approached parity with closed-source LLMs, ensemble models can provide marginal improvements, and fine-tuned SLMs can approach the performance of LLMs at much smaller footprint and running costs, making them feasible in many resource-constrained environments. © 2024 IEEE.
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Pages 162-165**Proceedings Title** Proc. - Conf. AI, Sci., Eng., Technol., AIxSET**Conference Name** Proceedings - 2024 Conference on AI, Science, Engineering, and Technology, AIxSET 2024**Date Added** 20/09/2026, 8:35:24 pm**Modified** 20/09/2026, 8:35:24 pm**Tags:**

Language model, Performance, Social media, Social media platforms, COVID-19, Sentiment classification, Public sentiments, Vaccines, Ensemble methods, Ensemble techniques, Model method, Open-source

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Evaluation of transformer models for financial targeted sentiment analysis in Spanish

Item Type Journal Article**Author** R. Pan**Author** J.A. García-Díaz**Author** F. Garcia-Sanchez**Author** R. Valencia-García

Abstract Nowadays, financial data from social media plays an important role to predict the stock market. However, the exponential growth of financial information and the different polarities of sentiment that other sectors or stakeholders may have on the same information has led to the need for new technologies that automatically collect and classify large volumes of information quickly and easily for each stakeholder. In this scenario, we conduct a targeted sentiment analysis that can automatically extract the main economic target from financial texts and obtain the polarity of a text towards such main economic target, other companies and society in general. To this end, we have compiled a novel corpus of financial tweets and news headlines in Spanish, constituting a valuable resource for the Spanish-focused research community. In addition, we have carried out a performance comparison of different Spanish-specific large language models, with MarIA and BETO achieving the best results. Our best result has an overall performance of 76.04%, 74.16%, and 68.07% in macro F1-score for the sentiment classification towards the main economic target, society, and other companies, respectively, and an accuracy of 69.74% for target detection. We have also evaluated the performance of multi-label classification models in this context and obtained a performance of 71.13% © Copyright 2023 Pan et al.

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Evolution of Deep Learning Models for Misinformation Detection in Social Media Textual Data: Background, Architectures, Datasets, and Emerging LLM Applications

Item Type Journal Article

Author Z. Elgammal

Author R. Alhajj

Abstract With the exponential growth in social media usage, the rapid spread of misinformation has become a critical global challenge. Recent advances in large language models (LLMs) have shown promising potential in automated misinformation detection. This survey provides a comprehensive review of LLM-based approaches for detecting misinformation in textual data on social media platforms. In this work, we analyze 70+ recent papers; to examine the evolution, implementation, and effectiveness of various LLM architectures in this domain. Our analysis reveals that BERT-based models dominate the field, appearing in approximately 85% of studies, with domain-specific variants like CT-BERT demonstrating superior performance in specialized contexts such as COVID-19

misinformation detection. We provide detailed comparisons of model architectures, implementation strategies, and performance metrics across different domains. Additionally, seven major datasets commonly used in this field were analyzed, examining their characteristics, limitations, and suitability for different detection tasks. The survey also addresses key challenges, including linguistic nuances, model interpretability, and ethical considerations. Our findings indicate that while LLM-based approaches achieve impressive accuracy metrics, significant challenges remain in cross-domain generalization and real-time detection. This survey concludes by identifying promising research directions and providing recommendations for robust model evaluation frameworks. © The Author(s) 2026

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: Z. Elgammal; Department of Computer Engineering, Istanbul Medipol University, Istanbul, Kavacik Mah., Ekinciler Cad., No:19, Beykoz, 34810, Turkey; email: ziad.elgammal@std.medipol.edu.tr

ExaASC: A General Target-Based Stance Detection Corpus in Arabic Language

Item Type Conference Paper

Author M.M. Jaziriyani

Author A. Akbari

Author H. Karbasi

Abstract Target-based Stance Detection is the task of finding a stance toward a target. Twitter is one of the primary sources of political discussions in social media and one of the best resources to analyze Stance toward entities. This work proposes a new method toward Target-based Stance detection by using the stance of replies toward a most important and arguing target in source tweet. This target is detected

with respect to the source tweet itself and not limited to a set of pre-defined targets which is the usual approach of the current state-of-the-art methods. Our proposed new attitude resulted in a new corpus called ExaASC for the Arabic Language, one of the low resource languages in this field. In the end, we used BERT to evaluate our corpus and reached a 70.69 Macro F-score. This shows that our data and model can work in a general Target-base Stance Detection system. The corpus is publicly available¹. © 2021 IEEE.

Date 2021

DOI 10.1109/ICCCKE54056.2021.9721486

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Examining media's coverage of COVID-19 vaccines and social media sentiments on vaccine manufacturers' stock prices

Item Type Journal Article

Author S.Y. Bai

Author E.W.J. Lee

Abstract Introduction: The COVID-19 pandemic caused a widespread public health and financial crisis. The rapid vaccine development generated extensive discussions in both mainstream and social media, sparking optimism in the global financial markets. This study aims to explore the key themes from mainstream media's coverage of

COVID-19 vaccines on Facebook and examine how public interactions and responses on Facebook to mainstream media's posts are associated with daily stock prices and trade volume of major vaccine manufacturers. Methods: We obtained mainstream media's coverage of COVID-19 vaccines and major vaccine manufacturers on Facebook from CrowdTangle, a public insights tool owned and operated by Facebook, as well as the corresponding trade volume and daily closing prices from January 2020 to December 2021. Structural topic modelling was used to analyze social media posts while regression analysis was conducted to determine the impact of Facebook reactions on stock prices and trade volume. Results: 10 diverse topics ranging from vaccine trials and their politicization (note: check that we use American spelling throughout), to stock market discussions were found to evolve over the pandemic. Although Facebook reactions were not consistently associated with vaccine manufacturers' stock prices, 'Haha' and 'Angry' reactions showed the strongest association with stock price fluctuations. In comparison, social media reactions had little observable impact on trading volume. Discussion: Topics generated reflect both actual events during vaccine development as well as its political and economic impact. The topics generated in this study reflect both the actual events surrounding vaccine development and its broader political and economic impact. While we anticipated a stronger correlation, our findings suggest a limited relationship between emotional reactions on Facebook and vaccine manufacturers' stock prices and trading volume. We also discussed potential technical enhancements for future studies, including the integration of large language models. Copyright © 2024 Bai and Lee.

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Export Date: 20 September 2026; Cited By: 4; Funding details: Nanyang Technological University, NTU, (020154-00001); Nanyang Technological University, NTU; Correspondence Address: S.Y. Bai; College of Computing and Data Science, Nanyang Technological University, Singapore, Singapore; email: sbai005@e.ntu.edu.sg

Excitements and concerns in the post-ChatGPT era: Deciphering public perception of AI through social media analysis

Item Type Journal Article

Author W. Qi

Author J. Pan

Author H. Lyu

Author J. Luo

Abstract As AI systems become increasingly prevalent in various aspects of daily life, gaining a comprehensive understanding of public perception towards these AI systems has become increasingly essential for several reasons such as ethical considerations, user experience, fear, disinformation, regulation, collaboration, and co-creation. In this study, we investigate how mass social media users perceive the recent rise of AI frameworks such as ChatGPT. We collect a total of 33,912 comments in 388 unique subreddits spanning from November 30, 2022 to June 8, 2023 using a list of AI-related keywords. We employ a combination of thematic and sentiment analysis, using advanced natural language processing techniques. Specifically, we use BERTopic to uncover the major themes regarding AI on Reddit. Our findings indicate that technology-focused subreddits primarily discuss the technical dimensions of AI, while non-technical subreddits more often address societal impacts, such as job displacement concerns. The disparity in focus between subreddits suggests a gap in the public understanding of AI. We leverage GPT-3.5 with zero-shot prompting and LIWC to analyze the sentiment and perception of AI among individual users. Through a comprehensive sentiment and emotion analysis, we discover that tech-centric communities exhibit greater polarization compared to non-tech communities when discussing AI topics. This suggests that individuals with a deeper understanding or familiarity with AI technologies might have more divided opinions, possibly reflecting a mix of optimism about technological advancements and skepticism about potential impacts. This research contributes to our broader understanding of public opinion surrounding artificial intelligence. © 2024 Elsevier Ltd

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Notes:

EID: 2-s2.0-85196561510

Export Date: 20 September 2026; Cited By: 51; Correspondence Address: H. Lyu; Rochester, 3504 Wegmans Hall, University of Rochester, 14627, United States; email: hlyu5@ur.rochester.edu; CODEN: TEINE

Explainable AI for Mental Health Detection from Social Media: A Comparative Study of Traditional Machine Learning and a Large Language Model

Item Type Conference Paper

Author C. Xie

Author D. Zhu

Author Z. Wang

Author H. Zhang

Author Z. Wei

Abstract Mental-health-related language on social media is increasingly studied as a source of early screening signals, but practical use depends not only on predictive performance but also on how readily model decisions can be interpreted. This study compares traditional machine learning (ML) models and a large language model (LLM) for seven-class classification of mental-health-related social media posts. Using 52,304 cleaned posts from the public Kaggle dataset Sentiment Analysis for Mental Health, we evaluated three ML baselines (LinearSVC, Random Forest, and XGBoost) on a stratified 80/20 split and evaluated Kimi K2 in a zero-shot setting on a stratified 500-post held-out subset. XGBoost achieved the strongest performance on the full test set (accuracy 77.6%, macro-F1 0.741), followed by SVM (76.8%, 0.734) and Random Forest (71.2%, 0.626). The zero-shot LLM achieved 68.4% accuracy and 0.619 macro-F1 without task-specific training. In an additional comparison on the same 500 posts, SVM reached 73.8% accuracy and 0.685 macro-F1, Random Forest reached 71.0% and 0.610, and Kimi K2 remained at 68.4% and 0.619. Explainability analysis combining SVM coefficients, SHAP, LIME, and LLM rationales showed a clear contrast between model-faithful lexical attributions and more fluent but less reliable free-text explanations. These results suggest that supervised ML remains the stronger option for stable, high-throughput screening on this dataset, while LLMs provide a useful zero-shot baseline and a more accessible explanation format. © 2026 IEEE.

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Tags:

Computational linguistics, Language model, Large language model, large language models, Mental health, Classification (of information), Learning systems, machine learning, Machine-learning, Sentiment analysis, Machine learning, Social media, Data mining, Contrastive Learning, social media, Explainable AI, Random forests, SHAP, Text classification, text classification, Mental health detection, mental health detection, LIME

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: C. Xie; University of Massachusetts Amherst, Amherst, United States; email: chen678x@gmail.com; Conference name: 14th IEEE International Conference on Healthcare Informatics, ICHI 2026; Conference date: 1 June 2026 through 3 June 2026; Conference code: 226950

Explainable AI with fine-tuned large language models for sustainable cultural heritage management

Item Type Journal Article

Author K. Zhou

Author W. Chen

Author W. Sheng

Author B. Hu

Author Y. Yang

Author L. Yang

Author H. Lu

Abstract The redevelopment of cultural heritage areas, especially in historical urban environments, requires a nuanced understanding of public perceptions to balance preservation with modernization. This study introduces an advanced AI-driven framework for Aspect Sentiment Quadruple Prediction (ASQP) to assess public perceptions of Lijiang Ancient Town, a UNESCO World Heritage site in China. We fine-tuned the large language model Qwen-14B using LoRA-based methods to augment sentiment data, effectively uncovering implicit emotional cues in social media content. The model integrates BERT, multi-layer BiLSTM, self-attention, CNN, and CRF for enhanced entity recognition and sentiment classification. Experimental results show that the enhanced model (Qwen-14B + ASQP) improved F1-score by 0.97% (from 75.42% to 76.39%) and Precision by 4.48% (from 76.14% to 80.62%) compared to the baseline. Analyzing data from platforms such as Weibo, Dazhong Dianping, and Xiaohongshu (2018–2024), the research uncovers factors influencing public perception, offering insights for heritage site management, urban planning, and the sustainable preservation of cultural heritage. © The Author(s) 2025.

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female, human, male, aged, article, social media, prediction, urban area, explainable artificial intelligence, SHAP, large language model, LoRA, Architectural heritage, city planning, Cultural heritage preservation, inheritance, Sustainable urban heritage management, United Nations Educational, Scientific and Cultural Organization

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Export Date: 20 September 2026; Cited By: 4; Funding details: National Natural Science Foundation of China, NNSF, (52278004); Correspondence Address: W. Chen; School of Architecture and Urban Planning, Chongqing University, Chongqing, China; email: jzx2007cw@126.com

Explicit Stance Detection in the Political Domain: A New Concept and Associated Dataset

Item Type Conference Paper
Author A.R. Caceres-Wright
Author N. Udhayasankar
Author G. Bunn
Author S.M. Shuster
Author K. Joseph
Editor Thomson R.
Editor Pyke A.
Editor Hariharan A.
Editor Renshaw S.
Editor Park P.
Editor Al-khateeb S.
Editor Burger A.

Abstract Stance detection, defined as the task of classifying an individual's attitude towards a target person or concept, offers the potential to understand political opinions at scale using social media data. However, recent studies have questioned the robustness and accuracy of current stance detection methods, highlighting issues such as generalizability in time and inconsistencies in annotations driven by subtle differences in annotation task design. We argue that central to these challenges is the unresolved question of what constitutes an expression of stance. To address this, the present work introduces a distinction between explicit and implicit stance expressions, and argue that a focus on explicit stance detection addresses many of the existing concerns with modern stance detection methods. To facilitate research on explicit stance detection, we then present a novel (and public) dataset of over 1000 tweets across 13 stance targets for explicit stance detection and evaluate baseline models to establish a foundation for future research in this area. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

Date 2024

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Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Exploiting contextual information to improve stance detection in informal political discourse with LLMs

Item Type Conference Paper

Author A.E. Sucu

Author Y. Zhou

Author M.A. Nascimento

Author T. Mullen

Editor Zhao J.

Editor Wang M.

Editor Liu Z.

Abstract This study investigates the use of Large Language Models (LLMs) for political stance detection in informal online discourse, where language is often sarcastic, ambiguous, and context-dependent. We explore whether providing contextual information, specifically user profile summaries derived from historical posts, can improve classification accuracy. Using a real-world political forum dataset, we generate structured profiles that summarize users' ideological leaning, recurring topics, and linguistic patterns. We evaluate seven state-of-the-art LLMs across baseline and context-enriched setups through a comprehensive cross-model evaluation. Our findings show that contextual prompts significantly boost accuracy, with improvements ranging from +17.5% to +38.5%, achieving up to 74% accuracy that surpasses previous approaches. We also analyze how profile size and post selection strategies affect performance, showing that strategically chosen political content yields better results than larger, randomly selected contexts. These findings underscore the value of incorporating user-level context to enhance LLM performance in nuanced political classification tasks. © 2025 Association for Computational Linguistics.

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Computational linguistics, Language model, Classification (of information), Natural language processing systems, Real-world, Context dependent, Linguistic patterns, State of the art, Contextual information, User profile, Classification accuracy, Political discourse, Structured profiles, User's profiles

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: T. Mullen; Khoury College of Computer Sciences, Northeastern University, Seattle Campus, United States; email: a.mullen@northeastern.edu; Conference name: 63rd Annual Meeting of the Association for Computational Linguistics, ACL 2025; Conference date: 27 July 2025 through 1 August 2025; Conference code: 213612

Exploration of Multi-corpus Learning for Hate Speech Classification in Low Resource Scenarios

Item Type Conference Paper

Author A.G. D'Sa

Author I. Illina

Author D. Fohr

Author A. Akbar

Editor Sojka P.

Editor Horák A.

Editor Kopeček I.

Editor Pala K.

Abstract The dramatic increase in social media has given rise to the problem of online hate speech. Deep neural network-based classifiers have become the state-of-the-art for automatic hate speech classification. The performance of these classifiers depends on the amount of available labelled training data. However, most hate speech corpora have a small number of hate speech samples. In this article, we

aim to jointly use multiple hate speech corpora to improve hate speech classification performance in low-resource scenarios. We harness different hate speech corpora in a multi-task learning setup by associating one task to one corpus. This multi-corpus learning scheme is expected to improve the generalization, the latent representations, and domain adaptation of the model. Our work evaluates multi-corpus learning for hate speech classification and domain adaptation. We show significant improvements in classification and domain adaptation in low-resource scenarios. © 2022, Springer Nature Switzerland AG.

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Notes:

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Exploring BERT-Based Models for Sentiment Analysis in the U.S. Airline Reviews

Item Type Conference Paper

Author X. Zheng

Author J. Huang

Author R. Sangi

Author B. Ihnaini

Abstract This study presents a comparative analysis of traditional models (CNN, LSTM) and transformer-based architectures (BERT, DistilBERT, GPT-2, RoBERTa, DeBERTa, and ModernBERT) for sentiment classification on the US Airline Twitter dataset. A systematic experimental framework was adopted, including comprehensive data preprocessing and evaluation using accuracy, precision, recall, and F1-score metrics. Experimental results demonstrate that transformer-based models outperform traditional models. Among them, ModernBERT achieves the best performance, obtaining 87.02% accuracy, while maintaining a reasonable inference cost. These findings highlight the effectiveness and practicality of transformer-based models in sentiment analysis and encourage further research on model optimization for low-resource environments and multi-label applications. © 2025 IEEE.

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Exploring Chinese Youths' Focus and Sentimental Tendencies toward Different Extreme Weather Events: A BERTopic and Zero-Shot GPT-4o-Mini Analysis

Item Type Journal Article

Author S. Che

Author L. Miller

Author Z. Wang

Abstract Extreme weather events increasingly intensify vulnerabilities in social, economic, and environmental systems. Typhoons damage infrastructure and disrupt mobility; cold waves threaten public health and energy demand; heatwaves strain electricity grids and heighten health risks; rainstorms often cause urban flooding and damage crops. Understanding how the public responds both sentimentally and cognitively to such events is crucial for enhancing climate communication and resilience strategies. This study advances current knowledge by employing a deep learning-based multimodel frame-work on Bilibili, a major Chinese social media platform popular among young users. The research integrates BERTopic for topic modeling and a zero-shot, prompt-based GPT-4o-mini model for sentiment classification, representing a rare application of this combined approach in Chinese climate discourse. It further contributes a comparative perspective by analyzing youth reactions to four major types of extreme weather. A total of 443 157 user comments on 182 videos posted by the nongovernmental organization (NGO) account "Chinese Weather Enthusiasts" in 2023 were examined. The results show pre-dominantly negative sentiment across all weather types, with rainstorms (51.90%) and heatwaves (49.63%) generating the most intense reactions. Typhoon discussions emphasized emergency preparedness and regional differences; cold waves triggered health concerns and nostalgia; heatwaves elicited frustration over physical discomfort; and rainstorms provoked criticism of infrastructure and public safety. These findings illuminate the sentimental and topical dimensions of Chinese youth's responses to extreme weather. By leveraging large-scale social media data, the study offers actionable insights for policy-makers, NGOs, and emergency planners seeking to design more targeted communication strategies, improve infrastructure, and strengthen social resilience to climate-related crises. © 2025 American Meteorological Society.

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Exploring Crisis-Driven Social Media Patterns: A Twitter Dataset of Usage During the Russo-Ukrainian War

Item Type Conference Paper

Author I. Lamprou

Author A. Shevtsov

Author D. Antonakaki

Author P. Pratikakis

Author S. Ioannidis

Editor Aiello L.M.

Editor Chakraborty T.

Editor Gaito S.

Abstract On 24 February 2022, Russia’s invasion of Ukraine, now known as the Russo-Ukrainian War, sparked extensive discussions on Online Social Networks (OSN). We initiate a data collection using the Twitter API to capture this dynamic environment. Next, we perform an analysis of the topics discussed and a detection of potential malicious activities. Our dataset consists of 127.2 million tweets originating from 10.9 million users. Given the dataset’s diverse linguistic composition and the absence of labeled data, we approach it as a zero-shot learning problem, employing various techniques that require no prior supervised training on the dataset. Our research covers several areas, including sentiment

analysis capturing the public's response to the distressing events of the war, topic analysis comparing narratives between social networks and traditional media, and examination of the correlation between message toxicity levels and Twitter suspensions. Furthermore, we explore the potential exploitation of social networks to acquire military-related information by belligerents, presenting a pipeline to classify such communications. The findings of this study provide fresh insights into the role of social media during conflicts, with broad implications for policy, security, and information dissemination. Finally, due to the recent Twitter API changes, we share anonymized data for any further research purposes. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Notes:

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Exploring Data Augmentation Strategies for Hate Speech Detection in Roman Urdu

Item Type Conference Paper

Author U. Azam
Author H. Rizwan
Author A. Karim
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Editor Cieri C.
Editor Declerck T.
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Editor Odijk J.
Editor Piperidis S.

Abstract In an era where social media platform users are growing rapidly, there has been a marked increase in hateful content being generated; to combat this; automatic hate speech detection systems are a necessity. For this purpose, researchers have recently focused their efforts on developing datasets, however, the vast majority of them have been generated for the English language, with only a few available for low-resource languages such as Roman Urdu. Furthermore, what few are available have small number of samples that pertain to hateful classes and these lack variations in topics and content. Thus, deep learning models trained on such datasets perform poorly when deployed in the real world. To improve performance the option of collecting and annotating more data can be very costly and time consuming. Thus, data augmentation techniques need to be explored to exploit already available datasets to improve model generalizability. In this paper, we explore different data augmentation techniques for the improvement of hate speech detection in Roman Urdu. We evaluate these augmentation techniques on two datasets. We are able to improve performance in the primary metric of comparison (F1 and Macro F1) as well as in recall, which is impertinent for human-in-the-loop AI systems. © European Language Resources Association (ELRA), licensed under CC-BY-NC-4.0.

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Language English

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Publisher European Language Resources Association (ELRA)

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Conference Name 2022 Language Resources and Evaluation Conference, LREC 2022

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Exploring Inflammatory Bowel Disease Discourse on Reddit Throughout the COVID-19 Pandemic Using OpenAI's GPT-3.5 Turbo Model: Classification Model Validation and Case Study

Item Type Journal Article

Author T. Babinski

Author S. Karley

Author M. Cooper

Author S. Shaik

Author Y. Ken Wang

Abstract Background: Inflammatory bowel disease (IBD) is a chronic autoimmune disorder with an increasing prevalence in the general population. Internet-based communities have become vital for communication among patients with IBD, especially throughout the COVID-19 pandemic. However, these internet-based patient-to-patient communications remain largely underexplored. Objective: This study aims to analyze community posts from 3 of the largest IBD support groups on Reddit between March 1, 2020, and December 31, 2022, using a pretrained transformer model, and to validate the classification system's results via comparison to human scoring. Methods: We collected posts (N=53,333) from subreddits r/CrohnsDisease, r/UlcerativeColitis, and r/IBD and classified them using OpenAI's GPT-3.5 Turbo model to determine sentiment, categorize topics, and identify demographic information and mentions of the COVID-19 pandemic. A subset of posts (n=397) was manually scored to measure interrater agreement between human raters and the GPT-3.5 Turbo model. Results: Fleiss κ and Gwet AC1 coefficients indicated a high level of agreement between raters, with values ranging from 0.53 to 0.91. The raters demonstrated almost perfect agreement on the classification of gender, with a Fleiss κ of 0.91 ($P<.001$). Medications (14,909/53,333) and symptoms (14,939/53,333) emerged as the most discussed topics, and most posts conveyed a neutral sentiment. While most users did not disclose their age, those who did primarily belonged to the 20-29 years (2392/4828) and 30-39 years (859/4828) age groups. Based on self-reported gender, we identified 1509 men and 1502 women among our IBD Reddit users. When comparing the users on the IBD subreddits to the general IBD population, there was a significant difference in gender distribution (N=3,090,011; $\chi^2=69.53$; $P<.001$; $\phi<0.001$). After an initial spike in posts within the first month,

most posts did not reference the COVID-19 pandemic. Conclusions: Our study showcases the potential of generative pretrained transformer models in processing and extracting insights from medical social media data. Future research can benefit from further subanalyses of our validated dataset or use OpenAI's model to analyze social media data for other conditions, particularly those for which patient experiences are challenging to collect. ©Tyler Babinski, Sara Karley, Marita Cooper, Salma Shaik, Y Ken Wang.

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adult, Adult, Article, female, Female, gender, human, Humans, male, Male, middle aged, machine learning, adolescent, pandemic, comparative study, demographics, ChatGPT, Reddit, social media, controlled study, sentiment analysis, Internet, case study, model, COVID-19, Pandemics, coronavirus disease 2019, epidemiology, large language model, SARS-CoV-2, Severe acute respiratory syndrome coronavirus 2, COVID-19 pandemic, interrater reliability, self report, validation study, Reddit discourse, child, topic analysis, autoimmune disorder, classification model, cohort analysis, Crohn disease, feasibility study, gastrointestinal, infant, inflammatory bowel disease, Inflammatory Bowel Diseases, newborn, OpenAI GPT 3.5 Turbo, population, scoring system, support group, ulcerative colitis

Notes:

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Exploring Multi-Agent Debate for Zero-Shot Stance Detection: A Novel Approach

Item Type Journal Article**Author** J. Ma**Author** C. Wang**Author** L. Rong**Author** B. Wang**Author** Y. Xu

Abstract Zero-shot stance detection aims to identify the stance expressed in social media text aimed at specific targets without relying on annotated data. However, due to insufficient contextual information and the inherent ambiguity of language, this task faces numerous challenges in low-resource scenarios. This work proposes a novel zero-shot stance detection method based on multi-agent debate (ZSMD) to address the aforementioned challenges. Specifically, we construct two debater agents representing the supporting and opposing stances. A knowledge enhancement module supplements the original tweet and target with relevant background knowledge, providing richer contextual support for argument generation. Subsequently, the two agents engage in debate over a predetermined number of rounds, employing rebuttal strategies such as factual verification, logical analysis, and sentiment analysis. If no consensus is reached within the specified rounds, a referee agent synthesizes the debate process and original input information to deliver the final stance determination. We evaluate ZSMD on two benchmark datasets, SemEval-2016 Task 6 and P-Stance, and compare it against strong zero-shot baselines such as MB-Cal and COLA. The experimental results show that ZSMD not only achieves higher accuracy than these baselines, but also provides deeper insights into subtle differences in opinion expression, highlighting the potential of structured argumentation in low-resource settings. © 2025 by the authors.

Date 2025**DOI** 10.3390/app15094612**URL** <https://www.scopus.com/pages/publications/105004903993?origin=resultslist>**Archive** Scopus**Language** English**Volume** 15**Publisher** Multidisciplinary Digital Publishing Institute (MDPI)**Publication** Applied Sciences (Switzerland)**ISSN** 20763417 (ISSN)**Issue** 9**Journal Abbr** Appl. Sci.**Date Added** 20/09/2026, 8:35:10 pm**Modified** 20/09/2026, 8:35:10 pm**Tags:**

Social media, Multi agent, Detection methods, Zero-shot stance detection, Multi-agent debate, Background knowledge, Contextual information, Contextual support, knowledge augment, Knowledge augment, multi-agent debate, zero-shot stance detection, ZSMD

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Exploring Sentiment Analysis in Health with CNN, GRU and Zero-Shot Learning on Imbalanced Textual Data.

Item Type Conference Paper

Author S. Hamza-Cherif

Author N. Settouti

Editor Benchikha F.

Editor Belala F.

Editor Boufaida Z.

Editor Ghemmaz W.

Editor Smaali S.

Abstract Sentiment analysis has recently gained popularity due to the emergence of social media and the semantic web, which provide platforms for individuals to express their thoughts and emotions. In the field of health, analyzing this type of data can have numerous benefits, particularly in psychology, where it can aid in automatically identifying a patient’s mental state. In this study, we present a method for categorizing emotions from text data in the health field. We evaluated various classifiers such as: Convolutional Neural Networks (CNN), Gated Recurrent Units (GRU), and Zero-Shot learning (ZSL) on the unbalanced EmoHD dataset. Our findings showed that the CNN model gives the best performance[1]. © 2023 CEUR-WS. All rights reserved.

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Conference Name CEUR Workshop Proceedings

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Hamza-Cherif; Biomedical Engineering Laboratory, University of Tlemcen, Algeria; email: souad.hamzacherif@univ-tlemcen.dz; Conference name: 12th Seminary of Computer Science Research at Feminine, RIF 2023; Conference code: 196383

Exploring Sentiment Analysis in Tigrigna: Insights from Social Media Texts

Item Type Conference Paper

Author H.G. Gebremeskel

Author C. Feng

Editor Mao X.-L.

Editor Ren Z.

Editor Yang M.

Abstract This study explores sentiment analysis for Tigrigna social media texts, a low-resource language with limited computational tools. We collected and annotated a dataset of Tigrigna posts and comments from social media platforms and evaluated several machine learning and deep learning models, including Naive Bayes, SVM, LSTM, and XLM-RoBERTa. Results show that XLM-RoBERTa, a transformer-based model, achieved the highest performance with an F1-score of 0.83, effectively handling Tigrigna's complex morphology and cultural nuances. Key challenges included data scarcity, dialectal variations, and idiomatic expressions unique to Tigrigna. Future work suggests expanding Tigrigna resources, exploring optimized models for resource-constrained environments, and developing applications for real-world sentiment monitoring. This research contributes to advancing NLP for low-resource languages, promoting inclusivity in sentiment analysis. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: C. Feng; School of Computer Science and Technology, Beijing Institute of Technology, Beijing, 100081, China; email: fengchong@bit.edu.cn; Conference name: 14th National CCF Conference on Natural Language Processing and Chinese Computing, NLPCC 2025; Conference date: 7 August 2025 through 9 August 2025; Conference code: 343639

Exploring the performance of generative models in detecting aggressive content in memes

Item Type Journal Article

Author P.C. de Queiroz Hermida

Author E.M.D. Santos

Abstract The spread of aggressive memes on social media is a major problem, demanding solutions for the automatic detection of these undesired memes. However, the line between freedom of expression and the dissemination of aggressive messages is very blurred and difficult to define. Moreover, the detection of aggressive elements in memes presents a significant challenge because memes are mostly multimodal, e.g., a composition of an image with a concise textual content which often lacks a direct connection or correlation. The advent of Generative Models presents an interesting alternative to unravel the inherent meaning of aggressive memes by allowing a fully multimodal analysis. This paper examines the use of Generative Models to detect aggressive memes. The proposed methodology groups aggressive memes from existing datasets according to levels of multimodal analysis difficulty and, together with the creation of prompts via zero-shot, tests and verifies how well these models perform in the task of detecting this type of content and where failures may occur. To validate the methodology, a series of experiments using three models and five datasets was conducted. According to manual annotation, these data were divided into three datasets representing three multimodality reasoning levels, which were submitted to the models to perform the task of identifying the presence of aggressive content. The results show the viability of using generative models in this task, as well

as highlight the better performance of some models compared to others. Moreover, a comparison with a non-generative model specially trained on the meme datasets shows that generative models can be competitive in the task of detecting memes with aggressive content as long as the multimodality reasoning level required is not very high. The findings can help academics, managers, and decision-makers develop more effective tools for detecting aggressive content while balancing freedom of expression. The research also serves as a reference for improving moderation systems and guiding future policies in online communication. © The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2025.

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Extracting Emotions from Customer Reviews Using Text Mining, Large Language Models and Fine-Tuning Strategies

Item Type Journal Article

Author S.-V. Oprea

Author A. Bâra

Abstract User-generated content, such as product and app reviews, offers more than just sentiment. It provides a rich spectrum of emotional expression that reveals users' experiences, frustrations and expectations. Traditional sentiment analysis, which typically classifies text as positive or negative, lacks the nuance needed to fully understand the emotional drivers behind customer feedback. In this research, we focus on fine-grained emotion classification using core emotions. By identifying specific emotions rather than sentiment polarity, we enable more actionable insights for e-commerce and app development, supporting strategies such as feature refinement, marketing personalization and proactive customer engagement. We leverage the Hugging Face Emotions dataset and adopt a two-phase modeling approach. In the first phase, we use a pre-trained DistilBERT model as a feature extractor and evaluate multiple classical classifiers (Logistic Regression, Support Vector Classifier, Random Forest) to establish performance baselines. In the second phase, we fine-tune the DistilBERT model end-to-end using the Hugging Face Trainer API, optimizing classification performance through task-specific adaptation. Training is tracked using the Weights & Biases (wandb) API. Comparative analysis highlights the substantial performance gains from fine-tuning, particularly in capturing informal or noisy language typical in user reviews. The final fine-tuned model is applied to a dataset of customers' reviews, identifying the dominant emotions expressed. Our results demonstrate the practical value of emotion-aware analytics in uncovering the underlying "why" behind user sentiment, enabling more empathetic decision-making across product design, customer support and user experience (UX) strategy. © 2025 by the authors.

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Extracting Symptoms of Complex Conditions From Online Discourse (Subreddit to Symptomatology): Lexicon-Based Approach

Item Type Journal Article

Author B. Hossain

Author S.M. Preum

Author M. Fazle Rabbi

Author R. Ara

Author M.E. Ali

Abstract Background: Millions of people affected with complex medical conditions with diverse symptoms often turn to online discourse to share their experiences. While some studies have explored natural language processing methods and medical information extraction tools, these typically focus on generic symptoms in clinical notes and struggle to identify patient-reported, disease-specific, subtle symptoms from online health discourse. Objective: We aimed to extract patient-reported, disease-specific symptoms shared on social media reflecting the lived experiences of thousands of affected individuals and explore the characteristics, prevalence, and occurrence patterns of the symptoms. Methods: We propose a lexicon-based symptom extraction (LSE) method to identify a diverse list of disease-specific, patient-reported symptoms. We initially used a large language model to accelerate the extraction of symptom-related key phrases that formed the lexicon. We evaluated the effectiveness of lexicon extraction against human annotation using a Jaccard index score. We then leveraged BERT-Base, BioBERT, and Phrase-BERT-based embeddings to learn representations of these symptom-related key phrases and cluster similar symptoms using k-means and hierarchical density-based spatial clustering of applications with noise (HDBSCAN). Among the different options explored in our experiments, BioBERT-based k-means clustering was found to be the most effective. Finally, we applied symptom normalization to eliminate duplicate and redundant entries in the comprehensive symptom list. Results: In a real-world polycystic ovary syndrome (PCOS) subreddit dataset, we found that LSE significantly outperformed state-of-the-art baselines, achieving at least 41% and 20% higher F1-scores (mean 86.10) than automatic medical extraction tools and large language models, respectively. Notably, the comprehensive list of 64 PCOS symptoms generated via LSE ensured extensive coverage of symptoms reported in 7 reputable eHealth forums. Analyzing PCOS symptomatology revealed 28 potentially emerging symptoms and 8 self-reported comorbidities co-occurring with PCOS. Conclusions: The comprehensive patient-reported, disease-specific symptom list can help patients and health practitioners resolve uncertainties surrounding the disease, eliminating the variability of PCOS symptoms prevailing in the community. Analyzing PCOS symptomatology across varied dimensions provides valuable insights for public health research. ©Bushra Hossain, Sarah M Preum, Md Fazle Rabbi, Rifat Ara, Mohammed Eunus Ali.

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Extracting the ‘Why’: Flood Cause Annotation and LLM Benchmarking from Multi-Platform Crisis Data

Item Type Conference Paper

Author J.A.D. Cruz

Author A. Asimov

Author J. Wagemaker

Author I. Hendrickx

Editor Rizza C.

Editor Chauhan A.

Editor Matser A.

Editor Kox J.

Editor Treurniet W.

Editor Wolbers J.

Abstract We present a framework for extracting and classifying flood cause information from crisis-related online media data. By harnessing data across multiple online platforms from both social media and online news sources, we can understand causes of floods ranging from environment factors like heavy rainfall to human-related factors like infrastructure failure. This task is operationally critical not only for disaster response but also for public accountability, yet it remains a challenging task for automated systems. Our key contributions are: (1) a comprehensive flood cause annotation protocol that operates at two levels of granularity i.e., document-level and sentence-level, with strict exclusion criteria for distinguishing causal statements from impacts; and (2) initial benchmarking experiments to evaluate the performance of zero-shot large language models (LLMs) on the classification task. Although the underlying corpus—covering 70 distinct flood events from 2022–2025 across News, X (formerly Twitter), YouTube, and Bluesky is not publicly released, we describe the data construction

methodology in full to support replication on comparable data. Our initial analysis on a stratified 10-event sample demonstrate that LLMs struggle: they have high recall but suffer from low precision in flood cause classification. © 2026, Information Systems for Crisis Response and Management, ISCRAM. All rights reserved.

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Fake News Detection Based on Large Language Models

Item Type Conference Paper

Author J. Ding

Author Z. Zhao

Abstract With the rapid advancement of social networks, the accessibility of information has significantly improved. However, alongside this progress, the proliferation of rumors and misinformation on social media platforms has become increasingly severe, substantially undermining the credibility of online information sources. Traditional methods for detecting fake news are often labor-intensive and time-consuming, with a high risk of misjudgment and erroneous classification. Against this backdrop, leveraging advanced large language models (LLMs) for automated fake news detection has become an area of considerable research interest. The rapid development of artificial intelligence, coupled with the vast and diverse news datasets available on the internet, provides favorable conditions for AI-driven misinformation detection. In this study, we utilize datasets collected from publicly available sources. After undergoing data cleaning and annotation enhancement, these datasets serve as the foundation for model training. For model selection, we employ a range of LLMs. To optimize the performance of smaller-scale models such as ChatGLM-6B, we apply low-rank adaptation (LoRA) for efficient fine-Tuning. The dataset is divided into training and validation sets, and model performance is evaluated using F1 Score and Accuracy Score as key metrics. This study aims to compare the effectiveness of various large language models in detecting fake and misleading news, ultimately identifying the most suitable model for enhancing the reliability of information dissemination on social media platforms. © 2025 IEEE.

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Fake News Detection in Hindi Using Feature Fusion

Item Type Conference Paper

Author R. Gabra

Author A. Gurushankar

Author R. Rajalakshmi

Editor Chakravarthi B.R.

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Editor García Cumbreñas M.A.

Editor Jiménez Zafra S.M.

Editor Kovács G.

Editor Eger S.

Editor Wahyu Pamungkas E.

Editor Dobrovoljc K.

Abstract As social media continues to shape modern communication, the rapid spread of fake news has become a critical issue. While existing solutions for fake news detection have made progress, there is a significant gap in addressing this problem for low-resource languages like Hindi. Most current research focuses on general algorithms without considering the potential role of sentiment analysis in improving detection accuracy. This paper introduces a novel approach that incorporates sentiment analysis into the identification of fake news, particularly in low-resource language contexts. Using the TALLIP Fake News Dataset, various models are evaluated for their performance. Our approach achieves an accuracy and F1 score of 89%, demonstrating a 3% improvement over existing methods. This research addresses an important gap in the detection of fake news, particularly in underrepresented languages, and has the potential to reduce the spread of misinformation, thereby mitigating its impact on social issues. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Fake News Detection: A comparative analysis of BERT and LSTM for Malay Language

Item Type Conference Paper
Author N.N.N.M. Zulhaidi
Author F.A. Bohani
Author M.R.M. Razali
Author M.H.A. Anuar
Abstract The rise of fake news, particularly on social media, threatens public trust and decision-making. This study focuses on detecting fake news in the Malay language using deep learning models, specifically Bidirectional Encoder Representations from Transformers (BERT) and Long ShortTerm Memory (LSTM). Using the Malay Fake News Classification dataset containing 37,592 articles, both models were trained and evaluated. BERT achieved superior performance (96.8% accuracy, 96.4% F1-score) compared to LSTM (89.65% accuracy, 92.22% F1-score). These findings demonstrate the potential of transformer-based models in improving fake news detection for low-resource languages. © 2026 IEEE.
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Fake News Detection: Multimodel Approaches and LLM-Based Verification

Item Type Conference Paper

Author S. Puraswani

Author S. Pathan

Author R. Bhagwat

Author A. Pandey

Author C. Patil

Editor Kaiser M.S.

Editor Xie J.

Editor Joshi A.

Abstract The wide-reaching growth of misinformation on online social media platforms creates significant challenges in today's information environment; with billions of users sharing content in various languages and formats, distinguishing fact from

fiction requires significantly more advanced detection methodologies than traditional methodologies. In this paper, we provide a comprehensive literature review of the advanced automatic detection and verification methods for fake news, and we highlight the transition from basic text classification methods to multimodal and reasoning-based systems. Specifically, we evaluate the relative effectiveness of various machine learning (ML) and deep learning (DL) approaches for recognizing misleading content, and we examine verification approaches using API-based evidence collection, web scraping, and large language models (LLMs) for contextualization. By drawing on the synthesis of results from several different studies, this review surfaces ongoing challenges, such as explainability and multilingual support and sets forth promising directions for developing more dependable, open, and effective misinformation detection systems. The purpose of this paper is to bind advances in technology to real-world dilemmas and to advance future studies on more effective countermeasures to misinformation. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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FakeThreads: Investigating Fake News Dissemination Patterns in Threads

Item Type Journal Article

Author A. Patidar

Author D. Sadhya

Abstract In the era of online presence, fake news is widely spread on social media platforms. Fake news refers to false or misleading information that is presented as legitimate news. Unfortunately, a large volume of this fake or misleading information is distributed on the Internet, which is a very serious concern. Among all existing social networks, one of the most rapidly evolving platforms is Threads. This work presents a comprehensive approach to the detection of fake news on Threads using crowd-sourced information. In our detection pipeline, data are initially collected from Threads, with posts labeled by cross-referencing reputable fact-checking platforms such as PolitiFact and Lead Stories. Subsequently, extensive data preprocessing and exploratory analysis are performed to understand engagement patterns, sentiment differences, and posting behaviors. To mitigate the inherent class imbalance between real and fake news posts, synthetic fake news samples are generated using a Large Language Model. For generating robust feature embeddings from the posts, Sentence-BERT is used. We evaluated multiple classification models on the scraped dataset, with Extreme Gradient Boosting (XGBoost) demonstrating the best accuracy of 91.22% (among traditional classifiers) and RoBERTa-base resulting in 91.93% accuracy (among Transformer-based models). Importantly, we were able to infer multiple interesting data patterns that are unique to Threads (e.g., bot behavior). Hence, this study integrates crowd-sourced data and modern natural language processing (NLP) techniques to develop an accurate fake news detection pipeline. © 2025 IEEE. All rights reserved.

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FanChuan: A Multilingual and Graph-Structured Benchmark For Parody Detection and Analysis

Item Type Conference Paper

Author Y. Zheng

Author S. Li

Author F. Wu

Author Y. Ziyi

Author L. Hongchao

Author Z. Hu

Author C. Xinjun

Author Z. Wang

Author J. Chen

Author S. Luan

Author J. Xu

Author L. Chen

Editor Che W.

Editor Nabende J.

Editor Shutova E.

Editor Pilehvar M.T.

Abstract Parody is an emerging phenomenon on social media, where individuals imitate a role or position opposite to their own, often for humor, provocation, or controversy. Detecting and analyzing parody can be challenging and is often reliant on context, yet it plays a crucial role in understanding cultural values, promoting subcultures, and enhancing self-expression. However, the study of parody is hindered by limited available data and deficient diversity in current datasets. To bridge this gap, we built seven parody datasets from both English and

Chinese corpora, with 14,755 annotated users and 21,210 annotated comments in total. To provide sufficient context information, we also collect replies and construct user-interaction graphs to provide richer contextual information, which is lacking in existing datasets. With these datasets, we test traditional methods and Large Language Models (LLMs) on three key tasks: (1) parody detection, (2) comment sentiment analysis with parody, and (3) user sentiment analysis with parody. Our extensive experiments reveal that parody-related tasks still remain challenging for all models, and contextual information plays a critical role. Interestingly, we find that, in certain scenarios, traditional sentence embedding methods combined with simple classifiers can outperform advanced LLMs, e.g., DeepSeek-R1 and GPT-o3, highlighting parody as a significant challenge for LLMs. Our code and data is available at <https://github.com/Lisaaa1017/> Fanchuan. © 2025 Association for Computational Linguistics.

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FarExStance: Explainable Stance Detection for Farsi

Item Type Conference Paper

Author M. Zarharan

Author M. Hashemi

Author M. Behroozrazegh

Author S. Eetemadi

Author M.T. Pilehvar

Author J. Foster

Editor Rambow O.

Editor Wanner L.

Editor Apidianaki M.

Editor Al-Khalifa H.

Editor Di Eugenio B.

Editor Schockaert S.

Abstract We introduce FarExStance, a new dataset for explainable stance detection in Farsi. Each instance in this dataset contains a claim, the stance of an article or social media post towards that claim, and an extractive explanation which provides evidence for the stance label. We compare the performance of a fine-tuned multilingual RoBERTa model to several large language models in zero-shot, few-shot, and parameter-efficient fine-tuned settings on our new dataset. On stance detection, the most accurate models are the fine-tuned RoBERTa model, the LLM Aya-23-8B which has been fine-tuned using parameter-efficient fine-tuning, and few-shot Claude-3.5-Sonnet. Regarding the quality of the explanations, our automatic evaluation metrics indicate that few-shot GPT-4o generates the most coherent explanations, while our human evaluation reveals that the best Overall Explanation Score (OES) belongs to few-shot Claude-3.5-Sonnet. The fine-tuned Aya-32-8B model produced explanations most closely aligned with the reference explanations. © 2025 Association for Computational Linguistics.

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Fast and Accurate Sentiment Analysis Using a Hybrid LLM Model With OpenVINO Optimization

Item Type Journal Article

Author B. Chandra Praveen

Author K. Sivasankaran

Abstract The rapid growth of social media and internet usage creates a vast amount of textual information, making sentiment analysis important for understanding the interpretation of people. Most sequence-based models capture long-range dependencies but have high latency, whereas transformer-based models are more accurate with higher computational complexity. To address these challenges, a hybrid RoBERTa-BiGRU model with an attention mechanism is proposed in this work. This hybrid model is a combination of the contextual learning capability of the transformer with the sequential modeling efficiency of Bi-GRU, with an attention mechanism to identify the relevant input to make predictions. This approach was evaluated on the IMDb dataset, the model has the accuracy of 96.02%, precision of 96.20%, recall of, 95.80% and F1-score of, 96.00% which is better than the conventional deep learning models. To provide efficient deployment, the trained model was converted to ONNX and optimized with OpenVINO in FP32 and FP16, and using the Neural Network Compression Framework (NNCF) further, we optimized it into INT8. The results showed that the optimized model demonstrated between 30-35% reduction in latency, significantly improved throughput, and inference can be 10-15 x faster than the baseline PyTorch version, with no significant decrease in accuracy. These results highlight that the use of a hybrid combination with an optimization technique is an effective and practical approach for real-time sentiment analysis applications. © 2013 IEEE.

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Few-Shot Cross-Lingual Stance Detection with Sentiment-Based Pre-training

Item Type Conference Paper

Author M. Hardalov

Author A. Arora

Author P. Nakov

Author I. Augenstein

Abstract The goal of stance detection is to determine the viewpoint expressed in a piece of text towards a target. These viewpoints or contexts are often expressed in many different languages depending on the user and the platform, which can be a local news outlet, a social media platform, a news forum, etc. Most research on stance detection, however, has been limited to working with a single language and on a few limited targets, with little work on cross-lingual stance detection. Moreover, non-English sources of labelled data are often scarce and present additional challenges. Recently, large multilingual language models have substantially improved the performance on many non-English tasks, especially such with a limited number of examples. This highlights the importance of model pre-training and its ability to learn from few examples. In this paper, we present the most comprehensive study of cross-lingual stance detection to date: we experiment with 15 diverse datasets in 12 languages from 6 language families, and with 6 low-resource evaluation settings each. For our experiments, we build on pattern-exploiting training (PET), proposing the addition of a novel label encoder to simplify the verbalisation procedure. We further propose sentiment-based

generation of stance data for pre-training, which shows sizeable improvement of more than 6% F1 absolute in few-shot learning settings compared to several strong baselines. Copyright © 2022, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Few-Shot Information Operation Detection Using Active Learning Approach

Item Type Conference Paper

Author M. Alizadeh

Author J.N. Shapiro

Editor Thomson R.

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Editor A. Pyke A.

Abstract Previous research suggested that supervised machine learning can be utilized to detect information operations (IO) on social media. Most of the related research assumes that the new data will always be available in the exact timing that models set to be updated. In practice, however, the detection and attribution of IO

accounts is time-consuming. There is thus a mismatch between the performance assessment procedures in existing work and the real-world problem they seek to solve. We bridge this gap by demonstrating how active learning approaches can extend the application of classifiers by reducing their dependence on new data. We evaluate the performance of an existing classifier when it gets updated according to five active learning strategies. Using state-sponsored information operation Twitter data, the results show that if querying from Twitter is possible, the best active learning strategy requires 5–10 times less tweets than the original model while only showing 1–3% reduction in the average monthly F1 scores across countries and prediction tasks. If querying from Twitter is not possible, the corresponding active learning strategy requires 5–10 times less tweets while showing 1–9% reduction in the average monthly F1 scores. Depending on the country, a hand-full to few hundred new ground-truth examples would suffice to achieve a reasonable performance. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Few-Shot Learning for Cross-Target Stance Detection by Aggregating Multimodal Embeddings

Item Type Journal Article

Author P.J. Khiabani

Author A. Zubiaga

Abstract Despite the increasing popularity of the stance detection task, existing approaches are predominantly limited to using the textual content of social media posts for the classification, overlooking the social nature of the task. The stance detection task becomes particularly challenging in cross-target classification scenarios, where even in few-shot training settings the model needs to predict the stance toward new targets for which the model has only seen a few relevant samples during training. To address the cross-target stance detection in social media by leveraging the social nature of the task, we introduce cross-target text-net (CT-TN), a novel model that aggregates multimodal embeddings derived from both textual and network features of the data. We conduct experiments in a few-shot cross-target scenario on six different combinations of source-destination target pairs. By comparing CT-TN with state-of-the-art cross-target stance detection models, we demonstrate the effectiveness of our model by achieving average performance improvements ranging from 11% to 21% across different baseline models. Experiments with different numbers of shots show that CT-TN can outperform other models after seeing 300 instances of the destination target. Further, ablation experiments demonstrate the positive contribution of each of the components of CT-TN toward the final performance. We further analyze the network interactions between social media users, which reveal the potential of using social features for cross-target stance detection. © 2014 IEEE.

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Few-Shot Learning Meets Large Language Models: Mining Medicine Interventions from Reddit

Item Type Conference Paper

Author C. Fu

Author M. Ritter

Author S. Diao

Author F. Jin

Abstract Extracting structured information from noisy social media discussions remains a significant challenge due to unstandardized language, data sparsity, and the need for precise alignment of entities such as pharmaceutical terms. Traditional Natural Language Processing pipelines often require extensive preprocessing and task-specific annotations, limiting their scalability. In this work, we propose a novel Large Language Model (LLM) framework based on instruction following few-shot learning to extract pharmaceutical interventions from Reddit discussions. Our framework leverages structured prompting to generate schema-constrained outputs, reducing post-processing complexity and enabling seamless normalization to biomedical ontologies. This design bridges the flexibility of Question Answering-style extraction (no predefined entity lists required) with the precision of structured approaches such as named entity recognition. In this study, we analyze Reddit discussions from the 'r/covid19positive' subreddit, spanning multiple phases of the pandemic, to evaluate our method. We further benchmark three state-of-the-art LLMs - GPT-3.5 Turbo, Gemini 1.5 Flash, and GPT-4o-Mini - highlighting their respective strengths and weaknesses in few-shot settings. Results demonstrate that our framework provides an effective and scalable solution for mining complex, user-generated text. © 2025 IEEE.

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Few-Shot Multimodal Sentiment Analysis Based on Dynamic Adjustment and Contrastive Learning

Item Type Journal Article

Author H. Wang

Author S. Liu

Author N. Xie

Abstract Multimodal sentiment analysis (MSA) is a crucial technique for understanding sentiment expression in social media, product reviews, and other multimedia content, and has been extensively studied in recent years. However, most of the existing MSA methods depend on large datasets. Collecting such data is costly and time-consuming, limiting the practical applicability of these models. To address this challenge, this paper proposes a few-shot multimodal sentiment analysis method based on dynamic adjustment and contrastive learning (DACL-FMSA). First, the method uses the BLIP model to generate semantic descriptions of images. These descriptions are then aligned with text inputs to bridge the semantic gap and enable more effective multimodal fusion. Second, based on the contrastive learning framework, the model's ability to capture emotional features is enhanced by generating diverse views of image and text data, thus improving performance in few-shot tasks. Finally, to further optimize the learning process, this study designs a dynamic learning rate adjustment method based on a long short-term memory network, which dynamically adjusts the learning rate according to gradient changes to accelerate model convergence and achieve better training results. The experimental results show that DACL-FMSA achieves significant performance improvements. It performed well across multiple benchmark datasets. For Twitter-15, Twitter-17, and MASAD, the accuracies were 61.88%, 54.00%, and 81.78%, respectively. The accuracies of MVSA-S and TumEmo were 67.89% and 54.42%, respectively. These results consistently demonstrate the effectiveness of the DACL-FMSA. © Fuji Technology Press Ltd. Creative Commons CC BY-ND: This is an Open Access article distributed under the terms of the Creative Commons Attribution-NoDerivatives 4.0 International License (<https://creativecommons.org/licenses/by-nd/4.0/>)

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Fin-STance: A Novel Deep Learning-Based Multi-Task Model for Detecting Financial Stance and Sentiment

Item Type Conference Paper

Author V.K. Singh

Author P. Mohankumar

Author A. Kamal

Abstract Stance detection has been gaining popularity in text mining and information retrieval-based research. Recognition of sentiment also plays an important role in the analysis of text whether the underline stances are in favor or against for a particular domain. However, there is extensive research in this direction, but considering its impact on the financial text has not been explored yet. This paper presents a new deep learning based multi-task approach for financial stance

detection and sentiment detection. A new model called Fin-Stance is introduced which performs two-classification tasks as stance detection and sentiment detection based on the financial data. It constitutes of an input, embedding, followed by shared and task-specific layers. The empirical evaluation of this study is conducted on a newly created dataset and it shows impressive results. Also, this study receives better results with relevant compared methods. © 2023 IEEE.

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: V.K. Singh; PayPal, Chennai, India; email: vishalksingh@paypal.com; Conference name: 14th International Conference on Computing Communication and Networking Technologies, ICCCNT 2023; Conference date: 6 July 2023 through 8 July 2023; Conference code: 194774

Financial Text Sentiment Classification Based on Baichuan2 Instruction Finetuning Model

Item Type Conference Paper

Author Y. Cao

Author L. Yang

Author C. Wei

Author H. Wang

Abstract Financial text analysis is a critical task in the domain of finance and investment, as it helps stakeholders make informed decisions based on textual data. This study explores the application of large language models for instruction finetuning financial text analysis. Specifically, we utilize the Baichuan2-7B model, a state-of-the-art language model, to investigate its performance in comparison to traditional models like Deberta and LLama2. In this research, we use a diverse dataset of financial documents, including news articles, quarterly reports, and social media posts, to evaluate the capabilities of Baichuan2-7B. Our findings reveal that Baichuan2-7B significantly outperforms Deberta and LLama2 in various aspects of financial text analysis, including sentiment analysis, event detection, and information extraction. © 2023 IEEE.

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Computational linguistics, Language model, Sentiment analysis, Investments, Critical tasks, Decision-based, Sentiment classification, Text analysis, Sentiment Classification, Baichuan2, Deberta, Informed decision, Instruct learning, Llama2

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Finding Offensiveness in Movie Reviews of Low Level Languages using Machine Learning

Item Type Conference Paper

Author Ch. Prathima
Author C. Nagajyothi
Author K. Rangaswamy
Author A. Hemanth
Author C.R. Reddy
Author L. Spoorthy

Abstract It is human behavior to look for others opinion before taking any decision. A lot of documents are available which express opinions on different issues. But the main challenge arises in analyzing these documents to produce useful knowledge. Tremendous works in the area of Classification is available for English language. However, there has been little work done for Indian languages. From the last few years, opinion-rich resources are booming in Kannada, Tamil, and Malayalam and hence there is a need to perform classification in Hindi. Classification of movie reviews could help in better rating of movies. This work proposes classification for low-resource languages like Tamil, Kannada, and Malayalam using Neuro-Fuzzy Technique. This report describes the design and implementation of the algorithm. Classification is a field of Natural Language Processing task which deals with classifying subjective information from the e-resources like blogs, reviews, forums etc. in the form text, image or, video to analyze the class or type of that resource. The purpose is to develop classification model for low-resource language like Tamil, Kannada, Malayalam with use KL-NF and BERT for finding offensiveness in a sentence. A language-independent model was developed capable of operating with any language and datasets of varying sizes. Training and testing utilized datasets from three languages. Following training, the model underwent testing, with results compared between KL-NF and BERT approaches. The accuracy comparison revealed that the BERT model outperformed KL-NF in multilingual classification of offensiveness, achieving an accuracy of 82.7% compared to KL-NF's 82.5%. This suggests that the BERT model offers superior predictive accuracy for multilingual classification compared to KL-NF. © 2024 IEEE.

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Findings from shared tasks on hate speech detection: Performance patterns for low-resource languages

Item Type Journal Article

Author K. Ghosh

Author S. Saha

Author T. Mandl

Author S. Modha

Abstract In the digital era, social media has emerged as a powerful channel for expressing opinions, but online platforms have also become a breeding ground for hate speech targeting individuals based on color, caste, gender, sexual orientation, and political ideologies. Despite growing interest in automatic hate speech detection, existing research remains predominantly focused on English, underscoring a critical need to extend efforts to under-resourced languages. To bridge this gap, the HASOC (Hate Speech and Offensive Content Identification) shared task has been promoting multilingual hate speech research. In this paper, we present a brief overview of these four shared tasks (Assamese, Bengali, Bodo and English), datasets, participating systems, and their performance across standard evaluation metrics—precision, recall, accuracy, and macro F1 score. In addition, we analyze the inter-system agreement using Cohen's κ and Fleiss' κ , and investigate item-level difficulty through hardness analyses. Our findings offer valuable insights into the challenges and progress in multilingual hate speech detection, particularly for low-resource languages. This paper also serves as a model for the analysis of other results of large-scale experimentation with text classification systems. © 2025 The Authors.

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Fine tuning large language models for hate speech detection in Hinglish and code mixed custom dataset through a socially responsible approach for safer digital platforms

Item Type Journal Article**Author** B.S. Rathore**Author** S. Chaurasia

Abstract This paper presents an extensive review and experimental framework for hate speech detection using large language models (LLMs). We explored traditional machine learning approaches and illustrated how state-of-the-art LLMs, such as BERT variants (DistilBERT and ModernBERT) and recent developments, such as DeepSeek R1, Llama 3, Gemma 3, Mistral, and Phi 3.5, enhance text classification performance. In this study, we evaluated the effectiveness of various large language models (LLMs) for Hindi–English code-mixed hate speech detection. We constructed a custom dataset using Reddit posts on social media, which were auto-labeled and subsequently verified and validated by expert annotators. LoRa adapters were used for model fine-tuning and in-domain performance evaluation of the model, with precision, recall, F1-score, and accuracy as metrics. To ensure a fair comparison, all the models were trained for the same number of epochs under the same training conditions. According to our findings, DeepSeek R1 outperformed larger general-purpose LLMs, such as Llama 3 and Gemma 3, with smaller margins. DeepSeek R1 surpassed the other models with the highest accuracy of 79 percent, demonstrating a better understanding of the complexity of code-mixed hate speech detection. These results indicate that lightweight, responsibly fine-tuned LLMs can strengthen moderation for multilingual, code-mixed communities without prohibitive computing costs. In practice, the approach offers a clear path towards safer, more inclusive platforms by pairing accuracy with bias-aware development and human oversight. © The Author(s) 2025.

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Fine-Grained Cultural Perception and Evaluation of Beijing's Capital Culture Integrating Large Language Models with Higher-Order Tensor Decomposition

Item Type Journal Article

Author S. Xi

Author Z. Ou

Author B. Meng

Author X. Li

Abstract Fine-grained urban cultural perception is critical for GIScience, yet traditional social media studies struggle with complex cultural semantics and heterogeneous factor integration. Addressing Beijing's "Capital Culture," this study couples LLM agents with higher-order tensor decomposition. Using 2019 full-sample geotagged Sina Weibo data, we developed a four-agent collaborative architecture with Chain-of-Thought prompting and human-in-the-loop mechanisms via a locally deployed Qwen3-32B model. A four-way tensor ("Cultural Type–Evaluation Aspect–Sentiment Polarity–Spatial Carrier") was constructed and integrated with kernel density estimation to characterize spatial differentiation. We address three questions: whether LLMs can reliably classify fine-grained cultural perceptions, how cultural types

associate with evaluation dimensions, sentiments, and spatial carriers, and whether tensor decomposition reveals latent patterns beyond marginal frequencies. The agentic workflow achieves over 90% accuracy in cultural and sentiment classification, and the tensor decomposition attains a 94.87% goodness-of-fit, successfully identifying latent patterns. Spatially, Beijing's capital culture exhibits an unbalanced hierarchical structure—"high coupling in the core area with differentiated expansion at the periphery." This study validates the transition from "data-driven" to "AI + data dual-driven" spatial analysis, providing a quantifiable pathway for LLM-supported urban cultural governance. © 2026 by the authors.

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Fine-Grained Emotion Classification in Reddit Mental Health Posts Using LLMs

Item Type Conference Paper

Author V.P. Orjuela-Llanos

Author I. Galpin

Editor Florez H.

Editor Peluffo-Ordoñez D.

Abstract This study evaluates the discourse surrounding mental health on Reddit through the lens of AI-based emotion analysis. Drawing from a comprehensive dataset comprising 7,899 posts and 54,754 comments on 10 mental health related conditions, this study combines NLP techniques such as BERTopic and lexical analysis, and the usage of GPT-4o-mini to classify emotions. The results point to clear emotional differences between posts and comments: while titles often express urgency (e.g., ‘help’), comments show empathy (e.g., ‘love’). GPT-4o-mini matched human emotion annotations with an accuracy of 52.6%, performing better with positive emotions (58%). Thematic analysis revealed that clinical diagnoses were dominant in the posts, but the comments contained supportive dialogue. This work illustrated that LLMs have potential as an avenue for monitoring mental health and well-being, while also indicating potential drawbacks with respect to ambiguous emotion classification. The study also offers practical information for the development of AI tools in the mental health space and outlines directions for future work. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Fine-grained Sexism Detection in Italian Newspapers

Item Type Conference Paper

Author F. Manzi

Author L. Weber-Genzel

Author B. Plank

Editor Dell'Orletta F.

Editor Lenci A.

Editor Montemagni S.

Editor Sprugnoli R.

Abstract In recent years, tasks revolving around hate speech detection have experienced a growing interest in the field of Natural Language Processing. Two main trends stand out in the context of sexism recognition: the focus on overt forms of sexism such as misogyny on social media and tackling the problem as a text classification task. The main objective of this work is to introduce a new approach to tackle sexism recognition as a sequence labelling task, operating on the token level rather than the document level. To achieve this goal, we introduce (i) the FGSDI (Fine-Grained Sexism Detection in Italian) corpus, containing Italian newspaper articles annotated with fine-grained linguistic markers of sexism, and (ii) a two-step pipeline that sequentially performs sexism detection on the sentence level and sexism classification on the token one. Our primary findings include that (i) tackling the task of sexism recognition as a sequence labelling task is possible, however, a large amount of labelled data is needed; (ii) leveraging few-shot learning for sexism detection proves to be an effective solution in scenarios where only a limited amount of data is available; (iii) the proposed pipeline approach allows for better results compared to the baseline by doubling the overall precision and achieving a better F1-score. © 2024 CEUR-WS. All rights reserved.

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Notes:

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Export Date: 20 September 2026; Cited By: 1; Conference name: 10th Italian Conference on Computational Linguistics, CLiC-it 2024; Conference date: 4 December 2024 through 6 December 2024; Conference code: 205363

Fine-Tuned Models for Hate Speech Detection: Assessing Generalization on Social Media

Item Type	Conference Paper
Author	S.-H. Wu
Author	T.T. Hsun
Abstract	With the rapid development of social media, the automatic detection of offensive language has become a crucial research topic in the field of Natural Language Processing (NLP). To address the challenges posed by diverse corpora and contextual variations in real-world data, this study proposes a classification system that integrates RoBERTa and Large Language Models (LLMs) for detecting offensive language on social media, with a particular focus on evaluating model performance and transferability in cross-dataset settings. Three social media corpora with distinct sources and annotation styles-Jigsaw Toxic Comment Classification Challenge, hate-speech-dataset-master, and Hate Speech and Offensive Language Dataset-are used for cross-training and evaluation. RoBERTa is fine-tuned using supervised learning on each dataset and tested on the other two to assess generalization across domains. Additionally, zero-shot classification is conducted using LLMs with prompt engineering to simulate inference without labeled data. To support large-scale experimentation, the system is equipped with high-stability mechanisms including auto-saving, resume-on-interruption, and timeout handling. The experimental results show that RoBERTa performs best when training and testing data are aligned, but experiences degradation in accuracy and F1-score in cross-dataset evaluations. In contrast, LLMs, despite no finetuning, maintain stable recall across datasets and achieve the highest F1-score of 0.9328 on dataset C, indicating strong contextual transferability and corpus adaptability. This research systematically examines model adaptability to varying linguistic environments through multi-corpus experiments, offering analytical insights that inform the design of offensive content detection systems and dataset selection in real-world applications. © 2025 IEEE.
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Extra	Journal Abbreviation: Proc. - IEEE Int. Conf. Inf. Reuse Integr. Data Sci., IRI
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Fine-Tuning Komodo Model with LoRA for Sentiment Analysis of YouTube Comments

Item Type Conference Paper

Author V.C. Mawardi

Author E.T. Alexander

Author D.J. Surjawan

Abstract The rapid development of digital technology has made online platforms such as YouTube increasingly significant in the education sector, serving not only as a medium for disseminating knowledge but also as an interactive space for exchanging opinions through the comment section. With more than 2.1 billion active users worldwide, YouTube comments provide a vast source of information about users' emotional responses and opinions, making them highly valuable for sentiment analysis. This research introduces a web application built with Streamlit, specifically developed to conduct sentiment analysis on YouTube comments. The system incorporates several key processes, including data scraping from YouTube using the API, data preprocessing, and sentiment classification. A Large Language Model, Komodo-7B, was employed and finetuned using a dataset comprising {2, 0 0 0} manually categorized comments into positive, neutral, and negative sentiments. The fine-tuning process utilized the Low-Rank Adaptation (LoRA) method, which effectively improves memory efficiency while maintaining high accuracy in classification tasks. The dataset was divided into training and validation sets with an 80: 20 ratio to ensure robust model evaluation. The model demonstrated a consistent improvement in accuracy

during training, achieving up to 91% accuracy by the third checkpoint, highlighting its potential for practical sentiment analysis applications on social media content. © 2025 IEEE.

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Sentiment analysis, Social networking (online), Large datasets, Fine tuning, Digital technologies, sentiment analysis, YouTube, Online platforms, Youtube comment, Education sectors, komodo-7b, Komodo-7b, lora, Lora, streamlit, Streamlit, youtube comments

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Fine-Tuning Multilingual Transformers for Multi-Label Discrimination Detection in Bangla Social Media Text

Item Type Conference Paper

Author F.M. Bhuiyan

Author M.F. Hossain

Author Md.S.H. Rizon

Author S. Farha

Abstract The increase in social media has been fuelling the proliferation of discriminatory speech against people on the grounds of their political affiliations, ethnicity, religion, gender, class and other characteristics. While there have been a lot of studies on hate speech detection in high-resource languages, there is very little research in Bangla, especially for multi-label discrimination detection. In this paper, we propose a transformer-based language classification model for Bangla and code-mixed Bangla-English text classifications into ten discrimination

categories based on the recently released BOISHOMMO dataset. We compare traditional TF-IDF with the Linear models against XLM-RoBERTa fine-tuned transformer. Experimental results indicate that the transformer performs significantly better than the baselines, which attain test micro-F1 of 0.483 and macro-F1 of 0.424 compared to being less than 0.41 for conventional methods. Per-label classification shows good performance on Political (F1 = 0.686) and Ethnicity (F1 = 0.596), but still the most challenging categories are low-resource categories such as Disability and Physical appearance. These results help highlight the potential of multilingual transformers while the importance of balanced data and data augmentation for the development of Bangla hate speech detection. © 2025 IEEE.

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Fine-Tuning SinLlama for Sentiment Analysis in Romanized Sinhala-English Code-Mixed Social Media Text

Item Type Conference Paper

Author V.S. Abeynayake

Author S.A.S. Lorensuhewa

Author M.A.L. Kalyani

Abstract Romanized Sinhala-English Code-mixed content dominates on Sri Lankan social media platforms. In this low-resource environment, traditional Machine Learning (ML) models have shown efficacy and robustness for NLP tasks like sentiment analysis. But the potential of current Large Language Models (LLMs) is still largely underexplored. In this study, we explore the performance of SinLlama, a Llama-3-8B model further pre-trained on a large Sinhala corpus, for sentiment analysis on Romanized Sinhala-English text. We evaluate the model performance against traditional Machine Learning baselines. Our experimental results show that the fine-tuned SinLlama model achieves an accuracy of 74.2% outperforming traditional ML baselines and demonstrating superior capability in understanding the semantic nuances of Romanized code-mixed text. © 2026 IEEE.

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FinHybridSA: A sentiment detection model for financial social media text used for financial decision-making

Item Type Conference Paper

Author Z. Yu

Abstract Sentiment analysis in financial texts remains challenging due to the prevalence of domain-specific language patterns on social media. To address these challenges, this paper proposes a hybrid semantic-symbolic framework, FinHybridSA, which combines contextual embeddings with interpretable domain features. The system consists of two modules: an intensity recognition module and an uncertainty detection module. These features are fused with the semantic representations of a large language model (BERT) and fed into a multilayer perceptron classifier. Experimental results show that FinHybridSA maintains superior performance with limited training data compared to benchmark models, while also exhibiting higher interpretability and robustness. Ablation experiments confirm that symbolic features can capture subtle emotional phenomena in financial social texts, thereby continuously improving classification accuracy. Our research demonstrates the effectiveness of hybrid representation learning as a resource-efficient alternative for sentiment analysis in financial texts. © 2026 Copyright held by the owner/author(s).

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Z. Yu; Faculty of Data Science, New York University Shanghai, Shanghai, China; email: 1964543937@qq.com; Conference name: 2026 International Conference on Artificial Intelligence and Fintech, IC-AIF 2026; Conference date: 9 January 2026 through 11 January 2026; Conference code: 223002

FinTNet: From Tweets to Trades

Item Type Conference Paper

Author D. Tavakoli

Author S. Haratizadeh

Abstract Recent research indicates a correlation between textual information, media sources, and stock price movements. Additionally, studies have highlighted the influence of inter-stock relationships on their respective behaviors. In this study, FinTNet is introduced as a novel graph neural network model that integrates financial text data, price information, and stock relationships for enhanced stock price prediction. Three text-based graphs, derived from Twitter text analysis and the exploration of stock relationships, are presented for prediction using a semi-supervised graph convolutional model. The predictive model concurrently incorporates textual data, stock prices, and indicators. For in-depth tweet content analysis, Large Language Models (LLMs), including Financial Bidirectional Encoder Representations from Transformers (FinBERT) and Gemini Pro, are employed. FinBERT transforms tweets into embedding vectors, yielding a dataset of approximately three million tweets with vectors of size 768. Additionally, about 85,000 are getting four different labels (trend direction, discussed time, amount of change in price, and sentiment), using the Gemini Pro model, forming an accessible labeled dataset. Experiments demonstrate that FinTNet achieves superior accuracy compared to baseline models by leveraging LLM-extracted textual features within carefully designed graph structures. © 2025 IEEE.

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DOI 10.1109/ICCCKE68588.2025.11273867

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: D. Tavakoli; University of Tehran, College of Interdisciplinary Science and Technology, Tehran, Iran; email: dorsa.tavakoli@ut.ac.ir; Conference name: 15th International Conference on Computer and Knowledge Engineering, ICCKE 2025; Conference date: 28 October 2025 through 29 October 2025; Conference code: 217882

FIRE 2025 - Proceedings of the 17th Annual Meeting of the Forum for Information Retrieval Evaluation

Item Type Journal Article

Editor Gangopadhyay S.

Editor Ganguly D.

Editor Sanyal D.K.

Abstract The proceedings contain 25 papers. The topics discussed include: towards ayurveda information retrieval – challenges and opportunities; from climate headlines to stock trends: a sentiment-aware financial model; automating exam evaluation using generative AI (GenAI); novel zero-shot framework for Sanskrit-English cross-lingual information retrieval; soft prompt improves direct search responses; cross-lingual transfer vs. minimal fine-tuning for Indic question answering: an empirical study on the Indic-QA benchmark; MDA GenUro: semantically inclined medical document annotation for genito-urinary and urological pathology; EventTempex: a tool for interpreting event-anchored temporal expressions; and prosodic features as universal ranking signals: cross-lingual speech summarization in Indic languages.

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Publisher Association for Computing Machinery, Inc

Publication FIRE 2025 - Proceedings of the 17th Annual Meeting of the Forum for Information Retrieval Evaluation

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FMSA: Few-Shot Multimodal Sentiment Analysis for Social Media via Integrated Prompt Learning and Vision-Language Models

Item Type Journal Article

Author X. Zhu

Author H. Feng

Author E. Cambria

Author X. Yu

Author J. Santamaria

Author X. Fan

Author R. Wang

Abstract Multimodal sentiment analysis (MSA) on social media is increasingly critical for understanding complex emotional expressions, yet it faces significant challenges in data-scarce environments where annotated multimodal content is limited. Here, we present FMSA, a novel framework that integrates prompt-based learning with advanced vision-language models to enable robust few-shot MSA. Our approach leverages an instruction-aware Query Transformer (Q-Former) to dynamically extract and align visual features with task-specific textual prompts, enhancing cross-modal fusion. We introduce a distributed consistency sampling strategy to construct representative few-shot datasets, ensuring statistical diversity under constrained conditions. Evaluated across six benchmark social media datasets - including MVSA-S, MVSA-M, and Twitter-Depression - FMSA outperforms state-of-the-art methods, achieving an accuracy of 63.47% and an F1 score of 57.34% on MVSA-S with full data, and 61.25% accuracy with 56.06% F1 in few-shot settings using just 1% of the data. By fine-tuning lightweight components while preserving pretrained model robustness, FMSA mitigates overfitting and delivers generalizable performance. We also release a curated few-shot dataset as a community resource. This framework advances MSA by offering an efficient, scalable solution for interpreting multimodal emotions in low-resource scenarios. © 2010-2012 IEEE.

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Food Neologisms and Word Formation Trends Identified on Social Media Posts Using LLMs for Hashtag Collection

Item Type Conference Paper

Author M. Panagiotou

Author K. Gkatzionis

Author E. Kaloudis

Editor Vezzani F.

Editor Anic A.O.

Editor Salgado A.

Editor Tallarico G.

Abstract Social media platforms can provide researchers with massive authentic data. In consumer studies, social media platforms have been recently used for linguistic and cultural data collection to gain insight into consumers' attitudes, responses, and expectations. The use of modern computational tools, such as Large Language Models and Generative Artificial Intelligence (GenAI) can automate the processes of data analysis, thus saving time and effort. The present study aims

at investigating into traditional and local food consumption (case studies: snail dishes, and cheeses of the North-Aegean Sea), and at identifying key concepts and the relevant specific words used by Greek consumers. Instagram posts on the cases under study were collected, and using GenAI applications, sentiment analysis was performed on the posts to identify positively related concepts. The hashtags collected revealed patterns in word formation on social media. GenAI applications were used in an attempt to automate analysis tasks. Hashtag coappearances in posts also revealed key concepts, food trends, as well as conceptual networks. The methodology can be transferred to a) specialized lexicography (e.g. to compile domain-specific word lists), b) linguistic and cultural studies (e.g. to study word formation patterns and conceptual networks in linguistic/cultural studies and in comparative studies), and c) terminology (e.g. to identify neologisms and check term usage over time). © 2025 Copyright for this paper by its authors.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: E. Kaloudis; Computer Simulation, Genomics and Data Analysis Laboratory, Department of Food Science and Nutrition, University of the Aegean, Myrina, Metropolitoe Ioakeim 2, Lemnos, 81400, Greece; email: stathiskaloudis@aegean.gr; Conference name: 1st International Workshop on Terminological Neologism Management, NeoTerm 2025; Conference date: 18 June 2025 through 18 June 2025; Conference code: 209435

Forecasting secondary risks in emergencies: A transferable early warning framework using event logic graphs from social media

Item Type Journal Article

Author S. Li

Author C. Tian

Author X. Du

Author Y. Li

Abstract Social media has become a critical arena for public opinion during emergencies, where the rapid evolution of events can generate unforeseen secondary risks. This study develops and validates a transferable early warning framework that applies Event Logic Graphs (ELGs) to move beyond retrospective analysis, toward proactive risk forecasting. First, this study proposes a novel hybrid Rule-Large Language Model framework to extract events and multiple types of logical relations from noisy social media data. Second, a comparative analysis is conducted by constructing separate ELGs for each event: a natural disaster and public health crisis. Finally, regarding the historical ELG as the knowledge base, the framework incorporates the event chain similarity calculations with context-aware filtering and sentiment enhanced re-ranking mechanism. Specifically, this mechanism ensures prediction validity by filtering context mismatched risks and capturing verified public sentiment to prioritize potential threats. The results present that the hybrid framework significantly outperforms the pure rule-based model in logic triples extraction. The analysis reveals that emergencies evolve into two different structural prototypes: natural disaster is manifested as “linear cascade” models, and public health crisis shows “cyclic feedback loop” characteristics, which provides a new computational perspective for agenda setting theory. In addition, the cross-event secondary risk prediction framework achieves satisfactory predictive performance across Top-3, 5, and 10 cutoffs, demonstrating that integrating context constraints and, more critically, credibility weighted sentiment analysis, achieves the most substantial improvement in identifying high-priority risks. This research contributes to the literature by proposing a robust method for building ELG with multiple logical relationships, helping emergency management shift from passive monitoring to active risk prevention and control. © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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From Ancient Manuscripts to Modern Social Media: Evolution of Tonality Analysis Methods for Low-Resource Languages

Item Type Journal Article

Author Z. Baishemirov

Author A. Kassymbayev

Author D. Yedilkhan

Author B. Amirgaliyev

Author B. Abdikenov

Abstract Recently, computational sentiment analysis has become an essential tool for detecting evaluative language in large text collections. However, its application to many low-resource language families and historical corpora remains largely unexplored. This paper reviews the evolution of sentiment analysis methods in the Turkic language family, with a particular focus on Chagatai, the classical predecessor of several modern Turkic languages. We outline the methods that have evolved since the advent of lexicon-based and rule-based approaches up to the present day with large language models, addressing longstanding problems in agglutinative morphology, data scarcity, orthographic instability, and multilingual lexical mixing. To examine the available options, we conducted a pilot experiment using multilingual models in a zero-shot setting on a curated Chagatai corpus. In the absence of ground-truth annotations, prediction stability was validated with ensemble consistency and inter-model agreement. The results show real promise but also distinct limitations when adapting traditional NLP technologies for historically remote, low-resource languages. Progress in the field will require cross-disciplinary work, systematic diachronic dataset deployment, and a nuanced adaptation of multilingual representation learning to handle linguistically rich, low-resource settings. © 2026 by the authors.

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From Annotation to Detection: Evaluating LLMs for Hate Speech and Stereotype Identification in Italian

Item Type Conference Paper

Author F. Mohammadi

Author S. Maghool

Author P. Ceravolo

Abstract Manual data annotation is often slow, expensive, and difficult to scale-especially for tasks that are subjective and context-dependent, like detecting hate speech and stereotypes. With recent progress in Large Language Models (LLMs), there is growing potential to automate this process. In this study, first, we explore the use of a committee of LLMs (GPT-4omini, Gemini-1.5-flash, and DeepSeek-R1) to generate annotations for Italian social media content. Then, the quality of LLM-generated labels is evaluated by following a teacher-student approach, where we trained a smaller student model (phi3.5-miniinstruct) on them and testing its performance against a humanlabeled dataset. The results indicate that fine-tuning the student model over LLM-generated labels, with careful dataset balancing and

hyperparameter tuning, can yield a significant improvement over the baseline (approximately 17% improvement), suggesting that committee-based LLM annotation can provide high-quality labels. Therefore, this work shows the proposed approach can be a reliable and scalable alternative to manual labeling by properly addressing challenges like class imbalance to be sure about the fairness and accuracy of the fine-tuned models. © 2025 IEEE.

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From Bag-of-Words to Transformers: A Comparative Study for Text Classification in Healthcare Discussions in Social Media

Item Type Journal Article

Author E. De Santis

Author A. Martino

Author F. Ronci

Author A. Rizzi

Abstract One notable paradigm shift in Natural Language Processing has been the introduction of Transformers, revolutionizing language modeling as Convolutional Neural Networks did for Computer Vision. The power of Transformers, along with many other innovative features, also lies in the integration of word embedding techniques, traditionally used to represent words in a text and to build classification systems directly. This study delves into the comparison of text representation techniques for classifying users who generate medical topic posts on Facebook discussion groups. Short and noisy social media texts in Italian pose challenges for user categorization. The study employs two datasets, one for word embedding model estimation and another comprising discussions from users. The main objective is to achieve optimal user categorization through different pre-processing and embedding techniques, aiming at high generalization performance despite class imbalance. The paper has a dual purpose, i.e., to build an effective classifier, ensuring accurate information dissemination in medical discussions and combating fake news, and to explore also the representational capabilities of various LLMs, especially concerning BERT, Mistral and GPT-4. The latter is investigated using the in-context learning approach. Finally, data visualization tools are used to evaluate the semantic embeddings with respect to the achieved performance. This investigation, focusing on classification performance, compares the classic BERT and several hybrid versions (i.e., employing different training strategies and approximate Support Vector Machines in the classification layer) against LLMs and several Bag-of-Words based embedding (notably, one of the earliest approaches in text classification). This research offers insights into the latest developments in language modeling, advancing in the field of text representation and its practical application for user classification within medical discussions. © 2017 IEEE.

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From Binary to Multi-Class: LLM-Judged Synthetic Annotation Applied to Hate Speech Detection

Item Type Journal Article**Author** A. Moreno-Cediel**Author** A. Garcia-Cabot**Author** E. Garcia-Lopez

Abstract The increasing prevalence of hate speech on social media platforms has spurred research aimed at mitigating this societal harm. However, the development of effective machine learning solutions is hindered by a lack of labelled hate speech data in languages beyond English, particularly when attempting granular, multi-class classification. This research aims to address this data scarcity by introducing a novel methodology leveraging the ‘Large Language Model as a judge’ paradigm to transform existing binary-labelled hate speech data into multi-class datasets. Our approach aims to generate balanced datasets and enables classification across seven identity groups: race, religion, origin, gender, sexuality, age, and disability. The methodology has been applied to the Spanish Hate Speech Superset, and it has been validated using the Measuring Hate Speech dataset, demonstrating significant efficacy and broad applicability. Specifically, our approach obtains a higher match rate with human labels and a lower number of mismatches when compared with prompt-only strategies. In addition, Cohen’s Kappa scores demonstrate that our approach exhibits a moderate strength of agreement with human annotators, outperforming prompt-only strategies’ scores by 5%. As a result of the application of the proposed strategy to the Spanish Hate Speech Superset dataset, a multi-class version is obtained, comprising 6325 hateful samples classified among the seven identity groups. The proposed strategy offers a versatile solution for nuanced classification tasks beyond hate speech, providing a valuable technique for detailed categorisation in various domains. Copyright © 2026 The Authors.

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From Comments to Insights: Comparing Local and Online LLM Models in Social Media Sentiment Evaluation

Item Type Conference Paper
Author R.A. Suindikovna
Author N.D. Gennadievych
Author A.D. Sadyrbekkyzy
Abstract Social media platforms such as YouTube, TikTok, and Reddit generate vast amounts of user-generated content, making sentiment analysis essential for understanding public opinion. Traditional sentiment analysis methods often struggle with sarcasm, informal language, and evolving trends, leading to the rise of Large Language Models (LLMs) as more effective alternatives. However, the choice between cloud-based LLMs (e.g., ChatGPT-4o, Gemini 2.0 Flash) and local LLMs (e.g., Meta LLaMA 3.2, DeepSeek-R1 7b) remains an open question, with each approach offering distinct advantages and limitations. This study compares the performance of online and local LLMs in sentiment analysis, focusing on discussions about AI-generated art and cancel culture. Through structured sentiment classification and comparative evaluation, we assess each model's accuracy, efficiency, and interpretative depth. Our findings indicate that cloud-based models excel in contextual comprehension, while local models provide enhanced privacy and structured data processing. The results highlight the trade-offs between computational power, data security, and adaptability, reinforcing the potential of hybrid approaches that integrate both cloud and local LLM capabilities for more effective sentiment analysis. The goal of this research is to provide a clear evaluation of different LLMs in sentiment analysis, offering practical insights into their strengths and weaknesses. By comparing models from multiple developers, we aim to determine the most effective deployment strategies for real-world applications, ensuring accurate and scalable sentiment evaluation in diverse online discussions. © 2025 IEEE.
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From human annotation to automation: LLM-in-the-loop active learning for Arabic sentiment analysis

Item Type Journal Article

Author D. Refai

Author A. Dalaq

Author D. Dalaq

Author I. Ahmad

Abstract Natural language processing (NLP), especially sentiment analysis, plays a vital role in marketing, customer service, and social media monitoring by providing insights into user opinions and emotions. However, progress in Arabic sentiment analysis remains limited due to the lack of large, high-quality labeled datasets. While active learning has proven effective in reducing annotation efforts in other languages, few studies have explored it in Arabic tasks. Likewise, the use of large language models

(LLMs) for assisting annotation and benchmarking to human labeling remains unexplored. In this paper, we propose an active learning framework for Arabic sentiment analysis to reduce annotation costs while maintaining high performance. We evaluate recurrent neural networks (RNN), long short-term memory (LSTM), and gated recurrent units (GRU), together with transformer-based Arabic language models AraBERT, MARBERT, and AraELECTRA, as non-active-learning baselines across three datasets: Hunger Station, AJGT, and MSAC. Two annotation strategies are compared: Human labeling and LLM-assisted labeling. Five LLMs are evaluated as annotators: GPT-4o, Claude 3.7 Sonnet, Gemini 2.5 Pro, DeepSeek Chat, and LLaMA 3 70B Instruct. Our results show that LLM-assisted active learning achieves performance comparable to human labeling while significantly reducing annotation requirements. Across the evaluated datasets, comparable performance is achieved using approximately 20% to 55% of the labeled training data, highlighting the effectiveness of LLM-assisted annotation in reducing labeling effort. Overall, active learning combined with LLM-based labeling offers a scalable and cost-effective solution for sentiment analysis in low-resource languages such as Arabic. © The Author(s), under exclusive licence to Springer-Verlag London Ltd., part of Springer Nature 2026.

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From Language Identification to Topic Modelling: A Fully Integrated Pipeline for Multilingual Aspect-Based Sentiment Analysis and Component-Level Insights

Item Type Journal Article

Author A. Şeker

Abstract The rapid expansion of multilingual digital platforms has made the accurate analysis of user-generated content across different languages and cultural contexts increasingly essential. However, existing methods struggle to maintain consistent performance due to linguistic diversity, morphological complexity, and structural variations in text. Many studies in the literature analysis stages as isolated components, which causes errors in early stages to propagate and negatively affect overall performance. To address these challenges, this study proposes an integrated and multilingual aspect-based sentiment analysis pipeline that encompasses language identification, machine translation, sentiment classification, and topic modelling. The proposed approach evaluates a comprehensive range of models, from statistical methods to transformer-based architectures and Large Language Models, using the M-ABSA and MARC datasets which comprise over one million reviews across 21 languages. The analyses not only assess the final model performance but also examine the relative contributions, limitations, and error-propagation effects of each component in the pipeline. The findings quantitatively reveal how linguistic diversity, noise, and contextual variability in real-world data influence analysis processes, while systematically comparing the behaviour of generative models and traditional approaches under such challenging conditions. Overall, the study underscores the necessity of an end-to-end, integrated analysis pipeline over fragmented solutions and provides a comprehensive methodological and practical contribution to the field of multilingual text processing. © 2026 The Author(s). Expert Systems published by John Wiley & Sons Ltd.

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From Lexicons to Large Language Models: A Comprehensive Survey of Sentiment Analysis Methods, Benchmarks, and Emerging Frontiers

Item Type Journal Article

Author S. De

Author A. Gosai

Author K. Thankachan

Author R.A. ZeinEldin

Author A.T. Almaktoom

Author M. Bayram

Author A.W. Mohamed

Abstract Sentiment analysis (SA) has evolved from a niche text-classification task into a central problem in natural language processing, spanning multiple domains, modalities, and languages. This survey provides a comprehensive review of sentiment analysis methods from their origins in lexicon-based approaches through classical machine learning, deep learning architectures, pre-trained transformers, and the current era of large language models (LLMs). We formalize the SA problem across multiple granularity levels (document, sentence, and aspect) and present a taxonomy that encompasses classification, regression, aspect-based sentiment analysis (ABSA), emotion detection, and stance detection tasks across diverse domains including movie reviews, product reviews, healthcare, finance, and social media. We review benchmark datasets spanning text-only corpora (IMDb, SST, SemEval series), multimodal benchmarks (CMU-MOSI, CMU-MOSEI, MELD), and domain-specific evaluation suites such as SentiEval. The methodological evolution is traced from VADER and SentiWordNet, through SVM and Naïve Bayes classifiers, CNN and LSTM architectures, BERT and its variants, to modern LLMs including GPT-4, Llama 3, and ModernBERT, with technical details of key architectures and their mathematical formulations. We provide dedicated analyses of chain-of-thought reasoning for implicit sentiment, multimodal fusion strategies, cross-lingual transfer methods, sarcasm and irony detection, explainability through SHAP and LIME, and

the emerging challenge of AI-generated fake reviews. A comparative analysis across paradigms reveals that while LLMs achieve strong zero-shot performance, fine-tuned smaller models remain competitive on standard benchmarks, a finding with significant implications for deployment efficiency. We identify persistent open challenges including domain drift, cultural bias, and the model variability problem, and outline future research directions encompassing reasoning-augmented SA, agentic workflows, federated learning, and real-time edge deployment. With coverage of over 130 references spanning two decades of research and 29 new references from 2024 and 2025, this survey provides a unified roadmap for both newcomers and researchers at the frontier of sentiment analysis. Copyright © 2026 The Authors. Published by Tech Science Press.

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From manual to AI: A systematic review of methodological evolution in social media content analysis

Item Type Conference Paper

Author D. Alsmadi

Author V. Prybutok

Abstract This systematic review examines methodological approaches to social media content analysis through analysis of 332 SCOPUS-indexed publications. We address three research questions: identifying predominant methodologies, tracing their temporal evolution, and evaluating AI's role in current and future research. Through systematic coding of abstracts, we classified methodologies into five categories: Manual/Qualitative, Statistical/Rule-Based, Machine Learning, Deep Learning, and AI/Foundation Models. Results show Manual/Qualitative approaches remain most prevalent (32.5%), followed by Machine Learning (18.7%), Deep Learning (7.5%), AI/Foundation Models (6.3%), and Statistical/Rule-Based methods (5.4%). Temporal analysis reveals progressive integration of computational approaches, with recent years showing increased adoption of transformer-based models, including BERT and large language models. The analysis of future research directions identified 59 papers (17.8%) that explicitly mentioned future work, with 91.5% proposing AI-enhanced methodologies. AI-related terms dominated future work discussions, with 135 general AI mentions, 29 machine learning references, and 13 LLM mentions. These findings indicate strong momentum toward AI integration while traditional qualitative methods maintain importance, suggesting optimal approaches combine computational efficiency with human interpretive capabilities for comprehensive social media content analysis. © 2026 IEEE.

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From Narratives to Diagnosis: A Machine Learning Framework for Classifying Sleep Disorders in Aging Populations: The sleepCare Platform

Item Type Journal Article**Author** C.A. Frantzidis

Abstract Background/Objectives: Sleep disorders are prevalent among aging populations and are often linked to cognitive decline, chronic conditions, and reduced quality of life. Traditional diagnostic methods, such as polysomnography, are resource-intensive and limited in accessibility. Meanwhile, individuals frequently describe their sleep experiences through unstructured narratives in clinical notes, online forums, and telehealth platforms. This study proposes a machine learning pipeline (sleepCare) that classifies sleep-related narratives into clinically meaningful categories, including stress-related, neurodegenerative, and breathing-related disorders. The proposed framework employs natural language processing (NLP) and machine learning techniques to support remote applications and real-time patient monitoring, offering a scalable solution for the early identification of sleep disturbances. Methods: The sleepCare consists of a three-tiered classification pipeline to analyze narrative sleep reports. First, a baseline model used a Multinomial Naïve Bayes classifier with n-gram features from a Bag-of-Words representation. Next, a Support Vector Machine (SVM) was trained on GloVe-based word embeddings to capture semantic context. Finally, a transformer-based model (BERT) was fine-tuned to extract contextual embeddings, using the [CLS] token as input for SVM classification. Each model was evaluated using stratified train-test splits and 10-fold cross-validation. Hyperparameter tuning via GridSearchCV optimized performance. The dataset contained 475 labeled sleep narratives, classified into five etiological categories relevant for clinical interpretation. Results: The transformer-based model utilizing BERT embeddings and an optimized Support Vector Machine classifier achieved an overall accuracy of 81% on the test set. Class-wise F1-scores ranged from 0.72 to 0.91, with the highest performance observed in classifying normal or improved sleep (F1 = 0.91). The macro average F1-score was 0.78, indicating balanced performance across all categories. GridSearchCV identified the optimal SVM parameters (C = 4, kernel = 'rbf', gamma = 0.01, degree = 2, class_weight = 'balanced'). The confusion matrix revealed robust classification with limited misclassifications, particularly between overlapping symptom categories such as stress-related and neurodegenerative sleep disturbances. Conclusions: Unlike generic large language model applications, our approach emphasizes the personalized identification of sleep symptomatology through targeted classification of the narrative input. By integrating structured learning with contextual embeddings, the framework offers a clinically meaningful, scalable solution for early detection and differentiation of sleep disorders in diverse, real-world, and remote settings. © 2025 by the author.

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Notes:

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Export Date: 20 September 2026; Cited By: 5; Correspondence Address: C.A. Frantzidis; School of Engineering and Physical Sciences, University of Lincoln, Lincoln, LN6 7TS, United Kingdom; email: cfrantzidis@lincoln.ac.uk

From Pixels to Concepts: How AI Interprets Visual Concepts in Social Media Videos? A Case Study of Mental Health YouTube Shorts

Item Type Journal Article

Author J.L. Liu

Author Y. Su

Author P. Seth

Abstract Visual information on social media is increasingly central to how people share and seek knowledge across health, political, and everyday domains. As video content proliferates, AI-assisted analysis offers new possibilities for social media research and information retrieval. This study examines the potential of multimodal large language models (MLLMs) to interpret conceptual content in user-generated videos. Using LLaVA-1.6 Mistral 7B, we analyzed 725 keyframes from 142 depression-related YouTube Shorts to evaluate MLLM performance on abstract social concepts. Focusing on four dimensions of video-based self-disclosure, we found that greater prompt detail did not necessarily improve human-AI alignment. Factors such as concept complexity and genre diversity also influenced outcomes. Our findings highlight both the promise and limits of MLLMs in visual content analysis and call for more human-centered evaluation and prompt design strategies when using AI in

multimodal contexts. Annual Meeting of the Association for Information Science & Technology | Nov. 14 – 18, 2025 | Washington, DC, USA.

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From Pre-Training to Meta-Learning: A Journey in Low-Resource-Language Representation Learning

Item Type Journal Article

Author D. Zaikis

Author I. Vlahavas

Abstract Language representation learning is a vital field in Natural Language Processing (NLP) that aims to capture the intricate semantics and contextual information of text. With the advent of deep learning and neural network architectures, representation learning has revolutionized the NLP landscape. However, the majority of research in this field has concentrated on resource-rich languages, putting Low-Resource

Languages (LRL) at a disadvantage due to their limited linguistic resources and the absence of pre-trained models. This paper addresses the significance of language representation learning in a low-resource language, Greek, and its impact on various downstream tasks that heavily rely on semantically and contextually enriched language representations. Accurate classification requires an understanding of nuanced linguistic cues and contextual dependencies. Effective representations bridge the gap between raw text data and classification models, encoding semantic meaning, syntactic structures, and contextual information. By leveraging various representation learning techniques using Transformer-based Language Models (LM), such as domain-adaptation and contrastive learning, we aim to enhance the performance of text classification in this LRL setting. We explore the challenges and opportunities in developing effective representations and propose a multi-stage LM pre-training and meta-learning approach to improve performance in classification downstream tasks. The proposed approach was evaluated on Greek expert-annotated texts from social media posts, news articles, press text clippings and internet articles such as blog posts and opinion pieces. The results show significant improvements in the classification effectiveness of each task in terms of micro-averaged F1-score in sentiment, irony, hate speech, emotion and three custom classification tasks. © 2013 IEEE.

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From Review to Practice: A Comparative Study and Decision-Support Framework for Sentiment Classification Models

Item Type Journal Article

Author K. Walji

Author A. Erraissi

Author A. ZAKRANI

Author M. Banane

Abstract Sentiment classification is a core task in natural language processing (NLP), enabling automated interpretation of opinionated text across domains, such as social media, e-commerce, and healthcare. While numerous models have been proposed—from classical machine learning algorithms to deep neural networks and transformer architectures—their adoption is often hindered by trade-offs in performance, interpretability, and computational cost. This paper presents a threefold contribution: 1) a structured review of over 30 peer-reviewed studies that compare sentiment classifiers across five analytical dimensions—accuracy, robustness, interpretability, efficiency, and context adaptability; 2) a lightweight empirical benchmark on the IMDb dataset, evaluating Naïve Bayes, linear SVM, and LSTM; and 3) a practitioner-oriented decision-support framework comprising a model selection flowchart and recommendation matrix. The experimental results show that SVM achieved the highest F1-score (0.8329), while Naïve Bayes provided strong performance with minimal training time, and LSTM underperformed under constrained conditions. We further highlight persistent challenges in benchmarking consistency, model explainability, and cross-lingual adaptability. The paper concludes with actionable future directions, including hybrid architectures, low-resource deployment strategies, and inclusive NLP systems for diverse user populations. To our knowledge, this is the first study that unifies systematic review, empirical validation, and practical decision tools in the field of sentiment classification. © (2025), (Science and Information Organization). All rights reserved.

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From Traditional Algorithms to Large Language Models: Evolution of AI in Financial Market Prediction

Item Type Conference Paper

Author P. Singh

Author S. Srivastav

Abstract This systematic review examines the evolution and current state of artificial intelligence applications in stock market prediction, analyzing over 100 research papers published between 2018-2025. The review synthesizes major AI approaches including traditional machine learning models, deep learning architectures, natural language processing for sentiment analysis, and emerging large language model integrations. Key findings reveal a clear paradigm shift from purely quantitative models to hybrid systems that combine technical analysis with real-time sentiment extraction from news and social media. Performance analysis across studies shows that multi-modal AI systems achieve significantly better directional accuracy (65-85%) compared to traditional methods (45-60%), with root mean squared errors reduced by 20-40% in short-term predictions. The review identifies three dominant architectural patterns: ensemble methods combining multiple ML algorithms, transformer-based models for sequence prediction, and hybrid systems integrating structured financial data with unstructured text analysis. Critical challenges include data quality issues, model interpretability concerns, computational overhead of real-time processing, and vulnerability to black swan events that break historical patterns. Emerging trends point toward integration of blockchain data, improved transparency mechanisms, and multi-timeframe forecasting capabilities. The review concludes that while AI has substantially enhanced stock prediction accuracy, successful implementation requires careful consideration of data preprocessing, model selection, and real-world deployment constraints. Future research directions emphasize the need for more robust evaluation frameworks, better handling of market volatility, and development of explainable AI systems for financial applications. © 2025 IEEE.

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From Tweets to Stance: An Unsupervised Framework for User Stance Detection on Twitter

Item Type Conference Paper
Author M. Gambini
Author C. Senette
Author T. Fagni
Author M. Tesconi
Editor Bifet A.
Editor Lorena A.C.
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Editor Gama J.

Editor Abreu P.H.

Abstract Current stance inference methods use topic-aligned training data, leaving many unexplored topics due to the lack of training data. Zero-shot approaches utilizing advanced pre-trained Natural Language Inference (NLI) models offer a viable solution when training data is unavailable. This work introduces the Tweets2Stance - T2S framework, an unsupervised stance detection framework based on Zero-Shot Learning. It detects a five-valued user's stance on social-political statements by analyzing their Twitter timeline. The ground-of-truth user's stance is obtained from Voting Advice Applications (VAAs), online tools that compare political preferences with party political stances. The T2S framework's generalization potential is demonstrated by measuring its performance (F1 and MAE scores) across nine datasets. These datasets were built by collecting tweets from competing parties' Twitter accounts in nine political elections held in different countries from 2019 to 2021. Through comprehensive experiments, an optimal setting was identified for each election. The results, in terms of F1 and MAE scores, outperformed all baselines and approached the best scores for each election. This showcases the ability of T2S to generalize across different cultural-political contexts. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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From Twitter to Reasoner: Understand Mobility Travel Modes and Sentiment Using Large Language Models

Item Type Conference Paper

Author K. Ruan

Author X. Wang

Author X. Di

Abstract Social media has become an important platform for people to express their opinions towards transportation services and infrastructure, which holds the potential for researchers to gain a deeper understanding of individuals' travel choices, for transportation operators to improve service quality, and for policymakers to regulate mobility services. A significant challenge, however, lies in the unstructured nature of social media data. In other words, textual data like social media is not labeled, and large-scale manual annotations are cost-prohibitive. In this study, we introduce a novel methodological framework utilizing Large Language Models (LLMs) to infer the mentioned travel modes from social media posts, and reason people's attitudes toward the associated travel mode, without the need for manual annotation. We compare different LLMs along with various prompting engineering methods in light of human assessment and LLM verification. We find that most social media posts manifest negative rather than positive sentiments. We thus identify the contributing factors to these negative posts and, accordingly, propose recommendations to traffic operators and policymakers. © 2024 IEEE.

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Fusion of heterogeneous attention mechanisms in multi-view convolutional neural network for text classification

Item Type Journal Article

Author Y. Liang

Author H. Li

Author B. Guo

Author Z. Yu

Author X. Zheng

Author S. Samtani

Author D.D. Zeng

Abstract The rapid proliferation of user generated content has given rise to large volumes of text corpora. Increasingly, scholars, researchers, and organizations employ text classification to mine novel insights for high-impact applications. Despite their prevalence, conventional text classification methods rely on labor-intensive feature engineering efforts that are task specific, omit long-term relationships, and are not suitable for the rapidly evolving domains. While an increasing body of deep learning and attention mechanism literature aim to address these issues, extant methods often represent text as a single view and omit multiple sets of features at varying levels of granularity. Recognizing that these issues often result in performance degradations, we propose a novel Spatial View Attention Convolutional Neural Network (SVA-CNN). SVA-CNN leverages an innovative and carefully designed set of multi-view representation learning, a combination of heterogeneous attention mechanisms and CNN-based operations to automatically extract and weight multiple granularities and fine-grained representations. Rigorously evaluating SVA-CNN against prevailing text classification methods on five large-scale benchmark datasets indicates its ability to outperform extant deep learning-based classification methods in both performance and training time for document classification, sentiment analysis, and thematic identification applications. To facilitate model reproducibility and extensions, SVA-CNN's source code is also available via GitHub. © 2020 Elsevier Inc.

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Fuzzy Ensemble of Large Language Models for Financial Sentiment Analysis

Item Type Conference Paper**Author** T. Stefanova**Author** S. Georgiev**Editor** Kahraman C.**Editor** Cebi S.**Editor** Oztaysi B.**Editor** Cevik Onar S.**Editor** Tolga C.**Editor** Ucal Sari I.**Editor** Otay I.

Abstract Integrating Large Language Models (LLMs) into financial sentiment analysis offers a promising avenue for understanding market dynamics. This research proposes a novel approach that combines the outputs of multiple LLMs to assess the sentiment of financial social media posts, simulating a fuzzy system to handle the inherent uncertainty and subjectivity in financial discourse. The methodology

involves selecting a diverse set of LLMs fine-tuned for financial contexts. Each model analyzes the same financial social media post, providing sentiment classifications. These outputs are then aggregated using fuzzy logic principles, assigning degrees of confidence to each model's prediction to compute a final sentiment score. To validate the effectiveness of this approach, our study employs a dataset of financial social media posts, comparing the ensemble model's performance against individual LLMs and traditional sentiment analysis methods. This research presents a method for financial sentiment analysis, integrating multiple LLMs and fuzzy logic, which would be useful for investors, financial analysts, and policymakers, as it allows for a more nuanced and accurate sentiment assessment. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Gaining insights into pet owner understanding/lived experience of canine chronic kidney disease using survey and social media data

Item Type Journal Article**Author** G. Tarrant**Author** L. Boyden**Author** T. Rai**Author** A. Wright**Author** A.J.C. Cook**Author** K. Wells

Abstract Introduction: Chronic kidney disease (CKD) is a common and progressive condition in dogs characterized by irreversible damage to one or both kidneys over an extended period leading to gradual decline in kidney function. Early diagnosis is crucial to improve quality of life and increase survival through medical interventions. Methods: This study investigated pet owner understanding of this condition using insights gained by comparing pet owner survey responses with bulk harvested social media discussions on canine CKD. We combined structured survey data (n = 132) with social media analysis spanning multiple platforms to understand owner perceptions of disease characteristics, clinical sign reporting, and pet owner experiences. Results: Both data sources highlighted increased urination and water consumption as primary pet owner concerns, with these clinical signs showing moderate positive correlation (Pearson correlation coefficient of $r = 0.66$). Although not explicitly investigated within the survey, social media data demonstrated pain as a significant concerning clinical sign and revealed the emotional toll of end-of-life care considerations. Further analysis also demonstrated significant associations between CKD diagnosis in dogs and both animal age ($p < 0.001$) and female gender ($p = 0.006$), while breed group and weight showed no significant correlations. Discussion: The complementary nature of structured surveys and social media analysis provided richer understanding of pet owner experiences, understanding and management of CKD. This combined methodological approach offers a model for investigating other chronic conditions in veterinary medicine. Copyright © 2025 Tarrant, Boyden, Rai, Wright, Cook and Wells.

Date 2025**DOI** 10.3389/fvets.2025.1506272**URL** <https://www.scopus.com/pages/publications/105005982672?origin=resultslist>**Archive** Scopus**Language** English**Volume** 12**Publisher** Frontiers Media SA**Publication** Frontiers in Veterinary Science**ISSN** 22971769 (ISSN)**Journal Abbr** Front. Vet. Sci.**Date Added** 20/09/2026, 8:35:19 pm**Modified** 20/09/2026, 8:35:19 pm**Tags:**

large language models, AI, Article, human, quality of life, marketing, demographics, social media, conceptual framework, data analysis, sentiment analysis, social media analysis, social media monitoring, data quality, correlation coefficient, data source, canine chronic kidney disease, canine chronic renal failure, canine CKD, canine CRF, chronic kidney failure, clinical practice, early diagnosis, fluid intake, kidney function, photography, statistical analysis, survey data, veterinary medicine

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GBERT: A hybrid deep learning model based on GPT-BERT for fake news detection

Item Type Journal Article

Author P. Dhiman

Author A. Kaur

Author D. Gupta

Author S. Juneja

Author A. Nauman

Author G. Muhammad

Abstract The digital era has expanded social exposure with easy internet access for mobile users, allowing for global communication. Now, people can get to know what is going on around the globe with just a click; however, this has also resulted in the issue of fake news. Fake news is content that pretends to be true but is actually false and is disseminated to defraud. Fake news poses a threat to harmony, politics, the economy, and public opinion. As a result, bogus news detection has become an emerging research domain to identify a given piece of text as genuine or fraudulent. In this paper, a new framework called Generative Bidirectional Encoder Representations from Transformers (GBERT) is proposed that leverages a combination of Generative pre-trained transformer (GPT) and Bidirectional Encoder Representations from Transformers (BERT) and addresses the fake news classification problem. This framework combines the best features of both cutting-edge techniques—BERT's deep contextual understanding and the generative capabilities of GPT—to create a comprehensive representation of a given text. Both GPT and BERT are fine-tuned on two real-world benchmark corpora and have attained 95.30 % accuracy, 95.13 % precision, 97.35 % sensitivity, and a 96.23 % F1 score. The statistical test results indicate the effectiveness of the fine-tuned framework for fake news detection and suggest that it can be a promising approach for eradicating this global issue of fake news in the digital landscape. © 2024 The Authors

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GeeSanBhava: Sentiment Tagged Sinhala Music Video Comment Data Set

Item Type Conference Paper

Author Y. De Mel

Author N. de Silva

Editor Nguyen N.T.

Editor Kozierkiewicz A.

Editor Dinh Duc Anh V.

Editor Nguyen Van S.

Editor Nunez M.

Editor Treur J.

Editor Vossen G.

Abstract This study introduces GeeSanBhava, a high-quality data set of Sinhala song comments extracted from YouTube manually tagged using Russell's Valence-Arousal model by three independent human annotators. The human annotators achieve a substantial inter-annotator agreement (Fleiss' kappa = 84.96%). The analysis revealed distinct emotional profiles for different songs, highlighting the importance of comment-based emotion mapping. The study also addressed the challenges of comparing comment-based and song-based emotions, mitigating biases inherent in user-generated content. A number of Machine learning and

deep learning models were pre-trained on a related large data set of Sinhala News comments in order to report the zero-shot result of our Sinhala YouTube comment data set. An optimized Multi-Layer Perceptron model, after extensive hyperparameter tuning, achieved a ROC-AUC score of 0.887. The model is a three-layer MLP with a configuration of 256, 128, and 64 neurons. This research contributes a valuable annotated dataset and provides insights for future work in Sinhala Natural Language Processing and music emotion recognition. The complete data set is publicly available at: <https://bit.ly/SinhalaYoutubeComments>.
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GemDetox at TextDetox CLEF 2025: Enhancing a Massively Multilingual Model for Text Detoxification on Low-resource Languages

Item Type Conference Paper

Author T.D.A. Dang

Author F.P. D'Elia

Editor Faggioli G.

Editor Ferro N.

Editor Rosso P.

Editor Spina D.

Abstract As social-media platforms emerge and evolve faster than the regulations meant to oversee them, automated detoxification might serve as a timely tool for moderators to enforce safe discourse at scale. We here describe our submission to the PAN 2025 Multilingual Text Detoxification Challenge, which rewrites toxic single-sentence inputs into neutral paraphrases across 15 typologically diverse languages. Building on a 12B-parameter Gemma-3 multilingual transformer, we apply parameter-efficient LoRA SFT fine-tuning and prompting techniques like few-shot and Chain-of-Thought. Our multilingual training corpus combines 3 600 human-authored parallel pairs, 21 600 machine-translated synthetic pairs, and model-generated pairs filtered by Jaccard thresholds. At inference, inputs are enriched with three LaBSE-retrieved neighbors and explicit toxic-span annotations. Evaluated via Style Transfer Accuracy, LaBSE-based semantic preservation, and xCOMET fluency, our system ranks first on high-resource and low-resource languages. Ablations show +0.081 joint score increase from few-shot examples and +0.088 from basic CoT prompting. ANOVA analysis identifies language resource status as the strongest predictor of performance ($\eta^2 = 0.667$, $p < 0.01$). Warning: This paper contains offensive and potentially triggering texts that only serve as illustrative examples. © 2025 Copyright for this paper by its authors.

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Generation of Training Data to Distinguish Adverse Events from Medical Conditions

Item Type	Conference Paper
Author	N. Allam
Author	P. Biragnet
Author	M.-C. Jaulent
Author	C. Bousquet
Editor	Hagglund M.
Editor	Lindskold L.
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Editor	Giordano N.
Abstract	To support pharmacovigilance activities in social media, innovative methods are required to detect named entities corresponding to drugs and adverse events. However, annotated resources are missing for French-language discussion forums, and manual annotation to create training datasets is a time-consuming and complex process. We propose an approach based on generative artificial intelligence to detect relevant training examples and annotate 200 user messages from forums based on our annotation guidelines. Two prompting strategies were implemented (zero-shot and few-shot) using a decoder-only large language model

(LLM). A comparison was performed between model-generated and manual annotations. Results using partial match evaluation yielded a score of 0.69-0.68 for zero-shot and few-shot prompting. While drug entity detection achieved F1-scores of 0.83, adverse event detection showed limitations (F1: 0.50-0.61), suggesting challenges in disambiguating adverse events from medical conditions in informal text. These findings suggest that while LLMs show promise for drug entity detection and initial filtering of pharmacovigilance relevant messages, the AE-Condition disambiguation task requires further refinement. © 2026 The Authors.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: N. Allam; Sorbonne Université, INSERM, Université Paris-Nord, Limics, Paris, France; email: nour.allam@sorbonne-universite.fr; Conference name: 36th Medical Informatics Europe Conference, MIE 2026; Conference date: 25 May 2026 through 28 May 2026; Conference code: 223971

Generative AI for Business Strategy: Social Media Analysis and Insights

Item Type Conference Paper

Author S. Lunawat

Author A. Kocharekar

Author I. Honmode

Author A. Lavate

Author P. Sarda

Abstract Under the extremely competitive marketplace scenario, social media websites perform the function of determining the attitude of the people towards brands and products. Organizations often rely on user-created content, such as reviews, opinions, and tweets, to assess product launch success, as well as engage in continuous monitoring of customer sentiments. However, the manual screening of this gargantuan amount of information is ineffective and could lead to delayed understandings, thus making the process more complex. It is crucial for businesses to respond to public opinion in a timely manner. This project presents a method that is designed to make the process automatic. Social media sentiment analysis for product performance evaluation by allowing product owners to input a brand designation, the tool fetches recent entries or comments from social media sites and performs a sentiment analysis to classify them as positive or negative, or neutral. Along with gathering sentiment outcomes, the tool determines significant concepts associated with both positive and negative provisions observations and derives a summary through a Large Language Model (LLM). The derived information offers product owners an in-depth perception of the attitudes of consumers. The approach will help organizations monitor the perception of the people towards their product launches, make informed decisions, and quickly address favorable and unfavorable feedback. It is designed to make the business competitive with timely consumer satisfaction knowledge, thus enhancing the product and enhancing customer engagement. © 2025 IEEE.

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Generative artificial intelligence in topicsentiment classification for Arabic text: a comparative study with possible future directions

Item Type Journal Article

Author F. Alderazi

Author A. Algosaiibi

Author M. Alabdullatif

Author H.F. Ahmad

Author A.M. Qamar

Author A. Albarrak

Abstract Social media platforms have become essential for disseminating news and expressing individual sentiments on various life topics. Arabic, widely used in the Middle East, presents unique challenges for sentiment analysis due to its complexity and multiple dialects. Motivated by the need to address these challenges, this article develops methods to overcome the lack of topic-based labeling techniques, compares different approaches for preparing extensive, annotated datasets, and analyzes the efficacy of machine learning (ML), deep learning (DL), and large language models (LLMs) in classifying Arabic textual data. Our research utilizes the topic-modeling technique to build a topic-based sentiment dataset of Arabic texts aimed at enhancing our understanding and processing capabilities. We present a comprehensive evaluation of dataset balancing techniques, including under-sampling, over-sampling, and using imbalanced datasets, providing insights into how these approaches impact classification outcomes. Additionally, we explore the influence of dataset sizes on the performance of various ML models, highlighting the importance of dataset scale in developing effective Arabic NLP applications. A further focus of our study is the comparative analysis of generative artificial intelligence (AI) models, including the emerging ChatGPT LLM, assessing their effectiveness in managing the complexities of Arabic language classification tasks. Our results show that support vector machines (SVM) achieved the highest performance, with F1-scores of 0.97 and 0.96 in classifying sentiment and topic, respectively, in Arabic tweets. This research not only benchmarks existing methodologies but also paves the way for more nuanced and robust models in the future, enhancing the application of generative AI in Arabic topic-based sentiment analysis. © (2024) Alderazi et al.

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GExSent: Gated Experts for Robust Sentiment Analysis Across Modalities

Item Type Conference Paper

Author S. Sethi

Author D. Saini

Author S. Singh

Author C. Dhiman

Author B. Bhushan

Abstract The field of multimodal sentiment analysis has gained huge attention through its broad applications in different forms of communication, from personal interactions to social media, rapid identification and understanding of public opinions and sentiments related to various social issues. Initially, traditional sentiment analysis relied on rule-based systems and predefined lexicons, which struggled to accurately interpret complex emotional cues. Recent advancements

have improved the robustness and adaptability of sentiment analysis across diverse datasets. In this study, a novel architecture is proposed that combines the powerful encoding capabilities of CLIP and Modern BERT. Furthermore, it introduces two feature enhancement modules-Hierarchical Gated Mixture of Experts (H-GMoE) and Gated Attention Mechanism (GAM)-to support task-specific learning and enhance the relevance of multimodal features. Experimental results on benchmark datasets, including MOSI, MOSEI, and Memotion, demonstrate superior performance when compared to current state-of-the-art methods. The code for this work is available at <https://github.com/Subhanshuthi/GExSENT-MOE.git>. © 2025 IEEE.

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GLMTopic: A Hybrid Chinese Topic Model Leveraging Large Language Models

Item Type Journal Article

Author W. Chen

Author W. Hussain

Author J. Chen

Abstract Topic modeling is a fundamental technique of content analysis in natural language processing, widely applied in domains such as social sciences and finance. In the era of digital communication, social scientists increasingly rely on large-scale social

media data to explore public discourse, collective behavior, and emerging social concerns. However, traditional models like Latent Dirichlet Allocation (LDA) and neural topic models like BERTopic struggle to capture deep semantic structures in short-text datasets, especially in complex non-English languages like Chinese. This paper presents Generative Language Model Topic (GLMTopic) a novel hybrid topic modeling framework leveraging the capabilities of large language models, designed to support social science research by uncovering coherent and interpretable themes from Chinese social media platforms. GLMTopic integrates Adaptive Community-enhanced Graph Embedding for advanced semantic representation, Uniform Manifold Approximation and Projection-based (UMAP-based) dimensionality reduction, Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) clustering, and large language model-powered (LLM-powered) representation tuning to generate more contextually relevant and interpretable topics. By reducing dependence on extensive text preprocessing and human expert intervention in post-analysis topic label annotation, GLMTopic facilitates a fully automated and user-friendly topic extraction process. Experimental evaluations on a social media dataset sourced from Weibo demonstrate that GLMTopic outperforms Latent Dirichlet Allocation (LDA) and BERTopic in coherence score and usability with automated interpretation, providing a more scalable and semantically accurate solution for Chinese topic modeling. Future research will explore optimizing computational efficiency, integrating knowledge graphs and sentiment analysis for more complicated workflows, and extending the framework for real-time and multilingual topic modeling. Copyright © 2025 The Authors. Published by Tech Science Press.

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Global Sentiment Toward Health AI at the Dawn of the ChatGPT Era: Empirical Analysis of Twitter (X) Discourse

Item Type Journal Article

Author L.M. Wass

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Author J. Vizoso

Author J.T. Wu

Author L. Lin

Abstract Background: Artificial intelligence (AI) is increasingly proposed for use in health and health care systems. Beyond technical performance, public perceptions and affective responses influence whether AI technologies are accepted and adopted in real-world contexts. Social media platforms such as X (formerly Twitter) provide large-scale, real-time insight into public discourse surrounding emerging technologies, yet remain underused for examining how health AI is discussed, evaluated, and emotionally framed. Objective: This study aimed to develop and apply large language model (LLM)-based methods for exploratory social listening on health AI. This is the first study to map large-scale sentiment, emotional expressions, and confidence-related signals in online discussions of applications of AI to health. Methods: We collected 786,750 English-language posts from X (Twitter) published between January 1 and December 5, 2023, using health- and AI-related Keywords. We benchmarked an LLM-based annotation framework by using OpenAI’s GPT-3.5-Turbo and GPT-4, comparing model classifications with trained human researchers. Annotations included overall sentiment and 6 evaluative domains frequently referenced in the literature surrounding attitudes toward health AI—usefulness, safety, privacy, ethics, quality, and trust. After cleaning, GPT-3.5-Turbo used the best-performing prompts to label 388,009 posts. A subset (n=268,347) was further analyzed using Emollama-7b, an open-source model fine-tuned from Meta’s LLaMA2-7B, for emotion detection, and latent Dirichlet allocation for thematic analysis. Comparisons were made across World Health Organization regions. Results: Compared against human annotations, optimized prompts achieved weighted F1-scores above 0.60 across evaluative domains and sentiment classification. Global discourse about health AI was 65.26% (95% CI 65.11%-65.4%) positive and 83.62% (95% CI 83.48%-83.76%) emotionally optimistic, although substantial regional variation was observed in sentiment (P<.001). The Eastern Mediterranean and South-East Asia regions expressed significantly higher levels of positive sentiment and evaluative agreement in the studied features of health AI, alongside frequent discussion of the tech industry and commercial development. In comparison, the Western Pacific region expressed lower confidence and significantly more mentions of research topics (19.27%, 95% CI 18.5%-20.07%). Privacy was the most prominent global concern, with 33.31% (95%

CI 32.98%-33.66%) of privacy-related posts expressing perceived risks. In the Region of the Americas, 18.19% (95% CI 17.92%-18.44%) of posts discussed algorithms and data governance, significantly higher than overall. Conclusions: This study offers the first systematic characterization of online health AI discourse at scale, mapping stances toward key features of AI, emotional tone, and discussion topics across regions. LLM-powered social listening is demonstrated as a feasible approach for identifying dominant narratives and regionally distinct concerns, capable of surfacing opinions absent from traditional media. This can extend to studying discourse on other evolving health technologies where public surveying is limited. While methodological refinement and multilingual expansion are needed, this framework can inform timely policy development, risk communication, and responsible health AI governance. ©Lily Minh Wass, Zhengdong Wu, José Vizoso, Joseph T Wu, Leesa Lin.

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GNoM: Graph Neural Network Enhanced Language Models for Disaster Related Multilingual Text Classification

Item Type Conference Paper

Author S. Ghosh

Author S. Maji

Author M.S. Desarkar

Abstract Online social media works as a source of various valuable and actionable information during disasters. These information might be available in multiple languages due to the nature of user generated content. An effective system to automatically identify and categorize these actionable information should be capable to handle multiple languages and under limited supervision. However, existing works mostly focus on English language only with the assumption that sufficient labeled data is available. To overcome these challenges, we propose a multilingual disaster related text classification system which is capable to work under monolingual, cross-lingual and multilingual lingual scenarios and under limited supervision. Our end-to-end trainable framework combines the versatility of graph neural networks, by applying over the corpus, with the power of transformer based large language models, over examples, with the help of cross-attention between the two. We evaluate our framework over total nine English, Non-English and monolingual datasets in monolingual, cross-lingual and multilingual lingual classification scenarios. Our framework outperforms state-of-the-art models in disaster domain and multilingual BERT baseline in terms of Weighted F1 score. We also show the generalizability of the proposed model under limited supervision. © 2022 ACM.

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Going negative across platforms and time: Assessing the supply and demand of negative campaigning

Item Type Journal Article

Author A.O. Larsson

Abstract This study investigates dynamics of negative campaigning by Norwegian political parties on Facebook, Instagram and Twitter from 2013 to 2024. Analysing 89,791 posts from nine major parties, it uses a novel content analysis approach combining Large Language Models (LLMs) and human annotation to categorize posts as negative, positive or neutral. The findings show an overall rise in negative campaigning across platforms, with significant changes following leadership shifts in the right-wing populist Progress Party (PP). The study examines engagement levels, measured by likes, for different content types. Initially, negative content attracted more engagement on Facebook. This trend, however, decreased over time, while Instagram saw increased engagement with negative content during later years. Twitter users appeared to have rather consistently favoured negative content, reflecting its reputation for incivility. The article enhances understanding of political communication strategies in a high-choice media environment, emphasizing the need to tailor campaign strategies to platform-specific user dynamics. © The Author(s) 2026. This article is distributed under the terms of the Creative Commons Attribution 4.0 License (<https://creativecommons.org/licenses/by/4.0/>) which permits any use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access pages (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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GORAG: Graph-based Online Retrieval Augmented Generation for Dynamic Few-shot Social Media Text Classification

Item Type Conference Paper**Author** Y. Wang**Author** H. Li**Author** F. Teng**Author** L. Chen

Abstract Text classification is vital for Web for Good applications like hate speech and misinformation detection. However, traditional models (e.g., BERT) often fail in dynamic few-shot settings where labeled data are scarce, and target labels frequently evolve. While Large Language Models (LLMs) show promise in few-shot settings, their performance is often hindered by increased input size in dynamic evolving scenarios. To address these issues, we propose GORAG, a Graph-based Online Retrieval-Augmented Generation framework for dynamic few-shot text classification. GORAG constructs and maintains a weighted graph of keywords and text labels, representing their correlations as edges. To model these correlations, GORAG employs an edge weighting mechanism to prioritize the importance and reliability of extracted information and dynamically retrieves relevant context using a tailored minimum-cost spanning tree for each input. Empirical evaluations show GORAG outperforms existing approaches by providing more comprehensive and precise contextual information. Our code is released at: <https://github.com/Wyb0627/GORAG>. © 2026 Owner/Author.

Date 2026**DOI** 10.1145/3774904.3793011**URL** <https://www.scopus.com/pages/publications/105038574048?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: WWW - Proc. ACM Web Conf.**Publisher** Association for Computing Machinery, Inc**ISBN** 979-840072307-0 (ISBN)**Pages** 9148-9159**Proceedings Title** WWW - Proc. ACM Web Conf.**Conference Name** WWW 2026 - Proceedings of the ACM Web Conference 2026**Date Added** 20/09/2026, 8:35:17 pm**Modified** 20/09/2026, 8:35:17 pm**Tags:**

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Government risk communication and response networks: Insights from typhoon news in China

Item Type Journal Article

Author L. Wang

Author S. Jiang

Abstract Risk communication plays a critical role in disaster warning and management, and social media is emerging as a powerful tool for information dissemination. While substantial research exists on risk communication, studies focusing on governments as the primary subject are limited, and there is a noticeable gap in examining inter-city and city-organization emergency responses. This study aims to address these gaps by exploring and evaluating China's governmental communication strategies during typhoon disasters. We collect typhoon-related news from the WeChat Official Accounts of 237 city governments over an entire year, alongside corresponding typhoon data for the same period. Based on this dataset, we propose an analytical framework for government risk communication. First, we design a three-stage analysis method (pre-disaster, mid-disaster, and post-disaster) based on the Disaster Management Cycle to analyze the dynamics of public engagement. Second, we develop city and organization extractors to construct city and organization response networks, offering fresh approaches to studying city collaboration and organizational coordination. Finally, leveraging large language models, we develop a title feature extractor and sentiment classification model, providing valuable insights for optimizing government media strategies. Overall, this study offers novel methods and a fresh perspective on government risk communication. © 2025 Elsevier Ltd

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GPPLA: Guided Polarity Prompt Learning for Enhanced Emotion Analysis in Low-sample Social Media Environments

Item Type Journal Article

Author R. Wang

Author H. Feng

Author E. Cambria

Author M.J. Piran

Author J. Santamaria

Author M. Ju

Author K. Shahzad

Author X. Zhu

Abstract Abstract: Accurate emotion categorization in social media is critical for applications ranging from mental-health monitoring to market intelligence, yet it remains hampered by two key challenges: the scarcity of high-quality labeled data and the complexity of multimodal content. Here, we introduce Guided Polarity Prompt Learning for Enhanced Emotion Analysis (GPPLA), a unified framework that addresses both challenges by injecting explicit polarity guidance into pre-trained language models and by leveraging self-supervised visual representation learning. First, we construct a compact few-shot dataset via stratified sampling to preserve label distributions under extreme annotation budgets. Next, we enrich image

representations through a rotation-prediction pretext task, and we cast text classification as masked-token prediction guided by a library of positive and negative exemplar prompts. Finally, we fuse image and text embeddings in a shared transformer, steered by polarity prompts that anchor the model's attention to emotional cues. Evaluated on four benchmark multimodal datasets, GPPLEA consistently outperforms state-of-the-art few-shot and full-data baselines, achieving up to a 2.3% absolute gain in accuracy under 1% training data. Our results demonstrate that guided polarity prompting not only amplifies learning from limited labels but also preserves robust generalization in real-world social media contexts. © The Author(s) 2026.

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GPT outperforms BERT and LIWC for stance and anger detection in German news articles and user comments

Item Type Journal Article

Author L. Mayrhofer

Author M. Foramitti
Author S. Fassnacht
Author J.K. Köhler
Author B. Todorova
Author C. Lamm
Author M. Martins

Abstract We live in an era of abundant qualitative language data that allows insights into societal trends and public attitudes, such as social media posts and news articles. Natural language processing (NLP) has become essential for analyzing this data, with recent advances in Large Language Models (LLMs) driving rapid progress. These models show strong potential for psychological research, though their validity in detecting attitudes varies across domains. This study evaluates GPT-3.5 and GPT-4 Turbo for identifying stances and expressions of anger in texts about climate activism, comparing them with traditional methods. The dataset included news articles and user comments from eight German outlets. A subset of 320 articles and 330 comments was manually annotated by three human raters for (1) stance toward Fridays for Future and (2) anger expression. GPT models outperformed BERT (Bidirectional Encoder Representations from Transformers) in stance detection and LIWC (Linguistic Inquiry and Word Count) in anger detection. Moreover, GPT achieved higher agreement with the human majority vote than individual raters agreed with each other. GPT-4 Turbo performed slightly better than GPT-3.5, though differences were minor and domain-specific. Overall, GPT models proved reliable and efficient for annotating nuanced German texts where conventional NLP tools often struggle. © 2026 Mayrhofer, Foramitti, Fassnacht, Köhler, Todorova, Lamm and Martins.

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GPT-4 as an X data annotator: Unraveling its performance on a stance classification task

Item Type Journal Article

Author C.R. Liyanage

Author R. Gokani

Author V. Mago

Abstract Data annotation in NLP is a costly and time-consuming task, traditionally handled by human experts who require extensive training to enhance the task-related background knowledge. Besides, labeling social media texts is particularly challenging due to their brevity, informality, creativity, and varying human perceptions regarding the sociocultural context of the world. With the emergence of GPT models and their proficiency in various NLP tasks, this study aims to establish a performance baseline for GPT-4 as a social media text annotator. To achieve this, we employ our own dataset of tweets, expertly labeled for stance detection with full inter-rater agreement among three annotators. We experiment with three techniques: Zero-shot, Few-shot, and Zero-shot with Chain-of-Thoughts to create prompts for the labeling task. We utilize four training sets constructed with different label sets, including human labels, to fine-tune transformer-based large language models and various combinations of traditional machine learning models with embeddings for stance classification. Finally, all fine-tuned models undergo evaluation using a common testing set with human-generated labels. We use the results from models trained on human labels as the benchmark to assess GPT-4's potential as an annotator across the three prompting techniques. Based on the experimental findings, GPT-4 achieves comparable results through the Few-shot and Zero-shot Chain-of-Thoughts prompting methods. However, none of these labeling techniques surpass the top three models fine-tuned on human labels. Moreover, we introduce the Zero-shot Chain-of-Thoughts as an effective strategy for aspect-based social media text labeling, which performs better than the standard Zero-shot and yields results similar to the high-performing yet expensive Few-shot approach. © 2024 Liyanage et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Graph embedding approaches for social media sentiment analysis with model explanation

Item Type Journal Article**Author** V.S. Anoop**Author** C. Subin Krishna**Author** U.H. Govindarajan

Abstract ChatGPT, the revolutionary chat agent launched in November 2022, is still an active topic of discussion among technology enthusiasts. This open-ended chatbot allows human-like conversations with it on almost all topics since it was trained on millions of documents and developed as a large language model. Since its inception, there have been several discussions and deliberations, especially on twitter and other social media handles, on the potential of ChatGPT and how the power of artificial intelligence is growing leaps and bounds in general. These platforms also witnessed several debates on the negative side of ChatGPT, such as adversely affecting the integrity and ethics and the biased training data. This work uses graph neural network embeddings with machine learning algorithms for classifying the user sentiments on ChatGPT. We have collected a total of 8202 tweets and manually labeled them into multiple classes such as positive, negative, and neutral. We make the models explainable using SHAP (SHapley Additive exPlanations), which is a game theoretical technique for explaining the output of any machine learning models. This paper also publishes our labeled dataset for other researchers to use and train advanced classification models. When our proposed approach was compared with

some chosen baselines, the proposed graph embedding-based machine learning classifiers were found to be outperforming in terms of precision, recall, and accuracy.
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Graph-Enhanced LLM Analysis of Multimodal Health Communities: A Computational Framework for Patient Discourse Understanding on TikTok

Item Type Conference Paper

Author T. Agboola

Author O. Ajao

Editor Danilova V.

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Abstract Social media platforms have become critical sources of patient-generated health data, yet existing computational approaches fail to capture the interconnected nature of online health discourse. We present a novel framework that integrates graph-based community detection with large language model analysis to understand patient narratives in multimodal social media content. Applied to

10,253 TikTok posts about JAK inhibitors (2020-2024), our approach constructs heterogeneous graphs representing user-content-medical entity relationships and applies community detection algorithms enhanced with context-aware LLM interpretation. Our comprehensive analysis of 10,253 posts (January 2020-September 2024) reveals five distinct patient communities characterized by different discourse patterns: treatment success narratives (873 nodes), medication guidance (642 nodes), side effect discussions (589 nodes), comparative treatment analysis (412 nodes), and dosage optimization (347 nodes). The Louvain algorithm significantly outperformed Girvan-Newman in modularity (0.9931 vs. 0.9928), conductance (0.0002 vs. 0.0006), and computational efficiency (0.14s vs. 54.24s). Temporal analysis demonstrates increasing community cohesion and evolving discourse patterns from cautious inquiry (2020-2021) to experience sharing and specialized sub-communities (2023-2024). This work contributes: (1) a scalable computational framework for multimodal health content analysis, (2) methodological innovations in graph-LLM integration, and (3) insights into platform-specific health communication patterns. The framework has applications in pharmacovigilance, computational social science, and AI-assisted health monitoring systems. © 2026 Association for Computational Linguistics.

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GuHPD: A Transformer-Driven Approach to Hostile Post Detection in Gujarati

Item Type Journal Article

Author J. Boda

Author K. Rana

Abstract In recent years, social media has become a prominent medium for public expression; however, it is increasingly exploited to disseminate hostility, particularly against individuals, communities, and religious groups. Religious hate speech can cause profound societal and psychological harm, underscoring the urgent need for automated detection systems. While considerable progress has been made in English-language hate speech detection, limited efforts have addressed low-resource languages such as Gujarati. To bridge this gap, this study presents the Gujarati Hostile Posts Detection (GuHPD) dataset, comprising approximately 14,800 manually annotated comments aimed at identifying hostile content in Gujarati. The dataset supports two core tasks: (i) binary classification to differentiate hostile and non-hostile posts, and (ii) multi-class classification to identify hostile subtypes, including hate speech, fake news, defamation, and offensive language. Annotation reliability was assessed using Fleiss' Kappa, indicating substantial agreement. Several transformer-based models were evaluated, with Multilingual BERT demonstrating the highest performance, achieving an accuracy of 0.93 for binary classification and 0.78 for multi-class classification. These findings demonstrate the utility of the GuHPD dataset in advancing hostile content detection for underrepresented languages and provide a benchmark for future research in regional NLP applications. © 2025 Seventh Sense Research Group®.

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Handling small and imbalanced datasets in government social media: A comparative study of machine learning and deep learning models

Item Type Journal Article

Author E. Manga

Author C. Marinagi

Author C. Skourlas

Author N. Karanikolas

Author D. Markopoulos

Abstract Analysis of social media content allows governments to identify and explore citizens' emotional responses to policies or services. Citizens' emotions, expressed in reviews and comments, provide valuable insights for more responsive and effective public administration. Using a real-world dataset of 148 citizen reviews collected from the U.S. Department of Transportation's Facebook page, and explicitly addressing the constraints of very small and imbalanced datasets, this study undertook a comparative evaluation of a wide range of classification models. The analysis included traditional machine learning algorithms, J4.8, Random Forest, Naive Bayes, BayesNet, Multilayer Perceptron, Logistic Regression, ZeroR, OneR and Nearest Neighbor, as well as more advanced models such as Gradient Boosted Decision Trees and the transformer-based BERT model. Model performance was assessed across multiple metrics, including Speed, Accuracy, macro and weighted F1-scores, Matthews Correlation Coefficient, and ROC AUC, to provide a descriptive comparison under low-resource conditions. The study's results indicate that greater model complexity does not necessarily translate into improved performance in this study. While the transformer-based BERT model underperformed, a result attributable to the limited size and imbalance of the dataset rather than to model inadequacy, the Gradient Boosted Decision Trees model achieved the highest observed performance, correctly classifying 93.9% of the 148 instances, with a true positive rate of 0.939 and a weighted F1-score of 0.933, suggesting strong effectiveness in low-resource, imbalanced classification scenarios. Traditional machine learning models such as BayesNet and Logistic Regression also demonstrated reliable performance, with 128 and 126 correctly classified instances respectively, offering a balance between predictive power and computational efficiency. These findings provide empirically grounded guidance for governments seeking effective and resource-efficient tools for analyzing public sentiment from social media data, particularly in settings characterized by limited data availability and class imbalance, thereby supporting more informed decision-making and responsive policy development. © 2026 Elsevier B.V.

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: E. Manga; School of Engineering, Department of Informatics and Computer Engineering, University of West Attica, Athens, 12243, Greece; email: emanga@uniwa.gr

HARE: Explainable Hate Speech Detection with Step-by-Step Reasoning

Warning: This paper contains examples of content that is offensive and may be upsetting

Item Type Conference Paper

Author Y. Yang

Author J. Kim

Author Y. Kim

Author N. Ho

Author J. Thorne

Author S.-Y. Yun

Abstract With the proliferation of social media, accurate detection of hate speech has become critical to ensure safety online. To combat nuanced forms of hate speech, it is important to identify and thoroughly explain hate speech to help users understand its harmful effects. Recent benchmarks have attempted to tackle this issue by training generative models on free-text annotations of implications in hateful text. However, we find significant reasoning gaps in the existing annotations schemes, which may hinder the supervision of detection models. In this paper, we introduce a hate speech detection framework, HARE, which harnesses the reasoning capabilities of large language models (LLMs) to fill these gaps in explanations of hate speech, thus enabling effective supervision of

detection models. Experiments on SBIC and Implicit Hate benchmarks show that our method, using model-generated data, consistently outperforms baselines, using existing free-text human annotations. Analysis demonstrates that our method enhances the explanation quality of trained models and improves generalization to unseen datasets. Our code is available at <https://github.com/joonkeekim/hare-hate-speech.git>. © 2023 Association for Computational Linguistics.

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Computational linguistics, Speech recognition, Detection models, Social media, Speech detection, Detection framework, Generative model, Text annotations, Reasoning capabilities, Annotation scheme, Free texts, Harmful effects

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Harnessing AI and social media to understand real-world patient experiences in SLE

Item Type Journal Article

Author S. Yang

Author C. Hawryluk

Author J. Liu

Author N. Eckert

Author J. Otoo

Author E.R. Vina

Author L. Yao

Abstract Objective: To apply large language models (LLMs) to Reddit posts referencing SLE to identify patient-expressed unmet medical needs, symptom experiences and healthcare challenges, demonstrating how artificial intelligence-enabled social media listening complements traditional patient-experience research. Methods: We extracted 4633 posts from 10 SLE-related Reddit communities using the public Reddit Application Programming Interface (October–November 2025). After removing duplicates, promotional content and posts with insufficient information, 2603 posts remained. We developed a thematic codebook through manual review of 300 posts and refined it iteratively. Two state-of-the-art LLMs (Gemini 3.0 and GPT-5.2) were evaluated for automated thematic labelling using per cent agreement, Cohen's κ and performance against a human-annotated reference set ($n=100$). The best-performing model was then used to quantify theme prevalence across the full dataset, followed by qualitative review of representative narratives. Results: GPT-5.2 demonstrated higher performance ($F1=0.844$) than Gemini 3.0 ($F1=0.811$), with substantial intermodel agreement across main themes (mean $\kappa=0.71$). Posts reflected multidimensional experiences. The most frequent subtheme was Advice Seeking (84.1%), followed by Emotional Coping (55.6%). Common symptom-related themes included Pain (37.2%), Other Symptom Presentations (37.6%), Fatigue (24.7%) and Acute or Worsening Flares (30.2%). Diagnostic uncertainty was prominent, including testing and results confusion (24.0%) and emotional impact of uncertainty and disease progression (33.0%). Qualitative review highlighted emotional and social burden, reliance on peer communities for advice and shared experiences, and difficulty managing complex treatment regimens. Conclusion: LLM-enabled social media listening offers a scalable method for synthesising large volumes of unstructured patient narratives, providing timely insights into lived experiences and unmet needs among individuals discussing lupus online. Findings align with established qualitative literature while highlighting persistent gaps in patient education, communication and care coordination. This analytical framework can be extended to other disease areas to support patient-centred care, measurement development and evidence generation relevant to therapeutic and health outcomes research. © Author(s) (or their employer(s)) 2026. Re-use permitted under CC BY-NC. No commercial re-use. See rights and permissions. Published by BMJ Group. This is an open access article distributed in accordance with the Creative Commons Attribution Non Commercial (CC BY-NC 4.0) license, which permits others to distribute, remix, adapt, build upon this work non-commercially, and license their derivative works on different terms, provided the original work is properly cited, appropriate credit is given, any changes made indicated, and the use is non-commercial. See: <https://creativecommons.org/licenses/by-nc/4.0/>.

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ISSN 20538790 (ISSN)**Issue** 2**Journal Abbr** Lupus Sci. Med.**Date Added** 20/09/2026, 8:35:14 pm**Modified** 20/09/2026, 8:35:14 pm**Tags:**

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: L. Yao; Polygon Health Analytics, Erie, United States; email: lixia@polygonhealthanalytics.com

Harnessing domain insights: A prompt knowledge tuning method for aspect-based sentiment analysis

Item Type Journal Article**Author** X. Sun**Author** K. Zhang**Author** Q. Liu**Author** M. Bao**Author** Y. Chen

Abstract Aspect-based sentiment analysis (ABSA) endeavours predict the sentiment polarity of specific aspects of a given review. Recently, prompt tuning has been widely explored and has achieved remarkable success in improving semantic comprehension in several NLP tasks. However, most existing methods consider semantic tuning for various tasks and overlook domain knowledge, such as common-sense background knowledge. This not only limits the model's ability to understand and apply domain knowledge but also often leads to the model's inability to fully utilise domain-specific information, resulting in poor semantic quality and inferior model performance. To bridge this gap, we conducted a systematic study of Prompt Tuning with Domain Knowledge (PTDK) for ABSA, which aimed to design efficient prompts that guide the model to learn the knowledge of specific aspects in ABSA. Specifically, we first fine-tune the Large Language Models (LLMs) using hard prompts, which enhance the

ability to extract enriched domain insights from the knowledge base. Additionally, we employed a Co-occurrence Gate to meticulously filter and refine the domain knowledge. This mechanism enhances the domain representation capability of the prompt template by selecting the most similar parts that are independently generated from a vast amount of domain knowledge for each comment. Simultaneously, a emphhybrid prompt template was constructed. This template integrates hard prompts and trainable soft prompts to compensate for the lack of specificity in hard prompts and to facilitate the integration of specific masks in various domain vectors. This hybrid strategy further enhances our ability to utilise domain-specific knowledge when performing ABSA. Experimental results on three public datasets – Restaurant, Laptop, and Twitter – demonstrate that our method consistently outperforms the current state-of-the-art baselines in all cases. The accuracies were 88.63%, 82.65%, and 81.65%, respectively, and F1-scores were 83.38%, 79.68%, and 80.36%, respectively. This translates into an average increase in accuracy of 0.97% and an enhancement in the F1-score of 1.03%. These enhancements not only validate the efficacy of our approach but also have substantial practical implications for real-world scenarios that require sophisticated sentiment analysis, such as the evaluation of customer feedback on e-commerce platforms. © 2024 Elsevier B.V.

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Harnessing Empathy and Ethics for Relevance Detection and Information Categorization in Climate and COVID-19 Tweets

Item Type Conference Paper

Author A. Upadhyaya

Author W. Nejdl

Author M. Fisichella

Abstract In this work, we aim to understand the general public perception of societal issues related to the current climate crisis and the COVID-19 pandemic on Twitter (X). Social media discussions on such matters often lead to misleading information, resulting in delays in initiatives proposed by governments or policymakers. Hence, we focus on extracting relevant information from the conversations on climate change and COVID that could be useful for authorities to curb the spread of potentially biased information by proposing the classification tasks of relevance detection (RD) and information categorization (IC). We first curate the datasets for the RD and IC tasks for the climate domain and extend the COVID-19 benchmark attention-worthy Twitter dataset for the IC task through manual annotation. We initially conduct experiments with LLMs and observe that LLMs can extract the relevant information in zero and few-shot settings based on multi-perspective reasoning in the form of cognitive empathy and ethical standards, but still perform worse than fine-tuned small language models. Based on the initial findings, we conclude that LLMs may not be the best extractor of relevant information, but induce cognitive empathy and ethical reasonings that can intuitively guide supervised models. To achieve this idea, we develop a cognitive empathy and ethical reasoning-based multi-tasking pipelined network for RD and IC tasks. Our proposed approach provides valuable insights that could be useful in real-world scenarios for governments, policymakers, and other researchers to decode the overall public outlook on societal issues. © 2024 ACM.

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Harnessing LLMs for Automated Video Content Analysis: An Exploratory Workflow of Short Videos on Depression

Item Type Conference Paper

Author J.L. Liu

Author Y. Wang

Author Y. Lyu

Author Y. Su

Author S. Niu

Author X.O. Xu

Author Y. Zhang

Editor Bernstein M.

Editor Bruckman A.

Editor Gadiraju U.

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Editor Ma X.

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Editor Zhang A.

Abstract Despite the growing interest in leveraging Large Language Models (LLMs) for content analysis, current studies have primarily focused on text-based content. In the present work, we explored the potential of LLMs in assisting video content analysis by conducting a case study that followed a new workflow of LLM-assisted multimodal content analysis. The workflow encompasses codebook design, prompt engineering, LLM processing, and human evaluation. We strategically crafted annotation prompts to get LLM Annotations in structured form and explanation prompts to generate LLM Explanations for a better understanding of LLM reasoning and transparency. To test LLM's video annotation capabilities, we analyzed 203 keyframes extracted from 25 YouTube short videos about depression. We compared the LLM Annotations with those of two human coders and found that LLM has higher accuracy in object and activity Annotations than emotion and genre Annotations. Moreover, we identified the potential and limitations of LLM's capabilities in annotating videos. Based on the findings, we explore opportunities and challenges for future research and improvements to the workflow. We also discuss ethical concerns surrounding future studies based on LLM-assisted video analysis. © 2024 Owner/Author.

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Harnessing Pre-Trained Sentence Transformers for Offensive Language Detection in Indian Languages

Item Type Conference Paper

Author A. Joshi

Author R. Joshi

Editor Ghosh K.

Editor Mandl T.

Editor Majumder P.

Editor Mitra M.

Abstract In our increasingly interconnected digital world, social media platforms have emerged as powerful channels for the dissemination of hate speech and offensive content. This work delves into the domain of hate speech detection, placing specific emphasis on three low-resource Indian languages: Bengali, Assamese, and Gujarati. The challenge is framed as a text classification task, aimed at

discerning whether a tweet contains offensive or non-offensive content. Leveraging the HASOC 2023 datasets, we fine-tuned pre-trained BERT and SBERT models to evaluate their effectiveness in identifying hate speech. Our findings underscore the superiority of monolingual sentence-BERT models, particularly in the Bengali language, where we achieved the highest ranking. However, the performance in Assamese and Gujarati languages signifies ongoing opportunities for enhancement. The goal of our team- 'Sanvadita' is to foster inclusive online spaces by countering hate speech proliferation. © 2023 Copyright for this paper by its authors.

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: A. Joshi; MKSSS Cummins College of Engineering for Women, Pune, Maharashtra, India; email: joshiananya20@gmail.com; Conference name: 15th Forum for Information Retrieval Evaluation, FIRE 2023; Conference date: 15 December 2023 through 18 December 2023; Conference code: 199455

Harnessing public sentiment discourse for early drought detection and water crisis response for strategic water management and resilient policy planning

Item Type Journal Article

Author S.-E.-E. Soomro

Author M.W. Boota

Author N.-E.-E. Soomro

Author G.-E.-Z. Soomro

Author J. Guo

Author C. Hu

Author J.A. Wahid

Abstract The extensive and gradual onset of drought prompts critical examination of the alterations and engagement among substantial demographics during the drought's advancement and the consequent effects of such shifts on drought detection. This research examines Fb data from 2016 to 2024 to investigate the role of online engagement in drought management. This study evaluates public discourse on drought-related matters through the analysis of five fundamental terms. The research employs topic modeling and sentiment analysis to assess regional awareness and utilizes machine learning techniques (Random Forest, Naive Bayes) in conjunction with Bag of Words to forecast drought progression. The research highlights the potential of Fb data in facilitating real-time drought management, offering significant hydrological insights. The study elucidates regional disparities in drought awareness through the examination of key terminology and sentiment, revealing that some regions exhibit a more rapid reaction to water scarcity, as indicated by Fb engagement. Furthermore, the incorporation of machine learning algorithms such as Random Forest and Naive Bayes facilitates a predictive paradigm for detecting prospective drought hotspots through online discourse analysis. The study confirmed that participation in online communities successfully ($p \leq 0.04$) alleviates the impact of drought and, on Facebook, significantly enhanced drought awareness across various regions of Pakistan ($p \leq 0.5$), as confirmed through statistical analysis with a paired t-test and regression analysis over labeled sentiment and topic-classified data. Fb engagement may function as a proactive indicator, assisting policymakers and hydrologists in optimizing water resource allocation in drought-prone areas, thus enhancing drought mitigation strategies. © 2025 Elsevier B.V.

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Harnessing public sentiment: A literature review of sentiment analysis in energy research

Item Type Journal Article

Author J.T. Ren

Abstract Public opinion is of great importance for the successful implementation of energy projects. Traditional methods for gauging this sentiment, however, are limited in scalability and reproducibility over time. This paper reviews the application of sentiment analysis in energy research, highlighting its potential to overcome these limitations by providing large-scale, quantifiable data across diverse energy technologies and geographies. By showcasing integrations of sentiment analysis with topic modeling, this review showcases approaches that enhance the understanding of public opinions and support more informed policy-making. This paper finds that researchers have leveraged a wide range of data sources, including social media, news reports, research papers, and transcripts of interviews and speeches, but have often relied on suboptimal models to perform the sentiment classification. In line with existing literature, our quantitative model comparison indicates that transformer-based and large language model approaches can improve classification accuracy by up to 20% compared to traditional lexicon-based methods. In particular, this study is the first to compare GPT model performance in an energy context. To support future research, this review provides practical guidance on model selection in sentiment analysis, lowering the technical barrier to adoption. By encouraging broader use of sentiment analysis, this work aims to improve public response forecasting, facilitating

the development of more effective energy policies and advancing energy technologies. © 2025

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Harnessing the Power of LLMs for Service Quality Assessment from User-Generated Content

Item Type Journal Article

Author T. Falatouri

Author D. Hrusecka

Author T. Fischer

Abstract Adopting Large Language Models (LLMs) creates opportunities for organizations to increase efficiency, particularly in sentiment analysis and information extraction tasks. This study explores the efficiency of LLMs in real-world applications, focusing on sentiment analysis and service quality dimension extraction from user-generated content (UGC). For this purpose, we compare the performance of two LLMs (ChatGPT 3.5 and Claude 3) and three traditional NLP methods using two datasets of customer reviews (one in English and one in Persian). The results indicate that LLMs can achieve notable accuracy in information extraction (76% accuracy for ChatGPT and 68% for Claude 3) and sentiment analysis (substantial agreement with human raters for ChatGPT and moderate agreement with human raters for Claude 3), demonstrating an improvement compared to other AI models. However, challenges

persist, including discrepancies between model predictions and human judgments and limitations in extracting specific dimensions from unstructured text. Whereas LLMs can streamline the SQ assessment process, human supervision remains essential to ensure reliability. © 2013 IEEE.

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Hashtag Segmentation: A Comparative Study Involving the Viterbi, Triangular Matrix and Word Breaker Algorithms

Item Type Journal Article

Author S.F. Abd-Hood

Author N. Omar

Abstract Microblogging and social networking sites such as Twitter, Facebook, and Instagram, are becoming increasingly popular, registering more than 500 million posts each day. Twitter uses hashtags that are dynamic, user-generated text, preceded by a pound (#) symbol, to retrieve similar posts or topics, to mark events or to tag channels.

Following segmentation, hashtags can be used for many Natural Language Processing (NLP) applications. These include sentiment analysis, text classification, named entity recognition, and sarcasm detection. This study delves into a comparison of three algorithms, namely the Viterbi, Triangular matrix and Word breaker algorithms, to determine the best among the three, for the segmentation of hashtags. These algorithms utilize different resources, to calculate the probability of the segmented parts, in order to rank the possible generated segmentations. For example, while the Viterbi and Triangular Matrix algorithms use two statistical corpora of unigram and bigram, the Word Breaker algorithm uses the n-gram language model. According to conducted experiment, the Viterbi algorithm is better for hashtag segmentation than the Triangular Matrix algorithm. This can be attributed to the manner in which the Viterbi algorithm conducts the backtracking. On the other hand, the Word Breaker algorithm, which can ascertain the meaningful tokens in the form of words, before proceeding with the segmentation of the remaining characters, is considered superior to both the Viterbi and Triangular Matrix algorithms, particularly when it comes to the detection of unknown words. Used together with the Good-Turing smoothing algorithm, the Word Breaker algorithm achieved 86.64% f1-score on a large language model. © 2021 J. Adv. Inf. Technol.

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Hate Speech Detection in Hindi Using Neural Networks

Item Type Conference Paper

Author A. Sorathiya

Author J. Mehta

Author J. Vithlani
Author M. Sorathiya
Author T. Mehta
Author H. Suresh
Author M. Sorathiya
Author M.A. Gubitra

Abstract The rise of social media platforms has facilitated rapid communication but also led to the widespread dissemination of hate speech, particularly in low-resource languages such as Hindi. This study presents a deep learning-based approach for detecting hate speech in Hindi using a Bidirectional Long Short-Term Memory (BiLSTM) architecture. A dataset of 15,000 annotated posts - sourced from Twitter, newspapers, and televised news - was curated, capturing both formal and informal language, including code-mixed Hindi-English content. To enhance robustness and generalization, the dataset was combined and split into three randomized train-test configurations (10k-5k), with the model trained and evaluated independently on each. Preprocessing steps included tokenization, padding, and label encoding, with text sequences passed through an embedding layer followed by stacked BiLSTM and dense layers. The model achieved consistent accuracy across all splits (72.67%-74.10%), demonstrating its stability under varied data distributions. The findings underscore the linguistic challenges of hate speech detection in Hindi and propose a multi-split evaluation framework as a reliable alternative to single-split benchmarks. This work contributes to the growing body of research on inclusive and context-aware content moderation systems for underrepresented languages, and lays the groundwork for future advancements involving transformer-based models and multi-label classification. The study also demonstrates the importance of evaluating models across multiple randomised data splits rather than a single fixed partition, and highlights the role of factors such as code-mixing, class imbalance, and annotation simplification in shaping model performance on Hindi text. © 2026 IEEE.

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URL <https://www.scopus.com/pages/publications/105046599725?origin=resultslist>

Archive Scopus

Language English

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Conference Name Proceedings of the 18th International Conference on Electronics, Computers and Artificial Intelligence, ECAI 2026

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Computational linguistics, Speech communication, Deep learning, Classification (of information), Language processing, Learning algorithms, Natural language processing, Natural language processing systems, Natural languages, Social networking (online), Neural-networks, Neural networks, Statistical tests, natural language processing, Short term memory, Text processing, Speech analysis, Speech detection, Low resource languages, hate speech detection, bidirectional LSTM, Bidirectional LSTM, Hate speech detection, low-resource languages, sequence modelling, Sequence models

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Sorathiya; Columbia University, MS Data Science, New York, United States; email: afreensorathiya6@gmail.com; Conference name: 18th International Conference on Electronics, Computers and Artificial Intelligence, ECAI 2026; Conference date: 2 July 2026 through 3 July 2026; Conference code: 226314

Hate Speech is not Free Speech: Explainable Machine Learning for Hate Speech Detection in Code-Mixed Languages

Item Type Conference Paper

Author S. Yadav

Author A. Kaushik

Author K. McDaid

Abstract The increase in connectivity provided by social media platforms comes with several disadvantages. It has become surprisingly easy for ill-intentioned individuals to stalk, harass and threaten others online on the basis of race, gender, religion, etc. Artificial intelligence models provide a valuable solution to the problem by automatically filtering such content. However, research for hate speech detection in low-resource languages is still nascent. To promote research in the area, several shared tasks are held in various languages and attract contributions with novel approaches. In this study, we use the English and Hindi code-mixed datasets provided by the HASOC Identification of Conversational Hate-Speech in Code-Mixed Languages (ICHCL) shared task to train and evaluate machine learning models and compare the performance of different vectorization techniques and model parameters. The Explainable Artificial Intelligence (XAI) technique Local Interpretable Model Agnostic Explanation (LIME) is also applied to the model results to ascertain if the model is behaving in a coherent and logical manner. The highest accuracy achieved is 67.61 % using Bernoulli Naive Bayes with count vectorizer and TF-IDF. The analysis suggests that many of the models may be heavily influenced by the presence of slur words while classifying a statement as hateful and offensive. © 2023 IEEE.

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Language English

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machine learning, Machine-learning, Sentiment analysis, Speech recognition, Machine learning, Explainable artificial intelligence, Internet of things, Word2vec, sentiment analysis, explainable artificial intelligence, Hate speech, Speech detection, hate speech, Free speech, LIME, Hinglish, Lime, Local interpretable model agnostic explanation, mix-code, Mix-code, word2vec

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HausaNLP at SemEval-2023 Task 12: Leveraging African Low Resource TweetData for Sentiment Analysis

Item Type Conference Paper

Author S.A. Salahudeen

Author F.I. Lawan

Author A.M. Wali

Author A.A. Imam

Author A.R. Shuaibu

Author A. Yusuf

Author N.B. Rabi

Author M. Bello

Author S.U. Adamu

Author S.M. Aliyu

Author M.S. Gadanya

Author S.A. Muaz

Author M.S. Ahmad

Author A. Abdullahi

Author A.Y. Jamoh

Editor Ojha A.K.

Editor Dogruoz A.S.
Editor Da San Martino G.
Editor Madabushi H.T.
Editor Kumar R.
Editor Sartori E.

Abstract We present the findings of SemEval-2023 Task 12, a shared task on sentiment analysis for low-resource African languages using Twitter dataset. The task featured three subtasks; subtask A is monolingual sentiment classification with 12 tracks which are all monolingual languages, subtask B is multilingual sentiment classification using the tracks in subtask A and subtask C is a zero-shot sentiment classification. We present the results and findings of subtask A, subtask B and subtask C. We also release the code on github. Our goal is to leverage low-resource tweet data using pretrained Afro-xlmr-large, AfriBERTa-Large, Bert-base-arabic-camelbert-da-sentiment (Arabic-camelbert), Multilingual-BERT (mBERT) and BERT models for sentiment analysis of 14 African languages. The datasets for these subtasks consists of a gold standard multi-class labeled Twitter datasets from these languages. Our results demonstrate that Afroxlmr-large model performed better compared to the other models in most of the languages datasets. Similarly, Nigerian languages: Hausa, Igbo, and Yoruba achieved better performance compared to other languages and this can be attributed to the higher volume of data present in the languages. © 2023 Association for Computational Linguistics.

Date 2023

DOI 10.18653/v1/2023.semeval-1.6

URL <https://www.scopus.com/pages/publications/85161065173?origin=resultslist>

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Language English

Extra Journal Abbreviation: Int. Workshop Semantic Eval., SemEval - Proc. Workshop

Publisher Association for Computational Linguistics

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HelaNER 2.0: A Novel Deep Neural Model for Named Entity Boundary Detection

Item Type Journal Article

Author Y.H.P.P. Priyadarshana

Author L. Ranathunga

Abstract Named entity recognition (NER) is a sequential labelling task in categorizing textual nuggets into specific types. Named entity boundary detection can be recognized as a prominent research area under the NER domain which has been heavily adapted for information extraction, event extraction, information retrieval, sentiment analysis etc. Named entities (NE) can be identified as per flat NEs and nested NEs in nature and limited research attempts have been made for nested NE boundary detection. NER in low resource settings has been identified as a current trend. This research work has been scoped down to unveil the uniqueness in NE boundary detection based on Sinhala related contents which have been extracted from social media. The prime objective of this research attempt is to enhance the approach of named entity boundary detection. Considering the low resource settings, as the initial step, the linguistic patterns, complexity matrices and structures of the extracted social media statements have been analyzed further. A dedicated corpus of more than 100,000 tuples of Sinhala related social media content has been annotated by an expert panel. As per the scientific novelties, NE head word detection loss function, which was introduced in HelaNER 1.0, has been further improved and the NE boundary detection has been further enhanced through tuning up the stack pointer networks. Additionally, NE linking has been improved as a by-product of the previously mentioned enhancements. Various experimentations have been conducted, evaluated and the outcome has revealed that our enhancements have achieved the state-of-art performance over the existing baselines © 2022, International Journal of Advanced Computer Science and Applications. All Rights Reserved.

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Notes:

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HiEn-mBERT+: A Lightweight Fine-Tuned Transformer for Hindi-English Code-Mixed Sentiment Analysis

Item Type Conference Paper**Author** S. Rai**Author** A. Kumari**Author** R.K. Behera

Abstract Social media platforms have emerged as crucial spaces for self-expression, particularly in linguistically diverse regions such as India. A major challenge in sentiment analysis arises from the widespread use of Hinglish, a code-mixed blend of Hindi and English, which introduces vocabulary ambiguity, non-standard grammar, and transliteration inconsistencies. These characteristics often impair the performance of traditional sentiment analysis models designed for monolingual text. A lightweight and efficient variant of multilingual BERT, termed Hindi English-mBERT (HiEn-mBERT+), has been developed to address this and fine-tuned specifically for code-mixed sentiment analysis. The model incorporates a novel preprocessing pipeline comprising custom transliteration, language tagging, and subword merging, enabling better handling of linguistic irregularities in Hinglish text. The importance of tailoring language models to code-mixed, low-resource environments is underscored, particularly for enhancing sentiment analysis in multilingual social media contexts. HiEn-mBERT+ has been evaluated on standard Hindi-English sentiment datasets and has shown superior performance over classical machine learning models, deep learning approaches, and standard transformer baselines, thereby demonstrating its robustness and effectiveness in real-world code-mixed scenarios. © 2025 IEEE.

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Computational linguistics, Codes (symbols), Deep learning, Performance, Learning systems, Sentiment analysis, Human computer interaction, Social networking (online), Social media, Data mining, Sentiment Analysis, Embeddings, Social Media, Social media platforms, Low resource languages, Low Resource Language, BERT model, BERT models, Blending, Embedding encoding, Embedding Encoding, Encodings, Non-standard grammar

Notes:

EID: 2-s2.0-105034984082

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Rai; Sikkim Manipal Institute of Technology, Dept of Computer Science and Engineering, Sikkim, India; email: simran_202200295@smit.smu.edu.in; Conference name: 3rd IEEE International Symposium on Sustainable Energy, Signal Processing and Cybersecurity, iSSSC 2025; Conference date: 6 November 2025 through 8 November 2025; Conference code: 220505

Hierarchical aspect-based sentiment analysis Using FinRoBERTa

Item Type Journal Article

Author S. Straleger

Author F. Frasincar

Abstract Investor sentiment embedded in financial news and social media strongly influences market dynamics, but automatically identifying sentiment tied to specific financial aspects remains a challenging task. This is largely due to the scarcity of annotated datasets and the complexity of hierarchical aspect structures in financial text. This work addresses the identification of aspect categories and sentiments from social media posts related to finance and investing, and introduces a new model based on Hierarchy-Aware Global Model (HiAGM) and RoBERTa, FinRoBERTa-TRC2, to tackle this problem. FinRoBERTa-TRC2 is obtained via domain-adaptive pre-training on a subset of the Thomson Reuters Text Research Collection 2 and fine-tuned on the Financial Opinion Mining and Question Answering dataset. The model first predicts a hierarchical aspect category, which is used to predict the associated aspect sentiment score. Our approach follows a three-stage pipeline: domain-adaptive pre-training of RoBERTa on financial corpora, hierarchical aspect categorization through HiAGM,

and final fine-tuning for sentiment prediction on FiQA. We find that domain-adaptive pre-training, in combination with an extensive hyperparameter search, leads to state-of-the-art performance. For aspect sentiment prediction, we find that this approach reduces the mean squared error by 5.6% compared to the current state-of-the-art on FiQA. © 2026 The Authors

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: F. Frasincar; Econometric Institute, Erasmus University Rotterdam, Rotterdam, P.O. Box 1738, NL-3000 DR, Netherlands; email: frasincar@ese.eur.nl; CODEN: DKENE

Hierarchical Attention Networks for Multilabel Sentiment Analysis in Spanish Reviews of Mexican Magic Towns

Item Type Conference Paper

Author E.F.T. Torres

Author E.D.S. Cárdenas

Editor Jimenez-Zafra S.

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Editor Valencia-Garc[◆]a R.

Abstract In this study, we address the task of joint classification of sentiment polarity, destination type, and Magic Town from Spanish-language tourist reviews, as proposed by the Rest-Mex 2025 challenge, part of the IberLEF 2025 evaluation forum [1, 2]. We hereby propose a lightweight, hierarchical attention-based model that leverages the inherent structure of review data. Specifically, the model posits that words form reviews, and reviews are grouped by town. The proposed approach is predicated on the Hierarchical Attention Network (HAN), with extensions to a multi-output setting that are concomitant with the preservation of efficiency and interpretability. Experimental results show that our model significantly outperforms the official baseline and the average of all participating systems, achieving a joint track score of 0.601, with macro-F1 scores of 0.541 for polarity, 0.943 for destination type, and 0.528 for town classification. It is noteworthy that the model demonstrates robust generalization capabilities, particularly in underrepresented classes and low-resource towns. In contrast to large-scale transformer-based models, our architecture can be trained efficiently without the use of GPUs or extensive pretraining, making it suitable for deployment in environments with limited resources. © 2025 Copyright for this paper by its authors.

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Archive Scopus

Language English

Extra Journal Abbreviation: CEUR Workshop Proc.

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Notes:

EID: 2-s2.0-105038707741

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: E.F.T. Torres; Mathematics Research Center (CIMAT), Valenciana, Jalisco S/N, Guanajuato, GTO, 36023, Mexico; email: ezau.torres@cimat.mx; Conference name: 2025 Iberian Languages Evaluation Forum, IberLEF 2025; Conference date: 23 September 2025 through 23 September 2025; Conference code: 217861

Hierarchical Multi-label Classification of Online Vaccine Concerns

Item Type Conference Paper

Author C. Qinyu Zhu

Author R. Stureborg

Author B. Dhingra

Editor Shaban-Nejad A.

Editor Michalowski M.

Editor Bianco S.

Abstract Vaccine concerns are an ever-evolving target, and can shift quickly as seen during the COVID-19 pandemic. Identifying longitudinal trends in vaccine concerns and misinformation might inform the healthcare space by helping public health efforts strategically allocate resources or information campaigns. We explore the task of detecting vaccine concerns in online discourse using large language models (LLMs) in a zero-shot setting without the need for expensive training datasets. Since real-time monitoring of online sources requires large-scale inference, we explore cost-accuracy trade-offs of different prompting strategies and offer concrete takeaways that may inform choices in system designs for current applications. An analysis of different prompting strategies reveals that classifying the concerns over multiple passes through the LLM, each consisting a boolean question whether the text mentions a vaccine concern or not, works the best. Our results indicate that GPT-4 can strongly outperform crowdworker accuracy when compared to ground truth annotations provided by experts on the recently introduced VaxConcerns dataset, achieving an overall F1 score of 78.7%. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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HindiMD: A Multi-domain Corpora for Low-resource Sentiment Analysis

Item Type Conference Paper

Author A. Ekbal

Author P. Bhattacharyya

Author T. Saha

Author A. Kumar

Author S. Srivastava

Editor Calzolari N.

Editor Bechet F.

Editor Blache P.

Editor Choukri K.

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Editor Declerck T.

Editor Goggi S.

Editor Isahara H.

Editor Maegaard B.

Editor Mariani J.

Editor Mazo H.

Editor Odijk J.

Editor Piperidis S.

Abstract Social media platforms such as Twitter have evolved into a vast information sharing platform, allowing people from a variety of backgrounds and expertise to share their opinions on numerous events such as terrorism, narcotics and many other social issues. People sometimes misuse the power of social media for their agendas, such as illegal trades and negatively influencing others. Because of this,

sentiment analysis has won the interest of a lot of researchers to widely analyze public opinion for social media monitoring. Several benchmark datasets for sentiment analysis across a range of domains have been made available, especially for high-resource languages. A few datasets are available for low-resource Indian languages like Hindi, such as movie reviews and product reviews, which do not address the current need for social media monitoring. In this paper, we address the challenges of sentiment analysis in Hindi and socially relevant domains by introducing a balanced corpus annotated with the sentiment classes, viz. positive, negative and neutral. To show the effective usage of the dataset, we build several deep learning based models and establish them as the baselines for further research in this direction. © European Language Resources Association (ELRA), licensed under CC-BY-NC-4.0.

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Extra Journal Abbreviation: Lang. Resour. Eval. Conf., LREC

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Deep learning, Social aspects, Sentiment analysis, Social networking (online), BERT, Deep Learning, Social media platforms, Low resource languages, Indian languages, Sentiment, Social media monitoring, Multi-domains, Indian Language, Information sharing platforms, Low-resource Language, Multi-domain

Notes:

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Homophobia and transphobia span identification in low-resource languages

Item Type Journal Article

Author P.K. Kumaresan

Author D.D. Kayande

Author R. Priyadharshini

Author P. Buitelaar

Author B.R. Chakravarthi

Abstract Online platforms have become prevalent because they promote free speech and group discussions. However, they also serve as platforms for hate speech, which can negatively impact the psychological well-being of vulnerable people. This is especially true for members of the LGBTQ+ community, who are often the targets of homophobia and transphobia in online environments. Our study makes three main contributions: (1) we developed a new dataset with span-level annotations for homophobia and transphobia in Tamil, English, and Marathi; (2) we employed advanced language models using BERT-based architectures, Conditional Random Field (CRF), and Bidirectional Long Short-Term Memory (BiLSTM) layers to enhance span-level detection of harmful content; and (3) we conducted benchmarking to evaluate the effectiveness of monolingual and multilingual models in detecting subtle forms of hate speech. The annotated dataset, which is collected from real-world social media (YouTube) content, provides diverse language contexts and enhances the representation of low-resource languages. The span-based detection approach enables models to detect subtle linguistic nuances, leading to more precise content moderation that accounts for cultural differences. The experimental results show that our models achieve effective span detection, which provides valuable information for creating inclusive moderation tools. Our research leads to the development of AI systems, and we aim to reduce the burden on moderators and improve the quality of online experiences for LGBTQ+ vulnerable. © 2025 The Author(s)

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Notes:

EID: 2-s2.0-105022089941

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P.K. Kumaresan; Data Science Institute, University of Galway, Ireland; email: P.Kumaresan1@universityofgalway.ie

How about electric vehicle? Sensing owners' experiences and attitudes through online short video

Item Type Journal Article

Author Q. Cui

Author Y. Zhang

Author H. Ma

Author K. Zhang

Author J. Peng

Author Z. Chen

Author P. Lin

Author Z. Lin

Abstract Understanding the experiences and attitudes of electric vehicle (EV) owners is crucial for driving EV sales and thus promoting a sustainable environment. Previous studies have used costly and inflexible crowdsourced questionnaires to assess EV owner satisfaction. This study adopts an innovative approach by analyzing 41 h of video interviews with 2101 EV owners from a short video platform. Utilizing the Latent Dirichlet Allocation (LDA) model, topics of concern for EV owners were identified, followed by sentiment analysis supported by a large language model (LLM) and random forest model. The results indicate that 1) "After-sales & Brand (19.75 %)," "Performance Experience (25.63 %)," "Functionality & Comfort (31.76 %)," and "Car Cost (22.86 %)" are the four main concerns of EV owners. Notably, concern for car costs is rising, while after-sales service is decreasing. 2) While EV ownership is mostly male, female owners exhibit a higher percentage of positive sentiment, with 47.16 % compared to 36.81 % for male owners. 3) "Functionality & Comfort" is the key satisfaction factor for both genders, then women prioritizing "After-sales and Brand" and men prioritizing "Performance Experience" as secondary concerns. This study proposes a new research framework and empirical case based on short-video data, offering valuable insights for potential consumers and automakers. © 2025 Elsevier Ltd

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Journal Abbr Transp. Policy

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How are You Really Doing? Dig into the Wheel of Emotions with Large Language Models

Item Type	Conference Paper
Author	M. Luca
Author	G. Lopez
Author	A. Longa
Author	J. Kaul
Abstract	This study explores the application of Large Language Models (LLMs) for Emotion Analysis on social media, focusing on predicting 80 diverse emotions. After comprehensive evaluations, fine-tuned Llama 70B emerges as the top-performing model across datasets, showcasing its capacity to capture a broad spectrum of emotions effectively. Comparisons with traditional baselines highlight the superior performance of LLMs, emphasizing their effectiveness in understanding and predicting emotions in social media posts. In this study, we have also investigated prompting strategies and we discussed the minimal impact that different prompts have on LLMs performance. The study's implications span social media analysis, sentiment analysis, and customer service, with potential future research exploring novel prompts, assessing model transferability, and addressing biases in emotion prediction. © 2024 IEEE.
Date	2024
DOI	10.1109/AIxB62249.2024.00020
URL	https://www.scopus.com/pages/publications/85215089344?origin=resultslist
Archive	Scopus
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Publisher Institute of Electrical and Electronics Engineers Inc.
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Proceedings Title Proc. - Artif. Intell. Bus., AIxB
Conference Name Proceedings - 2024 Artificial Intelligence for Business, AIxB 2024
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Tags:

Language model, Prediction models, Social networking (online), Social media, Emotion analysis, Comprehensive evaluation, component, Component, formatting, Formatting, insert, Insert, style, Style, styling, Styling, Broad spectrum

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Luca; Bruno Kessler Foudnation, Trento, Italy; email: mluca@fbk.eu; Conference name: 2024 IEEE International Conference of Artificial Intelligence for Business, AIxB 2024; Conference date: 30 September 2024 through 2 October 2024; Conference code: 204867

How Effective is the Shorts Transformation of Traditional Media? An Analysis from the Perspective of User-Generated Content

Item Type Conference Paper
Author P. Zhang
Author C.K. Xu
Editor Sun H.
Editor Fan H.
Editor Gao Y.
Editor Wang X.
Editor Liu D.
Editor Du B.
Editor Lu T.
Abstract The digital transformation of traditional media outlets has become a critical imperative in the evolving media landscape. As traditional media grapple with the challenges of adapting to the rise of digital and social media platforms, understanding the factors that shape their performance and user engagement on these emerging channels is of paramount importance. This study examines the comparative performance of a traditional media account with a government background (Hubei Daily) and a specialized new media account (Cover News) on Douyin platform, with a focus on the distribution of emotional sentiment and thematic diversity in user-generated content. We collected 19,343 comments from

“Hubei Daily,” with 18,792 valid (97.15%), and 36,737 comments from “Cover News,” with 28,910 valid (78.69%). Our research used a BERT fine-tuned model for sentiment analysis, TF-IDF for keyword extraction, keyword co-occurrence analysis, and a large language model for topic modeling. This approach allowed us to capture a wide range of discussion topics and understand the thematic focus. Findings show that the new media account covered broader topics, whereas the traditional media account had a more concentrated thematic focus. Typically, such events should generate negative emotions and discussions around accountability. However, the study indicates that users are more cautious in their comments on official media, suggesting ineffective digital transformation by traditional media. This underscores the need for traditional media to better understand and adapt to content preferences and engagement patterns of shorts platform users, highlighting the complexities of transitioning to emerging platforms and the importance of effective content strategies and meaningful user engagement. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

Date 2025

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: C.K. Xu; The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, 511453, China; email: coreyxu@hkust-gz.edu.cn; Conference name: 19th CCF Conference on Computer Supported Cooperative Work and Social Computing, ChineseCSCW 2024; Conference date: 12 July 2024 through 14 July 2024; Conference code: 328259

How effective is twitter (X) social media data for urban flood management?

Item Type Journal Article

Author S.-E.-E. Soomro

Author M.W. Boota

Author H.M. Zwain

Author G.-E.-Z. Soomro

Author X. Shi

Author J. Guo

Author Y. Li

Author M. Tayyab

Author M.H.A. Aamir Soomro

Author C. Hu

Author C. Liu

Author Y. Wang

Author J.A. Wahid

Author Y. Bai

Author S. Nazli

Author J. Yu

Abstract Social media has been an instant source of information for natural disasters, such as urban floods, throughout the world. Ex-post evaluation of the information is considered more important after a flood disaster with devastating consequences on critical infrastructure, the environment, and the well-being of the society. Research proposes an evaluation framework for integrating the disorganized online public opinion on urban flood disaster events towards emotional and conceptual characteristics for better ex-post analysis and public participation. Social media posts on an urban flood were acquired using a search engine, and then sentiment analysis, topic modeling, and spatial-temporal analysis were performed to generate measures of online public opinion about the urban flood crisis. Twitter (X) is the most popular microblogging service presently available. As per the methodology, we analyzed all tweets regarding the 2022 urban flood in the cosmopolitan city (Karachi) of Pakistan from users all over the world. The evaluation results demonstrated that the distribution patterns of post intensities and emotion polarity in response to the floods emphasized crucial aspects with contradictory emotions and highlighted the strategic implications. Experimental results on real datasets show relatively better performance than the baseline and state-of-the-art approaches and achieved the highest 91% score. Online public opinion is a valuable supplement to ex post-disaster evaluation as it helps the project (e.g., flood management) better perform and provides suggestions for future flood mitigation, especially public participation management. © 2024 Elsevier B.V.

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How to optimize large language model performance for human-centric post-occupancy evaluation of underground public space: From model selection to model tuning

Item Type Journal Article

Author C.-X. Ma

Author F.-L. Peng

Author Z.-J. Li

Author Y.-K. Qiao

Abstract Underground public space (UPS) is vital for three-dimensional urban development, and its post-occupancy evaluation (POE) forms a basis for human-centric planning, design and management. Although large language models (LLMs) provide powerful tools for POEs based on social media data (SMD), LLM application sare often constrained by lacking domain-specific knowledge. This study developed a framework to optimize LLM performance for SMD-based POEs. Model selection, parameter setting, prompt tuning and fine-tuning were examined for two specialized annotation tasks using 34 UPS cases and 14 LLMs. The methods proved satisfactory effects, achieving peak macro accuracy of approximately 0.95 and F1-scores over 0.83 for the two tasks. Reasoning models outperformed general-purpose models by

up to 25.74% (F1-scores) in complex tasks, though with lower consistency. Prompt tuning was highly effective but task-dependent with a peak increment of 21.99% in macro F1-scores. Parameter settings and system role modifying had minimal effects. Excessively long prompts may reduce performance. Fine-tuning significantly improved performance and reduced prompt dependency. POEs identified spatial form, wayfinding and operational management as high-priority factors for future design and renovation of UPS. The findings offered a practical and cost-effective optimization pathway for applying LLMs to other similar textual analysis in underground space studies. © The Author(s) 2026.

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How to Train Your Fact Verifier: Knowledge Transfer with Multimodal Open Models

Item Type Conference Paper

Author J. Lee

Author X. Lu
Author J. Hessel
Author F. Brahman
Author Y. Yu
Author Y. Bisk
Author Y. Choi
Author S. Gabriel
Editor Al-Onaizan Y.
Editor Bansal M.
Editor Chen Y.-N.

Abstract Given the growing influx of misinformation across news and social media, there is a critical need for systems that can provide effective real-time verification of news claims. Large language or multimodal model based verification has been proposed to scale up online policing mechanisms for mitigating spread of false and harmful content. While these can potentially reduce burden on human fact-checkers, such efforts may be hampered by foundation model training data becoming outdated. In this work, we test the limits of improving foundation model performance without continual updating through an initial study of knowledge transfer using either existing intra- and inter-domain benchmarks or explanations generated from large language models (LLMs). We evaluate open multimodal foundation models on twelve public benchmarks covering fact-checking, misinformation, toxicity and stance detection. Our results on two recent and widely used multi-modal fact-checking benchmarks, Mocheg and Fakeddit, indicate that knowledge transfer strategies can improve Fakeddit performance over the state-of-the-art by up to 1.7% and Mocheg performance by up to 2.9%. The code, model checkpoints, and dataset are available: <https://github.com/given131/fact-verifier-knowledge-transfer>. © 2024 Association for Computational Linguistics.

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Conference Name EMNLP 2024 - 2024 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2024

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Human and AI Alignment on Stance Detection: A Case Study of the United Healthcare CEO Assassination

Item Type Conference Paper

Author L. Hagen

Author A. Hagen

Author D. Tafmizi

Author C. Reddish

Author L. Li

Author A. Fox

Author N. DePaula

Abstract Media reporting and public opinion polls following the assassination of UnitedHealthcare CEO Brian Thompson suggested a surprising degree of support for the suspect Luigi Mangione, lack of empathy for the victim, and antipathy towards the health insurance industry. The goal of our project is to examine the social media discourse following the assassination to see whether the mainstream media reporting on this event was accurate and to examine key themes and conflicts in public discourse that emerged on social media. This poster reports the preliminary findings from the stance detection task using Large Language Models (LLMs) and human annotation. We report X users’ stance on Luigi Mangione (In-Favor, Neutral, and Against), and provide interpretation on “In-Favor” and “Against” stances. Further, we report evaluation results on human-human agreement, and human-AI agreement. Our findings and discussion contribute to developing better prompt design for fine-tuning and also guide social scientists in adopting LLMs for stance detection using social media data. Copyright © 2025 by the authors.

Date 2025

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Language English

Extra Journal Abbreviation: Proc. Int. Fla. Artif. Intell. Res. Soc. Conf., FLAIRS

Volume 38

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Conference Name Proceedings of the International Florida Artificial Intelligence Research Society Conference, FLAIRS

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Conference name: 38th International Florida Artificial Intelligence Research Society Conference, FLAIRS 2025; Conference date: 20 May 2025 through 23 May 2025; Conference code: 209341

Human Language Technologies - The Baltic Perspective - Proceedings of the 9th International Conference Baltic HLT 2020

Item Type Journal Article

Editor Utkā A.

Editor Vaicenoniene J.

Editor Kovalevskaitė J.

Editor Kalinauskaite D.

Abstract The proceedings contain 34 papers. The topics discussed include: error tagging in the Lithuanian learner corpus; the first corpus-driven lexical database of Lithuanian as L2; detailed error annotation for morphologically rich languages: Latvian use case; large language models for Latvian named entity recognition; pretraining and fine-tuning strategies for sentiment analysis of Latvian tweets; using privacy-transformed speech in the automatic speech recognition acoustic model training; automatic extraction of Lithuanian cybersecurity terms using deep learning approaches; targeted aspect-based sentiment analysis for Lithuanian social media reviews; similarities and differences of Lithuanian functional styles: a quantitative perspective; and evaluating multilingual BERT for Estonian.

Date 2020

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Human vs ChatGPT: Effect of Data Annotation in Interpretable Crisis-Related Microblog Classification

Item Type Conference Paper

Author T.H. Nguyen

Author K. Rudra

Abstract Recent studies have exploited the vital role of microblogging platforms, such as Twitter, in crisis situations. Various machine-learning approaches have been proposed to identify and prioritize crucial information from different humanitarian categories for preparation and rescue purposes. In crisis domain, the explanation of models' output decisions is gaining significant research momentum. Some previous works focused on human annotations of rationales to train and extract supporting evidence for model interpretability. However, such annotations are usually expensive, require much effort, and are not always available in real-time situations of a new crisis event. In this paper, we investigate the recent advances in large language models (LLMs) as data annotators on informal tweet text. We perform a detailed qualitative and quantitative evaluation of ChatGPT rationale annotations over a few-shot setup. ChatGPT annotations are quite close to humans but less precise in nature. Further, we propose an active learning-based interpretable classification model from a small set of annotated data. Our experiments show that (a). ChatGPT has the potential to extract rationales for the crisis tweet classification tasks, but the performance is slightly less than the model trained on human-annotated rationale data ($\sim 3-6\%$), (b). active learning setup can help reduce the burden of manual annotations and maintain a trade-off between performance and data size. © 2024 ACM.

Date 2024

DOI 10.1145/3589334.3648141

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Publisher Association for Computing Machinery, Inc
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Conference Name WWW 2024 - Proceedings of the ACM Web Conference
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Computational linguistics, Language model, Large language model, Economic and social effects, Performance, Classification (of information), Learning systems, Social networking (online), Twitter, Interpretability, twitter, large language model, Data annotation, Active Learning, Semi-supervised learning, interpretability, crisis events, Crisis events, semi-supervised learning, active learning, Micro-blog

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Hybrid CNN-BERT Model with Attention Mechanisms for Classifying Depression, Anxiety, and Stress in Malay Social Media Texts

Item Type Conference Paper
Author Z. Ahmad
Author M. Conway
Author M. Yusoff
Author R. Maskat
Abstract This study presents a hybrid deep learning approach for classifying Depression, Anxiety, and Stress (DAS) in Malay-language social media posts, addressing both the linguistic complexity and data limitations inherent in low-resource languages. The proposed model integrates semantic features from a fine-tuned Bidirectional Encoder Representations from Transformers (BERT) model with syntactic patterns captured through Term Frequency-Inverse Document Frequency (TF-IDF) and Convolutional Neural Network (CNN) layers. To enhance interpretability and performance on imbalanced datasets, two attention mechanisms, Bias-Corrected Weighted Attention (BCWA) and Class-Aware Attention Model (CAAM), are incorporated. BCWA corrects class bias by

recalibrating feature importance, while CAAM dynamically adjusts attention weights based on class-specific traits. Experimental evaluation on the MyDAS corpus demonstrates significant performance gains, with the hybrid model achieving a macro-averaged F1-score of 0.88, outperforming both standalone CNN and BERT baselines. This approach offers a robust solution for mental health text classification in low-resource contexts, with potential applications in real-time public health monitoring and early intervention systems. © 2025 IEEE.

Date 2025

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Language English

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Hybrid Confidence Model for Depression Detection

Item Type Conference Paper

Author A.K. Gandhi

Author C.S. Sreeja

Author S. Vinod

Abstract This work considers the detection of mental health from social media text, a challenging task due to noisy labels and subtle linguistic cues related to psychological distress. This paper presents a novel hybrid methodology that utilizes large language model (LLM) confidence scores for data filtering and combines classical machine learning with attention-based neural networks to identify depressive comments on Reddit. This approach utilizes Mistral LLM in an initial confidence thresholded labeling, as well as feature engineering that combines sentence embeddings and both sentiment analysis and LLM confidence scores. Comparing several ensembling methods such as logistic regression, XGBoost, hybrid meta-learning and learned attention based fusion networks. Results show that the confidence-guided data filtering greatly enhances model trust, achieving 93 percent accuracy (both precision and recall) with traditional ensemble method, and providing interpretable feature importance with attention-based fusion. To the best knowledge, this paper was the first study that systematically integrated LLM confidence scores as explicit features in mental health classification with improved performance and interpretability for clinical psychologists. © 2026 IEEE.

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DOI 10.1109/IC-AIDA68291.2026.11564315

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A.K. Gandhi; Christ (Deemed to be University), Department of Computer Science, Bangalore, India; email: kaushik.gandhi@msam.christuniversity.in; Conference name: 2026 International Conference on AI-Driven Innovations and Applications, IC-AIDA 2026; Conference date: 10 April 2026 through 11 April 2026; Conference code: 225161

Hybrid Deep Learning Approach for Hate Speech Detection in Low-Resource Languages

Item Type Conference Paper

Author L. Imchen

Author A. Ghosh

Author A. Jana

Editor Ullah A.

Editor Anwar S.

Abstract North East India is one example of an area with a linguistically distinguished population that has seen the rise of social media as a critical forum for public discourse in the modern internet age. Among different languages, Assamese is a language with limited resources due to its linguistic complexity and the absence of annotated data. However, the development of hate speech detection algorithms poses issues for Assamese. In this research, we provide a hybrid deep learning model that combines IndicBERT with a Convolutional Neural Network (CNN) and an attention mechanism to identify profanity and hate speech. A total of 2,347 hate/offensive samples and 1,689 non-hate samples, culled from social media sites such as Facebook and Twitter, comprise the HASOC 2023 Assamese dataset, which is used to train the model. We address the class imbalance issue by oversampling and enhancing model training through techniques such as AdamW, learning rate scheduling, weight decay, and mixed-precision training. Finally, our proposed method achieves a higher F1 score of 75% and also assesses its performance using precision, recall, AUC-ROC, and other metrics, demonstrating superior performance compared to prior benchmarks based on transformers. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Volume 1889 LNNS

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Hybrid ML-NLP Framework with Quantum Embeddings for Semantic Analysis and Multilingual Emotion Understanding

Item Type Conference Paper**Author** K. Kaur**Author** A. Avthankar**Author** G.R. Panigrahi**Author** H. Singh**Editor** Sharma V.**Editor** Kaur U.**Editor** Gandomi A.H.

Abstract Emotions and semantics across languages are important in digital communication to be used in social media analysis applications, customer feedback systems, and humancomputer interfaces. Conventional machine learning methods tend to have issues with the specifics of multilingual data, resulting in less than optimal results in emotion recognition and semantic understanding. To cope with these issues, the paper presents a hybrid ML-NLP model based on quantum embeddings. The framework takes advantage of transformerbased models to extract the initial features and uses quantum machine learning to achieve high classification accuracy. The dataset utilized in the present study is a mix of PEMO dataset of Punjabi text emotions based on the work of [1] pemo and GoEmotions dataset of English with an annotation of more than 58,000 samples in 27 emotion categories. We supplemented this with multilingual samples of SemEval-2018 Task 1, which left us with an evenly distributed corpus of 100,000 entries with support to the English, Punjabi and Hindi languages. This heterogeneous dataset is robust in the way of dealing with code-switched and diversified language forms. Driven by shortcomings of the current model, including lack of dealing

with ambiguity in semantics of low-resource languages, inability of classical embeddings to learn quantumlike superposition in emotional states, our model is an integration between BERT embeddings and quantum variational circuits. The approach is pre-training on semantic tasks, fine-tuning on quantum classifiers, and cross-lingual benchmark. According to the experimental results, an overall accuracy of 92%, precision and recall of over 90% across emotions have been proven. Quantum embeddings showed a 15 percent increase in semantic understanding compared to the classical approaches especially in the multilingual set-ups. This model exceeds such baselines as SVM and standard BERT by a wide margin, making it a promising solution in the realm of such applications in emotion-aware systems. © 2026 IEEE.

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Digital communication systems, Machine-learning, Speech recognition, Machine learning, Social networking (online), Semantics, Emotion Recognition, Embeddings, Interactive computer systems, Psychology computing, Hybrid framework, Transformer embedding, Semantic analysis, Quantum machine learning, Quantum machines, Emotion understanding, Hybrid Framework, Multilingual emotion understanding, Multilingual Emotion Understanding, Quantum computers, Quantum Machine Learning, Quantum theory, Semantic Analysis, Semantics understanding, Transformer Embeddings

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Hybrid sentiment analysis approach with domain-specific lexicons with active learning on Greek social media texts

Item Type Journal Article

Author K. Skoularikis

Author I. Savvas

Author G. Garani

Author G. Kakarontzas

Abstract This paper presents a hybrid sentiment analysis framework tailored for the Greek language, combining a novel lexicon construction methodology with an Active Learning (AL) strategy to enhance performance and minimize annotation effort. A domain-specific lexicon was developed using two original metrics—Scorew, capturing the sentiment polarity of words, and Weightw, reflecting their combined emotional and statistical importance. These metrics enabled the transformation of tweet-level annotations into semantically rich word-level representations. To evaluate the framework, four classification models (Naïve Bayes, Logistic Regression, SVM, and Decision Trees) were trained with and without lexicon-derived features. The results showed substantial improvements in accuracy and F1-score when incorporating the lexicon, especially for Logistic Regression and Decision Tree models. Furthermore, a dynamic Active Learning approach was employed, where the lexicon was reconstructed in each cycle using the evolving labeled dataset. This strategy achieved the same levels of accuracy and Macro F1-score, as with full-supervision framework, while using only 30% of the data. The findings demonstrate the efficacy of combining lexicon enhancement with uncertainty-based sampling, offering a cost-effective and scalable solution for sentiment analysis in low-resource languages such as Greek. © The Author(s) 2025.

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Export Date: 20 September 2026; Cited By: 1; Funding details: Hellenic Academic Libraries Link, ΣΕΑΒ; Correspondence Address: K. Skoularikis; Department of Digital Systems, University of Thessaly, Larisa, Greece; email: kyriskoul1@uth.gr

Hybrid sentiment analysis in healthcare: Multi-method approach

Item Type Journal Article

Author X. Tao

Author X. Song

Author T. Talaei Khoei

Abstract Sentiment analysis in healthcare discourse is vital for enhancing patient-provider communication, quality of care, and public health responsiveness. Current methods struggle with the nuanced emotional complexities inherent in healthcare narratives. This study proposes a hybrid sentiment analysis framework combining rule-based approaches, a fine-tuned RoBERTa model, and large language models LLMs, optimized for classifying six emotional dimensions: confidence, satisfaction, anxiety, anger, hope, and trust. Evaluated on over 100,000 Reddit posts from health-related subreddits, the hybrid model significantly outperformed individual methods, achieving a macro F1-score of 0.9348 compared to RoBERTa (0.9056) and LLM-based approaches (0.7532). This approach balances interpretability and semantic depth, enabling nuanced, real-time emotion tracking and early detection of shifts in public sentiment. Our findings offer critical implications for improving patient-centered care, clinical decision support, and ethical AI deployment in healthcare contexts. © 2026 The Authors.

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EID: 2-s2.0-105040014555

Export Date: 20 September 2026; Cited By: 0

HybridStack-BD: A Multilevel Stacking Ensemble Method with Hybrid Features for Bangla Drama Sentiment Analysis

Item Type Conference Paper**Author** M.S. Islam**Author** M.S. Anwar**Author** F.T.J. Lima**Author** H. Bushra

Abstract Sentiment analysis for low-resource languages like Bangla presents a significant challenge, primarily due to the scarcity of specialized datasets and the linguistic nuances of colloquial text. To address this, this study proposes a novel three-layer stacking ensemble underpinned by a synergistic hybrid feature architecture. This core innovation fuses deep contextual embeddings from advanced neural models, including M-BERT and Bi-LSTM, with classical statistical term-importance scores from TF-IDF, creating a uniquely rich feature representation. The effectiveness of our proposed model is verified on DramaSent-BD, a new purpose-built collection of 45,721 annotated user comments from the Bangla drama forum, which we also present as a contribution to the research community. The proposed architecture achieves a state-of-the-art accuracy of 97.40%, establishing a new performance benchmark by significantly outperforming both standalone models and a non-hybrid stacking baseline. This work thus presents a dual contribution: a highly effective and interpretable model for Bangla sentiment analysis and a valuable public dataset to facilitate future research. © 2025 IEEE.

Date 2025**DOI** 10.1109/STI69347.2025.11367561**URL** <https://www.scopus.com/pages/publications/105033965158?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Int. Conf. Sustain. Technol. Ind. 5.0, STI**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833158310-1 (ISBN)**Proceedings Title** IEEE Int. Conf. Sustain. Technol. Ind. 5.0, STI**Conference Name** 2025 IEEE 7th International Conference on Sustainable Technologies For Industry 5.0, STI 2025**Date Added** 20/09/2026, 8:35:30 pm**Modified** 20/09/2026, 8:35:30 pm**Tags:**

Feature extraction, Sentiment analysis, Benchmarking, Engineering research, Sentiment Analysis, Linguistics, Low resource languages, Stacking ensemble, Stacking Ensemble, Stackings, Multilevels, Hybrid features, Ensemble methods, Bangla language, Bangla Language, Hybrid Feature Extraction, Hybrid integrated circuits, Hybrid-feature extraction, M-BERT

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M.S. Islam; Shahjalal University of Science and Technology (SUST), Department of Computer Science and Engineering, Bangladesh; email: shymon-cse@sust.edu; F.T.J. Lima; North Western University, Department of Computer Science and Engineering, Bangladesh; email: fatematujjohralima2@gmail.com; Conference name: 7th International Conference on Sustainable Technologies For Industry 5.0, STI 2025; Conference date: 11 December 2025 through 12 December 2025; Conference code: 219995

Hyperparameter-Tuned LSTM for High-Performance Malay Sentiment Classification

Item Type Conference Paper

Author R. Alfred

Author S.S. Binti Faid

Author R. Pailus

Author K. Chekima

Author Z.H. Abdullah Sani

Author Y. Lim

Abstract Sentiment analysis is a vital task in natural language processing (NLP) that aims to determine the polarity of textual data as positive, negative, or neutral. While significant progress has been made in sentiment classification for high-resource languages, Malay language sentiment analysis remains underexplored, particularly in terms of optimizing deep learning performance. This paper addresses the performance limitations of Long Short-Term Memory (LSTM) models in Malay sentiment classification by proposing a comprehensive hyperparameter tuning strategy. The study investigates the effect of various hyperparameter combinations, namely epoch count, batch size, dropout rate, and learning rate, on LSTM performance across both domain-specific and general Malay Twitter datasets. A total of 48 hyperparameter configurations were tested using supervised datasets sourced from GitHub, covering both original and balanced (rearranged) sentiment distributions. The methodology includes data preprocessing, model implementation using TensorFlow and Keras, rigorous model evaluation using accuracy, precision, recall, and F1-score, and statistical validation through t-tests. The optimized model achieved a classification accuracy of 89.11% on domain-specific data and 72.14% on general (no-domain) data, demonstrating substantial improvement over baseline models. The novelty of this research lies in its systematic and large-scale exploration of hyperparameter settings for LSTM in a low-resource language context, combined with statistical validation to assess performance significance. The study also provides

comparative insights into the impact of domain specificity and data balancing on model performance. The main contribution is a practical, fine-tuned LSTM framework tailored for Malay sentiment analysis, which can serve as a foundation for future NLP applications in the Malay language and other low-resource languages. © 2025 IEEE.

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IberLEF 2022 - Proceedings of the Iberian Languages Evaluation Forum, co-located with the Conference of the Spanish Society for Natural Language Processing, SEPLN 2022

Item Type Journal Article**Editor** Montes-y-Gomez M.**Editor** Gonzalo J.**Editor** Rangel F.**Editor** Casavantes M.**Editor** Alvarez-Carmona M.A.**Editor** Bel-Enguix G.**Editor** Escalante H.J.**Editor** Freitas L.**Editor** Miranda-Escalada A.**Editor** Rodriguez-Sanchez F.**Editor** Rosa A.**Editor** Sobrevilla-Cabezudo M.A.**Editor** Taule M.**Editor** Valencia-Garcia R.

Abstract The proceedings contain 92 papers. The topics discussed include: deep learning Brasil at ABSAPT 2022: Portuguese transformer ensemble approaches; NILC at ABSAPT 2022: aspect extraction for Portuguese; Team PiLN at ABSAPT 2022: lexical and BERT strategies for aspect-based sentiment analysis in Portuguese; TeamUFPR at ABSAPT 2022: aspect extraction with CRF and BERT; UFSCar's Team at ABSAPT 2022: using syntax, semantics and context for solving the tasks; leveraging events sub-categories for violent-events detection in social media; Vicomtech at DA-VINCIS: detection of aggressive and violent incidents from social media in Spanish; ITAINNOVA@DA-VINCIS: a tale of transformers and simple optimization techniques; prompt based framework for violent event recognition in Spanish; and GAN-BERT: adversarial learning for detection of aggressive and violent incidents from social media.

Date 2022**URL** <https://www.scopus.com/pages/publications/85137354451?origin=resultslist>**Archive** Scopus**Language** English**Volume** 3202**Publisher** CEUR-WS**Publication** CEUR Workshop Proceedings**ISSN** 16130073 (ISSN)**Journal Abbr** CEUR Workshop Proc.**Date Added** 20/09/2026, 8:35:41 pm**Modified** 20/09/2026, 8:35:41 pm**Notes:**

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Identification and Classification of Textual Aggression in Social Media: Resource Creation and Evaluation

Item Type Conference Paper

Author O. Sharif

Author M.M. Hoque

Editor Chakraborty T.

Editor Shu K.

Editor Bernard H.R.

Editor Liu H.

Editor Akhtar M.S.

Abstract Recently, social media has gained substantial attention as people can share opinions, expressions, emotions and carry out meaningful interactions through it spontaneously. Unfortunately, with this rapid advancement, social media misuse has also been proliferated, which leads to an increase in aggressive, offensive and abusive activities. Most of these unlawful activities performed through textual communication. Therefore, it is monumental to create intelligent systems that can identify and classify these texts. This paper presents an aggressive text classification system in Bengali. To serve our purpose a corpus (hereafter we called, 'ATxtC') is developed using hierarchical annotation schema that contains 7591 annotated texts (3888 for aggressive and 3703 for non-aggressive). Furthermore, the proposed system can classify aggressive Bengali text into religious, gendered, verbal and political aggression classes. Data annotation obtained a 0.74 kappa score in coarse-grained and 0.61 kappa score in fine-grained categories, which ensures the data's acceptable quality. Several classification algorithms such as LR, RF, SVM, CNN and BiLSTM are implemented on AtxtC. The experimental result shows that the combined CNN and BiLSTM model achieved the highest weighted f1 score of 0.87 (identification task) and 0.80 (classification task). © 2021, Springer Nature Switzerland AG.

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Identifying COVID-19 Tweets Relevant to Low-Income Households Using Semi-supervised BERT and Zero-shot ChatGPT Models

Item Type Conference Paper

Author H. Li

Author D. Caragea

Author A. Mhatre

Author J. Ge

Author M. Liu

Abstract Understanding the COVID-19 pandemic impacts on low-income households can inform social services about the needs of vulnerable communities. Some recent works have studied such impacts through social media content analysis, and supervised machine learning models have been proposed to automatically classify COVID-19 tweets into different categories, such as income and economy impacts, social inequality and justice issues, etc. In this paper, we propose semi-supervised learning models based on BERT with Self-Training and Knowledge Distillation for identifying COVID-19 tweets relevant to low-income households by leveraging readily available unlabeled data in addition to limited amounts of labeled data. Furthermore, we explore ChatGPT's potential for annotating COVID-19 data and the performance of fine-tuned GPT-3 models. Our semi-supervised BERT model with Knowledge Distillation showed improvements compared to a supervised baseline model, while zero-shot ChatGPT showed good potential as a tool for annotating crisis data. However, our study suggests that the cost of fine-tuning large and expensive GPT-3 models may not be worth for some tasks. © 2023 Information Systems for Crisis Response and Management, ISCRAM. All rights reserved.

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Language English

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Li; Department of Computer Science, California State University, East Bay, United States; email: hongmin.li@csueastbay.edu; D. Caragea; Department of Computer Science, Kansas State University, United States; email: dcaragea@ksu.edu; Conference name: 20th Global Information Systems for Crisis Response and Management Conference, ISCRAM 2023; Conference date: 28 May 2023 through 31 May 2023; Conference code: 192193

Identifying Disinformation on the Extended Impacts of COVID-19: Methodological Investigation Using a Fuzzy Ranking Ensemble of Natural Language Processing Models

Item Type Journal Article

Author J.-A. Chen

Author W.-C. Chung

Author C.-L. Hung

Author C.-Y. Wu

Abstract Background: During the COVID-19 pandemic, the continuous spread of misinformation on the internet posed an ongoing threat to public trust and understanding of epidemic prevention policies. Although the pandemic is now under control, information regarding the risks of long-term COVID-19 effects and reinfection still needs to be integrated into COVID-19 policies. Objective: This study aims to develop a robust and generalizable deep learning framework for detecting misinformation related to the prolonged impacts of COVID-19 by integrating

pretrained language models (PLMs) with an innovative fuzzy rank-based ensemble approach. Methods: A comprehensive dataset comprising 566 genuine and 2361 fake samples was curated from reliable open sources and processed using advanced techniques. The dataset was randomly split using the scikit-learn package to facilitate both training and evaluation. Deep learning models were trained for 20 epochs on a Tesla T4 for hierarchical attention networks (HANs) and an RTX A5000 (for the other models). To enhance performance, we implemented an ensemble learning strategy that incorporated a reparameterized Gompertz function, which assigned fuzzy ranks based on each model's prediction confidence for each test case. This method effectively fused outputs from state-of-the-art PLMs such as robustly optimized bidirectional encoder representations from transformers pretraining approach (RoBERTa), decoding-enhanced bidirectional encoder representations from transformers with disentangled attention (DeBERTa), and XLNet. Results: After training on the dataset, various classification methods were evaluated on the test set, including the fuzzy rank-based method and state-of-the-art large language models. Experimental results reveal that language models, particularly XLNet, outperform traditional approaches that combine term frequency-inverse document frequency features with support vector machine or utilize deep models like HAN. The evaluation metrics—including accuracy, precision, recall, F1-score, and area under the curve (AUC)—indicated a clear performance advantage for models that had a larger number of parameters. However, this study also highlights that model architecture, training procedures, and optimization techniques are critical determinants of classification effectiveness. XLNet's permutation language modeling approach enhances bidirectional context understanding, allowing it to surpass even larger models in the bidirectional encoder representations from transformers (BERT) series despite having relatively fewer parameters. Notably, the fuzzy rank-based ensemble method, which combines multiple language models, achieved impressive results on the test set, with an accuracy of 93.52%, a precision of 94.65%, an F1-score of 96.03%, and an AUC of 97.15%. Conclusions: The fusion of ensemble learning with PLMs and the Gompertz function, employing fuzzy rank-based methodology, introduces a novel prediction approach with prospects for enhancing accuracy and reliability. Additionally, the experimental results imply that training solely on textual content can yield high prediction accuracy, thereby providing valuable insights into the optimization of fake news detection systems. These findings not only aid in detecting misinformation but also have broader implications for the application of advanced deep learning techniques in public health policy and communication. ©Jian-An Chen, Wu-Chun Chung, Che-Lun Hung, Chun-Ying Wu.

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Identifying Representation Bias in Large Language Models Used in Financial Sentiment Analysis

Item Type Conference Paper

Author A. Sabuncuoglu

Author C. Maple

Abstract Financial sentiment analysis is the task of evaluating and quantifying the emotions and opinions expressed in financial news, reports, or social media to help investors and institutions make informed decisions. Financial institutions have been actively exploring the use of large language models (LLMs) to analyse market sentiment signals for a more nuanced understanding of a broader context. However, issues such as the scale of training data, model complexity, and the potential for human oversight can introduce or even amplify bias in these systems. Representation bias is a common challenge for LLMs as training data fail to properly represent the target groups, hence causes harmful bias in general-purpose use. Therefore, replacing current solutions with LLMs in financial organisations requires a robust evaluation methodology to ensure fairness. This paper investigates a three-level bias evaluation approach that specifically focuses on representation bias and presents a baseline evaluation of the FinBERT model. Step 1 uses a synthetic dataset that explicitly reveals sources of bias, structured as probability- and embedding-based evaluation recipes. Step 2 evaluates the model against data released by another country (e.g. Indian News dataset) to assess its performance in relation to more implicit biases. Step 3 examines individual problematic samples using token-based interpretability methods (e.g. integrated gradients). This paper presents the application of this structured bias evaluation process and its results on the FinBERT model. The evaluation code and dataset

are available on GitHub (<https://github.com/asabuncuoglu13/faid-test-financial-sentiment-analysis>). © 2025 IEEE.

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Identifying Self-Disclosures of Use, Misuse and Addiction in Community-based Social Media Posts

Item Type Conference Paper

Author C. Yang

Author T. Chakrabarty

Author K.R. Hochstatter

Author M.N. Slavin

Author N. El-Bassel

Author S. Muresan

Editor Duh K.

Editor Gomez H.

Editor Bethard S.

Abstract In the last decade, the United States has lost more than 500, 000 people from an overdose involving prescription and illicit opioids¹ making it a national public

health emergency (USDHHS, 2017). Medical practitioners require robust and timely tools that can effectively identify at-risk patients. Community-based social media platforms such as Reddit allow self-disclosure for users to discuss otherwise sensitive drug-related behaviors. We present a moderate size corpus of 2500 opioid-related posts from various subreddits labeled with six different phases of opioid use: Medical Use, Misuse, Addiction, Recovery, Relapse, Not Using. For every post, we annotate span-level extractive explanations and crucially study their role both in annotation quality and model development.² We evaluate several state-of-the-art models in a supervised, few-shot, or zero-shot setting. Experimental results and error analysis show that identifying the phases of opioid use disorder is highly contextual and challenging. However, we find that using explanations during modeling leads to a significant boost in classification accuracy demonstrating their beneficial role in a high-stakes domain such as studying the opioid use disorder continuum. © 2024 Association for Computational Linguistics.

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Identifying Themes in Social Media Discussions of Eating Disorders: A Quantitative Analysis of How Meaningful Guidance and Examples Improve LLM Classification

Item Type Journal Article

Author A. Prasad

Author S.A. Shalmani

Author L. He

Author Y. Wang

Author S. McRoy

Abstract Background: Social media represents a unique opportunity to investigate the perspectives of people with eating disorders at scale. One forum alone, r/EatingDisorders, now has 113,000 members worldwide. In less than a day, where a manual analysis might sample a few dozen items, automatic classification using large language models (LLMs) can analyze thousands of posts. Methods: Here, we compare multiple strategies for invoking an LLM, including ones that include examples (few-shot) and annotation guidelines, to classify eating disorder content across 14 predefined themes using Llama3.1:8b on 6850 social media posts. In addition to standard metrics, we calculate four novel dimensions of classification quality: a Category Divergence Index, confidence scores (overall model certainty), focus scores (a measure of decisiveness for selected subsets of themes), and dominance scores (primary theme identification strength). Results: By every measure, invoking an LLM without extensive guidance and examples (zero-shot) is insufficient. Zero-shot had worse mean category divergence (7.17 versus 3.17). Whereas, few-shot yielded higher mean confidence, 0.42 versus 0.27, and higher mean dominance, 0.81 versus 0.46. Overall, a few-shot approach improved quality measures across nearly 90% of predictions. Conclusions: These findings suggest that LLMs, if invoked with expert instructions and helpful examples, can provide instantaneous high-quality annotation, enabling automated mental health content moderation systems or future clinical research. © 2025 by the authors.

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Notes:

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Identifying vulgarity in Bengali social media textual content

Item Type Journal Article

Author S. Sazed

Abstract The presence of abusive and vulgar language in social media has become an issue of increasing concern in recent years. However, research pertaining to the prevalence and identification of vulgar language has remained largely unexplored in low-resource languages such as Bengali. In this paper, we provide the first comprehensive analysis on the presence of vulgarity in Bengali social media content. We develop two benchmark corpora consisting of 7,245 reviews collected from YouTube and manually annotate them into vulgar and non-vulgar categories. The manual annotation reveals the ubiquity of vulgar and swear words in Bengali social media content (i.e., in two corpora), ranging from 20% to 34%. To automatically identify vulgarity, we employ various approaches, such as classical machine learning (CML) classifiers, Stochastic Gradient Descent (SGD) optimizer, a deep learning (DL) based architecture, and lexicon-based methods. Although small in size, we find that the swear/vulgar lexicon is effective at identifying the vulgar language due to the high presence of some swear terms in Bengali social media. We observe that the performances of machine leanings (ML) classifiers are affected by the class distribution of the dataset. The DL-based BiLSTM (Bidirectional Long Short Term Memory) model yields the highest recall scores for identifying vulgarity in both datasets (i.e., in both original and class-balanced settings). Besides, the analysis reveals that vulgarity is highly correlated with negative sentiment in social media comments. © Copyright 2021 Sazed

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Journal Abbr PeerJ Comput. Sci.

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Export Date: 20 September 2026; Cited By: 19; Correspondence Address: S. Sazzed; Computer Science, Old Dominion University, Norfolk, United States; email: ssazz001@odu.edu

IdSarcasm: Benchmarking and Evaluating Language Models for Indonesian Sarcasm Detection

Item Type Journal Article

Author D. Suhartono

Author W. Wongso

Author A. Tri Handoyo

Abstract Sarcasm detection in the Indonesian language poses a unique set of challenges due to the linguistic nuances and cultural specificities of the Indonesian social media landscape. Understanding the dynamics of sarcasm in this context requires a deep dive into language patterns and the socio-cultural background that shapes the use of sarcasm as a form of criticism and expression. In this study, we developed the first publicly available Indonesian sarcasm detection benchmark datasets from social media texts. We extensively investigated the results of classical machine learning algorithms, pre-trained language models, and recent large language models (LLMs). Our findings show that fine-tuning pre-trained language models is still superior to other techniques, achieving F1 scores of 62.74% and 76.92% on the Reddit and Twitter subsets respectively. Further, we show that recent LLMs fail to perform zero-shot classification for sarcasm detection and that tackling data imbalance requires a more sophisticated data augmentation approach than our basic methods. © 2013 IEEE.

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Impact of Translation on Sentiment Analysis of Bengali Rumor Texts: A Multi-Criteria Approach

Item Type Conference Paper

Author C.L. Karmaker

Author R.A. Aziz

Author K.Z. Thowai

Author N. Nath

Abstract Sentiment analysis of public perceptions on social media and digital platforms has grown significantly in recent years. While natural language processing (NLP) has advanced for resource-rich languages (e.g., English, Chinese), applying sentiment analysis to low-resource languages like Bengali remains challenging. The impact of translation quality on polarity detection (positive, negative, neutral), especially for Bengali, the fifth most spoken Indo-European language has not been thoroughly studied. This study proposes an integrated framework to analyze how translation affects polarity detection in Bengali rumor texts from Facebook. We collect rumor posts from Rumor Scanner, a fact-checking platform, along with their corresponding Facebook comments. To evaluate the translation's role, we compare machine translation (Google Translate) and manual translation in combination with lexiconbased tools (VADER, TextBlob) and a transformer-based model (BanglaBERT). Performance is assessed using accuracy, F1-score, translation quality, and processing time, with multicriteria decision-making (MCDM) techniques applied for comparative analysis. Results show that manual translation significantly outperforms automated methods in preserving sentiment

polarity, providing key insights for low-resource language processing. © 2025 IEEE.

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Implicit Stance Detection with Hashtag Semantic Enrichment

Item Type Journal Article

Author L. Dong

Author Z. Su

Author X. Fu

Author B. Zhang

Author G. Dai

Abstract Stance detection is a crucial task in natural language processing and social computing, focusing on classifying expressed attitudes towards specific targets based on the input text. Conventional methods predominantly view stance detection as a task of target-oriented, sentence-level text classification. On popular social media

platforms like Twitter, users often express their opinions through hashtags in addition to textual content within tweets. However, current methods primarily treat hashtags as data retrieval labels, neglecting to effectively utilize the semantic information they carry. In this paper, we propose a large language model knowledge-enhanced stance detection framework (LKESD) for stance detection. LKESD contains three main components: an instruction-prompted background knowledge acquisition module (IPBKA) that retrieves background knowledge of hashtags by providing handcrafted prompts to large language models (LLMs); a graph convolutional feature-enhancement module (GCFEM) is designed to extract the semantic representations of words that frequently co-occur with hashtags in the dataset by leveraging textual associations; an a knowledge fusion network (KFN) is proposed to selectively integrate graph representations and LLM features using a prompt-tuning framework. Extensive experimental results on three benchmark datasets demonstrate that our LKESD method outperforms 2.7% on all setups over compared methods, validating its effectiveness in stance detection tasks. © 2024 by the authors.

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Improved Multilingual Language Model Pretraining for Social Media Text via Translation Pair Prediction

Item Type Conference Paper

Author S. Mishra
Author A. Haghighi
Editor Xu W.
Editor Ritter A.
Editor Baldwin T.
Editor Rahimi A.

Abstract We evaluate a simple approach to improving zero-shot multilingual transfer of mBERT on social media corpus by adding a pretraining task called translation pair prediction (TPP), which predicts whether a pair of cross-lingual texts are a valid translation. Our approach assumes access to translations (exact or approximate) between source-target language pairs, where we fine-tune a model on source language task data and evaluate the model in the target language. In particular, we focus on language pairs where transfer learning is difficult for mBERT: those where source and target languages are different in script, vocabulary, and linguistic typology. We show improvements from TPP pretraining over mBERT alone in zero-shot transfer from English to Hindi, Arabic, and Japanese on two social media tasks: NER (a 37% average relative improvement in F1 across target languages) and sentiment classification (12% relative improvement in F1) on social media text, while also benchmarking on a non-social media task of Universal Dependency POS tagging (6.7% relative improvement in accuracy). Our results are promising given the lack of social media bitext corpus. Our code can be found at: <https://github.com/twitter-research/multilingual-alignment-tpp>. © 2021 Association for Computational Linguistics.

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Conference Name W-NUT 2021 - 7th Workshop on Noisy User-Generated Text, Proceedings of the Conference

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Improving Domain-Specific Data Question Answering with Deep and Cross-Lingual Transfer Learning

Item Type Conference Paper

Author M. Alsuhaibani

Author M.O. Beg

Editor Huang L.

Abstract In recent years, there has been significant development in the use of machine learning for various tasks involving natural language interpretation. Improving user experience is a top priority for many online services, including search engines and social media platforms. One such service is answering users' questions. Domain-specific data, such as sacred data, can present unique difficulties for natural language processing because of their intricate structures. In this paper, we employ domain-specific sacred data and a question-and-answer dataset to evaluate and contrast the optimal existing pre-trained deep transformer models for sacred data question answering. Based on the variations of the pre-trained deep transformer models, BERT models, we propose a multilingual domain and task-specific model. To improve the model predictions, we also undertake post-processing and cross-lingual transfer learning steps. On the official test set, our proposed model receives a Partial Reciprocal Rank (pRR) score of 62% recording promising results. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Improving Few-Shot Multi-modal Aspect-Level Sentiment Classification with Implicit In-context Learning

Item Type Conference Paper

Author A. Zhou

Author H. Wu

Author K. Peng

Author P. Liu

Author X. Li

Editor Lokoc J.

Editor Peška L.

Editor Zahálka J.

Editor Rudinac S.

Editor Kastner M.

Editor Chen J.

Editor Hu M.-C.

Editor Wu J.

Editor Sharma U.

Abstract Few-shot Multimodal Aspect-Level Sentiment Classification (MALSC) has garnered significant attention due to the proliferation of multimodal information on social media. In-Context Learning (ICL) with Large Vision-Language Models (LVLMs) offers a promising paradigm for MALSC. However, standard ICL is severely hampered by the high token cost of images, which restricts the number of contextual demonstrations. Circumventing this by replacing images with text captions often leads to critical information loss and error propagation. To address these limitations, we propose a novel framework based on Implicit In-Context Learning (I2CL). Our method compresses multiple image-text demonstrations into a single, compact “context vector” that encapsulates the essence of the task. During inference, this vector is directly injected into the LVLM’s activation space, guiding its reasoning without consuming any input tokens. This approach not only enables the model to leverage rich, raw visual information but also

fundamentally resolves the token bottleneck and caption-induced errors inherent in previous methods. Extensive experiments on the Twitter-2015 and Twitter-2017 benchmarks demonstrate that our I2CL framework significantly enhances both performance and stability for few-shot MALSC, outperforming state-of-the-art models. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Improving Large-scale Paraphrase Acquisition and Generation

Item Type Conference Paper

Author Y. Dou

Author C. Jiang
Author W. Xu
Editor Goldberg Y.
Editor Kozareva Z.
Editor Zhang Y.

Abstract This paper addresses the quality issues in existing Twitter-based paraphrase datasets, and discusses the necessity of using two separate definitions of paraphrase for identification and generation tasks. We present a new Multi-Topic Paraphrase in Twitter (MULTIPIT) corpus that consists of a total of 130k sentence pairs with crowdsourcing (MULTIPITCROWD) and expert (MULTIPITEXPERT) annotations using two different paraphrase definitions for paraphrase identification, in addition to a multi-reference test set (MULTIPITNMR) and a large automatically constructed training set (MULTIPITAUTO) for paraphrase generation. With improved data annotation quality and task-specific paraphrase definition, the best pre-trained language model fine-tuned on our dataset achieves the state-of-the-art performance of 84.2 F1 for automatic paraphrase identification. Furthermore, our empirical results also demonstrate that the paraphrase generation models trained on MULTIPITAUTO generate more diverse and high-quality paraphrases compared to their counterparts fine-tuned on other corpora such as Quora, MSCOCO, and ParaNMT. © 2022 Association for Computational Linguistics.

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Improving Sentiment Analysis in Social Media Using Deep Learning Techniques: A Comparison Study of Generative AI Model Approaches in Gujarati Language Context

Item Type Journal Article

Author V.S. Sharma

Author T. Nakrani

Author S. Sharma

Abstract Social media's explosive growth has drastically changed how people express their thoughts, feelings, and sentiments online. Low-resource languages like Gujarati are still not well studied, despite the significant advancements in sentiment analysis in widely spoken languages. In order to close this gap, this study compares generative AI and deep learning methods for sentiment analysis in the context of Gujarati Language. A dataset that captured a variety of linguistic nuances, colloquial expressions, and code-mixed content typical of user-generated posts was assembled from multiple social media platforms. To ascertain of their respective advantages and disadvantages, a number of models, including LSTM, BERT, RNN, CNN, GPT-2, Gemma, LLaMA, RoBERTa, and Word2Vec-based architectures, were put into practice and method assessed. We used cross validation and standard evaluation metrics to test how well the model performed. The results showed that transformer-based and generative models work better than traditional sequence and the embedding-based models when handling complex context and meaning. This study adds new data and shows that generative AI can make models more robust and easier to interpret, which helps advance sentiment analysis for supervised represented languages. Our findings show strong results for Gujarati and open the door for future research on sentiment analysis in other regional and low-resource languages used on social media. © 2026 by the authors.

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Improving sentiment analysis in tourism through LLM-enhanced irony detection

Item Type Journal Article**Author** W. Liu**Author** L. Wu**Author** H. Zhao

Abstract As user-generated content continues to proliferate across digital tourism platforms such as TripAdvisor, it has become a critical resource for evaluating service performance and tourist satisfaction within the tourism sector. Among these reviews, ironic ones often reveal underlying issues or dissatisfaction, reflecting more genuine and reliable human emotions to influence users' choices. However, conventional sentiment analysis methods often fail to recognize irony, which can express negative sentiments through a superficially positive tone. To address the challenge of accurately classifying sentiments in tourism reviews, particularly ironic ones that can mislead traditional sentiment analysis models, this study introduces an innovative sentiment analysis framework named TSI. Leveraging large language models (LLM) for data augmentation, our approach integrates transfer learning and domain-specific knowledge, enhancing detection accuracy through advanced natural language processing (NLP) techniques and deep learning technologies. Extensive experiments on travel-related datasets from Yelp and Tripadvisor demonstrate that TSI outperforms traditional models in both overall sentiment classification accuracy and the identification of negative sentiments, particularly in ironic reviews. This work advances theoretical understanding and practical implementation of sentiment analysis within the tourism sector, equipping businesses with refined tools for interpreting customer sentiments. A new dataset targeting ironic travel reviews is also released to support ongoing scholarly exploration. For better reproduction, the source code is available.¹ © 2025 Elsevier Ltd

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Inaccuracies, omissions, and bullshit: 2024 U.S. elections through the eyes of ChatGPT

Item Type Journal Article

Author H. Jang

Author S.C. McGregor

Author L. Neill

Abstract This study evaluated the accuracy, quality, and implications of outputs produced by generative artificial intelligences (AI) related to politics and electoral processes during the 2024 United States elections. We extend social media platform governance and content moderation theories to the study of AI, illustrating how large language models (LLMs), and the companies that own these technologies, shape and influence political information environments. Notably, we argue that the shift from social media to generative AI represents a fundamental evolution in platform power: a move from curative governance (providing information by sorting human speech) to generative governance (providing information by generating speech). Through a novel mixed-methods audit approach to studying ChatGPT-4o mini outputs regarding politics and elections collected less than two weeks before election day, we find that ChatGPT largely failed to provide users with accurate, reliable, and trustworthy information during a crucial election period. Our analysis identifies and categorizes potential democratic harms embedded in ChatGPT's responses—such as inaccuracies, hedging, bullshit, and omissions—that paint a picture of a technology ill-equipped for democratic elections. This study illustrates the need for new frameworks for accountability in generative AI, particularly as voluntary industry safeguards fail to protect democratic information environments. © 2026 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

Date 2026**DOI** 10.1080/1369118X.2026.2685121**URL** <https://www.scopus.com/pages/publications/105042527621?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Routledge**Publication** Information Communication and Society**ISSN** 1369118X (ISSN)**Journal Abbr** Inf. Commun. Soc.**Date Added** 20/09/2026, 8:35:10 pm**Modified** 20/09/2026, 8:35:10 pm**Tags:**

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Induce, Align, Predict: Zero-Shot Stance Detection via Cognitive Inductive Reasoning

Item Type Conference Paper**Author** B. Zhang**Author** J. Ma**Author** F. Niu**Author** L. Dong**Author** J. Cao**Author** G. Dai**Editor** Koenig S.**Editor** Jenkins C.**Editor** Taylor M.E.

Abstract Zero-shot stance detection (ZSSD) seeks to determine the stance of text toward previously unseen targets, a task critical for analyzing dynamic and polarized online discourse with limited labeled data. While large language models (LLMs) offer zero-shot capabilities, prompting-based approaches often fall short in handling complex reasoning and lack robust generalization to novel targets. Meanwhile, LLM-enhanced methods still require substantial labeled data and struggle to move beyond instance-level patterns, limiting their interpretability and adaptability. Inspired by cognitive science, we propose the Cognitive Inductive Reasoning Framework (CIRF), a schema-driven method that bridges linguistic

inputs and abstract reasoning via automatic induction and application of cognitive reasoning schemas. CIRF abstracts first-order logic patterns from raw text into multi-relational schema graphs in an unsupervised manner, and leverages a schema-enhanced graph kernel model to align input structures with schema templates for robust, interpretable zero-shot inference. Extensive experiments on SemEval-2016, VAST, and COVID-19-Stance benchmarks demonstrate that CIRF not only establishes new state-of-the-art results, but also achieves comparable performance with just 30% of the labeled data, demonstrating its strong generalization and efficiency in low-resource settings. © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Inference Gap in Domain Expertise and Machine Intelligence in Named Entity Recognition: Creation of and Insights from a Substance Use-related Dataset

Item Type Journal Article

Author S.K. Dey

Author J.M. Powell

Author A. Ismail

Author J. Perrone

Author A. Sarker

Abstract Nonmedical opioid use is an urgent public health challenge, with far-reaching clinical and social consequences that are often underreported in traditional healthcare settings. Social media platforms, where individuals candidly share first-person experiences, offer a valuable yet underutilized source of insight into these impacts. In this study, we present a named entity recognition (NER) framework to extract two categories of self-reported consequences from social media narratives related to opioid use: ClinicalImpacts (e.g., withdrawal, depression) and SocialImpacts (e.g., job loss). To support this task, we introduce RedditImpacts 2.0, a high-quality dataset with refined annotation guidelines and a focus on first-person disclosures, addressing key limitations of prior work. We evaluate both fine-tuned encoderbased models and state-of-the-art large language models (LLMs) under zero- and few-shot in-context learning settings. Our fine-tuned DeBERTa-large model achieves a relaxed tokenlevel F1 of 0.61 [95% CI: 0.43-0.62], consistently outperforming LLMs in precision, span accuracy, and adherence to task-specific guidelines. Furthermore, we show that strong NER performance can be achieved with substantially less labeled data, emphasizing the feasibility of deploying robust models in resource-limited settings. Our findings underscore the value of domain-specific fine-tuning for clinical NLP tasks and contribute to the responsible development of AI tools that may enhance addiction surveillance, improve interpretability, and support real-world healthcare decision-making. The best performing model, however, still significantly underperforms compared to inter-expert agreement (Cohen's kappa: 0.81), demonstrating that a gap persists between expert intelligence and current state-of-the-art NER/AI capabilities for tasks requiring deep domain knowledge. The dataset, annotation guidelines, appendix, and training scripts are publicly available to support future research.**https://github.com/SumonKantiDey/Reddit_Impacts_NER. © 2025 The Authors Open Access chapter published by World Scientific Publishing Company and distributed under the terms of the Creative Commons Attribution Non-Commercial (CC BY-NC) 4.0 License.

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Inferring Political Leaning on X (Twitter): A Zero-Shot Approach in an Italian Scenario

Item Type Conference Paper

Author C. Senette

Author M. Gambini

Author T. Fagni

Author V. Popa

Author M. Tesconi

Editor D'Angelo G.

Editor Luccio F.

Editor Palmieri F.

Abstract In recent years, there has been growing attention on predicting the political orientation of active social media users, aiding in political forecasts, modeling opinion dynamics, and understanding user polarization. Existing methods, primarily for X (Twitter) users, use content-based or a blend of content, network, and communication analysis. The latest research highlights that a user's political stance mainly hinges on their views on key political and social issues, prompting a shift towards detecting user stances through their content shared on social networks. This work investigates the use of an unsupervised stance-detection framework Tweets2Stance (T2S) based on zero-shot classification (ZSC) models [1] to predict users' stances toward a set of social-political statements using content-based analysis of their X (Twitter) timelines in an Italian scenario. The ground-truth user stances are drawn from Voting Advice Applications (VAAs), tools aiding citizens in identifying their political leanings by comparing their preferences with party stances. Leveraging the agreement levels of six parties on 20 statements from VAAs, the study aims to predict Party p's stance on each statement s using X (Twitter) Party account data. T2S, employing zero-shot learning, proves effective across various contexts beyond politics, showcasing a minimum MAE of 1.13 despite a general maximum F1 value of 0.4, demonstrating significant progress given the task complexity. Copyright © 2024 for this paper by its authors.

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Conference Name CEUR Workshop Proceedings
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Forecasting, Social networking (online), Social media, Zero-shot learning, X (twitter), X (Twitter), Opinion dynamics, User profile, Content-based, Forecast models, political leaning, Political leaning, Political orientation, unsupervised ML, Unsupervised ML, user stance detection, User stance detection, VAA, Voting advice application

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Infusing Knowledge from Wikipedia to Enhance Stance Detection

Item Type Conference Paper

Author Z. He

Author N. Mokhberian

Author K. Lerman

Editor Barnes J.

Editor De Clercq O.

Editor Barriere V.

Editor Tafreshi S.

Editor Alqahtani S.

Editor Sedoc J.

Editor Klinger R.

Editor Balahur A.

Abstract Stance detection infers a text author's attitude towards a target. This is challenging when the model lacks background knowledge about the target. Here, we show how background knowledge from Wikipedia can help enhance the

performance on stance detection. We introduce Wikipedia Stance Detection BERT (WS-BERT) that infuses the knowledge into stance encoding. Extensive results on three benchmark datasets covering social media discussions and online debates indicate that our model significantly outperforms the state-of-the-art methods on target-specific stance detection, cross-target stance detection, and zero/few-shot stance detection. © 2022 Association for Computational Linguistics.

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Injecting Commonsense Knowledge into Prompt Learning for Zero-Shot Text Classification

Item Type Conference Paper

Author J. Qian

Author Q. Chen

Author Y. Yue

Author K. Atkinson

Author G. Li

Abstract The combination of pre-training and fine-tuning has become a default solution to Natural Language Processing (NLP) tasks. The emergence of prompt learning breaks such routine, especially in the scenarios of low data resources. Insufficient labelled data or even unseen classes are frequent problems in text classification, equipping Pre-trained Language Models (PLMs) with task-specific prompts helps get rid of the dilemma. However, general PLMs are barely provided with commonsense knowledge. In this work, we propose a KG-driven verbalizer that leverages commonsense Knowledge Graph (KG) to map label words with predefined classes. Specifically, we transform the mapping relationships into semantic relevance in the commonsense-injected embedding space. For zero-shot text classification task, experimental results exhibit the effectiveness of our KG-driven verbalizer on a Twitter dataset for natural disasters (i.e. HumAID) compared with other baselines. © 2023 ACM.

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Innovative Framework based on Simulation and Generative AI for enhance Decision Making in Green Projects

Item Type Conference Paper

Author A.G. Bruzzone

Author A. Giovannetti

Author L. Cirillo

Author X. Sina

Author F. Ghisi

Author S.G. Ferrero

Editor Bruzzone A.

Editor Cimino A.

Editor Frydman C.

Editor Perez J.A.M.

Editor Zacharewicz G.

Abstract This paper introduces an advanced simulator designed to assist decision-makers in strategically implementing smart cities and green infrastructure initiatives. Thanks to the innovative Strategic Engineering approach, based on Modeling and Simulation, Data Analytics and Artificial Intelligence our Simulator has the capacity to model human behavior, capturing a spectrum of cognitive responses to urban and industrial development projects and green initiatives. By simulating public sentiment as well as propaganda of different parties involved in these action, over different channels (Broadcasting, Press and Media) it is possible to recreate a realistic scenario and evaluate impact of media campaigns on people and key players. The use of Generative AI and in particular of LLMs is functional to create real communication content, analyze and synthetize responses and update population parameters by a sentiment analysis. By simulating different scenarios, BRAINS2 (Behavioral Response Analysis and Integrated Network Simulator for Sustainability) provides insights into how each communication channel influences public opinion, either fostering support or inciting opposition to the proposed sustainability initiatives. © 2024 The Authors.

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Instruction Finetuning DeepSeek Based LLM Model for Financial Sentiment and Topic Classification

Item Type Conference Paper

Author Y. Zhang

Author Z. Liu

Author J. Ding

Author M. Luo

Abstract Financial text classification benchmarking has been widely used in quantitative trading systems development, and typical tasks involve financial sentiment classification and classification of financial news. Here, we look to the more recent application of advanced natural language processing (NLP) techniques. This study looks at the performance of the DeepSeek model (a large language model with 7B parameters) on both tasks. As stock and futures markets continue to grow in complexity, quantitative trading systems can not only process price data of financial products, but must also react in real-time to news and investor sentiment. In this research, we finetuned the DeepSeek Model using Low-Rank Adaptation (LoRA) on a Twitter based financial news dataset and financial sentiment classification dataset. We also implemented FlashAttention to speed up training time and use less GPU memory. In our experiments, we compared DeepSeek's performance against those of traditional non-LLM models including RoBERTa, DeBERTa and T5 and also large language models such as LLaMA2 and Baichuan2. The results indicated that DeepSeek outperformed the traditional models and the LLMs even with fewer training iterations. The results showed the benefits of the DeepSeek approach, as it produced better classification results with less data and fewer training epochs, thus overall providing a solid basis for future ethical and widely scalable applications of NLP in the media space. © 2025 Copyright held by the owner/author(s).

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Integrating Large Language Models and Graph Neural Networks for enhanced Arabic sentiment analysis

Item Type Journal Article

Author H. Iwidat

Abstract Morphological complexity and dialectal variation make Arabic one of the most demanding languages for automated sentiment analysis (SA). To tackle these limitations, this work proposes a hybrid framework that couples a Large Language Model with a Graph Neural Network (LLM–GNN), combining the complementary strengths of contextual text encoding and relational structure learning. Specifically, AraBERT v2 serves as the backbone encoder, while a Graph Convolutional Network (GCN) built on cosine-similarity edges captures inter-sentence dependencies that purely sequential architectures tend to overlook. The framework is benchmarked on the publicly available Arabic 100k Reviews dataset (99,999 authentic user-generated reviews balanced equally across Positive, Negative, and Mixed sentiment classes). Against four established baselines (fine-tuned AraBERT, AraBERT-BiLSTM, AraBERT-MLP, multilingual BERT and modern Arabic-centric LLMs such as

Jais-13B), the proposed model achieves an overall accuracy of 66.9% and a macro F1-score of 66.55%, representing gains of 7.6% and 4.4% over the strongest comparable baseline, respectively. Training curves indicate stable loss reduction from the earliest epochs, reflecting consistent optimization behavior throughout the 50-epoch schedule. A noted constraint is that graph construction operates at the mini-batch level, which limits the model's exposure to corpus-wide semantic relationships. Nevertheless, the results confirm that integrating graph-based relational reasoning with transformer-derived embeddings surfaces nuanced sentiment signals that sequential models routinely miss, pointing to practical utility in Arabic social media monitoring and large-scale customer review analysis. © The Author(s) 2026.

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Integrating Large Language Models in Political Discourse Studies on Social Media: Challenges of Validating an LLMs-in-the-loop Pipeline

Item Type Journal Article

Author F. Giglietto

Author G. Marino

Abstract The integration of Large Language Models (LLMs) into research workflows has the potential to transform the study of political content on social media. This essay discusses a validation protocol addressing three key aspects of LLM-integrated research: the versatility of LLMs as general-purpose models, the granularity and nuance in LLM-uncovered narratives, and the limitations of human assessment capabilities. The protocol includes phases for fine-tuning and validating a binary

political classifier, evaluating cluster coherence, and assessing machine-generated cluster label accuracy. We applied this protocol to validate an LLMs-in-the-loop research pipeline designed to analyze political content on Facebook during the Italian general elections of 2018 and 2022. Our approach classifies political links, clusters them by similarity, and generates descriptive labels for clusters. This methodology presents unique validation challenges, prompting a reevaluation of accuracy assessment strategies. By sharing our experiences, this essay aims to guide social scientists in employing LLM-based methodologies, highlighting challenges and advancing recommendations for colleagues intending to integrate these tools for political content analysis on social media. Copyright © 2024 Giada Marino, Fabio Giglietto.

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Integrating Neural Translation and Hybrid Deep Models for Multilingual Sentiment Classification

Item Type Conference Paper

Author R. Gauraha

Author A.K. Agrawal

Author P. Dubey

Abstract Sentiment analysis is one of the important tasks in natural language processing which finds its applications in business, healthcare and social media system by extracting opinions and feelings from textual data. Although the field of English sentiment analysis has grown, applying these techniques in multilingual settings is a challenging task since little data is available and languages and cultural contexts vary. Motivated by the scarce resources for non-English sentiment analysis, this paper takes translation-based approaches. All systems were evaluated on four benchmark datasets, Arabic (SemEval 2017), Chinese (Amazon Reviews Multi), French (DEFT 2017) and Italian (SENTIPOLC 2016), representing different domains. The combined method uses neural machine translation (Google NMT and LibreTranslate) and an ensemble of RoBERTa, mBERT, and GPT models. The novelty is to merge the translation pipelines with heterogeneous deep models using the ensemble learning, which results in language agnostic robustness. The performance is evaluated by Accuracy, Precision, Recall, F1-score and Specificity. Results show that the ensemble of these neural systems significantly surpasses its individual components, with Google NMT + Ensemble being the best balanced and overall top performing system on all corpus. © 2026 IEEE.

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INTEGRATING SEMI-SUPERVISED LEARNING AND TRANSFORMER MODELS FOR OFFENSIVE LANGUAGE DETECTION IN ENGLISH–TAMIL CODE-MIXED SOCIAL MEDIA

Item Type Journal Article

Author S. Rama

Author R. Mythili

Abstract In today's world, social media has evolved far bigger than the press media, becoming the dominant medium for public expression. Despite the extraordinary growth, social media involves the prevalence of offensive language and harmful content. As there is no real content moderation or compliance checks, social media has become an open market for trading offenses. lack of stringent content moderation and compliance checks, social media has become a place where individuals are able to express things that are objectionable. In terms of translation requirements, particularly in low-resource code-mixed language pairs, such as English–Tamil, detection of offensive verbiage is quite a challenge due to factors, such as variations in scripts, irregular spellings, and non-availability of reliable large annotated corpora. This paper describes automated annotation pipeline for the detection of offensive language in English–Tamil translations using the XLM-RoBERTa-large transformer model. The goal of the proposed system is to generate high-quality multi-class offensive-language labels—without manual annotation. This model has been optimized for multi-class classifications in seven categories: profanity, hate speech, sarcasm, threats, harassment/bullying, disrespect/dismissive, and non-offensive. We hypothesize that a semi-supervised, confidence-tiered annotation process, combined with lexicon-based category mapping, public datasets that are labelled, machine-translated resources, and synthetic data generation, can achieve annotation quality comparable to human labelling. The imbalance in classes is addressed through back-translation, generation of synthetic data, and optimization of class-weighted loss. Experimental results indicate that the model can achieve a Macro-F1 score of 92.4% and a Cohen's Kappa of 0.92, demonstrating near-human annotation quality. © (2026), (Little Lion Scientific). All rights reserved.

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INTEGRATING SEMI-SUPERVISED LEARNING AND TRANSFORMER MODELS FOR OFFENSIVE LANGUAGE DETECTION IN ENGLISH–TAMIL CODE-MIXED SOCIAL MEDIA

Item Type Journal Article

Author S. Rama

Author R. Mythili

Abstract In today's world, social media has evolved far bigger than the press media, becoming the dominant medium for public expression. Despite the extraordinary growth, social media involves the prevalence of offensive language and harmful content. As there is no real content moderation or compliance checks, social media has become an open market for trading offenses. Lack of stringent content moderation and compliance checks, social media has become a place where individuals are able to express things that are objectionable. In terms of translation requirements, particularly in low-resource code-mixed language pairs, such as English–Tamil, detection of offensive verbiage is quite a challenge due to factors, such as variations in scripts, irregular spellings, and non-availability of reliable large annotated corpora. This paper describes an automated annotation pipeline for the detection of offensive language in English–Tamil translations using the XLM-RoBERTa-large transformer model. The goal of the proposed system is to generate high-quality multi-class offensive-language labels—without manual annotation. This model has been optimized for multi-class classifications in seven categories: profanity, hate speech, sarcasm, threats, harassment/bullying, disrespect/dismissive, and non-offensive. We hypothesize that a semi-supervised, confidence-tiered annotation process, combined with lexicon-based category mapping, public datasets that are labelled, machine-translated resources, and synthetic data generation, can achieve annotation quality comparable to human labelling. The imbalance in classes is addressed through back-translation, generation of synthetic data, and optimization of class-weighted loss. Experimental results indicate that the model can achieve a Macro-F1 score of 92.4% and a Cohen's Kappa of 0.92, demonstrating near-human annotation quality. © Little Lion Scientific

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Pages 328-340**Publication** Journal of Theoretical and Applied Information Technology**ISSN** 19928645 (ISSN)**Issue** 1**Journal Abbr** J. Theor. Appl. Inf. Technol.**Date Added** 20/09/2026, 8:35:15 pm**Modified** 20/09/2026, 8:35:15 pm**Tags:**

Annotation, Low Resource Language, Multi-Class Classification, Offensive Language Labelling, Semi-Supervised Learning

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EID: 2-s2.0-105042648761

Export Date: 20 September 2026; Cited By: 0

Integrating Transformers and Knowledge Graphs for Twitter Stance Detection

Item Type Conference Paper**Author** T.H. Clark**Author** C. Conforti**Author** F. Liu**Author** Z. Meng**Author** E. Shareghi**Author** N. Collier**Editor** Xu W.**Editor** Ritter A.**Editor** Baldwin T.**Editor** Rahimi A.

Abstract Stance detection (SD) entails classifying the sentiment of a text towards a given target, and is a relevant sub-task for opinion mining and social media analysis. Recent works have explored knowledge infusion — supplementing the linguistic competence and latent knowledge of large pre-trained language models with structured knowledge graphs (KGs), yet few works have applied such methods to the SD task. In this work, we first perform stance-relevant knowledge probing on Transformers-based pre-trained models in a zero-shot setting, showing these models' latent real-world knowledge about SD targets and their sensitivity to context. We then propose novel knowledge-enriched stance detection models. We evaluate them on two Twitter stance datasets, achieving state-of-the-art performance on both. © 2021 Association for Computational Linguistics.

Date 2021**URL** <https://www.scopus.com/pages/publications/85138807732?origin=resultslist>**Archive** Scopus

Language English

Extra Journal Abbreviation: W-NUT - Workshop Noisy User-Gener. Text, Proc. Conf.

Publisher Association for Computational Linguistics (ACL)

ISBN 978-195408590-9 (ISBN)

Pages 304-312

Proceedings Title W-NUT - Workshop Noisy User-Gener. Text, Proc. Conf.

Conference Name W-NUT 2021 - 7th Workshop on Noisy User-Generated Text, Proceedings of the Conference

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Tags:

Computational linguistics, Language model, Sentiment analysis, Social networking (online), Real-world, Social media analysis, Opinion mining, Detection tasks, Zero-shot learning, Knowledge graph, Knowledge graphs, Subtask, Structured knowledge, Linguistic competences, World knowledge

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EID: 2-s2.0-85138807732

Export Date: 20 September 2026; Cited By: 11; Conference name: 7th Workshop on Noisy User-Generated Text, W-NUT 2021; Conference code: 182532

Intelligent model for target identification from Urdu threatening tweets in low-resource language using deep learning and fine-tuning of language models

Item Type Journal Article

Author G. Musa Raza

Author A. Naseem

Author S. Iqbal

Author T. Masood

Author B.S. Kim

Author T.N. Nguyen

Abstract The rise of social media sites like Twitter, which is one of the most widely used venues for expressing thoughts, is a result of the development of technology. However, these opinions may sometimes contain threatening text that can disturb other users. As a result, it's critical to spot harmful content on social media. Regrettably, there aren't many methods for this task in low-resource languages like Urdu, as opposed to high-resource languages like English and Dutch. To address this issue, a study proposes an intelligent model for detecting threatening language in Urdu. Machine learning (ML) classifiers such as Extreme Gradient Boosting (XGB), Random Forest (RF), and Bernoulli Naive Bayes (BNB) have been employed. Deep learning (DL) such as Convolutional neural networks (CNN), Long Short-Term Memory (LSTM), and Gated Recurrent Unit (GRU) have been used. Finally, BERT and RoBERTa have been utilized as transfer learning approaches. FastText has been

used as a feature vector for ML and DL classifiers. In comparison, BERT and RoBERTa have been fine-tuned on the benchmark dataset. The employed algorithms have been tested using Accuracy, Precision, Recall, and F1-score as evaluation metrics. The RoBERTa outperforms other models with 83% accuracy, 70% precision, 53% recall, and a 61% F1-score. This study highlights the importance of detecting target-based threatening language on social media and presents an effective approach for achieving this in a low-resource language such as Urdu, which can improve the safety and security of social media platforms. © 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

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Language English

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Journal Abbr J. Information. Telecommun.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: B.S. Kim; Department of Software and Communications Engineering, Hongik University, Sejong, 30016, South Korea; email: jsnbs@hongik.ac.kr

Intensity-Valued Emotions Help Stance Detection of Climate Change Twitter Data

Item Type Conference Paper

Author A. Upadhyaya

Author M. Fisichella

Author W. Nejdl

Editor Elkind E.

Abstract Our study focuses on the United Nations Sustainable Development Goal 13: Climate Action, by identifying public attitudes on Twitter about climate change. Public consent and participation is the key factor in dealing with climate crises. However, discussions about climate change on Twitter are often influenced by the polarised beliefs that shape the discourse and divide it into communities of climate change deniers and believers. In our work, we propose a framework that helps identify different attitudes in tweets about climate change (deny, believe, ambiguous). Previous literature often lacks an efficient architecture or ignores the characteristics of climate-denier tweets. Moreover, the presence of various emotions with different levels of intensity turns out to be relevant for shaping discussions on climate change. Therefore, our paper utilizes emotion recognition and emotion intensity prediction as auxiliary tasks for our main task of stance detection. Our framework injects the words affecting the emotions embedded in the tweet to capture the overall representation of the attitude in terms of the emotions associated with it. The final task-specific and shared feature representations are fused with efficient embedding and attention techniques to detect the correct attitude of the tweet. Extensive experiments on our novel curated dataset, two publicly available climate change datasets (ClimateICWSM-2023 and ClimateStance-2022), and a benchmark dataset for stance detection (SemEval-2016) validate the effectiveness of our approach. © 2023 International Joint Conferences on Artificial Intelligence. All rights reserved.

Date 2023

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Language English

Extra Journal Abbreviation: IJCAI Int. Joint Conf. Artif. Intell.

Volume 2023-August

Publisher International Joint Conferences on Artificial Intelligence

ISBN 10450823 (ISSN); 978-195679203-4 (ISBN)

Pages 6246-6254

Proceedings Title IJCAI Int. Joint Conf. Artif. Intell.

Conference Name IJCAI International Joint Conference on Artificial Intelligence

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Tags:

Artificial intelligence, Social networking (online), Key factors, Emotion recognition, Emotion Recognition, Climate change, Embeddings, Public attitudes, Main tasks, Efficient architecture, Emotion intensity, Feature representation, Intensity prediction, United Nations

Notes:

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Intent-Driven GraphRAG for LLM-Based Analysis of Online Conversations

Item Type Conference Paper

Author A. Ceni

Author A. Kika

Abstract The continuing development of large language models opens up significant opportunities for the analysis of online conversations. However, their usefulness is challenged when their responses are not grounded in the evidence of the social media data. Retrieval-Augmented Generation can help in this regard by providing text as evidence, but many questions related to social network analysis require relational information such as user interactions, communities, centrality measures, and diffusion patterns. In this paper, we propose an intent-driven GraphRAG framework that enables LLM-based analysis of online conversations and addresses the aforementioned limitation. The proposed framework transforms a social interaction graph into a post-centered knowledge graph by making users, posts, hashtags, URLs, domains, entities, communities, and sentiment first-class graph elements rather than simple attributes in a user-user interaction graph. Several analytical intents are derived from the literature, including topic overview, sentiment analysis, influential-user detection, community comparison, information diffusion, temporal trend analysis, and actionable-opportunity identification. User questions are mapped to these intents and predefined Cypher queries retrieve the appropriate information from Neo4j, which is then provided to the LLM as evidence. We evaluate the framework using a COVID-19 vaccine dataset constructed from X (formerly Twitter) by comparing three approaches: vanilla LLM prompting, text-based RAG, and the proposed GraphRAG method. The results of the comparative experiment show that GraphRAG achieves strong results primarily because of the structural nature of the questions. TextRAG provides a strong baseline for textual grounding, but remains limited for questions requiring structural reasoning. Vanilla LLMs provide ungrounded general responses or refuse to respond due to lack of evidence. These findings suggest that knowledge-graph-based retrieval can improve the reliability, interpretability, and analytical usefulness of LLMs for social media analysis. © 2026 IEEE.

Date 2026

DOI 10.1109/ECAI69016.2026.11613653

URL <https://www.scopus.com/pages/publications/105046595732?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Proc. Int. Conf. Electron., Comput. Artif. Intell., ECAI

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-833155818-5 (ISBN)

Proceedings Title Proc. Int. Conf. Electron., Comput. Artif. Intell., ECAI
Conference Name Proceedings of the 18th International Conference on Electronics, Computers and Artificial Intelligence, ECAI 2026
Date Added 20/09/2026, 8:35:07 pm
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Tags:

Language model, Large language model, Social networking (online), Social media, Information retrieval, Reliability analysis, Large Language Models, Graph theory, Social media analytics, Knowledge graph, Knowledge graphs, Graphic methods, Knowledge Graphs, Social Media Analytics, Continuing development, Graphrag, GraphRAG, Intent-driven retrieval, Intent-Driven Retrieval, Interaction graphs, Social Network Analysis

Notes:

EID: 2-s2.0-105046595732

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Ceni; University of Tirana, Faculty of Natural Sciences, Department of Informatics, Tirana, Albania; email: arber.ceni@unitir.edu.al; Conference name: 18th International Conference on Electronics, Computers and Artificial Intelligence, ECAI 2026; Conference date: 2 July 2026 through 3 July 2026; Conference code: 226314

'Inter-Inter-Coder' Analysis: A Multiset Approach for LLM Evaluation in Public Safety Analytics

Item Type Conference Paper

Author N.J. Elrod

Author J. Kropczynski

Author R. Grace

Abstract This study introduces a novel extension of Krippendorff's set-based 'multi-value' inter-coder analysis, eliminating the need for coder adjudication by evaluating the agreement of multiset sentiment responses. We use open-source large language models in an HPC environment to perform sentiment analysis on COVID-19 social media data, performing queries multiple times per-text, per-model; a set of human coder sentiment is included as well. We evaluate reliability through internal consistency (Shannon entropy) with respect to each coder, and validity by comparing coder response multisets to one another via Krippendorff's alpha and a Dice-Sorensen metric. LLMs showed higher levels of internal agreement than human coders, with all coders showing low multiset ('inter-inter-coder') agreement between one another. © 2025 IEEE.

Date 2025

DOI 10.1109/WF-PST65083.2025.00017

URL <https://www.scopus.com/pages/publications/105035617967?origin=resultslist>

Archive Scopus

Language English
Extra Journal Abbreviation: Proc. - IEEE World Forum Public Saf. Technol., WF-PST
Publisher Institute of Electrical and Electronics Engineers Inc.
ISBN 979-833158977-6 (ISBN)
Pages 66-73
Proceedings Title Proc. - IEEE World Forum Public Saf. Technol., WF-PST
Conference Name Proceedings - 2025 IEEE World Forum on Public Safety Technology, WF-PST 2025
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Tags:

Language model, Large language model, large language models, Safety engineering, Artificial intelligence, Public safety, artificial intelligence, Risk management, Sentiment analysis, Content analysis, Social media, Large datasets, social media, Reliability analysis, Open systems, public safety, content analysis, Open-source, inter-coder reliability, Inter-coder reliability, Multi-sets, Multi-value, multisets

Notes:

EID: 2-s2.0-105035617967

Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 IEEE World Forum on Public Safety Technology, WF-PST 2025; Conference date: 23 September 2025 through 25 September 2025; Conference code: 220584

Internal and External Verification Assisted Deep Clustering Network for Unsupervised Multimodal Fake News Detection

Item Type Journal Article
Author F. Wu
Author Z. Sheng
Author Y. Ji
Author X.-Y. Jing
Author G.-P. Jiang
Abstract The proliferation of fake news on social media platforms has evolved into a pervasive global societal issue. Its latent diffusion seriously distorts public perception and threatens the construction of the social trust system. Current fake news detection methods are highly dependent on manually annotated news data. However, social media information is characterized by real-time explosive nature and multimodal complexity, resulting in high annotation costs and poor timeliness of labels. To deal with the challenge of unsupervised multimodal fake news detection (UMFND), we propose an internal and external verification assisted deep clustering network (IEV-DCN) for UMFND, consisting of an internal and external verification (IEV) module, a joint optimization rectification (JOR) mechanism, and a density-based multimodal

clustering (DMC) module. The IEV module contains an internal verification (IV) part and an external verification (EV) part. The IV part employs multi-grained cross-modal attention mechanism to evaluate image-text semantic alignment, while the EV part leverages external knowledge corpus to retrieve relevant evidence of news samples, and implements evidence-guided preliminary assessment via multimodal large language model (MLLM). The JOR mechanism uses image-text consistency to correct the confidence of preliminary assessment, and generates pseudo labels for samples based on the corrected confidence. Furthermore, the DMC module distinguishes the quality of samples based on their densities, and uses the generated pseudo labels to constrain clustering and enhance the clustering effect. Experiments on widely used fake news datasets have shown that our approach outperforms state-of-the-art fake news detection works. © 2015 IEEE.

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DOI 10.1109/TBDATA.2026.3725875

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Publisher Institute of Electrical and Electronics Engineers Inc.

Publication IEEE Transactions on Big Data

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Journal Abbr IEEE Trans. Big Data

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Language model, Deep learning, Learning systems, Social networking (online), Semantics, Multi-modal, Clusterings, Interactive computer systems, Multimodal large language model, Error detection, Fake detection, Clustering networks, Deep clustering, Deep Clustering, Density-based, Joint optimization, Multimodal fake news detection, Multimodal Fake News Detection, Multimodal Large Language Model, Unsupervised learning, Unsupervised Learning

Notes:

EID: 2-s2.0-105048564421

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: F. Wu; Nanjing University of Posts and Telecommunications, College of Artificial Intelligence, Nanjing, 210023, China; email: wufei_8888@126.com

International Classification of Primary Care (ICPC-2) and search engines: an exploration of three algorithms for information retrieval to aid medical coding

Item Type Journal Article

Author V.A. de Almeida

Author E.J. van der Haring

Author K. van Boven

Author L.F. Lopez

Abstract Background: Medical coding is an essential process to collect structured data in healthcare. The International Classification of Primary Care, 2nd edition (ICPC-2), is more concise and appropriate for primary care compared to other classifications. However, healthcare professionals still struggle to correctly attribute codes in various clinical scenarios. Modern tools are necessary to help efficiently and accurately find the right codes. This study's goal was to evaluate and compare three different information retrieval algorithms for retrieving ICPC-2 codes. Methods: Three different strategies for information retrieval were compared. The strategies include BM25, Levenshtein distance, and semantic search with embeddings from Large Language Models (LLMs) from different providers. As embedding models, we included models from OpenAI (text-embedding-ada-002, text-embedding-3-small, and text-embedding-3-large), Cohere (embed-multilingual-v2.0, and embed-multilingual-v3.0 with the subtypes search_document, search_query, classification, and clustering), and Gemini (embedding-001 with the subtypes semantic_similarity, retrieval_query, retrieval_document, classification, and clustering). An official thesaurus for ICPC-2 codes in Brazilian Portuguese was used to develop a search engine. It was made publicly available and shared through social media with primary care professionals for data collection. A total of 11,868 queries were collected, of which 7,671 (64.6%) were unique. A random sample of 437 unique expressions (5.7% of the unique queries) was annotated with ICPC-2 through peer review. Annotation involved selecting the relevant ICPC-2 codes for each query. After this process, 398 entries (5.2% of the unique queries) were included in the evaluation dataset. Precision at k ($P@k$) and average precision at k ($AP@k$) were used as evaluation metrics, computed for each query and averaged over all queries. The analysis was conducted in a sample both weighted and non-weighted for query frequency. Analysis of variance (ANOVA) one-way and Tukey's tests were used for hypothesis testing. Results: The evaluation dataset with 398 queries encoded with ICPC-2 was used to evaluate the results of the three different information retrieval algorithms and of the different models. Semantic search with embeddings from LLMs outperformed BM25 and Levenshtein distance in all assessed metrics: $P@1$, $P@5$, $P@10$, $AP@1$, $AP@5$, $AP@10$ (P value <0.001). When comparing different embedding models, the OpenAI model performed better than the others in most metrics. The OpenAI model text-embedding-3-large showed at least one relevant result in the top 10 in 85.7% of the queries in the non-weighted sample and in 81.4% of the queries in the weighted sample. The BM25 algorithm combined with query preprocessing performed similarly to the semantic search in the $AP@5$ and $AP@10$ metrics. Conclusions: Semantic search with embeddings from LLMs seems to perform better than BM25 and Levenshtein distance for retrieving ICPC-2 codes in the Brazilian Portuguese language. It is a promising approach for aiding healthcare professionals in medical coding. The BM25 algorithm combined with query preprocessing is an interesting option that can perform similarly to semantic search in some metrics, although it has some additional limitations. © AME Publishing Company.

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Archive Scopus

Language English

Volume 8

Publisher AME Publishing Company

Publication Journal of Medical Artificial Intelligence

ISSN 26172496 (ISSN)

Journal Abbr J. Med. Artif. Intell.

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Tags:

natural language processing, electronic health records (EHRs), International Classification of Primary Care (ICPC), ranking algorithms

Notes:

EID: 2-s2.0-105007453254

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: V.A. de Almeida; Departamento de Medicina Legal, Faculdade de Medicina, Universidade de São Paulo, São Paulo, Av. Dr. Arnaldo, 455-Cerqueira César, SP, 01246-903, Brazil; email: vinicius.almeida@alumni.usp.br

International Conference on Communication and Applied Technologies, ICOMTA 2025

Item Type Journal Article

Editor López-López P.C.

Editor Vernier M.

Editor Freundt-Thurne Ú.

Editor Ibáñez D.

Abstract The proceedings contain 49 papers. The special focus in this conference is on Communication and Applied Technologies. The topics include: Using Virtual Reality to Improve Visualization and Comprehension Skills in a Multivariable Calculus Course; a Real-Time Political Sentiment Analysis System Using DeepSeek-R1 on Multiplatform Social Media Data; robertiño: A Low-Cost Robot for Executing Large Language Models and Convolutional Networks to Enhance Early Education in Underserved Communities; impact of Visual Merchandising on Promotional Strategies: A Neuromarketing Analysis with Eye Tracking at the Point-of-Sale; regulatory Frameworks for Digital Information Transparency: A Comparative Analysis Between Ecuador and Spain; adoption of Augmented Reality in Digital Commerce: Effect of Experience and Technological Attitude on Centennials' Purchase Intention; china's Roadmap for Developing a Successful Innovation Ecosystem; Look-up, Use and Emotions (LUE) Methodology: Technological Hybridization for Active Learning in Higher Education; does the Level of Pre-service Teachers' Digital Competence in Teacher Digital Competence Influence the Acceptance of Using Educational Robotics?; Perception of Using ChatGPT in Academic Writing in Peruvian University Students; Educational Background, ICT Engagement and Media Literacy: Exploring Relationships Among Undergraduate Early Cohorts of Two Universities in Peru; evaluating University Students' Digital Competence Through Task-Based Performance Tests; boomers to Zoomers: Intergenerational Service-Learning in the Age of Media and Information Literacy;

integrating Technological Tools in Oral and Written Communication for Higher Education Through a Systematic Review; Virtual Reality as a Resource in Educational Methodologies: Experiences and Perceptions of English Students at Keep Going VR; Digital Transformation of Government Procedures: New Preferences for Mexicans in the Post-COVID Era?.

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Volume 458 SIST

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Publication Smart Innovation, Systems and Technologies

ISSN 21903018 (ISSN); 978-303209910-5 (ISBN)

Journal Abbr Smart Innov. Syst. Technol.

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Export Date: 20 September 2026; Cited By: 0; Conference name: International Conference on Communication and Applied Technologies, ICOMTA 2025; Conference date: 2 September 2025 through 4 September 2025; Conference code: 345249

International Conference on Computational Techniques and Applications, ICCTA 2021

Item Type Journal Article

Editor Mandal J.K.

Editor Hsiung P.

Editor Sankar Dhar R.

Abstract The proceedings contain 64 papers. The special focus in this conference is on Computational Techniques and Applications. The topics include: A Universal Dependency Treebank for Definitely Endangered Low-Resource Kangri Language; An Observation of Energy Density for PKL, Aloe Vera, Myrobalan, Lemon, and Tomato Electrochemical Cell; review on Various Techniques for Load Frequency Control in Deregulated Power Structures; investigations on the Impact of Soiling on Bifacial Gain; classification and Area Computation Modelling of Remote Sensing Images Using Histogram and Convolutional Neural Network; performance Analysis of Lead-Free Perovskite Solar Cells; analysis of Unsupervised Statistical Machine Translation Using Cross-Lingual Word Embedding for English–Hindi; a Ternary Sentiment Classification of Bangla Text Data using Support Vector Machine and Random Forest Classifier; third Wave Prediction Analysis for Kerala in India; preface; skin Cancer Detection Using Computer Vision; Comparative Study on Sentiment Analysis of Human Speech using DNN and CNN; RNN-Based Deep Learning Model for Generating Caption of an Image; object Detection in Railway Track using Deep Learning Techniques; An SVM-Based Approach to Predicting

Level of Job Anxiety in Corporate Professionals using Linguistic Markers on Twitter; markerless Location-Based Augmented Reality Application for Showcasing Deals; development of a Web-Based Corona Emergency Portal; coronavirus Detection Using Computer Vision; a Novel Distributed Database Architectural Model for Mobile Cloud Computing; dual-Band Stop Filter with Controllable Stop-Bands Based on Defect in Shunt Radial Stub; Calibration Techniques in ASIC and FPGA Based Time-to-Digital Converters; a Comparative Study of Machine Learning Algorithms for Anomaly-Based Network Intrusion Detection System; Dual-Element CPW-Fed MIMO Antenna for ISM Band Application.

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Interpretable stance detection in social media via topic-guided transformers

Item Type Journal Article

Author R. Muthusami

Author K. Saritha

Author K.S. Rao

Author P. Sugapriya

Author G. Saveetha

Abstract Stance detection in social media is a critical task in computational social science, supporting the analysis of ideological polarization, public opinion, and sociopolitical discourse. However, the brevity, noise, and contextual ambiguity of user-generated content pose significant challenges for traditional NLP models, while large language models (LLMs), despite their impressive zero-shot capabilities, often act as opaque black boxes with limited interpretability and domain adaptability. To overcome these limitations, we propose an interpretable hybrid framework that combines BERTopic for unsupervised semantic topic discovery with RoBERTa for sentiment-informed stance classification. Our approach explicitly fuses latent topical structure, sentiment polarity, and deep contextual embeddings, enabling both high predictive accuracy and transparent explanatory insights. Extensive evaluations on SemEval-2016 Task 6 and

the COVID-19 Stance Dataset demonstrate that our model achieves macro-F1 scores of 78.4% and 77.2%, surpassing competitive baselines such as TextCNN, BiLSTM-Attention, fine-tuned BERT, and CT-BERT. Topic coherence metrics (NPMI, UCI, UMass) further confirm BERTopic's superiority over Structural Topic Modeling (STM), underscoring its effectiveness in short-text settings. Beyond quantitative results, we enhance interpretability through topic-stance heatmaps, qualitative case studies, and a human-centered evaluation, while ablation analysis validates the contribution of each pipeline component. Finally, we discuss ethical and societal risks, cost considerations, and real-world deployment implications, and outline future directions including multilingual expansion, real-time stance monitoring, and human-in-the-loop integration. This work advances explainable social AI by bridging performance and interpretability in stance detection, making it a powerful tool for opinion mining, policy analysis, and misinformation tracking. © The Author(s) 2025.

Date 2025

DOI 10.1007/s44163-025-00635-9

URL <https://www.scopus.com/pages/publications/105023180685?origin=resultslist>

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Volume 5

Publisher Springer Nature

Publication Discover Artificial Intelligence

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Issue 1

Journal Abbr Discov. Artif. Intell.

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Behavioral research, Economic and social effects, Social aspects, Sentiment analysis, Human computer interaction, Social networking (online), Semantics, Topic Modeling, Social sciences computing, Social media analysis, Topic modeling, Interpretability, Opinion mining, Stance detection, Text mining, Bertopic, BERTopic, RoBERTa, Computational social science, Explainable NLP, Sentiment-aware classification

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EID: 2-s2.0-105023180685

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: R. Muthusami; Department of Computer Applications, Dr. Mahalingam College of Engineering and Technology, Pollachi, Tamil Nadu, India; email: r.muthusami@gmail.com

Interpretable zero-shot stance detection with proactive content intervention

Item Type Journal Article

Author A. Upadhyaya

Author W. Nejdl

Author M. Fisichella

Abstract Zero-Shot Stance Detection (ZSSD) identifies an author's stance towards unseen targets. Existing works have mainly focused on contrastive, meta, adversarial learning, or data augmentation but face issues like data scarcity, generalizability, and lack of coherence between text and targets. Moreover, stance detection must be interpretable to ensure transparency. Recent works with large language models (LLMs) aim to enhance unseen target knowledge or generate explanations but often rely excessively on explicit reasoning or provide coarse explanations, overlooking implicit cues and complicating interpretation. To address these challenges, we propose a novel interpretable multi-stage ZSSD framework. Stage 1 decodes explanations (rationales) justifying the stance while Stage 2 provides the final stance label, thus providing inherent interpretability in predicting stances. Extensive experiments prove that our approach outperforms other baselines with an average improvement in F1 scores of 27.99% with LLMs and 23.60% without LLMs for SemEval and 14.62% with LLMs and 25.24% without LLMs for VAST datasets for the ZSSD task, benefiting from the proposed pipeline architecture and interpretable design. Furthermore, to mitigate the harmful effects of offensive content and promote a more respectful online environment, we integrate an intervention module that leverages the contextual insights derived from our ZSSD framework with the ethics-based text generation power of LLMs to develop interventions. Automatic and human evaluation of LLM-generated interventions based on various proposed criteria provide insights into how LLMs perceive similar information from different perspectives, which can help foster morally sound and respectful online discourse. © 2025 The Authors

Date 2025

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Publication Information Processing and Management

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Issue 6

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Tags:

Language model, Large language model, Ethical aspects, Large language models, Interpretability, Adversarial learning, Data augmentation, Data scarcity, Decoding, Detection framework, Intervention, Rationale decoding, Zero-shot stance detection

Notes:

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Export Date: 20 September 2026; Cited By: 6; Correspondence Address: A. Upadhyaya; L3S Research Center, Leibniz University Hannover, Hannover, Appelstraße 9A, 30167, Germany; email: upadhyaya@l3s.de; CODEN: IPMAD

Intervention in Health Misinformation Using Large Language Models for Automated Detection, Thematic Analysis, and Inoculation: Case Study on COVID-19

Item Type Journal Article

Author S. Malek

Author C. Griffin

Author R.D. Fraleigh

Author R. Lennon

Author V. Monga

Author L. Shen

Abstract Background: The rapid growth of social media as an information channel has enabled the swift spread of inaccurate or false health information, significantly impacting public health. This widespread dissemination of misinformation has caused confusion, eroded trust in health authorities, led to noncompliance with health guidelines, and encouraged risky health behaviors. Understanding the dynamics of misinformation on social media is essential for devising effective public health communication strategies. Objective: This study aims to present a comprehensive and automated approach that leverages large language models (LLMs) and machine learning techniques to detect misinformation on social media, uncover the underlying causes and themes, and generate refutation arguments, facilitating control of its spread and promoting public health outcomes by inoculating people against health misinformation. Methods: We use 2 datasets to train 3 LLMs, namely, BERT, T5, and GPT-2, to classify documents into 2 categories: misinformation and nonmisinformation. In addition, we use a separate dataset to identify misinformation topics. To analyze these topics, we applied 3 topic modeling algorithms—Latent Dirichlet Allocation, Top2Vec, and BERTopic—and selected the optimal model based on performance evaluated across 3 metrics. Using a prompting approach, we extract sentence-level representations for the topics to uncover their underlying themes. Finally, we design a prompt text capable of identifying misinformation themes effectively. Results: The trained BERT model demonstrated exceptional performance, achieving 98% accuracy in classifying misinformation and nonmisinformation, with a 44% reduction in false-positive rates for artificial intelligence-generated misinformation. Among the 3 topic modeling approaches used, BERTopic outperformed the others, achieving the highest metrics with a Coherence Value of 0.41, Normalized Pointwise Mutual Information of −0.086, and Inverse Rank-Biased Overlap of 0.99. To address the issue of unclassified documents, we developed an algorithm to assign each document to its closest topic. In addition, we proposed a novel method using prompt engineering to generate sentence-level representations for each topic, achieving a 99.6% approval rate as “appropriate” or “somewhat appropriate” by 3 independent raters. We further designed a prompt text to identify themes of misinformation topics and developed another prompt capable of detecting misinformation themes with 82% accuracy. Conclusions: This study presents a

comprehensive and automated approach to addressing health misinformation on social media using advanced machine learning and natural language processing techniques. By leveraging LLMs and prompt engineering, the system effectively detects misinformation, identifies underlying themes, and provides explanatory responses to combat its spread. The proposed method was tested on an English language COVID-19–related dataset and has not been evaluated on real-world online social media data; the experiments were conducted offline. © Samira Malek, Christopher Griffin, Robert D Fraleigh, Robert Lennon, Vishal Monga, Lijiang Shen.

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large language models, Article, artificial intelligence, human, Humans, machine learning, interpersonal communication, Communication, ChatGPT, algorithm, Machine Learning, social media, language, controlled study, Social Media, natural language processing, topic modeling, sentiment analysis, thematic analysis, case study, deep neural network, COVID-19, benchmarking, coronavirus disease 2019, dimensionality reduction, disinformation, epidemiology, generative pretrained transformer, health behavior, health outcome, Language, large language model, Large Language Models, medical information, misinformation, performance indicator, prevalence, prompt engineering, SARS-CoV-2, Severe acute respiratory syndrome coronavirus 2, vaccination

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Investigating Code Generation Performance of ChatGPT with Crowdsourcing Social Data

Item Type Conference Paper

Author Y. Feng

Author S. Vanam

Author M. Cherukupally

Author W. Zheng

Author M. Qiu

Author H. Chen

Editor Shahriar H.

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Editor Majumder AKM.J.A.

Editor Kashiwazaki H.

Editor Yang J.-J.

Editor Takemoto M.

Editor Sakib N.

Editor Banno R.

Editor Ahamed S.I.

Abstract The recent advancements in Artificial Intelligence, particularly in large language models and generative models, are reshaping the field of software engineering by enabling innovative ways of performing various tasks, such as programming, debugging, and testing. However, few existing works have thoroughly explored the potential of AI in code generation and users' attitudes toward AI-assisted coding tools. This knowledge gap leaves it unclear how AI is transforming software engineering and programming education. This paper presents a scalable crowdsourcing data-driven framework to investigate the code generation performance of generative large language models from diverse perspectives across multiple social media platforms. Specifically, we utilize ChatGPT, a popular generative large language model, as a representative example to reveal its insights and patterns in code generation. First, we propose a hybrid keyword word expansion method that integrates words suggested by topic modeling and expert knowledge to filter relevant social posts of interest on Twitter and Reddit. Then we collect 316K tweets and 3.2K Reddit posts about ChatGPT's code generation, spanning from Dec. 1, 2022 to January 31, 2023. Our data analytics show that ChatGPT has been used in more than 10 programming languages, with Python and JavaScript being the two most popular, for a diverse range of tasks such as code debugging, interview preparation, and academic assignment solving. Surprisingly, our analysis shows that fear is the dominant emotion associated with ChatGPT's code generation, overshadowing emotions of happiness, anger, surprise, and sadness. Furthermore, we construct a ChatGPT prompt and corresponding code dataset by analyzing the screen-shots of ChatGPT code generation shared on social media. This dataset enables us to evaluate the quality of the generated code, and we have released this dataset to the public. We believe the insights gained from our work will provide valuable guidance for future research on AI-powered code generation. © 2023 IEEE.

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Export Date: 20 September 2026; Cited By: 119; Correspondence Address: Y. Feng; University of North Texas, United States; email: yunhe.feng@unt.edu; Conference name: 47th Annual IEEE Computers, Software, and Applications Conference, COMPSAC 2023; Conference date: 26 June 2023 through 30 June 2023; Conference code: 191323; CODEN: PSICD

Investigating the Classification of Living Kidney Donation Experiences on Reddit and Understanding the Sensitivity of ChatGPT to Prompt Engineering: Content Analysis

Item Type Journal Article

Author J. Nielsen

Author X. Chen

Author L. Davis

Author A. Waterman

Author M. Gentili

Abstract Background: Living kidney donation (LKD), where individuals donate one kidney while alive, plays a critical role in increasing the number of kidneys available for those experiencing kidney failure. Previous studies show that many generous people are interested in becoming living donors; however, a huge gap exists between the number of patients on the waiting list and the number of living donors yearly. Objective: To bridge this gap, we aimed to investigate how to identify potential living

donors from discussions on public social media forums so that educational interventions could later be directed to them. Methods: Using Reddit forums as an example, this study described the classification of Reddit content shared about LKD into three classes: (1) present (presently dealing with LKD personally), (2) past (dealt with LKD personally in the past), and (3) other (LKD general comments). An evaluation was conducted comparing a fine-tuned distilled version of the Bidirectional Encoder Representations from Transformers (BERT) model with inference using GPT-3.5 (ChatGPT). To systematically evaluate ChatGPT's sensitivity to distinguishing between the 3 prompt categories, we used a comprehensive prompt engineering strategy encompassing a full factorial analysis in 48 runs. A novel prompt engineering approach, dialogue until classification consensus, was introduced to simulate a deliberation between 2 domain experts until a consensus on classification was achieved. Results: BERT and GPT-3.5 exhibited classification accuracies of approximately 75% and 78%, respectively. Recognizing the inherent ambiguity between classes, a post hoc analysis of incorrect predictions revealed sensible reasoning and acceptable errors in the predictive models. Considering these acceptable mismatched predictions, the accuracy improved to 89.3% for BERT and 90.7% for GPT-3.5. Conclusions: Large language models, such as GPT-3.5, are highly capable of detecting and categorizing LKD-targeted content on social media forums. They are sensitive to instructions, and the introduced dialogue until classification consensus method exhibited superior performance over stand-alone reasoning, highlighting the merit in advancing prompt engineering methodologies. The models can produce appropriate contextual reasoning, even when final conclusions differ from their human counterparts. © Joshua Nielsen, Xiaoyu Chen, LaShara Davis, Amy Waterman, Monica Gentili.

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prompt engineering, generative artificial intelligence, kidney donation, living donor, transplant

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Investigating the Impact of Language-Adaptive Fine-Tuning on Sentiment Analysis in Hausa Language Using AfriBERTa

Item Type Conference Paper

Author S.A. Sani

Author S.H. Muhammad

Author D. Jarvis

Editor Hettiarachchi H.

Editor Ranasinghe T.

Editor Rayson P.

Editor Mitkov R.

Editor Gaber M.

Editor Premasiri D.

Editor Tan F.A.

Editor Uyangodage L.

Abstract Sentiment analysis (SA) plays a vital role in Natural Language Processing (NLP) by identifying sentiments expressed in text. Although significant advances have been made in SA for widely spoken languages, low-resource languages such as Hausa face unique challenges, primarily due to a lack of digital resources. This study investigates the effectiveness of Language-Adaptive Fine-Tuning (LAFT) to improve SA performance in Hausa. We first curate a diverse, unlabeled corpus to expand the model's linguistic capabilities, followed by applying LAFT to adapt AfriBERTa specifically to the nuances of the Hausa language. The adapted model is then fine-tuned on the labeled NaijaSenti sentiment dataset to evaluate its performance. Our findings demonstrate that LAFT gives modest improvements, which may be attributed to the use of formal Hausa text rather than informal social media data. Nevertheless, the pre-trained AfriBERTa model significantly outperformed models not specifically trained on Hausa, highlighting the importance of using pre-trained models in low-resource contexts. This research emphasizes the necessity for diverse data sources to advance NLP applications for low-resource African languages. We will publish the code and the data set to encourage further research and facilitate reproducibility in low-resource NLP. © 2025 Association for Computational Linguistics.

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Investigating the Impact of Multimodality and External Knowledge in Aspect-level Complaint and Sentiment Analysis

Item Type Conference Paper

Author A. Singh

Author A. Verma

Author R. Jain

Author S. Saha

Abstract Automated complaint analysis is vital for generating critical insights, which in turn enhance customer satisfaction, product quality, and overall business performance. Nevertheless, conventional methods frequently fail to capture the nuances of aspect-level complaints and inadequately utilize external knowledge, thus creating a gap in effective complaint detection and analysis. In response to this issue, we proactively explore the role of external knowledge and multimodality in this domain. This leads to the development of MGasD (Multimodal Generative framework for aspect-based complaint and sentiment Detection), a multimodal knowledge-infused unified framework. MGasD diverges from traditional methods by reframing the complaint detection problem as a multimodal text-to-text generation task. Significantly, our research includes the development of a novel aspect-level dataset. Annotated for both complaint and sentiment categories across diverse domains such as books, electronics, edibles, fashion, and miscellaneous, this dataset provides a comprehensive

platform for the concurrent study of complaints and sentiment. This resource facilitates a more robust understanding of consumer feedback. Our proposed methodology establishes a benchmark performance in the novel aspect-based complaint and sentiment detection tasks based on extensive evaluation. We also demonstrate that our model consistently outperforms all other baselines and state-of-the-art models in both full and few-shot settings. © 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Investigating the Impact of Noise Injection on BERT for Offensive Arabic Text Detection

Item Type Conference Paper

Author K.E.A. Abdulgawad

Author N. Ben Saud

Editor Albaji A.O.

Abstract Large language models are being trained on big data mostly gathered from the Internet, where not all this data are true and correct data. There is a lot of false and misleading data, and the percentage of these false data increases in social media content. This paper investigates the influence of adding noise for Arabic social media comments content on the accuracy and classification results of BERT model which is one of the first and famous LLM models. Our study utilizes a BERT-based classification model trained on a real-world comment dataset to classify the sentiment. The model's performance has been analyzed and the sensitivity to noise across different noise percentages to the dataset, exploring the impact of noisy data on the performance of the BERT model, including its sensitivity to false or inaccurate data, and analyzing how the effects of noisy data differ between balanced and imbalanced datasets. We employ various evaluation metrics, like accuracy, precision, recall, and F1 score, to assess the model's performance. The findings from our research offer insights into the performance and noise sensitivity of BERT in text classification tasks. Our findings demonstrate that BERT is generally resistant to noise at low-medium levels, however on higher noise injection level the model performance affected negatively. Also, we found that imbalanced datasets are particularly sensitive to false positives due to the majority of negative examples, while balanced datasets sustained similar impact on both false positives and false negatives. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Investigating the impact of sentiments on stock market using digital proxies: Current trends, challenges, and future directions

Item Type Journal Article

Author T. Gupta

Author S. Devji

Author A.K. Tripathi

Abstract Social media and online news have emerged as significant sources of market sentiment, influencing stock market dynamics globally. With the growing availability of digital data, the current research focus is on leveraging advanced computational techniques for sentiment-driven stock market prediction. The era of financial forecasting has been revolutionized by integrating cutting-edge technologies such as Machine Learning, Deep Learning, and Large Language Models. In this paper, a comprehensive survey of 108 research articles has been undertaken to explore the recent advancements in these technologies, with a focus on utilizing sentiment data extracted from social media platforms and news sources. The technology-wise state-of-the-art findings, current trends, challenges, and literature gaps in this domain are analyzed, and potential future directions are proposed to address these gaps. Additionally, publicly available benchmark datasets for social media and news sentiment indices are compiled and analyzed, with insights into their limitations and potential improvements. A comparative evaluation of prediction methods across heterogeneous user-generated datasets is performed, identifying the most effective techniques for various data types and problem formulations. Recommendations are offered for selecting suitable techniques based on the nature of the data and the specific problem formulation. By incorporating the latest advancements in the field of sentiment analysis and stock market prediction, this work provides actionable insights for researchers and practitioners, advancing the understanding and development of sentiment-driven financial forecasting. © 2025 Elsevier Ltd

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Item Type Conference Paper

Author M. Stanley

Author S. Datta

Author A. Kumar

Author A.R. Khudabukhsh

Editor Koenig S.

Editor Jenkins C.

Editor Taylor M.E.

Abstract A significant gap exists in datasets regarding post-COVID-19 vaccination experiences, particularly “vaccine buyer’s re-morse”. Understanding the prevalence and nature of vaccine regret, whether based on personal or vicarious experiences, is vital for addressing vaccine hesitancy and refining public health communication. In this paper, we curate a novel dataset from a large YouTube news corpus capturing COVID-19 vaccination experiences, and construct a benchmark subset focused on vaccine regret, annotated by a politically diverse panel to account for the subjective and often politicized nature of the topic. We utilize large language models (LLMs) to identify posts expressing vaccine regret, analyze the reasons behind this regret, and quantify its occurrence in both first and second-person accounts. This paper aims to (1) quantify the prevalence of vaccine regret; (2) identify common reasons for this sentiment; (3) analyze differences between first-person and vicarious experiences; and (4) assess potential biases introduced by different LLMs. We find that while vaccine buyer’s remorse appears in only $< 2\%$ of public discourse, it is disproportionately concentrated in vaccine-skeptic influencer communities and is predominantly expressed through first-person narratives citing adverse health events. © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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April 2023. The fine-tuned LLM model clearly outperforms the base LLM model as well as a bidirectional encoder representations from transformers (BERT) model combination, which was chosen as a benchmark and comprises entity recognition from the BERT base model and sentiment analysis from FinBERT. Classification of LLM sentence embeddings with a novel approach using uniform manifold approximation and projection (UMAP) dimensionality reduction and hierarchical density-based spatial clustering of applications with noise (HDBSCAN) also generates investment signals that outperform the S&P 500 index on a risk-adjusted basis over the period from June 2018 to June 2023. All investment strategies are demonstrated to be unique by means of a standard regression against the Fama–French investment factors. ©Infopro Digital Limited 2025. All rights reserved. © 2025 Infopro Digital Risk (IP) Limited.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: D. Romoff; Enterprise Risk Management, School of Professional Studies, Columbia University, New York, 303 Lewisohn Hall, 10027, United States; email: [dj2132@columbia.edu](mailto:djr2132@columbia.edu)

IPATH: A Large-Scale Pathology Image–Text Dataset from Instagram for Vision–Language Model Training

Item Type Journal Article

Author S. Mirhosseini

Author T. Rai
Author P. Diaz-Santana
Author R. La Ragione
Author N. Bacon
Author K. Wells

Abstract Recent advancements in artificial intelligence (AI) have revealed important patterns in pathology images imperceptible to human observers that can improve diagnostic accuracy and decision support systems. However, progress has been limited due to the lack of publicly available medical images. To address this scarcity, we explore Instagram as a novel source of pathology images with expert annotations. We curated the IPATH dataset from Instagram, comprising 45,609 pathology image–text pairs rigorously filtered and curated for domain quality using classifiers, large language models, and manual filtering. To demonstrate the value of this dataset, we developed a multimodal AI model called IP-CLIP by fine-tuning a pretrained CLIP model using the IPATH dataset. We evaluated IP-CLIP on seven external histopathology datasets using zero shot classification and linear probing, where it consistently outperformed the original CLIP model. Furthermore, IP-CLIP matched or exceeded several recent state-of-the-art pathology vision–language models, despite being trained on a substantially smaller dataset. We also assessed image–text alignment on a 5k held-out IPATH subset using image–text retrieval, where IP-CLIP surpassed CLIP and other specialized models. These results demonstrate the effectiveness of the IPATH dataset and highlight the potential of leveraging social media data to develop AI models for medical image classification and enhance diagnostic accuracy. © The Author(s) under exclusive licence to Society for Imaging Informatics in Medicine 2026.

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Irony and Sarcasm Detection in Turkish Texts: A Comparative Study of Transformer-Based Models and Ensemble Learning

Item Type Journal Article**Author** M. Eser**Author** M. Bilgin

Abstract Irony and sarcasm are forms of expression that emphasize the inconsistency between what is said and what is meant. Correctly classifying such expressions is an important text mining problem, especially on user-centered platforms such as social media. Due to the increasing prevalence of implicit expressions, this topic has become a significant area of research in Natural Language Processing (NLP). However, the simultaneous detection of ironic and sarcastic expressions is highly challenging, as both types of implicit sentiments often convey closely related meanings. To address the detection of irony and sarcasm, this study compares the performance of transformer-based models and an ensemble learning method on Turkish texts, using five textual datasets—monogram, bigram, trigram, quadrigram, and omnigram—that share the same textual content but differ in context length. To improve classification performance, an ensemble learning approach based on the Artificial Rabbit Optimization (ARO) algorithm was implemented, combining the outputs of the models to produce final predictions. The experimental results indicate that as the context width of the datasets increases, the models achieve better predictions, leading to improvements across all performance metrics. The ensemble learning method outperformed individual models in all metrics, with performance increasing as the context expanded, achieving the highest success in the omnigram dataset with 76.71% accuracy, 74.64% precision, 73.29% sensitivity, and 73.96% F-Score. This study demonstrates that both model architecture and data structure are decisive factors in text classification performance, showing that community methods can make significant contributions to the effectiveness of deep learning solutions in low-resource languages. © 2025 by the authors.

Date 2025**DOI** 10.3390/app152312498**URL** <https://www.scopus.com/pages/publications/105024904283?origin=resultslist>**Archive** Scopus**Language** English**Volume** 15**Publisher** Multidisciplinary Digital Publishing Institute (MDPI)**Publication** Applied Sciences (Switzerland)**ISSN** 20763417 (ISSN)**Issue** 23**Journal Abbr** Appl. Sci.

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“It’s a complicated issue”: exploring perspectives on Reddit from 2016 to 2024 about the American Psychiatric Association’s Goldwater Rule and the ethics of public-professional mental health commentary

Item Type Journal Article

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Author A. Ventriglio

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Abstract Introduction: The American Psychiatric Association’s (APA) Goldwater Rule prohibits its member-psychiatrists from offering professional opinions on an individual’s mental health without direct evaluation or consent. While prior work has accentuated expert perspectives, little is known about how the public interprets and evaluates this professional guideline, despite ongoing discussions about its applicability in contemporary psychiatry. Accordingly, this study addresses this knowledge gap by exploring public discourse about the Rule on Reddit from 2016 to 2024, a period when sociopolitical and technological developments intensified debates about its practical and ethical relevance. Methods: A dataset comprising 4,679 records referencing the Goldwater Rule from June 2016 to November 2024 was derived from Reddit’s Application Programming Interface via a Python script and relevant keywords. Subsequently, these were annotated and grouped into four coarse-grained categories (i.e., “Positive”, “Neutral”, “Negative” and “Irrelevant”) using a multi-phase human and Large Language Model-driven review process. After filtering out “Irrelevant” content, this yielded a corpus of 566 “Positive”, “Negative” and

“Neutral” Reddit comments, from which recurring discursive patterns were then thematically analyzed. Results: Throughout the observation period, “Positive” sentiments (42.23%) constituted the most common singular category by content volume. Specifically, users cited the Rule’s role in upholding psychiatric credibility and protecting against potential abuses. In turn, “Neutral” responses (36.57%) represented a nuanced spectrum of ambivalence, suggested modifications or clarifications, and reiterations of the status quo, with most unique users contributing to this category. Notably, “Positive” and “Negative” (21.20%) content recurrently surfaced in political dialogues and was particularly polarized surrounding President Donald Trump; here, users portrayed the Rule as a necessary safeguard against partisanship or, alternatively, as outdated and overly restricting psychiatric input into societal debates. More broadly, many Redditors discussed the Rule’s practical boundaries in areas such as digital media and telehealth. Conclusions: Although the Rule retained much “Positive” support, the combined prevalence of “Neutral” and “Negative” sentiments highlights the varied (and often contested) public interpretations of this ethical policy, mirroring similar debates within the psychiatric discipline. Ultimately, through wider engagement, the APA may benefit from appraising how the Rule can remain attentive to societal expectations in an era of digital advances and sociopolitical divisions. © The Author(s) 2026.

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Iterative Large Language Model–Guided Sampling and Expert-Annotated Benchmark Corpus for Harmful Suicide Content Detection: Development and Validation Study

Item Type Journal Article

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Abstract Background: Harmful suicide content on the internet poses significant risks, as it can induce suicidal thoughts and behaviors, particularly among vulnerable populations. Despite global efforts, existing moderation approaches remain insufficient, especially in high-risk regions such as South Korea, which has the highest suicide rate among Organisation for Economic Co-operation and Development countries. Previous research has primarily focused on assessing the suicide risk of the authors who wrote the content rather than the harmfulness of content itself which potentially leads the readers to self-harm or suicide, highlighting a critical gap in current approaches. Our study addresses this gap by shifting the focus from assessing the suicide risk of content authors to evaluating the harmfulness of the content itself and its potential to induce suicide risk among readers. Objective: This study aimed to develop an artificial intelligence (AI)–driven system for classifying online suicide-related content into 5 levels: illegal, harmful, potentially harmful, harmless, and non–suicide-related. In addition, the researchers construct a multimodal benchmark dataset with expert annotations to improve content moderation and assist AI models in detecting and regulating harmful content more effectively. Methods: We collected 43,244 user-generated posts from various online sources, including social media, question and answer (Q&A) platforms, and online communities. To reduce the workload on human annotators, GPT-4 was used for preannotation, filtering, and categorizing content before manual review by medical professionals. A task description document ensured consistency in classification. Ultimately, a benchmark dataset of 452 manually labeled entries was developed, including both Korean and English versions, to support AI-based moderation. The study also evaluated zero-shot and few-shot learning to determine the best AI approach for detecting harmful content. Results: The multimodal benchmark dataset showed that GPT-4 achieved the highest F1-scores (66.46 for illegal and 77.09 for harmful content detection). Image descriptions improved classification accuracy, while directly using raw images slightly decreased performance. Few-shot learning significantly enhanced detection, demonstrating that

small but high-quality datasets could improve AI-driven moderation. However, translation challenges were observed, particularly in suicide-related slang and abbreviations, which were sometimes inaccurately conveyed in the English benchmark. Conclusions: This study provides a high-quality benchmark for AI-based suicide content detection, proving that large language models can effectively assist in content moderation while reducing the burden on human moderators. Future work will focus on enhancing real-time detection and improving the handling of subtle or disguised harmful content. © Kyumin Park, Myung Jae Baik, YeongJun Hwang, Yen Shin, HoJae Lee, Ruda Lee, Sang Min Lee, Je Young Hannah Sun, Ah Rah Lee, Si Yeun Yoon, Dong-ho Lee, Jihyung Moon, JinYeong Bak, Kyunghyun Cho, Jong-Woo Paik, Sungjoon Park.

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Japanese-Language AI Agent System for Human Papillomavirus Vaccine Infection and Public Communication: Development and Feasibility Evaluation

Item Type Journal Article

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Author D. Ma

Author Q. Niu

Author Z. Zhang

Author M. Nagai-Tanima

Author T. Aoyama

Abstract Background: Human papillomavirus (HPV) vaccine hesitancy remains a significant public health challenge in Japan, where proactive vaccination recommendations were suspended between 2013 and 2021. The resulting information gap between medical institutions and vaccine-hesitant populations is exacerbated by misinformation on social media platforms. Traditional public health communication strategies cannot address individual queries while simultaneously monitoring population-level discourse. Objective: This study aimed to develop and conduct a feasibility evaluation of a dual-purpose artificial intelligence agent system that delivers verified HPV vaccine information to the public through a conversational interface while generating infoveillance reports for medical institutions based on user interactions and social media discourse. Methods: We implemented a system with 3 components: a vector database integrating 139,803 documents, including academic papers, Japanese government sources, news media, and social media posts; a retrieval-augmented generation chatbot using a ReAct agent architecture with iterative multitool orchestration across 5 specialized knowledge sources; and an automated report generation system with modules for news analysis, research synthesis, social media sentiment analysis, including stance classification and topic modeling, and user interaction pattern identification. System performance was assessed using both automated and manual evaluation protocols on a scale from 0 to 5. Results: The entire system functioned as expected. For single-turn evaluation, the chatbot achieved mean scores of 4.83 (SD 0.67; 95% CI 4.71-4.93) for relevance, 4.89 (SD 0.53; 95% CI 4.79-4.97) for routing, 4.50 (SD 1.29; 95% CI 4.27-4.70) for reference quality, 4.90 (SD 0.62; 95% CI 4.78-4.99) for correctness, and 4.88 (SD 0.54; 95% CI 4.78-4.96) for professional identity, with an overall mean of 4.80. Multiturn evaluation yielded higher mean scores: 4.94 for context memory (SD 0.25; 95% CI 4.84-5.00) and an overall mean of 4.98, with topic centering and identity achieving 5.00. The report generation system achieved high scores across all sections: 4.83 for completeness (SD 0.37; 95% CI 4.73-4.94), 4.88 for correctness (SD 0.33; 95% CI 4.77-4.96), and 4.12 for helpfulness (SD 0.48; 95% CI 3.98-4.27). Reference validity achieved perfect scores (5.00) across all periods, with citation correctness averaging 4.21 (SD 0.58; 95% CI 3.96-4.46). Conclusions: This feasibility study demonstrated that an integrated artificial intelligence agent system can support both public HPV vaccine communication and social media infoveillance in a Japanese-language context. Prospective deployment with real users is needed to assess actual public health impact. © Junyu Liu, Siwen Yang, Dexiu Ma, Qian Niu, Zequn Zhang, Momoko Nagai-Tanima, Tomoki Aoyama. Originally published in JMIR Infodemiology (<https://infodemiology.jmir.org>), 21.May.2026. This is an open-access article distributed under the terms of the Creative Commons Attribution License (<https://creativecommons.org/licenses/by/4.0/>), which permits unrestricted use, distribution, and reproduction in any medium, provided the original work, first published in JMIR Infodemiology, is properly cited. The complete bibliographic information, a link to the original publication on <https://infodemiology.jmir.org/>, as well as this copyright and license information must be included.

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Just a Scratch: Enhancing LLM Capabilities for Self-harm Detection through Intent Differentiation and Emoji Interpretation

Item Type Conference Paper

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Author G.V. Singh

Author S. Choudhury

Author A. Ekbal

Editor Che W.

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Abstract Self-harm detection on social media is critical for early intervention and mental health support, yet remains challenging due to the subtle, context-dependent nature of such expressions. Identifying self-harm intent aids suicide prevention by enabling timely responses, but current large language models (LLMs) struggle to interpret implicit cues in casual language and emojis. This work enhances LLMs' comprehension of self-harm by distinguishing intent through nuanced language-emoji interplay. We present the Centennial Emoji Sensitivity Matrix (CESM-100)-a curated set of 100 emojis with contextual self-harm interpretations-and the Self-Harm Identification aNd intent Extraction with Supportive emoji sensitivity (SHINES) dataset, offering detailed annotations for

self-harm labels, casual mentions (CMs), and serious intents (SIs). Our unified framework: a) enriches inputs using CESM-100; b) fine-tunes LLMs for multi-task learning-self-harm detection (primary) and CM/SI span detection (auxiliary); c) generates explainable rationales for self-harm predictions. We evaluate the framework on three state-of-the-art LLMs-Llama 3, Mental-Alpaca, and MentalLlama-across zero-shot, few-shot, and fine-tuned scenarios. By coupling intent differentiation with contextual cues, our approach commendably enhances LLM performance in both detection and explanation tasks, effectively addressing the inherent ambiguity in self-harm signals. The SHINES dataset, CESM-100 and codebase are publicly available at: <https://www.iitp.ac.in/%7eai-nlp-ml/resources.html#SHINES>. © 2025 Association for Computational Linguistics.

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KALD: A Knowledge Augmented multi-contrastive learning model for low resource abusive Language Detection

Item Type Journal Article

Author R. Song

Author F. Giunchiglia

Author Y. Li

Author J. Li

Author J. Wang

Author H. Xu

Abstract Warning: This paper contains insulting statements that may cause discomfort for readers. With the development of online social media, quite a few methods focus on automatic Abusive Language Detection (ALD), which requires numerous annotations as the basis for reliable classifier training. However, the labor-intensive, expensive, and time-consuming data labeling process brings difficulties to the acquisition of the annotations. Although some studies have improved the model performance in the absence of labeled data by studying cross-domain generalization and semi-supervised learning, there is still a lack of specific research on making full use of prior knowledge to improve detection effectiveness in the context of limited resources. To solve this problem, we propose a Knowledge Augmented abusive Language Detection framework (KALD), to fully utilize three kinds of prior knowledge: lexical knowledge, sample knowledge, and category knowledge. First, lexicon knowledge is injected into the language model to promote its focus on abusive keyword by context reconstruction. Meanwhile Lexicon-based data augmentation is used to obtain reasonable positive samples necessary for contrastive learning. Subsequently Joint optimization of multi-contrastive learning is applied to encourage language models to learn stable sample-level and in-class representations. The following tasks are performed on the four public datasets to verify the validity of the proposed method (a) ALD (b) semi-supervised ALD And (c) cross-domain abusive language generalization. For semi-supervised ALD, the proposed framework has an average improvement of 2.19% with different sample size settings compared to the most advanced baseline approach and 3.58% compared to the basic language model. For cross-domain abusive language generalization, the proposed framework has an average improvement of 2.58% and 3.42% compared with the most advanced baseline approach and the basic language model, separately. © 2025 Elsevier B.V.

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KE-MLLM: A Knowledge-Enhanced Multi-Sensor Learning Framework for Explainable Fake Review Detection

Item Type Journal Article

Author J. Chen

Author J. Liu

Author Y. Liang

Author M. Zhou

Abstract The proliferation of fake reviews on e-commerce and social platforms has severely undermined consumer trust and market integrity, necessitating robust and interpretable real-time detection mechanisms with multi-sensor data fusion capabilities. While traditional machine learning approaches have shown promise in identifying fraudulent reviews, they often lack transparency and fail to leverage the rich contextual knowledge embedded in large-scale datasets. In this paper, we propose KE-MLLM (Knowledge-Enhanced Multimodal Large Language Model), a unified framework that integrates knowledge-enhanced prompting with parameter-efficient fine-tuning for explainable fake review detection. Our approach employs LoRA (Low-Rank Adaptation) to fine-tune lightweight large language models (LLaMA-3-8B) on review text, while incorporating multimodal behavioral sensor signals including temporal patterns, user metadata, and social network characteristics for comprehensive anomaly sensing. To address the critical need for interpretability in fraud detection systems, we implement a Chain-of-Thought (CoT) reasoning module that generates human-understandable explanations for classification decisions, highlighting linguistic anomalies, sentiment inconsistencies, and behavioral red flags. We enhance the model's discriminative capability through a knowledge distillation strategy that transfers domain-specific expertise from larger teacher models while maintaining computational efficiency suitable for edge sensing devices. Extensive experiments on two benchmark datasets—YelpChi and Amazon Reviews from the DGL Fraud Dataset—show that KE-MLLM achieves strong performance, reaching an F1-score of 94.3% and an AUC-ROC of 96.7% on YelpChi and outperforming the strongest baseline in our comparison by 5.8 and 4.2 percentage points, respectively. Furthermore, human evaluation indicates that the generated explanations achieve 89.5% consistency with expert annotations, suggesting that the framework can improve the interpretability and practical usefulness of automated fraud detection systems. The proposed framework provides a useful step toward more accurate and interpretable fake review detection and offers a practical reference for building more transparent and accountable AI systems in high-stakes applications. © 2026 by the authors.

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KITA: Knowledge-Infused Targeted Augmentation for Zero-Shot Stance Detection

Item Type Journal Article
Author S. Wang
Author Y. Jin
Author R. Yan

Abstract Zero-shot stance detection (ZSSD) focuses on effectively transferring stance recognition models from source to unseen target domains, especially when target domain data are scarce or absent. Existing research relies predominantly on contrastive learning strategies to enhance model generalization or incorporates external knowledge resources to compensate for semantic-information deficits. However, these approaches often suffer from two critical issues: (1) the inherent high sparsity and heterogeneity of social-media data and (2) insufficient attention to domain relevance and implicit semantic targets within user-generated content. To address these limitations, this paper proposes a data-augmentation framework based on internal knowledge expansion to improve model performance in ZSSD tasks. Within the framework, an implicit target-mining model is designed to discover relevant instances from the training data that exhibit explicit or implicit semantic tendencies similar to the target sample with an unsupervised scheme, providing contextual supplementation and enhancing target representation capabilities. Meanwhile, a stance-classification module integrated with a semantic filtering mechanism can effectively suppress the interference of irrelevant or noisy information on stance determination to ensure the augmented information can be used effectively. Extensive experiments on three mainstream benchmark datasets demonstrate that the proposed framework achieves superior accuracy and robustness compared to existing state-of-the-art methods, thereby validating its effectiveness and generalizability for ZSSD tasks. © 2026 World Scientific Publishing Company.

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Knowledge-Augmented Interpretable Network for Zero-Shot Stance Detection on Social Media

Item Type Journal Article

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Author Z. Huang

Author A. Li

Author Y. Li

Author B. Zhang

Author H. Huang

Abstract Stance detection on social media has become increasingly important for understanding public opinions on controversial issues. Existing methods often require large amounts of labeled data to learn target-independent transferable knowledge, which is infeasible under zero-shot settings where the target is unseen. Furthermore, most current stance detection models, primarily based on end-to-end deep learning architectures, lack transparency and may produce counter-intuitive and uninterpretable predictions. In this article, we propose a novel knowledge-augmented interpretable network (KAI) to enable zero-shot stance detection (ZSSD). First, we introduce an unsupervised approach based on large language models (LLM-KE) to elicit analysis perspectives, which is target-independent knowledge shared across different targets. This transferable knowledge bridges connections between seen and unseen targets. Second, we develop a bidirectional knowledge-guided neural production system (Bi-KGNPS) that effectively integrates such transferable knowledge through an iterative knowledge-variable binding process to guide stance predictions. Extensive experiments on benchmark datasets demonstrate KAI achieves new state-of-the-art performance on ZSSD. Moreover, our approach also delivers strong results on conventional in-target and cross-target stance detection. With the dual benefits of knowledge-augmented accuracy and model interpretability, this work represents an important advance toward practical stance detection systems that can generalize to emerging topics of interest. The proposed KAI framework provides an interpretable approach to effectively transfer knowledge across domains for zero-shot learning. © 2014 IEEE.

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Knowledge-Aware Neural Networks for Medical Forum Question Classification

Item Type Conference Paper**Author** S. Roy**Author** S. Chakraborty**Author** A. Mandal**Author** G. Balde**Author** P. Sharma**Author** A. Natarajan**Author** M. Khosla**Author** S. Sural**Author** N. Ganguly

Abstract Online medical forums have become a predominant platform for answering health-related information needs of consumers. However, with a significant rise in the number of queries and the limited availability of experts, it is necessary to automatically classify medical queries based on a consumer's intention, so that these questions may be directed to the right set of medical experts. Here, we develop a novel medical knowledge-aware BERT-based model (MedBERT) that

explicitly gives more weightage to medical concept-bearing words, and utilize domain-specific side information obtained from a popular medical knowledge base. We also contribute a multi-label dataset for the Medical Forum Question Classification (MFQC) task. MedBERT achieves state-of-the-art performance on two benchmark datasets and performs very well in low resource settings. © 2021 ACM.

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KonkaniFood: A Code-Mixed Marathi-English Dataset for Sentiment Analysis Using Multilingual Transformer-Based Models

Item Type Conference Paper

Author A. Phadke

Author S. Kohle

Author P. Gupta

Abstract Given the increasing use of multilingual content on social media, there is a growing need for effective multilingual sentiment analysis tools for code-mixed languages. This paper introduces KonkaniFood, the publicly available dataset for Marathi-English code-mixed text, specifically focusing on user-generated YouTube comments about Konkani food. This dataset includes 1,039 comments manually annotated as positive, negative, and neutral. To assess its usefulness, we fine-tuned three pre-trained multilingual transformer models - IndicBERT, mBERT, and MuRIL - for sentiment classification. In our experiments, the MuRIL model performed best with 79% accuracy. We also demonstrated the accuracy of the transformer-based multi-language model in sentiment analysis using precision, recall, F1-score, the confusion matrix, and a detailed comparison of training-validation loss curves, etc. This research provides a valuable resource and benchmark for advancing sentiment analysis in low-resource, code-mixed Indian languages. © 2025 IEEE.

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Codes (symbols), Classification (of information), Sentiment analysis, Social media, Data mining, Sentiment Analysis, User-generated, YouTube, Computer hardware description languages, MuRIL, Indian languages, Positive/negative, Analysis tools, Code mixed, Code Mixed, Marathi-english, Marathi-English

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L3Cube-HingCorpus and HingBERT: A Code Mixed Hindi-English Dataset and BERT Language Models

Item Type Conference Paper

Author R. Nayak

Author R. Joshi

Editor Jha G.N.

Editor Devi S.L.

Editor Bali K.

Editor Ojha A.Kr.

Abstract Code-switching occurs when more than one language is mixed in a given sentence or a conversation. This phenomenon is more prominent on social media platforms and its adoption is increasing over time. Therefore code-mixed NLP has been extensively studied in the literature. As pre-trained transformer-based architectures are gaining popularity, we observe that real code-mixing data are scarce to pre-train large language models. We present L3Cube-HingCorpus, the first large-scale real Hindi-English code mixed data in a Roman script. It consists of 52.93M sentences and 1.04B tokens, scraped from Twitter. We further present HingBERT, HingMBERT, HingRoBERTa, and HingGPT. The BERT models have been pre-trained on codemixed HingCorpus using masked language modelling objectives. We show the effectiveness of these BERT models on the subsequent downstream tasks like code-mixed sentiment analysis, POS tagging, NER, and LID from the GLUECoS benchmark. The HingGPT is a GPT2 based generative transformer model capable of generating full tweets. Our models show significant improvements over currently available models pre-trained on multiple languages and synthetic code-mixed datasets. We also release L3Cube-HingLID Corpus, the largest code-mixed Hindi-English language identification(LID) dataset and HingBERT-LID, a production-quality LID model to facilitate capturing of more code-mixed data using the process outlined in this work. The dataset and models are available at <https://github.com/l3cube-pune/code-mixed-nlp>. © European Language Resources Association (ELRA), licensed under CC-BY-NC-4.0.

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URL <https://www.scopus.com/pages/publications/85145960369?origin=resultslist>

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Language English

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Publisher European Language Resources Association (ELRA)

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Pages 7-12

Proceedings Title Workshop Indian Lang. Data: Resour. Eval., WILDRE - held conjunction Int. Conf. Lang. Resour. Eval., LREC - Proc.

Conference Name 6th Workshop on Indian Language Data: Resources and Evaluation, WILDRE 2022 - held in conjunction with the International Conference on Language Resources and Evaluation, LREC 2022 - Proceedings

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Computational linguistics, Codes (symbols), Sentiment analysis, Social networking (online), BERT, GPT, Modeling languages, MBERT, Hindi, XLM-RoBERTa, Hinglish, English, Code mixed, code switch, Code switch, HingBERT, Hingcorpus, HingCorpus

Notes:

EID: 2-s2.0-85145960369

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Label knowledge-guided heterogeneous graph contrastive learning for semi-supervised short text sentiment classification

Item Type Journal Article

Author M. Wu

Abstract Semi-supervised classification demonstrates effective performance in categorizing short-length texts, such as social media posts and online reviews, through the utilization of limited labeled data. Consequently, semi-supervised short text sentiment classification has emerged as a significant research domain within semi-supervised short text classification. However, existing sentiment classification methods predominantly rely on extensive labeled datasets for implementation and typically treat textual labels as discrete symbolic representations (e.g., categorical identifiers for classification tasks). This conventional method results in oversight of two critical linguistic dimensions: the inherent linguistic characteristics embedded within labels themselves and the underlying semantic correlations between labels and textual content. To address the limitations above, this study proposes a novel Label Knowledge-guided Heterogeneous Graph Contrastive Learning (LKG-HGCL) framework for semi-supervised short text sentiment classification. Specifically, we incorporate both label-related terms and their semantic expansions as label knowledge to construct a short text-label knowledge heterogeneous graph, explicitly modeling dynamic interactions between label semantics and short texts. By performing heterogeneous graph contrastive learning through multi-relational edge augmentation, adaptive feature augmentation, heterogeneous graph encoding, and the various contrastive learning modes, the model significantly enhances its capability to capture critical label semantics while generating optimized short text embeddings. The framework establishes robust associations between label knowledge and limited labeled and large amounts of unlabeled short texts, thereby effectively improving semi-supervised learning performance in sentiment analysis. Extensive experiments on three benchmark datasets demonstrate that the proposed LKG-HGCL method outperforms state-of-the-art semi-supervised approaches in classification accuracy and Macro-F1 metrics. © The Author(s) 2025.

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Language Independent Stance Detection: Social Interaction-based Embeddings and Large Language Models

Item Type Journal Article

Author J.F. de Landa

Author R. Agerri

Abstract The large majority of the research performed on stance detection has been focused on developing more or less sophisticated text classification systems, even when many benchmarks are based on social network data such as Twitter. This paper aims to take on the stance detection task by placing the emphasis not so much on the text itself but on the interaction data available on social networks. More specifically, we propose a new method to leverage social information such as friends and retweets by generating Relational Embeddings, namely, dense vector representations of interaction pairs. Our experiments on seven publicly available datasets and four different languages (Basque, Catalan, Italian, and Spanish) show that combining our relational embeddings with discriminative textual methods helps to substantially improve performance, obtaining state-of-the-art results for six out of seven evaluation settings, outperforming strong baselines based on Large Language Models, or other popular interaction-based approaches such as DeepWalk or node2vec. ©2025 Sociedad Española para el Procesamiento del Lenguaje Natural.

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Language-Agnostic Method for Sentiment Analysis of Twitter

Item Type Conference Paper

Author A.R. Jafari

Author R. Farahbakhsh

Author M. Salehi

Author N. Crespi

Editor Swaroop A.

Editor Polkowski Z.

Editor Correia S.D.

Editor Virdee B.

Abstract With the different events and crises that we are witnessing these days, Twitter plays an essential role in sharing thoughts, opinions, and news worldwide in various languages. Understanding the sentiment of user-generated content has garnered much interest in both industrial and academic communities in recent studies. Due to the limited availability of data from low-resource languages, the focus on multilingual resources is a limiting and challenging issue of sentiment analysis task. Considering the importance of pre-processing in the implementation of a sentiment analysis system, we propose a method consisting

of two steps for the pre-processing of tweets in different languages i) a language-agnostic step to replace or remove some elements in the Twitter data structure and ii) a text-normalization step based on the main high-resource language. In addition, we used machine translation techniques to translate low-resource language texts into the main language. We evaluated sentiment classification approaches based on four deep models: an RNN model and three BERT-based architectures, namely vanilla-version, a language-specific, and a large-scale pre-trained model for Twitter. The results show that our method had better accuracy when using a large-scale BERT-based pre-trained model. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd 2024.

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Language-Based Mixture of Transformers for Sexism Identification in Social Networks

Item Type Conference Paper

Author A. Petrescu

Author C.-O. Truică
Author E.-S. Apostol
Editor Carrillo-de-Albornoz J.
Editor García Seco de Herrera A.
Editor Gonzalo J.
Editor Plaza L.
Editor Mothe J.
Editor Piroi F.
Editor Rosso P.
Editor Spina D.
Editor Faggioli G.
Editor Ferro N.

Abstract In this paper, we present an enhanced approach for sexism identification that employs a Mixture of Transformers (MoT) framework, which leverages the language performance of individual models. Our method incorporates straightforward yet effective preprocessing modules, which are integrated with state-of-the-art Transformer architectures. We systematically compare the effectiveness of general-purpose, task-specific, and data source-specific models, evaluating both English and multilingual variants for the EXIST 2024 shared tasks. This refined approach addresses key limitations observed in earlier methods, leading to improved results across all evaluated tasks. Notably, our model demonstrates superior performance in soft-label evaluations compared to hard-label assessments. We introduce three distinct types of model mixtures, each of which achieved optimal results on training data and exhibited strong generalization to unseen data. Furthermore, the proposed architecture is designed for easy upgrades, maintains reasonable resource requirements, and delivers robust overall performance in competitive settings. Our findings underline the value of combining multiple Transformer models tailored to specific language and task requirements, paving the way for more adaptable and effective natural language processing solutions. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Performance, Classification (of information), Network architecture, Natural language processing systems, Social networking (online), Social media analysis, Social Media Analysis, Text classification, Text Classification, Text processing, Data-source, Individual modeling, Learning with disagreement, Learning with Disagreements, Mixture of transformer, Mixture of Transformers, Preprocessing modules, Sexism identification, Sexism Identification, State of the art

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Language, Culture, and Ideology: Personalizing Offensiveness Detection in Political Tweets with Reasoning LLMs

Item Type Conference Paper

Author D. Pihulski

Author J. Kocoń

Abstract We explore how large language models (LLMs) assess offensiveness in political discourse when prompted to adopt specific political and cultural perspectives. Using a multilingual subset of the MD-Agreement dataset centered on tweets from the 2020 US elections, we evaluate several recent LLMs including DeepSeek-R1, o4-mini, GPT-4.1-mini, Qwen3, Gemma, and Mistral - tasked with judging tweets as offensive or nonoffensive from the viewpoints of varied political personas (farright, conservative, centrist, progressive) across English, Polish, and Russian contexts. Our results show that larger models with explicit reasoning abilities (e.g., DeepSeek-R1, o4-mini) are more consistent and sensitive to ideological and cultural variation, while smaller models often fail to capture subtle distinctions. We find that reasoning capabilities significantly improve both the personalization and interpretability of offensiveness judgments, suggesting that such mechanisms are key to adapting LLMs for nuanced sociopolitical text classification across languages and ideologies. © 2025 IEEE.

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Speech communication, Classification (of information), Social media, Large datasets, Stance detection, Text processing, Speech detection, Tweets, stance detection, Cultural bias, Frame analysis, Frame detection, frame detection and analysis, hate-speech detection, Hate-speech detection, language/cultural bias analysis, Language/cultural bias analyze, NLP tool for social analyze, NLP tools, NLP tools for social analysis, quantitative analyses of news and/or social media, Quantitative analyze of news and/or social medium

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Large Language Model Cascades and Persona-Based In-Context Learning for Multilingual Sexism Detection

- Item Type** Conference Paper
- Author** L. Tian
- Author** N. Huang
- Author** X. Zhang
- Editor** Goeuriot L.
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Editor Soulier L.

Editor Galuščáková P.

Editor García Seco de Herrera A.

Abstract This paper presents an approach for detecting and categorising sexism in social media posts using large language models (LLMs) and ensemble methods. The sEXism Identification in Social neTworks (EXIST) shared task, part of CLEF 2023, consists of three sub-tasks: Sexism Identification, Source Intention, and Sexism Categorisation. We formulate sexism detection in English and Spanish as text classification problems. A distinctive feature of the EXIST datasets is that, in addition to that each social media post is assigned a hard label from majority vote from all annotations, the soft labels – the labels by all annotators of different gender and age profiles – are also included. We propose cascade strategies to leverage LLMs for learning from hard labels. Our hard label-based system Mario is ranked first for the hard label evaluation on both sexism identification (Task 1) and source intention classification (Task 2) of the EXIST 2023 shared task. Our experiments show that more advanced base LLMs (e.g. Llama-3) can further improve the performance of Mario. To learn from soft labels for sexism identification, we further propose fine-grained in-context learning strategies based on personas of different age and gender profiles. Our experiments show that our few-shot persona-based in-context learning strategy leveraging Llama-3 can achieve reasonable performance for soft label prediction for sexism identification, and outperforms previous approaches of directly fine-tuning ensemble of BERT-based models. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Large Language Model Enhanced Fuzzy Logic Fusion Framework for Stance Detection

Item Type Conference Paper

Author B. Zhang

Author G. Dai

Author J. Ma

Author H. Lin

Author H. Huang

Editor Huang X.

Editor Mao Q.

Abstract Analyzing public attitudes on social media is crucial for opinion-mining systems. Stance detection aims to identify the attitude of an opinionated text towards a given target. Despite advances in conventional stance detection methods, handling sentences with ambiguous stances remains a significant challenge. Meanwhile, fuzzy logic theory can improve knowledge representation under uncertainty, but learning logical rules from samples requires a large corpus of annotated data, which is impractical for stance detection tasks involving numerous targets. Inspired by the impressive zero-shot performance of large language models (LLM) on commonsense reasoning tasks, we propose the LLM-enhanced Fuzzy Logic Fusion (LLM-FLF) Framework, which synergizes the strengths of LLMs and fuzzy logic to improve performance in complex reasoning tasks. LLM-FLF consists of two main components: Logical Schema Graph construction and Logic-integrated fuzzy relational graph convolutional network. Our approach achieves state-of-the-art results across zero-shot, cross-target, and in-target settings. Notably, it demonstrates a substantial performance gain, outperforming the best baseline methods by 5.1% and 5.5% on Semeval-2016 and VAST benchmark datasets in the most challenging zero-shot scenario. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Language model, Social media, Public attitudes, Fuzzy logic, Chain-of-thought, Stance detection, Convolutional networks, Fuzzy graph, Fuzzy graph convolutional network, Stance Detection, Chain-of-Thought, Fuzzy-Logic, Fuzzy Graph Convolutional Network, Image representation, Network theory (graphs), Reasoning tasks

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Large Language Model Enhanced Logic Tensor Network for Stance Detection

Item Type Journal Article

Author G. Dai

Author J. Liao

Author S. Zhao

Author X. Fu

Author X. Peng

Author H. Huang

Author B. Zhang

Abstract Social media platforms, rich in user-generated content, offer a unique perspective on public opinion, making stance detection an essential task in opinion mining. However, traditional deep neural networks for stance detection often suffer from limitations, including the requirement for large amounts of labeled data, uninterpretability of prediction results, and difficulty in incorporating human intentions and domain knowledge. This paper introduces the First-Order Logic Aggregated Reasoning framework (FOLAR), an innovative approach that integrates first-order logic (FOL) with large language models (LLMs) to enhance the interpretability and efficacy of stance detection. FOLAR comprises three key components: a Knowledge Elicitation module that generates FOL rules using a chain-of-thought prompting method, a Logic Tensor Network (LTN) that encodes these rules for stance detection, and a Multi-Decision Fusion mechanism that aggregates LTNs' outputs to minimize biases and improve robustness. Our experiments on standard benchmarks demonstrate the effectiveness of FOLAR, showing it as a promising solution for explainable and accurate stance detection. The source code will be made publicly available to foster further research. © 2024

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Large Language Model-based Multimodal Knowledge Distillation: A Review of Methodologies and Applications

Item Type Conference Paper
Author U.B.P. Shamika
Author B.T.G.S. Kumara
Author D.M.K.N. Seneviratna
Author R.M.K.T. Rathnayaka
Editor Ariyaratne M.K.A.
Editor Hettige B.
Editor Rathnayake H.U.W.

Abstract Large language models (LLMs) have rapidly become essential to the development of multimodal AI, enabling the creation of systems capable of understanding and combining information from text, images, audio, and video. Despite their impressive capabilities, these models are often difficult to implement in practice due to their high processing and memory requirements. Knowledge distillation (KD) has emerged as a promising solution: a large teacher model transfers its knowledge to a smaller student model, creating leaner, faster, and more practical systems without loss of accuracy. This article provides a systematic review of recent work on multimodal KD for LLMs, focusing on both the technical methods and their applications. Based on the PRISMA guidelines, 46 studies published between 2024 and 2025 were analyzed, covering approaches such as the teacher-student framework, parametrically efficient methods like low-rank adaptation (LoRA), pre-trained pipelines, and compression strategies including pruning and quantization. This review highlights that KD offers clear efficiency advantages, but challenges remain in areas such as handling noise or data diversity, reliance on synthetic labels, and evaluation practices often limited to accuracy alone. Applications in healthcare, social media analytics, and audiovisual segmentation demonstrate the potential of KD, but also reveal its shortcomings in terms of robustness and reproducibility. By mapping these methods into a structured taxonomy and identifying their limitations, this review opens prospects for the reproducibility, customization, and scalable implementation of multimodal KD with LLM. © 2025 IEEE.

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Language model, Large language model, Model-based OPC, Classification (of information), Large Language Model, Distillation, Knowledge distillation, Knowledge Distillation, Multi-modal, Students, Text classification, Text Classification, Text processing, Teaching, Text images, Pretrained model, Low rank adaptation, Low Rank Adaptation, Pretrained Model, Reproducibilities

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Large Language Model-Guided Disentangled Belief Representation Learning on Polarized Social Graphs

Item Type Conference Paper

Author J. Li

Author R. Han

Author C. Sun

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Author Y. Yan

Author H. Tong

Author T. Abdelzaher

Abstract The paper advances belief representation learning in polarized networks - the mapping of social beliefs espoused by users and posts in a polarized network into a disentangled latent space that separates (the members and beliefs of) each side. Our prior work embeds social interaction data, using non-negative variational graph auto-encoders, into a disentangled latent space. However, the interaction graphs alone may not adequately reflect similarity and/or disparity in beliefs, especially for those graphs with sparsity and outlier issues. In this paper, we investigate the impact of limited guidance from Large Language Models (LLMs) on the accuracy of belief separation. Specifically, we integrate social graphs with LLM-based soft labels as a novel weakly-supervised interpretable graph representation learning framework. This framework combines the strengths of graph-and text-based information, and is shown to maintain the interpretability of learned representations, where different axes in the latent space denote association with different sides of the divide. An evaluation on six real-world Twitter datasets illustrates the effectiveness of the proposed model at solving stance detection problems, demonstrating 5.9%-6.5% improvements in the accuracy, F1 score, and purity metrics, without introducing a significant computational overhead. An ablation study is also discussed to study the impact of different components of the proposed architecture. © 2024 IEEE.

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Large Language Model–Assisted Annotation Framework for Cross-Platform Analysis of Online Autism Communities: Implications for Parent Education and Digital Support

- Item Type** Journal Article
- Author** Y. Xu
 - Author** J. Ma
 - Author** Y. Hu
 - Author** Y. Liu
 - Author** Y. Chen
 - Author** W. Feng
 - Author** C. Zhang
 - Author** L. Zhang
 - Author** X. Zhang
 - Author** R. Huang

Abstract Background: Online health communities (OHCs) are important channels for families of children with autism spectrum disorder to obtain health information and psychosocial support. Differences between an open forum platform and the physician-patient consultation platforms may shape caregiver decisions, yet comparative evidence from China remains limited. A large language model (LLM) provides a scalable approach for systematic content annotation in large OHC datasets. Objective: This study proposes and validates a standardized LLM-assisted annotation framework under a unified classification schema and compares topic distributions and poster identities across an open forum platform (Baidu Tieba) and physician-patient consultation platforms (Chunyu Doctor and Haodf). Methods: We implemented an LLM-assisted annotation framework. A unified taxonomy of topics and poster identities was first developed through human open coding. Poster identities in Baidu Tieba were annotated through a double-blind manual procedure. For topic classification, interannotator and human-LLM agreement were evaluated on a manually labeled subset to benchmark models of varying sizes. The best-performing LLM was selected for full-dataset topic annotation, followed by statistical and cross-platform analysis. Results: When metrics were arithmetically averaged across all annotation tasks, the best-performing LLM achieved agreement levels comparable to human annotation (accuracy=79.18%, SD 0.20%; κ =0.736, SD 0.003; F1-score=0.727, SD 0.006), approaching interannotator agreement (accuracy=81.65%; κ =0.767; F1-score=0.758), demonstrating strong stability and scalability. Full-dataset analysis yielded 3 main findings. First, model performance increased with parameter scale but plateaued beyond 14B, indicating diminishing marginal returns from further scaling. Second, clear cross-platform differences in poster identity were observed: the open forum platform was dominated by family members of patients (caregivers: 2377/3516, 67.61%), with substantial participation from commercial rehabilitation practitioners (commercial posters: 427/3516, 12.14%), resulting in a more heterogeneous participation structure. Third, topic distributions reflected both shared high demand for resource-related information and differentiated help-seeking pathways: both platform types demonstrated consistently high demand for resource recommendation and evaluation; the open forum platform was primarily characterized by diagnosis-related discussions (1183/7535, 15.70%), whereas the physician-patient consultation platforms were centered on intervention-related consultation (2864/7687, 37.26%). Conclusions: The LLM-assisted annotation framework proposed in this study enables reliable large-scale annotation of OHC data while maintaining high human-LLM agreement and operational stability. Midsized models (eg, 14B) demonstrated favorable cost-performance efficiency. The findings reveal 2 key aspects: the open forum platform exhibits a complex participation structure, and the influence of commercially affiliated actors should not be overlooked; users on both platform types show sustained demand for resource-related information but follow different help-seeking pathways, emphasizing diagnostic exploration and professional intervention, respectively. These results suggest that platform structure and governance mechanisms may shape caregivers' information access and decision-making. The framework provides a transparent, reproducible, and cost-effective approach for OHC research. All data were deidentified and handled in accordance with relevant platform policies and ethical standards. ©Yifan Xu, Jianhao Ma, Yujia Hu, Yixue Liu, Yu Chen, Wei Feng, Changwei Zhang, Lei Zhang, Xuening Zhang, Ruochen Huang.

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Large Language Models as Kuwaiti Annotators

Item Type Journal Article

Author H. Alostad

Abstract Stance detection for low-resource languages, such as the Kuwaiti dialect, poses a significant challenge in natural language processing (NLP) due to the scarcity of annotated datasets and specialized tools. This study addresses these limitations by evaluating the effectiveness of open large language models (LLMs) in automating stance detection through zero-shot and few-shot prompt engineering, with a focus on the potential of open-source models to achieve performance levels comparable to those of closed-source alternatives. We also highlight the critical distinctions between zero- and few-shot learning, emphasizing their significance for addressing the challenges posed by low-resource languages. Our evaluation involved testing 11 LLMs on a manually labeled dataset of social media posts, including GPT-4o, Gemini Pro 1.5, Mistral-Large, Jais-30B, and AYA-23. As expected, closed-source models such as GPT-4o, Gemini Pro 1.5, and Mistral-Large demonstrated superior performance, achieving maximum F1 scores of 95.4%, 95.0%, and 93.2%, respectively, in few-shot scenarios with English as the prompt template language. However, open-source models such as Jais-30B and AYA-23 achieved competitive results, with maximum F1 scores of 93.0% and 93.1%, respectively, under the same conditions. Furthermore, statistical analysis using ANOVA and Tukey's HSD post hoc tests revealed no significant differences in overall performance among GPT-4o,

Gemini Pro 1.5, Mistral-Large, Jais-30B, and AYA-23. This finding underscores the potential of open-source LLMs as cost-effective and privacy-preserving alternatives for low-resource language annotation. This is the first study comparing LLMs for stance detection in the Kuwaiti dialect. Our findings highlight the importance of prompt design and model consistency in improving the quality of annotations and pave the way for NLP solutions for under-represented Arabic dialects. © 2025 by the author.

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Large Language Models Can Enable Inductive Thematic Analysis of a Social Media Corpus in a Single Prompt: Human Validation Study

Item Type Journal Article

Author M.S. Deiner

Author V. Honcharov

Author J. Li

Author T.K. Mackey

Author T.C. Porco

Author U. Sarkar

Abstract Background: Manually analyzing public health–related content from social media provides valuable insights into the beliefs, attitudes, and behaviors of individuals, shedding light on trends and patterns that can inform public understanding, policy decisions, targeted interventions, and communication strategies. Unfortunately, the time and effort needed from well-trained human subject matter experts makes extensive manual social media listening unfeasible. Generative large language models (LLMs) can potentially summarize and interpret large amounts of text, but it is unclear to what extent LLMs can glean subtle health-related meanings in large sets of social media posts and reasonably report health-related themes. Objective: We aimed to assess the feasibility of using LLMs for topic model selection or inductive thematic analysis of large contents of social media posts by attempting to answer the following question: Can LLMs conduct topic model selection and inductive thematic analysis as effectively as humans did in a prior manual study, or at least reasonably, as judged by subject matter experts? Methods: We asked the same research question and used the same set of social media content for both the LLM selection of relevant topics and the LLM analysis of themes as was conducted manually in a published study about vaccine rhetoric. We used the results from that study as background for this LLM experiment by comparing the results from the prior manual human analyses with the analyses from 3 LLMs: GPT4-32K, Claude-instant-100K, and Claude-2-100K. We also assessed if multiple LLMs had equivalent ability and assessed the consistency of repeated analysis from each LLM. Results: The LLMs generally gave high rankings to the topics chosen previously by humans as most relevant. We reject a null hypothesis ($P<.001$, overall comparison) and conclude that these LLMs are more likely to include the human-rated top 5 content areas in their top rankings than would occur by chance. Regarding theme identification, LLMs identified several themes similar to those identified by humans, with very low hallucination rates. Variability occurred between LLMs and between test runs of an individual LLM. Despite not consistently matching the human-generated themes, subject matter experts found themes generated by the LLMs were still reasonable and relevant. Conclusions: LLMs can effectively and efficiently process large social media–based health-related data sets. LLMs can extract themes from such data that human subject matter experts deem reasonable. However, we were unable to show that the LLMs we tested can replicate the depth of analysis from human subject matter experts by consistently extracting the same themes from the same data. There is vast potential, once better validated, for automated LLM-based real-time social listening for common and rare health conditions, informing public health understanding of the public’s interests and concerns and determining the public’s ideas to address them. ©Michael S Deiner, Vlad Honcharov, Jiawei Li, Tim K Mackey, Travis C Porco, Urmimala Sarkar. Originally published in JMIR Infodemiology (<https://infodemiology.jmir.org>), 29.08.2024.

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Large language models for newspaper sentiment analysis during COVID-19: The Guardian

Item Type Journal Article

Author R. Chandra

Author B. Zhu

Author Q. Fang

Author E. Shinjikashvili

Abstract During the COVID-19 pandemic, news media coverage encompassed topics such as viral transmission, allocation of medical resources, and government response measures. Studies have analysed sentiment on social media platforms during COVID-19 to understand the public's response to the rising cases and the government strategies implemented to control the spread of the virus. Sentiment analysis can enable an understanding of the dynamics in the social and psychological well-being of a group during the pandemic. Apart from social media, newspapers have played a vital role in disseminating information, including information from the government, experts, and the public about various topics. A study of sentiment analysis of newspaper sources during COVID-19 for selected countries can give an overview of how the media covered the pandemic. In this study, we selected The Guardian newspaper and conducted a sentiment analysis during various stages of COVID-19 including initial transmission, lockdowns and vaccination. We employed novel large language models (LLMs) and refined them with expert-labelled sentiment analysis data. We also provide an analysis of sentiments experienced pre-pandemic for comparison. The results indicate that during the early pandemic stages, public sentiment prioritised urgent crisis response, later shifting focus to addressing the impact on health and the economy. In comparison with related studies about social

media sentiment analyses, we found a discrepancy between The Guardian, with a dominance of negative sentiments (sad, annoyed, anxious and denial), suggesting that social media offered a more diverse emotional expression during the pandemic. We found a grim narrative in The Guardian with overall dominance of negative sentiments, pre- and during COVID-19 across news sections including Australia, UK, World News, and Opinion. © 2025 The Authors

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Large language models identify immigration attitudes in online discourse regardless of language

Item Type Journal Article

Author A. Nasuto

Author S. Iacus

Author F. Rowe

Author D. Jain

Abstract Large language models (LLMs) offer new opportunities for scalable analysis of online discourse. Yet their use in multilingual social science research remains constrained by model size, cost and linguistic bias. We develop a lightweight, open-source LLM framework using fine-tuned LLaMA 3.2–3B models to classify immigration-related tweets across 13 languages. Unlike prior work relying on BERT-

style models or translation pipelines, we combine topic classification with stance detection and demonstrate that LLMs fine-tuned in just one or two languages can generalize topic understanding to unseen languages. Capturing ideological nuance, however, benefits from multilingual fine-tuning. Our approach corrects pretraining biases with minimal data from under-represented languages and avoids reliance on proprietary systems. With 19–96 faster inference and up to 2,017 cost savings compared to commercial LLMs, our method supports real-time analysis of billions of tweets. This scale-first framework enables inclusive, reproducible research on public attitudes across linguistic and cultural contexts. © The Author(s) 2026.

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Large Language Models in an App: Conducting a Qualitative Synthetic Data Analysis of How Snapchat’s “My AI” Responds to Questions About Sexual Consent, Sexual Refusals, Sexual Assault, and Sexting

Item Type Journal Article

Author T.L. Marcantonio

Author G. Avery

Author A. Thrash

Author R.M. Leone

Abstract Comprehensive sexual health education for young people often remains largely inaccessible, leaving gaps in knowledge about sexual consent, refusals, sexual assault, and sexting. Snapchat's My AI, tailored to its users, offers accessible and potentially stigma-free responses to sexual health queries, showing promise in addressing these gaps. This study aimed to evaluate My AI's effectiveness in handling questions related to sexual communication, assault, and sexting, and its response consistency across users. Collaborating with young adults and leveraging prior literature, our team formulated questions on sexual consent, refusals, sexual assault, and sexting submitted to My AI by 15 researchers. Each team member opened a chat with My AI in Snapchat and asked it a set of questions in the same order. We then conducted a qualitative content analysis to assess for consistency across users. My AI responses emphasized the importance of clear, honest consent and refusal communication through physical contact or direct conversation. With potential experiences of sexual assault or sexting, it advised consultation with a trusted caregiver. While My AI provided similar responses across researchers, variations were noted. My AI's responses mirrored common sexual health education messages and the responses used language reflective of sympathy toward potential assault victims or those who were asked for sexual images. My AI's potential to enhance accessibility to sexual health information could empower young people in making informed sexual health decisions. However, variability in responses may limit impact and comprehension, underscoring the importance of educators complementing these tools. © 2024 The Society for the Scientific Study of Sexuality.

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Large Language Models in equity markets: applications, techniques, and insights

Item Type Journal Article

Author A. Jadhav

Author V. Mirza

Abstract Recent breakthroughs in Large Language Models (LLMs) have the potential to disrupt equity investing by enabling sophisticated data analysis, market prediction, and automated trading. This paper presents a comprehensive review of 84 research studies conducted between 2022 and early 2025, synthesizing the state of LLM applications in stock investing. We provide a dual-layered categorization: first, by financial applications such as stock price forecasting, sentiment analysis, portfolio management, and algorithmic trading; second, by technical methodologies, including prompting, fine-tuning, multi-agent frameworks, reinforcement learning, and custom architectures. Additionally, we consolidate findings on the datasets used, ranging from financial statements to multimodal data (news, market trends, earnings transcripts, social media), and systematically compare general-purpose vs. finance-specialized LLMs used in research. Our analysis identifies key research trends, commonalities, and divergences across studies, evaluating both their empirical contributions and methodological innovations. We highlight the strengths of existing research, such as improved sentiment extraction and the use of reinforcement learning to factor market feedback, alongside critical gaps in scalability, interpretability, and real-world validation. Finally, we propose directions for future research, emphasizing hybrid modeling approaches, architectures that factor reasoning and large context windows, and robust evaluation frameworks to advance AI-driven financial strategies. By mapping the intersection of LLMs and equity markets, this review provides a foundation and roadmap for future research and practical implementation in the financial sector. Copyright © 2025 Jadhav and Mirza.

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Large Language Models meet moral values: A comprehensive assessment of moral abilities

Item Type Journal Article

Author L. Bulla

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Author M. Mongiovi

Author A. Gangemi

Abstract Automatic moral classification in textual data is crucial for various fields including Natural Language Processing (NLP), social sciences, and ethical AI development. Despite advancements in supervised models, their performance often suffers when faced with real-world scenarios due to overfitting to specific data distributions. To address these limitations, we propose leveraging state-of-the-art Large Language Models (LLMs) trained on extensive common-sense data for unsupervised moral classification. We introduce an innovative evaluation framework that directly compares model outputs with human annotations, ensuring an assessment of model performance. Our methodology explores the effectiveness of different LLM sizes and prompt designs in moral value detection tasks, considering both multi-label and binary classification scenarios. We present experimental results using the Moral Foundation Reddit Corpus (MFRC) and discuss implications for future research in ethical AI development and human-computer interaction. Experimental results demonstrate that GPT-4 achieves superior performance, followed by GPT-3.5, Llama-70B, Mixtral-8x7B, Mistral-7B and Llama-7B. Additionally, the study reveals significant variations in model performance across different moral domains, particularly between everyday morality and political contexts. Our work provides meaningful insights into the use of zero-shot and few-shot models for moral value detection and discusses the potential and limitations of current technology in this task. © 2025

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Large Language Models Outperform Expert Coders and Supervised Classifiers at Annotating Political Social Media Messages

Item Type Journal Article

Author P. Törnberg

Abstract Instruction-tuned Large Language Models (LLMs) have recently emerged as a powerful new tool for text analysis. As these models are capable of zero-shot annotation based on instructions written in natural language, they obviate the need of large sets of training data—and thus bring potential paradigm-shifting implications for using text as data. While the models show substantial promise, their relative performance compared to human coders and supervised models remains poorly understood and subject to significant academic debate. This paper assesses the strengths and weaknesses of popular fine-tuned AI models compared to both conventional supervised classifiers and manual annotation by experts and crowd workers. The task used is to identify the political affiliation of politicians based on a single X/Twitter message, focusing on data from 11 different countries. The paper finds that GPT-4 achieves higher accuracy than both supervised models and human coders across all languages and country contexts. In the US context, it achieves an accuracy of 0.934 and an inter-coder reliability of 0.982. Examining the cases where the models fail, the paper finds that the LLM—unlike the supervised models—correctly annotates messages that require interpretation of implicit or unspoken references, or reasoning on the basis of contextual knowledge—capacities that have traditionally been understood to be distinctly human. The paper thus contributes to our understanding of the revolutionary implications of LLMs for text analysis within the social sciences. © The Author(s) 2024. This article is distributed under the terms of the Creative Commons Attribution 4.0 License (<https://creativecommons.org/licenses/by/4.0/>) which permits any use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access page (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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Large Language Models Powered Aspect-Based Sentiment Analysis for Enhanced Customer Insights

Item Type Journal Article

Author M. Água

Author N. António

Author P. Carrasco

Author C. Rassal

Abstract In the age of social networks, user-generated content has become vital for organizations in tourism and hospitality. Traditional sentiment analysis methods often struggle to process large volumes of data and capture implicit sentiments. This study examines the potential of Aspect-Based Sentiment Analysis (ABSA) using Large Language Models (LLMs) to enhance sentiment analysis. By employing GPT-4o via ChatGPT, we benchmark three approaches: a fuzzy logic-based method, manual human analysis, and a new ChatGPT-based analysis. We analyze a dataset of 500 all-inclusive hotel reviews, comparing these methods to assess ChatGPT's effectiveness in identifying nuanced language and handling subjectivity. The findings reveal a high similarity between ChatGPT and human analysis, showcasing ChatGPT's ability to interpret complex sentiments and automate sentiment classification tasks. This study highlights the potential of LLMs in transforming customer feedback analysis,

providing deeper insights, and improving responsiveness in the hospitality industry. These results contribute to academia by presenting a framework for using LLMs in ABSA and guiding future applications and development. © 2025 University of Algarve. All rights reserved.

Date 2025

DOI 10.18089/tms.202501011

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Archive Scopus

Short Title Análise de Sentimentos por Aspectos com Modelos de Linguagem de Grande Escala para Melhor Compreensão dos Clientes

Language English

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Export Date: 20 September 2026; Cited By: 10; Funding details: Fundação para a Ciência e a Tecnologia, FCT, (UIDB/04152/2020); Fundação para a Ciência e a Tecnologia, FCT

Large Language Models' Accuracy in Emulating Human Experts' Evaluation of Public Sentiments about Heated Tobacco Products on Social Media: Evaluation Study

Item Type Journal Article

Author K. Kim

Author S. Kim

Abstract Background: Sentiment analysis of alternative tobacco products discussed on social media is crucial in tobacco control research. Large language models (LLMs) are artificial intelligence models that were trained on extensive text data to emulate the linguistic patterns of humans. LLMs may hold the potential to streamline the time-consuming and labor-intensive process of human sentiment analysis. Objective: This study aimed to examine the accuracy of LLMs in replicating human sentiment

evaluation of social media messages relevant to heated tobacco products (HTPs). Methods: GPT-3.5 and GPT-4 Turbo (OpenAI) were used to classify 500 Facebook (Meta Platforms) and 500 Twitter (subsequently rebranded X) messages. Each set consisted of 200 human-labeled anti-HTPs, 200 pro-HTPs, and 100 neutral messages. The models evaluated each message up to 20 times to generate multiple response instances reporting its classification decisions. The majority of the labels from these responses were assigned as a model's decision for the message. The models' classification decisions were then compared with those of human evaluators. Results: GPT-3.5 accurately replicated human sentiment evaluation in 61.2% of Facebook messages and 57% of Twitter messages. GPT-4 Turbo demonstrated higher accuracies overall, with 81.7% for Facebook messages and 77% for Twitter messages. GPT-4 Turbo's accuracy with 3 response instances reached 99% of the accuracy achieved with 20 response instances. GPT-4 Turbo's accuracy was higher for human-labeled anti- and pro-HTP messages compared with neutral messages. Most of the GPT-3.5 misclassifications occurred when anti- or pro-HTP messages were incorrectly classified as neutral or irrelevant by the model, whereas GPT-4 Turbo showed improvements across all sentiment categories and reduced misclassifications, especially in incorrectly categorized messages as irrelevant. Conclusions: LLMs can be used to analyze sentiment in social media messages about HTPs. Results from GPT-4 Turbo suggest that accuracy can reach approximately 80% compared with the results of human experts, even with a small number of labeling decisions generated by the model. A potential risk of using LLMs is the misrepresentation of the overall sentiment due to the differences in accuracy across sentiment categories. Although this issue could be reduced with the newer language model, future efforts should explore the mechanisms underlying the discrepancies and how to address them systematically. ©Kwanho Kim, Soojong Kim.

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Publication Journal of Medical Internet Research

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Journal Abbr J. Med. Internet Res.

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large language models, Article, artificial intelligence, human, Humans, machine learning, Artificial Intelligence, ChatGPT, social media, GPT, LLM, controlled study, Social Media, natural language processing, sentiment analysis, public opinion, large language model, accuracy, NLP, language model, OpenAI, sentiment, prevention, tobacco, alternative, evaluation, generative pre-trained transformer, heated tobacco products, nicotine, Tobacco Products

Notes:

EID: 2-s2.0-86000186978

Export Date: 20 September 2026; Cited By: 8; Correspondence Address: S. Kim; Department of Communication, University of California Davis, Davis, 1 Shields Ave, Kerr Hall #361, 95616, United States; email: sjokim@ucdavis.edu

LAStER: Lexical Data Augmentation with Stacking Ensembles for Robust Emotion Detection in Low-Resource Corpora

Item Type Journal Article

Author E. Noersasongko

Author F. Budiman

Author C. Supriyanto

Abstract Emotion classification is a crucial task in Natural Language Processing (NLP) with applications in sentiment analysis, social media monitoring, and user modeling. Conventional augmentation methods often fail to identify nuanced emotional cues. We propose Lexical Data Augmentation with Stacking Ensembles for Robust Emotion Detection (LAStER), which integrates emotional lexicons with ensemble learning techniques. We augmented the ISEAR and the English datasets from SemEval-2025 Task 11, with explicit emotional indicators from NRC Emotion Lexicon. Our findings indicate that LAStER significantly enhances performance in multi-label contexts, with the stacking configuration utilized on the SemEval dataset attaining a subset accuracy of 0.229 and a macro F1-score of 0.324. The stacking configuration utilized for the single-label attained an accuracy of 0.350 and an F1-score of 0.182. The results demonstrate that the integration of lexical augmentation with stacking ensembles is an efficacious method for emotion identification, and suggest that LAStER operates competently with multi-label datasets. © This article is licensed under a Creative Commons Attribution-ShareAlike 4.0 International License. License details: <https://creativecommons.org/licenses/by-sa/4.0/>

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Language English

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Journal Abbr Int. J. Intelligent Eng. Syst.

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Sentiment analysis, Natural language processing, Benchmarking, Emotion classification, Data augmentation, ISEAR, Lexicon-based augmentation, SemEval

Notes:

EID: 2-s2.0-105022289751

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Wagino; Faculty of Computer Science, Universitas Dian Nuswantoro, Semarang, Indonesia; email: p41202300064@mhs.dinus.ac.id

Late breaking papers from the 27th International Conference on Human-Computer Interaction, HCI International 2025

Item Type Journal Article

Editor Coman A.

Editor Vasilache S.

Editor Moallem A.

Abstract The proceedings contain 29 papers. The special focus in this conference is on Human-Computer Interaction. The topics include: Multimodal Speed Dating Dataset: Behavioral Cues, Psychological Ratings, and Empirical Findings; cross-Cultural Interaction on Social Media Health Platforms: Challenges and Opportunities from Africa; Exploring HIV Conversations: Social Media Interactions and Social Institutions in the LGBTQIA+ Community in the Philippines; content Themes and Self-presentation of Virtual Influencers - Exploring the Impact on User Engagement and Sentiment; analysis of Combined Impact of Information and Emotional Content on User Engagement Based on Social Media Data; Retrieval-Augmented Generation for Secure Environments: Theoretical Foundations, Agent Architectures, and Local Deployment with FAIRD; An Investigation into the Use of LLM-Based Features for Multimodal Estimation of Elderly Engagement in Active Listening; SNS Search System that Uses a Large Language Model to Display Posts on an Interactive Map from Various Perspectives; ChatGPT Usability. An Exploratory Study in Higher Education; Assessing AI-Based System Acceptance Through the Design of a Trustworthiness Estimation Tool for Machine Learning Models; A Comparison of the Coverage of Chat GPT in China Daily and The New York Times Based on Frame Theory and Content Analysis; representating Negative State of a Robot's Abdominal/Thoracic Breathing Accompanied Movement System; towards a Formal Methodology for Evaluating Information Consumer Experience at Work; unlocking Multilingualism: Thoughts Shaped by Tools?; Estimating Subjective Evaluation on Drinking Experiences Using fNIRS Sensing; internet of Things (IoT) for Weather Monitoring Systems in Rural Zones: Literature Review; key Factors in Patient Satisfaction: A Random Forest Strategy; method for Evaluating Cyber Risk Management Frameworks; evaluating the Impact of Insider Threats.

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Publication Lecture Notes in Computer Science
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Journal Abbr Lect. Notes Comput. Sci.
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Export Date: 20 September 2026; Cited By: 0; Conference name: Late breaking papers from the 27th International Conference on Human-Computer Interaction, HCI International 2025; Conference date: 22 June 2025 through 27 June 2025; Conference code: 346229

LaTeCH-CLfL 2024 - 8th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature, Proceedings of the Workshop

Item Type Journal Article
Editor Bizzoni Y.
Editor Degaetano-Ortlieb S.
Editor Kazantseva A.
Editor Szpakowicz S.
Abstract The proceedings contain 28 papers. The topics discussed include: evaluating in-context learning for computational literary studies: a case study based on the automatic recognition of knowledge transfer in German drama; coreference in long documents using hierarchical entity merging; metaphorical framing of refugees, asylum seekers and immigrants in UKs left and right-wing media; computational analysis of dehumanization of Ukrainians on Russian social media; compilation of a synthetic Judeo-French corpus; detecting structured language alternations in historical documents by combining language identification with Fourier analysis; multi-word expressions in English scientific writing; modeling Moravian memoirs: ternary sentiment analysis in a low resource setting; applying information-theoretic notions to measure effects of the plain English movement on English law reports and scientific articles; and historical portrayal of Greek tourism through topic modeling on international newspapers.
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URL <https://www.scopus.com/pages/publications/85189781059?origin=resultslist>
Archive Scopus
Language English
Publisher Association for Computational Linguistics (ACL)
Publication LaTeCH-CLfL 2024 - 8th Joint SIGHUM Workshop on Computational Linguistics for Cultural Heritage, Social Sciences, Humanities and Literature, Proceedings of the Workshop
ISSN 979-889176069-1 (ISBN)
Journal Abbr LaTeCH-CLfL - Jt. SIGHUM Workshop Comput. Linguist. Cult. Herit., Soc. Sci., Humanit. Lit., Proc. Workshop

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Learning Faithful Attention for Interpretable Classification of Crisis-Related Microblogs under Constrained Human Budget

Item Type Conference Paper**Author** T.H. Nguyen**Author** K. Rudra

Abstract The recent widespread use of social media platforms has created convenient ways to obtain and spread up-to-date information during crisis events such as disasters. Time-critical analysis of crisis data can help human organizations gain actionable information and plan for aid responses. Many existing studies have proposed methods to identify informative messages and categorize them into different humanitarian classes. Advanced neural network architectures tend to achieve state-of-the-art performance, but the model decisions are opaque. While attention heatmaps show insights into the model's prediction, some studies found that standard attention does not provide meaningful explanations. Alternatively, recent works proposed interpretable approaches for the classification of crisis events that rely on human rationales to train and extract short snippets as explanations. However, the rationale annotations are not always available, especially in real-time situations for new tasks and events. In this paper, we propose a two-stage approach to learn the rationales under minimal human supervision and derive faithful machine attention. Extensive experiments over four crisis events show that our model is able to obtain better or comparable classification performance ($\sim 86\%$ Macro-F1) to baselines and faithful attention heatmaps using only 40-50% human-level supervision. Further, we employ a zero-shot learning setup to detect actionable tweets along with actionable word snippets as rationales. © 2023 ACM.

Date 2023**DOI** 10.1145/3543507.3583861**URL** <https://www.scopus.com/pages/publications/85159375858?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: ACM Web Conf. - Proc. World Wide Web Conf., WWW**Publisher** Association for Computing Machinery, Inc**ISBN** 978-145039416-1 (ISBN)**Pages** 3959-3967**Proceedings Title** ACM Web Conf. - Proc. World Wide Web Conf., WWW

Conference Name ACM Web Conference 2023 - Proceedings of the World Wide Web Conference, WWW 2023

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Network architecture, Social networking (online), Twitter, Budget control, Neural network architecture, Social media platforms, Interpretability, Zero-shot learning, Under-constrained, Classification, Crisis events, Micro-blog, Crisis Events, Critical analysis, Heatmaps, Time-critical

Notes:

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'Learning to code in a virtual world': A preliminary comparative analysis of discourse and learning in two online programming communities

Item Type Conference Paper

Author S. Sengupta

Abstract Software programming is increasingly becoming a community-driven effort, with online discussion channels becoming vital resources for learning and knowledge sharing. This study explores differences in the discourse patterns of two popular online programming communities (Stack Overflow and r/Askprogramming) to provide preliminary insights into the type of learning practices these collectives support and scaffold. A three-step content analysis framework that investigates a sample of 8639 and 6126 contributions from Stack Overflow and r/Askprogramming respectively is presented. Preliminary results indicate that differences emerge in the scope of topics and the nature of responses the communities provide. While Stack Overflow is more task-specific, r/Askprogramming supports a greater sense of bonding and camaraderie among community members in addition to task-specific discussions. These results provide insights into the type of practices these communities support, which can be essential in considering how online communities that support learning activities should be designed. © 2020 Owner/Author.

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Pages 389-394

Proceedings Title Proc. ACM Conf. Comput. Support. Coop. Work CSCW

Conference Name Proceedings of the ACM Conference on Computer Supported Cooperative Work, CSCW

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: S. Sengupta; Syracuse University, Syracuse, United States; email: susengup@syr.edu; Conference name: 3rd ACM Conference on Computer-Supported Cooperative Work and Social Computing, CSCW 2020 Companion; Conference date: 17 October 2020 through 21 October 2020; Conference code: 164204

Leave the world(view) behind, but keep the words: The effect of conspiracism on writing

Item Type Journal Article

Author A. Miani

Author I. Adornetti

Author D. Altavilla

Author V. Deriu

Author A. Chiera

Author F. Ferretti

Abstract We investigate whether a stable predisposition to interpret events as the result of conspiracies—conspiracism—is associated with the spontaneous generation of conspiratorial content in writing when interpreting ambiguous information. Across two studies (N=385), participants watched the apocalyptic thriller *Leave the World Behind* and wrote an essay interpreting its meaning. Each essay was rated by a Large Language Model for its conspiratorial narrative content, that is, the degree to which the text contains claims that the public is being pervasively lied to about aspects of reality, enabling some groups to enact a harmful, self-serving agenda. Contrary to our preregistered hypothesis, we did not find an association between participants' conspiracism and their essays' level of conspiratorial narrative content. Exploratory

linguistic analyses revealed that conspiracism was associated with greater use of conspiracy-related vocabulary (e.g., deception, government), a disproportionate use of sophisticated words, and increased syntactic complexity. These results suggest that conspiracism may emerge more readily at the lexical level rather than through fully structured narratives. We discuss potential methodological and theoretical factors contributing to these unexpected results, including the roles of context, perceived relevance, motivation, and collective social dynamics. We also consider the possibility that conspiracism may not directly translate into conspiratorial narratives. If so, we recommend comparative research on online vs offline conspiratorial writing to clarify whether conspiracy theories emerge spontaneously from genuine beliefs or are constructed strategically, detached from genuinely held beliefs. © 2026 Miani et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Less but Better: Parameter-Efficient Fine-Tuning of Large Language Models for Personality Detection

Item Type Conference Paper

Author L. Shen

Author Y. Long

Author X. Cai

Author G. Chen

Author I. Razzak

Author S. Jameel

Abstract Personality detection automatically identifies an individual's personality from various data sources, such as social media texts. However, as the parameter scale of language models continues to grow, the computational cost becomes increasingly difficult to manage. Fine-tuning also grows more complex, making it harder to justify the effort and reliably predict outcomes. We introduce a novel parameter-efficient fine-tuning framework, PersLLM, to address these challenges. In PersLLM, a large language model (LLM) extracts high-dimensional representations from raw data and stores them in a dynamic memory layer. PersLLM then updates the downstream layers with a replaceable output network, enabling flexible adaptation to various personality detection scenarios. By storing the features in the memory layer, we eliminate the need for repeated complex computations by the LLM. Meanwhile, the lightweight output network serves as a proxy for evaluating the overall effectiveness of the framework, improving the predictability of results. Experimental results on key benchmark datasets like Kaggle and Pandora show that PersLLM significantly reduces computational cost while maintaining competitive performance and strong adaptability. © 2025 IEEE.

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Proceedings Title Proc Int Jt Conf Neural Networks

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Large Language Models, Text Classification, Parameter-Efficient Fine-Tuning, Personality Detection

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Leveraging Chain-of-Thought to Enhance Stance Detection with Prompt-Tuning

Item Type Journal Article

Author D. Ding

Author X. Fu

Author X. Peng

Author X. Fan

Author H. Huang

Author B. Zhang

Abstract Investigating public attitudes towards social media is crucial for opinion mining systems to gain valuable insights. Stance detection, which aims to discern the attitude expressed in an opinionated text towards a specific target, is a fundamental task in opinion mining. Conventional approaches mainly focus on sentence-level classification techniques. Recent research has shown that the integration of background knowledge can significantly improve stance detection performance. Despite the significant improvement achieved by knowledge-enhanced methods, applying these techniques in real-world scenarios remains challenging for several reasons. Firstly, existing methods often require the use of complex attention mechanisms to filter out noise and extract relevant background knowledge, which involves significant annotation efforts. Secondly, knowledge fusion mechanisms typically rely on fine-tuning, which can introduce a gap between the pre-training phase of pre-trained language models (PLMs) and the downstream stance detection tasks, leading to the poor prediction accuracy of the PLMs. To address these limitations, we propose a novel prompt-based stance detection method that leverages the knowledge acquired using the chain-of-thought method, which we refer to as PSDCOT. The proposed approach consists of two stages. The first stage is knowledge extraction, where instruction questions are constructed to elicit background knowledge from a VLPLM. The second stage is the multi-prompt learning network (M-PLN) for knowledge fusion, which learns model performance based on the background knowledge and the prompt learning framework. We evaluated the performance of PSDCOT on publicly available benchmark datasets to assess its effectiveness in improving stance detection performance. The results demonstrate that the proposed method achieves state-of-the-art results in in-domain, cross-target, and zero-shot learning settings. © 2024 by the authors.

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Tags:

chain-of-thought, stance detection, prompt-tuning

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Leveraging Deep Active Learning and Large Language Models for Cost-Efficient Categorization of User-Generated Content

Item Type Conference Paper

Author R. Jindal

Author A. Malhotra

Editor Swaroop A.

Editor Virdee B.

Editor Correia S.D.

Editor Polkowski Z.

Abstract Due to the increase in the online user base and their Internet usage and activity, a humungous amount of content is generated daily on the Internet, especially on online social networks. Though this user-generated content can be collected easily from these websites and online social networks using their corresponding Web APIs, however, annotating and obtaining ground truth labels for every user post (which is required to develop any social-computing system) still remains a challenging and cost-intensive process. Data annotation is even more difficult for healthcare applications where these ground truth labels must be provided by domain specialists. These challenges related to data annotation of user-generated content inhibit the research and development of social-computational linguistic systems for any real-world use case. One of the crucial applications of user-generated content is public healthcare surveillance. The recent research in this

domain has mainly utilized deep learning neural networks, especially pre-trained Transformer-based Large Language Models, e.g., BERT. BERT has now become the state-of-the-art for most of the NLP tasks. However, the accuracy and reliability of any machine learning driven decision-making system heavily depends on the quality of training data. Any supervised learner is only as good as the annotated data used as input to train them. Most of the previous research studies have primarily worked on improving the classification accuracy via algorithmic enhancements. In contrast, our research emphasizes on reducing the data annotation costs for cost-efficient categorization of user-generated content to develop decision-making systems for any social-computational task, e.g., sentiment analysis, opinion mining, content moderation, topic identification, etc. In this research, through our proof of concept using Deep Active Learning and Large Language Models, we demonstrate that it is possible to achieve high/comparable accuracy with as few as 10% of samples from the entire dataset compared to using the entire dataset for model development. For our POC, we have used mental healthcare datasets to demonstrate the use of BERT with Active Learning Loop to detect various mental health disorders from online user-generated content. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

Date 2025

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Archive Scopus

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Volume 1300 LNNS

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Conference Name Lecture Notes in Networks and Systems

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Computational linguistics, Language model, Behavioral research, Deep learning, Deep neural networks, Decision making, Learning systems, Sentiment analysis, Language processing, Learning algorithms, Natural language processing, Natural languages, Social networking (online), Supervised learning, Data mining, Social sciences computing, Public health, Health care, Online systems, E-learning, User-generated, User-generated content, Active learning, Active Learning, Social Network Analysis, Arts computing, Cost benefit analysis, Transformer and large language model, Computational cost, Social network analysis, Transformers and Large Language Models

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Malhotra; Department of Computer Science Engineering, Delhi Technological University, Delhi, India; email: anshumalhotra5@gmail.com; Conference name: 5th International Conference on Data Analytics and Management, ICDAM 2024; Conference date: 14 June 2024 through 15 June 2024; Conference code: 330679

Leveraging Large Language Models and Weak Supervision for Social Media Data Annotation: An Evaluation Using COVID-19 Self-reported Vaccination Tweets

Item Type Conference Paper

Author R. Tekumalla

Author J.M. Banda

Editor Mori H.

Editor Asahi Y.

Editor Coman A.

Editor Vasilache S.

Editor Rauterberg M.

Abstract The COVID-19 pandemic has presented significant challenges to the healthcare industry and society as a whole. With the rapid development of COVID-19 vaccines, social media platforms have become a popular medium for discussions on vaccine-related topics. Identifying vaccine-related tweets and analyzing them can provide valuable insights for public health researchers and policymakers. However, manual annotation of a large number of tweets is time-consuming and expensive. In this study, we evaluate the usage of Large Language Models, in this case GPT-4 (March 23 version), and weak supervision, to identify COVID-19 vaccine-related tweets, with the purpose of comparing performance against human annotators. We leveraged a manually curated gold-standard dataset and used GPT-4 to provide labels without any additional fine-tuning or instructing, in a single-shot mode (no additional prompting). © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Conference Name Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics)

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Computational linguistics, Language model, Large language model, Policy makers, Social networking (online), GPT, Large language models, Social media platforms, COVID-19, Data annotation, Manual annotation, social media data, Vaccines, Weak supervision, weak supervision, Healthcare industry, Social media datum

Notes:

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Export Date: 20 September 2026; Cited By: 12; Correspondence Address: J.M. Banda; Georgia State University, Atlanta, 30328, United States; email: jbanda@gsu.edu; Conference name: 25th International Conference on Human-Computer Interaction, HCII 2023; Conference date: 23 July 2023 through 28 July 2023; Conference code: 297769

LEVERAGING LARGE LANGUAGE MODELS FOR RUMOURS DETECTION IN SOCIAL MEDIA

Item Type Journal Article**Author** P. Borase**Author** S. Kolhe

Abstract The rapid increase in the content generated by users on the social media has significantly accelerated the dissemination of rumors and misinformation, creating serious societal challenges. This paper introduces a robust and scalable framework for real-time rumor detection powered by Large Language Models, like BERT, RoBERTa, and GPTs-4. The proposed system combines Natural Language Processing techniques with sentiment analysis, stance detection, and automated fact-checking to enhance contextual understanding and assess credibility more effectively. Data is collected from various social media platforms like Twitter, Facebook, and Reddit, along with benchmark datasets like PHEME and FakeNewsNet. Experimental results show that LLMs significantly outperform traditional machine learning models, with GPT-4 achieving an accuracy of up to 94.7%. An ablation study further highlights the contribution of each module to the overall performance of GPT-4 for Rumor Detection. This framework offers a very comprehensive and flexible approach for countering misinformation across a range of languages and social environments. © 2026 by authors and Galileo Institute of Technology and Education of the Amazon (ITEGAM).

Date 2026**DOI** 10.5935/jetia.v12i59.3539**URL** <https://www.scopus.com/pages/publications/105046071588?origin=resultslist>**Archive** Scopus**Language** English**Volume** 12

Publisher Galileo Institute of Technology and Education of the Amazon (ITEGAM)

Pages 1468-1475

Publication Journal of Engineering and Technology for Industrial Applications

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Journal Abbr J. Eng. Technol. Ind. Appl.

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Tags:

Natural Language Processing, Social Media, Large Language Models, GPT-4, Rumor Detection

Notes:

EID: 2-s2.0-105046071588

Export Date: 20 September 2026; Cited By: 0

Leveraging Large Language Models for Sentiment Analysis in E-Commerce Product Reviews

Item Type Conference Paper

Author K. Thiyagarajan

Abstract Sentiment analysis has gained significant attention due to its ability to extract insights from unstructured data, such as reviews, social media posts, and customer feedback. Large language models (LLMs), with their advanced natural language understanding capabilities, have revolutionized sentiment analysis by providing more accurate and nuanced results. This paper explores the application of LLMs in sentiment analysis within the e-commerce domain, where customer feedback is abundant and crucial for business decisions. We demonstrate the effectiveness of LLMs in analyzing sentiments in product reviews, comparing their performance with traditional machine learning models. Experimental results show that LLMs significantly outperform traditional models in terms of accuracy and contextual understanding. The study also discusses the potential challenges and limitations of LLMs in sentiment analysis, as well as future research directions in this area. © 2025 IEEE.

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DOI 10.1109/AIRC64931.2025.11077508

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Archive Scopus

Language English

Extra Journal Abbreviation: Int. Conf. Artif. Intell., Robot., Control, AIRC

Publisher Institute of Electrical and Electronics Engineers Inc.

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Pages 234-239

Proceedings Title Int. Conf. Artif. Intell., Robot., Control, AIRC
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Tags:

Computational linguistics, Language model, Large language model, Deep learning, Sales, Classification (of information), Learning systems, Sentiment analysis, Natural Language Processing, Language processing, Learning algorithms, Natural language processing, Natural languages, Electronic commerce, Deep Learning, Reviews, Information management, Customer feedback, Sentiment Analysis, E-commerce, Large Language Models, Text classification, Text Classification, Product reviews, Customer Feedback

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K. Thiyagarajan; Independent Researcher, Austin, United States; email: kailash.thiyagarajan@ieee.org; Conference name: 6th International Conference on Artificial Intelligence, Robotics, and Control, AIRC 2025; Conference date: 7 May 2025 through 9 May 2025; Conference code: 210525

Leveraging Large Language Models for Suicide Detection on Social Media with Limited Labels

Item Type Conference Paper

Author V. Nguyen

Author C. Pham

Editor Ding W.

Editor Lu C.-T.

Editor Wang F.

Editor Di L.

Editor Wu K.

Editor Huan J.

Editor Nambiar R.

Editor Li J.

Editor Ilievski F.

Editor Baeza-Yates R.

Editor Hu X.

Abstract The increasing frequency of suicidal thoughts highlights the importance of early detection and intervention. Social media platforms, where users often share personal experiences and seek help, could be utilized to identify individuals at

risk. However, the large volume of daily posts makes manual review impractical. This paper explores the use of Large Language Models (LLMs) to automatically detect suicidal content in text-based social media posts. We propose a novel method for generating pseudo-labels for unlabeled data by prompting LLMs, along with traditional classification fine-tuning techniques to enhance label accuracy. To create a strong suicide detection model, we develop an ensemble approach involving prompting with Qwen2-72B-Instruct, and using fine-tuned models such as Llama3-8B, Llama3.1-8B, and Gemma2-9B. We evaluate our approach on the dataset of the Suicide Ideation Detection on Social Media Challenge, a track of the IEEE Big Data 2024 Big Data Cup. Additionally, we conduct a comprehensive analysis to assess the impact of different models and fine-tuning strategies on detection performance. Experimental results show that the ensemble model significantly improves the detection accuracy, by 5% points compared with the individual models. It achieves a weight F1 score of 0.770 on the public test set, and 0.731 on the private test set, providing a promising solution for identifying suicidal content in social media. Our analysis shows that the choice of LLMs affects the prompting performance, with larger models providing better accuracy. Our code and checkpoints are publicly available at <https://github.com/khanhvynguyen/Suicide-Detection-LLMs>. © 2024 IEEE.

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DOI 10.1109/BigData62323.2024.10825313

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Language English

Extra Journal Abbreviation: Proc. - IEEE Int. Conf. Big Data, BigData

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Pages 8550-8559

Proceedings Title Proc. - IEEE Int. Conf. Big Data, BigData

Conference Name Proceedings - 2024 IEEE International Conference on Big Data, BigData 2024

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Language model, Large language model, large language models, Economic and social effects, Classification (of information), Social media, Fine tuning, Social media analysis, Self-supervised learning, Prompt engineering, prompt engineering, Text classification, social media analysis, text classification, Semi-supervised learning, semi-supervised learning, suicide detection, Suicide detection, Limited label, limited labels

Notes:

EID: 2-s2.0-85218067792

Export Date: 20 September 2026; Cited By: 8; Correspondence Address: V. Nguyen; Northeastern University, Boston, United States; email: nguyen.vy7@northeastern.edu; Conference name: 2024 IEEE International Conference on Big Data, BigData 2024; Conference date: 15 December 2024 through 18 December 2024; Conference code: 206131

Leveraging Large Language Models for Synthetic Data Generation to Enhance Adverse Drug Event Detection in Tweets

Item Type Conference Paper

Author A. Yazdani

Author H. Rouhizadeh

Author A. Bornet

Author D. Teodoro

Editor Andrikopoulou E.

Editor Gallos P.

Editor Arvanitis T.N.

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Editor Benis A.

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Editor Chatzistergos P.

Editor Dejaco A.

Editor Dusseljee-Peute L.

Editor Mohasseb A.

Editor Natsiavas P.

Editor Nakkas H.

Editor Scott P.

Abstract Adverse drug event (ADE) detection in social media texts poses significant challenges due to the informal nature of the text and the limited availability of annotations. The scarcity of ADE named entity recognition (NER) datasets for social media hinders the development of robust ADE detection models for this type of corpus. In this paper, we leveraged the generative capabilities of large language models (LLMs) to create synthetic data, addressing this dataset gap. Specifically, we generated 17,000 tweets with ADE annotations and pre-trained NER models on this synthetic data. Our evaluations on an out-of-sample collection of 915 manually annotated tweets revealed that these models outperform state-of-the-art lexico-based and massively pre-trained open NER models. We also show that fine-tuning our synthetically pre-trained models on human-annotated data surpasses the current state-of-the-art in ADE detection on tweets. These findings suggest that synthetic data generated by LLMs can enhance ADE detection performance, offering a promising avenue to explore in response to the scarcity of annotated ADE datasets. The synthetic dataset is available at <https://huggingface.co/datasets/anthonyyazdaniml/synthetic-ner-ade-tweets-v1>. © 2025 The Authors.

Date 2025

DOI 10.3233/SHTI250465**URL** <https://www.scopus.com/pages/publications/105005816901?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Stud. Health Technol. Informatics**Volume** 327**Publisher** IOS Press BV**ISBN** 09269630 (ISSN); 978-164368596-0 (ISBN)**Pages** 778-782**Proceedings Title** Stud. Health Technol. Informatics**Conference Name** Studies in Health Technology and Informatics**Date Added** 20/09/2026, 8:35:12 pm**Modified** 20/09/2026, 8:35:12 pm**Tags:**

Computational linguistics, Language model, Large language model, human, Humans, Natural Language Processing, Natural language processing systems, Social networking (online), Social media, Large datasets, Synthetic data, social media, procedures, diagnosis, Social Media, adverse drug reaction, data mining, Data Mining, Drug-Related Side Effects and Adverse Reactions, natural language processing, pharmacovigilance, large language model, Large Language Models, Named entity recognition, Synthetic data generations, Tweets, Synthetic Data Generation, Zero-Shot Learning, Adverse Drug Reaction Reporting Systems, Pharmacodynamics, Recognition models, Adverse drug event, Adverse Drug Events, Events detection, Metamap, MetaMap, Named Entity Recognition

Notes:

EID: 2-s2.0-105005816901

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: A. Yazdani; Department of Radiology and Medical Informatics, Faculty of Medicine, University of Geneva, Geneva, Switzerland; email: Anthony.Yazdani@unige.ch; Conference name: 35th Medical Informatics Europe Conference, MIE 2025; Conference date: 19 May 2025 through 21 May 2025; Conference code: 209956

Leveraging Large Language Models in Tourism: A Comparative Study of the Latest GPT Omni Models and BERT NLP for Customer Review Classification and Sentiment Analysis

Item Type Journal Article**Author** K.I. Roumeliotis**Author** N.D. Tselikas**Author** D.K. Nasiopoulos

Abstract In today's rapidly evolving digital landscape, customer reviews play a crucial role in shaping the reputation and success of hotels. Accurately analyzing and classifying the sentiment of these reviews offers valuable insights into customer satisfaction, enabling businesses to gain a competitive edge. This study undertakes a comparative analysis of traditional natural language processing (NLP) models, such as BERT and advanced large language models (LLMs), specifically GPT-4 omni and GPT-4o mini, both pre- and post-fine-tuning with few-shot learning. By leveraging an extensive dataset of hotel reviews, we evaluate the effectiveness of these models in predicting star ratings based on review content. The findings demonstrate that the GPT-4 omni family significantly outperforms the BERT model, achieving an accuracy of 67%, compared to BERT's 60.6%. GPT-4o, in particular, excelled in accuracy and contextual understanding, showcasing the superiority of advanced LLMs over traditional NLP methods. This research underscores the potential of using sophisticated review evaluation systems in the hospitality industry and positions GPT-4o as a transformative tool for sentiment analysis. It marks a new era in automating and interpreting customer feedback with unprecedented precision. © 2024 by the authors.

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Sales, Tourism, Decision making, Sentiment analysis, Natural language processing systems, Customer satisfaction, Contrastive Learning, Customer feedback, sentiment analysis, Recommender systems, User-generated, few-shot learning, Few-shot learning, user-generated content, Customer review, User-generated content, customer feedback, customer review classification, Customer review classification, decision-making systems, Decision-making systems, forecasting models, Forecasting models, gpt-4 omni, Gpt-4 omni, Hotel review, hotel reviews, Tourism recommendation system, tourism recommendation systems

Notes:

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Leveraging LLM GPT-3.5 for Sentiment Analysis: A Comparative Study Across PC and Google Colab

Item Type Conference Paper

Author B.N. Lavanya

Author K.V. Anitha Rathnam

Author A. Appaji

Author K. Kiran

Author P.D. Shenoy

Author K.R. Venugopal

Abstract The aim of this study is to inspect sentiment analysis models that have been trained on Twitter corpus by utilising the Large Language Model (LLM) gpt-3-5-turbo-16k version of the Generative Pretrained Transformer (GPT 3.5) model. The Bidirectional long short-term memory (BiLSTM), Convolutional Neural Networks (CNN), Gated Recurrent Unit (GRU) and Recurrent Neural Network (RNN) are the trained models, which are used to perform computational tasks on the Google Colab and Personal PC. On the other hand, the training accuracies varied from 51.98% to 99.88% and the test accuracies varied from 50.00% to 75.00%. Every model has its accuracy, recall and f1-score metrics which will show the performance score of that specific model across the platforms. The refined aspects, such as time complexity, memory consumed, and CPU/GPU allocated to find the computational resource expenses on training and testing phases of each model and the issues that were pondered. The study is the first step for the sentiment analysis systems decision makers by showing the best deployment modes via presenting the sentiments classifiers that are used along with different architectures and computing configurations. © 2024 IEEE.

Date 2024

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Archive Scopus

Language English

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ISBN 979-835038592-2 (ISBN)

Proceedings Title Proc. CONECCT - IEEE Int. Conf. Electron., Comput. Commun. Technol.

Conference Name Proceedings of CONECCT 2024 - 10th IEEE International Conference on Electronics, Computing and Communication Technologies

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Language model, Large language model, Google+, Distributed computer systems, Sentiment analysis, Convolutional neural network, Convolutional neural networks, Neural-networks, CNN, GPT, LLM, Long short-term memory, Comparatives studies, RNN, Gated recurrent unit, Bi-LSTM, Electric transformer testing, GRU, Personal computers

Notes:

EID: 2-s2.0-85205823890

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Leveraging LLMs for Integrated Sentiment and Topic Analysis on African Social Media

Item Type Conference Paper

Author H. Sibitenda

Author R. Hu

Author E. Rundensteiner

Author A. Diattara

Author A. Traore

Author C. Ba

Editor Wani M.A.

Editor Angelov P.

Editor Luo F.

Editor Ogihara M.

Editor Wu X.

Editor Precup R.-E.

Editor Ramezani R.

Editor Gu X.

Abstract Leveraging AI to analyze key topics on African social media can enhance public governance. Our study analyzes social media discourse within African society on development concerns by (1) evaluating AI techniques for sentiment, topic, and theme extraction, comparing the accuracy of these methods with human annotations, and (2) extracting key insights from the data to provide policymakers with actionable recommendations for sustainable development. For this study, we utilized a data corpus of 22,036 posts from Twitter and YouTube, all focused on development issues in Africa. We applied topic modeling to extract relevant topics from the corpus and used similarity analysis, powered by Large Language Models, to link these topics to prevalent development themes. Additionally, we leveraged unsupervised models such as VADER and Large Language Models to extract sentiment related to the identified topics. To validate these model-generated sentiments, we conducted a small crowdsourced study to gather

human-annotated labels as ground truth. Our sentiment analysis findings show improvements with models like TextBlob, VADER, and Llama. Fine-tuning, particularly with BERT, achieved an impressive F1 score of 0.988. Meanwhile, Llama demonstrated strong precision (0.72) and balanced accuracy (0.55) in capturing contextual sentiment. We identified 304 topics using BERTopic and Llama, with robust coherence (0.81 C-v) and divergence (0.58 IRBO). In theme analysis, the One-vs-Rest classification with ensemble voting performed exceptionally well, with 'Poverty' achieving the highest F1 score of 0.89. Our results suggest that African policymakers prioritize addressing corruption, unemployment, drought, and instability, while closely monitoring the positive impacts of policy interventions. © 2024 IEEE.

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EID: 2-s2.0-105000969041

Export Date: 20 September 2026; Cited By: 1; Conference name: 23rd IEEE International Conference on Machine Learning and Applications, ICMLA 2024; Conference date: 18 December 2024 through 20 December 2024; Conference code: 207395

Leveraging LLMs for User Rating Prediction from Textual Reviews: A Hospitality Data Annotation Case Study

Item Type Journal Article

Author P. Nnanna

Author O. Amujo

Author C.P. Ezenkwu

Author E. Ibeke

Abstract The proliferation of user-generated content in today's digital landscape has further increased dependence on online reviews as a source for decision-making in the hospitality industry. There has been an increasing interest in automating this decision-support mechanism through recommender systems. However, this process often requires a large amount of labelled corpus to train an effective algorithm, necessitating the use of human annotators for developing training data, where this is lacking. Although the manual annotation can be helpful in enriching the training corpus, it can, on the one hand, introduce errors and annotator bias, including subjectivity and cultural bias, which can affect the quality of the data and fairness in the model. This paper examines the alignment of ratings derived from different annotation sources and the original ratings provided by customers, which are treated as the ground truth. The paper compares the predictions from Generative Pre-trained Transformer (GPT) models against ratings assigned by Amazon Mechanical Turk (MTurk) workers. The GPT 4o annotation outputs closely mirror the original ratings, given its strong positive correlation (0.703) with the latter. The GPT-3.5 Turbo and MTurk showed weaker correlations (0.663 and 0.15, respectively) than GPT 4o. The potential cause of the large difference between original ratings and MTurk (largely driven by human perception) lies in the inherent challenges of subjectivity, quantitative bias, and variability in context comprehension. These findings suggest that the use of advanced models such as GPT-4o can significantly reduce the potential bias and variability introduced by Amazon MTurk annotators, thus improving the prediction accuracy of ratings with actual user sentiment as expressed in textual reviews. Moreover, with the per-annotation cost of an LLM shown to be thirty times cheaper than MTurk, our proposed LLM-based textual review annotation approach will be cost-effective for the hospitality industry. © 2025 by the authors.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: C.P. Ezenkwu; School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, Garthdee Road, AB10 7AQ, United Kingdom; email: p.ezenkwu@rgu.ac.uk; E. Ibeke; School of Computing, Engineering and Technology, Robert Gordon University, Aberdeen, Garthdee Road, AB10 7AQ, United Kingdom; email: e.ibeke@rgu.ac.uk

Leveraging Multilingual Transformer for Multiclass Sentiment Analysis in Code-Mixed Data of Low-Resource Languages

Item Type Journal Article

Author M.K. Nazir

Author C.N. Faisal

Author M.A. Habib

Author H. Ahmad

Abstract The widespread use of online social media has enabled users to express their thoughts, feelings, opinions, and sentiments in their preferred languages. These diverse perspectives offer valuable insights for data-driven decision-making. While extensive sentiment analysis approaches have been developed for resource-rich languages like English and Chinese, low-resource languages such as Roman Urdu and Roman Punjabi, especially in code-mixed contexts, have been largely neglected due to the lack of datasets and limited research on their unique morphological structures and grammatical complexities. This study aims to present a novel approach for multiclass sentiment analysis of low-resource, code-mixed datasets using multilingual transformers. Specifically, a dataset comprising Roman Urdu, Roman Punjabi, and English comments was collected. After applying traditional natural language preprocessing techniques, transformer-based libraries were used for tokenization and embedding. Subsequently, the Multilingual Bidirectional Encoder Representations from Transformers (mBERT) model was optimized and trained for multiclass sentiment analysis on the code-mixed data. The evaluation results showed a significant improvement in accuracy (+22.55%), precision (+21.06%), recall (+22.55%), and F-measure (+25.50%) compared to benchmark algorithms. Additionally, the proposed model outperformed other transformer-based models, as well as deep learning and machine learning algorithms in sentiment extraction from code-mixed data. These findings highlight the potential of the proposed approach for sentiment analysis in low-resource, code-mixed languages. © 2025 The Authors.

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Notes:

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Export Date: 20 September 2026; Cited By: 33; Correspondence Address: H. Ahmad; National Textile University, Department of Computer Science, Faisalabad, 37610, Pakistan; email: haseeb_ad@hotmail.com

Leveraging RoBERTa and Large Language Models for Early Mental Health Signal Detection Through English and Hinglish Social Media Text

Item Type Conference Paper

Author S. Singh

Author M. Chandra

Author D. Awasthi

Editor Hassanien A.E.

Editor Anand S.

Editor Jaiswal A.

Editor Kumar P.

Abstract Mental health concerns such as depression, stress ideation are increasingly expressed on social media platforms, often in English and Hinglish (Hindi–English code-mixed text). We propose a hybrid AI framework that integrates a fine-tuned RoBERTa encoder with a Large Language Model (LLM) based semantic refinement module to detect early-stage mental-health signals across

these languages. The system performs language identification and transliteration-aware preprocessing, applies a RoBERTa classifier for fast largescale screening, and routes low-confidence or high-risk samples to the LLM for contextual reasoning. Labels include Depression, Stress, and Neutral/Non-risk; the ensemble outputs both a categorical label and a risk score (Low/Medium/ High). Experimental evaluation on a platform social media corpus (Reddit pages) shows the hybrid RoBERTa–LLM approach yields consistent improvements over baseline ML and standalone transformer models, particularly on Hinglish code-mixed posts. Cross-language F1 increases of ~4–8% points are observed compared to RoBERTa alone, and the ensemble maintains scalability by keeping RoBERTa as the dominant inference path. Privacy, ethical annotation procedures are integrated into system design to support responsible deployment in public-health surveillance and early intervention pipelines. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Language English

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Volume 2023 LNNS

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Singh; Department of Computer Science and Engineering, Institute of Engineering and Technology, Lucknow, Sitapur Road, Uttar Pradesh, 226021, India; email: shachisinghparmar@gmail.com; Conference name: 9th International Conference on Intelligent Computing and Communication, ICICC 2026; Conference date: 6 February 2026 through 7 February 2026; Conference code: 355219

Leveraging Sentiment Analysis of Consumer Feedback for Data-Driven CRM in Digital Game Ecosystems

Item Type Journal Article

Author S. Poddar

Author A. Deepansh

Author G. Poornalatha

Abstract Companies in the digital age rely on Customer Relationship Management (CRM) to build loyalty and keep users engaged. Traditionally built around structured inputs such as transactional logs or survey results, CRM systems are now challenged by the overwhelming influx of unstructured, experience-driven feedback from digital forums, review platforms, and social media. This sudden shift introduces feedback that is both high in emotional expressiveness and rich in feature-level granularity, posing significant interpretation challenges for legacy CRM workflows. This problem is obvious in the gaming industry, where users actively and continuously respond to game updates, service breakdowns, or balance issues through real-time public discourse. The domain-oriented tools are needed to manage this type of feedback to parse multi-dimensional opinion data. The proposed framework is specifically designed for gaming-related CRM applications. The current framework considers various online user reviews to process them through a dual pipeline architecture, considering the Removal of Personally Identifiable Information (PII), linguistic normalization, and ontology-driven tagging. The rule-based approach was used in the product pipeline using feature proximity. A hybrid NLP approach was employed for the service path. The sentiment tags were initialized by zero-shot RoBERTa, and a high-confidence-based fine-tuning was carried out using Sentiment140. Final predictions were reassigned and evaluated against pseudo-gold standards. Various interactive dashboards were designed to visualize the insights obtained from the framework, which assists CRM teams in acting on player sentiment with feature-level and time-sensitive clarity. © 2013 IEEE.

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Tags:

Behavioral research, Artificial intelligence, machine learning, Machine-learning, Sentiment analysis, Artificial Intelligence, Language processing, Natural language processing, Natural languages, Social networking (online), Data mining, Service industry, Pipelines, Reddit, Reviews, Information management, Feedback, natural language processing, Computer games, sentiment analysis, Linguistics, User-generated, user-generated content, Aspect-based sentiment analyze, User-generated content, aspect-based sentiment analysis, customer relationship management, Customer relationship management, gaming industry, Gaming industry, Ontology, Ontology-based, ontology-based extraction, Ontology-based extraction, Product service, product-service feedback, Product-service feedback, Public relations, review mining, Review mining, SDG 9, Steam

Notes:

EID: 2-s2.0-105031567125

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: G. Poornalatha; Manipal Institute of Technology, Manipal Academy of Higher Education, Manipal, 576104, India; email: poornalatha.g@manipal.edu

Leveraging Training Dynamics and Self-Training for Text Classification

Item Type Conference Paper

Author T. Sosea

Author C. Caragea

Editor Goldberg Y.

Editor Kozareva Z.

Editor Zhang Y.

Abstract The effectiveness of pre-trained language models in downstream tasks is highly dependent on the amount of labeled data available for training. Semi-supervised learning (SSL) is a promising technique that has seen wide attention recently due to its effectiveness in improving deep learning models when training data is scarce. Common approaches employ a teacher-student self-training framework, where a teacher network generates pseudo-labels for unlabeled data, which are then used to iteratively train a student network. In this paper, we propose a new self-training approach for text classification that leverages training dynamics of unlabeled data. We evaluate our approach on a wide range of text classification tasks, including emotion detection, sentiment analysis, question classification and grammaticality, which span a variety of domains, e.g, Reddit, Twitter, and online forums. Notably, our method is successful on all benchmarks, obtaining an average increase in F1 score of 3.5% over strong baselines in low resource settings. We make our code available at <https://github.com/tsosea2/AUM-ST>. © 2022 Association for Computational Linguistics.

Date 2022

DOI 10.18653/v1/2022.findings-emnlp.287

URL <https://www.scopus.com/pages/publications/85149861710?origin=resultslist>

Archive Scopus

Language English**Extra** Journal Abbreviation: Find. Assoc. Comput. Linguist.: EMNLP**Publisher** Association for Computational Linguistics (ACL)**ISBN** 978-195942943-2 (ISBN)**Pages** 4779-4791**Proceedings Title** Find. Assoc. Comput. Linguist.: EMNLP**Conference Name** Findings of the Association for Computational Linguistics: EMNLP 2022**Date Added** 20/09/2026, 8:35:44 pm**Modified** 20/09/2026, 8:35:44 pm**Tags:**

Computational linguistics, Language model, Deep learning, Classification (of information), Learning systems, Sentiment analysis, Social networking (online), Learning models, Students, Iterative methods, Training data, Text classification, Labeled data, Personnel training, Semi-supervised learning, Down-stream, Self-training, Unlabeled data, Teachers'

Notes:

EID: 2-s2.0-85149861710

Export Date: 20 September 2026; Cited By: 15; Funding details: UIC Discovery Partners Institute; National Science Foundation, NSF, (1802358, 1823292, 2107518); National Science Foundation, NSF; Department of Primary Industries, DPI; Conference name: 2022 Findings of the Association for Computational Linguistics: EMNLP 2022; Conference date: 7 December 2022 through 11 December 2022; Conference code: 186900

Leveraging Transformer-Based Models for Cyberbullying Detection in the Moroccan Arabic Dialect

Item Type Conference Paper**Author** H. Laachari**Author** A. Kobbane**Author** H. Tembine**Editor** Alrami B.**Editor** Suwais K.**Editor** El Maliani A.D.**Editor** El Kamili M.**Editor** Ibrahimi K.**Editor** Kobbane A.

Abstract Cyberbullying is an increasing threat on social media, with serious consequences for mental health, particularly among young people. Despite growing global efforts to address this problem, developing accurate detection systems remains challenging for low-resource languages in both text and audio modalities, such as Arabic and its dialects. This paper presents a binary-labeled dataset for

cyberbullying detection in Moroccan Darija. The dataset merges over 4,000 newly collected YouTube comments with the OMCD (Offensive Moroccan Comments Dataset) corpus of 8,024 comments, originally labeled for offensive language. To better reflect the nature of cyberbullying, the entire corpus was reannotated from scratch, clearly distinguishing between bullying and non-bullying content, including subtle forms like sarcasm, shaming, and indirect aggression. Several Arabic transformer models were fine-tuned and evaluated using standard classification metrics. The results show that dialectspecific models, particularly DarijaBERT, achieve the best performance, underlining the importance of context-aware annotation and in-domain pre-training for cyberbullying detection in lowresource settings. © 2025 IEEE.

Date 2025

DOI 10.1109/WINCOM65874.2025.11313421

URL <https://www.scopus.com/pages/publications/105034194941?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Int. Conf. Wirel. Networks Mob. Commun., WINCOM

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 27699994 (ISSN); 979-833159878-5 (ISBN)

Proceedings Title Int. Conf. Wirel. Networks Mob. Commun., WINCOM

Conference Name International Conference on Wireless Networks and Mobile Communications, WINCOM

Issue 2025

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Tags:

Mental health, Natural Language Processing, Language processing, Natural language processing, Natural language processing systems, Natural languages, Social networking (online), Health risks, Social media, Computer crime, Audio systems, Transformer, Transformers, Text processing, Labeled data, Low resource languages, Electric transformers, Cyber bullying, Arabic dialects, Cyberbullying classification, Low resource language

Notes:

EID: 2-s2.0-105034194941

Export Date: 20 September 2026; Cited By: 1; Funding details: Centre National pour la Recherche Scientifique et Technique, CNRST; Conference name: 12th International Conference on Wireless Networks and Mobile Communications, WINCOM 2025; Conference date: 25 November 2025 through 27 November 2025; Conference code: 218841

Leveraging Zero-Shot Learning in Large Language Models for Sentiment Analysis: A Comparative Study on the Indonesian Language

Item Type Conference Paper**Author** I. Prawira

Abstract This research assesses the performance of large language models (LLMs), specifically GPT-3.5 and GPT-4o mini, in sentiment analysis of Indonesian compared to traditional models like IndoBERT and DistilBERT. Using the IndoLAM dataset, we benchmarked these models on precision, recall, and F1 score metrics, focusing on their zero-shot learning capabilities. Our results show that GPT-3.5 achieved the highest F1 score of 0.83, reflecting a strong balance between precision (0.81) and recall (0.86). GPT-4o Mini excelled in precision (0.88) but had a lower recall (0.72), resulting in an F1 score of 0.76. In contrast, traditional models, particularly DistilBERT, exhibited weaker performance with an F1 score of 0.59. Additionally, we applied these models to analyze sentiment during the 2024 Indonesian presidential election using unlabeled Twitter data, demonstrating their effectiveness in real-world zero-shot scenarios. The study highlights the superiority of LLMs in contextual understanding and their capability to handle complex linguistic tasks with minimal labeled data, contributing to the discourse on the practical applications of LLMs in sentiment analysis, especially in linguistically diverse contexts. © 2024 IEEE.

Date 2024**DOI** 10.1109/ICIMCIS63449.2024.10956237**URL** <https://www.scopus.com/pages/publications/105004416283?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. - Int. Conf. Informatics, Multimed., Cyber Inf. Syst. ICIMCIS**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833153324-3 (ISBN)**Pages** 614-619**Proceedings Title** Proc. - Int. Conf. Informatics, Multimed., Cyber Inf. Syst. ICIMCIS**Conference Name** Proceedings - 6th International Conference on Informatics, Multimedia, Cyber and Information System, ICIMCIS 2024**Date Added** 20/09/2026, 8:35:30 pm**Modified** 20/09/2026, 8:35:30 pm

Tags:

Language model, Large language model, Performance, Sentiment analysis, Emotion Recognition, Sentiment Analysis, Comparatives studies, Large Language Models (LLMs), Labeled data, Zero-shot learning, F1 scores, Traditional models, Zero-Shot Learning, Indonesian Language, Indonesian languages, Election Sentiment Analysis, Election sentiment analyze

Notes:

EID: 2-s2.0-105004416283

Export Date: 20 September 2026; Cited By: 1; Conference name: 6th International Conference on Informatics, Multimedia, Cyber and Information System, ICIMCIS 2024; Conference date: 20 November 2024 through 21 November 2024; Conference code: 208360

LightFSA: A Lightweight Financial Sentiment Analysis Model

Item Type Conference Paper

Author Z. Liu

Author J. Li

Author C.-W. Sham

Author S.L. Ma

Author C. Fu

Editor Huang D.-S.

Editor Zhang Q.

Editor Pan Y.

Editor Zhang C.

Abstract Financial sentiment analysis involves interpreting information from financial articles, news and social media to understand market trends and guide investment decisions. Numerous studies have used AI techniques, particularly large language models, to improve sentiment classification in the financial domain. Although these approaches have shown promising results, existing LLMs face two critical challenges: their large size, which results in high inference times and significant resource consumption, and their inability to fully account for the unique characteristics of financial sentiment classification. Unlike general sentiment analysis, financial sentiment requires robust reasoning and domain-specific expertise, which current models often do not adequately incorporate. To address these issues, this study proposes a lightweight chain-of-thought-based model for financial sentiment classification. We designed zero-shot chain-of-thought prompts tailored specifically for financial sentiment, integrating contextual reasoning into the classification process. This approach effectively improves accuracy by 2%–5% on financial sentiment classification tasks. In addition, we applied an activation-aware weight (AWQ) quantization algorithm to selectively reduce the model parameters. Experimental results demonstrate that our model significantly reduces memory usage by 90.3% and inference time by 87.9% compared to the baseline model, with only minimal impact on accuracy. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

Date 2025

DOI 10.1007/978-981-96-9914-8_28

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Archive Scopus

Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

Volume 15856 LNCS

Publisher Springer Science and Business Media Deutschland GmbH

ISBN 03029743 (ISSN); 978-981969913-1 (ISBN)

Pages 332-343

Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science

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Language model, Large language model, Classification (of information), Sentiment analysis, Large Language Model, Investments, Social media, Chains, Sentiment classification, News media, Analysis models, Quantisation, Financial sentiment classification, Financial Sentiment Classification, Quantization, Supervised fine tune, Supervised Fine Tune

Notes:

EID: 2-s2.0-105011817038

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: Z. Liu; University of Auckland, Auckland, 1010, New Zealand; email: zliu604@aucklanduni.ac.nz; Conference name: 21st International Conference on Intelligent Computing, ICIC 2025; Conference date: 26 July 2025 through 29 July 2025; Conference code: 335459

Lightweight Approach for Multi-Modal Irony Detection by Image Caption Generation

Item Type Conference Paper

- Author B.N. Koc
- Author R.F. Cekineli
- Author P. Karagoz
- Editor Xu C.-Z.
- Editor U L.H.
- Editor Cheng X.
- Editor Gao J.
- Editor Polese G.
- Editor Mei H.
- Editor Boniol P.
- Editor Tatsubori M.
- Editor Zhao C.
- Editor Zhou D.
- Editor Hu X.

Abstract In this work, we tackle the multi-modal irony detection problem and propose a lightweight approach that combines pretrained multi-modal and text-based smaller language models. The proposed approach makes use of generated image captions and original text content to fine-tune a transformer based model for text classification. In the proposed solution, Vision-Language Model (VLM) is used for generating image captions, without any fine-tuning. On two benchmark datasets, the experiments demonstrate the benefit of the proposed method against the baseline solutions. © 2025 IEEE.

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DOI 10.1109/BigData66926.2025.11402099

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Language English

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Proceedings Title Proc. IEEE Int. Conf. Big Data, BigData
Conference Name Proceedings of the IEEE International Conference on Big Data, BigData
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Computational linguistics, Language model, Large language model, Visual languages, Classification (of information), Computer vision, Social media, Multi-modal, Social Media, Large Language Models, Text processing, Sarcasm detection, Sarcasm Detection, Detection problems, Image caption, Multimodal irony detection, Multimodal Irony Detection, Vision-language model, Vision-Language Models

Notes:

EID: 2-s2.0-105037812892

Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 IEEE International Conference on Big Data, BigData 2025; Conference date: 8 December 2025 through 11 December 2025; Conference code: 220864

LinkNER: Linking Local Named Entity Recognition Models to Large Language Models using Uncertainty

Item Type Conference Paper
Author Z. Zhang
Author Y. Zhao
Author H. Gao
Author M. Hu
Abstract Named Entity Recognition (NER) serves as a fundamental task in natural language understanding, bearing direct implications for web content analysis, search engines, and information retrieval systems. Fine-tuned NER models exhibit satisfactory performance on standard NER benchmarks. However, due to limited fine-tuning data and lack of knowledge, it performs poorly on unseen entity recognition. As a result, the usability and reliability of NER models in web-related applications are compromised. Instead, Large Language Models (LLMs) like GPT-4 possess extensive external knowledge, but research indicates that they lack specialty for NER tasks. Furthermore, non-public and large-scale weights make tuning LLMs difficult. To address these challenges, we propose a framework that combines small fine-tuned models with LLMs (LinkNER) and an uncertainty-based linking strategy called RDC that enables fine-tuned models to

complement black-box LLMs, achieving better performance. We experiment with both standard NER test sets and noisy social media datasets. LinkNER enhances NER task performance, notably surpassing SOTA models in robustness tests. We also quantitatively analyze the influence of key components like uncertainty estimation methods, LLMs, and in-context learning on diverse NER tasks, offering specific web-related recommendations. © 2024 ACM.

Date 2024

DOI 10.1145/3589334.3645414

URL <https://www.scopus.com/pages/publications/85194085596?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: WWW - Proc. ACM Web Conf.

Publisher Association for Computing Machinery, Inc

ISBN 979-840070171-9 (ISBN)

Pages 4047-4058

Proceedings Title WWW - Proc. ACM Web Conf.

Conference Name WWW 2024 - Proceedings of the ACM Web Conference

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Computational linguistics, Language model, Large language model, large language models, Performance, Uncertainty analysis, Natural language processing systems, Benchmarking, Uncertainty, Information extraction, Information retrieval, Search engines, Robustness, robustness, Named entity recognition, Information retrieval systems, Recognition models, information extraction, Natural language understanding, uncertainty estimation, Uncertainty estimation

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LLM Based Sentiment Analysis in Private Social Media: Case Study of My Tel-U App

Item Type Conference Paper

Author F. Alfiansyah

Author M.D. Purbolaksono

Author A.A. Gozali

Abstract My Tel-U is a private social media application used by the Telkom University academic community to access academic information and express opinions through its timeline feature. Posts shared on the timeline often reflect users' sentiments toward campus services or the environment. However, previous sentiment analysis approaches have relied on small datasets and traditional methods that struggle to handle informal and mixed-language Indonesian texts. This study aims to address these limitations by evaluating the effectiveness of Large Language Models (LLMs), including BERT, RoBERTa, and IndoBERT, in comparison to traditional models like Support Vector Machine (SVM), for sentiment analysis on imbalanced and noisy data from My Tel-U. The research was conducted through three experimental scenarios, which include examining the effect of different oversampling techniques (SMOTE, random oversampling, and no oversampling) on IndoBERT, the performance comparison among LLMs, and comparison between LLMs and SVM using TF-IDF and Word2Vec representations. This study is among the first to systematically explore the use of Indonesian LLMs for sentiment classification in private educational social media platforms. Results show that IndoBERT outperforms all other models with an F1-Macro score of 0.8423, achieving strong performance without requiring oversampling. This indicates that pretraining on an Indonesian corpus significantly enhances contextual understanding. In contrast, English-based LLMs and traditional SVM models demonstrate lower performance due to their limitations in handling Indonesian linguistic structures. These findings demonstrate the practical potential of integrating language-specific LLMs into institutional feedback and monitoring systems, enabling universities to better understand and respond to user sentiments. © 2025 IEEE.

Date 2025

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Archive Scopus

Language English

Extra Journal Abbreviation: Int. Conf. Data Sci. Its Appl., ICoDSA

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Pages 1100-1105

Proceedings Title Int. Conf. Data Sci. Its Appl., ICoDSA

Conference Name 2025 International Conference on Data Science and Its Applications, ICoDSA 2025

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Computational linguistics, Language model, Large language model, large language models, Performance, Support vector machines, Support vectors machine, Sentiment analysis, Social networking (online), Social media, sentiment analysis, Over sampling, imbalanced data, Imbalanced data, indobert, Indobert, my tel-u, My tel-u

Notes:

EID: 2-s2.0-105018086625

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: F. Alfiansyah; Telkom University, Faculty of Informatics, Bandung, Indonesia; email: fahrialfiansyah@student.telkomuniversity.ac.id; Conference name: 8th International Conference on Data Science and Its Applications, ICoDSA 2025; Conference date: 3 July 2025 through 5 July 2025; Conference code: 212871

LLM GPT-3.5 study for sentiment analysis across Utkarsh server, Ohio supercomputer, Google Colab and PC

Item Type Journal Article

Author L. B N

Author A.R. K V

Author A. Appaji

Author K. K

Author P.D. Shenoy

Author V. K R

Abstract The major objective of the present research is to inspect sentiment analysis models that have been trained on Twitter corpus by utilising the Large Language Model (LLM) gpt-3-5-turbo-16k version of the Generative Pretrained Transformer (GPT 3.5) model. Such trained models include the Bidirectional long short-term memory neural network (BiLSTM), Convolutional Neural Networks, Gated Recurrent Unit and Recurrent Neural Network which were used to perform computational tasks on the Ohio Supercomputer and Utkarsh Server, in comparison to work conducted on Google Colab and a Personal Computer (PC). This research also looks at the performance as well as the computational aspects of these models in terms of accuracy, recall, F1-score, time/memory complexity and resource requirements (CPU/GPU) throughout the training and testing phases of each model. The training accuracies in this case were concentrated between 49.91 % and 99.98 % while those of the testing accuracies of the models accounts for about 50.00–75.00 %. For instance, models including BiLSTM and RNN usually exhibit more time complexity because of the nature of the models (sequential computation), on the contrary, CNNs are less time-consuming and are more effective in terms of storage modifying layered architecture. The use of supercomputers and specialized servers reduces training time, but resource constraints on platforms such as personal computers or Colab cause considerable divergence. © 2024 The Author(s)

Date 2024

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Language model, Large language model, Convolutional neural network, Convolutional neural networks, Language processing, Natural language processing, Natural language processing systems, Natural languages, Neural-networks, Generative pretrained transformer, Computer circuits, Long short-term memory, Natural language processing (NLP), Convolutional neural networks (CNN), Recurrent neural network, Short term memory, Gated recurrent unit, Multilayer neural networks, Bidirectional long short-term memory (BiLSTM), Large language model (LLM), Electric transformer testing, Associative storage, Bidirectional long short-term memory, Gated recurrent unit (GRU), Generative pretrained transformer (GPT), Personal computers, Recurrent neural network (RNN), Supercomputers

Notes:

EID: 2-s2.0-85209244481

Export Date: 20 September 2026; Cited By: 7; Funding details: CDAC; Correspondence Address: A. Appaji; Department of Medical Electronics Engineering, B.M.S. College of Engineering, Bangalore, India; email: abhishek.ml@bmsce.ac.in

LLM Safety in Judicial AI: A Stress Test of Social Media Influence on Real-World Judgments

Item Type	Conference Paper
Author	Y. Xie
Author	Y. He
Author	X. Yang
Author	X. Gai
Author	P. Hui
Editor	Koenig S.
Editor	Jenkins C.
Editor	Taylor M.E.
Abstract	Integrating Large Language Models (LLMs) into judicial decision-making demands rigorous safety examination against non-legal influences. This paper presents a novel stress test where we evaluate LLM-generated labor dispute outcomes by introducing social media sentiment as an external pressure, critically comparing them against 10,000 real-world court judgments from China Judgments Online (CJOL). Our findings reveal significant LLM safety vulnerabilities: models exhibit inherent deviations from real rulings, and public opinion substantially amplifies these discrepancies, leading to unstable and often inflated compensation predictions. Furthermore, these safety risks are compounded across low-skilled occupational categories and emotionally charged topics. This study uncovers critical threats to judicial integrity and public trust,

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Language	English
Extra	Journal Abbreviation: Proc. AAAI Conf. Artif. Intell.
Volume	40
Publisher	Association for the Advancement of Artificial Intelligence

Pages 39468-39476

Conference Name Proceedings of the AAAI Conference on Artificial Intelligence

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Language model, Safety engineering, Artificial intelligence, Economic and social effects, Decisions makings, Public opinions, Social aspects, Decision making, Sentiment analysis, Social networking (online), Social media, Real-world, Laws and legislation, Occupational risks, Court judgement, External pressures, Industrial relations, Labor dispute, Stress test, Vulnerability models

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P. Hui; The Hong Kong University of Science and Technology (Guangzhou), Guangzhou, China; email: panhui@hkust-gz.edu.cn; Conference name: 40th AAAI Conference on Artificial Intelligence, AAAI 2026; Conference date: 20 January 2026 through 27 January 2026; Conference code: 349419

LLM-Assisted Sentiment Analysis for Indonesia's Coretax Policy: A Knowledge Distillation and Human-in-the-Loop Approach

Item Type Conference Paper

Author N. Ashari

Author U.O. Sihaloho

Author N.A. Sari

Abstract The lack of high-quality labeled datasets remains a major challenge for sentiment analysis in low-resource languages such as Indonesian, particularly in specialized domains like fiscal policy. This study investigates the effectiveness of Large Language Models (LLMs) as automated annotators within a teacher-student knowledge distillation framework. Using social media data from X related to Indonesia's Coretax system, three training scenarios were evaluated: AI-labeled data, human-labeled data, and a hybrid approach. The results show that GPT-4o achieves substantial agreement with human annotators, with a Cohen's Kappa score of 0.61. Furthermore, the student model IndoBERT trained on the combined dataset outperforms other configurations, achieving a Macro F1-score of 0.64 and a Macro ROC-AUC of 0.84. These findings indicate that while LLMs cannot fully replace human judgment, they significantly enhance scalability and enable near real-time policy evaluation in low-resource settings through effective human-AI collaboration. © 2026 IEEE.

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Archive Scopus

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Language model, Large language model, Information systems, Sentiment analysis, Data mining, Distillation, Knowledge distillation, Knowledge Distillation, Sentiment Analysis, Information use, High quality, Human-in-the-loop, Large Language Models, Labeled data, Coretax policy, Coretax Policy, Human-in-the-Loop, Indonesia, Labeled dataset, Personnel training, Teaching

Notes:

EID: 2-s2.0-105048344847

Export Date: 20 September 2026; Cited By: 0; Conference name: 27th International Seminar on Intelligent Technology and Its Applications, ISITIA 2026; Conference date: 22 July 2026 through 24 July 2026; Conference code: 226960

LLM-Based Classification and Topic Modeling of Generative AI Ethical Risk Discourse on Social Media

Item Type Journal Article

Author S. Kim

Author C. Yoon

Author S.H. Kim

Author B.G. Lee

Abstract Identifying sparse, context-dependent target discourse such as ethical risk discourse on generative AI in noisy social media data is a long-standing challenge in computational text analysis. Beyond classification, interpreting the resulting topic structure transparently poses an additional methodological challenge. To address these problems, this study proposes a reproducible two-stage framework. First, four LLMs (GPT-4.1, GPT-3.5-turbo, Claude Sonnet 4.6, and Gemini 2.5 Pro) are compared on a human-annotated validation set using a zero-shot prompt grounded in five ethical risk categories, and the best-performing model is selected to classify the full keyword-prescreened corpus. Validity is evaluated against human annotations and supervised baselines. Second, BERTopic is applied to the classified corpus, and sub-topics are linked to higher-level categories through an explicit mapping protocol. The study draws on approximately 500,000 ChatGPT-related English-language tweets from January to March 2023. Only 36.4% of the validation sample constituted ethical risk discourse, confirming substantial pre-screening noise. GPT-4.1, selected as the final classification model, achieved the strongest performance (accuracy = 0.899, F1 = 0.859, Cohen's κ = 0.780), significantly outperforming conventional supervised baselines. BERTopic yielded 33 non-outlier sub-topics; within this subset, Societal and democratic risks (36.16%) and Technical safety (20.90%) were the largest categories. These findings should be interpreted not as a direct measure of users' risk perceptions in 2023, but as the category distribution within the structured subset excluding outliers, after keyword screening and classification under the current ethical risk taxonomy used in this study. Within this scope, societal and institutional concerns were at least as prominent as those related to technical failures. © 2026 by the authors.

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LLM-based Hierarchical Label Annotation for Foodborne Illness Detection on Social Media

Item Type Conference Paper

Author D. Zhang

Author R. Hu

Author D. Tao

Author H. Feng

Author E. Rundensteiner

Editor Ding W.

Editor Lu C.-T.

Editor Wang F.

Editor Di L.

Editor Wu K.

Editor Huan J.
Editor Nambiar R.
Editor Li J.
Editor Ilievski F.
Editor Baeza-Yates R.
Editor Hu X.

Abstract Foodborne illnesses pose a threat to public health, leading to morbidity, mortality, and economic burden annually. Social media, while providing a rich timely source for training AI models for surveillance, requires effective tools for annotation. While Large Language Models (LLMs) have shown promise for generating simple labels, here hierarchical labels composed of entity types like food type and symptom (at individual word level) and the foodborne illness event (at complete post level) are required. For this, we introduce ICL2FID, the first LLM-based hierarchical labeling framework designed to annotate social media posts for foodborne illness detection at two levels using only a few demonstration examples. To utilize the interconnection between post and word levels, ICL2FID instructs the LLM to leverage information from one level when predicting the other level. To combat model hallucination and cyclic dependencies, a verification step improves evidence propagation between interconnected word and post-level labeling tasks. Strategies for custom selection of demonstration examples are designed reducing biases and increasing representation. We compare ICL2FID against traditional supervised learning and other LLM methods, demonstrating that it not only achieves superior accuracy but does so at a fraction of the cost and time. These findings highlight ICL2FID's potential as a viable alternative for hierarchical label generation in scenarios with limited resources and huge data sets. Code is available at <https://github.com/zdy93/ICL2FID>. © 2024 IEEE.

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LLM-Based Sentiment Analysis of RPG Genre Game Reviews on Steam

Item Type Conference Paper**Author** E.D. Sakti**Author** M.D. Purbolaksono**Author** A.A. Gozali

Abstract The rapid growth of the digital gaming industry has led to a significant increase in user-generated content, particularly reviews on platforms like Steam. Analyzing the sentiment of these reviews offers valuable insights into user experiences and provides developers with data to support decision making. This study examines the effectiveness of three prominent large language models (LLMs), BERT, RoBERTa, and DistilBERT, in classifying sentiment within RPG game reviews sourced from Steam. The models were evaluated across performance metrics including accuracy, precision, recall, and F1-score, under varying learning rates. Experimental results reveal that RoBERTa outperforms BERT and DistilBERT, achieving the highest F1-score of 97.50% at a learning rate of $2e-5$. BERT follows with a peak F1-score of 95.50%, while DistilBERT achieves a competitive score of 95.30%. The impact of text preprocessing was also analyzed, showing that excessive transformations such as lemmatization and stopword removal may negatively affect performance. These findings provide important insights for optimizing sentiment analysis methods in the context of user reviews and contribute to advancing natural language processing (NLP) applications on digital platforms. The study also introduces a domain-specific sentiment lexicon analysis to reflect on the contextual ambiguity of common gaming terms, offering a valuable foundation for future research in sentiment analysis for online gaming communities. © 2025 IEEE.

Date 2025**DOI** 10.1109/ICoDSA67155.2025.11157177**URL** <https://www.scopus.com/pages/publications/105018090626?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Int. Conf. Data Sci. Its Appl., ICoDSA**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833159854-9 (ISBN)**Pages** 1244-1249**Proceedings Title** Int. Conf. Data Sci. Its Appl., ICoDSA

Conference Name 2025 International Conference on Data Science and Its Applications, ICoDSA 2025

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Computational linguistics, Language model, Artificial intelligence, Model-based OPC, User experience, Decision making, Sentiment analysis, Learning algorithms, Data mining, sentiment analysis, F1 scores, Bert, bert, distilbert, Distilbert, Game reviews, Learning rates, roberta, Robertum, Steam review, steam reviews

Notes:

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LLM-Supported Content Analysis of Motivated Reasoning on Climate Change

Item Type Journal Article

Author Y. Kim

Author Q. Liu

Author J. Hemsley

Abstract Public discourse around climate change remains polarized despite scientific consensus on anthropogenic climate change (ACC). This study examines how “believers” and “skeptics” of ACC differ in their YouTube comment discourse. We analyzed 44,989 comments from 30 videos using a large language model (LLM) as a qualitative annotator, identifying ten distinct topics. These annotations were combined with social network analysis to examine engagement patterns. A linear mixed-effects model showed that comments about government policy and natural cycles generated significantly lower interaction compared to misinformation, suggesting these topics are ideologically settled points within communities. These patterns reflect motivated reasoning, where users selectively engage with content that aligns with their identity and beliefs. Our findings highlight the utility of LLMs for large-scale qualitative analysis and highlight how climate discourse is shaped not only by content, but by underlying cognitive and ideological motivations. Annual Meeting of the Association for Information Science & Technology | Nov. 14 – 18, 2025 | Washington, DC, USA.

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Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: Y. Kim; Syracuse University, United States; email: ykim72@syr.edu

LLMs Cannot (Yet) Match the Specificity and Simplicity of Online Finance Communities in Long Form Question Answering

Item Type Conference Paper

Author K.-F. Kahl

Author T. Buz

Author R. Biswas

Author G. de Melo

Editor Al-Onaizan Y.

Editor Bansal M.

Editor Chen Y.-N.

Abstract Retail investing is on the rise, and a growing number of users are relying on online finance communities to educate themselves. However, as Large Language Models (LLMs) are increasingly viewed as powerful question-answering (QA) tools, users have shifted away from interacting in communities towards discourse with AI-driven conversational interfaces. Such AI tools are currently constrained by the availability of labelled data providing domain-specific financial knowledge. Therefore, in this work, we curate a QA preference dataset called SOCIALFINANCEQA for fine-tuning and aligning LLMs, extracted from more than 7.4 million submissions and 82 million comments from 2008 to 2022 in

Reddit's 15 largest finance communities. Additionally, we propose the novel framework SOCIALQA-EVAL as a generally applicable method to evaluate generated QA responses. We evaluate various LLMs fine-tuned on this dataset, using traditional metrics, LLM-based evaluation, and human annotation. Our results demonstrate the value of high-quality Reddit data, with even state-of-the-art LLMs improving on producing simpler and more specific responses. © 2024 Association for Computational Linguistics.

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Language	English
Extra	Journal Abbreviation: EMNLP - Conf. Empir. Methods Nat. Lang. Process., Find. EMNLP
Publisher	Association for Computational Linguistics (ACL)
ISBN	979-889176168-1 (ISBN)
Pages	2028-2053
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Conference Name	EMNLP 2024 - 2024 Conference on Empirical Methods in Natural Language Processing, Findings of EMNLP 2024
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Computational linguistics, Language model, Fine tuning, High quality, Labeled data, Domain specific, Question Answering, Human annotations, State of the art, Conversational interface, Model-based evaluation, Question answering

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LLMs vs Transformer Models: Which Approach is Crucial in Improving Arabic Sentiment Analysis?

Item Type	Conference Paper
Author	A. Debbah
Author	S. Lagrini
Author	S. Zenati
Author	S. Debboub
Editor	Ensari T.

Editor Bennour A.

Editor Bouaziz B.

Editor Mahdi W.

Editor Rida I.

Abstract Sentiment analysis is a primary task in Natural Language Processing (NLP), where the goal is to classify emotions from text. This paper investigates and compares two prominent approaches for Arabic sentiment analysis: encoder-based models, which utilize Transformer Embeddings such as AraBERT and CAMELBERT, and encoder-decoder-based models, which involve the Fine-Tuning of large language Models (LLMs) like ArabianGPT-0.1B and AraGPT2. We explore how each approach handles the complexities of Arabic morphology and semantics to capture nuanced sentiment in the text. Empirical evaluation on LABR dataset (a large-scale Arabic Book Reviews dataset) shows that the Fine-Tuning of LLMs consistently outperforms the Embedding-based models offering superior accuracy and generalizability. These findings underscore the potential of Fine-tuned LLMs for real-world applications, including social media monitoring, opinion mining, and customer feedback analysis in Arabic-speaking contexts. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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LocalTweets to LocalHealth: A Mental Health Surveillance Framework Based on Twitter Data

Item Type Conference Paper

Author V. Deshpande

Author M. Lee

Author Z. Yao

Author Z. Zhang

Author J.B. Gibbons

Author H. Yu

Editor Calzolari N.

Editor Kan M.-Y.

Editor Hoste V.

Editor Lenci A.

Editor Sakti S.

Editor Xue N.

Abstract Prior research on Twitter (now X) data has provided positive evidence of its utility in developing supplementary health surveillance systems. In this study, we present a new framework to surveil public health, focusing on mental health (MH) outcomes. We hypothesize that locally posted tweets are indicative of local MH outcomes and collect tweets posted from 765 neighborhoods (census block groups) in the USA. We pair these tweets from each neighborhood with the corresponding MH outcome reported by the Center for Disease Control (CDC) to create a benchmark dataset, LocalTweets. With LocalTweets, we present the first population-level evaluation task for Twitter-based MH surveillance systems. We then develop an efficient and effective method, LocalHealth, for predicting MH outcomes based on LocalTweets. When used with GPT3.5, LocalHealth achieves the highest F1-score and accuracy of 0.7429 and 79.78%, respectively, a 59% improvement in F1-score over the GPT3.5 in zero-shot setting. We also utilize LocalHealth to extrapolate CDC's estimates to proxy unreported neighborhoods, achieving an F1-score of 0.7291. Our work suggests that Twitter data can be effectively leveraged to simulate neighborhood-level MH outcomes. © 2024 ELRA Language Resource Association: CC BY-NC 4.0.

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Logic Augmented Multi-Decision Fusion Framework for Stance Detection on Social Media

Item Type Journal Article

Author B. Zhang

Author J. Ma

Author X. Fu

Author G. Dai

Abstract Stance detection in social media has become increasingly crucial for understanding public opinions on controversial issues. While large language models (LLMs) have shown promising results in stance detection, existing methods face challenges in reconciling inconsistent predictions and logical reasoning processes across different LLMs. To address these limitations, we propose LogiMDF, a Logic Augmented Multi-Decision Fusion framework that effectively integrates multiple LLMs' decision processes through a unified logical framework. Our approach first employs zero-shot prompting to extract first-order logic (FOL) rules representing each LLM's prediction rationale, then constructs a Logical Fusion Schema (LFS) to bridge different LLMs' knowledge representations. We further develop a Multi-view Hypergraph Convolutional Network (MvHGCN) that effectively models and encodes the integrated logical knowledge. Extensive experiments on benchmark datasets demonstrate that LogiMDF significantly outperforms existing methods, achieving state-of-the-art performance in stance detection tasks. The results confirm that our framework effectively leverages the complementary strengths of multiple LLMs while maintaining consistent logical reasoning across different targets. © 2025

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Long COVID Discourse in Canada, the United States, and Europe: Topic Modeling and Sentiment Analysis of Twitter Data

Item Type Journal Article

Author A.G.T. AbuRaed

Author E.A. Prikryl

Author G. Carenini

Author N.Z. Janjua

Abstract Background: Social media serves as a vast repository of data, offering insights into public perceptions and emotions surrounding significant societal issues. Amid the COVID-19 pandemic, long COVID (formally known as post-COVID-19 condition) has emerged as a chronic health condition, profoundly impacting numerous lives and livelihoods. Given the dynamic nature of long COVID and our evolving understanding of it, effectively capturing people's sentiments and perceptions through social media becomes increasingly crucial. By harnessing the wealth of data available on social platforms, we can better track the evolving narrative surrounding long COVID and the collective efforts to address this pressing issue. Objective: This study aimed to investigate people's perceptions and sentiments around long COVID in

Canada, the United States, and Europe, by analyzing English-language tweets from these regions using advanced topic modeling and sentiment analysis techniques. Understanding regional differences in public discourse can inform tailored public health strategies. Methods: We analyzed long COVID–related tweets from 2021. Contextualized topic modeling was used to capture word meanings in context, providing coherent and semantically meaningful topics. Sentiment analysis was conducted in a zero-shot manner using Llama 2, a large language model, to classify tweets into positive, negative, or neutral sentiments. The results were interpreted in collaboration with public health experts, comparing the timelines of topics discussed across the 3 regions. This dual approach enabled a comprehensive understanding of the public discourse surrounding long COVID. We used metrics such as normalized pointwise mutual information for coherence and topic diversity for diversity to ensure robust topic modeling results. Results: Topic modeling identified five main topics: (1) long COVID in people including children in the context of vaccination, (2) duration and suffering associated with long COVID, (3) persistent symptoms of long COVID, (4) the need for research on long COVID treatment, and (5) measuring long COVID symptoms. Significant concern was noted across all regions about the duration and suffering associated with long COVID, along with consistent discussions on persistent symptoms and calls for more research and better treatments. In particular, the topic of persistent symptoms was highly prevalent, reflecting ongoing challenges faced by individuals with long COVID. Sentiment analysis showed a mix of positive and negative sentiments, fluctuating with significant events and news related to long COVID. Conclusions: Our study combines natural language processing techniques, including contextualized topic modeling and sentiment analysis, along with domain expert input, to provide detailed insights into public health monitoring and intervention. These findings highlight the importance of tracking public discourse on long COVID to inform public health strategies, address misinformation, and provide support to affected individuals. The use of social media analysis in understanding public health issues is underscored, emphasizing the role of emerging technologies in enhancing public health responses. ©Ahmed Ghassan Tawfiq AbuRaed, Emil Azuma Prikryl, Giuseppe Carenini, Naveed Zafar Janjua.

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LSE-FRGCN: A Logical Schema-Enhanced Fuzzy Relational Graph Convolutional Network for Stance Detection

Item Type Journal Article

Author H. Huang

Author L. Dong

Author J. Ma

Author Y. Yang

Author G. Dai

Author B. Zhang

Abstract Understanding public attitudes expressed on social media platforms forms the foundation of opinion-mining applications, especially within the consumer electronics domain. Stance detection, as a fundamental subtask, aims to identify the viewpoint or position conveyed in subjective texts with regard to specific targets, such as products, features, or brands. Although there have been notable improvements in traditional stance detection methodologies, accurately interpreting texts with inherently ambiguous positions remains a challenging. Fuzzy logic offers a theoretical framework for representing uncertainty in knowledge. However, existing fuzzy-based and LLM-fuzzy approaches typically assume static and homogeneous uncertainty patterns, and their practical application is further hindered by the need for extensive labeled datasets to extract logical rules, which is unrealistic given the diversity of stance detection targets. Drawing on the strong zero-shot reasoning abilities of large language models (LLMs), we introduce a novel architecture termed the LLM-enhanced Fuzzy Relational Graph Convolutional Network with Logical Schema Enhancement (LSE-FRGCN). Unlike prior frameworks that rely on fixed fuzzification schemes, LSE-FRGCN generalizes fuzzy reasoning from static fuzzification to an adaptive and learnable uncertainty modeling paradigm that is conditioned on relational graph structure. This design combines the representational flexibility of large language models with uncertainty-aware relational reasoning for stance detection. Specifically, LSE-FRGCN is structured around two key modules.

The first module generates a logical schema graph that abstracts first-order logical rationales into transferable conceptual structures. The second module employs a fuzzy relational graph convolutional network to integrate logical information under adaptive uncertainty modeling. This generalization enables robust stance inference in zero-shot, cross-target, and low-resource scenarios. Experiments show that our method consistently achieves superior results across these settings. In particular, on the challenging zero-shot benchmarks SemEval-2016 and VAST, our approach surpasses the strongest baselines by margins of 5.1% and 5.5%, respectively. © 2026 IEEE.

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M2SA: Multimodal and Multilingual Model for Sentiment Analysis of Tweets

Item Type Conference Paper

Author G. Thakkar

Author S. Hakimov

Author M. Tadić

Editor Calzolari N.

Editor Kan M.-Y.

Editor Hoste V.

Editor Lenci A.

Editor Sakti S.

Editor Xue N.

Abstract In recent years, multimodal natural language processing, aimed at learning from diverse data types, has garnered significant attention. However, there needs to be more clarity when it comes to analysing multimodal tasks in multi-lingual contexts. While prior studies on sentiment analysis of tweets have predominantly focused on the English language, this paper addresses this gap by transforming an existing textual Twitter sentiment dataset into a multimodal format through a straightforward curation process. Our work opens up new avenues for sentiment-related research within the research community. Additionally, we conduct baseline experiments utilising this augmented dataset and report the findings. Notably, our evaluations reveal that when comparing unimodal and multimodal configurations, using a sentiment-tuned large language model as a text encoder performs exceptionally well. © 2024 ELRA Language Resource Association: CC BY-NC 4.0.

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MAC-RAG: A Multi-Agent Multi-Emotional Comment Classification Framework Based on Retrieval-Augmented Generation

Item Type Conference Paper

Author J. Wu

Abstract In the era of information explosion on social media, efficient and accurate sentiment classification of comments is crucial for public opinion monitoring and user feedback analysis. Traditional Large Language Models (LLMs) often face challenges such as insufficient context understanding and hallucinations when handling Multi-Emotional Comment Classification tasks, limiting their reliability. We introduces a novel collaborative multi-agent framework MAC-RAG, designed to enhance LLM performance and robustness in this task through Retrieval-Augmented Generation (RAG) technology. The framework decomposes the classification task into two specialized agent processes: a Retrieval Agent and a Reasoning Agent. We constructed a dataset containing 1,000 multi-emotional comments and conducted extensive comparative experiments on Qwen and DeepSeek models. Experimental results demonstrate that incorporating RAG strategies significantly improves classification performance. Specifically, the Qwen model combined with a threshold-filtered optimized retrieval strategy achieved an accuracy of 42.67%, validating the effectiveness of retrieval augmentation in resolving complex sentiment classification problems. © 2026 IEEE.

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MACD: Multilingual Abusive Comment Detection at Scale for Indic Languages

Item Type Conference Paper

Author V. Gupta

Author S. Roychowdhury

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Editor Oh A.

Abstract Social media platforms were conceived to act as online 'town squares' where people could get together, share information and communicate with each other peacefully. However, harmful content borne out of bad actors are constantly plaguing these platforms slowly converting them into 'mosh pits' where the bad actors take the liberty to extensively abuse various marginalised groups. Accurate and timely detection of abusive content on social media platforms is therefore very important for facilitating safe interactions between users. However, due to the small scale and sparse linguistic coverage of Indic abusive speech datasets, development of such algorithms for Indic social media users (one-sixth of global population) is severely impeded. To facilitate and encourage research in this important direction, we contribute for the first time MACD - a large-scale

(150K), human-annotated, multilingual (5 languages), balanced (49% abusive content) and diverse (70K users) abuse detection dataset of user comments, sourced from a popular social media platform - ShareChat2. We also release AbuseXLMR, an abusive content detection model pretrained on large number of social media comments in 15+ Indic languages which outperforms XLM-R and MuRIL on multiple Indic datasets. Along with the annotations, we also release the mapping between comment, post and user id's to facilitate modelling the relationship between them. We share competitive monolingual, cross-lingual and few-shot baselines so that MACD can be used as a dataset benchmark for future research. Dataset, code and AbuseXLMR are available at: <https://github.com/ShareChatAI/MACD>. © 2022 Neural information processing systems foundation. All rights reserved.

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Machine Learning and Deep Learning Approaches Based Sentiment Analysis for Multilingual Text

Item Type Conference Paper

Author S.K. Singh

Author A. Koyal

Author U. Saghir

Author N.K. Ojha

Editor Tomar G.S.

Editor Shrivastava N.

Abstract The rapid increase in user-generated content on social media sites, blogs and product ratings has created an increased need for automated sentiment analysis systems to analyze sentiments in different languages. However, most existing models primarily focus on English, failing to embrace the diversity of languages used in online communication across the globe. The goal of this research is to investigate the gap through multilingual sentiment analysis using both machine learning and deep learning methods. The primary focus of this research is to make a comparison of different machine learning and deep learning techniques on three different languages, Hindi, English, and German, to conduct sentiment analysis and show the results. In this paper the performance of traditional models such as SVM, Naïve Bayes, Logistic Regression, Random Forest with advanced neural architectures including BiLSTM and BERT models on datasets that include English, Hindi, and German text has been evaluated. Also compared the overall performance of each language with the existing state-of-the-art respective methods. Implemented both ML and DL approaches along with hybrid-models to handle the language specific nuances. The datasets were used EmolnHindi dataset from IIT Patna for Hindi language; Amazon Review and Sentiment140 dataset for English; and Amazon Review for German language. The experiments demonstrate that deep learning models significantly outperform the traditional models while multilingual BERT models show substantial generalization across languages. For English sentiment, DistilBERT achieved the highest accuracy while for Hindi sentiment-Random Forest achieved highest accuracy of 89.11% in ML, and 85.62% in DL and BiLSTM. For German sentiment, for BERT and LSTM, accuracy was 79%. © 2026 IEEE.

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Machine Learning and Deep Learning Approaches for Fake News Detection: A Systematic Review of Techniques, Challenges, and Advancements

Item Type Journal Article

Author O. Bashaddadh

Author N. Omar

Author M. Mohd

Author M.N.A. Khalid

Abstract In response to the escalating threat of fake news on social media, this systematic literature review analyzes the recent advancements in machine learning and deep learning approaches for automated detection. Following the PRISMA guidelines, we examined 90 peer-reviewed studies published between 2020 and 2024 to evaluate the model effectiveness, identify limitations, and highlight emerging trends. Our analysis shows that deep learning models, particularly transformer-based architectures such as BERT, consistently outperform traditional machine learning methods, often achieving a high accuracy (Acc), precision (P), recall (R), and F1-score (F1). For instance, a BERT-based model reported up to 99.9% accuracy on the Kaggle fake news dataset and above 98% accuracy on other public datasets, including ISOT, Fake-or- Real, and D3. Similarly, theGANMmodel demonstrated robust performance on the FakeNewsNet dataset by integrating text and social features. Transfer learning and multimodal models that incorporate user behaviour and network information significantly improve detection in diverse, low-resource environments. However, challenges persist in terms of the dataset quality, model interpretability, domain generalisability, and realtime deployment. This review also underscores the limited adoption of few-shot and zero-shot learning techniques, highlighting a promising direction for future research on handling emerging misinformation using minimal training data. To support practical deployment, we advocate the development of explainable, multilingual, and lightweight models with greater emphasis on human-centred evaluation and ethical considerations. Our findings provide a foundation for researchers and practitioners to build scalable, trustworthy, and context-aware fake

news detection systems for global use. © 2025 The Authors.

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Machine Learning and Deep Learning Sentiment Analysis Models: Case Study on the SENT-COVID Corpus of Tweets in Mexican Spanish

Item Type Journal Article

Author H. Gomez-Adorno

Author G. Bel-Enguix

Author G. Sierra

Author J.-C. Barajas

Author W. Álvarez

Abstract This article presents a comprehensive evaluation of traditional machine learning and deep learning models in analyzing sentiment trends within the SENT-COVID Twitter corpus, curated during the COVID-19 pandemic. The corpus, filtered by COVID-19 related keywords and manually annotated for polarity, is a pivotal resource for conducting sentiment analysis experiments. Our study investigates various approaches, including classic vector-based systems such as word2vec, doc2vec, and diverse phrase modeling techniques, alongside Spanish pre-trained BERT models. We assess the performance of readily available sentiment analysis libraries for Python users, including TextBlob, VADER, and Pysentimiento. Additionally, we implement and evaluate traditional classification algorithms such as Logistic Regression, Naive Bayes, Support Vector Machines, and simple neural networks like Multilayer Perceptron. Throughout the research, we explore different dimensionality reduction techniques. This methodology enables a precise comparison among classification methods, with BETO-uncased achieving the highest accuracy of 0.73 on the test set. Our findings underscore the efficacy and applicability of traditional machine learning and deep learning models in analyzing sentiment trends within the context of low-resource Spanish language scenarios and emerging topics like COVID-19. © 2024 by the authors.

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Machine translation of restaurant reviews: New corpus for domain adaptation and robustness

Item Type Conference Paper

Author A. Bérard

Author I. Calapodescu

Author M. Dymetman

Author C. Roux

Author J.-L. Meunier

Author V. Nikoulina

Abstract We share a French-English parallel corpus of Foursquare restaurant reviews, and define a new task to encourage research on Neural Machine Translation robustness and domain adaptation, in a real-world scenario where better-quality MT would be greatly beneficial. We discuss the challenges of such user-generated content, and train good baseline models that build upon the latest techniques for MT robustness. We also perform an extensive evaluation (automatic and human) that shows significant improvements over existing online systems. Finally, we propose task-specific metrics based on sentiment analysis or translation accuracy of domain-specific polysemous words. © 2019 Association for Computational Linguistics.

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Machine Unlearning of Multilingual Transformers for Kazakh Text Classification

Item Type Conference Paper

Author M. Bolatbek

Author S. Mussiraliyeva

Editor Laborde R.

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Editor Gimenez P.-F.

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Editor Yanai N.

Editor Shukla A.

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Abstract This paper explores the application of machine unlearning techniques to multilingual natural language processing, focusing on Kazakh-language text classification using the XLM-RoBERTa transformer model. The study implements Gradient Ascent Forgetting to selectively degrade the model's performance on a targeted subset of training data while preserving its general classification capabilities. A dataset of 10,000 annotated texts across categories including bullying, violent extremism, nationalist extremism, racism, and neutral content was used for training and evaluation. Experimental results demonstrate that after applying unlearning, the model's accuracy on the forgotten class decreased significantly, while overall performance on other classes remained relatively stable, confirming the method's effectiveness in achieving controlled forgetting. The findings highlight Gradient Ascent Forgetting as a practical and computationally feasible approach for fulfilling data deletion requirements without complete retraining. Limitations and future research directions are discussed, including the trade-off between forgetting precision and model stability, optimization of unlearning parameters, and the extension of this methodology to additional low-resource languages. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Manual and automatic paragraph-level analysis of climate change framing in academic journal editorials

Item Type Journal Article

Author N. Badullovich

Author M. Stede

Author B. Aktaş

Author N. Mirzakhmedova

Author P. Saint-Dizier

Abstract Framing is an extensively used research method in the context of climate change communication research. Frames are an essential part of communication by which a speaker can put boundaries on complex issues and create a tailored lens through which to communicate while remaining true to reality. Though there is a rich literature looking at frames present in a variety of media and social media content, rather little attention has been given so far to scientific editorials. We extend upon previously published research on climate change related editorials from the journals Nature and Science by extending it with new data, new coding structures, new analyses, and results on automatic classification. Specifically, we show that applying frame analysis not as regularly done on text level but on the level of paragraphs (and moving to the level of individual sentences when necessary) leads to a more nuanced account of the underlying communication strategies. Next, in addition to analysing customary issue frames, we also code the rhetorical dimensions present in the paragraphs, informed by Entman's four frame functions. In analysing the resulting data, we find a pervasive use of the Governance and Scientific issue frames in both journals, and for the rhetorical dimension a dominance of frames that represent the Describe Problem and the Moral Judgement perspectives. Finally, as an add-on to the qualitative work, we assess the capabilities of automatic text classification methods for our three tasks of determining (i) the degree of paragraph topicality, (ii) the issue frames, and (iii) the rhetorical frames. We compare various supervised and unsupervised methods and find that a RoBERTa model achieves decent performance

on the frequent classes, while the rare classes pose problems. Similarly, zero-shot learning with LLMs is not yet able to provide a reliable classification, showcasing the results close to those of an SVM baseline. At the end of the paper, we situate our approach in the quest for even more fine-grained analyses and computational models of framing. © The Author(s) 2025.

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MAOffens: Moroccan Arabic Offensive Language Dataset

Item Type Conference Paper

Author R. Zarnoufi

Author M. Hajhouj

Author W. Bachri

Author H. Jaafar

Author M. Abik

Editor Hdioud B.

Editor Aouragh S.L.

Abstract Moroccan Arabic (MA) dialect is a low resource language. To perform any NLP task, we have to develop the necessary resources from scratch. This paper introduces our work on MAOffens, the first MA dataset for offensive language detection. The dataset will serve to build predictive models to detect offensive

content widely present on social media and hence help ensure online safety. We built the dataset with a mixture of comments in Arabic and Latin scripts to cover offensiveness in both cases. The resulting dataset consists of 23k comments totally balanced. The dataset is open to the public (<https://huggingface.co/datasets/randa/maoffens>). We evaluated the annotation and classification power of the dataset through various classifier architectures. Our best performing classifier was based on a MA transformer model. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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Mapping Tourist Sentiments Through Lexicon-Based Analysis of Social Media Reviews: The Case of Salak Sibetan Agritourism

Item Type Journal Article

Author P. Dewanti

Author P.A.G. Permana

Author I.G.A.W. Upadani

Author E.L. Maramento

Author A.J.G. Borreros

Abstract Salak Sibtan, Bali's emblematic snake fruit cultivated in Sibtan Village, Karangasem, has gained increasing digital visibility through user-generated content across social media platforms. This study applies a bilingual lexicon-based sentiment analysis framework integrating Indonesian and English sentiment lexicons, explicit negation handling, domain-specific agritourism vocabulary, and a mean-based sentiment scoring function to classify 500 online reviews collected from Facebook, Instagram, Shopee, TikTok, and Twitter/X. The methodology adapts Indonesian and English lexicons with negation handling, domain-specific refinement for agritourism terminology, and mean-based scoring to improve neutrality discrimination. Results indicate an overall sentiment distribution of 40% positive, 40% neutral, and 20% negative. Positive reviews emphasize taste quality (manis legit, renyah, fresh) and cultural authenticity, while negative feedback highlights packaging issues, inconsistent quality, and pricing concerns. Image-centric platforms (Instagram and TikTok) exhibit higher proportions of positive sentiment emphasizing taste quality and authenticity, whereas transaction-oriented platforms (Shopee and Twitter/X) show more neutral and negative expressions related to logistics, packaging, and pricing. Beyond sentiment measurement, the study demonstrates how lexicon-based methods can capture platform-specific evaluative behavior within heritage agritourism contexts, offering methodological insights for multilingual sentiment analysis in low-resource domains and strategic implications for sustainable destination communication aligned with Sibtan's FAO GIAHS recognition. © 2026, The Author(s).

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Mask-guided rationales meet LLMs: Hybrid encoder–LLM question answering framework for grounded, faithfulness-regularized explainable hate speech detection

Item Type Journal Article

Author A. Mitra

Author S. Mondal

Author A. Bhattacharya

Author J. Paul

Author J. Sil

Abstract Effective hate speech moderation must address explainability and bias alongside raw classification accuracy. We argue that model decisions should be accompanied by human-readable rationales—short spans of the input text that justify the assigned label—ensuring that predictions are auditable and robust. In this paper, we present Ex-HateQA, an Explainable Hate Speech Detection framework featuring Question Answering-style rationale span generation. Ex-HateQA is a parameter-efficient hybrid encoder–LLM architecture that predicts (label, rationale) tuples by formulating explanation generation as an extractive question answering (QA) task. Ex-HateQA follows the principle of parameter-efficient adaptation strategy rather than the absolute size of the underlying 7B LLM backbone. The LLM remains frozen and is updated exclusively through low-rank adapters (LoRA/QLoRA), yielding a minimal trainable parameter budget (approximately <1% of the backbone) while preserving high generation quality for extractive, copy-only rationales. The encoder pathway performs Mask-Guided Rationale Extraction (MGRE) and multi-label prediction, while the LLM pathway generates rationale spans under a strict copy-only constraint, ensuring that the output is either a verbatim substring of the post or <unknown>. It is worth mentioning that here rather than claiming an absolute theorem of causal faithfulness, Ex-HateQA is designed to encourage decision–evidence consistency: sufficiency and comprehensiveness losses regularize the encoder toward counterfactually informative rationales, and an optional alignment objective encourages agreement between MGRE token salience and the LLM's extractive spans. We further introduce LLM-aware probes—masked prompting for counterfactual sensitivity analysis and optional rationale-tagged prompting—and optimize the model using contrastive (negative) questions to mitigate reliance on spurious triggers. Across two rigorous benchmarks, HateXplain (single-label) and CAVES (multi-label), Ex-HateQA advances the state of the art while preserving a low-latency label-screening path and a selectively invoked rationale-generation path. On HateXplain, Ex-HateQA achieves an F1 score of 0.7445 (a 7.2% improvement over the best encoder baseline) and leads span generation with a ROUGE-2 score of 0.4235 and a BLEU score of 0.5169. Furthermore, it attains the highest subgroup robustness on GMB-Sub. and GMB-BNSP, indicating a significant reduction in bias toward target communities. On CAVES, the framework reaches an F1 score of ≈ 0.55 (an approximate 16.8% improvement over the strongest encoder baseline) and establishes new state-of-the-art results for ROUGE-1/2/L (e.g., ROUGE-L = 0.4243), while matching the highest BLEU scores. Explainability analyses corroborate these empirical gains: plausibility substantially improves (e.g., Token F1 ≈ 50.3 , AUPRC ≈ 70.3), while external evaluation under an independent classifier and human judgment shows more modest but directionally consistent gains in decision–evidence consistency. We therefore interpret these results as conservative evidence of improved rationale quality rather than as a standalone proof of strong causal faithfulness. Ablation studies validate the efficacy of MGRE pre-finetuning, contrastive QA

prompting, and copy-only constraints, while demonstrating robust performance in low-resource and non-English settings. Operationally, we interpret the measured label+evidence latency as an on-demand explanation budget within a cascaded human-in-the-loop moderation pipeline, rather than as the cost of synchronous explanation generation for every incoming post at ingestion scale. We therefore position Ex-HateQA as a framework with deterministic extractive grounding, empirically supported faithfulness-oriented behavior, and a deployment scope centered on cascaded moderation workflows rather than universal firehose-scale full-explanation serving. Warning: This paper contains samples that may be upsetting. © 2026 Elsevier B.V.

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MaskTrack: Auto-Labeling and Stable Tracking for Video Object Segmentation

Item Type Journal Article

Author Z. Chen

Author L. Zhang

Author P. Hu

Author H. Lu

Author Y. He

Abstract Video object segmentation (VOS) has witnessed notable progress due to the establishment of video training datasets and the introduction of diverse, innovative network architectures. However, video mask annotation is a highly intricate and labor-intensive task, as meticulous frame-by-frame comparisons are needed to ascertain the positions and identities of targets in the subsequent frames. Current VOS benchmarks often annotate only a few instances in each video to save costs, which, however, hinders the model's understanding of the complete context of the video scenes. To simplify video annotation and achieve efficient dense labeling, we introduce a zero-shot auto-labeling strategy based on the segment anything model (SAM), enabling it to densely annotate video instances without access to any manual annotations. Moreover, although existing VOS methods demonstrate improving performance, segmenting long-term and complex video scenes remains challenging due to the difficulties in stably discriminating and tracking instance identities. To this end, we further introduce a new framework, MaskTrack, which excels in long-term VOS and also exhibits significant performance advantages in distinguishing instances in complex videos with densely packed similar objects. We conduct extensive experiments to demonstrate the effectiveness of the proposed method and show that without introducing image datasets for pretraining, it achieves excellent performance on both short-term (86.2% in YouTube-VOS val) and long-term (68.2% in LVOS val) VOS benchmarks. Our method also surprisingly demonstrates strong generalization ability and performs well in visual object tracking (VOT) (65.6% in VOTS2023) and referring VOS (RVOS) (65.2% in Ref YouTube VOS) challenges. © 2012 IEEE.

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Performance, Object detection, Image segmentation, article, Benchmarking, social media, controlled study, Labelings, benchmarking, YouTube, videorecording, Image annotation, Auto-labeling, Footage counters, Masktrack, MaskTrack, Segment anything model, segment anything model (SAM), Video object segmentation, video object segmentation (VOS), Video objects segmentations, Video scene, Video training

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Measuring online public opinion polarization: Comparing and optimizing metrics using LLMs on social media text datasets

Item Type Journal Article

Author X. Chen

Author J. Lin

Author Z. Zhong

Abstract The development of social media has facilitated the emergence of opinion-based groups, online public opinion polarization attracting significant scholarly attention. Despite the proliferation of polarization metrics, little is known about their differences and consistency. Recent advancements in large language models (LLMs), such as GPT-4, offer new opportunities to measure polarization using large-scale social media text. This study reviews 14 commonly used polarization metrics and constructs a dataset of 603,591 Weibo posts covering 20 high-profile public events. Using GPT-4-mini and Baidu's sentiment analysis API, we compute and compare 14 polarization metrics. Results show that metrics sharing similar distributional characteristics tend to align, with the proportion of negative sentiment demonstrating the strongest correlation with expert evaluations. Furthermore, the composite weighted metric constructed by Lasso regression outperforms all individual metrics in both numerical value and ranking consistency. Future research may consider utilizing this more comprehensive metric. © 2026 Elsevier B.V.

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Language English

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Journal Abbr Online Soc. Netw. Med.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Fundamental Research Funds for the Central Universities; Correspondence Address: Z. Zhong; School of Journalism and CommunicationGuangdong Social Science Research Base: Center for Internet and Governance Research, Guangzhou, Sun Yat-sen University, Guangdong, 510006, China; email: zhzhijin@mail.sysu.edu.cn

Measuring Social Media Polarization Using Large Language Models and Heuristic Rules

Item Type Conference Paper

Author J. Chowdhury

Author R. Rashid

Author G. Terejanu

Editor An A.

Editor Cuzzocrea A.

Editor Hu H.

Abstract Understanding affective polarization in online discourse is crucial for evaluating the societal impact of social media interactions. This study presents a novel framework that leverages large language models (LLMs) and domain-informed heuristics to systematically analyze and quantify affective polarization in discussions on divisive topics such as climate change and gun control. Unlike most prior approaches that relied on sentiment analysis or predefined classifiers, our method integrates LLMs to extract stance, affective tone, and agreement patterns from large-scale social media discussions. We then apply a rule-based scoring system capable of quantifying affective polarization even in small conversations consisting of single interactions, based on stance alignment, emotional content, and interaction dynamics. Our analysis reveals distinct polarization patterns that are event dependent: (i) anticipation-driven polarization, where extreme polarization escalates before well-publicized events, and (ii) reactive polarization, where intense affective polarization spikes immediately after sudden, high-impact events. By combining AI-driven content annotation with domain-informed scoring, our framework offers a scalable and interpretable

approach to measuring affective polarization. The source code is publicly available at: <https://github.com/hasanjawad001/llm-social-media-polarization>. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2026

DOI 10.1007/978-3-032-14107-1_35

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Language English

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Pages 429-444

Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science

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Tags:

Language model, Large language model, Artificial intelligence, Sentiment analysis, Social networking (online), Social media, Climate change, Large Language Models, Stance detection, Stance Detection, Polarization, Affective polarization, Affective Polarization, AI for social impact, AI for Social Impact, Climate models, Heuristic rules, Social impact, Social Media Discourse, Social medium discourse, Societal impacts

Notes:

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Export Date: 20 September 2026; Cited By: 0; Funding details: Army Research Office, ARO, (W911NF-22-1-0035); Army Research Office, ARO; Correspondence Address: G. Terejanu; Department of Computer Science, University of North Carolina at Charlotte, Charlotte, 28223, United States; email: gabriel.terejanu@charlotte.edu; Conference name: 17th International Conference on Social Networks Analysis and Mining, ASONAM 2025; Conference date: 25 August 2025 through 28 August 2025; Conference code: 346829

Meme Classification using ModernBERT

Item Type Conference Paper

Author M. Ibrahim

Editor Jimenez-Zafra S.

Editor Marmol-Romero A.

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Editor Hernandez-Farías D.
Editor Lopes E.
Editor Lopes E.
Editor Rodrigo A.
Editor Valencia-García R.

Abstract The rapid proliferation of memes on social media necessitates robust automated systems for detecting harmful content, particularly in linguistically diverse contexts like Mexican Spanish. This paper presents a benchmark study for the DIMEMEX shared task at IberLEF 2025, leveraging ModernBERT, a transformer model enhanced with rotary positional embeddings (RoPE) and FlashAttention, to classify memes into three categories (hate speech, inappropriate content, neither) and subcategorize hate speech into classism, sexism, racism, and others. Our hierarchical multi-task learning framework achieved macro-F1 scores of 0.44 on Subtask 1 and 0.26 on Subtask 2, underscoring the challenges of fine-grained classification in low-resource, culturally nuanced settings. This work demonstrates ModernBERT's potential for long-context understanding while highlighting the need for multimodal approaches. Future research should prioritize synthetic data generation to address label scarcity, integrate vision-language architectures for joint text-image modeling, and refine ethical safeguards against cultural bias. © 2025 Copyright for this paper by its authors.

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Archive Scopus

Language English

Extra Journal Abbreviation: CEUR Workshop Proc.

Volume 4098

Publisher CEUR-WS

ISBN 16130073 (ISSN)

Proceedings Title CEUR Workshop Proc.

Conference Name CEUR Workshop Proceedings

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Automation, Classification (of information), Learning systems, Transformer modeling, Social media, Modeling languages, Embeddings, Text classification, Text Classification, Text processing, Transformer models, Subtask, Meme classification, Automated systems, Benchmark study, Meme Classification, Modernbert, ModernBERT, Three categories

Notes:

EID: 2-s2.0-105038704025

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Ibrahim; Computer Engineering Department, Cairo University, Giza, 1 Gamaa Street, 12613, Egypt; email: michael.nawar@eng.cu.edu.eg; Conference name: 2025 Iberian Languages Evaluation Forum, IberLEF 2025; Conference date: 23 September 2025 through 23 September 2025; Conference code: 217861

MemoSen: A Multimodal Dataset for Sentiment Analysis of Memes

Item Type	Conference Paper
Author	E. Hossain
Author	O. Sharif
Author	M.M. Hoque
Editor	Calzolari N.
Editor	Bechet F.
Editor	Blache P.
Editor	Choukri K.
Editor	Cieri C.
Editor	Declerck T.
Editor	Goggi S.
Editor	Isahara H.
Editor	Maegaard B.
Editor	Mariani J.
Editor	Mazo H.
Editor	Odijk J.
Editor	Piperidis S.
Abstract	Posting and sharing memes have become a powerful expedient of expressing opinions on social media in recent days. Analysis of sentiment from memes has gained much attention to researchers due to its substantial implications in various domains like finance and politics. Past studies on sentiment analysis of memes have primarily been conducted in English, where low-resource languages gain little or no attention. However, due to the proliferation of social media usage in recent years, sentiment analysis of memes has become a crucial research issue in low resource languages. The scarcity of benchmark dataset is a significant barrier in performing multimodal sentiment analysis research in resource-constrained languages like Bengali. This paper presents a novel multimodal dataset (named MemoSen) for Bengali containing 4368 memes with three annotated sentiment labels positive, negative, and neutral. A detailed annotation guideline is provided to facilitate further resource development in this domain. Additionally, a set of

experiments are carried out on MemoSen by constructing twelve unimodal (i.e., visual, textual) and ten multimodal (image+text) models. The evaluation exhibits that the integration of multimodal information significantly improves (about 1.2%) the meme sentiment classification compared to the unimodal counterparts and thus elucidate the novel aspects of multimodality. © European Language Resources Association (ELRA), licensed under CC-BY-NC-4.0.

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URL <https://www.scopus.com/pages/publications/85144382971?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Lang. Resour. Eval. Conf., LREC

Publisher European Language Resources Association (ELRA)

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Pages 1542-1554

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Conference Name 2022 Language Resources and Evaluation Conference, LREC 2022

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Tags:

Modal analysis, Classification (of information), Sentiment analysis, Social networking (online), Social media, Multi-modal dataset, Low resource languages, Multi-modal fusion, Multimodal fusion, Code-mixing, Bengalis, Unimodal, Media usage, Meme, Memes

Notes:

EID: 2-s2.0-85144382971

Export Date: 20 September 2026; Cited By: 36; Conference name: 13th International Conference on Language Resources and Evaluation Conference, LREC 2022; Conference date: 20 June 2022 through 25 June 2022; Conference code: 184830

Mental Health Disorder Prediction with PEFT-Enhanced Large Language Model

Item Type Conference Paper

Author S. Nagpal

Author S. Sabharwal

Editor Hassanien A.E.

Editor Anand S.

Editor Jaiswal A.

Editor Kumar P.

Abstract Timely identification of mental health disorders remains a major challenge due to limited clinical resources, slow assessment workflows, and barriers to seeking help. Social media offers an alternative source of real-life behavioral indicators,

as users frequently express emotions and distress in natural, unprompted language. This study evaluates the effectiveness of parameter-efficient fine-tuning (PEFT) techniques for adapting the large language model (LLM) to mental health disorder prediction. A large, multi-source, publicly available social media dataset of 52,681 de-identified posts is utilized, with annotations covering different major mental-disorder categories. Four parameter-efficient fine-tuning strategies were evaluated: Low-Rank Adaptation (LoRA), Quantized LoRA (QLoRA), weight-decoupled Low-Rank Adaptation (DoRA), and Quantized DoRA (QDoRA). These approaches were examined alongside zero-shot and two-shot prompting settings. Experiments were conducted using the Mistral-7B-Instruct LLM and focused on six multi-class disorder prediction tasks. The results indicate a significant improvement in utilizing the PEFT approach for fine-tuning LLMs on mental health prediction tasks, where LoRA yields 87.1%, DoRA yields 87.8%, QDoRA yields 87.4%, QLoRA yields 85.7%. In contrast, prompting-only baselines yield considerably lower performance, with accuracies ranging from 60.9% to 64.3%. PEFT outperforms all baselines in distinguishing between similar mental health classes like Depression, Anxiety, and Stress. The QDoRA achieves the highest detection rate for Suicidal content, reaching 79.1%. The results suggest PEFT can be used for efficient LLM adaptation for mental health prediction tasks with high accuracy, making it suitable for developing scalable and optimized mental health tools. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2026

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URL <https://www.scopus.com/pages/publications/105047335402?origin=resultslist>

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Language English

Extra Journal Abbreviation: Lect. Notes Networks Syst.

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Pages 471-483

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Conference Name Lecture Notes in Networks and Systems

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Tags:

Language model, Large language model, Mental health, Forecasting, Large datasets, Fine tuning, Fine-Tuning, Low-rank adaptation, LLM, LoRA, Tuning, Diseases, Decoupled low-rank adaptation, Disorder predictions, DoRA, Mental disorder prediction, Mental disorders, QLoRA, Quantized LoRA

Notes:

EID: 2-s2.0-105047335402

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Priyanka; Department of Computer Science and Engineering, Netaji Subhas University of Technology, Delhi, India; email: priyanka.phd21@nsut.ac.in; Conference name: 9th International Conference on Intelligent Computing and Communication, ICICC 2026; Conference date: 6 February 2026 through 7 February 2026; Conference code: 355219

Merchant Review Analysis in Food Delivery Apps: LSTM-Based Sentiment and Dual-Model Topic Classification with LLM and Logistic Regression

Item Type Conference Paper

Author T.H. Muliawati

Author F.M. Humaira

Author R. Asmara

Author A. Fariza

Abstract Technological advancements, particularly in natural language processing and deep learning, have enabled the automatic extraction of insights from user-generated content such as user reviews. In the context of food delivery services, merchant applications like GoBiz, Shopee Partner, and Grab Merchant offer various features and policies that often cause confusion and dissatisfaction among business users. This study proposes a sentiment analysis and topic classification approach using user reviews from the Google Play Store to help merchants evaluate these merchant applications. The system employs a Long Short-Term Memory (LSTM) model for sentiment analysis and uses logistic regression and zero-shot for topic classification. Data preprocessing involved data reduction, data labeling, and text preprocessing, followed by data augmentation through rule-based and generative paraphrasing techniques. The sentiment model for GoBiz achieved the highest accuracy of 92.55% and a weighted F1-score of 94%, while the Shopee Partner model produced the most contextually relevant sentiment predictions. The system also categorized topics into key areas such as service, technical issues, platform policies, etc., providing meaningful insights to support application evaluation and future improvement. © 2025 IEEE.

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URL <https://www.scopus.com/pages/publications/105018102524?origin=resultslist>

Archive Scopus

Language English

Extra Journal Abbreviation: Int. Electron. Symp., IES

Publisher Institute of Electrical and Electronics Engineers Inc.

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Conference Name 2025 International Electronics Symposium, IES 2025

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Tags:

Classification (of information), Sentiment analysis, Language processing, Natural languages, Data mining, Engineering research, Technological advancement, Long short-term memory, Logistic regression, sentiment analysis, LSTM, Short term memory, Logistics regressions, Dual model, food delivery, Food delivery, topic classification, Topic Classification, User reviews

Notes:

EID: 2-s2.0-105018102524

Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 International Electronics Symposium, IES 2025; Conference date: 5 August 2025 through 7 August 2025; Conference code: 212992

Meta-learning contrastive fusion intelligence for domain-adaptive content categorization using adaptive DNN

Item Type Journal Article

Author J.S. Venkatakrishnan

Author S.M.R. Moorthy

Author A. Shanmugam

Abstract The rapid growth of heterogeneous digital text sources, including social media, news streams, and scientific literature, poses significant challenges to traditional content categorization models due to domain shift, vocabulary drift, and semantic inconsistencies. Existing deep learning approaches often rely on domain-specific patterns and static representations, resulting in degraded performance when applied to unseen or cross-domain data. Moreover, these methods lack scalability in real-world scenarios characterized by domain drift and limited labeled data. To address these challenges, this study proposes a meta-learning contrastive fusion intelligence (MCFI) framework for domain-adaptive content categorization. The framework integrates domain context-aware normalization (DCAN) for robust preprocessing, binary particle swarm optimization (BPSO) for selecting domain-invariant features, and a hybrid architecture combining contrastive learning-enhanced bidirectional encoder representations from transformers (CL-BERT) with capsule network-enhanced transformer (Caps-Trans). A meta-learning strategy is employed to learn domain-invariant representations and enable rapid adaptation to new domains, while contrastive learning enhances inter-domain separability. A domain-adaptive decision layer further refines feature contributions dynamically. Experimental results on multiple benchmark datasets demonstrate that the proposed MCFI framework consistently outperforms state-of-the-art methods in terms of accuracy, F1-score, and generalization, providing a scalable and effective solution for cross-domain text classification. © 2026, Institute of Advanced Engineering and Science. All rights reserved.

Date 2026

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Contrastive learning, Categorization, Domain-adaptive content, Feature engineering, Meta-learning, Swarm intelligence

Notes:

EID: 2-s2.0-105043126256

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: J.S. Venkatakrishnan; Department of Computer Science, Kongunadu Arts and Science College, Coimbatore, India; email: janani.sivapriyaphd@gmail.com

MetaAdapt: Domain Adaptive Few-Shot Misinformation Detection via Meta Learning

Item Type Conference Paper

Author Z. Yue

Author H. Zeng

Author Y. Zhang

Author L. Shang

Author D. Wang

Abstract With emerging topics (e.g., COVID-19) on social media as a source for the spreading misinformation, overcoming the distributional shifts between the original training domain (i.e., source domain) and such target domains remains a non-trivial task for misinformation detection. This presents an elusive challenge for early-stage misinformation detection, where a good amount of data and annotations from the target domain is not available for training. To address the data scarcity issue, we propose MetaAdapt, a meta learning based approach for domain adaptive few-shot misinformation detection. MetaAdapt leverages limited target examples to provide feedback and guide the knowledge transfer from the source to the target domain (i.e., learn to adapt). In particular, we train the initial model with multiple source tasks and compute their similarity scores to the meta task. Based on the similarity scores, we rescale the meta gradients to adaptively learn from the source tasks. As such, MetaAdapt can learn how to adapt the

misinformation detection model and exploit the source data for improved performance in the target domain. To demonstrate the efficiency and effectiveness of our method, we perform extensive experiments to compare MetaAdapt with state-of-the-art baselines and large language models (LLMs) such as LLaMA, where MetaAdapt achieves better performance in domain adaptive few-shot misinformation detection with substantially reduced parameters on real-world datasets. © 2023 Association for Computational Linguistics.

Date 2023

DOI 10.18653/v1/2023.acl-long.286

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Archive Scopus

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Volume 1

Publisher Association for Computational Linguistics (ACL)

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Conference Name Proceedings of the Annual Meeting of the Association for Computational Linguistics

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Computational linguistics, Performance, Social media, Large dataset, Metalearning, Learning-based approach, COVID-19, Data scarcity, Knowledge management, Target domain, Learn+, Emerging topics, Non-trivial tasks, Similarity scores

Notes:

EID: 2-s2.0-85174403217

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MG-SIN: Multigraph Sparse Interaction Network for Multitask Stance Detection

Item Type Journal Article

Author H. Chai

Author J. Cui

Author S. Tang

Author Y. Ding

Author X. Liu

Author B. Fang

Author Q. Liao

Abstract Stance detection on social media aims to identify if an individual is in support of or against a specific target. Most existing stance detection approaches primarily rely on modeling the contextual semantic information in sentences and neglect to explore the pragmatics dependency information of words, thus degrading performance. Although several single-task learning methods have been proposed to capture richer semantic representation information, they still suffer from semantic sparsity problems caused by short texts on social media. This article proposes a novel multigraph sparse interaction network (MG-SIN) by using multitask learning (MTL) to identify the stances and classify the sentiment polarities of tweets simultaneously. Our basic idea is to explore the pragmatics dependency relationship between tasks at the word level by constructing two types of heterogeneous graphs, including task-specific and task-related graphs (tr-graphs), to boost the learning of task-specific representations. A graph-aware module is proposed to adaptively facilitate information sharing between tasks via a novel sparse interaction mechanism among heterogeneous graphs. Through experiments on two real-world datasets, compared with the state-of-the-art baselines, the extensive results exhibit that MG-SIN achieves competitive improvements of up to 2.1% and 2.42% for the stance detection task, and 5.26% and 3.93% for the sentiment analysis task, respectively. © 2023 IEEE.

Date 2025

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ISSN 2162237X (ISSN)

Issue 2

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Notes:

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MGKM at StanceEval2024: Fine-Tuning Large Language Models for Arabic Stance Detection

Item Type Conference Paper

Author M. Alghaslan

Author K. Almutairy

Editor Habash N.

Editor Bouamor H.

Editor Eskander R.

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Editor Farha I.A.

Editor Abdelali A.

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Editor Hamed I.

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Editor Alhafni B.

Editor Antoun W.

Editor Khalifa S.

Editor Haddad H.

Editor Zitouni I.

Editor AlKhamissi B.

Editor Almatham R.

Editor Mrini K.

Abstract Social media platforms have become essential in daily life, enabling users to express their opinions and stances on various topics. Stance detection is a task that identifies the viewpoint expressed in text toward a target subject. Despite the growing importance of Arabic tweets in shaping public opinion, there is a lack of research on stance detection in this domain. In this work, we evaluate the effectiveness of fine-tuning three Large Language Models (LLMs) in detecting target-specific stances in the MAWQIF dataset (Alturayef et al., 2022). The LLMs assessed are ChatGPT-3.5turbo, Meta-Llama-3-8B-Instruct, and Falcon-7B-Instruct. Our findings demonstrate that fine-tuning substantially enhances the stance detection capabilities of LLMs in Arabic tweets. GPT-3.5-Turbo exhibits the highest performance among the evaluated models, achieving a macro-F1 score of 82.93. Our work ranked second in the StanceEval2024

leaderboard, on a blind test set (AlturayEIF et al., 2024). ©2024 Association for Computational Linguistics.

Date	2024
URL	https://www.scopus.com/pages/publications/85204297149?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: ArabicNLP - Arabic Nat. Lang. Process. Conf., Proc. Conf.
Publisher	Association for Computational Linguistics (ACL)
ISBN	979-889176132-2 (ISBN)
Pages	816-822
Proceedings Title	ArabicNLP - Arabic Nat. Lang. Process. Conf., Proc. Conf.
Conference Name	ArabicNLP 2024 - 2nd Arabic Natural Language Processing Conference, Proceedings of the Conference
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Modified	20/09/2026, 8:35:26 pm

Tags:

Computational linguistics, Language model, Performance, Public opinions, Detection capability, Social networking (online), Fine tuning, Social media platforms, Tweets, F1 scores, Daily lives, Blind test, Detecting target

Notes:

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Export Date: 20 September 2026; Cited By: 3; Conference name: 2nd Arabic Natural Language Processing Conference, ArabicNLP 2024; Conference code: 202260

Miko: Multimodal Intention Knowledge Distillation from Large Language Models for Social-Media Commonsense Discovery

Item Type	Conference Paper
Author	F. Lu
Author	W. Wang
Author	Y. Luo
Author	Z. Zhu
Author	Q. Sun
Author	B. Xu
Author	H. Shi
Author	S. Gao
Author	Q. Li
Author	Y. Song
Author	J. Li

Abstract Social media has become ubiquitous for connecting with others, staying updated with news, expressing opinions, and finding entertainment. However, understanding the intention behind social media posts remains challenging due to the implicit and commonsense nature of these intentions, the need for cross-modality understanding of both text and images, and the presence of noisy information such as hashtags, misspelled words, and complicated abbreviations. To address these challenges, we present MIKO, a Multimodal Intention Knowledge Distillation framework that collaboratively leverages a Large Language Model (LLM) and a Multimodal Large Language Model (MLLM) to uncover users' intentions. Specifically, our approach uses an MLLM to interpret the image, an LLM to extract key information from the text, and another LLM to generate intentions. By applying MIKO to publicly available social media datasets, we construct an intention knowledge base featuring 1,372K intentions rooted in 137,287 posts. Moreover, We conduct a two-stage annotation to verify the quality of the generated knowledge and benchmark the performance of widely used LLMs for intention generation, and further apply MIKO to a sarcasm detection dataset and distill a student model to demonstrate the downstream benefits of applying intention knowledge. © 2024 ACM.

Date 2024

DOI 10.1145/3664647.3681339

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Archive Scopus

Language English

Extra Journal Abbreviation: MM - Proc. ACM Int. Conf. Multimed.

Publisher Association for Computing Machinery, Inc

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Language model, Large language model, Visual languages, Social media, Multi-modal, social media, Modeling languages, User's intentions, large language model, Cross modality, Hashtags, intention knowledge distillation, Intention knowledge distillation, large vision language model, Large vision language model

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Mimicking the Mavens: Agent-Based Opinion Synthesis and Emotion Prediction for Social Media Influencers

Item Type Journal Article

Author Q. Wei

Author R. Xue

Author H. Xiao

Author Y. Zhang

Author L. Ye

Author Y. Wang

Abstract Understanding influencers' perspectives and predicting public sentiment are crucial for event assessment and guidance in computational social systems, enabling more informed decision-making. However, this task is inherently challenging due to the unstructured, context-sensitive, and heterogeneous nature of online communication. To address these challenges, we propose a novel intelligent computational framework, Multi-domain Opinion Leader Agents Emotion Prediction (MOAEP). Our framework comprises three key components: (1) An Automatic Question Generation (AQG) module employing “Who, What, Where, When, Why, and How” (5W1H) questioning to systematically explore topic dimensions; (2) A Multi-domain Opinion Leader Agents (MOA) module that integrates enhanced Large Language Models (LLMs) with Retrieval-Augmented Generation (RAG) to produce domain-specific responses; and (3) An emotion prediction engine that synthesizes agent interactions to forecast collective emotional responses, enabling proactive social computing analysis that surpasses conventional post-event methods. Experimental results demonstrate the framework's efficacy: the AQG module generates high-fidelity outputs, while the influencer agents maintain consistent performance, achieving an average “Generative Pretrained Transformer 4” (GPT-4) evaluation score of 6.85 (on a 0–10 scale) across multiple dimensions. In a social media conflict case study, “Russia-Ukraine War”, our framework successfully predicts key influencers' perspectives and aligns emotional forecasts with observed real-world sentiment trends. These findings underscore the potential of MOAEP to provide actionable insights for decision-making in computational social science. © 2020 Tsinghua University Press.

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MindVote: When AI Meets the Wild West of Social Media Opinion

Item Type Conference Paper

Author X. Mao

Author E.X. Tao

Author L. Wang

Editor Koenig S.

Editor Jenkins C.

Editor Taylor M.E.

Abstract Large Language Models (LLMs) are increasingly used as scalable tools for pilot testing, predicting public opinion distributions before deploying costly surveys. However, the prevailing paradigm for evaluating these models relies on traditional structured surveys—a methodology misaligned with the more realistic scenarios like social media where opinions are rich in digital contexts. By design, surveys strip away the social and cultural context that shapes public opinion, and LLM benchmarks built on this paradigm inherit these critical limitations. To

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Misspellings in natural language processing: A survey of recent literature

Item Type Journal Article

Author G. Sperduti

Author A. Moreo

Abstract This survey provides an overview of the challenges of misspellings in natural language processing (NLP). Misspellings are ubiquitous in digital communication, and even if humans can generally interpret misspelt text, NLP models frequently struggle to handle it: this causes a decline in performance in common tasks like text classification and machine translation. In this paper, we reconstruct a history of misspellings as a scientific problem. We then discuss the latest advancements to address the challenge of misspellings in NLP. Main strategies to mitigate the effect of misspellings include data augmentation, double step, character-order agnostic, and tuple-based methods, among others. This survey also examines dedicated data challenges and competitions to spur progress in the field. Critical safety and ethical concerns are also examined, for example, the voluntary use of misspellings to inject malicious messages and hate speech on social networks. The survey also explores psycholinguistic perspectives on how humans process misspellings, potentially informing innovative computational techniques for text normalisation and representation. Additionally, the survey explores the challenges that misspellings pose in multilingual contexts. Finally, the misspelling-related challenges and opportunities associated with modern large language models are also analysed, including benchmarks, datasets and performances of the most prominent language models against misspellings. This survey provides a comprehensive review of recent research on misspellings and aims to serve as a valuable resource for researchers seeking to get up to speed on this problem within the rapidly evolving landscape of NLP. © The Author(s), 2026. Published by Cambridge University Press.

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Language English

Publisher Cambridge University Press

Publication Natural Language Processing

ISSN 29770424 (ISSN)

Journal Abbr Nat Lang Process.

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data augmentation, hate speech detection, Misspellings, text normalisation, user-generated content

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MITIGATING BIAS IN HATE SPEECH DETECTION WITH A SMALL NUMBER OF EXPERT ANNOTATIONS: A PROMPT-BASED LEARNING APPROACH

Item Type Journal Article**Author** D. Wei**Author** M. Chau**Author** Z. Li

Abstract Hate speech is a major problem on social media platforms. Automatic hate speech detection methods relying on machine learning models, which learn from manually labeled datasets, have been proposed in both academia and industry. However, there is increasing evidence that hate speech detection datasets labeled by general annotators (e.g., amateurs or MTurk workers) contain systematic bias, as they cannot effectively consider language use differences among different speakers. When such biased datasets are used to train machine learning models, the resulting models will also be biased. Unlike general annotators, experts can produce much less biased annotations. However, expert annotations cannot be efficiently obtained in large quantities. This paper bridges the gap by adopting a weakly supervised learning method for hate speech detection using a small number of expert annotations. We propose a novel design that uses contrastive learning and prompt-based learning based on large language models, incorporating a group estimator, a pair generator, and knowledge injection. Using real-world Twitter posts written by African American English speakers and other racial groups as an example, extensive experiments were conducted to demonstrate the superior performance of the proposed method. The proposed approach was also evaluated on data in the LGBTQ+ community and achieved consistent results. The study has important academic and practical implications for hate speech detection and large language models. © 2025 University of Minnesota. All rights reserved.

Date 2025**DOI** 10.25300/MISQ/2025/18416**URL** <https://www.scopus.com/pages/publications/105026338772?origin=resultslist>**Archive** Scopus**Language** English**Volume** 49**Publisher** University of Minnesota**Pages** 1483-1512**Publication** MIS Quarterly: Management Information Systems**ISSN** 02767783 (ISSN)

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Mitigating Hallucinations in Financial LLMs via Residual-Quantization Retrieval and Weighted Re-ranking

Item Type Conference Paper

Author G. Wang

Author Y. Sun

Editor Shen W.

Editor Barthes J.-P.

Editor Luo J.

Editor Shi Y.

Editor Zhu H.

Editor Zhang J.

Editor Peng K.

Editor Hou L.

Abstract The introduction of information technology has led to a dramatic increase in data volume within the financial industry, and large language models have shown strong capabilities in extracting key information from financial texts. However, these models face challenges in the financial domain, including a lack of Chinese financial text datasets and the tendency to generate hallucinated answers when addressing complex financial knowledge questions. To address these limitations, we construct a financial question-answering dataset based on the open-source financial evaluation dataset OpenFinData, leveraging GLM-4 and the principle of question equivalence. We also integrate heterogeneous data from financial news

websites, social media platforms, and company annual reports, which is then cleaned, preprocessed, and consolidated into a comprehensive financial text corpus. To mitigate hallucination in responses to complex and time-sensitive financial knowledge questions, we propose a retrieval-augmented approach combining financial knowledge retrieval with large language models. Experiments show that the retrieval-augmented large language model outperforms in both financial sentiment analysis and financial interpretation tasks. © 2026 IEEE.

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MLESA-Net: Attention-Enhanced Multi-Label Emotion and Sentiment Analysis on Nepali Reddit Corpus

Item Type Journal Article

Author N. Kumar Rouniyar

Author B. Sah

Author N. Kumar Subhashis Behera

Abstract Emotion and sentiment analysis of low-resource, code-mixed languages such as Nepali presents significant challenges owing to limited annotated corpora, morphological complexity, and mixed-script social media content. We proposed a Multi-Head Attention framework named MLESA-Net based on deep learning models such as BiLSTM, CNN-LSTM and multilingual transformers such as XLM-RoBERTa is evaluated on two Nepali Reddit corpora. We have evaluated the proposed architecture on the benchmark datasets such as NepEMO, MELD, GoEmotions and our newly introduced dataset named NEMoSen corpora. On NepEMO, our proposed model MLESA-Net achieves the best emotion scores such as Mic-F1 is 0.6991 and Mac-F1 is 0.6851, while XLM-RoBERTa performs better on the GoEmotions dataset having 0.462 and Mic-F1 as 0.554. The performance of mBERT is comparatively better on the MELD data, with Mic-F1 and Mac-F1 scores as 0.347 and 0.503 respectively. On our smaller NEMoSen dataset, Linear SVM outperforms all deep learning and transformer models (Micro-F1 = 0.6309), confirming that large pretrained models require proportionally larger training corpora to generalise effectively on code mixed low-resource text. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: N. Kumar Subhashis Behera; School of Computer Science, KIIT Deemed to be University, Bhubaneshwar, Odisha, 751024, India; email: nksbehera@gmail.com

MLLMGUARD: A Multi-dimensional Safety Evaluation Suite for Multimodal Large Language Models

Item Type Conference Paper

- Author** T. Gu
- Author** Z. Zhou
- Author** K. Huang
- Author** D. Liang
- Author** Y. Wang
- Author** H. Zhao
- Author** Y. Yao
- Author** X. Qiao
- Author** K. Wang
- Author** Y. Yang
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Abstract Powered by remarkable advancements in Large Language Models (LLMs), Multimodal Large Language Models (MLLMs) demonstrate impressive capabilities in manifold tasks. However, the practical application scenarios of MLLMs are intricate, exposing them to potential malicious instructions and thereby posing safety risks. While current benchmarks do incorporate certain safety considerations, they often lack comprehensive coverage and fail to exhibit the necessary rigor and robustness. For instance, the common practice of employing GPT-4V as both the evaluator and a model to be evaluated lacks credibility, as it tends to exhibit a bias toward its own responses. In this paper, we present MLLMGUARD, a multidimensional safety evaluation suite for MLLMs, including a bilingual image-text evaluation dataset, inference utilities, and a lightweight evaluator. MLLMGUARD's assessment comprehensively covers two languages (English and Chinese) and five important safety dimensions (Privacy, Bias, Toxicity, Truthfulness, and Legality), each with corresponding rich subtasks. Focusing on these dimensions, our evaluation dataset is primarily sourced from platforms such as social media, and it integrates text-based and image-based red teaming techniques with meticulous annotation by human experts. This can prevent inaccurate evaluation caused by data leakage when using open-source datasets and ensures the quality and challenging nature of our benchmark. Additionally, a fully automated lightweight evaluator termed GUARDRANK is developed, which achieves significantly higher evaluation accuracy than GPT-4. Our evaluation results across 13 advanced models indicate that MLLMs still have a substantial journey ahead before they can be considered safe and responsible. © 2024 Neural information processing systems foundation.

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Modeling Romanized Hindi and Bengali: Dataset Creation and Multilingual LLM Integration

Item Type Conference Paper

Author K. Gharami

Author Q.S. Muhtaseem

Author D. Gupta

Author L. Elluri

Author S.S. Moni

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Editor U L.H.

Editor Cheng X.

Editor Gao J.

Editor Polese G.

Editor Mei H.

Editor Boniol P.

Editor Tatsubori M.

Editor Zhao C.

Editor Zhou D.

Editor Hu X.

Abstract The development of robust transliteration techniques to enhance the effectiveness of transforming Romanized scripts into native scripts is crucial for Natural Language Processing tasks, including sentiment analysis, speech recognition, information retrieval, and intelligent personal assistants. Despite significant advancements, state-of-the-art multilingual models still face challenges in handling Romanized script, where the Roman alphabet is adopted to represent the phonetic structure of diverse languages. Within the South Asian context, where the use of Romanized script for Indo-Aryan languages is widespread across social media and digital communication platforms, such usage continues to pose significant challenges for cutting-edge multilingual models. While a limited number of transliteration datasets and models are available for Indo-Aryan languages, they generally lack sufficient diversity in pronunciation and spelling variations, adequate code-mixed data for large language model (LLM) training, and low-resource adaptation. To address this research gap, we introduce a novel transliteration dataset for two popular Indo-Aryan languages, Hindi and Bengali, which are ranked as the 3rd and 7th most spoken languages worldwide. Our dataset comprises nearly 1.8 million Hindi and 1 million Bengali transliteration pairs. In addition to that, we pre-train a custom multilingual seq2seq LLM based on Marian architecture using the developed dataset. Experimental results demonstrate significant improvements compared to existing relevant models in terms of BLEU and CER metrics. © 2025 IEEE.

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Monitoring african geopolitics: a multilingual sentiment and public attention framework: Monitoring african geopolitics: a.: A. M. Naira et al.

Item Type Journal Article

Author N. Abdou Mohamed

Author I. Benelallam

Author Y. Rahmani

Abstract In this paper, we present a framework for assessing geopolitical news based on local sentiment and public attention. Our approach uses data from social media and local online press in Kenya, Nigeria, Senegal, and South Africa, considering both local languages and colonial languages (French and English). We focus on four main topics: Foreign Relations, Institutional Stability, Conflicts, and Nature and Pandemics, using specific keywords to retrieve relevant data. We construct a Pre-trained Multilingual BERT Model, fine-tuned for tasks like text classification and sentiment analysis, emphasizing African low-resource languages. Our experiments compare different embedding approaches, showing better performance with our domain-specific model, ToumBERT, compared to the multilingual BERT-base model. The proposed Geopolitical Risk measurement methodology employs three indices: Novelty, Severity and Reach while Novelty is related to the newness of a geopolitical controversial topic, the Severity, to the roughness of comments in front of this controversy and the Reach to its media coverage. Low scores in these indices signal potential political issues in specific topics, regions and dates. To facilitate monitoring the progression of the constructed metrics, we have implemented a dashboard that visualizes their temporal evolution. This dashboard supports multiple filters (dates, topics, countries, etc.) and allows direct access to the original documents as the primary information source. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2024.

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Monitoring Opioid-Related Social Media Chatter Using Natural Language Processing and Large Language Models: Temporal Analysis

Item Type Journal Article

Author G. Sidorov

Author M. Ahmad

Author P. Basile

Author M. Waqas

Author R. Orji

Author I. Batyrshin

Abstract Background: Opioid overdose is a global public health emergency, with the United States experiencing high rates of morbidity and mortality due to prescription and illicit opioid use. Traditional public health monitoring systems often fail to provide real-time insights, limiting their capacity for early detection and intervention. Social media platforms, especially Reddit, offer a promising alternative for timely toxicovigilance due to the abundance of user-generated, real-time content. Objective: This study aimed to explore the use of Reddit as a real-time, high-volume source for toxicovigilance and develop an automated system that can classify and analyze opioid-related social media posts to detect behavioral patterns and monitor the evolution of public discourse on opioid use. Methods: To investigate evolving social media discourse around opioid use, we collected a large-scale dataset from Reddit spanning 6 years, from January 1, 2018, to December 30, 2023. Using a comprehensive opioid lexicon—including formal drug names, street slang, common misspellings, and abbreviations—we filtered relevant posts for further analysis. A subset of these data was manually annotated according to well-defined annotation guidelines into 4 categories: self-misuse, external misuse, information, and unrelated, with distributions of 37.21%, 27.25%, 27.57%, and 7.97%, respectively. To automate the classification of opioid-related chatter, we developed a robust natural language processing pipeline leveraging classical machine learning algorithms, deep learning models, and transformer-based architecture, and fine-tuned a state-of-the-art large language model (LLM; OpenAI GPT-3.5 Turbo). In the final stage, the trained LLM was deployed on an unlabeled dataset comprising 74,975 additional Reddit chatter posts. This enabled a detailed temporal analysis of opioid-related discussions, aligned

with 6 years of opioid-related death records from the Centers for Disease Control and Prevention (CDC). For this study, self-misuse and external misuse were merged into a misuse category for direct comparison with the CDC's mortality data, examining whether trends in social media discourse on opioid misuse reflect patterns in real-world mortality statistics. Results: The fine-tuned GPT-3.5 Turbo model achieved the highest classification accuracy of 0.93, outperforming the baseline (random forest 0.85) by representing a performance improvement of 9.14% over the machine learning model. The temporal analysis of the unlabeled data revealed evolving trends in opioid-related discussions, indicating shifts in user behavior and overdose-related chatter over time. To quantify this relationship, we calculated the Pearson correlation coefficient between misuse-related posts and CDC death records ($r=0.854$). This correlation was statistically significant ($P<.001$), indicating a strong positive relationship between web-based discussions and CDC mortality data. Conclusions: This study demonstrates the potential of integrating advanced natural language processing techniques and LLMs with social media data to support real-time public health surveillance. Reddit provides a valuable platform for identifying emerging trends in opioid use and overdose risk. The proposed system offers a proactive tool for researchers, clinicians, and policymakers to better understand and respond to the opioid crisis. © Grigori Sidorov, Muhammad Ahmad, Pierpaolo Basile, Muhammad Waqas, Rita Orji, Ildar Batyrshin.

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MoralBERT: A Fine-Tuned Language Model for Capturing Moral Values in Social Discussions

Item Type Conference Paper

Author V. Preniqi

Author I. Ghinassi

Author J. Ive

Author C. Saitis

Author K. Kalimeri

Abstract Moral values play a fundamental role in how we evaluate information, make decisions, and form judgements around important social issues. Controversial topics, including vaccination, abortion, racism, and sexual orientation, often elicit opinions and attitudes that are not solely based on evidence but rather reflect moral worldviews. Recent advances in Natural Language Processing (NLP) show that moral values can be gauged in human-generated textual content. Building on the Moral Foundations Theory (MFT), this paper introduces MoralBERT, a range of language representation models fine-tuned to capture moral sentiment in social discourse. We describe a framework for both aggregated and domain-adversarial training on multiple heterogeneous MFT human-annotated datasets sourced from Twitter (now X), Reddit, and Facebook that broaden textual content diversity in terms of social media audience interests, content presentation and style, and spreading patterns. We show that the proposed framework achieves an average F1 score that is between 11% and 32% higher than lexicon-based approaches, Word2Vec embeddings, and zero-shot classification with large language models such as GPT-4 for in-domain inference. Domain-adversarial training yields better out-of domain predictions than aggregate training while achieving comparable performance to zero-shot learning. Our approach contributes to annotation-free and effective morality learning, and provides useful insights towards a more comprehensive understanding of moral narratives in controversial social debates using NLP. © 2024 ACM.

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Morphology-Driven Meta-Adapter for Low-Resource Mongolian Sentiment Analysis

Item Type Conference Paper

Author H. Zhu

Author Y. Ji

Author L. Shi

Author Z. Bao

Author Q.-D.-E.-J. Ren

Author N. Wu

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Author S. Sun

Editor Huang D.-S.

Editor Zhang C.

Editor Zhang Q.

Editor Pan Y.

Abstract This paper presents the Mongolian Meta-Adapter (MMA), An innovative methodological system developed to simultaneously mitigate lexical formation intricacies and annotation scarcity in Mongolian affective computing. MMA integrates three key components: a rule-based morphological tokenizer, a parameter-efficient meta-learning scheme, and cross-lingual adversarial training. To preserve the semantic consistency of Mongolian's agglutinative structure, we develop a rule-guided morphological parser with auto-annotation, achieving 92.7% segmentation accuracy. A morphology-attentive gating module further emphasizes root-level sentiment signals and leads to a 5.2% improvement in Macro-F1. For efficient adaptation, we combine low-rank adapters (LoRA, rank 8) with dynamically generated meta-tasks, reducing trainable parameters to less than 10% of the full model and improving generalization under limited supervision. In addition, adversarial training with a gradient reversal layer enhances cross-domain robustness, reducing the performance drop from news to social media by 27%. Evaluated on a 7,000-sample, 8-category Mongolian

sentiment dataset we constructed, MMA achieves 83.5% Macro-F1, outperforming both XLM-RoBERTa full fine-tuning (74.4%) and conventional meta-learning baselines (78.3%). Our work introduces the first morphology-aware sentiment analysis system for Mongolian and provides a transferable approach for other low-resource agglutinative languages. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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MOST FREQUENTLY ASKED QUESTIONS BY OLDER ADULTS IN GERIATRIC REHABILITATION: EVALUATING LARGE LANGUAGE MODELS AS A SOURCE OF INFORMATION

Item Type Journal Article

Author U. Sözlü
Author S.M. Günay
Author S.D. Türe
Author G. Özkazanç

Abstract Introduction: Older adults in geriatric rehabilitation are increasingly turning to internet-based resources and large language models for health-related information outside of clinical follow-up. The aim of this study was to compare the reliability, clinical accuracy, quality, usefulness, and readability of ChatGPT-5.2, Gemini 3, and DeepSeek V3.2 responses to patient questions regarding geriatric rehabilitation. Materials and Method: In this cross-sectional comparative content analysis, 24 predefined questions on geriatric rehabilitation were developed from YouTube comments, relevant literature, and clinical experience. Each question was submitted to three large language models under standardized conditions. Anonymized responses were independently evaluated by two experienced physiotherapists for reliability, clinical accuracy, quality, and usefulness, with disagreements resolved by consensus with a specialist physician. Readability was assessed using the Flesch Reading Ease scores. Results: No statistically significant difference was found among the three large language models in terms of reliability scores ($p = 0.097$). However, significant differences were observed for clinical accuracy, quality, usefulness, readability, and text characteristics (all $p = 0.001$). ChatGPT-5.2 and DeepSeek V3.2 showed the highest clinical accuracy scores, while ChatGPT-5.2 was superior in terms of quality and usefulness. For readability, ChatGPT-5.2 and Gemini 3 outperformed DeepSeek V3.2. Conclusion: Although all three large language models generally produced reliable content, ChatGPT-5.2 and DeepSeek V3.2 showed stronger clinical accuracy performance. Nevertheless, because of the risk of incorrect information being generated, the use of large language models by the older population should preferably be done under expert supervision. © 2026, Turkish Geriatrics Society. All rights reserved.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: U. Sözlü; Tokat Gaziosmanpasa University, Department of Physical Therapy and Rehabilitation, Tokat, Turkey; email: ugur.sozlu@gop.edu.tr

MOTIF: A text-aware and disentangled Mamba-based architecture for multimodal sentiment analysis

Item Type Journal Article

Author B. Hu

Author Y. Li

Abstract Understanding human sentiment is a fundamental goal of affective computing and plays a crucial role in applications such as human-computer interaction, healthcare, and social media analysis. However, multimodal sentiment analysis (MSA) presents enduring challenges due to the heterogeneous, noisy, and temporally complex nature of textual, acoustic, and visual inputs. To address these issues, we propose MOTIF, a lightweight yet expressive framework that integrates disentangled representation learning, hierarchical temporal modeling, and text-guided cross-modal fusion. At its core, MOTIF introduces the BiMamba-X layer, a dual-branch temporal encoder combining bidirectional Mamba with GRU and convolutional refinement to jointly capture long-range dependencies and local emotional dynamics. To reduce redundancy and enhance interpretability, a feature disentanglement module decomposes each modality into shared and modality-specific subspaces, supervised by reconstruction, alignment, consistency, and orthogonality constraints. We further design a Text-aware Modality Enhancement (TAME) mechanism and a Text-guided Query Mamba (TQ-Mamba) module to inject linguistic priors into audio-visual streams and enable fine-grained semantic fusion. In addition, a Text-Guided Asymmetric Temporal Modeling strategy updates only the text pathway during sequence encoding, stabilizing training and anchoring cross-modal interactions. A composite loss function jointly optimizes disentanglement and task-specific prediction. Extensive experiments on the CMU-MOSI and CMU-MOSEI benchmarks show that MOTIF consistently outperforms state-of-the-art models across the majority of classification and regression metrics, while maintaining high parameter efficiency. Ablation studies further confirm the complementary roles of each module, particularly BiMamba-X and the text-guided fusion strategy. We also provide visualization analyses to illustrate the interpretability of the learned representations and cross-modal interactions. © 2025 Elsevier B.V.

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MT2-CSD and LLM-CRAN: A new dataset and an LLM-based multi-semantic knowledge fusion model for conversational stance detection

Item Type Journal Article

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Author G. Dai

Author Y. Lu

Author J. Liao

Author X. Li

Author J. Jiang

Author H. Huang

Author B. Zhang

Abstract In contemporary social media, automatic stance detection is pivotal for opinion mining, as it synthesizes and examines user perspectives on contentious topics to uncover prevailing trends and sentiments. Traditional stance detection research often targets individual instances, thereby limiting its capacity to model multi-party discussions typical in real social media scenarios. This shortcoming largely stems from the scarcity of datasets that authentically capture the dynamics of social media

interactions, hindering advancements in conversational stance detection. In this paper, we introduce MT2-CSD, a comprehensive dataset for multi-target, multi-turn conversational stance detection. To the best of our knowledge, MT2-CSD is the largest dataset available for this purpose, comprising 24,457 annotated instances and exhibiting substantial conversational depth, thereby presenting new challenges for stance detection. To address these challenges, we propose the Large Language Model enhanced Conversational Relational Attention Network (LLM-CRAN), which uses LLM-derived relational knowledge to improve conversational understanding. We conduct extensive experiments to evaluate the efficacy of LLM-CRAN on the MT2-CSD dataset. The experimental results indicate that LLM-CRAN significantly outperforms strong baseline models in the task of conversational stance detection. The dataset and code for MT2-CSD are available at <https://github.com/nfq729/MT-CSD>. © 2026 Published by Elsevier Ltd.

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MUCSD@DravidianLangTech2023: Predicting Sentiment in Social Media Text using Machine Learning Techniques

Item Type Conference Paper

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Editor Chakravarthi B.R.

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Editor Sherly E.

Abstract Social media users utilize online platforms to express their thoughts, sentiments, and views through posts/comments. Identifying such sentiments expressed in reviews or comments on a given concept/topic is known as Sentiment Analysis (SA). SA has considerable applications including customer service, social media monitoring, product reviews analysis, and so on. Content creators, bloggers, and researchers can evaluate public opinions, obtain feedback, and make informed choices by analyzing reviews and comments on social media platforms. Online users often express their sentiments using mixing words/scripts of more than one language leading to code-mixed texts. Analysing the code-mixed text to predict sentiments is challenging and lack of resources for code-mixed low-resource languages enhances the complexity further due to complexities of code-mixed texts. To address the challenges of predicting sentiments in code-mixed low-resource languages, in this paper, we - team MUCSD, describe Machine Learning (ML) models submitted to "Sentiment Analysis in Tamil and Tulu" shared task at DravidianLangTech@RANLP 2023. The proposed methodology makes use of ML models: i) Linear Support Vector Classifier (LinearSVC), ii) Logistic Regression (LR), and iii) Ensemble model (LR, Decision Tree (DT), and Support Vector Machine (SVM)) with hard voting, trained with Term Frequency-Inverse Document Frequency (TF-IDF) of word unigrams, to perform SA in Tamil and Tulu languages. Among these models, LinearSVC model performed better with

macro F1-scores of 0.189 and 0.508 and obtained 8th and 9th rank for Tamil and Tulu code-mixed texts respectively. © DravidianLangTech 2023 - 3rd Workshop on Speech and Language Technologies for Dravidian Languages, associated with 14th International Conference on Recent Advances in Natural Language Processing, RANLP 2023 - Proceedings.

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Multi-Agent Adjudication for Transtheoretical Model-Based Classification of Vaping Cessation Stages in Reddit Vaping Discourse

Item Type Conference Paper

Author L. Aust

Author A. Fu

Author D. Hu

Author E. Kasson

Author D. Wu

Author P. Cavazos-Rehg

Author L.-S. Chen

Author H. Liu

Author M. Huang

Abstract Electronic cigarettes, i.e., vaping, remain a major public-health concern, and Reddit posts provide naturalistic information about vaping cessation. In this study, we evaluated whether a set of refined prompting strategies and a multi-agent consensus workflow improve automated behavioral stage classification of vaping cessation posts on Reddit. First, clinicians annotated sample Reddit posts using the original five Transtheoretical Model (TTM) stages including (1) precontemplation, (2) contemplation, (3) preparation, (4) action, and (5) maintenance, plus what they added a new stage, namely "no stage" for this study. Ultimately, the first three stages were deterministically collapsed into pre-Action while Action, Maintenance, and No Stage were retained. We then evaluated six large language models (LLMs) including GPT-5, GPT-5.1, OpenAI o1, DeepSeek R1, DeepSeek V3, and LLaMA-4 Maverick, across five prompting strategies, consisting of a basic prompt, a basic prompt with TTM definitions, a basic prompt with a clinical annotator role, a structured prompt with role, definitions, and few-shot examples, and a chain-of-thought prompt, as well as a four-agent workflow, i.e., extractor, classifier, critic, and judge. Using 1,184 vaping-related Reddit posts, the best configuration is DeepSeek R1 with agentic adjudication, achieving macro-F1 = 0.862 and weighted F1 = 0.875. Structured prompts and agentic adjudication outperformed minimal prompts, showing that simplified TTM staging and clearer decision rules can support large-scale annotation of vaping cessation discourse. © 2026 IEEE.

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Multi-Agent VLMs Guided Self-Training with PNU Loss for Low-Resource Offensive Content Detection

Item Type Conference Paper

Author H. Wang

Author D. Ji

Author J. Lu

Author L. Zhu

Author H. Zhang

Author H. Wu

Author L. Liu

Author P. Shu

Author R.K.-W. Lee

Editor Koenig S.

Editor Jenkins C.

Editor Taylor M.E.

Abstract Accurate detection of offensive content on social media demands high-quality labeled data; however, such data is often scarce due to the low prevalence of offensive instances and the high cost of manual annotation. To address this low-resource challenge, we propose a self-training framework that leverages abundant unlabeled data through collaborative pseudo-labeling. Starting with a lightweight classifier trained on limited labeled data, our method iteratively assigns pseudo-labels to unlabeled instances with the support of Multi-Agent Vision-Language Models (MA-VLMs). Unlabeled data on which the classifier and MA-VLMs agree are designated as the Agreed-Unknown set, while conflicting samples form the Disagreed-Unknown set. To enhance label reliability, MA-VLMs simulate dual perspectives, moderator and user, capturing both regulatory and subjective viewpoints. The classifier is optimized using a novel Positive-Negative-Unlabeled (PNU) loss, which jointly exploits labeled, Agreed-Unknown, and Disagreed-Unknown data while mitigating pseudo-label noise. Experiments on benchmark datasets demonstrate that our framework substantially outperforms baselines under limited supervision and approaches the performance of large-scale models. © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Item Type Journal Article**Author** K. Rauniyar**Author** S. Poudel**Author** S. Shiwakoti**Author** S. Thapa**Author** J. Rashid**Author** J. Kim**Author** M. Imran**Author** U. Naseem

Abstract In today's social media-dominated landscape, digital platforms wield substantial influence over public opinion, particularly during crucial political events such as electoral processes. These platforms become hubs for diverse discussions, encompassing topics, reforms, and desired changes. Notably, in times of government dissatisfaction, they serve as arenas for anti-establishment discourse, highlighting the need to analyze public sentiment in these conversations. However, the analysis of such discourse is notably scarce, even in high-resource languages, and entirely non-existent in the context of the Nepali language. To address this critical gap, we present Nepal Anti Establishment discourse Tweets (NAET), a novel dataset comprising 4,445 multi-aspect annotated Nepali tweets, facilitating a comprehensive understanding of political conversations. Our contributions encompass evaluating tweet relevance, sentiment, and satire, while also exploring the presence of hate speech, identifying its targets, and distinguishing directed and non-directed expressions. Additionally, we investigate hope speech, an underexplored aspect crucial in the context of anti-establishment discourse, as it reflects the aspirations and expectations from new political figures and parties. Furthermore, we set NLP-based baselines for all these tasks. To ensure a holistic analysis, we also employ topic modeling, a powerful technique that helps us identify and understand the prevalent themes and patterns emerging from the discourse. Our research thus presents a comprehensive and multi-faceted perspective on anti-establishment election discourse in a low-resource language setting. The dataset is publicly available, facilitating in-depth analysis of political tweets in Nepali discourse and further advancing NLP research for the Nepali language through labeled data and baselines for various NLP tasks. The dataset for this work is made available at <https://github.com/rkritesh210/NAET>. © 2013 IEEE.

Date 2023**DOI** 10.1109/ACCESS.2023.3342154**URL** <https://www.scopus.com/pages/publications/85179834026?origin=resultslist>**Archive** Scopus**Language** English**Volume** 11**Publisher** Institute of Electrical and Electronics Engineers Inc.**Pages** 143092-143115**Publication** IEEE Access**ISSN** 21693536 (ISSN)**Journal Abbr** IEEE Access**Date Added** 20/09/2026, 8:35:44 pm**Modified** 20/09/2026, 8:35:44 pm**Tags:**

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Multi-class sentiment classification on Bengali social media comments using machine learning

Item Type Journal Article

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Author M. Tasneem

Author A.K. Das

Abstract Multi-class Sentiment Analysis (SA) is an important field of computational linguistics that extracts multiple opinions expressed in a text using NLP and text-mining techniques. Existing research on multi-class SA in the Bengali language is directed towards ternary classification with unsatisfactory classification performance. Moreover, obtaining a higher performance score is challenging due to the peculiarities of Bengali text, lack of ground truth datasets, and low resources of preprocessing tools. Moreover, no research has shown that deep learning algorithms perform higher on four types of sentiments. Therefore, we proposed a supervised deep learning classifier based on CNN and LSTM to conduct multi-class SA on Bengali social media comments labelled as sexual, religious, political, and acceptable. The study aims to achieve maximum accuracy using the proposed model and provide a comparative analysis with the baseline models. Six machine learning models with two different feature extraction techniques were considered baseline models. The performance of our proposed CLSTM architecture can greatly improve the performance of SA with 85.8% accuracy and 0.86 F1 scores on a labelled dataset of 42,036 Facebook comments. A web application based on the proposed model and the highest-performing baseline model was built to detect the real-life sentiment of social media comments. © 2023

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Multi-class Short Text Classification Using Ensemble of Deep Learning Classifier

Item Type Conference Paper

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Author E. Hossain

Author M.M. Hoque

Author M.A. Rahaman

Editor Vasant P.

Editor Weber G.-W.

Editor Marmolejo-Saucedo J.A.

Editor Munapo E.

Editor Thomas J.J.

Abstract With the substantial outgrowth of e-commerce, social media and online news portals have witnessed a great wave in expressing views through short text. Most textual contents are unstructured and messy forms, which are impractical and cumbersome to organize or manipulate by human experts. Therefore, developing an automatic short text classification model concerning low-resource languages, including Bengali, is critical. Moreover, the crucial barrier to classifying short text in Bengali is the unavailability of text corpora, scarcity of linguistics tools, a limited number of words in the text, and a lack of dependencies between the

words. This paper presents a short text classification model using the ensemble of four base deep learning classifiers (Neural Network (NN), Convolutional Neural Network (CNN), Bidirectional Long Short Term Memory (BiLSTM), and Bidirectional Gated Recurrent Unit (BiGRU)). Additionally, a corpus of around 0.13 million Bengali texts is developed for short text classification into six categories (e.g., international, national, sports, amusement, technology, and politics). The evaluation results on the developed corpus demonstrated that the proposed method outperformed all the baselines machine learning and deep learning models by obtaining the highest weighted f1-score of 84.4 %. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Multi-label disaster tweet classification for human value identification via IDAP-APE: Example prompt construction enabled by inaccuracy-driven active-prompt and automatic prompt engineer

Item Type Journal Article

Author L. FU

Author L. ZHANG

Abstract Developing effective multi-label classification methods that accurately identify human values in disaster tweets is crucial for understanding shifting priorities in dynamic, disaster-affected environments. Large language models (LLMs), when guided by example prompts consisting of task instructions with representative exemplars, can potentially improve tweet analysis. This study proposes a prompting framework, the IDAP-APE, for multi-label disaster tweet classification. This framework proposes the IDAP, a new exemplar selection mechanism that identifies informative, annotation-worthy exemplars for in-context learning based on composite classification inaccuracy aggregated from three metrics using a rank-based strategy; and extends our previously developed Genetic Algorithm (GA)-driven, Definition-integrated, Multistep APE (GADM-APE) framework for multi-label classification. We conducted quantitative analyses examining (1) the effectiveness of extending the Matthews Correlation Coefficient (MCC) for prompt optimization in multi-label classification; (2) the optimal exemplar size; and (3) the impact of rank-based aggregation for inaccuracy estimation. The IDAP-APE was evaluated on two disaster datasets against three baseline prompting methods, seven machine learning algorithms, and five fine-tuned language models. The results show that it outperforms all baselines and demonstrates consistent performance across two disaster events. This study advances disaster informatics by enabling real-time identification of human values from social media data, facilitating inclusive and human-centered decision making. © 2026 The Authors.

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Multi-Layered Prompting of Small Language Models for Twitter Sentiment Analysis

Item Type Conference Paper

Author S. Suen

Author R. Satapathy

Author K. Kwok

Author E. Cambria

Abstract This study examines the capacity of Small Language Models (SLMs) for complex sentiment analysis tasks through a novel multi-layered prompting strategy. The approach guides model reasoning across four layers: lexical, contextual, pragmatic, and cultural. The SLMs are also prompted to generate structured rationales based on these layers. 6 SLMs from three model families (DeepSeek R1, LLaMA and Qwen) are evaluated on a curated dataset testing sentiment polarity, subjectivity, and irony detection. High-capacity LLM judges (Claude and GPT4o) evaluate the quality of the structured rationales generated by the baseline using a structured. Comparative analysis reveal mixed effectiveness of the multi-layered prompting approach: while certain models, such as the Qwen family, show improved performance and rank the highest in explanation quality, other SLM families demonstrate limited benefits or even degraded performance. Furthermore, while most Small Language Models (SLMs) underperform across sentiment tasks relative to traditional machine learning baselines, the Qwen family stands out, achieving comparable performance and notably higher irony detection rate (0.446 for Qwen 38 B compared to 0.277). These findings underscore the model-dependent nature of prompting strategies and highlight both the potential and current limitations of SLMs in interpreting culturally embedded sentiment. © 2025 IEEE.

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Language model, Large language model, large language models, Artificial intelligence, Performance, Learning systems, Sentiment analysis, Statistical tests, sentiment analysis, Prompt engineering, prompt engineering, Multi-layered, Guide models, High-capacity, Model reasonings, Three models

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Multi-level embeddings for processing Arabic social media contents

Item Type Journal Article

Author L. Moudjari

Author F. Benamara

Author K. Akli-Astouati

Abstract Embeddings are very popular representations that allow computing semantic and syntactic similarities between linguistic units from text co-occurrence matrix. Units can vary from character n-grams to words, including more coarse-grained units such as sentences and documents. Recently, multi-level embeddings combining representations from different units have been proposed as an alternative to single-level embeddings to account for the internal structure of words (i.e., morphology) and help systems to generalise well over out of vocabulary words. These representations, either pre-trained or learned, have shown to be quite effective, outperforming word-level baselines in several NLP tasks such as machine translation, part of speech tagging and named entity recognition. Our aim here is to contribute to this line of research proposing for the first time in Arabic NLP an in-depth study of the impact of various subwords configurations ranging from character to character n-grams (including word) for social media text classification. We propose several neural architectures to learn character, subword and word embeddings, as well as a combination of these three levels, exploring different composition functions to obtain the final representation of a given text. To evaluate the effectiveness of these representations, we perform extrinsic evaluations on three text classification tasks (sentiment analysis, emotion detection and irony detection) while accounting for different Arabic varieties (Modern Standard Arabic, dialects (Levantine and Maghrebi)). For each task, we experiment with well-known dialect-agnostic and dialect-specific datasets, including those that have been recently used in shared tasks to better compare our results with those reported in previous studies on the same datasets. The results show that the multi-level embeddings we propose outperform current static and contextualised embeddings as well as best performing state of the art models in sentiment and emotion detection. In addition, we achieve competitive results in irony detection. Our models are also the most productive across dialects observing that different dialects require different composition configurations. We finally show that these performances tend to increase when coupling the multi-level representations with task-specific features. © 2021 Elsevier Ltd

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Multi-modal fake news detection: A comprehensive survey on deep learning technology, advances, and challenges

Item Type Journal Article

Author J. Lv

Author Y. Gao

Author L. Li

Author L. Shi

Author S. Li

Abstract The rapid development of social media and networks has dramatically accelerated the spread of multimodal fake news, posing significant threats to social stability, the public's right to information, and the credibility of news organizations. News modalities have advanced from simple text or images to combined video and audio formats. Integrating these heterogeneous data streams poses greater challenges for fake news detection. In recent years, the rapid advancement of Artificial Intelligence (AI) technology, including Large Language Models (LLMs), social network analysis, and external knowledge integration, has provided new insights and methods for detecting fake news. Nevertheless, there is a clear need for a thorough survey of multi-modal fake news detection. To address this, we have undertaken a comprehensive review to complement and consolidate existing research over the past five years. We systematically organize the key technologies and methods in this field and analyze the advantages of different technical approaches. Subsequently, we

summarize mainstream fake news detection datasets and evaluation metrics, and analyze existing research directions. We discuss the challenges in this field, such as insufficient data annotation and model interpretability issues. Through this survey, we aim to provide researchers with a comprehensive understanding of fake news detection technologies and promote further development in this field. © The Author(s) 2025.

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Export Date: 20 September 2026; Cited By: 20; Correspondence Address: L. Li; Computer School, Beijing Information Science and Technology University, Beijing, 100192, China; email: lili_bistu@bistu.edu.cn

Multi-Source Data Integration for Enhanced Scam Token Detection in Solana Blockchain

Item Type Conference Paper

Author R.G. Kurnia

Author A.N. Fajar

Abstract The decentralized finance (DeFi) sector on the Solana blockchain has experienced significant growth accompanied by a sharp increase in fraudulent activities, with over 98.6% of tokens on major Solana aggregator (PumpFun) exhibiting fraud characteristics. Most of the time, existing detection tools only look at on-chain measurements. This means that important off-chain signals, such as social media sentiment, bot activity, and web presence, go unexamined. This paper presents a multi-source data integration framework that combines five heterogeneous data sources, comprising on-chain metrics from RugCheck API and DexScreener API with offchain signals from X, Telegram, and web scraping, to enable

comprehensive scam token detection. The framework employs a modular workflow-based architecture with an ETL pipeline that normalizes, enriches, and engineers features from both structured blockchain data and unstructured social data. A Large Language Model serves as the classification engine, leveraging its reasoning capabilities to perform multidimensional assessment across all integrated data sources. Comparative analysis shows that the multi-source approach addresses key limitations of conventional single-source detection tools, including improved detection of social engineering-based fraud, broader coverage of analyses, and explainable classification outputs. The framework architecture, data pipeline design, and prompt specifications are presented along with design validation across three representative token scenarios. © 2026 IEEE.

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Multi-Task Transformer-Based Aspect Sentiment Analysis for Indonesian Energy and Mining Discourse

Item Type Conference Paper

Author S.A. Nugraha

Author J.L. Tedja

Author H.M. Sutedjo

Author N.M.K.S. Rasyid

Author R. Limantara

Author C. Andreas

Abstract Public discussions on Indonesia's energy and mining sectors increasingly reflect concerns over sustainability, governance, and industrial transition, with social media serving as a key platform for public expression. However, informal language and mixed sentiment in social media text pose challenges for fine-grained sentiment analysis. This study proposes a Transformer-based multi-task learning framework for Aspect-Based Sentiment Analysis of Indonesian energy and mining discourse on social media. The framework jointly models aspect classification and sentiment polarity prediction within a unified architecture, enabling shared contextual representations and reducing error propagation. Four pretrained Transformer backbones, IndoBERT, mBERT, DistilBERT, and XLMRoBERTa, are evaluated under the same multi-task setting. Experimental results show that the proposed approach achieves stable optimization and strong generalization, with IndoBERT consistently outperforming multilingual and lightweight models. Aspect-level evaluation indicates robust performance for economic, technological, and industrial aspects, while environmental sustainability remains more challenging due to implicit and subjective sentiment expressions. These findings demonstrate the effectiveness of multi-task learning and language-specific pretraining for aspect-level sentiment analysis in low-resource, domain-specific social media environments. © 2026 IEEE.

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Learning systems, Sentiment analysis, Energy, Sustainable development, Social networking (online), Social media, Data mining, Transformer, Transformers, Multitask learning, Aspect-based sentiment analyze, Indonesia, Electric transformers, Aspect-Based Sentiment Analysis, Energy and mining, Energy and Mining, Indonesian Language, Indonesian languages, Multi tasks, Multi-task learning, Multi-Task Learning

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Multichannel 2D-CNN Attention-Based BiLSTM Method for Low-Resource Ewe Sentiment Analysis

Item Type Journal Article**Author** V.K. Agbesi**Author** W. Chen**Author** C.C. Ukwuoma**Author** N.A. Kuadey**Author** C.C.M. Agbesi**Author** C.J. Ejayi**Author** E.S.A. Gyarteng**Author** G.W. Muoka**Author** A.M. Kuadey

Abstract The unavailability of an annotated dataset for a low-resource Ewe language makes it difficult to develop an automated system to appropriately evaluate public opinion on events, news, policies, and regulations. In this study, we collected and preprocessed a low-resourced document-level Ewe sentiment dataset based on social media comments. We used three features learned by word embeddings (Global vectors, word-to-vector, and FastText) rather than hand-crafted features. We further proposed a novel method termed MC2D-CNN+BiLSTM-Attn to detect the exact sentiment feature from the Ewe dataset. Extensive experiments indicate that the proposed method efficiently classifies various sentiments and is superior to benchmark deep learning methods. Results show that in detecting the precise sentiments from raw Ewe textual context, the BiLSTM incorporating Glove outperforms Word2Vec and FastText embedding with an accuracy of 0.727. Furthermore, Attn +BiLSTM and multichannel convolutional neural network methods incorporating the Word2Vec embedding layer perform better than Glove and FastText embedding with an accuracy of 0.848 and 0.896. In contrast, our proposed method with the same Word2Vec embedding recorded 0.949. © The Author(s) 2023.

Date 2025**DOI** 10.47852/bonviewJDSIS32021512**URL** <https://www.scopus.com/pages/publications/105020710056?origin=resultslist>**Archive** Scopus**Language** English**Volume** 3**Publisher** Bon View Publishing Pte Ltd**Pages** 67-77**Publication** Journal of Data Science and Intelligent Systems**ISSN** 29723841 (ISSN)

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Multilingual clickbait detection with BERT: An empirical evaluation of training strategies

Item Type Journal Article**Author** C. Liebeskind**Author** B. Lewandowska-Tomaszczyk**Author** N. Vanetik

Abstract Clickbait detection is an important component of automated content analysis systems used in large-scale web and social media platforms. While prior work has primarily focused on monolingual settings, practical deployments often require robust performance across multiple languages, including low-resource scenarios. This paper presents a controlled empirical study of transformer-based clickbait detection across nine typologically diverse languages. Using publicly available datasets normalized to a common binary label scheme, we evaluate three training strategies under a unified protocol: (1) monolingual fine-tuning with native BERT models, (2) cross-lingual transfer using with multilingual BERT models, and (3) joint multilingual training with multilingual BERT models. We complement these supervised settings with zero-shot and few-shot prompting using Gemini. Beyond the strategy ranking, we report three error-analysis findings that explain why differences arise. First, false negatives are consistently associated with longer headlines across all languages and training strategies, indicating a universal structural bottleneck in headline-only classification. Second, languages with very high inter-class semantic similarity (cosine >0.95 for Bengali and Urdu) impose a performance ceiling that no training strategy fully overcomes. Third, error clusters are language-specific, which helps explain the strong asymmetry observed in cross-lingual transfer. Together, these findings provide practical guidance for deploying clickbait detection in multilingual environments and demonstrate that error-aware evaluation reveals substantially more than accuracy

comparisons alone. © 2026

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: C. Liebeskind; Department of Computer Science, Jerusalem College of Technology, Jerusalem, 21 Havaad Haleumi St., 91160, Israel; email: cliebesk@g.jct.ac.il

Multilingual Cyber Threat Detection in Tweets/X Using ML, DL, and LLM: A Comparative Analysis

Item Type Journal Article

Author S.A. Murad

Author A. Dahal

Author N. Rahimi

Abstract Cyber threat detection has become an important area of focus in the digital age of today due to the growing spread of fake information and harmful content on social media platforms such as Twitter (now “X”). These cyber threats, often disguised within tweets, pose significant risks to individuals, communities, and even nations, emphasizing the need for effective detection systems. While previous research has explored tweet-based threats, much of the work is limited to specific languages, domains, or locations or relies on single-model approaches, reducing their applicability to diverse real-world scenarios. To address these gaps, our study focuses on multilingual tweet cyber threat detection using a variety of advanced models. The research was conducted in three stages: 1) we collected and labeled tweet datasets in four languages—English, Chinese, Russian, and Arabic—employing both manual

and polarity-based labeling methods to ensure high-quality annotations; 2) each dataset was analyzed individually using ML and DL models to assess their performance on distinct languages; and 3) finally, we combined all four datasets into a single multilingual dataset and applied deep learning (DL) and large language model (LLM) architectures to evaluate their efficacy in identifying cyber threats across various languages. Our results show that among machine learning models, random forest (RF) attained the highest performance, and the bidirectional long short-term memory (BiLSTM) architecture consistently surpassed other DL and LLM architectures across all datasets. These findings underline the effectiveness of BiLSTM in multilingual cyber threat detection. © 2025 IEEE. All rights reserved.

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Multilingual Emotion Detection in Social Media Using an Attention-Enhanced and Explainable Machine Learning Framework

Item Type Journal Article

Author M. Ahmad
Author A. Siddique
Author R. Orji
Author S. Olatinwo
Author G. Sidorov

Abstract Emotions play a central role in human experience, influencing decisions, behaviors, and social interactions. Despite substantial advances in natural language processing, detecting emotions in multilingual and code-mixed texts remains challenging due to the linguistic diversity of social media users, particularly in low-resource languages. While prior work has focused mainly on monolingual approaches, with limited exploration of multilingual scenarios. This study addresses multilingual emotion detection by creating a machine learning framework enhanced with a custom attention mechanism and fine-tuning strategy. The proposed custom attention mechanism is devised to capture emotionally salient information and improve classification accuracy. Two complementary approaches are explored: 1) a joint multilingual model based on a universal emotion classifier for multiple languages; and 2) a translation-based model that standardizes all inputs into a single target language. Additionally, explainable artificial intelligence (XAI) techniques are implemented to interpret the model's decisions and enhance transparency. To obtain a proof-of-concept, two existing datasets are transformed into three new multilingual resources with four additional emotion categories. Experiments are conducted using both conventional and advanced machine learning models—integrated with the custom attention mechanism—to evaluate the system's effectiveness for multilingual emotion recognition tasks. The proposed model achieved superior performance, with accuracy-scores of 93.1%, 79.6%, and 93.3% for augmented English, augmented Urdu, and multilingual datasets, respectively. © 2014 IEEE.

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Multilingual Named Entity Recognition in Arabic and Urdu Tweets Using Pretrained Transfer Learning Models

Item Type Journal Article

Author F. Ullah

Author M. Ahmad

Author G. Sidorov

Author I. Batyrshin

Author E.M.F. Riverón

Author A. Gelbukh

Abstract The increasing use of Arabic and Urdu on social media platforms, particularly Twitter, has created a growing need for robust Named Entity Recognition (NER) systems capable of handling noisy, informal, and code-mixed content. However, both languages remain significantly underrepresented in NER research, especially in social media contexts. To address this gap, this study makes four key contributions: (1) We introduced a manual entity consolidation step to enhance the consistency and accuracy of named entity annotations. In the original datasets, entities such as person names and organization names were often split into multiple tokens (e.g., first name and last name labeled separately). We manually refined the annotations to merge these segments into unified entities, ensuring improved coherence for both training and evaluation. (2) We selected two publicly available datasets from GitHub—one in Arabic and one in Urdu—and applied two novel strategies to tackle low-resource challenges: a joint multilingual approach and a translation-based approach. The joint approach involved merging both datasets to create a unified multilingual corpus, while the translation-based approach utilized automatic translation to generate cross-lingual datasets, enhancing linguistic diversity and model generalizability. (3) We presented a comprehensive and reproducible pseudocode-driven framework that integrates translation, manual refinement, dataset merging, preprocessing, and multilingual model fine-tuning. (4) We designed, implemented, and evaluated a customized XLM-RoBERTa model integrated with a novel attention mechanism, specifically optimized for the morphological and syntactic complexities of Arabic and Urdu. Based on the experiments, our proposed model (XLM-RoBERTa) achieves 0.98 accuracy across Arabic, Urdu, and multilingual datasets. While it shows a 7–8% improvement over traditional baselines (RF), it also achieves a 2.08% improvement over a deep learning (BiLSTM = 0.96), highlighting the effectiveness of our cross-lingual, resource-efficient approach for NER in low-resource, code-mixed social media text. © 2025 by the authors.

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Multilingual Sentiment Analysis on Twitter Data Towards Enhanced Policy Making

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Author G. Manias
Author A. Kiourtis
Author A. Mavrogiorgou
Author D. Kyriazis
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Editor Macintyre J.
Editor Cortez P.

Abstract The great expansion in the usage and popularity of social media platforms during the last decades has led to the production of an enormous real-time volume of social texts and posts, including tweets, that are being produced by users. These collections of social data can be potentially useful and provide useful insights to policymakers to adjust new user-centric policies and regulations. However, extracting and analyzing valuable information and knowledge out of these data is a challenging task as concerns the high multilingualism that describes these data. Thus, both the research and the business communities focus on the utilization of multilingual approaches and solutions to enhance the policy making procedures. To investigate a portion of these challenges this research work performs a comparative analysis of two multilingual sentiment analysis approaches. In this context, three multilingual deep learning classifiers and a zero-shot classification approach were utilized and compared. Their comparison has unveiled insightful outcomes and has a two-fold interpretation. Multilingual deep learning classifiers that have pre-trained and evaluated in monolingual data achieve high performances and transfer inference when applied afterwards in multilingual data. However, the zero-shot classification approach fails to achieve high accuracies in monolingual data as in contrary to when applied on multilingual data. © 2022, IFIP International Federation for Information Processing.

Date 2022

DOI 10.1007/978-3-031-08337-2_27

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Language English

Extra Journal Abbreviation: IFIP Advances in Information and Communication Technology

Volume 647 IFIP

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Pages 325-337

Proceedings Title IFIP Advances in Information and Communication Technology

Conference Name IFIP Advances in Information and Communication Technology

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Tags:

Deep learning, Classification (of information), Decision making, Sentiment analysis, Social networking (online), Transfer learning, Transformer, Transformers, Shot classification, Zero-shot classification, Multilingual sentiment analysis, Learning classifiers, Classification approach, Multilingual classifier, Multilingual classifiers, Multilingual sentiment analyse, Policy making

Notes:

EID: 2-s2.0-85133218063

Export Date: 20 September 2026; Cited By: 13; Correspondence Address: G. Manias; University of Piraeus, Piraeus, Greece; email: gmanias@unipi.gr; Conference name: 18th IFIP WG 12.5 International Conference on Artificial Intelligence Applications and Innovations, AIAI 2022; Conference date: 17 June 2022 through 20 June 2022; Conference code: 279479

Multilingual text categorization and sentiment analysis: a comparative analysis of the utilization of multilingual approaches for classifying twitter data

Item Type Journal Article

Author G. Manias

Author A. Mavrogiorgou

Author A. Kiourtis

Author C. Symvoulidis

Author D. Kyriazis

Abstract Text categorization and sentiment analysis are two of the most typical natural language processing tasks with various emerging applications implemented and utilized in different domains, such as health care and policy making. At the same time, the tremendous growth in the popularity and usage of social media, such as Twitter, has resulted on an immense increase in user-generated data, as mainly represented by the corresponding texts in users' posts. However, the analysis of these specific data and the extraction of actionable knowledge and added value out of them is a challenging task due to the domain diversity and the high multilingualism that characterizes these data. The latter highlights the emerging need for the implementation and utilization of domain-agnostic and multilingual solutions. To investigate a portion of these challenges this research work performs a comparative analysis of multilingual approaches for classifying both the sentiment and the text of an examined multilingual corpus. In this context, four multilingual BERT-based classifiers and a zero-shot classification approach are utilized and compared in terms of their accuracy and applicability in the classification of multilingual data. Their comparison has unveiled insightful outcomes and has a twofold interpretation. Multilingual BERT-based classifiers achieve high performances and transfer inference when trained and fine-tuned on multilingual data. While also the zero-shot approach presents a novel technique for creating multilingual solutions in a faster, more efficient, and scalable way. It can easily be fitted to new languages and new tasks while achieving relatively good results across many languages. However, when efficiency and scalability are less important than accuracy, it seems that this model, and zero-shot models in general, can not be compared to fine-tuned and trained multilingual BERT-based classifiers. © 2023, The Author(s).

Date 2023

DOI 10.1007/s00521-023-08629-3

URL <https://www.scopus.com/pages/publications/85158086967?origin=resultslist>

Archive Scopus

Language English

Volume 35

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Pages 21415-21431

Publication Neural Computing and Applications
ISSN 09410643 (ISSN)
Issue 29
Journal Abbr Neural Comput. Appl.
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Classification (of information), Sentiment analysis, Social networking (online), Transfer learning, Transformer, Comparative analyzes, Transformers, Shot classification, Zero-shot classification, Zero-shot learning, Multilingual approach, Text categorization, Multilingual classifier, Multilingual classifiers, Multilingual text categorization

Notes:

EID: 2-s2.0-85158086967

Export Date: 20 September 2026; Cited By: 80; Funding details: Horizon 2020 Framework Programme, H2020, (870675); Horizon 2020 Framework Programme, H2020; Correspondence Address: G. Manias; University of Piraeus, Piraeus, Greece; email: gmanias@unipi.gr

Multimodal Affect Perception With Large Language Model Enhancement Network

Item Type Journal Article
Author K. Yang
Author Y. Luo
Author Z. Zhang
Author C.L.P. Chen
Author T. Zhang

Abstract Multimodal Sentiment Analysis (MSA) plays a vital role in understanding emotional content from social media and multimedia data. However, existing methods often rely on large-scale labeled datasets, leading to high annotation costs and poor adaptability. They also suffer from modality imbalance and suboptimal feature fusion. To address these issues, we propose MapleNet - a Multimodal Affect Perception framework enhanced by Large Language Models. MapleNet integrates a prototype-guided fusion strategy and a dynamic modality balancing mechanism to improve alignment and collaboration between text and image features. Specifically, a shared-space encoder combined with prompt optimization ensures semantic consistency across modalities. Within the prototype learning framework, the model dynamically adjusts modality-specific learning by aligning features with class prototypes, thus mitigating imbalance and uncovering complementary affective cues. In addition, MapleNet employs a similarity-based sample retrieval module to construct contextual prompts, enriching sentiment understanding in few-shot settings. Experiments on six benchmark datasets show that MapleNet consistently outperforms state-of-the-art methods, especially under few-shot conditions, achieving superior accuracy and generalization. © 2025

IEEE.
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Archive Scopus
Language English
Volume 17
Publisher Institute of Electrical and Electronics Engineers Inc.
Pages 132-146
Publication IEEE Transactions on Affective Computing
ISSN 19493045 (ISSN)
Issue 1
Journal Abbr IEEE Trans. Affective Comput.
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Tags:

Computational linguistics, Learning systems, Sentiment analysis, Large datasets, Fine tuning, Semantics, Multi-modal, Contrastive Learning, Balancing, few-shot learning, Few-shot learning, Labeled data, Multi-modal pre-training, Multimedia systems, multimodal pre-training, Multimodal sentiment analysis, Multimodal sentiment analyze, Pre-training, prompt-based fine-tuning, Prompt-based fine-tuning, prototypical learning, Prototypical learning

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Multimodal Aspect-Based Sentiment Analysis Method Based on Image Perception Enhancement

Item Type Journal Article

Author J. Wang

Author W. Zhang

Author H. Yan

Author Z. Song

Abstract Multimodal aspect-based sentiment analysis (MABSA) aims to integrate multimodal information to accurately identify the sentiment tendencies of specific aspects. However, existing methods generally fail to pay attention to the differences in sentiment expression of various aspect words in images and texts, neglecting the promoting or interfering effects of image sentiment expression on the sentiment judgment of aspect words. Images have not been finely semantically aligned with specific aspect words in texts, and semantic conflicts arise when the sentiments expressed in images and texts are inconsistent. To address these issues, this paper proposes a multimodal aspect-based sentiment analysis method based on image perception enhancement (BIPE). Based on aspect sentiment labels, the method automatically annotates image sentiment labels using a label distribution voting mechanism. A weakly supervised semantic segmentation method is introduced to capture the foreground information of the target in the original image. According to the foreground information of the target in the image, aspect words are guided to construct aspect-supplementary phrases. Meanwhile, to improve the accuracy of image description, a large language model is used to generate text descriptions for the original image. A joint optimization learning module is proposed, which includes a cross-modal aspect sentiment classification task, a visual sentiment classification task, and a contrastive learning strategy. Through joint optimization training, the features of text and image are fused and modelled, providing sentiment supervision for the visual encoder and guiding the learning of sentiment features, while optimizing the cross-modal feature space and enhancing the discrimination between sentiment consistent and inconsistent samples. Experiments are conducted on two public datasets. On the Twitter-2015 dataset, the accuracy and F1-score are increased by 0.59 and 0.87 percentage points, respectively, and on the Twitter-2017 dataset, the accuracy and F1-score are increased by 0.10 and 1.45 percentage points, respectively. Experimental results show that BIPE outperforms most comparison models, can better identify multi-aspect sentiment differences, and has the ability to adapt to sentiment conflict samples. It effectively improves the performance of multimodal aspect-based sentiment analysis and provides new research ideas for this task. © (2026), (Journal of Computer Engineering and Applications Beijing Co., Ltd.; Science Press). All rights reserved.

Date 2026

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Short Title 基于图像感知增强的多模态方面级情感分析方法

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aspect- supplementary phrase, contrastive learning, image quality enhancement, image sentiment labelling, joint optimization, multimodal aspect- based sentiment analysis, semantic segmentation, 图像情感标签, 图像质量增强, 多模态方面级情感分析, 对比学习, 方面补充短语, 联合优化, 语义分割

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: Z. Song; College of Information and Cyber Security, People's Public Security University of China, Beijing, 100032, China; email: songzhenfeng@ppsuc.edu.cn

Multimodal Misinformation Detection using Large Vision-Language Models

Item Type Conference Paper

Author S. Tahmasebi

Author E. Müller-Budack

Author R. Ewerth

Abstract The increasing proliferation of misinformation and its alarming impact have motivated both industry and academia to develop approaches for misinformation detection and fact checking. Recent advances on large language models (LLMs) have shown remarkable performance in various tasks, but their potential in misinformation detection remains relatively underexplored. Most of existing state-of-the-art approaches either do not consider evidence and solely focus on claim related features or assume the evidence is provided. Few approaches consider evidence retrieval as part of the misinformation detection but rely on fine-tuning models. In this paper, we investigate the potential of LLMs for misinformation detection in a zero-shot setting. We incorporate an evidence retrieval component as it is crucial to gather pertinent information from various sources to detect the veracity of claims. To this end, we propose a novel re-ranking approach for multimodal evidence retrieval using both LLMs and large vision-language models (LVLM). The retrieved evidence samples (images and texts) serve as the input for an LVLM-based approach for multimodal fact verification (LVLM4FV). To enable a fair evaluation, we address the issue of incomplete ground truth in an existing evidence retrieval dataset by annotating a more complete set of evidence samples for both image and text retrieval. Our experimental results on two datasets demonstrate the superiority of the proposed approach in both evidence retrieval and fact verification tasks, with a better generalization capability. © 2024 Owner/Author.

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Language	English
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Publisher	Association for Computing Machinery
ISBN	21550751 (ISSN); 979-840070436-9 (ISBN)
Pages	2189-2199
Proceedings Title	Int Conf Inf Knowledge Manage
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Tags:

Language model, Performance, Visual languages, Social media, Fine tuning, Multi-modal, social media, Modeling languages, Image annotation, Re-ranking, State-of-the-art approach, multimodal misinformation detection, Multimodal misinformation detection, News analytic, news analytics, Ranking approach

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Multimodal Sarcasm Dataset Generation for a Low-Resource Language: Swahili

Item Type	Conference Paper
Author	E.K. Kamau
Author	N. Abbas
Editor	Bramer M.
Editor	Stahl F.
Abstract	Natural Language Processing (NLP) for low-resource languages like Swahili remains constrained by the scarcity of annotated datasets, restricting progress in machine translation, sentiment analysis, and speech recognition. Although human annotation is valuable, it is costly, time-consuming, and difficult to scale. This study investigates the use of GPT-4o to generate a Swahili multimodal dataset containing sarcastic, positive, negative, and neutral captions, centered on environmental themes. The dataset was evaluated using LLM-as-a-judge methods, automated metrics, and human validation. Captions exhibited strong grammatical fluency and contextual alignment, with a sentiment classification accuracy of ~ 79%. Claude and Gemini models rated grammatical correctness and creativity at ~ 98% and ~ 80%, respectively. Lexical diversity was high, especially

in sarcastic captions, evidenced by BERTScore F1 (0.8681) and BLEU (0.0172). Human evaluation confirmed coherence and relevance but highlighted difficulties in reflecting culturally grounded sarcasm, such as idiomatic phrasing and regional irony. These findings expose limitations of general-purpose LLMs in modelling sociolinguistic nuance. This study affirms that LLM-generated synthetic data is a scalable and cost-effective alternative to manual annotation. Future work should focus on fine-tuning LLMs with culturally specific corpora to support real-world applications, including content moderation, environmental advocacy, and social media sentiment analysis. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2026

DOI 10.1007/978-3-032-11442-6_17

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Archive Scopus

Language English

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Volume 16302 LNAI

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Conference Name Lecture Notes in Computer Science

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Notes:

EID: 2-s2.0-105023588085

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: E.K. Kamau; University of Leeds, Leeds, United Kingdom; email: kamau.eugene@gmail.com; Conference name: 45th SGAI International Conference on Artificial Intelligence, AI 2025; Conference date: 16 December 2025 through 18 December 2025; Conference code: 344179

MultiPICO: Multilingual Perspectivist Irony Corpus

Item Type Conference Paper

Author S. Casola

Author S. Frenda
Author S.M. Lo
Author E. Sezerer
Author A. Uva
Author V. Basile
Author C. Bosco
Author A. Pedrani
Author C. Rubagotti
Author V. Patti
Author D. Bernardi
Editor Ku L.-W.
Editor Martins A.F.T.
Editor Srikumar V.

Abstract Recently, several scholars have contributed to the growth of a new theoretical framework in NLP called perspectivism. This approach aims to leverage data annotated by different individuals to model diverse perspectives that affect their opinions on subjective phenomena such as irony. In this context, we propose MultiPICO, a multilingual perspectivist corpus of ironic short conversations in different languages and linguistic varieties extracted from Twitter and Reddit. The corpus includes sociodemographic information about its annotators. Our analysis of the annotated corpus shows how different demographic cohorts may significantly disagree on their annotation of irony and how certain cultural factors influence the perception of the phenomenon and the agreement on the annotation. Moreover, we show how disaggregated annotations and rich annotator metadata can be exploited to benchmark the ability of large language models to recognize irony, their positionality with respect to sociodemographic groups, and the efficacy of perspective-taking prompting for irony detection in multiple languages. © 2024 Association for Computational Linguistics.

Date 2024

DOI 10.18653/v1/2024.acl-long.849

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Language English

Extra Journal Abbreviation: Proc. Annu. Meet. Assoc. Comput Linguist.

Volume 1

Publisher Association for Computational Linguistics (ACL)

ISBN 0736587X (ISSN); 979-889176094-3 (ISBN)

Pages 16008-16021

Proceedings Title Proc. Annu. Meet. Assoc. Comput Linguist.

Conference Name Proceedings of the Annual Meeting of the Association for Computational Linguistics

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Conference name: 62nd Annual Meeting of the Association for Computational Linguistics, ACL 2024;
Conference date: 11 August 2024 through 16 August 2024; Conference code: 202222

Multitask Analysis of Spanish Travel Reviews: Sentiment, Destination, and Topic Classification with RoBERTa and LLaMA Ensembles

Item Type	Conference Paper
Author	C. Minutti-Martinez
Author	B. Escalante-Ramirez
Author	J. Olveres-Montiel
Editor	Jimenez-Zafra S.
Editor	Marmol-Romero A.
Editor	Raez A.
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Editor	Gonzalez-Barba J.
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Editor	Alvarez-Carmona M.
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Editor	Mellado E.
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Editor	Bel-Enguix G.
Editor	Gil B.
Editor	Butt S.
Editor	Hernandez-Far◈as D.
Editor	Lopes E.
Editor	Lopes E.
Editor	Rodrigo A.
Editor	Valencia-Garc◈a R.
Abstract	The recent advancements in Large Language Models (LLMs) have significantly improved sentiment analysis by enabling better understanding of nuanced emotions and contextual information. This has led to their widespread adoption across various domains, including healthcare, finance, social media monitoring, market research, and public policy. In the tourism sector, sentiment analysis of visitor reviews plays a key role in understanding tourist satisfaction, influencing travel decisions, improving service quality, supporting reputation management, and enabling personalized travel recommendations. Despite the development of fine-tuned models across different languages, most LLMs are originally trained

on English corpora, potentially affecting their performance in other languages such as Spanish. Therefore, evaluating their effectiveness in Spanish-language contexts is essential. In this work, we present the system developed by team AxoloTux for the Rest-Mex 2025 challenge (Researching Sentiment Evaluation in Text for Mexican Magical Towns) at IberLEF 2025. The task involves analyzing Spanish-language reviews of Mexican tourist destinations to predict sentiment polarity, classify the type of attraction, and identify the associated Magical Town. Our approach involves fine-tuning multiple RoBERTa variants, along with LLaMA 3.2 models (1B and 3B parameters), and combining them using an ensemble strategy to enhance robustness. Models were trained and validated on 70% of the dataset, while the remaining 30% was held out as a test set. Ensemble weights were determined based on validation performance. This strategy achieved a final track score of 0.7226 on the test set, securing second place in the competition, closely trailing the top score of 0.7254. © 2025 Copyright for this paper by its authors.

Date 2025

URL <https://www.scopus.com/pages/publications/105038734999?origin=resultslist>

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Volume 4098

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Conference Name CEUR Workshop Proceedings

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Notes:

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Multitasking of sentiment detection and emotion recognition in code-mixed Hinglish data

Item Type Journal Article

Author S. Ghosh

Author A. Priyankar

Author A. Ekbal

Author P. Bhattacharyya

Abstract As the number of non-native English speakers on social media has skyrocketed in recent years, sentiment and emotion analysis on regional languages and code-mixed data has gained traction. Despite extensive research on English, the area of Hindi–English code-mixed texts is still relatively new and understudied. We create an emotion annotated Hindi–English (Hinglish) code-mixed dataset by performing emotion annotation on the benchmark SentiMix dataset to solve this problem and enable future researchers to contribute to this domain. We propose an end-to-end transformer-based multitask framework for sentiment detection and emotion recognition from the SentiMix code-mixed dataset. We fine-tune the pre-trained cross-lingual embedding model, XLMR, using task-specific data to further exploit the efficacy of transfer learning to improve the overall efficiency of our methods. Our proposed multi-task solution outperforms the state-of-the-art single-task and multitask baselines by a considerable margin, implying that the auxiliary task (i.e. emotion recognition) increases the efficiency of the primary task (i.e. sentiment detection) in a multi-task environment. It should be noted that the reported findings were obtained without the use of any ensemble techniques, thereby adhering to a model of effective and production-ready NLP. © 2022 Elsevier B.V.

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Export Date: 20 September 2026; Cited By: 62; Funding details: CDAC; Digital India Corporation; Ministry of Electronics and Information technology, Meity, (P-264); Ministry of Electronics and Information technology, Meity; Correspondence Address: A. Ekbal; Department of Computer Science and Engineering, IIT Patna, India; email: asif@iitp.ac.in; CODEN: KNSYE

My Boli: Code-mixed Marathi-English Corpora, Pretrained Language Models and Evaluation Benchmarks

Item Type Conference Paper

Author T. Chavan

Author O. Gokhale

Author A. Kane

Author S. Patankar

Author R. Joshi

Editor Park J.C.

Editor Arase Y.

Editor Hu B.

Editor Lu W.

Editor Wijaya D.

Editor Purwarianti A.

Editor Krisnadhi A.A.

Abstract The research on code-mixed data is limited due to the unavailability of dedicated code-mixed datasets and pre-trained language models. In this work, we focus on the low-resource Indian language Marathi which lacks any prior work in code-mixing. We present L3Cube-MeCorpus, a large code-mixed Marathi-English (Mr-En) corpus with 10 million social media sentences for pretraining. We also release L3Cube-MeBERT and MeRoBERTa, code-mixed BERT-based transformer models pre-trained on MeCorpus. Furthermore, for benchmarking, we present three supervised datasets MeHate, MeSent, and MeLID for downstream tasks like codemixed Mr-En hate speech detection, sentiment analysis, and language identification respectively. These evaluation datasets individually consist of manually annotated ~12,000 Marathi-English code-mixed tweets. Ablations show that the models trained on this novel corpus significantly outperform the existing state-of-the-art BERT models. This is the first work that presents artifacts for code-mixed Marathi research. All datasets and models are publicly released at <https://github.com/l3cube-pune/MarathiNLP>. © 2023 Asian Federation of Natural Language Processing.

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Language English

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Conference Name IJCNLP-AAACL 2023 - 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics, Findings of the Association for Computational Linguistics: IJCNLP-AAACL 2023

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Notes:

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Export Date: 20 September 2026; Cited By: 8; Correspondence Address: T. Chavan; Pune Institute of Computer Technology, L3Cube, India; email: chavanttanmay1402@gmail.com; O. Gokhale; Georgia Institute of Technology, L3Cube, United States; email: ogokhale3@gatech.edu; A. Kane; Georgia Institute of Technology, L3Cube, United States; email: adityakane1@gmail.com; S. Patankar; Georgia Institute of Technology, L3Cube, United States; email: spatankar34@gatech.edu; Conference name: 13th International Joint Conference on Natural Language Processing and the 3rd Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics: Findings of the Association for Computational Linguistic, IJCNLP-AAACL 2023; Conference date: 1 November 2023 through 4 November 2023; Conference code: 197967

NaijaSenti: A Nigerian Twitter Sentiment Corpus for Multilingual Sentiment Analysis

Item Type Conference Paper

Author S.H. Muhammad

Author D.I. Adelani

Author S. Ruder

Author I.S. Ahmad

Author I. Abdulmumin

Author B.S. Bello

Author M. Choudhury

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Editor Odijk J.
Editor Piperidis S.

Abstract Sentiment analysis is one of the most widely studied applications in NLP, but most work focuses on languages with large amounts of data. We introduce the first large-scale human-annotated Twitter sentiment dataset for the four most widely spoken languages in Nigeria-Hausa, Igbo, Nigerian-Pidgin, and Yorùbá-consisting of around 30,000 annotated tweets per language, including a significant fraction of code-mixed tweets. We propose text collection, filtering, processing, and labeling methods that enable us to create datasets for these low-resource languages. We evaluate a range of pre-trained models and transfer strategies on the dataset. We find that language-specific models and language-adaptive fine-tuning generally perform best. We release the datasets, trained models, sentiment lexicons, and code to incentivize research on sentiment analysis in under-represented languages. © European Language Resources Association (ELRA), licensed under CC-BY-NC-4.0.

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Computational linguistics, Large-scales, Sentiment analysis, Language processing, Natural language processing, Natural languages, Social networking (online), Large dataset, natural language processing, sentiment analysis, Low-resource, Spoken languages, low-resource, Nigerians, Large amounts of data, twitter corpus, Twitter corpus

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Natural Language Explanations for Suicide Risk Classification Using Large Language Models

Item Type Conference Paper

Author W. Stern

Author S.J. Goh

Author N. Nur

Author P.J. Aragon

Author T. Mercer

Author S. Bhattacharyya

Author C. Sen

Author V.M. Nguyen

Editor Stanojevic M.

Abstract In recent years, automated machine learning has assisted mental health researchers in analyzing large amounts of data. Despite this, many individuals and organizations hesitate to use these methods, as they are black-box methods. Our work uses off-the-shelf generative large language models (LLM) to generate natural language explanations for suicide risk from users' Reddit posts. We benchmark various language models utilizing annotations and explanations by psychology experts. Generated explanations can accurately explain the risk of suicide using evidence from users' posts. We optimized this method for low-resource settings by leveraging pre-existing general instruction-tuned and quantized models. We conduct user studies with experts to compare the explanations and predictions generated by our approach with human expert perception and report some interesting findings for further improving our approach. We find that LLMs can effectively classify and respond with helpful reasoning of a suicidal risk diagnosis. © 2024 Copyright for this paper by its authors.

Date 2024

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Language English
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Notes:

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: N. Nur; Electrical Engineering and Computer Science, Florida Institute of Technology, Melbourne, 32901, United States; email: nurn@fit.edu; Conference name: 1st Workshop on Machine Learning for Cognitive and Mental Health, ML4CMH 2024; Conference date: 26 February 2024 through 26 February 2024; Conference code: 198032

Natural language processing for mental health interventions: a systematic review and research framework

Item Type Journal Article

Author M. Malgaroli

Author T.D. Hull

Author J.M. Zech

Author T. Althoff

Abstract Neuropsychiatric disorders pose a high societal cost, but their treatment is hindered by lack of objective outcomes and fidelity metrics. AI technologies and specifically Natural Language Processing (NLP) have emerged as tools to study mental health interventions (MHI) at the level of their constituent conversations. However, NLP's potential to address clinical and research challenges remains unclear. We therefore conducted a pre-registered systematic review of NLP-MHI studies using PRISMA guidelines (osf.io/s52jh) to evaluate their models, clinical applications, and to identify biases and gaps. Candidate studies (n = 19,756), including peer-reviewed AI conference manuscripts, were collected up to January 2023 through PubMed,

PsycINFO, Scopus, Google Scholar, and ArXiv. A total of 102 articles were included to investigate their computational characteristics (NLP algorithms, audio features, machine learning pipelines, outcome metrics), clinical characteristics (clinical ground truths, study samples, clinical focus), and limitations. Results indicate a rapid growth of NLP MHI studies since 2019, characterized by increased sample sizes and use of large language models. Digital health platforms were the largest providers of MHI data. Ground truth for supervised learning models was based on clinician ratings (n = 31), patient self-report (n = 29) and annotations by raters (n = 26). Text-based features contributed more to model accuracy than audio markers. Patients' clinical presentation (n = 34), response to intervention (n = 11), intervention monitoring (n = 20), providers' characteristics (n = 12), relational dynamics (n = 14), and data preparation (n = 4) were commonly investigated clinical categories. Limitations of reviewed studies included lack of linguistic diversity, limited reproducibility, and population bias. A research framework is developed and validated (NLPxMHI) to assist computational and clinical researchers in addressing the remaining gaps in applying NLP to MHI, with the goal of improving clinical utility, data access, and fairness. © 2023, Springer Nature Limited.

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DOI 10.1038/s41398-023-02592-2

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Negative Sentiment Analysis on Chinese Social Media with LLMs

Item Type	Conference Paper
Author	L. Huang
Author	Y. Tian
Author	J. Duan
Author	M. Zhang
Author	N. Li
Author	R. Sheng
Author	Z. Wang

Author Y. Liu
Author Y. Liao
Editor Yin G.
Editor Liu X.
Editor Su J.
Editor Yang Y.

Abstract In recent years, social media has rapidly become an essential platform for daily life and emotional expression. However, with the rise of social pressures, particularly during the COVID-19 pandemic, public mental health has been significantly impacted. Traditional methods for detecting negative sentiment on social media have several limitations, such as difficulty in accurately capturing subtle emotional nuances, lack of adaptability to diverse contexts, and inefficiency in handling large-scale data. These shortcomings hinder timely intervention and effective psychological support. To address these challenges, this study proposes a detection method based on large models to improve the accuracy and management of negative sentiment detection on social media. We collected and analyzed a large amount of user text data from the Weibo platform, identifying seven major negative sentiments and their corresponding features. Based on this, a high-quality dataset of 800 entries was constructed, combining expert manual screening with automatic annotation techniques. We fine-tuned large language models like Baichuan, Qwen, and GPT on the llama-factory platform and systematically evaluated their performance in detecting negative sentiment. Leveraging the advanced capabilities of large language models in semantic understanding and generation, our method overcomes the limitations of traditional techniques, better capturing emotional differences, adapting to diverse contexts, and efficiently handling large-scale data. The key contributions of this research include summarizing and annotating seven major negative sentiments and their features, constructing a high-quality abnormal psychological dataset, and systematically evaluating the effectiveness of large language models in this field. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Neural Network based Approach for Content Generation: Comprehensive Analysis of LLM Models

Item Type	Conference Paper
Author	S. Marathe
Author	V.S. Gaikwad
Author	A. Khodke
Author	K. Shardul
Author	R. Kuvar
Editor	Stephen J.
Editor	Sharma P.
Editor	Chaba Y.
Editor	Abraham K.U.
Editor	P.K A.
Editor	Mohammad N.
Editor	Thomas G.
Editor	Srikiran S.
Abstract	This paper presents a detailed comparative analysis of Large Language Models (LLMs) for automated content creation on social media, focusing on GPT-3.5, LLaMA 3.1, Mixtral 8x7B, Mistral Nemo 12B, and Gemma 2 (9B). As social media continues to evolve, creating platform-specific, engaging, and scalable content has become critical for both personal and business application. The study examines the selection criteria for these models, including scalability, support for multiple languages, and ethical content moderation, while highlighting the importance of fine-tuning and prompt engineering. Furthermore, this paper evaluates the performance of LLMs on both short and long-form content, exploring their effect on user engagement, retention, and factual accuracy. Using evaluation metrics such as ROUGE, sentiment analysis, and readability scores, the paper aims to highlight the strengths and weaknesses of each model, providing a basis for optimizing content creation for social media platforms. © Grenze Scientific Society, 2026.

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Notes:

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Neural Synchronization and Analysis-Grounded Computational Model for Fine-Grained Sentiment Understanding

Item Type Conference Paper

Author Y. Yin

Author B. Chen

Author S. Li

Author Z. Yin

Author C. Zhou

Author H. Wang

Editor Taniguchi T.

Editor Leung C.S.A.

Editor Kozuno T.

Editor Yoshimoto J.

Editor Mahmud M.

Editor Doborjeh M.

Editor Doya K.

Abstract Aspect-based sentiment analysis (ABSA) aims to identify and extract opinion targets and their associated sentiments from user-generated content, enabling businesses to gain valuable insights. However, capturing the intricate interplay between aspects and their contexts remains a challenge, especially for deep learning models that struggle to effectively integrate syntactic and semantic information. Drawing inspiration from the principles of neural information processing, we propose the Neural Aspect Sentiment Analysis (Neural-ASA) framework to address these challenges. Central to Neural-ASA is the neural syncretic encoder (NSE), which leverages neural synchronization and oscillatory dynamics to capture the complex interactions between words and their syntactic relationships, forming rich, context-aware representations. Furthermore, the neural attentive sentiment aggregator (NASA) emulates the brain's attentional processes to selectively focus on sentiment-bearing aspects while ignoring irrelevant information. Experimental evaluations on the SemEval dataset demonstrate Neural-ASA's superiority over state-of-the-art ABSA methods and large language models, achieving impressive performance while maintaining interpretability and computational efficiency. Neural-ASA offers valuable insights into the potential of brain-inspired approaches for sentiment analysis, paving the way for more neurologically-grounded NLP models. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

Date 2026

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: B. Chen; University of Shanghai for Science and Technology, Shanghai, China; email: chenbin1991@usst.edu.cn; Conference name: 32nd International Conference on Neural Information Processing, ICONIP 2025; Conference date: 20 November 2025 through 24 November 2025; Conference code: 342999

New benchmark dataset and fine-grained cross-modal fusion framework for Vietnamese multimodal aspect-category sentiment analysis

Item Type Journal Article

Author Q.H. Nguyen

Author M.-V.T. Nguyen

Author K. Van Nguyen

Abstract The emergence of multimodal data on social media platforms presents new opportunities to better understand user sentiments toward a given aspect. However, existing multimodal datasets for aspect-category sentiment analysis (ACSA) often focus on textual annotations, neglecting fine-grained information in images. Consequently, these datasets fail to fully exploit the richness inherent in multimodal. To address this, we introduce a new Vietnamese multimodal dataset, named ViMACSA, which consists of 4876 text-image pairs with 14,618 fine-grained annotations for both text and image in the hotel domain. Additionally, we propose a fine-grained cross-modal fusion framework (FCMF) that effectively learns both intra- and inter-modality interactions and then fuses these information to produce a unified multimodal representation. Experimental results show that our framework outperforms SOTA models on the ViMACSA dataset, achieving the highest F1 score of 79.73%. We also explore characteristics and challenges in Vietnamese multimodal sentiment analysis, including misspellings, abbreviations, and the complexities of the Vietnamese language. This work contributes both a benchmark dataset and a new framework that leverages fine-grained multimodal information to improve multimodal aspect-category sentiment analysis. Our dataset is available for research purposes. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2024.

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Notes:

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New Forms of (Mis)communication and Artificial Intelligence: Exploring Alienation, Empowerment and Accessibility

Item Type Conference Paper

Author N. Di Leo

Author S. Arace

Author M. Ciletti

Editor Toto G.A.

Abstract The rapid advancement of Large Language Models (LLMs) has revolutionized Artificial Intelligence (AI), particularly in Natural Language Processing and generation. This technological leap has profound implications for human-machine interaction and communication at large. This study examines the impact of Artificial Intelligence on communication, exploring its potential for empowerment and accessibility alongside risks of alienation, while also investigating how AI representations in popular culture influence public perceptions. The research combines theoretical analysis with an exploratory study of social media discourse. Sentiment analysis and topic modeling were applied to a dataset of Reddit posts and comments (n = 1739) mentioning AI published in the last year. Findings reveal a complex landscape of opinions, with a slight tendency towards positive sentiment but significant concerns about AI's societal impact. The study identifies key themes in public discourse, including AI's reasoning capabilities, human-AI comparisons, and practical applications. Results highlight the need for enhanced AI literacy, ethical guidelines, and inclusive design in AI development to address both opportunities and challenges in AI-mediated communication. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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News vs. Social Media: Sentiment Impact on Stock Performance of Big Tech Companies

Item Type Journal Article

Author H. Kim-Hahm

Author A.S. Abou-Zaid

Author A. Mohd

Abstract With the growing prominence of large technology firms and the shift in news dissemination driven by social media, scholars have increasingly examined how public discourse about these companies shapes financial markets. Focusing on Apple, Amazon, and Microsoft during the transitional period of January 2015–January 2020, this study evaluates attention and sentiment across traditional news media, social media, and web search in relation to stock market outcomes. We use relatively fine-grained weekly data to link media attention and sentiment to stock returns, volatility, and trading volume. To compare media sentiment across sources, we apply FinBERT-based sentiment analysis, drawing on advances in domain-specific language modeling tailored to financial texts. Results show that social media sentiment (Twitter), exerts a consistently positive and significant influence, while the effects of traditional news media (New York Times) and web search activity (Google Trends) are more irregular. The impact also varies across firms: Twitter sentiment is strongly related to trading volume and volatility for Amazon and Microsoft, but appears less influential for Apple, whose large trading base may dilute the effect. These findings offer a historical baseline for media–finance interactions and highlight how text analysis

illuminates the pre-COVID era of big technology firms. © 2025 by the authors.

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Notes:

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Correspondence Address: H. Kim-Hahm; School of Business, Eastern Illinois University, Charleston,
61920, United States; email: hkimhahm@eiu.edu

NLNDE at SemEval-2023 Task 12: Adaptive Pretraining and Source Language Selection for Low-Resource Multilingual Sentiment Analysis

Item Type Conference Paper

Author M. Wang

Author H. Adel

Author L. Lange

Author J. Strötgen

Author H. Schütze

Editor Ojha A.K.

Editor Dogruoz A.S.

Editor Da San Martino G.

Editor Madabushi H.T.

Editor Kumar R.

Editor Sartori E.

Abstract This paper describes our system developed for the SemEval-2023 Task 12 “Sentiment Analysis for Low-resource African Languages using Twitter Dataset”. Sentiment analysis is one of the most widely studied applications in natural

language processing. However, most prior work still focuses on a small number of high-resource languages. Building reliable sentiment analysis systems for low-resource languages remains challenging, due to the limited training data in this task. In this work, we propose to leverage language-adaptive and task-adaptive pretraining on African texts and study transfer learning with source language selection on top of an African language-centric pretrained language model. Our key findings are: (1) Adapting the pretrained model to the target language and task using a small yet relevant corpus improves performance remarkably by more than 10 F1 score points. (2) Selecting source languages with positive transfer gains during training can avoid harmful interference from dissimilar languages, leading to better results in multilingual and cross-lingual settings. In the shared task, our system wins 8 out of 15 tracks and, in particular, performs best in the multilingual evaluation. © 2023 Association for Computational Linguistics.

Date 2023

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NLP in cybersecurity: Advances, applications, and future directions for threat analysis and dark web intelligence

Item Type Journal Article

Author A. Alsirhani

Author S. Ali

Author M. Humayun

Author S.U. Rehman

Author M.S. Alzahrani

Abstract The rapid evolution of cyber threats requires advanced techniques for analyzing large amounts of textual data from threat reports, security bulletins, and dark web forums. Natural Language Processing (NLP) has emerged as a powerful tool for automating the extraction of actionable intelligence from unstructured cybersecurity data. This review paper explores the role of NLP in cybersecurity, focusing on its applications in processing threat reports and dark web data to identify, predict, and mitigate cyber risks. We begin by discussing the challenges in cybersecurity text analysis, including noisy data, multilingual content, and adversarial language obfuscation. Next, we review key NLP techniques—such as named entity recognition (NER), sentiment analysis, topic modeling, and deep learning-based approaches—that enhance threat intelligence by classifying malware, detecting phishing attempts, and uncovering threat actor communications. We also examine how large language models (LLMs) such as BERT and GPT are being adapted for cybersecurity tasks. Furthermore, the paper highlights the ethical and privacy concerns associated with dark web data scraping and the risks of AI-powered cyberattacks. We compare existing NLP-based cybersecurity frameworks and identify gaps in real-time threat detection and cross-domain adaptability. Finally, we outline future research directions, including multimodal threat analysis, adversarial NLP defenses, and explainable AI for security analysts. This review serves as a comprehensive resource for researchers and practitioners, bridging the gap between NLP advancements and cybersecurity applications while addressing emerging challenges in the field. © 2026 The Author(s)

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NollySenti: Leveraging Transfer Learning and Machine Translation for Nigerian Movie Sentiment Classification

Item Type Conference Paper

Author I. Shode

Author D.I. Adelani

Author J. Peng

Author A. Feldman

Abstract Africa has over 2000 indigenous languages but they are under-represented in NLP research due to lack of datasets. In recent years, there have been progress in developing labelled corpora for African languages. However, they are often available in a single domain and may not generalize to other domains. In this paper, we focus on the task of sentiment classification for cross-domain adaptation. We create a new dataset, NollySenti-based on the Nollywood movie reviews for five languages widely spoken in Nigeria (English, Hausa, Igbo, Nigerian-Pidgin, and Yorùbá). We provide an extensive empirical evaluation using classical machine learning methods and pre-trained language models. Leveraging transfer learning, we compare the performance of cross-domain adaptation from Twitter domain, and cross-lingual adaptation from English language. Our evaluation shows that transfer from English in the same target domain leads to more than 5% improvement in accuracy compared to transfer from Twitter in the same language. To further mitigate the domain difference, we leverage machine translation (MT) from English to other Nigerian languages, which leads to a further improvement of 7% over cross-lingual evaluation. While MT to low-resource languages are often of low quality, through human evaluation, we show that most of the translated sentences preserve the sentiment of the original English reviews. © 2023 Association for Computational Linguistics.

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Novel Method for Sentiment Analysis in Social Media Data Using Hybrid Deep Learning Model

Item Type Journal Article

Author H. Vanam

Author J.R. Raj

Abstract It is common practise to employ a contextual mining approach called sentiment analysis (SA) to glean subjective but potentially helpful information from textual data. In order to recognise, analyse, and extract answers, states, or emotions from the data, it employs Natural Language Processing (NLP), language understanding, forensics, and cognitive science. Significant progress towards a better SA model may be made using the features analysis method. In recent years, feature extractions have made extensive use of GloVe and Word2vec embedding models. They require a huge corpora of text data for training and creating accurate vectors, but they ignore emotional and contextual information in the text. Out-of-Vocabulary (OOV) words are not considered while creating these vectors, hence some information may be lost

if these methods are used. The limited availability of annotated data is another obstacle to sentiment categorization. Misclassification may occur when there is a discrepancy between the review and the label. In this research, we offer a hybrid SA model capable of overcoming the challenges posed by missing ratings and reviews due to noise, out-of-context phrases, and context. To investigate and conduct sentiment and appropriate analysis, this study proposes an Autoencoder Bi-directional Recurrent Neural Network (ABRNN) based on Bi-directional Encoding from Transformers (BET). The reviews are initially categorised by their polarity ratings using the zero-shot categorization. Then, the data is fed into a pre-trained BET system to extract embeddings based on the semantics and context of sentences. The neural network, made up of expanded and hybrid LSTM, was then fed the acquired contextual embedded vectors. For extracting both global and local spatial contextual characteristics from the embedded data, this approach employs expanded approach in place of traditional approach. The complete phrase sequencing is performed with the help of Hybrid Long Short-Term Memory (HLSTM). Measures of accuracy, precision, recall, f1-score, and area under the curve (AUC) are used to assess the ABRNN model on four separate text datasets from different domains. Because of this, ABRNN may be utilised well for SA processes on social media evaluations, with no loss of data. © 2023, Penerbit Akademia Baru. All rights reserved.

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OASIS 2024 - Proceedings of the 2024 Workshop on Open Challenges in Online Social Media, Held in conjunction with the 35th ACM Conference on Hypertext and Social Media, HT 2024

Item Type Journal Article

Abstract The proceedings contain 6 papers. The topics discussed include: the monetization of toxicity: analyzing YouTube content creators and controversy-driven engagement; using explainable AI (XAI) for identification of subjectivity in hate speech annotations for low-resource languages; a comparative study of hybrid models in health misinformation text classification; characterizing non-consensual intimate image abuse on telegram groups and channels; on the use of heterogeneous graph neural networks for detecting malicious activities: a case study with cryptocurrencies; and AWESOME: analysis framework for WEb3 social media.

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Occupational therapy intervention for children with developmental coordination disorder: A survey of occupational therapists in Australia

Item Type Journal Article

Author J. Hunt

Author J.G. Zwicker

Author A. Raynor

Author E. Godecke

Abstract Introduction: Developmental coordination disorder (DCD), a common neurodevelopmental condition, is associated with persistent motor and psychosocial difficulties impacting participation and quality of life. International guidelines recommend activity-/participation-focused, task-specific interventions; however, little is known about how Australian occupational therapists translate these recommendations into clinical practice. This study aimed to describe current Australian occupational therapy intervention for children with (or suspected to have) DCD and identify factors that shape occupational therapists' intervention choices. Methods: A cross-sectional online survey was distributed to Australian occupational therapists via professional organisations/social media September–October 2020. Eligible participants were registered occupational therapists with $\geq 15\%$ children in their caseload. Quantitative data were analysed using descriptive statistics and regressions. Open-ended responses were examined using inductive content analysis. Consumer and Community Engagement: No consumers (e.g., parents or children) or community service providers were involved in the design, implementation, analysis, or dissemination of this research. This study focused on describing occupational therapists' self-reported practice, and the survey was developed and piloted exclusively with occupational therapists. Results: One hundred seventy-five occupational therapists met inclusion criteria; 146 completed intervention questions. Most occupational therapists provided individual therapy alongside consultation with parents/educators/other health professionals, and 95% reported providing home/school programmes. Weekly, ongoing sessions were common. Most occupational therapists (97%) used some form of activity-/participation-oriented intervention. Only around half used CO-OP, while two thirds used task-specific approaches; however, more occupational therapists used process-/impairment-oriented interventions targeting body structure and function (77%) and/or interventions aimed at achieving developmental milestones (72%). Six key considerations shaping interventions choices were identified: family circumstances and relationships, practical supports, child factors, access/funding, child/family goals, and occupational therapist/peer experience. Conclusion: Australian occupational therapists integrate research evidence, clinical experience, and contextual factors when making intervention decisions for children with DCD. While many use approaches consistent with best practice guidelines, non-recommended practices remain common. Targeted DCD-specific training, improved support for evaluating practice evidence, and addressing funding barriers may align everyday practice with current guidelines. © 2026 The Author(s). Australian Occupational Therapy Journal published by John Wiley & Sons Australia, Ltd on behalf of Occupational Therapy Australia.

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Offensive language detection in low resource languages: A use case of Persian language

Item Type Journal Article

Author M. Mozafari

Author K. Mnassri

Author R. Farahbakhsh

Author N. Crespi

Abstract THIS ARTICLE USES WORDS OR LANGUAGE THAT IS CONSIDERED PROFANE, VULGAR, OR OFFENSIVE BY SOME READERS. Different types of abusive content such as offensive language, hate speech, aggression, etc. have become prevalent in social media and many efforts have been dedicated to automatically detect this phenomenon in different resource-rich languages such as English. This is mainly due to the comparative lack of annotated data related to offensive language in lowresource languages, especially the ones spoken in Asian countries. To reduce the vulnerability among social media users from these regions, it is crucial to address the problem of offensive language in such low-resource languages. Hence, we present a new corpus of Persian offensive language consisting of 6,000 out of 520,000 randomly sampled microblog posts from X (Twitter) to deal with offensive language detection in Persian as a lowresource language in this area. We introduce a method for creating the corpus and annotating it according to the annotation practices of recent efforts for some benchmark datasets in other languages which results in categorizing offensive language and the target of offense as well. We perform extensive experiments with three classifiers in different levels of annotation with a number of classical Machine Learning (ML), Deep learning (DL), and transformer- based neural networks including monolingual and multilingual pre-trained language models. Furthermore, we propose an ensemble model integrating the aforementioned models to boost the performance of our offensive language detection

task. Initial results on single models indicate that SVM trained on character or word n-grams are the best performing models accompanying monolingual transformer-based pre-trained language model Pars- BERT in identifying offensive vs non-offensive content, targeted vs untargeted offense, and offensive towards individual or group. In addition, the stacking ensemble model outperforms the single models by a substantial margin, obtaining 5% respective macro F1-score improvement for three levels of annotation. © 2024 Mozafari et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Offensive Language Detection in Spanish Social Media: Testing from Bag-of-Words to Transformers Models

Item Type Journal Article

Author J.M. Molero

Author J. Perez-Martin

Author A. Rodrigo

Author A. Penas

Abstract Social networks allow us to communicate with people around the world. However, some users usually take advantage of anonymity for writing offensive comments to others, which might affect those who receive offensive messages or discourage the use of these networks. However, it is impossible to manually check every message. This has promoted several proposals for automatic detection systems. Current state-of-the-art systems are based on the transformers' architecture and most of the work has been focused on the English language. However, these systems do not pay too much attention to the unbalanced nature of data, since there are fewer offensive comments than non-offensive in a real environment. Besides, these previous works have not studied the impact on the final results of pre-processing or the corpora used for pre-training the models. In this work, we propose and evaluate a series of automatic methods aimed at detecting offensive language in Spanish texts addressing the unbalanced nature of data. We test different learning models, from those based on classical Machine Learning algorithms using Bag-of-Words as data representation to those based in large language models and neural networks such as transformers, paying more attention to minor classes and the corpora used for pre-training the transformer-based models. We show how transformer-based models continue obtaining the best results, but we improved previous results by a 6,2% by adding new steps of pre-processing and using models pre-trained with Spanish social-media data, setting new state-of-the-art results. © 2013 IEEE.

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OffensiveLang: A Community-Based Implicit Offensive Language Dataset

Item Type Journal Article

Author A. Das

Author M. Rahgouy

Author D. Feng

Author Z. Zhang

Author T. Bhattacharya

Author N. Raychawdhary

Author F. Jamshidi

Author V. Jain

Author A. Chadha

Author M.J. Sandage

Author L. Pope

Author G.V. Dozier

Author C.D. Seals

Abstract The widespread presence of hateful languages on social media has resulted in adverse effects on societal well-being. As a result, addressing this issue with high priority has become very important. Hate speech or offensive languages exist in both explicit and implicit forms, with the latter being more challenging to detect. Current research in this domain encounters several challenges. Firstly, the existing datasets primarily rely on the collection of texts containing explicit offensive keywords, making it challenging to capture implicitly offensive contents that are devoid of these keywords. Secondly, common methodologies tend to focus solely on textual analysis, neglecting the valuable insights that community information can provide. In this research paper, we introduce a novel dataset OffensiveLang, a community based implicit offensive language dataset generated by ChatGPT 3.5 containing data for 38 different target groups. Despite limitations in generating offensive texts using ChatGPT due to ethical constraints, we present a prompt-based approach that effectively generates implicit offensive languages. To ensure data quality, we evaluate the dataset with human. Additionally, we employ a prompt-based zero-shot method with ChatGPT and compare the detection results between human annotation and ChatGPT annotation. We utilize existing state-of-the-art models to see how effective they are in detecting such languages. The dataset is available here: <https://github.com/AmitDasRup123/OffensiveLang>. © 2024 The Authors.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: A. Das; University of North Alabama, Florence, 35632, United States; email: adas@una.edu

On Left and Right: Understanding the Discourse of Presidential Election in Social Media Communities

Item Type Conference Paper

Author L. Zhou

Author J. Tao

Author K. Wang

Editor Bui T.X.

Abstract As a promising platform for political discourse, social media becomes a battleground for presidential candidates as well as their supporters and opponents. Stance detection is one of the key tasks in the understanding of political discourse. However, existing methods are dominated by supervised techniques, which require labeled data. Previous work on stance detection is largely conducted at the post or user level. Despite that some studies have considered online political communities, they either only select a few communities or assume the stance coherence of these communities. Political party extraction has rarely been addressed explicitly. To address the limitations, we developed an unsupervised learning approach to political party extraction and stance detection from social media discourse. We also analyzed and compared (sub)communities with respect to their characteristics of political stances and parties. We further explored (sub)communities' shift in political stance after the 2020 US presidential election. © 2023 IEEE Computer Society. All rights reserved.

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On the development of an information system for monitoring user opinion and its role for the public

Item Type Journal Article

Author V. Karyukin

Author G. Mutanov

Author Z. Mamykova

Author G. Nassimova

Author S. Torekul

Author Z. Sundetova

Author M. Negri

Abstract Social media services and analytics platforms are rapidly growing. A large number of various events happen mostly every day, and the role of social media monitoring tools is also increasing. Social networks are widely used for managing and promoting brands and different services. Thus, most popular social analytics platforms aim for business purposes while monitoring various social, economic, and political problems

remains underrepresented and not covered by thorough research. Moreover, most of them focus on resource-rich languages such as the English language, whereas texts and comments in other low-resource languages, such as the Russian and Kazakh languages in social media, are not represented well enough. So, this work is devoted to developing and applying the information system called the OMSystem for analyzing users' opinions on news portals, blogs, and social networks in Kazakhstan. The system uses sentiment dictionaries of the Russian and Kazakh languages and machine learning algorithms to determine the sentiment of social media texts. The whole structure and functionalities of the system are also presented. The experimental part is devoted to building machine learning models for sentiment analysis on the Russian and Kazakh datasets. Then the performance of the models is evaluated with accuracy, precision, recall, and F1-score metrics. The models with the highest scores are selected for implementation in the OMSystem. Then the OMSystem's social analytics module is used to thoroughly analyze the healthcare, political and social aspects of the most relevant topics connected with the vaccination against the coronavirus disease. The analysis allowed us to discover the public social mood in the cities of Almaty and Nur-Sultan and other large regional cities of Kazakhstan. The system's study included two extensive periods: 10-01-2021 to 30-05-2021 and 01-07-2021 to 12-08-2021. In the obtained results, people's moods and attitudes to the Government's policies and actions were studied by such social network indicators as the level of topic discussion activity in society, the level of interest in the topic in society, and the mood level of society. These indicators calculated by the OMSystem allowed careful identification of alarming factors of the public (negative attitude to the government regulations, vaccination policies, trust in vaccination, etc.) and assessment of the social mood. © 2022, The Author(s).

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On the effectiveness of limited-data large language model fine-tuning for Arabic

Item Type Journal Article

Author M. Alkaoud

Abstract This paper presents an investigation into fine-tuning large language models (LLMs) for Arabic natural language processing (NLP) tasks. Although recent multilingual LLMs have made remarkable progress in zero-shot and few-shot settings, specialized models such as fine-tuned BERT variants continue to define state-of-the-art (SOTA) performance in many Arabic tasks. We demonstrate that by fine-tuning a general-purpose LLM (GPT-4o mini) on only a small subset (3.0%–7.5%) of the training samples, we exceed previous best reported results in sentiment analysis (ArSAS) and sarcasm detection (ArSarcasm), while achieving performance statistically comparable to the SOTA in news categorization (ASND). This study highlights that LLMs, when properly adapted, can outperform established models without relying on full-scale annotated training sets. Furthermore, our analysis with the open-source Gemma-3-27B model confirms the generalizability of our data-efficient method. Notably, this approach enabled the model to achieve performance statistically comparable to SOTA on all three tasks, although the proprietary GPT-4o mini maintained an overall performance advantage. We further compare GPT-4o with GPT-4o mini to examine the impact of model size on fine-tuning. GPT-4o outperforms GPT-4o mini across all sample sizes but by small margins (<1%). Notably, GPT-4o fine-tuned on 100 samples achieves marginally better performance than GPT-4o mini fine-tuned on 500 samples, indicating that larger models require fewer labeled examples. Additionally, we find that fine-tuning performance follows predictable scaling, with GPT-4o mini's performance growth function accurately estimating GPT-4o's results (error < 0.005). This enables efficient performance estimation for larger models using smaller ones. Our findings emphasize the practical benefits of fine-tuning LLMs for Arabic NLP, while demonstrating predictable scaling laws that can guide efficient model selection and adaptation. © 2025 Mohamed Alkaoud. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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On the Impact of Language Nuances on Sentiment Analysis With Large Language Models: Paraphrasing, Sarcasm, and Emojis

Item Type Journal Article

Author N. Bhargava

Author M.I. Radaideh

Author O.H. Kwon

Author A. Verma

Author M.I. Radaideh

Abstract This research investigates how textual nuances like emojis and sarcasm impact the performance of Large Language Models (LLMs) in sentiment analysis, especially when using social media data. To improve data quality, the study explores text paraphrasing techniques and addresses the scarcity of labeled sarcasm data by creating a human-labeled dataset of 5,929 tweets, enabling a more robust evaluation of LLMs in handling sarcasm. The results show that when topic-specific datasets related to nuclear power are used to finetune LLMs, these models are not able to comprehend accurate sentiment in the presence of sarcasm due to less diverse text, requiring external interventions like sarcasm removal to boost model accuracy. Sarcasm removal led to up to 21% improvement in sentiment accuracy, as LLMs trained on nuclear power-related content struggled with sarcastic tweets, achieving only 30% accuracy. In contrast, LLMs trained on dataset of 690,000 general tweets, covering a broader range of topics, showed considerable improvements in predicting sentiment for sarcastic tweets (60% accuracy), indicating that incorporating general text data can enhance sarcasm detection. The study also utilized adversarial text paraphrasing, showing that creating synthetic text variants by making minor changes

significantly increased model robustness and accuracy for sarcastic tweets (approximately 85%). Additionally, text paraphrasing of tweets with fragmented language transformed around 40% of approximately 363,000 tweets, improving LLMs sentiment analysis accuracy by 6%. Finally, emojis did not significantly affect sentiment analysis for nuclear power-related content, suggesting that emojis may only reinforce, rather than reveal, sentiment in certain contexts. © 2013 IEEE.

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On the reliability and validity of detecting approval of political actors in tweets

Item Type Conference Paper

Author I. Sen

Author F. Flöck

Author C. Wagner

Abstract Social media sites like Twitter possess the potential to complement surveys that measure political opinions and, more specifically, political actors' approval.

However, new challenges related to the reliability and validity of social-media-based estimates arise. Various sentiment analysis and stance detection methods have been developed and used in previous research to measure users' political opinions based on their content on social media. In this work, we attempt to gauge the efficacy of untargeted sentiment, targeted sentiment, and stance detection methods in labeling various political actors' approval by benchmarking them across several datasets. We also contrast the performance of these pretrained methods that can be used in an off-the-shelf (OTS) manner against a set of models trained on minimal custom data. We find that OTS methods have low generalizability on unseen and familiar targets, while low-resource custom models are more robust. Our work sheds light on the strengths and limitations of existing methods proposed for understanding politicians' approval from tweets. © 2020 Association for Computational Linguistics

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One for “All”: A unified model for fine-grained sentiment analysis under three tasks

Item Type Journal Article

Author H.-Y. Lu

Author J. Yang

Author C. Hu

Author W. Fang

Abstract Background. Fine-grained sentiment analysis is used to interpret consumers' sentiments, from their written comments, towards specific entities on specific aspects. Previous researchers have introduced three main tasks in this field (ABSA, TABSA, MEABSA), covering all kinds of social media data (e.g., review specific, questions and answers, and community-based). In this paper, we identify and address two common challenges encountered in these three tasks, including the low-resource problem and the sentiment polarity bias. Methods. We propose a unified model called PEA by integrating data augmentation methodology with the pre-trained language model, which is suitable for all the ABSA, TABSA and MEABSA tasks. Two data augmentation methods, which are entity replacement and dual noise injection, are introduced to solve both challenges at the same time. An ensemble method is also introduced to incorporate the results of the basic RNN-based and BERT-based models. Results. PEA shows significant improvements on all three fine-grained sentiment analysis tasks when compared with state-of-the-art models. It also achieves comparable results with what the baseline models obtain while using only 20% of their training data, which demonstrates its extraordinary performance under extreme low-resource conditions. © Copyright 2021 Lu et al.

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OneLove beyond the field - A few-shot pipeline for topic and sentiment analysis during the FIFA World Cup in Qatar

Item Type Conference Paper

Author C. Rauchegger

Author S.M. Wang

Author P. Delobelle

Editor de Araujo P.H.L.

Editor Baumann A.

Editor Gromann D.

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Editor Wiegand M.

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Abstract The FIFA World Cup in Qatar was discussed extensively in the news and on social media. Due to news reports with allegations of human rights violations, there were calls to boycott it. Wearing a OneLove armband was part of a planned protest activity. Controversy around the armband arose when FIFA threatened to sanction captains who wear it. To understand what topics Twitter users Tweeted about and what the opinion of German Twitter users was towards the OneLove armband, we performed an analysis of German Tweets published during the World Cup using in-context learning with LLMs. We validated the labels on human annotations. We found that Twitter users initially discussed the armband's impact, LGBT rights, and politics; after the ban, the conversation shifted towards politics in sports in general, accompanied by a subtle shift in sentiment towards neutrality. Our evaluation serves as a framework for future research to explore the impact of sports activism and evolving public sentiment. This is especially useful in settings where labeling datasets for specific opinions is unfeasible, such as when events are unfolding. ©2024 Association for Computational Linguistics.

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Opinion Mining for Marathi Language Using Machine Learning

Item Type Conference Paper

Author S.S. Pardeshi

Author G.K. Salve

Author S.S. Thakur

Author R.J. Sutar

Editor Kaiser M.S.

Editor Xie J.

Editor Rathore V.S.

Abstract Sentiment analysis for low-resource languages, including Marathi, has garnered increased interest, driven by the rising volume of digital content generated in regional languages. This survey paper provides a comprehensive review of 30 research papers focusing on opinion mining and sentiment analysis in the Marathi language and other related domains. The reviewed works are categorized based on their primary contributions, including model development, methodology proposals, and surveys of existing techniques. Various approaches such as lexicon-based methods, classical machine learning models, and deep learning techniques have been explored, highlighting the evolving trends and challenges in handling Marathi language text, including issues with transliteration and code-mixing. Besides the survey, this paper introduces an innovative approach for sentiment analysis of transliterated Marathi text. Our approach involves manually curating a sentiment wordlist using a Marathi-to-English dictionary and assigning weighted sentiment scores. We collected user-generated content from social media platforms such as Instagram, Twitter, and YouTube, which was then pre-

processed and analyzed based on the occurrence of sentiment words. The sentences were labeled and used to train machine learning models, specifically Logistic Regression (LR) and Support Vector Machines (SVM). The models' performance was assessed using standard evaluation metrics such as accuracy, precision, recall, and F1-score. Future work will explore additional pre-processing techniques and advanced models to optimize sentiment classification in transliterated Marathi. By combining a detailed literature review with our practical implementation, this paper aims to provide insights into the state-of-the-art in Marathi sentiment analysis and set the stage for future research to address existing gaps and challenges. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Opportunities and challenges of integrating ChatGPT in education: Sentiment analysis and topic modeling

Item Type Journal Article

Author S. Teerakapibal

Author P. Kusawat

Abstract Since its recent debut, ChatGPT has become a global sensation and significantly impacted the field of education. Both educational researchers and practitioners have identified opportunities as well as risks associated with the use of this novel tool in educational settings. Despite the ongoing debate, there is still no research exploring occupational differences in the perception of ChatGPT in education. In this paper, we analyzed Twitter data using topic modeling and sentiment analysis to investigate how ChatGPT is perceived and discussed differently in different occupations. Our study found diverse topics discussed, including its use in schools, impact on exams, academic integrity concerns, and response accuracy evaluations. While most tweets were positive or neutral, concerns about integrity and response accuracy were evident. Analysis revealed sentiment and topic variations among users' occupations. These findings emphasize the opportunities and challenges of integrating ChatGPT in education, necessitating continued monitoring and informed policy-making for responsible utilization. © 2025 Taylor & Francis Group, LLC.

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Optimal strategies to perform multilingual analysis of social content for a novel dataset in the tourism domain

Item Type Journal Article

Author M. Masson

Author R. Agerri

Author C. Sallaberry

Author M.-N. Bessagnet

Author A. Le Parc Lacayrelle

Author P. Roose

Abstract The rising influence of social media platforms in various domains, including tourism, has highlighted the growing need for efficient and automated Natural Language Processing (NLP) strategies to take advantage of this valuable resource. However, the transformation of multilingual, unstructured, and informal texts into structured knowledge still poses significant challenges, most notably the never-ending requirement for manually annotated data to train deep learning classifiers. In this work, we study different NLP techniques to establish the best ones to obtain competitive performances while keeping the need for training annotated data to a minimum. To do so, we built the first publicly available multilingual dataset (French, English, and Spanish) for the tourism domain, composed of tourism-related tweets. The dataset includes multilayered, manually revised annotations for Named Entity Recognition (NER) for Locations and Fine-grained Thematic Concepts Extraction mapped to the Thesaurus of Tourism and Leisure Activities of the World Tourism Organization, as well as for Sentiment Analysis at the tweet level. Extensive experimentation comparing various few-shot and fine-tuning techniques with modern language models demonstrate that modern few-shot techniques allow us to obtain competitive results for all three tasks with very little annotation data: 5 tweets per label (15 in total) for Sentiment Analysis, 30 tweets for Named Entity Recognition of Locations and 1K tweets annotated with fine-grained thematic concepts, a highly fine-grained sequence labeling task based on an inventory of 315 classes. We believe that our results, grounded in a novel dataset, pave the way for applying NLP to new domain-specific applications, reducing the need for manual annotations and circumventing the complexities of rule-based, ad-hoc solutions. © 2025 The Author(s)

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Optimising Text Classification in Social Networks via Deep Learning-Based Dimensionality Reduction

Item Type Journal Article

Author J.A. Diaz-Garcia

Author A. Morales-Garzón

Author K. Gutiérrez-Batista

Author M.J. Martin-Bautista

Abstract Text classification is essential for handling the large volume of user-generated textual content in social networks. Nowadays, dense word representation techniques, especially those yielded by large language models, capture rich semantic and contextual information from text that is useful for classification tasks, but generates high-dimensional vectors that hinder the efficiency and scalability of the classification algorithms. Despite this, limited research has explored effective dimensionality reduction techniques to balance representation quality with computational demands. This study presents a deep learning-based framework for enhancing text classification in social networks, focusing on computational performance, by compressing high-dimensional text representations into a low-dimensional space while retaining essential features for text classification. To demonstrate the feasibility of the proposal, we conduct a benchmarking study using traditional dimensionality reduction techniques on two widely used benchmark datasets. The findings reveal that our approach can substantially improve the efficiency of text classification in social networks without compromising—and, in some cases, enhancing—the predictive performance. © 2025 by the authors.

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Optimization of machine learning models for sentiment analysis in social media

Item Type Journal Article**Author** J.G. Brandão**Author** A.P. Castro Junior**Author** V.M.G. Pacheco**Author** C.G. Rodrigues**Author** O.M.O. Belo**Author** A.P. Coimbra**Author** W.P. Calixto

Abstract The primary goal of this paper is to develop optimized classical machine learning models for sentiment analysis of social media posts. The methodology includes the application of k-fold cross-validation and feature selection using the χ^2 independence test to prevent overfitting and improve classifier accuracy. Classical algorithms such as naïve Bayes, support vector machines, decision trees, and k-nearest neighbors are applied, along with preprocessing techniques, across five distinct datasets for model construction and evaluation. These optimized classical models are compared with

recurrent neural network architectures, including LSTM and GRU, to evaluate the relative efficiency of both approaches. In total, 34 classification models were generated, with hyperparameter optimization for classical methods performed by grid search. The highest accuracy achieved was 82.45% for data without preprocessing and 78.83% for fully preprocessed data, both with the naïve Bayes algorithm. After hyperparameter optimization, some models achieved an accuracy greater than 90%. An analysis of variance indicated statistically significant differences among the models, confirming that feature selection and hyperparameter optimization are important factors in classifier performance. The proposed approach proved efficient for analyzing unstructured textual data, allowing the development of optimized models even with limited labeled data, and providing valuable insights into user opinions across different contexts. © 2024 Elsevier Inc.

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Optimized Identification of Sentence-Level Multiclass Events on Urdu-Language-Text Using Machine Learning Techniques

Item Type Journal Article

Author S. Ali
Author U. Jamil
Author M. Younas
Author B. Zafar
Author M. Kashif Hanif

Abstract In today's digital world, social media platforms generate a plethora of unstructured information. However, for low-resource languages like Urdu, there is a scarcity of well-structured data for specific tasks such as event classification. Urdu, a language prominent in South Asia, has boasted a complex morphological structure with unique features but has lacked standard linguistic resources like datasets. Long-text classification has demanded more effort than short-text classification due to its expansive vocabulary, information redundancy, and noise. Text processing has been the latest trend in research, with many machine learning and deep learning techniques widely used for it. Multiclass classification has been utilized to classify different languages for various purposes. In this research, a multiclass classification for the Urdu language was performed using a text dataset taken from five different social media platforms including Geo News, Samaa News, Dawn News, Express News, and Urdu Blogs totaling 103,771 sentences. We used sentence-level classification to categorize sentences including terrorist attacks, national news, sports, entertainment, politics, safety, earthquakes, fraud and corruption, sexual assault, weather, accidents, forces, inflation, murder and death, education, and international news. Deep learning, transformer-based and machine learning classifiers are used for event classification. The SMFCNN classifier achieved the greatest accuracy of 88.29%. We incorporated transformer-based models, with the proposed XLM-R+ model demonstrating superior performance with an accuracy of 89.8%. Our results were compared to previously reported techniques that used traditional models, highlighting the significant improvements offered by our approaches. The novelty of this research lies in the inclusion of 16 event categories to broaden coverage and the implementation of the SMFCNN and transformer-based algorithms. This study highlights the potential of deep learning and transformer-based models in enhancing the accuracy and generalizability of multiclass classification in low-resource languages Urdu. © 2013 IEEE.

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Optimized Sentiment Classification in Low-Resource Language Processing using Element-Wise Attention Gated Recurrent Unit with Bayesian Optimization

Item Type Conference Paper

Author P.K. Pareek

Author A.S. Al-Akayshee

Author J.R. Gottam

Author K.M. Monica

Author T. Saravanan

Abstract In recent days, Low-Resource Language Processing (LRLP) in sentiment classification has gained attention because of limited annotated data and tools, demanding efficient models for accurate emotional analysis in underrepresented languages. Traditional models for LRLP had faced challenges in capturing long-term dependencies in text. Therefore, this research proposed Element-Wise Attention Gated Recurrent based Bayesian Optimization (EWAG based BO) for LRLP in sentiment classification. Initially, data is collected from Arabic Sentiment Twitter Corpus dataset and preprocessed by using stemming and lemmatization which normalize words to their root and dictionary forms by reducing vocabulary size. Then, feature extraction is done by using SentiBERT which generates the context-aware embeddings by capturing sentiment-specific semantics. After that, sentiment classification is done by EWAG which enhances GRU through applying element-wise attention to focus on crucial input features that improved model's ability in capturing context and increase classification accuracy. To further improve performance, BO is utilized hyperparameter tuning that efficiently explores search space to find optimal model settings. The proposed EWAG-BO effectively achieved better results when compared with existing LRLP-Improved Deep Learning Hunt Prey Optimization (LRLP-IDLHPO) in terms of accuracy (99.50%), precision (99.20%) and recall (99.18%) respectively. ©2025 IEEE.

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Optimizing Airline Review Sentiment Analysis: A Comparative Analysis of LLaMA and BERT Models through Fine-Tuning and Few-Shot Learning

Item Type Journal Article

Author K.I. Roumeliotis

Author N.D. Tselikas

Author D.K. Nasiopoulos

Abstract In the rapidly evolving landscape of natural language processing (NLP) and sentiment analysis, improving the accuracy and efficiency of sentiment classification models is crucial. This paper investigates the performance of two advanced models, the Large Language Model (LLM) LLaMA model and NLP BERT model, in the context of airline review sentiment analysis. Through fine-tuning, domain adaptation, and the application of few-shot learning, the study addresses the subtleties of sentiment expressions in airline-related text data. Employing predictive modeling and comparative analysis, the research evaluates the effectiveness of Large Language Model Meta AI (LLaMA) and Bidirectional Encoder Representations from

Transformers (BERT) in capturing sentiment intricacies. Fine-tuning, including domain adaptation, enhances the models' performance in sentiment classification tasks. Additionally, the study explores the potential of few-shot learning to improve model generalization using minimal annotated data for targeted sentiment analysis. By conducting experiments on a diverse airline review dataset, the research quantifies the impact of fine-tuning, domain adaptation, and few-shot learning on model performance, providing valuable insights for industries aiming to predict recommendations and enhance customer satisfaction through a deeper understanding of sentiment in user-generated content (UGC). This research contributes to refining sentiment analysis models, ultimately fostering improved customer satisfaction in the airline industry. Copyright © 2025 The Authors.

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Optimizing BERT Models with Fine-Tuning for Indonesian Twitter Sentiment Analysis

Item Type Journal Article

Author L. Afuan

Author N. Hidayat

Author H. Hamdani

Author H. Ismanto

Author B.C. Purnama

Author D.I. Ramdhani

Abstract Twitter has emerged as a critical platform for capturing public sentiment, offering a valuable source for sentiment analysis. This study presents a comparative evaluation of two BERT (Bidirectional Encoder Representations from Transformers) models—baseline and fine-tuned—targeted at analyzing Indonesian-language tweets. Employing the CRISP-DM framework, the methodology encompasses automated data crawling, comprehensive text pre-processing (including case folding, cleaning, tokenisation, normalisation, and data augmentation), and model development using the IndoBERT-base-p1 architecture. The experimental results reveal that the fine-tuned BERT model achieves significantly improved performance over the non-optimized model, with accuracy, precision, recall, and F1-score values reaching 91%, 0.91, 0.90, and 0.91, respectively. These findings indicate the fine-tuned model's superior ability to capture linguistic subtleties and contextual sentiment features within informal social media text. Furthermore, the model is deployed in a web-based application for real-time sentiment classification, demonstrating its practical applicability. This study underscores the effectiveness of Fine-tuning in enhancing BERT-based sentiment analysis for low-resource languages. It highlights its potential for informing decision-making in digital communication, marketing, and policy research contexts. © 2025, Innovative Information Science and Technology Research Group. All rights reserved.

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Tags:

BERT, Sentiment Analysis, Social Media, Fine-tuning, Indonesian Twitter, Text Mining

Notes:

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Export Date: 20 September 2026; Cited By: 6; Funding details: Indonesian Ministry of Education, Culture; Lembaga Penelitian dan Pengabdian kepada Masyarakat, Universitas Riau, LPPM; Lembaga Penelitian dan Pengabdian kepada Masyarakat, LPPM; Jenderal Soedirman University, UNSOED; Correspondence Address: L. Afuan; Department of Informatics, Universitas Jenderal Soedirman, Indonesia; email: lasmedi.afuan@unsoed.ac.id

Optimizing Election Result Prediction Through Fine-Tuned Transformer Models

Item Type Conference Paper**Author** S. Singhal**Author** G. Pruthi**Author** A. Kumar**Author** L. Kapoor**Author** V. Bhatia

Abstract Predicting election results accurately is crucial for understanding political dynamics and making informed decisions. In recent years, Transformer-Based models have shown remarkable performance in various Natural Language Processing tasks. However, applying these models to Election Result Prediction remains a challenging task due to the complexity of political landscapes and the inherent unpredictability of voter behavior. The pervasive use of Social Media Platforms, exemplified by Twitter, has established itself as a significant channel for users to express and disseminate their sentiments and viewpoints. This widespread adoption has created novel opportunities to quantify and evaluate the popularity and public sentiment concerning political leaders. This paper introduces an innovative approach that harnesses Sentiment Analysis techniques to forecast the outcome of the 2022 Mid-Term Elections in the United States. The proposed methodology comprises of approximately 42,000 tweets and employing diverse Sentiment Analysis approaches to assess the sentiment towards each political candidate. By harnessing the copious user-generated content available on platforms like Twitter, this work uncovers valuable insights that can augment the precision of Election Result Predictions. The performance of the proposed approach is evaluated by computing sentiment scores for individual candidates and employing them to prognosticate poll results and it has been found that the proposed approach is optimal approach amongst the competent approaches for predicting election results based on sentiment analysis of social media. The findings of this paper contribute to the domain of political forecasting by demonstrating the efficacy of Sentiment Analysis in leveraging Social Media data for Election Result Prediction. © 2023 IEEE.

Date 2023**DOI** 10.1109/NMITCON58196.2023.10276268**URL** <https://www.scopus.com/pages/publications/85175398931?origin=resultslist>**Archive** Scopus**Language** English

Extra Journal Abbreviation: Int. Conf. Netw., Multimed. Inf. Technol., NMITCON
Publisher Institute of Electrical and Electronics Engineers Inc.
ISBN 979-835030082-6 (ISBN)
Proceedings Title Int. Conf. Netw., Multimed. Inf. Technol., NMITCON
Conference Name 2023 International Conference on Network, Multimedia and Information Technology, NMITCON 2023
Date Added 20/09/2026, 8:35:19 pm
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Tags:

Deep learning, Sentiment analysis, Forecasting, Social networking (online), BERT, Deep Learning, Twitter, Sentiment Analysis, Transformer, Transformers, Shot classification, Zero-shot classification, Zero-shot learning, Zero-Shot Classification, Bag of words, Bag of Words, Rule based, Rule-Based, US election, US Election

Notes:

EID: 2-s2.0-85175398931

Export Date: 20 September 2026; Cited By: 4; Conference name: 2023 IEEE International Conference on Network, Multimedia and Information Technology, NMITCON 2023; Conference date: 1 September 2023 through 2 September 2023; Conference code: 193544

Optimizing News Topic Classification with Instructional Fine-Tuning of Chatglm3

Item Type Conference Paper

Author C. Ye

Author X. Shi

Abstract In the rapidly evolving landscape of natural language processing (NLP), the deployment of large language models has been at the forefront of innovation, particularly in the realm of text classification. This study probes into the utilization of advanced NLP techniques, emphasizing the instructional fine-tuning of the ChatGLM3-6B model - a variant of the General Language Model with 6.2 billion parameters - for the specific application of news topic classification. As the volume and velocity of news data burgeon, the imperative for models to not only accurately classify but also to efficiently process text becomes paramount. Our research undertakes a meticulous fine-tuning of ChatGLM3-6B using the QLoRA framework, which is renowned for its quantization efficiency and adaptability in processing, on the Twitter Financial News dataset. This dataset, characterized by its multi-label text classification challenge and inherent data imbalance, provides a rigorous testing ground for our model. We benchmarked ChatGLM3-6B against established models such as Roberta-Base, Roberta-Large, and the DeBERTa-V2 variants, measuring performance in terms of accuracy - a critical metric in the domain of news classification. The findings of our study are compelling, demonstrating that the instructionally fine-tuned ChatGLM3-6B

model not only achieves but exceeds the classification accuracy of traditional models, registering an accuracy of 88.15%. This superior performance can be attributed to the strategic combination of the model's inherent capabilities and the fine-tuning process that was rigorously tailored to the complexities and nuances inherent in financial news texts. Moreover, we closely monitored the training stability and the model's ability to generalize, which is graphically evidenced by the consistent decline in training loss over approximately 1,500 iterations, without any indication of overfitting. This attests to the robustness of the chosen optimization strategies, including an empirically determined learning rate, batch size, and gradient accumulation strategy. Conclusively, this study not only elucidates the potential of fine-tuned large language models like ChatGLM3-6B in the efficient and accurate categorization of news topics but also propels the conversation forward regarding the scalability of these models. Our findings suggest significant implications for the automation of news categorization processes, paving the way for future research into the ethical and scalable applications of NLP in media and journalism. © 2024 ACM.

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Language English

Extra Journal Abbreviation: ACM Int. Conf. Proc. Ser.

Publisher Association for Computing Machinery

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Conference Name ACM International Conference Proceeding Series

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Tags:

Language model, Large language model, Language processing, Natural language processing, Natural Language Processing (NLP), Natural language processing systems, Natural languages, Large datasets, Fine tuning, Large Language Models (LLMs), Text classification, Text Classification, Financial data processing, Topic Classification, Chatglm3, ChatGLM3, Instruction fine- tuning, Instruction Fine-Tuning, News topic classification, News Topic Classification, News topics

Notes:

EID: 2-s2.0-85201426681

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: C. Ye; Chuanzhi Education Group, Beijing, China; email: yechunyang2011@163.com; Conference name: 3rd International Conference on Computer, Artificial Intelligence and Control Engineering, CAICE 2024; Conference date: 26 January 2024 through 28 January 2024; Conference code: 201572

Optimizing Patient Feedback With Generative Adversarial Network Leveraging Knowledge Distillation to Improve Healthcare

Item Type Journal Article

Author M. Fahim-Ul-Islam

Author A. Chakrabarty

Author M. Hasan

Author A.H. Efaz

Author X. Wang

Author M.J. Piran

Abstract Despite progress in global healthcare systems, the utilization of domestic healthcare facilities remains limited in several regions, with a considerable proportion of people pursuing treatment overseas. This development highlights the necessity for systematic incorporation of patient-centered input, an essential element for enhancing accountability, transparency, and quality in local healthcare. Our research seeks to address this deficiency by establishing a system that collects and analyzes patient feedback to inform and improve healthcare policies and practices, particularly in areas with elevated demand for medical services. We offer an effective platform for viewing reviews from different hospitals, especially in places where people routinely visit for medical services. Therefore, we built our primary 'Dhaka Private Hospitals Review Dataset,' considering gathering and evaluating patient opinions methodically. We further employ transformer-based generative adversarial learning to evaluate sentiment analysis using knowledge distillation (KD) to boost model efficiency. Our proposed GANBERT architecture includes two optimized student models gated recurrent unit-based Contextualized BERT (GC-BERT) and LSTM-based Contextualized BERT (LC-BERT) with enhanced generators and discriminators. Our GC-BERT enhances execution time by 1.27% to 24.27%, while LC-BERT improves by 14.13% to 23.30%, showing superior advancements compared to other contemporary models. Each model with reductions ranging from 82.50% to 99.99% parameters, making them lightweight and efficient compared to other teacher models in the KD process. Instead of using contextual word representations which demand more space and complexity for reviewing patient feedback, we utilize the single static pretrained and low-dimensional word embedding space approach integrating student models. © 2013 IEEE.

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Language English

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Issue 8

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Behavioral research, Article, human, Humans, Sentiment analysis, Healthcare, Knowledge distillation, BERT, algorithm, social media, performance, Feedback, procedures, training, deep learning, feature extraction, Health care, Students, Transformer, environment, sentiment analysis, Hospitals, Transformers, Medical services, federated learning, healthcare, benchmarking, accuracy, mental health, Teaching, medical informatics, architecture, knowledge, Knowledge management, logistic regression analysis, long short term memory network, sensitivity analysis, adaptation, adversarial learning, Adversarial networks, dynamics, experience, feedback system, few shot learning, generative adversarial network, Generative Adversarial Networks, generative artificial intelligence, Generative Artificial Intelligence, health care, Healthcare facility, Healthcare systems, hospital rating, Hospital rating, information processing, knowledge distillation, mathematical model, Medical Informatics, memory, methodology, morphology, patient care, patient feedback, selenium, Student Modeling

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Optimizing Social Media Sentiment Analysis: RoBERTa Framework for Evaluating Public Perception of DeepSeek

Item Type Conference Paper

Author A. Santosa

Author G.A. Prasanna

Author M. Rizky

Author A.S. Prabowo

Editor Wibowo F.W.

Abstract In the past few years, Artificial Intelligence (AI) has become a widely discussed topic by people. AI has affected various aspects of human life. DeepSeek is one of the AIs that has received a lot of attention. This study introduces an optimized framework for social media sentiment analysis to find out people's views of DeepSeek using Tweet data from X (formerly Twitter) taken in January 2025. Using a RoBERTa-based model, fine-tuned for social media text, and featuring an optimized preprocessing stage. One important result is that including stopwords during preprocessing significantly boosts sentiment classification accuracy to

87.5%. The sentiment analysis results show that people's opinions are very diverse and polarized, with a sentiment distribution of 41.1% being Positive, 36.8% being Neutral, and 22.1% being Negative. Word cloud analysis shows different views on DeepSeek. Some people appreciate DeepSeek's open-source accessibility while others express concerns about security and trust. These insights show that people's views on AI are complicated, making it much more vital to be transparent in AI development, set performance benchmarks, and build trust in AI. © 2025 IEEE.

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Archive Scopus

Language English

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Conference Name Proceedings - 2025 4th International Conference on Electronics Representation and Algorithm: Artificial Intelligence: Creating Tomorrow's World Today, ICERA 2025

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Behavioral research, Artificial intelligence, Network security, Information systems, public perception, Public perception, Sentiment analysis, Social networking (online), Benchmarking, Social media, Data mining, social media, Twitter, sentiment analysis, Sentiment classification, Tweets, RoBERTa, zero-shot learning, DeepSeek, Word clouds, Deepseek, Classification accuracy, Cloud analysis, Human lives

Notes:

EID: 2-s2.0-105013464930

Export Date: 20 September 2026; Cited By: 2; Conference name: 4th International Conference on Electronics Representation and Algorithm, ICERA 2025; Conference code: 210850

Optimizing Transformer-Based Models for Financial Sentiment in Indonesian Stock Tweets

Item Type Conference Paper

Author R.D. Tanjaya

Author G.E. Simanjuntak

Author K.S. Nugroho

Author M. Isnani

Author B. Pardamean

Abstract The rise of informal financial discourse on Indonesian social media poses challenges for sentiment classification due to noisy language and limited annotated data. This study aims to evaluate and improve sentiment classification performance on Indonesian financial tweets using transformer-based models, including IndoBERT-p1, IndoBERT-p2, IndoBERTweet, BERT-Multi, and IndoBERT uncased. Four preprocessing strategies were explored as baseline configurations: raw with emoji, raw without emoji, cleaned with emoji, and cleaned without emoji. Experimental results identified IndoBERTweet with emoji configuration as the best performing combination, achieving an F1 macro score of 0.9710 on the training set and 0.8471 on the validation set, although signs of overfitting were observed. To address this, hyperparameters were fine-tuned and specific model layers were customized, which reduced overfitting and resulted in F1 macro scores of 0.8572 on training and 0.8056 on validation. The model was further tested with and without the introduction of noise in the test data, yielding scores of 0.8177 and 0.8106, respectively. These findings demonstrate the model's robustness to domain specific noise and emphasize the importance of targeted preprocessing and domain aware fine tuning for effective sentiment analysis in low resource financial contexts. © 2025 IEEE.

Date 2025

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Archive Scopus

Language English

Extra Journal Abbreviation: Proc. - IEEE Inf. Technol. Int. Semin., ITIS

Publisher Institute of Electrical and Electronics Engineers Inc.

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Pages 214-219

Proceedings Title Proc. - IEEE Inf. Technol. Int. Semin., ITIS

Conference Name Proceeding - 2025 IEEE 11th Information Technology International Seminar, ITIS 2025

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Tags:

Speech communication, Classification (of information), Sentiment analysis, Fine tuning, Electronic trading, Sentiment classification, Tuning, Tweets, fine-tuning, Finance, Overfitting, Financial sentiment analyze, financial sentiment analysis, indoBERTweet, Indobertweet, Indonesian stock market tweet, indonesian stock market tweets, Optimizing transformers, Robustness evaluation, robustness evaluation introduction, Robustness evaluation introduction

Notes:

EID: 2-s2.0-105032366536

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: R.D. Tanjaya; Bina Nusantara University, School of Computer Science, Computer Science Department, Jakarta, Indonesia; email: richard.tanjaya@binus.ac.id; Conference name: 11th Information Technology International Seminar, ITIS 2025; Conference date: 8 October 2025 through 10 October 2025; Conference code: 218865

Out-of-Fold Stacking for Reliable Transformer Ensembles in Indonesian Depression/Anxiety Indicator Detection

Item Type Conference Paper

Author M.A. Perdana

Author B.S. Hantono

Author R. Ferdiana

Abstract Transformer-based models are widely used for depression/anxiety indicator detection from social media text, often further enhanced through ensemble learning. While stacking ensembles are commonly adopted, the validation strategy used to train the meta-learner is frequently overlooked and may lead to unreliable performance gains. In particular, conventional stacking approaches that train the meta-learner on predictions from base models trained on the same data risk meta-level overfitting. This paper presents a controlled empirical comparison between conventional stacking and stacking with strict Out-of-Fold (OOF) meta-training for Indonesian depression/anxiety indicator detection from Twitter data. Results show that non-OOF stacking fails to consistently outperform the strongest individual model, whereas OOF stacking consistently produces the best-performing ensemble across multiple runs. Although the absolute gains are modest, the improvement is systematic and reproducible. These findings highlight the importance of validation discipline in stacking ensembles and demonstrate that OOF meta-training yields more reliable ensemble performance for low-resource mental health text classification. © 2026 IEEE.

Date 2026

DOI 10.1109/ICCRAIDS67816.2026.11519635

URL <https://www.scopus.com/pages/publications/105041605567?origin=resultslist>

Archive Scopus

Language English

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Conference Name Proceedings - 2026 International Conference on Current Research in Artificial Intelligence and Data Science, ICCRAIDS 2026

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Mental health, Classification (of information), Learning systems, Social networking (online), Social media, E-learning, Mental Health, Depression/anxiety indicator detection, Depression/Anxiety Indicator Detection, Error detection, Indonesian NLP, Meta-learner, Meta-training, Out-of-fold learning, Out-of-Fold Learning, Reliable transformers, Stacking ensemble, Stacking Ensemble, Stackings

Notes:

EID: 2-s2.0-105041605567

Export Date: 20 September 2026; Cited By: 0; Conference name: 2026 International Conference on Current Research in Artificial Intelligence and Data Science, ICCRAIDS 2026; Conference date: 10 April 2026 through 11 April 2026; Conference code: 224130

Overview of the CLEF-2024 CheckThat! Lab Task 2 on Subjectivity in News Articles

Item Type	Conference Paper
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Author	J.M. Struß
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Author	F. Ruggeri
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Author	A. Barrón-Cedeno
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Author	F. Alam
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Author	D. Dimitrov
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Author	A. Galassi
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Author	G. Pachov
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Author	I. Koychev
--------	------------

Author	P. Nakov
--------	----------

Author	M. Siegel
--------	-----------

Author	M. Wiegand
--------	------------

Author	M. Hasanain
--------	-------------

Author	R. Suwaileh
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Author	W. Zaghouani
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Editor	Faggioli G.
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Editor	Ferro N.
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Editor	Galuscakova P.
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Editor	de Herrera A.G.S.
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Abstract	We present an overview of Task 2 of the seventh edition of the CheckThat! lab at the 2024 iteration of the Conference and Labs of the Evaluation Forum (CLEF). The task focuses on subjectivity detection in news articles and was offered in five languages: Arabic, Bulgarian, English, German, and Italian, as well as in a multilingual setting. The datasets for each language were carefully curated and annotated, comprising over 10,000 sentences from news articles. The task challenged participants to develop systems capable of distinguishing between subjective statements (reflecting personal opinions or biases) and objective ones
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(presenting factual information) at the sentence level. A total of 15 teams participated in the task, submitting 36 valid runs across all language tracks. The participants used a variety of approaches, with transformer-based models being the most popular choice. Strategies included fine-tuning monolingual and multilingual models, and leveraging English models with automatic translation for the non-English datasets. Some teams also explored ensembles, feature engineering, and innovative techniques such as few-shot learning and in-context learning with large language models. The evaluation was based on macro-averaged F1 score. The results varied across languages, with the best performance achieved for Italian and German, followed by English. The Arabic track proved particularly challenging, with no team surpassing an F1 score of 0.50. This task contributes to the broader goal of enhancing the reliability of automated content analysis in the context of misinformation detection and fact-checking. The paper provides detailed insights into the datasets, participant approaches, and results, offering a benchmark for the current state of subjectivity detection across multiple languages. © 2024 Copyright for this paper by its authors.

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URL <https://www.scopus.com/pages/publications/85201612702?origin=resultslist>

Archive Scopus

Language English

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Volume 3740

Publisher CEUR-WS

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Proceedings Title CEUR Workshop Proc.

Conference Name CEUR Workshop Proceedings

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Tags:

Classification (of information), Fine tuning, Contrastive Learning, Translation (languages), Zero-shot learning, News articles, F1 scores, Fact-checking, Feature engineerings, misinformation detection, Automatic translation, fact-checking, Factual information, Misinformation detection, Sentence level, subjectivity classification, Subjectivity classifications

Notes:

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Overview of the CLEF-2025 CheckThat! Lab Task 1 on Subjectivity in News Articles

Item Type	Conference Paper
Author	F. Ruggeri
Author	A. Muti
Author	K. Korre
Author	J.M. Struß
Author	M. Siegel
Author	M. Wiegand
Author	F. Alam
Author	Md.R. Biswas
Author	W. Zaghouani
Author	M. Nawrocka
Author	B. Ivasiuk
Author	G. Razvan
Author	A. Mihail
Editor	Faggioli G.
Editor	Ferro N.
Editor	Rosso P.
Editor	Spina D.
Abstract	We present an overview of Task 1 of the eighth edition of the CheckThat! lab at the 2025 edition of the Conference and Labs of the Evaluation Forum (CLEF). The task required participants to determine whether individual sentences from news articles expressed subjective viewpoints, such as opinions or personal bias, or presented objective, fact-based information. The task was offered in nine languages: Arabic, Bulgarian, English, German, Italian, Greek, Polish, Romanian, and Ukrainian, as well as in a multilingual setting. We curated datasets for each language, comprising roughly 14,000 sentences sourced from diverse news outlets. Participants were tasked with developing classification systems to identify subjectivity (personal opinions or biases) and objectivity (factual information) at the sentence level. A total of 22 teams participated in the task, submitting 436 valid runs across all language tracks. Most systems were based on transformer models, with approaches ranging from fine-tuning language-specific

and multilingual encoders to applying English-centric models in combination with machine translation. Several teams also experimented with ensemble techniques, handcrafted features, and in-context learning using large language models. Systems were evaluated using macro-averaged F1 score to ensure equal weighting of subjective and objective classes. Performance varied considerably by language: German, Italian, English and Romanian yielded the highest results. In contrast, Greek and Ukrainian emerged as the most challenging languages, with no team surpassing the 0.65 and 0.51 F1 score marks, respectively. Task 1 offers a valuable benchmark for the development and evaluation of multilingual subjectivity detection systems. This paper presents an overview of Task 1, including datasets, system strategies, and outcomes, contributing to broader research efforts aimed at improving the transparency and trustworthiness of automated content analysis. © 2025 Copyright for this paper by its authors. Use permitted under Creative Commons License Attribution 4.0 International (CC BY 4.0).

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Language English

Extra Journal Abbreviation: CEUR Workshop Proc.

Volume 4038

Publisher CEUR-WS

ISBN 16130073 (ISSN)

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Proceedings Title CEUR Workshop Proc.

Conference Name CEUR Workshop Proceedings

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Computational linguistics, Classification (of information), Natural language processing systems, News articles, F1 scores, Fact-checking, misinformation detection, fact-checking, Factual information, Misinformation detection, Sentence level, subjectivity classification, Subjectivity classifications, Classification system, Objective facts, Romanians

Notes:

EID: 2-s2.0-105019038240

Export Date: 20 September 2026; Cited By: 7; Correspondence Address: F. Ruggeri; University of Bologna, Italy; email: federico.ruggeri6@unibo.it; Conference name: 26th Working Notes of the Conference and Labs of the Evaluation Forum, CLEF 2025; Conference date: 9 September 2025 through 12 September 2025; Conference code: 213051

Parameter-Efficient Hate Speech Detection: A Comprehensive Evaluation of LoRA-Adapted LLMs Across 18 Architectures

Item Type Conference Paper

Author M. Abusaqer

Author J. Saquer

Abstract The proliferation of hate speech on social media platforms necessitates automated detection systems that balance accuracy with computational efficiency. This study evaluates 18 LoRA-adapted Large Language Models (LLMs) across three model families (Llama, Phi, Qwen) at scales ranging from 0.5B to 14B parameters for hate speech detection. Using a dataset of 48,049 samples, we investigate the relationship between model architecture, size, instruction tuning, and detection performance while focusing on parameter-efficient fine-tuning approaches. Our results demonstrate that Phi-4 achieves state-of-the-art performance of 91.84% accuracy and 92.79% F1-score despite using only 0.15% of its parameters for task-specific adaptation. Smaller models like Llama-3.2-1B show competitive performance (91.53% accuracy, 92.69% F1-score) with significantly faster training times. We find that architectural design often outweighs parameter count, with smaller models frequently outperforming larger counterparts within the same family. The inconsistent benefits of instruction tuning across architectures highlight the importance of task-specific adaptation for hate speech detection. These findings offer valuable insights for developing efficient and effective hate speech detection systems across diverse deployment scenarios, from high-performance servers to resource-constrained environments. © 2025 Knowledge Systems Institute Graduate School. All rights reserved.

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Language English

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Conference Name Proceedings of the International Conference on Software Engineering and Knowledge Engineering, SEKE

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Tags:

Language model, Large language model, Speech communication, Architecture, Computational efficiency, Language processing, Natural language processing, Natural language processing systems, Natural languages, Transformer modeling, Fine tuning, Low-rank adaptation, natural language processing, Knowledge engineering, LLMs, Text classification, Speech analysis, Speech detection, parameter-efficient fine-tuning, Parameter-efficient fine-tuning, Tuning, text classification, hate speech detection, Hate speech detection, Llama, Knowledge management, transformer models, Phi, Qwen, low-rank adaptation

Notes:

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Patient Experiences in the Cochlear Implant Reddit Community: Comparing Human and Large Language Model Categorization

Item Type Journal Article

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Abstract Purpose: Although some work has leveraged automated analyses of online communities to gain cochlear implant (CI) patient insights, there remains a gap in comparing human versus automated analysis of the nuanced, real-world experiences patients share outside clinical settings. This study characterizes experiences within the r/Cochlearimplants Reddit community and compares human to large language model (LLM) performance in annotating posts. Method: Using reflexive thematic analysis, 996 publicly available r/Cochlearimplants posts (October 2024–June 2025) were manually coded and consolidated into themes. Three LLMs—OpenAI o3, Gemini 2.5 Pro, and Claude Sonnet 4—were prompted with the posts and human-generated codebook to perform post categorization. Model performance was evaluated against human coding using Cohen’s kappa, percent agreement, sensitivity,

specificity, positive predictive value (PPV), negative predictive value (NPV), and time. Results: Five themes emerged. Community engagement and support (N=944, 94.8%) frequently involved eliciting advice (N=721, 72.4%), seeking shared experiences (N=249, 25.0%), and sharing negative experiences (N=247, 24.8%). Other themes included the medical/surgical journey (N=463, 46.5%), device/technical issues (N=343, 34.4%), daily life/adjustments (N=236, 23.7%), and media/outreach (7.2%, N=72). OpenAI o3 and Gemini 2.5 Pro achieved the highest interrater reliability with human annotators ($\kappa=.35$ and $\kappa=.34$, respectively). OpenAI o3 had higher sensitivity (46.7%) but lower specificity (90.4%) than Gemini 2.5 Pro, which had the highest specificity (93.4%) but lower sensitivity (38.0%). Claude Sonnet 4 showed the lowest agreement ($\kappa=.25$) and PPV (30.9%). Compared to human annotation requiring 52 hr across all annotators, each LLM required less than 20 min. Conclusions: Reddit posts revealed rich discourse across CI topics. LLMs demonstrated fair agreement with human coders and can quickly aid in large-scale qualitative analysis. Although careful model selection and human expertise remain essential for accurate interpretation, LLM annotation shows potential for real-time monitoring of patient concerns to inform counseling, rehabilitation strategies, and iterative device design. © 2026, American Speech-Language-Hearing Association. All rights reserved.

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PEKIN: Prompt-Based External Knowledge Integration Network for Rumor Detection on Social Media

Item Type Conference Paper

Author Z. Hu

Author H. Liu

Author K. Li

Author Y. Wang

Author Z. Liu

Author X. Zhang

Editor Khanna S.

Editor Cao J.

Editor Bai Q.

Editor Xu G.

Abstract Pretrained language models (PLMs) and additional features have been used in rumor detection with excellent performance. However, on the one hand, some recent studies find one of its critical challenges is the significant gap of objective forms in pretraining and fine-tuning, which restricts taking full advantage of knowledge in PLMs. On the other hand, text contents are condensed and full of knowledge entities, but existing methods usually focus on the textual contents and social contexts, and ignore external knowledge of text entities. In this paper, to address these limitations, we propose a Prompt-based External Knowledge Integration Network (PEKIN) for rumor detection, which incorporates both prior knowledges of rumor detection tasks and external knowledge of text entities. For one thing, unlike the conventional “pretrain, finetune” paradigm, we propose a prompt-based method, which brings prior knowledge to help PLMs understand the rumor detection task and better stimulate the rich knowledge distributed in PLMs. For another, we identify entities mentioned in the text and then get these entities’ annotations from a knowledge base. After that, we use these annotations contexts as external knowledge to provide complementary information. Experiments on three datasets showed that PEKIN outperformed all compared models, significantly beating the old state-of-the-art on Weibo dataset. © 2022, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Perception study of urban green spaces in Singapore urban parks: Spatio-temporal evaluation and the relationship with land cover

Item Type Journal Article

Author W. Zhang

Author Y. Su

Abstract In the current era of increasing urbanization, urban green spaces play a crucial role in enhancing human well-being. However, quantifying public perceptions from text data at spatio-temporal scales remains challenging, and the relationship between urban green space perception and spatial-physical attributes requires further exploration. This study systematically examines public perceptions of urban green spaces within Singapore's urban parks from 2018 to 2022. Utilizing Twitter data, it applies large language models to conduct textual content analysis related to urban green space. The findings reveal a positive trend, with individuals expressing favorable perceptions and satisfaction towards urban green spaces in Singapore. Specifically, this study demonstrates that people's perceptions of urban green spaces are influenced by vegetation density. Higher vegetation density heightens people's awareness of spatial presence, while shrub and grassland may lead to neglect of urban green spaces as individuals focus more on themselves. Additionally, due to the spatial heterogeneity of the area, there is no clear correlation between all land covers and public satisfaction with urban green spaces in Singapore. The results also indicate a significant decrease in public perception in 2020, followed by a subsequent recovery. This fluctuation is attributed to the substantial impact of the COVID-19 pandemic, suggesting that external socio-political, economic, and public health events can impact public green space needs and spatial perceptions. In conclusion, this study contributes to the understanding of urban green spaces by effectively analyzing textual content extracted from social media data using large language models. The insights gained contribute valuable to the following discussions regarding the planning and design of urban green spaces and urban parks. © 2024 Elsevier GmbH

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Performance Analysis of Classical Machine Learning Algorithms and Transformer Models on Multilingual Sentiment Data

Item Type Conference Paper

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Author B. Sarkar

Author S. Roy

Author P. Routh

Author P. Chowdhury

Author D. Banerjee

Author S. Bal

Editor Hassanien A.E.

Editor Anand S.

Editor Jaiswal A.

Editor Kumar P.

Abstract The rapid use of user-generated information on social media platforms has made sentiment analysis crucial to understanding public opinion on a variety of different fields. This work bridges the gap in comparative research between transformer-based methods and traditional machine learning methods on multilingual sentiment data for low-resource Indian languages. It provides an

analysis of both types of models using common preprocessing techniques on a recently created multilingual sentiment dataset containing English, Hindi, Bengali, and Assamese. While IndicBERT and XLM-R transformer models perform very well for morphologically rich languages, scoring almost full marks for the Hindi, Bengali, and Assamese languages, the classical models Logistic Regression and SVM perform exceptionally well, especially for English. Along with model complexity and computing costs, the research also presents a detailed comparison of various methods with respect to practical deployment consequences. Experiments show high accuracy for the classical machine learning models that classify English sentiment. IndicBERT and MuRIL models, which are transformer-based models, achieve very high accuracy and F1-scores for Hindi, Bengali, and Assamese. It also defines that multilingual transformer models perform better for low-resource and morphologically rich Indian languages. Overall, transformer-based models offer better cross-lingual generalization, while classical models show effectiveness for the high-resource languages. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Bal; Department of Computer Science and Engineering, Techno India University, West Bengal, Kolkata, West Bengal, 700091, India; email: sauvikbal@gmail.com; Conference name: 9th International Conference on Intelligent Computing and Communication, ICICC 2026; Conference date: 6 February 2026 through 7 February 2026; Conference code: 355219

Performance and Reproducibility of Large Language Models in Named Entity Recognition: Considerations for the Use in Controlled Environments

Item Type Journal Article

Author J. Dietrich

Author A. Hollstein

Abstract Introduction: Recent artificial intelligence (AI) advances can generate human-like responses to a wide range of queries, making them a useful tool for healthcare applications. Therefore, the potential use of large language models (LLMs) in controlled environments regarding efficacy, reproducibility, and operability will be of paramount interest. Objective: We investigated if and how GPT 3.5 and GPT 4 models can be directly used as a part of a GxP validated system and compared the performance of externally hosted GPT 3.5 and GPT 4 against LLMs, which can be hosted internally. We explored zero-shot LLM performance for named entity recognition (NER) and relation extraction tasks, investigated which LLM has the best zero-shot performance to be used potentially for generating training data proposals, evaluated the LLM performance of seven entities for medical NER in zero-shot experiments, selected one model for further performance improvement (few-shot and fine-tuning: Zephyr-7b-beta), and investigated how smaller open-source LLMs perform in contrast to GPT models and to a small fine-tuned T5 Base. Methods: We performed reproducibility experiments to evaluate if LLMs can be used in controlled environments and utilized guided generation to use the same prompt across multiple models. Few-shot learning and quantized low rank adapter (QLoRA) fine-tuning were applied to further improve LLM performance. Results and Conclusion: We demonstrated that zero-shot GPT 4 performance is comparable with a fine-tuned T5, and Zephyr performed better than zero-shot GPT 3.5, but the recognition of product combinations such as product event combination was significantly better by using a fine-tuned T5. Although Open AI launched recently GPT versions to improve the generation of consistent output, both GPT variants failed to demonstrate reproducible results. The lack of reproducibility together with limitations of external hosted systems to keep validated systems in a state of control may affect the use of closed and proprietary models in regulated environments. However, due to the good NER performance, we recommend using GPT for creating annotation proposals for training data as a basis for fine-tuning. © The Author(s) 2024.

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Performing Sentiment Analysis Using Natural Language Models for Urban Policymaking: An analysis of Twitter Data in Brussels

Item Type Journal Article

Author F. Tori

Author S. Tori

Author I. Keseru

Author V. Ginis

Abstract Objectives: Mobility is a core challenge to transition towards sustainability. Cities are, therefore, rethinking their mobility to reduce negative externalities such as (greenhouse) gas emissions or congestion. When trying to implement sustainable urban mobility plans, there is often resistance from citizens. This can indicate a disconnect between the public and policymakers due to a lack of participation, coupled with the fact that current data-collection methods often used (such as travel surveys) are limited in scope. Advances in big data analysis and user-generated content provide opportunities to gain deeper insights into citizens' perceptions of mobility policy changes. This paper explores how sentiment analysis through deep learning can be used in transport planning. Materials and Methods: In this research, we analyse the sentiments of citizens towards recent changes in mobility policy in Brussels, Belgium, through Twitter data. We analyse 1998 tweets about changing mobility policy in Brussels between July 18th, 2019 (forming of the last Brussels regional government), and December 31st, 2022 (starting date of the analyses). For our analysis, we employ two pre-trained language models: XLM-T and GPT4. Results: Our results show that the sentiment with regard to the new mobility interventions is, as reflected by Twitter posts, not overwhelmingly negative. Furthermore, we find that the performance scores of XLM-T when no domain-specific fine-tuning has occurred (zero-shot evaluation) is fairly low (0.48). Once the model is trained on our domain-specific data, it reaches an accuracy of 0.67. When using GPT4, the model reaches an accuracy of 0.66. Additionally, GPT4 seems better suited at identifying mismatched tweets, i.e. tweets using vocabulary that has a different sentiment than the one the tweets expresses (e.g., sarcasm). This might indicate that large language models might be better suited to obtain implicit sentiments expressed in a text. Conclusions: From a machine learning perspective,

our experiments highlight the difficulty of recognising contextual sentiment (in this case, a sentiment towards changes in mobility policies), which may differ from the sentiment reflected in the vocabulary used. This is especially important if these two sentiments do not correspond, a problem both models struggled with. Additionally, we show that GPT4 can provide additional information when performing sentiment analysis by prompting it to attribute scores to texts. This paper opens new perspectives on understanding and addressing public sentiment in urban mobility policies. The advancements in language models, and the effective integration of user-generated content, can provide policymakers with a more comprehensive understanding of public sentiment, facilitating the implementation of certain policies. Clinical Relevance: None. © The Author(s) 2024.

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PERSEID - Perspectivist Irony Detection: A CALAMITA Challenge

Item Type Conference Paper

Author V. Basile

Author S. Casola

Author S. Frenda

Author S.M. Lo

Editor Dell'Orletta F.

Editor Lenci A.

Editor Montemagni S.

Editor Sprugnoli R.

Abstract Works in perspectivism and human label variation have emphasized the need to collect and leverage various voices and points of view in the whole Natural Language Processing pipeline. PERSEID places itself in this line of work. We consider the task of irony detection from short social media conversations in Italian collected from Twitter (X) and Reddit. To do so, we leverage data from MultiPICO, a recent multilingual dataset with disaggregated annotations and annotators' metadata, containing 1000 Post, Reply pairs with five annotations each on average. We aim to evaluate whether prompting LLMs with additional annotators' demographic information (namely gender only, age only, and the combination of the two) results in improved performance compared to a baseline in which only the input text is provided. The evaluation is zero-shot; and we evaluate the results on the disaggregated annotations using f1. © 2024 CEUR-WS. All rights reserved.

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Personalized Emotion Analysis Using In-Context Learning

Item Type Conference Paper

Author S. Kakegawa

Author K. Yamada

Author H. Masuda

Abstract Emotion analysis models are typically trained on corpora, whose labels are aggregated across annotators, implicitly reflecting a population-averaged scale. However, aggregation masks individual variability in emotional expressions, leading to mismatches between model predictions and user's subjectivity. To address this problem, we propose a parameter-free method for personalized emotion and sentiment analysis using in-context learning. Given few posts labeled by the user, the proposed method first elicits baseline predictions and rationales from a large language model (LLM). Then, it constructs corrective demonstrations by pairing each post with the LLM initial response and user's ground-truth label. Finally, triplets containing the initial response, reasoning, and ground truth are prepended to the prompt for classifying emotion and its intensity in unseen posts. Exposing the model to any discrepancies between its reasoning and user's labels promotes guided self-correction and alignment with the user's emotions. Experiments on WRIME-Ver2, a Japanese social media dataset, show that providing five user examples with a chain of thought reduces the mean absolute error by 22% for emotions and 11% for sentiment polarity compared with zero-shot inference. In addition, the proposed method surpasses a five-shot baseline and is comparable to or even outperforms averages across reader annotations. An analysis of prediction distributions reveals that the proposed method recalibrates the model usage of the emotion rating scale, improving discrimination between neutral, low-intensity, and high-intensity emotions. Because the method requires few labeled examples and no parameter updates, it offers a practical path to personalized emotion analysis for dialogue systems and social media applications. © 2026 IEEE.

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Notes:

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Pilot Testing Transformers-based Models for Multi-class Classification of Greek Text

Item Type	Conference Paper
Author	P.S. Giannaris
Author	N. Doukas
Author	J. Borges
Abstract	Text data are generated on a daily basis through various social media platforms. From a computational perspective, these text data can be analyzed using machine learning algorithms. The Bidirectional Encoder Representations from Transformers (BERT) is a model specifically tailored for natural language processing tasks, such as text classification, which requires understanding of word context. GreekBERT is a monolingual, pre-trained transformers-based language model developed for the analysis of Greek text. This paper investigates the fine-tuning of GreekBERT and of other transformers-based models, including few-shot and zero-shot learning techniques, for the multi-class classification of Greek tweets, with the objective of facilitating further informatics research. To fine-tune the transformers models, we perform experiments utilizing a training dataset of 2,000 tweets based on eleven categories, while evaluating on a distinct set of 120 tweets. The experimental results indicate issues such as mislabeled tweets, suboptimal evaluation metrics, with accuracy levels at or below 35%, and a significant class imbalance. This experimentation underscores the challenges associated with adapting pre-trained language models for specific downstream tasks and highlights the need for ongoing experimentation and optimization of the models. © 2024 IEEE.
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Extra	Journal Abbreviation: Proc. - Int. Conf. Dependable Syst., Serv. Technol.: Trustworthy AI, Internet Robot., Big Data for a Safe Secure World, DESSERT
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PinLID: a dataset for Pinglish language identiftcation based on code-mixing sentence on unstructured resources

Item Type Journal Article

Author A. Ghafouri

Author H. Naderi

Author M. Firouzmandi

Abstract Language identification is a major task in natural language processing. It serves as an initial and effective stage in critical tasks such as information extraction, sentiment analysis, and question answering. Most research on language identification has focused on monolingual contexts, performing poorly with texts containing code-mixing. Identifying the language in social media texts, such as those on Twitter, poses challenges due to high levels of code-mixing. Consequently, creating an accurate language identification tool for code-mixed texts is essential for intelligent systems that rely on natural language processing, such as advanced search engines and question-answering systems. Recently, significant research has been conducted in non-Persian languages in this field. However, no substantial efforts have been made to recognize languages in code-mixed Persian texts. In this paper, we introduce a dataset called PinLID, collected from tweets with Persian-English code-mixing, labeled at both the sentence and token levels using a supervised learning approach to language identification. We evaluated the dataset using various machine learning classification algorithms, including the classical SVM method, the multilingual BERT language model, XLM-RoBERTa, ParsBERT, AriaBERT, and PersianLLaMA: Persian Large Language Model. The testing yielded results as high as 99.59% F1 score at both the sentence and token levels in the test data. © The Author(s), under exclusive licence to Springer Nature B.V. 2024.

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PolCSBD: A contextual dataset of code-mixed Bengali and Banglish for political counter-speech detection

Item Type Journal Article

Author C.M. Mehedi

Author A.A. Emon

Author M.A. Mozumder

Author S. Akther

Author F. Tasnim

Author M.B. Uddin

Abstract The detection of counter-speech on social media is heavily dependent on the conversational context, particularly in low-resource and code-mixed languages. This article presents a dataset of 10,021 contextual comment pairs extracted from political discussions on Bangladeshi YouTube channels. The dataset captures native Bengali script, fully Romanized Bengali ("Banglish"), and hybrid code-mixed sentences. Each instance consists of a parent comment and a corresponding nested reply, annotated manually by native speakers to indicate the presence or absence of counter-speech. Replies that actively dispute, correct, or challenge the parent text with a logical counter-narrative are labeled as '1' (Counter-Speech), while replies indicating

agreement, off-topic remarks, or non-argumentative insults are labeled as '0' (Non-Counter Speech). The dataset underwent rigorous preprocessing, including the removal of URLs, user tags, and emojis, alongside the normalization of English characters. This dataset provides a foundational resource for training and evaluating natural language processing (NLP) models in understanding contextual political discourse in South Asian code-mixed environments. By preserving the direct link between a baseline statement and its response, the data allows researchers to move beyond isolated sentence classification and study actual dialogic structures. It specifically targets the acute shortage of resources for analyzing informal, phonetically typed regional dialects. Consequently, machine learning practitioners and social scientists can utilize this collection to benchmark text classifiers, test large language models (LLMs) on South Asian discourse, and explore algorithmic moderation strategies in highly polarized online spaces. © 2026 The Author(s).

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Computational linguistics, Speech communication, Codes (symbols), Classification (of information), Learning systems, Sentiment analysis, Language processing, Learning algorithms, Natural language processing, Natural languages, Social networking (online), Benchmarking, Data mining, Speech analysis, Speech detection, Social media minings, Social media mining, Code-mixing, Bengalis, Algorithmic languages, Banglish, Counter-speech detection

Notes:

EID: 2-s2.0-105044604521

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M.B. Uddin; Department of Computer Science and Engineering, Chittagong University of Engineering and Technology, Chattogram, 4349, Bangladesh; email: mohammedborhan15@gmail.com

Political-RAG: using generative AI to extract political information from media content

Item Type Journal Article

Author M. Arslan

Author S. Munawar

Author C. Cruz

Abstract In the digital era, media content is crucial for political analysis, providing valuable insights through news articles, social media posts, speeches, and reports. Natural Language Processing (NLP) has transformed Political Information Extraction (IE), automating tasks such as event extraction and sentiment analysis. Traditional NLP methods, while effective, are often task-specific and require specialized expertise. In contrast, Large Language Models (LLMs) powered by Generative Artificial Intelligence (GenAI) offer a more integrated solution. However, domain-specific challenges persist, which led to the development of the Retrieval-Augmented Generation (RAG) framework. RAG enhances LLMs by incorporating external data retrieval, addressing issues related to data availability. To demonstrate RAG's capabilities, we introduce the Political-RAG system, designed to extract political event information from media content, including Twitter data and news articles. Initially developed for event extraction, the Political-RAG system lays the foundation for developing various complex Political IE tasks. These include detecting hate speech, analyzing conflicts, assessing political bias, and evaluating social trends, sentiment, and opinions. © 2024 The Author(s). Published with license by Taylor & Francis Group, LLC.

Date 2025

DOI 10.1080/19331681.2024.2417263

URL <https://www.scopus.com/pages/publications/85207520168?origin=resultslist>

Archive Scopus

Language English

Volume 22

Publisher Routledge

Pages 479-494

Publication Journal of Information Technology and Politics

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Journal Abbr J. Inf. Technol. Polit.

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Natural Language Processing, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Information Extraction (IE), Political events extraction

Notes:

EID: 2-s2.0-85207520168

Export Date: 20 September 2026; Cited By: 8; Funding details: Agence Nationale de la Recherche, ANR; Correspondence Address: M. Arslan; University of the West of England, Bristol, United Kingdom; email: muhammad.arslan@u-bourgogne.fr

PolitiKweli: A Swahili-English Code-Switched Twitter Political Misinformation Classification Dataset

Item Type Conference Paper

Author C. Amol

Author L. Wanzare

Author J. Obuhuma

Editor Chakravarthi B.R.

Editor B B.

Editor García Cumbresas M.A.

Editor Jiménez Zafra S.M.

Editor Subramanian M.

Editor Shanmugavadivel K.

Editor Nakov P.

Abstract In the age of freedom of speech, users of the various social media platforms post millions of unverified messages, resulting in misinformation. Despite these platforms' set policies against misinformation, there is an alarming rise in misleading news dissemination. On political matters, misinformation online can result in defamation and in extreme cases, violence offline. Misinformation classification involves classifying text as fake or fact. Most of the existing studies address misinformation classification for posts in a single language only. Among most bilingual or multilingual social media users, code-switching such as Swahili-English, is common practice. This poses a threat to code-switching being used to spread misinformation. There is need for more research in low-resource languages such as Swahili, especially their use to spread misinformation. This study curated the PolitiKweli (dataset: <https://github.com/jayneamol/kweli>) dataset, a Swahili-English code-switched misinformation classification dataset, containing 6,345 Swahili-English code-switched texts, 22,954 English texts and 211 Swahili texts. The texts are labeled as fake, fact or neutral as compared to fact-checked Twitter dataset also created as part of this study. The paper discusses the dataset curation process including data collection, data processing and data annotation. It also highlights the challenges during the annotation. The study develops a benchmark classification model based on pretrained language model BERT that achieves an F-score of 0.62. The results of the experiment show promising results on the usefulness of the Swahili-English code-switched misinformation classification models. When applied, the classification model can be able to flag instances of misinformation on social media platforms. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

Date 2024

DOI 10.1007/978-3-031-58495-4_1

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Archive Scopus

Language English

Extra Journal Abbreviation: Commun. Comput. Info. Sci.

Volume 2046 CCIS

Publisher Springer Science and Business Media Deutschland GmbH

ISBN 18650929 (ISSN); 978-303158494-7 (ISBN)

Pages 3-17

Proceedings Title Commun. Comput. Info. Sci.
Conference Name Communications in Computer and Information Science
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Tags:

Offline, Codes (symbols), Classification (of information), Data acquisition, Social networking (online), Data handling, Social media platforms, low-resource language, Text processing, Low resource languages, Bilinguals, code-switching, Code-switching, Fake detection, Classification models, Classification datasets, Freedom of speech, misinformation classification, Misinformation classification, swahili, Swahilus

Notes:

EID: 2-s2.0-85192379165

Export Date: 20 September 2026; Cited By: 3; Correspondence Address: C. Amol; Maseno University, Maseno, Kenya; email: cynthia@maseno.ac.ke; Conference name: 2nd International Conference on Speech and Language Technologies for Low-Resource Languages, SPELLL 2023; Conference date: 6 December 2023 through 8 December 2023; Conference code: 311549

PolyGnosis: An AI Framework for Forecasting Public Opinion from Polymarket Dynamics

Item Type Conference Paper

Author D. Wang

Author T. Ma

Author H. Duan

Abstract This study investigates prediction markets, specifically the blockchain-based Polymarket, as a novel supervision signal for modeling public opinion dynamics. We develop a theoretical framework showing that, if Polymarket participants possess superior information, historical market signals must contain learnable structure predictive of future public sentiment. To instantiate this framework, we construct a dataset that pairs a rich set of Polymarket features with large language model (LLM)-derived, model-ready scores of online social media opinions. The empirical results show that AI models are able to extract robust predictive signals from market dynamics, clearly outperforming chance-level benchmarks. In conclusion, our study reveals prediction markets as a valuable, real-time supervision source for training AI systems to forecast public-opinion shifts. © 2026 IEEE.

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DOI 10.1109/INFOCOM59046.2026.11571417

URL <https://www.scopus.com/pages/publications/105044556688?origin=resultslist>

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Language English

Extra Journal Abbreviation: Proc IEEE INFOCOM
Publisher Institute of Electrical and Electronics Engineers Inc.
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Proceedings Title Proc IEEE INFOCOM
Conference Name Proceedings - IEEE INFOCOM
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Tags:

Language model, Commerce, Information systems, Block-chain, Blockchain, Public opinions, Social aspects, Sentiment analysis, Prediction models, Forecasting, Social networking (online), Theoretical framework, Public sentiments, Opinion dynamics, Polymarket, Prediction markets, Prediction Markets, Public opinion forecasting, Public Opinion Forecasting, Social forecasting, Social Forecasting

Notes:

EID: 2-s2.0-105044556688

Export Date: 20 September 2026; Cited By: 0; Funding details: Key Field Projects of Ordinary Universities in Guangdong Province, (2025ZDZX3050); Correspondence Address: H. Duan; Shenzhen MSU-BIT University, Artificial Intelligence Research Institute, Shenzhen, China; email: duanhaihan@smbu.edu.cn; Conference name: 2026 IEEE Conference on Computer Communications, INFOCOM 2026; Conference date: 18 May 2026 through 21 May 2026; Conference code: 225400; CODEN: PINFE

PolyHope-M at RANLP2025 Subtask-1 Binary Hope Speech Detection: Spanish Language Classification Approach with Comprehensive Learning using Transformer, Traditional ML, and DL

Item Type Conference Paper
Author Md. Julkar Naeen
Author S.K. Das
Author S.A. Khushbu
Author S.S. Ramit
Author A.P. Alo
Editor Angelova G.
Editor Kunilovskaya M.
Editor Escribe M.
Editor Mitkov R.
Abstract This paper presents our system for the RANLP 2025 shared task on multilingual binary sentiment classification for Task-2 Spanish datasets for domains including social media and customer reviews. We experimented with various models from traditional machine learning approaches—Naive Bayes and LightGBM—to deep learning architectures like LSTM. Among them, the transformer-based XLM-

RoBERTa model performed best with an F1 of 0.85, demonstrating its promise for multilingual sentiment work. Basic text preprocessing techniques were used for data quality assurance and improving model performance. Our comparison reflects the superiority of transformer-based models over the traditional methods in binary sentiment classification for multilingual and low-resource environments. This study enables the development of cross-lingual sentiment classification by establishing strong baselines and paying close attention to model performance in joint task settings. © 2025 Incoma Ltd. All rights reserved.

Date 2025

DOI 10.26615/978-954-452-098-4-090

URL <https://www.scopus.com/pages/publications/105034059946?origin=resultslist>

Archive Scopus

Language English

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Proceedings Title Int. Conf. Recent Adv. Nat. Lang. Proces., RANLP

Conference Name International Conference Recent Advances in Natural Language Processing, RANLP

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Computational linguistics, Deep learning, Classification (of information), Learning systems, Sentiment analysis, Social media, Speech detection, Sentiment classification, Customer review, Modeling performance, Barium compounds, Classifiers, Subtask, Machine learning approaches, Data quality, Spanish language, Classification approach, Comprehensive learning, Quality assurance

Notes:

EID: 2-s2.0-105034059946

Export Date: 20 September 2026; Cited By: 0; Conference name: 15th International Conference on Recent Advances in Natural Language Processing - Natural Language Processing in the Generative AI Era, RANLP 2025; Conference date: 8 September 2025 through 10 September 2025; Conference code: 220364

Portrayals of Chinese companies in American and British economic news tweets during China's macroeconomic transitions 2007–2023

Item Type Journal Article

Author M. Ye

Author E. Friginal

Abstract This study investigates how Chinese companies are portrayed in American and British economic news tweets, as well as their relationship to Chinese economic fluctuations. The analysis included a corpus of 55,394 tweets (934,155 words) from well-known media outlets between 2007 and 2023. It also incorporated China's quarterly Gross Domestic Product (GDP) and monthly Purchasing Managers' Index (PMI) to contextualise the tweets in terms of their actual economic performance. Using van Dijk's ingroup and outgroup ideologies, this project examined the ideological depiction through sentiments and emotions. RoBERTa-based transformer models were used to analyse sentiments and emotions, whereas Large Language Models (LLMs) were used for evaluative target annotation. Positive and negative sentiments were found to be significantly ($p < 0.01$) correlated with China's macroeconomic indices. The representation of Chinese companies trended between ingroup and outgroup portrayals. Positive sentiment diminished as the economy transitioned from expansion to contraction, while negative sentiment increased. American news tweets were most positive during economic balance and most negative during downturns, while British news tweets were most positive during stability or early recovery and most negative when a downturn was predicted. As the economy shifted from growth to decline, positive sentiment emphasised corporate external opportunities alongside corporate strengths, with both evaluative targets reinforcing ingroup representation. Negative sentiment shifted from corporate weaknesses to contextual threats, both strengthening outgroup representation. Moreover, certain emotions had a significant ($p < 0.0001$) influence on sentiment swings and ideological transitions. These findings emphasise how economic changes and journalistic factors influence the ideological representation of Chinese firms in economic news on social media. © The Author(s) 2024.

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DOI 10.1057/s41599-024-03958-7

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Archive Scopus

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: M. Ye; The Department of English and Communication, The Hong Kong Polytechnic University, Hong Kong; email: meng-mary.ye@polyu.edu.hk

Potentials of ChatGPT for Annotating Vaccine Related Tweets

Item Type Conference Paper

Author M.R. Biswas

Author F. Mohsen

Author Z. Shah

Author W. Zaghouni

Abstract This study evaluates ChatGPT's performance in annotating vaccine-related Arabic tweets by comparing its annotations with human annotations. A dataset of 2,100 tweets representing various factors contributing to vaccine hesitancy was examined. Two domain experts annotated the data, with a third resolving conflicts. ChatGPT was then employed to annotate the same dataset using specific prompts for each factor. The ChatGPT annotations were evaluated through zero-shot, one-shot, and few-shot learning tests, with an average accuracy of 82.14%, 83.85%, and 85.57%, respectively. Precision averaged around 86%, minimizing false positives. The average recall and F1-score ranged from 0.74 to 0.80 and 0.65 to 0.93, respectively. AUC for zero-shot, one-shot, and few-shot learning was 0.79, 0.80, and 0.83. In cases of ambiguity, both human annotators and ChatGPT faced challenges. These findings suggest that ChatGPT holds promise as a tool for annotating vaccine-related tweets. © 2023 IEEE.

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DOI 10.1109/SNAMS60348.2023.10375433

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Language English

Extra Journal Abbreviation: Proc. - Int. Conf. Soc. Networks Anal., Manag. Secur., SNAMS

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 979-835031890-6 (ISBN)

Proceedings Title Proc. - Int. Conf. Soc. Networks Anal., Manag. Secur., SNAMS

Conference Name Proceedings - 2023 10th International Conference on Social Networks Analysis, Management and Security, SNAMS 2023

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Language model, Large language model, Performance, ChatGPT, Twitter, large language model, Zero-shot learning, Data annotation, Human annotations, Vaccine hesitancy, Vaccines, vaccine hesitancy, Bioinformatics, Domain experts, Two domains

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EID: 2-s2.0-85183458277

Export Date: 20 September 2026; Cited By: 2; Funding details: Hamad Bin Khalifa University, Qatar Foundation; College of Science and Engineering, University of Minnesota; Conference name: 10th International Conference on Social Networks Analysis, Management and Security, SNAMS 2023; Conference date: 21 November 2023 through 24 November 2023; Conference code: 196145

Pre-Trained and Large Language Models for Sentiment Analysis of Spanish Social Media in the 2024 Mexican Elections

Item Type Conference Paper

Author J. Valverde-Rebaza

Author L. Cañete-Sifuentes

Author F. Castro-Zenteno

Author E. Jimenez-Garibaldy

Author J. Trejo

Author A.M. del Campo

Editor Martínez-Villaseñor L.

Editor Vázquez R.A.

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Editor Montes Rivera M.

Editor Zapotecas-Martínez S.

Editor Barrón-Estrada M.L.

Editor Mezura Montes E.

Editor Gomez Chavez A.

Abstract The growing influence of social media on electoral processes underscores the need for robust sentiment analysis methods capable of handling informal, noisy, and politically charged discourse. This challenge is particularly acute in Spanish, where resources and benchmarks remain limited compared to English. This study makes two main contributions. First, we introduce a novel human-annotated corpus of 2,000 posts from Facebook, Instagram, X, and YouTube during the 2024 Mexican presidential elections. The annotation, performed independently by multiple native speakers and consolidated into a reliable ground truth, ensures consistency across positive, negative, and neutral categories. Second, we propose an evaluation framework that benchmarks pre-trained transformer models, including BERT, BETO, RoBERTa, and pysentimiento, against large language models (LLMs) such as GPT-3.5, GPT-4o-mini, o1-mini, Llama-3, and Llama-3.1, under zero-shot and few-shot prompting. Results show that recent LLMs consistently surpass both pre-trained transformers and open-source alternatives, achieving stronger and more stable performance across platforms. To support future research, we release both the annotated dataset and the complete evaluation results. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2026

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Language English

Extra Journal Abbreviation: Lect. Notes Comput. Sci.

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Conference Name Lecture Notes in Computer Science

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Notes:

EID: 2-s2.0-105035839283

Export Date: 20 September 2026; Cited By: 0; Funding details: Department of Computer Science, Mexico City Campus; Instituto Tecnológico y de Estudios Superiores de Monterrey, ITESM; Correspondence Address: L. Cañete-Sifuentes; School of Engineering and Sciences, Tecnológico de Monterrey, Monterrey, Mexico; email: leonardo.c@tec.mx; Conference name: International Workshops on Advances in Computational Intelligence, MICA I 2025; Conference date: 3 November 2025 through 7 November 2025; Conference code: 349929

Predicting emotion intensity in Polish political texts: comparing supervised models and large language models in a low-resource language

Item Type Journal Article

Author H. Plisiecki

Author P. Koc

Author M. Flakus

Author A. Pokropek

Abstract This study explores the use of large language models (LLMs) to predict emotion intensity in Polish political texts, a low-resource language context. The research compares the performance of state-of-the-art LLMs against a supervised model trained on an annotated corpus of 10,000 social media texts, evaluated for the intensity of emotions by expert judges. The findings indicate that while the supervised model generally outperforms LLMs, offering higher accuracy and lower variance, LLMs present a viable alternative, especially given the high costs associated with data annotation. The study highlights the potential of LLMs in low-resource language settings and underscores the need for further research on emotion intensity prediction and its application across different languages and continuous features. The implications suggest a nuanced decision-making process to choose the right approach to emotion prediction for researchers and practitioners based on resource availability and the specific requirements of their tasks. © The Author(s), under exclusive licence to Springer Nature B.V. 2025.

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Tags:

Large language models (LLMs), Low-resource languages, Emotion annotation, Emotion intensity prediction, Polish political texts, Supervised models

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Export Date: 20 September 2026; Cited By: 0; Funding details: Narodowe Centrum Nauki, NCN, (SONATA BIS-10, UMO-020/38/E/HS6/00302); Narodowe Centrum Nauki, NCN; Correspondence Address: H. Plisiecki; Institute of Psychology, Polish Academy of Sciences, Warsaw, St. Stefana Jaracza 1, 00-378, Poland; email: hplisiecki@gmail.com

Predicting IoT Privacy Concerns From Tweets Through BERTweet-Based Fusion of Psycholinguistic Big Five and Risk Features: Toward Integration With IoT Security Frameworks

Item Type Journal Article

Author A. Alshaqhaa

Author R. Hassan

Author F. Qamar

Author A.H. Ibrahim

Abstract The rapid expansion of the Internet of Things (IoT) into everyday environments has heightened concerns about how personal data is perceived, shared, and protected. While much research on IoT privacy has concentrated on technical safeguards, individuals' attitudes toward privacy are equally shaped by psychological and linguistic factors. This study investigates how privacy concerns expressed in social media discourse can be predicted by integrating psycholinguistic patterns with personality and risk-related features. We introduce a deep learning framework, BERTweet, that fuses textual representations with auxiliary features derived from the Big Five personality traits and general risk propensity. Our dataset comprises 85,112 tweets (a consensus subset where GPT-4.1, Grok, and DeepSeek all agreed), drawn

from five publicly available corpora. Privacy annotations were generated through a multi-step process that began with zero-shot classification and was finalised via majority voting across the three large language models. The proposed fusion architecture dynamically balances contributions from linguistic, personality, and risk dimensions. Evaluation on a held-out test set of 8,512 tweets shows that the BERTweet-Large fusion model achieved 96.8% accuracy, with strong F1 performance for sensitive (low-privacy, high-disclosure) tweets (97.6%) and non-sensitive (high-privacy, low-disclosure) tweets (95.2%). We benchmark this model against BERT, RoBERTa, and DeBERTa backbones under identical training conditions to substantiate the choice of BERTweet for tweet-level privacy prediction. Unlike approaches that rely on explicit metadata or direct tagging, our method reveals the implicit ways privacy concerns emerge in everyday language use. The findings highlight the value of combining psycholinguistic, Big Five, and risk features in deep learning systems, offering a scalable approach to building privacy-aware models that address the complexities of IoT and social media ecosystems. The study therefore contributes a psychologically grounded, empirically validated framework that links tweet-level language, inferred traits, and risk propensity to IoT privacy concern, and provides a transparent benchmark for future privacy-aware language models. © 2013 IEEE.

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Priority Assessment and Decision Support Research for Old Residential Community Renovation: A Case Study of Hangzhou

Item Type Journal Article

Author T. Ma

Author S. Ma

Author H. Long

Author Y. Wang

Author W. Li

Author X. Wang

Abstract In the context of China's urban transition from large-scale expansion to intensive regeneration, the establishment of a scientific priority assessment mechanism for the renovation of old residential communities has become a critical yet challenging task in urban governance. This study aimed to address the prevailing challenges in contemporary decision-making processes, including the complexity of quantifying public needs and the absence of a comprehensive city-wide perspective. To this end, a novel data-driven priority evaluation framework was proposed using an empirical case study of Hangzhou. Methodologically, this study introduces key innovations. First, a Graph Neural Network (GNN) model incorporating spatial semantics was developed in order to identify the construction era of residential communities, achieving a high-precision recognition rate of 97.6 and effectively overcoming the limitations of traditional image classification. Second, a Large Language Model (LLM) is employed to perform a fine-grained sentiment analysis of geotagged social media texts, and a geographically weighted approach is applied to spatially quantify public sentiment at the neighborhood scale. Third, moving beyond conventional facility-oriented evaluation, an integrated three-dimensional indicator system was constructed, covering problem urgency, implementation feasibility, and potential benefits. Priority assessment was conducted using an XGBoost model, and its robustness was strengthened through Monte Carlo simulation and full-gradient sensitivity analysis, ensuring reliable and stable ranking outcomes across all 2188 old residential communities under study. The main findings are as follows: (1) a distinct scale-dependent spatial heterogeneity in renovation priorities. While high-priority communities are clustered in central urban districts, the average priority score is notably higher in certain peripheral counties—an "inversion" phenomenon that reflects the underlying tension between spatial equity and renewal efficiency, shaped by historical investment gaps, diminishing marginal returns, and regional economic disparities. (2) A dual-scale spatial clustering analysis conducted at both the community and administrative district levels identified four renovation archetypes and four corresponding policy zones, offering a refined understanding of priority differentiation and enabling a targeted governance approach characterized by

categorized renovation and zonal coordination. (3) Building on these spatial patterns, this study has proposed a synergistic governance paradigm that integrates differentiated renovation strategies with macro-level resource allocation, thereby enhancing the precision and effectiveness of policy implementation. By establishing a "data-driven, model-supported, policy-adaptive" technical pathway, this research contributes to shifting urban renewal practices from experience-based judgment to evidence-based decision-making. The framework not only provides a scientific basis for optimizing renovation sequencing and resource distribution but also translates complex spatial diagnostics into operable policy tools, supporting more equitable, efficient, and sustainable urban regeneration in high-density cities. © 2026, Editorial Committee of Tropical Geography. All rights reserved.

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Archive Scopus

Short Title 老旧小区改造优先级评估机制与决策支持研究 ——以杭州市为例

Language English

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Journal Abbr Tropic. Geogr.

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Notes:

EID: 2-s2.0-105047093845

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Ma; College of Civil Engineering and Architecture, Zhejiang University, Hangzhou, 310058, China; email: mashuang@zju.edu.cn

Proceedings - 17th IEEE International Conference on Semantic Computing, ICSC 2023

Item Type Journal Article

Abstract The proceedings contain 61 papers. The topics discussed include: retrieving users' opinions on social media with multimodal aspect-based sentiment analysis; coherence based document clustering; conditional cross correlation network for video question answering; does noise really matter? investigation into the influence of noisy labels on BERT-based question answering system; robust zero-shot intent detection via contrastive transfer learning; utilizing priming to identify optimal class ordering to alleviate catastrophic forgetting; an approach for fusing two training-datasets with partially overlapping classes; semantic validation and interpretability of object detection; modeling recommendation and project deployment automation for semantic knowledge graphs; ontology mediated document retrieval for exploratory big data analytics; learning from the evidence: impact evaluations, ontology and policy; and improving zero-shot inference with unsupervised key-sentences extraction.

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Language English

Publisher Institute of Electrical and Electronics Engineers Inc.

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Proceedings - 28th Pan-Hellenic Conference on Progress in Computing and Informatics with International Participation, PCI 2024

Item Type Journal Article

Editor Karanikolas N.N.

Editor Varlamis I.

Editor Sgarbas K.

Abstract The proceedings contain 74 papers. The topics discussed include: analyzing the freedom of information requests using topic modeling; towards user-driven open government data ecosystems; analysis of the performance of transformers and ensemble schemes for Twitter US airlines sentiment analysis; architecturing the Greek data pile for LLM development; evaluating large language models in interaction with open government data; the role of ChatGPT in modern human

resource management: a decision support perspective for human resource allocation problems; a convolutionally-informed XLM-RoBERTa for classifying short text inquires in Greek school network; summarizing university students' qualitative feedback on online exams: a microservice-oriented approach; and an intelligent chatbot in Greek using machine learning technology.

Date 2025

URL <https://www.scopus.com/pages/publications/105010757241?origin=resultslist>

Archive Scopus

Language English

Publisher Association for Computing Machinery, Inc

Publication Proceedings - 28th Pan-Hellenic Conference on Progress in Computing and Informatics with International Participation, PCI 2024

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Journal Abbr Proc. - Pan-Hellenic Conf. Prog. Comput. Informatics Int. Particip., PCI

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Proceedings - 2024 International Conference on Computational Linguistics and Natural Language Processing, CLNLP 2024

Item Type Journal Article

Abstract The proceedings contain 21 papers. The topics discussed include: PSSOP: a prompt-style approach for few-shot text classification; research on the comprehensive evaluation method of topic model; text mining domestic extremism topics on multiple social media platforms; multimodal named entity recognition model based on cross-modal feature enhancement mechanism; public opinion analysis in short videos of emergencies using pre-trained language models; unsupervised Chinese word segmentation based on minimum information content; research on the comprehensive evaluation method of topic model; correcting factual errors in LLMs via inference paths based on knowledge graph; and incorporating feature interaction into relation contrastive learning for zero-shot relation extraction.

Date 2024

URL <https://www.scopus.com/pages/publications/85211341668?origin=resultslist>

Archive Scopus

Language English

Publisher Institute of Electrical and Electronics Engineers Inc.

Publication Proceedings - 2024 International Conference on Computational Linguistics and Natural Language Processing, CLNLP 2024

ISSN 979-835039209-8 (ISBN)

Journal Abbr Proc. - Int. Conf. Comput. Linguist. Nat. Lang. Process., CLNLP

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Export Date: 20 September 2026; Cited By: 0; Conference name: 2024 International Conference on Computational Linguistics and Natural Language Processing, CLNLP 2024; Conference date: 19 July 2024 through 21 July 2024; Conference code: 204126

Proceedings of the 15th International Conference on Recent Advances in Natural Language Processing - Natural Language Processing in the Generative AI Era, RANLP 2025

Item Type Journal Article

Editor Angelova G.

Editor Kunilovskaya M.

Editor Escribe M.

Editor Mitkov R.

Abstract The proceedings contain 166 papers. The topics discussed include: harnessing open-source LLMs for tender named entity recognition; on the limitations of large language models (LLMs): false attribution; candidate profile summarization - a RAG approach with synthetic data generation for tech jobs; PersianSciQA: a new dataset for bridging the language gap in scientific question answering; multilingual pre-training meets supervised neural machine translation: a reproducible evaluation on English–French and Finnish translation; advancing clinical translation in Nepali through fine-tuned multilingual models; advancing active learning with ensemble strategies; evaluating large language models on sentiment analysis in Arabic dialects; and from posts to predictions: a user-aware framework for faithful and transparent detection of mental health risks on social media.

Date 2025

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Archive Scopus

Language English

Publisher Incoma Ltd

Publication International Conference Recent Advances in Natural Language Processing, RANLP

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Journal Abbr Int. Conf. Recent Adv. Nat. Lang. Proces., RANLP

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Proceedings of the 2020 4th International Symposium on Computer Science and Intelligent Control, ISCSIC 2020

Item Type Journal Article

Abstract The proceedings contain 42 papers. The topics discussed include: towards real-time anomaly detection and calibration for large parking lot in urban complex; infrared pedestrian detection based on GAN data augmentation; pipeline voter based on moving accumulator sign filter; VAE-GAN based zero-shot outlier detection; objective assessment for classification of muscle spasticity level; automatic text summarization on social media; dynamic pricing strategy for logistics revenue management using data mining technology; intrinsic evaluation of word embeddings for Turkish; and semantic similarity of XML documents based on structural and content analysis.

Date 2020

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Archive Scopus

Language English

Publisher Association for Computing Machinery

Publication ACM International Conference Proceeding Series

ISSN 978-145038889-4 (ISBN)

Journal Abbr ACM Int. Conf. Proc. Ser.

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Export Date: 20 September 2026; Cited By: 0; Conference name: 4th International Symposium on Computer Science and Intelligent Control, ISCSIC 2020; Conference date: 17 November 2020 through 19 November 2020; Conference code: 167082

Profile-Level Positivity Assessment Using Multimodal Feature Aggregation From Social Media Content

Item Type Journal Article

Author M. Gorraiz-Bengoechea

Author A. Gonzalez-Santocildes

Author I. Pastor

Author A. Estévez

Author G. Aonso-Diego

Abstract Accurately evaluating positivity in social media profiles has important implications for advertising, mental health monitoring, and responsible AI applications. However, the multimodal nature of Instagram, where images, captions, hashtags, and emojis co-occur, challenges conventional unimodal sentiment analysis tools. We present a multimodal system that integrates computer vision and natural language processing to assess Instagram profiles. The architecture comprises two blocks: (i) feature extraction via facial-emotion analysis, object detection, chromatic cues, and linguistic processing; and (ii) profile-level inference, which fuses aggregated features to produce an overall positivity score together with semantically meaningful profile tags. Experiments on public Instagram profiles (images, captions, and comments), including an expert-annotated subset of comments, compare unimodal baselines with transformer-based embeddings and OpenAI large language models. The multimodal pipeline improves over unimodal variants, and the OpenAI approach achieves higher concordance with expert judgements than RoBERTa/Bi-LSTM baselines. Beyond technical contributions, we discuss privacy, transparency, and misuse risks associated with affective computing on social media. The proposed framework contributes to the advancement of multimodal sentiment analysis in social media and highlights its applicability to areas such as marketing analytics, well-being monitoring, and ethically aligned AI systems. © 2013 IEEE.

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Profiling Cryptocurrency Influencers with Few-Shot Learning Using Data Augmentation and ELECTRA

Item Type Conference Paper

Author M. Siino

Author M. Tesconi

Author I. Tinnirello

Editor Aliannejadi M.

Editor Faggioli G.

Editor Ferro N.

Editor Vlachos M.

Abstract With this work we propose an application of the ELECTRA Transformer, fine-tuned on two augmented version of the same training dataset. Our team developed the novel framework for taking part at the Profiling Cryptocurrency Influencers with Few-shot Learning task hosted at PAN@CLEF2023. Our proposed strategy consists of an early data augmentation stage followed by a fine-tuning of ELECTRA. At the first stage we augment the original training dataset provided by the organizers using backtranslation. Using this augmented version of the training dataset, we perform a fine tuning of ELECTRA. Finally, using the fine-tuned version of ELECTRA, we inference the labels of the samples provided in the test set. To develop and test our model we used a two-ways validation on the training set. Firstly, we evaluate all the metrics on the augmented training set, and then we evaluate on the original training set. The metrics we considered span from accuracy to Macro F1, to Micro F1, to Recall and Precision. According to the official evaluator, our best submission reached a Macro F1 value equal to 0.3762. © 2023 Copyright for this paper by its authors.

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URL <https://www.scopus.com/pages/publications/85175610529?origin=resultslist>

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Language English

Extra Journal Abbreviation: CEUR Workshop Proc.

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Proceedings Title CEUR Workshop Proc.

Conference Name CEUR Workshop Proceedings

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Classification (of information), Fine tuning, Twitter, Electronic money, Text classification, Text processing, few-shot learning, Few-shot learning, text classification, data augmentation, Data augmentation, Training sets, Training dataset, author profiling, Author profiling, Cryptocurrency influencer, cryptocurrency influencers, electra, Electra

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Export Date: 20 September 2026; Cited By: 20; Conference name: 24th Working Notes of the Conference and Labs of the Evaluation Forum, CLEF-WN 2023; Conference date: 18 September 2023 through 21 September 2023; Conference code: 193170

Prompt-based fine-tuning with multilingual transformers for language-independent sentiment analysis

Item Type Journal Article

Author F. Ullah

Author S. Faizullah

Author I.U. Khan

Author T. Alghamdi

Author T.A. Syed

Author A.B. Alkhodre

Author M.S. Ayub

Author A. Karim

Abstract In the era of global digital communication, understanding user sentiment across multiple languages is a critical challenge with wide-ranging applications in opinion mining, customer feedback analysis, and social media monitoring. This study advances the field of language-independent sentiment analysis by leveraging prompt-based fine-tuning with state-of-the-art transformer models. The performance of classical machine learning approaches, hybrid deep learning architectures, and multilingual transformer models is evaluated across eight typologically diverse languages: Arabic, English, French, German, Hindi, Italian, Portuguese, and Spanish. Baseline models are established using traditional machine learning approaches such as Support Vector Machines (SVM) and Logistic Regression, with feature extraction methods like TF-IDF. A hybrid deep learning model is introduced, combining Long Short-Term Memory (LSTM) and Convolutional Neural Networks (CNNs) to capture local and sequential text patterns. Building on these, pre-trained multilingual transformer models, specifically BERT-base-multilingual and XLM-RoBERTa, are fine-tuned for language-independent sentiment classification tasks. The key contribution lies in the implementation of prompt-based fine-tuning strategies for language independent sentiment analysis. Using (1) prefix prompts and (2) cloze-style prompts, a unified framework is established that employs templates designed in one language and evaluates their performance on data from the remaining (n-1) languages. Experimental results demonstrate that transformer models, particularly

XLM-RoBERTa with prompt-based fine-tuning outperform both classical and deep learning methods. With only 32 training examples per class, prefix prompts produce results comparable to standard fine-tuning, which typically uses 70-80% of the data for training. This highlights the potential of prompt-based learning for scalable, multilingual sentiment analysis in diverse language settings. © The Author(s) 2025.

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PromptMix: A Class Boundary Augmentation Method for Large Language Model Distillation

Item Type Conference Paper

Author G. Sahu

Author O. Vechtomova

Author D. Bahdanau

Author I.H. Laradji

Editor Bouamor H.

Editor Pino J.

Editor Bali K.

Abstract Data augmentation is a widely used technique to address the problem of text classification when there is a limited amount of training data. Recent work often tackles this problem using large language models (LLMs) like GPT3 that can generate new examples given already available ones. In this work, we propose a method to generate more helpful augmented data by utilizing the LLM's abilities to follow instructions and perform few-shot classifications. Our specific PromptMix method consists of two steps: 1) generate challenging text augmentations near class boundaries; however, generating borderline examples increases the risk of false positives in the dataset, so we 2) relabel the text augmentations using a prompting-based LLM classifier to enhance the correctness of labels in the generated data. We evaluate the proposed method in challenging 2-shot and zero-shot settings on four text classification datasets: Banking77, TREC6, Subjectivity (SUBJ), and Twitter Complaints. Our experiments show that generating and, crucially, relabeling borderline examples facilitates the transfer of knowledge of a massive LLM like GPT3.5-turbo into smaller and cheaper classifiers like DistilBERTbase and BERTbase. Furthermore, 2-shot PromptMix outperforms multiple 5-shot data augmentation methods on the four datasets. Our code is available at <https://github.com/ServiceNow/PromptMix-EMNLP-2023>. © 2023 Association for Computational Linguistics.

Date 2023

DOI 10.18653/v1/2023.emnlp-main.323

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Computational linguistics, Language model, Classification (of information), Distillation, Training data, Text classification, Text processing, Shot classification, Data augmentation, Zero-shot learning, Knowledge management, Classification datasets, Augmentation methods, Class boundary, False positive, Relabelling

Notes:

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Prosoul: A framework to identify propaganda from online urdu content

Item Type Journal Article

Author S. Kausar

Author B. Tahir

Author M.A. Mehmood

Abstract Today, the rapid dissemination of information on digital platforms has seen the emergence of information pollution such as misinformation, disinformation, fake news, and different types of propaganda. Information pollution has become a serious threat to the online digital world and has posed several challenges to social media platforms and governments around the world. In this article, we propose Propaganda Spotting in Online Urdu Language (ProSOUL) - a framework to identify content and sources of propaganda spread in the Urdu language. First, we develop a labelled dataset of 11,574 Urdu news to train the machine learning classifiers. Next, we develop the Linguistic Inquiry and Word Count (LIWC) dictionary to extract psycholinguistic features of Urdu text. We evaluate the performance of different classifiers by varying ngram, News Landscape (NELA), Word2Vec, and Bidirectional Encoder Representations from Transformers (BERT) features. Our results show that the combination of NELA, word n-gram, and character n-gram features outperform with 0.91 accuracy for Urdu text classification. In addition, Word2Vec embedding outperforms BERT features in classification of the Urdu text with 0.87 accuracy. Moreover, we develop and classify large scale Urdu content repositories to identify web sources spreading propaganda. Our results show that ProSOUL framework performs best for propaganda detection in the online Urdu news content compared to the general web content. To the best of our knowledge, this is the first study on the detection of propaganda content in the Urdu language. © 2020 Institute of Electrical and Electronics Engineers Inc.. All rights reserved.

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Notes:

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Export Date: 20 September 2026; Cited By: 39; Correspondence Address: M.A. MEHMOOD; Al-Khawarizmi Institute of Computer Science, University of Engineering and Technology, Lahore, 54890, Pakistan; email: amir.mehmood@kics.edu.pk

Protecting Vulnerable Voices: Synthetic Dataset Generation for Self-disclosure Detection

Item Type Conference Paper**Author** S. Jangra**Author** S. De**Author** N. Sastry**Author** S. Fadaei**Editor** An A.**Editor** Cuzzocrea A.**Editor** Hu H.

Abstract Social platforms such as Reddit have a network of communities of shared interests, with a prevalence of posts and comments from which one can infer users' Personal Information Identifiers (PIIs). While such self-disclosures can lead to rewarding social interactions, they pose privacy risks and the threat of online harms. Research into the identification and retrieval of such risky self-disclosures of PIIs is hampered by the lack of open-source labeled datasets. Important hindrances to sharing high-quality labelled data include high annotation costs and privacy risks associated with the release of datasets containing self-disclosive text, especially when users include vulnerable populations. To foster reproducible research into PII-revealing text detection, we develop a novel methodology to create synthetic equivalents of PII-revealing data that can be safely shared. Our contributions include creating a taxonomy of 19 PII-revealing categories for vulnerable populations and the creation and release of a synthetic PII-labeled multi-text span dataset generated from 3 text generation Large Language Models (LLMs), Llama2-7B, Llama3-8B, and zephyr-7b-beta, with sequential instruction prompting to resemble the original Reddit posts. The utility of our methodology to generate this synthetic dataset is evaluated with three metrics: First, we require reproducibility equivalence, i.e., results from training a model on the synthetic data should be comparable to those obtained by training the same models on the original posts. Second, we require that the synthetic data be unlinkable to the original users, through common mechanisms such as Google Search. Third, we wish to ensure that the synthetic data be indistinguishable from the original, i.e., trained humans should not be able to tell them apart. We release our dataset and code at <https://netsys.surrey.ac.uk/datasets/synthetic-self-disclosure/> to foster reproducible research into PII privacy risks in online social media. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Language model, Large language model, Data privacy, Social networking (online), Large datasets, Population statistics, Privacy risks, Synthetic data, Search engines, Large Language Models, Labeled data, Personal information, Personal information identifier, Personal Information Identifiers, Privacy leak, Privacy Leaks, Self-disclosure, Synthetic datasets, Vulnerable population, Vulnerable Populations

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EID: 2-s2.0-105029841702

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Psychological reactance to vaccine mandates on Twitter: a study of sentiments in the United States

Item Type Journal Article

Author P.-H. Hsieh

Abstract This study examines the relationship between vaccine mandates and public sentiment toward vaccines and health officials on Twitter. I analyzed 6.6 million vaccine-related tweets from July 2021 to February 2022 in the United States. Leveraging a large language model, BERT, I identified tweets discussing vaccine mandates even when lacking explicit keywords. Compared to non-mandate tweets, those mentioning mandates exhibit greater negativity, anger, and freedom-related language. Furthermore, increased state-level discussion of mandates correlates with rising levels of negativity and anger toward both vaccines and public health officials. Finally, greater disparity in vaccination progress across counties within a state is associated with increased anger in tweets directed toward both. © The Author(s) 2025.

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Public Health Discussions on Social Media: Evaluating Automated Sentiment Analysis Methods

Item Type Journal Article

Author L.M. Gandy

Author L.V. Ivanitskaya

Author L.L. Bacon

Author R. Bizri-Baryak

Abstract Background: Sentiment analysis is one of the most widely used methods for mining and examining text. Social media researchers need guidance on choosing between manual and automated sentiment analysis methods. Objective: Popular sentiment

analysis tools based on natural language processing (NLP; VADER [Valence Aware Dictionary for Sentiment Reasoning], TEXT2DATA [T2D], and Linguistic Inquiry and Word Count [LIWC-22]), and a large language model (ChatGPT 4.0) were compared with manually coded sentiment scores, as applied to the analysis of YouTube comments on videos discussing the opioid epidemic. Sentiment analysis methods were also examined regarding ease of programming, monetary cost, and other practical considerations. Methods: Evaluation methods included descriptive statistics, receiver operating characteristic (ROC) curve analysis, confusion matrices, Cohen κ , accuracy, specificity, precision, sensitivity (recall), F1-score harmonic mean, and the Matthews correlation coefficient. An inductive, iterative approach to content analysis of the data was used to obtain manual sentiment codes. Results: A subset of comments were analyzed by a second coder, producing good agreement between the 2 coders' judgments ($\kappa=0.734$). YouTube social media about the opioid crisis had many more negative comments (4286/4871, 88%) than positive comments (79/662, 12%), making it possible to evaluate the performance of sentiment analysis models in an unbalanced dataset. The tone summary measure from LIWC-22 performed better than other tools for estimating the prevalence of negative versus positive sentiment. According to the ROC curve analysis, VADER was best at classifying manually coded negative comments. A comparison of Cohen κ values indicated that NLP tools (VADER, followed by LIWC's tone and T2D) showed only fair agreement with manual coding. In contrast, ChatGPT 4.0 had poor agreement and failed to generate binary sentiment scores in 2 out of 3 attempts. Variations in accuracy, specificity, precision, sensitivity, F1-score, and MCC did not reveal a single superior model. F1-score harmonic means were 0.34-0.38 (SD 0.02) for NLP tools and very low (0.13) for ChatGPT 4.0. None of the MCCs reached a strong correlation level. Conclusions: Researchers studying negative emotions, public worries, or dissatisfaction with social media face unique challenges in selecting models suitable for unbalanced datasets. We recommend VADER, the only cost-free tool we evaluated, due to its excellent discrimination, which can be further improved when the comments are at least 100 characters long. If estimating the prevalence of negative comments in an unbalanced dataset is important, we recommend the tone summary measure from LIWC-22. Researchers using T2D must know that it may only score some data and, compared with other methods, be more time-consuming and cost-prohibitive. A general-purpose large language model, ChatGPT 4.0, has yet to surpass the performance of NLP models, at least for unbalanced datasets with highly prevalent (7:1) negative comments. ©Lisa M Gandy, Lana V Ivanitskaya, Leeza L Bacon, Rodina Bizri-Baryak.

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Notes:

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Export Date: 20 September 2026; Cited By: 20; Correspondence Address: L.M. Gandy; Department of Computer Science, College of Sciences and Liberal Arts, Kettering University, Flint, 1700 University Ave 2300 AB, 48504, United States; email: lgandy@kettering.edu

Public Opinion Classification on Government Policy Using Social Media: An Exploration of ChatGPT's Capabilities and Limitations

Item Type Conference Paper

Author T. Babad-Falk

Author S.A. Chun

Abstract Gauging the public's sentiments and opinions toward policies is a critical task for policy makers. Social media posts offer a wealth of information for such a task but also pose unique challenges for achieving reasonable accuracy and transparency. While a well-trained machine learning model can offer accurate classifications of public perspectives, it cannot offer its reasoning for that classification or any further abstraction, which makes it difficult to use such solutions in practice. ChatGPT offers a possible solution; if the LLM has the requisite accuracy, the ability to explain the reasoning behind its decisions is built in. In this poster, we demonstrate ChatGPT's potential for such a task by documenting its accuracy and reasoning on opinion classification for posts regarding government policy in zero-shot and few-shot scenarios and compare the results to those of well-respected Natural Language Inference (NLI) models and ground truth human labels. Copyright © 2025 by the authors.

Date 2025

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Language English

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Tags:

Public policy, Public opinions, Social aspects, Classification (of information), Decision making, Learning systems, Sentiment analysis, Machine learning, Policy makers, Natural languages, Critical tasks, Social networking (online), Social media, Machine learning models, Language inference, Inference models, Reasonable accuracy, Wealth of information

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Public Opinion Mining Using Large Language Models on COVID-19 Related Tweets

Item Type Conference Paper

Author V. Tran

Author T. Matsui

Editor Binh H.T.T.

Editor Hoang V.T.

Editor Nguyen L.M.

Editor Le S.V.

Editor Vu T.D.

Editor Pham D.T.

Abstract Social media platforms have emerged as a significant source of public opinion, offering a massive user-generated data in which user-opinions are valuable if obtainable. Large language models (LLMs) have been in the spotlight recently with suggestions on the emergent abilities to solve tasks that are not explicitly trained for. Thus, this study explores the potential of utilizing LLMs for opinion mining on social media data by asking LLMs difficult questions, instead of simply asking whether the text's sentiment polarity is either positive, negative, or neutral. This study compares the LLM response statistics and the corresponding public surveys related to COVID-19, including the intention to take vaccination and the stress check. The results indicate that it is promising, but also challenging, to utilize LLMs for the tasks. © 2023 IEEE.

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Conference Name Proceedings - International Conference on Knowledge and Systems Engineering, KSE
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Tags:

Computational linguistics, Language model, Large language model, Public opinions, Social aspects, Sentiment analysis, Large Language Model, Social networking (online), Social media, Data mining, Twitter, Sentiment Analysis, Social Media, Social media platforms, COVID-19, Opinion mining, User-generated, Opinion Mining, Social media datum

Notes:

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Export Date: 20 September 2026; Cited By: 2; Funding details: Research Organization of Information and Systems, ROIS; Japan Society for the Promotion of Science, JSPS, (JP23K16954); Ministry of Education, Culture, Sports, Science and Technology, MEXT; Conference name: 15th International Conference on Knowledge and Systems Engineering, KSE 2023; Conference date: 18 October 2023 through 20 October 2023; Conference code: 194303

Public Versus Academic Discourse on ChatGPT in Health Care: Mixed Methods Study

Item Type Journal Article

Author P. Baxter

Author M.-H. Li

Author J. Wei

Author N. Koizumi

Abstract Background: The rapid emergence of artificial intelligence–based large language models (LLMs) in 2022 has initiated extensive discussions within the academic community. While proponents highlight LLMs’ potential to improve writing and analytical tasks, critics caution against the ethical and cultural implications of widespread reliance on these models. Existing literature has explored various aspects of LLMs, including their integration, performance, and utility, yet there is a gap in understanding the nature of these discussions and how public perception contrasts with expert opinion in the field of public health. Objective: This study sought to explore how the general public’s views and sentiments regarding LLMs, using OpenAI’s ChatGPT as an example, differ from those of academic researchers and experts in the field, with the goal of gaining a more comprehensive understanding of the future role of LLMs in health care. Methods: We used a hybrid sentiment analysis approach, integrating the Syuzhet package in R (R Core Team) with GPT-3.5, achieving an 84% accuracy rate in sentiment classification. Also, structural topic modeling was applied to identify and analyze 8 key discussion topics, capturing both optimistic and critical perspectives on LLMs. Results: Findings revealed a

predominantly positive sentiment toward LLM integration in health care, particularly in areas such as patient care and clinical decision-making. However, concerns were raised regarding their suitability for mental health support and patient communication, highlighting potential limitations and ethical challenges.

Conclusions: This study underscores the transformative potential of LLMs in public health while emphasizing the need to address ethical and practical concerns. By comparing public discourse with academic perspectives, our findings contribute to the ongoing scholarly debate on the opportunities and risks associated with LLM adoption in health care. © Patrick Baxter, Meng-Hao Li, Jiaxin Wei, Naoru Koizumi.

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Tags:

large language models, social media, natural language processing, sentiment analysis, discourse, ethics, medical, health knowledge, attitudes, practice, structural topic modeling

Notes:

EID: 2-s2.0-105010884628

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M.-H. Li; Schar School of Policy and Government, George Mason University, Arlington, 3351 Fairfax Dr, 22201, United States; email: mlil1@gmu.edu

PyFin-sentiment: Towards a machine-learning-based model for deriving sentiment from financial tweets

Item Type Journal Article

Author M. Wilksch

Author O. Abramova

Abstract Responding to the poor performance of generic automated sentiment analysis solutions on domain-specific texts, we collect a dataset of 10,000 tweets discussing the topics of finance and investing. We manually assign each tweet its market sentiment, i.e., the investor's anticipation of a stock's future return. Using this data, we show that all existing sentiment models trained on adjacent domains struggle with accurate market sentiment analysis due to the task's specialized vocabulary.

Consequently, we design, train, and deploy our own sentiment model. It outperforms all previous models (VADER, NTUSD-Fin, FinBERT, TwitterRoBERTa) when evaluated on Twitter posts. On posts from a different platform, our model performs on par with BERT-based large language models. We achieve this result at a fraction of the training and inference costs due to the model's simple design. We publish the artifact as a python library to facilitate its use by future researchers and practitioners.

© 2023 The Author(s)

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Issue 1

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Notes:

EID: 2-s2.0-85150288650

Export Date: 20 September 2026; Cited By: 23; Correspondence Address: M. Wilksch; University of Potsdam, Potsdam, Karl-Marx-Straße 67, 14482, Germany; email: wilksch@uni-potsdam.de

QBERT-LSTM: Quantum Intelligence-Based Mental Health Sentiment Analysis Using Web Scraping

Item Type Journal Article

Author N.S. Shirin

Author M.D.A. Islam

Author M.A. Abin

Author M. Abouyoussef

Abstract Social media has become a significant resource for evaluating public attitudes regarding mental health, as individuals may express their thoughts through postings, comments, and conversations. Sentiment analysis is a promising method for studying emotional patterns expressed in mental-health-related text. Traditional machine learning for sentiment analysis faces challenges like transparency, scarce labeled data, and language limitations. This paper presents a sentiment classification (positive, negative, neutral) framework applied to mental-health-related Reddit posts. Repeated

emotional patterns can offer helpful insights into sentiment trends in mental health discussions, but they shouldn't be seen as proof of clinical disorders. In this paper, we propose a hybrid framework for sentiment classification from text, termed QBERT-LSTM, which integrates quantum-enhanced bidirectional encoder representations from transformers (QBERT) with long short-term memory (LSTM) networks. In the proposed approach, transformer models excel at capturing global context, while the LSTM enhances sequential patterns and temporal features. The quantum component uses a parameterized circuit with angle encoding and variational gates to perform nonlinear transformations on a quantum simulator (PennyLane). Subsequently, the model is tested on more intricate circuits comprising 2, 4, 6, 8, 10, and 12 qubits to assess its consistency. A novel Reddit dataset of mental health-related subreddits (anxiety, panic attacks, and depression) is constructed, with sentiment labels derived from the VADER sentiment analyzer. We thoroughly assessed the performance of our proposed model against deep learning models and cutting-edge transformer architectures, achieving up to 98% accuracy in the best-performing configuration on the generated dataset (mean accuracy 0.97 ± 0.0059 across five random seeds). © 2013 IEEE.

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Language model, Large language model, Behavioral research, Deep learning, Mental health, Learning systems, Machine-learning, Sentiment analysis, deep learning, Long short-term memory, Memory architecture, Signal encoding, Mathematical transformations, multilingual input via translation, Multilingual sentiment analyze, quantum machine learning, Quantum machine learning, Quantum machines, Qubits, transformer-based models

Notes:

EID: 2-s2.0-105046266580

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: N.S. Shirin; Department of Computer Science and Engineering, Jagannath University, Dhaka, 1100, Bangladesh; email: b190305031@cse.jnu.ac.bd

QMPN: A Quality-Aware Memory Prompting Network for Few-Shot Multimodal Aspect-Based Sentiment Analysis

Item Type Journal Article

Author L. Pan

Author T. Geng

Author Y. Liu

Abstract Multimodal aspect-based sentiment analysis (MABSA) predicts the sentiment polarity associated with a specified aspect by jointly exploiting textual and visual information. Existing models may be sensitive to limited prompting examples, cross-modal noise, and unreliable generated context. This paper proposes QMPN, Quality-Aware Memory Prompting Network, that stores sample-specific prompts derived from a small support set, retrieves relevant prompting evidence for each query, and uses the retrieved prompts to guide aspect-aware context generation. A task-oriented quality gate, learned indirectly through the sentiment classification objective, controls the contribution of the generated context to the final prediction. Under the fixed protocol used in this study, QMPN employs 50 labeled support instances for prompt-memory construction and achieves 78.6% accuracy and 74.8% macro-F1 on Twitter-2015, and 72.0% accuracy and 70.5% macro-F1 on Twitter-2017. Relative to the variant without context generation, the complete model improves accuracy/macro-F1 by 3.31/2.83 percentage points on Twitter-2015 and 3.84/3.22 percentage points on Twitter-2017. Ablation and parameter analyses further show the contributions of memory retrieval, adaptive prompt selection, context generation, and quality-aware fusion. Because the evaluation is limited to two historical Twitter benchmarks and a single fixed seed, the reported results should be interpreted within this experimental scope rather than as evidence of universal cross-domain generalization. © 2026 by the authors.

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Notes:

EID: 2-s2.0-105048764076

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: T. Geng; School of Computer Science, Zhongyuan University of Technology, Zhengzhou, 450007, China; email: gengtong@zut.edu.cn

Quantifying cultural ecosystem services in urban parks using social media and large language models: Insights for Beijing’s Garden City initiative

- Item Type** Journal Article
- Author** S. Zheng
- Author** Y. Ren
- Author** C. Zhu
- Author** X. Wang
- Author** M. Zhao
- Author** S. Wang
- Author** J. Wang
- Author** T. Yang

Abstract Urban parks are vital components of sustainable cities, providing cultural ecosystem services (CES) that enhance residents' well-being. Yet understanding how people perceive these intangible benefits remains challenging. This study refines existing CES research by integrating large-scale social media analytics with large language models (LLMs) to capture and interpret public perceptions of parks in Beijing's core districts. Drawing on more than 46,000 user-generated comments, the LLM-based framework identifies six CES categories (i.e., historical & cultural, aesthetic, recreational, social interaction, educational, and spiritual values) and evaluates their sentiment tendencies. The results reveal that aesthetic value is the most positively perceived dimension, while recreational, educational, and social interaction values show moderate emotional intensity, indicating potential areas for experiential enhancement. Historical & cultural and spiritual values, though less frequently mentioned, generate strong positive affect when recognised. Integrating these findings through Importance-Performance Analysis highlights clear renewal priorities for Beijing's Garden City initiative. Methodologically, the study demonstrates how LLMs can classify Chinese-language social media text with high accuracy (F1 = 0.8945) and minimal manual annotation, offering a scalable, interpretable, and cost-effective approach for CES assessment. The combined use of social media data, semantic modelling, and performance analysis provides transferable evidence for guiding park management and renewal in other high-density urban contexts. © 2026 The Authors.

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RacismoBR: A Manually Annotated Dataset for Racist Discourse Detection in Brazilian Portuguese

Item Type	Conference Paper
Author	J.V. Vaz
Author	F. Benevenuto
Author	M.A. Gonçalves
Abstract	Racist discourse on social media appears both through explicit attacks and subtle, context-dependent forms, remaining a challenge for Natural Language Processing. We introduce RacismoBR, a culturally grounded dataset for detecting racist discourse in Brazilian Portuguese, manually annotated exclusively by Black researchers to ensure sociolinguistic validity and epistemic representativeness. We conduct a controlled evaluation of binary racism classification in our dataset considering several classification modeling paradigms: classical machine learning, supervised Transformer-based (Small) Language Models, and Large Language models under in-context, few-shot learning. Results show that GPT-4.1 and BERTimbau yield the highest Macro-F1 scores; however, Wilcoxon signed-rank tests reveal no statistically significant differences across models, mostly due to high variability. Across paradigms, classifiers consistently display higher precision for non-racist content and higher recall for racist content. A qualitative analysis highlights persistent difficulties with implicit, euphemized, and context-dependent racism. These findings indicate that culturally grounded annotation plays a more decisive role than architectural sophistication alone in advancing racism detection. Dataset URL: https://github.com/joaovitorvaz/RacismoBR-Dataset . © 2026 Association for Computational Linguistics.
Date	2026
URL	https://www.scopus.com/pages/publications/105041072547?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Int. Conf. Comput. Process. Portuguese, PROPOR
Volume	1
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Conference Name 17th International Conference on Computational Processing of Portuguese, PROPOR 2026
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Computational linguistics, Language model, Classification (of information), Learning systems, Machine-learning, Language processing, Learning algorithms, Natural language processing systems, Natural languages, Social media, Large datasets, Modeling languages, Annotated datasets, Classification models, Context dependent, In contexts, Modeling paradigms

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Rage against the AI machine: Can generative AI understand Donald Trump-related posts as well as humans?

Item Type Journal Article

Author D. Murthy

Author J. Shermak

Author S. Pathania

Author K. Vinton

Author K. Whipple

Author L. Willard

Author R. Madison

Abstract Global socio-political discourse on social media is often heavily colored by its highly volatile and divisive nature. Nonetheless, social media remains central to political communication, becoming its own public square with the unique distinction of also intersecting with the multiverse of public squares across nations and platforms. While the sociological value of studying this discourse is undeniable, successfully deciphering the meaning of large corpora of polarized political content remains elusive. AI machine learning methods can tackle large datasets but are limited in their abilities to decipher highly convoluted and nuanced meanings associated with language. Using a historical dataset of Donald Trump-related tweets from 2016 to 2017 (immediately before and after his inauguration in January 2017), this study evaluated ChatGPT compared with humans in effectively discerning sentiment from controversial political discourse. Using human coding of sentiment and 10 sociolinguistic attributes, we found that ChatGPT and humans achieved similarly low

levels of agreement—not because AI has reached human-level proficiency, but because both struggled considerably with this type of polarized political content. Human coding itself fell well short of acceptable reliability standards, meaning AI’s comparable performance reflects a shared deficiency rather than a shared strength. The agreement between human and machine coders ($\kappa = 0.308$) fell well short of expectations and should not be interpreted as evidence of AI adequacy. We highlight the most problematic areas for machine learning such as its inability to analyze tweets because of special characters, tweet length, and links. For humans, things like misspellings and abbreviations brought confusion. Our findings urge caution in deploying a large language model (LLM)-based sentiment analysis of politically divisive discourse in the near future. © The Author(s) 2026. This article is distributed under the terms of the Creative Commons Attribution-NonCommercial 4.0 License (<https://creativecommons.org/licenses/by-nc/4.0/>) which permits non-commercial use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access page (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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Journal Abbr Convergence

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large language models, machine learning, ChatGPT, social media, sentiment analysis, political sociology

Notes:

EID: 2-s2.0-105042385525

Export Date: 20 September 2026; Cited By: 1; Correspondence Address: D. Murthy; School of Journalism and Media, Moody College of Communication and Department of Sociology, Department of Journalism and Media, University of Texas at Austin, Austin, Dealey 300 W. Dean Keeton St., 3rd Floor, 78712, United States; email: dhiraj.murthy@austin.utexas.edu

RankLLM: A Multi-Criteria Decision-Making Method for LLM Performance Evaluation in Sentiment Analysis

Item Type Conference Paper

Author H. Xue

Author B. Zhao

Author H. Xie

Author Z. Sun

Editor Sun M.

Editor Duan P.

Editor Liu Z.

Editor Xu R.

Editor Sun W.

Editor Chen Y.

Editor Tian Z.

Editor Liu Z.

Abstract Large Language Models (LLMs) have made significant advancements in sentiment analysis, yet their quality and reliability vary widely. Existing LLM evaluation studies are limited in scope, lack a comprehensive framework for integrating diverse capabilities, and fail to quantify the impact of prompt design on performance. To address these gaps, this paper introduces a set of LLM evaluation criteria with detailed explanations and mathematical formulations, aiding users in understanding LLM limitations and selecting the most suitable model for sentiment analysis. Using these criteria, we apply the Technique for Order Preference by Similarity to an Ideal Solution (TOPSIS), a classic decision-making method, to rank the performance of LLMs in sentiment analysis. We evaluated six popular LLMs on three Twitter datasets covering different topics and analyzed the impact of prompt design by assessing model-prompt combinations. Additionally, a validation experiment on a publicly available annotated dataset further confirms our ranking results. Finally, our findings offer valuable insights into the evaluation and selection of LLMs for sentiment analysis. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: H. Xie; Beijing Institute of Mathematical Sciences and Applications (BIMSA), Beijing, 101408, China; email: haihua.xie@bimsa.cn; Conference name: 24th China National Conference on Chinese Computational Linguistics, CCL 2025; Conference date: 11 August 2025 through 14 August 2025; Conference code: 342859

RANLPStud 2023 - Proceedings of the 8th Student Research Workshop, associated with the 14th International Conference on Recent Advances in Natural Language Processing, RANLP 2023

Item Type Journal Article

Editor Hardalov M.

Editor Kancheva Z.

Editor Velichkov B.

Editor Nikolova-Koleva I.

Editor Slavcheva M.

Abstract The proceedings contain 11 papers. The topics discussed include: detecting ChatGPT: a survey of the state of detecting ChatGPT-generated text; unsupervised calibration through prior adaptation for text classification using large language models; controllable active-passive voice generation using prefix tuning; age-specific linguistic features of depression via social media; trigger warnings: a computational approach to understanding user-tagged trigger warnings; evaluating hallucinations in large language models for Bulgarian language; leveraging probabilistic graph models in nested named entity recognition for polish; crowdsourcing veridicality annotations in Spanish: can speakers actually agree?; exploring low-resource neural machine translation for Sinhala-Tamil language pair; and prompting ChatGPT to draw morphological connections for new word comprehension.

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RATSD: Retrieval Augmented Truthfulness Stance Detection from Social Media Posts Toward Factual Claims

Item Type Conference Paper

Author Z. Zhu

Author Z. Zhang

Author H. Zhang

Author C. Li

Editor Chiruzzo L.

Editor Ritter A.

Editor Wang L.

Abstract Social media provides a valuable lens for assessing public perceptions and opinions. This paper focuses on the concept of truthfulness stance, which evaluates whether a textual utterance affirms, disputes, or remains neutral or indifferent toward a factual claim. Our systematic analysis fills a gap in the existing literature by offering the first in-depth conceptual framework encompassing various definitions of stance. We introduce RATSD (Retrieval Augmented Truthfulness Stance Detection), a novel method that leverages large language models (LLMs) with retrieval-augmented generation (RAG) to enhance the contextual understanding of tweets in relation to claims. RATSD is evaluated on TSD-CT, our newly developed dataset containing 3,105 claim-tweet pairs, along with existing benchmark datasets. Our experiment results demonstrate that RATSD outperforms state-of-the-art methods, achieving a significant increase in Macro-F1 score on TSD-CT. Our contributions establish a foundation for advancing research in misinformation analysis and provide valuable tools for understanding public perceptions in digital discourse. © 2025 Association for Computational Linguistics.

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Reading between modalities: multimodal hate speech detection in low-resource Indonesian social media

Item Type Journal Article
Author E.W. Pamungkas
Author A.F. Syafiandini
Author D. Purworini
Author W. Widayat
Author D.G.P. Putri
Author I. Amal
Author M. Song
Abstract Hate speech on social media poses a serious threat to online safety and social harmony, especially in multilingual and low-resource contexts such as Indonesia. The challenge is further amplified by the increasing prevalence of multimodal contents that combine both text and images, which are often used to convey harmful messages more subtly. However, most existing research focuses only on text-based detection, leaving a gap in understanding how visual information contributes to hate speech utterance. This study aims to comprehensively investigate hate speech detection on Indonesian social media within a multimodal setting. We constructed a novel multimodal hate speech dataset that includes tweet text and accompanying images collected from X, with annotations provided by experts. For evaluation, we explore five hate speech detection approaches: unimodal baselines, early fusion, late fusion, multimodal large language models (MLLMs), and textualization-based fusion. Additionally, we also employed traditional machine learning and transformer-based models for more comprehensive assessment. Our experiment results show that textual

information alone remains highly predictive, with a strong baseline F1-score of 0.803 achieved by XLM-RoBERTa. MLLMs such as Gemini 2.5 Flash and GPT-5.2 demonstrate competitive zero-shot performance (up to $F1 = 0.799$), though with variability across languages and prompting strategies. Our experiment also uncovered that combining raw image and text via early or late fusion yields only marginal gains. In contrast, our proposed textualization approach by transforming images into descriptive text using GPT-4o and concatenating them with tweet text achieves the best overall result with an F1-score of 0.833. These findings suggest that structured textual representations of visual content can significantly enhance multimodal classification performance while improving interpretability. This work provides valuable insights for future research on hate speech detection, especially in resource-scarce and linguistically diverse regions. © The Author(s), under exclusive licence to Springer Nature Singapore Pte Ltd. 2026.

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Real-Time Financial Sentiment Extraction Using Large Language Model (LLM) Architectures

Item Type Journal Article

Author L.H. Palivela

Author S. Athinarapu

Author G.R. Basireddy

Author T.V.K. Teja

Abstract The volume of financial data continues to increase and society moves towards a real-time analytical platform, this project reports on the design and implementation of a Large Language Model (LLM)-driven system for financial news sentiment analysis. It leverages both domain specific and general transformer models such as FinBERT, FinBERT-tone, ProsusAI FinBERT, Twitter-RoBERTa, ELECTRA, and DistilBERT—to classify news sentiment as Positive, Negative, or Neutral. A hybrid processing pipeline has been designed where LLMs will first enhance and summarize the financial news articles to better situational context. Once the text is enhanced, it is sent to the relevant fine-tuned transformer models. The outputs are brought together through a multi-agent ensemble process that functionally simulates a diverse set of analysts and improves decision robustness. Evaluation on benchmark datasets shows the ensemble outperforms individual models in accuracy and consistency. This system enables automated financial insight extraction, supports investment decisions, and gauges market sentiment. Copyright © 2026 by IGI Global Scientific Publishing.

Date 2026

DOI 10.4018/979-8-3693-6690-5.ch006

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Pages 215-254

Publication Harnessing Large Language Models for Enhanced Business Analytics

ISSN 979-836936692-9 (ISBN); 979-836936690-5 (ISBN); 979-836936691-2 (ISBN)

Journal Abbr Harnessing Large Language Models for Enhanced Business Analytics

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Tags:

Language model, Real- time, Architecture, Modeling architecture, Sentiment analysis, Investments, Transformer modeling, Data mining, Extraction, Financial news, Multi agent systems, Software architecture, Pipeline processing systems, Design and implementations, Driven system, Financial data, Model-driven, Project reports

Notes:

EID: 2-s2.0-105042385428

Export Date: 20 September 2026; Cited By: 0

Real-Time Twitter Sentiment Analysis with a Hybrid Transformer Framework

Item Type Conference Paper

Author S.P. Bellamkonda

Author S. Veeramachineni

Author S. Kattamuri

Author E. Ajith Jubilson

Abstract Real-time sentiment analysis on online social platforms is a major task for understanding public opinion in changing environments. This paper presents a hybrid transformer-based framework for real-time sentiment analysis on Twitter (now X). The framework combines a fine-tuned, Twitterspecific RoBERTa model with lexicon-based VADER scoring and zero-shot BART-MNLI classification in one evaluation framework. This framework was trained and validated using the Sentiment140 benchmark dataset, which has 1.6 million labeled tweets and was assessed on 500 live-stream tweets collected via the X API. Evaluated model performance using five-fold crossvalidation, macro-averaged F1-score, and significance testing. The experimental results show that the fine-tuned RoBERTa model achieved 91.8% accuracy, which is better than VADER (72.4%) and BART-MNLI (90.1%). Additionally, on real-time data, it showed an 89.3% consistency rate with human annotations. From these results it can be inferred that improving real-time sentiment monitoring can be achieved by fine-tuning transformers for particular domains and using complementary evaluation methods. The proposed framework offers a solid foundation for scalable social media analytics and supports applications in public opinion analysis, monitoring of events, and decision-making systems. © 2026 Bharati Vidyapeeth's Institute of Computer Applications and Management (BVICAM).

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Language English

Extra Journal Abbreviation: Proc. Int. Conf. Comput. Sustain. Glob. Dev., INDIACom

Publisher Institute of Electrical and Electronics Engineers Inc.

ISBN 978-938054464-9 (ISBN)

Proceedings Title Proc. Int. Conf. Comput. Sustain. Glob. Dev., INDIACom

Conference Name Proceedings of the 2026 13th International Conference on Computing for Sustainable Global Development, INDIACom 2026

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Tags:

Artificial intelligence, Real- time, Social aspects, Decision making, Sentiment analysis, Transformer modeling, Social networking (online), Data mining, Tweets, RoBERTa, transformer models, Hybrid sentiment analysis, Hybrid sentiment analyze, BART, Real-time text analytic, Real-Time Text Analytics, Text analytics, Twitter data, Twitter Data, VADER

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Export Date: 20 September 2026; Cited By: 0; Conference name: 20th INDIACom; 13th International Conference on Computing for Sustainable Global Development; Conference date: 8 April 2026 through 10 April 2026; Conference code: 224224

ReFACT: A Benchmark for Scientific Confabulation Detection with Positional Error Annotations

- Item Type** Conference Paper
- Author** Y. Wang
- Author** M. Preiß
- Author** M. Bugueño
- Author** J.V. Hoffbauer
- Author** A. Ghajar
- Author** T. Buz
- Author** G. de Melo
- Editor** Demberg V.
- Editor** Inui K.
- Editor** Marquez Villodre L.
- Abstract** The mechanisms underlying scientific confabulation in Large Language Models (LLMs) remain poorly understood. We introduce Reddit False And Correct Texts (ReFACT), a benchmark of 1,001 expert-annotated question–answer pairs with span-level error annotations derived from Reddit’s r/AskScience. Evaluating 9 state-of-the-art LLMs reveals two critical limitations. First, models exhibit a dominant salient distractor failure mode: 61% of incorrect span predictions are semantically unrelated to actual errors. Crucially, this pattern persists across all model scales (1B to 70B), indicating a fundamental semantic grounding deficit that scaling alone fails to resolve. Second, we find that comparative judgment is paradoxically harder than independent detection—even GPT-4o’s F1 score drops from 0.67 to 0.53 when comparing answers side-by-side. These findings directly challenge the reliability of LLM-as-Judge paradigms for scientific factuality. Code and data are released at <https://github.com/ddz5431/ReFACT>. © 2026 Association for Computational Linguistics.
- Date** 2026
- DOI** 10.18653/v1/2026.eacl-long.381
- URL** <https://www.scopus.com/pages/publications/105040548721?origin=resultslist>
- Archive** Scopus
- Language** English
- Extra** Journal Abbreviation: EACL - Conf. Eur. Chapter Assoc. Comput. Linguist., Proc. Conf., Vol. 1 - Long Pap.
- Volume** 1
- Publisher** Association for Computational Linguistics (ACL)
- ISBN** 979-889176380-7 (ISBN)
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Proceedings Title EACL - Conf. Eur. Chapter Assoc. Comput. Linguist., Proc. Conf., Vol. 1 - Long Pap.

Conference Name EACL 2026 - 19th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference, Vol. 1 - (Long Papers)

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Refining Bengali Hate Speech Detection: Multi-label Classification Using RNN and LSTM

Item Type Conference Paper

Author M. Islam

Author M.S. Hossain

Author M.M. Islam

Author M.A.H. Murad

Author S.S. Niloy

Author M.S. Islam

Author T.B. Islam

Editor Iwendi C.

Editor Boulouard Z.

Editor Kryvinska N.

Abstract The Internet and social media have made significant progress and are now accessible to a wide audience. With this rapid growth there has also been a corresponding increase in negativity to a certain degree. The main goal is to eliminate the issue of Cyberbullying, which often occurs via the use of Offensive language. Although several options exist, many of them are currently under development. Nevertheless, the issue is always changing, particularly for low-resource languages like Bengali. It is crucial to discover more advanced and enhanced solutions since there are always constraints with present methods and opportunities for further enhancement. This work presents a novel approach to detecting foul language by using a combination of two models in a multi-step

solution. Applying two distinct classifiers enables a more detailed method for identifying offensive language. Between the two models, one may ascertain the character of the statement by categorising it as either offensive or non-offensive. The second approach employs multi-label classification to categorise objectionable words into four distinct categories: “Sexual”, “Dishonour”, “Incursive”, and “Neutral” hence advancing the classification process. Several models were used in the process, while Recurrent Neural Networks (RNN) exhibited superior outcomes and were used for both models. The accuracy for both the offensive and non-offensive models was 87.37%. In terms of Multi-label classification, the accuracy achieved was 67%. Although the performance of the multi-label classification model was comparatively low, however the combination of these two models demonstrates remarkable performance in terms detecting Offensive Bengali language. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

Date 2025

DOI 10.1007/978-3-031-94623-3_21

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Conference Name Lecture Notes in Networks and Systems

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Refining GPT-3 Embeddings with a Siamese Structure for Technical Post Duplicate Detection

Item Type Conference Paper

Author X. Wu

Author H. Li

Author N. Yoshioka

Author H. Washizaki

Author F. Khomh

Abstract One goal of technical online communities is to help developers find the right answer in one place. A single question can be asked in different ways with different wordings, leading to the existence of duplicate posts on technical forums. The question of how to discover and link duplicate posts has garnered the attention of both developer communities and researchers. For example, Stack Overflow adopts a voting-based mechanism to mark and close duplicate posts. However, addressing these constantly emerging duplicate posts in a timely manner continues to pose challenges. Therefore, various approaches have been proposed to detect duplicate posts on technical forum posts automatically. The existing methods suffer from limitations either due to their reliance on handcrafted similarity metrics which can not sufficiently capture the semantics of posts, or their lack of supervision (i.e., leveraging existing duplicate annotations) to improve the performance. Additionally, the efficiency of these methods is hindered by their dependence on pair-wise feature generation, which can be impractical for large amount of data. In this work, we attempt to employ and refine the GPT-3 embeddings for the duplicate detection task. We assume that the GPT-3 embeddings can accurately represent the semantics of the posts. In addition, by training a Siamese-based network based on the GPT-3 embeddings, we obtain a latent embedding that accurately captures the duplicate relation in technical forum posts. Our experiment on a benchmark dataset confirms the effectiveness of our approach and demonstrates superior performance compared to baseline methods. When applied to the dataset we constructed with a recent Stack Overflow dump, our approach attains a Top-1, Top-5, and Top-30 accuracy of 23.1 %, 43.9 %, and 68.9 %, respectively. With a manual study, we confirm our approach's potential of finding unlabelled duplicates on technical forums. We released our dataset and code in our supplementary package to promote further studies¹Supplementary material package: <https://github.com/mooselab/suppmaterial-PostDupGPT3>. © 2024 IEEE.

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Language English

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Language model, Large language model, Performance, Large Language Model, Benchmarking, Semantics, Siamese network, Embeddings, Duplicate detection, Duplicate Detection, On-line communities, Community IS, Siamese Network, Stack overflow, Technical Forum, Technical forums

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Relation Extraction from Real-World Unstructured Text in the Domain of Dementia

Item Type Conference Paper

Author S. Suravee

Author D. Yadav

Author K. Yordanova

Editor Tonkin E.L.

Editor Tourte G.J.L.

Editor Yordanova K.

Abstract Relation extraction (RE) identifies semantic relations between named entities and is crucial for tasks such as knowledge graph construction and question answering. RE can be leveraged for automatic relationship annotation by employing trained RE models to identify and annotate semantic relationships between ontology-driven entities in unstructured, conversational texts. This paper examines RE in real-world conversational texts collected from an online dementia forum comprising 45,216 sentences, where language is often ambiguous and departs from standard medical terminology. We present a unified RE pipeline based on the domain-specific annotations and conduct a comparative evaluation of two model types: (i) a neural network-based architecture combining a bidirectional long-short-term memory (Bi-LSTM) model with static GloVe: Global Vectors for Word Representation (GloVe) and a conditional random field (CRF) classifier, and (ii) a transformer-based language model, BERT-Large, followed by a CRF layer. The results show that the use of GloVe embedding in the Bi-LSTM model achieves a significantly higher F1 score of 0.81, compared to 0.64 with the BERT-Large model. Despite lacking contextualised embeddings, the GloVe embedding captures relation patterns more effectively in this low-resource, domain-specific setting. These results demonstrate that lightweight models can outperform large pre-trained transformers in challenging real-world scenarios. The output of the RE task is a set of structured annotations, which can be integrated into

downstream knowledge bases. Therefore, the investigated RE models have the potential to be applied as automatic annotators of informal texts from the domain of dementia, transforming the text into richly annotated resources by systematically annotating entity relationships without human intervention. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Notes:

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Research of Tibetan-Chinese Bilingual Internet Public Opinion Analysis Based on BERT+BiGRU+CNN

Item Type Conference Paper

Author Y. Shen

Author W. Suonan

Author N. Qun

Author Y. Liang

Abstract To address the current challenges in Tibetan-Chinese bilingual online public opinion analysis, such as scarce language resources, significant cultural disparities, and insufficient feature extraction of traditional models, this study proposes an integrated deep learning model combining BERT, BiGRU, and CNN with an attention mechanism for Tibetan-Chinese bilingual processing. A hierarchical feature extraction framework is designed: context-aware bilingual word embeddings are generated via the pre-trained BERT model, the BiGRU network captures long-range temporal dependencies, and the multi-scale CNN focuses on local semantic features. Experiments demonstrate that the integrated model with the attention mechanism achieves an accuracy of 96.7% in sentiment classification tasks on Chinese social media data, representing a 13.3% improvement compared to the standalone BERT model. For sentiment classification tasks on Tibetan data, it reaches an accuracy of 88.4%, which is a 9.5% increase relative to the standalone Tibetan BERT tokenizer. This research enables in-depth joint analysis of Tibetan-Chinese bilingual public opinion and holds innovative significance for low-resource language processing and cross-cultural public opinion analysis. ©2025 IEEE.

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Research on Multimodal Sentiment Analysis Models for Social Media Based on Multimodal Large Language Models

Item Type Journal Article

Author L. Yueqi

Author W. Xiwei

Author W. Qiuyue

Abstract [Purpose/significance] In contemporary social media, users' opinions and emotional expressions are increasingly presented in multimodal forms. To address the challenges of incomplete feature representation and semantic inconsistency across modalities in multimodal sentiment analysis tasks, this study investigates the application and performance of multimodal large language models in social media multimodal sentiment analysis tasks. [Method/process] Specifically, two models are constructed, a multi-modal sentiment analysis model based on llama3.2-vision, fine-tuned using LoRA, and another multimodal sentiment analysis model based on GPT-4o combined with prompt strategy optimization. The effectiveness of the proposed models was evaluated through experiments on the public datasets MVSA-Single and MVSA-Multiple. [Result/conclusion] Experimental results show that the sentiment analysis models under both schemes significantly outperform the baseline models in terms of accuracy, precision, and recall. The study provides a new analytical perspective and theoretical framework for multimodal sentiment analysis tasks on social media in the context of generative artificial intelligence, which is of great significance for multimodal sentiment monitoring and guidance in cyberspace. © (2025), (Information studies: Theory and Application). All rights reserved.

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Research on sentiment analysis method of opinion mining based on multi-model fusion transfer learning

Item Type Journal Article

Author Z. Zhao

Author W. Liu

Author K. Wang

Abstract With the popularity of social media, opinion mining has gradually become a popular research field. Among these fields, sentiment analysis mining is an important research direction in the field of opinion mining. It aims to reveal the public's sentiment tendency, and attitude towards specific topics or events by analyzing text data generated by users on online platforms and digital media. However, the large amount of opinion data usually lacks effective annotation, which limits the learning and construction of opinion models. Therefore, focusing on the problem of the scarcity of labeled data in opinion analysis, this paper proposes a mining method for public opinion sentiment analysis based on multi-model fusion transfer learning, that can make full use of the limited labeled data to improve the learning efficiency of sentiment features by integrating the advantages of different models. Additionally, it introduces a transfer learning strategy to enable the models of the target domains to perform better in the absence of labeled data. Furthermore, the attention mechanism is combined to strengthen the acquisition of key features of the emotional colors and improve the accuracy of sentiment analysis. Specifically, the paper uses the ERNIE model to generate dynamic representations of the text word vectors in the dataset. It also uses TextCNN and BiGRU to construct a joint model for extracting local and overall features of the text word vectors. The parameters of the feature layer of the trained model are migrated to the target domain through transfer learning. The attention mechanism is combined with the model to identify the extreme elements of the sentiment. Finally, the local and overall features are fused to achieve comprehensive mining of public opinion and emotional information. This method can effectively improve the accuracy and generalization of public opinion analysis in cases of data scarcity. In the experimental part, the paper conducts comparisons and analyses in eight aspects: word embedding model, model combination, attention mechanism, transfer learning, source domain dataset, target domain dataset, model training, and baseline model. The four indicators, namely accuracy, precision, recall, and F1-measure are used to evaluate the performance of the method. The experiments

are thorough and detailed, demonstrating the effective improvement of opinion mining performance. © 2023, Springer Nature Switzerland AG.

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Research on Toxic Speech Detection Based on Large Language Models

Item Type Conference Paper

Author W. Li

Author Y. Gao

Author Y. Zhang

Author L. Yang

Author R. Gao

Abstract Toxic Speech (TS) is widely spread in social media and is increasingly harmful to the online networking environment. At the same time, words and their variants containing implicit toxic semantics are influenced by the humanistic and social environment of the language and are extremely rich in form and shape, and the lack of background knowledge related to the "discourse body" of existing research has led to the confusion of Implicit Toxic Speech (ITS) with Non-Toxic Speech

(NTS). Therefore, this paper proposes the Prompt-enhanced Attention Head Fusion (PAHF) toxic text classification model based on the generative language model. We optimize the hidden layer vectors by intervening in the hidden layer information through prompt, then filter the vectors by combining linear probes to locate those vectors in the hidden layer information that are strongly correlated with the encoding of the background knowledge, and finally construct an attention fusion mechanism to fuse the filtered features, which provides a higher detection accuracy compared with the existing algorithms. © 2024 IEEE.

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Notes:

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RESEMO: A Benchmark Chinese Dataset for Studying Responsive Emotion from Social Media Content

Item Type Conference Paper**Author** B. Hu**Author** M. Zhang**Author** C. Xie**Author** Y. Tian**Author** Y. Song**Author** Z. Mao**Editor** Ku L.-W.**Editor** Martins A.**Editor** Srikumar V.

Abstract On social media platforms, users' emotions are triggered when they encounter particular content from other users, where such emotions are different from those that spontaneously emerged, owing to the “responsive” nature. Analyzing the aforementioned responsive emotions from user interactions is a task of significant importance for understanding human cognition, the mechanisms of emotion generation, behavior on the Internet, etc. Performing the task with artificial intelligence generally requires human-annotated data to help train a well-performing system, while existing data resources do not cover this specific area, with none of them focusing on responsive emotion analysis. In this paper, we propose a Chinese dataset named RESEMO for responsive emotion analysis, including 3,813 posts with 68,781 comments collected from Weibo, the largest social media platform in China. RESEMO contains three types of human annotations with respect to responsive emotions, namely, responsive relationship, responsive emotion cause, and responsive emotion category. Moreover, to test this dataset, we build large language model (LLM) baseline methods for responsive relation extraction, responsive emotion cause extraction, and responsive emotion detection, which show the potential of the proposed RESEMO being a benchmark for future studies on responsive emotions. © 2024 Association for Computational Linguistics.

Date 2024**DOI** 10.18653/v1/2024.findings-acl.970**URL** <https://www.scopus.com/pages/publications/85205326741?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Proc. Annu. Meet. Assoc. Comput Linguist.**Publisher** Association for Computational Linguistics (ACL)**ISBN** 0736587X (ISSN); 979-889176099-8 (ISBN)**Pages** 16375-16387**Proceedings Title** Proc. Annu. Meet. Assoc. Comput Linguist.**Conference Name** Proceedings of the Annual Meeting of the Association for Computational Linguistics**Date Added** 20/09/2026, 8:35:19 pm**Modified** 20/09/2026, 8:35:19 pm**Tags:**

Computational linguistics, Social media, Social media platforms, Emotion analysis, Media content, Data resources, Emotion generation, Human cognition, Specific areas, User emotions, User interaction

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: Y. Song; University of Science and Technology of China, Hefei, China; email: clksong@gmail.com; Conference name: Findings of the 62nd Annual Meeting of the Association for Computational Linguistics, ACL 2024; Conference date: 11 August 2024 through 16 August 2024; Conference code: 202475

Residential sentiments and spatial environments in urban villages: An X-minute community life circle analysis

Item Type Journal Article**Author** L. Song**Author** L. Wang**Author** M.-P. Kwan**Author** Y. Zhang

Abstract Urban villages, a unique residential typology of China's rapid urbanization, and their residents' sentiments have become a prominent issue in the development and management of Shenzhen. While numerous studies have investigated the impact of the urban environment on residents' sentiments, most have focused on the general population and used administrative units, neglecting environmental impacts across spatial scales based on daily mobility. To address these research gaps, this study employed multi-source big data and a large language model to assess residents' sentiments and evaluate environmental features within 5-, 10-, and 15-min community life circles in the urban villages of Shenzhen. Hierarchical regression and ridge regression were used to explore the key environmental features affecting urban village residents' sentiments. The findings indicated that the sentiment scores of urban village residents were significantly lower than those of the general population, while low sentiment scores were concentrated in suburban areas. Further, the environmental variables within the 10-min community life circle had the greatest influence on sentiments for urban villagers. Specifically, subway stations and green spaces contributed positively to residents' sentiments, while industrial parks and building density/height had a negative impact. These findings advance urban research by integrating activity-space theory with big data analytics, deepening the understanding of how spatial environments shape marginalized groups' sentiments. The results also provide an empirical foundation for optimizing urban renewal policies to improve the quality of life and mental well-being of urban village residents through the development of more livable and healthier urban environments. © 2026 The Authors.

Date 2026**DOI** 10.1016/j.apgeog.2026.103945**URL** <https://www.scopus.com/pages/publications/105030451642?origin=resultslist>**Archive** Scopus**Language** English**Volume** 189**Publisher** Elsevier Ltd**Publication** Applied Geography**ISSN** 01436228 (ISSN)

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Sentiment analysis, China, social media, urban development, greenspace, Social media data, Community life circles, Guangdong, life cycle analysis, Shenzhen, suburban area, Urban villages, urbanization, village

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Rethinking Ground Truth: A Case Study on Human Label Variation in MLLM Benchmarking

Item Type Conference Paper

Author T. Ruiz

Author T. Agustoslu

Author C. Schwemmer

Editor Xu C.-Z.

Editor U L.H.

Editor Cheng X.

Editor Gao J.

Editor Polese G.

Editor Mei H.

Editor Boniol P.

Editor Tatsubori M.

Editor Zhao C.

Editor Zhou D.

Editor Hu X.

Abstract Human Label Variation (HLV), i.e. systematic differences among annotators' judgments, remains underexplored in benchmarks despite rapid progress in large language model (LLM) development. We address this gap by introducing an evaluation protocol for multimodal large language model (MLLM) benchmarking that explicitly accounts for two conditions: (1) human label agreement and (2) disagreement. We apply this protocol to two state-of-the-art MLLM families

(Gemma 3, Qwen 2.5 VL) using non-aggregated human annotations from a social media content classification dataset. Across tasks, we find that larger models tend to perform best on high-agreement subsets, yet often underperform medium-sized models when human disagreement is high, indicating that parameter count alone does not determine sensitivity to ambiguity and subjectivity. These results show that benchmarks based solely on consensus labels can overstate model capabilities in such domains and that incorporating human label variation yields more realistic and robust assessments of MLLMs in content moderation pipelines. © 2025 IEEE.

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Retrieval-augmented few-shot large language models (LLMs) for financial sentiment classification on social media

Item Type Journal Article

Author J. Ling

Author K.H. Gan

Abstract Financial sentiment classification on social media is challenging due to rapidly evolving language, including slang, sarcasm, and shifting market targets. Models that perform well on financial news often degrade on tweet-style text because of domain shift and temporal drift, resulting in unstable accuracy and frequent recalibration requirements. This work addresses sentiment analysis under such non-stationarity by calibrating a frozen large language model (LLM) at inference time using a lightweight external memory, enabling adaptation to contemporaneous market language without modifying model parameters. We construct an external exemplar memory composed of previously annotated financial tweets. For each incoming tweet, both the query and all memory items are encoded using a finance-specific sentence encoder based on FinBERT embeddings. Cosine similarity is then used to retrieve the top-k most relevant labeled exemplars ($k = 5$). These exemplars are combined with the query into a standardized few-shot prompt that asks the frozen LLM to classify sentiment as Positive, Neutral, or Negative. In this design, retrieval acts as a non-parametric inference-time prior, locally constraining the model's decision space toward contemporaneous market semantics while avoiding permanent parameter updates. We evaluate zero-shot prompting, few-shot prompting with random exemplars, and relevance-based retrieval-augmented prompting across multiple tweet datasets, including standard benchmarks and a more recent 2024 out-of-distribution tweet set collected after the reported training cutoff of Large Language Model Meta AI (LLaMA)-3. Downstream utility is evaluated by converting daily sentiment predictions into next-day long-short trading signals. Across all models and datasets, relevance-based retrieval consistently improves accuracy and macro-averaged F1 score (Macro-F1) relative to non-retrieval baselines. On a strong GPT-5 configuration, retrieval yields a modest but statistically significant improvement of approximately 1 percentage point, while on LLaMA-3 the gains are substantially larger, ranging from 10-14 percentage points with comparable Macro-F1 increases. In a 2015 trading backtest, retrieval-augmented prompting achieves an annualized return of 19.24%, outperforming non-retrieval Generative Pre-trained Transformer 5 (GPT-5) (14.33%) as well as supervised domain baselines such as Financial Bidirectional Encoder Representations from Transformers (FinBERT) (13.98%) and FinLLaMA (11.59%), while the Standard & Poor's 500 (S&P 500) index records -0.73% over the same period. These results demonstrate that inference-time exemplar retrieval can provide effective calibration under linguistic drift, matching or approaching fine-tuned systems while reducing the need for continual retraining and model upkeep. Copyright 2026 Ling and Gan Distributed under Creative Commons CC-BY 4.0. <http://www.creativecommons.org/licenses/by/4.0/>

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ISSN 23765992 (ISSN)**Journal Abbr** PeerJ Comput. Sci.**Date Added** 20/09/2026, 8:35:12 pm**Modified** 20/09/2026, 8:35:12 pm**Tags:**

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K.H. Gan; School of Computer Sciences, Universiti Sains Malaysia, Penang, Malaysia; email: khgan@usm.my

Revisiting Sentiment Analysis for Software Engineering in the Era of Large Language Models

Item Type Journal Article**Author** T. Zhang**Author** I.C. Irsan**Author** F. Thung**Author** D. Lo

Abstract Software development involves collaborative interactions where stakeholders express opinions across various platforms. Recognizing the sentiments conveyed in these interactions is crucial for the effective development and ongoing maintenance of software systems. For software products, analyzing the sentiment of user feedback, e.g., reviews, comments, and forum posts can provide valuable insights into user satisfaction and areas for improvement. This can guide the development of future updates and features. However, accurately identifying sentiments in software engineering datasets remains challenging. This study investigates bigger large language models (bLLMs) in addressing the labeled data shortage that hampers fine-tuned smaller large language models (sLLMs) in software engineering tasks. We conduct a comprehensive empirical study using five established datasets to assess three open source bLLMs in zero-shot and few-shot scenarios. Additionally, we compare them with fine-tuned sLLMs, using sLLMs to learn contextual embeddings of text from software platforms. Our experimental findings demonstrate that bLLMs exhibit state-of-the-art performance on datasets marked by limited training data and imbalanced distributions. bLLMs can also achieve excellent performance under a zero-shot setting. However, when ample training data are available or the dataset exhibits a more balanced distribution, fine-tuned sLLMs can still achieve superior results. © 2025 Copyright held by the owner/author(s).

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ReZG: Retrieval-augmented zero-shot counter narrative generation for hate speech

Item Type Journal Article

Author S. Jiang

Author W. Tang

Author X. Chen

Author R. Tang

Author H. Wang

Author W. Wang

Abstract The proliferation of hate speech (HS) on social media poses a serious threat to societal security. Automatic counter narrative (CN) generation, as an active strategy for HS intervention, has garnered increasing attention in recent years. Existing methods for automatically generating CNs mainly rely on re-training or fine-tuning pre-trained language models (PLMs) on human-curated CN corpora. Unfortunately,

the annotation speed of CN corpora cannot keep up with the growth of HS targets, while generating specific and effective CNs for unseen targets remains a significant challenge for the model. To tackle this issue, we propose Retrieval-Augmented Zero-shot Generation (ReZG) to generate CNs with high-specificity for unseen targets. Specifically, we propose a multi-dimensional hierarchical retrieval method that integrates stance, semantics, and fitness, extending the retrieval metric from single dimension to multiple dimensions suitable for the knowledge that refutes HS. Then, we implement an energy-based constrained decoding mechanism that enables PLMs to use differentiable knowledge preservation, countering, and fluency constraint functions instead of in-target CNs as control signals for generation, thereby achieving zero-shot CN generation. With the above techniques, ReZG can integrate external knowledge flexibly and improve the specificity of CNs. Experimental results show that ReZG exhibits stronger generalization capabilities and outperforms strong baselines with significant improvements of 2.0%+ in the relevance and 4.5%+ in the countering success rate metrics. © 2024 Elsevier B.V.

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RideKE: Leveraging Low-Resource, User-Generated Twitter Content for Sentiment and Emotion Detection in Kenyan Code-Switched Dataset

Item Type Conference Paper

Author N.A. Etori

Author M.L. Gini

Editor De Clercq O.

Editor Barriere V.

Editor Barnes J.

Editor Klinger R.

Editor Sedoc J.

Editor Tafreshi S.

Abstract Social media has become a crucial open-access platform for individuals to express opinions and share experiences. However, leveraging low-resource language data from Twitter is challenging due to scarce, poor-quality content and the major variations in language use, such as slang and code-switching. Identifying tweets in these languages can be difficult as Twitter primarily supports high-resource languages. We analyze Kenyan code-switched data and evaluate four state-of-the-art (SOTA) transformer-based pretrained models for sentiment and emotion classification, using supervised and semi-supervised methods. We detail the methodology behind data collection and annotation, and the challenges encountered during the data curation phase. Our results show that XLM-R outperforms other models; for sentiment analysis, XLM-R supervised model achieves the highest accuracy (69.2%) and F1 score (66.1%), XLM-R semi-supervised (67.2% accuracy, 64.1% F1 score). In emotion analysis, DistilBERT supervised leads in accuracy (59.8%) and F1 score (31%), mBERT semi-supervised (accuracy 59% and F1 score 26.5%). AfriBERTa models show the lowest accuracy and F1 scores. All models tend to predict neutral sentiment, with Afri-BERT showing the highest bias and unique sensitivity to empathy emotion. © 2024 Association for Computational Linguistics.

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RoBERTa and VADER Sentiment Analysis: A Comparative Approach

Item Type Conference Paper
Author R. Bharadwaj
Author S. Shendurkar
Author T. Kadam
Author U. Patekar
Author S. Waghule
Editor Bhateja V.
Editor Lin H.
Editor Simic M.
Editor Tang J.
Editor Sivakumar Reddy V.

Abstract Social media is very popular worldwide with 4.62 million active users. It has become a part of our daily lives to connect with friends and new people. It will continuously inform us about new trends, and events and keep us updated. It is a platform where we can express ourselves. The advantages of social media are that we can stay connected with family and friends even if they live far away from us. We can share photos, videos, and updates about our daily lifestyle. By using social media we can learn new things through watching videos. Social media is the biggest source of Entertainment. While social media has many advantages, it is important to be aware of the potential risks, such as privacy concerns, spam

messages, negative messages, threatening comments, and hate speech that spreads hate among the communities. In this research, we will implement an AI model using the RoBERTa and Vader modes. RoBERTa is a large language model which is very effective in natural language processing tasks such as text classification. Also, Vader is a sentiment analysis model that is useful for identifying the positive and negative sentiments in the text. The models will be trained on a dataset of labeled social media posts. The labels will be useful to indicate whether the post is harmful or not. To overcome these concerns, we implement NLP-powered Thread: WebApp with RoBERTa and VADER using AI models that are RoBERTa and Vader. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2024.

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Robust Arabic tweet NER via label-aware data augmentation and AraBERTv2

Item Type Journal Article

Author B. Ghazoui

Author I.E. Bazi

Author I. Essadik

Author B.A. Benali

Author H. Moussa

Abstract Named entity recognition (NER) is vital for turning unstructured social media text into structured information. However, Arabic tweets pose distinct challenges; informality, brevity, dialectal variation, and inconsistent orthography. This study targets those challenges by coupling targeted data augmentation with a transformer model, bert-base-arabertv2. We design a lightweight augmentation pipeline—synonym replacement, name and location replacement, and deletion of third-person Arabic names—to expand linguistic variety and reduce overfitting under limited annotation. The approach is simple, but deliberate: preserve labels when substituting entities with type-consistent alternatives; remove corresponding tags when deleting names; and keep tweet semantics intact where possible. We then fine-tune bert-base-arabertv2 on the combined original and augmented data and evaluate on a held-out set of tweets. The result is a substantial gain in overall performance: F1=0.93 with augmentation versus 0.72 without. These findings indicate that controlled, label-aware augmentation can improve robustness and generalization for Arabic tweet NER, where data scarcity and linguistic variability otherwise degrade accuracy. Beyond empirical gains, our work offers a practical recipe—clear augmentation heuristics and a standard transformer backbone—that can be replicated and adapted to similar low-resource, noisy domains. This contributes to more reliable Arabic social media analysis and downstream information extraction. © 2026 Institute of Advanced Engineering and Science. All rights reserved.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: B. Ghazoui; LGS Laboratory, Faculty of Sciences and Technics, Sultan Moulay Slimane University, Beni Mellal, 23000, Morocco; email: brahim.ghazoui@usms.ac.ma

Robust Stance Detection: Understanding Public Perceptions in Social Media

Item Type Conference Paper

Author N. Kim

Author D. Mosallanezhad

Author L. Cheng

Author M.V. Mancenido

Author H. Liu

Editor Aiello L.M.

Editor Chakraborty T.

Editor Gaito S.

Abstract The abundance of social media data has presented opportunities for accurately determining public and group-specific stances around policy proposals or controversial topics. In contrast with sentiment analysis which focuses on identifying prevailing emotions, stance detection identifies precise positions (i.e., supportive, opposing, neutral) relative to a well-defined topic, such as perceptions toward specific global health interventions during the COVID-19 pandemic. Traditional stance detection models, while effective within their specific domain (e.g., attitudes towards masking protocols during COVID-19), often lag in performance when applied to new domains and topics due to changes in data distribution. This limitation is compounded by the scarcity of domain-specific, labeled datasets, which are expensive and labor-intensive to create. A solution we present in this paper combines counterfactual data augmentation with contrastive learning to enhance the robustness of stance detection across domains and topics of interest. We evaluate the performance of current state-of-the-art stance detection models, including a prompt-optimized large language model, relative to our proposed framework succinctly called STANCE-C3 (domain-adaptive Cross-target STANCE detection via Contrastive learning and Counterfactual generation). Empirical evaluations demonstrate STANCE-C3's consistent improvements over the baseline models with respect to accuracy across domains and varying focal topics. Despite the increasing prevalence of general-purpose models such as generative AI, specialized models such as STANCE-C3 provide utility in safety-critical domains wherein precision is highly valued, especially when a nuanced understanding of the concerns of different population segments could result in crafting more impactful public policies. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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RSD-15K: A Large-Scale User-Level Annotated Dataset for Suicide Risk Detection on Social Media

Item Type Conference Paper

Author S. Zheng

Author Y. Tao

Author T. Zhou

Abstract In recent years, cognitive and mental health (CMH) disorders have increasingly become an important challenge for global public health, especially the suicide problem caused by multiple factors such as social competition, economic pressure and interpersonal relationships among young and middle-aged people. Social media, as an important platform for individuals to express emotions and seek help, provides the possibility for early detection and intervention of suicide risk. This paper introduces a large-scale dataset containing 15,000 user-level posts. Compared with existing datasets, this dataset retains complete user posting time sequence information, supports modeling the dynamic evolution of suicide risk, and we have also conducted comprehensive and rigorous annotations on these datasets. In the benchmark experiment, we systematically evaluated the performance of traditional machine learning methods, deep learning models, and fine-tuned large language models. The experimental results show that our dataset can effectively support the automatic assessment task of suicide risk. Considering the sensitivity of mental health data, we also discussed the privacy protection and ethical use of the dataset. In addition, we also explored the potential applications of the dataset in mental health testing, clinical psychiatric auxiliary treatment, etc., and provided directional suggestions for future research work. ©2025 IEEE.

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Tags:

Large-scales, Data privacy, Risk assessment, Deep learning, Mental health, Learning systems, Machine-learning, Machine learning, Language processing, Learning algorithms, Natural language processing, Natural languages, Social networking (online), Benchmarking, Health risks, Social media, Large datasets, Statistical tests, Public health, natural language processing, mental health, suicide detection, Suicide detection, dataset annotation, Dataset annotation, User levels

Notes:

EID: 2-s2.0-105015525891

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: S. Zheng; Department of Computing, The Hong Kong Polytechnic University, Hong Kong; email: 21097982d@connect.polyu.hk; Conference name: 41st IEEE International Conference on Data Engineering Workshops, ICDEW 2025; Conference date: 19 May 2025 through 23 May 2025; Conference code: 211443

RuArg-2022: Argument Mining Evaluation

Item Type Conference Paper
Author E. Kotelnikov
Author N. Loukachevitch
Author I. Nikishina
Author A. Panchenko
Abstract Argumentation analysis is a field of computational linguistics that studies methods for extracting arguments from texts and the relationships between them, as well as building argumentation structure of texts. This paper is a report of the organizers on the first competition of argumentation analysis systems dealing with Russian language texts within the framework of the Dialogue conference. During the competition, the participants were offered two tasks: stance detection and argument classification. A corpus containing 9,550 sentences (comments on social media posts) on three topics related to the COVID-19 pandemic

(vaccination, quarantine, and wearing masks) was prepared, annotated, and used for training and testing. The system that won the first place in both tasks used the NLI (Natural Language Inference) variant of the BERT architecture, automatic translation into English to apply a specialized BERT model, retrained on Twitter posts discussing COVID-19, as well as additional masking of target entities. This system showed the following results: for the stance detection task an F1-score of 0.6968, for the argument classification task an F1-score of 0.7404. We hope that the prepared dataset and baselines will help to foster further research on argument mining for the Russian language. © 2022 ABBYY PRODUCTION LLC. All rights reserved.

Date 2022

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Archive Scopus

Short Title RuArg-2022: соревнование по анализу аргументации

Language English

Extra Journal Abbreviation: Komp'ut. Lingvist. Intellekt. Tehnol.

Volume 2022

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Conference Name Komp'juternaja Lingvistika i Intellektual'nye Tehnologii

Issue 21

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COVID-19, Stance Detection, Argumentation Mining, Premise Classification

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Rubric-conditioned large language model labeling: Agreement, uncertainty, and label consistency in subjective text annotation

Item Type Journal Article

Author J. Liu

Abstract Large-scale annotation of online text is central to research on computer-mediated behavior, yet subjective categories often show substantial disagreement even among trained human raters. Large language models (LLMs) have recently been proposed as annotators, but their behavior as rubric-conditioned labeling protocols is still poorly understood. Using HateXplain, a corpus of social media posts labeled by three crowdworkers on a three-level scale of social norm-violating language, we ask: (a) how closely a lightweight LLM matches human reference constructions under a fixed rubric, (b) how model-reported uncertainty relates to disagreement, and (c) how alternative labeling regimes function as supervision for downstream classifiers. We re-annotate all posts with gpt-4o-mini using a fixed prompt that returns hard labels and a class-probability vector. First, agreement with high-consensus human labels falls near the empirical range observed across pairwise human annotations, with disagreements concentrated at boundaries between adjacent severity levels. Second, Shannon entropy of the probability vector is strongly right-skewed and is higher on posts where the protocol diverges from unanimous human labels. Third, across machine-learning and deep-learning models, supervision from LLM-generated labels yields performance close to that obtained under unanimous human labels and higher than simple majority-vote labels, while covering the full dataset; soft-label training using human vote distributions and model probability vectors shows the same pattern under probability-level evaluation. These gains are therefore interpreted as greater internal consistency (learnability) under a fixed rubric. Overall, rubric-conditioned LLM labeling can support scalable, reproducible annotation, and uncertainty signals can serve as cues for human-AI workflows. © 2026

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Language English

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Publication Computers in Human Behavior

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Journal Abbr Comput. Hum. Behav.

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Language model, Large language model, Behavioral research, Deep learning, Classification (of information), Learning systems, Uncertainty analysis, Human computer interaction, Social networking (online), Entropy, Uncertainty, Large language models, Labelings, Text classification, Text processing, Probability distributions, Labels, C (programming language), Annotation and labeling, Inter-rater agreement, Inter-rater agreements, Label noise, Online social interaction, Social interactions, Subjective text classification, Uncertainty and entropy

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EID: 2-s2.0-105033569924

Export Date: 20 September 2026; Cited By: 1; CODEN: CHBEE

SAGU-Net: Gate-Level Lexicon–Neural Fusion via Sentiment-Aware Gated Units for Social Media Sentiment Analysis

Item Type Journal Article**Author** L. Zhao**Author** K. Huang**Author** X. Ma**Author** H. Zhu**Author** C. Yuan**Author** Y. Su

Abstract Social media sentiment analysis demands systems that are simultaneously accurate, scalable, and interpretable. Lexicon-based methods offer transparency but ignore context, while pre-trained language models (PLMs) capture contextual semantics yet encode sentiment only implicitly. Existing integration strategies inject lexicon signals at the input, attention, or feature layer—all outside the recurrent gating mechanism that controls how affective evidence accumulates over a sequence. We propose the SAGU-Net, a framework built around the Sentiment-Aware Gated Unit (SAGU), a gated recurrent unit (GRU) variant with a dedicated sentiment gate conditioned on external lexicon signals. A complementary Context-Adaptive Sentiment Scoring (CASS) module transforms static polarity scalars into context-dependent vectors via learned projections over PLM representations, bridging the gap between discrete lexicon scores and continuous embeddings. The sentiment gate activations provide token-level explainability without post hoc attribution. On a 12,700-sample Chinese social media corpus of intellectual property co-branding reviews (Fleiss' (Formula presented.)) and two public benchmarks, the SAGU-Net achieves 93.62% accuracy and 93.21% Macro-F1, outperforming nine baselines and matching or exceeding LoRA-fine-tuned large language models (GPT-5, Claude Sonnet 4.6, DeepSeek V3.2, Qwen3.5) while requiring three to four orders of magnitude fewer parameters. Ablation confirms the sentiment gate as the single most impactful component. © 2026 by the authors.

Date 2026**DOI** 10.3390/app16104994**URL** <https://www.scopus.com/pages/publications/105040133214?origin=resultslist>**Archive** Scopus**Language** English**Volume** 16**Publisher** Multidisciplinary Digital Publishing Institute (MDPI)**Publication** Applied Sciences (Switzerland)**ISSN** 20763417 (ISSN)**Issue** 10**Journal Abbr** Appl. Sci.**Date Added** 20/09/2026, 8:35:02 pm

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Tags:

Computational linguistics, Language model, Article, artificial intelligence, machine learning, Sentiment analysis, Social networking (online), Social media, demographics, Data mining, Semantics, ChatGPT, Contrastive Learning, social media, deep learning, Recurrent neural networks, artificial neural network, natural language processing, attention, Explainability, sentiment analysis, convolutional neural network, benchmarking, Chinese, k nearest neighbor, learning, contrastive learning, architecture, complementary context adaptive sentiment scoring, computer model, Contextual semantics, embedding, English (language), explainability, feeding, Gate levels, gated recurrent unit, Gated recurrent unit, gated recurrent unit network, hybrid, knowledge, language model, learning algorithm, lexicon neural fusion, Lexicon-based, lexicon–neural fusion, Lexicon–neural fusion, lifestyle, pre-trained language model, Pre-trained language model, sentiment aware gated unit, support vector machine, validation process, word processing, working memory

Notes:

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Sarcasm Detection in Roman Urdu Text: A Comprehensive Study Using Machine Learning and Large Language Model

Item Type Conference Paper

Author M. Hafeez

Author N. Hussain

Author A. Qasim

Author M. Zain

Author G. Mehak

Author O. Kolesnikova

Author G. Sidorov

Author A. Gelbukh

Editor Martínez-Villaseñor L.

Editor Vázquez R.A.

Editor Ochoa-Ruiz G.

Abstract Sarcasm detection is important in sentiment analysis and social media analysis, as the literal meaning does not agree with the filtered sentiment. This paper is specifically about sarcasm detection in the Roman Urdu language. The dataset was originally introduced for Urdu sarcasm detection and was alternatively

transliterated into Roman Urdu using systematic phonetic mapping. We evaluated twelve popular machine-learning models with GloVe and Word2Vec embeddings and fine-tuned several large language models (LLMs) such as LLaMA 2 (7B), LLaMA 3 (8B), and Mistral. Experimental results reveal that Gradient Boosting and Support Vector Machines showed the best performance with F1-scores of 96.62% and 95.32% respectively, using the GloVe-based embeddings. For the LLM model that reached the best result among all classifiers, it was LLaMA 3, which was also the best among all nine evaluated models LLaMA 3 with 97.15% F1-Score, followed by Mistral with 96.32%, then LLaMA 2 with 95.43%. This work highlights the potential of Roman Urdu for advanced sarcasm detection and compares the performance of traditional machine learning techniques with state-of-the-art large language models (LLMs). © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Proceedings Title Lect. Notes Comput. Sci.

Conference Name Lecture Notes in Computer Science

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Computational linguistics, Language model, Large language model, Performance, Support vector machines, Classification (of information), Machine-learning, Sentiment analysis, Machine learning, Language processing, Natural language processing, Natural Language Processing (NLP), Natural languages, Data mining, Embeddings, Adaptive boosting, Machine Learning (ML), Large Language Models (LLMs), F1 scores

Notes:

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Sarcasm Detection in the Era of AI: A Systematic Review of Techniques, Datasets, Performance, and Future Outlook

Item Type Conference Paper

Author F. Santosa

Author E. Utami

Author F.W. Wibowo

Abstract Automated sarcasm detection is a critical yet challenging task in Natural Language Processing (NLP), essential for accurately interpreting sentiment in user-generated content. Due to its implicit, often context-dependent nature, sarcasm requires models to understand nuances that go beyond literal meaning. This paper presents a systematic literature review of 57 peer-reviewed studies published between 2017 and 2025, offering a comprehensive analysis of the field. It examines the evolution of techniques from traditional machine learning to advanced deep learning, with a strong emphasis on Transformer-based and hybrid architectures. The review also explores the datasets commonly used, performance benchmarks, and the strengths and weaknesses reported across studies. Key challenges include limited contextual understanding, resource scarcity for low-resource languages, and the intrinsic ambiguity of sarcasm. Emerging opportunities lie in multimodal detection incorporating visual cues, explainable AI, and applications in areas like mental health and computational social science. By synthesizing current knowledge and identifying critical gaps, this review serves as a valuable resource for researchers and practitioners while outlining a clear research agenda for the future of sarcasm detection in the AI era. © 2025 IEEE.

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Conference Name International Conference on Computer, Control, Informatics and its Applications, IC3INA

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Deep learning, Performance, Deep neural networks, Learning systems, Sentiment analysis, Natural Language Processing, Language processing, Learning algorithms, Natural language processing, Natural languages, Neural-networks, Social sciences computing, Systematic Review, Deep Learning, Sentiment Analysis, User-generated, Sarcasm detection, Sarcasm Detection, Neural Network

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Export Date: 20 September 2026; Cited By: 0; Conference name: 12th International Conference on Computer, Control, Informatics and its Applications, IC3INA 2025; Conference date: 15 October 2025 through 16 October 2025; Conference code: 219332

Sarcasm-Aware Neural Translation for Informal Social Media Texts in Low-Resource Languages

Item Type Conference Paper**Author** S.B. Kurada**Author** A.M. Agrawal**Author** V. Gandamalla**Author** K. Goel**Author** B.A. Dhirwani**Author** P. Bansal**Author** G.C. Injarapu

Abstract The performance of integrated social media translators remains inconsistent when dealing with texts that contain sarcasm. One contributing factor is the informal nature of language commonly used on these platforms, which often confuses standard translation models. This study explores the challenge of interpreting sarcasm in machine translation, with a particular focus on the Telugu language. We propose fine-tuned transformer-based models for two tasks: interpreting sarcasm within English texts and translating sarcastic English into accurate Telugu. We evaluate both direct and indirect approaches one translating sarcastic English directly into honest Telugu, and the other converting sarcastic English into honest English before translating into Telugu. Our methodology combines natural language processing techniques for both sarcasm interpretation and translation, addressing a gap in existing literature that often overlooks the need for sarcasm explanation. Furthermore, we highlight the essential role of human annotation in overcoming persistent challenges in open class neural machine translation, especially for underrepresented languages. This work aims to contribute to the development of more effective models for handling sarcasm and related complexities in low-resource language translation. © 2025 IEEE.

Date 2025**DOI** 10.1109/ICACT67549.2025.11351510**URL** <https://www.scopus.com/pages/publications/105033620862?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: IEEE Int. Conf. Adv. Comput. Technol., ICACT**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-833159002-4 (ISBN)**Pages** 407-411**Proceedings Title** IEEE Int. Conf. Adv. Comput. Technol., ICACT

Conference Name 2025 IEEE International Conference on Advanced Computing Technologies, ICACT 2025

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Computational linguistics, Performance, Language processing, Natural language processing, Natural language processing systems, Natural languages, Transformer modeling, Social networking (online), Social media, Neural networks, natural language processing, Low resource languages, transformer models, Computer aided language translation, Machine translations, Neural machine translation, low resource languages, neural machine translation, Sarcasm interpretation, Telugu

Notes:

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Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 IEEE International Conference on Advanced Computing Technologies, ICACT 2025; Conference date: 1 September 2025 through 2 September 2025; Conference code: 219800

SAViOR: Sentiment, Sarcasm, Abuse, and Vulgarity in Online Realities (Memes)

Item Type Conference Paper

Author K. Tewari

Author S. Chanda

Author A. Namdeo

Author S. Pal

Editor Ghosh K.

Editor Mandl T.

Editor Pal S.

Editor Majumdar S.

Editor Chakraborty A.

Abstract Meme-based communication on social media often encodes sentiment, sarcasm, vulgarity, and abuse in subtle and context-dependent ways, posing significant challenges for automated detection systems. These challenges are further amplified in low-resource Indian languages, where code-mixing, transliteration, and the scarcity of annotated data complicate modeling efforts. As part of the HASOC-meme shared task at FIRE 2025, we developed and evaluated two independent runs under the team name IReL. Run 1 employed XLM-RoBERTa fine-tuned separately for each language (Bangla, Hindi, Gujarati, and Bodo), leveraging multilingual transformer embeddings to capture semantic nuances in noisy meme texts. Run 2 applied a zero-shot approach using ChatGPT specifically for Bodo, addressing the severe lack of annotated resources for this language. Both approaches processed text-based meme content exclusively and

were evaluated on the official leaderboard using the macro-F1 metric. Our systems achieved macro-F1 scores of 0.5082 (Bangla), 0.4630 (Hindi), 0.5920 (Gujarati), and 0.5011 (Bodo), demonstrating the efficacy of multilingual transformers for high-resource languages and the viability of zero-shot large language models for severely underrepresented languages. These results highlight the strengths of transformer-based architectures in handling linguistic diversity while exposing limitations in handling subtle contextual cues in memes. © 2025 Copyright for this paper by its authors.

Date 2025

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Language English

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Volume 4173

Publisher CEUR-WS

ISBN 16130073 (ISSN)

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Sentiment analysis, Social networking (online), Social media, Data mining, Semantics, Modeling languages, Sentiment Analysis, Social Media, Sarcasm detection, Sarcasm Detection, Context dependent, Indian languages, XLM-RoBERTa, Macroinvertebrates, Detection system, Abuse detection, Abuse Detection, Automated detection, Meme classification

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: K. Tewari; Indian Institute of Technology (BHU), Varanasi, India; email: krishnatewari.rs.cse24@itbhu.ac.in; Conference name: 2025 - Forum for Information Retrieval Evaluation; Conference date: 17 December 2025 through 20 December 2025; Conference code: 222801

SCaLAR NITK at Touché: Comparative Analysis of Machine Learning Models for Human Value Identification

Item Type Conference Paper

Author K. Praveen

Author R.K. Darshan

Author C.T. Reddy

Author M. Anand Kumar

Editor Faggioli G.

Editor Ferro N.

Editor Galuscakova P.

Editor de Herrera A.G.S.

Abstract This study delves into task of detecting human values in textual data by making use of Natural Language Processing (NLP) techniques. With the increasing use of social media and other platforms, there is an abundance in data that is generated. Finding human values in these text data will help us to understand and analyze human behavior in a better way, because these values are the core principle that influence human behavior. Analyzing these human values will help not only in research but also for practical applications such as sentiment evaluation, market analysis and personalized recommendation systems. The study tries to evaluate the performance of different existing models along with proposing novel techniques. Models used in this study range from simple machine learning model like SVM, KNN and Random Forest algorithms for classification using embeddings obtained from BERT till transformer models like BERT and RoBERTa for text classification and Large Language Models like Mistral-7b. The task that has be performed is a multilabel, multitask classification. QLoRA quantization method is used for reducing the size of weights of the model which makes it computationally less expensive for training and Supervised Fine Tuning (SFT) trainer is used for fine tuning LLMs for this specific task. It was found that LLMs performed better compared to all other models. © 2024 Copyright for this paper by its authors.

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Human behaviors, Machine learning, Natural language processing systems, Fine tuning, BERT, Adversarial machine learning, Contrastive Learning, Machine learning models, Random forests, Recommender systems, RoBERTa, Mistral, SVM, QLoRA, Human values, Human Values, SFT trainer, Supervised fine tuning trainer

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Scaling Public Health Text Annotation: Zero-Shot Learning vs. Crowdsourcing for Improved Efficiency and Labeling Accuracy

Item Type	Conference Paper
Author	K. Kazari
Author	Y. Chen
Author	Z. Shakeri
Abstract	Public health researchers are increasingly interested in using social media data to study health-related behaviors, but manually labeling this data can be labor-intensive and costly. This study explores whether zero-shot labeling using large language models (LLMs) can match or surpass conventional crowd-sourced annotation for Twitter posts related to sleep disorders, physical activity, and sedentary behavior. Multiple annotation pipelines were designed to compare labels produced by domain experts, crowd workers, and LLM-driven approaches under varied prompt-engineering strategies. Our findings indicate that LLMs can rival human performance in straightforward classification tasks and significantly reduce labeling time, yet their accuracy diminishes for tasks requiring more nuanced domain knowledge. These results clarify the trade-offs between automated scalability and human expertise, demonstrating conditions under which LLM-based labeling can be efficiently integrated into public health research without undermining label quality. © 2025 IEEE.
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URL	https://www.scopus.com/pages/publications/105023808798?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Proc. Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. EMBS
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	1557170X (ISSN); 979-833158618-8 (ISBN)
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Language model, human, Humans, Economic and social effects, machine learning, Crowdsourcing, Social networking (online), Social media, Machine Learning, social media, public health, Public health, procedures, Social Media, Labelings, Labour-intensive, information processing, Text annotations, Scalings, Public Health, crowdsourcing, Data Curation, Labeling accuracies, Physical activity, Sleep disorders, Twitter posts

Notes:

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SCRAG: Social Computing-Based Retrieval Augmented Generation for Community Response Forecasting in Social Media Environments

Item Type Conference Paper

Author D. Sun

Author Y. Lyu

Author J. Li

Author Y. Chen

Author T. Wang

Author T. Kimura

Author T. Abdelzaher

Abstract This paper introduces SCRAG, a prediction frame-work inspired by social computing, designed to forecast community responses to real or hypothetical social media posts. SCRAG can be used by public relations specialists (e.g., to craft messaging in ways that avoid unintended misinterpretations) or public figures and influencers (e.g., to anticipate social responses), among other applications related to public sentiment prediction, crisis management, and social what-if analysis. While large language models (LLMs) have achieved remarkable success in generating coherent and contextually rich text, their reliance on static training data and susceptibility to hallucinations limit their effectiveness at response forecasting in dynamic social media environments. SCRAG overcomes these challenges by integrating LLMs with a Retrieval-Augmented Generation (RAG) technique rooted in social computing. Specifically, our framework retrieves (i) historical responses from the target community to capture their ideological, semantic, and emotional makeup, and (ii) external knowledge from sources such as news articles to inject time-sensitive context. This information is then jointly used to forecast the responses of the target community to new posts or narratives. Extensive experiments across six scenarios on the X platform

(formerly Twitter), tested with various embedding models and LLMs, demonstrate over 10% improvements on average in key evaluation metrics. A concrete example further shows its effectiveness in capturing diverse ideologies and nuances. Our work provides a social computing tool for applications where accurate and concrete insights into community responses are crucial. © 2025 IEEE.

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Self-Reported Health Outcomes in Metabolic Health YouTube Comments: Cross-Sectional Study and Rule-Based Natural Language Processing Framework Development and Validation

Item Type Journal Article

Author R. Ribeiro

Author A. Zutshi

Abstract Background: YouTube is increasingly used for healthcasting, the sharing of evidence-based dietary and lifestyle interventions by domain experts. In the metabolic health domain, channels focused on therapeutic carbohydrate restriction have accumulated audiences of millions. A distinctive feature is the comment section, where viewers share first-person accounts of health changes, constituting a unique source of real-world outcome data at scale. However, extracting structured health information from unstructured comments presents computational challenges. Objective: This observational, cross-sectional study aims to develop and validate a precision-optimized computational framework for extracting self-reported health outcomes from healthcasting YouTube comments and to characterize the prevalence, distribution across health aspects, and channel-level variation of reported outcomes across a large-scale metabolic health corpus. Methods: This study analyzed 43,111 unique YouTube comments from 110 videos across 11 therapeutic carbohydrate restriction-focused healthcasting channels (37,458 unique authors; data span November 2013 to January 2026; collected via YouTube data application programming interface version 3). The methodology comprised 3 construction phases and 5 validation studies. The construction phases were (1) exploratory corpus characterization, (2) iterative development of a 35-aspect hierarchical health outcome ontology, and (3) precision-optimized rule-based classification, validated through precision validation (stratified sample of $n=500$), recall estimation ($n=510$), external validation on 5 held-out channels ($n=12,653$ comments), large language model-assisted interrater reliability assessment, and transformer baseline comparison against Bidirectional Encoder Representations from Transformers (BERT) and Robustly Optimized BERT Pretraining Approach (ROBERTa) classifiers. A supplementary aspect-based sentiment analysis contextualized the positive-only design. Results: The framework identified 1790 positive health outcome reports (1790/43,111, 4.15% prevalence), achieving 97.6% (488/500) precision (95% CI 95.7%-98.6%) and estimated 56.2% recall (95% CI 43.4%-67.9%). The reports described 6674 positive outcomes, distributed across 35 health aspects and 18 named disease conditions extending beyond weight loss: pain and inflammation reduction (1137/6674, 17%), type 2 diabetes improvement (977/6674, 14.6%), skin health (784/6674, 11.8%), and psychological well-being (731/6674, 11%). Over half (3355/6674, 50.3%) spanned multiple research objectives. Significant channel-level variation was observed ($\chi^2_{210}=927.5$; $P < .001$), with positive outcome rates ranging from 1.32% to 10.40% (odds ratio 8.68, 95% CI 7.10-10.61). Transformer baselines achieved higher recall but lower precision, confirming their advantage for high-confidence corpus generation. A supplementary aspect-based sentiment analysis indicated a positive-to-negative ratio of approximately 4.6:1 ($n=1003$), with negative experiences (59/495, 11.9%) predominantly involving gastrointestinal and cardiovascular concerns. Conclusions: This study presents, to our knowledge, the first validated, rule-based framework for extracting self-reported metabolic health outcomes from healthcasting YouTube comments at corpus scale. Unlike existing recall-oriented social media health classifiers, the precision-optimized design achieves the confidence threshold required for outcomes research without manual review. These findings demonstrate that expert-led health content comment sections constitute a scalable, complementary data source for monitoring real-world engagement with dietary interventions, with implications for public health surveillance, platform design, and health communication research. © Ricardo Ribeiro, Aneesh Zutshi.

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Self-supervised contrastive learning for robust multilingual NLP in cross-border social media analysis

Item Type Journal Article

Author A. Alghamdi

Abstract Social media platforms have experienced rapid growth which creates a massive amount of multilingual cross-border text data. This data presents major difficulties to natural language processing because it includes multiple languages and informal speech patterns. This paper presents a self-supervised contrastive learning system which enables multilingual sentiment analysis by using pretrained multilingual transformer models to create language-neutral understanding of text. The proposed framework combines text augmentation methods with contrastive learning goals to create a system which detects semantic relationships between languages. The proposed framework generates positive sample pairs through text augmentation

methods while it obtains negative samples from tweets that do not share any semantic relationship to other tweets. The proposed framework establishes multilingual embedding connections through these sample pairs. The proposed framework uses a linear softmax classifier to evaluate its learned representations which it trained to classify sentiment. The proposed model achieved an accuracy of 91% with a 90% precision rate and an 89% recall rate which resulted in an 89.5% F1-score according to experiments conducted in various languages. The results demonstrate that contrastive representation learning effectively captures language-independent semantic features, which enhance the proposed framework's ability to generalize across multiple languages and code-mixed social media content. The performance improvements occur because This approach learns strong embedding representations from its unlabelled training data while its ability to process noise decreased. The proposed framework displays strong performance across mixed-language environments and noisy social media data through additional evaluations. © The Author(s), under exclusive licence to Springer Nature Switzerland AG 2026.

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SEM-PDPL: Semantic Exposure Graphs for Privacy-Law-Informed Risk Assessment of Public Social-Media Data

Item Type Journal Article

Author H. Ismail

Abstract Public social-media content often contains self-disclosed personal attributes that appear low-risk in isolation but become privacy-relevant when linked across posts, platform accounts, or user-level traces. Existing research has advanced privacy-sensitive content detection, de-anonymization analysis, social-media research ethics, and privacy-compliance workflows; however, limited work operationalizes how personal-data disclosures combine structurally and how these structures can be translated into auditable governance actions. This paper proposes SEM-PDPL, a computational, privacy-law-informed risk-assessment framework for modeling public social-media exposure as semantic exposure graphs and mapping graph patterns to controls aligned with the United Arab Emirates Personal Data Protection Law (PDPL) and compatible with GDPR principles. SEM-PDPL combines governance scoping; a PDPL-informed disclosure taxonomy; hybrid extraction using rule-based methods; named-entity recognition; fine-tuned BERT; and schema-constrained large language model annotation, followed by graph construction at post, platform, corpus, and persona levels. The framework is evaluated on a synthetic multi-platform corpus of 1095 posts generated for 150 personas across 290 platform accounts. Results show that, within this controlled synthetic corpus, fine-tuned BERT provides the strongest extraction performance among six evaluated methods, achieving a macro-F1 of 0.975. Graph analysis shows that exposure density increases with aggregation, rising from 0.275 at post level to 1.000 at corpus level, and from 0.859 at platform level to 0.967 at persona level. Across all graph resolutions, quasi-identifiers emerge as the dominant weighted-degree and betweenness node, indicating that ordinary location, employer, school, and demographic cues often function as bridges connecting sensitive categories such as health and biometric data to identifying information. These findings indicate that, within this controlled corpus, privacy risk in public social-media data is not only attribute-based but also structurally graph-shaped. SEM-PDPL contributes an explainable and reproducible framework for identifying exposure hubs, sensitive bridges, and aggregation risks before applying masking, minimization, exclusion, retention, or review controls. The framework does not automate legal compliance; rather, it provides evidence-based decision support for privacy-aware social-media analytics. © 2026 by the author.

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Semantic-Aware Youtube Trend Analysis Using Large Language Model-Based Intelligence Engines

Item Type Conference Paper

Author T. Sharma

Author A. Vajpayee

Abstract The fast increase in user-generated videos on YouTube has complicated trend identification and content intelligence, which are still more and more difficult due to the high volume of data, multimodal signals, and quickly changing viewer behavior. Conventional analytics techniques that mainly rely on statistical measures like the number of views, likes, and watch time are incapable of revealing the deeper semantic, contextual, and temporal dynamics that are at the core of viral trends. To overcome these shortcomings, this article introduces an LLM-powered YouTube Intelligence Engine intended for the automated analysis and discovery of trend videos. The suggested model combines large language models with multimodal data sources such as video metadata, transcripts, comments, engagement patterns, and temporal signals to understand semantic trends, provide context, and make predictions. The platform utilizes topic clustering based on LLM rationale, sentiment interpretation, narrative development tracking, and assessment of trend momentum, thus enabling a better, more explainable, and also more adaptable analysis of the latest YouTube trends. Evaluation through experiments on actual YouTube data sets has shown that the proposed method is superior to existing machine learning and rule-based analytics in terms of trend detection, accuracy, interpretability, and the capability of identifying trends at a very early stage. The findings suggest that LLM-based intelligence engines have a great promise for allowing content creators, marketers, researchers, and platform analysts to make data-driven decisions and rapidly optimize their strategies for content-wise. © 2026 IEEE.

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SentiGOLD: A Large Bangla Gold Standard Multi-Domain Sentiment Analysis Dataset and Its Evaluation

Item Type Conference Paper
Author M.E. Islam
Author L. Chowdhury
Author F.A. Khan
Author S. Hossain
Author M.S. Hossain
Author M.M.O. Rashid
Author N. Mohammed
Author M.R. Amin

Abstract In this study, we present a Bangla multi-domain sentiment analysis dataset, named as SentiGOLD, developed using 70,000 samples, which was compiled from a variety of sources and annotated by a gender-balanced team of linguists. This dataset was created in accordance with a standard set of linguistic conventions that were established after multiple meetings between the Government of Bangladesh and a nationally recognized Bangla linguistics committee. Although there are standard sentiment analysis datasets available for English and other rich languages, there are not any such datasets in Bangla, especially because, there was no standard linguistics framework agreed upon by national stakeholders. Senti-GOLD derives its raw data from online video comments, social media posts and comments, blog posts and comments, news and numerous other sources. Throughout the development of this dataset, domain distribution and class distribution were rigorously maintained. SentiGOLD was created using data from a total of 30 domains (e.g. politics, entertainment, sports, etc.) and was labeled using 5 classes (e.g. strongly negative, weakly negative, neutral, weakly positive, and strongly positive). In order to maintain annotation quality, the national linguistics committee approved an annotation scheme to ensure a rigorous Inter Annotator Agreement (IAA) in a multi-annotator annotation scenario. This procedure yielded an IAA score of 0.88 using Fleiss' kappa method, which is elaborated upon in the paper. A protocol for intra- and cross-dataset evaluation was utilized in our efforts to develop a classification system as a standard. The cross-dataset evaluation was performed on the SentNoB dataset, which contains noisy Bangla text samples, thereby establishing a demanding test scenario. We also performed cross-dataset testing by employing zero-shot experiments, and our best model produced competitive performance, which exemplify our dataset's generalizability. Our top model attained a macro f1 of 0.62 (intra-dataset) for 5 classes establishing the benchmark for SentiGOLD, and 0.61 (cross-dataset from SentNoB) for 3 classes which stands comparable to the current state-of-the-art. Our fine-tuned sentiment analysis model\footnote{https://sentiment.bangla.gov.bd} can be accessed online. © 2023 ACM.

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Sentiment Analysis and Extraction of Recommendations in Cultural and Tourism Contexts

Item Type Conference Paper

Author N. Gkarpis

Author G. Trichopoulos

Author G. Caridakis

Abstract This research aims to explore the application of Aspect-Based Sentiment Analysis (ABSA) using Large Language Models (LLMs) on user-generated content related to culture and tourism. We propose a pipeline that collects and preprocesses review data, extracts domain-specific aspects (e.g., accommodation, service, cultural experience) and identifies sentiment polarity per aspect using LLMs. The results will be structured in machine-readable formats (e.g., JSON) and evaluated against baseline models such as BERT. The goal is to assess the effectiveness of LLMs in fine-grained sentiment understanding and their potential use in intelligent recommendation systems for tourism and heritage sectors. © 2025 IEEE.

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Sentiment Analysis During Covid-19 Pandemic: A Comparative Study

Item Type Conference Paper

Author A. Bhattacharya

Author A. Majumdar

Author S. Kanjilal

Author S.S. Bhowmick

Editor Das N.

Editor Ozyirmidokuz E.K.

Editor Mandal S.

Editor Özbakir L.

Editor Bhattacharjee D.

Abstract In recent times, social media comments and tweets have proven to be a rich resource for the understanding of people's opinions. The research done in this domain mainly focuses on sentiment analysis of tweets for simplistic multi-class classification. Conventional techniques are used to determine the polarity of sentiment namely positive, negative, and neutral. In this paper, we conduct a comprehensive comparative analysis of various models that are used to perform sentiment analysis, focusing primarily on tweets during the pandemic period of COVID-19 for multi-label emotion classification and emotion detection. Our work focuses on performing sentiment analysis on multi-label multi-class tweets taken from Twitter by using different models like LSTM, Naive Bayes, ANNs, and Transformers. The aim of this study is to give a comparative viewpoint of the various machine learning paradigms for multi-label classification for sentiment analysis. Different pre-processing techniques like GloVe, Word2Vec, CountTokenizer and Transformers as well as standalone Transformer models have been used and the results have been compared based on the model's classification performance and precision as evaluation metrics. Our survey shows that pre-

trained transformer based Large Language Models have performed the best with a maximum precision score of 71% during classification. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Sentiment analysis for code-mixed low-resource languages: a systematic review of approaches, techniques, applications, challenges, and future directions

Item Type Journal Article

Author M.K. Nazir

Author M. Bilal

Author S.C. Shongwe

Abstract With the rapid proliferation of user-generated content across internet-based platforms, sentiment analysis has become an essential tool for understanding public opinion. However, sentiment analysis on code-mixed, low-resource languages presents significant challenges due to limited annotated data, linguistic complexity, and resource constraints. This systematic literature review (SLR) comprehensively examines recent developments in sentiment analysis for code-mixed and low-resource languages, with a focus on studies published between 2017 and 2025. The review investigates a wide range of approaches, including deep learning models, transfer learning strategies, and pre-trained language models, evaluating their effectiveness in low-resource and code-mixed contexts. Findings indicate that transfer learning and transformer-based architectures are increasingly preferred due to their superior performance and reduced reliance on large annotated datasets. Social media is identified as the dominant data source, followed by reviews from e-commerce and entertainment domains. Despite these advancements, significant challenges persist, including the scarcity of high-quality labeled data, high computational costs, and limited generalization across languages and domains. This review not only synthesizes current methodologies and applications but also proposes a conceptual framework for future research. It offers actionable insights for scholars seeking to develop efficient, scalable sentiment analysis systems tailored to code-mixed and low-resource language scenarios. © The Author(s) 2026.

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Sentiment Analysis for Egyptian Arabic-English Code-Switched Data Using Traditional Neural Models and Advanced Language Models

Item Type	Conference Paper
Author	A. Sherif
Author	C. Sabty
Editor	Karpov A.
Editor	Delić V.
Abstract	As code-switching on social media becomes more common, the amount of bilingual users and code-switched data is rapidly increasing. This presents unique challenges for sentiment analysis, which has traditionally focused on monolingual text. Sentiment analysis involves categorizing sentiments from comments, reviews, or tweets into positive, negative, or neutral. To address the scarcity of research in code-switched sentiment analysis, this paper contributes the first code-switched corpus for Egyptian Arabic-English sentiment analysis (EESA), featuring 4,100 annotated YouTube comments. Another contribution is the implementation of various sentiment analysis models, including traditional neural models (BiLSTM-Attention and Hybrid-Transformer) and the utilization of advanced language models (Gemini and GPT). Traditional models, using non-contextual, contextual, and character embeddings with additional word features, achieved an ensemble test F1-score of 92.54%. Advanced models, evaluated in zero-shot and fine-tuned configurations, showed significant potential. Gemini-1.5 and GPT-4o performed well as zero-shot models, and the fine-tuned GPT-3.5 model achieved the highest test F1-score of 92.67%. This work compares traditional and advanced models, highlighting advancements in code-switched sentiment analysis. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.
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Sentiment Analysis for Low-Resource Languages Using a Hybrid XLM-Roberta and Bi-LSTM Model

Item Type Conference Paper

Author G. Bharathi Mohan

Author V. Sampreetha

Author K.U. Meghana

Abstract In this study, we tackle the challenges of sentiment analysis in low-resource languages by analyzing social media datasets with natural language processing (NLP). Previous research has highlighted several limitations, including difficulties in managing informal language, limited use of emoticons, and accuracy limits in traditional models such as CNN and LSTM. Furthermore, the absence of fine-tuning for large-language models has hindered performance. Issues such as semantic complexity, translation errors, and generally poor outcomes in low-resource languages have also persisted. To address these challenges, we introduce a hybrid LLM model that combines the strengths of XLM-R and Bi-LSTM, which has been proven to deliver more accurate results compared to traditional approaches. By fine-tuning these large language models for better adaptation to specific contexts and streamlining the complexity of our methods, particularly in low-resource settings. Our findings classify sentiments into distinct categories such as Not offensive, Offensive Untargeted, and various forms of targeted insults. Ultimately, this study not only contributes to the design of effective sentiment analysis frameworks using hybrid LLM models but also discusses their implications for business and management. Improved sentiment analysis can provide valuable insights that inform decision-making and enhance customer engagement strategies, making it a vital tool in today's data-driven world. We significantly improve multilingual performance with an accuracy of 62%. © 2025 IEEE.

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Sentiment Analysis from Kannada Text

Item Type Conference Paper

Author L.S. Badiger

Author S.B. Donkanavar

Author S. Chikkamath

Editor Fong S.

Editor Dey N.

Editor Joshi A.

Abstract Sentiment analysis is a vital task in NLP. That identifies the emotions in the text. Many studies concentrate only on the English language. There is an insufficient resource in Kannada language emotion analysis. This paper examines the sentiment in Kannada text using a manually prepared dataset. The datasets are divided into three classes positive, negative, and neutral. The processing techniques like context cleaning, tokenization, and sequence padding are used. The model uses RNN-LSTM which is efficient in handling the sequential data. The embedding layer is used to represent the words, the LSTM layer is used to get the context of the sentence, the Dropout layer is used to reduce the over fitting of the model and the dense layer is used to classify the sentiments into categories.

The effectiveness of the model was measured using evaluation metrics like precision, recall and f1 score. By predicting sentiments for Kannada text, the paper also exhibits practical use of the model. This study demonstrates that LSTM-based models work well for sentiment analysis in Kannada. It also emphasizes the importance of creating and using manual datasets for low resource languages. The findings would be helpful to further research, and the outcomes can be directly applied in many areas, including social media monitoring, customer feedback analysis, and regional language processing. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: L.S. Badiger; School of Electronics and Communication Engineering, KLE Technological University, Hubli, India; email: badigerl387@gmail.com; Conference name: 10th International Conference on ICT for Sustainable Development, ICT4SD 2025; Conference date: 17 July 2025 through 19 July 2025; Conference code: 341139

Sentiment Analysis in African Languages: Evaluating Generative AI and Afrocentric Language Models

Item Type Conference Paper

Author K.R. Mabokela

Author T. Celik

Author M. Primus

Editor Gerber A.

Abstract Sentiment analysis for low-resource African languages such as Sepedi, Sesotho, and Setswana (Sotho-Tswana family) and isiZulu and isiXhosa (Nguni family) remains underexplored due to data scarcity and linguistic complexities. This study investigates the application of generative AI models—including large multilingual language models (e.g., GPT-4, BLOOM) and Afrocentric language models (AfroLM and AfroXLMR)—to sentiment classification in these languages. The study leverages a multilingual Twitter corpus covering five African languages and fine-tuned models using cross-lingual and language-specific strategies. The results show that Afrocentric transformer-based models achieve the best performance (e.g., Afro-XLMR reaching over 75% F1 score on average) for sentiment detection in these languages, outperforming other multilingual language models. GPT-4 demonstrates a reasonable zero-shot learning accuracy but still lags behind fine-tuned domain-specific models. For instance, the fine-tuned Afro-XLMR model achieved over 75% F1-score ($\approx 80\%$ accuracy) on isiZulu and isiXhosa, whereas GPT-4 reached only about 52–55% accuracy in zero-shot mode. The study investigated how the complex word structures (morphological richness) and the mixing of languages (code-switching) in the Nguni and Sotho-Tswana language families, especially with English, create significant difficulties. These results suggest that generative AI holds a potential promise for sentiment analysis within under-resourced African languages. Finally, this work emphasises the value of creating language-specific models and proposes future research directions for natural language processing (NLP) in African languages. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: K.R. Mabokela; Applied Information Systems, University of Johannesburg, Johannesburg, South Africa; email: krmabokela@gmail.com; Conference name: 46th Annual Conference of the South African Institute of Computer Scientists and Information Technologists, SAICSIT 2025; Conference date: 17 July 2025 through 18 July 2025; Conference code: 335669

Sentiment Analysis in Assamese: Unveiling Emotions in Native Texts

Item Type Conference Paper

Author D. Sarkar

Author S.S. Ahmed

Editor Nayak S.K.

Editor Kalita P.

Editor Changmai P.

Abstract Sentiment Analysis (SA) refers to the process of detecting and classifying opinions conveyed within a given piece of text, audio, or other forms of communication, which is one of the main sub-field of Natural Language Processing (NLP). There is numerous amount of SA task in the language like English. Because of the rapid growth of social media, the use of various regional languages from different regions of India is increasing dramatically. Nowadays, people often prefer to leave comments or reviews in their own language. So, the Sentiment analysis in regional language is becoming very important. Due to the unavailability of resources, a very little amount of sentiment analysis task is done in low-resource language like Assamese. Our primary objective is to develop an Assamese-language dataset for sentiment analysis and to analyze different ML (Machine Learning) classifier on the dataset. We have created a dataset of 8000 sentences and performance of the classifiers such as Logistic Regression classifier (LR), SVC, MultinomialNB, Decision Tree classifier(DT), Random Forest classifier (RF), Extra Tree Classifier on the dataset are evaluated. The SVC (Support Vector Classification) outperformed all other classifiers, achieving an F1 score of 67.9%. Given the absence of various sentiment analysis tools, such as POS taggers, in low-resource languages, there is significant potential for improvement in this area. Developing and enhancing these tools could greatly benefit the processing and analysis of text in these languages, making sentiment analysis more accurate and accessible. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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Sentiment Analysis in Low-Resource Settings: A Comprehensive Review of Approaches, Languages, and Data Sources

Item Type Journal Article

Author Y. Aliyu

Author A. Sarlan

Author K. Usman Danyaro

Author A.S.B.A. Rahman

Author M. Abdullahi

Abstract The field of low-resource sentiment analysis has seen significant developments in recent years. This research review SLR evaluates the approaches and data sources utilized in low-resource sentiment analysis by deep learning. The primary aim is to discover suitable approaches for future sentiment analysis in low-resource. Our studies explore various languages, models, and data sources expressing a desire to create effective approaches. Our emphasis lies in the critical evaluation of the approaches and the datasets utilized, to identify areas where further research is needed. Our analysis study adds to the existing body of literature reviews, encompassing multilingual low-resource sentiment analysis research spanning from 2018 to 2023. The findings indicate that the transfer learning approach is the most frequently used, followed by word embedding learning and machine translation systems. Additionally, the study shows that social media is the most used platform for data collection, followed by product reviews, movies, and hotels. There has been a significant surge in the adoption of pre-trained transformers, indicating a growing interest in exploring the potential of these models for low-resource languages within the natural language processing (NLP) community. This trend is largely attributed to the novel nature of these models and their feature of being non-labour intensive.

However, the scarcity of annotated datasets for such languages remains a major hurdle. finally, these research findings are relevant and informative for any researcher working in the field of low-resource multilingual sentiment analysis. The study introduces a conceptual framework for performing sentiment analysis in low-resource. The study provides a valuable resource for future researchers. © 2013 IEEE.

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Sentiment Analysis in Mexican Spanish: A Comparison Between Fine-Tuning and In-Context Learning with Large Language Models

Item Type Journal Article

Author T. Bernal-Beltrán

Author M.A. Paredes-Valverde

Author M.D.P. Salas-Zárate

Author J.A. García-Díaz

Author R. Valencia-García

Abstract The proliferation of social media has made Sentiment Analysis an essential tool for understanding user opinions, particularly in underrepresented language variants such as Mexican Spanish. Recent advances in Large Language Models have made effective sentiment analysis through in-context learning techniques, reducing the need for supervised training. This study compares the performance of zero and few-shot with traditional fine-tuning approaches of tourism-related texts in Mexican Spanish. Two annotated datasets from the REST-MEX 2022 and 2023 shared tasks were used for this purpose. Results show that fine-tuning, particularly with the MarIA model, achieves the best overall performance. However, modern LLMs that use in-context learning strategies, such as Mixtral 8x7B for zero-shot and Mistral 7B for few-shot, demonstrate strong potential in low-resource settings by closely approximating the accuracy of fine-tuned models, suggesting that in-context learning is a viable alternative to fine-tuning for sentiment analysis in Mexican Spanish when labeled data is limited. These approaches can enable intelligent, data-driven digital services with applications in tourism platforms and urban information systems that enhance user experience and trust in large-scale socio-technical ecosystems. © 2025 by the authors.

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Sentiment analysis in public health: a systematic review of the current state, challenges, and future directions

Item Type Journal Article

Author I. Villanueva-Miranda

Author Y. Xie

Author G. Xiao

Abstract Introduction: Sentiment analysis, using natural language processing to understand opinions in text, is increasingly relevant for public health given the volume of online health discussions. Effectively using this approach requires understanding its methods, applications, and limitations. This systematic review provides a comprehensive overview of sentiment analysis in public health, examining methodologies, applications, data sources, challenges, evaluation practices, and ethical considerations. Methods: We conducted a systematic review following PRISMA guidelines, searching academic databases through Semantic Scholar and screening studies for relevance. A total of 83 papers analyzing the use of sentiment analysis in public health contexts were included. Results: The review identified a trend toward the use of advanced deep learning methods and large language models (LLMs) for a wide range of public health applications. However, challenges remain, particularly related to interpretability and resource demands. Social media is the predominant data source, which raises concerns about data quality, bias, linguistic complexity, and ethical issues. Discussion: Sentiment analysis offers the potential for gaining public health insights but faces significant methodological, data-related, and ethical challenges. Reliable and ethical application demands rigorous validation, improved model interpretability, the development of ethical frameworks, and continued research to support responsible development and deployment. Copyright © 2025 Villanueva-Miranda, Xie and Xiao.

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Sentiment Analysis of Arabic Social Media Comments Related to the 2023 Moroccan Earthquake Using Machine Learning Approaches

Item Type Conference Paper

Author A.A. Ichou

Author A. Ouacha

Editor Abdellaoui Alaoui E.A.

Editor Merras M.

Editor Nayyar A.

Abstract Sentiment analysis plays an important role in measuring public's emotions and social media moods using affective tone evaluation of textual material. Sentiment analysis of text in Arabic is particularly challenging due to the diverse morphology of the language, right-to-left writing system, and dialectal variabilities. It is postings from Arabic social media regarding the 2023 Moroccan earthquake that are the subject matter of this research. We collected a large corpus of social media tweets, conducted preprocessing, and applied common machine learning models coupled with neural networks for sentiment analysis. Our results indicate that common models, particularly the Support Vector Classifier (SVC), are superior to deep learning techniques. These findings indicate the effectiveness of machine learning in Arabic sentiment analysis and contribute to the creation of Natural Language Processing (NLP) solutions for low-resource languages. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Pages 301-305**Proceedings Title** Adv. Sci. Tech. Inno.**Conference Name** Advances in Science, Technology and Innovation**Date Added** 20/09/2026, 8:35:03 pm**Modified** 20/09/2026, 8:35:03 pm**Tags:**

Sentiment analysis, Social media analytics, Arabic NLP

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Sentiment Analysis of ChatGPT on Indonesian Text using Hybrid CNN and Bi-LSTM

Item Type Journal Article**Author** V.R. Prasetyo**Author** M.F. Naufal**Author** K. Wijaya

Abstract This study explores sentiment analysis on Indonesian text using a hybrid deep learning approach that combines Convolutional Neural Networks (CNN) and Bidirectional Long Short-Term Memory (Bi-LSTM). Due to the complex linguistic structure of the Indonesian language, sentiment classification remains challenging, necessitating advanced methods to capture both local patterns and sequential dependencies. The primary objective of this research is to improve sentiment classification accuracy by leveraging a hybrid model that integrates CNN for feature extraction and Bi-LSTM for contextual understanding. The dataset consists of 800 manually labeled samples collected from social media platforms, preprocessed using case folding, stop word removal, and lemmatization. Word embeddings are generated using the Word2Vec CBOW model, and the classification model is trained using a hybrid architecture. The best performance was achieved with 32 Bi-LSTM units, a dropout rate of 0.5, and L2 regularization, which was evaluated using Stratified K-Fold cross-validation. Experimental results demonstrate that the hybrid model outperforms conventional deep learning approaches, achieving 95.24% accuracy, 95.09% precision, 95.15% recall, and 95.99% F1 score. These findings highlight the effectiveness of hybrid architectures in sentiment analysis for low-resource languages. Future work may explore larger datasets or transfer learning to enhance generalizability. © 2025, Ikatan Ahli Informatika Indonesia. All rights reserved.

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Sentiment Analysis of ChatGPT Tweets Using Machine Learning Techniques

Item Type Conference Paper

Author W.S. Ismail

Editor Hachimi H.

Editor Abdo A.A.

Editor Benmamoun Z.

Abstract Large language models like ChatGPT by OpenAI have gained significant attention for their impressive performance across various tasks. They have become highly popular among adopters. Gaining an understanding of the sentiment of these adopters is crucial, as it can offer valuable information about the potential outcomes of the technology, whether it will thrive or falter, and the advantages and drawbacks it possesses. In this research, data mining techniques are utilized to analyze Twitter data and automatically categorize their sentiment polarity. A dataset of 15,000 English tweets is gathered in natural language from ChatGPT users, and text mining and data mining techniques are applied to convert them into a useful format. The objective is to build a sentiment classifier capable of predicting whether a tweet conveys positive, negative, or neutral sentiments. To achieve this, we utilize two widely used data mining classifiers: K-Nearest Neighbor and Naive Bayes. These classifiers are applied to analyze the sentiment of the tweets, and their overall accuracy, precision, and recall values are assessed, providing a comprehensive comparison between the two methods. © 2023 IEEE.

Date 2023**DOI** 10.1109/ICOA58279.2023.10308835**URL** <https://www.scopus.com/pages/publications/85179131399?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: Int. Conf. Optim. Appl., ICOA - Proc.**Publisher** Institute of Electrical and Electronics Engineers Inc.**ISBN** 979-835031254-6 (ISBN)**Proceedings Title** Int. Conf. Optim. Appl., ICOA - Proc.**Conference Name** 2023 9th International Conference on Optimization and Applications, ICOA 2023 - Proceedings**Date Added** 20/09/2026, 8:35:25 pm**Modified** 20/09/2026, 8:35:25 pm**Tags:**

Language model, Performance, Nearest neighbor search, Sentiment analysis, Machine learning, Natural Language Processing, Language processing, Learning algorithms, Natural language processing, Natural languages, Social networking (online), Data mining, Machine learning techniques, ChatGPT, Twitter, Sentiment Analysis, data mining, Data-mining techniques

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Sentiment analysis of classical Chinese literature: An unsupervised deep learning model with BERT and graph attention networks

Item Type Journal Article**Author** X. Yu**Author** J. Wang

Abstract Sentiment analysis has become a transformative technology in various contexts, particularly in Natural Language Processing (NLP), social media analytics, and literary analysis, as it can extract information from a wide range of texts. The advancements in deep learning, particularly with transformer models such as BERT and graph-based models like GATs, have enabled faster progress in analyzing complex language structures. However, the issue lies in incorporating these technologies into classical Chinese literature, which involves delicate syntax, semantics, and emotions that are difficult to harness using traditional methods. The existing methods, which rely on strictly labeled data or unsupervised learning

methods that do not effectively manage contextual dependencies, are very limited in analyzing historical or philosophical texts that abound in metaphor and implicit sentiment. To minimize the limitations, this paper proposes an unsupervised deep learning framework that integrates BERT embeddings, sentiment lexicon enrichment, and graph attention networks (GATs) for sentiment analysis in classical Chinese literature. Firstly, the BERT-based model extracts contextualised embeddings from a raw text, providing a deep understanding of semantics. Secondly, embedding includes sentiment-specific data from the NTUSD lexicon, thus injecting it with emotional information. Thirdly, a graph-based formulation is developed, in which words are represented as nodes, and the relations between them are defined using GATs to modify the features of nodes based on their significance in the context. Finally, unsupervised sentiment labelling, or K-Means clustering, is used to classify sentiment. The experimental results demonstrate the proposed model's efficiency – an accuracy of 0.95, precision of 0.97, recall of 0.96, and F1-score of 0.91 in several runs. These results surpass those of the traditional approach, which includes SentiCNN, MLT-ML4, and BERT-LLSTM-DL, which achieve an accuracy score of 0.90 to 0.95. Additionally, the comparison with large-scale foundation models (such as ChatGPT-4o and DeepSeek R1) in zero-shot prompt-based classification further validates the domain-adapted advantage of our model in the classical Chinese text processing. These results demonstrate that the proposed model significantly enhances the handling of the intricate linguistic features and cultural nuances in classical Chinese texts, providing a robust solution for sentiment analysis in low-resource domains. © 2025 Yu, Wang. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Sentiment Analysis of Code-Mixed Telugu-English Data Leveraging Syllable and Word Embeddings

Item Type Journal Article

Author U.R. Rayala

Author K. Seshadri

Author N.B. Sristy

Abstract Learning the inherent meaning of a word in Natural Language Processing (NLP) has motivated researchers to represent a word at various levels of abstraction, namely character-level, morpheme-level, and subword-level vector representations. Syllable-Aware Word Embedding (SAWE) can effectively handle agglutinative and fusion-based NLP tasks. However, research attempts on assessing the SAWE on such extrinsic NLP tasks has been scanty, especially for low-resource languages in the context of code-mixing with English. A model to learn SAWE to extract semantics at fine-grained subunits of a word is proposed in this article, and the representative ability of the embeddings is assessed through sentiment analysis of code-mixed Telugu-English review corpora. Multilingual societies and advancements in communication technologies have accounted for the prolific usage of mixed data, which renders the State-of-the-Art (SOTA) sentiment analysis models developed based on monolingual data ineffective. Social media users in the Indian subcontinent exhibit a tendency to mix English and their respective native language (using the phonetic form of English) in expressing their opinions or sentiments. A code-mixing scenario provides flexibility to borrow words from a foreign language, usage of shorthand notations, elongation of vowels, and usage of words without following syntactic/grammatical rules, which renders the sentiment analysis of code-mixed data challenging to perform. Deep neural architectures like Long Short-Term Memory and Gated Recurrent Unit networks have been shown to be effective in solving several NLP tasks, such as sequence labeling, named entity recognition, and machine translation. In this article, a framework to perform sentiment analysis on a code-mixed Telugu-English review corpus is implemented. Both word embedding and SAWE are input to a unified deep neural network that contains a two-level Bidirectional Long Short-Term Memory/Gated Recurrent Unit network with Softmax as the output layer. The proposed model leverages the advantages of both word embedding and SAWE, which enable the proposed model to outperform existing SOTA code-mixed sentiment analysis models on a Telugu-English code-mixed dataset of the International Institute of Information Technology-Hyderabad and a dataset curated by the authors. The improvement realized by the proposed model on these datasets is [3% increase in F1-score and 2% increase in accuracy] and [7% increase in F1-score and 5% in accuracy], respectively, in comparison with the best-performing SOTA model. © 2023 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Sentiment Analysis of Code-Switched Taglish Telecom Feedback Using Multilingual BERT

Item Type Conference Paper

Author A.D.A. Carillo

Author J.C. Bustillo

Abstract The increasing use of code-switched Taglish in social media presents significant challenges for sentiment analysis due to its mixed-language structure, informal expressions, and context-dependent meanings. Despite recent advances in multilingual natural language processing, limited studies have investigated sentiment classification of Taglish telecommunications-related feedback, particularly in social media environments where customer opinions are frequently expressed through code-switching. This study addresses by evaluating the effectiveness of Multilingual Bidirectional Encoder Representations from Transformers (mBERT) and comparing its performance with a traditional Support Vector Machine (SVM) baseline for sentiment classification. A dataset consisting of 2,232 telecommunications-related comments was collected from the official Facebook pages of Globe, Smart, and DITO and manually annotated into positive, negative, and neutral sentiment categories. The comments underwent preprocessing and were divided using a stratified train-validation-test split. Model

performance was evaluated using accuracy, precision, recall, and F1-score metrics. The results showed that mBERT significantly outperformed SVM, achieving an overall accuracy of 91.6% and demonstrating superior capability in capturing contextual and semantic relationships within code-switched Taglish text. The model achieved strong classification performance for positive and negative sentiments, while neutral comments remained more challenging because of sentiment ambiguity and mixed emotional cues. The findings highlight the effectiveness of transformer-based multilingual models for analyzing lowresource, code-switched languages and provide valuable insights into public perceptions of telecommunications services in the Philippines. Future research should explore larger datasets, multiple social media platforms, and advanced multilingual transformer architectures to further improve sentiment classification performance in Taglish and other low-resource multilingual contexts. © 2026 IEEE.

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Sentiment Analysis of COVID-19 Vaccination Responses From Twitter Using Ensemble Learning

Item Type Conference Paper

Author Q. Ismail

Author R. Obeidat

Author K. Alissa

Author E. Al-Sobh

Editor Quwaider M.

Abstract Despite the evidence that shows the benefits and safety of immunizations, the widespread vaccine-related misinformation and conspiracy theories online have fueled a general vaccine hesitancy, and coronavirus disease (COVID-19) vaccinations are no exception. COVID-19 vaccine hesitancy is considered a global threat to public health that undermines the efforts to control the COVID-19 pandemic. Twitter and other social media platforms allow people to exchange information and express concerns and emotions on COVID-19-related issues. This research aims to understand people's sentiment on COVID-19 vaccines from data collected from Twitter. Analyzing the public attitude toward the vaccines helps the authorities to make better decisions and reach the intended herd immunity. In this paper, we utilize the state-of-the-art transformer-based classification models, RoBERTa and BERT, along with multiple task-specific versions, to classify people's opinions about COVID-19 vaccinations into positive, negative, and neutral. A Twitter dataset that consists of people's opinions about vaccines is used to train and evaluate the presented models. Two ensemble learning techniques that aggregate the individual classifiers are presented for further performance improvement: majority voting and stacking with Support Vector Machine (SVM) as meta-learner. The results also show that applying ensemble learning significantly outperforms the individual classifiers using all evaluation measures. We also found that ensembling with stacking has an advantage over simple majority voting. © 2022 IEEE.

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Sentiment analysis of cultural differences in online comments on popular news

Item Type Journal Article

Author K. Hordiienko

Author L. Kormaníková

Abstract The rapid growth of online communication through social networks has created new opportunities for understanding public opinion on socially relevant issues. This research examines how sentiment analysis (SA) can reveal cultural differences, specifically analyzing Czech and Ukrainian online comments on news topics including the war in Ukraine, political discussions, public health issues (tick-borne diseases, COVID-19), LGBTQ+ community matters, and natural disasters. By comparing three large language models (GPT-3.5-Turbo, Twitter-XLM-RoBERTa, and Zephyr 7B) with native speaker evaluations, we assess whether AI-based sentiment analysis can accurately capture culturally-specific emotional expressions in medium-resource languages. Our dataset comprises 6,085 comments (2,999 Czech from X/Twitter, 3,086 Ukrainian from Telegram) collected during 2023, focusing on socially relevant news coverage. We employed a hybrid methodology combining machine learning analysis with expert validation by native speakers. The study addresses a critical gap in cross-cultural sentiment analysis research, as no previous studies have compared Czech and Ukrainian linguistic patterns in this context. Results demonstrate significant performance differences among models depending on language: GPT-3.5-Turbo achieved highest accuracy for Czech ($p < 0.001$), while all models performed comparably for Ukrainian. Both populations showed predominantly negative sentiment (Czech: 69.93%, Ukrainian: 68.93% via GPT-3.5), reflecting shared emotional responses to crisis events. © 2025 Hordiienko, Kormaníková, published by Sciendo.

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Sentiment Analysis of English-Marathi-Konkani Code-Mixed Social Media Text: A Multilingual Approach

Item Type Conference Paper

Author A. Phadte

Author M.L. Dhore

Abstract With the rise of social media, multilingual and informal communication has become increasingly prevalent, particularly in code-mixed languages such as English, Konkani, and Marathi. Analyzing sentiment in such text presents unique challenges due to inconsistent spellings, informal linguistic structures, and the scarcity of labeled datasets. This paper introduces a novel approach to sentiment classification of English-Konkani-Marathi code-mixed text, leveraging both machine learning and deep learning techniques. The proposed method directly processes raw text without requiring manual feature engineering, enabling efficient and scalable sentiment analysis. Experimental evaluations demonstrate that BiLSTM significantly outperforms traditional machine learning classifiers, achieving superior sentiment classification accuracy. The effectiveness of our approach is assessed using precision, recall, F1-score, accuracy, and additional novel evaluation metrics, highlighting its robustness in handling code-mixed sentiment analysis. © 2025 IEEE.

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Sentiment Analysis of Imbalanced Dataset Through Data Augmentation and Generative Annotation Using DistilBERT and Low-Rank Fine-Tuning

Item Type Journal Article

Author H. Nekkouei Nasrabadi

Author M.H. Moattar

Abstract Sentiment analysis on social media data often suffers from severe class imbalance, which can negatively affect the performance of machine learning models. In this paper, we propose a framework that leverages large language models and lightweight transformer fine tuning to improve sentiment classification on imbalanced datasets. First, GPT-4, a multimodal large language model, is used to generate synthetic tweets through paraphrasing and back translation, with Italian serving as an intermediate language, to increase data diversity. In addition, GPT-4 is employed to annotate tweets with positive reasons by generating semantic counterparts to the 10 predefined negative categories in the Twitter US Airline Sentiment dataset. This process enables the creation of meaningful positive annotations derived from existing category structures, thereby improving dataset balance and interpretability. The augmented data are then encoded using DistilBERT to obtain sentence embeddings, while low rank adaptation (LoRA) is applied for efficient fine tuning with reduced computational cost. Finally, a SoftMax classifier is used to predict sentiment labels (positive, neutral, and negative). Experimental results on the Twitter US Airline Sentiment dataset—evaluated rigorously using a held-out test set and 10-fold cross-validation—demonstrate that the proposed framework achieves strong classification performance while maintaining low training complexity, highlighting the effectiveness of combining Large Language Model (LLM) based data augmentation with parameter-efficient transformer adaptation. © 2026 The Author(s). Applied AI Letters published by John Wiley & Sons Ltd.

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Sentiment Analysis of Low-Resource Language Literature Using Data Processing and Deep Learning

Item Type Journal Article

Author A. Ali

Author M. Khan

Author K. Khan

Author R.U. Khan

Author A. Aloraini

Abstract Sentiment analysis, a crucial task in discerning emotional tones within the text, plays a pivotal role in understanding public opinion and user sentiment across diverse languages. While numerous scholars conduct sentiment analysis in widely spoken languages such as English, Chinese, Arabic, Roman Arabic, and more, we come to grappling with resource-poor languages like Urdu literature which becomes a challenge. Urdu is a uniquely crafted language, characterized by a script that amalgamates elements from diverse languages, including Arabic, Parsi, Pashtu, Turkish, Punjabi, Saraiki, and more. As Urdu literature, characterized by distinct character sets and linguistic features, presents an additional hurdle due to the lack of accessible datasets, rendering sentiment analysis a formidable undertaking. The limited availability of resources has fueled increased interest among researchers,

prompting a deeper exploration into Urdu sentiment analysis. This research is dedicated to Urdu language sentiment analysis, employing sophisticated deep learning models on an extensive dataset categorized into five labels: Positive, Negative, Neutral, Mixed, and Ambiguous. The primary objective is to discern sentiments and emotions within the Urdu language, despite the absence of well-curated datasets. To tackle this challenge, the initial step involves the creation of a comprehensive Urdu dataset by aggregating data from various sources such as newspapers, articles, and social media comments. Subsequent to this data collection, a thorough process of cleaning and preprocessing is implemented to ensure the quality of the data. The study leverages two well-known deep learning models, namely Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN), for both training and evaluating sentiment analysis performance. Additionally, the study explores hyperparameter tuning to optimize the models' efficacy. Evaluation metrics such as precision, recall, and the F1-score are employed to assess the effectiveness of the models. The research findings reveal that RNN surpasses CNN in Urdu sentiment analysis, gaining a significantly higher accuracy rate of 91%. This result accentuates the exceptional performance of RNN, solidifying its status as a compelling option for conducting sentiment analysis tasks in the Urdu language. © 2024 Tech Science Press. All rights reserved.

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Correspondence Address: A. Aloraini; Department of Information Technology, College of Computer,
Qassim University, Buraydah, P.O.Box 1162, Saudi Arabia; email: a.aloraini@qu.edu.sa

Sentiment Analysis of Malayalam-English Code-Mixed Data Using GPT-Based Augmentation

Item Type Conference Paper

Author A.A. Joshy

Author S. James

Abstract Code-mixing is a linguistic phenomenon that blends multiple languages during a conversation. Malayalees often express their opinions on social media using a mix of English and Malayalam. Sentiment analysis(SA) of these opinions is crucial for real-time feedback monitoring. Developing end-to-end deep learning models for code-mixed(CM) data requires access to high-quality and sufficiently large training datasets. The current gold standard annotated dataset for Malayalam-English CM data is the DravidianCodeMix(DCM) dataset containing around 20,000 YouTube comments. However, due to the high variability in CM data, and a significantly larger vocabulary, this data is insufficient for the models to achieve appreciable performance. Mining CM data is challenging and tedious, necessitating the adoption of various data augmentation techniques. The rise of large language models motivated us to explore the generation of high-quality synthetic CM data using ChatGPT. All the different forms of code-mixing were explicitly induced in the generation process using fewshot prompting. The SA classifier showed significant improvement after augmentation, giving an F1-score of 0.76, outperforming the highest published benchmark relatively by 20.6%. Further, human evaluation confirmed the naturalness of the synthetic reviews, suggesting ChatGPT as a cost-effective, high-quality synthetic Malayalam-English CM data generator. © 2025 IEEE.

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Sentiment analysis of Mizo using lexical features in low resource based models

Item Type Journal Article

Author M. Lalthangmawii

Author T.D. Singh

Abstract Sentiment analysis is a vital area of natural language processing (NLP) for interpreting emotions in user-generated content. Although significant progress has been made for widely spoken languages, low-resource languages such as Mizo remain underexplored. This study addresses this gap by developing the first comprehensive sentiment analysis framework for Mizo language. We created a meticulously annotated data set that captures positive, negative, and neutral sentiments. Using classical machine learning models enhanced with lexicon features and transfer learning with XLM-RoBERTa, we demonstrate the feasibility of sentiment analysis in low-resource settings. Our approach achieves an accuracy of 82% with Logistic Regression and 78% with XLM-RoBERTa, which establishes a benchmark for future research in Mizo sentiment analysis. © 2025 The Author(s)

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Sentiment analysis of Nepali social media text with a hybrid deep learning model

Item Type Journal Article

Author S. Sitoula

Author T.B. Shahi

Author S. Wibowo

Author A. Neupane

Abstract Sentiment analysis (SA) has been a well-researched area in text mining, and significant progress has been made recently. Both traditional machine learning (ML) and recent deep learning (DL) methods have shown significant accomplishments in the SA for resource-rich languages. Various DL models, such as convolutional neural networks (CNNs), long short-term memory (LSTM), Bidirectional LSTM (Bi-LSTM), and pre-trained language models (PLMs), have been successfully used for SA in resource-rich settings. However, these models often struggle with low-resource languages like Nepali, primarily due to the limited availability of pre-trained models for handling complex language structures such as rich morphology, short texts, and unbalanced datasets. We propose a hybrid DL model that leverages contextual features from a PLM and spatial features from a convolutional module. For this, pre-trained multilingual embeddings from XLM-RoBERTa are utilised to comprehend the language context, which is then gated with a 1D convolution for local spatial pattern extraction. Finally, the softmax activation is applied with a dense layer to detect positive, negative, and neutral sentiments. We evaluate our model on three diverse Nepali SA datasets (D1, D2, and D3) and find the highest accuracy of 74.77% (D1), 79.52% (D2) and 55.82% (D3) compared to the state-of-the-art (SOTA) SA models for Nepali text. © The Author(s) 2025.

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Sentiment Analysis of Noisy Malay Text using a Large Language Model

Item Type Conference Paper

Author K.I. Khalip

Author K.M.N.K. Khalif

Author M.K.B.M. Aziz

Author A. Gegov

Abstract Due to the informality of social media, Malay user-generated content sentiment analysis is difficult. Existing methods struggle to capture cultural and contextual details. This study proposes publishing an open-source annotated dataset, fine-tuning an open-source large language model (LLM), and using an open-source chatbot interface to create a robust sentiment analysis model for noisy Malay text. The research addresses three main issues: lack of labelled Malay social media data, insufficient generic Malay language models, and lack of practical sentiment analysis tools. Its three goals are to create a diverse dataset with accurate sentiment labels, parameter-efficiently fine-tune an LLM, and export the model for an interactive chatbot. The process involves collecting social media data using Contextual Lexical Adaptation, pre-processing and analysing it, fine-tuning the TinyLlama LLM using LoRA, and comparing it to traditional models. Real-world applications, such as sentiment analysis of Malaysian tweets, will be shown using a locally deployed chatbot interface for fine-tuned model inference. This study lays the groundwork for practical sentiment analysis, benefiting businesses, researchers, and politicians seeking data-driven insights. This research aims to revolutionise open-source Malay sentiment analysis by addressing current limitations through an integrated approach. © 2025 IEICES/Kyushu University.

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Sentiment Analysis of Product Reviews Using Machine Learning and Pre-Trained LLM

Item Type Journal Article

Author P.S. Ghatora

Author S.E. Hosseini

Author S. Pervez

Author M.J. Iqbal

Author N. Shaukat

Abstract Sentiment analysis via artificial intelligence, i.e., machine learning and large language models (LLMs), is a pivotal tool that classifies sentiments within texts as positive, negative, or neutral. It enables computers to automatically detect and interpret emotions from textual data, covering a spectrum of feelings without direct human intervention. Sentiment analysis is integral to marketing research, helping to gauge consumer emotions and opinions across various sectors. Its applications span analyzing movie reviews, monitoring social media, evaluating product feedback, assessing employee sentiments, and identifying hate speech. This study explores the application of both traditional machine learning and pre-trained LLMs for automated

sentiment analysis of customer product reviews. The motivation behind this work lies in the demand for more nuanced understanding of consumer sentiments that can drive data-informed business decisions. In this research, we applied machine learning-based classifiers, i.e., Random Forest, Naive Bayes, and Support Vector Machine, alongside the GPT-4 model to benchmark their effectiveness for sentiment analysis. Traditional models show better results and efficiency in processing short, concise text, with SVM in classifying sentiment of short length comments. However, GPT-4 showed better results with more detailed texts, capturing subtle sentiments with higher precision, recall, and F1 scores to uniquely identify mixed sentiments not found in the simpler models. Conclusively, this study shows that LLMs outperform traditional models in context-rich sentiment analysis by not only providing accurate sentiment classification but also insightful explanations. These results enable LLMs to provide a superior tool for customer-centric businesses, which helps actionable insights to be derived from any textual data. © 2024 by the authors.

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Sentiment analysis of research attention: the Altmetric proof of concept

Item Type Journal Article

Author C. Areia

Author M. Taylor

Author M. Garcia

Author J. Hernandez

Abstract Traditional bibliometric approaches to research impact assessment have predominantly relied on citation counts, overlooking the qualitative dimensions of how research is received and discussed. Altmetrics have expanded this perspective by capturing mentions across diverse platforms, yet most analyses remain limited to quantitative measures, failing to account for sentiment. This study aimed to introduce a novel artificial intelligence-driven sentiment analysis framework designed to evaluate the tone and intent behind research mentions on social media, with a primary focus on X (formerly Twitter). Our approach leverages a bespoke sentiment classification system, spanning seven levels from strong negative to strong positive, to capture the nuanced ways in which research is endorsed, critiqued, or debated. Using a machine learning model trained on 5,732 manually curated labels (ML2024) as a baseline (F1 score = 0.419), we developed and refined a Large Language Model (LLM)-based classification system through three iterative rounds of expert evaluation. The final AI-driven model demonstrated improved alignment with human assessments, achieving an F1 score of 0.577, significantly enhancing precision and recall over traditional methods. These findings underscore the potential of advanced AI methodologies in altmetric analysis, offering a richer, more context-aware understanding of research reception. This study laid the foundation for integrating sentiment analysis into Altmetric platforms, providing researchers, institutions, and policymakers with deeper insights into the societal discourse surrounding scientific outputs. Copyright © 2025 Areia, Taylor, Garcia and Hernandez.

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Sentiment analysis of the 2024 Paris Olympics using RoBERTa language model

Item Type Journal Article

Author M.G. Yazdi

Author H. Rafieizadeh

Author P. Tajasob

Abstract Purpose – This study investigates public views of the Paris 2024 Olympics by analyzing tweets with the aim of elucidating public perceptions and identifying the factors that influence engagement, sentiment, and discourse related to this mega event. Design/methodology/approach – Two advanced models, ChatGPT and Roberta, were used to compare sentiment analysis on the dataset, which included 30,300 valid tweets. This research analyzed by RoBERTa the trends and fluctuations of the sentiment in engagement, regarding cultural, social, and geopolitical aspects. Sentiment trends were analyzed by country, sport, gender, user engagement, etc. Findings – 91.1% of the tweets were either neutral or positive, showing a generally favorable public opinion. Certain events, such as the basketball semifinals and the javelin finals, received high engagement, while the opening ceremony garnered mixed reactions, underscoring the importance of cultural and social sensitivities in event design. Country-specific analysis showed positive feedback from the USA and India, national pride and athletic success, and negative sentiment from Russia, possibly because of the country's exclusion from this Olympics. The top three sports with positive reactions were basketball, athletics, and swimming. On the other hand, boxing was criticized for fairness debates. A gender-based sentiment analysis showed positive sentiment toward female athletes in 57% of the tweets. Negative tweets, although lesser in number, received more likes and retweets. Originality/value – This paper investigates sentiment fluctuations during the Paris 2024 Olympics using advanced language models, exploring the convergence of sports, culture, and public perception to provide actionable insights. © Emerald Publishing Limited

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Sentiment analysis of the Algerian social movement inception

Item Type Journal Article

Author M. Laifa

Author D. Mohdeb

Abstract Purpose: This study provides an overview of the application of sentiment analysis (SA) in exploring social movements (SMs). It also compares different models for a SA task of Algerian Arabic tweets related to early days of the Algerian SM, called Hirak. Design/methodology/approach: Related tweets were retrieved using relevant hashtags followed by multiple data cleaning procedures. Foundational machine learning methods such as Naive Bayes, Support Vector Machine, Logistic Regression (LR) and Decision Tree were implemented. For each classifier, two feature extraction techniques were used and compared, namely Bag of Words and Term Frequency–Inverse Document Frequency. Moreover, three fine-tuned pretrained transformers AraBERT and DziriBERT and the multilingual transformer XLM-R were used for the comparison. Findings: The findings of this paper emphasize the vital role social media played during the Hirak. Results revealed that most individuals had a positive attitude toward the Hirak. Moreover, the presented experiments provided important insights into the possible use of both basic machine learning and transfer learning models to analyze SA of Algerian text datasets. When comparing machine learning models with transformers in terms of accuracy, precision, recall and F1-score, the results are fairly similar, with LR outperforming all models with a 68 per cent accuracy rate. Originality/value: At the time of writing, the Algerian SM was not thoroughly investigated or discussed in the Computer Science literature. This analysis makes a limited but unique contribution to understanding the Algerian Hirak using artificial intelligence. This study proposes what it considers to be a unique basis for comprehending this event with the goal of generating a foundation for future studies by comparing different SA techniques on a low-resource language. © 2023, Emerald Publishing Limited.

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Export Date: 20 September 2026; Cited By: 7; Correspondence Address: M. Laifa; Department of Computer Science, University of Mohamed El Bachir El Ibrahimi, Bordj Bou Arreridj, Algeria; email: meriem.laifa@univ-bba.dz

Sentiment analysis of the United States public support of nuclear power on social media using large language models

Item Type Journal Article

Author O.H. Kwon

Author K. Vu

Author N. Bhargava

Author M.I. Radaideh

Author J. Cooper

Author V. Joynt

Author M.I. Radaideh

Abstract This study utilized large language models (LLMs) to analyze public sentiment in the United States (US) regarding nuclear power on social media, focusing on X/Twitter, considering climate change challenges and advancements in nuclear power technology. Approximately, 1.26 million nuclear tweets from 2008–2023 were examined to fine-tune LLMs for sentiment classification. We found the crucial role of accurate data labeling for model performance, with potential implications for a 15% improvement, achieved through high-confidence labels. LLMs demonstrated better performance compared to traditional machine learning classifiers, with reduced susceptibility to overfitting and up to 96% classification accuracy. LLMs are used to segment the US public tweets into policy and energy-related categories, revealing that 68% are politically themed. Policy tweets tended to convey negative sentiment, often reflecting opposing political perspectives and focusing on nuclear deals and international relations. Energy-related tweets covered diverse topics with predominantly neutral to positive sentiment, indicating broad support for nuclear power in 48 out of 50 US states. The US public positive sentiments toward nuclear power stemmed from its high power density, reliability regardless of weather conditions, environmental benefits, application versatility, and recent innovations and advancements in both fission and fusion technologies. Negative sentiments primarily focused on waste management, high capital costs, and safety concerns. The neutral campaign highlighted global nuclear facts and advancements, with varying tones leaning towards positivity or negativity. An interesting neutral theme was the

advocacy for the combined use of renewable and nuclear energy to attain net-zero goals. © 2024 The Author(s)

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Sentiment Analysis of Twitter Data on Quantum Computing: An Exploratory Silver-Label Baseline Study

Item Type Journal Article

Author F. Mehmood

Author A.A. Alsanad

Author M.A. Akbar

Author V. Leiva

Author C. Castro

Abstract Quantum software engineering is advancing rapidly in parallel with equally ambitious hardware roadmaps. However, systematic evidence on how online audiences perceive these advances remains scarce. We present an exploratory baseline of Twitter sentiment toward quantum computing, using automated (silver-standard) labels for benchmarking. Six months of English-language tweets containing the hashtag #Quantum (December 1, 2022 and May 31, 2023) were processed, with #Quantum treated as a proxy for online discourse on quantum computing. We then applied a transparent natural language processing (NLP) methodology combining two zero-shot lexicon-based tools (TextBlob and the Valence Aware Dictionary and sEntiment Reasoner [VADER]) with three lightweight supervised classifiers (multinomial naïve Bayes, Rocchio, and perceptron). Following standard preprocessing and a stratified 70/30 train–test split, we do not aim to measure definitive public opinion; rather, our primary contribution is to establish a transparent and reproducible baseline for future benchmarking. In this context, the multinomial naïve Bayes classifier attained a macro F1-score of 0.88 on the 30% hold-out set when benchmarked against the TextBlob silver labels. This score captures internal agreement rather than accuracy against human annotation. All five methods converged on a largely—though not universally—positive sentiment orientation ($\approx 78\%$ – 81% of nonneutral tweets, depending on the tool). Grounded in the technology acceptance model (TAM) and the unified theory of acceptance and use of technology (UTAUT), we interpret our results as indicating the constructs of curiosity and perceived usefulness, rather than unequivocal adoption readiness. These constructs were not operationalized and serve only as interpretative lenses. By documenting every preprocessing step and model configuration, and making tweet identifiers and code available upon request, the study delivers a reproducible benchmark against which future work can (i) extend the query vocabulary, (ii) incorporate neutral and fine-grained emotions, (iii) apply cross-validation protocols, and (iv) evaluate advanced transformer models on manually annotated data. Addressing these four points is essential before making any definitive claims about public opinion. Copyright © 2025 Faisal Mehmood et al. IET Software published by John Wiley & Sons Ltd.

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Sentiment Analysis of User Opinions on ChatGPT Using Machine Learning Techniques

Item Type Conference Paper

Author S.N.H. Bukhari

Author J.K. Wani

Author S. Farooq

Author S. Mushtaq

Author R. Manzoor

Editor Anand S.

Editor Ella Hassanien A.

Editor Jaiswal A.

Editor Kumar P.

Abstract Technological progress has significantly enhanced convenience in daily life. Among these advancements, Natural Language Processing (NLP) has emerged as a groundbreaking innovation, particularly in the realm of large language models like Chat Generative Pre-trained Transformer (ChatGPT), leading to transformative improvements across diverse fields. Social networking platforms generate vast amounts of user-generated content containing rich sentiments, such as tweets, status updates, and blog posts. Analyzing the sentiments embedded in this data is highly valuable for understanding collective opinions on emerging technologies, brands, businesses, and more. Sentiment analysis provides organizations and stakeholders with a fast and efficient tool to monitor public sentiments, enabling them to identify areas of concern and address issues effectively. The proposed study examines the linguistic feature effectiveness to determine the tweet sentiment related to ChatGPT. The utility of lexical resources as well as features designed to capture the creative and informal language usually found in microblogging is then evaluated. The study utilized the Valence Aware Dictionary for Sentiment Reasoning (VADER) for sentiment analysis that

automates the process of labeling and polarity of tweets is classified (negative, positive or neutral). The word cloud was generated as well as topic modeling was performed using Latent Dirichlet Allocation (LDA). Analyzing user comments for ChatGPT, policymakers and related stakeholders can obtain important insights ultimately enabling them to optimize and improve the systems. © 2026, Springer Science and Business Media Deutschland GmbH. All rights reserved.

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Sentiment Analysis on Indonesian Military Law Debate Using Machine Learning and IndoBERT

Item Type Conference Paper

Author R. Winata

Author A. Willson

Author W. Tjen

Author E.D. Madyatmadja

Abstract In the digital era, social media platforms have become influential channels for expressing public opinion on national governance issues. One such controversy in Indonesia is the proposed amendment to the National Armed Forces Law (RUU TNI), which has reignited concerns over the potential revival of the military dual-function (Dwifungsi ABRI) role. This study investigates public sentiment surrounding the RUU TNI by analyzing data collected from Twitter. Tweets containing relevant hashtags were obtained through web scraping and preprocessed using Natural Language Processing (NLP) techniques. The sentiment classification was conducted using three machine learning algorithms Naïve Bayes, Support Vector Machine (SVM), and Random Forest and further enhanced by an ensemble approach incorporating IndoBERT, a transformer based model trained on Indonesian language corpora. Among the evaluated models, SVM demonstrated the highest classification accuracy, marginally outperforming Random Forest and Naïve Bayes. The sentiment distribution revealed a predominant negative response from the public, with sarcasm identified as a frequent linguistic feature, as indicated by the ironic usage of typically positive terms such as "bagus" and "semangat." These findings underscore the importance of contextual language understanding in sentiment analysis and the limitations of keyword-based models in detecting nuanced expressions like sarcasm. The study contributes to the field of computational social science by offering insights into political discourse analysis using machine learning and language models tailored for low-resource languages. Future research should focus on enhancing sarcasm detection capabilities and leveraging deep learning architectures to improve sentiment interpretation in complex socio-political contexts. © 2025 IEEE.

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Correspondence Address: R. Winata; Bina Nusantara University, School of Information Systems,
Information Systems Department, Jakarta, Indonesia; email: ryan.winata001@binus.ac.id; Conference
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2025; Conference date: 15 August 2025 through 16 August 2025; Conference code: 213113

Sentiment Analysis on Indonesian-Javanese-English Code-Mixed Texts via Parameter-Efficient Fine-Tuning and Selective Layer Unfreezing

Item Type Conference Paper**Author** A.F. Hidayatullah**Author** A.Y. Al-Sabahi

Abstract This study presents a parameter-efficient fine-tuning framework for sentiment analysis on Indonesian-Javanese-English code-mixed texts using Transformer-based models. Motivated by the growing prevalence of multilingual expression on social media and the challenges of limited computational resources in low-resource settings, we explore the effectiveness of Low-Rank Adaptation, a Parameter-Efficient Fine-Tuning technique, in combination with Selective Layer Unfreezing. These strategies are designed to reduce the number of trainable parameters while retaining strong model performance. To evaluate this approach, we fine-tuned five pre-trained Transformer models: IndoBERT, IndoBERTweet, M-BERT, XLM-RoBERTa-base, and XLM-RoBERTa-large. Our experiments showed that Low-Rank Adaptation was particularly effective for smaller, domain-adapted models such as IndoBERTweet, which achieved the highest F1 score of 91.12%. The combination of Low-Rank Adaptation and Selective Layer Unfreezing further improved performance in larger multilingual models, such as XLM-RoBERTa large, increasing the F1 score from 80.65 to 86.29%. Across all configurations, the number of trainable parameters was kept under one percent. These findings demonstrate that the integration of Parameter-Efficient Fine-Tuning and Selective Layer Unfreezing provides an effective and efficient solution for sentiment analysis in code-mixed and computationally constrained environments. © 2025 IEEE.

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Sentiment Analysis Through Prompt Engineering with GPT-4 on the SmSA IndoNLU Dataset

Item Type Conference Paper

Author Z. Mardiah

Author H.R. Wicaksono

Author R.B. Hadiprakoso

Abstract This research investigates the application of prompt engineering techniques for sentiment analysis using GPT-4 and BERT models on the IndoNLU SmSA dataset. The BERT model is fine-tuned on the dataset, while GPT-4 is employed using various prompt engineering techniques, including zero-shot, one-shot, few-shot, chain-of-thought, and a fusion technique that integrates structured reasoning with factual identification. Both models are evaluated using accuracy, precision, recall, and F1-score. The results reveal that the fusion technique for GPT -4 outperforms other methods, achieving an accuracy of 96.8% and an F1-score of 0.968, significantly surpassing the performance of the fine-tuned BERT model (accuracy: 90.6%). This superior performance is attributed to the fusion technique's ability to guide the model through step-by-step reasoning, enhancing its capacity to handle complex language features like sarcasm and irony. Practical implications of these findings include the potential for more accurate and reliable sentiment analysis in low-resource languages such as Indonesian, with applications in business intelligence, social media monitoring, and customer feedback analysis. Additionally, the study highlights the importance of prompt design in improving model performance and suggests that combining multiple prompt techniques can significantly enhance sentiment classification accuracy. The research also underscores the need for robust security measures, successfully mitigating prompt injection attacks by reducing vulnerability rates from 25.3% to 3.8%. The implications for enhancing AI-driven sentiment analysis in low-

resource languages, with opportunities for future research to explore local models and hybrid prompt strategies. © 2024 IEEE.

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Sentiment Analysis Using Large Language Models: A Case Study of GPT-3.5

- Item Type** Journal Article
- Author** F. Nadi
- Author** H. Naghavipour
- Author** T. Mehmood
- Author** A.B. Azman
- Author** J.A.P. Nagantheran
- Author** K.S.K. Ting
- Author** N.M.I.B.N. Adnan
- Author** R.A.P. Sivarajan
- Author** S.A.P. Veerah

Author R.F. Rahmat

Abstract Sentiment analysis, which utilizes machine learning, natural language processing, and computational linguistics, has been developed to comprehend the emotions and viewpoints of individuals on social media platforms. This area of research has undergone extensive investigation, leading to the integration of diverse algorithms and techniques specifically tailored to this objective. A recent breakthrough in the field of general AI is exemplified by Large Language Models, which have surpassed numerous research domains as quintessential achievements. This study examines the performance of GPT-3.5 in sentiment analysis. We evaluate its performance on established benchmark datasets as well as a collection of reviews extracted from Google Maps. The findings indicate that the Large Language Models can outperform traditional methods in the literature, and they do not require pre-processing, unlike traditional methods. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2024.

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Sentiment Analysis Using Natural Language Processing

Item Type Conference Paper

Author R. Shettar

Author V. Ramya

Abstract Social media platforms, especially Twitter, have become powerful channels for users to express opinions on diverse topics ranging from products and politics to social issues. Valuable insights could be derived from the extracted unstructured information and analyzed by applying sentiment analysis techniques. In this paper, a comprehensive study on Twitter reviews is done by categorizing user sentiments into three classes: positive, neutral, and negative. This work is divided into two experimental segments. In the first segment, dataset comprising of 27,453 tweets is evaluated and three techniques are applied, namely, a rule-based method (VADER), a traditional machine learning pipeline using TF-IDF with Logistic Regression, and a deep learning model based on the transformer architecture, RoBERTa. Among these, RoBERTa achieved the highest accuracy of 72.13%, outperforming VADER (63.13%) and TF-IDF + Logistic Regression (69.14%), demonstrating the effectiveness of transformer-based contextual embeddings for sentiment classification. In the second segment, experiment is conducted with a reduced dataset of 5,000 tweets to simulate a low-resource environment. A comparison of Logistic Regression, Support Vector Classifier (SVC), Random Forest (RF), and Gradient Boosting (GB) performance is explored. Logistic Regression and Random Forest achieved the highest accuracy of 75.8% and 75.9% respectively, highlighting their robustness even on limited data. The results show the effectiveness of both transformer-based and classical models, depending on the dataset size and complexity. © 2025 IEEE.

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Sentiment Analysis Using XLM-R Transformer and Zero-shot Transfer Learning on Resource-poor Indian Language

Item Type Journal Article

Author A. Kumar

Author V.H.C. Albuquerque

Abstract Sentiment analysis on social media relies on comprehending the natural language and using a robust machine learning technique that learns multiple layers of representations or features of the data and produces state-of-the-art prediction results. The cultural miscellanies, geographically limited trending topic hash-tags, access to aboriginal language keyboards, and conversational comfort in native language compound the linguistic challenges of sentiment analysis. This research evaluates the performance of cross-lingual contextual word embeddings and zero-shot transfer learning in projecting predictions from resource-rich English to resource-poor Hindi language. The cross-lingual XLM-RoBERTa classification model is trained and fine-tuned using the English language Benchmark SemEval 2017 dataset Task 4 A and subsequently zero-shot transfer learning is used to evaluate the classification model on two Hindi sentence-level sentiment analysis datasets, namely, IITP-Movie and IITP-Product review datasets. The proposed model compares favorably to state-of-the-art approaches and gives an effective solution to sentence-level (tweet-level) analysis of sentiments in a resource-poor scenario. The proposed model compares favorably to state-of-the-art approaches and achieves an average performance accuracy of 60.93 on both the Hindi datasets. © 2021 Association for Computing Machinery.

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Sentiment Analysis with Deep Learning: Benchmarking CNNs vs. Prompt-Tuned BERT Models

Item Type Journal Article

Author B. Huang

Author F. Song

Abstract With the growth of user-generated content on platforms like social networks and e-commerce, effective sentiment analysis has become increasingly important. This paper presents a comparative study of deep learning models for sentiment analysis, focusing on both Convolutional Neural Networks (CNNs) and prompt-based Bidirectional Encoder Representations from Transformers (BERT) models. We evaluated three CNN variants and multiple prompt strategies for BERT (hand-crafted, adaptive, and hybrid) across five diverse datasets. The results show that the hybrid Gated Recurrent Units_Attention Mechanism (GRU_ATT) BERT model outperforms all others. Prompt-based approaches, particularly adaptive and hybrid designs, offer strong performance and surpass CNN-Non-Static. Moreover, dataset characteristics, such as input length, class distribution, and language structure, significantly impact the performance of the models. Lastly, practical guidelines are proposed for selecting appropriate models based on dataset characteristics. © 2025 by the authors.

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sentiment analysis, Bidirectional Encoder Representations from Transformers (BERT), Convolutional Neural Network (CNN), prompting

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: F. Song; School of Computer Science, University of Guelph, Guelph, Canada; email: fsong@uoguelph.ca

Sentiment and Interest Detection in Social Media using GPT-based Large Language Models

Item Type Conference Paper

Author M.A. Al Asad

Author H.M. Imran

Author M. Alamin

Author T.A. Abdullah

Author S.I. Chowdhury

Abstract In the ever-expanding realm of social media, deciphering sentiments and identifying relevant topics from various textual content is challenging. This paper comprehensively investigates sentiment analysis and interest/topic detection using cutting-edge language models, ChatGPT3.5 and gpt4All. The methodology encompasses data collection, meticulous text pre-processing, innovative prompt design, and the exploration of zero-shot, one-shot, and few-shot learning techniques. We unveil the nuances of model performance in sentiment analysis and interest/topic detection through a detailed comparative analysis. Our findings highlight the power of ChatGPT3.5 in achieving a substantial accuracy enhancement in sentiment analysis compared to gpt4All. Moreover, we delve into the intricacies of interest detection, demonstrating the complexities of linguistic structures and model biases. We offer a holistic view of social entities' preferences by categorizing topics into distinct domains. A web portal developed using Google's Flutter SDK facilitates the visualization of user-friendly sentiment and interest outcomes. This research contributes to the understanding of sentiment analysis and interest detection and underscores the evolving capabilities of AI and NLP in navigating the dynamic landscape of social media content. © 2023 ACM.

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Pages 209-214

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Conference Name ACM International Conference Proceeding Series

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Computational linguistics, Language model, Learning systems, Sentiment analysis, Social networking (online), Social media, Topic Modeling, social media, Modeling languages, Cutting edges, topic modeling, sentiment analysis, Zero-shot learning, Textual content, Gpt3.5, GPT3.5, Gpt4ALL, interest detection, Interest detection, Portals, Topic detection

Notes:

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Sentiment Classification of Fine-Tuned mBERT for Myanmar Cosmetic Domain

Item Type	Conference Paper
Author	N.Y.T. Thein
Author	K.M. Soe
Abstract	Nowadays, sentiment analysis is the crucial task for the feedback of customer review on social media, business organization, and very popular between the market researcher. It becomes an important in making decision for business owners because they want to know the feedback of their products. Sentiment analysis classifies the polarities into positive, negative, or neutral classes. Recent advances in deep learning, particularly transformer-based models, have significantly enhanced the performance of sentiment analysis across various languages. This paper explores sentiment classification using the Multilingual Bidirectional Encoder Representations from Transformers (mBERT) on the user review comment that are collected from YouTube channels, Facebook and Tiktok applications written in Myanmar language. Different from BERT transformer, mBERT support 104 languages including Myanmar language. By leveraging the pre-trained mBERT architecture, the study demonstrates significant improvements in classifying Myanmar text into sentiment classification task. Experimental results show the effectiveness of fine-tuning for low-resource languages like Myanmar. The model achieves state-of-the-art performance on a manually annotated dataset by comparing the accuracy on both BERT and mBERT on customized cosmetic dataset. © 2025 IEEE.
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URL	https://www.scopus.com/pages/publications/105007855464?origin=resultslist
Archive	Scopus
Language	English
Extra	Journal Abbreviation: Proc. IEEE Int. Conf. Comput. Appl., ICCA
Publisher	Institute of Electrical and Electronics Engineers Inc.
ISBN	979-833153458-5 (ISBN)

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Tags:

Sentiment analysis, Neural-networks, BERT, Emotion Recognition, Engineering education, sentiment analysis, neural network, mBERT, Sentiment classification, Customer review, Myanmars, Multilingual bidirectional encoder representation from transformer, Cosmetic review, cosmetic reviews, Myanmar language, Myanmar Language, neural sentiment, Neural sentiment

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: N.Y.T. Thein; Ministry of Science and Technology, Department of Advanced Science and Technology, Naypyidaw, Myanmar; email: nwetyintunthein@ucsy.edu.mm; Conference name: 22nd IEEE International Conference on Computer Applications, ICCA 2025; Conference code: 209174

Sentiment Classification of Low Resource Language Tweets Using Machine Learning Algorithms

Item Type Conference Paper

Author K. Chattu

Author D. Sumathi

Abstract Sentiment analysis is a way of figuring out if a piece of writing or a sentence is positive or negative. Over the past few years, it has garnered a significant amount of interest in the English language. As a result of the rapid expansion of social media in the modern day, an increasing amount of content can be found in languages apart from English. Telugu represents one of the many native languages for which there is comparatively little information on social media. Finding labeled data of sentences in Telugu for sentiment analysis is becoming increasingly challenging. This study provides Sentikanna, a standard Telugu speech corpus built to assist Telugu Sentiment Analysis across multiple domains, including Recipe, Tourism, and Movie reviews. Sentikanna is a corpus including preprocessed and manually annotated Telugu sentences from a variety of sources, annotated according to annotation rules. All the scripts written in the Telugu language have been used. Apart from the creation of the corpora, polarities have been assigned to the sentences. In total, 3113 Reviews are annotated that have been extracted from various domains. Proposed dataset is used to evaluate different sentiment analysis methods and its performances. Later, sentiment evaluation algorithms using support vector machines (SVM) and Naive Bayes (NB) were applied to the proposed dataset in addition to Sentirama data samples. Performance evaluation has been done and it has been observed that SVM and NB showed promising results on proposed dataset comparatively. © 2023 IEEE.

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Support vector machines, Support vectors machine, Machine-learning, Sentiment analysis, Language processing, Learning algorithms, Natural language processing, Natural languages, Social networking (online), Social media, Machine Learning, Sentiment Analysis, Naive Bayes, Sentiment classification, support vector machine, Barium compounds, Naive bayes, Sodium compounds, Telugu language, Telugu Language

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Sentiment Classification on Indodax Using Term Frequency, FastText, and Neural Attention Models

Item Type Journal Article

Author D. Hartama

Author G. Riski

Abstract The rapid growth of mobile-based investment platforms such as Indodax has triggered a surge in user-generated reviews that reflect public perception and sentiment. This study aimed to develop and evaluate sentiment classification models that can accurately classify Indonesian user reviews on the Indodax app into negative, neutral, and positive sentiments. A dataset of 11,000 reviews was collected via web scraping from the Google Play Store. Reviews were preprocessed, labeled using a lexicon-based unsupervised method, and balanced using oversampling. Two models were built: a Bidirectional LSTM (BiLSTM) with attention mechanism using

FastText embeddings, and a Feedforward Neural Network (FFNN) using a hybrid feature vector combining TF-IDF and FastText. The evaluation was performed using accuracy, classification report, confusion matrix, and PCA visualization. The FFNN model outperformed the BiLSTM-Attention model with an accuracy of 97.07% compared to 96.00%. Both models demonstrated strong performance in classifying three sentiment classes, though the FFNN showed better separation in PCA space and higher macro-average metrics. This study demonstrates the effectiveness of combining statistical and semantic feature representations for sentiment classification in Indonesian text. The proposed approach is particularly valuable for low-resource languages and informal user-generated content. © 2025, Ikatan Ahli Informatika Indonesia. All rights reserved.

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Sentiment Extraction in Financial Texts from Decentralized Markets Using RoBERTa Language Model

Item Type Conference Paper

Author K. Shaalan

Author H. Yousuf

Author A.Q. Saeed

Author A. Elnekiti

Author M. Elersy

Author A.M. Al-Zoghby

Author T.M. Ghazal

Abstract Sentiment analysis in financial texts plays a vital role in understanding market trends, especially in decentralized finance (DeFi) markets where information flow is rapid and largely unstructured. The emergence of large language models has significantly enhanced the ability to extract meaningful insights from unstructured data sources. However, existing sentiment analysis models often underperform on decentralized market texts due to their reliance on traditional financial datasets and their inability to capture domain-specific language, slang, and context. These models typically struggle with the informal, volatile, and jargon-rich nature of DeFi communications found in forums, social media, and blockchain-related discussions. To address these limitations, this paper propose Decentralized Finance Sentiment Extraction using a RoBERTa-based Neural Transformer (DeFiSENT). This framework incorporates a domain-adapted preprocessing pipeline tailored to DeFi language. It fine-tunes the RoBERTa model on curated, labeled datasets from decentralized platforms such as Reddit, Twitter, and DAO channels. Additionally, DeFiSENT integrates context-aware tokenization and class imbalance strategies to enhance performance in detecting sentiment across imbalanced and noisy text data. The proposed method can be deployed in applications such as real-time crypto sentiment dashboards, DeFi asset risk monitoring, and automated trading signal generation, offering a robust tool for both researchers and investors. Experimental evaluations demonstrate that DeFiSENT significantly outperforms baseline models in accuracy, F1-score, and generalization across unseen DeFi datasets. It effectively captures nuanced sentiments and outliers in dynamic financial discourse, providing timely and interpretable sentiment insights within decentralized finance ecosystems. © 2025 IEEE.

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Sentiment Informed Sentence BERT-Ensemble Algorithm for Depression Detection

Item Type Journal Article

Author B. Ogunleye

Author H. Sharma

Author O. Shobayo

Abstract The World Health Organisation (WHO) revealed approximately 280 million people in the world suffer from depression. Yet, existing studies on early-stage depression detection using machine learning (ML) techniques are limited. Prior studies have applied a single stand-alone algorithm, which is unable to deal with data complexities, prone to overfitting, and limited in generalization. To this end, our paper examined the performance of several ML algorithms for early-stage depression detection using two benchmark social media datasets (D1 and D2). More specifically, we incorporated sentiment indicators to improve our model performance. Our experimental results showed that sentence bidirectional encoder representations from transformers (SBERT) numerical vectors fitted into the stacking ensemble model achieved comparable F1 scores of 69% in the dataset (D1) and 76% in the dataset (D2). Our findings suggest that utilizing sentiment indicators as an additional feature for depression detection yields an improved model performance, and thus, we recommend the development of a depressive term corpus for future work. © 2024 by the authors.

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Sentiment lexicon integrated meta-training for sentiment analysis in data-scarce settings

Item Type Journal Article

Author H. Kim

Author S. Cho

Abstract Sentiment analysis is a fundamental task in natural language processing with broad applications. While pre-trained language models have achieved remarkable success in sentiment analysis, these approaches typically require large training data. In sentiment analysis, labeled data may be scarce as the problem expands to new areas, such as novel modes of expression on social media, emerging topics, and reviews on new products. Moreover, given the dynamic nature of real-world applications, continuously updating the data with new information is impractical. This highlights the need for efficient models that perform well with limited labeled data. In this paper, we propose Sentiment Lexicon Integrated Meta-training (SLIM), a novel framework designed to optimize pre-trained language models for sentiment analysis in data-scarce settings. SLIM integrates the knowledge of a sentiment lexicon into self-supervised meta-training tasks. Specifically, we propose three methods, SLIMboolean, SLIMmixed, and SLIMPolar, to generate sentiment-aware meta-training tasks by exploiting the semantic relationship between words. SLIMboolean distinguishes sentiment-bearing expressions from neutral ones, SLIMmixed captures variations among sentiment expressions, and SLIMPolar explicitly contrasts opposing

sentiment polarities. We then meta-train several SLIM models using these task generation methods along with two model initialization settings. Experimental results on 11 benchmark datasets show that SLIM significantly improves performance in few-shot settings, demonstrating the capability of SLIM models to adapt effectively with minimal training examples. When combined with pre-existing sentiment knowledge, our model outperforms recent meta-learning models and sentiment knowledge-enhanced pre-trained language models. Furthermore, we conduct comprehensive ablation studies to examine the impact of specific design choices within the SLIM framework. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025.

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Sentiment Showdown - Sentence Transformers stand their ground against Language Models: Case of Sentiment Classification using Sentence Embeddings

Item Type Conference Paper

Author A. Siddhanta

Author A.K. Bhagat

Editor Shakshuki E.

Editor Yasar A.-U.-H.

Abstract Understanding customer sentiment is crucial for business success. This study leverages the popular Twitter US Airline Sentiment dataset to develop a cutting-edge sentence transformer architecture using pre-trained transformer models like Masked and Permuted Pre-training for Language Understanding (MPNet) and Robustly Optimized Bidirectional Encoder Representations from Transformers (BERT) approach (RoBERTa). By fine-tuning these models with specific loss functions, we created semantically rich sentence embeddings that excel in sentiment prediction. Our approach, particularly the RoBERTa-Large-based sentence transformer, fine-tuned with the CosineSimilarity loss function and combined with the Extreme Gradient Boosting (XGBoost) Classifier, achieved an excellent overall accuracy of 88.4% and high recall rates even for minority sentiment classes without any data augmentation. We further validated our method on the classic benchmark dataset for sentiment classification - Internet Movie Database (IMDb) reviews, attaining an impressive accuracy of 95.9%. We also compared our model with the advanced Meta-Llama-3-8B large language model with respect to performance, training time, and inference time. The study demonstrates that fine-tuned sentence transformer models can compete with, or exceed, existing sentiment classification techniques, including large language models, albeit with reduced computational load while maintaining generalizability, even with diverse datasets. © 2025 Elsevier B.V.. All rights reserved.

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Export Date: 20 September 2026; Cited By: 2; Correspondence Address: A. Siddhanta; Georgia State University, Atlanta, United States; email: agni.siddhanta@gmail.com; Conference name: 16th International Conference on Ambient Systems, Networks and Technologies Networks, ANT 2025 / 8th International Conference on Emerging Data and Industry 4.0, EDI40 2025; Conference date: 22 April 2025 through 24 April 2025; Conference code: 208474

Sentiment-Annotated Hibapress: A Moroccan News Arabic Dataset (SAHMNAD) predicted using Fine-Tuned Arabic Language Models and Zero-Shot LLMs

Item Type Conference Paper**Author** A. Jannani**Author** T. Amzil**Author** N. Sael**Author** S. Bouhsissin**Editor** Benhala B.**Editor** Raihani A.**Editor** Qbadou M.**Editor** Boukili B.

Abstract This study presents the Sentiment-Annotated Hibapress, a Moroccan News Arabic Dataset (SAHMNAD), a Moroccan news headlines dataset for sentiment analysis, sourced from a national digital news website, Hibapress, with user engagement metrics such as likes and dislikes as sentiment indicators. Unlike prior research focused on social media, this news headline data enables structured sentiment classification in mainly Modern Standard Arabic, addressing linguistic challenges unique to Moroccan digital news. We fine-tune Arabic-specific pretrained language models, including Asafaya-BERT and DarijaBERT, for sentiment classification and compare them with multilingual and general-purpose models. Asafaya-BERT achieves the highest accuracy of 90.74% with an F1 score of 90.58%, significantly outperforming zero-shot large language models such as BLOOM-560m with 78.76% and XLM-R with 66.04%, as well as multilingual models like DistilBERT with 88.65%. The results highlight the effectiveness of fine-tuned models for Moroccan Arabic sentiment classification. This research underscores the necessity of localized Arabic Natural Language Processing (NLP) models and explores large language models for zero-shot sentiment classification. Future directions include integrating reader comments, multimodal sentiment analysis, and cross-lingual approaches to further enhance sentiment classification in Moroccan digital media. © 2025 IEEE.

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Language model, Large language model, Sentiment analysis, Emotion Recognition, Sentiment Analysis, Large Language Models, Sentiment classification, Pre-trained language model, Digital news, Arabic languages, Hibapress, Pre-trained Language Models, SAHMNAD, Sentiment-annotated hibapress: A moroccan news arabic dataset

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Sentiment-Based Prediction of Election Outcomes Using Multilingual Social Media Comments

Item Type Conference Paper

Author R. Jaiswal

Author T. Saha

Author S. Bal

Author S. Mishra

Editor Bhattacharya A.

Editor Dutta S.

Editor Razzak M.A.

Editor Chakrabarti A.

Abstract Social media has been increasingly used in recent years to voice public opinion during election periods. Within this research, an attempt was made to propose a two-phase, sentiment-oriented paradigm for election outcome prediction using multilingual social media data, which specifically addresses code-mixed YouTube comments for the 2024 Indian General Election. The approach was framed to combine lexicon-based methods (from VADER, TextBlob) and supervised machine learning classifiers (SVM, Naïve Bayes) for sentiment classification,

leading to the creation of a domain-adapted VADER model for constituency-level seat prediction. Over 200,000 comments were analyzed, and a cumulative seat prediction accuracy of 94% compared to an actual election outcome was reported. By this method, it has been shown that sentiment-based election prediction is feasible and useful, especially in language-rich and low-resource environments. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Sentiment-Enhanced Opinion Sentence Recognition Using AgriBERT-SentiDPCNN and Multi-Opinion Summarization via LLMs for Agricultural Scientific Literature

Item Type Conference Paper

Author M. Zhang

Author Y. Wang

Author Y. Wang

Author Z. Zhang

Abstract Opinion sentences in scientific literature are characterized by complex reasoning, technical terminology, and implicit argumentation, setting them apart from the subjective statements typically found in product reviews and social media. These sentences often convey nuanced perspectives on research findings, methodologies, and theoretical discussions, making their accurate recognition and summarization essential for a comprehensive understanding of academic research. However, existing opinion mining techniques are largely designed for nonacademic contexts and thus prove inadequate for processing scientific texts. To fill this gap, we propose AgriBERTSentiDPCNN, a sentiment-enhanced opinion sentence recognition model tailored to the agricultural domain. This model integrates AgriBERT, a pre-Trained language model fine-Tuned for agricultural texts, with a Sentiment-Enhanced Deep Pyramid Convolutional Neural Network (SentiDPCNN) that employs a weighted sentiment feature fusion mechanism to effectively combine sentiment and textual features. Furthermore, we propose a multi-opinion summarization pipeline in which we first aggregate the opinions of multiple authors on the same research themes and then employ prompt-based Large Language Models (LLMs) to generate comprehensive summaries of these aggregated opinions. Comparative experiments demonstrate the superior performance of our model in opinion sentence recognition, with an F1-score of 90.19%, while ablation studies validate the effectiveness of the integrated sentiment features and the DPCNN module. Additionally, case studies underscore the practical effectiveness and value of the multi-opinion summarization process. © 2025 IEEE.

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Computational linguistics, Language model, Large language model, large language models, Sentiment analysis, Agriculture, Convolutional neural network, Data mining, Features fusions, Agribusiness, multi-opinion summarization, Multi-opinion summarization, opinion sentence recognition, Opinion sentence recognition, Scientific literature, Sentence recognition, sentiment feature fusion, Sentiment feature fusion, Sentiment features

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SentimentGPT: Leveraging GPT for Advancing Sentiment Analysis

Item Type Conference Paper

Author K. Kheiri

Author H. Karimi

Editor Ding W.

Editor Lu C.-T.

Editor Wang F.

Editor Di L.

Editor Wu K.

Editor Huan J.

Editor Nambiar R.

Editor Li J.

Editor Ilievski F.

Editor Baeza-Yates R.

Editor Hu X.

Abstract This study thoroughly examines various Generative Pretrained Transformer (GPT) methodologies in sentiment analysis, specifically in the context of Task 4 on the SemEval 2017 dataset. Three primary strategies are employed: 1) prompt engineering using the advanced GPT-3.5 Turbo, 2) fine-tuning GPT models, and 3) an inventive approach to embedding classification. The research yields detailed comparative insights among these strategies and individual GPT models, revealing their unique strengths and potential limitations. Additionally, the study compares these GPT-based methodologies with other current, high-performing models previously used with the same dataset. The results illustrate the significant superiority of the GPT approaches in terms of predictive performance, with more than 22% in the F1-score compared to the state-of-the-art. Further, the paper sheds light on common challenges in sentiment analysis tasks, such as understanding context and detecting sarcasm. It underscores the enhanced capabilities of the GPT models to handle these complexities effectively. These findings highlight the promising potential of GPT models in sentiment analysis, setting the stage for future research in this field. The code can be found at <https://github.com/DSAatUSU/SentimentGPT> © 2024 IEEE.

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Language model, Large language model, 'current, Economic and social effects, Machine-learning, Sentiment analysis, Transformer modeling, Program processors, Fine tuning, Machine Learning, Sentiment Analysis, Embeddings, Large Language Models, Social Media Mining, Social media minings, Chat-generative pretrained transformer, Chat-GPT

Notes:

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Export Date: 20 September 2026; Cited By: 10; Correspondence Address: K. Kheiri; Utah State University, Logan, United States; email: kiana.kheiri@usu.edu; Conference name: 2024 IEEE International Conference on Big Data, BigData 2024; Conference date: 15 December 2024 through 18 December 2024; Conference code: 206131

SentimentPulse: A Concurrent Multi-Platform System for Social Media Sentiment Monitoring with LLM-Based Interpretation

Item Type Journal Article

Author G.A. León-Paredes

Author E.C. Villa-Quishpi

Author J.E. Márquez-Chávez

Author E.F. Zhigue-Granda

Abstract Monitoring public perception on social media is increasingly important for detecting reputational risks and communication opportunities in rapidly evolving digital environments. However, operational sentiment monitoring remains challenging due to platform fragmentation, heterogeneous data formats, and the need to generate interpretable reports quickly when specialized analysts are not available. This study presents SentimentPulse, a web-based system for multi-platform sentiment monitoring driven by a user-defined query. The system integrates concurrent data extraction from X, Facebook, LinkedIn, and Instagram with large language model (LLM) based sentiment classification and automated executive storytelling generation. The architecture combines parallel scraping processes for data acquisition with multithreaded LLM inference to improve throughput, while structured persistence enables job tracking and cross-platform analysis. The system operates under practical constraints, including platform-specific access limitations, dynamic content availability, and dependency on external LLM services, which may introduce

variability in response times and outputs. The evaluation is conducted under controlled experimental conditions using fixed query limits and asynchronous execution settings, and results should be interpreted within these operational boundaries. Experimental results from two anonymized case studies demonstrate the effectiveness and operational performance of the approach. In the first case study, the system processed 1032 social media interactions and produced a sentiment distribution of 49.0% positive, 36.6% negative, and 14.2% neutral, with a manual validation accuracy of 0.88. In the second case study, the pipeline processed 1121 records, with parallel scraping accounting for the majority of the runtime and LLM inference achieving a throughput of 12.08 items per second. These results show that combining concurrent multi-platform extraction with LLM-based interpretation enables practical and interpretable social listening workflows, while highlighting the importance of considering system-level constraints when deploying such solutions in real-world environments. © (2026), (Science and Information Organization), All Rights Reserved.

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Export Date: 20 September 2026; Cited By: 0

SentiUzNet: A Domain-Independent Sentiment Lexicon for Uzbek and Its Application in Machine Learning Models

Item Type Conference Paper

Author S. Allanazarova

Author I. Rabbimov

Author D. Elova

Editor Adali E.

Abstract This paper presents the structure and principal components of the linguistic resources required for sentiment analysis in the Uzbek language. The research aims to identify and develop effective approaches for constructing a linguistic database - referred to as SentiUzNet - and to establish a foundational sentiment lexicon tailored specifically to the characteristics of the Uzbek language. In particular, the paper discusses key principles for annotating words with sentiment polarity and subjectivity scores, as well as methodological foundations for building a lexicographic database to support automated emotional analysis of texts. A significant part of the research focuses on experimenting with large-scale user-generated content, specifically social media comments written in Uzbek. These datasets were used to train and evaluate sentiment analysis models, thereby allowing an assessment of their performance and practical applicability. The results of this research represent one of the first comprehensive attempts to facilitate automatic sentiment detection in the Uzbek language and are expected to contribute substantially to the advancement of natural language processing technologies in under-resourced linguistic settings. © 2025 IEEE.

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Conference Name International Conference on Computer Science and Engineering, UBMK

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Allanazarova; Tashkent State University of Uzbek Language and Literature Named After Alisher Navoi, Dept of Computational Linguistics and Digital Technologies, Tashkent, Uzbekistan; email: allanazarova.sabosh@gmail.com; Conference name: 10th International Conference on Computer Science and Engineering, UBMK 2025; Conference date: 17 September 2025 through 21 September 2025; Conference code: 214243

SGPT: An Improved Sentiment Classifier Using Generative Pretrained Transformer

Item Type Conference Paper

Author Y. Prasad

Author M. Cahay

Editor Noor A.

Editor Saroha K.

Editor Pricop E.

Editor Sen A.

Editor Trivedi G.

Abstract Sentiment analysis has numerous applications across various domains such as Social Media Monitoring, Customer Feedback Analysis, Market Research, Financial Markets, Product Reviews, Political Analysis, Employee Feedback, Healthcare, Tourism, and Education. Industries use sentiment analysis to develop informing policy and product design for wider acceptance and improved revenue. Classification methods, from keyword distribution to advanced models like Recurrent Neural Networks and Transformers, have been employed to categorize sentiments into Positive, Negative, and Neutral. Transformer-based models like OpenAI's chatGPT (a chatbot using Generative Pretrained Transformer (GPT)-based Large Language Model) and Google's BARD (a chatbot based on Bidirectional Encoder Representations from Transformers (BERT) framework) struggle with local context and bias issues. In this paper, we combine Latent Semantic Analysis (LSA) with Continuous Bag of Words (CBOW), GPT and BERT features to predict the sentiments, addressing local context to enhance the model. We name our proposed method as SGPT. The proposed SGPT method demonstrates significant improvement over state-of-the-art sentiment classification approaches on benchmark finance sentiment datasets. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

Date 2025

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Volume 1073

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EID: 2-s2.0-105000404008

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Y. Prasad; Indian Institute of Technology Jammu, Jammu, India; email: yamuna.prasad@iitjammu.ac.in; Conference name: 4th International Conference on Emerging Trends and Technologies on Intelligent Systems, ETTIS 2024; Conference date: 27 March 2024 through 28 March 2024; Conference code: 327929

SHE: Sentiment Hashtag Embedding Through Multitask Learning

Item Type Journal Article**Author** L.G. Singh**Author** A. Anil**Author** S.R. Singh

Abstract Recent studies have shown the importance of utilizing hashtags for sentiment analysis task on social media data. However, as the hashtag generation process is less restrictive, it throws several challenges, such as hashtag normalization, topic modeling, and semantic similarity. Recently, researchers have tried to address the above-mentioned challenges through representation learning. However, most of the studies on hashtag embedding try to capture the semantic distribution of hashtags and often fail to capture the sentiment polarity. Furthermore, generating a task-specific hashtag embedding can distort its semantic representation, which is undesirable for sentiment representation of hashtag. Therefore, this article proposes a semisupervised sentiment hashtag embedding (SHE) model, which is capable of preserving both semantic as well as sentiment distribution of the hashtags. In particular, SHE leverages a multitask learning approach using an autoencoder and a convolutional neural network-based classifier. To assess the efficacy of hashtag embedding, we compare the performance of SHE against suitable baselines for three different tasks, namely, hashtag sentiment classification, tweet sentiment classification, and retrieval of semantically similar hashtags. It is evident from various experimental results that SHE outperforms the majority of the baselines with significant margins. © 2014 IEEE.

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Sheetpedia: A 300K-Spreadsheet Corpus for Spreadsheet Intelligence and LLM Fine-Tuning

Item Type Conference Paper

Author Z. Tian

Author Z. Han

Author H. Wang

Author L. Liao

Abstract Spreadsheets are widely used for data analysis and reporting, yet their complex structure and formula logic pose significant challenges for AI systems. We introduce Sheetpedia, a large-scale corpus of over 290,000 diverse spreadsheets (from 324,000+ workbooks) compiled from enterprise email archives and online forums. We detail a rigorous collection and preprocessing pipeline (integrating the Enron email spreadsheet archive and the Fuse web corpus, plus a new crawl

of Excel forums) to standardize formats, filter languages, and remove duplicates. Sheetpedia provides extensive coverage of real formulas and annotations - addressing a gap left by prior table datasets (e.g. web tables used in TURL or Text-to-SQL in Spider) which often lack formula semantics. We present comprehensive corpus statistics, highlighting rich formula diversity and a majority (78%+) of English content. To demonstrate the corpus's utility, we fine-tune large language models on Sheetpedia for two novel spreadsheet understanding tasks: Natural Language to Semantic Range (NL2SR) and Natural Language to Formula (NL2Formula). Using a rejection-sampling data generation strategy, our fine-tuned models achieve up to 97.5% accuracy on NL2SR and 71.7% on NL2Formula - substantially outperforming baseline approaches. Sheetpedia (to be released publicly) fills a crucial need for a large, high-quality spreadsheet benchmark, enabling more effective spreadsheet intelligence and natural language interfaces for spreadsheet tools. © 2025 Neural information processing systems foundation. All rights reserved.

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Archive	Scopus
Language	English
Extra	Journal Abbreviation: Adv. neural inf. proces. syst.
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SiaKey: A Method for Improving Few-shot Learning with Clinical Domain Information

Item Type	Conference Paper
Author	Z. Li
Author	K. Thaker

Author D. He

Abstract Supervised Natural Language Processing (NLP) models can achieve high accuracy, but they often require a significant amount of annotated data for training, which can be expensive and time-consuming. This is especially true for clinical NLP, where annotations on large-scale electronic health records (EHRs) and online posts necessitate specialists with clinical expertise. On the other hand, fine-tuning Pretrained Language Models (PLMs) may yield poor performance due to limited training data. Few-Shot Learning (FSL) methods offer a promising solution as they can significantly improve clinical NLP with only a small amount of labeled data. In this paper, we introduce a novel FSL technique named SiaKey, which utilizes Siamese Networks and integrates Keyphrases Extraction and Domain Knowledge, for the task of online post classification. This task is challenging since online posts typically contain a greater amount of irrelevant information compared to traditional EHRs. By incorporating Keyphrases using domain knowledge, we extract essential information and reduce distractions, enhancing the classification process. To evaluate SiaKey's performance, we conducted tests with 5, 10, 15, and 20-shot learning on health-related online post-classification tasks. The results of our experiments demonstrate SiaKey's effectiveness in capturing text features, showcasing its superior performance compared to BioBERT on similar FSL tasks. © 2023 IEEE.

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Classification (of information), Learning systems, Language processing, Natural language processing systems, Natural languages, Social networking (online), Social media, Data mining, social media, Extraction, few-shot learning, Few-shot learning, Domain Knowledge, Domain knowledge, clinical domain knowledge, Clinical domain knowledge, Electronic health, Health records, Key-phrases extractions, keyphrases extraction, Post classification

Notes:

EID: 2-s2.0-85179510193

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: Z. Li; University of Pittsburgh, School of Computing and Information, Pittsburgh, United States; email: zh1163@pitt.edu; Conference name: 2023 IEEE-EMBS International Conference on Biomedical and Health Informatics, BHI 2023; Conference date: 15 October 2023 through 18 October 2023; Conference code: 194485

SiAraSent: From Features to Deep Transformers for Large-Scale Arabic Sentiment Analysis

Item Type Journal Article

Author O. Almousa

Author Y. Tashtoush

Author A. AlSobeh

Author P. Zahariev

Author O. Darwish

Abstract Sentiment analysis of Arabic text, particularly on social media platforms, presents a formidable set of unique challenges that stem from the language's complex morphology, its numerous dialectal variations, and the frequent and nuanced use of emojis to convey emotional context. This paper presents SiAraSent, a hybrid framework that integrates traditional text representations, emoji-aware features, and deep contextual embeddings based on Arabic transformers. Starting from a strong and fully interpretable baseline built on Term Frequency–Inverse Definition Frequency (TF–IDF)-weighted character and word N-grams combined with emoji embeddings, we progressively incorporate SinaTools for linguistically informed preprocessing and AraBERT for contextualized encodings. The framework is evaluated on a large-scale dataset of 58,751 Arabic tweets labeled for sentiment polarity. Our design works within four experimental configurations: (1) a baseline traditional machine learning architecture that employs TF-IDF, N-grams, and emoji features with an Support Vector Machine (SVM) classifier; (2) an Large-language Model (LLM) feature extraction approach that leverages deep contextual embeddings from the pre-trained AraBERT model; (3) a novel hybrid fusion model that concatenates traditional morphological features, AraBERT embeddings, and emoji-based features into a high-dimensional vector; and (4) a fully fine-tuned AraBERT model specifically adapted for the sentiment classification task. Our experiments demonstrate the remarkable efficacy of our proposed framework, with the fine-tuned AraBERT architecture achieving an accuracy of 93.45%, a significant 10.89% improvement over the best traditional baseline. © 2026 by the authors.

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SICon 2024 - 2nd Workshop on Social Influence in Conversations, Proceedings of the Workshop

Item Type Journal Article**Editor** Hale J.**Editor** Chawla K.**Editor** Garg M.

Abstract The proceedings contain 11 papers. The topics discussed include: observing the southern US culture of honor using large-scale social media analysis; should we respect LLMs? a cross-lingual study on the influence of prompt politeness on LLM performance; personality differences drive conversational dynamics: a high-dimensional NLP approach; redefining proactivity for information seeking dialogue; leveraging large language models for code-mixed data augmentation in sentiment analysis; balancing transparency and accuracy: a comparative analysis of rule-based and deep learning models in political bias classification; and principles for AI-assisted social influence and their application to social mediation.

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Situation-Aware Public Opinion Analysis: A Multi-Platform Benchmark with Grounded LLM Reasoning

Item Type Conference Paper

Author S. Yu

Abstract Understanding and predicting public opinion dynamics is crucial for social risk assessment and crisis management. However, existing benchmarks predominantly focus on isolated tasks such as sentiment classification, lacking multi-platform coverage, temporal evolution modeling, and situation-level annotations. Meanwhile, Large Language Models (LLMs) often produce unreliable predictions in this domain due to hallucinations and the absence of contextual grounding. To bridge this gap, we introduce Situation-Aware Public Opinion Analysis, a novel task that extends beyond stance detection to encompass four critical dimensions: Urgency, Sensitivity, Influence Scope, and Severity. We construct a large-scale, multi-modal benchmark comprising 1.75 million records (including 1.04M text posts, 699K images, and 5.2K videos) across 15,955 trending topics from four major Chinese social media platforms (Weibo, Zhihu, Douyin, and Tieba). The dataset features hierarchical annotations at both post-level (stance) and event-level (situation indicators), ensuring comprehensive coverage. To guarantee annotation quality at scale, we propose a Multi-LLM Collaborative Annotation framework that achieves 85% accuracy. Furthermore, we present Grounded LLM Reasoning, a retrieval-augmented framework that anchors LLM predictions in high-confidence stance signals and semantically similar historical events. Extensive experiments demonstrate that our grounded reasoning approach significantly outperforms zero-shot and standard retrieval baselines, achieving up to 90% accuracy on situation awareness prediction, providing a robust foundation for reliable and interpretable public opinion analysis. © 2026 Copyright held by the owner/author(s).

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Language model, Large language model, Risk assessment, Risk management, Human engineering, Public opinions, Social aspects, Sentiment analysis, Forecasting, Situation awareness, Benchmarking, Large datasets, Information retrieval, Large Language Models, Benchmark datasets, Multi-platform, Retrieval-augmented generation, Opinion analysis, Retrieval-Augmented Generation, Benchmark Dataset, Public Opinion Analysis, Public opinion analyze, Situation Awareness, Situation-aware

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: S. Yu; Southern University of Science and Technology, Shenzhen, Guangdong, China; email: 12333461@mail.sustech.edu.cn; Conference name: 2026 2nd International Conference on Artificial Intelligence and Computational Intelligence, AICI 2026; Conference date: 6 February 2026 through 8 February 2026; Conference code: 227081

SLANG-GraphRAG: Multi-Layered Retrieval with Domain-Specific Knowledge for Low Resource Social Media Conversations

Item Type Conference Paper

Author I. Wuraola

Author D. Marciniak

Author N. Dethlefs

Abstract Emotion classification on social media is especially difficult when texts include informal, culturally grounded language like slang. Standard NLP benchmarks often miss these nuances, particularly in low-resource settings. We present SLANG-GraphRAG, a retrieval-augmented framework that integrates a culture-specific slang knowledge graph into large language models via one-shot prompting. Using multiple retrieval strategies, we incorporate slang definitions, regional usage, and conversational context. Our results show that incorporating structured cultural knowledge into the retrieval process leads to significant improvements, improving accuracy by up to 31% and F1 score by 28%, outperforming traditional and unstructured retrieval methods. To better evaluate model behavior, we propose a probabilistic metric that reflects the distribution of

human annotations, providing a more nuanced measure of performance. This highlights the value of culturally sensitive applications and more balanced evaluation in subjective NLP tasks. ©2026 Association for Computational Linguistics.

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Publisher	Association for Computational Linguistics (ACL)
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Proceedings Title	Conf. Eur. Chapter Assoc. Comput. Linguis., Findings EACL
Conference Name	19th Conference of the European Chapter of the Association for Computational Linguistics, Findings of EACL 2026
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Computational linguistics, Language model, Natural language processing systems, Social networking (online), Social media, Information retrieval, Emotion classification, Knowledge graphs, Cultural knowledge, Domain Knowledge, Domain-specific knowledge, Low-resource settings, Multi-layered, Retrieval process, Retrieval strategies

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Smart City Concern: An AI-Driven Multimodal Civic Grievance Management System

Item Type	Conference Paper
Author	R.P. Kumar
Author	P.S. Benny
Author	C. Manideep
Author	G. Sushmitha
Author	M. Athfaan

Abstract Manual Urban civic grievance systems are usually based on manual processing and resulting to delayed routing, misclassification and lack of transparency. A smart city concierge system is a multimodal AI-driven civic grievance intelligence that is proposed in this paper. site that combines web-based complaints. Social media monitoring via X(formerly Twitter). The system leverages transformer-based zero-shot classification for automated department routing, Named Entity Recognition for structured information extraction, sentiment-driven priority scoring, and OCR-based image text extraction for multimodal analysis. Duplicate complaints are detected using TF-IDF and cosine similarity to reduce redundancy. A scalable layered architecture using Flask and MongoDB ensures efficient ticket management and analytics. Experimental evaluation demonstrates improved classification accuracy, reduced manual workload, and enhanced response efficiency, making the proposed research suitable for practical smart city governance. © 2026 IEEE.

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DOI 10.1109/ICTMIM68190.2026.11507401

URL <https://www.scopus.com/pages/publications/105041434476?origin=resultslist>

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Language English

Extra Journal Abbreviation: Proc. Int. Conf. Trends Mater. Sci. Inventive Mater., ICTMIM

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Pages 1894-1901

Proceedings Title Proc. Int. Conf. Trends Mater. Sci. Inventive Mater., ICTMIM

Conference Name Proceedings of the 6th International Conference on Trends in Material Science and Inventive Materials, ICTMIM 2026

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Tags:

Artificial intelligence, Automation, Big data, Information systems, Sentiment analysis, Natural Language Processing, Smart city, Language processing, Natural language processing, Natural languages, Transformer modeling, Transformer Models, Social networking (online), Sentiment Analysis, MongoDB, Smart City, Shot classification, Zero-shot classification, Civic grievance management, Civic Grievance Management, Duplicate detection, Duplicate Detection, Government data processing, Image-based analysis, ImageBased Analysis, Multimodal analysis, Multimodal Analysis, Optical character recognition, Optical Character Recognition (OCR), Optical-, Social Media Mining, Social media minings, Urban governance, Urban governance automation, Urban Governance Automation, Urban growth, Websites, X(formerly twitter) API integration, X(formerly Twitter) API Integration, Zero-Shot Classification

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Export Date: 20 September 2026; Cited By: 0; Conference name: 6th International Conference on Trends in Material Science and Inventive Materials, ICTMIM 2026; Conference date: 8 April 2026 through 10 April 2026; Conference code: 223974

Smartphone Apps for Domestic Violence Prevention: A Systematic Review

Item Type Journal Article

Author M. Sumra

Author S. Asghar

Author K.S. Khan

Author J.M. Fernández-Luna

Author J.F. Huete

Author A. Bueno-Cavanillas

Abstract Smartphone applications or apps are increasingly being produced to help with protection against the risk of domestic violence. There is a need to formally evaluate their features. Objective: This study systematically reviewed app-based interventions for domestic violence prevention, which will be helpful for app developers. Methods: We overviewed all apps concerning domestic violence awareness and prevention without language restrictions, collating information about features and limitations. We conducted searches in Google, the Google Play Store, and the App Store (iOS) covering a 10-year time period (2012–2022). We collected data related to the apps from the developers' descriptions, peer reviewed research articles, critical reviews in blogs, news articles, and other online sources. Results: The search identified 621 potentially relevant apps of which 136 were selected for review. There were five app categories: emergency assistance (n = 61, 44.9%), avoidance (n = 29, 21.3%), informative (n = 29, 21.3%), legal information (n = 10, 7.4%), and self-assessment (n = 7, 5.1%). Over half the apps (n = 97, 71%) were released in 2020–22. Around a half were from north-east America (n = 63, 46.3%). Where emergency alerts existed, they required triggering by the potential victim. There was no automation. Content analysis showed 20 apps with unique features, including geo-fences, accelerometer-based alert, shake-based alert, functionality under low resources, alert auto-cancellation, anonymous communication, and data encryption. None of the apps deployed artificial intelligence to assist the potential victims. Conclusions: Apps currently have many limitations. Future apps should focus on automation, making better use of artificial intelligence deploying multimedia (voice, video, image capture, text and sentiment analysis), speech recognition, and pitch detection to aid in live analysis of the situation and for accurately generating emergency alerts. © 2023 by the authors.

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Publisher MDPI

Publication International Journal of Environmental Research and Public Health

ISSN 16617827 (ISSN)**Issue** 7**Journal Abbr** Int. J. Environ. Res. Public Health**Date Added** 20/09/2026, 8:35:22 pm**Modified** 20/09/2026, 8:35:22 pm**Tags:**

artificial intelligence, automation, human, prevention and control, United States, Artificial Intelligence, systematic review, social media, Review, sentiment analysis, content analysis, information processing, mobile application, Mobile Applications, text analysis, peer review, accuracy assessment, automatic speech recognition, avoidance behavior, domestic violence, Domestic Violence, emergency treatment, encryption, gender relations, health promotion, image processing, information system, intimate partner violence, legal aspect, mobile phone, North America, partner violence, pitch discrimination, prevention, reminder system, resource limited setting, self evaluation, smartphone, Smartphone, smartphone apps, victim, violence against women, violence prevention, womens health, womens status

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Export Date: 20 September 2026; Cited By: 13; Funding details: Ministerio de Educación, Cultura y Deporte, MECD; Correspondence Address: J.M. Fernández-Luna; Department of Computer Science and Artificial Intelligence, University of Granada, Granada, 18071, Spain; email: jmfluna@decsai.ugr.es

SMATM: Social Media Account Topic Modeling with Multi-Modal Data and Hashtag Weighting

Item Type Journal Article**Author** J. Song**Author** X. Zhang**Author** S. Cui**Author** N. Xu**Author** D. Chen

Abstract With the rapid growth of global social media users, platforms have become crucial venues for shaping public opinions. However, most existing multimodal topic models focus on large-scale content analysis and often fail to capture nuanced patterns in individual user content. To address this gap, we propose Social Media Account Topic Modeling (SMATM), a novel account-level multimodal topic modeling framework designed to enhance the analysis of individual users' multimodal data and extract fine-grained personalized themes. SMATM innovatively enhances the ability of topic models to capture social media account data characteristics by introducing a label weighting module, employs a flexible parameter learning module to improve the model's adaptability to users' sparse content, and proposes novel evaluation metrics

that leverage large language models' (LLMs) understanding of complex contexts to enhance model interpretability. Experiments conducted on a multimodal account-level dataset collected from social media marketing scenarios validated the effectiveness of the SMATM model in account-level topic extraction. Compared with existing baseline models and ablation models, SMATM achieved significant leads in both topic consistency and diversity evaluation metrics. Particularly in terms of interpretability, the SMATM framework elevated theme accuracy from a baseline of 0.1864 to 0.6059. This 3.25-fold enhancement underscores the model's superior capability for analyzing individual user behavior and multimodal data. Visualization of cross-industry account themes further confirms that SMATM produces semantically consistent and relevant topics by effectively integrating textual and visual information. Overall, the SMATM model represents a substantial advance in social media content analysis, user profiling, and personalized recommendations. © 2026 by the authors.

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Issue 2

Journal Abbr Technol.

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Language model, Behavioral research, Modal analysis, Model-based OPC, Social networking (online), Social media, Data mining, Marketing, Topic Modeling, Multi-modal, Data visualization, Social media analysis, social media analysis, Large language model'-based topic modeling metric, LLM-based topic modeling metric, multi-modal account topic modeling, Multi-modal account topic modeling, multi-modal data, Multi-modal data, SMATM, Social medium account topic modeling

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Natural Science Foundation of China, NNSFC, (72474198); National Natural Science Foundation of China, NNSFC; Correspondence Address: J. Song; State Key Laboratory of Media Convergence and Communication, Communication University of China, Beijing, 100024, China; email: songjinbao@cuc.edu.cn

Smile, be happy :) emoji embedding for visual sentiment analysis

Item Type Conference Paper

Author Z. Al-Halah

Author A. Aitken

Author W. Shi

Author J. Caballero

Abstract Due to the lack of large-scale datasets, the prevailing approach in visual sentiment analysis is to leverage models trained for object classification in large datasets like ImageNet. However, objects are sentiment neutral which hinders the expected gain of transfer learning for such tasks. In this work, we propose to overcome this problem by learning a novel sentiment-aligned image embedding that is better suited for subsequent visual sentiment analysis. Our embedding leverages the intricate relation between emojis and images in large-scale and readily available data from social media. Emojis are language-agnostic, consistent, and carry a clear sentiment signal which make them an excellent proxy to learn a sentiment aligned embedding. Hence, we construct a novel dataset of 4 million images collected from Twitter with their associated emojis. We train a deep neural model for image embedding using emoji prediction task as a proxy. Our evaluation demonstrates that the proposed embedding outperforms the popular object-based counterpart consistently across several sentiment analysis benchmarks. Furthermore, without bells and whistles, our compact, effective and simple embedding outperforms the more elaborate and customized state-of-the-art deep models on these public benchmarks. Additionally, we introduce a novel emoji representation based on their visual emotional response which support a deeper understanding of the emoji modality and their usage on social media. © 2019 IEEE.

Date 2019

DOI 10.1109/ICCVW.2019.00550

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Language English

Extra Journal Abbreviation: Proc. - Int. Conf. Comput. Vis. Workshop, ICCVW

Publisher Institute of Electrical and Electronics Engineers Inc.

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Pages 4491-4500

Proceedings Title Proc. - Int. Conf. Comput. Vis. Workshop, ICCVW

Conference Name Proceedings - 2019 International Conference on Computer Vision Workshop, ICCVW 2019

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Classification (of information), Sentiment analysis, Computer vision, Social networking (online), Transfer learning, Large dataset, Object classification, Embeddings, Emotional response, Emotion classification, State of the art, Image analysis, Visual sentiment analysis, Emoji, Large-scale datasets, Prediction tasks, Signaling, Zero shot learning

Notes:

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SMTCE: A Social Media Text Classification Evaluation Benchmark and BERTology Models for Vietnamese

Item Type Conference Paper

Author L.T. Nguyen

Author K. Van Nguyen

Author N.L.-T. Nguyen

Abstract Text classification is a typical natural language processing or computational linguistics task with various interesting applications. As the number of users on social media platforms increases, data acceleration promotes emerging studies on Social Media Text Classification (SMTCE) or social media text mining on these valuable resources. In contrast to English, Vietnamese, one of the low-resource languages, is still not concentrated on and exploited thoroughly. Inspired by the success of the GLUE, we introduce the Social Media Text Classification Evaluation (SMTCE) benchmark, as a collection of datasets and models across a diverse set of SMTCE tasks. With the proposed benchmark, we implement and analyze the effectiveness of a variety of multilingual BERT-based models (mBERT, XLM-R, and DistilmBERT) and monolingual BERT-based models (PhoBERT, viBERT, vELECTRA, and viBERT4news) for tasks in the SMTCE benchmark. Monolingual models outperform multilingual models and achieve state-of-the-art results on all text classification tasks. It provides an objective assessment of multilingual and monolingual BERT-based models on the benchmarks, which will benefit future studies about BERTology in the Vietnamese language. © 2022 PACLIC 36 (2022) Organizing Committee and PACLIC Steering Committee All rights reserved.

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Language English

Extra Journal Abbreviation: Proc. Pac. Asia Conf. Lang., Inf. Comput. PACLIC.

Publisher Association for Computational Linguistics

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Proceedings Title Proc. Pac. Asia Conf. Lang., Inf. Comput. PACLIC.

Conference Name PACLIC - Pacific Asia Conference on Language Information and Computation

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Computational linguistics, Classification (of information), Language processing, Natural language processing systems, Natural languages, Social networking (online), Social media, Social media platforms, Text classification, Text mining, Text-mining, Low resource languages, Vietnamese, Classification tasks, Classification evaluation

Notes:

EID: 2-s2.0-105037961254

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Social Group Differences in the Social Media Discussion about ChatGPT and Bing Chat

Item Type Conference Paper

Author M. Weber

Editor Aiello L.C.

Editor Mejova Y.

Editor Seneviratne O.

Editor Sun J.

Editor Kaiser S.

Editor Staab S.

Abstract This paper investigates how social media discussions evolved after the release and adoption of large language models by a wider public, with a focus on ChatGPT and the integration of a GPT model into Bing. The study aims to explore how social factors impact the way in which NLP technologies are perceived in social media posts. Using the official Twitter API, we collected a dataset of English and German tweets posted between November 30, 2022, and February 19, 2023. The study employs sentiment analysis and demographic prediction, the results reveal that tweets mentioning 'Bing' (related to the integration of a GPT model) were more likely to be negative compared to tweets about 'ChatGPT', with female users relatively more likely to express negative sentiment. The sentiment of tweets varied by language and whether an account belonged to an organization or not. This study provides insights into how social factors shape the discourse around NLP technologies on Twitter. © 2024 Copyright held by the owner/author(s)

Date 2024

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Language English

Extra Journal Abbreviation: Proc. ACM Web Sci. Conf., WebSci

Publisher Association for Computing Machinery, Inc

ISBN 979-840070334-8 (ISBN)**Pages** 114-118**Proceedings Title** Proc. ACM Web Sci. Conf., WebSci**Conference Name** Proceedings of the 16th ACM Web Science Conference, WebSci 2024**Date Added** 20/09/2026, 8:35:35 pm**Modified** 20/09/2026, 8:35:35 pm**Tags:**

Language model, Social groups, Economic and social effects, Sentiment analysis, Social networking (online), Social media, Negative sentiments, Demographic predictions, Group differences, Social factor

Notes:

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: M. Weber; Faculty of Social Science, Goethe University, Frankfurt, Germany; email: m.weber@soz.uni-frankfurt.de; Conference name: 16th ACM Web Science Conference, WebSci 2024; Conference date: 21 May 2024 through 24 May 2024; Conference code: 199712

Social image presentation of virtual sports in social media through the analysis of Twitter data

Item Type Journal Article**Author** W. Du**Author** X. Shen**Author** S. Zhou**Author** T. Ma**Author** C. Xu

Abstract Virtual sports have become an important site for shaping public value alignment through its social representation on digital platforms, particularly within social media environments. The current study examines public discourse on X(Twitter) surrounding virtual sports during the period from the 2021 olympic virtual series to the 2023 olympic esports week. Informed by framing theory, our methodology integrates computational approaches to analyze a dataset of 15,585 tweets. We employed topic modeling to identify salient thematic patterns and conducted sentiment analysis leveraging large language models to evaluate affective dimensions within public discourse. The study identifies six core thematic dimensions that characterize the social image of virtual sports. Although overall sentiment leans positive, the distribution across themes is uneven. Notably, the themes related to event-based mobilization exhibit a marked tendency toward negative sentiment. Overall, the study offers a systematic account of how the social image of virtual sports is presented in social media discourse, providing a valuable reference point for the future construction and communication of virtual sports' public identity. © The Author(s) 2026.

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Tags:

human, Humans, psychology, Sentiment analysis, interpersonal communication, article, Social media, social media, Social Media, Topic modeling, sentiment analysis, large language model, digital media, Digital Media, mobilization, Social image, sport, Sports, Virtual sports

Notes:

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Social media polarization during conflict: insights from an ideological stance dataset on Reddit comments

Item Type Journal Article

Author H.J. Ali

Author A. Abrar

Author S.M.H. Hossain

Author M.F. Mridha

Abstract In politically sensitive scenarios such as wars, social media often becomes a platform for polarized discourse and expressions of strong ideological stances. While prior studies have explored ideological stance detection in general contexts, limited attention has been given to conflict-specific settings. This study addresses this gap by analyzing 9969 Reddit comments related to the Israel–Palestine conflict collected between October 2023 and August 2024. The comments were manually annotated into three stance classes—Pro-Israel, Pro-Palestine, and Neutral—with high inter-annotator agreement (Fleiss’ $\kappa = 0.93$). We conduct a comprehensive comparative evaluation of neural networks, pre-trained language models, and large language

models (LLMs), along with multiple prompt engineering strategies and an LLM majority voting ensemble. Model performance is assessed using accuracy, precision, recall, and F1-score. Among the tested approaches, the Scoring and Reflective Re-read prompting strategy applied to Mixtral 8x7B achieves the best overall performance across evaluation metrics. Beyond model performance, the findings provide empirical insights into how ideological polarization manifests in online discourse surrounding the Israel–Palestine conflict, offering a computational perspective on the distribution of Pro-Israel, Pro-Palestine, and Neutral narratives. The dataset used in this study is publicly available to support further research on ideological stance detection in politically sensitive contexts. © The Author(s) 2026.

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Notes:

EID: 2-s2.0-105044604793

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Abrar; Department of Computer Science and Engineering, Islamic University of Technology, Dhaka, Gazipur, Bangladesh; email: ajwadabrar@iut-dhaka.edu

Social Media Sentiment Analysis

Item Type Journal Article

Author J.Y.M. Nip

Author B. Berthelier

Abstract Social media sentiment analysis is the computational detection and extraction of human subjective evaluation of objects embedded on social media. Previous sentiment analysis was conducted on isolated written texts, and typically classified sentiment into positive, negative, and neutral states. Social media sentiment analysis has included multi-modal texts, temporal dynamics, interactions, network relationships, and sentiment propagation. Specific emotions and sentiment intensity are also detected. © 2024 by the authors.

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large language models, artificial intelligence, machine learning, social media, deep learning, natural language processing, aspect-based sentiment analysis, knowledge graphs, multi-modal analysis, sentiment

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Export Date: 20 September 2026; Cited By: 13; Correspondence Address: J.Y.M. Nip; Discipline of Media and Communications, School of Art, Communication and English, Faculty of Arts and Social Sciences, The University of Sydney, Sydney, 2006, Australia; email: joyce.nip@sydney.edu.au

Social Media Topic Classification on Greek Reddit

Item Type Journal Article

Author C. Mastrokostas

Author N. Giarelis

Author N. Karacapilidis

Abstract Text classification (TC) is a subtask of natural language processing (NLP) that categorizes text pieces into predefined classes based on their textual content and thematic aspects. This process typically includes the training of a Machine Learning (ML) model on a labeled dataset, where each text example is associated with a

specific class. Recent progress in Deep Learning (DL) enabled the development of deep neural transformer models, surpassing traditional ML ones. In any case, works of the topic classification literature prioritize high-resource languages, particularly English, while research efforts for low-resource ones, such as Greek, are limited. Taking the above into consideration, this paper presents: (i) the first Greek social media topic classification dataset; (ii) a comparative assessment of a series of traditional ML models trained on this dataset, utilizing an array of text vectorization methods including TF-IDF, classical word and transformer-based Greek embeddings; (iii) a fine-tuned GREEK-BERT-based TC model on the same dataset; (iv) key empirical findings demonstrating that transformer-based embeddings significantly increase the performance of traditional ML models, while our fine-tuned DL model outperforms previous ones. The dataset, the best-performing model, and the experimental code are made public, aiming to augment the reproducibility of this work and advance future research in the field. © 2024 by the authors.

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Notes:

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Export Date: 20 September 2026; Cited By: 6; Correspondence Address: N. Giarelis; Industrial Management and Information Systems Lab, MEAD, University of Patras, Rio Patras, 26504, Greece; email: giarelis@ceid.upatras.gr

Social Sentiment Analysis of College Students Based on Semi Supervised Machine Learning in Online Ideological Education

Item Type Journal Article

Author Q.-L. Feng

Author Y.-F. Zhu

Abstract In online ideological education, social emotional analysis of college students is of great significance in improving educational effectiveness. This article proposes an analysis method based on semi supervised machine learning to improve emotion recognition ability in limited labeled data. Firstly, collect text data from college students' social media platforms, and after data cleaning and segmentation processing, extract TF-IDF features as model inputs. Secondly, combining label propagation algorithm to automatically generate labels for unlabeled data, using support vector machine (SVM) to classify and model emotion categories, achieving the classification of positive, negative, and neutral emotions. Experimental verification shows that this method can effectively utilize a small amount of annotated data, with improved classification accuracy and F1 score compared to traditional methods. The research results provide technical support for emotional analysis in online ideological education, which is helpful for precise intervention and the development of personalized education plans. © 2026 ISSN 2414-8105.

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Notes:

EID: 2-s2.0-105048747378

Export Date: 20 September 2026; Cited By: 0

Socio-Technical Constraints and Cross-Platform Model Generalization: Text Classification in Brazilian Public Security

Item Type Conference Paper**Author** S. Rocha**Author** C.H.V. Silva**Author** M.J. Da Silva**Author** J.T. Neto**Author** C.H.G. Ferreira**Author** G.D. Gonçalves

Abstract Stance detection, framed as a text classification task, in social media discourse on sensitive topics such as public security faces critical challenges in low-resource settings (LRS), where data, model, and computational scarcity hinder the development of robust NLP systems. This study addresses these challenges in the Brazilian public security domain by introducing a hybrid annotation strategy that combines human judgment with Large Language Model (LLM) validation to efficiently create high-quality labeled datasets. We conduct a systematic evaluation of text classification models for stance detection, including traditional classifiers (SVM, Random Forest) and the transformer-based fine-tuned, on user comments from X and YouTube, focusing on both intra-platform and cross-platform generalization. Our results reveal a critical asymmetry in transferability. Models trained on X generalize significantly better to YouTube (F1 drops of 1%-17%) than the reverse (drops up to 21%). This finding demonstrates that the socio-technical architecture of platforms - specifically, linguistic constraints imposed by design features such as character limits - shapes the quality and transferability of learned representations. By providing empirical evidence on the interplay between platform design and model generalizability, our work offers a new perspective for building scalable NLP systems in resource-constrained environments. © 2026 Copyright held by the owner/author(s).

Date 2026**DOI** 10.1145/3795766.3799777**URL** <https://www.scopus.com/pages/publications/105040832918?origin=resultslist>**Archive** Scopus**Language** English**Extra** Journal Abbreviation: WebSci - Proc. ACM Web Science Conf.**Publisher** Association for Computing Machinery, Inc**ISBN** 979-840072504-3 (ISBN)**Pages** 58-63**Proceedings Title** WebSci - Proc. ACM Web Science Conf.**Conference Name** WebSci 2026 - Proceedings of the 18th ACM Web Science Conference**Date Added** 20/09/2026, 8:35:13 pm**Modified** 20/09/2026, 8:35:13 pm

Tags:

Computational linguistics, Natural language processing systems, Large datasets, Sociotechnical, Public security, YouTube, Text classification, Text processing, Labeled data, Classification tasks, Model generalization, Cross-platform, NLP systems, Platform models, Technical constraints

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Socrates or Smartypants: Testing Logic Reasoning Capabilities of Large Language Models with Logic Programming-Based Test Oracles

Item Type Conference Paper

Author Z. Xu

Author J. Ding

Author Y. Lou

Author K. Zhang

Author D. Gong

Author Y. Li

Editor Koenig S.

Editor Jenkins C.

Editor Taylor M.E.

Abstract Large Language Models (LLMs) have achieved significant progress in language understanding and reasoning. Evaluating and analyzing their logical reasoning abilities has therefore become essential. However, existing datasets and benchmarks are often limited to overly simplistic, unnatural, or contextually constrained examples. In response to the growing demand, we introduce SMARTYPAT-BENCH, a challenging, naturally expressed, and systematically labeled benchmark derived from real-world high-quality Reddit posts containing subtle logical fallacies. Unlike existing datasets and benchmarks, it provides more detailed annotations of logical fallacies and features more diverse data. To further scale up the study and address the limitations of manual data collection and labeling, such as fallacy-type imbalance and labor-intensive annotation, we

introduce SMARTYPAT, an automated framework powered by logic programming-based oracles. SMARTYPAT utilizes Prolog rules to systematically generate logically fallacious statements, which are then refined into fluent natural-language sentences by LLMs, ensuring precise fallacy representation. Extensive evaluation demonstrates that SMARTYPAT produces fallacies comparable in subtlety and quality to human-generated content and significantly outperforms baseline methods. Finally, experiments reveal insights into LLM capabilities, highlighting that while excessive reasoning steps hinder fallacy detection accuracy, structured reasoning enhances fallacy categorization performance. © 2026, Association for the Advancement of Artificial Intelligence (www.aaai.org). All rights reserved.

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Computational linguistics, Language model, Artificial intelligence, Natural language processing systems, Benchmarking, Real-world, Growing demand, Software testing, Logical reasoning, Language understanding, Logic programming, Logic reasoning, Logic-programming, PROLOG (programming language), Reasoning ability, Reasoning capabilities, Test oracles

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Span Extraction Aided Improved Code-mixed Sentiment Classification

Item Type Conference Paper

Author S. Ramaneswaran

Author S. Benhur

Author S. Ghosh

Abstract Sentiment classification is a fundamental NLP task of detecting the sentiment polarity of a given text. In this paper we show how solving sentiment span extraction as an auxiliary task can help improve final sentiment classification performance in a low-resource code-mixed setup. To be precise, we don't solve a simple multi-task learning objective, but rather design a unified transformer framework that exploits the bidirectional connection between the two tasks simultaneously. To facilitate research in this direction we release gold-standard human-annotated sentiment span extraction dataset for Tamil-english code-switched texts. Extensive experiments and strong baselines show that our proposed approach outperforms sentiment and span prediction by 1.27% and 2.78% respectively when compared to the best performing MTL baseline. We also establish the generalizability of our approach on the Twitter Sentiment Extraction dataset. We make our code and data publicly available on GitHub 1 © 2022 COLING. All Rights Reserved.

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Spell Correction for the Minangkabau Language Using BERT-Based Embeddings

Item Type Journal Article

Author D. Soyusiawaty

Author A. Fadlil

Abstract Spell checking is an essential component of natural language processing, as it directly influences applications such as sentiment analysis, text classification, and machine translation. Developing a reliable system for low-resource languages like Minangkabau is challenging due to frequent spelling variations and limited annotated data. This study proposes a contextual spell correction model using pre-trained IndoBERT and multilingual BERT (mBERT) embeddings applied without additional training. The method masks misspelled words, extracts the contextual embedding of the [MASK] token, and compares it with candidate embeddings generated through dictionary filtering and Levenshtein Distance. Evaluation was conducted on the Spell Error Corpus for Minangkabau Language (SPEML), which includes insertion errors, deletion errors, substitution errors, transposition errors, punctuation errors, real-word errors, and loanword errors. Results show that mBERT consistently outperformed IndoBERT, achieving an average F1-score of 0.83 compared to 0.75. Statistical validation using paired t-test and Wilcoxon signed-rank test further confirmed that the performance difference between the two models was significant. Both models reached perfect scores (1.0) in real-word and loanword categories, and strong results in insertion_medium (0.97 for mBERT and 0.95 for IndoBERT). The lowest performance occurred in deletion_short (0.52 for IndoBERT) and long words cases (0.57 for mBERT). In addition, a small-scale external validation using 100 Twitter/X sentences was conducted to assess the applicability of the proposed method to real-world social media text. Overall, the findings confirm the effectiveness of contextual embeddings for Minangkabau spelling correction while highlighting challenges in long misspelled words, deletion errors, and informal real-world text. © 2026, Ikatan Ahli Informatika Indonesia. All rights reserved.

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Stance Detection in Arabic Tweets: A Machine Learning Framework for Identifying Extremist Discourse

Item Type Journal Article

Author A.K. Alkhraji

Author A.M. Azmi

Abstract Terrorism remains a critical global challenge, and the proliferation of social media has created new avenues for monitoring extremist discourse. This study investigates stance detection as a method to identify Arabic tweets expressing support for or opposition to specific organizations associated with extremist activities, using Hezbollah as a case study. Thousands of relevant Arabic tweets were collected and manually annotated by expert annotators. After extensive preprocessing and feature extraction using term frequency-inverse document frequency (tf-idf), we implemented traditional machine learning (ML) classifiers—Support Vector Machines (SVMs) with multiple kernels, Multinomial Naïve Bayes, and Weighted K-Nearest Neighbors. ML models were selected over deep learning (DL) approaches due to (1) limited annotated Arabic data availability for effective DL training; (2) computational efficiency for resource-constrained environments; and (3) the critical need for interpretability in counterterrorism applications. While interpretability is not a core focus of this work, the use of traditional ML models (rather than DL) makes the system inherently more transparent and readily adaptable for future integration of interpretability techniques. Comparative experiments using FastText word embeddings and tf-idf with supervised classifiers revealed superior performance with the latter approach. Our best result achieved a macro F-score of 78.62% using SVMs with the RBF kernel, demonstrating that interpretable ML frameworks offer a viable and resource-efficient approach for monitoring extremist discourse in Arabic social media. These findings highlight the potential of such frameworks to support scalable and explainable counterterrorism tools in low-resource linguistic settings. © 2025 by the authors.

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Stance Detection on Social Media with Background Knowledge

Item Type Conference Paper**Author** A. Li**Author** B. Liang**Author** J. Zhao**Author** B. Zhang**Author** M. Yang**Author** R. Xu**Editor** Bouamor H.**Editor** Pino J.**Editor** Bali K.

Abstract Identifying users' stances regarding specific targets/topics is a significant route to learning public opinion from social media platforms. Most existing studies of stance detection strive to learn stance information about specific targets from the context, in order to determine the user's stance on the target. However, in real-world scenarios, we usually have a certain understanding of a target when we express our stance on it. In this paper, we investigate stance detection from a novel perspective, where the background knowledge of the targets is taken into account for better stance detection. To be specific, we categorize background knowledge into two categories: episodic knowledge and discourse knowledge, and propose a novel Knowledge-Augmented Stance Detection (KASD) framework. For episodic knowledge, we devise a heuristic retrieval algorithm based on the topic to retrieve the Wikipedia documents relevant to the sample. Further, we construct a prompt for ChatGPT to filter the Wikipedia documents to derive episodic knowledge. For discourse knowledge, we construct a prompt for ChatGPT to paraphrase the hashtags, references, etc., in the sample, thereby injecting discourse knowledge into the sample. Experimental results on four benchmark datasets demonstrate that our KASD achieves state-of-the-art performance in in-target and zero-shot stance detection. ©2023 Association for Computational Linguistics.

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Stay Tuned: Improving Sentiment Analysis and Stance Detection Using Large Language Models

Item Type Journal Article

Author M. Griswold

Author M.W. Robbins

Author M.S. Pollard

Abstract Sentiment analysis and stance detection are key tasks in text analysis, with applications ranging from understanding political opinions to tracking policy positions. Recent advances in large language models (LLMs) offer significant potential to enhance sentiment analysis techniques and to evolve them into the more nuanced task of detecting stances expressed toward specific subjects. In this study, we evaluate lexicon-based models, supervised models, and LLMs for stance detection using two corpuses of social media data—a large corpus of tweets posted by members

of the U.S. Congress on Twitter and a smaller sample of tweets from general users—which both focus on opinions concerning presidential candidates during the 2020 election. We consider several fine-tuning strategies to improve performance—including cross-target tuning using an assumption of congressmembers’ stance based on party affiliation—and strategies for fine-tuning LLMs, including few shot and chain-of-thought prompting. Our findings demonstrate that: 1) LLMs can distinguish stance on a specific target even when multiple subjects are mentioned, 2) tuning leads to notable improvements over pretrained models, 3) cross-target tuning can provide a viable alternative to in-target tuning in some settings, and 4) complex prompting strategies lead to improvements over pretrained models but underperform tuning approaches. © The Author(s), 2025. Published by Cambridge University Press on behalf of The Society for Political Methodology.

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Stereotype-Strategy Interplay: Warmth and Competence as Moderators of Persuasion Effectiveness in Online Communities

Item Type Conference Paper

Author Y. Wang

Author Y. Yang

Author N. Yao

Author J. Zhu

Author H. Chen

Editor Shen W.
Editor Barthes J.-P.
Editor Luo J.
Editor Shi Y.
Editor Zhu H.
Editor Zhang J.
Editor Peng K.
Editor Hou L.

Abstract Online communities are pivotal collaborative environments where successful persuasion is essential for shaping opinions and fostering constructive discourse. Existing persuasion research typically overlooks how the stereotype content of the subject under discussion systematically shapes the effectiveness of the persuasion strategies. To bridge this gap, we propose a novel framework that investigates how stereotype content, modeled by the dimensions of warmth and competence, moderates the effectiveness of persuasion strategies. Leveraging a multi-stage pipeline powered by large language models (LLMs), we annotate a large-scale dataset from Reddit's r/ChangeMyView, coding stereotype profiles in original posts and fine-grained persuasion strategies across 72,627 comments from 1,953 discussions. Through both machine and human evaluations, we examine how persuasion strategy effectiveness differs across stereotypic quadrants (high vs. low on warmth and competence) based on Stereotype Content Model (SCM). Our analysis reveals that persuasion effectiveness is systematically shaped by the two dimensions of SCM. Specifically, the strategy combinations are more effective when the target is characterized by low competence and high warmth. These findings provide a predictive, psychologically-grounded framework for achieving context-aware and precise persuasion in digital discourse. © 2026 IEEE.

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Stock Price Prediction with Heterogeneous Data Fusion

Item Type Conference Paper

Author F. Karadaş

Author B. Eravcı

Author A.M. Özbayoğlu

Editor van den Herik H.J.

Editor Rocha A.

Editor Steels L.

Abstract Accurate stock price forecasting in volatile financial markets requires integrating heterogeneous data with advanced architectures. This study proposes a multimodal framework for stock price prediction that integrates historical financial data with textual sentiment from news and social media. The framework’s performance is benchmarked across two core empirical comparisons: a domain-specific sentiment model (FinBERT) versus a general-purpose large language model (ChatGPT-4o), and a classic forecasting architecture (LSTM) versus a modern state-space model (Mamba). We employ a dual-evaluation methodology, assessing performance with both standard statistical metrics R^2 , MAE) and a practical market simulation with initial capital to assess real-world profitability. The framework also incorporates an influence weighted sentiment score (IWSS), a metric that quantifies sentiment by weighting it with user engagement and influence to better capture market impact. Results demonstrate that models incorporating sentiment, particularly the IWSS, yield superior trading returns. We find that the generalist ChatGPT-4o performs competitively against the specialized FinBERT and the findings provide a robust, data-driven framework for investors seeking to improve their decision-making processes by strategically incorporating multimodal data analytics into financial forecasting. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Stress Identification in Telugu Using Large Language Models

Item Type Conference Paper

Author T. Durairaj

Author R. Rohan

Author H.S. Allan

Author S. Sivakumar

Author N. Jayagupta

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Editor Kovács G.

Editor Eger S.

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Editor Dobrovoljc K.

Abstract Stress, particularly mental stress, is a state of mind that results from one's perception of the external or internal environment, which in turn leads to the psychological response of anxiety or distress. Often, social media is used as a platform to express this kind of stress in the form of extreme statements. It is crucial to identify and moderate such content on social media platforms, as the virtual space is accessed by millions of people from around the world. This paper develops and evaluates machine learning models to identify stressed and non-stressed comments from social media texts. We utilized various Large Language Models (LLMs) like Bidirectional Encoder Representations from Transformers (BERT), Robustly Optimized BERT Pretraining Approach (RoBERTa), Flan-T5 and Bidirectional Auto-Regressive Transformers (BART). We also experimented with traditional models to prove the superiority of LLMs over traditional models such as K-Nearest Neighbour, Logistic Regression, Decision Tree Classifier, Support Vector Machine and also ensemble models including Random Forest, Meta Stacking, Gradient Boosting and Bootstrap Aggregating. Additionally, we examined the impact of different vectorization techniques, including TF-IDF and Word2Vec on the traditional models. These models were applied and evaluated on a dataset consisting of Telugu language comments, offering important insights into the effectiveness of the approaches used. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Style and Stance Alignment in Reddit Discourse

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Author	S.A.A. Jordao
Author	C.S. Santos
Author	F. Santos
Author	M.J.P. Dantas
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Abstract	This work investigates whether writing style is associated with stance alignment in large scale online discourse. We construct a Reddit based pipeline that transforms r/AskReddit questions and comments interactions into stance units using large language models (LLMs), stylistic embeddings, and semantic embeddings. The study begins with 169,053 original AskReddit questions, categorizes them into 48 semantic classes, and filters them for 'stance suitability'. The full question corpus contains 176,932 rows; in the target stance evaluation stage, 22,238 original questions from 10 selected semantic classes are paired with up to 200 top level comments each and annotated by an LLM, yielding 277,474 comment level model outputs and 352,407 stance instances. Style is represented with answer text embeddings trained to capture writing style, while semantic context is represented with embeddings of question summaries, stance targets, and canonical opinions. We analyze 8,000,000 randomly sampled stance instance pairs by measuring style similarity, topic similarity, and stance label agreement. Across topic similarity thresholds from 0.50 to 0.95, pairs with high stylistic similarity consistently show higher model annotated stance agreement than the

topic similar baseline, with lifts of approximately 6% to 12% points. A parallel augmentation analysis shows that synthetic questions produced by the tested models under generalize real class diversity, motivating the use of original questions for the main stance analysis. These findings suggest that writing style carries a weak but measurable signal related to stance alignment under semantically controlled conditions. © 2026 IEEE.

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Suicidal Text Detection Using Machine Learning and Large Language Model

Item Type Conference Paper

Author H. Jain

Author A. Kashyap

Author S. Sivakumar

Author V.D. Chakravarthy

Author P. Selvaraju

Author F. Yakub

Abstract The growing prevalence of suicidal ideation in online communities calls for robust and accessible early detection systems. This study presents a machine learning-based approach for identifying suicidal intent from user-generated text, specifically Reddit posts. A curated dataset of approximately 348,000 posts—consolidated to 232,074 unique entries—was collected from the r/SuicideWatch, r/depression, and r/teenagers subreddits using the Pushshift API. The aim of classification was to differentiate the suicidal and non-suicidal content. Several types of models were tried Gaussian Naive Bayes (GNB) Logistic Regression, and a fine-tuned BERT transformer. Extraction of features was carried out on CountVectorizer as well as the Term Frequency-Inverse Document Frequency (TF-IDF). In the models, Bidirectional Encoder Representations of Transformers (BERT) recorded the closest results of 97.68 % and 97.66 % accuracy and F1-scores, respectively, as compared to the traditional classifiers. The paper evidence shows the possibility of the NLP and deep learning techniques combination as scalable and accurate mental health support methods. ©2025 IEEE.

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Supervised Graph Contrastive Learning for Few-Shot Node Classification

Item Type Conference Paper

Author Z. Tan

Author K. Ding

Author R. Guo

Author H. Liu

Editor Amini M.-R.

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Abstract Graphs present in many real-world applications, such as financial fraud detection, commercial recommendation, and social network analysis. But given the high cost of graph annotation or labeling, we face a severe graph label-scarcity problem, i.e., a graph might have a few labeled nodes. One example of such a problem is the so-called few-shot node classification. A predominant approach to this problem resorts to episodic meta-learning. In this work, we challenge the status quo by asking a fundamental question whether meta-learning is a must for few-shot node classification tasks. We propose a new and simple framework under the standard few-shot node classification setting as an alternative to meta-learning to learn an effective graph encoder. The framework consists of supervised graph contrastive learning with novel mechanisms for data augmentation, subgraph encoding, and multi-scale contrast on graphs. Extensive experiments on three benchmark datasets (CoraFull, Reddit, Ogbn) show that the new framework significantly outperforms state-of-the-art meta-learning based methods. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Sustainable Artificial Intelligence-Based Twitter Sentiment Analysis on COVID-19 Pandemic

Item Type Journal Article

Author T. Vaiyapuri

Author S.K. Jagannathan

Author M.A. Ahmed

Author K.C. Ramya

Author G.P. Joshi

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Author G. Lee

Abstract The COVID-19 outbreak is a disastrous event that has elevated many psychological problems such as lack of employment and depression given abrupt social changes. Simultaneously, psychologists and social scientists have drawn considerable attention towards understanding how people express their sentiments and emotions during the pandemic. With the rise in COVID-19 cases with strict lockdowns, people expressed their opinions publicly on social networking platforms. This provides a deeper knowledge of human psychology at the time of disastrous events. By applying user-produced content on social networking platforms such as Twitter, the sentiments and views of people are analyzed to assist in introducing awareness campaigns and health intervention policies. The modern evolution of artificial intelligence (AI) and natural language processing (NLP) mechanisms has revealed remarkable performance in sentimental analysis (SA). This study develops a new Marine Predator Optimization with Natural Language Processing for Twitter Sentiment Analysis (MPONLP-TSA) for the COVID-19 Pandemic. The presented MPONLP-TSA model is focused on the

recognition of sentiments that exist in the Twitter data during the COVID-19 pandemic. The presented MPONLP-TSA technique undergoes data preprocessing to convert the data into a useful format. Furthermore, the BERT model is used to derive word vectors. To detect and classify sentiments, a bidirectional recurrent neural network (BiRNN) model is utilized. Finally, the MPO algorithm is exploited for optimal hyperparameter tuning process, and it assists in enhancing the overall classification performance. The experimental validation of the MPONLP-TSA approach can be tested by utilizing the COVID-19 tweets dataset from the Kaggle repository. A wide comparable study reported a better outcome of the MPONLP-TSA method over current approaches. © 2023 by the authors.

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Swahili and Code-Switched English-Swahili Political Hate Speech Detection Textual Dataset

Item Type Journal Article

Author N.O. Onyango

Author L. Wanzare

Author J. Obuhuma

Abstract The growth of online platforms coupled with users' ability to post and comment anonymously has resulted in an alarming spread of offensive and hateful content especially during election period. Efforts to monitor and detect hate speech, including computer based automated hate speech detection models remain unsatisfactory. This is primarily due to the fact that most annotated hate speech datasets focus on English, despite the growing use of low-resourced languages like Swahili and code-switched English-Swahili on Kenyan social media platforms such as Twitter (now X). Currently, there is only one annotated publicly available hate speech dataset having Swahili and code-switched English-Swahili texts. However, the dataset lacks language annotation distinguishing Swahili from English-Swahili. This poses a challenge, as the dataset also contains pure English tweets, making it difficult for researchers who require tweets in a specific language. Moreover, the dataset lacks hate target annotations, such as politics, disability, religion, ethnicity, social-status, nationality, gender, and others. This is a challenge since hate target labels help in comprehensive analysis of the hate stereotypes. This study therefore fills this gap by curating Swahili and code-switched English-Swahili hate speech dataset and annotating with hate class, target and language. Data preparation involved combining the Politikweli, AfriSenti, and Hate_Speech_Kenya datasets, resulting in a collection of 101,014 tweets. Hate target labels include nationality, social status, politics, disability, ethnicity, gender, religion and other. Multilingual index of 0.72 was achieved for code switching. Randolph's free marginal kappa score of 0.435 and 0.552 was achieved for the first and second pilot dataset respectively with a significant difference of 0.117. The overall inter annotator score using Randolph's multirater kappa was 0.70 showing substantial agreement. As a result, this study provides hate speech textual dataset having a total of 19369 tweets labeled as hate, offensive and neither. To assess the dataset's suitability, experiments were conducted using a non-linear SVM and a Bi-directional Long Short-Term Memory (Bi-LSTM) model. SwahBERT model, a variant of BERT pre-trained on Swahili texts was fine-tuned on the code-switched data and achieved an F1-score of 0.89. Bi-LSTM model with fastText embeddings achieved an f1-score, accuracy, recall and precision of 0.8 on the test set showing the ability of fastText embeddings in capturing the nuances in code-switched texts. The recall for the hate speech class improved from 0.57 to 0.76 when the Synthetic Minority Oversampling Technique (SMOTE) was applied, demonstrating the dataset's practical applicability. Results for SVM trained on code-switched part of the dataset shows that traditional models performs better on a relatively smaller dataset. These results suggest that the dataset could be reused to develop machine learning models for moderation of bias and hate speech contents from social media especially in low-resource languages. This can help in promoting good mental health among social media users. The dataset can also serve as a benchmark for evaluating NLP techniques in low-resource settings. © 2025 Chinese Academy of Sciences.

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Sympathy as a Lens for Human-Robot Interaction: Analysing YouTube Responses to Robot Abuse

Item Type Conference Paper**Author** V. Tolj**Author** C. Neef**Author** B. Bruno**Editor** Torre I.**Editor** Baillie L.**Editor** Smart W.D.**Editor** Graaf M.D.**Editor** Gombolay M.

Abstract When witnessing the abuse of others, humans generally exhibit emotional responses. A large number of studies in Human-Robot-Interaction (HRI) have shown that humans also react with sympathy when robots are abused, but most of these insights come from controlled laboratory studies using short videos and student samples. To complement existing research with observations drawn from real-world online discussions, this paper presents a sentiment analysis of 103,413 YouTube comments on videos depicting abuse of animal-like, humanoid, and cart-shaped robots. To validate our sentiment classification, we analysed the comments using a lexicon-based tool, two fine-tuned language models, and three general-purpose state-of-the-art large language models (LLMs). The comparison yielded interesting results: LLMs generally classified science-fiction-related comments, e.g., references to dystopian TV shows, as negative, while lexicon and fine-tuned models mainly labelled them as neutral. The six models agreed on the

classification of a total of 27,427 comments, which we used to explore the sentiment expressions occurring across videos featuring robots with different physical forms. Our findings provide large-scale, ecologically valid insights into how emotional responses to robot abuse are expressed and analysed in online video platforms. © 2026 Owner/Author.

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SynSym: A Synthetic Data Generation Framework for Psychiatric Symptom Identification

Item Type Conference Paper

Author M. Kang

Author J. Kim

Author H. Jeon

Author S. Hwang

Author J. An
Author Y. Kim
Author H. Kwak
Author J. An
Author J. Han

Abstract Psychiatric symptom identification on social media aims to infer fine-grained mental health symptoms from user-generated posts, allowing a detailed understanding of users' mental states. However, the construction of large-scale symptom-level datasets remains challenging due to the resource-intensive nature of expert labeling and the lack of standardized annotation guidelines, which in turn limits the generalizability of models to identify diverse symptom expressions from user-generated text. To address these issues, we propose SynSym, a synthetic data generation framework for constructing generalizable datasets for symptom identification. Leveraging large language models (LLMs), SynSym constructs high-quality training samples by (1) expanding each symptom into sub-concepts to enhance the diversity of generated expressions, (2) producing synthetic expressions that reflect psychiatric symptoms in diverse linguistic styles, and (3) composing realistic multi-symptom expressions, informed by clinical co-occurrence patterns. We validate SynSym on three benchmark datasets covering different styles of depressive symptom expression. Experimental results demonstrate that models trained solely on the synthetic data generated by SynSym perform comparably to those trained on real data, and benefit further from additional fine-tuning with real data. These findings underscore the potential of synthetic data as an alternative resource to real-world annotations in psychiatric symptom modeling, and SynSym serves as a practical framework for generating clinically relevant and realistic symptom expressions. © 2026 Owner/Author.

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System design and algorithm optimization of large language model based on knowledge graph in rumour detection

Item Type Journal Article

Author J. Shi

Abstract The widespread dissemination of misinformation makes finding new methods that combine unstructured text mining and relational data critically important. This work introduces KG-RumorDetect, a framework that combines Knowledge Graphs (KGs) with Large Language Models (LLMs) for improved rumour detection. Transformer-based language modelling was employed and sophisticated KG integration, graph neural networks, TransE, DistMult and RotatE KG embeddings, multi-modal fusion, and end-to-end joint learning. To further enhance performance and efficiency, the author apply optimisation techniques such as knowledge distillation, model pruning, quantisation, mixed-precision training, dynamic learning rate gradient clipping, and sparse attention with gradual weight adjustment. Extensive experiments confirm that KG-RumorDetect achieves high detection accuracy while significantly reducing computational cost, making it a reliable and efficient option for real-time misinformation detection. Experimental evaluations on benchmark datasets, including Propagation of Health Misinformation in Social Media (PHEME) and Credibility Corpus with Annotations for Events in Social Media (CREDBANK), demonstrate that KG-RumorDetect outperforms standard LLM-based rumour detection models. The proposed model achieves 91.2% accuracy, a 9% improvement over the baseline, with a harmonic mean of the precision and recall (F1-score) of 90.1% and an Area Under the Receiver Operating Characteristic Curve (AUC) of 0.94. The precision-recall analysis highlights that KG-RumorDetect maintains high precision even at high recall levels, reducing false positives by 20% compared to the baseline. Furthermore, training time is reduced by 16% and memory usage by 11%, making the model more suitable for real-time applications. Copyright 2026 Shi Distributed under Creative Commons CC-BY 4.0

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Systematic review of recent advances in multimodal sentiment analysis

Item Type Journal Article

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Author N.A. Shelke

Author K. Anwar

Abstract Recently, multimodal sentiment analysis (MSA) has gained significant traction due to its wide-range of applications in social media monitoring, healthcare, e-commerce, content creation, and business research. Unlike classical unimodal approaches, MSA integrates verbal and non-verbal characteristics such as text, speech, facial expressions, gestures, and physiological signals to enable a comprehensive understanding of human sentiments, particularly in human–computer interaction systems. Despite several existing reviews, many either focus on limited modalities or overlook the latest advancements, such as transformer-based and large language models (LLMs). This review presents a comprehensive and critical overview of MSA, with an integrated modalities, fusion techniques, classification, and current research in joint embedding and LLMs. It performs a comparative assessment of benchmark corpora, metrics, and model performance. The study also provides a critical analysis of current methods, highlighting their strengths, limitations, and

future directions in the field. It also extends the discussion to practical applications and long-standing issues, which will frame the future research agenda in MSA. The Literature was carefully reviewed and selected from top academic databases using systematic search strategies. This survey aims to help researchers, students, and practitioners understand the history of MSA's development over the years and explore current research directions in the rapidly emerging field. © The Author(s) 2025.

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Target-Aware Bilingual Stance Detection in Social Media Using Transformer Architecture

Item Type Journal Article

Author A.R.W. Sait

Author Y. Alkhurayyif

Abstract Stance detection has emerged as an essential tool in natural language processing for understanding how individuals express agreement, disagreement, or neutrality toward specific targets in social and online discourse. It plays a crucial role in bilingual and multilingual environments, including English-Arabic social media ecosystems, where differences in language structure, discourse style, and data availability pose significant challenges for reliable stance modelling. Existing approaches often struggle with target awareness, cross-lingual generalization, robustness to noisy user-generated text, and the interpretability of model decisions. This study aims to build a reliable, explainable target-aware bilingual stance-detection framework that

generalizes across heterogeneous stance formats and languages without retraining on a dataset specific to the target language. Thus, a unified dual-encoder architecture based on mDeBERTa-v3 is proposed. Cross-language contrastive learning offers an auxiliary training objective to align English and Arabic stance representations in a common semantic space. Robustness-oriented regularization is used to mitigate the effects of informal language, vocabulary variation, and adversarial noise. To promote transparency and trustworthiness, the framework incorporates token-level rationale extraction, enables fine-grained interpretability, and supports analysis of hallucination. The proposed model is tested on a combined bilingual test set and two structurally distinct zero-shot benchmarks: MT-CSD and AraStance. Experimental results show consistent performance, with accuracies of 85.0% and 86.8% and F1-scores of 84.7% and 86.8% on the zero-shot benchmarks, confirming stable performance and realistic generalization. Ultimately, these findings reveal that effective bilingual stance detection can be achieved via explicit target conditioning, cross-lingual alignment, and explainability-driven design. © 2026 by the authors.

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Target-Oriented Dynamic Denosing Curriculum Learning for Multimodal Stance Detection

Item Type Conference Paper

Author Z. Suo

Author S. Pan

Editor Ide I.

Editor Kompatsiaris I.

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Editor Yanai K.

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Editor Nitta N.

Editor Riegler M.

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Abstract Multimodal Stance Detection aims to classify public opinions on specific targets in social media, incorporating both text and image data. However, prior studies have overemphasized the significance of images, neglecting the presence of irrelevant images in the dataset. Moreover, previous research has shown that employing the Chain of Thought approach with large language models can introduce noise from the generated text as well as noise from the image modality into the text input modality. These noises can degrade the performance of multimodal models. Additionally, both image and text modalities exhibit complex data patterns, resulting in significant disparities in training difficulty across the dataset. To address these issues, We proposed Target-Oriented Dynamic Denoising Curriculum Learning(TODDCL), which effectively measures different types of noise and automatically tackles the noise in both image and text modalities based on their relevance to the target, in an escalating complexity order. Experimental results on five benchmark datasets demonstrate that our proposed TODDCL method achieves state-of-the-art performance in multimodal stance detection. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2025.

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Target-Oriented Relation Alignment for Cross-Lingual Stance Detection

Item Type Conference Paper

Author R. Zhang

Author N. Xu

Author H. Yang

Author Y. Tian

Author W. Mao

Abstract Stance detection is an important task in text mining and social media analytics, aiming to automatically identify the user's attitude toward a specific target from text, and has wide applications in a variety of domains. Previous work on stance detection has mainly focused on monolingual setting. To address the problem of imbalanced language resources, cross-lingual stance detection is proposed to transfer the knowledge learned from a high-resource (source) language (typically English) to another low-resource (target) language. However, existing research on cross-lingual stance detection has ignored the inconsistency in the occurrences and distributions of targets between languages, which consequently degrades the performance of stance detection in low-resource languages. In this paper, we first identify the target inconsistency issue in cross-lingual stance detection, and propose a fine-grained Target-oriented Relation Alignment (TaRA) method for the task, which considers both target-level associations and language-level alignments. Specifically, we propose the Target Relation Graph to learn the in-language and cross-language target associations. We further devise the relation

alignment strategy to enable knowledge transfer between semantically correlated targets across languages. Experimental results on the representative datasets demonstrate the effectiveness of our method compared to competitive methods under variant settings. © 2023 Association for Computational Linguistics.

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TaSC-LLM: A Large Language Model–Enabled Business Intelligence Framework for Topic Analytics in Live-Streaming E-Commerce Systems

Item Type Journal Article

Author G. Peng

Author X. Wang

Author R. Zhang

Author Y. Liu

Author J. Yao

Author J. Li

Author J. Wu

Abstract In live-streaming e-commerce systems, massive volumes of user-generated danmaku constitute a critical yet underutilized source of business intelligence. However, transforming such unstructured, noisy, and highly context-dependent textual data into structured and actionable knowledge remains a fundamental challenge for enterprise information systems. To address this issue, this study proposes TaSC-LLM, an LLM-enabled topic recognition method for constructing interpretable topic measurements from unstructured user-generated content. The proposed framework integrates topic taxonomy construction and zero-shot classification into a unified semantic reasoning pipeline. Unlike conventional topic modeling or supervised classification approaches, TaSC-LLM leverages chain-of-thought reasoning, multi-stage taxonomy induction, sliding window context modeling, and self-consistency verification to eliminate reliance on predefined label spaces and annotated training data. This design allows the system to dynamically construct and update topic taxonomies while ensuring interpretability, robustness, and cross-scenario adaptability. Empirical evaluation on three large-scale live-streaming e-commerce danmaku datasets shows that TaSC-LLM achieves strong taxonomy coverage, classification accuracy, and agreement with expert annotations. The findings suggest that LLM-based reasoning can help convert unstructured user-generated text into interpretable topic measures for downstream empirical and managerial analysis. While the present evaluation is conducted offline, TaSC-LLM provides a methodological foundation for future business applications that can be further examined under multi-session, multi-platform, and deployment-oriented conditions. © 2026 by the authors.

Date 2026

DOI 10.3390/systems14080941

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Export Date: 20 September 2026; Cited By: 0; Funding details: High-Level Scientific Research Foundation of Hebei Province; National Natural Science Foundation of China, NSFC, (72272140, 72334006, L2424116); National Natural Science Foundation of China, NSFC; Correspondence Address: X. Wang; School of Economics and Management, University of Chinese Academy of Sciences, Beijing, 100190, China; email: wangxiaoxi@bimsa.cn

Teaching AI to Decode Vaccine Hesitancy Narratives: A Few-Shot Learning and Topic Modeling Approach

Item Type Journal Article

Author M.E. Kabir

Author S.H. Tanim

Author D.D. Sellnow

Author G.L.P. Luteria

Author L. Rennert

Abstract Vaccine hesitancy—which can be defined as a delay in acceptance or the refusal to get vaccinated—has substantially increased over the past decade. This study introduces a computational and qualitative approach designed to efficiently classify stance and uncover narratives in social media discourse without relying on extensive manual annotation. Using 298,356 COVID-19 vaccine-related X posts geolocated to South Carolina (June 2021–May 2022), zero-shot and few-shot learning with instruction-tuned large language models (Mistral-7B, Meta-Llama-3.1, and DeepSeek-7B) was applied for stance detection while Latent Dirichlet Allocation (LDA) was used for topic modeling. The topic modeling identified five dominant themes in vaccine hesitant conversations: skepticism of vaccine efficacy, comparative framing, scientific justification, disapproval of regulations, and distrust. Temporal analysis revealed that skepticism peaked during late 2021, coinciding with booster campaigns and mandate debates. These findings suggest that vaccine hesitancy is influenced through complex rhetorical strategies rather than misinformation alone. These underlying narratives often frame skepticism as rational and evidence-based, using scientific language and statistical reasoning to challenge the effectiveness of vaccines. © 2026 by the authors.

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Publication Big Data and Cognitive Computing

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Issue 5

Journal Abbr Big Data Cogn. Computing

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: D.D. Sellnow; Social Media Listening Center, Department of Communication, Clemson University, Clemson, 29634, United States; email: dsellno@clemson.edu

Team ISCL_WINTER at SemEval-2023 Task 12:AfriSenti-SemEval: Sentiment Analysis for Low-resource African Languages using Twitter Dataset

Item Type Conference Paper

Author A. Hancharova

Author J. Wang

Author M. Kumar

Editor Ojha A.K.

Editor Dogruoz A.S.

Editor Da San Martino G.

Editor Madabushi H.T.

Editor Kumar R.

Editor Sartori E.

Abstract This paper presents a study on the effectiveness of various approaches for addressing the challenge of multilingual sentiment analysis in low-resource African languages. The study focuses on Task 12 of the SemEval-2023 Competition, which aims to promote interest in these languages and develop efficient models for their analysis. The approaches evaluated in the study include Support Vector Machines (SVM), translation, and an ensemble of pre-trained multilingual sentimental model methods. The paper provides a detailed analysis of the performance of each approach based on experimental results. In our findings, we suggest that the ensemble method is the most effective with an F1 Score of 0.68 on the final testing. This system ranked 19 out of 33 participants in the competition. © 2023 Association for Computational Linguistics.

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DOI 10.18653/v1/2023.semeval-1.149

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Extra Journal Abbreviation: Int. Workshop Semantic Eval., SemEval - Proc. Workshop

Publisher Association for Computational Linguistics

ISBN 978-195942999-9 (ISBN)**Pages** 1085-1089**Proceedings Title** Int. Workshop Semantic Eval., SemEval - Proc. Workshop**Conference Name** 17th International Workshop on Semantic Evaluation, SemEval 2023 -
Proceedings of the Workshop**Date Added** 20/09/2026, 8:35:42 pm**Modified** 20/09/2026, 8:35:42 pm**Tags:**

Performance, Support vector machines, Support vectors machine, Sentiment analysis, Semantics,
African languages, F1 scores, Machine translations, Ensemble methods, Model method

Notes:

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Export Date: 20 September 2026; Cited By: 1; Conference name: 17th International Workshop on
Semantic Evaluation, SemEval 2023, co-located with the 61st Annual Meeting of the Association for
Computational Linguistics, ACL 2023; Conference date: 13 July 2023 through 14 July 2023;
Conference code: 192857

Telemedicine contraceptive counselling through WhatsApp in South African abortion care: a mixed-methods acceptability study

Item Type Journal Article**Author** M. Weinryb**Author** L. Knight**Author** E.C. Larsson**Author** M. Endler

Abstract Introduction: While telemedicine contraceptive counselling may enhance access and uptake, particularly in low-resource settings, the evidence for its acceptability as an alternative to in-person counselling remains limited. We therefore assessed the acceptability of PlanUrFam, a comprehensive WhatsApp-based contraceptive counselling tool, among women seeking abortion care in South Africa. Methods: We conducted a mixed-methods pilot study at two public health clinics. Women seeking abortion care (n = 31) used PlanUrFam for contraceptive counselling and were followed-up at two weeks (n = 28). The primary outcome was a composite measure acceptability, informed by the Theoretical Framework of Acceptability (TFA). The secondary outcome was choice of long-acting reversible contraception (LARC) and preference for future counselling. In-depth interviews with a nested subgroup (n = 8) explored experiences of the tool using the TFA and content analysis. Results: There was high acceptability (89%, n = 25/28) across framework domains. Qualitative results confirmed that PlanUrFam was acceptable, easy to use, understand and reduced judgement experienced in in-person care. Comprehensive information increased perceived confidence and agency in contraceptive decision-making. After using PlanUrFam, most participants chose LARC methods (64%, n = 18/28), chose a different contraceptive than previously used (96%, n = 27/28), and preferred either

PlanUrFam alone (54%) or in combination with in-person counselling (36%) for future contraceptive care. Discussion: This pilot demonstrates high acceptability of PlanUrFam in abortion care and supported informed decision-making and choice of LARC. The effectiveness of PlanUrFam or similar interventions compared to standard care should be explored in randomised controlled trials measuring method initiation, continuation, recurrent pregnancy and abortion. © 2025 The Author(s). Published by Informa UK Limited, trading as Taylor & Francis Group.

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Tags:

adult, Adult, Article, female, Female, human, Humans, psychology, young adult, Young Adult, patient preference, adolescent, Adolescent, mHealth, social media, conceptual framework, decision making, procedures, qualitative research, controlled study, Telemedicine, content analysis, experience, mobile application, Mobile Applications, pregnancy, patient attitude, follow up, pilot study, abortion, abortion care, Abortion, Induced, contraception, Contraception, Contraception Behavior, contraceptive agent, contraceptive behavior, counseling, Counseling, counselling, e-counseling, family planning, Family Planning Services, female sterilization, first trimester abortion, futurology, gestagen, gestational age, housing, induced abortion, intervention acceptability, interview, long-acting reversible contraception, Long-Acting Reversible Contraception, multiple choice test, nurse, opportunity cost, oral contraceptive agent, Patient Acceptance of Health Care, Pilot Projects, Pregnancy, program acceptability, public health service, Qualitative Research, reproductive history, second trimester abortion, South Africa, South African, teleconsultation, telemedicine, unemployment

Notes:

EID: 2-s2.0-105008079527

Export Date: 20 September 2026; Cited By: 2; Correspondence Address: M. Weinryb; Karolinska Institutet, Institutionen för kvinnors och barns hälsa, Stockholm, Tomtebodavägen 18A, 171 77, Sweden; email: Maja.weinryb@ki.se; CODEN: ECRCF

Temporal-semantics: an era-aware Context-Gated framework for suicide risk detection on social media

Item Type Journal Article

Author S. Sisodia

Author T. Bansal

Author S. Bansal

Author K. Malik

Abstract Early detection of depression and suicide risk from textual data is a critical challenge in mental health informatics, where false negatives can have severe real-world consequences. Transformer-based language models such as SentenceTransformer models based on MPNet have shown strong performance in affective text classification, yet they largely operate on isolated textual signals and overlook the broader contextual and relational patterns through which mental health distress often manifests. Graph-based representations offer a natural way to capture such relationships, but prior studies suggest that naive fusion of text and graph features frequently leads to limited or unstable improvements. In this work, we introduce a Context-Gated Graph Fusion framework that combines MPNet-based semantic sentence representations with era-aware graph centrality features through a learnable multiplicative gating mechanism. Rather than treating textual and relational information as equally informative, the proposed gate allows the model to rely on graph context only when it is supported by semantic evidence in the text. This design reflects a clinically intuitive assumption: contextual risk factors are meaningful only when grounded in linguistic expression. We evaluate the proposed approach on a dataset of 6,037 social media posts annotated with fine-grained emotional states and suicide-risk labels. The model achieves a Suicide Sensitivity (Recall) of 77.56%, outperforming standard text–graph concatenation baselines (66.67%) while maintaining stable F1 score, as well as transformer–GNN hybrid models and LLM-based baselines in zero-shot and few-shot settings. Cross-dataset evaluation on the Kaggle Suicide Watch dataset shows that the model generalizes well to unseen data. Additionally, ablation experiments (both encoder ablation and component-wise ablation) show that these gains arise from the gating mechanism itself, rather than the inclusion of graph features alone. For ensuring model robustness, stratified 5-fold cross-validation (average recall of 77.06%), McNemar’s statistical significance test (), and a temporal “Time Capsule” analysis (revealing a 16.93% amplification of suicide risk in the post-pandemic era) were conducted. Together, these results highlight the value of context-aware fusion for safety-critical mental health prediction and support the practical relevance of the proposed framework. © The Author(s), under exclusive licence to Springer-Verlag GmbH Austria, part of Springer Nature 2026.

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Publisher Springer

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Issue 1**Journal Abbr** Netw. Model. Anal. Health Informatics Bioinformatics**Date Added** 20/09/2026, 8:35:17 pm**Modified** 20/09/2026, 8:35:17 pm**Tags:**

Computational linguistics, Risk perception, Risk assessment, Classification (of information), Learning systems, Human computer interaction, Language processing, Natural language processing, Natural language processing systems, Natural languages, Pattern recognition, Health risks, Risk predictions, Semantics, Statistical tests, Modeling languages, Transformer, Transformers, Text processing, Graphic methods, Temporal models, Ablation, Context-gated fusion, Context-Gated fusion, Graph-based learning, MPNet, Sentence transformer, Sentence transformers, Suicide risk prediction, Temporal modeling

Notes:

EID: 2-s2.0-105042118902

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: Sneha; Department of AI & DS, Indira Gandhi Delhi Technical University for Women, Delhi, India; email: sneha14166@gmail.com

Text Classification with Limited Training Data: Suicide Risk Detection on Social Media

Item Type Conference Paper**Author** S. Pasch**Author** J. Cutura**Editor** Ding W.**Editor** Lu C.-T.**Editor** Wang F.**Editor** Di L.**Editor** Wu K.**Editor** Huan J.**Editor** Nambiar R.**Editor** Li J.**Editor** Ilievski F.**Editor** Baeza-Yates R.**Editor** Hu X.

Abstract In this work, we evaluate the effectiveness of several machine learning models for text classification on small datasets, focusing on a collection of Reddit posts labeled for suicidal behavior. Unlike with larger datasets, where fine-tuning complex models is typically effective, we demonstrate that carefully engineered

prompts can achieve superior classification accuracy when training data is limited. Our findings highlight the potential of prompt-based approaches for effective use in resource-constrained scenarios, offering insights for researchers tackling similar small dataset challenges in text classification. © 2024 IEEE.

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Extra Journal Abbreviation: Proc. - IEEE Int. Conf. Big Data, BigData

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Social media, Large datasets, Fine tuning, Adversarial machine learning, Contrastive Learning, Machine learning models, Text classification, NLP, Small data set, GenAI, Risk detections, Limited training data, suicide detection, Suicide detection

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Export Date: 20 September 2026; Cited By: 6; Funding details: Kaiqi and Leong; Hong Kong Polytechnic University, PolyU; DSTI School of Engineering; Correspondence Address: S. Pasch; Hankuk University of Foreign Studies, Social Science and Ai, Seoul, 107 Imun-ro, Dongdaemun-gu, South Korea; email: stefan.pasch@outlook.com; Conference name: 2024 IEEE International Conference on Big Data, BigData 2024; Conference date: 15 December 2024 through 18 December 2024; Conference code: 206131

Text Enrichment with Japanese Language to Profile Cryptocurrency Influencers

Item Type Conference Paper

Author F. Lomonaco

Author M. Siino

Author M. Tesconi

Editor Aliannejadi M.

Editor Faggioli G.

Editor Ferro N.

Editor Vlachos M.

Abstract From a few-shot learning perspective, we propose a strategy to enrich the latent semantic of the text provided in the dataset provided for the Profiling Cryptocurrency Influencers with Few-shot Learning, the task hosted at PAN@CLEF2023. Our approach is based on data augmentation using the backtranslation forth and back to and from Japanese language. We translate samples in the original training dataset to a target language (i.e. Japanese). Then we translate it back to English. The original sample and the backtranslated one are then merged. Then we fine-tuned two state-of-the-art Transformer models on this augmented version of the training dataset. We evaluate the performance of the two fine-tuned models using the Macro and Micro F1 accordingly to the official metric used for the task. After the fine-tuning phase, ELECTRA and XLNet obtained a Macro F1 of 0.7694 and 0.7872 respectively on the original training set. Our best submission obtained a Macro F1 equal to 0.3851 on the official test set provided. © 2023 Copyright for this paper by its authors.

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Notes:

EID: 2-s2.0-85175657413

Export Date: 20 September 2026; Cited By: 15; Conference name: 24th Working Notes of the Conference and Labs of the Evaluation Forum, CLEF-WN 2023; Conference date: 18 September 2023 through 21 September 2023; Conference code: 193170

Text2Topic: Multi-Label Text Classification System for Efficient Topic Detection in User Generated Content with Zero-Shot Capabilities

Item Type Conference Paper

Author F. Wang

Author M. Beladev

Author O. Kleinfeld

Author E. Frayerman

Author T. Shachar

Author E. Fainman

Author K.L. Assaraf

Author S. Mizrachi

Author B. Wang

Editor Wang M.

Editor Zitouni I.

Abstract Multi-label text classification is a critical task in the industry. It helps to extract structured information from large amount of textual data. We propose Text to Topic (Text2Topic), which achieves high multi-label classification performance by employing a Bi-Encoder Transformer architecture that utilizes concatenation, subtraction, and multiplication of embeddings on both text and topic. Text2Topic also supports zero-shot predictions, produces domain-specific text embeddings, and enables production-scale batch-inference with high throughput. The final model achieves accurate and comprehensive results compared to state-of-the-art baselines, including large language models (LLMs). In this study, a total of 239 topics are defined, and around 1.6 million text-topic pairs annotations (in which 200K are positive) are collected on approximately 120K texts from 3 main data sources on Booking.com. The data is collected with optimized smart sampling and partial labeling. The final Text2Topic model is deployed on a real-world stream processing platform, and it outperforms other models with 92.9% micro mAP, as well as a 75.8% macro mAP score. We summarize the modeling choices which are extensively tested through ablation studies, and share detailed in-production decision-making steps. © 2023 Association for Computational Linguistics.

Date 2023

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Language English

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Conference Name EMNLP 2023 - 2023 Conference on Empirical Methods in Natural Language Processing, Proceedings of the Industry Track

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Computational linguistics, Classification (of information), Decision making, Critical tasks, Embeddings, Throughput, User-generated, Text processing, Multi-label classifications, Multi-label text classification, Textual data, Classification performance, Large amounts, Structured information, Text classification systems

Notes:

EID: 2-s2.0-85184661096

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The 3SD few-shot method for detecting stances on Twitter: A case study of Brazilian Stances on COVID vaccination

Item Type	Conference Paper
Author	K. Becker
Author	J.D. Yepez Rojas
Author	A.M. De Souza
Abstract	Stance detection (SD) is used in social networks to analyze public perception of specific topics or targets. 3SD (Semantic Search Stance Detection) is a Few-Shot SD method designed for Twitter. It leverages semantic search and pre-trained models on a small set of labeled tweets (query set) to create a more comprehensive dataset for training an SD model using supervised learning. Configured with the proper pre-trained model (e.g. trained in a specific language, multi-lingual), 3SD is language-independent. This paper develops a case study on the stances of Brazilians about the COVID vaccination topic. We tackled a strategy based on topic modeling to create a query-set for this new topic, and show that our results outperform baselines, including Large Language Models. The experiments evaluate the relationship between performance and query set size, as well as the ability of 3SD to filter relevant tweets while considering artificial engagement and mislabeled instances due to the crawling process. © 2024 IEEE.
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URL	https://www.scopus.com/pages/publications/105007134578?origin=resultslist
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Language	English
Extra	Journal Abbreviation: Proc. - IEEE/WIC Int. Conf. Web Intell. Intell. Agent Technol., WI-IAT
Publisher	Institute of Electrical and Electronics Engineers Inc.
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Public perception, Detection models, Case-studies, Twitter, Detection methods, Tweets, Few-shot, few-shot, Language independents, pre-trained model, Pre-trained model, semantic search, Semantic search, Specific languages, stance, Stance

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The Challenge of Performing Ontology-driven Entity Extraction in Real-world Unstructured Textual Data from the Domain of Dementia

Item Type Conference Paper

Author S. Suravee

Author C.O. Schmidt

Author K. Yordanova

Editor Angelova G.

Editor Kunilovskaya M.

Editor Escribe M.

Editor Mitkov R.

Abstract Named entity recognition allows the automated extraction of structured domain-related information from unstructured textual data. Our study explores the task of ontology-driven entity recognition, a sequence labelling process for custom named entity recognition for the domain of dementia, specifically from unstructured forum texts where unprofessional caregivers of people with dementia discuss the challenges they face related to agitation. The targeted corpus is loosely structured, contains ambiguous sentences and vocabulary that does not match the agitation-related medical vocabulary. To address the above challenges, we propose a pipeline that involves the following steps: 1) development of an annotation codebook; 2) annotation of a textual corpus collected from dementia forums, consisting of 45,216 sentences (775 questions and 5571 answers); 3) data augmentation to reduce the imbalance in the corpus; 4) training of a bidirectional LSTM model and a transformer model; 5) comparison of the results with those from few shot- and zero-shot based prompt engineering techniques using a

pretrained large language model (LLaMa 3). The results showed that LLaMa 3 was more robust than traditional neural networks and transformer models in detecting underrepresented entities. Furthermore, the study demonstrates that data augmentation improves the entity recognition task when fine-tuning deep learning models. The paper illustrates the challenges of ontology-driven entity recognition in real-world datasets and proposes a roadmap to addressing them that is potentially transferable to other real-world domains. © 2025 Incoma Ltd. All rights reserved.

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Computational linguistics, Learning systems, Natural language processing systems, Transformer modeling, Data mining, Real-world, Neural networks, Ontology's, Extraction, Labeling, Named entity recognition, Data augmentation, Sequence Labeling, Textual data, Ontology, Neurodegenerative diseases, Automated extraction, Entity extractions, Entity recognition

Notes:

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The Effects of Demographic Instructions on LLM Personas

Item Type Conference Paper
Author A.F.M. de Paula
Author J.S. Culpepper
Author A. Moffat
Author S.P. Cherumanal

Author F. Scholer

Author J. Trippas

Abstract Social media platforms must filter sexist content in compliance with governmental regulations. Current machine learning approaches can reliably detect sexism based on standardized definitions, but often neglect the subjective nature of sexist language and fail to consider individual users' perspectives. To address this gap, we adopt a perspectivist approach, retaining diverse annotations rather than enforcing gold-standard labels or their aggregations, allowing models to account for personal or group-specific views of sexism. Using demographic data from Twitter, we employ large language models (LLMs) to personalize the identification of sexism. Our empirical results show that OpenAI's LLMs (GPT-3.5, GPT-4, and GPT-4o) and two open-source LLMs (Mistral and Qwen) exhibit higher Krippendorff's alpha label agreement with female annotators than with male annotators. As well, each LLM presents higher Krippendorff's alpha agreement with a specific annotator age group. We then sought to counter these trends by providing "persona" instructions as part of the LLM prompt, with somewhat surprising outcomes, highlighting the potential of user-centered perspectivist methods to improve content moderation systems. © 2025 Copyright held by the owner/author(s).

Date 2025

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Language model, Large language model, large language models, 'current, Artificial intelligence, Human engineering, Learning systems, Human computer interaction, Social networking (online), Population statistics, Bias, Open source software, Open systems, Social media platforms, bias, Evaluation, Governmental regulations, perspectivism, Perspectivism, Population dynamics, sexism detection, Sexism detection, Unbiased method, unbiased methods

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The Efficacy of Large Language Models and Crowd Annotation for Accurate Content Analysis of Political Social Media Messages

Item Type Journal Article

Author J. Stromer-Galley

Author B. McKernan

Author S. Zaman

Author C. Maganur

Author S. Regmi

Abstract Systematic content analysis of messaging has been a staple method in the study of communication. While computer-assisted content analysis has been used in the field for three decades, advances in machine learning and crowd-based annotation combined with the ease of collecting volumes of text-based communication via social media have made the opportunities for classification of messages easier and faster. The greatest advancement yet might be in the form of general intelligence large language models (LLMs), which are ostensibly able to accurately and reliably classify messages by leveraging context to disambiguate meaning. It is unclear, however, how effective LLMs are in deploying the method of content analysis. In this study, we compare the classification of political candidate social media messages between trained annotators, crowd annotators, and large language models from Open AI accessed through the free Web (ChatGPT) and the paid API (GPT API) on five different categories of political communication commonly used in the literature. We find that crowd annotation generally had higher F1 scores than ChatGPT and an earlier version of the GPT API, although the newest version, GPT-4 API, demonstrated good performance as compared with the crowd and with ground truth data derived from trained student annotators. This study suggests the application of any LLM to an annotation task requires validation, and that freely available and older LLM models may not be effective for studying human communication. © The Author(s) 2025.

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The Growing Impact of Natural Language Processing in Healthcare and Public Health

Item Type Journal Article

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Author R. Balkrishnan

Abstract Natural Language Processing (NLP) is a subset of Artificial Intelligence, specifically focused on understanding and generating human language. NLP technologies are becoming more prevalent in healthcare and hold potential solutions to current problems. Some examples of existing and future uses include: public sentiment analysis in relation to health policies, electronic health record (EHR) screening, use of speech to text technology for extracting EHR data from point of care, patient communications, accelerated identification of eligible clinical trial candidates through automated searches and access of health data to assist in informed treatment decisions. This narrative review aims to summarize the current uses of NLP in healthcare, highlight successful implementation of computational linguistics-based approaches, and identify gaps, limitations, and emerging trends within the subfield of NLP in public health. The online databases Google Scholar and PubMed were scanned for papers published between 2018 and 2023. Keywords “Natural Language Processing, Health Policy, Large Language Models” were utilized in the initial search. Then, papers were limited to those written in English. Each of the 27 selected papers was subject to careful analysis, and their relevance in relation to NLP and healthcare respectively is utilized in this review. NLP and deep learning technologies scan large datasets, extracting valuable insights in various realms. This is especially significant in healthcare where huge amounts of data exist in the form of unstructured text. Automating labor intensive and tedious tasks with language processing algorithms, using text analytics systems and machine learning to analyze social media data and extracting insights from unstructured data allows for better public sentiment analysis, enhancement of risk prediction models, improved patient communication, and informed treatment decisions. In the recent past, some studies have applied NLP tools to social media posts to evaluate public sentiment regarding COVID-19 vaccine use. Social media data also has the capacity to be harnessed to develop pandemic prediction models based on reported symptoms. Furthermore, NLP has the potential

to enhance healthcare delivery across the globe. Advanced language processing techniques such as Speech Recognition (SR) and Natural Language Understanding (NLU) tools can help overcome linguistic barriers and facilitate efficient communication between patients and healthcare providers. © The Author(s) 2024.

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The Insight-Inference Loop: Efficient Text Classification via Natural Language Inference and Threshold-Tuning

Item Type Journal Article

Author S. Chausson

Author M. Fourcade

Author D.J. Harding

Author B. Ross

Author G. Renard

Abstract Modern computational text classification methods have brought social scientists tantalizingly close to the goal of unlocking vast insights buried in text data—from centuries of historical documents to streams of social media posts. Yet three barriers still stand in the way: the tedious labor of manual text annotation, the technical complexity that keeps these tools out of reach for many researchers, and, perhaps most critically, the challenge of bridging the gap between sophisticated algorithms and the deep theoretical understanding social scientists have already developed about human interactions, social structures, and institutions. To counter these limitations, we propose an approach to large-scale text analysis that requires substantially less human-labeled data, and no machine learning expertise, and efficiently integrates the social scientist into critical steps in the workflow. This approach, which allows the detection of statements in text, relies on large language models pre-trained for natural language inference, and a “few-shot” threshold-tuning algorithm rooted in active learning principles. We describe and showcase our approach by analyzing tweets collected during the 2020 U.S. presidential election campaign, and benchmark it against various computational approaches across three datasets. © The Author(s) 2025. This article is distributed under the terms of the Creative Commons Attribution 4.0 License (<https://creativecommons.org/licenses/by/4.0/>) which permits any use, reproduction and distribution of the work without further permission provided the original work is attributed as specified on the SAGE and Open Access page (<https://us.sagepub.com/en-us/nam/open-access-at-sage>).

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The IPATH Dataset: Training CLIP Models with Histopathology-based Image-Caption Pairs from Instagram

Item Type Conference Paper

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Author T. Rai

Author P. Diaz-Santana

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Author N. Bacon

Author K. Wells

Editor Tomaszewski J.E.

Editor Ward A.D.

Abstract Recent advancements in artificial intelligence (AI) have revealed important patterns in pathology images imperceptible to human observers that can improve diagnostic accuracy and decision support systems. However, progress has been limited due to the lack of publicly available medical/veterinary images. To address this scarcity, we explore Instagram as a novel source of pathology images with expert annotations. We curated the IPATH dataset from Instagram, comprising 45,609 pathology image-text pairs, using a combination of classifiers, large language models, and manual filtering. To demonstrate the value of this dataset, we developed a multimodal AI model called IP-CLIP by fine-tuning a pre-trained CLIP model using the IPATH dataset. IP-CLIP outperformed the original CLIP model in a zero-shot classification task on an external histopathology dataset, achieving an F1 score of 0.71 compared to the baseline model's 0.31. These findings demonstrate the effectiveness of the IPATH dataset and underscore the potential of leveraging social media data to develop robust AI models for medical/veterinary image classification and enhanced diagnostic accuracy. © 2025 SPIE.

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THE METHOD FOR INTEGRATING LARGE LANGUAGE MODELS INTO ALGORITHMS FOR FOCUSED MONITORING OF OPEN SOCIAL MEDIA DATA

Item Type Journal Article

Author F. Andrey

Author D. Igor

Author V. Ivan

Abstract The relevance of the study is determined by the importance and complexity of performing rapid summarization of a vast array of user-generated content on social networks. It is proposed to reduce the complexity of the problem by using robotic algorithms and their automated intelligent focusing on specific platforms, data availability, and data volumes. The paper examines the ability of large language models (LLMs) to generate high-quality, coherent, and context-sensitive annotations (summaries) that are suitable for the dynamic nature of unstructured, noisy social network data. The features of the RAG LLM technology for summarizing social network publications are presented. The main disadvantage of language models is instability and the difficulty of tracking the results to confirm factual accuracy. The authors propose a hybrid method for summarizing social media posts over a given period of time. The method involves a complex and variable combination of classical methods for extracting data from their repositories, as well as the abstractive and generative capabilities of large language models. Large language models are used to vectorize the analyzed data. The application of clustering algorithms to the obtained vector representations made it possible to increase the stability and quality of the results. Within the RAG technology, the capabilities of large language models are expanded by means of intelligent search in the MongoDB database used to store the original data. The paper presents three pipelines, each of which is a variant of the method implementation, and has advantages and disadvantages in various application

conditions. The metrics used to evaluate the pipelines are given, and a comparative analysis is performed. Overall, the method allows us to reduce the confabulations of a large language model and obtain annotations of publications for different time periods in real-time. The proposed method is used in practice in the open social media data monitoring system developed by the authors. Copyright (c) Igor Olegovich Datyev, Andrey Mikhailovich Fedorov, Ivan Gennadievich Vishnyakov. This work is licensed under a Creative Commons Attribution 4.0 International License .

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ФОКУСИРОВАННОГО МОНИТОРИНГА ОТКРЫТЫХ ДАННЫХ
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The Moral Foundations Weibo Corpus

Item Type Conference Paper

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Author M. Hu

Author J. Wei

Author B. Ihnaini

Editor Peled-Cohen L.

Editor Calderon N.

Editor Lissak S.

Editor Reichart R.

Abstract Moral sentiments expressed in natural language significantly influence both online and offline environments, shaping behavioral styles and interaction patterns, including social media self-presentation, cyberbullying, adherence to social norms, and ethical decision-making. To effectively measure moral sentiments in natural language processing texts, it is crucial to utilize large, annotated datasets that provide nuanced understanding for accurate analysis and model training. However, existing corpora, while valuable, often face linguistic limitations. To address this gap in the Chinese language domain, we introduce the Moral Foundation Weibo Corpus. This corpus consists of 25,671 Chinese comments on Weibo, encompassing six diverse topic areas. Each comment is manually annotated by at least three systematically trained annotators based on ten moral categories derived from a grounded theory of morality. To assess annotator reliability, we present the kappa test results, a gold standard for measuring consistency. Additionally, we apply several the latest large language models to supplement the manual annotations, conducting analytical experiments to compare their performance and report baseline results for moral sentiment classification. © 2024 Association for Computational Linguistics.

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The Quality of Suicide-Related Stories Generated by Large Language Models: A Content Analysis

Item Type Journal Article

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Abstract Background: Suicide-related media is known to influence suicide rates. Large language models (LLMs), a form of generative artificial intelligence (AI), are increasingly being used as a writing tool. However, the quality of LLM-generated suicide-related content has yet to be assessed. Aims: We aimed to examine suicide-related outputs from three LLMs (GPT-4, Grok, ERNIE) to characterize output quality. Methods: We provided the LLMs with 11 prompts to write different types of suicide-related content common in public media, each in five writing/content styles (broadsheet news report, tabloid news report, adult fiction, teen fiction, social media influencer). We used 3×2 chi square tests to compare characteristics of the outputs particularly related to adherence to responsible media guidelines for suicide reporting and overarching narratives. Results: AI outputs were generated from March 12 to July 11, 2024. A total of 147, 263, and 143 responses from GPT-4, Grok, and ERNIE were analyzed, respectively. Willingness to respond to suicide-related prompts and narrative content varied substantially across the three LLMs, with Grok producing more responses than the other two (GPT-4 = 53%, Grok = 96%, ERNIE = 52%). GPT-4 was more likely to generate emotionally supportive and antistigma messaging. Grok more frequently included harmful details such as suicide methods and romanticized portrayals. ERNIE emphasized male suicide, social support, and societal/community solutions. The proportion of outputs communicating stories of hope and recovery from a suicide crisis was both relatively low and consistent across the LLMs (16–22%). Limitations: This study examined specific prompts posed to three specific LLMs during a single epoch of time. Conclusions: LLMs produce a variety of suicide-related story content when prompted. Some stories, particularly those generated by Grok, were inconsistent with responsible media guidelines and only a minority of stories focused on hope and recovery. Engagement with AI companies to promote safer and more accurate suicide-related content is warranted as the use of LLMs continues to expand. 2026, Hogrefe Publishing

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: M. Sinyor; Department of Psychiatry, University of Toronto, Toronto, 2075 Bayview Avenue, M4N 3M5, Canada; email: mark.sinyor@sunnybrook.ca

The Reliability and Quality of Videos as Guidance for Gastrointestinal Endoscopy: Cross-Sectional Study

Item Type Journal Article

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- Author** Y. Qiu
- Author** Y. Liu
- Author** W. Xu
- Author** W. Ning
- Author** P. Shi
- Author** Z. Yuan
- Author** F. Wang
- Author** Y. Shi

Abstract Background: Gastrointestinal endoscopy represents a useful tool for the diagnosis and treatment of gastrointestinal diseases. Video platforms for spreading endoscopy-related knowledge may help patients understand the pros and cons of endoscopy on the premise of ensuring accuracy. However, videos with misinformation may lead to adverse consequences. Objective: This study aims to evaluate the quality of gastrointestinal endoscopy-related videos on YouTube and to assess whether large language models (LLMs) can help patients obtain information from videos more efficiently. Methods: We collected information from YouTube videos about 3 commonly used gastrointestinal endoscopes (gastroscope, colonoscopy, and capsule endoscopy) and assessed their quality (rated by the modified DISCERN Tool, mDISCERN), reliability (rated by the Journal of the American Medical Association), and recommendation (rated by the Global Quality Score). We tasked LLM with summarizing the video content and assessed it from 3 perspectives: accuracy, completeness, and readability. Results: A total of 167 videos were included. According to the indicated scoring, the quality, reliability, and recommendation of the 3 gastrointestinal endoscopy-related videos on YouTube were overall unsatisfactory, and the quality of the videos released by patients was particularly poor. Capsule

endoscopy yielded a significantly lower Global Quality Score than did gastroscopy and colonoscopy. LLM-based summaries yielded accuracy scores of 4 (IQR 4-5), completeness scores of 4 (IQR 4-5), and readability scores of 2 (IQR 1-2).
Conclusions: The quality of gastrointestinal endoscope-related videos currently on YouTube is poor. Moreover, additional regulatory and improvement strategies are needed in the future. LLM may be helpful in generalizing video-related information, but there is still room for improvement in its ability. © 2025 JMIR Publications Inc.. All rights reserved.

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The Temporal Persistence of Generative Language Models in Sentiment Analysis

Item Type Conference Paper

Author P. Medina-Alias

Author Ö. Şimşek

Editor Aliannejadi M.

Editor Faggioli G.

Editor Ferro N.

Editor Vlachos M.

Abstract Pre-trained transformer-based language models (PLMs) have revolutionised text classification tasks, but their performance tends to deteriorate on data distant in time from the training dataset. Continual supervised re-training help address this issue but it is limited by the availability of newly labelled samples. This paper explores the longitudinal generalisation abilities of large generative PLMs, such as GPT-3 and T5, and smaller encoder-only alternatives for sentiment analysis in social media. We investigate the impact of time-related variations in data, model size, and fine-tuning on the classifiers' performance. Through competitive evaluation in the CLEF-2023 LongEval Task 2, we compare results from fine-tuning, few-shot learning, and zero-shot learning. Our analysis reveals the superior performance of large generative models over the benchmark RoBERTa and highlights the benefits of limited exposure to training data in achieving robust predictions on temporally distant test sets. The findings contribute to understanding how to build more temporally robust transformer-based text classifiers, reducing the need for continuous re-training with annotated data. © 2023 Copyright for this paper by its authors.

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The Unspoken Words: Leveraging Transformer-based Approaches for Detecting Depression from Social Media

Item Type Conference Paper

Author F. Symoom

Author K.M. Tanzil Islam

Author J. Hossain

Author M.H. Imam Bijoy

Author A. Haque

Abstract Mental health issue is a topic of great concern in the present world, yet in spite of being so widely spoken language in Bengali, there are not many notable works addressing this problem. This paper focuses on detecting and classifying depressive content in Bengali text by leveraging linguistic patterns and emotional cues. A dataset of 3,002 Bengali social media texts has been created to identify signs of depression. The dataset is categorized into two classes: Depressive and Non-Depressive, enabling the detection of depression from social media content. This paper presents a comprehensive evaluation of detecting depression for Bengali text, encompassing five machine learning techniques, three deep learning models, and five transformer-based approaches. Among the evaluated models, Bangla-BERT, our proposed transformer-based method, achieved the highest accuracy of 0.96. The model demonstrates exceptional capability in detecting depression by effectively understanding the contextual nuances of sentences. The findings highlight the potential of transformer-based architectures, particularly Bangla-BERT, in advancing sentiment analysis tasks for low-resource languages (LRLs) like Bengali. © 2025 IEEE.

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The Vulnerable Identities Recognition Corpus (VIRC) for Hate Speech Analysis

Item Type Conference Paper

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Author A. Longo

Author M.A. Stranisci

Author V. Patti

Author C. Badenes-Olmedo

Editor Dell'Orletta F.

Editor Lenci A.

Editor Montemagni S.

Editor Sprugnoli R.

Abstract This paper presents the Vulnerable Identities Recognition Corpus (VIRC), a novel resource designed to enhance hate speech analysis in Italian and Spanish news headlines. VIRC comprises 880 headlines, manually annotated for vulnerable identities, dangerous discourse, derogatory expressions, and entities. Our experiments reveal that recent large language models (LLMs) struggle with the fine-grained identification of these elements, underscoring the complexity of detecting hate speech. VIRC stands out as the first resource of its kind in these languages, offering a richer annotation scheme compared to existing corpora. The insights derived from VIRC can inform the development of sophisticated detection tools and the creation of policies and regulations to combat hate speech on social media, promoting a safer online environment. Future work will focus on expanding the corpus and refining annotation guidelines to further enhance its comprehensiveness and reliability. © 2024 CEUR-WS. All rights reserved.

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Threatening Expression and Target Identification in Under-Resource Languages Using NLP Techniques

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Abstract In recent decades, hate speech on social media platforms has been on the rise. It is highly desired to control this kind of material because it initiates unrest and harms to the society. Literature describes several forms of the hate speech and it is quite challenging to differentiate between these forms and to design an automated detection system, especially for under-resource languages. In this study, we propose a robust framework for threatening expressions and its target identification in Urdu (Nastaliq style) language. The proposed methodology presents each step in detail like data collection & annotation, cleaning & pre-processing step, and fine-tuning of Robustly Optimized Bidirectional Encoder Representations from Transformer (Urdu-RoBERTa) with grid search technique for hyper-parameters optimization. The study exploits the strength of a pre-trained Urdu-RoBERTa as a transfer learning technique with grid search fine-tuning. The proposed framework is compared with state-of-the art baseline and ten comparable models and it outperformed all for both tasks (threatening expression and target identification). Furthermore, the proposed framework obtained benchmark performance and improved the f1-score with substantial margin. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Threatening URDU Language Detection from Tweets Using Machine Learning

Item Type Journal Article

Author A. Mehmood

Author M.S. Farooq

Author A. Naseem

Author F. Rustam

Author M.G. Villar

Author C.L. Rodríguez

Author I. Ashraf

Abstract Technology's expansion has contributed to the rise in popularity of social media platforms. Twitter is one of the leading social media platforms that people use to share their opinions. Such opinions, sometimes, may contain threatening text, deliberately or non-deliberately, which can be disturbing for other users. Consequently, the detection of threatening content on social media is an important task. Contrary to high-resource languages like English, Dutch, and others that have several such approaches, the low-resource Urdu language does not have such a luxury. Therefore, this study presents an intelligent threatening language detection for the Urdu language. A stacking model is proposed that uses an extra tree (ET) classifier and Bayes theorem-based Bernoulli Naive Bayes (BNB) as the based learners while logistic regression (LR) is employed as the meta learner. A performance analysis is carried out by deploying a support vector classifier, ET, LR, BNB, fully connected network, convolutional neural network, long short-term memory, and gated recurrent unit. Experimental results indicate that the stacked model performs better than both machine learning and deep learning models. With 74.01% accuracy, 70.84% precision, 75.65% recall, and 73.99% F1 score, the model outperforms the existing benchmark study. © 2022 by the authors.

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TI-Prompt: Towards a Prompt Tuning Method for Few-shot Threat Intelligence Twitter Classification

Item Type Conference Paper**Author** Y. You**Author** Z. Jiang**Author** K. Zhang**Author** J. Jiang**Author** X. Wang**Author** Z. Zhang**Author** S. Wang**Author** H. Feng**Editor** Va Leong H.**Editor** Sarvestani S.S.**Editor** Teranishi Y.**Editor** Cuzzocrea A.**Editor** Kashiwazaki H.**Editor** Towey D.**Editor** Yang J.-J.**Editor** Shahriar H.

Abstract Obtaining the latest Threat Intelligence (TI) via Twitter has become one of the most important methods for defenders to catch up with emerging cyber threats. Existing TI Twitter classification works mainly based on supervised learning methods. Such approaches require large amounts of annotated data and are difficult to be transferred to other TI Twitter classification tasks. This paper proposes a prompt-based method for classifying TI on Twitter, named TI-Prompt. TI-Prompt lever-ages the prompt-tuning method with two templates in different TI Twitter classification tasks. TI-Prompt also uses a semantic similarity-based approach to automatically enrich the prompt verbalizer without expert knowledge and a verbalizer refinement method to calibrate the verbalizer based on the

training data. We evaluate TI-Prompt with binary and multi-classification tasks on two Twitter Threat Intelligence datasets. Evaluation results show that the proposed TI-Prompt improves 5-10% over the best performance of previous supervised learning methods under the few-shot settings. Compared to the general prompt-tuning methods, the proposed prompt-tuning templates can also improve the classification performance by 2-5%. Meanwhile, the proposed verbalizer enrichment method and refinement method improve classification accuracy by 1-4% compared with the general single-word verbalizer prompt method. Therefore, TI-Prompt can be extended to other Threat Intelligence classification tasks without requiring large amounts of training data, significantly reducing the annotation cost. © 2022 IEEE.

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Tibetan-BERT-wwm: A Tibetan Pretrained Model with Whole Word Masking for Text Classification

Item Type Journal Article

Author Y. Liang

Author H. Lv

Author Y. Li

Author L. Duo

Author C. Liu

Author Q. Zhou

Abstract Social networks contributed massive text data generated by users in it, which were crucial in information explosion. These unstructured and ambiguous expressed data may result in difficulty in obtaining available contextual information from it, which can help us gain more accurate insights into user-generated content, user preferences, and topic dynamics within social networks. By training on a large-scale unsupervised corpus and fine-tuning parameters using a limited amount of supervised data, the pretrained language model can effectively capture rich contextual information and achieve excellent performance in numerous downstream tasks of natural language processing (NLP). For the low-resource language such as Tibetan, the distributed representation results of dynamic changes obtained from pretrained language models can effectively alleviate the problem of insufficient labeled data. In order to achieve more effective contextual information and word-level semantic information in Tibetan social media, we collected a large amount of Tibetan language corpus and trained a Tibetan pretrained language model, named as Tibetan-BERT-wwm, by using the whole word masking strategy. Additionally, we apply the model to analyze textual data from social networks to assess its efficacy in capturing user sentiments and news topic in Tibetan social media. In this study, accuracy, precision, recall, and F1 score were used to evaluate its performance. The results showed that the macro-F1 of Tibetan-BERT-wwm in the public dataset TNCC document and title are 75.55% and 64.17%, the self-built sentiment analysis dataset is 70.98%. Compared with other pretrained language models, the Tibetan-BERT-wwm model can capture the semantic information of Tibetan well and improve the Tibetan classification effect. © 2014 IEEE.

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TIGSEN: Building a Low-Resource Dataset and Benchmarking for Tigrigna Sentiment Analysis with Cross-Lingual Transfer Learning Approaches

Item Type Conference Paper

Author H.G. Gebremeskel

Author C. Feng

Author A.M. Abera

Editor Xu J.

Editor Tu Z.

Editor Chen K.

Editor Guo Y.

Abstract Sentiment analysis in low-resource languages faces several challenges. This paper addresses the challenges of extremely low-resource and methodologically understudied sentiment analysis for Tigrigna, an official language spoken in Eritrea and the Tigray region of Ethiopia. We introduce TIGSEN, the first large-scale, multi-domain benchmark dataset for Tigrigna sentiment analysis, comprising 68,596 annotated text samples from official social media, news, and review forums. Created through a rigorous native-speakers annotation protocol, the dataset is designed to enable robust model training and evaluation. To prove its utility and to establish a strong, reproducible baseline for the community, we propose and evaluate a systematic cross-transfer learning framework. This methodology deliberately leverages annotated data from high-resource to linguistically related languages, English and Amharic, to overcome the limitations of Tigrigna's small data pool. Our experiments show that models fine-tuned directly on TIGSEN achieved competitive performance. At the same time, the proposed cross-transfer framework yields a significant performance gain, achieving an accuracy of 87.6% over a strong multilingual baseline. This result validates TIGSEN as a learnable and challenging benchmark and provides a

practical blueprint for resource amplification in low-resource settings. We publicly release the TIGSEN dataset, annotation guidelines, and benchmarking code to serve as a foundational resource and a catalyst for future LLM research in Tigrigna NLP. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2026.

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TigXMM: A Multimodal Cross-Lingual Transfer LLM Method for Low-Resource Sentiment Analysis: Multimodal Low-Resource Tigrigna Sentiment Analysis

Item Type Conference Paper

Author H.G. Gebremeskel

Author C. Feng

Author A.M. Abera

Author X. Zhao

Abstract Sentiment analysis in low-resource languages remains underexplored, particularly for Tigrigna, where communication frequently combines text, emojis, and memes. We introduce TigXMM, a crosslingual, multilingual and multimodal LLM method for Tigrigna sentiment analysis, along with the first multimodal sentiment dataset for this language. The dataset, collected from social media, integrates text, emojis, and meme content to capture real-world communication patterns. We benchmark widely used multilingual models, including mBERT, AfriBERTa, XLM-RoBERTa, XLNet, BLOOMZ, and LLaMA, and highlight their limitations in processing multimodal signals. To address these challenges, we design an LLM-based cross-lingual transfer model with multimodal adapters for text, emoji, and meme fusion using hybrid attention and additive hierarchical strategies. Experimental results demonstrate that our approach consistently improves sentiment classification performance: from 78.4% accuracy on text-only inputs to 81.2% with emojis, 86.3% with memes, and 89.7% when combining all modalities, achieving state-of-the-art performance for Tigrigna sentiment analysis. Beyond performance gains, this work contributes the first multimodal dataset and a reproducible framework, providing open resources to advance sentiment analysis for underrepresented African languages. © 2026 Copyright held by the owner/author(s).

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TinyLLM Efficacy in Low-Resource Language: An Experiment on Bangla Text Classification Task

Item Type Conference Paper

Author F.N. Dehan

Author M. Fahim

Author A.K.M.M. Rahman

Author M.A. Amin

Author A.A. Ali

Editor Antonacopoulos A.

Editor Chaudhuri S.

Editor Chellappa R.

Editor Liu C.-L.

Editor Bhattacharya S.

Editor Pal U.

Abstract Delving into the realm of Bangla text analysis, our study ventures to unlock the potential of both Large and Tiny Language Models across a range of classification tasks, from deciphering sentiment to detecting sarcasm, emotion, hate speech, and fake news. In a linguistic landscape where resources are scarce, we fill a crucial gap by meticulously evaluating model performance. Our findings unveil Gemma-2B and Bangla-BERT as top performers, with Gemma-2B excelling in detecting hate speech and sarcasm, while BanglaBERT shines in sentiment analysis and emotion detection. Notably, TinyLlama emerges as a standout, showcasing exceptional prowess in fake news detection. We emphasize the importance of selecting models attuned to the intricacies of Bangla text, with Gemma-2B, TinyLlama, and BanglaBERT exhibiting notable accuracy improvements, surpassing other contenders. Furthermore, we uncover performance disparities influenced by dataset origins, with Bangla Language Models adept at capturing social media sentiments, and Large Language Models excelling in identifying misinformation and abusive language in formal sources. Our comparison with ChatGPT's zero-shot prompting underscores the necessity for advanced NLP methodologies. By spotlighting TinyLLM, we showcase the potential of advanced NLP in Bangla text classification, paving the way for broader advancements in NLP research. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2025.

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TL; DR: Text Normalization for Social Media Corpus

Item Type Conference Paper
Author G. Feoktistov
Author D. Morozov
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Abstract Text annotation is the process of enriching text with metalinguistic information. The most common types of annotation include morphological annotation, which involves assigning grammatical tags to words; syntactic annotation, which identifies syntactic relationships within sentences; and lemmatization, which determines the base form (lemma) of each word. Text annotation is a fundamental task in computational linguistics and plays a crucial role in both linguistic research and practical applications of natural language processing (NLP). Given the scale of modern text corpora (exceeding 1 billion words), manual annotation is virtually unfeasible, making automated annotation tools the only viable solution. However, most of these tools have been developed, trained, and tested on texts with standard orthography. As a result, their performance can significantly degrade when applied to non-standard texts, such as those from social media. This issue can be mitigated through automated text normalization prior to annotation. This task is related to automated spelling correction but is not equivalent to it, and remains significantly less studied. For instance, text normalization requires not only correcting mistyping and spelling errors but also restoring abbreviations commonly used in online communication. To explore approaches to text normalization, we compiled a corpus of social media sentences and manually paired them with their normalized versions. Using this corpus, we compared various text normalization methods, leveraging pre-trained language models. We examined both fine-tuning and prompt-based approaches. Our study has identified the most effective strategies in this domain, providing valuable insights for future research and applications. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

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Topic-Aware Summarization of Lived Health Care Experiences: Large Language Model Evaluation Study

Item Type Journal Article

Author M. Bilalpur

Author M.E. Hamm

Author Y.J. Lee

Author N.G. Norman

Author K.M. McTigue

Author Y. Wang

Abstract Background: Existing work to understand adults' health care experiences has focused on the analysis of patient feedback provided as written responses to after-visit surveys or social media discourse. Often, such written feedback has been studied using natural language processing techniques, such as topic detection and sentiment analysis, to provide coarse-grained insights. Storytelling is a powerful form of communication and may provide insights into factors contributing to gaps in health care outcomes and avenues for improvement. In addition, studying health care experiences using natural language processing techniques has been limited to patients. The experiences of stakeholders, such as caregivers and health care providers, remain underexplored. Objective: We extract fine-grained insights from health care experiences through narratives collected from patients, caregivers, and health care providers using large language models (LLMs). Topic detection, together with hierarchical summarization of long-form stories from individuals, offers fine-grained insights. Furthermore, the study demonstrates that generated summaries can be evaluated using the LLM-as-a-judge framework and validates the outcomes through comparisons with 2 domain experts. Methods: Fifty automatically transcribed stories of African American experiences were used to identify topics in their experiences using the latent Dirichlet allocation (LDA) technique. Stories about a given topic were summarized using an open-source LLM-based hierarchical summarization approach. Topic summaries were generated by summarizing across story summaries for each story that addressed a given topic. The generated topic summaries were rated for fabrication, accuracy, comprehensiveness, and usefulness by the GPT-4 model; its reliability was validated against the original story summaries by 2 domain experts. Results: Whisper-based automatic transcription of audio narrations achieved a Levenshtein score of 6%. Twenty-six topics were identified using LDA and labeled using the LLM in the 50 African American stories. The GPT-4 ratings suggest that topic summaries were free from fabrication, highly accurate, comprehensive, and useful. The reliability of GPT ratings compared to expert assessments showed moderate-to-high agreement (Bennett S-score of 0.65 or higher). Our approach identified African American experience-relevant topics, such as health behaviors, interactions with medical team members, caregiving, and symptom management, among others. Such insights could help researchers learn

from unstructured datasets in an efficient manner—leveraging the communicative power of storytelling. Conclusions: The use of LDA and LLMs to identify and summarize the experiences of African American individuals suggests a variety of possible avenues for health research and possible clinical improvements to support patients and caregivers, thereby improving health outcomes. © Maneesh Bilalpur, Megan E Hamm, Young Ji Lee, Natasha G Norman, Kathleen M Mctigue, Yanshan Wang.

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Toward a Programmable Humanizing Artificial Intelligence Through Scalable Stance-Directed Architecture

Item Type Journal Article

Author Y.M. Cetinkaya

Author Y. Lee

Author E. Kulah

Author I.H. Toroslu

Author M.A. Cowan

Author H. Davulcu

Abstract The rise of harmful online content underscores the urgent need for artificial intelligence (AI) systems to effectively detect, filter, and foster safer and healthier communication. This article introduces a novel approach to mitigating toxic content generation propensities of large language models (LLMs) by fine-tuning them with a programmable stance-directed focus on core human values and the common good. We propose a streamlined keyword coding and processing pipeline that generates weakly labeled data to train AI models to avoid toxicity and champion civil discourse. We also developed a toxicity classifier and an aspect-based sentiment analysis model to assess and control the effectiveness of a humanizing AI model. We evaluate the proposed pipeline using a contentious real-world X (formerly Twitter) dataset on U.S. race relations. Our approach successfully curbs the toxic content generation propensity of an unrestricted LLM by a significant 85%. © 1997-2012 IEEE.

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Towards a cyberbullying detection approach: fine-tuned contrastive self-supervised learning for data augmentation

Item Type Journal Article

Author L.M. Al-Harigy

Author H.A. Al-Nuaim

Author N. Moradpoor

Author Z. Tan

Abstract Cyberbullying on social media platforms is pervasive and challenging to detect due to linguistic subtleties and the need for extensive data annotation. We introduce a Deep Contrastive Self-Supervised Learning (DCSSL) model that integrates a Natural Language Inference (NLI) dataset, a fine-tuned sentence encoder, and data augmentation to enhance the understanding of cyberbullying's nuanced semantics and offensiveness. The DCSSL model effectively captures contextual dependencies and the varied semantic implications inherent in cyberbullying instances, addressing the limitations of manual data annotation processes when compared against established models such as BERT and Bi-LSTM. Our proposed model registers a significant improvement, achieving a macro average F1 score of 0.9231 on cyberbullying datasets, highlighting its applicability in environments where manual annotation is impractical or unavailable. © The Author(s) 2024.

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Towards a Sentiment Analyser for Low-resource Languages

Item Type Conference Paper

Author D. Indriani

Author A.H. Nasution

Author W. Monika

Author S. Nasution

Editor Pattnaik P.K.

Editor Sain M.

Editor Al-Absi A.A.

Editor Kumar P.

Abstract Twitter is one of the top influenced social media which has a million number of active users. It is commonly used for microblogging that allows users to share messages, ideas, thoughts and many more. Thus, millions of interactions such as short messages or tweets are flowing around among the twitter users discussing various topics that have been happening worldwide. This research aims to analyse a sentiment of the users towards a particular trending topic that has been actively and massively discussed at that time. We chose a hashtag #kpujangancurang that was the trending topic during the Indonesia presidential election in 2019. We use the hashtag to obtain a set of data from Twitter to analyse and investigate further the positive or the negative sentiment of the users from their tweets. This research utilizes RapidMiner tool to generate the Twitter data and comparing Naive Bayes, k-nearest neighbor, decision tree and multi-layer perceptron classification methods to classify the sentiment of the Twitter data. There are overall 200 labeled data in this experiment. Overall, Naive Bayes and multi-layer perceptron classification outperformed the other two methods on 11 experiments with different size of training-testing data split. The two classifiers are potential to be used in creating sentiment analyser for low-resource languages with small corpus. © 2021, The Editor(s) (if applicable) and The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd.

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Towards Agile Text Classifiers for Everyone

Item Type Conference Paper

Author M. Mozes

Author J. Hoffmann

Author K. Tomanek

Author M. Kouate

Author N. Thain

Author A. Yuan

Author T. Bolukbasi

Author L. Dixon

Abstract Text-based safety classifiers are widely used for content moderation and increasingly to tune generative language model behavior—a topic of growing concern for the safety of digital assistants and chatbots. However, different policies require different classifiers, and safety policies themselves improve from iteration and adaptation. This paper introduces and evaluates methods for agile text classification, whereby classifiers are trained using small, targeted datasets that can be quickly developed for a particular policy. Experimenting with 7 datasets from three safety-related domains, comprising 15 annotation schemes, led to our key finding: prompt-tuning large language models, like PaLM 62B, with a labeled dataset of as few as 80 examples can achieve state-of-the-art performance. We argue that this enables a paradigm shift for text classification, especially for models supporting safer online discourse. Instead of collecting millions of examples to attempt to create universal safety classifiers over months or years, classifiers could be tuned using small datasets, created by individuals or small organizations, tailored for specific use cases, and iterated on and adapted in the time-span of a day. © 2023 Association for Computational Linguistics.

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Towards efficient optimization of multi-agent social simulation via large language models

Item Type Journal Article

Author K. Zhang

Author X. Yu

Author H. Peng

Author Z. Yang

Author Y. Tian

Author H. Jin

Author T. Feng

Author H. Lin

Abstract Public opinion on social media during major societal events is highly vulnerable to manipulation by misinformation and extreme emotions, posing severe threats to mainstream values and public trust. While most existing studies focus on passive detection of negative content, systematic approaches for proactively optimizing opinion guidance remain scarce. To address this issue, we propose a Social Influence Text Generation Network, namely SITGNet, a novel multi-agent social simulation framework powered by large language models (LLMs). SITGNet populates a digital social sandbox with cognitively sophisticated agents, each possessing distinct profiles, memories, and behavioral patterns grounded in empirical data. This enables the high-fidelity reproduction of opinion dynamics, capturing emergent phenomena from complex user interactions and psychological transitions. Within this platform, we introduce Guiding Agents, a specialized class of agents designed to proactively

steer discourse. These agents leverage a multi-stage retrieval-augmented generation (RAG) pipeline to synthesize information from external knowledge bases, grounding their outputs in factual evidence to generate nuanced and influential content. Through extensive, large-scale simulations, we systematically evaluate various intervention strategies. Our results demonstrate that a hybrid deployment of Guiding Agents can effectively suppress negative sentiment propagation, alleviate group polarization, and guide opinion trajectories toward rational outcomes. In addition to building the simulation framework, this study treats Social Influence Text Generation as a key modeling objective, enabling systematic evaluation of text-based opinion guidance strategies. © The Author(s), under exclusive licence to Springer-Verlag GmbH Germany, part of Springer Nature 2026.

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Towards Robust Stance Detection: A Hybrid Architecture for Target-Specific Stance Detection in Roman Urdu

Item Type Conference Paper

Author N. Iftikhar

Author S. Safdar

Author M. Wasim

Abstract Stance detection is the measure of detecting the point of view about a target. The task has become significant due to an increase in the application of social media discourse and the dissemination of fake information. This paper suggests a hybrid deep learning model, which combines a pre-trained T5 transformer encoder, a BiLSTM and an attention mechanism. The framework is able to encompass contextual semantics and sequential dependencies in textual data. The T5 model can give rich contextualised embeddings encoding subtle patterns in language, and the BiLSTM gives bidirectional dependencies needed to interpret the underlying stance. An attention layer is incorporated to dynamically weigh the most informative tokens, enabling the model to concentrate on the part that is essential for stance detection. This architecture is designed to be target-aware, robust to linguistic noise, and suitable for both high and low-resource datasets. We assess our framework on benchmark datasets, including P-STANCE, SRQ, SemEval-2016, and AMUSED, achieving competitive performance and demonstrating its effectiveness in various social media and political discourse settings. The results indicate that combining pre-trained language models with sequence-aware attention-based architectures represents a promising direction for advancing stance detection, particularly in noisy and informal text environments such as Roman Urdu. ©2025 IEEE.

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Towards Rumor Detection With Multi-Granularity Evidences: A Dataset and Benchmark

Item Type Journal Article

Author Z. Yang

Author J. Lin

Author Z. Guo

Author Y. Li

Author X. Li

Author Q. Li

Author W. Liu

Abstract Social media serves as a real-time collecting and disseminating center of users' ideas, opinions, and experiences. The deliberate disinformation and rumors propagate rapidly online due to their exaggerated facts, controversial opinions, divisive perspectives, and stunning expressions. Rumor detection approaches typically use social media posts with rumor or non-rumor labels for training and testing without disclosing the rationale behind decision-makings. On one hand, collecting evidence data to verify claims relies on expert efforts. On the other hand, verifying the truthfulness of confusing claims with distracting and lengthy evidences is still challenging. In this paper, we contribute a rumor detection dataset with multi-granularity evidences, denoted as the RD-E dataset, which includes response, fact-check, article, sourcing data and generated evidence by large language models, supporting models to verify the truthfulness of claims on social media. A number of 32,892 claims from 4,525 public individuals and organizations are annotated to 6 kinds of labels, including true, mostly true, half true, mostly false, false, pants on fire, covering a wide range of topics, e.g., politics, economy, society, technology, and health. In the experiments, seven rumor detection models have been investigated and customized on four predefined subtasks for comparisons. © 2024 IEEE.

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Towards Safer Communities: Detecting Aggression and Offensive Language in Code-Mixed Tweets to Combat Cyberbullying

Item Type Conference Paper

Author N. Nafis

Author D. Kanojia

Author N. Saini

Author R. Murthy

Editor Chung Y.-L.

Editor Mostafazadeh Davani A.

Editor Nozza D.

Editor Rottger P.

Editor Talat Z.

Abstract Cyberbullying is a serious societal issue widespread on various channels and platforms, particularly social networking sites. Such platforms have proven to be exceptionally fertile grounds for such behavior. The dearth of high-quality training data for multilingual and low-resource scenarios, data that can accurately capture the nuances of social media conversations, often poses a roadblock to this task. This paper attempts to tackle cyberbullying, specifically its two most common manifestations - aggression and offensiveness. We present a novel, manually annotated dataset of a total of 10, 000 English and Hindi-English code-mixed tweets, manually annotated for aggression detection and offensive

language detection tasks¹. Our annotations are supported by inter-annotator agreement scores of 0.67 and 0.74 for the two tasks, indicating substantial agreement. We perform comprehensive fine-tuning of pre-trained language models (PTLMs) using this dataset to check its efficacy. Our challenging test sets show that the best models achieve macro F1-scores of 67.87 and 65.45 on the two tasks, respectively. Further, we perform cross-dataset transfer learning to benchmark our dataset against existing aggression and offensive language datasets. We also present a detailed quantitative and qualitative analysis of errors in prediction, and with this paper, we publicly release the novel dataset, code, and models. © 2023 Association for Computational Linguistics.

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Towards sentiment and Temporal Aided Stance Detection of climate change tweets

Item Type Journal Article

Author A. Upadhyaya

Author M. Fisichella

Author W. Nejdl

Abstract Climate change has become one of the most significant crises of our time. Public opinion on climate change is influenced by social media platforms such as Twitter, often divided into believers and deniers. In this paper, we propose a framework to classify a tweet's stance on climate change (denier/believer). Existing approaches to stance detection and classification of climate change tweets either have paid little attention to the characteristics of deniers' tweets or often lack an appropriate architecture. However, the relevant literature reveals that the sentimental aspects and time perspective of climate change conversations on Twitter have a major impact on public attitudes and environmental orientation. Therefore, in our study, we focus on exploring the role of temporal orientation and sentiment analysis (auxiliary tasks) in detecting the attitude of tweets on climate change (main task). Our proposed framework STASY integrates word- and sentence-based feature encoders with the intra-task and shared-private attention frameworks to better encode the interactions between task-specific and shared features. We conducted our experiments on our novel curated climate change CLiCS dataset (2465 denier and 7235 believer tweets), two publicly available climate change datasets (ClimateICWSM-2022 and ClimateStance-2022), and two benchmark stance detection datasets (SemEval-2016 and COVID-19-Stance). Experiments show that our proposed approach improves stance detection performance (with an average improvement of 12.14% on our climate change dataset, 15.18% on ClimateICWSM-2022, 12.94% on ClimateStance-2022, 19.38% on SemEval-2016, and 35.01% on COVID-19-Stance in terms of average F1 scores) by benefiting from the auxiliary tasks compared to the baseline methods. © 2023 Elsevier Ltd

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Toxic comment classification and rationale extraction in code-mixed text leveraging co-attentive multi-task learning

Item Type Journal Article

Author K.B. Nelatoori

Author H.B. Kommanti

Abstract Detecting toxic comments and rationale for the offensiveness of a social media post promotes moderation of social media content. For this purpose, we propose a Co-Attentive Multi-task Learning (CA-MTL) model through transfer learning for low-resource Hindi-English (commonly known as Hinglish) toxic texts. Together, the cooperative tasks of rationale/span detection and toxic comment classification create a strong multi-task learning objective. A task collaboration module is designed to leverage the bi-directional attention between the classification and span prediction tasks. The combined loss function of the model is constructed using the individual loss functions of these two tasks. Although an English toxic span detection dataset exists, one for Hinglish code-mixed text does not exist as of today. Hence, we developed a dataset with toxic span annotations for Hinglish code-mixed text. The proposed CA-MTL model is compared against single-task and multi-task learning models that lack the co-attention mechanism, using multilingual and Hinglish BERT variants. The F1 scores of the proposed CA-MTL model with HingRoBERTa encoder for both tasks are significantly higher than the baseline models. Caution: This paper may contain words disturbing to some readers. © The Author(s), under exclusive licence to Springer Nature B.V. 2024.

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ToxiFusion: A Hybrid Semantic-Sequential Neural Architecture for Multi-Label Toxic Comment Classification

Item Type Conference Paper

Author S. Saha

Author S. Ghosh

Author K.M.N. Asib

Abstract Despite being the seventh-largest spoken language in the world, Bengali remains lightly explored in the field of toxic text detection compared to other high-resource languages. The growing influence of social media platforms has facilitated the emergence of toxic text content largely because of the anonymity they offer. This serves as a major factor in the development of effective toxic text detection methods in Bengali. This work presents a comprehensive benchmarking study on a manually annotated Bengali dataset, evaluating a wide array of models ranging from traditional machine learning to modern transformer architectures. Our experiments demonstrate that transformer-based models significantly outperform other approaches. Specifically, a fusion model combining BanglaBERTlarge with a Bidirectional Gated Recurrent Unit (Bi-GRU) classifier achieved state-of-the-art performance, yielding a weighted precision of 0.91, recall of 0.92, and a weighted F1-score of 0.92. The strong results from multiple transformers, including MBERT, MuRIL, and IndicBERT, further confirm their efficacy for this critical natural language processing task in a low-resource language context. © 2025 IEEE.

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Tracing Belief-Driven Thoughts with Theory-of-Mind Agents: An Opinion Analysis Framework

Item Type Conference Paper

Author J. Wen

Author Y. Ning

Author H. Kang

Author X. Miao

Author T. Qian

Abstract Opinion Analysis (OA) is a critical task in Natural Language Processing (NLP), aimed at identifying stance, sentiment, or hate speech toward specific targets in social media text. With the advancement of Large Language Models (LLMs), researchers have begun exploring LLM-based agent frameworks for OA. However, existing LLM-based agent frameworks rely on heuristic simulations and lack the capacity for emotional resonance, which is achieved by understanding others' internal thoughts, intentions, and beliefs during interactions - a cognitive ability rooted in Theory of Mind (ToM). To bridge this gap, we introduce OpinionToM, a multi-agent framework that shifts from heuristic simulation to ToM-driven cognitive reasoning. This shift requires a formal process capable of capturing key aspects of ToM, particularly the uncertainty each agent has about the hidden mental states of others. To this end, we model multi-agent reasoning in OA as a Social Partially Observable Markov Decision Process (Social POMDP). The Social POMDP is designed to track agents' continuous belief states by dynamically generating beliefs and adjusting corresponding weights based on observations. The weight adjustment mechanism draws inspiration from Bayesian inverse planning, leveraging LLMs as a computational backend to perform approximate probabilistic inference over competing beliefs, conditioned on the agents' received perceptions. We evaluate

OpinionToM on six benchmarks across three distinct opinion analysis tasks, demonstrating significant performance improvements compared to baselines. We release the code and dataset at <https://github.com/wenjt/OpinionToM>. © 2026 Owner/Author.

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Tracking the Spread of Pollen on Social Media Using Pollen-Related Messages From Twitter: Retrospective Analysis

Item Type Journal Article

Author M. Pérez-Pérez

Author M.F. Gonzalez

Author F.J. Rodriguez-Rajo

Author F. Fdez-Riverola

Abstract Background: Allergy disorders caused by biological particles, such as the proteins in some airborne pollen grains, are currently considered one of the most common chronic diseases, and European Academy of Allergy and Clinical Immunology forecasts indicate that within 15 years 50% of Europeans will have some kind of allergy as a consequence of urbanization, industrialization, pollution, and climate change. Objective: The aim of this study was to monitor and analyze the dissemination of information about pollen symptoms from December 2006 to January 2022. By conducting a comprehensive evaluation of public comments and trends on Twitter, the research sought to provide valuable insights into the impact of pollen on sensitive individuals, ultimately enhancing our understanding of how pollen-related information spreads and its implications for public health awareness. Methods: Using a blend of large language models, dimensionality reduction, unsupervised clustering, and term frequency-inverse document frequency, alongside visual representations such as word clouds and semantic interaction graphs, our study analyzed Twitter data to uncover insights on respiratory allergies. This concise methodology enabled the extraction of significant themes and patterns, offering a deep dive into public knowledge and discussions surrounding respiratory allergies on Twitter. Results: The months between March and August had the highest volume of messages. The percentage of patient tweets appeared to increase notably during the later years, and there was also a potential increase in the prevalence of symptoms, mainly in the morning hours, indicating a potential rise in pollen allergies and related discussions on social media. While pollen allergy is a global issue, specific sociocultural, political, and economic contexts mean that patients experience symptomatology at a localized level, needing appropriate localized responses. Conclusions: The interpretation of tweet information represents a valuable tool to take preventive measures to mitigate the impact of pollen allergy on sensitive patients to achieve equity in living conditions and enhance access to health information and services. © 2024 JMIR Publications Inc.. All rights reserved.

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Article, human, Humans, social media, Twitter, LLM, text mining, information dissemination, public health, awareness, procedures, data analysis, cloud computing, workflow, Social Media, adverse event, data mining, natural language processing, sentiment analysis, thematic analysis, Information Dissemination, knowledge base, k means clustering, dimensionality reduction, epidemiology, large language model, medical information, prevalence, medical informatics, knowledge, symptom, data extraction, semantic web, comprehension, statistical analysis, symptomatology, social network analysis, conversation, access to information, allergen, Allergens, bidirectional encoder representations from transformers model, clinical evaluation, data interpretation, graph analysis, knowledge reconstruction, online social network, pollen, Pollen, pollen allergy, public health message, respiratory allergies, respiratory tract allergy, Retrospective Studies, retrospective study, Rhinitis, Allergic, Seasonal, sex difference, term frequency inverse document frequency, trend study

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Transfer Learning for Code-Mixed Data: Do Pretraining Languages Matter?

Item Type Conference Paper

Author K. Tatariya

Author H. Lent

Author M. de Lhoneux

Editor Barnes J.

Editor De Clercq O.

Editor Klinger R.

Abstract Monolinguals make up a minority of the world's speakers, and yet most language technologies lag behind in handling linguistic behaviours produced by bilingual and multilingual speakers. A commonly observed phenomenon in such communities is code-mixing, which is prevalent on social media, and thus requires attention in NLP research. In this work, we look into the ability of pretrained language models to handle code-mixed data, with a focus on the impact of languages present in pretraining on the downstream performance of the model as measured on the task of sentiment analysis. Ultimately, we find that the pretraining language has little effect on performance when the model sees code-

mixed data during downstream finetuning. We also evaluate the models on code-mixed data in a zero-shot setting, after task-specific finetuning on a monolingual dataset. We find that this brings out differences in model performance that can be attributed to the pretraining languages. We present a thorough analysis of these findings that also looks at model performance based on the composition of participating languages in the code-mixed datasets. © 2023 Association for Computational Linguistics.

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Notes:

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Transfer Learning-Based Automatic Sentiment Annotation of a Twitter-Based Arabic Mental Illness (AMI) Dataset

Item Type Journal Article

Author A. Diwali

Author K. Saeedi

Author K. Dashtipour

Author M. Gogate

Author Z. Hussain

Author A. Howard

Author A. Hussain

Abstract Sentiment analysis, crucial for discerning emotional tones in text, relies on manual annotation to train machine learning models and is considered the gold standard for creating annotated corpora. However, this process is time-consuming, labour-intensive, and prone to biases. This paper proposes an automatic annotation approach for the Twitter-based Arabic Mental Illness (AMI) dataset, which encompasses both Modern Standard Arabic and Dialectal Arabic. The approach leverages transfer learning with existing manually annotated datasets and three advanced Arabic language models to automate annotation, thereby enriching Arabic as a low-resource language with labelled sentiment data. Validation was conducted by comparing the automatically generated annotations to manual annotation on the same dataset, achieving strong inter-annotator agreement with a Cohen's Kappa statistic of $k = 0.8457$. Additionally, various baseline models were evaluated on the AMI dataset, identifying AraBERT as the top performer with the highest F1 score and accuracy. © 2025 The Author(s). Expert Systems published by John Wiley & Sons Ltd.

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Transformer based multilingual joint learning framework for code-mixed and english sentiment analysis

Item Type Journal Article

Author A. Ekbal

Abstract In recent times, there has been tremendous growth in the number of multi-lingual users on social media platforms. Consequently, the code-mixing phenomenon, i.e., mixing of more than one language, has become ubiquitous in Internet content. In this paper, we present a shared-private, multi-lingual, multi-task model coupled with a transformer-based pre-trained encoder for sentiment analysis of code-mixed and English languages. Our model is tailored for multitasking that transfers the knowledge between code-mixed and English sentiment tasks. We consider code-mixed sentiment analysis as the primary task and enhance its performance by English sentiment analysis (auxiliary task) by sharing knowledge between them. We fine-tune the Bidirectional Encoder Representation using Transformer (BERT) encoder in a shared-private fashion to obtain the shared and task-specific features using the multi-task objective function. We evaluate our proposed framework using three benchmark datasets for the Hindi-English (Hinglish), Punjabi-English (Punglish) code-mixed and English sentiment tasks. Experiment results justify that our proposed multi-task framework improves the performance of our primary task in comparison to the state-of-art single-task systems. © The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2023.

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Transformer-based Arabic Stance Detection with Multi-Head Attention Fusion and Class-Weighted Loss

Item Type Conference Paper

Author M. Al-Smadi

Editor Erenli K.

Editor Jararweh Y.

Abstract The rapid growth of social media content necessitates the development of intelligent technologies capable of analyzing writers' opinions, stances, and sentiments toward specific topics and events. Stance detection in Arabic social media is a challenging task due to dialectal variation, noisy orthography, and lack of publicly available and comprehensive datasets with balanced label distributions. Recent advances in pretrained language models such as AraELECTRA have improved performance; however, these models often struggle with robustness to class imbalance and limited lexical cues. In this paper, we introduce a hybrid architecture that fuses AraELECTRA representations with character-level TF-IDF features through a multi-head attention mechanism and task-specific classification layers. The model is further optimized using class-weighted training to address imbalanced label distributions. We evaluate our model on the MAWQIF Arabic stance detection dataset. Experimental results demonstrate that the proposed architecture achieves strong performance, with macro-F1=0.9347 outperforming prior state-of-the-art models. Ablation study confirm that multi-head attention and class weighting are the most influential components, while TF-IDF features provide complementary but smaller gains. © 2025 IEEE.

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Notes:

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Transformer-based deep learning models for the sentiment analysis of social media data

Item Type Journal Article

Author S. Tabinda Kokab

Author S. Asghar

Author S. Naz

Abstract Sentiment analysis (SA) is a widely used contextual mining technique for extracting useful and subjective information from text-based data. It applies on Natural Language Processing (NLP), text analysis, biometrics, and computational linguistics to identify, analyse, and extract responses, states, or emotions from the data. The features analysis technique plays a significant role in the development and improvement of a SA model. Recently, GloVe and Word2vec embedding models have been widely used for feature extractions. However, they overlook sentimental and contextual information of the text and need a large corpus of text data for training and generating exact vectors. These techniques generate vectors for just those words that are included in their vocabulary and ignore Out of Vocabulary Words (OOV), which can lead to information loss. Another challenge for the classification of sentiments is that of the lack of readily available annotated data. Sometimes, there is a contradiction between the review and their label that may cause misclassification. The aim of this paper is to propose a generalized SA model that can handle noisy data, OOV words, sentimental and contextual loss of reviews data. In this research, an effective Bi-directional Encoder Representation from Transformers (BERT) based Convolution Bi-directional Recurrent Neural Network (CBRNN) model is proposed with for exploring the syntactic and semantic information along with the sentimental and contextual analysis of the data. Initially, the zero-shot classification is used for labelling the reviews by calculating their polarity scores. After that, a pre-trained BERT model is employed for obtaining sentence-level semantics and contextual features from that data and generate embeddings. The obtained contextual embedded vectors were then passed to the neural network, comprised of dilated convolution and

Bi-LSTM. The proposed model uses dilated convolution instead of classical convolution to extract local and global contextual semantic features from the embedded data. Bi-directional Long Short-Term Memory (Bi-LSTM) is used for the entire sequencing of the sentences. The CBRNN model is evaluated across four diverse domain text datasets based on accuracy, precision, recall, f1-score and AUC values. Thus, CBRNN can be efficiently used for performing SA tasks on social media reviews, without any information loss. © 2022

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Notes:

EID: 2-s2.0-85130846957

Export Date: 20 September 2026; Cited By: 254; Correspondence Address: S. Tabinda Kokab; Department of Computer Science, COMSATS University Islamabad, Islamabad, 44000, Pakistan; email: tabi.syed19@gmail.com

Transformer-Based Detection of Satire, Sarcasm, and Irony in Romanian Text

Item Type Conference Paper

Author G. Bara

Author E. Stetco

Author D. Bran

Abstract Detecting satire, sarcasm, and irony remains challenging due to their reliance on contextual incongruity and implicit meaning. While extensive resources exist for English, Romanian remains under-resourced in figurative language processing. This paper presents a Transformer-based approach for detecting satire, sarcasm, and irony in Romanian text. We introduce a manually curated corpus of over 77,000 documents collected from journalistic and social media sources and annotated through a semi-automatic pipeline with human validation. We discuss

the limitations of an initial multi-class formulation and evaluate three independent binary classifiers fine-tuned from a multilingual Transformer model. Experimental results show that the binary one-vs-rest strategy achieves macro F1 scores above 0.84 and ROC-AUC values exceeding 0.93. The findings indicate substantial semantic overlap between figurative categories and show that independent modeling supports robust category-specific detection in a low-resource setting. © 2026 IEEE.

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Transformers Unveiled: A Comprehensive Evaluation of Emotion Detection in Text Transcription

Item Type Conference Paper

Author S. Ali

Author B. Naz

Author S. Narejo

Author S. Ali

Author J.K. Pabani

Abstract The fast growth of digital communication via smartphones and social media has made textual data as one of the most essential forms of people's expressions. Accurate emotion discovery from text is thus important for applications in customer sentiment analysis and interactive technologies, be it a web, desktop, or mobile application, an intelligent robot, or minor connected devices. This paper is intended to evaluate the performance of five state-of-the-art transformer models, including BERT, ALBERT, DistilBERT, Transformer XL, and RoBERTa, for multiclass emotion detection using the IEMOCAP dataset. From the results, it is observed that DistilBERT has the highest accuracy of 79.2%, implying an effective approach in balancing model size and performance. These research show that smaller models like DistilBERT often outperform larger models by reducing their overfitting and computational demands. The findings also highlights the need for task-specific fine-tuning and optimization in emotion detection. © 2024 IEEE.

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Transforming social media sentiment analysis with generative large language models: A case study from the cultured meat debate

Item Type Journal Article

Author T. Chen

Author L.Y.-F. Su

Author Y.M.M. Ng

Author M.J. Stasiewicz

Author Y.-C. Wang

Abstract Social media platforms can serve as important data sources for discourse about novel foods. However, prevailing methods for analyzing the food-related and other sentiments expressed on such platforms by their users have drawbacks including high cost, time intensiveness, and/or low accuracy. To address those drawbacks, this paper presents a series of sentiment-analysis frameworks leveraging the generative large language model (LLM) GPT-4o. Using a human-annotated subset of more than 230,000 cultured-meat tweets from a 12.5-year period, we compared five methods: i.e., three GPT-4o configurations (zero-shot with basic instructions, zero-shot with detailed guidelines, and fine-tuning), alongside VADER and BERT. Among them, fine-tuned GPT-4o demonstrated superior accuracy, while zero-shot GPT-4o, with or without detailed guidelines, achieved performance comparable to BERT without requiring any model training. Applying the LLM framework using fine-tuned GPT-4o to the full dataset as a case study captured meaningful dynamics within online discourse about cultured meat, including its overall balance of positivity, negativity, neutrality, and mixed views, and how regulatory milestones might have affected it. Overall, our rigorously evaluated, highly scalable generative LLM frameworks for sentiment analysis were able to accurately and efficiently process online discourse about a novel food – outperforming widely used lexicons (VADER) and close to or equaling the performance of fine-tuned deep-learning models (BERT) – without requiring extensive machine-learning expertise or even pre-labeled datasets in some instances. As such, they provide food and consumer scientists with easy-to-deploy and time-efficient computational tools for analyzing online narratives. © 2026 The Authors

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Translate and Classify: Improving Sequence Level Classification for English-Hindi Code-Mixed Data

Item Type Conference Paper

Author D. Gautam

Author K. Gupta

Author M. Shrivastava

Editor Solorio T.

Editor Chen S.

Editor Black A.W.

Editor Diab M.

Editor Sitaram S.

Editor Soto V.

Editor Yilmaz E.

Editor Srinivasan A.

Abstract Code-mixing is a common phenomenon in multilingual societies around the world and is especially common in social media texts. Traditional NLP systems, usually trained on monolingual corpora, do not perform well on code-mixed texts. Training specialized models for code-switched texts is difficult due to the lack of large-scale datasets. Translating code-mixed data into standard languages like English could improve performance on various code-mixed tasks since we can use transfer learning from state-of-the-art English models for processing the translated data. This paper focuses on two sequence-level classification tasks for English-Hindi code mixed texts, which are part of the GLUECoS benchmark - Natural Language Inference and Sentiment Analysis. We propose using various pre-trained models that have been fine-tuned for similar English-only tasks and have shown state-of-the-art performance. We further fine-tune these models on the translated code-mixed datasets and achieve state-of-the-art performance in

both tasks. To translate English-Hindi code-mixed data to English, we use mBART, a pre-trained multilingual sequence-to-sequence model that has shown competitive performance on various low-resource machine translation pairs and has also shown performance gains in languages that were not in its pre-training corpus. ©2021 Association for Computational Linguistics

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Trinity at SemEval-2023 Task 12: Sentiment Analysis for Low-resource African Languages using Twitter Dataset

Item Type Conference Paper

Author S. Rathi

Author S. Pande

Author H. Atkare

Author R. Tangsali

Author A. Vyawahare

Author D. Kadam

Editor Ojha A.K.
Editor Dogruoz A.S.
Editor Da San Martino G.
Editor Madabushi H.T.
Editor Kumar R.
Editor Sartori E.

Abstract This paper presents a summary of our findings obtained on sentiment analysis of 3 African languages among the 17 languages mentioned in the shared task. We carried out a sentiment analysis on Hausa, Yoruba, and Swahili. The models used here for the mentioned task were logistic regression, SVM, RandomForest, and mBERT along with a few data-preprocessing and oversampling techniques. The performance of the models used was evaluated by considering weighted average and macro average F1 scores as metrics. The best set of scores obtained on the languages Hausa, Yoruba and Swahili are (76.53, 76.55), (74.83, 73.15) and (57.79, 48.59) respectively. © 2023 Association for Computational Linguistics.

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TurkishBERTweet: Fast and reliable large language model for social media analysis

Item Type Journal Article

Author A. Najafi

Author O. Varol

Abstract Turkish is one of the most spoken languages in the world; however, it is still among the low-resource languages. Wide us of this language on social media platforms such as Twitter, Instagram, or Tiktok and strategic position of the country in the world politics makes it appealing for the social network researchers and industry. To address this need, we introduce TurkishBERTweet, the first large scale pre-trained language model for Turkish social media built using over 894 million Turkish tweets. The model shares the same architecture as RoBERTa-base model with smaller input length, making TurkishBERTweet lighter than the most used model, called BERTurk, and can have significantly lower inference time. We trained our model using the same approach for RoBERTa model and evaluated on two tasks: Sentiment Classification and Hate Speech Detection. We demonstrate that TurkishBERTweet outperforms the other available alternatives on generalizability and its lower inference time gives significant advantage to process large-scale datasets. We also show custom preprocessors for social media can acquire information from platform specific entities. We also conduct comparison with the commercial solutions like OpenAI and Gemini, and other available Turkish LLMs in terms of cost and performance to demonstrate TurkishBERTweet is scalable and cost-effective. © 2024 Elsevier Ltd

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Twitter Sentiment Analysis in Resource Limited Language

Item Type Conference Paper

Author R. Gupta

Author S. Agarwal

Author S. Garg

Author R. Kaushal

Editor Sachdeva S.

Editor Watanobe Y.

Abstract Sentiment analysis is essential for understanding public opinion and user feedback in various languages. However, a language barrier often limits the use of existing models which are primarily pre-trained on English. Therefore, previous approaches have focused on building language-specific models for non-English languages. In this work, we investigate the efficacy of low resource language specific models (like GreekBERT) and compare their performance with RoBERTa model for predicting sentiments in Greek and English language tweets. More specifically, we explore whether Greek tweets translated to English and fed to RoBERTa model performs better than Greek tweets directly fed to GreekBERT model. We find the RoBERTa model performs well not only for the English tweets but also for the non-English tweets (Greek) translated to English. We present a detailed summary of model performance for sentiment classification of non-English (Greek) tweets. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2024.

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Twitter Stance Detection via Neural Production Systems

Item Type Conference Paper

Author B. Zhang

Author D. Ding

Author G. Xu

Author J. Guo

Author Z. Huang

Author X. Huang

Abstract Stance detection is an important task, which aims to classify the attitude of an opinionated text toward a given target. In this paper, we develop an interpretable neural production system for stance detection (NPS4SD). NPS4SD is an end-to-end deep learning model, which consists of a set of knowledge rules that are applied by binding with specific entities. NPS4SD consists of two main components: a pretrained model for learning the text representation and a variable binding network (VBN) to bind the knowledge rules with text entities. Extensive experiments are conducted to evaluate the effectiveness of the proposed NPS4SD model on three real-world datasets with in-domain, cross-target and zero-shot setups. Experimental results demonstrate that NPS4SD achieves substantially better performance than the strong competitors for the stance detection task. © 2023 IEEE.

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Performance, Deep neural networks, Learning systems, Learning models, deep neural network, Stance detection, Zero-shot learning, stance detection, Neural production system, Production system, Text representation, End to end, Knowledge rules, neural production system, Real-world datasets, Variable binding

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Twitter-Based Sentiment Analysis for the 2025 Bihar Elections

Item Type Conference Paper

Author J. Pandey

Author P. Rajput

Author R. Agrawal

Editor Kumar B.

Editor Agrawal R.

Editor Javed Quraishi S.

Editor Gandhi P.

Abstract —Social media platforms have transformed political discourse by enabling immediate public responses to significant events. This study examines voter sentiment regarding the 2025 Bihar Assembly Elections by analysing over 1,100 election-related tweets collected via the Twitter Application Programming Interface (API). In contrast to previous research that primarily utilized large datasets or conventional models, this work addresses a low-resource electoral context and compares three sentiment classification methods: Random Forest, Support Vector Machine (SVM), and the transformer-based Bidirectional Encoder Representations from Transformers (BERT) model. Sentiment was initially annotated using the Valence Aware Dictionary for Sentiment Reasoning (VADER), and party-specific trends were assessed using the Net Sentiment Score (NSS) to facilitate comprehensive analysis. The findings indicate that BERT

achieves an accuracy of 80 percent, outperforming traditional models and demonstrating the effectiveness of transformer-based approaches in data-constrained political environments. Furthermore, sentiment analysis reveals a relative online preference for the Bharatiya Janata Party (BJP), suggesting that social media analytics can complement traditional polling methods by providing timely, data-driven insights into electoral dynamics. ©2025 IEEE.

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UEF-HOCUrdu: Unified Embeddings Ensemble Framework for Hate and Offensive Text Classification in Urdu

Item Type Journal Article

Author K. Ullah

Author M. Aslam

Author M.U.G. Khan

Author F.S. Alamri

Author A.R. Khan

Abstract Hate speech and other forms of hostile communication on social media have several implications such as; fostering violence, promoting social divide, and negative psychological effects. Since such toxic language is becoming more and more common, it is imperative to have a proper way of identifying it, especially in low resource language like Urdu. To meet this challenge, this research proposed a new ensemble based multi-classification model and generated new dataset of 36,000 Urdu tweets categorized as 'Hate', 'Offensive' and 'Neither'. This study sought to create a model that not only achieves a high classification accuracy but also overcome key challenges inherent in natural language processing, namely, high dimensionality, sparsity, overfitting, OOV words and dialectal variations. For this purpose, an extensive comparison of different learning algorithms were conducted. As a result, the most efficient models, namely FastText, XLM-RoBERTa, ULMFiT, and XGBoost were incorporated in the proposed ensemble approach to achieve the best results in both classification and mitigation of NLP issues. To further enhance the confidence in proposed model, a stratified 5-fold cross-validation technique has been utilized. The ensemble model performed the best and achieved macro F1 score of 0.94, complemented by comprehensive labeled dataset focusing on hate and offensive speech in Urdu. By addressing key research gaps, this research provides a valuable foundation for future work and benchmarking in Urdu hate speech multi-classification tasks. © 2013 IEEE.

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Uncovering the role of reviewer heterogeneity in customer needs identification and customer satisfaction improvement: A perspective from target customers

Item Type Journal Article

Author J. Lin

Author L. Li

Author C. Wang

Author X. Jiang

Author S. Xiao

Abstract Identifying customer needs from User-Generated Content (UGC) is fundamental for marketing strategies, product improvement, and innovation. However, for products targeted at a specific customer segment, UGC involves reviews from both target and nontarget customers. Moreover, customer needs from these two segments can differ substantially, particularly for experience goods. Prior studies that do not consider the reviewer heterogeneity fail to acknowledge these differences. To fill this gap, we propose an integrated mixed-methods framework for identifying customer needs from two distinct customer segments and examining their differences within the experience goods. The framework leverages generative Large Language Model (LLM) with innovative adaptations to capture customer needs and integrates Multivariate Analysis of Variance (MANOVA), Conjoint analysis, and Kano model to discern customer needs differences. Using cosmetic reviews, our framework shows superior performance over benchmark methods in capturing nuanced customer needs and observes divergent customer needs between target and nontarget customers. The observed distinctions between these two segments highlight the importance of reviewer heterogeneity in uncovering customer needs and offer informed guidance for improving satisfaction through product improvement, reducing mismatched purchases, and negative reviews management. © 2025 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Export Date: 20 September 2026; Cited By: 4; Correspondence Address: L. Li; School of Economics and Management, Tongji University, Shanghai, 200092, China; email: cassie_@tongji.edu.cn; CODEN: IPMAD

Understanding Audience Engagement with Japan Travel Itinerary Videos Across Social Media Platforms

Item Type Conference Paper**Author** S.S.A. Hussain**Author** K. Fujisaki**Author** T. Yai**Editor** Patuleia M.

Abstract Social media plays a central role in shaping destination image and tourist behavior. Beyond providing information and entertainment, it influences how viewers explore places, plan trips, and share their experiences after visiting destinations. While some prior studies have examined the role of social media in shaping tourists' choices, most have focused on individual social media platforms. Very few studies have compared how engagement differs across social media platforms such as YouTube, Instagram, and TikTok. This study addresses this gap by analyzing engagement on one popular 14 days Japan travel itinerary video from each platform, specifically YouTube, TikTok, and Instagram. After cleaning, filtering, removing non-English comments and quality checks, the final dataset used for analysis consisted of 525 YouTube comments, 804 TikTok comments, and 451 Instagram comments. Sentiment analysis was done using RoBERTa and emotional tone was identified using the DistilRoBERTa-based J-Hartmann emotion model. Hugging Face's facebook/bart-large-mnli zero-shot model was used to assign comments to manually defined candidate labels, and final topic grouping was completed through manual mapping. Results show that YouTube generated predominantly positive sentiment and strong joy-based emotions, with viewers asking detailed trip-planning and itinerary-related questions and expressing intentions to visit Japan and follow the itinerary. In

contrast, TikTok and Instagram showed largely neutral sentiment, dominated by emojis, tagging friends and family, and shorter emotional responses. Zero-shot topic analysis further confirmed this difference, showing that YouTube comments centered on information-seeking, trip planning, and past travel reflections, while TikTok and Instagram were more focused on short reactions, visual appreciation, destination praise, and social sharing. The findings show that YouTube platform supports detailed trip planning and engagement, while TikTok and Instagram encourage quick, emotion-based reactions. These insights can help influencers, destination management organizations (DMOs), and relevant policy makers design content that both guides travel decisions and inspires interest in destinations. © ICTR 2026. All rights reserved.

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Understanding Citizens' Response to Social Activities on Twitter in US Metropolises During the COVID-19 Recovery Phase Using a Fine-Tuned Large Language Model: Application of AI

Item Type Journal Article

Author R. Saito

Author S. Tsugawa

Abstract Background: The COVID-19 pandemic continues to hold an important place in the collective memory as of 2024. As of March 2024, >676 million cases, 6 million deaths, and 13 billion vaccine doses have been reported. It is crucial to evaluate sociopsychological impacts as well as public health indicators such as these to understand the effects of the COVID-19 pandemic. Objective: This study aimed to explore the sentiments of residents of major US cities toward restrictions on social activities in 2022 during the transitional phase of the COVID-19 pandemic, from the peak of the pandemic to its gradual decline. By illuminating people's susceptibility to COVID-19, we provide insights into the general sentiment trends during the recovery phase of the pandemic. Methods: To analyze these trends, we collected posts (N=119,437) on the social media platform Twitter (now X) created by people living in New York City, Los Angeles, and Chicago from December 2021 to December 2022, which were impacted by the COVID-19 pandemic in similar ways. A total of 47,111 unique users authored these posts. In addition, for privacy considerations, any identifiable information, such as author IDs and usernames, was excluded, retaining only the text for analysis. Then, we developed a sentiment estimation model by fine-tuning a large language model on the collected data and used it to analyze how citizens' sentiments evolved throughout the pandemic. Results: In the evaluation of models, GPT-3.5 Turbo with fine-tuning outperformed GPT-3.5 Turbo without fine-tuning and Robustly Optimized Bidirectional Encoder Representations from Transformers Pretraining Approach (RoBERTa)—large with fine-tuning, demonstrating significant accuracy (0.80), recall (0.79), precision (0.79), and F1-score (0.79). The findings using GPT-3.5 Turbo with fine-tuning reveal a significant relationship between sentiment levels and actual cases in all 3 cities. Specifically, the correlation coefficient for New York City is 0.89 (95% CI 0.81-0.93), for Los Angeles is 0.39 (95% CI 0.14-0.60), and for Chicago is 0.65 (95% CI 0.47-0.78). Furthermore, feature words analysis showed that COVID-19-related keywords were replaced with non-COVID-19-related keywords in New York City and Los Angeles from January 2022 onward and Chicago from March 2022 onward. Conclusions: The results show a gradual decline in sentiment and interest in restrictions across all 3 cities as the pandemic approached its conclusion. These results are also ensured by a sentiment estimation model fine-tuned on actual Twitter posts. This study represents the first attempt from a macro perspective to depict sentiment using a classification model created with actual data from the period when COVID-19 was prevalent. This approach can be applied to the spread of other infectious diseases by adjusting search keywords for observational data. ©Ryuichi Saito, Sho Tsugawa.

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Understanding Open Source Large Language Models: An Exploratory Study

Item Type	Conference Paper
Author	S. Sowe
Author	Y. Mou
Author	D. Cheng
Author	L. Kong
Author	A.T. Neumann
Author	S. Decker
Editor	Jararweh Y.
Editor	Jansen J.
Editor	Alsmirat M.
Abstract	Prompted by the increasing dominance of proprietary Large Language Models (LLMs), such as OpenAI's GPT-4 and Google's Gemini, concerns about data privacy, accessibility and bias have led to a growing advocacy for OSLLMs. This study investigates Open Source Large Language Models (OSLLMs), exploring their characteristics, openness, and community interactions. Our research aims to define OSLLMs (license, openness, community engagement). Utilizing data from the Hugging Face platform, we examine the popularity metrics, license distribution, artefact accessibilities and community engagement of LLM projects. Findings reveal a skewed distribution of model usage, with a few models dominating downloads and likes. Apache 2.0 and MIT are the most common licenses among top models, highlighting a preference for flexible usage terms. However, a significant portion of models lack specified licenses, posing potential legal challenges. Openness analysis shows that nearly half of the examined models share their training code and datasets, with standardized evaluation metrics common across repositories. Community engagement analysis indicates that engineer users are more active than general users, contributing significantly to technical discussions. Sentiment analysis of forum interactions reveals varying user attitudes, with licensed models generally receiving more positive feedback.

This study underscores the potential of OSLLMs to democratize AI access and foster innovation, while also highlighting areas for improvement in community engagement and model openness. © 2024 IEEE.

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Understanding Political Polarization With Generative AI: A Large-Scale Analysis of Voter Preferences on Social Media

Item Type Journal Article

Author L. Belcastro

Author C. Cosentino

Author F. Marozzo

Abstract In an era where users continuously express opinions across digital platforms, social media provides a large-scale lens on how people frame issues, justify choices, and react to events. Traditional approaches typically combine classifiers with unsupervised models to produce analytical outputs (e.g., spreadsheets, aggregate statistics), leaving human experts to interpret these outputs, connect signals across them, and translate them into explanations. The advent of Large Language Models (LLMs) has reshaped this process by enabling end-to-end pipelines that can both annotate large-scale text and generate coherent narrative reports that synthesize the evidence. This study presents an AI-based methodology for large-scale analysis of social media opinions during political events to estimate users' propensity to support different candidates or factions. We classify and enrich posts with multidimensional annotations (e.g., candidate support, topics, sentiment) using a combination of traditional models and LLMs, inferring missing attributes from profile metadata and textual cues. We then aggregate annotations at the user level to build representative samples, validate their distributions against official population statistics, and generate concise reports that summarize the main motivations and issues driving support for each candidate or faction. We apply this methodology to X/Twitter data from swing states in the 2020 and 2024 U.S. presidential elections. For each state-candidate pair, the pipeline generates a report, which we compare against a baseline in which an LLM analyzes raw posts without any preprocessing or enrichment. The enriched configuration consistently outperforms the baseline across semantic similarity metrics, LLM-as-a-judge scores, and human preferences. Moreover, the motivations inferred from pre-election social-media data align closely with post-election expert summaries from sources such as Wikipedia and major newspapers, indicating that our approach can translate large-scale digital corpora into robust and interpretable political insight. © 2026 The Authors.

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Export Date: 20 September 2026; Cited By: 1; Correspondence Address: F. Marozzo; University of Calabria, Department of Informatics, Modeling, Electronics and Systems Engineering (DIMES), Rende, 87036, Italy; email: fabrizio.marozzo@unical.it

UniSent: Universal adaptable sentiment lexica for 1000+ languages

Item Type Conference Paper

Author E. Asgari

Author F. Braune

Author B. Roth

Author C. Ringlstetter

Author M.R.K. Mofrad

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Editor Goggi S.

Editor Isahara H.

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Editor Mariani J.

Editor Mazo H.

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Editor Piperidis S.

Abstract In this paper, we introduce UniSent universal sentiment lexica for 1000+ languages. Sentiment lexica are vital for sentiment analysis in absence of document-level annotations, a very common scenario for low-resource languages. To the best of our knowledge, UniSent is the largest sentiment resource to date in terms of the number of covered languages, including many low resource ones. In this work, we use a massively parallel Bible corpus to project sentiment information from English to other languages for sentiment analysis on Twitter data. We introduce a method called DomDrift to mitigate the huge domain mismatch between Bible and Twitter by a confidence weighting scheme that uses domain-specific embeddings to compare the nearest neighbors for a candidate sentiment word in the source (Bible) and target (Twitter) domain. We evaluate the quality of UniSent in a subset of languages for which manually created ground truth was available, Macedonian, Czech, German, Spanish, and French. We show that the quality of UniSent is comparable to manually created sentiment resources when it is used as the sentiment seed for the task of word sentiment prediction on top of embedding representations. In addition, we show that emoticon sentiments could be reliably predicted in the Twitter domain using only UniSent and monolingual embeddings in German, Spanish, French, and Italian. With the publication of this paper, we release the UniSent sentiment lexica at <http://>

language-lab.info/unisent. © European Language Resources Association (ELRA),
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Bioengineering, University of California, Berkeley, 94720, United States; email: inquiries@language-lab.info;
Conference name: 12th International Conference on Language Resources and Evaluation,
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Unmasking Public Sentiment: A Sample Efficient Approach to Analyzing Twitter Opinion on U.S. Aid to Ukraine

Item Type Conference Paper

Author S. Ghosh

Author N. Dewitt

Author S. Ghosh

Author D. McCoy

Editor Bui T.X.

Abstract The U.S. aid to Ukraine is a bipartisan topic of extreme socio-political
importance. While several organizations have conducted surveys to understand
the public stance on this topic, there is no research to date that analyses public
opinion on social media, possibly due to the lack of annotated data. Therefore,

this research compares several sample-efficient methods (including in-context learning) to analyze tweet sentiments with minimal training data. First, we collect 11,289 tweets about the U.S. aid to Ukraine and mapped them to U.S. states. Next, we explore three different approaches to sentiment analysis: tool-based, embedding-based, and prompt-based. Our results indicate that GPT-4 Few Shot improves accuracy by 121.8% and 77.5% over TextBlob and Vader, respectively. Our geospatial analysis shows that Indiana has the most negative normalized net sentiment (NNS) of -0.83, while Vermont has the most positive NNS of +0.33. Finally, we perform a detailed thematic analysis to identify the common arguments that support or oppose the aid. We highlight that our results do not correlate with media surveys, possibly due to the presence of echo chambers and algorithmic biases. © 2025 IEEE Computer Society. All rights reserved.

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Unpacking Generative AI in Education: Computational Modeling of Teacher and Student Perspectives in Social Media Discourse

Item Type Journal Article

Author P. DeVito

Author A. Vallala

Author S. McMahon

Author Y. Hinda

Author B. Thaw

Author H. Zhuang

Author H. Kalva

Abstract Generative AI (GAI) technologies are quickly reshaping the educational landscape. As adoption accelerates, understanding how students and educators perceive these tools is essential. This study presents one of the most comprehensive analyses to date of stakeholder discourse dynamics on GAI in education using social media data. Our dataset includes 1199 Reddit posts and 13 959 corresponding top-level comments. We apply sentiment analysis, topic modeling, and author classification. To support this, we propose and validate a modular framework that leverages prompt-based large language models (LLMs) for analysis of online social discourse, and we evaluate this framework against classical natural language processing (NLP) models. Our GPT-4o pipeline consistently outperforms prior approaches across all tasks. For example, it achieved 90.6% accuracy in sentiment analysis against gold-standard human annotations. Topic extraction uncovered 12 latent topics in the public discourse with varying sentiment and author distributions. Teachers and students convey optimism about GAI's potential for personalized learning and productivity in higher education. However, key differences emerged: students often voice distress over false accusations of cheating by AI detectors, while teachers generally express concern about job security, academic integrity, and institutional pressures to adopt GAI tools. These contrasting perspectives highlight the tension between innovation and oversight in GAI-enabled learning environments. Our findings suggest a need for clearer institutional policies, more transparent GAI integration practices, and support mechanisms for both educators and students. More broadly, this study demonstrates the potential of LLM-based frameworks for modeling stakeholder discourse within online communities. © 2014 IEEE.

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Unraveling the multimodal features and audience responses to thinspiration videos: A computational analysis of Chinese TikTok (Douyin)

Item Type Journal Article**Author** Y. Tao**Author** S. Sun**Author** J. He

Abstract Short-video social media platforms have emerged as powerful channels for disseminating body image-related content, particularly the thin ideal. Yet much of the existing research has focused on individual-level usage patterns rather than the platform-level mechanisms that shape content exposure. A platform-level perspective, however, is crucial for understanding how body image messages are algorithmically curated and how multimodal features influence audience responses. To address this gap, we developed a computational framework comprising body iconography, communication strategies, and audiovisual aesthetics to investigate thinspiration videos on Douyin (Chinese TikTok). Specifically, integrating computer vision, natural language processing, large language models, and audio signal processing, we extracted textual, visual, and auditory features from 757 trending thinspiration videos tagged with #thin and linked these features to nearly three million associated comments to assess large-scale audience responses. Results showed that trending thinspiration videos promoted the thin-ideal, along with the fit-ideal and beauty-ideal, often with gendered portrayals of the human body. Multimodal features such as body size, evaluative rhetoric (thin-praise, fat-stigma), and specific aesthetic patterns were significantly associated with engagement metrics (likes, comments, reshares) or the prevalence of appearance-focused comments. Gender differences were pronounced: female-focused videos featured smaller body sizes, higher beauty scores, and greater objectification, whereas male-focused videos emphasized muscularity and informational framing. This study advances understanding of body image formation in algorithmically mediated environments and offers a scalable methodological framework for multimodal content analysis. The findings offer insights for content creators, platform regulators, and policymakers to promote healthier digital body image environments. © 2026 Elsevier Ltd. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

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Unsupervised Aspect-Based Sentiment Analysis Through LLM: A Case Study of an Unlabeled Portuguese Beer Database

Item Type Journal Article

Author D.M. De Almeida Eiras

Author A.C.D. Brito

Abstract This study investigates the application of Large Language Models (LLMs) to unsupervised Aspect-Based Sentiment Analysis (ABSA) in Portuguese, focusing on consumer reviews of Brazilian beers. We construct a novel domain-specific dataset comprising nearly 60,000 filtered reviews collected from a major Brazilian beer forum, along with a manually annotated gold-standard subset containing 1,712 labeled Beer Characteristics (BC), categories, and sentiment polarities. Two LLM families - one monolingual (Sabiá-3) and one multilingual (GPT-4o mini) - are systematically evaluated under zero-shot, one-shot, and few-shot prompting strategies. Results show that the best configuration achieves F1-scores of 0.857 for aspect extraction, 0.826 for category identification, and 0.552 for sentiment classification, with sentiment detection proving the most sensitive to prompt design. Using the optimal model and prompt configuration, a large-scale annotated dataset of over 880,000 extracted BC instances is generated, enabling longitudinal analysis of consumer perceptions from 2008 to 2025. Findings indicate that positive attributes such as refreshing mouthfeel and persistent foam dominate consumer praise. The proposed pipeline demonstrates that LLM-based ABSA can effectively uncover fine-grained consumer preferences in low-resource languages without extensive labeled data, offering a scalable and cost-effective tool for market intelligence and product

development in the brewing industry. © 2013 IEEE.

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Unsupervised Self-Training for Sentiment Analysis of Code-Switched Data

Item Type Conference Paper

Author A. Gupta

Author S. Menghani

Author S.K. Rallabandi

Author A.W. Black

Editor Solorio T.

Editor Chen S.

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Editor Sitaram S.

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Abstract Sentiment analysis is an important task in understanding social media content like customer reviews, Twitter and Facebook feeds etc. In multilingual communities around the world, a large amount of social media text is characterized by the presence of code-switching. Thus, it has become important to build models that can handle code-switched data. However, annotated code-switched data is scarce and there is a need for unsupervised models and algorithms. We propose a general framework called Unsupervised Self-Training and show its applications for the specific use case of sentiment analysis of code-switched data. We use the power of pre-trained BERT models for initialization and fine-tune them in an unsupervised manner, only using pseudo labels produced by zero-shot transfer. We test our algorithm on multiple code-switched languages and provide a detailed analysis of the learning dynamics of the algorithm with the aim of answering the question - 'Does our unsupervised model understand the Code-Switched languages or does it just learn its representations?'. Our unsupervised models compete well with their supervised counterparts, with their performance reaching within 1-7% (weighted F1 scores) when compared to supervised models trained for a two class problem. ©2021 Association for Computational Linguistics

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Unveiling international tourist perceptions with a new human-AI collaborative approach: case of Uzbekistan

Item Type Journal Article

Author W. Sun

Author Z. Yin

Author Y. Chen

Author W. Sun

Author S. Weng

Author Y. Wang

Abstract Focusing on tourism destinations in Uzbekistan, this study constructed a human-AI collaborative framework to analyze user-generated content (UGC) from Tripadvisor. By employing the DeepSeek large language model (LLM) with zero-shot learning (ZSL) and prompt engineering, combining human collaboration and validation, this study constructed a tourist perception evaluation index system and conducted sentiment-semantic analysis. First, ZSL combined with prompt engineering was used to automatically extract perception-related evaluation features that tourists focused on through the LLM. Second, by manually refining and filtering the indicators, this study constructed a hierarchical tourist perception evaluation index system comprising four primary indicators and twenty-four secondary indicators. Furthermore, based on the human-AI collaborative approach, each review was assigned sentiment scores across twenty-four evaluation dimensions. Finally, the visualized analytical results were produced to support government decision-making. This study establishes a technical framework for tourism sentiment analysis and governance, providing a practical, low-cost, and efficient tool for destination management. © 2026 Asia Pacific Tourism Association.

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Unveiling Maternity and Infant Care Conversations: A Chinese Dialogue Dataset for Enhanced Parenting Support

Item Type Conference Paper

Author B. Xu

Author L. Li

Author J. Wang

Author X. Qiao

Author E. Yu

Author Y. Qian

Author L. Zong

Author H. Lin

Editor Kwok J.

Abstract The rapid development of large language models has greatly advanced human-computer dialogue research. However, applying these models to specialized fields like maternity and infant care often leads to subpar performance due to a lack of domain-specific datasets. To address this problem, we have created MicDialogue, a Chinese dialogue dataset for maternity and infant care. MicDialogue involves a wide range of specialized topics, including gynecological health, pediatric care, pregnancy preparation, emotional counseling and other related topics. This dataset is curated from two types of Chinese social media: short videos and blog posts. Short videos capture real-time interactions and pragmatic dialogue patterns, while blog posts offer comprehensive coverage of various topics within the domain. We have also included detailed annotations for topics, diseases, symptoms, and causes, enabling in-depth research. Additionally, we developed a knowledge-driven benchmark model using LLM-based prompt learning and multiple knowledge graphs to address diverse dialogue topics. Experiments validate MicDialogue's usability, providing benchmarks for future research and essential data for fine-tuning language models in maternity and infant care. © 2025 International Joint Conferences on Artificial Intelligence. All rights reserved.

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Urban traffic evaluation with social media data: A consensus-based LLM negotiation paradigm

Item Type Journal Article

Author M. Li

Author C. Fan

Author H. Yan

Author P. Wu

Author Z. Su

Author W. Ma

Abstract Evaluating urban traffic systems from the perspective of travelers is critical for traffic planning and operations management, and social media data has been proven to be a rich and timely source for extracting travel experiences. However, it is challenging to evaluate urban traffic using social media data due to difficulties in obtaining a high-quality dataset that reveals travel experiences and conducting precise evaluations based on the collected posts. With recent advancements in large language models (LLMs), this paper establishes a consensus-based LLM negotiation paradigm to address these challenges. Firstly, we propose an LLM-driven automated data collection and processing pipeline for social media data. Specifically, the understanding and reasoning abilities of LLMs facilitate filtering the obtained raw data with high accuracy. More importantly, LLMs do not require any manual labeling of data or extensive pre-training compared to traditional machine learning methods. As for the evaluation system, we leverage the LLMs to cluster and summarize the key indicators for urban traffic evaluation. Furthermore, we propose a consensus-based

LLM negotiation mechanism in which two distinct LLMs serve as the generator and discriminator, respectively. They iteratively evaluate and refine their assessments for each social media post until consensus is reached. In this way, the text data is classified into its associated indicators, with scores assigned based on sentiment analysis. Lastly, we propose the LLM-augmented analytic hierarchy process (AHP), by which the weights of each indicator are determined by LLMs with different role-playing prompts. The effectiveness of the proposed framework is validated in Shanghai and Beijing using social media posts collected from Weibo. The obtained dataset by LLMs is significantly enhanced in both quality and quantity. Moreover, the negotiation mechanism of LLMs can provide not only reliable and robust evaluation results, but also identify key issues behind the evaluation results and offer targeted and precise suggestions. This paper contributes to the use of LLMs in future traffic evaluation and operations. © 2026 Elsevier Ltd

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Use of large language models as a scalable approach to understanding public health discourse

Item Type Journal Article

Author L. Espinosa

Author M. Salathé

Abstract Online public health discourse is becoming more and more important in shaping public health dynamics. Large Language Models (LLMs) offer a scalable solution for analysing the vast amounts of unstructured text found on online platforms. Here, we explore the effectiveness of Large Language Models (LLMs), including GPT models and open-source alternatives, for extracting public stances towards vaccination from social media posts. Using an expert-annotated dataset of social media posts related to vaccination, we applied various LLMs and a rule-based sentiment analysis tool to classify the stance towards vaccination. We assessed the accuracy of these methods through comparisons with expert annotations and annotations obtained through crowdsourcing. Our results demonstrate that few-shot prompting of best-in-class LLMs are the best performing methods, and that all alternatives have significant risks of substantial misclassification. The study highlights the potential of LLMs as a scalable tool for public health professionals to quickly gauge public opinion on health policies and interventions, offering an efficient alternative to traditional data analysis methods. With the continuous advancement in LLM development, the integration of these models into public health surveillance systems could substantially improve our ability to monitor and respond to changing public health attitudes. © 2024 Espinosa, Salathé. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

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Use of Large Language Models to Assess the Likelihood of Epidemics From the Content of Tweets: Infodemiology Study

Item Type Journal Article

Author M.S. Deiner

Author N.A. Deiner

Author V. Hristidis

Author S.D. McLeod

Author T. Doan

Author T.M. Lietman

Author T.C. Porco

Abstract Background: Previous work suggests that Google searches could be useful in identifying conjunctivitis epidemics. Content-based assessment of social media content may provide additional value in serving as early indicators of conjunctivitis and other systemic infectious diseases. Objective: We investigated whether large language models, specifically GPT-3.5 and GPT-4 (OpenAI), can provide probabilistic assessments of whether social media posts about conjunctivitis could indicate a regional outbreak. Methods: A total of 12,194 conjunctivitis-related tweets were obtained using a targeted Boolean search in multiple languages from India, Guam (United States), Martinique (France), the Philippines, American Samoa (United States), Fiji, Costa Rica, Haiti, and the Bahamas, covering the time frame from January 1, 2012, to March 13, 2023. By providing these tweets via prompts to GPT-3.5 and GPT-4, we obtained probabilistic assessments that were validated by 2 human raters. We then calculated Pearson correlations of these time series with tweet volume and the occurrence of known outbreaks in these 9 locations, with time series bootstrap used to compute CIs. Results: Probabilistic assessments derived from GPT-3.5 showed correlations of 0.60 (95% CI 0.47-0.70) and 0.53 (95% CI 0.40-0.65) with the 2 human raters, with higher results for GPT-4. The weekly averages of GPT-3.5 probabilities showed substantial correlations with weekly tweet volume for 44% (4/9) of the countries, with correlations ranging from 0.10 (95% CI 0.0-0.29) to 0.53 (95% CI 0.39-0.89), with larger correlations for GPT-4. More modest correlations were found for correlation with known epidemics, with substantial correlation only in American Samoa (0.40, 95% CI 0.16-0.81). Conclusions: These findings suggest that GPT prompting can efficiently assess the content of social media posts and indicate possible disease outbreaks to a degree of accuracy comparable to that of humans. Furthermore, we found that automated content analysis of tweets is related to tweet volume for conjunctivitis-related posts in some locations and to the occurrence of actual epidemics. Future work may improve the sensitivity and specificity of these methods for disease outbreak detection. © 2024 JMIR Publications Inc.. All rights reserved.

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User eXperience Perception Insights Dataset (UXPID): Synthetic User Feedback from Public Industrial Forums

Item Type Journal Article

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Author J. Joosten

Author C. Ulan uulu

Author N.M. Martins Pacheco

Author F. Ries

Author F. Petridis

Author J. Bosch

Author H.H. Olsson

Abstract Customer feedback in industrial forums offers rich but underexplored insights into real-world product experience. Yet systematic analysis remains challenging due to unstructured, domain-specific content and the scarcity of high-quality labeled datasets. This paper presents the User eXperience Perception Insights Dataset (UXPID), a collection of 7130 synthesized and anonymized user feedback branches extracted from a public industrial automation forum. Each JSON record contains multi-post comments enriched with metadata and annotated by a large language model (LLM) for UX insights, user expectations, severity ratings, sentiment, and topic classifications. UXPID is designed to facilitate research in user requirements, user experience (UX) analysis, and AI-driven feedback processing, particularly where

privacy and licensing restrictions limit access to real-world data. It supports the training and evaluation of transformer-based models for tasks such as issue detection, sentiment analysis, and requirements extraction in technical forums, providing a valuable resource for advancing NLP methods within industrial product support and software engineering domains. © The Author(s) 2026.

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Using Language Models for assessment of users' satisfaction with their partner in Persian

Item Type Conference Paper

Author Z. Habibzadeh

Author M. Asadpour

Editor Hettiarachchi H.

Editor Ranasinghe T.

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Abstract Sentiment analysis, the process of gauging user attitudes and emotions through their textual data, including social media posts and other forms of communication, is a valuable tool for informed decision-making. In other words, by determining whether a statement conveys positivity, negativity, or neutrality, sentiment analysis offers insights into public sentiment regarding a product, individual, event, or other significant topics. This research focuses on the effectiveness of sentiment analysis techniques, using Machine Learning (ML) and Natural Language Processing (NLP) especially pre-trained language models for Persian, in assessing users' satisfaction with their partner, using data collected from X (formerly Twitter). Our motivation stems from traditional in-person surveys, which periodically analyze societal challenges in Iran. The limitations of these surveys led us to explore Artificial Intelligence (AI) as an alternative solution for addressing contemporary social issues. We collected Persian tweets and utilized data annotation techniques to label them according to our research question, forming the dataset. Our goal also was to provide a benchmark of Persian tweets on this specific topic. To evaluate our dataset, we employed several classification methods, including classical ML models, Deep Neural Networks, and pre-trained language models for Persian. Following a comprehensive evaluation, our results show that BERTweet-FA (one of the pre-trained language models for Persian) emerged as the best performer among the classifiers for assessing users' satisfaction. This point indicates the ability of language models to understand conversational Persian text and perform sentiment analysis, even in a low-resource language like Persian. © 2025 Association for Computational Linguistics.

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Computational linguistics, Language model, Economic and social effects, Deep neural networks, Decisions makings, Sentiment analysis, Natural language processing systems, Social media, Users' satisfactions, User attitudes, Tweets, Persians, Textual data, User emotions, Informed decision

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Export Date: 20 September 2026; Cited By: 0; Conference name: 1st Workshop on Language Models for Low-Resource Languages, LoResLM 2025 - co-located at the 31st International Conference on Computational Linguistics, COLING 2025; Conference code: 207335

Using Large Language Models and Prompt Strategies to Annotate Healthcare Posts on Social Media

Item Type Conference Paper

Author L. Aust

Author A. Fu

Editor Nah F.F.-H.

Editor Siau K.L.

Abstract Using youth vaping as a use case, this paper explores the feasibility of applying the latest large language models (LLMs) to automatically annotate social media posts, especially on Reddit. Youth vaping poses serious health risks, and social media offers key insights into user behavior. This study evaluates the capacity of the latest LLMs—including GPT-o1, GPT-o3-mini, Gemini 2.0 Flash, Gemma 2, LLaMA 3.3, DeepSeek R1, and Grok-2—to detect user quit intentions and cessation stages from social media posts. Model outputs were benchmarked against human annotations using various prompting strategies. Preliminary results indicate OpenAI's GPT-o1 is the best-performing model, followed by LLAMA 3.3; the worst was Gemma 2. Although the current preliminary results demonstrate the potential of LLMs, none of them can replace human annotators yet. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2026

DOI 10.1007/978-3-032-13167-6_7

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Language model, Large language model, Behavioral research, 'current, Social networking (online), Health risks, Social media, User behaviors, Annotations, Large Language Models, Social media analytics, Human annotations, Annotation, Social Media Analytics, Model outputs, Vaping cessation, Vaping Cessation

Notes:

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: L. Aust; University of South Carolina, Columbia, United States; email: LAUST@email.sc.edu; Conference name: Late breaking papers from the 27th International Conference on Human-Computer Interaction, HCI International 2025; Conference date: 22 June 2025 through 27 June 2025; Conference code: 346229

Using large language models for extracting and pre-annotating texts on mental health from noisy data in a low-resource language

Item Type Journal Article

Author S. Koltcov

Author A. Surkov

Author O. Koltsova

Author V. Ignatenko

Abstract Recent advancements in large language models (LLMs) have opened new possibilities for developing conversational agents (CAs) in various subfields of mental healthcare. However, this progress is hindered by limited access to high-quality training data, often due to privacy concerns and high annotation costs for low-resource languages. A potential solution is to create human-AI annotation systems that utilize extensive public domain user-to-user and user-to-professional discussions on social media. These discussions, however, are extremely noisy, necessitating the adaptation of LLMs for fully automatic cleaning and pre-classification to reduce human annotation effort. To date, research on LLM-based annotation in the mental health domain is extremely scarce. In this article, we explore the potential of zero-shot classification using four LLMs to select and pre-classify texts into topics representing psychiatric disorders, in order to facilitate the future development of CAs for disorder-specific counseling. We use 64,404 Russian-language texts from online discussion threads labeled with seven most commonly discussed disorders: depression, neurosis, paranoia, anxiety disorder, bipolar disorder, obsessive-compulsive disorder, and borderline personality disorder. Our research shows that while preliminary data filtering using zero-shot technology slightly improves classification, LLM fine-tuning makes a far larger contribution to its quality. Both standard and natural language inference (NLI) modes of fine-tuning increase classification accuracy by more than three times compared to non-fine-tuned training with preliminarily filtered data. Although NLI fine-tuning achieves slightly higher accuracy (0.64) than the standard approach, it is six times slower, indicating a need for further experimentation with NLI hypothesis engineering. Additionally, we demonstrate that lemmatization does not affect classification quality and that multilingual models using texts in their original language perform slightly better than English-only models using automatically translated texts. Finally, we introduce our

dataset and model as the first openly available Russian-language resource for developing conversational agents in the domain of mental health counseling. © 2024 Koltcov et al.

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Notes:

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Using Large Language Models for sentiment analysis of health-related social media data: empirical evaluation and practical tips

Item Type Journal Article

Author L. He

Author S. Omranian

Author S. McRoy

Author K. Zheng

Abstract Health-related social media data generated by patients and the public provide valuable insights into patient experiences and opinions toward health issues such as vaccination and medical treatments. Using Natural Language Processing (NLP) methods to analyze such data, however, often requires high-quality annotations that

are difficult to obtain. The recent emergence of Large Language Models (LLMs) such as the Generative Pre-trained Transformers (GPTs) has shown promising performance on a variety of NLP tasks in the health domain with little to no annotated data. However, their potential in analyzing health-related social media data remains underexplored. In this paper, we report empirical evaluations of LLMs (GPT-3.5-Turbo, FLAN-T5, and BERT-based models) on a common NLP task of health-related social media data: sentiment analysis for identifying opinions toward health issues. We explored how different prompting and fine-tuning strategies affect the performance of LLMs on social media datasets across diverse health topics, including Healthcare Reform, vaccination, mask wearing, and healthcare service quality. We found that LLMs outperformed VADER, a widely used off-the-shelf sentiment analysis tool, but are far from being able to produce accurate sentiment labels. However, their performance can be improved by data-specific prompts with information about the context, task, and targets. The highest performing LLMs are BERT-based models that were fine-tuned on aggregated data. We provide practical tips for researchers to use LLMs on health-related social media data for optimal outcomes. We also discuss future work needed to continue to improve the performance of LLMs for analyzing health-related social media data with minimal annotations. ©2024 AMIA - All rights reserved.

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Notes:

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Using Large Language Models to Classify Public Sentiment Toward Nuclear Power

Item Type Conference Paper

Author O.H. Kwon

Author K. Vu

Author J. Cooper

Author V. Joynt

Author M.I. Radaideh

Abstract Understanding public sentiment regarding nuclear power is a key component in achieving an equitable shift to clean energy, which encompasses the integration of nuclear energy. In this study, we navigate through social media to obtain data as an indicator of public sentiment related to nuclear power. We present sentiment analysis findings based on a dataset of 500,000 tweets collected from X (Twitter). A novel approach is utilized to label the data, employing seven different libraries to determine the sentiment label. This method highlighted the variability and bias among these libraries in labeling the same text. Consequently, the data was categorized into low and high confidence tweets depending on the level of agreement among the seven labeling tools. Then, the performance of various models, including Large Language Models (LLMs), neural networks (e.g., Long Short-Term Memory), and classical models (e.g., Random Forests), is evaluated in sentiment classification using these labeled data. Notably, LLMs like GPT and BERT showed superior performance when trained solely on high confidence data labels, as well as when dealing with both high and low confidence data. Their accuracy ranged from 81% for both low and high confidence data to 95% for high confidence data, surpassing all other models. Additionally, the analysis revealed a tendency toward neutral to negative sentiments regarding nuclear power within the dataset, with fewer instances of positive sentiment. Further investigation is needed to understand the underlying reasons behind these sentiments, such as potential political factors, an area that requires further investigation. © 2024 Pacific Basin Nuclear Conference, PBNC 2024. All rights reserved.

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Language model, Large language model, Economic and social effects, Performance, BERT, GPT, Long short-term memory, Decision trees, Labeled data, Sentiment classification, Tweets, Public sentiments, Power, Sentiment Classification, Clean energy, High confidence, Large Language Model (LLM), Nuclear Power

Notes:

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Using large language models to investigate cultural ecosystem services perceptions: A few-shot and prompt method

Item Type Journal Article

Author H. Luo

Author Z. Zhang

Author Q. Zhu

Author N.E. Houda Ben Ameer

Author X. Liu

Author F. Ding

Author Y. Cai

Abstract The advancement of generative AI has profoundly impacted various aspects of society, including scientific research, but its application in landscape research remains underexplored. In this study, large language models are applied to analyze cultural ecosystem services, which are a key connection between humans and nature, reflecting the intangible benefits that ecosystems provide. Social media texts from the Lushan Scenic Area, known for its rich cultural ecosystem services, were analyzed. The methodology involved adapting the model using few-shot learning to classify cultural ecosystem services and associated sentiments. Prompts were specifically designed to optimize model performance. The validation process compared the performance of three base models (GLM-4-0520, ERNIE-4.0-8K, and Moonshot-v1-8k) alongside five prompts. The cultural ecosystem services within the study area were subsequently analyzed based on model outputs. The findings indicated superior performance by the Moonshot-v1-8k model, achieving 82.2 % micro-F1 and 80.3 % macro-F1. The implementation of chain-of-thought prompts and cultural ecosystem services definition prompts enhanced micro-F1 and macro-F1 by up to 6.3 % and 3.3 %, respectively. Within the Lushan Scenic Area, aesthetic services were identified as the most frequently perceived, while recreational services received the most negative sentiments. A marked increase in public interest in physical health was observed following the COVID-19 pandemic. This study highlights the potential of large language models to advance the analysis of cultural ecosystem services and landscape perceptions. By offering a novel approach to text analysis, the findings contribute valuable insights for landscape management and underscore the utility of AI technologies. © 2025 Elsevier B.V.

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Using Large Language Models to Support Content Analysis: A Case Study of ChatGPT for Adverse Event Detection

Item Type Journal Article

- Author** E.C. Leas
- Author** J.W. Ayers
- Author** N. Desai
- Author** M. Dredze
- Author** M. Hogarth
- Author** D.M. Smith

Abstract This study explores the potential of using large language models to assist content analysis by conducting a case study to identify adverse events (AEs) in social media posts. The case study compares ChatGPT’s performance with human annotators’ in detecting AEs associated with delta-8-tetrahydrocannabinol, a cannabis-derived product. Using the identical instructions given to human annotators, ChatGPT closely approximated human results, with a high degree of agreement noted: 94.4% (9436/10,000) for any AE detection (Fleiss κ =0.95) and 99.3% (9931/10,000) for serious AEs (κ =0.96). These findings suggest that ChatGPT has the potential to replicate human annotation accurately and efficiently. The study recognizes possible limitations, including concerns about the generalizability due to ChatGPT’s training data, and prompts further research with different models, data sources, and content analysis tasks. The study highlights the promise of large language models for

enhancing the efficiency of biomedical research. © 2024 JMIR Publications Inc.. All rights reserved.

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Notes:

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Using LLMs for the multidimensional perception assessment of recreation and leisure spaces: a case study of Hangzhou, China

Item Type Journal Article

Author Z. Bai

Author Y. Wu

Author X. Kang

Author X. Kong

Author J. Zhang

Abstract Urban recreation and leisure spaces (RLSs) are vital venues for citizens to relax, socialize, and experience urban life. Public perception of RLSs represents a crucial component of urban studies. Traditional Natural Language Processing (NLP)-based

methods for analyzing RLS perceptions from social media data face limitations in semantic understanding, multidimensional sentiment differentiation, and spatial recognition. To overcome these limitations, we propose and validate a novel Large Language Model (LLM)-based analytical framework for multidimensional perception assessment of urban recreation and leisure spaces. Through comparative evaluation of three mainstream LLMs (DeepSeek-R1, Kimi, and QWEN), we selected the best-performing Qwen model (F1-score = 0.899) to construct a multidimensional Aspect-Based Sentiment Analysis (ABSA) framework. Taking Hangzhou, China as a case study, we collected note data from the "REDnote" platform and employed the LLM to identify seven perception dimensions and determine sentiment polarities. Building upon the ABSA perception evaluation data generated by the LLM, we further conducted spatial distribution analysis, keyword co-occurrence network analysis, and place-activity bipartite network analysis to reveal perception patterns and their underlying causes. The findings indicate that: a) Hangzhou's recreation and leisure spaces excel in "soft power" dimensions (cultural atmosphere, spatial aesthetics) but require optimization in "hard power" aspects (accessibility, functional configuration); b) Hangzhou's leisure resources are unevenly distributed, with perception evaluations exhibiting a core-periphery structure; c) open outdoor spaces are more positively perceived than enclosed commercial complexes, with "city walking" emerging as the dominant positive activity. This research confirms the transformative value of LLM as core analytical engines rather than preprocessing tools, providing a new research paradigm and practical guidance for the planning and management of urban recreation and leisure spaces. © The Author(s) 2026.

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Notes:

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Using LLMs to Establish Implicit User Sentiment of Software Desirability

Item Type Conference Paper

Author S. Weitzl-Harms

Author J.D. Hastings

Author J. Lum

Editor Wani M.A.

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Editor Luo F.

Editor Ogihara M.

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Editor Gu X.

Abstract This study explores the use of LLMs for providing quantitative zero-shot sentiment analysis of implicit software desirability, addressing a critical challenge in product evaluation where traditional review scores, though convenient, fail to capture the richness of qualitative user feedback. Innovations include establishing a method that 1) works with qualitative user experience data without the need for explicit review scores, 2) focuses on implicit user satisfaction, and 3) provides scaled numerical sentiment analysis, offering a more nuanced understanding of user sentiment, instead of simply classifying sentiment as positive, neutral, or negative. Data is collected using the Microsoft Product Desirability Toolkit (PDT), a well-known qualitative user experience analysis tool. For initial exploration, the PDT metric was given to users of two software systems. PDT data was fed through several LLMs (Claude Sonnet 3 and 3.5, GPT4, and GPT4o) and through a leading transfer learning technique, Twitter-Roberta-Base-Sentiment, and Vader, a leading sentiment analysis tool. Each system was asked to evaluate the data in two ways, by looking at the sentiment expressed in the PDT word/explanation pairs; and by looking at the sentiment expressed by the users in their grouped selection of five words and explanations, as a whole. Numerical analysis is used to provide insights into the magnitude of sentiment to drive high quality decisions regarding product desirability. Each LLM is asked to provide its confidence (low, medium, high) in its sentiment score, along with an explanation of its score. All LLMs tested were able to statistically detect user sentiment from the users' grouped data, whereas TRBS and Vader were not. The confidence and explanation of confidence provided by the LLMs assisted in understanding user sentiment. This study adds deeper understanding of evaluating user experiences, toward the goal of creating a universal tool that quantifies implicit sentiment. © 2024 IEEE.

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Using Masked Language Modeling to Enhance BERT-Based Aspect-Based Sentiment Analysis for Affective Token Prediction

Item Type Conference Paper
Author W. Jin
Author B. Zhao
Author C. Liu
Author H. Zhang
Author M. Jiang
Editor Iliadis L.
Editor Papaleonidas A.
Editor Angelov P.
Editor Jayne C.

Abstract Aspect-based sentiment analysis is a challenging yet critical task for recognizing emotions in text, with various applications in social media, commodity reviews, and movie comments. Many researchers are working on developing more powerful sentiment analysis models. Most existing models use the pre-trained language models based fine-tuning paradigm, which only utilizes the encoder parameters of pre-trained language models. However, this approach fails to effectively leverage the prior knowledge revealed in pre-trained language models. To address these issues, we propose a novel approach, Target Word Transferred Language Model for aspect-based sentiment analysis (WordTransABSA), which investigates the potential of the pre-training scheme of pre-trained language models. WordTransABSA is an encoder-decoder architecture built on top of the Masked Language Model of Bidirectional Encoder Representation from Transformers. During the training procedure, we reformulate the previous generic fine-tuning models as a “Masked Language Model” task, which follows the original BERT pre-training paradigm. WordTransABSA takes full advantage of the versatile linguistic knowledge of Pre-trained Language Model, resulting in competitive accuracy compared with recent baselines, especially in data-insufficient scenarios. We have made our code publicly available on GitHub (<https://github.com/albert-jin/WordTransABSA>). © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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Utilizing LLMs and ML Algorithms in Disaster-Related Social Media Content

Item Type Journal Article

Author V. Linardos

Author M. Drakaki

Author P. Tzionas

Abstract In this research, we explore the use of Large Language Models (LLMs) and clustering techniques to automate the structuring and labeling of disaster-related social media content. With a gathered dataset comprising millions of tweets related to various disasters, our approach aims to transform unstructured and unlabeled data into a structured and labeled format that can be readily used for training machine learning algorithms and enhancing disaster response efforts. We leverage LLMs to preprocess and understand the semantic content of the tweets, applying several semantic properties to the data. Subsequently, we apply clustering techniques to identify emerging themes and patterns that may not be captured by predefined categories, with these patterns surfaced through topic extraction of the clusters. We proceed with manual labeling and evaluation of 10,000 examples to evaluate the LLMs' ability to understand tweet features. Our methodology is applied to real-world data for disaster events, with results directly applicable to actual crisis situations. © 2025 by the authors.

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VaxiBERT: A BERT-Based Classifier for Vaccine Tweets with Multi-Label Annotations

Item Type Conference Paper

Author S. Bithel

Author S. Verma

Author R. Singh

Editor Ghosh K.

Editor Mandl T.

Editor Majumder P.

Editor Mitra M.

Abstract Vaccination has long been seen as an essential component of public health, providing a critical line of defense against infectious diseases. Our primary objective is to build a robust multi-label classification system capable of categorizing individual social media posts, specifically tweets, based on the numerous vaccine-related concerns stated by their writers. These reservations cover many issues, including misgivings about necessity, safety, and political intentions. We discuss our approach and evaluation results, shedding light on the intricate interplay between feeling, society, and science in the arena of the vaccine debate, using cutting-edge models such as Covid-Twitter-BERT and OpenLLaMA-7B. Our best-submitted run achieved a 0.67 macro-F1 Score and 0.70 Jaccard score. Github Code: https://github.com/shivangibithel/VaxiBERT_AISoMe2023. © 2023 Copyright for this paper by its authors.

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Notes:

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VideoConviction: A Multimodal Benchmark for Human Conviction and Stock Market Recommendations

Item Type Conference Paper

Author M. Galarnyk

Author V. Kejriwal

Author A. Shah

Author Y. Bhardwaj

Author N.W. Meyer

Author A. Krishnan

Author S. Chava

Abstract Social media has amplified the reach of financial influencers known as "finfluencers," who share stock recommendations on platforms like YouTube. Understanding their influence requires analyzing multimodal signals like tone, delivery style, and facial expressions, which extend beyond text-based financial analysis. We introduce VideoConviction, a multimodal dataset with 6,000+ expert annotations, produced through 457 hours of human effort, to benchmark multimodal large language models (MLLMs) and text-based large language models (LLMs) in financial discourse. Our results show that while multimodal inputs improve stock ticker extraction (e.g., extracting Apple's ticker AAPL), both MLLMs and LLMs struggle to distinguish investment actions and conviction-the strength of belief conveyed through confident delivery and detailed reasoning-often misclassifying general commentary as definitive recommendations. While high-conviction recommendations perform better than low-conviction ones, they still underperform the popular S&P 500 index fund. An

inverse strategy-betting against finfluencer recommendations-outperforms the S&P 500 by 6.8% in annual returns but carries greater risk (Sharpe ratio of 0.41 vs. 0.65). Our benchmark enables a diverse evaluation of multimodal tasks, comparing model performance on both full video and segmented video inputs. This enables deeper advancements in multimodal financial research. Our code, dataset, and evaluation leaderboard are available under the CC BY-NC 4.0 license. © 2025 Copyright held by the owner/author(s).

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Extra Journal Abbreviation: Proc. ACM SIGKDD Int. Conf. Knowl. Discov. Data Min.

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Conference Name Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: A. Shah; Georgia Institute of Technology, Atlanta, United States; email: ashah482@gatech.edu; Conference name: 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, KDD 2025; Conference date: 3 August 2025 through 7 August 2025; Conference code: 211085

ViGoEmotions: A Benchmark Dataset For Fine-grained Emotion Detection on Vietnamese Texts

Item Type Conference Paper

Author H.Q. Tran

Author N.T. Pham

Author S.T. Luu
Author K. Van Nguyen
Editor Demberg V.
Editor Inui K.
Editor Marquez Villodre L.

Abstract Emotion classification plays a significant role in emotion prediction and harmful content detection. Recent advancements in NLP, particularly through large language models (LLMs), have greatly improved outcomes in this field. This study introduces ViGoEmotions - a Vietnamese emotion corpus comprising 20,664 social media comments in which each comment is classified into 27 fine-grained distinct emotions. To evaluate the quality of the dataset and its impact on emotion classification, eight pretrained Transformer-based models were evaluated under three preprocessing strategies: preserving original emojis with rule-based normalization, converting emojis into textual descriptions, and applying ViSoLex, a model-based lexical normalization system. Results show that converting emojis into text often improves the performance of several BERT-based baselines, while preserving emojis yields the best results for ViSoBERT and CafeBERT. In contrast, removing emojis generally leads to lower performance. ViSoBERT achieved the highest Macro F1-score of 61.50% and Weighted F1-score of 63.26%. Strong performance was also observed from CafeBERT and PhoBERT. These findings highlight that while the proposed corpus can support diverse architectures effectively, preprocessing strategies and annotation quality remain key factors influencing downstream performance. © 2026 Association for Computational Linguistics.

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URL <https://www.scopus.com/pages/publications/105040580904?origin=resultslist>

Archive Scopus

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Extra Journal Abbreviation: EACL - Conf. Eur. Chapter Assoc. Comput. Linguist., Proc. Conf., Vol. 1 - Long Pap.

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Publisher Association for Computational Linguistics (ACL)

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Proceedings Title EACL - Conf. Eur. Chapter Assoc. Comput. Linguist., Proc. Conf., Vol. 1 - Long Pap.

Conference Name EACL 2026 - 19th Conference of the European Chapter of the Association for Computational Linguistics, Proceedings of the Conference, Vol. 1 - (Long Papers)

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Computational linguistics, Performance, Classification (of information), Emotion Recognition, Emotion detection, Psychology computing, Emotion classification, Fine grained, Benchmark datasets, Vietnamese, F1 scores, Normalisation, Quality control, Content detection, Emotion predictions

Notes:

EID: 2-s2.0-105040580904

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ViTASA: New benchmark and methods for Vietnamese targeted aspect sentiment analysis for multiple textual domains

Item Type Journal Article

Author K.Q. Tran

Author Q.P.-M. Huynh

Author O.T.-H. Le

Author K. Van Nguyen

Author N.L.-T. Nguyen

Abstract Targeted Aspect Sentiment Analysis (TASA) has gained substantial attraction in recent years, fostering diverse studies and technological advancements. However, the development of TASA resources for Vietnamese has been limited. This paper introduces ViTASA, a comprehensive, high-quality dataset designed to catalyze advancements in Vietnamese TASA. ViTASA encompasses over 500,000 target-aspect pairs from social media comments across three key domains: mobile, restaurant, and hotel, thereby addressing critical gaps in existing datasets. Additionally, ViTASA integrates a novel multi-task evaluation framework, posing new challenges and enabling robust model assessments. We present ViTASD, an innovative BERT-based approach optimized for the linguistic features of Vietnamese. Comparative analyses demonstrate that ViTASD significantly outperforms existing state-of-the-art methods, including CG-BERT, QACG-BERT, BERT-pair-QA, BERT-pair-NLI, and a range of zero-shot learning models like Gemma, Llama, Mistral and Qwen. Notably, ViTASD achieves superior macro F1-scores of 61.77%, 41.12%, and 52.64% in the mobile, restaurant, and hotel domains respectively. This study not only highlights the challenges inherent in Vietnamese sentiment analysis but also lays a robust foundation for future research endeavors in this area. In a commitment to advancing TASA technology and enhancing the reliability of digital media analyses, we have made the ViTASA dataset, model checkpoints, and source code openly accessible on GitHub¹. © 2025 Elsevier Ltd

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Sentiment analysis, Social media, Contrastive Learning, Technological advancement, Reliability analysis, High quality, Opinion mining, Low-resource NLP, Transformer-based, Vietnamese, Zero-shot learning, Multi-head attention, Hotels, Motels, Targeted aspect sentiment analysis, Targeted aspect sentiment analyze

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EID: 2-s2.0-105001360968

Export Date: 20 September 2026; Cited By: 4; Correspondence Address: K. Van Nguyen; Faculty of Information Science and Engineering, University of Information Technology, Ho Chi Minh City, Viet Nam; email: kietnv@uit.edu.vn; CODEN: CSPLE

Voices of the Stacks: A Multifaceted Inquiry into Academic Librarians' Tweets

Item Type Conference Paper

Author S. Ghosh

Author J. Thajudeen

Abstract Twitter has emerged as an important forum for discussion among academic librarians. In this research, we take a mixed-methods approach to study the thematic content and sentiment of tweets authored by academic librarians in the United States, Canada, and the United Kingdom. We found differences in the semantic content and themes present in the data from each country that point to differences in how librarians in each country engage on Twitter. While more work remains to be done, we cast new light on how members of professional communities use social media. Our qualitative analysis identified 11 thematic categories in academic librarians' Twitter discussions, focusing on professional topics. UK librarians exhibited a higher frequency of labor- and employment-related terms compared to their US and Canadian counterparts. Sentiment ratios for US and Canadian tweets were similar, while the UK displayed nearly double the positive-to-negative tweet ratio. We also present a methodological intervention comparing two different sentiment analysis methods, VADER, and Zero-Shot Learning (ZSL), to classify posts by academic librarians. ZSL significantly outperformed the off-the-shelf classifier, highlighting how accurate prediction is possible without annotated training data. Annual Meeting of the Association for Information Science & Technology | Oct. 27 – 31, 2023 | London, United Kingdom. Author(s) retain copyright, but ASIS&T receives an exclusive publication license.

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Classification (of information), Sentiment analysis, Thematic analysis, Social networking (online), Social media, Mixed method, Semantics, Sentiment Analysis, Social Media, Libraries, Zero-shot learning, Zero-Shot Learning, Academic librarian, Academic Librarians, High frequency HF, Professional community, Qualitative analysis, Semantic content, Thematic Analysis, United kingdom

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EID: 2-s2.0-85174501374

Export Date: 20 September 2026; Cited By: 7; Correspondence Address: S. Ghosh; San José State University, United States; email: souvick.ghosh@sjsu.edu

vSTREAM: Video Summarization Through Retrieval of Audio and Metadata

Item Type Conference Paper
Author P. Lohumi
Author M. Dash
Author M.S. Arya
Author T. Pant
Editor Virdee B.
Editor Duarte Correia S.
Editor Swaroop A.
Editor Thakur N.
Editor Shetty S.D.
Abstract We present vSTREAM (Video Summarization Through Retrieval of Audio and Metadata), an end-to-end, zero-annotation pipeline that generates concise yet context-rich summaries for short-form social-media videos. vSTREAM operates in five stages: (1) automatic video download; (2) audio transcription with the Whisper ASR model; (3) uniform sampling of N visual frames; (4) extraction of post-level metadata such as hashtags and descriptions; and (5) construction of a

single, multimodal prompt that is submitted to a Multimodal Large Language Model (GPT-4o). The prompt-engineering scheme fuses transcript, images, and metadata, enabling the LLM to understand temporal content while grounding events in the wider social-media context. We evaluate vSTREAM on more than 150 TikTok, Instagram, and YouTube clips spanning across different genres. Pair-wise, blind comparisons performed by OpenAI's o1 model show that vSTREAM's metadata-augmented summaries are preferred over transcript-plus-image baselines in most cases, demonstrating that inexpensive, automatically harvested hashtags readily improve summary quality. Because the approach relies only on freely available media, open-source ASR, lightweight frame sampling, and off-the-shelf LLMs, it scales readily and can be adopted without additional human annotation or domain-specific tuning. © The Author(s), under exclusive license to Springer Nature Switzerland AG 2026.

Date 2026

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Language English

Extra Journal Abbreviation: Lect. Notes Networks Syst.

Volume 1930 LNNS

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Conference Name Lecture Notes in Networks and Systems

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Context-Aware, Visual languages, Social media, Metadata, Multi-modal, Information extraction, Information retrieval, Extraction, Interactive computer systems, Multimodal, Video recording, Audio visual search, Audio-visual, Context-aware video summarization, Social media metadata, Social medium metadata, Video summarization, Visual search

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: P. Lohumi; AI Research, Accrete.ai, Mumbai, India; email: pratyush.lohumi@accrete.ai; Conference name: 2nd International Conference on Artificial Intelligence and Networks, ICAIN 2025; Conference date: 6 October 2025 through 7 October 2025; Conference code: 353049

WASSA 2023 - 13th Workshop on Computational Approaches to Subjectivity, Sentiment and Social Media Analysis, Proceedings of the Workshop

Item Type Journal Article**Editor** Barnes J.**Editor** De Clercq O.**Editor** Klinger R.

Abstract The proceedings contain 61 papers. The topics discussed include: PESTO: a post-user fusion network for rumor detection on social media; sentimental matters - predicting literary quality by sentiment analysis and stylometric features; instruction tuning for few-shot aspect-based sentiment analysis; you are what you read: inferring personality from consumed textual content; UNIDECOR: a unified deception corpus for cross-corpus deception detection; discourse mode categorization of Bengali social media health text; emotion and sentiment guided paraphrasing; emotions in spoken language - do we need acoustics?; understanding emotion valence is a joint deep learning task; Czeching the news: article trustworthiness dataset for Czech; a dataset for explainable sentiment analysis in the German automotive industry; fluency matters! controllable style transfer with syntax guidance; and ChatGPT for suicide risk assessment on social media: quantitative evaluation of model performance, potentials and limitations.

Date 2023**URL** <https://www.scopus.com/pages/publications/85174821971?origin=resultslist>**Archive** Scopus**Language** English**Publisher** Association for Computational Linguistics (ACL)**Publication** Proceedings of the Annual Meeting of the Association for Computational Linguistics**ISSN** 0736587X (ISSN); 978-195942987-6 (ISBN)**Journal Abbr** Proc. Annu. Meet. Assoc. Comput Linguist.**Date Added** 20/09/2026, 8:35:41 pm**Modified** 20/09/2026, 8:35:41 pm**Notes:**

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WASSA 2024 - 14th Workshop on Computational Approaches to Subjectivity, Sentiment, and Social Media Analysis, Proceedings of the Workshop

Item Type Journal Article**Editor** De Clercq O.**Editor** Barriere V.**Editor** Barnes J.**Editor** Klinger R.**Editor** Sedoc J.**Editor** Tafreshi S.

Abstract The proceedings contain 53 papers. The topics discussed include: enhanced financial sentiment analysis and trading strategy development using large language models; modeling complex interactions in long documents for aspect-based sentiment analysis; hierarchical adversarial correction to mitigate identity term bias in toxicity detection; a systematic analysis on the temporal generalization of language models in social media; entity-level sentiment: more than the sum of its parts; polarization of autonomous generative ai agents under echo chambers; know thine enemy: adaptive attacks on misinformation detection using reinforcement learning; the model arena for cross-lingual sentiment analysis: a comparative study in the era of large language models; emotion identification for French in written texts: considering modes of emotion expression as a step towards text complexity analysis; and comparing tools for sentiment analysis of Danish literature from hymns to fairytales: low-resource language and domain challenges.

Date 2024

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Language English

Publisher Association for Computational Linguistics (ACL)

Publication WASSA 2024 - 14th Workshop on Computational Approaches to Subjectivity, Sentiment, and Social Media Analysis, Proceedings of the Workshop

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Watch and learn: leveraging expert knowledge and language for surgical video understanding

Item Type Journal Article

Author D. Gastager

Author G. Ghazaei

Author C. Patsch

Abstract Purpose: Automated surgical workflow analysis is a common yet challenging task with diverse applications in surgical education, research, and clinical decision-making. Although videos are commonly collected during surgical interventions, the lack of annotated datasets hinders the development of accurate and comprehensive workflow analysis solutions. We introduce a novel approach for addressing the sparsity and heterogeneity of annotated training data inspired by the human learning procedure of watching experts and understanding their explanations. Methods: Our

method leverages a video-language model trained on alignment, denoising, and generative tasks to learn short-term spatio-temporal and multimodal representations. A task-specific temporal model is then used to capture relationships across entire videos. To achieve comprehensive video-language understanding in the surgical domain, we introduce a data collection and filtering strategy to construct a large-scale pretraining dataset from educational YouTube videos. We then utilize parameter-efficient fine-tuning by projecting downstream task annotations from publicly available surgical datasets into the language domain. Results: Extensive experiments in two surgical domains demonstrate the effectiveness of our approach, with performance improvements of up to 7% in phase segmentation tasks, 5% in zero-shot phase segmentation, and comparable capabilities to fully supervised models in few-shot settings. Harnessing our model's capabilities for long-range temporal localization and text generation, we present the first comprehensive solution for dense video captioning (DVC) of surgical videos, addressing this task despite the absence of existing DVC datasets in the surgical domain. Conclusion: We introduce a novel approach to surgical workflow understanding that leverages video-language pretraining, large-scale video pretraining, and optimized fine-tuning. Our method improves performance over state-of-the-art techniques and enables new downstream tasks for surgical video understanding. © CARS 2025.

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Weakly Supervised Deep Learning for Arabic Tweet Sentiment Analysis on Education Reforms: Leveraging Pre-Trained Models and LLMs with Snorkel

Item Type Journal Article

Author A. Alotaibi

Author F. Nadeem

Author M. Hamdy

Abstract This study introduces a novel approach to sentiment classification of Arabic tweets regarding educational reforms in Saudi Arabia. The complexity of the Arabic language, with its numerous dialects, poses challenges for natural language processing tasks, particularly when large volumes of data require manual annotation. To overcome the limitations of traditional labeling methods, we developed a weakly supervised learning framework that combines LLMs (GPT-3.5) and pre-trained language models (MarBERT and XLM-RoBERTa) to generate high-quality weakly labeled training data using the Snorkel framework. We fine-tuned the AraBERT model with this weakly labeled data for sentiment classification. Our experimental results demonstrated the effectiveness of the proposed approach, achieving 83% precision, 76% recall, and an 85% F1 score in classifying tweets as positive, negative, or neutral. Comparative analysis showed that GPT-3.5 outperformed Llama 2 in prompting tasks, and our weakly supervised model surpassed baseline machine learning methods. These findings highlight the potential of weakly supervised learning in analyzing public opinion on Arabic social media platforms without relying on large, labeled datasets. © 2013 IEEE.

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WeiboDRP: Sentiment Analysis on Chinese Social Media with Multi-Level Comment Structures

Item Type Conference Paper

Author W. Ding

Author H. Bhuiyan

Editor Long C.

Editor Shi J.

Editor Huang X.

Editor Yiu M.L.

Editor Liang Y.

Editor Choi B.

Editor Song Y.

Abstract Social media provides valuable insights into public sentiment, but analyzing sentiment accurately remains challenging due to the widespread presence of multi-level comment structures (MLCS) in social media texts. Existing research lacks dedicated datasets and systematic evaluation metrics for MLCS-based sentiment analysis tasks. This study proposes WeiboDRP, a large-scale dataset explicitly designed for Chinese sentiment analysis with MLCS annotation. Benchmark evaluations using dictionary methods, fine-tuned pre-trained models, and zero-shot large language models on this dataset reveal that DeepSeek-R1 achieves the best overall performance. Additionally, this study proposes a novel evaluation metric called the Hierarchical Consistency Score (HCS), which effectively measures models' ability to capture and utilize hierarchical contextual information in MLCS. The effectiveness of HCS is verified through ablation experiments. Its cross-domain generalizability is further demonstrated on a publicly available benchmark dataset. These findings highlight the importance of effectively understanding and leveraging the contextual information provided by MLCS to improve classification performance in social media-based sentiment analysis tasks. © The Author(s), under exclusive license to Springer Nature Singapore Pte Ltd. 2027.

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Export Date: 20 September 2026; Cited By: 0; Correspondence Address: W. Ding; Faculty of Information Technology, Monash University, Melbourne, 3800, Australia; email: wdin0014@student.monash.edu; Conference name: Workshop on AI and Data Science for Digital Finance, AI4DF 2026, Workshop on Decision Making and Optimization in Financial Technologies, DMO-FinTech 2026, Workshop on Research and Applications of Foundation Models for Data Mining and Affective Computing, RAFDA 2026, Workshop on Autonomous Machine Intelligence, AMI-PAKDD 2026, Workshop on Resource-Efficient Foundation Models for Large-Scale Data Mining, EFMxDM 2026, Workshop on Generative AI and Computational Journalism, JEN-AI 2026, held in conjunction with the 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining, PAKDD 2026; Conference date: 9 June 2026 through 12 June 2026; Conference code: 356649

When Automated Assessment Meets Automated Content Generation: Examining Text Quality in the Era of GPTs

Item Type Journal Article

Author M. Bevilacqua

Author K. Oketch

Author R. Qin

Author W. Stamey

Author X. Zhang

Author Y. Gan

Author K. Yang

Author A. Abbasi

Abstract The use of machine learning (ML) models to assess and score textual data has become increasingly pervasive in an array of contexts including natural language processing, information retrieval, search and recommendation, and credibility assessment of online content. A significant disruption at the intersection of ML and

text are text-generating large-language models (LLMs) such as generative pre-trained transformers (GPTs). We empirically assess the differences in how ML-based scoring models trained on human content assess the quality of content generated by humans versus GPTs. To do so, we propose an analysis framework that encompasses essay scoring ML models, human- and ML-generated essays, and a statistical model that parsimoniously considers the impact of type of respondent, prompt genre, and the ML model used for assessment model. A rich testbed is utilized that encompasses 18,460 human-generated and GPT-based essays. Results of our benchmark analysis reveal that LLMs and transformer pretrained language models (PLMs) more accurately score human essay quality as compared to CNN/RNN and feature-based ML methods. Interestingly, we find that LLMs and transformer PLMs tend to score GPT-generated text 10-20% higher on average, relative to human-authored documents. Conversely, traditional deep learning and feature-based ML models score human text considerably higher. Further analysis reveals that even though the LLMs and transformer PLMs are exclusively fine-tuned on human text, they more prominently attend to certain tokens appearing only in GPT-generated text, possibly (in part) due to familiarity/overlap in pre-training. Our framework and results have implications for text classification settings where automated scoring of text is likely to be disrupted by generative AI. © 2025 Copyright held by the owner/author(s).

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Which Matters More: Model or Language? An Empirical Study in English-Bangla Mental Health Classification

Item Type Conference Paper

Author A. Islam

Author I.A. Rafi

Author S. Mondal

Author S.A.S. Rahman

Author G.R. Alam

Abstract We present an empirical comparison of classical baselines and pretrained transformer encoders for mental health status classification across English and Bangla social media text. Using two public datasets-an English multi-class corpus and a Bangla binary corpus-we evaluate TF-IDF with Logistic Regression and Random Forest against BERT, RoBERTa, DeBERTa, and BanglaBERT under a matched setup with a stratified eighty twenty split and macro F1 for model selection. In English, RoBERTa achieves 81.6% accuracy with a macro F1 of 78.8, while TF-IDF with Logistic Regression reaches 77.3% accuracy. In Bangla, BanglaBERT attains 88.3% accuracy with a macro F1 of 88.3, and classical baselines surpass several non-Bangla encoders. Findings highlight the value of language models and the importance of classical machine learning models in classifying mental status across different languages. © 2025 IEEE.

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Computational linguistics, Speech communication, Information systems, Mental health, Classification (of information), Learning systems, Machine learning, Natural language processing systems, Social networking (online), Social media, Data mining, BERT, social media, Logistic regression, Random forests, Text classification, Text processing, Low-resource NLP, Depression detection, Cross-lingual, RoBERTa, Signal encoding, Bengalis, Banglabert, BanglaBERT, English, low-resource NLP, depression detection, Bangla (bengali), Bangla (Bengali), crosslingual evaluation, Crosslingual evaluation, DeBERTa-v3, mental health text classification, Mental health text classification, Transformer encoder, transformer encoders

Notes:

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Export Date: 20 September 2026; Cited By: 0; Conference name: 2025 5th International Conference on Robotics, Automation, and Artificial Intelligence, RAAI 2025; Conference date: 18 December 2025 through 20 December 2025; Conference code: 221324

Why are you traveling? Inferring trip profiles from online reviews and domain-knowledge

Item Type Journal Article
Author L.G.S. Félix
Author W. Cunha
Author C.M.V. de Andrade
Author M.A. Gonçalves
Author J.M. Almeida
Abstract This paper addresses the task of inferring trip profiles (TPs), which consists of determining the profile of travelers engaged in a particular trip given a set of possible categories. TPs may include working trips, leisure journeys with friends, or family vacations. Travelers with different TPs typically have varied plans regarding destinations and timing. TP inference may provide significant insights for numerous tourism-related services, such as geo-recommender systems and tour planning. We focus on TP inference using TripAdvisor, a prominent tourism-centric social media platform, as our data source. Our goal is to evaluate how effectively we can automatically discern the TP from a user review on this platform. A user review encompasses both textual feedback and domain-specific data (such as a user's previous visits to the location), which are crucial for accurately characterizing the trip. To achieve this, we assess various feature sets (including text and domain-specific) and implement advanced machine learning models, such as neural Transformers and open-source Large Language Models (Llama 2, Bloom). We examine two variants of the TP inference task—binary and multi-class. Surprisingly, our findings reveal that combining domain-specific features with TF-IDF-based representation in an LGBM model performs as well as more complex Transformer and LLM models, while being much more efficient and interpretable. © 2024 Elsevier B.V.

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Tags:

Language model, Large language model, Tourism, Machine-learning, Machine learning, Adversarial machine learning, Contrastive Learning, Large language models, Recommender systems, Text classification, Domain specific, Leisure, Online reviews, User reviews, Domain knowledge, Online domains, Trip purpose

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Why Text-Based Emotion Recognition on Social Media Remains Limited: A Data- and Context-Centric Analysis

Item Type Conference Paper

Author K.S.H. Arvanaghi

Author A. Alhouei

Author N.M. Chahardeh

Abstract This paper presents a unified data- and context-centric analysis of text-based emotion recognition on social media, focusing on research published between 2020 and 2025. Although recent advances in neural language models have substantially improved benchmark performance, real-world robustness and cross-domain generalization remain limited. We examine the field along three structural dimensions: (1) dataset construction and annotation practices, including subjectivity and annotator disagreement; (2) platform bias and cross-domain

generalization across heterogeneous social media environments; and (3) context-related failure modes and evaluation limitations that obscure systematic errors. Across these axes, we identify a shared bottleneck rooted in dataset fragmentation, context omission, and reliance on aggregate metrics. Evidence from cross-dataset evaluations reveals significant performance degradation under domain shift, indicating that models frequently capture platform-specific cues rather than transferable emotional representations. Based on this analysis, we outline concrete research directions, including disagreement-aware learning frameworks, minimal context-rich annotation protocols, and standardized multi-faceted evaluation procedures incorporating calibration and cross-dataset testing. Our findings suggest that advancing emotion recognition requires principled dataset design and evaluation reform to enable robust emotion understanding in heterogeneous web environments. ©2026 IEEE.

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Proceedings Title Int. Conf. Web Res., ICWR

Conference Name International Conference on Web Research, ICWR

Issue 2026

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Notes:

EID: 2-s2.0-105041083923

Export Date: 20 September 2026; Cited By: 0; Correspondence Address: N.M. Chahardeh; Dept. of Computer Engineering SR.C., Islamic Azad University, Tehran, Iran; email: Niloofer.Mirzaeichahardeh@iau.ir; Conference name: 12th International Conference on Web Research, ICWR 2026; Conference date: 22 April 2026 through 23 April 2026; Conference code: 224090

Why They Link: An Intent Taxonomy for Including Hyperlinks in Social Posts

Item Type Conference Paper

Author F. Lan

Author A. Aljebreen

Author E. Dragut

Abstract URLs serve as bridges between social media platforms and the broader web, linking user-generated content to external information resources. On Twitter (X), approximately one in five tweets contains at least one URL, underscoring their central role in information dissemination. While prior studies have examined the motivations of authors who share URLs, such author-centered intentions are difficult to observe in practice. To enable broader downstream use, this work investigates reader-centered interpretations - how users perceive the intentions behind hyperlinks included in posts. We develop an intent taxonomy for including hyperlinks in social posts through a hybrid approach that begins with a bottom-up, data-driven process using large-scale crowdsourced annotations, and is then refined using large language model (LLM) assistance to generate descriptive category names and precise definitions. The final taxonomy¹ comprises 6 top-level categories and 26 fine-grained intention classes, capturing diverse communicative purposes. Applying this taxonomy, we annotate and analyze 1,000 user posts, revealing that advertising, arguing, and sharing are the most prevalent intentions. We further compare our taxonomy with existing taxonomies and demonstrate its utility in a microblog retrieval task by incorporating intent as an additional feature. Overall, our taxonomy provides a foundation for intent-aware information retrieval and NLP applications, enabling more accurate retrieval, recommendation, and interpretation of social media content. © 2026 Copyright held by the owner/author(s).

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Publisher Association for Computing Machinery, Inc

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Conference Name WebSci 2026 - Proceedings of the 18th ACM Web Science Conference

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Social networking (online), Social media, Twitter, Information retrieval, Taxonomies, Social Media, Social media platforms, User-generated, Tweets, Information dissemination, Down-stream, External informations, Hybrid approach, Hyperlinked post, Hyperlinked Posts, Hyperlinks, Hypertext systems, Information resource, Information Retrieval, Intent taxonomy, Intent Taxonomy

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Export Date: 20 September 2026; Cited By: 0; Funding details: National Science Foundation, NSF, (III-2107213, 2026513); Correspondence Address: E. Dragut; Temple University, Philadelphia, United States; email: edragut@temple.edu; Conference name: 18th ACM Web Science Conference, WebSci 2026; Conference date: 26 May 2026 through 29 May 2026; Conference code: 223714

WordTransABSA: Enhancing Aspect-based Sentiment Analysis with masked language modeling for affective token prediction

Item Type Journal Article**Author** W. Jin**Author** B. Zhao**Author** Y. Zhang**Author** J. Huang**Author** H. Yu

Abstract In recent years, Aspect-based Sentiment Analysis (ABSA) has been a crucial yet challenging task in recognizing textual emotions from text. ABSA has numerous application across various fields, such as social media, commodity review, and movie comment, making it an attractive area of research. Many researchers are working to develop more powerful sentiment analysis models. Currently, most existing ABSA models use the generic pre-trained language models (PLMs) based fine-tuning paradigm, which only utilizes the encoder parameters while discarding the decoder parameters of PLMs. However, this approach fails to leverage the prior knowledge revealed in PLMs effectively. To address these issues, we investigate the potential of the initial pre-training scheme of PLMs to conduct ABSA and thus propose a novel approach in this paper, namely Target Word Transferred ABSA (WordTransABSA). In WordTransABSA, we propose “Word Transferred LM”, a novel sequence-level optimization strategy that transferred target words in sentence into pivot tokens to stimulate better PLM semantic understanding capability. Given a sentence with aspect terms as input, WordTransABSA generates contextually appropriate semantics and predicts the affective tokens on the corresponding positions of the aspect terms. The final sentiment polarity of each aspect term is determined through several sentiment identification strategies that we selected. WordTransABSA takes full advantage of the versatile linguistic knowledge of Pre-trained Language Model, resulting in competitive accuracy compared with recent baselines. The WordTransABSA demonstrates its superiority and effectiveness through extensive experiments in both data-sufficient (full-data supervised learning) and data-insufficient (few-shot learning) scenarios. We have made our code publicly available on GitHub: <https://github.com/albert-jin/WordTransABSA>. © 2023 Elsevier Ltd

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Notes:

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Working with Few Samples: Methods that Help Analyze Social Attitude and Personal Emotion

Item Type Conference Paper
Author M. Liang
Author J. Zhou
Author Y. Sun
Author L. He
Editor Shen W.
Editor Barthes J.-P.
Editor Luo J.
Editor Shi Y.
Editor Zhang J.
Abstract In the past decade, sentiment analysis on social media has attracted great attention and has been used in many studies in CSCW and related fields. Recently, with the rapid development of machine learning, using machine learning methods to analyze sentiment has become an efficient experiment framework. Now, the existing sentiment analysis methods in machine learning are mainly based on

supervised learning, and they need enough training data to ensure high accuracy. They encounter a common problem that they cannot recognize and calculate the emotion of samples with unseen labels, which don't belong to the training set. However, most data collected from social media is unstructured and unlabeled, which challenges the effectiveness and usability of existing methods. In our study, we first refer to existing sentiment analysis methods and zero-shot learning for addressing the problem. After that, we propose two zero-shot sentiment analysis methods and design an experiment to compare our methods and strong baselines. In conclusion, our methods obtain better results and we try to apply these methods to future social media research. © 2021 IEEE.

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Language English

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Publisher Institute of Electrical and Electronics Engineers Inc.

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Proceedings Title Proc. IEEE Int. Conf. Comput. Support. Coop. Work Des., CSCWD

Conference Name Proceedings of the 2021 IEEE 24th International Conference on Computer Supported Cooperative Work in Design, CSCWD 2021

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Sentiment analysis, Machine learning, Social networking (online), Social media, Interactive computer systems, sentiment analysis, Training data, social computing, High-accuracy, zero-shot learning, Training sets, Machine learning methods

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Export Date: 20 September 2026; Cited By: 5; Conference name: 24th IEEE International Conference on Computer Supported Cooperative Work in Design, CSCWD 2021; Conference date: 5 May 2021 through 7 May 2021; Conference code: 169216

Workshop on AI and Data Science for Digital Finance, AI4DF 2026, Workshop on Decision Making and Optimization in Financial Technologies, DMO-FinTech 2026, Workshop on Research and Applications of Foundation Models for Data Mining and Affective Computing, RAFDA 2026, Workshop on Autonomous Machine Intelligence, AMI-PAKDD 2026, Workshop on Resource-Efficient Foundation Models for Large-Scale Data Mining, EFMxDM 2026, Workshop on Generative AI and Computational Journalism, JEN-AI 2026, held in conjunction with the 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining, PAKDD 2026

Item Type Journal Article

Editor Long C.

Editor Shi J.

Editor Huang X.

Editor Yiu M.L.

Editor Liang Y.

Editor Choi B.

Editor Song Y.

Abstract The proceedings contain 39 papers. The special focus in this conference is on Decision Making and Optimization in Financial Technologies. The topics include: AI Adoption Phases in Business Intelligence: From Outsourcing to Human-Centered Systems; WeiboDRP: Sentiment Analysis on Chinese Social Media with Multi-Level Comment Structures; fidelity Assessment of Large Language Models Across Cross-Domain Benchmarks; backdooring Societal Bias into Instruction-Tuned Large Language Models Via Virtual Prompt Injection; Reflect-SQL: A Self-reflection Based Framework for Text-to-SQL; HS-GPBoost: A Spatial-Feature Framework for Explainable County-Level Health Prediction; textual Anomaly Detection: A Systematic Evaluation of Representations and Algorithms; Entity Matching with LLMs at Scale: Accuracy, Cost, and Reasoning Effects; bid Optimization for Search Advertising in Non-stationary Environment; Rethinking NL2VIS: A Comparative Survey of LLM-Based Natural-Language-to-Visualisation Frameworks; CMAS4G2: A Locally Deployed Cooperative Multi-agent System for Gherkin Acceptance Test Generation; SCTS: Self-Critique Tree Search for Out-of-Distribution Vulnerability Detection; Invariant Offline Constrained Reinforcement Learning for Circular Aquaculture Under Heterogeneous Environments: Case Study on AIoT Enabled Shrimp Farming in the Mekong Delta; World Models as Constraints: Heterogeneous Rail Robot Coordination via CA-JEPA and LLM-Based Planning; an Empirical Analysis of Distributional Effects in Learning-Based Routing Systems; Learn to Bid: Near-Optimal Knapsack Bidding Under Budget and CPA Constraints with Sparse Delayed Feedback; a Benchmark for Structured Multihop Reasoning in Cross-Institution University Admission Advisory Question Answering; learning What Can Be Picked: Active Reachability Estimation for Efficient Robotic Fruit Harvesting; latticeGuard: A Multi-branch Adaptive Defense Framework for Graph Neural Networks; brand Evaluation System from Big Data on the Amazon E-Commerce Platform; FinRL-X: An AI-Native Modular Infrastructure for Quantitative Trading; habitatAgent: An End-to-End Multi-agent System for Housing Consultation; portfolio Selection with Adaptive Conformal Prediction; Toward Reliable Evaluation of LLM-Based Financial Multi-agent Systems: Taxonomy, Coordination Primacy, and Cost Awareness.

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ISSN 03029743 (ISSN); 978-981922013-7 (ISBN)
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X-MuTeST: A Multilingual Benchmark for Explainable Hate Speech Detection and A Novel LLM-consulted Explanation Framework

Item Type Conference Paper

Author M.Z.U. Rehman

Author S.K.R. Kasu

Author S.R. Koppula

Author S.R.R. Chirra

Author S.S. Singh

Author N. Kumar

Editor Koenig S.

Editor Jenkins C.

Editor Taylor M.E.

Abstract Hate speech detection on social media faces challenges in both accuracy and explainability, especially for underexplored Indic languages. We propose a novel explainability-guided training framework, X-MuTeST (eXplainable Multilingual haTe Speech deTection), for hate speech detection that combines high-level semantic reasoning from large language models (LLMs) with traditional attention-enhancing techniques. We extend this research to Hindi and Telugu alongside English by providing benchmark human-annotated rationales for each word to justify the assigned class label. The X-MuTeST explainability method

Language model, Speech communication, Artificial intelligence, Social media, Semantics, Text processing, Speech analysis, Speech detection, Personnel training, High level languages, Bigrams, Class labels, High level semantics, Prediction probabilities, Semantic reasoning, Training framework, Tri grams

Notes:

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Export Date: 20 September 2026; Cited By: 1; Funding details: HOLD; Correspondence Address: N. Kumar; Indian Institute of Technology Indore, India; email: nagendra@iiti.ac.in; Conference name: 40th AAAI Conference on Artificial Intelligence, AAAI 2026; Conference date: 20 January 2026 through 27 January 2026; Conference code: 349419

XLM-T: Multilingual Language Models in Twitter for Sentiment Analysis and Beyond

Item Type Conference Paper

Author F. Barbieri

Author L.E. Anke

Author J. Camacho-Collados

Editor Calzolari N.

Editor Bechet F.

Editor Blache P.

Editor Choukri K.

Editor Cieri C.

Editor Declerck T.

Editor Goggi S.

Editor Isahara H.

Editor Maegaard B.

Editor Mariani J.

Editor Mazo H.

Editor Odijk J.

Editor Piperidis S.

Abstract Language models are ubiquitous in current NLP, and their multilingual capacity has recently attracted considerable attention. However, current analyses have almost exclusively focused on (multilingual variants of) standard benchmarks, and have relied on clean pre-training and task-specific corpora as multilingual signals. In this paper, we introduce XLM-T, a model to train and evaluate multilingual language models in Twitter. In this paper we provide: (1) a new strong multilingual baseline consisting of an XLM-R (Conneau et al., 2020) model pre-trained on millions of tweets in over thirty languages, alongside starter code to subsequently fine-tune on a target task; and (2) a set of unified sentiment analysis Twitter datasets in eight different languages and a XLM-T model fine-tuned on them. © European Language Resources Association (ELRA), licensed

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Publisher European Language Resources Association (ELRA)

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Computational linguistics, Language model, 'current, Sentiment analysis, Social networking (online), Twitter, sentiment analysis, Pre-training, language models, Multilinguality, Current analysis, multilinguality, Starters, T-a models, T-model

Notes:

EID: 2-s2.0-85144409197

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XLNet with Data Augmentation to Profile Cryptocurrency Influencers

Item Type Conference Paper

Author M. Siino

Author I. Tinnirello

Editor Aliannejadi M.

Editor Faggioli G.

Editor Ferro N.

Editor Vlachos M.

Abstract In this work we propose an application of XLNet to address the task hosted at PAN@CLEF2023 related to Profiling Cryptocurrency Influencers with Few-shot Learning. For our proposed approach we made use of XLNet fine-tuned on an augmented version of the training set provided for the competition. Given the few-shot learning perspective of the task we found useful to employ a data augmentation strategy similar to one proposed in a previous edition of a PAN task. The augmentation is performed augmenting each sample in the training dataset with its corresponding backtranslated version to a target language. The target languages we used for our two submissions were German and Italian. After

the fine-tuning of the XLNet we predict the labels for the unlabeled test set. After fine-tuning the XLNet model we evaluated it on the original non-augmented training set. We evaluated all the F1 with regards to each label, and then we reported the Macro F1 across all the labels provided. Our results prove that on the original training set our approach can obtain a maximum Macro F1 of 0.6937 and a maximum accuracy of 0.6893. © 2023 Copyright for this paper by its authors.

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Language English

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Volume 3497

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Zero-Shot Conversational Stance Detection: Dataset and Approaches

Item Type Conference Paper

Author Y. Ding

Author K. He

Author B. Li

Author L. Zheng

Author H. He

Author F. Li

Author C. Teng

Author D. Ji
Editor Che W.
Editor Nabende J.
Editor Shutova E.
Editor Pilehvar M.T.

Abstract Stance detection, which aims to identify public opinion towards specific targets using social media data, is an important yet challenging task. With the increasing number of online debates among social media users, conversational stance detection has become a crucial research area. However, existing conversational stance detection datasets are restricted to a limited set of specific targets, which constrains the effectiveness of stance detection models when encountering a large number of unseen targets in real-world applications. To bridge this gap, we manually curate a large-scale, high-quality zero-shot conversational stance detection dataset, named ZS-CSD, comprising 280 targets across two distinct target types. Leveraging the ZS-CSD dataset, we propose SITPCL, a speaker interaction and target-aware prototypical contrastive learning model, and establish the benchmark performance in the zero-shot setting. Experimental results demonstrate that our proposed SITPCL model achieves state-of-the-art performance in zero-shot conversational stance detection. Notably, the SITPCL model attains only an F1-macro score of 43.81%, highlighting the persistent challenges in zero-shot conversational stance detection. Our data and code are available at <https://github.com/whuyzding/ZS-CSD>. © 2025 Association for Computational Linguistics.

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Zero-shot cross-lingual stance detection via adversarial language adaptation

Item Type Journal Article

Author A. Bharathi

Author A. Zubiaga

Abstract Stance detection has been widely studied as the task of determining if a social media post is positive, negative or neutral towards a specific issue, such as support towards vaccines. Research in stance detection has often been limited to a single language and, where more than one language has been studied, research has focused on few-shot settings, overlooking the challenges of developing a zero-shot cross-lingual stance detection model. This article makes the first such effort by introducing a novel approach to zero-shot cross-lingual stance detection, multilingual translationaugmented bidirectional encoder representations from Transformers (BERT) (MTAB), aiming to enhance the performance of a cross-lingual classifier in the absence of explicit training data for target languages. Our technique employs translation augmentation to improve zero-shot performance and pairs it with adversarial learning to further boost model efficacy. Through experiments on datasets labeled for stance towards vaccines in four languages-English, German, French, Italian, we demonstrate the effectiveness of our proposed approach, showcasing improved results in comparison to a strong baseline model as well as ablated versions of our model. Our experiments demonstrate the effectiveness of model components, not least the translation-augmented data as well as the adversarial learning component, to the improved performance of the model. We have made our source code accessible on GitHub: <https://github.com/amcs18pd05/MTAB-cross-lingual-vaccine-stance-detection-2>. © Copyright 2025 A. and Zubiaga. Distributed under Creative Commons CC-BY 4.0. <https://creativecommons.org/licenses/by/4.0/>

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Notes:

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Zero-shot Explainable Mental Health Analysis on Social Media by Incorporating Mental Scales

Item Type Conference Paper

Author W. Li

Author Y. Zhu

Author X. Lin

Author M. Li

Author Z. Jiang

Author Z. Zeng

Abstract Traditional discriminative approaches in mental health analysis are known for their strong capacity but lack interpretability and demand large-scale annotated data. The generative approaches, such as those based on large language models (LLMs), have the potential to get rid of heavy annotations and provide explanations but their capabilities still fall short compared to discriminative approaches, and their explanations may be unreliable due to the fact that the generation of explanation is a black-box process. Inspired by the psychological assessment practice of using scales to evaluate mental states, our method which is called Mental Analysis by Incorporating Mental Scales (MAIMS), incorporates two procedures via LLMs. First, the patient completes mental scales, and second, the psychologist interprets the collected information from the mental scales and makes informed decisions. Experimental results show that MAIMS outperforms other zero-shot methods. MAIMS can generate more rigorous explanation based on the outputs of mental scales.¹ © 2024 Copyright held by the owner/author(s). Publication rights licensed to ACM.

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Zero-shot prompt-based classification: topic labeling in times of foundation models in German Tweets

Item Type Conference Paper

Author S. Munker

Author K. Kugler

Author A. Rettinger

Editor Zhao J.

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Abstract Filtering and annotating textual data are routine tasks in many areas, including social media and news analytics. Automating these tasks enables scaling analyses with respect to speed and breadth while reducing manual effort. Recent advancements in Natural Language Processing, particularly the success of large foundation models, provide new tools for automating annotation processes through text-to-text interfaces with written guidelines, eliminating the need for training samples. This work assesses these advancements in a real-world setting by empirically testing them on German Twitter data about social and political European crises. We compare prompt-based results with human annotations and established classification approaches, including Naive Bayes and BERT-based

fine-tuning with domain adaptation. Despite hardware limitations during model selection, our prompt-based approach achieves comparable performance to fine-tuned BERT without requiring annotated training data. These findings highlight the ongoing paradigm shift in NLP toward task unification and the elimination of pre-labeled training data requirements. © 2025 Association for Computational Linguistics.

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Zero-shot stance detection: Paradigms and challenges

Item Type Journal Article

Author E. Allaway

Author K. McKeown

Abstract A major challenge in stance detection is the large (potentially infinite) and diverse set of stance topics. Collecting data for such a set is unrealistic due to both the expense of annotation and the continuous creation of new real-world topics (e.g., a new politician runs for office). Furthermore, stancetaking occurs in a wide range of languages and genres (e.g., Twitter, news articles). While zero-shot stance detection in English, where evaluation is on topics not seen during training, has received increasing attention, we argue that this attention should be expanded to multilingual and multi-genre settings. We discuss two paradigms for English zero-shot stance detection evaluation, as well as recent work in this area. We then discuss recent work on multilingual and multi-genre stance detection, which has focused primarily on non-zero-shot settings. We argue that this work should be expanded to multilingual and multi-genre zero-shot stance detection and propose best practices to systematize and stimulate future work in this direction. While domain adaptation techniques are well-suited for work in these settings, we argue that increased care should be taken to improve model explainability and to conduct robust evaluations, considering not only empirical generalization ability but also the understanding of complex language and inferences. Copyright © 2023 Allaway and McKeown.

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Zero-Shot Text Classification via Knowledge Graph Embedding for Social Media Data

Item Type Journal Article

Author Q. Chen

Author W. Wang
Author K. Huang
Author F. Coenen

Abstract The idea of 'citizen sensing' and 'human as sensors' is crucial for social Internet of Things, an integral part of cyber-physical-social systems (CPSSs). Social media data, which can be easily collected from the social world, has become a valuable resource for research in many different disciplines, e.g., crisis/disaster assessment, social event detection, or the recent COVID-19 analysis. Useful information, or knowledge derived from social data, could better serve the public if it could be processed and analyzed in more efficient and reliable ways. Advances in deep neural networks have significantly improved the performance of many social media analysis tasks. However, deep learning models typically require a large amount of labeled data for model training, while most CPSS data is not labeled, making it impractical to build effective learning models using traditional approaches. In addition, the current state-of-the-art, pretrained natural language processing (NLP) models do not make use of existing knowledge graphs, thus often leading to unsatisfactory performance in real-world applications. To address the issues, we propose a new zero-shot learning method which makes effective use of existing knowledge graphs for the classification of very large amounts of social text data. Experiments were performed on a large, real-world tweet data set related to COVID-19, the evaluation results show that the proposed method significantly outperforms six baseline models implemented with state-of-the-art deep learning models for NLP. © 2014 IEEE.

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Zero-Shot Visual Emotion Recognition by Exploiting BERT

Item Type Conference Paper

Author H. Kang

Author D. Hazarika

Author D. Kim

Author J. Kim

Editor Arai K.

Abstract The explosive growth of multimedia has attracted many people to express their opinions through social media like Flickr and Facebook. As a result, social media has become the rich source of data for analyzing human emotions. Many earlier studies have been conducted to automatically assess human emotions due to their wide range of applications such as education, advertisement, and entertainment. Recently, many researchers have been focusing on visual contents to find out clues for evoking emotions. In literature, this type of study is called visual sentiment analysis. Although a great performance has been achieved by many earlier studies on visual emotion analysis, most of them are limited to classification tasks with pre-determined emotion categories. In this paper, we aim to recognize emotion classes that do not exist in the training set. The proposed model is trained by mapping the visual features to the emotional semantic representation embedded by the BERT language model. By evaluating the model on a cross-domain affective dataset, we achieved 66% accuracy for predicting the unseen emotions not included in the training set. © 2023, The Author(s), under exclusive license to Springer Nature Switzerland AG.

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